NSX

Use **NSX** as a network administrator for networking and automation for virtual machines, containers, or bare metal applications.

> **Important:**
> Starting with NSX 9, NSX is only available through VCF BOM and a standalone upgrade or installation of NSX is not supported. You must use the VCF BOM and follow the recommended process to upgrade NSX 4.x to VCF 9 or later. For more information about upgrading to VCF 9.1.x or later, see Paths to Building 9.1.x Environments.

In This Chapter

- NSX Manager
- Tier-0 Gateways
- Tier-1 Gateway
- Segments
- NSX DHCP
- Host Switches
- Transport Zones and Profiles
- Host Profile integration with NSX
- Host Transport Nodes
- Installing NSX Edge
- Setting up Network Connectivity
- Virtual Private Cloud in NSX
- Virtual Private Network (VPN)
- Network Address Translation (NAT)
- NSX Native Load Balancer
- Ethernet VPN (EVPN)
- IP Address Management (IPAM)
- Networking Settings
- vDefend Firewall with Advanced Threat Prevention
- Inventory
- Multisite and NSX Federation
- NSX Multi-tenancy
- System Monitoring
- Network Monitoring
- Authentication and Authorization
- Certificates in NSX
- Integration of Kubernetes Clusters with Antrea CNI
- Backing Up and Restoring NSX Manager or Global Manager
- Operations and Management
- Scale Out NSX Manager
- Troubleshooting Issues
- NSX Feature and Edition Information
- Related Resources

Getting to Know VCF

To apply this documentation, you must be acquainted with the VCF Product Overview, VCF Design, and VCF Release Notes.

This documentation is written for experienced Windows or Linux system administrators who are familiar with virtual machine technology, networking, and security operations.

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See Administration SDKs, APIs, and CLI.

NSX Manager

NSX Manager provides a web-based user interface where you can manage your NSX environment. It also hosts the API server that processes API calls.

If you modify NSX Edge nodes, NSX Edge clusters, or transport zones, it can take up to 5 minutes for those changes to become visible in the NSX Manager interface. To synchronize changes immediately, use `POST /policy/api/v1/infra/sites/default/enforcement-points/default?action=reload`.

> **Note:**
>
> Beginning with VCF 9.0, the NSX Manager interface provides a single mode, Policy mode, for configuring resources. The Manager mode and Manager API provided by NSX 4.x and earlier are no longer supported.

Policy API

The NSX Manager provides a Policy API to automate management activities. The Policy API is part of the NSX REST APIs and contains URIs that begin with `/policy/api`.

Policy API methods support partial patching of objects. This feature needs to be explicitly enabled. If enabled, you can provide the partial payload for updating the existing object using PATCH methods.

To enable the feature, use the Partial Patch Config method.

```
PATCH /policy/api/v1/system-config/nsx-partial-patch-config

 { "enable_partial_patch": "true" }
```

The default is 'false'.

For more information about using the Policy API, see the NSX API Guide.

Security of NSX Manager

NSX Manager is a restricted system and has features designed to ensure the integrity of the system and to keep the system secure.

Details of the NSX Manager security features:

- NSX Manager supports session timeout and automatic user logout. NSX Manager does not support session lock. Initiating a session lock can be a function of the workstation operating system being used to access NSX Manager. Upon session termination or user logout, users are redirected to the login page.
- NSX Manager has one active local built-in user account called **admin**, which has access rights to all resources, but does not have rights to the operating system to install software. NSX upgrade files are the only files allowed for installation. Rights of this account cannot be edited and neither the account can be deactivate or deleted.
- There are three default inactive local user accounts: audit and two guest user accounts. In the Enterprise environment, the **guestuser1** and the **guestuser2** user accounts are available. You can delete, activate, or deactivate them. The admin in the Enterprise environment, as well as any user with the Enterprise Admin role, can add up to 10 more guest user accounts in addition to admin, audit, and two guest users.
- NSX Manager enforces approved authorizations for controlling the flow of management information within the network device based on information flow control policies.
- NSX Manager initiates session auditing upon startup.
- NSX Manager uses its internal system clock to generate time stamps for audit records.
- All passwords in the system (databases, configuration files, log files, and so on) get encrypted using a strong one-way hashing algorithm with a salt. During authentication, when the user enters the password it gets obfuscated. You can

configure the password complexity using the UI, API and CLI. You can reset the node the authentication policy and the password complexity configuration back to their default system settings, if desired.

- FIPS compliance
  - NSX Manager uses FIPS 140 approved algorithms for authentication to a cryptographic module.
  - NSX Manager generates unique session identifiers using a FIPS 140 approved random number generator.
  - NSX Manager uses a FIPS 140 approved cryptographic algorithm to protect the confidentiality of remote maintenance and diagnostic sessions.
  - NSX Manager authenticates SNMP messages using a FIPS-validated Keyed-Hash Message Authentication Code (HMAC).
- An audit log gets generated for events such as logon, logoff, and access to resources. Each audit log contains the timestamp, source, result, and a description of the event. For more information, see Log Messages and Error Codes.
- NSX Manager recognizes only system-generated session identifiers and invalidates session identifiers upon administrator logout or other session termination.
- You can use NSX Manager to manage all NSX certificates except for transport node certificates. You can also use the Certificate Analyzer, Results and Recovery (CARR) script to resolve certificate issues, as described in Broadcom knowledge base article 369034.
- Authentication mechanisms implemented on NSX follow security best practices and are resistant to replay attacks. The secure practices are deployed systematically. For example, sessions IDs and tokens on NSX Manager for each session are unique and expire after the user logs out or after a period of inactivity. Also, every session has a time record and the session communications are encrypted to prevent session hijacking.

You can view and change the session timeout value with the following CLI commands:

- The command `get service http` displays a list of values including session timeout.
- To change the session timeout value, run the following commands:

```
set service http session-timeout <timeout-value-in-seconds>
restart service ui-service
```

Licensing in NSX Manager

Starting with VMware Cloud Foundation 9, product components are licensed automatically after you license the vCenter instance connected to the component. You no longer license NSX manually.

For more information, see the following documentation pages.

Documentation Reference	Description
"Licensing Overview" in the *Licensing* guide.	Overview of core licensing procedures and concepts for VMware Cloud Foundation.
"vDefend Firewall Prerequisites" in the VMware vDefend Firewall documentation.	Description of core licensing procedures for VMware vDefend.
"Licensing Overview" in the License Hub for vDefend and Avi documentation.	Overview of the License Hub and core licensing concepts for VMware vDefend and VMware Avi Load Balancers.

View Monitoring Dashboards

The NSX Manager interface provides numerous monitoring dashboards showing details regarding system health, networking and security, and compliance reporting. This information is displayed or accessible throughout the NSX Manager interface, but can be accessed together in the **Home** > **System Health** and **Home** > **Monitoring Dashboards** pages.

You can access the monitoring dashboards from the Home page of the NSX Manager interface. From each dashboard, you can click through and access the source pages from which the dashboard data is drawn.

1. In the NSX Manager interface, click **Home** if you are not already on the Home page.
2. Click **System Health**, or click **Monitoring Dashboards** and select the desired category of dashboards from the drop-down menu. The dashboard graphics are color-coded, with the color code key displayed directly above the dashboards.

3. To access a deeper level of detail, click the title of the dashboard, or one of the elements of the dashboard, if activated.

The following sections describe the default dashboards and their sources.

System Health Dashboard

The system health dashboard collects extended health metrics for the management cluster, host transport nodes, and edge transport nodes in a single dashboard. You can hover over or click on certain elements in the dashboard to view more detailed information.

To access the system health dashboard, select **Home** > **System Health**.

You can also retrieve extended system health metrics through the NSX API.

> **Note:**
>
> The extended system health dashboard is not available for the Global Manager in an NSX Federation. Instead, you can access separate monitoring dashboards containing basic health metrics for a Global Manager.

Dashboard Section	Sources	Description
Management Cluster	**System** > **Appliances** **Home** > **Alarms**	Shows the status of the NSX Manager cluster and utilization of its CPU, memory, and disk storage resources. You can hover over or click points on the CPU and memory graphs to view average metrics at one-minute intervals for the past 15 minutes. This section also summarizes the number of open alarms and the status of services in the cluster. This section displays an alert if the management cluster is unstable or a management node is down. Click the section title to view the **Appliance Details** for the cluster. Here you can monitor more granular details, such as the status of individual services and utilization metrics for individual disk partitions. Click the **Up** link for a healthy service to view further details about the service uptime and average CPU and memory usage over the past 5 minutes.
Hosts	**System** > **Fabric** > **Hosts** > **Clusters**	Shows the health status of host transport nodes within host clusters. You can filter the view of clusters based on status, and hover over a cluster to view a snapshot of the status details. Click a cluster to access full details on the **Clusters** page.
Edges	**System** > **Fabric** > **Nodes** > **Edge Clusters**	Shows the health status of edge transport nodes within edge clusters. You can filter the view of clusters based on status, and hover over a cluster to view a snapshot of the status details. Click a cluster to access full details on the **Edge Clusters** page.
Compute Manager Reachability	**System** > **Fabric** > **Compute Manager**	Shows the connectivity status of registered compute managers and whether they can be reached by the NSX Manager cluster.
Backups	**System** > **Backup & Restore**	Shows the health status of NSX backups, if configured. It is strongly recommended that you configure scheduled backups that are stored remotely to an SFTP site.

Networking & Security Dashboards

Table 1028: Networking & Security Dashboards

Dashboard	Sources	Description
Security	**Inventory** > **Groups** **Security** > **Distributed Firewall**	Shows the status of groups and security policies. A group is a collection of workloads, segments, segment ports, IP addresses, MAC addresses, and so on where security policies, including East-West firewall rules, may be applied.
Gateways	**Networking** > **Tier-0 Gateways** **Networking** > **Tier-1 Gateways**	Shows the status of Tier-0 and Tier-1 gateways.
Segments	**Networking** > **Segments**	Shows the status of network segments.

Dashboard	Sources	Description
Load Balancers	**Networking** > **Load Balancing**	Shows the status of the load balancer VMs.
Virtual Services	**Networking** > **Network Services** > **Avi Load Balancer/Load Balancing**	Shows the availability and scalability for virtual services applications by monitoring their health and distributing traffic.
VPNs	**Networking** > **VPN**	Shows the status of virtual private networks.

Compliance Report Dashboard

Table 1029: Compliance Report Dashboard

Column	Description
Non-Compliance Code	Displays the specific non-compliance code.
Description	Specific cause of non-compliance status.
Resource Name	The NSX resource (node, switch, and profile) in non-compliance.
Resource Type	Resource type of cause.
Affected Resources	Number of resources affected. Click the number value to view a list.

You can also add widgets to configure custom monitoring dashboards using NSX REST APIs. See the latest version of the *NSX REST API Guide* at for API details. See the Compliance Status Report Codes for more information about each compliance report code.

Tier-0 Gateways

An NSX gateway provides optimized distributed routing as well as centralized routing and services such as NAT, Load Balancer, DHCP server and so on. In a single tier (only Tier-0) routing topology, the Tier-0 gateway is connected to segments southbound providing E-W routing and is also connected to physical infrastructure to provide N-S connectivity. This gateway is referred to as a Tier-0 Gateway.

An Edge node can support only one tier-0 gateway or logical router. When you create a tier-0 gateway or logical router, make sure you do not create more tier-0 gateways or logical routers than the number of Edge nodes in the NSX Edge cluster.

> **Note:**
> When connecting tier-0 uplinks to multi-chassis port-channel topologies such as vPC (virtual PortChannel) or VSS (Virtual Switching System) from Cisco, or MLAG (Multi-Chassis Link Aggregation) from Arista, be sure to consult with the network provider to understand the limitations of the topology when it is being used for transit routing.

Add an NSX Tier-0 Gateway

A tier-0 gateway has router links to tier-1 gateways and external interfaces to networks north of the tier-0 gateway.

- If you plan to configure multicast, refer to Configuring Multicast on an NSX Tier-0 or Tier-1.
- If you plan to configure the gateway DHCP server, refer to Attach an NSX DHCP Profile to a Tier-0 or Tier-1 Gateway.

If you are adding a tier-0 gateway from Global Manager in NSX Federation, see Add a Tier-0 Gateway from Global Manager.

You can configure the HA (high availability) mode of a tier-0 gateway to be active-active or active-standby. The following stateful centralized services are only supported in active-standby mode:

- SNAT
- DNAT
- Stateful gateway firewall. For more details, see the VMware vDefend Firewall documentation
- VPN

Tier-0 and tier-1 gateways support the following addressing configurations for all interfaces (external interfaces, service interfaces and downlinks) in both single tier and multi-tiered topologies:

- IPv4 only
- IPv6 only
- Dual Stack - both IPv4 and IPv6

To use IPv6 or dual stack addressing, enable **IPv4 and IPv6** as the L3 Forwarding Mode in **Networking** > **Networking Settings** > **Global Networking Config**.

You can configure the tier-0 gateway to support EVPN (Ethernet VPN). For more information about configuring EVPN, see Ethernet VPN (EVPN). Note that this tier-0 gateway EVPN solution along with its Inline and Route Server modes is diffrent from the Distributed VXLAN Connectivity solution. The Distributed VXLAN Connectivity solution works without a tier-0 gateway or NSX Edge nodes but requires an EVPN Route Controller instead.

If you configure route redistribution for the tier-0 gateway, you can select from two main groups of sources when you set the Route Re-distribution: tier-0 gateway subnets and advertised tier-1 gateway subnets.

The sources in the tier-0 subnets group are:

Source Type	Description
Connected Interfaces and Segments	Redistribute all subnets configured on Interfaces and routes related to tier-0 segments, tier-0 DNS Forwarder IP, tier-0 IPsec Local IP, tier-0 NAT types. Redistribute subnets configured on segments connected to tier-0. Selecting this option, selects all the following sources: • Servcie Interface Subnet • Loopback Interface Subnet • External Interface Subnet • Connected Segments
Static Routes	Redistribute static routes that you have configured on the tier-0 gateway.
NAT IP	Redistribute NAT IPs owned by tier-0 and discovered from NAT rules that are configured on the tier-0 gateway.
IPsec Local IP	Redistribute local IPsec endpoint IP address for establishing VPN sessions. Redistribute IPsec subnets.
DNS Forwarder IP	Redistribute listener IP for DNS queries from clients and also used as source IP used to forward DNS queries to upstream DNS server. Redistribute DNS forwarder subnets.
EVPN TEP IP	Redistribute EVPN tunnel local endpoint subnets on tier-0 gateway.
Inter VRF Static	Redistribute IPs advertised by tier-0 gateway or tier-0 gateway VRF instances.
Router Link	Redistribute router link port subnets on tier-0 gateways.
Transit Gateway Static	Select this option to enable VPC subnet advertisement by the tier-0 gateway towards the upstream physical network environment. This option ensures that static routes learned by a centralized transit gateway are included in the VPC subnet advertisement. This option ensures that the tier-0 gateway redistributes all VPC routes learned through the centralized transit gateway and advertises them to the upstream physical router.

The sources in the advertised tier-1 group are:

Source Type	Description
Connected Interfaces & Segments	• Redistribute subnets configured on segments and advertised from the connected tier-1 gateway. • Redistribute subnets configured in NSX VPC and advertised from the connected NSX VPC. • NSX VPC advertises all its public subnets to the connected tier-0 gateway.
Static Routes	Redistribute all subnets and static routes advertised by tier-1 gateways or NSX VPCs.
NAT IP	Redistribute NAT IP addresses owned by the tier-1 gateway or NSX VPC and discovered from NAT rules that are configured on the tier-1 gateway or NSX VPC.
LB VIP	Redistribute IP address of the load balancing virtual server.
LB SNAT IP	Redistribute IP address or a range of IP addresses used for source NAT by the load balancer.
DNS Forwarder IP	Redistribute Listener IP for DNS queries from clients and also used as source IP used to forward DNS queries to upstream DNS server.
IPsec Local Endpoint	Redistribute IP address of the IPsec local endpoint.

Proxy ARP is automatically enabled on a tier-0 gateway when a NAT rule or a NSX native load balancer VIP uses an IP address from the subnet of the tier-0 gateway external interface. By enabling proxy ARP, hosts on the overlay segments and hosts on a VLAN segment can exchange network traffic together without implementing any change in the physical networking fabric.

For a detailed example of a packet flow in a proxy ARP topology, refer to the *NSX Reference Design Guide* on the Broadcom Communities portal.

Proxy ARP is supported on a tier-0 gateway in an active-standby configuration, and it responds to ARP queries for the external and service interface IPs. Proxy ARP also responds to ARP queries for service IPs that are in an IP prefix list that is configured with the `Permit` action.

Proxy ARP is also supported on a tier-0 gateway in an active-active configuration. However, all the NSX Edge nodes in the active-active tier-0 configuration must have direct reachability to the network on which proxy ARP is required. In other words, you must configure the external interface and the service interface on all the NSX Edge nodes that are participating in the tier-0 gateway for the proxy ARP to work.

You can find out the total number of routes for a tier-0 gateway with the following APIs. For more information about the APIs, refer to the *NSX API Guide*.

```
GET /policy/api/v1/infra/tier-0s/{tier-0-id}/number-of-routes
GET /policy/api/v1/global-infra/tier-0s/{tier-0-id}/number-of-routes
```

Inter-SR iBGP routing is also supported on active-active tier-0 VRF gateways within a single site without NSX Federation. To enable this feature, you must first turn on the **Multi-VRF Inter SR** toggle under the **Additional Settings** section while configuring a tier-0 gateway. Turning on this toggle runs MP-BGP between NSX Edges that is required for inter-node route sync to handle asymmetric VRF connectivity and inter-site connectivity for stretched VRFs in a NSX Federation setup. Additionally, to enable inter-SR iBGP routing for a tier-0 VRF gateway, you must also turn on the **Inter SR iBGP** toggle under the BGP section while configuring the tier-0 VRF gateway.

When you turn on the **Multi-VRF Inter SR** toggle:

- There will be disruption in inter-SR and inter-site traffic.
- You cannot turn the toggle off.

Perform the following steps to add a Tier-0 gateway.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Tier-0 Gateways.**
3. Click **Add Gateway** > **Tier-0**.
4. Enter a name for the gateway.
5. Select an HA (high availability) mode.

   The default mode is active-active and stateful is disabled. In the active-active mode, traffic is load balanced across all members. In active-standby mode, all traffic is processed by an elected active member. If the active member fails, a new member is elected to be active.
6. If the HA mode is active-standby, select a failover mode.

Option	Description
Preemptive	If the preferred node fails and recovers, it will preempt its peer and become the active node. The peer will change its state to standby.
Non-preemptive	If the preferred node fails and recovers, it will check if its peer is the active node. If so, the preferred node will not preempt its peer and will be the standby node.

7. Select an NSX Edge cluster.
8. Click Set to add **DHCP Config** on the gateway.
9. Click **Additional Settings**.
   a) Turn the **Multi-VRF Inter SR** toggle on if you require asymmetric traffic failures mitigation support on tier-0 VRF gateways to be handled. Turning the toggle on runs MP-BGP between NSX Edges for inter node route sync redundancy, which handles asymmetric traffic failures on tier-0 VRF gateways.
   b) In the **Internal Transit Subnet** field, enter a subnet/CIDR.

      This is the subnet used for communication between SR and SR components within this tier-0 gateway. The default is 169.254.0.0/24. Once you create a tier-0 gateway, Internal Transit Subnet can not be modified.
   c) In the **T0-T1 Transit Subnets** field, enter one or more subnets/CIDRs.

      These subnets/CIDRs are used for communication between this tier-0 gateway and all tier-1 gateways that are linked to it. After you create this tier-0 gateway and link a tier-1 gateway to it, you will see the actual IP address assigned to the link on the tier-0 gateway side and on the tier-1 gateway side. The address is displayed in **Additional Settings** > **Router Links** on the tier-0 gateway page and the tier-1 gateway page. The default is 100.64.0.0/16.

      After the tier-0 gateway is created, you can change the **T0-T1 Transit Subnets** by editing the tier-0 gateway. Note that this will cause a brief disruption in traffic.
   d) In the **Inter VRF Transit Subnet** field, enter a subnet/CIDR.
   e) In the **Forwarding Up Timer** field, enter a time.

      Forwarding up timer defines the time in seconds that the tier-0 gateway must wait before sending the up notification after the first BGP, BFD, or OSPF session comes up. This timer (previously known as forwarding delay) minimizes downtime in case of fail-overs for active-active or active-standby configurations of tier-0 gateways on NSX Edge that use dynamic routing (BGP or OSPF). It should be set to the number of seconds an external router (TOR) takes to advertise all the routes toNSX Edge after the first BGP/OSPF session comes up. The default value of this timer is 5 seconds. It should be increased to a larger value for environments with a larger scale of routes learned onNSX Edge. In general, a tier-0 gateway must have more than one Edge nodes. If redundant Edge nodes are not available, this timer should be set to 0.
   f) In the **Transit Gateway Transit Subnet** field, enter a Transit Subnet CIDR if later you want to configure a Centralized Transit Gateway (CTGW) with an independent HA. Any CTGW which you create that attaches to this Tier-0 is assigned a unique subnet from this block. If you do not set any value in this field, the system assigns the subnet to the CTGW from the default IP block global-tgw-transit-subnet-block.
10. Click **Route Distinguisher for VRG Gateways** to configure a route distinguisher admin address.

    This is only needed for EVPN in Inline mode. Admin address is used to auto-assign a route distinguisher.

11. To assign a unique route distinguisher from the route distinguisher pool to each Edge Node for EVPN VRF, turn on the **Route Distinguisher per Edge** switch. Note that for the in-line mode of EVPN, the route distinguisher is assigned from either a manually configured pool or it is auto-assigned. For the route-server mode of EVPN, the route distinguisher is assigned only from a manually configured pool.
12. Optional: Add one or more tags.
13. Click **Save**.
14. For IPv6, under **Additional Settings**, you can select or create an **ND Profile** and a **DAD Profile**.

    These profiles are used to configure Stateless Address Autoconfiguration (SLAAC) and Duplicate Address Detection (DAD) for IPv6 addresses.
15. Optional: Click **EVPN Settings** to configure EVPN.
    a) Select an EVPN mode.

       The options are:

       - **Inline** - In this mode, EVPN handles both data plane and control plane traffic.
       - **Route Server** - Available only if this gateway's HA mode is active-active. In this mode, EVPN handles control plane traffic only.
       - **No EVPN**

    b) If EVPN mode is **Inline**, select an EVPN/VXLAN VNI pool or create a new pool by clicking the menu icon (3 dots).
    c) If EVPN mode is **Route Server**, select an **EVPN Tenant** or create a new EVPN tenant by clicking the menu icon (3 dots).
    d) In the **EVPN Tunnel Endpoint** field click **Set** to add EVPN local tunnel endpoints.

       For the tunnel endpoint, select an Edge node and specify an IP address.

       Optionally, you can specify the MTU.

       Note:
       Ensure that the external interface has been configured on the NSX Edge node that you select for the EVPN tunnel endpoint.
    e) To facilitate L3 ECMP on the DCGW towards the Edge nodes, turn on the **Enable RD per Edge node** flag in the parent tier-0 gateway.
16. To configure route redistribution, click **Route Redistribution** and **Set**.

    Select one or more of the sources:

    - Tier-0 subnets: **Static Routes**, **NAT IP**, **IPsec Local IP**, **DNS Forwarder IP**, **Router Link**, **Connected Interfaces & Segments**.

      Under **Connected Interfaces & Segments**, you can select one or more of the following: **Service Interface Subnet**, **External Interface Subnet**, **Loopback Interface Subnet**, **Connected Segment**.
    - Tier-0 subnets: **Static Routes**, **NAT IP**, **IPsec Local IP**, **DNS Forwarder IP**, **EVPN TEP IP**, **Connected Interfaces & Segments**.

      Under **Connected Interfaces & Segments**, you can select one or more of the following: **Service Interface Subnet**, **External Interface Subnet**, **Loopback Interface Subnet**, **Connected Segment**.
    - Advertised tier-1 subnets: **DNS Forwarder IP**, **Static Routes**, **LB VIP**, **NAT IP**, **LB SNAT IP**, **IPsec Local Endpoint**, **Connected Interfaces & Segments**.

      Under **Connected Interfaces & Segments**, you can select **Service Interface Subnet** and/or **Connected Segment**.
17. To configure interfaces, click **Interfaces and GRE Tunnels** and then **Set** for **External and Service Interfaces**.
    a) Click **Add Interface**.
    b) Enter a name.
    c) Select a location. All onboarded sites will be available for selection.
    d) Select a type.

       If the HA mode is active-standby, the choices are **External**, **Service**, and **Loopback**. If the HA mode is active-active, the choices are **External** and **Loopback**.

e) Enter an IP address in CIDR format.
f) Select a segment.
g) If the interface type is not **Service**, select an NSX Edge node.
h) Optional: If the interface type is not **Loopback**, enter an MTU value.
i) Optional: If the interface type is **External**, you can enable multicast by setting **PIM** (Protocol Independent Multicast) to **Enabled**.

   You can also configure the following:

   - **IGMP Join Local** - Enter one or more IP addresses. IGMP join is a debugging tool used to generate (*,g) join to Rendezvous Point (RP) and get traffic forwarded to the node where the join is issued. For more information, refer to About IGMP Join.
   - **Hello Interval (seconds)** - Default is 30. The range is 1 - 180. This parameter specifies the time between Hello messages. After the **Hello Interval** is changed, it takes effect only after the currently scheduled PIM timer expires
   - **Hold Time (seconds)** - The range is 1 - 630. Must be greater than **Hello Interval**. The default is 3.5 times **Hello Interval**. If a neighbor does not receive a Hello message from this gateway during this time interval, the neighbor will consider this gateway unreachable.

j) Optional: Add tags and select an ND profile.
k) Optional: If the interface type is **External**, for **URPF Mode**, you can select **Strict** or **None**.

   Asymmetric routing or forwarding is not recommended. Therefore, **URPF Mode** is set to **Strict** by default.

   For symmetric routing, ensure that the configuration between tier-0 gateway and northbound routers is such that the same set of prefixes are advertised from each edge node within a tier-0 gateway on a given site. Also, the same set of prefixes must be learned from TORs on all edge nodes of a tier-0 gateway on a given site.

   If you have BGP between the tier-0 gateway and northbound routers and you are using route-maps or filters, ensure that this configuration is consistent for inbound and outbound directions for all edge nodes.

   For a federation environment with primary and secondary sites, you should advertise longer AS paths for BGP advertisements on secondary site BGP neighbors to resolve asymmetric forwarding.

   If there is a need for asymmetric routing, set **URPF Mode** to **None**.
l) Optional: After you create an interface, you can download the aggregate of ARP proxies for the gateway by clicking the menu icon (three dots) for the interface and selecting **Download ARP Proxies**.

   You can also download the ARP proxy for a specific interface by expanding a gateway and then expanding **Interfaces**. Click an interface and click the menu icon (three dots) and select **Download ARP Proxy**.

   Note:
   You cannot download the ARP proxy for loopback interfaces.

18. Optional: If the HA mode is active-standby, click **Set** next to **HA VIP Configuration** to configure HA VIP.

    With HA VIP configured, the tier-0 gateway is operational even if one external interface is down. The physical router interacts with the HA VIP only. HA VIP is intended to work with static routing and not with BGP.

    a) Click **Add HA VIP Configuration**.
    b) Enter an IP address and subnet mask.

       The HA VIP subnet must be the same as the subnet of the interface that it is bound to.

c) Select a location. All onboarded sites will be available for selection.
d) Ensure that VIP configuration is enabled for HA mode.
e) Select two interfaces from two different Edge nodes.

19. Click **Routing** to add IP prefix lists, community lists, static routes, and route maps.
20. Click **Multicast** to configure multicast routing.
21. Click **BGP** to configure BGP.
22. Click **OSPF** to configure OSPF.
23. Optional: To download the routing table or forwarding table, do the following:
    a) Click the menu icon (three dots) and select a download option.
    b) Enter values for **Transport Node**, **Network**, and **Source** as required.
    c) Click **Download** to save the .CSV file.
24. Optional: To download the ARP table from a linked tier-1 gateway, do the following:
    a) From the **Linked Tier-1 Gateways** column, click the number.
    b) Click the menu icon (3 dots) and select **Download ARP Table**.
    c) Select an edge node.
    d) Click **Download** to save the .CSV file.

The new gateway is added to the list. For any gateway, you can modify its configurations by clicking the menu icon (3 dots) and select **Edit**. For the following configurations, you do not need to click **Edit**. You only need to click the expand icon (right arrow) for the gateway, find the entity and click the number next to it. Note that the number must be non-zero. If it is zero, you must edit the gateway.

- In the **Interfaces** section: **External and Service Interfaces**.
- In the **Routing** section: **IP Prefix Lists**, **Static Routes**, **Static Route BFD Peer**, **Community Lists**, **Route Maps**.
- In the **BGP** section: **BGP Neighbors**.

If NSX Federation is configured, this feature of reconfiguring a gateway by clicking on an entity is applicable to gateways created by the Global Manager (GM) as well. Note that some entities in a GM-created gateway can be modified by the Local Manager, but others cannot. For example, **IP Prefix Lists** of a GM-created gateway cannot be modified by the Local Manager. Also, from the Local Manager, you can edit existing **External and Service Interfaces** of a GM-created gateway but you cannot add an interface.

Create an NSX IP Prefix List

An IP prefix list contains single or multiple IP addresses that are assigned access permissions for route advertisement. The IP addresses in this list are processed sequentially. IP prefix lists are referenced through BGP neighbor filters or route maps with in or out direction.

Verify that you have a tier-0 gateway configured.

See Add an NSX Tier-0 Gateway.

For example, you can add the IP address 192.168.100.0/27 to the IP prefix list and deny the route from being redistributed to the northbound router. You can also append an IP address with less-than-or-equal-to (le) and greater-than-or-equal-to (ge) modifiers to grant or limit route redistribution. For example, 192.168.100.0/27 ge 28 le 30 modifiers match subnet masks greater than or equal to 28-bits and less than or equal to 30-bits in length.

> **Note:**
> The default action for a route is **Deny**. When you create a prefix list to deny or permit specific routes, be sure to create an IP prefix with no specific network address (select **Any** from the dropdown list) and the **Permit** action if you want to permit all other routes.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Tier-0 Gateways**.
3. To edit a tier-0 gateway, click the menu icon (three dots) and select **Edit**.
4. Click **Routing**.
5. Click **Set** next to **IP Prefix List**.
6. Click **Add IP Prefix List**.
7. Enter a name for the IP prefix list.
8. Click **Set** to add IP prefixes.
9. Click **Add Prefix**.
   a) Enter an IP address in CIDR format.
   For example, 192.168.100.0/27.
   b) Optional: Set a range of IP address numbers in the **ge** or **le** modifiers.

   For example, set **ge** to 28 and **le** to 30.
   c) Select **Deny** or **Permit** from the drop-down menu.
   d) Click **Add**.
10. Repeat the previous step to specify more prefixes.
11. Click **Save**.

Create a BGP Community List

You can create BGP community lists so that you can configure route maps based on community lists.

Community lists are user-defined lists of community attribute values. These lists can be used for matching or manipulating the communities attribute in BGP update messages.
Both the BGP Communities attribute (RFC 1997) and the BGP Large Communities attribute (RFC 8092) are supported. The BGP Communities attribute is a 32-bit value split into two 16-bit values. The BGP Large Communities attribute has 3 components, each 4 octets in length.

In route maps we can match on or set the BGP Standard Communities or BGP Large Communities attribute. Using this feature, network operators can implement network policy based on the BGP Communities attribute.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Tier-0 Gateways**.
3. To edit a tier-0 gateway, click the menu icon (three dots) and select **Edit**.
4. Click **Routing**.
5. Click **Set** next to **Community List**.
6. Click **Add Community List**.
7. Enter a name for the community list.
8. Specify a list of communities. For a BGP standard community, use the ASN:value format, for example, 65000:500. For a BGP large community, use the format Global Administrator:Local Data Part 1:Local Data Part 2, for example, 65001:10:41. Note that a specific list cannot have both BGP standard communities and BGP large communities. It must contain only BGP standard communities, or only BGP large communities.

   In addition, you can select one or more of the following predefined BGP standard communities. Note that they cannot be added if the list contains large communities.

   - NO_EXPORT_SUBCONFED - Do not advertise to EBGP peers.
   - NO_ADVERTISE - Do not advertise to any peer.
   - NO_EXPORT - Do not advertise outside BGP confederation

9. Click **Save**.

Configure a Static Route

You can configure a static route on the tier-0 gateway to external networks. After you configure a static route, there is no need to advertise the route from a tier-0 to tier-1 gateway, because tier-1 gateways automatically have a static default route towards their connected tier-0 gateway.

Recursive static routes are supported.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Tier-0 Gateways**.
3. To edit a tier-0 gateway, click the menu icon (three dots) and select **Edit**.
4. Click **Routing**.
5. Click **Set** next to **Static Routes**.
6. Click **Add Static Route**.
7. Enter a name and network address in CIDR format. Static routes based on IPv4 and IPv6 are supported. IPv6 prefixes can only have an IPv6 next hop.
8. Click **Set Next Hops** to add next-hop information.
9. Click **Add Next Hop**.
10. Enter an IP address or select **NULL**.

    If **NULL** is selected, the route is called a device route.
11. Specify the administrative distance.
12. Select a scope from the drop-down list. A scope can be an interface, a gateway, an IPSec session, or a segment.
13. Click **Add**.

Check that the static route is configured properly.

Create a Route Map

A route map consists of a sequence of IP prefix lists, BGP path attributes, and an associated action. The tier-0 gateway scans the sequence for an IP address match. If there is a match, the tier-0 gateway performs the action and scans no further.

- Verify that an IP prefix list or a community list is configured. See Create a BGP Community List.
- For details about using regular expressions to define route-map match criteria for community lists, see Using Regular Expressions to Match Community Lists When Adding Route Maps.

Route maps can be referenced at the BGP neighbor level and for route redistribution.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Tier-0 Gateways**.
3. To edit a tier-0 gateway, click the menu icon (three dots) and select **Edit**.
4. Click **Routing**.
5. Click **Set** next to **Route Maps**.
6. Click **Add Route Map**.
7. Enter a name and click **Set** to add match criteria.
8. Click **Add Match Criteria** to add one or more match criteria.
9. For each match criterion, select **IP Prefix** or **Community List** and click **Set** to specify one or more match expressions.
   a) If you selected **Community List**, specify match expressions that define how to match members of BGP community lists. For each BGP community list, the following match options are available:
      - **MATCH ANY** - perform the set action in the route map if any of the communities in the community list is matched.
      - **MATCH ALL** - perform the set action in the route map if all the communities in the community list are matched regardless of the order.
      - **MATCH EXACT** - perform the set action in the route map if all the communities in the community list are matched in the exact same order.
      - **MATCH COMMUNITY REGEXP** - perform the set action in the route map if all the standard communities associated with the NRLI match the regular expression.
      - **MATCH LARGE COMMUNITY REGEXP** - perform the set action in the route map if all the large communities associated with the NRLI match the regular expression.

      You should use the match criterion MATCH_COMMUNITY_REGEX to match routes against BGP standard communities, and use the match criterion MATCH_LARGE_COMMUNITY_REGEX to match routes against BGP large communities. If you want to permit routes containing either the BGP standard community or BGP large community value, you must create two match criteria. If the match expressions are given in the same match criterion, only the routes containing both the standard and large communities will be permitted.

      For any match criterion, the match expressions are applied in an AND operation, which means that all match expressions must be satisfied for a match to occur. If there are multiple match criteria, they are applied in an OR operation, which means that a match will occur if any one match criterion is satisfied.
10. Set BGP attributes.

BGP Attribute	Description
AS-path Prepend	Prepend a path with one or more AS (autonomous system) numbers to make the path longer and therefore less preferred.
MED	Multi-Exit Discriminator indicates to an external peer a preferred path into an AS (inbound).
Weight	Set a weight to influence path selection. The range is 0 - 65535.
Community	Specify a list of communities. For a BGP standard community, use the ASN:value format, for example, 65000:500. For a BGP large community, use the format Global Administrator:Local Data Part 1:Local Data Part 2, for example, 65001:10:41. In addition, you can select one or more of the following predefined BGP standard communities. Note that they cannot be added if the list contains large communities. • NO_EXPORT_SUBCONFED - Do not advertise to EBGP peers. • NO_ADVERTISE - Do not advertise to any peer. • NO_EXPORT - Do not advertise outside BGP confederation
Local Preference	Use this value to influence the egress path (outbound). The path with the highest value is preferred.

11. In the Action column, select **Permit** or **Deny**.

    You can permit or deny IP addresses matched by the IP prefix lists or community lists from being advertised.

12. Click **Save**.

Using Regular Expressions to Match Community Lists When Adding Route Maps

You can use regular expressions to define the route-map match criteria for community lists. BGP regular expressions are based on POSIX 1003.2 regular expressions.

The following expressions are a subset of the POSIX regular expressions.

Expression	Description
.*	Matches any single character.
*	Matches 0 or more occurrences of pattern.
+	Matches 1 or more occurrences of pattern.
?	Matches 0 or 1 occurrence of pattern.
^	Matches the beginning of the line.
$	Matches the end of the line.
_	This character has special meanings in BGP regular expressions. It matches to a space, comma, AS set delimiters { and } and AS confederation delimiters (and). It also matches to the beginning of the line and the end of the line. Therefore this character can be used for an AS value boundaries match. This character technically evaluates to (^|[,{}()]|$).

Here are some examples for using regular expressions in route maps:

Expression	Description
^101	Matches routes having community attribute that starts with 101.
^[0-9]+	Matches routes having community attribute that starts with a number between 0-9 and has one or more instances of such a number.
.*	Matches routes having any or no community attribute.
.+	Matches routes having any community value.
^$	Matches routes having no/null community value.

Configure BGP

To enable communication between VMs in your virtualized environment and the outside world, you can configure an external or internal BGP (eBGP or iBGP) connection between a tier-0 gateway and a router in your physical infrastructure.

When configuring BGP, you must configure a local Autonomous System (AS) number for the tier-0 gateway. You must also configure the peer AS number. External BGP neighbors must be directly connected and in the same subnet as the tier-0 gateway uplink. If they are not in the same subnet, you must increase the eBGP multi-hop value from 1 (the default) to the appropriate value.

BGPv6 is supported for single hop and multi-hop. Redistribution, prefix list, and route maps are supported with IPv6 prefixes.

RFC-5549 enables BGPv6 sessions to exchange IPv4 routes with an IPv6 next hop. To minimize the number of BGP sessions and IPv4 addresses, you can exchange both IPv4 and IPv6 routes over a BGP session. Support for encoding and processing an IPv4 route with an IPv6 next hop is negotiated as part of the capability exchange in the BGP OPEN message. If both sides of a peering session support the capability, IPv4 routes are advertised with an IPv6 next hop. Multi-protocol BGP (MP-BGP) is used to advertise the Network Layer Reachability Information (NLRI) of a IPv4 address family using the next hop of an IPv6 address family.

A tier-0 gateway in active-active mode supports Inter-SR (service router) iBGP. If NSX Edge node #1 is unable to communicate with a northbound physical router, traffic is re-routed to NSX Edge node #2 in the active-active NSX Edge cluster. If NSX Edge node #2 is able to communicate with the physical router, traffic between NSX Edge node #1 and the

physical router is not affected because the Inter-SR iBGP link is used as an alternative path. A route learned by an NSX Edge node from a northbound router always takes precedence over the same route learned over Inter-SR iBGP. It is not possible to change this preference.

The Inter-SR iBGP feature has the following capabilities and restrictions:

- Redistribution, prefix lists, and route maps are supported.
- The Longest Prefix Match rule is used.
- Local prefixes are disregarded to prevent the tromboning effect of unnecessary hops between BGP peers.
- Route reflectors are not supported.
- BGP confederation is not supported.
- BGP attribute filtering is not supported

NSX provides support for:

- **BGP Autonomous System Number (ASN) per tier-0 VRF Gateway and BGP neighbor**: You can configure a different BGP ASN per tier-0 VRF Gateway and also per BGP neighbor. For more information, see BGP Autonomous System Number per Tier-0 VRF Gateway and BGP neighbor.
- **Inter-VRF Routing**: You can configure inter-VRF routing using easier workflows by importing and exporting routes between VRFs. For more information, see Inter-VRF Routing.
- **Dynamic BGP Neighbor Discovery**: You can configure a range of BGP neighbors using CIDR notation. The NSX Edge node peers dynamically with every BGP neighbor within the IP range specified by the CIDR value. The following restrictions apply:
  - L2VPN EVPN route filtering is not supported.
  - When designating a BGP neighbor IP range, you must enter the CIDR value based on the first IP address in the range. For example, specify 100.102.0.0/27 instead of 100.102.0.2/27.
  - The following CIDR subnet values are not allowed:
    - /32 subnet for IPv4
    - /128 subnet for IPv6
  - A neighbor IP range cannot overlap with another configured neighbor. For example, if you have already configured a static BGP neighbor with IP address 4.1.44.1, you cannot configure a new dynamic neighbor with IP range 4.1.44.0/24. Similarly, if you have already configured a dynamic BGP neighbor with IP range 4.1.41.0/24, a new BGP neighbor range of 4.1.41.0/8 is not allowed.
  - To modify the CIDR value of a configured dynamic BGP neighbor range, you must first deactivate that BGP neighbor configuration. Similarly, to change a static BGP neighbor to a dynamic BGP neighbor range, you must first deactivate the static BGP neighbor configuration.
  - For BGP neighbor-specific graceful restart options, **Graceful Restart and Helper** is not supported.
- **Autonomous-System-Wide Unique BGP Identifier**: NSX supports RFC6286, to relax the uniqueness of router ID between BGP peers across AS. BGP will form neighborship across AS even if the routers have the same router ID.

  Note:
  This new behavior of accepting the same BGP router ID will be the default in the system and cannot be changed.

Determining BGP router ID (RID) for the tier-0 gateway

The BGP router ID (RID) for the tier-0 gateway is determined as follows:

- **Loopback interface is present**: BGP uses the loopback interface IP as the RID. If the loopback interface has an IPv6 address, BGP does not use it as RID. It will take the highest IPv4 interface IP.
- **Loopback interface is not present**: BGP uses the highest non-loopback interface IP address as the RID.
  - If RID is the highest interface IP and loopback interface is present, the RID can be changed to the loopback interface IP address by:
    - Redeploying or replacing the NSX Edge node
    - Modifying or deleting the highest interface IP address
  - If RID is the highest interface IP and loopback interface is present, the RID does not change when:
    - Disabling and enabling BGP
    - Rebooting the edge node, enabling maintenance mode on the edge node, or restarting the routing process

The RID, whether it is a loopback interface IP or the highest interface IP address, will not change under the following condtions:

- Clearing BGP neighbors
- A soft restart or hard restart of BGP adjacency from a remote site

Supported BGP Capabilities

As defined in https://datatracker.ietf.org/doc/html/rfc2842, a BGP speaker determines the capabilities supported by its peer by examining the list of capabilities present in the Capabilities Optional Parameter in the OPEN message that the speaker receives from the peer. NSX supports the following capabilities:

Capability Code	Capability Description	Address Families Supported	Advertised Support from Tier-0 Gateway	Supported by Tier-0 Gateway when Received from Peer	Default Behavior	Configurable
1	Multiprotocol extensions, with: • AFI=1, SAFI=1 : IPv4 Unicast • AFI=2, SAFI=1 : IPv6 Unicast • AFI=25, SAFI=70 : L2VPN EVPN	IPv4 Unicast IPv6 Unicast L2VPN EVPN	Yes	Yes	IPv4 Unicast address family is enabled and advertised by default when an IPv4 neighbor is configured, or manually added in the Route Filter settings under the BGP neighbor configuration. IPv6 Unicast address family is enabled and advertised by default when an IPv6 neighbor is configured, or manually added in the Route Filter settings under the BGP neighbor configuration. L2VPN EVPN address family is enabled and advertised when configured in the Route Filter settings under the BGP neighbor configuration. The IPv4 Unicast address family is mandatory in NSX and automatically enabled when adding L2VPN EVPN address family.	Yes
2	Route refresh	IPv4 Unicast IPv6 Unicast L2VPN EVPN	Yes	Yes	Advertised by default	No

Capability Code	Capability Description	Address Families Supported	Advertised Support from Tier-0 Gateway	Supported by Tier-0 Gateway when Received from Peer	Default Behavior	Configurable
5	Extended next hop encoding	IPv6 Unicast	Yes	Yes	Not advertised by default. To enable this capability you must provide an IPv4 address family along with the IPv6 address family for the IPv6 BGP peer IP address.	Yes
64	Graceful restart	IPv4 Unicast IPv6 Unicast L2VPN EVPN	Yes	Yes	Not advertised by default (Edge node by default is a helper)	Yes
65	Support for 4-octet AS number	IPv4 Unicast IPv6 Unicast L2VPN EVPN	Yes	Yes	Advertised by default	No
69	ADD-Path, with: • AFI=1/2/25, SAFI=1/70 • Send/ Receive=1 (Send Only) • Send/ Receive=2 (Receive Only) • Send/ Receive=3 (Both)	IPv4 Unicast IPv6 Unicast L2VPN EVPN	Yes (Receive only)	Yes (both Send and Receive)	The receive-only capability is supported and advertised by default. When the Edge node receives the same BGP prefix multiple times but with the same metric, if ECMP is enabled, all paths will be installed and active. When the Edge node receives the same BGP prefix multiple times with different metrics (for example, a larger ASPATH length) the best path route will be installed and active. The less preferred paths will be kept in the BGP routing table to improve control plane convergence.	No
73	FQDN	IPv4 Unicast IPv6 Unicast L2VPN EVPN	Yes	Yes	Advertised by default	No

Capability Code	Capability Description	Address Families Supported	Advertised Support from Tier-0 Gateway	Supported by Tier-0 Gateway when Received from Peer	Default Behavior	Configurable
128	Route refresh (Cisco)	IPv4 Unicast IPv6 Unicast L2VPN EVPN	Yes	Yes	Advertised by default	No

BGP Down Events

If a BGP Down or Dynamic BGP Down event occurs, an alarm is raised. For details about the alarm, see the Routing Events table in the NSX Event Catalog. In this release, additional information is provided regarding the reason for state changes of the BGP peer.

You can also use either the PUT or PATCH method with the following APIs to disable a specific BGP peer. This will prevent alarms from being generated when that peer does not come up.

```
/policy/api/v1/infra/tier-0s/{tier-0-id}/locale-services/{locale-service-id}/bgp/neighbors/{neighbor-id}
/policy/api/v1/global-infra/tier-0s/{tier-0-id}/locale-services/{locale-service-id}/bgp/neighbors/{neighbor-id}
```

To disable a BGP peer, set the parameter `enabled` to `false`. For example:

```
PATCH https://<nsx-mgr>/policy/api/v1/infra/tier-0s/T0-1/locale-services/default/bgp/neighbors/peer-1
{
 "enabled": "false"
}
```

You can also use the GET method to view the value of the `enabled` and other parameters.

> **Note:**
> If the tier-0 gateway firewall is configured with a default `Drop` or `Reject` rule, you must manually add an `Allow` rule for BGP and for BFD if it is configured.
>
> **CAUTION:**
> Note the following scenarios when there are connection failures involving BGP or BFD:
>
> - With only BGP configured, if all BGP neighbors go down, the service router's state will be down.
> - With only BFD configured, if all BFD neighbors go down, the service router's state will be down.
> - With BGP and BFD configured, if all BGP and BFD neighbors go down, the service router's state will be down.
> - With BGP and static routes configured, if all BGP neighbors go down, the service router's state will be down.
> - With only static routes configured, the service router's state will always be up unless the node is experiencing a failure or in a maintenance mode.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Tier-0 Gateways**.
3. To edit a tier-0 gateway, click the menu icon (three dots) and select **Edit**.
4. Click **BGP**.
   a) Enter the local AS number (ASN).

      In active-active mode, the default ASN value, 65000, is already filled in. In active-standby mode, there is no default ASN value.
   b) Click the **BGP** toggle to activate or deactivate BGP.

      In active-active mode, **BGP** is activated by default. In active-standby mode, **BGP** is deactivated by default.
   c) If this tier-0 gateway is in active-active mode, click the **Inter SR iBGP** toggle to activate or deactivate Inter-SR iBGP. This feature is activated by default.

      If the tier-0 gateway is in active-standby mode, this feature is not available.

d) Click the **ECMP** toggle to activate or deactivate Equal-Cost Multi-Path (ECMP), which allows load sharing across equal-cost paths.

The implementation of ECMP is based on the 5-tuple of the protocol number, source and destination address, and source and destination port.

e) Click the **Multipath Relax** toggle to activate or deactivate load sharing across multiple paths that differ only in AS-path attribute values but have the same AS-path length.

Note:
ECMP must be enabled for **Multipath Relax** to work.

f) Specify the tier-0 gateway **Graceful Restart** option for BGP. You can choose from **Deactivate**, **Helper Only**, and **Graceful Restart and Helper**.

You can optionally change the **Graceful Restart Timer** and **Graceful Restart Stale Timer**.

By default, the BGP Graceful Restart mode is set to **Helper Only**. Helper mode is useful for eliminating and/or reducing the disruption of traffic associated with routes learned from a neighbor capable of Graceful Restart. The neighbor must be able to preserve its forwarding table while it undergoes a restart.

For EVPN on a tier-0 gateway, the **Graceful Restart and Helper** mode is not supported.

BGP Graceful Restart is not recommended for an active/active tier-0 gateway configuration in which all NSX Edge nodes actively process traffic. Enabling BGP Graceful Restart can delay convergence since it instructs BGP neighbors to retain stale routes even after BFD has detected a failure, preventing the immediate redirection of traffic to remaining healthy NSX Edge nodes.

Note:

Unless overridden by neighbor-specific configuration, the tier-0 gateway configuration applies to all BGP neighbors.

BGP Graceful restart is not supported when a tier-0 gateway has only one BGP peer configured on a specific NSX Edge node, because the NSX Edge node will go down by design when that single BGP peer goes down. If you specify a BGP Graceful Restart mode for a configuration that has only one BGP neighbor, the BGP Graceful Restart mode will not take effect.

5. Configure **Route Aggregation** by adding IP address prefixes.
   a) Click **Set** for Route Aggregation.
   b) Click **Add Prefix**.
   c) Enter an IP address prefix in CIDR format.
   d) For the option **Summary - Only**, select **Yes** or **No**.
6. Click **Apply**.

   You must save the global BGP configuration before you can configure BGP neighbors.
7. Configure **BGP Neighbors**.
   a) Click **Set** for **BGP Neighbors**.
   b) Click **Add BGP Neighbor**.
   c) Specify the IP address(es) of the neighbor. Use a single IP address to add a single static BGP neighbor, or use CIDR notation to dynamically discover BGP neighbors within the specified IP range.
   d) Enable or disable **BFD**.
   e) Enter a value for **Remote AS number**.

      For iBGP, enter the same AS number as the one in step 4a. For eBGP, enter the AS number of the remote BGP peer.
   f) Under **Route Filter**, click **Set** to add one or more route filters.

      For **IP Address Family**, you can select **IPv4**, **IPv6**, or **L2VPN EVPN** (for static BGP neighbors only). The following combinations are supported:

      - **IPv4** and **IPv6**
      - **IPv4** and **L2VPN EVPN**

The combination of **IPv6** and **L2VPN EVPN** is not supported.

For the RFC 5549 feature, ensure that you provide an IPv4 address family along with the IPv6 address family for the IPv6 BGP peer IP address.

For **Out Filter** and **In Filter**, click **Configure** and select filters, then click **Save**.

For **Maximum Routes**, you can specify a value between `1` and `1,000,000`. When the number of BGP routes received from the peer reaches 75% of the configured limit, or when it exceeds the configured limit for the first time, an alarm is raised. In addition, the output of NSX CLI command `get bgp neighbor` shows if the number of prefixes received exceeds the configured limit. Note that the tier-0 gateway will continue to accept routes from the BGP neighbor even after the **Maximum Routes** limit is reached.

Note:
If you configure a BGP neighbor with one address family, for example, **L2VPN EVPN**, and then later add a second address family, the established BGP connection will be reset.

g) Activate or deactivate the **Allow as-in** feature. This feature is deactivated by default.

The allow as-in feature enables a router to accept BGP updates containing its own Autonomous System number (ASN) in the AS_PATH attribute, bypassing the default loop-prevention mechanism of BGP. This feature applies to all the address families and cannot be applied to specific address families.

h) In the **Source Addresses** field, you can select a source address to establish a peering session with a neighbor using this specific source address from a specific NSX Edge node. If you do not select any, the gateway will automatically set up a BGP peering session with all the BGP neighbors on each NSX Edge node If an NSX Edge node does not have an interface in the subnet of the neighbor, the BGP session configured on this NSX Edge node will remain down.

i) Enter a value for **Max Hop Limit**. The default value is 1. You must configure this setting, along with configuring static routing or an Interior Gateway Protocol (IGP), to enable multi-hop BGP.

j) Optionally configure the BGP neighbor-specific **Graceful Restart** mode. By default, the tier-0 gateway's **Graceful Restart** mode is selected. If you configure a different mode, that mode takes precedence over the mode set for tier-0 gateway.

Option	Description
Deactivate	Graceful Restart is deactivated for this neighbor.
Helper Only	Graceful Restart is configured as **Helper Only** for this neighbor.
Graceful Restart and Helper	Graceful Restart is configured as **Graceful Restart and Helper** for this neighbor. **Note:** **Graceful Restart and Helper** is not supported as a neighbor-specific mode for dynamic neighbor ranges or for EVPN.

Graceful Restart is not supported when a tier-0 gateway has only one BGP peer since the tier-0 service router will go down by design when that single BGP peer goes down. If you specify a Graceful Restart mode for a configuration that has only one BGP neighbor, the Graceful Restart mode will not take effect.

k) Enable **Configure Neighbor Local AS** if required.

- Choose 'Yes' if you want to override the Local AS for this neighbor
- Choose 'No' if you do not want to override the Local AS for this neighbor

Note:
Neighbor Local AS and **AS Path Modifier** fields appear after you enable **Configure Neighbor Local AS**.

l) Enter the replaced local AS number, (ASN) which you want to override the Local ASN for this neighbor, in **Neighbor Local AS**.

m) Choose the AS path option in **AS Path Modifier**:

- **Default**: BGP prepends neighbor's local ASN to the AS path for both outgoing and incoming route advertisements from the peer neighbor. You can modify the default prepend action on the AS path in both inbound and outbound direction.
- **No Prepend**: The local router does not prepend the incoming advertisement from the peer with neighbor's local ASN. The advertised AS path only prepends the BGP local ASN of the router.
- **No Prepend Replace AS**: The local routes are advertised with the neighbor's local ASN instead of the BGP's local ASN to peer router.

n) Click **Timers & Password**.

o) Enter a value in milliseconds for **BFD Interval**.

For an NSX Edge node deployed in VM form factor, the minimum interval value is 500. For a bare-metal NSX Edge node, the minimum interval value is 50. The default value is 500.

p) Enter a value for **BFD Multiplier**. The default value is 3.

q) Enter a value, in seconds, for **Hold Down Time** and **Keep Alive Time**.

The **Keep Alive Time** specifies how frequently KEEPALIVE messages will be sent. The value can be between 0 and 65535. Zero means no KEEPALIVE messages will be sent.

The **Hold Down Time** specifies how long the tier-0 gateway will wait for a KEEPALIVE message from a neighbor before considering the neighbor dead. The value can be 0 or between 3 and 65535. Zero means no KEEPALIVE messages are sent between the BGP neighbors and the neighbor will never be considered unreachable.

Hold Down Time must be at least three times the value of the **Keep Alive Time**.

r) Enter a password.

A password is required when you configure MD5 authentication between BGP peers. Blank space characters are not allowed in BGP passwords.

8. Click **Save**.
9. After a CIDR range for BGP neighbors is added, you can monitor the status of discovered neighbors.
   a) Click the number from the **BGP Neighbors** field.
   b) Expand the BGP neighbor, and click **View Discovered Neighbors**. From the dynamic list that appears, you can view the status details of discovered neighbors. You can also use the options menu for a neighbor to download its advertised and learned routes.
10. Optional: After a BGP neighbor is added, you can download its advertised and learned routes.
    a) Click the number from the **BGP Neighbors** field.
    b) From the **Set BGP Neighbors** dialog box, click the options menu (three dots) of a BGP neighbor and select **Download Advertised Routes** or **Download Learned Routes**.
    c) If you have configured a dynamic range of BGP neighbors, select a discovered neighbor from the drop-down menu to download its advertised or learned routes.
11. To edit the CIDR value of a dynamic BGP neighbor or to change an existing static BGP neighbor to a dynamic neighbor, you must first deactivate the existing neighbor and save the deactivated configuration.
    a) Enter edit mode for the existing neighbor configuration. Deselect the **Enable Neighbor** toggle and click **Save**.
    b) Reenter edit mode and specify the new CIDR value. Select the **Enable Neighbor** toggle and click **Save**.

BGP Autonomous System Number (ASN) per Tier-0 VRF Gateway and per BGP Neighbor

You can configure a different BGP ASN per Tier-0 VRF gateway and per BGP neighbor.

BGP Autonomous System Number (ASN) per Tier-0 VRF Gateway

Tier-0 VRF gateways can have a different ASN other than the default/parent tier-0 gateway local Autonomous System (AS). A separate ASN is important for service providers and multi-tenant topologies where end-customers provide their own BGP ASN.

When a tier-0 VRF gateway is created and configured for BGP, you can define an ASN within the VRF in the **Local AS** field. Else, you can leave the field blank to inherit the ASN from the parent tier-0 gateway. To learn more, see Deploy an NSX VRF-Lite with BGP and Configure BGP.

> **Note:**
>
> BGP ASN per tier-0 VRF gateway supports active-active and active-standby NSX Edge node HA modes. EVPN is currently not supported.

Changes in BGP ASN will cause the already established BGP connections to flap in the following situations until the session is re-established:

- When the parent tier-0 gateway ASN changes, all child tier-0 VRF gateways that inherit the ASN from the parent will change, causing all neighbors to flap.
- When the child tier-0 VRF gateway has its own ASN different from the parent tier-0 gateway:
  - If the parent tier-0 gateway ASN is changed, the ASN of the child tier-0 VRF gateway will not change, and no flap should occur.
  - If the child tier-0 VRF gateway ASN is changed, all neighbors of that VRF will flap regardless of whether the neighbor has a local AS set. The parent tier-0 gateway connections in other VRFs will not flap.
- Any change to the ASN for the neighbor local AS option causes that connection to flap. No other neighbors should be affected.

BGP Per Neighbor ASN Override on Outgoing BGP Updates

NSX provides BGP mechanisms for seamless ASN migration.

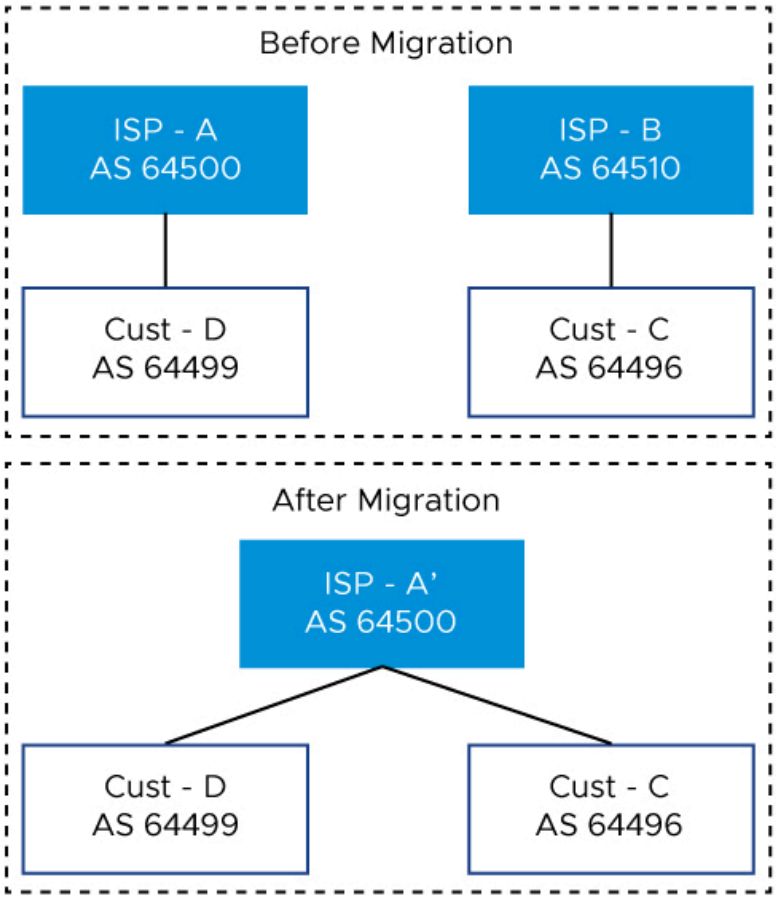
When an ISP acquires a network that belongs to a different autonomous system (AS), its BGP peers must be moved to the acquiring ISP's AS. This can be complicated and time-consuming. With the BGP mechanisms, you can configure BGP neighbors in Tier-0 gateways and Tier-0 VRF gateways with a different local ASN than the gateway's ASN. Using a neighbor's local ASN permits the routing devices in an acquired network to appear to belong to the former AS (Local AS).

BGP neighbors can be configured to have their ASN replaced in outgoing updates. Outgoing BGP updates use AS Path prepending. It can be used to influence the best path algorithm by making it longer or shorter, as well as to enable routers to migrate to new ASNs without having to upgrade all the peers simultaneously.

For more information, see RFC7795.

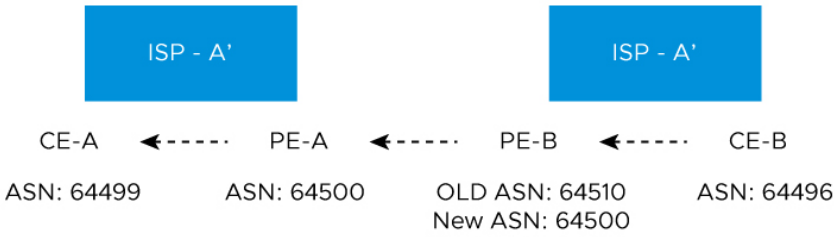
> **Note:**
> This is an eBGP feature. A neighbor's local AS must be different from the tier-0 gateway's local AS. Changes in BGP local AS value will cause the already established neighbor BGP connections to flap until the session is re-established.

Let’s take an example based on merging two ISPs to one ASN of ISP A:

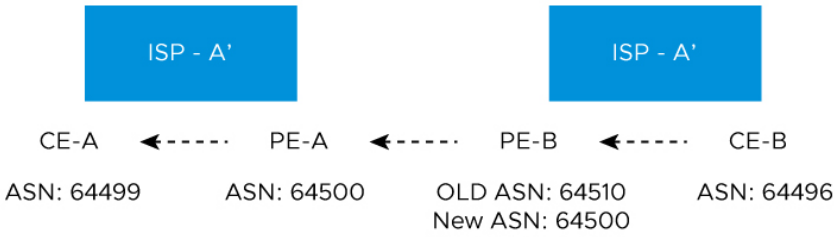


There are three AS path modifier options. **Default Prepend** is the default when Neighbor Local AS is enabled and defined.

- **Default Prepend**: By default, AS Path prepends all received BGP updates with the old ASN while advertising it to other BGP speakers (A').

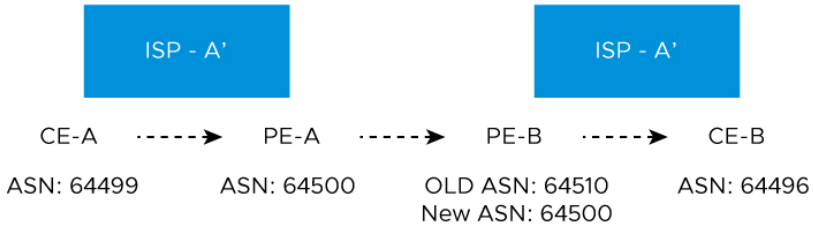


  A route received on CE-A from CE-B will have an AS PATH 64500 64510 64496.
- **No Prepend**: The local router will not prepend the old ASN to the incoming advertisement from that peer. AS Path will be prepended only with router's local AS.



A route received on CE-A from CE-B will have an AS PATH 64500 64496.

- **No Prepend Replace AS**: Local routes will be advertised with the old ASN instead of the router's local AS, and they will appear to be running with the old ASN.



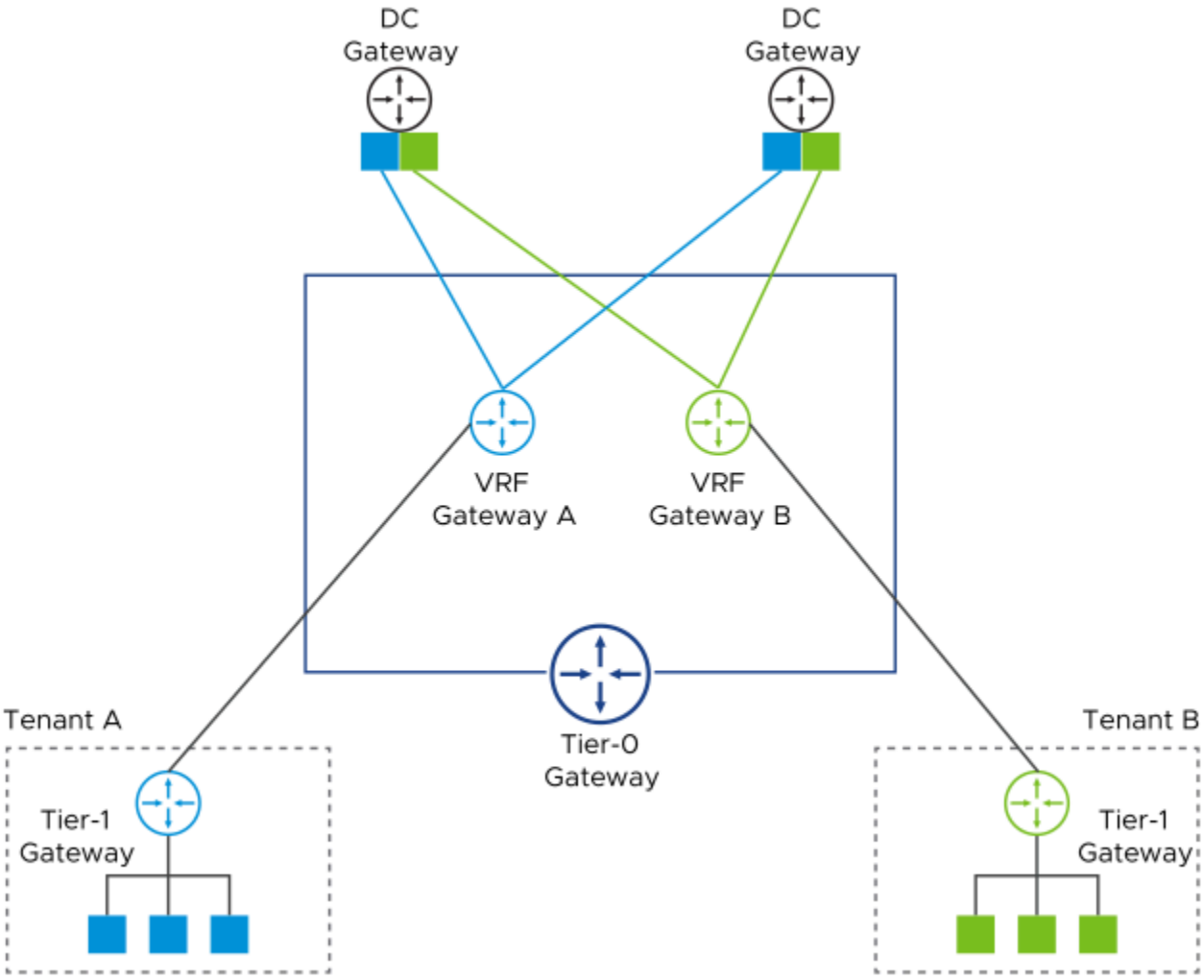
A route received on CE-B from CE-A will have an AS PATH 64510 64499.

Tier-0 VRF Gateways

Virtual routing and forwarding (VRF) makes it possible to instantiate isolated routing and forwarding tables within a router. VRFs are supported by deploying tier-0 VRF gateways. A tier-0 VRF gateway must be linked to a parent tier-0 gateway and inherits some of the tier-0 gateway settings, such as HA mode, NSX Edge cluster, internal transit subnet, T0-T1 transit subnets, and BGP routing configuration.

Multiple tier-0 VRF gateway instances can be created under the same parent tier-0 gateway, which allows the separation of segments and tier-1 gateways into multiple isolated tenants. With tier-0 VRF gateways, tenants can use overlapping IP addresses without any interference or communication with each other.

Tier-0 VRF gateways can be used to connect tenant networks to external routers using static routes or BGP [RFC4364]. This is also known as VRF-Lite.



Tier-0 VRF gateways can also be deployed with EVPN. For more information, see Ethernet VPN (EVPN).

Note:

EVPN Inline and Route Server modes are not supported with the Multi-VRF Inter SR feature.

NSX Federation Support

- You can configure a tier-0 VRF gateway on a Global Manager in an NSX Federation. The Global VRF can have its own locale operation mode and fallback preference and uses the same HA mode as tier-0 gateway.
- You can also stretch a tier-0 VRF gateways to multiple sites and enable inter-SR routing for an active-active tier-0 gateway.

Inter-SR routing is also supported on VRF gateways within a single site. You can also stretch a VRF gateway to multiple sites in an NSX Federation setup and enable Inter-SR routing on them. To enable this feature, you must first turn on the **Multi-VRF Inter SR** toggle under the Additional Settings section while configuring a tier-0 gateway. Turning on this toggle runs MP-BGP between NSX Edge nodes, which is required for inter-node route sync to handle asymmetric VRF connectivity and inter-site connectivity for stretched VRFs. If the **Multi-VRF Inter SR** toggle is off, inter-SR routing cannot be enabled for a VRF gateway on this tier-0 gateway and the tier-0 VRF gateway cannot be stretched to multiple sites. Note that in an NSX Federation setup, the **Multi-VRF Inter SR** toggle can be turned on the NSX Global Manager only when all sites where the tier-0 gateway spans are on the same version.

When you turn on the **Multi-VRF Inter SR** toggle:

- There will be a traffic disruption.

- You cannot turn the toggle off.

	Multi-VRF Inter SR	Inter SR iBGP
Single Site		
Off	You can create a tier-0 VRF gateway.	Inter SR routing will be available only for an active/active tier-0 gateway, but not on the VRF gateway connected to it.
On	You can create a tier-0 VRF gateway.	Inter SR routing will be available for an active/active tier-0 gateway and the VRF gateway connected to it.
Federation		
Off	You can create a single-site tier-0 VRF gateway on the NSX Global Manager for one of the locations of the parent tier-0 gateway.	Inter SR routing will be available across the sites for an active/active tier-0 VRF gateway, but not within the same site.
On	You can create a tier-0 VRF gateway from the NSX Global Manager and stretch it to multiple locations that are subsets of the parent tier-0 gateway. Note: It can be enabled on NSX Global Manager only when all Local Managers are on the same version.	Inter SR routing will be available across the sites for an active/active tier-0 VRF gateway and also within the same site.

Note that even though a tier-0 VRF gateway has an HA mode, it does not have a mechanism to respond to a communication failure that is independent of the parent tier-0 gateway's mechanism. If a tier-0 VRF gateway loses connectivity to a BGP neighbor but the criteria for the parent tier-0 gateway to fail over is not met, the individual tier-0 VRF gateway will not fail over. The only time a tier-0 VRF gateway will fail over is when the parent tier-0 gateway does a failover.

Deploy an NSX VRF-Lite with BGP

- The parent tier-0 gateway must be created before the tier-0 VRF gateway instance.
- The parent tier-0 gateway must have an external interface before you create an external interface on the tier-0 VRF gateway.
- VLAN tagging (802.1q) is used to differentiate traffic among VRFs. The external interface on the tier-0 VRF gateway must be connected to a trunk segment with the corresponding access VLAN ID defined in the segment VLAN range.

1. With admin privileges, log in to NSX Manager.
2. Configure the VLAN trunk segment.
   a) Select **Networking** > **Segments**.
   b) Click **Add Segments**.
   c) Enter a name for the segment.
   d) In **Connected Gateway**, set the type of connectivity for the segment as **None**.
   e) Select a VLAN transport zone.
   f) Expand the **Additional Settings** category.
   g) In **VLAN**, enter a list or range of VLAN IDs allowed in the trunk segment.
   h) Click **Save**.
3. Create the parent tier-0 gateway.

   The parent tier-0 gateway must be created before the tier-0 VRF gateway instance. For more information about configuring a tier-0 gateway, see Add an NSX Tier-0 Gateway.

4. Create the tier-0 VRF gateway.
   a) Select **Networking** > **Tier-0 Gateway**.
   b) Click **Add Gateway** > **VRF**.
   c) Enter a name for the gateway.
   d) Select a tier-0 gateway in **Connect to Tier-0 Gateway**.

      Note:
      Some advanced configurations are inherited from the parent tier-0, such as HA mode, edge cluster, internal transit subnet, T0-T1 transit subnets.
   e) If you are creating a VRF gateway on a Global Manager, In the **Location** field, select the location for which you want the VRF gateway.
   f) If you are configuring a VRF gateway on a Global Manager, you can click **Add Location** to add sites for VRF stretching. Locations are a subset of the parent tier-0 gateway. Note that this button is available only if you had turned on the **Multi-VRF Inter SR** toggle on for the tier-0 that this VRF will connect to.

      Note:
      Note that you can select the primary/secondary roles for a VRF independent of tier-0 primary/secondary mode.
   g) Click **VRF Settings**.

      Note:
      The VRF settings are optional for regular VRF-Lite deployments, but are mandatory for EVPN use cases. For EVPN use cases, see Ethernet VPN (EVPN).
   h) Under **L3 VRF Settings**, specify a **Route Distinguisher**.

      If the connected tier-0 gateway has **RD Admin Address** configured, the **Route Distinguisher** is automatically populated. Enter a new value if you want to override the assigned Route Distinguisher.
   i) Click **Save** and then **Yes** to continue configuring the VRF gateway.
5. Configure the external interfaces on the VRF gateway.
   a) Expand the **Interfaces and GRE Tunnels** category.
   b) In **External and Service Interfaces**, click **Set** or the hyperlinked number.
   c) Click **Add Interface**.
   d) Enter a name for the interface.
   e) Enter the IP address and mask for the external interface.
   f) In **Type**, select **External**.
   g) In **Connected To(Segment)**, select the trunk segment created from Step 2.
   h) Select an edge node.
   i) Enter the **Access VLAN ID** from the list as configured for the segment.
   j) Click **Save** and then **Close**.
6. Configure BGP neighbor for VRF-Lite.
   a) Click **BGP**.
   b) Click the **BGP** toggle to enable BGP.
   c) Turn on the **Inter SR iBGP** toggle to enable inter SR routing. Note that this toggle can be enabled only for an active-active VRF gateway and if you had turned on the **Multi-VRF Inter SR** toggle on for the tier-0 that this VRF connects to.
   d) Enter **Local AS** number.

      Note:
      Leave this field blank to inherit the Local AS number from the parent tier-0 gateway.

      You can configure the other advanced BGP settings such as ECMP.

e) In the **BGP Neighbors** field, click **Set** > **Add BGP Neighbor**.
f) Enter the neighbor IP address.
g) Enable **BFD** if required.
h) Enter the **Remote AS number** of the neighbor.
i) Enter the source IP address.

There should be one or more addresses of created external interfaces or loopback.

j) Under **Route Filter**, click **Set** > **Add Route Filter** to enable **IP Address Family**, filters based on prefix lists, and maximum routes received from the BGP neighbor.
k) Enable or disable the **Allow as-in** feature.

This is disabled by default. With this feature enabled, BGP neighbors can receive routes with the same AS, for example, when you have two locations interconnected using the same service provider. This feature applies to all the address families and cannot be applied to specific address families.

l) In the **Source Addresses** field, you can select a source address to establish a peering session with a neighbor using this specific source address. If you do not select any, the gateway will automatically choose one.
m) Enter a value for **Max Hop Limit**.
n) In the **Graceful Restart** field, you can optionally select **Disable**, **Helper Only**, or **Graceful Restart and Helper**.

Note:
For EVPN, only the **Helper Only** mode is supported.

o) Enable **Configure Neighbor Local AS** if required.

- Choose 'Yes' if you want to override the Local AS for this neighbor
- Choose 'No' if you do not want to override the Local AS for this neighbor

Note:
Neighbor Local AS and **AS Path Modifier** fields appear after you enable **Configure Neighbor Local AS**.

p) Enter the replaced local AS number, (ASN) which you want to override the Local ASN for this neighbor, in **Neighbor Local AS**.
q) Choose the AS path option in **AS Path Modifier**:

- **Default**: BGP prepends neighbor's local ASN to the AS path for both outgoing and incoming route advertisements from the peer neighbor. You can modify the default prepend action on the AS path in both inbound and outbound direction.
- **No Prepend**: The local router does not prepend the incoming advertisement from the peer with neighbor's local ASN. The advertised AS path only prepends the BGP local ASN of the router.
- **No Prepend Replace AS**: The local routes are advertised with the neighbor's local ASN instead of the BGP's local ASN to peer router.

r) Click **Add** and then **Apply**.
s) Click **Save** and then **Close**.

7. Re-distribute the routes in the VRF gateway and announce to the BGP neighbors.
   a) Click **Route Re-distribution**.
   b) In the **Route Re-distribution** field, click **Set** > **Add Route Re-distribution**.
   c) Enter a name for the redistribution policy.
   d) Click **Set** to select available sources, such as tier-0 connected interfaces and segments and then click **Apply**.
   e) Click **Add** and then click **Apply**.
8. Make sure that your segments or tier-1 gateways are connected to the tier-0 VRF gateway.

VRF Route Leaking

By default, the data plane traffic between VRF instances is isolated in NSX. By configuring VRF route leaking, traffic can be exchanged between VRF instances. Static routes must be configured on the tier-0 VRF instances to allow traffic to be exchanged.

Two topology options are supported in NSX:

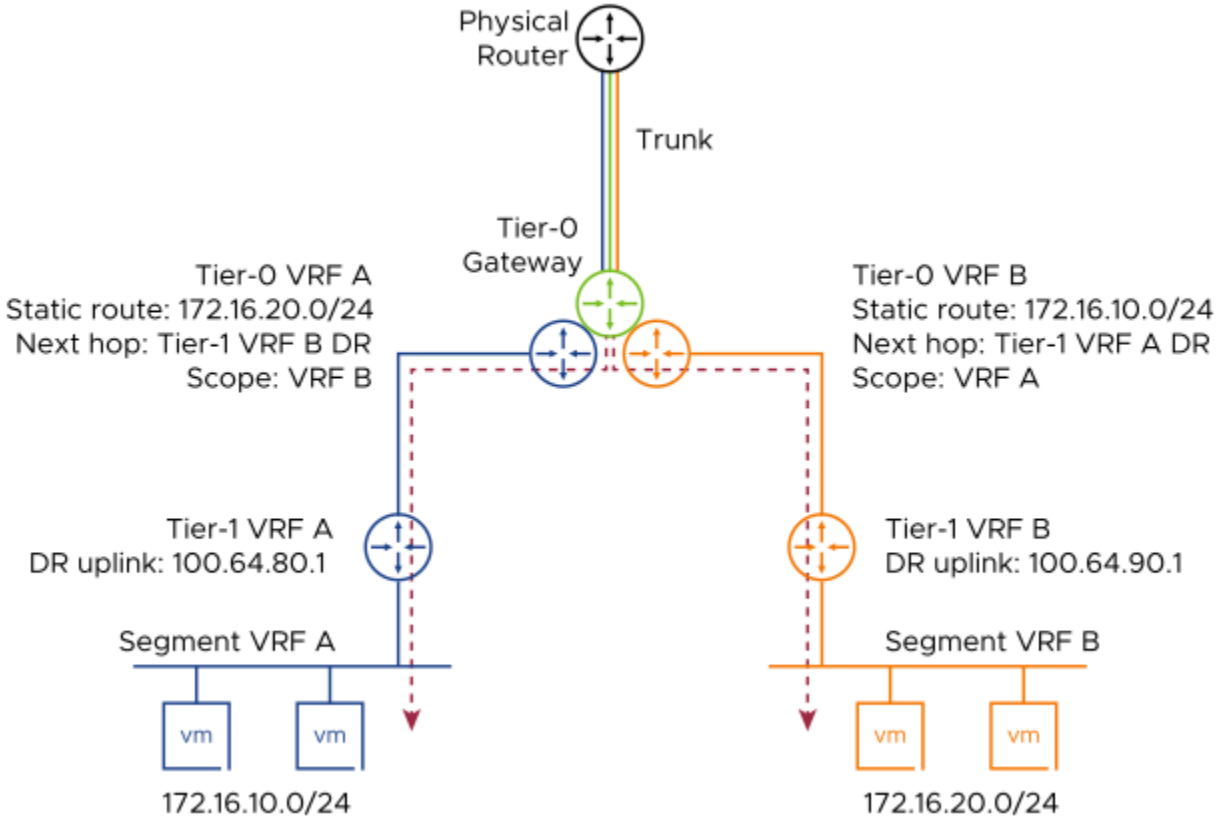
- Local VRF-to-VRF route leaking
- Northbound VRF leaking

Multi-tier routing architecture is required for traffic to be exchanged in a VRF leaking topology since static routes pointing to tier-1 distributed router (DR) uplinks are required.

Local VRF-to-VRF Route Leaking

Tier-1 DR IP addresses can be checked by using both the Edge node CLI or **Network Topology** in NSX Manager.

The diagram depicts a sample topology for the local VRF-to-VRF route leaking option.



The configuration workflow for the sample topology is as follows:

Tier-0 VRF A Configuration Workflow

1. Select **Networking** > **Tier-0 Gateway**.
2. For tier-0 VRF A, click the menu icon (three dots) and select **Edit**.
3. Click **Routing**.
4. In the **Static Routes** field, click **Set** > **Add Static Route** and configure the static route:
   a. Enter a name for the static route.
   b. In the **Network** field, enter a prefix.
      For example, `172.16.20.0/24`
5. In the **Next Hops** column, click **Set** > **Set Next Hops** and define the next hops for the static route:
   a. Enter the IP address of the tier-1 DR uplink in VRF B.
      For example, `100.64.90.1`

      Note:

      Tier-1 DR IP addresses can be checked by using both the Edge node CLI or **Network Topology** in NSX Manager.
   b. Enter the **Admin Distance** of `1`.
   c. Enter the scope.
      For example, `VRF-B`

Tier-0 VRF B Configuration Workflow

1. Select **Networking** > **Tier-0 Gateway**.
2. For tier-0 VRF B, click the menu icon (three dots) and select **Edit**.
3. Click **Routing**.
4. In the **Static Routes** field, click **Set** > **Add Static Route** and configure the static route:
   a. Enter a name for the static route.
   b. In the **Network** field, enter a prefix.
      For example, `172.16.10.0/24`
5. In the **Next Hops** column, click **Set** > **Set Next Hops** and define the next hops for the static route:
   a. Enter the IP address of the tier-1 DR uplink in VRF A.
      For example, `100.64.80.1`

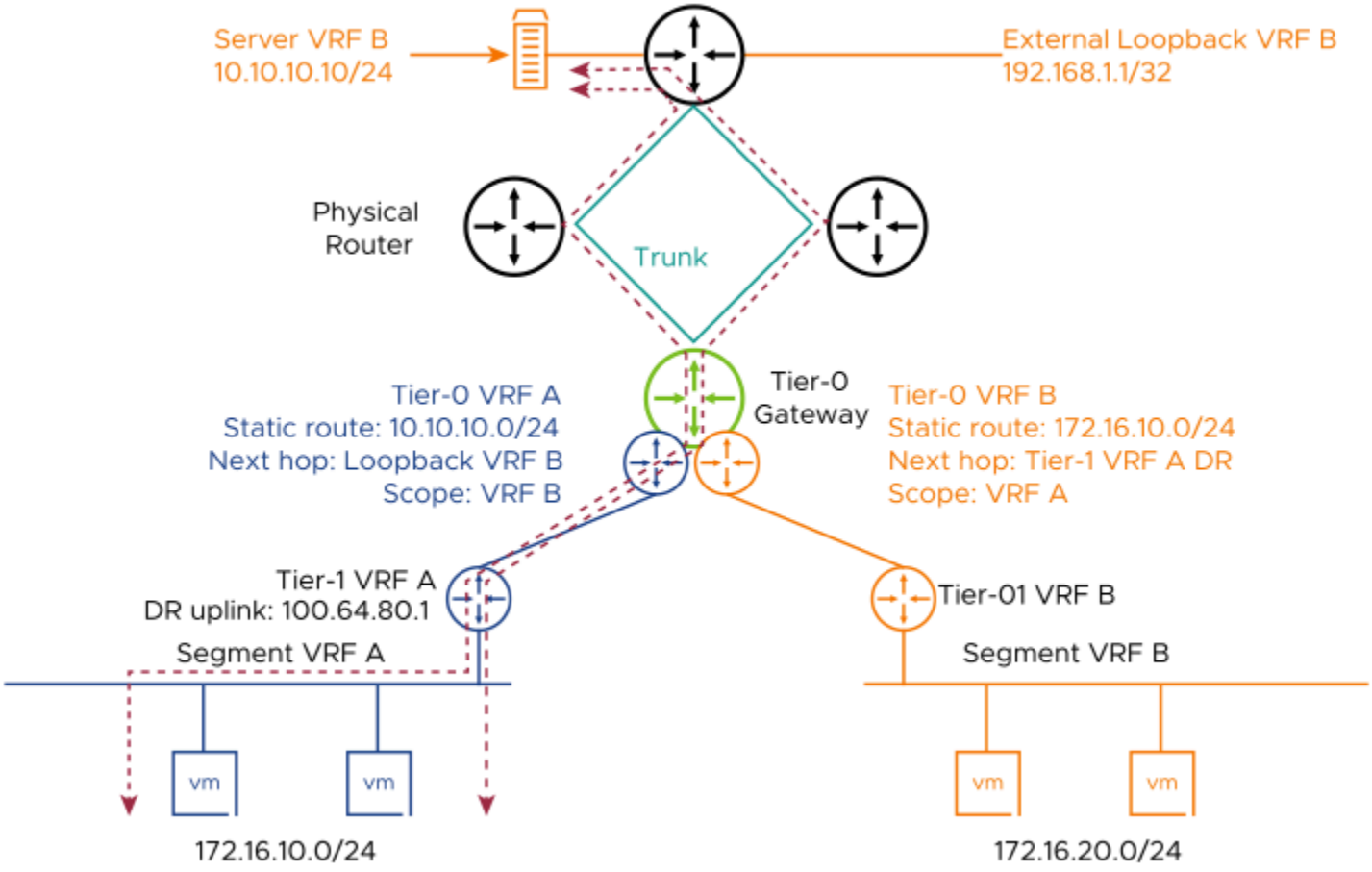
      Note:

      Tier-1 DR IP addresses can be checked by using both the Edge node CLI or **Network Topology** in NSX Manager.
   b. Enter the **Admin Distance** of `1`.
   c. Enter the scope.
      For example, `VRF-A`

Northbound VRF Route Leaking

For this topology option, the northbound static route should have a next hop as an external IP address reachable in the destination VRF routing table. It is not recommended to point static routes directly to connected IP address uplinks as the static route would fail if an outage occurs on that link or neighbor. A loopback or virtual IP address host route (/32) can be advertised in the network in the destination VRF. Since the host route is advertised by both top of rack switches, two ECMP routes are installed in the tier-0 VRF. A return static route should be created in the destination VRF pointing to the tier-1 DR uplink IP address as the next hop.

The following diagram depicts a sample topology for the northbound VRF route leaking option.


Server VRF B
10.10.10.10/24
External Loopback VRF B
192.168.1.1/32
Physical
Router
Trunk
Tier-0 VRF A
Static route: 10.10.10.0/24
Next hop: Loopback VRF B
Scope: VRF B
Tier-0
Gateway
Tier-0 VRF B
Static route: 172.16.10.0/24
Next hop: Tier-1 VRF A DR
Scope: VRF A
Tier-1 VRF A
DR uplink: 100.64.80.1
Segment VRF A
Tier-01 VRF B
Segment VRF B
vm
vm
vm
vm
172.16.10.0/24
172.16.20.0/24

The configuration workflow for the sample topology is as follows:

Tier-0 VRF A Configuration Workflow

1. Select **Networking** > **Tier-0 Gateway**.
2. For tier-0 VRF A, click the menu icon (three dots) and select **Edit**.
3. Click **Routing**.
4. In the **Static Routes** field, click **Set** > **Add Static Route** and configure the static route:
   a. Enter a name for the static route.
   b. In the **Network** field, enter a prefix.
      For example, `10.10.10.0/24`
5. In the **Next Hops** column, click **Set** > **Set Next Hops** and define the next hops for the static route:
   a. Enter the IP address of the tier-1 DR uplink in VRF B.
      For example, `192.168.1.1`
   b. Enter the **Admin Distance** of `1`.
   c. Enter the scope.
      For example, `VRF-B`

Tier-0 VRF B Configuration Workflow

1. Select **Networking** > **Tier-0 Gateway**.
2. For tier-0 VRF B, click the menu icon (three dots) and select **Edit**.
3. Click **Routing**.
4. In the **Static Routes** field, click **Set** > **Add Static Route** and configure the static route:
   a. Enter a name for the static route.
   b. In the **Network** field, enter a prefix.
      For example, `172.16.10.0/24`
5. In the **Next Hops** column, click **Set** > **Set Next Hops** and define the next hops for the static route:
   a. Enter the IP address of the tier-1 DR uplink in VRF A.
      For example, `100.64.80.1`
   b. Enter the **Admin Distance** of `1`.
   c. Enter the scope.
      For example, `VRF-A`

Inter-VRF Routing

The Inter-VRF Routing feature allows you to import and export routes between VRFs.

You can import and export routes between VRF gateways defined on the tier-0 gateway using the following methods:

1. Inter-VRF import and export policies:
   – Inter-VRF Route Advertisement and In-Prefix Filter - Advertise routes that are not BGP, such as Static, Connected, NAT, and all other sources that are available as inter-VRFstatic routes on connected gateway.
   – Inter-VRF BGP Route Leaking - Enable BGP route leak for IPv4/IPv6 or both address families.
2. Manual Inter-VRF static route pointing to target VRF.

Inter-VRF routing can be configured:

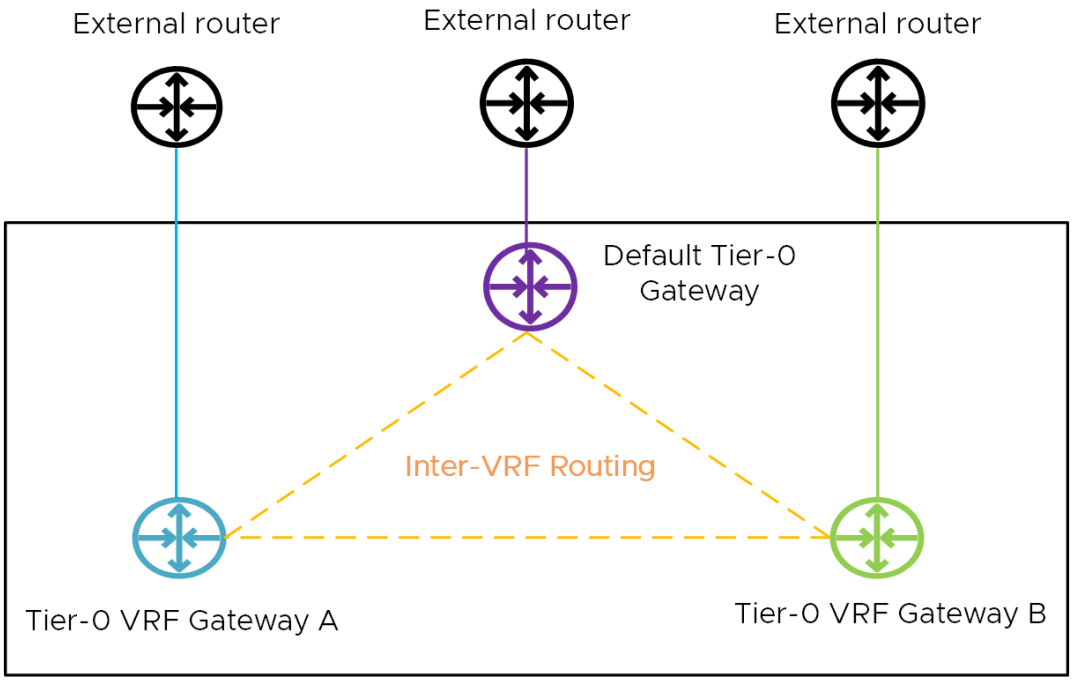
- From Default/Parent tier-0 gateway to tier-0 VRF gateway
- From tier-0 VRF gateway to Default/Parent tier-0 gateway
- From tier-0 VRF gateway to another tier-0 VRF gateway

Configure an Inter-VRF Route Advertisement and In-Prefix Filter

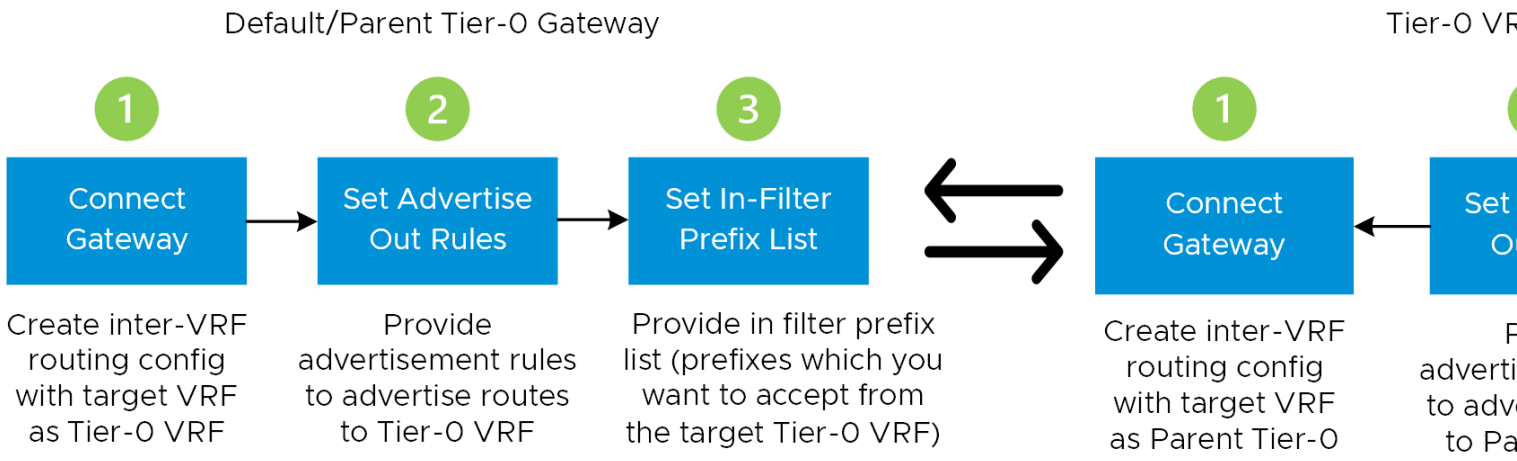
With an inter-VRF route advertisement and in-prefix filter, you can configure advertisement rules to advertise routes on the source and filter on prefixes on target connected gateway to create inter-VRF routing between two VRFs or between a tier-0 gateway and tier-0 VRF gateway.

Ensure that the IP prefix list is created, see Create an IP Prefix List.

Inter-VRF forwarding is enabled when Inter-VRF routing is configured on either end between two tier-0 VRF gateways or between a tier-0 VRF gateway and its parent Tier-0 gateway.



Configuration workflow example:



You can enable inter-VRF routing by configuring:

- **Connected Gateway** - To attach the Target VRF to the source parent tier-0 gateway or tier-0 VRF gatway.
- **Set Advertisement Out Rules** - To advertise the connected subnets of the source to the target VRF. You can allow or deny the subnets that you want to advertise to the target VRF i.e., you want the target VRF to learn.
- **Set In Filter Prefix Lists** - To select prefixes that you want to accept or deny from the target VRF.

> **Note:**
> NSX assigns an IPv4 and IPv6 address to each interface. IPv4 subnet is configured in the **Inter VRF Transit Subnet** field of the default tier-0 gateway. For assigning IPv6 addresses to inter-VRF interfaces, NSX allocates one IPv6 subnet. A /64 IPv6 subnet is allocated from the tier-1-tier-0 transit subnets configured on the default tier-0 gateway.

1. With admin privileges, log in to NSX Manager.
2. Connect the tier-0 gateway:
   a) Select **Networking**.
   b) Select either the default/parent Tier-0 gateway or the Tier-0 VRF gateway, which you would like to configure.
   c) Click on the ellipsis button (three dots) and select **Edit**.
   d) Click **Routing**.
   e) Click **Set** next to **Inter VRF Routing**.
   f) Click **Add Inter VRF Routing**.
   g) In **Connected Gateway**, select the target VRF from the drop-down menu.
   h) Click **Save**.

   Note:
   Wait for the Status to change to ‘Success’.
3. Set Advertisement Out Rules:
   a) Click on the ellipsis button (three dots) and select **Edit**.
   b) Click **Set** below **Advertisement Out Rules**.
   c) Click **Add Advertisement Out Rules**.
      - Enter **Name** - Requires validation for Name.
      - Enter **Subnets** - Specify subnets that are used to assign addresses to logical links connecting parent tier-0 gatewayand child tier-0 VRF gateway.

      Note:
      Leave the **Subnet** field blank to set it as 'any'. If subnet is 'any', Route Advertisement Type filters must be applied.
      - Apply filter – Click on the (i) icon for more information about it. Toggle to 'Yes' if you would like to apply the filters.

      Note:
      Disable the filters to advertise the configured aggregate prefix subnets.
      - Advertisement Action - Action to advertise filtered routes to the connected Tier0 gateway.
        – Allow: Enables the advertisement
        – Deny: Disables the advertisement
      - Prefix Operator - Prefix operator to filter subnets.
        – GE prefix operator: Filters all the routes with prefix length greater than or equal to the subnets configured.
        – EQ prefix operator: Filters all the routes with prefix length equal to the subnets configured.

      Note:
      Prefix operation is ignored if no route advertisement type is specified.
      - Select route advertisement types to be filtered.
   d) Click **Save**.

   Note:
   Wait for the Status to change to ‘Success.’
4. Set In Filter Prefix Lists:
   a) Click on the ellipsis button (three dots) and select **Edit**.
   b) Click **Set** below **In Filter Prefix Lists**
   c) Click **Add In Filter Prefix Lists**.
   d) Select the Prefix List that you had created in IP Prefix Lists window.
   e) Click **Add**.
   f) Click **Save**.

   Note:
   Wait for the Status to change to ‘Success’.

   You may also download the advertised routes, by selecting **Download Advertised Network** from the ellipsis menu (three dots) next to the configured target VRF.

Configure Inter-VRF BGP Route Leaking

With inter-VRF BGP route leaking, you can enable BGP route leaking for IPv4/IPv6 or both address families.

- Enable BGP and provide local AS on both source and target VRF.
- On source VRF, configure redistribution rules or learn external routes through BGP neighbours (To add prefixes to BGP table). See Configure BGP.
- Use route maps to import/export filter. See Create a Route Map.

Inter-VRF BGP route leaking uses BGP VPN to leak BGP routes. NSX internally configures route targets (RTs) for route import and export.

Note:
Inter-VRF BGP Route Leaking is supported on both IPv4 and IPv6.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking**.
3. Select either the default/parent Tier-0 Gateway or the Tier-0 VRF Gateway, which you would like to configure.
4. Click on the ellipsis button (three dots) and select **Edit**.
5. Click **Routing**.
6. Click **Set** next to **Inter VRF Routing**.
7. Click on the ellipsis button (three dots) and select **Edit**.
8. Click **Set** below **BGP Route Leaking**.
   a) Click on the ellipsis button (three dots) and select **Edit**.
   b) Toggle the **BGP Route Leaking** button to 'On' for IPv4 and/or IPv6.
   c) Assign a route map in the **In Filter** field for IPv4 and IPv6. The 'In filter' is used to determine which network or a range of networks are permitted or denied in the direction of In.
   d) Assign a route map in the **Out Filter** field for IPv4 and IPv6. The 'Out filter' is used to determine which network or a range of networks are permitted or denied in the direction of Out.
9. Click **Save**.

Note:

BGP route leaking reduces the number of routes supported for the NSX Edge, since the BGP VPN table has to contain an additional copy of the route for import and export.

Create an Inter-VRF Static Route with Next Hop as the Destination VRF

You can create a manual inter-VRF static route with next hop as the destination VRF.

Inter-VRF routing must be configured on both VRFs: source and destination.

You can allow a datapath to route traffic to another VRF as the next hop scope using inter-vrf static routes. You can configure destination VRF path, for example 'Tier-0 VRF', as scope for route next hop. This will relax validation to require IP address, as internally it will translate the scope to the next hop IP.

Note:
The recommended way to configure inter-VRF routing is to set scope as destination VRF and keep the next hop IP blank.

By default, static routes are tagged as 'INTER_VRF_STATIC', are not advertised to another VRF, and are not included in the BGP table. Static route only works if you have configured inter-VRF-routing with the destination VRF. Else, system will show an error.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Tier-0 Gateways.**.
3. Select either the default/parent Tier-0 Gateway or the Tier-0 VRF Gateway, which you would like to configure.
4. Click on the ellipsis button (three dots) and select **Edit**.
5. Click **Routing**.
6. Click **Set** next to **Static Routes**.
7. Click **Add Static Route** and configure the static route:
   a) Enter a name for the static route.
   b) In the **Network** field, enter an IP prefix.
8. Click **Set** below **Next Hops**.
9. Click **Set Next Hops** and define the next hops for the static route:
   a) Leave the IP address blank.
   b) Specify the administrative distance.
   c) Enter the destination VRF as scope.
10. Click **Add**.
11. Click **Apply**.

    Note:
    Wait for the Status to change to ‘Success’.

Configure OSPF

OSPF (Open Shortest Path First) is an interior gateway protocol (IGP) that operates within a single autonomous system (AS).

The OSPF feature has the following capabilities and restrictions:

- Only OSPFv2 is supported.
- The tier-0 gateway can be active-active or active-standby (preemptive and non-preemptive).
- Only the default VRF is supported.
- You can configure a single area on a tier-0 gateway with a maximum of two tier-0 uplinks per NSX Edge node.
- Backbone, normal area, and NSSA (not-so-stubby area) are supported.
- No redistribution is supported between BGP and OSPF.
- OSPF and BGP can be used together in the case of BGP multi-hop where the peer IP is learned through OSPF.
- The same redistribution features supported for BGP are supported for OSPF (tier-0 gateway uplinks, downlinks, loopbacks, tier-1 gateway downlinks, etc.). Depending on the area type, redistribution for all these networks will result in the NSX Edge node generating Type 5 external LSA (link-state advertisement) or Type 7 external LSA with Type 2 metric only (External Type 2 or NSSA External Type 2 routes). The NSX Edge node itself can learn any type of LSA.
- MD5 and plain password authentication are supported on the area configuration.
- NSX Federation is not supported.
- Route summarization for External Type 2 and NSSA External Type 2 routes is supported.
- The interface running OSPF can be broadcast or numbered point-to-point (/31).
- BFD for OSPF is supported.
- For OSPF graceful restart, only the helper mode is supported.
- Redistribution route maps are supported. Only the matching of prefix lists is applicable. No set actions.
- OSPF ECMP is supported up to maximum of 8 paths.
- Default Originate is supported.
- NAT with OSPF protocol stack is not supported.
- The VIP configured with the tier-0 gateway does not support OSPF.

OSPF router ID (RID) determination

The OSPF router ID (RID) is determined as follows:

- **Loopback interface is present**: OSPF uses the loopback interface IP as the RID. If the loopback interface has an IPv6 address, OSPF does not use it as RID. It will take the highest IPv4 interface IP.
- **Loopback interface is not present**: OSPF uses the highest interface IP address as the RID.
  - If RID is the highest interface IP and loopback interface is present, the RID can be changed to the loopback interface IP address by:
    - Redeploying or replacing the edge transport node
    - Modifying or deleting the highest interface IP address
  - If RID is the highest interface IP and loopback interface is present, the RID does not change when:
    - Disabling and enabling OSPF
    - Rebooting the edge node, enabling maintenance mode on the edge node, or restarting the routing process

The RID, whether it is a loopback interface IP or the highest interface IP address, will not change under the following condtions:

- Clearing OSPF neighbors
- A soft restart or hard restart of OSPF adjacency from a remote site

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Tier-0 Gateway**.
3. Click **Edit** from the three-dot menu of the tier-0 gateway on which you want to configure OSPF.

   If you do not plan to use BGP, turn off the **BGP** toggle under the **BGP** section.
4. Expand the **OSPF** section and click the **OSPF** toggle to enable OSPF.
5. In the **Area Definition** field, click **Set** to add an area definition.

   You can add only one area definition.
   a) Click **Add Area Definition**.
   b) Enter an area ID.

      The value must be a number or 4 numbers in IPv4 format (for example, 1.2.3.4).
   c) In the **Type** column, select **Normal** or **NSSA**.

      An OSPF NSSA (not-so-stubby area) allows external routes to be redistributed into the area by a local ASBR and flooded as Type 7 LSAs, while still blocking Type 5 external LSAs from other areas.
   d) In the **Authentication** column, select **None**, **Password**, or **MD5**.
   e) If **Authentication** is set to **MD5**, enter a key ID. The Key ID identifies which MD5 secret key is used for authenticating OSPF packets within the area. It can be any value between 1 to 255.
   f) In the **Password** column, enter a password if **Authentication** is set to **Password** or **MD5**.

      The MD5 password can have a maximum of 16 characters, and the plain-text password a maximum of 8 characters.
   g) Click **Save**.
6. In the **Graceful Restart** field, select either **Deactivated** or **Helper Only**.
7. Click the **ECMP** toggle to activate or deactivate Equal-Cost Multi-Path (ECMP), which allows load sharing across equal-cost paths. By default, the toggle is activated. The implementation of ECMP is based on the 5-tuple of the protocol number, source and destination address, and source and destination port.

   ECMP (equal-cost multi-path) routing allows packet forwarding to occur over multiple best paths. It can provide fault tolerance for failed paths.
8. In the **Route Summarization** field, click **Set** to add a summary address.

   Route summarization can reduce the number of LSAs that are flooded into an area. You can summarize one or more ranges of IP addresses and send routing information about these addresses in a single LSA.

a) Click **Add Prefix**. Enter an IP address prefix in CIDR format.
b) In the **Advertise** column, select **Yes** or **No** to indicate whether to advertise the summary route.
   The default is **Yes**.
c) Repeat the steps above to add more prefixes.

9. Click the **Default Route Originate** toggle to enable or disable default route originate.
   Enable this to redistribute the default route in OSPF.
10. In the **OSPF Configured Interfaces** field, click **Set** to configure OSPF on existing external interfaces. Note that the **Set** link is available only when the interfaces on the tier-0 have been configured previosuly.
    a) Click **Configure Interface**.
    b) In the **Interface** column, select an interface from the dropdown list.
    c) In the **Area ID** column, select an area ID from the dropdown list.
    d) In the **Network Type** column, select **Broadcast** or **P2P**.
    e) In the **OSPF** column, set the toggle to **Enabled** (the default setting).
    f) Click the **BFD** toggle to enable or deactivate BFD (default is deactivated).
    g) If BFD is enabled, select a BFD profile.
    h) To change the **OSPF Hello Interval**, enter a new value.
       The default is 10 seconds. This parameter specifies the time between Hello messages.
    i) To change the **OSPF Dead Interval**, enter a new value.
       The default is 40 seconds. If a Hello message is not received within this time interval, the neighbor is considered unavailable.
    j) Click **Save**.
    k) Repeat the steps above to configure more interfaces.
11. Click **Save**.
12. Click **Route Re-distribution** to expand the section.
13. Click **Set** in the **Route Re-distribution** field.
    - Select one or more of the sources:
      – Tier-0 subnets: Static Routes, NAT IP, IPsec Local IP, DNS Forwarder IP, Router Link, Connected Interfaces & Segments. Under Connected Interfaces & Segments, you can select one or more of the following: Service Interface Subnet, External Interface Subnet, Loopback Interface Subnet, Connected Segment.
      – Tier-0 subnets: Static Routes, NAT IP, IPsec Local IP, DNS Forwarder IP, EVPN TEP IP, Connected Interfaces & Segments. Under Connected Interfaces & Segments, you can select one or more of the following: Service Interface Subnet, External Interface Subnet, Loopback Interface Subnet, Connected Segment.
      – Advertised tier-1 subnets: DNS Forwarder IP, Static Routes, LB VIP, NAT IP, LB SNAT IP, IPsec Local Endpoint, Connected Interfaces & Segments. Under Connected Interfaces & Segments, you can select Service Interface Subnet and/or Connected Segment.
    - Add **OSPF** to the **Destination Protocol** of any rule that will redistribute routes into OSPF
14. Turn on the **Via OSPF** toggle to enable OSPF.

    If you do not plan to use BGP, do not turn the **Via BGP** toggle on. Also, turn off the **BGP** toggle under the **BGP** section.

After you configure OSPF, in the **OSPF Neighbors** field, you can click **View** to see information about OSPF neighbors. The information displayed includes **Neighbor IP Address**, **Interface**, **Source**, **Edge Node**, **Priority**, and **State**.

Note: If a neighbor is not reachable, an alarm about the neighbor will be raised. If the neighbor is no longer in the network, simply acknowledge the alarm but do not resolve it. If you resolve the alarm, it will be raised again.

Configure BFD

BFD (Bidirectional Forwarding Detection) is a protocol that can detect forwarding path failures.

Bidirectional Forwarding Detection (BFD) works with BGP and OSPF to detect link or neighbor failures much faster than either protocol can on its own. BGP and OSPF register each neighbor with BFD, which monitors the forwarding path with

fast hello packets and immediately notifies the routing protocol when a failure is detected. The protocol can then tear down the session without waiting for its own timers to expire. BFD can also be used with static routes to remove a route from the forwarding table when its next hop becomes unreachable.

Note:

Both IPv4 and IPv6 are supported.

CAUTION:
Note the following scenarios when there are connection failures involving BGP or BFD:

- With only BGP configured, if all BGP neighbors go down, the service router's state will be down.
- With only BFD configured, if all BFD neighbors go down, the service router's state will be down.
- With BGP and BFD configured, if all BGP and BFD neighbors go down, the service router's state will be down.
- With BGP and static routes configured, if all BGP neighbors go down, the service router's state will be down.
- With only static routes configured, the service router's state will always be up unless the node is experiencing a failure or in a maintenance mode.

For more information about BFD for BGP, see Configure BGP. For more information about BFD for OSPF, see Configure OSPF.

For more information about static routes, see Configure a Static Route.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Tier-0 Gateways**.
3. To edit a tier-0 gateway, click the menu icon (three dots) and select **Edit**.
4. Click **Routing** and **Set** for **Static Route BFD Peer**.
5. Click **Add Static Route BFD Peer**.
6. Select a **BFD profile**. If the system default BFD profile named `<default-external-gw-bfd-profile>` is selected, it has a default Interval value of 500 ms that cannot be changed. A customizable profile can also be added. See Add an NSX BFD Profile.
7. Enter the peer IP address and optionally the source addresses.
8. Click **Save**.

Configure Multicast Routing on a Tier-0 Gateway

IP multicast routing enables a virtual machine (source) to send a single copy of data to a single multicast address. Data is then distributed to a group of recipients using a special form of IP address called the IP multicast group address. You can configure multicast routing on a tier-0 gateway for an IPv4 network to enable multicast routing.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Tier-0 Gateways**.
3. To edit a tier-0 gateway, click the menu icon (three dots) and select **Edit**.
4. Click the **Multicast** toggle to enable multicast.
5. In the **Replication Multicast Range** field, enter an address range in CIDR format.

   Replication Multicast Range is a range of multicast group addresses (GENEVE outer destination IP) that is used in the underlay to replicate workload/tenant multicast group addresses. It is recommended that there is no overlap between the Replication Multicast Range and workload/tenant multicast group addresses.
6. In the **IGMP Profile** drop-down list, select an IGMP profile.
7. In the **PIM Profile** drop-down list, select a PIM profile.

Configure IPv6 Layer 3 Forwarding

IPv4 layer 3 forwarding is enabled by default. You can also configure IPv6 layer 3 forwarding.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Global Networking Settings**.
3. Click the **Global Networking Config** tab.
4. Edit the Global Gateway Configuration and select **IPv4 and IPv6** for the **L3 Forwarding Mode**.
   IPv6 only is not supported.
5. Click **Save**.
6. Select **Networking** > **Tier-0 Gateways**.
7. Edit a tier-0 gateway by clicking the menu icon (three dots) and select **Edit**.
8. Go to **Additional Settings**.
   a) There are no configurable IPv6 addresses for **Internal Transit Subnet**. The system automatically uses IPv6 link local addresses.
   b) Enter a /48 IPv6 subnet for **T0-T1 Transit Subnets**.
9. Go to **Interfaces** and add an interface for IPv6.

Create NSX SLAAC and DAD Profiles for IPv6 Address Assignment

When using IPv6 on a tier-0 gateway interface, you can set up Stateless Address Autoconfiguration (SLAAC) for the assignment of IP addresses. SLAAC enables the addressing of a host, based on a network prefix advertised from a local network router, through router advertisements. Duplicate Address Detection (DAD) ensures the uniqueness of IP addresses.

Navigate to **Networking** > **Global Networking Settings**, click the **Global Gateway Config** tab and select `IPv4 and IPv6` as the **L3 Forwarding Mode**

1. From your browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address>.
2. Select **Networking** > **Tier-0 Gateways**.
3. To edit a tier-0 gateway, click the menu icon (three dots) and select **Edit**.
4. Click **Additional Settings**.
5. To create an **ND Profile** (SLAAC profile), click the menu icon (three dots) and select **Create New**.
   a) Enter a name for the profile.
   b) Select a mode:
      - **Disabled** - Router advertisement messages are disabled.
      - **SLAAC with DNS Through RA** - The address and DNS information is generated with the router advertisement message.
      - **SLAAC with DNS Through DHCP** - The address is generated with the router advertisement message and the DNS information is generated by the DHCP server.
      - **DHCP with Address and DNS through DHCP** - The address and DNS information is generated by the DHCP server.
      - **SLAAC with Address and DNS through DHCP** - The address and DNS information is generated by the DHCP server. This option is only supported by NSX Edge and not by ESX hosts.
   c) Enter the reachable time and the retransmission interval for the router advertisement message.
   d) Enter the domain name and specify a lifetime for the domain name. Enter these values only for the `SLAAC with DNS Through RA` mode.
   e) Enter a DNS server and specify a lifetime for the DNS server. Enter these values only for the `SLAAC with DNS Through RA` mode.
   f) Enter the values for router advertisement:
      - **RA Interval** - The interval of time between the transmission of consecutive router advertisement messages.
      - **Hop Limit** - The lifetime of the advertised routes.
      - **Router Lifetime** - The lifetime of the router.
      - **Prefix Lifetime**- The lifetime of the prefix in seconds.

- **Prefix Preferred Time** - The time that a valid address is preferred.

6. To create a **DAD Profile**, click the menu icon (three dots) and select **Create New**.
   a) Enter a name for the profile.
   b) Select a mode:
      - **Loose** - A duplicate address notification is received but no action is taken when a duplicate address is detected.
      - **Strict** - A duplicate address notification is received and the duplicate address is no longer used.
   c) Enter the **Wait Time (seconds)** that specifies the interval of time between the NS packets.
   d) Enter the **NS Retries Count** that specifies the number of NS packets to detect duplicate addresses at intervals defined in **Wait Time (seconds)**

State Synchronization of Tier-0 Gateways

Connection information of the traffic running on a given tier-0 gateway SR (Service Router) is synchronized to its peer tier-0 gateway SR in active-standby or stateful active-active HA modes.

The state synchronization between the Active SR and Standby SR is triggered whenever there is a change in the local state of TCP SYN packet, after a certain interval, or if a local state is created or deleted. During state synchronization, NSX Edge uses a proprietary protocol to replicate CCP packets between Active and Standby SRs on the tier-0 gateway.

Note the following points about state synchronization:

- State synchronization is supported for Gateway Firewall, Identity Firewall, NAT, IPSec VPN, and DHCP.
- If new sessions were going through a tier-0 gateway SR just before a failover, it might happen that those sessions were not synchronized on the associated tier-0 gateway SR and potentially affect the traffic for those sessions.
- In active-standby mode, state synchronization is supported for Gateway Firewall, Identity Firewall, NAT, IPSec VPN, and DHCP.
- In active-active mode, state synchronization is supported for Gateway Firewall, Identity Firewall, and NAT. IPSec VPN is not supported.
- If new sessions were going through a tier-0 SR just before a failover, it might happen that those sessions were not synchronized on the associated tier-0 SR and potentially affect the traffic for those sessions.

Changing the HA Mode of a Tier-0 Gateway

You can change the high-availability (HA) mode of a tier-0 gateway in certain circumstances.

Changing the HA mode is allowed only if there is no more than one service router running on the tier-0 gateway, that is, uplinks on only one NSX Edge node configured. However, you can have more than one uplink on the same NSX Edge node configured.

After you set the HA mode from active-active to active-standby, you can set the failover mode. The default is non-preemptive.

HA mode change is not allowed if the following services or features are configured.

- DNS Forwarder
- IPSec VPN
- L2 VPN
- HA VIP
- Stateful Gateway Firewall
- SNAT, DNAT, NO_SNAT, or NO_DNAT
- Reflexive NAT applied on an interface
- Service Insertion
- VRF
- Service Interface

Configure the ARP Limit of a Tier-0 or Tier-1 Gateway or Logical Router

You can configure the ARP limit of a tier-0 or tier-1 gateway or logical router on each ESX host using the API. The limit specifies the maximum number of ARP entries per transport node at each gateway or logical router.

To read or set the global ARP limit, use the following API methods and parameter:

Method	URI	Parameter
`GET`, `PUT`, `PATCH`	`/policy/api/v1/infra/global-config`	`arp_limit_per_gateway` (range: 5000 - 50000, default: 50000)

To read or set the ARP limit for a specific tier-0 or tier-1 gateway, use the following API methods and parameter. If the limit is not set, the global ARP limit will apply.

Method	URI	Parameter
`GET`, `PUT`, `PATCH`	`/policy/api/v1/infra/tier-0s/<tier-0-id>`	`arp_limit` (range: 5000 - 50000, no default)
`GET`, `PUT`, `PATCH`	`/policy/api/v1/infra/tier-1s/<tier-1-id>`	`arp_limit` (range: 5000 - 50000, no default)

Note that updating the ARP limit using Manager `GlobalConfig` API is not allowed.

Stateful Services on Tier-0 and Tier-1 on NSX

To meet the demands of stateful services such as more bandwidth and throughput, you can configure Tier-0 and Tier-1 gateways in Active-Active (A-A) configuration. Stateful services are required for next generation firewall, Layer 7 rules, URL filtering or TLS decryption.

You can scale-out or scale-in the number of service routers by adding NSX Edge nodes to the cluster.

> **CAUTION:**
> As you scale-in or scale-out NSX Edge nodes, you might see loss of traffic packets for existing flows.

The supported stateful gateway services are:

- Gateway Firewall L3-L4
- APP-ID (L7)
- User-ID
- URL Filtering
- TLS Inspection
- IDS/IPS
- Malware Detection and Sandboxing
- NAT
- DHCP Relay Server
- DHCP Server

The unsupported services are:

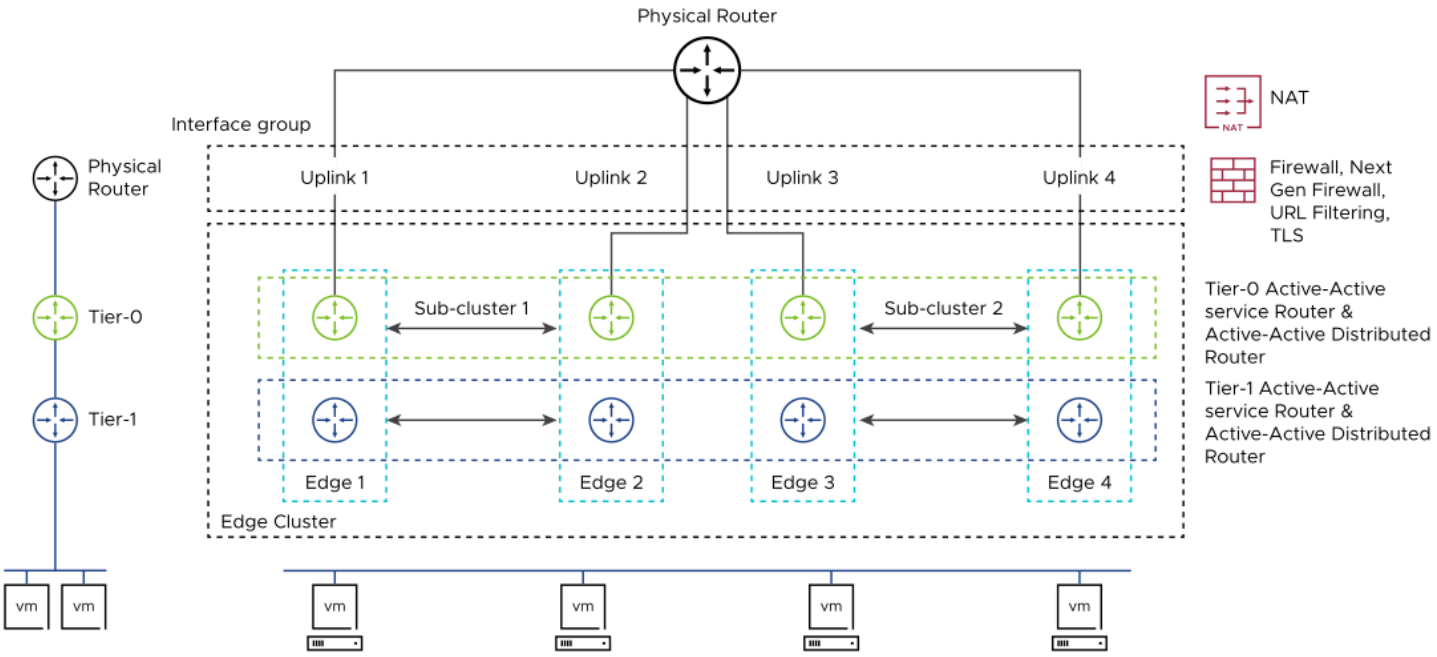
- FQDN Analysis
- L2VPN
- IPSecVPN
- Gateway Network Introspection
- Local DHCP Server
- Service Interface

In your existing topology, if Tier-1 gateway is in active-standby (A-S) mode, you cannot reconfigure it in A-A HA mode and it cannot share the same NSX Edge cluster with A-A stateful Tier-0 gateways. As a workaround, deploy that Tier-1

gateway in active-standby mode on a separate cluster. Then, deploy Tier-0 gateway on another NSX Edge cluster. If your environment requires a Tier-1 gateway, configure it in A-A HA mode. See Supported Topologies.

Key Concepts of Stateful Services with Active-Active Gateways

Understand the key concepts required to configure stateful services with an active-active gateway.



Interface Groups

Use an interface group to group equivalent external (uplink) interfaces across service routers. Equivalent interfaces refer to interfaces that have the same inbound policies such as firewall rules, NAT rules, and so on, and equivalent network reachability. NSX only supports external interface. You must only create homogenous interface groups, such as an interface group comprising of only external interfaces.

To define an interface group, meet the following conditions:

- Only one interface link from each service router of the NSX Edge cluster must participate in the interface group.
- Every interface link must be part of a single interface group.
- Every interface group must have the same number of interfaces from each service router.

> **Note:**
> An interface group is required to create a stateful active-active group. If these conditions are not met, NSX raises a status alarm on the tier-0 gateway or the tier-1 gateway UI screen.

Interface groups allow traffic flows to continue without disruption even when the original uplink interface that supported packet transmission fails because the peer uplink interface takes over its traffic. So, if Uplink 1 or NSX Edge 1 fails, then

the interface group decides where the next packet must be punted to, which is the peer shadow port on the peer edge node on the same sub-cluster.

You can have more than one interface groups, where every interface group is dedicated to a specific requirement of the NSX Edge node. For example, one interface group can serve internet traffic while another group can serve the DMZ.

On tier-1 gateway, a default interface group is automatically created. On the tier-0 gateway, you must manually create an interface group.

To create an interface group, go to the **Interfaces and Interfaces groups** section on your tier-0 gateway. Alternatively, run the API `PUT /infra/tier-0s/<name>/locale-services/<location>/interface-groups/<group-name>`

```
{
    "resource_type": "Tier0InterfaceGroup",
    "id": "uplinkgroup",
    "display_name": "uplinkgroup",
    "path": "/infra/tier-0s/Tier0Gateway1/locale-services/Tier0LocalServices-1/interface-groups/uplinkgroup",
    "relative_path": "uplinkgroup",
    "parent_path": "/infra/tier-0s/Tier0Gateway1/locale-services/Tier0LocalServices-1",
    "unique_id": "c5b2a758-7040-410b-a35d-298a16b55df0",
    "realization_id": "c5b2a758-7040-410b-a35d-298a16b55df0",
    "marked_for_delete": false,
    "overridden": false,
    "members": [
        {
            "interface_path": "/infra/tier-0s/Tier0Gateway1/locale-services/Tier0LocalServices-1/interfaces/
tier0_interface1"
        },
        {
            "interface_path": "/infra/tier-0s/Tier0Gateway1/locale-services/Tier0LocalServices-1/interfaces/
tier0_interface2"
        }
    ],
}
```

External Interface

Interface connecting to the physical infrastructure. It supports static routing and BGP. In previous releases, this interface was referred to as uplink interface. This interface can also be used to extend a VRF (Virtual routing and forwarding instance) from the physical networking fabric into the NSX domain.

Sub-clusters

When you configure stateful services on NSX Edge nodes, NSX automatically creates sub-clusters on the given NSX Edge cluster. So, NSX Edge a cluster of four NSX Edge nodes becomes two sub-clusters, where each sub-cluster is a pair of NSX Edge nodes.

All service routers on NSX Edge nodes participating in an interface group are converted into pairs.

For example, in sub-cluster 1, if NSX Edge node 1 goes down, all ingress or egress traffic on NSX Edge node 1 is switched to NSX Edge node 2. So, NSX Edge node 1 and NSX Edge node 2 function as the original NSX Edge node and peer NSX Edge node respectively in the sub-cluster. During the failover process, NSX Edge node 2 takes over the backplane IP address that NSX Edge node 1 was serving to ensure no traffic is lost and traffic flow is maintained. When the failed NSX Edge node 1 comes back up, the initial state is restored, where all traffic is redirected back to NSX Edge node 1.

Failure Domain Considerations in Stateful Active-Active Gateway Deployments

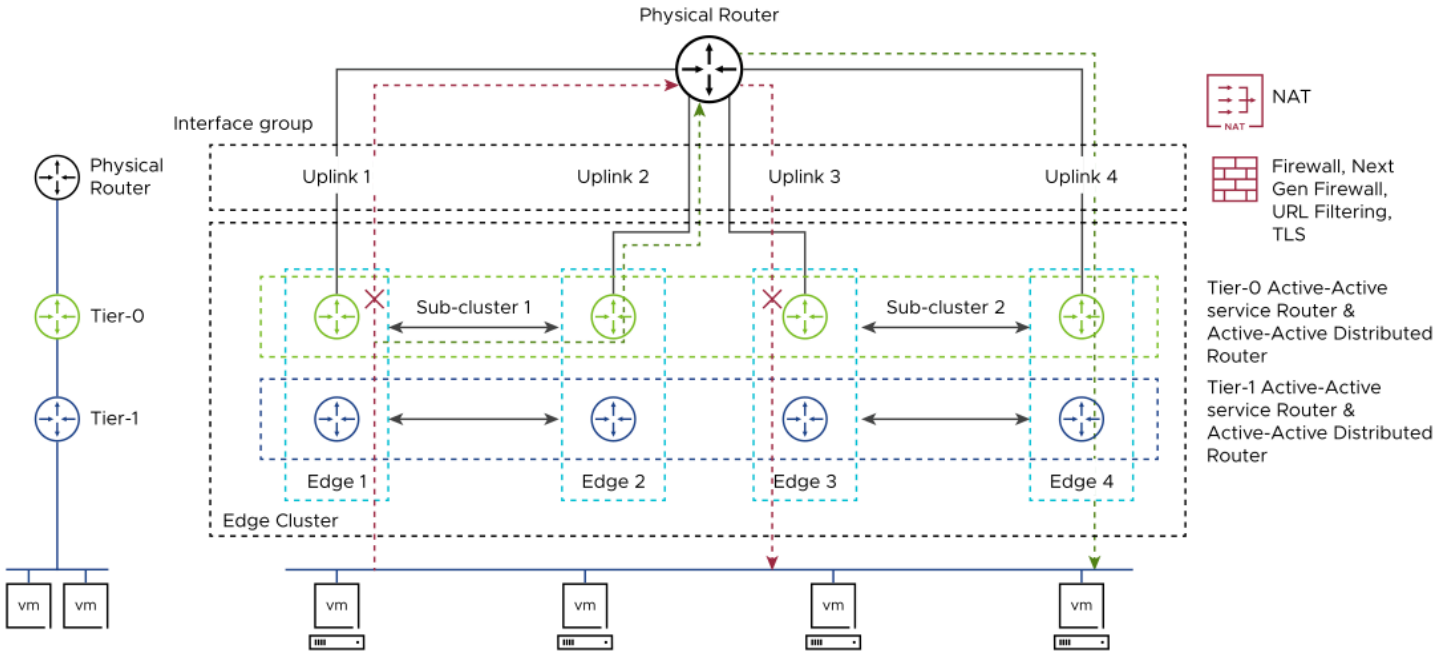
Configure failure domains to ensure that both NSX Edge nodes selected for a sub-cluster do not belong to the same failure domain. For more information, see Configure Failure Domains.

To ensure failure domain functions as per design, meet these conditions:

- Configure each NSX Edge with a failure domain and deploy one NSX Edge node in each failure domain. Both NSX Edge nodes of a sub-cluster must not belong to the same failure domain.
- Ensure that both NSX Edge nodes of a sub-cluster remain as part of the same sub-cluster. To ensure that these nodes are automatically paired in the same sub-cluster, follow a specific sequence when referencing these nodes to failure domains. For example, in a sub-cluster, first reference NSX Edge node 1 to a failure domain and then reference NSX Edge node 2 to a different failure domain. So, when NSX Edge node 1 comes back up after a failure, the failure domain where the node was referenced to allows it to rejoin the same sub-cluster.

Understanding Traffic Flows

Traffic flows on stateful tier-0 and tier-1 gateways configured in stateful active-active HA mode.



South-North Traffic Flow

1. Based on a deterministic hash, an incoming packet from a southbound VM is punted to the backplane of the Edge-2.
2. Edge-2 determines that Edge-4 is actively managing the traffic flows and forwards the flow out through the external interfaces (which are part of the interface group).
3. An IP hash is performed, based on external server destination IP, and traffic is punted from Edge-2 to Edge-4. The packet is further forwarded to tier-0 gateway service router (SR), where SNAT changes the source IP address to translated IP address.
4. After the flow reaches Edge-4 tier-0 SR, the shadow port forwards the NAT traffic to the uplink interface and then sent out to the physical router.

5. If tier-0 SR on Edge-4 fails, NSX punts traffic to its backup node in the sub-cluster, Edge-3, where SNAT changes the source IP address to translated IP address. The backup interface on Edge-1 takes over the backplane IP and the uplink IP of tier-0 gateway before beginning to process traffic. The backup interface on Edge-3 is operationally `Up` and the shadow interface on Edge-4 is `Down`.
6. All traffic flows processed by firewall and NAT rules are synchronized on the tier-0 SR on Edge-3.
7. When Edge-4 comes back up, the flow is resynchronized back to Edge-4. When the shadow port comes back up, NSX punts traffic to it.

North-South Traffic Flow

1. A packet from a northbound VM is hashed by the physical router using its own hashing algorithm to send the packet to Edge-3, based on an ECMP routing choice. The tier-0 gateway is running on Edge-3.
2. Edge-3 determines that Edge-4 is actively managing the traffic flow and forwards the flow to Edge-4. The flow is managed by the shadow interface of Edge-4.
3. An IP hash is performed, based on external server source IP, traffic is punted from Edge-3 tier-0 SR to Edge-4 tier-0 SR, where NAT is enabled. The source IP is changed to the translated IP address.
4. The packet is sent from Edge-4 Tier-0 SR to Edge 4 tier-0 DR and then to tier-1 gateway, finally reaching the destination VM.
5. If tier-0 service router on Edge-4 fails, NSX punts traffic to its peer node (sub-cluster 2), which is Edge-3. NAT enabled on Edge-3 changes the source IP address to translated IP address.
6. Before beginning to process traffic, the backup shadow port on Edge-3 manages the traffic flow. Now, the backup shadow port on Edge-3 is operationally `Up` and the shadow port on Edge-4 is `Down`.
7. All traffic flows processed by firewall and NAT rules are synchronized on the tier-0 SR on Edge-3.
8. When Edge-4 comes back up, the flow is resynchronized back to it. The shadow port on Edge-4 comes back up and manages the punted traffic.

Sub-cluster Failure

If both the NSX Edge nodes in a sub-cluster go down, the sub-cluster goes down.

- Existing flows are disrupted causing traffic loss.
- New flows are punted to the other sub-cluster.
- When the failed sub-cluster comes back up again, the flows return to the original sub-cluster.

If a sub-cluster goes down for any reason, then the other sub-cluster in the NSX Edge cluster takes over.

Single NSX Edge Node Failure

Upon failure of an NSX Edge node, the following events happen:

1. Interface links of the NSX Edge node fail.
2. The shadow port on the failed NSX Edge node is in `Down` state.
3. The backup port of the peer node in the sub-cluster takes over.
4. The firewall and the NAT states are synchronized on the peer NSX Edge node.
5. The backup port on the peer node provides connectivity to new traffic flows.
6. When interface links of the failed node comes back up, the firewall and the NAT states are resynchronized with the shadow port of the active NSX Edge node.
7. NSX punts back traffic flows to the original NSX Edge node.

Supported Topologies

These are the supported topologies for stateful services on a tier-0 gateway or tier-1 gateway in active-active HA mode.

Greenfield Deployment Considerations

In new deployments, note the following considerations when building one of the supported topologies for stateful services on a NSX Edge cluster:

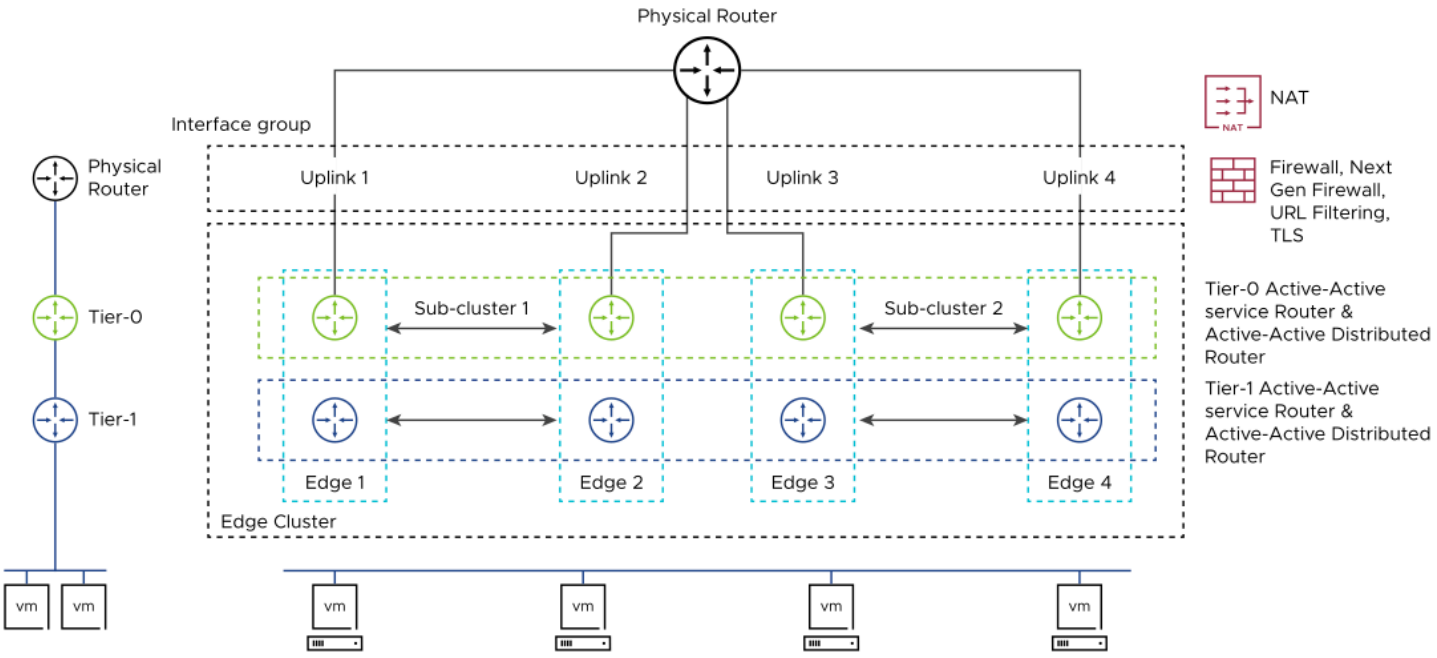
- Tier-1 gateway in stateful active-active HA mode running stateful services must be connected to a tier-0 gateway in stateful active-active HA mode and must be hosted on the same NSX Edge cluster.
- Tier-1 gateway in active-standby HA mode can be connected to a tier-0 gateway in stateful active-active HA mode, but tier-1 gateways must be hosted on a different NSX Edge cluster.

Brownfield Deployment Considerations

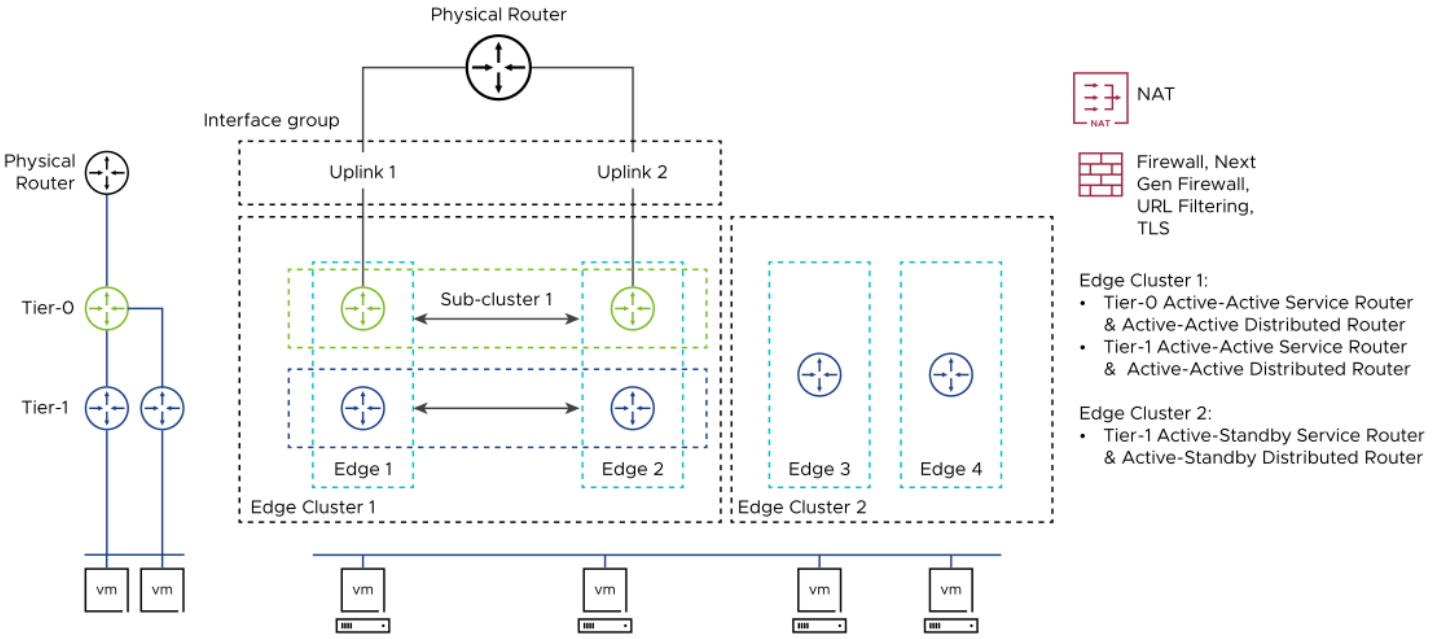
In existing deployments, note the following considerations when building one of the supported topologies for stateful services on a NSX Edge cluster:

- An existing tier-1 gateway in active-standby HA mode cannot be converted to a tier-1 gateway in active-active HA mode. You must create a new tier-1 gateway in active-active HA mode.
- Tier-0 gateway in active-standby HA mode cannot be converted to a tier-0 gateway in active-active HA mode.
- Tier-1 gateway in Distributed Only mode can be converted to a tier-1 gateway in stateful active-active HA mode, as long the tier-1 gateway is not attached to the tier-0 gateway.

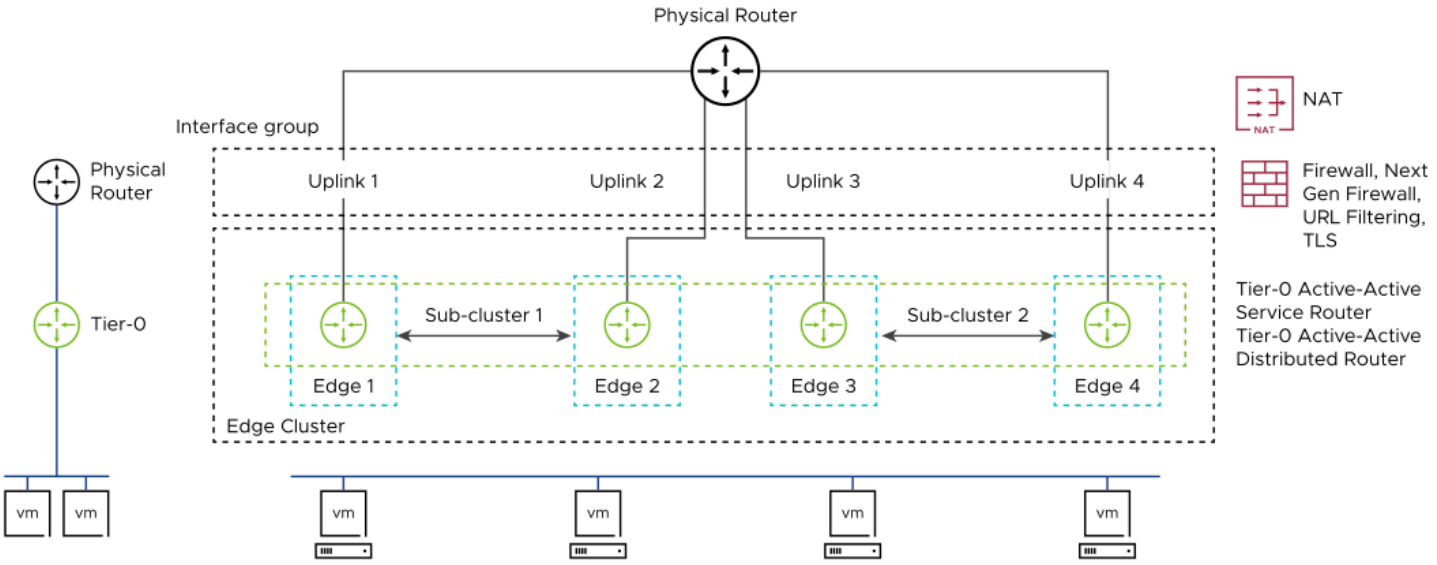
Tier-0 and Tier-1 Gateways in Stateful Active-Active HA Mode



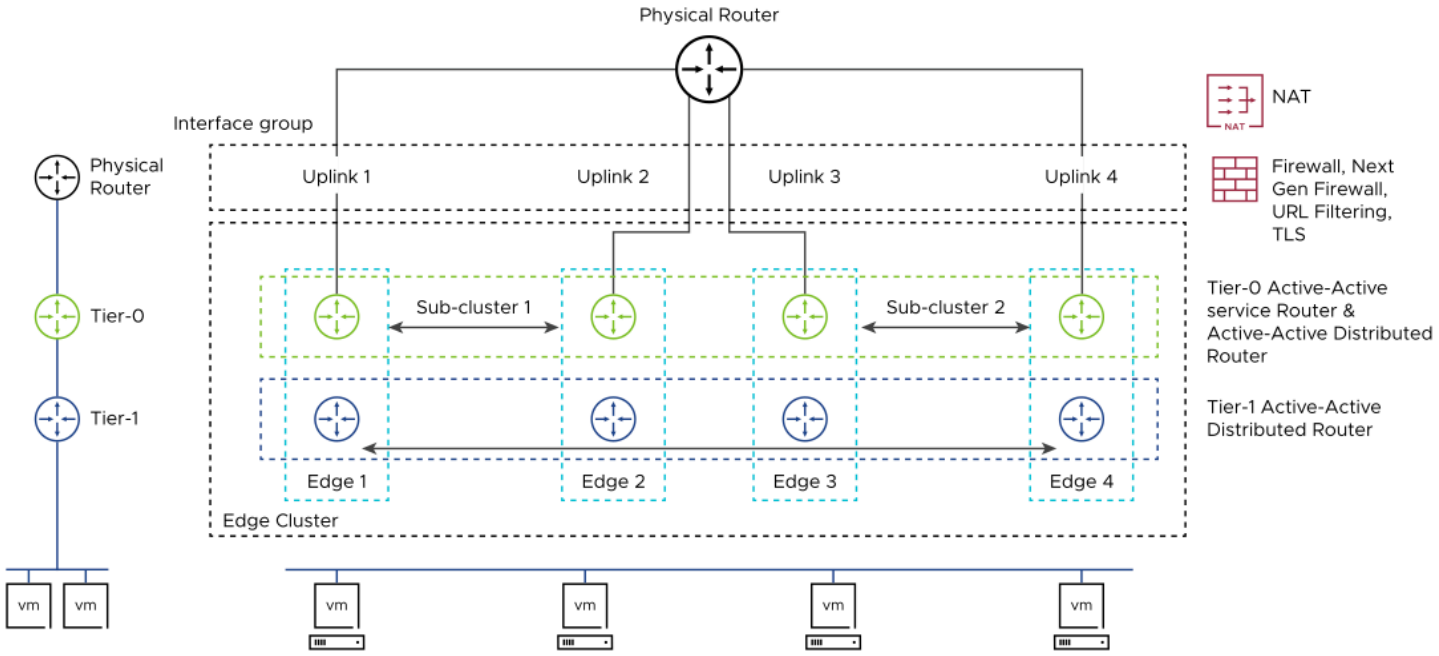
Tier-0 and Tier-1 Gateways in Stateful Active-Active Mode and Tier-1 Gateway in Stateful Active-Standby HA Mode



Tier-0 Gateway in Stateful Active-Active HA mode (no Tier-1 gateways)



Tier-0 Gateway in Stateful Active-Active HA mode and Tier-1 Gateway in Distributed Router only mode



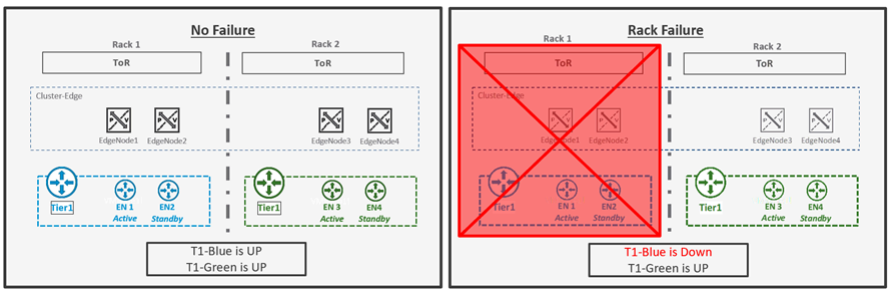
Configure Failure Domains

A failure domain is a logical grouping of NSX Edge nodes within an NSX Edge cluster. Failure domains complement auto placement algorithm and guarantee service availability in case of a failure affecting multiple NSX Edge nodes.

In a failure domain, active and standby instances of a tier-1 gateway service router or members of a sub-cluster always run in different failure domains. Without a failure domain, a tier-1 gateway service router could be auto placed on NSX Edge nodes that are in the same rack. So, if rack1 fails, both active and standby instance of this tier-1 SR fail as well.

Without failure domains configured:

Figure 400:

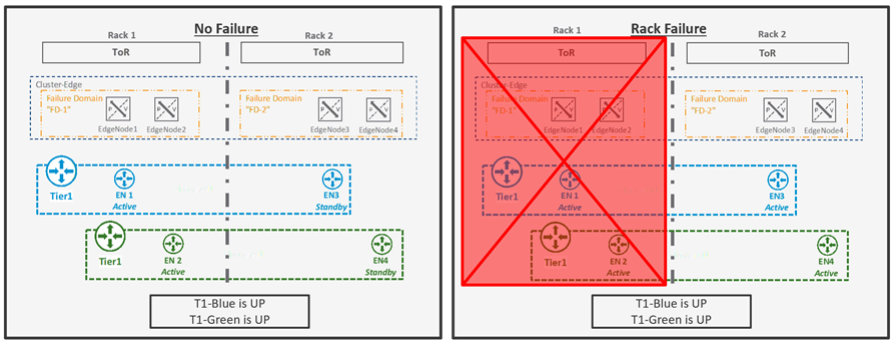


- In an NSX Edge cluster consisting of four NSX Edge nodes (EdgeNode1, EdgeNode2, EdgeNode3, EdgeNode4), any new tier-1 gateways in active-standby HA mode are automatically placed in any two of those four NSX Edge nodes.

- However, high availability cannot be achieved if Tier-1 A/S is deployed in Rack1 and Tier-2 A/S is deployed in Rack2. If Rack1 fails, Tier-1 A/S on EdgeNode1 and EdgeNode2 are lost as they are in the same rack.

With failure domains configured:

Figure 401:



- EdgeNode1 and EdgeNode2 are configured to be a part of failure domain FD-1, while EdgeNode3 and EdgeNode4 are in failure domain FD-2. When a new tier-1 gateway in active-standby HA mode is created and if the active service router of that tier-1 gateway is hosted on EdgeNode1 for example, then the standby service router of the same tier-1 gateway will be instantiated in failure domain FD-2, which includes EdgeNode3 or EdgeNode4.

- After configuring failure domains on an NSX Edge cluster, any new tier-1 gateway and its active and standby service routers are correctly placed in different failure fomains to ensure high availabilty in case of a rack failure.

1. Using the API, create failure domains for the each NSX Edge node that you will add to the stateful A-A cluster, for example, in FailureDomain1 include FD1-EdgeNode1 and FD1-EdgeNode2 and in FailureDomain2 include FD2-EdgeNode3 and FD2-EdgeNode4. Set the parameter `preferred_active_edge_services` to true for NSX Edge nodes in both failure domains. The `preferred_active_edge_services` is useful only when a Tier-1 gateway is created in preemptive failover mode.

```
POST /api/v1/failure-domains
{
"display_name": "FD1-EdgeNode1",
"preferred_active_edge_services": "true"
"display_name": "FD1-EdgeNode2",
"preferred_active_edge_services": "true"
}

POST /api/v1/failure-domains
{
"display_name": "FD2-EdgeNode3",
"preferred_active_edge_services": "true"
"display_name": "FD2-EdgeNode4",
"preferred_active_edge_services": "true"
}
```

2. Using the API, associate each NSX Edge node with the failure domain for the site. First call the `GET /api/v1/transport-nodes/<transport-node-id>` API to get the data about the NSX Edge node. Use the result of the GET API as the input for the `PUT /api/v1/transport-nodes/<transport-node-id>` API, with the additional property, `failure_domain_id`, set appropriately. For example,

```
GET /api/v1/transport-nodes/<transport-node-id>
Response:
{
    "resource_type": "TransportNode",
    "description": "Updated NSX configured Test Transport Node",
    "id": "77816de2-39c3-436c-b891-54d31f580961",
    ...
}
```

```
PUT /api/v1/transport-nodes/<transport-node-id>
{
    "resource_type": "TransportNode",
    "description": "Updated NSX configured Test Transport Node",
    "id": "77816de2-39c3-436c-b891-54d31f580961",
    ...
    "failure_domain_id": "<UUID>",
}
```

3. Using the API, configure the NSX Edge cluster to allocate nodes based on failure domain. First call the `GET /api/v1/edge-clusters/<edge-cluster-id>` API to get the data about the Edge cluster. Use the result of the GET API as the input for the `PUT /api/v1/edge-clusters/<edge-cluster-id>` API, with the additional property, `allocation_rules` set appropriately. For example,

```
GET /api/v1/edge-clusters/<edge-cluster-id>
Response:
{
    "_revision": 0,
    "id": "bf8d4daf-93f6-4c23-af38-63f6d372e14e",
    "resource_type": "EdgeCluster",
    ...
}
```

```
PUT /api/v1/edge-clusters/<edge-cluster-id>
{
    "_revision": 0,
    "id": "bf8d4daf-93f6-4c23-af38-63f6d372e14e",
    "resource_type": "EdgeCluster",
    ...
    "allocation_rules": [
        {
            "action":
                {
                 "enabled": true,
                 "action_type": "AllocationBasedOnFailureDomain"
                }
        }
    ],
}
```

The NSX Edge nodes are referenced to different failure domains. You can now use them to create an NSX Edge cluster and configure a tier-0 gateway in active-standby stateful HA mode.

Configure Stateful Services on NSX Tier-0 and Tier-1 Gateways in Active-Active HA mode

Configure tier-0 and tier-1 gateways in active-active stateful high-availability (HA) mode on an NSX Edge cluster and enable stateful services.

- If there are odd number of NSX Edge nodes in the NSX Edge cluster, it leads to the scenario where one sub-cluster does not have a backup node. On failure of that single NSX Edge node, traffic is disrupted. NSX triggers an alarm that you must resolve to correctly configure stateful services. Ensure an NSX Edge cluster consists of an even number of NSX Edge nodes. For example, in an NSX Edge cluster of four nodes, NSX forms two sub-clusters, where each sub-cluster contains two nodes. One node in each sub-cluster is the backup node of the active NSX Edge node.
- Ensure the NSX Edge nodes you will use as part of the NSX Edge cluster are referenced to different failure domains.

The topology considered for this procedure uses tier-0 gateways and tier-1 gateways both in active-active stateful HA mode.

1. With admin privileges, log in to NSX Manager.
2. Go to **Networking** > **Tier-0 Gateways**.
3. From the **Add Gateway** drop-down menu, click **Tier-0**.
4. Enter the name of the tier-0 gateway.
5. In the HA Mode field, select **Active Active** and enable **Stateful**.

   Note:
   Once you enable the tier-0 gateway to be stateful, you cannot edit the HA mode.
6. Select the NSX Edge cluster and click **Save**.
7. Click **Yes** to continue to edit the Tier-0 gateway.
8. Expand the **Interfaces and Interface Groups** section and in the **External** field click **Set**.
9. In the **Set Interfaces** window, click **Add Interface**.
10. Enter the name, select the segment the interface is connected to and NSX Edge node. Enter any other optional details.
11. Click **Save** to complete adding the interface.
12. After you add interfaces, go to the **Interface Groups** field, click **Set**.
13. In the **Set Interface Groups** window, click **Add Interface Group**.

    Important:

    Create an interface group consisting of one external interface from each tier-0 gateway service router of the NSX Edge cluster. Ensure that one external interface from every service router belongs to the group and that each external interface belongs to only one group. Each interface participating in the interface group must be equivalent. External interfaces are considered equivalent when they are reachable on the network and when they share the same firewall, NAT and other network layer 4-7 policies.

    An interface group allows multiple segments to be grouped into a single group which is connected to an NSX Edge cluster.

    If the interface group does not have an external interface from each service router, then it can result in traffic loss. NSX triggers an alarm when this requirement is not met.
14. Click **Close Editing** to update the tier-0 gateway in stateful active-active HA mode.
15. After deploying the tier-0 gateway in stateful active-active HA mode, deploy tier-1 gateways in stateful active-active HA mode on the same NSX Edge cluster where the tier-0 gateway is configured. When you scale out or scale in a tier-0 gateway, which means new sub-clusters of NSX Edge nodes are added or removed, associated tier-1 gateways also follow the same behavior.
16. Create a locale service on tier-0 gateways.

```
PUT https://<policy-mgr>/policy/api/v1/infra/tier-0s/<tier-0-id>/locale-services/<locale_service>
 {
   "route_redistribution_types": [ "TIER0_STATIC", "TIER0_NAT" ],
   "edge_cluster_path": "/infra/sites/default/enforcement-point/nsx/edge-clusters/<95196903-6b8a-4276-
 a7c4-387263e834fd>",
   "preferred_edge_paths": [ "/infra/sites/default/enforcement-point/nsx/edge-clusters/<95196903-6b8a-4276-
 a7c4-387263e834fd>/edge-nodes/<940f1f4b-0317-45d4-84e2-b8c2394e7405>"],
   "_revision": 0
 }
```

17. Deploy tier-1 gateways in in stateful active-active HA mode and select the same NSX Edge cluster used for the previously configured tier-0 gateway in stateful active-active HA mode.
18. Create a locale service on tier-1 gateway.

    Without creating a locale service, the gateway is a DR-only gateway.

```
PUT https://<policy-mgr>/policy/api/v1/infra/tier-1s/cgw/locale-services/<locale_service>
 {
   "edge_cluster_path": "/infra/sites/default/enforcement-point/nsx/edge-clusters/<95196903-6b8a-4276-
 a7c4-387263e834fd>",
   "preferred_edge_paths": [ "/infra/sites/default/enforcement-point/nsx/edge-clusters/<95196903-6b8a-4276-
 a7c4-387263e834fd>/edge-nodes/<940f1f4b-0317-45d4-84e2-b8c2394e7405>"],
   "_revision": 0
 }
```

19. Create an SNAT rule for the service router on the Tier-0 A-A Stateful gateway. It is mandatory to enter the translated IP.
20. Go to **Networking** > **NAT** and click **Add NAT Rule**.
21. From the **Action** drop-down list, select **SNAT** and enter source and destination IP.
22. In the **Translated IP | Port** field, enter the IP that the source IP must be translated to.
23. Click **Save**.
24. Verify the HA status on tier-0/1 gateway service routers. Verify that a pair of NSX Edge nodes form a sub-cluster. Both are active. The peer node only takes over and processes traffic on the failure of the active node.

    On a Tier-0 node> `# get high-availability status`

```
Service Router
UUID            : 073a9fda-7a11-4d59-80c3-a7ea5371d265
state           : Active
type            : TIER0
mode            : Stateful A/A
failover mode   : Preemptive
rank            : 0
service count   : 0
service score   : 0
HA ports state
    UUID        : de647a80-d27c-46ee-a251-b35a3cead0d0
    op_state    : Up
    addresses   : 169.254.0.2/25;fe80::50:56ff:fe56:5300/64
Sub-cluster Information
    UUID            : c8db92e7-21da-453d-9853-2648849e7bda
    Peer SR UUID    : daaca25b-9028-4e31-b9b7-35bae481e60a
    Peer Node UUID  : 68668f1c-0330-11ec-84cf-00505682699c
Peer Routers
    Node UUID   : 9fe732b6-0330-11ec-ae4e-005056821b5a
    HA state    : Active
    Node UUID   : 8486560a-0330-11ec-902b-00505682411d
    HA state    : Active
    Node UUID   : 68668f1c-0330-11ec-84cf-00505682699c
    HA state    : Active
```

You can run stateful services on tier-0 gateways in stateful active-active HA mode.

- To scale out an NSX Edge cluster, add an even number of NSX Edge nodes.

  Note:

  If you add an odd number of NSX Edge nodes, the newly added NSX Edge node does not have a backup node. If the newly added NSX Edge node fails, then traffic is disrupted. NSX Manager raises an alarm if you add an odd number of NSX Edge nodes to an NSX Edge cluster.
- To scale in an NSX Edge cluster, remove an even number of NSX Edge nodes from the NSX Edge cluster.

Add a GRE Tunnel to a Tier-0 and Tier-0 VRF Gateway

You can add GRE (Generic Routing Encapsulation) tunnels for NSX tier-0 and tier-0 VRF gateways to connect on-premises and cloud networks.

For the tier-0 or tier-0 VRF gateway, ensure that:

- Either external or loopback interfaces are configured as the GRE tunnel source IP address (local endpoint).
- Routing is configured for reachability between the GRE source and remote endpoints.
- For BGP over GRE configuration, ensure that proper filters are installed on the BGP neighbor over GRE. Remote GRE endpoint network should not be learned over BGP. Otherwise, it results in a routing loop.

NSX supports the following GRE tunnel deployment options:

- Active-standby edge clusters
- Active-active stateless
- Static routes
- BGP
- BFD
- IPv4 and IPv6 workloads

There are some limitations for the GRE functionality in NSX:

- No support for stateful services over GRE (Layer 4/7).
- GRE is not supported on tier-0 gateways in stateful active-active HA mode.
- Multicast over GRE is not supported.
- NSX Federation is not supported.
- OSPF (Open Shortest Path First) over GRE is not supported.
- IPv6 addresses cannot be used for outer IP addresses of the GRE tunnel (source and destination IP addresses).
- PMTU discovery and MTU updates are not supported.
- TCP MSS clamping is not supported.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Tier-0 Gateways**.
3. Expand the **Interfaces and GRE Tunnels** category.
4. For **GRE Tunnels**, click **Set** and then **Add GRE Tunnel**.
5. Enter a name for the GRE tunnel.
6. Click **Set** for **Tunnel Addresses** and configure the tunnel addresses:
   a) Click **Add Tunnel Address**.
   b) Select an NSX Edge node.
   c) Select the source IP address of the GRE tunnel. This address is either an existing external interface or a previously configured loopback interface per NSX Edge node.
   d) Enter the subnets.
   e) Click **Add** and then **Apply**.
7. Enter the destination IP address for the GRE tunnel.
8. Optional: Configure the GRE keepalive settings:
   a) Use the **Keep Alive** toggle to enable or deactivate GRE keepalive signals to be periodically sent to the remote GRE endpoint.

      GRE keepalive signals are used to check the health of the stateless tunnel between the GRE tunnel endpoints and whether the GRE tunnel should remain up.

b) Use the **Keep Alive Ack** toggle to enable or deactivate GRE keepalive packets to be sent back to their source GRE endpoint.

After the decapsulation of incoming keepalive packets, the NSX Edge datapath sends back the keepalive packets to the source GRE endpoint with an acknowledgment (ack) packet. This setting is enabled by default.

c) Use the **Dead Multiplier** field to specify the number of consecutive failures or times when the GRE keepalive is not received before making the GRE tunnel to go down.

Note:
The minimum is 3 and the maximum is 5.

d) Use the **Keep Alive Timer** field to specify the number of seconds for the intervals at which GRE keepalive signals are sent.

Note:
The minimum is 2 and the maximum is 120.

9. Click **Save**.
10. Optional: After the GRE tunnel is created, you can view statistics for the GRE tunnel:
    a) Click **Tunnel Statistics** to view a summary of the received and transmitted traffic.
    b) Click **Keep Alive Statistics** to view details of the GRE keepalive signals and GRE keepalive acknowledgment, including timestamps.
    c) Click **Tunnel Connectivity Status** to view the connectivity related information.

Note:
To display the **Last Up Time** and **Last Down Time** data, the **Keep Alive** and **Keep Alive Ack** toggles must be enabled.

Tier-1 Gateway

A tier-1 gateway has downlink connections to segments and uplink connections to tier-0 gateways.

You can configure route advertisements and static routes on a tier-1 gateway. Recursive static routes are supported.

Add an NSX Tier-1 Gateway

A tier-1 gateway is typically connected to a tier-0 gateway in the northbound direction and to segments in the southbound direction.

- If you are adding a tier-1 gateway from Global Manager in NSX Federation, refer to Add a Tier-1 Gateway from Global Manager.
- If you plan to configure the gateway DHCP server, refer to Attach an NSX DHCP Profile to a Tier-0 or Tier-1 Gateway.

Tier-0 and tier-1 gateways support the following addressing configurations for all interfaces (external interfaces, service interfaces and downlinks) in both single tier and multi-tiered topologies:

- IPv4 only
- IPv6 only
- Dual Stack - both IPv4 and IPv6

To use IPv6 or dual stack addressing, enable **IPv4 and IPv6** as the L3 Forwarding Mode in **Networking** > **Networking Settings** > **Global Networking Config**.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Segment Connectivity** > **Tier-1 Gateways**.
3. Click **Add Tier-1 Gateway**.
4. Enter a name for the gateway.
5. In the **HA Mode** field, select a high availability (HA) mode: You can select from the following options:
   - Active Active: In the active-active stateful mode, traffic is load balanced across all members and includes support of stateful services such as NAT. This HA mode is available only when the linked tier-0 gateway is configured in stateful active-active mode. To use this HA mode, you must assign an NSX Edge cluster to the tier-1 gateway.
   - Active Standby: In the active-standby mode, traffic is load balanced by an elected active member. If the active member fails, a new member is elected as the active member. To use this HA mode, you must assign an NSX Edge cluster to the tier-1 gateway.
   - Distributed Only: Select this mode to configure a Distributed Router-only (DR) gateway. This HA mode does not require the assignment of an NSX Edge cluster to the tier-1 gateway. This HA mode uses the fully distributed routing architecture of NSX and is the recommended option unless you require stateful network services, such as NAT, on the tier-1 gateway.
6. Select a tier-0 gateway to connect to this tier-1 gateway to create a multi-tier topology.
7. Select an NSX Edge cluster if you want this tier-1 gateway to host stateful services such as NAT, load balancer, or firewall.

   If an NSX Edge cluster is selected, a service router will always be created (even if you do not configure stateful services), affecting the north/south traffic pattern.
8. After you select an NSX Edge cluster, a toggle gives you the option to select NSX Edge nodes.
9. If you selected an NSX Edge cluster, select a failover mode or accept the default.

Option	Description
Preemptive	If the preferred NSX Edge node fails and recovers, it will preempt its peer and become the active node. The peer will change its state to standby.
Non-preemptive	If the preferred NSX Edge node fails and recovers, it will check if its peer is the active node. If so, the preferred node will not preempt its peer and will be the standby node. This is the default option.

10. Optional: Add **DHCP Config** on the gateway.
11. If you plan to configure a load balancer on this gateway, select an **Edges Pool Allocation Size** setting according to the size of the load balancer.

    The options are **Routing**, **LB Small**, **LB Medium**, **LB Large**, and **LB XLarge**. The default is **Routing** and is suitable if no load balancer will be configured on this gateway. This parameter allows the NSX Manager to place the tier-1 gateway on the Edge nodes in a more intelligent way. With this setting the number of load balancing and routing functions on each node is taken into consideration. Note that after you create the gateway, you can change this setting if you have not configured a load balancer.
12. Optional: Click the **Enable StandBy Relocation** toggle to enable or disable standby relocation.

    Standby relocation means that if the Edge node where the active or standby logical router is running fails, a new standby logical router is created on another Edge node to maintain high availability. If the Edge node that fails is running the active logical router, the original standby logical router becomes the active logical router and a new standby logical router is created. If the Edge node that fails is running the standby logical router, the new standby logical router replaces it.
13. Optional: Click **Route Advertisement**.

    Select one or more of the following:
    - **All Static Routes**
    - **All NAT IP's**
    - **All DNS Forwarder Routes**

- **All LB VIP Routes**
- **All Connected Segments and Service Ports**
- **All LB SNAT IP Routes**
- **All IPSec Local Endpoints**

14. Click **Save**.
15. Optional: Click **Route Advertisement**.
    a) In the **Set Route Advertisement Rules** field, click **Set** to add route advertisement rules.
16. Optional: Click **Additional Settings**.
    a) For IPv6, you can select or create an **ND Profile** and a **DAD Profile**.

       These profiles are used to configure Stateless Address Autoconfiguration (SLAAC) and Duplicate Address Detection (DAD) for IPv6 addresses.
    b) Select an **Ingress QoS Profile** and an **Egress QoS Profile** for traffic limitations.

       These profiles are used to set information rate and burst size for permitted traffic. See Add a Gateway QoS Profile for more information on creating QoS profiles.

    If this gateway is linked to a tier-0 gateway, the **Router Links** field shows the link addresses.
17. Optional: Click **Service Interfaces** and **Set** to configure connections to segments. Required in some topologies such as VLAN-backed segments or one-arm load balancing.
    a) Click **Add Interface**.
    b) Enter a name and IP address in CIDR format.

       If you configure multicast on this gateway, you must not configure tier-1 addresses as static RP address in the PIM profile.
    c) Select a segment.
    d) In the **MTU** field, enter a value between 64 and 9000.
    e) For **URPF Mode**, you can select **Strict** or **None**.

       URPF (Unicast Reverse Path Forwarding) is a security feature.
    f) Add one or more tags.
    g) In the **ND Profile** field, select or create a profile.
    h) Click **Save**.
    i) Optional: After you create an interface, you can download the ARP proxies for the gateway by clicking the menu icon (three dots) for the interface and selecting **Download ARP Proxies**.

       You can also download the ARP proxy for a specific interface by expanding a gateway and then expanding **Service Interfaces**. Click an interface and click the menu icon (three dots) and select **Download ARP Proxy**.
18. Optional: Click **Static Routes** and **Set** to configure static routes.
    a) Click **Add Static Route**.
    b) Enter a name and a network address in the CIDR or IPv6 CIDR format.
    c) Click **Set Next Hops** to add next hop information.
    d) Click **Save**.
19. Optional: Click **Multicast** and then the toggle to enable multicast.

    You must select an Edge cluster for this gateway. Also, this gateway must be linked to a tier-0 gateway that has multicast enabled.

20. Optional: If the tier-1 gateway is connected to a tier-0 gateway, you can download the ARP table of the tier-0 gateway. Do the following:
    a) Click the tier-0 gateway from the **Linked Tier-0 Gateway** column.
    b) Click the menu icon (3 dots) and select **Download ARP Table**.
    c) Select an edge node.
    d) Click **Download** to save the .CSV file.

The new gateway is added to the list. For any gateway, you can modify its configurations by clicking the menu icon (3 dots) and select **Edit**. To reconfigure service interfaces or static routes, you do not need to click **Edit**. You only need to click the expand icon (right arrow) for the gateway, expand the **Service Interfaces** or **Static Routes** section, and click the number that is shown. Note that the number must be non-zero. If it is zero, you must edit the gateway.

If NSX Federation is configured, this feature of reconfiguring a gateway by clicking on an entity is applicable to gateways created by the Global Manager (GM) as well. Note that some entities in a GM-created gateway can be modified by the Local Manager, but others cannot. For example, **Static Routes** of a GM-created gateway cannot be modified by the Local Manager. Also, from the Local Manager, you can edit existing **Service Interfaces** of a GM-created gateway but you cannot add an interface.

State Synchronization of Tier-1 Gateways

Connection information of the traffic running on a given tier-1 SR (Service Router) is synchronized to its peer tier-1 SR in active-standby or stateful active-active HA modes.

The synchronization between the Active SR and Standby SR is triggered whenever there is a change in the local state of TCP SYN packet, after a certain interval, or if a local state is created or deleted. During synchronization, Edge uses proprietary protocol to replicate CCP packets between Active and Standby SRs.

> **Note:**
> State synchronization is not supported for TLS Inspection and IDPS.

Note the following about state synchronization:

- State synchronization is supported for Gateway Firewall, Identity Firewall, NAT, IPSec VPN, DHCP, FQDN analysis, and URL filtering.
- If new sessions were going through a tier-1 SR just before a failover, it might happen that those sessions were not synchronized on the associated tier-1 SR and potentially affect the traffic for those sessions.
- In active-standby mode, state synchronization is supported for Gateway Firewall, Identity Firewall, NAT, IPSec VPN, DHCP, FQDN analysis, and URL filtering.
- In active-active mode, state synchronization is supported for Gateway Firewall, Identity Firewall, NAT, FQDN analysis, and URL filtering. FQDN analysis is only supported with a single sub-cluster. IPSec VPN is not supported.
- If new sessions were going through a tier-1 SR just before a failover, it might happen that those sessions were not synchronized on the associated tier-1 SR and potentially affect the traffic for those sessions.

Segments

In NSX, segments are virtual layer 2 domains. A segment was earlier called a logical switch.

There are two types of segments in NSX:

- VLAN-backed segments
- Overlay-backed segments

A VLAN-backed segment is a layer 2 broadcast domain that is implemented as a traditional VLAN in the physical infrastructure. This means that traffic between two VMs on two different hosts but attached to the same VLAN-backed segment is carried over a VLAN between the two hosts. The resulting constraint is that you must provision an appropriate VLAN in the physical infrastructure for those two VMs to communicate at layer 2 over a VLAN-backed segment.

In an overlay-backed segment, traffic between two VMs on different hosts but attached to the same overlay segment have their layer 2 traffic carried by a tunnel between the hosts. NSX instantiates and maintains this IP tunnel without the need for any segment-specific configuration in the physical infrastructure. As a result, the virtual network infrastructure

is decoupled from the physical network infrastructure. That is, you can create segments dynamically without any configuration of the physical network infrastructure.

The default number of MAC addresses learned on an overlay-backed segment is 2048. The default MAC limit per segment can be changed through the API field `remote_overlay_mac_limit` in `MacLearningSpec`.

For more information see the `MacSwitchingProfile` in the *NSX API Guide*.

Segment Profiles

Segment profiles include Layer 2 networking configuration details for segments and segment ports. NSX Manager supports several types of segment profiles.

The following types of segment profiles are available:

- QoS (Quality of Service)
- IP Discovery
- Spoof Guard
- Segment Security
- MAC Discovery

> **Note:**
> You cannot edit or delete the default segment profiles. If you require alternate settings from what is in the default segment profile you can create a custom segment profile. By default all custom segment profiles except the segment security profile will inherit the settings of the appropriate default segment profile. For example, a custom IP discovery segment profile by default will have the same settings as the default IP discovery segment profile.

Each default or custom segment profile has a unique identifier. You use this identifier to associate the segment profile to a segment or a segment port.

A segment or segment port can be associated with only one segment profile of each type. You cannot have, for example, two QoS segment profiles associated with a segment or segment port.

If you do not associate a segment profile when you create a segment, then the NSX Manager associates a corresponding default system-defined segment profile. The children segment ports inherit the default system-defined segment profile from the parent segment.

When you create or update a segment or segment port you can choose to associate either a default or a custom segment profile. When the segment profile is associated or disassociated from a segment the segment profile for the children segment ports is applied based on the following criteria.

- If the parent segment has a profile associated with it, the child segment port inherits the segment profile from the parent.
- If the parent segment does not have a segment profile associated with it, a default segment profile is assigned to the segment and the segment port inherits that default segment profile.
- If you explicitly associate a custom profile with a segment port, then this custom profile overrides the existing segment profile.

> **Note:**
> If you have associated a custom segment profile with a segment, but want to retain the default segment profile for one of the child segment port, then you must make a copy of the default segment profile and associate it with the specific segment port.

You cannot delete a custom segment profile if it is associated to a segment or a segment port. You can find out whether any segments and segment ports are associated with the custom segment profile by going to the Assigned To section of the Summary view and clicking on the listed segments and segment ports.

Understanding QoS Segment Profile

QoS provides high-quality and dedicated network performance for preferred traffic that requires high bandwidth. The QoS mechanism does this by prioritizing sufficient bandwidth, controlling latency and jitter, and reducing data loss for preferred

packets even when there is a network congestion. This level of network service is provided by using the existing network resources efficiently.

How QoS Segment Profiles Work

For this feature, shaping and traffic marking -- namely, CoS and DSCP -- is supported. The Layer 2 Class of Service (CoS) allows you to specify priority for data packets when traffic is buffered in the segment due to congestion. The Layer 3 Differentiated Services Code Point (DSCP) detects packets based on their DSCP values. CoS is always applied to the data packet irrespective of the trusted mode.

NSX trusts the DSCP setting applied by a virtual machine or modifying and setting the DSCP value at the segment level. In each case, the DSCP value is propagated to the outer IP header of encapsulated frames. This enables the external physical network to prioritize the traffic based on the DSCP setting on the external header. When DSCP is in the trusted mode, the DSCP value is copied from the inner header. When in the untrusted mode, the DSCP value is not preserved for the inner header.

You can use the QoS segment profile to configure the average ingress and egress bandwidth values to set the transmit limit rate. The peak bandwidth rate is used to support burst traffic a segment is allowed to prevent congestion on the northbound network links. These settings do not guarantee the bandwidth but help limit the use of network bandwidth. The actual bandwidth you will observe is determined by the link speed of the port or the values in the switching profile, whichever is lower.

The QoS segment profile settings are applied to the segment and inherited by the child segment port.

Limitations

- QoS segment policies do not have any effect on tier-1 gateway service interfaces, even when the interface is connected to a segment with a QoS segment profile applied.
- DSCP settings work only on tunneled traffic. These settings do not apply to traffic inside the same hypervisor.

Create an NSX QoS Segment Profile

You can define the DSCP value and configure the ingress and egress settings to create a custom QoS switching profile.

- Familiarize yourself with the QoS switching profile concept. See Understanding QoS Segment Profile.
- Identify the network traffic you want to prioritize.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Segments** > **Segment Profiles**.
3. Click **Add Segment Profile** to add a new profile.
4. Click
   
   and click **Edit**.
5. Complete the QoS switching profile details.

Option	Description
Name	Name of the profile.
Mode	Select either a **Trusted** or **Untrusted** option from the Mode drop-down menu. When you select the Trusted mode the inner header DSCP value is applied to the outer IP header for IP/IPv6 traffic. For non IP/IPv6 traffic, the outer IP header takes the default value. Trusted mode is supported on an overlay-based logical port. The default value is 0. Untrusted mode is supported on overlay-based and VLAN-based logical port. For the overlay-based logical port, the DSCP value of the outbound IP header is set to the configured value

Option	Description
	irrespective to the inner packet type for the logical port. For the VLAN-based logical port, the DSCP value of IP/IPv6 packet will be set to the configured value. The DSCP values range for untrusted mode is between 0 to 63. **Note:** DSCP settings work only on tunneled traffic. These settings do not apply to traffic inside the same hypervisor.
Priority	Set the DSCP priority value. The priority values range from 0 to 63.
Class of Service	Set the CoS value. CoS is supported on VLAN-based logical port. CoS groups similar types of traffic in the network and each type of traffic is treated as a class with its own level of service priority. The lower priority traffic is slowed down or in some cases dropped to provide better throughput for higher priority traffic. CoS can also be configured for the VLAN ID with zero packet. The CoS values range from 0 to 7, where 0 is the best effort service.
Ingress	Set custom values for the outbound network traffic from the VM to the logical network. You can use the average bandwidth to reduce network congestion. The peak bandwidth rate is used to support burst traffic and the burst size is based on the duration with peak bandwidth. You set burst duration in the burst size setting. You cannot guarantee the bandwidth. However, you can use the Average, Peak, and Burst Size settings to limit network bandwidth. For example, if the average bandwidth is 30 Mbps, peak bandwidth is 60 Mbps, and the allowed duration is 0.1 second, then the burst size is 60 * 1000000 * 0.10/8 = 750000 Bytes. The default value 0 disables rate limiting on the ingress traffic.
Ingress Broadcast	Set custom values for the outbound network traffic from the VM to the logical network based on broadcast. For example, when you set the average bandwidth for a logical switch to 3000 Kbps, peak bandwidth is 6000 Kbps, and the allowed duration is 0.1 second, then the burst size is 6000 * 1000 * 0.10/8 = 75000 Bytes. The default value 0 disables rate limiting on the ingress broadcast traffic.
Egress	Set custom values for the inbound network traffic from the logical network to the VM. The default value 0 disables rate limiting on the egress traffic.

If the ingress, ingress broadcast, and egress options are not configured, the default values are used.

6. Click **Save**.

Understanding IP Discovery Segment Profile

IP Discovery uses DHCP and DHCPv6 snooping, ARP (Address Resolution Protocol) snooping, ND (Neighbor Discovery) snooping, and VM Tools to learn MAC and IP addresses.

Note:
IP discovery methods for IPv6 are disabled in the default IP discovery segment profile. To enable IP discovery for IPv6 for segments, you must create an IP discovery profile with the IPv6 options enabled and attach the profile to the segments. In addition, make sure that distributed firewall allows IPv6 Neighbor Discovery packets between all workloads (allowed by default).

The discovered MAC and IP addresses are used to achieve ARP/ND suppression, which minimizes traffic between VMs connected to the same segment. The number of IPs in the ARP/ND suppression cache for any given port is determined by the settings in the port's IP Discovery profile. The relevant settings are ARP Binding Limit, ND Snooping Limit, Duplicate IP Detection, ARP ND Binding Limit Timeout, and Trust on First Use (TOFU).

The discovered MAC and IP addresses are also used by the SpoofGuard and distributed firewall (DFW) components. DFW uses the address bindings to determine the IP address of objects in firewall rules.

DHCP/DHCPv6 snooping inspects the DHCP/DHCPv6 packets exchanged between the DHCP/DHCPv6 client and server to learn the IP and MAC addresses.

ARP snooping inspects the outgoing ARP and GARP (gratuitous ARP) packets of a VM to learn the IP and MAC addresses.

VM Tools is software that runs on an ESX-hosted VM and can provide the VM's configuration information including MAC and IP or IPv6 addresses. This IP discovery method is available for VMs running on ESX hosts only.

ND snooping is the IPv6 equivalent of ARP snooping. It inspects neighbor solicitation (NS) and neighbor advertisement (NA) messages to learn the IP and MAC addresses.

Duplicate address detection checks whether a newly discovered IP address is already present on the realized binding list for a different port. This check is performed for ports on the same segment. If a duplicate address is detected, the newly discovered address is added to the discovered list, but is not added to the realized binding list. All duplicate IPs have an associated discovery timestamp. If the IP that is on the realized binding list is removed, either by adding it to the ignore binding list or by disabling snooping, the duplicate IP with the oldest timestamp is moved to the realized binding list. The duplicate address information is available through an API call.

By default, the discovery methods ARP snooping and ND snooping operate in a mode called trust on first use (TOFU). In TOFU mode, when an address is discovered and added to the realized bindings list, that binding remains in the realized list forever. TOFU applies to the first 'n' unique <IP, MAC, VLAN> bindings discovered using ARP/ND snooping, where 'n' is the binding limit that you can configure. You can disable TOFU for ARP/ND snooping. The methods will then operate in trust on every use (TOEU) mode. In TOEU mode, when an address is discovered, it is added to the realized bindings list and when it is deleted or expired, it is removed from the realized bindings list. DHCP snooping and VM Tools always operate in TOEU mode.

When using the default IP discovery profile (TOFU enabled), IP address changes are not allowed.

Note:

TOFU is not the same as SpoofGuard, and it does not block traffic in the same way as SpoofGuard. For more information, see Understanding SpoofGuard Segment Profile.

For Linux VMs, the ARP flux problem might cause ARP snooping to obtain incorrect information. The problem can be prevented with an ARP filter.

For each port, NSX Manager maintains an ignore bindings list, which contains IP addresses that cannot be bound to the port.

Create an NSX IP Discovery Segment Profile

NSX has several default IP Discovery segment profiles. You can also create additional ones.

Familiarize yourself with the IP Discovery segment profile concepts. See Understanding IP Discovery Segment Profile.

IP Discovery is the central infrastructure in NSX which determines the set of IP addresses that are associated with a port in the system. IP Discovery policies are applied via the Segment IP Discovery Profile which is configurable from the Policy Manager. It can be associated with a segment, segment port or a group. See Configure IP Discovery Segment Profile on

Groups. When a segment or segment port is created, it is initially assigned a Default Segment IP Discovery Profile with a predefined set of policies.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Segments** > **Segment Profiles**.
3. Click **Add Segment Profile** and select **IP Discovery**.
4. Specify the IP Discovery segment profile details.

Option	Description
Name	Enter a name.
ARP Snooping	For an IPv4 environment. Applicable if VMs have static IP addresses.
ARP Binding Limit	The maximum number of IPv4 IP addresses that can be bound to a port. The minimum value allowed is 1 and the maximum is 256. The default is 1.
ARP ND Binding Limit Timeout	The timeout value, in minutes, for IP addresses in the ARP/ ND binding table if TOFU is disabled. If an address times out, a newly discovered address replaces it.
DHCP Snooping	For an IPv4 environment. Applicable if VMs have IPv4 addresses.
DHCP Snooping - IPv6	For an IPv6 environment. Applicable if VMs have IPv6 addresses.
VM Tools	Available for ESX-hosted VMs only.
VM Tools - IPv6	Available for ESX-hosted VMs only.
ND Snooping	For an IPv6 environment. Applicable if VMs have static IP addresses.
ND Snooping Limit	The maximum number of IPv6 addresses that can be bound to a port.
Trust on First Use	Applicable to ARP and ND snooping.
Duplicate IP Detection	For all snooping methods and both IPv4 and IPv6 environments.

5. Click **Save**.

Configure IP Discovery Segment Profile on Groups

Configuring IP Discovery segment profiles on a group allows a Security Administrator to configure IP discovery profile parameters, and apply them to group members.

Configuring The following static and dynamic group members are supported:

- Segment
- Segment Port
- VM
- Groups
- Mix of the above

Profiles on groups only apply if the default profile is applied to the segment or segment port:

Custom Group Profile	Custom Profile on Segment (S) and Segment Port(SP)	Effective Profile on Port
Custom	Default (S), Default (SP)	Custom
Custom 1	Default (S), Custom 2 (SP)	Custom 2
Custom 1	Custom 2 (S), Default (SP)	Custom 2
Custom 1	Custom 2 (S), Custom 3 (SP)	Custom 3

Each time a profile is applied to a group, a sequence number is specified. If a member is present in multiple groups, the group with the lower sequence number has higher priority.

Discovery Profile Binding Map API

Method	API	Resource Type
PUT, PATCH, GET, DELETE	`/infra/domains/<domain-id>/groups/<group-id>/discovery-profile-binding-maps/<binding-map-id>`	DiscoveryProfileBindingMap
GET	`/infra/domains/<domain-id>/groups/<group-id>/discovery-profile-binding-maps`	DiscoveryProfileBindingMapListResult

Parameters for DiscoveryProfileBindingMap

Field	Type	Description
profile_path	Policy Path	Required
sequence_number	Integer	Required. Sequence number is used to resolve conflicts when two profiles are applied to the same segment or segment port. The low sequence number has higher precedence.

API for Segments and Ports

Method	API	Resource Type

GET	`/infra/tier-1s/<tier-1-id>/segments/<segment-id>/effective-profiles`
	`/infra/segments/<segment-id>/effective-profiles`
	`/infra/segments/<segment-id>/effective-profiles`
	`/infra/segments/<segment-id>/ports/<port-id>`
		EffectiveProfilesResponse

Example Request

POST https://{{policy-ip}}/policy/api/v1/infra/domains/default/groups/TestGroup/discovery-profile-binding-maps/ipdmap

```
{
    "profile_path" : "/infra/ip-discovery-profiles/ip-discovery-custom-profile-1",
    "sequence_number" :"10"
}
```

Understanding SpoofGuard Segment Profile

SpoofGuard helps prevent a form of malicious attack called "web spoofing" or "phishing." A SpoofGuard policy blocks traffic determined to be spoofed.

SpoofGuard is a tool that is designed to prevent virtual machines in your environment from sending traffic with an IP address it is not authorized to send traffic from. In the instance that a virtual machine's IP address does not match the IP address on the corresponding logical port and segment address binding in SpoofGuard, the virtual machine's vNIC

is prevented from accessing the network entirely. SpoofGuard can be configured at the port or segment level. There are several reasons SpoofGuard might be used in your environment:

- Preventing a rogue virtual machine from assuming the IP address of an existing VM.
- Ensuring the IP addresses of virtual machines cannot be altered without intervention – in some environments, it's preferable that virtual machines cannot alter their IP addresses without proper change control review. SpoofGuard facilitates this by ensuring that the virtual machine owner cannot simply alter the IP address and continue working unimpeded.
- Guaranteeing that distributed firewall (DFW) rules will not be inadvertently (or deliberately) bypassed – for DFW rules created utilizing IP sets as sources or destinations, the possibility always exists that a virtual machine could have its IP address forged in the packet header, thereby bypassing the rules in question.

NSX SpoofGuard configuration covers the following:

- MAC SpoofGuard - authenticates MAC address of packet
- IP SpoofGuard - authenticates MAC and IP addresses of packet
- Dynamic Address Resolution Protocol (ARP) inspection, that is, ARP and Gratuitous Address Resolution Protocol (GARP) SpoofGuard and Neighbor Discovery (ND) SpoofGuard validation are all against the MAC source, IP Source and IP-MAC source mapping in the ARP/GARP/ND payload.

At the port level, the allowed MAC/VLAN/IP allow-list is provided through the Address Bindings property of the port. When the virtual machine sends traffic, it is dropped if its IP/MAC/VLAN does not match the IP/MAC/VLAN properties of the port. The port level SpoofGuard deals with traffic authentication, i.e. is the traffic consistent with VIF configuration.

At the segment level, the allowed MAC/VLAN/IP allow-list is provided through the Address Bindings property of the segment. This is typically an allowed IP range/subnet for the segment and the segment level SpoofGuard deals with traffic authorization.

Traffic must be permitted by port level AND segment level SpoofGuard before it will be allowed into segment. Enabling or disabling port and segment level SpoofGuard, can be controlled using the SpoofGuard segment profile.

Create an NSX Spoof Guard Segment Profile

When SpoofGuard is configured, if the IP address of a virtual machine changes, traffic from the virtual machine may be blocked until the corresponding configured port/segment address bindings are updated with the new IP address.

Enable SpoofGuard for the port group(s) containing the guests. When enabled for each network adapter, SpoofGuard inspects packets for the prescribed MAC and its corresponding IP address.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Segments** > **Segment Profiles**.
3. Click **Add Segment Profile** and select **Spoof Guard**.
4. Enter a name.
5. To enable port level SpoofGuard, set **Port Bindings** to **Enabled**.
6. Click **Save**.

Understanding Segment Security Segment Profile

Segment security provides stateless Layer2 and Layer 3 security by checking the ingress traffic to the segment and dropping unauthorized packets sent from VMs by matching the IP address, MAC address, and protocols to a set of allowed addresses and protocols. You can use segment security to protect the segment integrity by filtering out malicious attacks from the VMs in the network.

Note that the default segment security profile has the DHCP settings `Server Block` and `Server Block - IPv6` enabled. This means that a segment that uses the default segment security profile will block traffic from a DHCP server to a DHCP client. If you want a segment that allows DHCP server traffic, you must create a custom segment security profile for the segment.

Create a Segment Security Segment Profile

You can create a custom segment security segment profile if the settings of the default profile do not meet your needs.

Familiarize yourself with the segment security segment profile concept. See Understanding Segment Security Segment Profile.

> **Note:**
> All the features described on this page are only applicable to the ports where workloads are connected. They are not applicable to NSX Edge interfaces.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Segments** > **Segment Profiles**.
3. Click **Add Segment Profile** and select **Segment Security**.
4. Complete the segment security profile details.

Option	Description
Name	Name of the profile.
BPDU Filter	Toggle the **BPDU Filter** button to enable BPDU filtering. Disabled by default. When the BPDU filter is enabled, all of the traffic to BPDU destination MAC address is blocked. The BPDU filter when enabled also disables STP on the logical switch ports because these ports are not expected to take part in STP.
BPDU Filter Allow List	Click the destination MAC address from the BPDU destination MAC addresses list to allow traffic to the permitted destination. You must enable **BPDU Filter** to be able to select from this list.
DHCP Filter	Toggle the **Server Block** button and **Client Block** button to enable DHCP filtering. Both are disabled by default. DHCP Server Block blocks traffic from a DHCP server to a DHCP client. Packets whose UDP destination port number is 68 are blocked. Note that it does not block traffic from a DHCP server to a DHCP relay agent and DHCP Server replying to a DHCP relay agent must have DHCP Client Block disabled. DHCP Client Block prevents a VM from acquiring a DHCP IP address by blocking DHCP requests. Packets whose UDP destination port number is 67 are blocked.
DHCPv6 Filter	Toggle the **Server Block - IPv6** button and **Client Block - IPv6** button to enable DHCP filtering. Both are disabled by default. DHCPv6 Server Block blocks traffic from a DHCPv6 server to a DHCPv6 client. Packets whose UDP destination port number is 546 are blocked. Note that it does not block traffic from a DHCPv6 server to a DHCPv6 relay agent and DHCPv6 Server replying to a DHCPv6 relay agent must have DHCPv6 Client Block disabled. DHCPv6 Client Block prevents a VM from acquiring a DHCPv6 IP address by blocking DHCPv6 requests. Packets whose UDP destination port number is 547 are blocked.
Block Non-IP Traffic	Toggle the **Block Non-IP Traffic** button to allow only IPv4, IPv6, ARP, and BPDU traffic. The rest of the non-IP traffic is blocked. The permitted IPv4, IPv6, ARP, GARP and BPDU traffic is based on other policies set in address binding and SpoofGuard configuration. By default, this option is disabled to allow non-IP traffic to be handled as regular traffic.
RA Guard	Toggle the **RA Guard** button to filter out ingress IPv6 router advertisements. ICMPv6 type 134 packets are filtered out. This option is enabled by default.
Rate Limits	Set a rate limit for broadcast and multicast traffic. This option is enabled by default. Rate limits can be used to protect the workloads and VMs from events such as broadcast storms. To avoid any connectivity problems, the minimum rate limit value must be >= 10 pps.

5. Click **Save**.

Understanding MAC Discovery Segment Profile

The MAC discovery segment profile supports two functionalities: MAC learning and MAC address change.

The MAC address change feature allows a VM to change its MAC address. A VM connected to a port can run an administrative command to change the MAC address of its vNIC and still send and receive traffic on that vNIC. This feature is supported on ESX only. In the default MAC discovery segment profile, this property is enabled.

MAC learning provides network connectivity to deployments where multiple MAC addresses get configured behind one vNIC, for example, in a nested hypervisor deployment where an ESX VM runs on an ESX host and multiple VMs run inside the ESX VM. Without MAC learning, when the vNIC of the ESX VM connects to a segment port, its MAC address is static. VMs running inside the ESX VM do not have network connectivity because their packets have different source MAC addresses. With MAC learning, the vSwitch inspects the source MAC address of every packet coming from the vNIC, learns the MAC address and allows the packet to proceed. If a MAC address that is learned is not used for a certain period of time, it is removed. This time period is not configurable. The field **MAC Learning Aging Time** displays the pre-defined value, which is 600.

MAC Learning will not learn a MAC address if it is already a known static MAC address on the host on the same VLAN or VNI. For example, the MAC address belongs to another VM's vNIC, a vmknic, or a VDR (virtual distributed router) port.

Note: A VDR port is always configured to send and receive traffic on any possible VNI (similar to how a trunk VLAN port behaves when it is configured on 0-4094). So the usage of a VDR port MAC address on any overlay segment through MAC learning is not possible.

MAC learning also supports unknown unicast flooding. Normally, when a packet that is received by a port has an unknown destination MAC address, the packet is dropped. With unknown unicast flooding enabled, the port floods unknown unicast traffic to every port on the switch that has MAC learning and unknown unicast flooding enabled. This property is enabled by default, but only if MAC learning is enabled.

The number of MAC addresses that can be learned is configurable. The maximum value is 4096, which is the default. You can also set the policy for when the limit is reached. The options are:

- **Drop** - Packets from an unknown source MAC address are dropped. Packets inbound to this MAC address will be treated as unknown unicast. The port will receive the packets only if it has unknown unicast flooding enabled.
- **Allow** - Packets from an unknown source MAC address are forwarded although the address will not be learned. Packets inbound to this MAC address will be treated as unknown unicast. The port will receive the packets only if it has unknown unicast flooding enabled.

If you enable MAC learning or MAC address change, to improve security, configure SpoofGuard as well.

Create an NSX MAC Discovery Segment Profile

You can create a MAC discovery segment profile to manage MAC addresses.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Segments** > **Segment Profiles**.
3. Click **Add Segment Profile** and select **MAC Discovery**.
4. Complete the MAC discovery profile details.

Option	Description
Name	Name of the profile.
MAC Change	Enable or disable the MAC address change feature. The default is disabled.
MAC Learning	Enable or disable the MAC learning feature. The default is disabled.
MAC Limit Policy	Select **Allow** or **Drop**. The default is **Allow**. This option is available if you enable MAC learning
Unknown Unicast Flooding	Enable or disable the unknown unicast flooding feature. The default is enabled. This option is available if you enable MAC learning
MAC Limit	Set the maximum number of MAC addresses. The default is 4096. This option is available if you enable MAC learning
MAC Learning Aging Time	For information only. This option is not configurable. The pre-defined value is 600.

5. Click **Save**.

Understanding Industrial vSwitch Segment Profile

The Industrial vSwitch (IvS) is an EDP (Enhanced Data Path) Dedicated mode switch that can facilitate real-time PROFINET communication between PROFINET devices in an industrial Ethernet network.

The IvS aims to deliver deterministic, low-latency, and highly available connectivity between vPLCs and IO devices in industrial networks. It is not meant to support high-performance, high-throughput workloads and networks.

The IvS Segment Profile can only be enabled on VLAN Transport Zones. It can be configured to enable support for PROFINET protocol on vNICs connected to applications that require PROFINET and PROFINET-RT, such as virtual programmable logic controllers (vPLCs). VLANs/segments configured with the IvS Segment Profile, can have the following capabilities:

- Allows for the vPLCs to be discovered by other PROFINET devices using Discovery and Basic Configuration Protocol (DCP).
- Enables the option to retain Priority Code Point (PCP) value from the 802.1Q VLAN tag in PROFINET RT packets sent by the vPLCs and PROFINET IO devices.
- Enables packet latency measurement for TX/RX traffic between the vNICs and the physical uplinks, and therefore allows the administrator to monitor and alert on the critical latency metrics for PROFINET traffic to and from the vPLCs.
- Enables Parallel Redundancy Protocol (PRP) to be configured on the segment and provides for the vPLC to behave like a vDAN in a PRP domain.

> **Note:**
>
> The default IvS Segment Profile has all settings disabled and does not impact the operations of vNICs attached with such profile.

Create an Industrial Virtual Switch Segment Profile

You can create an Industrial vSwitch (IvS) Segment Profile to support these switch types.

For a complete description of configuring an IvS for NSX, see Configure an Industrial vSwitch.

1. With admin privileges, log in to the NSX Manager.
2. Navigate to **Networking** > **Segments** > **Profiles** > **Segment Profiles**.
3. Click **Add Segment Profile** and select **Industrial vSwitch** and enter Segment Profile name.
4. Complete the segment IvS profile details:

Options	Description
Parallel Redundancy Protocol	Enables PRP for the Segment and allows the connected vNICs to behave as vDANs.
Latency Measurement	Turns on or off the vNIC to the uplink latency measurement. This requires the vmxnet driver version 1.9 or newer.
Sampling Rate (Number of packets)	Indicates the frequency to sample packets for latency measurement. A value of 10 means to sample every 10th packet. The default Sampling Rate is 1.
Full Network Path Latency	Set to On for full network path latency. If set to Off, then latency through the guest OS is not measured.
Retain VLAN Priority Field	If set to On, the priority field or Priority Code Point (PCP) priority tag will be retained after VLAN operation.

Options	Description
DCP Multicast Address	Discovery and Configuration Protocol (DCP) is part of the PROFINET protocol suite. It is used to discover IO devices, Identity device information, and configure device settings, such as the device name and IP address. The default configuration uses a couple of multicast macs - 01:0e:cf:00:00:00, 01:0e:cf:00:00:01. If non-empty set MAC addresses are provided as input, then it will override these default multicast MACs.

5. Click **Save**.

Create an NSX Segment

You can add two kinds of segments: overlay-backed segments and VLAN-backed segments.

Segments are created as part of a transport zone. There are two types of transport zones: VLAN transport zones and overlay transport zones. A segment created in a VLAN transport zone is a VLAN-backed segment, and a segment created in an overlay transport zone is an overlay-backed segment.

- Starting with NSX 3.1.1, version 4 DHCP relay is supported on a VLAN-backed segment through the Service Interface. Only one DHCP v4 relay or service is supported on a segment.
- For a standalone segment that is not connected to a gateway, only local DHCP server is supported.
- For a VLAN segment requiring DHCP server, only local DHCP server is supported. Gateway DHCP is not supported on VLAN.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Segment Connectivity** > **Tier-1 Gateways**.
3. Click **Add Segment**.
4. Click **Add Segment**.
5. Enter a name for the segment.
6. Select the type of connectivity for the segment.

Connectivity	Description
None	Select this option when you do not want to connect the segment to any upstream gateway (tier-0 or tier-1). Typically, you want to add a standalone segment in the following scenarios: • When you want to create a local testing environment for users that are running workloads on the same subnet. • When east-west connectivity with users on the other subnets is not necessary. • When north-south connectivity to users outside the data center is not necessary. • When you want to configure guest VLAN tagging.
Tier-1	Select this option when you want to connect the segment to a tier-1 gateway.
Tier-0	Select this option when you want to connect the segment to a tier-0 gateway.

Note:
You can change the connectivity of a gateway-connected segment from one gateway to another gateway (same or different gateway type). In addition, you can change the connectivity of segment from "None" to a tier-0 or tier-1 gateway. The segment connectivity changes are permitted only when the gateways and the connected segments are in the same transport zone. However, if the segment has DHCP configured on it, some restrictions and caveats apply on changing the segment connectivity. For more information, see Scenarios: Impact of Changing Segment Connectivity on NSX DHCP.

7. Enter the Gateway IP address of the subnet in a CIDR format. A segment can contain an IPv4 subnet, or an IPv6 subnet, or both.
   - If a segment is not connected to a gateway, subnet is optional.
   - If a segment is connected either to a tier-1 or tier-0 gateway, subnet is required.

Subnets of one segment must not overlap with the subnets of other segments in your network. A segment is always associated with a single virtual network identifier (VNI) regardless of whether it is configured with one subnet, two subnets, or no subnet.

8. Select a transport zone, which can be an overlay or a VLAN.

   To create a VLAN-backed segment, add the segment in a VLAN transport zone. Similarly, to create an overlay-backed segment, add the segment in an overlay transport zone.

9. Optional: To configure DHCP on the segment, click **Set DHCP Config**.

   For a detailed information about DHCP configuration, see Configure NSX DHCP Service.

10. In the **VLAN** field, specify the VLAN IDs.

    - If the transport zone is of type VLAN, you must configure a VLAN ID or a range of VLAN IDs.
      - The configuration of a single VLAN ID defines the VLAN ID of the VLAN backing the segment in the physical infrastructure.
      - The configuration of a range of VLAN IDs enables guest VLAN tagging for the segment and defines the VLAN IDs for the VLAN tags allowed on the segment.
    - If the transport zone is of type overlay, you can optionally enter a range of VLAN IDs, which enables guest VLAN tagging on the segment and defines the VLAN IDs for the VLAN tags allowed on the segment.

      Note:
      The VLAN IDs must be within the allowable range specified by the transport zone.

11. Optional: Select an uplink teaming policy for the segment.

    This drop-down menu displays the named teaming policies, if you have added them in the VLAN transport zone. If no uplink teaming policy is selected, the default teaming policy is used.

    - Named teaming policies are not applicable to overlay segments. Overlay segments always follow the default teaming policy.
    - For VLAN-backed segments, you have the flexibility to override the default teaming policy with a selected named teaming policy. This capability is provided so that you can steer the infrastructure traffic from the host to specific VLAN segments in the VLAN transport zone. Before adding the VLAN segment, ensure that the named teaming policy names are added in the VLAN transport zone.

12. Optional: Enter the fully qualified domain name.

    DHCPv4 server and DHCPv4 static bindings on the segment automatically inherit the domain name from the segment configuration as the Domain Name option.

13. If you want to use Layer 2 VPN to extend the segment, click the **L2 VPN** text box and select an L2 VPN server or client session.

    You can select more than one.

14. In **VPN Tunnel ID**, enter a unique value that is used to identify the segment.

15. In the **Metadata Proxy** field, select a metadata proxy from the drop-down list, or click the menu icon (3 dots) to create one.

16. Click **Save**.

17. To add segment ports, click **Yes** when prompted if you want to continue configuring the segment.
    a) Click **Set** from the **Ports / Interfaces** column.
    b) Click **Add Segment Port**.
    c) Enter a port name.
    d) For **ID**, enter the VIF UUID of the VM or server that connects to this port.
    e) Select a type: **Child**, or **Static**.

       Leave this text box blank except for use cases such as containers or VMware HCX. If this port is for a container in a VM, select **Child**. If this port is for a bare metal container or server, select **Static**.
    f) Enter a context ID.

       Enter the parent VIF ID if **Type** is **Child**, or transport node ID if **Type** is **Static**.

g) Enter a traffic tag.

   Enter the VLAN ID in container and other use cases.

h) Select an address allocation method: **IP Pool**, **MAC Pool**, **Both**, or **None**.

i) Specify tags.

j) Apply address binding by specifying the IP (IPv4 address, IPv4 subnet, IPv6 address, or IPv6 subnet) and MAC address of the logical port to which you want to apply address binding. For example, for IPv6, 2001::/64 is an IPv6 subnet, 2001::1 is a host IP, whereas 2001::1/64 is an invalid input. You can also specify a VLAN ID.

   Manual address bindings, if specified, override the auto discovered address bindings.

k) Select segment profiles for this port.

18. To select segment profiles, click **Segment Profiles**.

19. Click **Save**.

20. Optional: You can click the menu icon (3 dots) for the following options to save specific information about the segment in a CSV file:

    - **Download MAC Table**: Select the source which can be the Central Control Plane or a specific transport node for the associated MAC addresses.
    - **Download VTEP Table**: Select the source which can be the Central Control Plane or a specific transport node for the associated VTEPs.
    - **Download ARP Table**: Select the edge node to save the ARP table.

      Note:
      This option is only available if the segment is connected to a gateway.
    - **Download ARP Proxy**: Save the aggregate of the ARP proxy for the segment.

      Note:
      This option is only available if the segment is connected to a gateway.

21. Optional: You can view more information about the segment by expanding the segment and clicking the following options on the right:

    - **View Statistics**: Contains the following tabs:
      – **Local Port**: Displays the traffic details for the local port.
      – **Interface Statistics**: Displays the data details for specific edge nodes.

      Note:
      This tab is only available if the segment is connected to the gateway.
      – **DHCP Statistics**: Displays the DHCP server packet counts and DHCP pool usage statistics. This tab is available only when you have configured Segment DHCP server on the segment.

        If you have configured both DHCPv4 and DHCPv6 servers on a segment, the **DHCP Statistics** tab will display only the DHCPv4 packet counts and the DHCPv4 pool usage statistics. DHCPv6 packet counts and DHCPv6 pool usage statistics are currently not supported.
    - **View Related Groups**: Displays the groups associated with the segment.
    - **View DAD Status**: Displays Duplicate Address Detection (DAD) status for the segment.

      Note:
      This tab is only available if the segment is connected to the gateway.

The new segment is added to the list. For any segment, you can modify its configurations by clicking the menu icon (3 dots) and select **Edit**. To reconfigure ports, you do not need to click **Edit**. You only need to click the expand icon (right arrow) for the segment and click the number in the **Ports** column. Note that the number must be non-zero. If it is zero, you must edit the segment.

If NSX Federation is configured, this feature of reconfiguring a segment by clicking on an entity is applicable to segments created by the Global Manager (GM) as well. Note that from the Local Manager, you can create ports for a GM-created segment because you cannot create segment ports from the Global Manager.

Edge Bridging: Extending Overlay Segments to VLAN

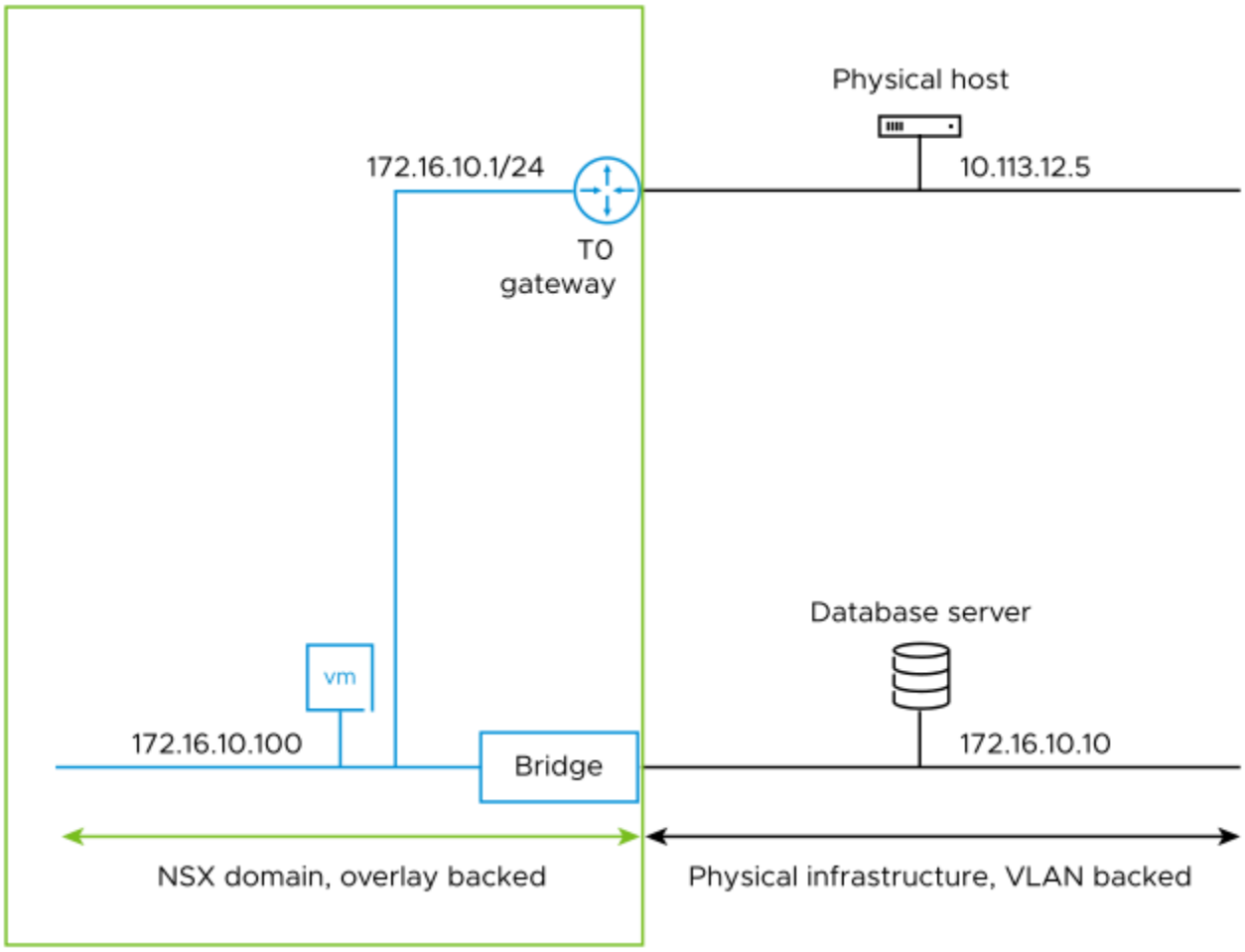
Workloads attached to overlay segments typically communicate at layer 3 with physical devices outside of the NSX domain, through tier-0 gateways instantiated on NSX Edge. However, there are some scenarios where layer 2 connectivity is required between virtual machines in NSX and physical devices.

Some examples are:

- Migration from physical to virtual, or virtual to virtual.
- Integration of a physical appliance that provides services to a segment, like an external load balancer.
- Connection to a database server that requires layer 2 adjacency to its virtual machine clients.

For that purpose, on the top of the gateway service, NSX Edge can also run a bridge service. The following diagram represents those two options: the virtual machine in the bottom left corner has layer 3 connectivity through a gateway to the physical host, and layer 2 connectivity through a bridge to the database server. It is possible to both route and bridge a segment. In fact, it is possible to use the tier 0 gateway in this diagram as a default gateway for the database server.

Figure 402: NSX VM Bridge and Gateway Communication



The NSX Edge bridge, like the gateway, is supported for long term deployments, even if it is often used as a temporary solution during migrations.

The bridge functionality extends an overlay segment into a VLAN, identified by a VLAN ID on an uplink of the NSX Edge where the bridge is running. Typically, two redundant active and standby bridges get deployed on separate edges as part of the same edge cluster. There is no active/active redundancy possible. Setting up the bridge functionality involves the following configuration steps:

- Make sure that the NSX Edge is suitable for hosting the bridge service. The bridge adds a few constraints to the deployment of an edge in a VM form factor.

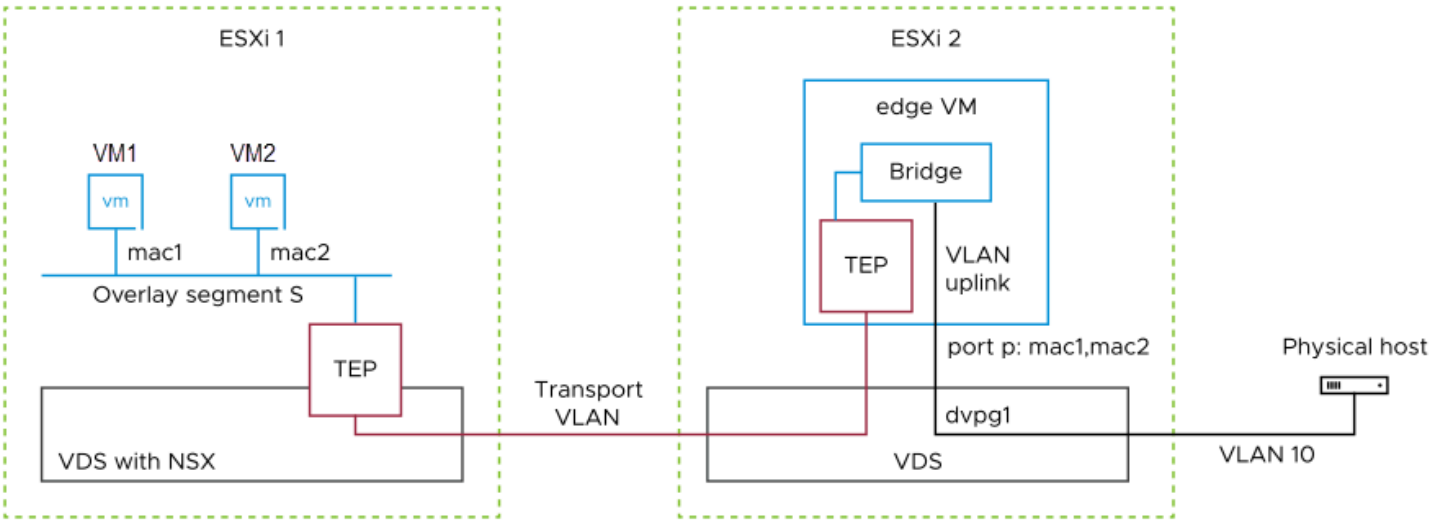
- Identify the NSX Edges that run the bridge service. A bridge profile statically designates the edge responsible for running the active bridge and optionally designates a second edge hosting the standby bridge.
- Lastly associate an overlay segment to a VLAN ID or IDs and a bridge profile. This results in the creation of the appropriate active/standby bridges on the edges specified in the bridge profile, that extend at layer 2 the overlay segment to the VLAN or VLANs identified by the VLAN IDs.

Configure an Edge VM for Bridging

If you plan to run a bridge on an NSX Edge VM, use this section to understand the specific configuration to perform in the vSphere infrastructure.

As an example, our scenario includes two virtual machines, VM1 and VM2, on transport node ESX 1 attached to an overlay segment S. The VMs can communicate at layer 2 with the physical host on the right side of the diagram thanks to a bridge instantiated on the edge VM running on ESX host 2. The TEP (tunnel end point) on ESX 1 encapsulates the traffic from VM1/VM2 and forwards it to the TEP of the edge VM. Then the bridge unencapsulates the traffic and sends it tagged with VLAN ID 10 on its VLAN uplink. Then the traffic gets switched to the physical host.

Figure 403: Edge VM Bridging



As you can see on the diagram, the VLAN uplink of the bridge is linked to port p, which is attached to distributed portgroup dvpg1. This port p is injecting into dvpg1 traffic with the source mac addresses mac1 and mac2 of virtual machined VM1 and VM2. Port p must also accept traffic with destination mac addresses mac1 and mac2, so that the physical host can reach VM1 and VM2. When a bridge is running on the edge VM, the port of this edge VM behaves in a non-standard way as far as the vSphere switching infrastructure is concerned. That means that dvpg1 will need some additional configuration to accommodate the edge VM. The following section lists the different options based on your environment.

Option 1: Edge VM is on a VSS portgroup

This option is for when the Edge VM is connected to a VSS (vSphere Standard Switch). You must enable promiscuous mode and forged transmit.

- Set promiscuous mode on the portgroup.
- Allow forged transmit on the portgroup.
- Run the following command to enable reverse filter on the ESX host where the Edge VM is running:

  ```
  esxcli system settings advanced set -o /Net/ReversePathFwdCheckPromisc -i 1
  ```

  Then disable and enable promiscuous mode on the portgroup with the following steps:

  - Edit the portgroup's settings.

  - Disable promiscuous mode and save the settings.
  - Edit the portgroup's settings again.
  - Enable promiscuous mode and save the settings.
- Do not have other port groups in promiscuous mode on the same host sharing the same set of VLANs.
- Avoid running other VMs attached to the portgroup in promiscuous mode on the same host, as the traffic gets replicated to all those VMs and affect performance.

Option 2a: Edge VM is on a VDS 6.6.0 (or later) portgroup

This option is for when the Edge VM is connected to a VDS (vSphere Distributed Switch). You must be running ESX 6.7 or later, and VDS 6.6.0 or later.

- Enable MAC learning with the option “allow unicast flooding” on the distributed portgroup.
  Starting with vSphere 8.0, you can enable the Mac Learning UI option in the distributed portgroup configuration. For previous releases, you need to use the VIM API `DVSMacLearningPolicy` and setting `allowUnicastFlooding` to `true`.

  Note:

  If the MAC learning feature is available for your release and VDS version, it is highly recommended over setting forged transmit and promiscuous mode.

  The only exception is when all of the following conditions are met:

  - You bridge a segment to VLAN 0.
  - You use a distributed router on this segment.
  - The edge VM is on the same VDS prepared for NSX.

  In this case, do not use the MAC learning option. Instead, use Option 2b as a special case, if you cannot use the edge VM on regular vSphere VDS.

Option 2b: Edge VM is on a VDS 6.5.0 (or earlier) portgroup

This option is for when the Edge VM is connected to a VDS (vSphere Distributed Switch). You enable promiscuous mode and forged transmit.

- Set promiscuous mode on the distributed portgroup.
- Allow forged transmit on the distributed portgroup.
- Run the following command to enable reverse filter on the ESX host where the Edge VM is running:
  ```
  esxcli system settings advanced set -o /Net/ReversePathFwdCheckPromisc -i 1
  ```
  Then disable and enable promiscuous mode on the distributed portgroup with the following steps:
  - Edit the distributed portgroup's settings.
  - Disable promiscuous mode and save the settings.
  - Edit the distributed portgroup's settings again.
  - Enable promiscuous mode and save the settings.
- Do not have other distributed port groups in promiscuous mode on the same host sharing the same set of VLANs.
- Avoid running other VMs attached to the distributed portgroup in promiscuous mode on the same host, as the traffic gets replicated to all those VMs and affects performance.

Option 3: Edge VM is connected to an NSX segment

If the Edge is deployed on a host with NSX installed, it can connect to a VLAN segment and use MAC Learning, which is the preferred configuration option.

- Create a new MAC Discovery segment profile by navigating to **Networking** > **Segments** > **Profiles**.
  - Click **Add Segment Profile** > **MAC Discovery**.
  - Enable **MAC Learning**. This will also enable **Unknown Unicast Flooding**. Keep the flooding option enabled for bridging to work in all scenarios.
  - Click **Save**.

- Edit the segment used by the Edge by navigating to **Networking** > **Segments**.
  - Click the menu icon (3 dots) and select **Edit**.
  - Expand the **Segment Profiles** section, then set the **MAC Discovery** profile to the one created above.

    Note:
    If you bridge a segment to VLAN 0 and you use a distributed router on this segment, the gateway might not route VLAN 0 traffic when using MAC learning. In this scenario, avoid option 3. Avoid option 2a if the edge VM is attached to the distributed portgroup of a VDS prepared for NSX for vSphere.

Create an NSX Edge Bridge Profile

The edge bridge profile is a template for instantiating bridges. In the template, you define a primary edge, an optional backup edge from the same edge cluster as the primary, and a failover mode, preemptive or non-preemptive.

The preference is to use the primary edge for running the active bridge, the bridge forwarding traffic between overlay segment and VLAN. The standby bridge, that typically runs on the backup edge, does not forward any traffic.

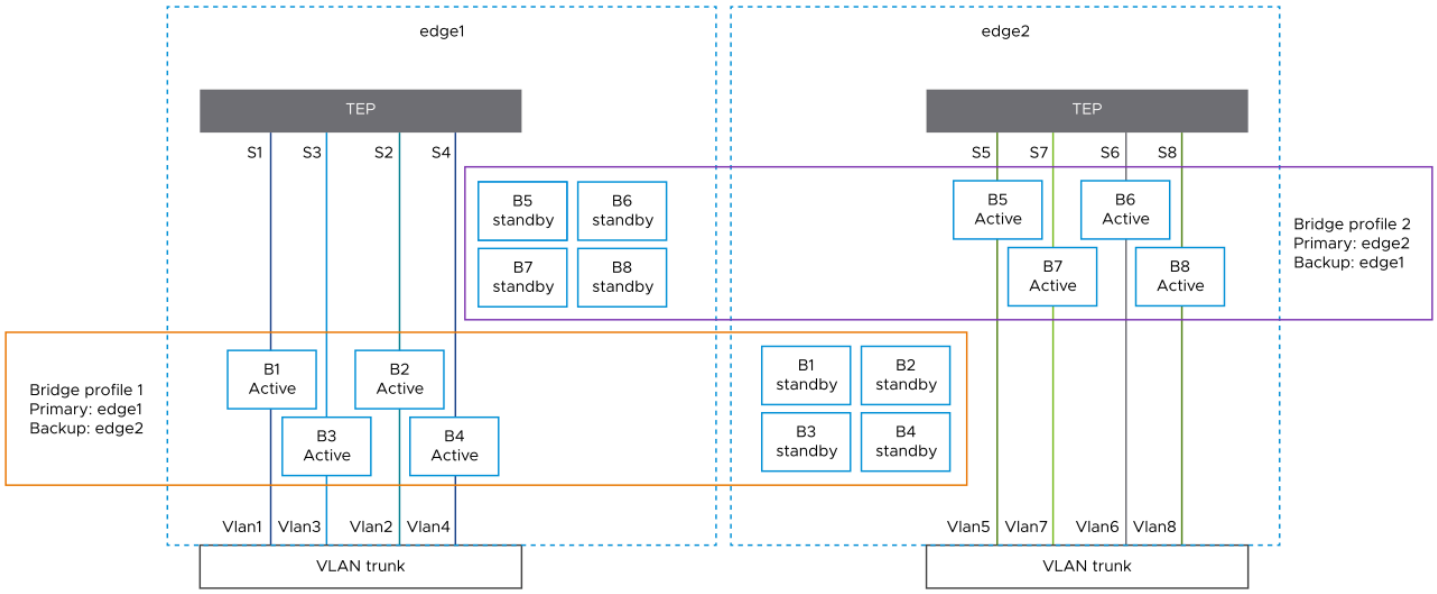
You can instantiate multiple bridges from the same bridge profile. As a result, in most cases, few edge bridge profiles are required. For example, if you plan to use two edges (edge1 and edge2) for bridging, you might want to create two edge bridge profiles:

- Profile 1 with edge1 as primary and edge2 as backup
- Profile 2 with edge2 as primary and edge1 as backup

You can then create an arbitrary number of bridges using edge1 as primary (respectively backup), by associating them to the profile 1 (respectively profile 2). Those two profiles are enough to load share the bridged traffic between the two edges, on a per segment basis. The Few Bridge Profiles for Many Bridges diagram represents an example of bridging eight segments across two edges, using two edge bridge profiles.

This diagram shows bridge overlay segments S1 to VLAN 1, segment S2 to VLAN 2, and so on. Segments S1 to S4 are using bridge profile 1, resulting in active bridges on edge1, standby on edge2. Segment S5 to S8 are using bridge profile 2, leading to active bridges on edge2, standby on edge1. This diagram shows load sharing of the bridging functionality, on a per segment basis.

Figure 404: Few Bridge Profiles for Many Bridges



Depending on the availability of the edges and the failover mode selected for the bridge profiles, the active bridges might be running on the backup edges.

When both edges in the bridge profile are available, the active bridge is typically running on the primary edge. If the active bridge or the primary edge fails, the standby bridge on the backup edge takes over the active role and starts forwarding traffic between overlay segment and VLAN.

A bridge switchover, moving the active bridge to a different edge, is an operation that results in traffic loss. The bridge that is becoming active synchronizes the mac addresses that were learned on the previously active bridge and starts flooding RARP packets, using those mac addresses as source mac addresses. This mechanism is necessary to update the mac address tables of the physical infrastructure.

For example, what if a failure occurs on the primary edge and the bridge running on the backup edge is already active? In preemptive mode, when the failure is recovered on the primary edge, a bridge switchover is triggered and the bridge on the primary edge becomes active again.

The benefit of the preemptive mode is that the system is attempting to forward the bridged traffic along a deterministic path. If you take the example of the Few Bridge Profiles for Many Bridges figure, with a preemptive mode, you are sure that the bridge traffic gets distributed on a per segment basis as soon as both edges are available, thus providing more bandwidth.

The drawback of the preemptive mode is that there is a disruptive bridge convergence when the bridge on the primary edge recovers and becomes active again.

In non-preemptive mode, the bridge on the primary edge recovers from failure as a standby bridge. The benefit of this mode is that there is no additional traffic disruption when the primary recovers. The preemptive mode is the best option in terms of bandwidth, thanks to its load sharing. The drawback of the non-preemptive mode is that bridge traffic flow is non-deterministic and can be sub-optimal. In the example shown in the Few Bridge Profiles for Many Bridges figure, after a failed edge recovers, the bridge traffic still flows through a unique edge, with no load sharing.

You can manually trigger a bridge switchover. To manually trigger a bridge switchover from the CLI of the edge currently hosting the standby bridge, enter: `set bridge <uuid> state active`.

Use this command only in non-preemptive mode. If you use it in preemptive mode, it returns an error.

For more information on set or get bridge commands, see the *NSX Command-Line Interface Reference*.

Ensure that you have an NSX Edge cluster with two NSX Edge transport nodes.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Segments** > **Profiles**.
3. Click **Edge Bridge Profiles**.
4. Click **Add Edge Bridge Profile**.
5. Enter a name for the Edge bridge profile and optionally a description.
6. Select an NSX Edge cluster.
7. Select a primary node.
8. Select a backup node.
9. Select a failover mode.
   The options are **Preemptive** and **Non-Preemptive**.
10. Click **Save**.

Create a bridge-backed segment. See Extend an NSX Overlay Segment to a VLAN or a Range of VLANs.

Extend an NSX Overlay Segment to a VLAN or a Range of VLANs

After you have identified the edges on which you want the bridging functionality to be performed and created the appropriate edge bridge profile, the final step is to edit the segment configuration and specify the edge bridge profile to which you want to associate with the segment and the VLAN ID or range of VLAN IDs to which to bridge your segment. This step instantiates one or two bridges on the edges identified in the edge bridge profile.

When you configure a bridge with a single VLAN ID, a frame received on the overlay segment by the bridge gets decapsulated and forwarded on the VLAN uplink of the bridge with an added 802.1Q tag corresponding to this VLAN ID.

When you create the bridge specifying a VLAN ID range, you must configure the overlay segment being bridged for Guest VLAN Tagging (GVT). This means that the encapsulated frames already carry an 802.1Q tag. When the bridge receives an encapsulated frame carrying a VLAN tag on its overlay interface, it first checks that VLAN ID in the tag belongs to the VLAN range configured for the bridge. After confirmation, it forwards the frame on the VLAN uplink of the bridge carrying the original 802.1Q tag that was received on the overlay. Otherwise, it drops the frame.

> **Note:**
> If needed, you can configure multiple bridges on the same segment, but:
>
> - The same segment cannot be bridged twice on the same edge.
> - The bridge does not have any loop detection or prevention. If you configure multiple bridges to the same bridging domain on the VLAN side, it results in a permanent bridging loop.

Configuring a Bridge-Backed Segment

Prerequisites

- You have identified an overlay segment you want to bridge.

- You have an edge bridge profile specifying one or two edges attached to the overlay transport zone of your segment.
- If you are using edge VMs, you have checked the configuration requirements in Configure an Edge VM for Bridging.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Segments**.
3. Click the menu icon (three dots) of the overlay segment that you want to configure layer 2 bridging on and select **Edit**.
4. Expand **Additional Settings** and in the **Edge Bridges** field, click **Set**.
5. Click **Add Edge Bridge**.
6. Select an Edge bridge profile.
7. Select a VLAN transport zone to identify the VLAN uplinks used by the bridge.
8. Enter a VLAN ID or a VLAN ID range (specify VLAN ranges and not individual VLANs).
9. Optional: Select a teaming policy.

   If there are multiple VLAN uplinks on the edge NVDS attached to the VLAN transport zone selected in the previous steps, use a failover order named teaming policy to identify the exact uplink on which VLAN bridged traffic gets forwarded. The uplinks of a VM edge do not fail, so the teaming policy only needs a single uplink. If you do not enter a specific teaming policy and there are multiple VLAN uplinks, the first one configured on the edge NVDS is used.
10. Click **Add**.
11. Click **Apply**.

Create an NSX Metadata Proxy Server

A metadata proxy server enables VMs to retrieve metadata from an OpenStack Nova API server.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Segments** > **Profiles** > **Metadata Proxies**.
3. Click **Add Metadata Proxy**.
4. Enter a name for the metadata proxy server.
5. In the **Server Address** field, enter the URL and port for the Nova server.

   The valid port range is 3000 - 9000.
6. Select an Edge cluster.
7. Optional: Select Edge nodes.

   If you select any Edge node, you cannot enable **Standby Relocation** in the next step.
8. Optional: Enable **Standby Relocation**.

   Standby relocation means that if the Edge node running the metadata proxy fails, the metadata proxy will run on a standby Edge node. You can only enable standby relocation if you do not select any Edge node.
9. In the **Shared Signature Secret** field, enter the secret that the metadata proxy will use to access the Nova server.
10. Optional: Select a certificate for encrypted communication with the Nova server.
11. Optional: Select a cryptographic protocol.

    The options are TLSv1, TLSv1.1, and TLSv1.2. TLSv1.1 and TLSv1.2 are supported by default.

Distributed Port Groups

A distributed port group specifies port configuration options for each member port on a vSphere distributed switch. Distributed port groups define how a connection is made to a network.

Distributed Port Group Creation in NSX

When you install Distributed Security to a vSphere Distributed Switch (VDS), the Distributed Virtual port groups (DVPG) and DVports of the VDS are discovered and objects are automatically created to represent them in NSX.

For details, see Activate NSX on Distributed Virtual Port Groups (DVPGs).

> **Important:**
>
> The objects created in NSX Manager for the DVPGs are called *distributed port groups*. They are not called *segments* in the UI.
>
> The objects created in NSX Manager for the DVports are called *distributed ports*. They are not called *segment ports* in the UI.

Also, the following events occur during the Distributed Security installation:

- The VLAN tags for the DVPG are automatically discovered and shown in NSX Manager. The VLAN tags can only be edited in vCenter.
- The default segment profiles are applied to the distributed port groups. You can later switch them to custom profiles.
- Only connected DVports from vCenter are discovered by NSX. Free DVports are not discovered.

After the Distributed Security installation, you can view these objects by navigating to **Networking** > **Segments**, and then selecting the **Distributed Port Groups** tab.

The distributed port group and distributed port objects are kept in sync between vCenter and NSX. This means that if DVPGs or DVports are created or removed in vCenter, then those changes are automatically made to the respective distributed port groups or distributed ports in NSX Manager. If changes are made in vCenter while connectivity is lost between vCenter and NSX, those changes are automatically processed and reflected in NSX Manager when connectivity is restored.

Available Actions for Distributed Port Groups and Distributed Ports

You can perform the following actions for distributed port groups and distributed ports in NSX Manager:

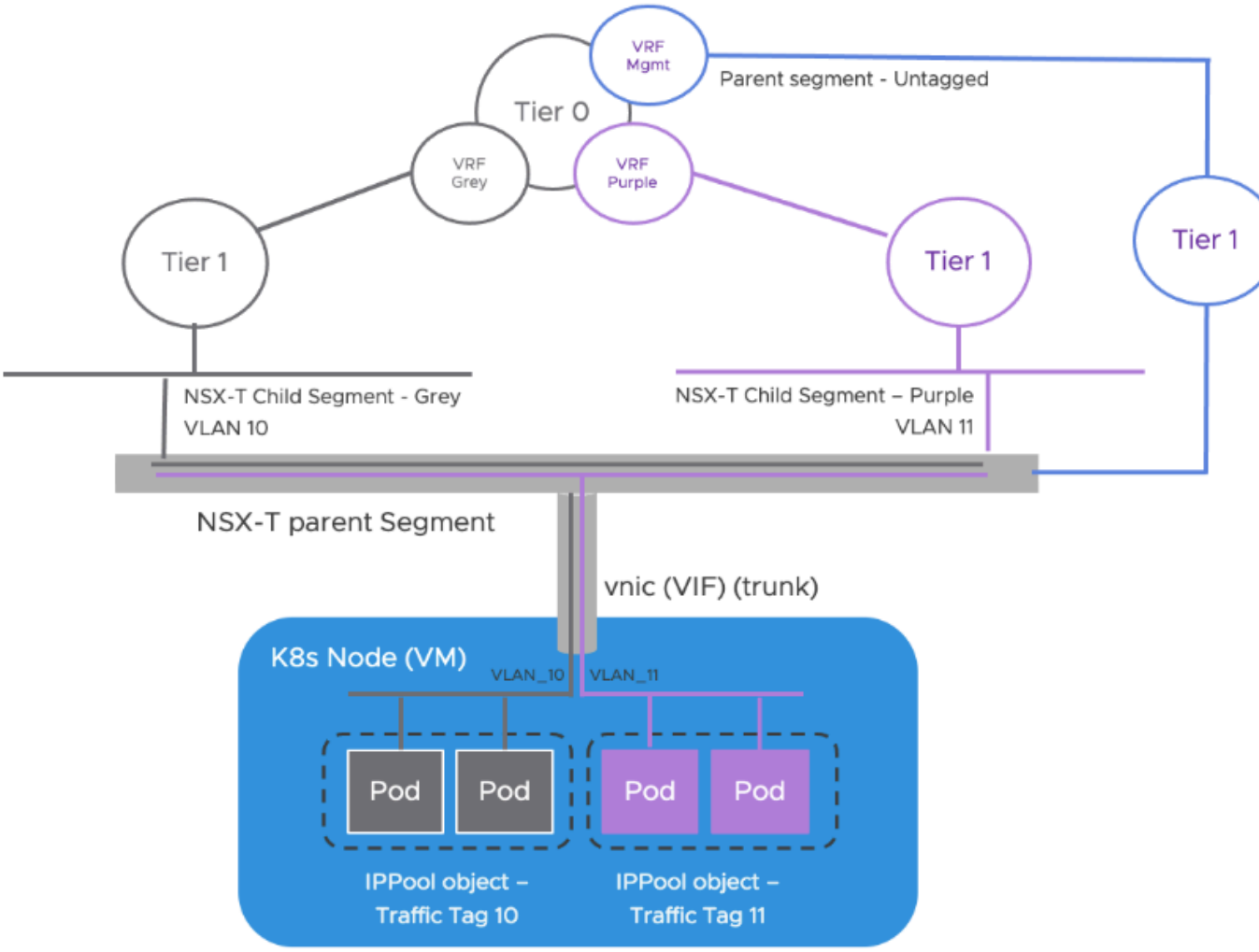
Object	Available Actions
Distributed port groups	• Apply SpoofGuard. • Apply IP Discovery. • Apply switch security profile. • Add and remove tags which allows the distributed port group to be added to dynamic NSGroups. • Add and remove the distributed port group from static NSGroups.
Distributed ports	• Add and remove tags which allows distributed ports to be added to dynamic NSGroups. • Add and remove distributed ports from static NSGroups. • Manage address bindings.

Creating a Child Segment

You can create child segments to enable advanced IPAM, routing, and NAT for Kubernetes workloads with Antrea. Child segments are designed to support Antrea features such as flexible IPAM of routable pods and Antrea Egress.

> **Note:**
> This feature is only available through the API.

The following diagram shows how child segments are connected to other



components.

A parent segment is a regular NSX segment. No configuration is required to make a segment a parent segment. A child segment is associated with one or more parent segments and is assigned a VLAN ID for each of its parents. A parent segment will only send a child segment traffic that is tagged with the specific VLAN ID.

Properties of child and parent segments:

- A child segment can be associated with several parent segments.
- A parent segment can have numerous child segments.
- Each child segment linked to a parent must have a distinct VLAN ID, and no two child segments connected to the same parent can share a VLAN ID.
- A child segment connecting to multiple parents can use the same or different VLAN IDs for these connections.
- A child segment cannot be the parent of another child segment.
- A child segment can be either an overlay segment or a VLAN segment.
- Virtual Machines (VMs) can connect to child segments. When they do, a segment port is created, similar to when a VM is connected to a standard segment.

- No segment port is created on the child segment for the parent-child connection.
- Child segments and parent segments are independent. That means:
  - They can connect to the same or different tier-1 or tier-0 gateways.
  - They can have different configurations or switching profiles.
  - Configurations on a parent segment are not inherited by its child segments.
  - A group that includes a parent segment does not automatically include its child segments.
  - Services configured on a parent segment, such as a Distributed Firewall, do not extend to child segments.

For VMs connected to a child segment, all segment-related features are supported. For container traffic passing through a child segment, the following table lists the supported and unsupported features:

Feature	Supported
Distributed Firewall	No
IP Discovery segment profile	No
MAC Discovery segment profile	No
Segment Security segment profile	No
SpoofGuard segment profile	No
QoS segment profile	No
Port Mirroring	No
IPFix	No
Uplink Teaming	Yes
Mirror configuration (uplink span)	Yes
NSGroup	No
Latency	No

Child Segment Usage with Antrea

Each routable subnet (for example, IP pools for routable Pods, external IP pools for Egresses) configured in Antrea is mapped to a child segment and the segments of the Kubernetes node VMs are the parents of the child segment. OVS (Open vSwitch) on the Kubernetes node VM will tag traffic with the correct VLAN ID before sending the packets through the VNIC.

Workflow to create a connection between a parent segment and a child segment:

- Create the parent segment. You can use the API or NSX Manager UI.
- Create the child segment. You can use the API or NSX Manager UI.
- Use the API to create a connection binding map on the child segment by specifying the parent segment path and the VLAN ID.

Workflow to delete a child segment:

- Use the API to delete all the existing connection binding maps of the child segment.
- Delete the child segment. You can use the API or NSX Manager UI.

Workflow to delete a parent segment:

- Use the API to delete all the existing connection binding maps that connect this parent segment to all its child segments.
- Delete the parent segment. You can use the API or NSX Manager UI.

API

The following API calls can be used to manage child segments. For more information, see the *NSX API Guide*.

```
GET /policy/api/v1/infra/segments/{segment-id}/segment-connection-binding-maps
```

```
GET /policy/api/v1/orgs/{org-id}/projects/{project-id}/infra/segments/{segment-id}/segment-connection-binding-maps
DELETE /policy/api/v1/infra/segments/{segment-id}/segment-connection-binding-maps/{map-id}
DELETE /policy/api/v1/orgs/{org-id}/projects/{project-id}/infra/segments/{segment-id}/segment-connection-binding-maps/{map-id}
GET /policy/api/v1/infra/segments/{segment-id}/segment-connection-binding-maps/{map-id}
GET /policy/api/v1/orgs/{org-id}/projects/{project-id}/infra/segments/{segment-id}/segment-connection-binding-maps/{map-id}
PATCH /policy/api/v1/infra/segments/{segment-id}/segment-connection-binding-maps/{map-id}
PATCH /policy/api/v1/orgs/{org-id}/projects/{project-id}/infra/segments/{segment-id}/segment-connection-binding-maps/{map-id}
PUT /policy/api/v1/infra/segments/{segment-id}/segment-connection-binding-maps/{map-id}
PUT /policy/api/v1/orgs/{org-id}/projects/{project-id}/infra/segments/{segment-id}/segment-connection-binding-maps/{map-id}
GET /policy/api/v1/infra/tier-1s/{tier-1-id}/segments/{segment-id}/segment-connection-binding-maps
GET /policy/api/v1/orgs/{org-id}/projects/{project-id}/infra/tier-1s/{tier-1-id}/segments/{segment-id}/segment-connection-binding-maps
DELETE /policy/api/v1/infra/tier-1s/{tier-1-id}/segments/{segment-id}/segment-connection-binding-maps/{map-id}
DELETE /policy/api/v1/orgs/{org-id}/projects/{project-id}/infra/tier-1s/{tier-1-id}/segments/{segment-id}/segment-connection-binding-maps/{map-id}
GET /policy/api/v1/infra/tier-1s/{tier-1-id}/segments/{segment-id}/segment-connection-binding-maps/{map-id}
GET /policy/api/v1/orgs/{org-id}/projects/{project-id}/infra/tier-1s/{tier-1-id}/segments/{segment-id}/segment-connection-binding-maps/{map-id}
PATCH /policy/api/v1/infra/tier-1s/{tier-1-id}/segments/{segment-id}/segment-connection-binding-maps/{map-id}
PATCH /policy/api/v1/orgs/{org-id}/projects/{project-id}/infra/tier-1s/{tier-1-id}/segments/{segment-id}/segment-connection-binding-maps/{map-id}
PUT /policy/api/v1/infra/tier-1s/{tier-1-id}/segments/{segment-id}/segment-connection-binding-maps/{map-id}
PUT /policy/api/v1/orgs/{org-id}/projects/{project-id}/infra/tier-1s/{tier-1-id}/segments/{segment-id}/segment-connection-binding-maps/{map-id}
```

When creating a connection binding map, the map ID that you specify in the PUT API must be unique. Example of using `PUT /policy/api/v1/infra/segments/{segment-id}/segment-connection-binding-maps/{map-id}` to create a binding map:

API request:

```
PUT https://<nsx-manager>/policy/api/v1/infra/segments/test2/segment-connection-binding-maps/map-id-1
{
    "vlan_traffic_tag": "1002",
    "segment_path": "/infra/segments/openshift-segment"
}
```

API response:

```
{
  "vlan_traffic_tag" : 1002,
  "segment_path" : "/infra/segments/openshift-segment",
  "resource_type" : "SegmentConnectionBindingMap",
  "id" : "map-id-1",
  "display_name" : "map-id-1",
  "path" : "/infra/segments/test2/segment-connection-binding-maps/map-id-1",
  "relative_path" : "map-id-1",
  "parent_path" : "/infra/segments/test2",
  "remote_path" : "",
  "unique_id" : "7ac8c0fb-a0e1-471a-bed6-d15f1b85e0c6",
  "realization_id" : "7ac8c0fb-a0e1-471a-bed6-d15f1b85e0c6",
  "owner_id" : "210584d9-6329-452e-bb01-0133945ab675",
  "marked_for_delete" : false,
  "overridden" : false,
  "_system_owned" : false,
```

```
  "_protection" : "NOT_PROTECTED",
  "_create_time" : 1712718210160,
  "_create_user" : "admin",
  "_last_modified_time" : 1712718210160,
  "_last_modified_user" : "admin",
  "_revision" : 0
}
```

Traceflow

Traceflow supports the following scenarios related to child segments:

- The source is a container on a child segment and the destination is a container on a child segment.
- The source is a container on a child segment and the destination is a regular port.
- The source is a parent port. There is no child segment because the Antrea containers are running inside a VM.

To perform a traceflow of a packet that will pass through a child segment, you must use the API. To set up a traceflow, use the API call `PUT https://<mgr-ip>/policy/api/v1/infra/traceflows/<traceflow-id>`. For example,

```
PUT https://<mgr-ip>/policy/api/v1/infra/traceflows/<traceflow-id>
{
  "segment_port_path": "/infra/segments/parent/ports/default:658c6c45-286f-4a23-832f-a646253b200b",
  "connected_segment_path_as_source": "/infra/segments/child",
  "packet": {
    "eth_header": {
      "src_mac": "00:50:56:ad:28:d5",
      "dst_mac": "00:50:56:ad:c0:a4",
      "eth_type": 2048
    },
    "ip_header": {
      "src_ip": "192.168.100.10",
      "dst_ip": "192.168.100.20",
      "protocol": 1,
      "ttl": 64,
      "flags": 0
    },
    "transport_header": {
      "icmp_echo_request_header": {
        "id": 0,
        "sequence": 0
      }
    },
    "payload": "",
    "resource_type": "FieldsPacketData",
    "frame_size": 128,
    "routed": false,
    "transport_type": "UNICAST"
  },
  "timeout": 10,
  "is_transient": true
}
```

Note that the parameter `connected_segment_path_as_source` points to the child segment. If this parameter is not set, the packet will not pass through the child segment.

For more information, see the *NSX API Guide*.

Child and Parent Segment Statistics

Statistics for a child segment will only include statistics for VMs that are connected to the segment. They will not include statistics for Antrea containers.

For a parent segment, the ingress statistics will include statistics for Antrea containers. The egress statistics will not include statistics for Antrea containers.

NSX DHCP

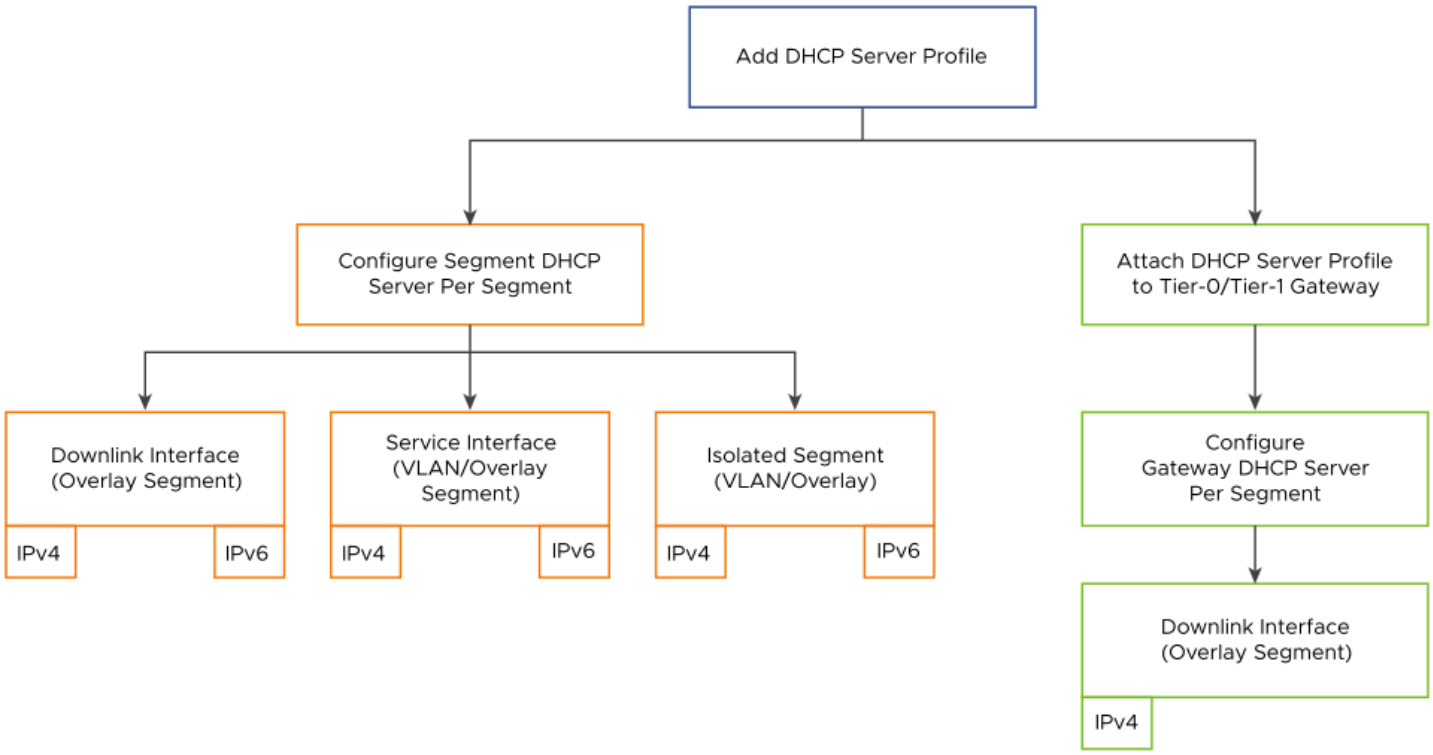
You can configure NSX DHCP service on each segment regardless of whether the segment is connected to a gateway. Both DHCP for IPv4 (DHCPv4) and DHCP for IPv6 (DHCPv6) servers are supported.

NSX supports the following types of DHCP configuration on a segment:

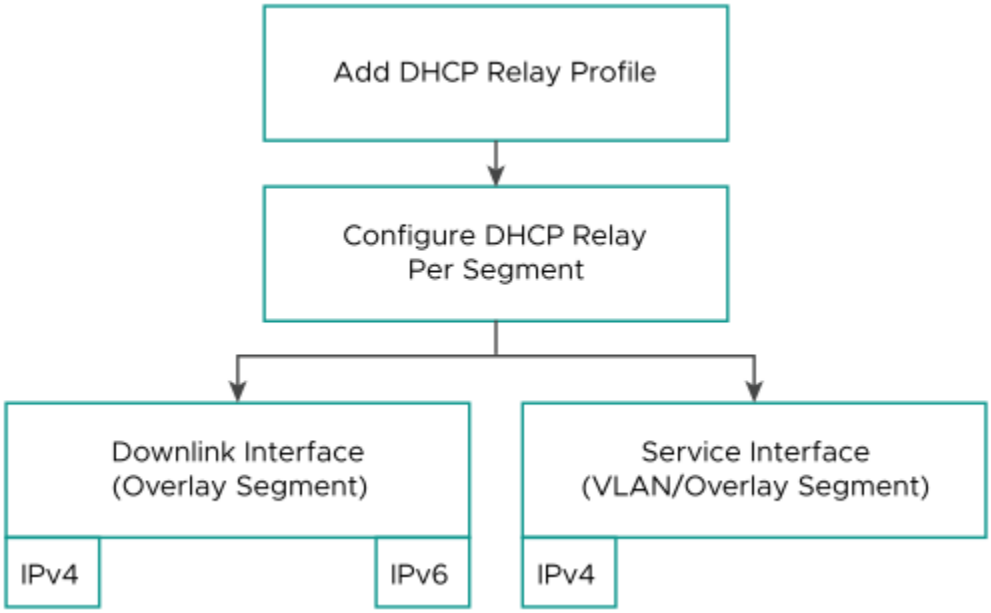
- Segment DHCP server
- Gateway DHCP server (supported only for IPv4 subnets in a segment)
- DHCP Relay

High-level Overview of Configuration

The following figure shows the high-level overview of DHCP server configuration in NSX.



The following figure shows the high-level overview of DHCP Relay configuration in NSX.



Supported DHCP Configuration Types

The following figure shows an example of the various scenarios for Segment DHCP server, Gateway DHCP server, and DHCP Relay in an NSX network.

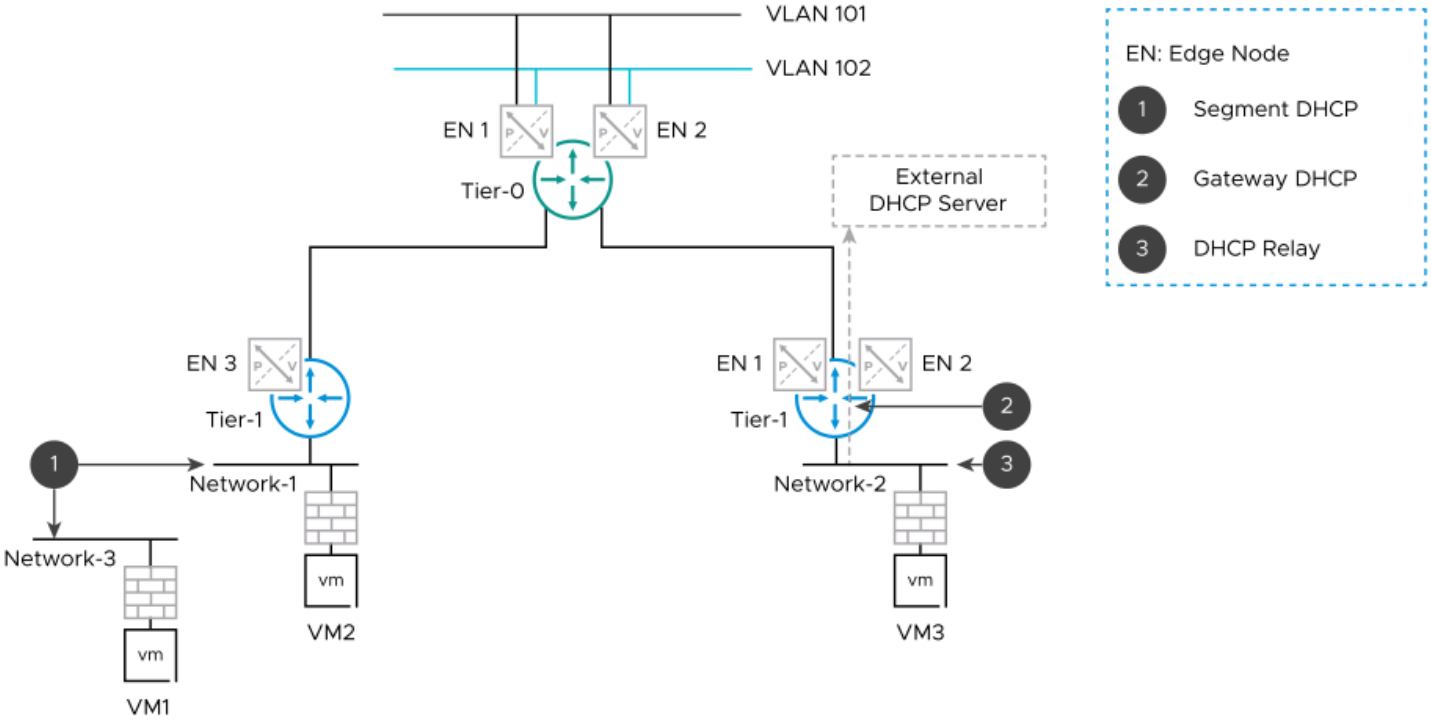


Table 1030: Types of DHCP Configuration in NSX

DHCP Configuration Type	Description
Segment DHCP server	Select this option to create a Segment DHCP server that has an IP address on the segment. A Segment DHCP server provides a dynamic IP assignment service only to the VMs that are attached to the segment. The DHCP server IP address must belong to the subnet that is configured on the segment. Also, the server IP address must be different from the Gateway IP address of the segment. Segment DHCP server is local to the segment and not available to the other segments in the network. You can configure all DHCP settings, including DHCP ranges, DHCP options, and static bindings on the segment. For isolated segments, which are not connected to a gateway, Segment DHCP server configuration type is selected by default. You can configure DHCPv6 in the IPv6 subnet of a segment with a Segment DHCP server.
DHCP Relay	Select this option to relay the DHCP client requests to the external DHCP servers. The external DHCP servers can be in any subnet, outside the SDDC, or in the physical network. DHCP Relay service is local to the segment and is not available to the other segments in the network. When you use a DHCP Relay on a segment, you cannot configure DHCP settings, DHCP options, and static bindings on the segment.
Gateway DHCP server	Gateway DHCP server is analogous to a central DHCP server that dynamically assigns IP and other network configuration to the VMs on all the segments that are connected to the gateway and using Gateway DHCP server. By default, segments that are connected to a tier-1 or tier-0 gateway use Segment DHCP server. If needed, you can choose to configure a Gateway DHCP server or a DHCP Relay on the segment. To configure Gateway DHCP server on a segment, a DHCP server profile must be attached to the gateway. If the IPv4 subnet of a segment uses a Gateway DHCP server, you cannot configure DHCPv6 in the IPv6 subnet of the same segment because Gateway DHCPv6 server is not supported. In this case, the IPv6 subnet cannot support any DHCPv6 server configuration, including the IPv6 static bindings.

Configure NSX DHCP Service

You can configure NSX DHCP service on each segment regardless of whether the segment is connected to a gateway. Both DHCP for IPv4 (DHCPv4) and DHCP for IPv6 (DHCPv6) servers are supported.

NSX DHCP Configuration Settings: Reference

Use this reference documentation to understand the various considerations that you must keep in mind while configuring the NSX DHCP service and to obtain a detailed guidance about the configuration settings on the **Set DHCP Config** page.

The following note mentions the NSX DHCP configuration types that are supported or not supported based on how overlay or VLAN segments are connected:

Note:

- On an isolated segment that is not connected to a gateway, only Segment DHCP server is supported.
- Segments that are configured with an IPv6 subnet can have either a Segment DHCPv6 server or a DHCPv6 relay. Gateway DHCPv6 server is not supported.
- If a segment contains an IPv4 subnet and an IPv6 subnet, you can configure both DHCPv4 and DHCPv6 servers on the segment.
- DHCPv4 relay is supported on a VLAN segment through the service interface of a tier-0 or tier-1 gateway. Only one DHCPv4 relay service is supported on a VLAN segment.

- For a VLAN segment requiring a DHCP server, only Segment DHCP server is supported. Gateway DHCP server is not supported on a VLAN segment.

When a Segment DHCP server or DHCP Relay is configured on a VLAN segment, one of the following configurations is required:

- For VDS or VSS, **Forged Transmits** must be set to **Accept**.
- For NSX segments or VDS 6.6 and later, **MAC Learning** must be set to **Enabled**.

In vSphere 7.0 or later, the **Forged Transmits** option is set to **Reject**, by default. To enable this option on the host, do the following steps:

1. Log in to the vSphere Client UI with **admin** privileges.
2. Go to **Hosts and Clusters** and click a host in the cluster.
3. Navigate to **Configure** > **Networking** > **Virtual Switches**, and then click **Edit**.
4. On the **Edit Settings** window, click **Security**. From the **Forged Transmits** drop-down menu, select **Accept**.

To learn about Forged Transmits, see the *vSphere Security* documentation.

Important:
After a segment has DHCP service configured on it, some restrictions and caveats apply on changing the connectivity of the segment. For more information, see Scenarios: Impact of Changing Segment Connectivity on NSX DHCP.

The following sections provide guidance about the configuration settings on the **Set DHCP Config** page.

DHCP Type

- When a segment is connected to a gateway, Segment DHCP server is selected by default. If needed, you can select Gateway DHCP server or DHCP Relay from the drop-down menu.
- If you select the DHCP type as Gateway DHCP server, the DHCP profile that is attached to the gateway is autoselected. The name and server IP address are fetched automatically from that DHCP profile and displayed in a read-only mode.
- For an isolated segment, which is not connected to a gateway, Segment DHCP server is selected by default.

DHCP Profile

- When you are configuring a Segment DHCP server or a DHCP Relay on the segment, you must select a DHCP profile from the drop-down menu. If no profiles are available in the **DHCP Profile** drop-down menu, click  and create a DHCP profile. After the profile is created, it is automatically attached to the segment.
- When a segment is using a Gateway DHCP server, ensure that an edge cluster is selected either in the gateway, or DHCP server profile, or both. If an edge cluster is unavailable in either the profile or the gateway, an error message is displayed when you save the segment.
- When a segment is using a Segment DHCP server, ensure that the DHCP server profile contains an edge cluster. If an edge cluster is unavailable in the profile, an error message is displayed when you save the segment.

IPv4 Server or IPv6 Server Settings

This section explains the configuration settings in the **IPv4 Server** tab page and the **IPv6 Server** tab page.

DHCP Server Address	• If you are configuring a Segment DHCP server, server IP address is required. A maximum of two server IP addresses are supported. One IPv4 address and one IPv6 address. For an IPv4 address, the prefix length must be <= 30, and for an IPv6 address, the prefix length must be <= 126. The server IP addresses must belong to the subnets that you have specified in this segment. The DHCP server IP address must not overlap with the IP addresses in the DHCP ranges and DHCP static binding. The DHCP server profile might contain server IP addresses, but these IP addresses are ignored when you configure a Segment DHCP server on the segment.

- After a Segment DHCP server is created, you can edit the server IP addresses on the **Set DHCP Config** page. However, the new IP addresses must belong to the same subnet that is configured in the segment.
- If you are configuring a Gateway DHCP server, the **DHCP Server Address** text box is not editable. The server IP addresses are fetched automatically from the DHCP profile that is attached to the connected gateway.
- The Gateway DHCP server IP addresses in the DHCP server profile can be different from the subnet that is configured in the segment. In this case, the Gateway DHCP server connects with the IPv4 subnet of the segment through an internal relay service, which is autocreated when the Gateway DHCP server is created. The internal relay service uses any one IP address from the subnet of the Gateway DHCP server IP address.
- The IP address used by the internal relay service acts as the default gateway on the Gateway DHCP server to communicate with the IPv4 subnet of the segment.
- After a Gateway DHCP server is created, you can edit the server IP addresses in the DHCP profile of the gateway. However, you cannot change the DHCP profile that is attached to the gateway. When a DHCP server profile is used in your network, preferably avoid editing the server IP addresses in the DHCP server profile. It might cause a failure while renewing or releasing the IP addresses that are leased to the DHCP clients.

DHCP Ranges

- IP ranges, CIDR subnet, and IP addresses are allowed. IPv4 addresses must be in a CIDR /32 format, and IPv6 addresses must be in a CIDR /128 format. You can also enter an IP address as a range by entering the same IP address in the start and the end of the range. For example, 172.16.10.10-172.16.10.10.
- IP addresses in the DHCP ranges must belong to the subnet that is configured on the segment. That is, DHCP ranges cannot contain IP addresses from multiple subnets.
- IP ranges must not overlap with the DHCP server IP address and the DHCP static binding IP addresses.
- IP ranges in the DHCP IP pool must not overlap each other.
- Number of IP addresses in any DHCP range must not exceed 65536.
- The following types of IPv6 addresses are not permitted in DHCP for IPv6 ranges:
  - Link Local Unicast addresses (FE80::/64)
  - Multicast addresses (FF00::/8)
  - Unspecified address (0:0:0:0:0:0:0:0)
  - Address with all F (F:F:F:F:F:F:F:F)

  CAUTION:
  After a DHCP server is created, you can update existing ranges, append new IP ranges, or delete existing ranges. However, it is a good practice to avoid deleting, shrinking, or expanding the existing IP ranges. For example, do not try to combine multiple smaller IP ranges to create a single large IP range. When you modify existing ranges after the DHCP service is running, it might cause the DHCP clients to lose network connection and result in a temporary traffic disruption.

Excluded Ranges (Only for DHCPv6)

Enter IPv6 addresses or a range of IPv6 addresses that you want to exclude for dynamic IP assignment to DHCPv6 clients.

In IPv6 networks, the DHCP ranges can be large. Sometimes, you might want to reserve certain IPv6 addresses, or multiple small ranges of IPv6 addresses from the large DHCP range for static binding. In such situations, you can specify excluded ranges.

Lease Time

Default value is 86400 seconds. Valid range of values is 60–4294967295.

Preferred Time (Only for DHCPv6)

Preferred time is the length of time that a valid IP address is preferred. When the preferred time expires, the IP address becomes deprecated. If no value is entered, preferred time is autocalculated as (lease time * 0.8).

Lease time must be > preferred time.

Valid range of values is 60–4294967295. Default is 69120 seconds.

DNS Servers

A maximum of two DNS servers are permitted. When not specified, no DNS is assigned to the DHCP client.

Domain names (Only for DHCPv6)

One or more domain names are supported.

	DHCPv4 server configuration automatically fetches the domain name that you specified in the segment configuration.
SNTP Servers (Only for DHCPv6)	A maximum of two SNTP servers are permitted. DHCPv6 server does not support NTP.

DHCP Options (Only for DHCPv4)

DHCP Options for IPv6 are not supported.

Each classless static route option in DHCP for IPv4 can have multiple routes with the same destination. Each route includes a destination subnet, subnet mask, next hop router. For information about classless static routes in DHCPv4, see RFC 3442 specifications. You can add a maximum of 127 classless static routes on a DHCPv4 server.

In addition to the Generic Option 121 (classless static route), NSX supports other Generic Options that are described in the following table. The Generic Options, which are not listed in this table are also accepted without any validation, but they do not take effect.

Table 1031: Supported Generic Options

Code	Name	Value Type	Example Value
2	Time Offset	Integer - seconds offset from UTC Allowed values: -43200–43200 Maximum items: 1	28800
13	Boot File Size	Number of blocks. One block is 512 bytes. Integer values: 1–65535 Maximum items: 1	1385
19	Forward On/Off	IP forwarding Allowed values: [0, 1] 1 for on, 0 for off Maximum items: 1	0
26	MTU Interface	MTU for a given interface. Allowed values: 68–65535 Maximum items: 1	9600
28	Broadcast Address	IP address Maximum items: 1	10.10.10.255
35	ARP Timeout	Integer (seconds) Allowed values: 0–4294967295	360
40	NIS Domain	Text Maximum: 255 characters	vmware.com
41	NIS Servers	IP addresses in a preferred order Maximum items: 63	10.10.10.10
42	NTP Servers	IP addresses in a preferred order Maximum items: 63	10.10.10.11
44	NETBIOS Name Server	IP addresses in a preferred order Maximum items: 63	10.10.10.12
45	NETBIOS Dist Server	IP addresses in a preferred order Maximum items: 63	10.10.10.13

Code	Name	Value Type	Example Value
46	NETBIOS Node Type	Integer encoding of node type Allowed values: [1, 2, 4, 6] Maximum items: 4 1 = B-node - broadcast no WINS 2 = P-node - WINS only 4 = M-node - broadcast then WINS 8 = H-node - WINS then broadcast	2
47	NETBIOS Scope	String encoded according to RFC 1001/1002 Maximum: 255 characters	
58	Renewal Time	N/A - based on the lease time between 0–4294967295 Maximum items: 1	300
59	Rebinding Time	N/A - based on the lease time between 0–4294967295 Maximum items: 1	300
64	NIS+ Domain Name	Text (domain name)	vmware.com
65	NIS+ Server Address	IP addresses in a preferred order	10.10.10.10
66	Server Name	Text (server domain name) Maximum: 255 characters	10.10.10.253
67	Bootfile Name	Text (file name) Maximum: 255 characters	`/tftpboot/pxelinux/pxelinux.bin`
117	Name Service Search	Not natively supported with API Allowed values: [0, 6, 41, 44, 65] Maximum items: 5	6
119	Domain Search	One or more domain names. Each domain name must be enclosed in quotes and separated by commas.	vmware.com
150	TFTP server address	IP address	10.10.10.10
209	PXE Configuration File	Maximum: 255 characters	`configs/common`
210	PXE Path Prefix	Maximum: 255 characters	`/tftpboot/pxelinux/files/`
211	PXE Reboot Time	Allowed values: 0–4294967295	1800

DHCP Static Bindings

In a typical network environment, you have VMs that run services, such as FTP, email servers, application servers, and so on. You might not want the IP address of these VMs to change in your network. In this case, you can bind a static IP address to the MAC address of each VM (DHCP client). The static IP address must belong to the subnet (if any) that is configured on the segment, and it must not overlap with the DHCP IP ranges and DHCP server IP address.

DHCP static bindings are supported when you are configuring either a Segment DHCP server or a Gateway DHCP server on the segment. When a segment is using a DHCP Relay, you cannot configure static bindings.

On a DHCP for IPv4 server, static bindings are supported regardless of whether the segment uses a Segment DHCP or a Gateway DHCP configuration. On a DHCP for IPv6 server, static bindings are supported only when the segment uses a Segment DHCP configuration.

Static Binding Options Common to DHCPv4 and DHCPv6 Server

The following table describes the static binding options that are common to DHCP for IPv4 and DHCP for IPv6 servers.

Option	Description
Name	Enter a unique display name to identify each static binding. The name must be limited to 255 characters.
MAC Address	Required. Enter the MAC address of the DHCP client to which you want to bind a static IP address. The following validations apply to MAC address in static bindings: • MAC address must be unique in all the static bindings on a segment that uses a Segment DHCP server. • MAC address must be unique in all the static bindings across all the segments that are connected to the gateway and which use the Gateway DHCP server. For example, consider that you have 10 segments connected to a tier-1 gateway. You use a Gateway DHCP server for four segments (Segment1 to Segment4), and a Segment DHCP server for the remaining six segments (Segment5 to Segment10). Assume that you have a total of 20 static bindings across all the four segments (Segment1 to Segment4), which use the Gateway DHCP server. In addition, you have five static bindings in each of the other six segments (Segment5 to Segment10), which use a Segment DHCP server. In this example: • The MAC address in each of the 20 static bindings must be unique across all the segments (Segment1 to Segment4) that use the Gateway DHCP server. • The MAC address in the five static bindings must be unique on each segment (Segment5 to Segment10) that use a Segment DHCP server.

Option	Description
IP Address	• Required for IPv4 static binding. Enter a single IPv4 address to bind to the MAC address of the client. • Optional for IPv6 static binding. Enter a single Global Unicast IPv6 address to bind to the MAC address of the client. When no IPv6 address is specified for static binding, Stateless Address Autoconfiguration (SLAAC) is used to auto-assign an IPv6 address to the DHCPv6 clients. Also, you can use Stateless DHCP to assign other DHCP configuration options, such as DNS, domain names, and so on, to the DHCPv6 clients. For more information about Stateless DHCP for IPv6, read the RFC 3376 specifications. The following types of IPv6 addresses are not permitted in IPv6 static binding: • Link Local Unicast addresses (FE80::/64) • Multicast IPv6 addresses (FF00::/8) • Unspecified address (0:0:0:0:0:0:0:0) • Address with all F (F:F:F:F:F:F:F:F) The static IP address must belong to the subnet (if any) that is configured on the segment, and it must be outside the DHCP ranges that you have configured on the segment.
Lease Time	Optional. Enter the amount of time in seconds for which the IP address is bound to the DHCP client. When the lease time expires, the IP address becomes invalid and the DHCP server can assign the address to other DHCP clients on the segment. Valid range of values is 60–4294967295. Default is 86400.
Description	Optional. Enter a description for the static binding.
Tags	Optional. Add tags to label static bindings so that you can quickly search or filter bindings, troubleshoot and trace binding-related issues, or do other tasks. For more information about adding tags and use cases for tagging objects, see NSX Tags.

Static Binding Options (Only in DHCPv4 Server)

The following table describes the static binding options that are available only in a DHCP for IPv4 server.

DHCP For IPv4 Option	Description
Gateway Address	Enter the default gateway IP address that the DHCP for IPv4 server must provide to the DHCP client.
Host Name	Enter the host name of the DHCP for IPv4 client so that the DHCPv4 server can always bind the client (host) with the same IPv4 address each time. The host name must be limited to 63 characters. The following validations apply to host name in static bindings: • Host name must be unique in all the static bindings on a segment that uses a Segment DHCP server. • Host name must be unique in all the static bindings across all the segments that are connected to the gateway and which use the Gateway DHCP server. For example, consider that you have 10 segments connected to a tier-1 gateway. You use a Gateway DHCP server for four segments (Segment1 to Segment4), and a Segment DHCP server for the remaining six segments (Segment5 to Segment10). Assume that you have a total of 20 static bindings across all the four segments (Segment1 to Segment4), which use the Gateway DHCP server. In addition, you have five static bindings in each of the other six segments (Segment5 to Segment10), which use a Segment DHCP server. In this example: • The host name in each of the 20 static bindings must be unique across all the segments (Segment1 to Segment4) that use the Gateway DHCP server. • The host name in the five static bindings must be unique on each segment (Segment5 to Segment10) that use a Segment DHCP server.
DHCP Options	Optional. Click **Set** to configure DHCP for IPv4 Classless Static Routes and other Generic Options.

Some additional notes for DHCPv4 static binding:

- IPv4 static bindings automatically inherit the domain name that you configured on the segment.
- To specify DNS servers in the static binding configuration, add the **Generic Option** (Code 6 - DNS Servers).
- To synchronize the system time on DHCPv4 clients with DHCPv4 servers, use NTP. DHCP for IPv4 server does not support SNTP.
- If DHCP options are not specified in the static bindings, the DHCP options from the DHCPv4 server on the segment are automatically inherited in the static bindings. However, if you have explicitly added one or more DHCP options in the static bindings, these DHCP options are not autoinherited from the DHCPv4 server on the segment.

Static Binding Options (Only in DHCPv6 Server)

The following table describes the static binding options that are available only in a DHCP for IPv6 server.

DHCP for IPv6 Option	Description
DNS Servers	Optional. Enter a maximum of two domain name servers to use for the name resolution. When not specified, no DNS is assigned to the DHCP client.
SNTP Servers	Optional. Enter a maximum of two Simple Network Time Protocol (SNTP) servers. The clients use these SNTP servers to synchronize their system time to that of the standard time servers.
Preferred Time	Optional. Enter the length of time that a valid IP address is preferred. When the preferred time expires, the IP address becomes deprecated. If no value is entered, preferred time is auto-calculated as (lease time * 0.8). Lease time must be > preferred time. Valid range of values is 60–4294967295. Default is 69120.
Domain Names	Optional. Enter the domain name to provide to the DHCPv6 clients. Multiple domain names are supported in an IPv6 static binding. When not specified, no domain name is assigned to the DHCP clients.

Some additional notes for DHCPv6 static binding:

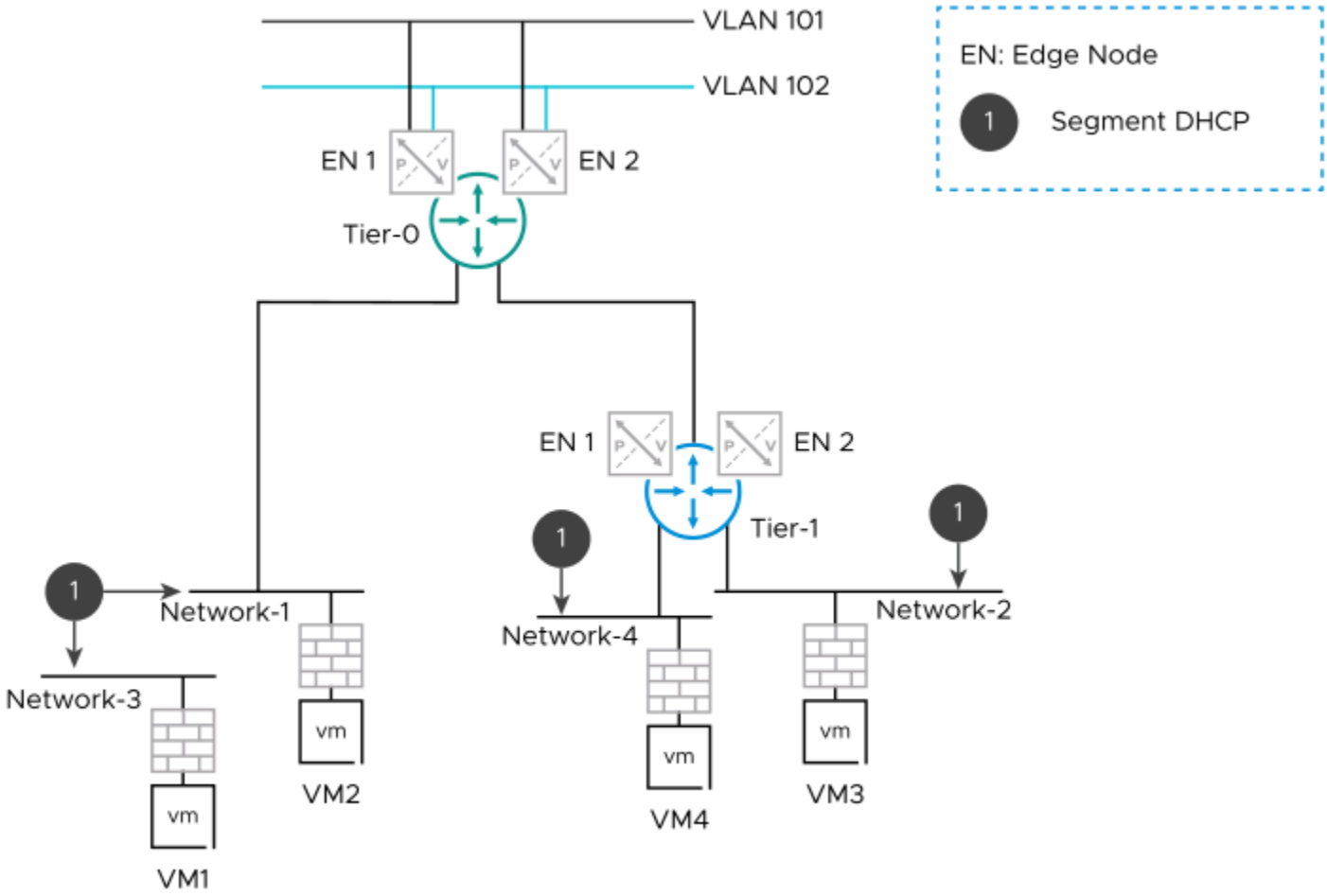
- Gateway IP address configuration is unavailable in IPv6 static bindings. IPv6 client learns about its first-hop router from the ICMPv6 router advertisement (RA) message.
- NTP is not supported in DHCPv6 static bindings.

Configure Segment DHCP Server on an NSX Segment

A Segment DHCP server provides a dynamic IP assignment service only to the VMs that are attached to the NSX segment. NSX supports Segment DHCP server configuration on the downlink interface and the service interface.

You can configure a Segment DHCPv4 server, or a Segment DHCPv6 server, or both, on the segment.

The following figure shows a sample network topology that has a Segment DHCP server configured on four networks.



In this network topology, a Segment DHCP server is configured on the following networks:

- Network-2 is connected to the downlink interface of tier-1 gateway.
- Network-1 is connected to the service interface of tier-0 gateway.
- Network-4 is connected to the service interface of tier-1 gateway.
- Network-3 is an isolated segment, which is not connected to any gateway.

1. From your browser, log in with **admin** privileges to an NSX Manager at https://*nsx-manager-ip-address*.
2. Select **Networking** > **Segments**.
3. Find the segment where you want to configure the DHCP service. Next to the segment name, click
   
   and then click **Edit**.
4. Click **Set DHCP Config**.
5. From the **DHCP Type** drop-down menu, select **Segment DHCP Server**.
6. Required: From the **DHCP Profile** drop-down menu, select a DHCP server profile. If no profile is available in the drop-down menu, click
   
   and then click **Create New** to add a DHCP server profile. After the profile is created, it is automatically attached to the segment.
   For more information about creating a DHCP server profile, see Add an NSX DHCP Server Profile.

7. Click the **DHCP Config** toggle button to enable DHCP configuration on the segment.

   If you are configuring a Segment DHCPv4 server and a Segment DHCPv6 server, ensure that you enable the **DHCP Config** toggle button in both the **IPv4 Server** and **IPv6 Server** tabs.

8. Specify the following DHCP configuration settings:

   - DHCP Server Address
   - DHCP Ranges
   - Optional: Excluded Ranges (only for DHCPv6)
   - Optional: Lease Time
   - Optional: Preferred Time (only for DHCPv6)
   - Optional: Domain Names (only for DHCPv6)
   - Optional: DNS Servers
   - Optional: SNTP Servers (only for DHCPv6)
   - Optional: DHCP Options (only for DHCPv4)
   - Optional: Static Bindings

   For a detailed information about these DHCP configuration settings, see the reference documentation at NSX DHCP Configuration Settings: Reference.

9. Click **Apply**.

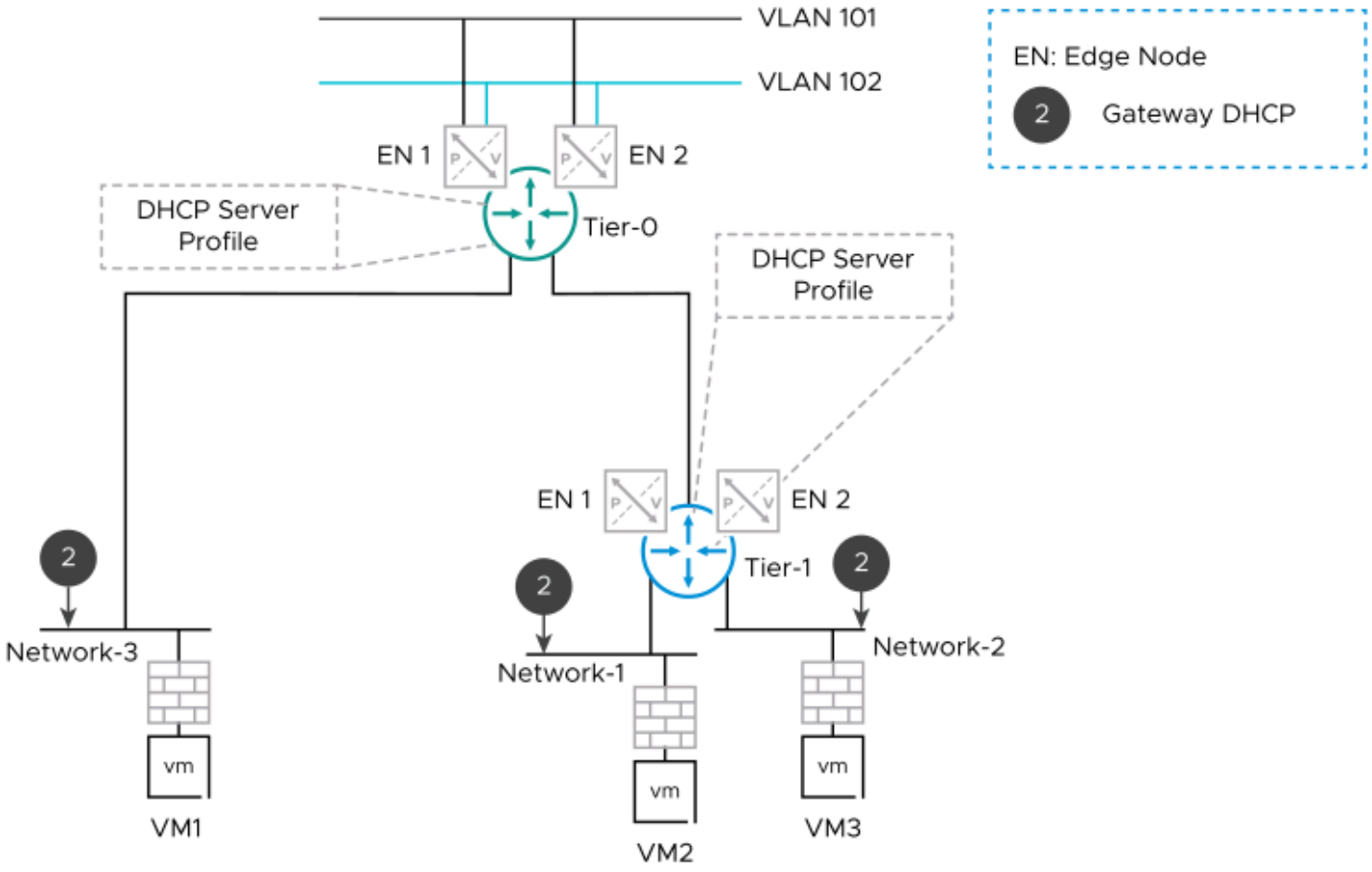
Configure Gateway DHCP Server on an NSX Segment

A Gateway DHCP server is attached to a tier-0 or tier-1 gateway, and it provides DHCP service to the networks (overlay segments), which are directly connected to the gateway and configured to use a Gateway DHCP server.

Ensure that you have specified the Gateway IP address of the IPv4 subnet in the segments that are directly connected to the tier-0 or tier-1 gateway.

In this case, the DHCP server that is created on the tier-0 or tier-1 gateway will have an internal relay so that the connected segments can forward traffic to the DHCP servers, which you specified in the DHCP server profile.

The following figure shows a sample network topology with Gateway DHCP servers that are servicing networks, which are directly connected to the tier-0 and tier-1 gateway.



In this network topology, a Gateway DHCP server is configured on the following networks:

- Network-1 is connected to the service interface of the tier-1 gateway.
- Network-2 is connected to the downlink interface of the tier-1 gateway.
- Network-3 is connected to the downlink or service interface of the tier-0 gateway.

1. From your browser, log in with **admin** privileges to an NSX Manager at https://*nsx-manager-ip-address*.
2. Select **Networking** > **Segments**.
3. Find the segment where you want to configure the DHCP service. Next to the segment name, click
   ,
   and then click **Edit**.
4. Click **Set DHCP Config**.
5. In the **DHCP Type** drop-down menu, select **Gateway DHCP Server**.
6. Required: Ensure that a DHCP server profile is attached to the gateway.
   If a profile is set on the gateway, the name of the profile is displayed in a read-only mode.
   If a DHCP server profile is not set on the gateway, do these steps:
   a) Click the information icon next to **DHCP Profile**, and then click the gateway name to navigate to the gateway page.
   b) Next to the gateway name, click
      ,
      and then click **Edit**.
   c) Next to **DHCP Config**, click **Set**.
      The **Set DHCP Configuration** window opens.

d) In the **Type** drop-down menu, select **DHCP Server**.
e) Select a DHCP server profile to attach to this gateway and click **Save**.
f) Close the edit mode on the gateway and return to the edit mode on the Segments page.
g) Click **Set DHCP Config** and continue the remaining steps in this procedure.

7. Click the **DHCP Config** toggle button to enable DHCP configuration on the segment.

   Note:
   You can configure only Gateway DHCPv4 server on a segment. Gateway DHCPv6 server is not supported.

8. Observe that the **DHCP Server Address** is fetched automatically from the DHCP profile and displayed on the **IPv4 Server** > **Settings** page.
9. Specify the following DHCP configuration settings:

   - DHCP Ranges
   - Optional: Lease Time
   - Optional: DNS Servers
   - Optional: DHCP Options
   - Optional: Static Bindings

   For a detailed information about these DHCP configuration settings, see the reference documentation at NSX DHCP Configuration Settings: Reference.

10. Click **Apply**.

Configure DHCP Relay on an NSX Segment

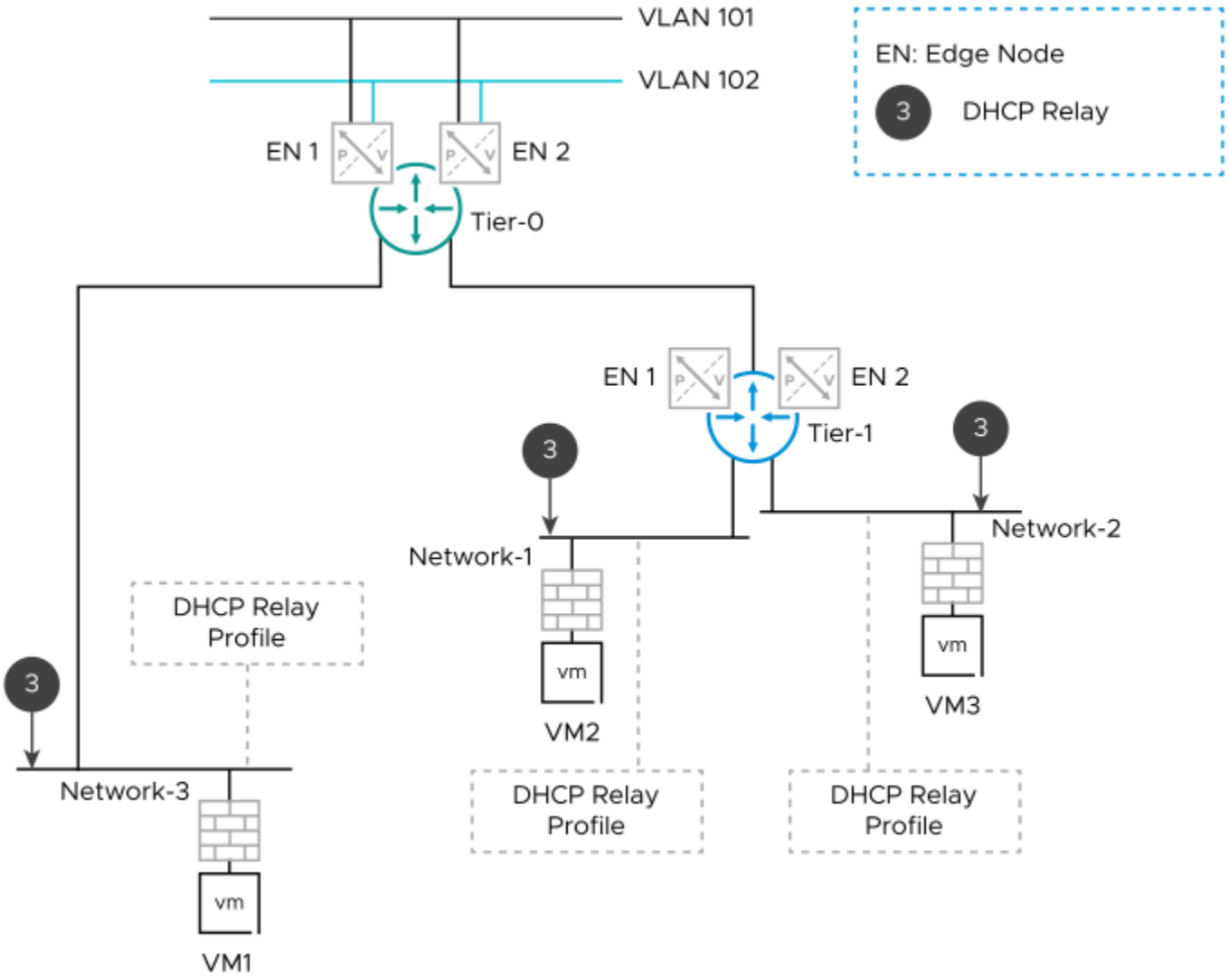
In a DHCP Relay configuration, the DHCP messages are forwarded to the external DHCP servers. The external DHCP servers can be in any subnet, outside the SDDC, or in the physical network.

DHCP Relay configuration is supported in the following scenarios:

- When an overlay segment is connected to the downlink interface of a tier-0 or tier-1 gateway. In this case, the DHCP messages can be relayed either to DHCPv4 servers or DHCPv6 servers.
- When an overlay or a VLAN segment is connected to the service interface of a tier-0 or tier-1 gateway that is configured in an active-standby mode. In this case, the DHCP messages are relayed only to DHCPv4 servers.

  Note:
  When you use a DHCP Relay on a segment, you cannot configure DHCP settings, DHCP options, and static bindings on the segment.

The following figure shows a sample network topology that has a DHCP Relay configured on three networks.



In this network topology, a DHCP Relay is configured on the following networks:

- Network-1 is connected to the service interface of the tier-1 gateway.
- Network-2 is connected to the downlink interface of the tier-1 gateway.
- Network-3 is connected either to the downlink or service interface of the tier-0 gateway.

1. From your browser, log in with **admin** privileges to an NSX Manager at https://*nsx-manager-ip-address*.
2. Configure a DHCP Relay on an overlay segment that is connected to the downlink interface of a tier-0 or tier-1 gateway.
   a) Navigate to **Networking** > **Segments**.
   b) Find the overlay segment where you want to configure the DHCP Relay. Next to the segment name, click
   and then click **Edit**.
   c) Click **Set DHCP Config**.
   d) From the **DHCP Type** drop-down menu, select **DHCP Relay**.
   e) From the **DHCP Profile** drop-down menu, select a DHCP relay profile. If no profile is available in the drop-down menu, click

and then click **Create New** to add a DHCP relay profile. After the profile is created, it is automatically attached to the segment.

f) Click **Apply**.

3. Configure a DHCP Relay on a segment that is connected to the service interface of a tier-0 or tier-1 gateway, which is configured in an active-standby mode.
   a) Navigate to **Networking** > **Segments**.
   b) Add a segment in either a VLAN or an overlay transport zone. Do not connect this segment to any gateway. Also, do not set any DHCP configuration on this segment, such as DHCP server address, DHCP ranges, and static bindings.
   For example, assume that you have added a segment in the VLAN transport zone with name as `My-VLAN-Segment`.
   c) Navigate to **Networking** > **Tier-0 Gateways** or **Networking** > **Tier-1 Gateways**.
   d) Find the gateway where you want to connect this VLAN segment to the service interface. Click  , and then click **Edit**.
   e) Expand the **Interfaces** or **Service Interfaces** section, and click the link to open the **Set Interfaces** page.
   f) Click **Add Interface**.
   g) In the **Name** text box, enter a name for this interface.
   For example, specify the name as `Connect-to-VLAN`.
   h) (Only for tier-0 gateway): From the **Type** drop-down menu, select **Service**.
   This step is not applicable to a tier-1 gateway.
   i) Enter the **IP Address/Mask** in a CIDR format.
   For example, enter `172.16.10.1/24`.
   j) From the **Connect To (Segment)** drop-down menu, select the **My-VLAN-Segment**, which you created earlier.
   k) From the **DHCP Profile** drop-down menu, select a DHCP relay profile.
   l) Click **Save** and then click **Close**.

Attach an NSX DHCP Profile to a Tier-0 or Tier-1 Gateway

To use Gateway DHCP server for a dynamic IP assignment, you must attach an NSX DHCP server profile to a tier-0 or tier-1 gateway.

A DHCP server profile is added in the network. For more details, refer to Add an NSX DHCP Profile.

You can attach a DHCP profile to a gateway only when the segments connected to that gateway do not have a Segment DHCP server or a DHCP relay configured on them. If a Segment DHCP server or DHCP relay exists on the segment, the UI throws an error when you try to attach a DHCP profile to the gateway. You must disconnect the segments from the gateway, and then attach a DHCP profile to the gateway.

1. From your browser, log in with **admin** privileges to an NSX Manager at `https://nsx-manager-ip-address`.
2. Go to **Networking** > **Tier-0 Gateways** or **Networking** > **Tier-1 Gateways**.
3. Edit the appropriate gateway.
4. Next to **DHCP Config**, click **Set**.
   The **Set DHCP Configuration** window opens.
5. Select one of the following options from the **Type** drop-down menu:
   - No Dynamic IP Address Allocation
   - DHCP Server – If there are segments under this gateway using Gateway DHCP Server, a DHCP server is created.

6. If you selected the **DHCP Server** option, select an existing DHCP server profile, or click
   
   to add a new DHCP server profile to attach to this gateway. For more details, see Add an NSX DHCP Server Profile.
7. Click **Save**.

Navigate to **Networking** > **Segments**. On each segment that is connected to this gateway, configure the DHCP settings, DHCP options, and static bindings.

For more information, see Configure NSX DHCP Service.

View NSX Gateway DHCP Statistics

After a Gateway DHCP server is in use, you can view the DHCP pool usage statistics of segments that are directly connected to the tier-0 or tier-1 gateway and using Gateway DHCP server.

- DHCP server profile is attached to the tier-0 or tier-1 gateway.
- Segments that are connected to the gateway are configured to use a Gateway DHCP server.
- DHCP settings are configured on the segments that are directly connected to the gateway.
- The server runtime status is up and the Gateway DHCP server is in use.

1. From your browser, log in to an NSX Manager at https://*nsx-manager-ip-address*.
2. Navigate to **Networking** > **Tier-0 Gateways** or **Networking** > **Tier-1 Gateways**.
3. Find the gateway whose Gateway DHCP server statistics you want to view.
4. Expand the gateway configuration settings, and then click the link next to **DHCP**.
5. In the pop-up window, click **View Statistics**.
   Gateway DHCPv4 server statistics are displayed.

   In the DHCP Server Packets section, a pie chart displays the breakup of the DHCP packet counts. These packet counts represent the count of the various DHCP message types. The number at the center of the pie chart represents the sum of all DHCP packet counts.

   The DHCP Pool Statistics section displays the pool usage statistics of segments that are directly connected to the gateway and using Gateway DHCP server. For example, this section shows statistics, such as the size of the DHCP pool, the number of IP addresses used from the pool, and the allocation percentage.

   Note:
   If you have configured a Segment DHCP server on a gateway-connected segment, the statistics of the Segment DHCP server are not displayed on the **DHCP Statistics** page of the gateway. DHCP statistics are displayed only for those segments that are configured to use a Gateway DHCP server.

   For example, assume that you have four segments connected to the downlink interfaces of a tier-1 gateway. Segments 1 and 2 are using a Segment DHCP server, whereas segments 3 and 4 are using the Gateway DHCP server. In this case, the **DHCP Statistics** page displays the Gateway DHCP server statistics only for segments 3 and 4.

   To view segment DHCP statistics, you must navigate to the **Segments** page. For more information, see View NSX Segment DHCP Statistics.
6. Optional: To reset DHCP packet counts, click **Reset Packet Counter**.
   The DHCP packet counts that are displayed next to the pie chart are reset. The DHCP pool statistics are not impacted.

View NSX Segment DHCP Statistics

After a Segment DHCP server is in use, you can view the DHCP pool usage statistics on the **Segments** page.

- DHCP settings are configured on the segment.
- The server runtime status is up and the Segment DHCP server is in use.

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Select **Networking** > **Segments**.
3. Find the segment whose DHCP statistics you want to view.
4. Expand the segment configuration settings, and then click **View Statistics**.
   The **Segment Statistics** page opens.
5. Click the **DHCP Statistics** tab.

   In the DHCP Server Packets section, a pie chart displays the breakup of the DHCP packet counts. The packet counts represent the count of the various DHCP message types. The number at the center of the pie chart represents the sum of all DHCP packet counts.

   The DHCP Pool Statistics section displays the pool usage statistics. For example, this section shows statistics, such as the size of the DHCP pool, the number of IP addresses used from the pool, and the allocation percentage.

   Important:
   If you have configured both DHCPv4 and DHCPv6 servers on a segment, the **DHCP Statistics** page will display only the DHCPv4 packet counts and the DHCPv4 pool usage statistics. DHCPv6 packet counts and DHCPv6 pool usage statistics are currently not supported.
6. Optional: To reset DHCP packet counts, click **Reset Packet Counter**.
   The DHCP packet counts that are displayed next to the pie chart are reset. The DHCP pool statistics are not impacted.

Scenarios: Selection of Edge Cluster for NSX DHCP Service

DHCP server runs as a service (service router) in the edge nodes of an NSX Edge cluster.

Isolated segments that are not connected to a gateway can use only a Segment DHCP server. Segments that are connected to a gateway on the downlink interface can use either a Segment DHCP server, DHCP Relay, or Gateway DHCP server.

Regardless of whether a segment uses a Segment DHCP server or a Gateway DHCP server, DHCP server always runs as a service router in the edge transport nodes of an edge cluster. If the segment uses a Segment DHCP server, the DHCP service is created in the edge cluster that you specified in the DHCP profile. However, if the segment uses a Gateway DHCP server, the edge cluster in which the DHCP service is created depends on the combination of the following factors:

- Is an edge cluster specified in the gateway?
- Is an edge cluster specified in the DHCP profile of the gateway?
- Is the edge cluster in the gateway and in the DHCP profile same or different?
- Is the tier-1 routed segment connected to a tier-0 gateway?

The following scenarios explain how the edge cluster is selected for creating the DHCP service.

Scenario 1: Isolated Segment Uses Segment DHCP Server

Scenario Description:

- An edge cluster (Cluster1) is created with four edge nodes: N1, N2, N3, N4.
- A segment with None connectivity is added in the overlay transport zone.
- Segment uses a Segment DHCP server, by default.

The DHCP server profile configuration is as follows:

- Profile Type: `DHCP Server`
- Edge Cluster: `Cluster1`
- Edges: `Autoallocated`

In this scenario, any two edge nodes from Cluster1 are autoallocated to create the DHCP service, and DHCP high availability (HA) is automatically configured. One of the edge nodes in Cluster1 runs in active mode and the other edge runs in passive mode.

Note:

- If you manually allocate the edge nodes in the DHCP profile, the edge node that is added first becomes the active edge. The second edge node takes the passive role.
- If you select only one edge node in the DHCP profile, DHCP HA is not configured.

Scenario 2: Tier-1 Routed Segment Uses Gateway DHCP and Different Edge Clusters in Gateway and DHCP Profile

Consider that you have two edge clusters in your network (Cluster1 and Cluster2). Both clusters have four edge nodes each:

- Cluster1 edge nodes: N1, N2, N3, N4
- Cluster2 edge nodes: N5, N6, N7, N8

Scenario Description:

- Segment is connected to a tier-1 gateway.
- Tier-1 gateway is not connected to a tier-0 gateway.
- DHCP server profile in the tier-1 gateway uses Cluster1.
- Tier-1 gateway uses Cluster2.
- Segment is configured to use the Gateway DHCP server.

The DHCP server profile in the tier-1 gateway has the following configuration:

- Profile Type: `DHCP Server`
- Edge Cluster: `Cluster1`
- Edges: `N1,N2` (allocated manually in the given sequence)

The tier-1 gateway configuration is as follows:

- Edge Cluster: `Cluster2`
- Edges: `N5,N6` (allocated manually in the given sequence)

In this scenario, DHCP service runs on the edge nodes of Cluster2. As Cluster2 contains multiple edge nodes, DHCP HA is autoconfigured. However, the manually allocated edges N5 and N6 on the gateway are ignored for DHCP HA. Any two nodes from Cluster2 are randomly autoallocated for DHCP HA.

This scenario also applies when the segment is directly connected to a tier-0 gateway, and there is no tier-1 gateway in your network topology.

CAUTION:
You can change the edge cluster on the Gateway DHCP server after the DHCP server is created. However, this action causes all the existing DHCP leases that are assigned to the DHCP clients to be lost.

To summarize, the main points of this scenario are as follows:

- When you use a Gateway DHCP server and set different edge clusters in the gateway DHCP profile and tier-1 gateway, then DHCP service is always created in the edge cluster of the gateway.
- The edge nodes are randomly allocated from the edge cluster of the tier-1 gateway for DHCP HA configuration.
- If no edge cluster is specified in the tier-1 gateway, the edge cluster in the DHCP profile of the tier-1 gateway (Cluster1) is used to create the DHCP service.

Scenario 3: Tier-1 Routed Segment Uses Segment DHCP Server and Different Edge Clusters in Gateway and DHCP Profile

In this scenario, a segment is connected to a tier-1 gateway, but you use a Segment DHCP server on the segment. Consider that you have three edge clusters in your network (Cluster1, Cluster2, Cluster 3). Each cluster has two edges nodes each.

- Cluster1 edge nodes: N1, N2

- Cluster2 edge nodes: N3, N4
- Cluster3 edge nodes: N5, N6

Scenario Description:

- Segment is connected to a tier-1 gateway.
- Tier-1 gateway is connected to a tier-0 gateway (optional).
- DHCP profile on the gateway uses Cluster1.
- Gateway uses Cluster2.
- Segment is configured to use Segment DHCP server.
- DHCP server profile on the segment uses Cluster3.

The DHCP server profile on the gateway is as follows:

- Profile Name: `ProfileX`
- Profile Type: `DHCP Server`
- Edge Cluster: `Cluster1`
- Edges: `N1,N2` (allocated manually in the given sequence)

The tier-1 gateway configuration is as follows:

- Edge Cluster: `Cluster2`
- Edges: `N3,N4`(allocated manually in the given sequence)

The profile on the Segment DHCP server is as follows:

- Profile Name: `ProfileY`
- Profile Type: `DHCP Server`
- Edge Cluster: `Cluster3`
- Edges: `N5,N6`(allocated manually in the given sequence)

In this scenario, because the segment is configured to use a Segment DHCP server, the edge cluster (Cluster2) in the connected tier-1 gateway is ignored to create the DHCP service. DHCP service runs in the edge nodes of Cluster3 (N5, N6). DHCP HA is also configured. N5 becomes the active edge node and N6 becomes the standby edge.

If edge nodes are not manually allocated from Cluster3, any two nodes from this cluster are autoallocated for creating the DHCP service and configuring DHCP HA. One of the edge nodes becomes an active edge and the other node becomes the standby edge. If only one edge node is allocated manually from Cluster3, DHCP HA is not configured.

This scenario also applies when the segment is directly connected to a tier-0 gateway, and there is no tier-1 gateway in your network topology.

Scenario 4: Tier-1 Routed Segment Uses Gateway DHCP and Same Edge Clusters in Gateway and DHCP Profile

Consider that you have a single edge cluster (Cluster1) in your network with four edge nodes: N1, N2, N3, N4.

Scenario Description:

- Segment is connected to a tier-1 gateway.
- Tier-1 gateway is connected to a tier-0 gateway (optional)
- Gateway and DHCP profile on the gateway use the same edge cluster (Cluster1).
- Segment is configured to use Gateway DHCP server.

The DHCP server profile on the gateway is as follows:

- Profile Type: `DHCP Server`
- Edge Cluster: `Cluster1`
- Edges: `N1,N2`(allocated manually in the given sequence)

The tier-1 gateway configuration is as follows:

- Edge Cluster: `Cluster1`
- Edges: `N3,N4` (allocated manually in the given sequence)

In this scenario, as the gateway DHCP profile and gateway use a similar edge cluster (Cluster1), DHCP service is created in the edge nodes N1 and N2 of the gateway DHCP profile. The edge nodes N3 and N4 that you specified in the connected tier-1 gateway are ignored for creating the DHCP service.

If edge nodes are not manually set in the DHCP profile, any two nodes from Cluster1 are autoallocated for creating the DHCP service and configuring DHCP HA. One of the edge nodes becomes an active edge and the other edge becomes the standby edge.

To summarize, the main points of this scenario are as follows:

- When you use a Gateway DHCP server and specify similar edge clusters in the DHCP profile and connected gateway, then DHCP service is created in the edge nodes of the DHCP profile.
- The edges nodes that you manually specified in the connected gateway are ignored.

Scenario 5: Tier-1 Routed Segment is Connected to Tier-0 Gateway and No Edge Cluster is Set in Tier-1 Gateway

In this scenario, a segment is connected to a tier-1 gateway, and the tier-1 gateway is connected to a tier-0 gateway. Consider that you have three edge clusters in your network (Cluster1, Cluster2, Cluster 3). Each cluster has two edges nodes each.

- Cluster1 edge nodes: N1, N2
- Cluster2 edge nodes: N3, N4
- Cluster3 edge nodes: N5, N6

Scenario Description:

- Segment is directly connected to a tier-1 gateway.
- Tier-1 gateway is connected to a tier-0 gateway.
- DHCP server profile is specified on both tier-1 and tier-0 gateways.
- DHCP profile on tier-1 gateway uses Cluster1.
- DHCP profile on tier-0 gateway uses Cluster2.
- No edge cluster is selected in tier-1 gateway.
- Tier-0 gateway uses Cluster3.
- Segment is configured to use a Gateway DHCP server.

In this scenario, because the tier-1 gateway has no edge cluster specified, NSX falls back on the edge cluster of the connected tier-0 gateway. DHCP service is created in the edge cluster of tier-0 gateway (Cluster3). Any two edge nodes from this edge cluster are autoallocated for creating the DHCP service and configuring DHCP HA.

To summarize, the main points of this scenario are as follows:

- When a tier-1 gateway has no edge cluster specified, NSX falls back on the edge cluster of the connected tier-0 gateway to create the DHCP service.
- If no edge cluster is detected in the tier-0 gateway, DHCP service is created in the edge cluster of the tier-1 gateway DHCP profile.

Scenarios: Impact of Changing Segment Connectivity on NSX DHCP

After you save a segment with DHCP configuration, you must be careful about changing the connectivity of the segment.

Segment connectivity changes are allowed only when the segments and gateways belong to the same transport zone.

The following scenarios explain the segment connectivity changes that are allowed or disallowed, and whether DHCP is impacted in each of these scenarios.

Scenario 1: Move a Routed Segment with Gateway DHCP Server to a Different Gateway

Consider that you have added a segment and connected it either to a tier-0 or tier-1 gateway. You configured Gateway DHCP server on this segment, saved the segment, and connected workloads to this segment. DHCP service is now used by the workloads on this segment.

Later, you decide to change the connectivity of this segment to another tier-0 or tier-1 gateway, which is in the same transport zone. This change is allowed. However, when you save the segment, an information message alerts you that changing the gateway connectivity impacts the existing DHCP leases, which are assigned to the workloads.

Scenario 2: Move a Routed Segment with Segment DHCP Server or Relay to a Different Gateway

Consider that you have added a segment and connected it either to a tier-0 or tier-1 gateway. You configured Segment DHCP server or DHCP Relay on this segment, saved the segment, and connected workloads to this segment. DHCP service is now used by the workloads on this segment.

Later, you decide to change the connectivity of this segment to another tier-0 or tier-1 gateway, which is in the same transport zone. This change is allowed. As the DHCP server is local to the segment, the DHCP configuration settings, including ranges, static bindings, and DHCP options are retained on the segment. The DHCP leases of the workloads are retained and there is no loss of network connectivity.

After the segment is moved to a new gateway, you can continue to update the DHCP configuration settings and other segment properties. You can change the DHCP type and DHCP profile of a routed segment after moving the segment to a different gateway.

Scenario 3: Move a Standalone Segment with Segment DHCP Server to a Tier-0 or Tier-1 Gateway

Consider that you have added a segment with None connectivity in your network. You have configured Segment DHCP server on this segment, saved the segment, and connected workloads to this segment. DHCP service is now used by the workloads on this segment.

Later, you decide to connect this segment either to a tier-0 or tier-1 gateway, which is in the same transport zone. This change is allowed. As a Segment DHCP server existed on the segment, the DHCP configuration settings, including ranges, static bindings, and DHCP options are retained on the segment. The DHCP leases of the workloads are retained and there is no loss of network connectivity.

After the segment is connected to the gateway, you can continue to update the DHCP configuration settings, and other segment properties. However, you cannot select a different DHCP type and the DHCP profile in the segment. For example, you cannot change the DHCP type from a Segment DHCP server to a Gateway DHCP server or a DHCP Relay. In addition, you cannot change the DHCP server profile in the segment. But you can edit the properties of the DHCP profile, if needed.

Scenario 4: Move a Standalone Segment Without DHCP Configuration to a Tier-0 or Tier-1 Gateway

Consider that you have added a segment with None connectivity in your network. You have not configured DHCP on this segment, saved the segment, and connected workloads to this segment.

Later, you decide to connect this segment either to a tier-0 or tier-1 gateway, which is in the same transport zone. This change is allowed. As no DHCP configuration existed on the segment, the segment automatically uses the Gateway DHCP server after it is connected to the gateway. The DHCP profile attached to this gateway gets autoselected in the segment.

Now, you can specify the DHCP configuration settings, including ranges, static bindings, and DHCP options on the segment. You can also edit the other segment properties, if necessary. However, you cannot change the DHCP type from a Gateway DHCP server to a Segment DHCP server or a DHCP Relay.

Remember, you can configure only a Gateway DHCPv4 server on the segment. Gateway DHCPv6 server is not supported.

Scenario 5: Move a Segment with Tier-0 or Tier-1 Connectivity to None Connectivity

Consider that you have added a segment to a tier-0 or tier-1 gateway in your network. You have configured Gateway DHCP server or DHCP Relay on this segment, saved the segment, and connected workloads to this segment. DHCP service is now used by the workloads on this segment.

Later, you decide to change the connectivity of this segment to None. This change is not allowed.

In this scenario, the following workaround can help:

1. Temporarily disconnect the existing segment from the gateway or delete the segment.
   a. Navigate to **Networking** > **Segments**.
   b. Click the vertical ellipses next to the segment, and then click **Edit**.
   c. Turn off the **Gateway Connectivity** option to disconnect the segment temporarily from the gateway.
2. Add a new segment and do not connect it to any gateway.
3. Configure a Segment DHCP server on this standalone segment, if needed.

Host Switches

A host switch managed object is a virtual network switch that provides networking services to the various hosts in the network. It is instantiated on every host that participates in NSX networking.

A host switch is a virtual switch that implements the networking data plane on various ESX hosts. The NSX host switch is based on the vSphere Distributed Switch (VDS), centrally managed by vCenter. The VDS switch is enhanced with some NSX capabilities and is centrally managed by the NSX Manager.

Managing NSX on a vSphere Distributed Switch

Configure and run NSX on a vSphere Distributed Switch (VDS).

You use a VDS switch to prepare ESX host nodes as transport nodes. Configure the vDefend Distributed Firewall functionality on VDS in data centers and workloads where segmentation, visibility, or advanced security capabilities are desired. This ensures distributed firewall capabilities work on a VM managed by whether it is managed by vCenter.

However, to prepare an NSX Edge VM as a transport node, you can only use a N-VDS switch. You can connect a NSX Edge VM to a VDS based on the topology in your network.

After you prepare a cluster of transport node hosts with VDS as the host switch, you can do the following:

- Manage NSX transport nodes on a VDS switch.
- Realize a segment created in NSX as an NSX Distributed Virtual port group in vCenter.
- Migrate VMs between VDS port groups.

Configuring a vSphere Distributed Switch

When a transport node is configured on a VDS host switch, some network parameters can only be configured in vCenter. The created VDS switch can be configured to centrally manage networking for NSX hosts.

Configuring a VDS switch for NSX networking requires objects to be configured on NSX and in vCenter.

- In vSphere:
- Create a VDS switch.
  - Set MTU to at least 1600. However, a minimum MTU of 1700 bytes is recommended.
  - Add ESX hosts to the switch. These hosts are later prepared as NSX transport nodes.
  - Assign uplinks to physical NICs.

    Note:

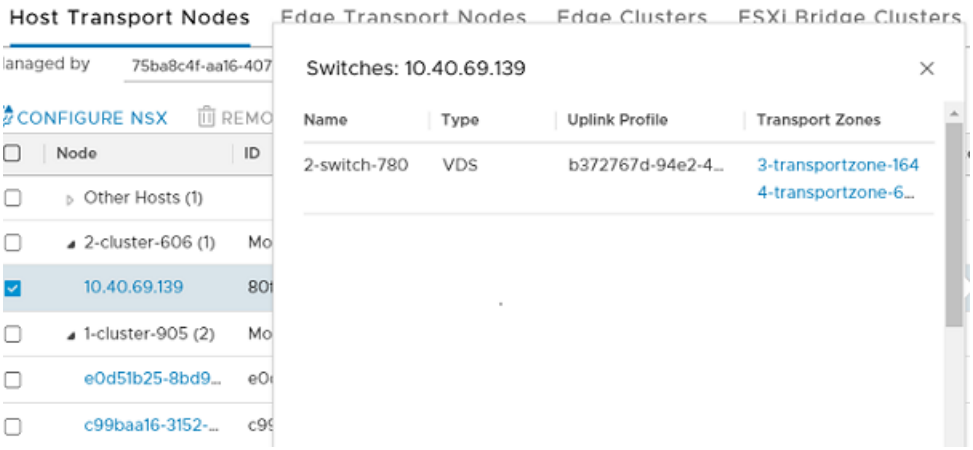
    If after creating the VDS switch you change the MTU value in vSphere, you must then access the NSX UI and complete a transport node resync for all hosts using this VDS. This can be done at the cluster level by reapplying the same Transport Node Profile (TNP), or at the host level by issuing a transport node resync.
- In NSX:
  - When configuring a transport node, map uplinks created in NSX uplink profile with uplinks in VDS.

The following parameters can only be configured in a vCenter on a VDS backed host switch:

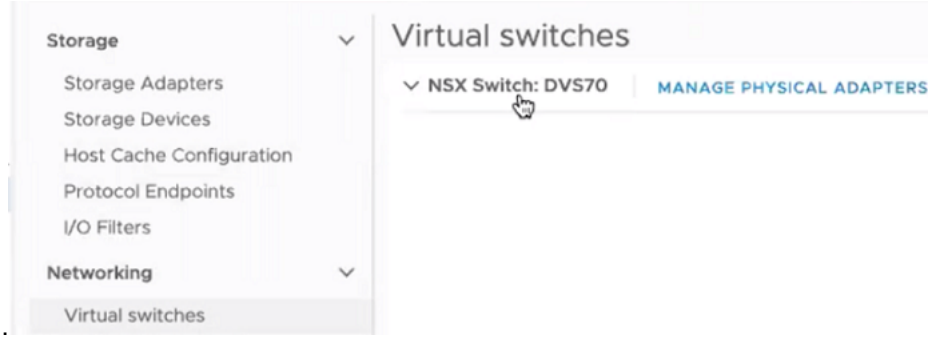
Configuration	VDS	NSX	Description
MTU	In vCenter, set an MTU value on the switch. **Note:** A VDS switch must have an MTU of 1600 or higher. Minimum MTU of 1700 bytes is recommended. In vCenter, select VDS, click **Actions** > **Settings** > **Edit Settings**.	Any MTU value set in an NSX uplink profile is overriden.	As a host transport node that is prepared using VDS as the host switch, the MTU value needs to be set on the VDS switch in vCenter.
Uplinks/LAGs	In vCenter, configure Uplinks/LAGs on a VDS switch. In vCenter, select VDS, click **Actions** > **Settings** > **Edit Settings**.	When a transport node is prepared, the teaming policy on NSX is mapped to uplinks/ LAGs configured on a VDS switch.	As a host transport node that is prepared using VDS as the host switch, the uplink or LAG are configured on the VDS switch. During configuration, NSX requires teaming policy be configured for the transport node. This teaming policy is mapped to the uplinks/LAGs configured on the VDS switch.
NIOC	Configure in vCenter. In vCenter, select VDS, click **Actions** > **Settings** > **Edit Settings**.	NIOC configuration is not available when a host transport node is prepared using a VDS switch.	As a host transport node that is prepared using VDS as the host switch, the NIOC profile can only be configured in vCenter.
Link Layer Discovery Protocol (LLDP)	Configure in vCenter. In vCenter, select VDS, click **Actions** > **Settings** > **Edit Settings**.	LLDP configuration is not available when a host transport node is prepared using a VDS switch.	As a host transport node that is prepared using VDS as the host switch, the LLDP profile can only be configured in vCenter.
Add or Manage Hosts	Manage in vCenter. In vCenter, go to **Networking** > **VDS Switch** > **Add and Manage Host**.	Prepared as transport nodes in NSX.	Before preparing a transport node using a VDS switch, that node must be added to the VDS switch in vCenter.

Note:
NIOC profiles, Link Layer Discovery Protocol (LLDP) profile, and Link Aggregation Group (LAG) for these virtual machines are managed by VDS switches and not by NSX. As a vSphere administrator, configure these parameters from vCenter UI or by calling VDS API commands.

After preparing a host transport node with VDS as a host switch, the host switch type displays VDS as the host switch. It displays the configured uplink profile in NSX and the associated transport zones.



In vCenter, the VDS switch used to prepare NSX hosts is created as an NSX Switch.



Managing NSX Distributed Virtual Port Groups

A transport node prepared with VDS as a host switch ensures that segments created in NSX is realized as an NSX Distributed Virtual port group on a VDS switch and Segment in NSX .

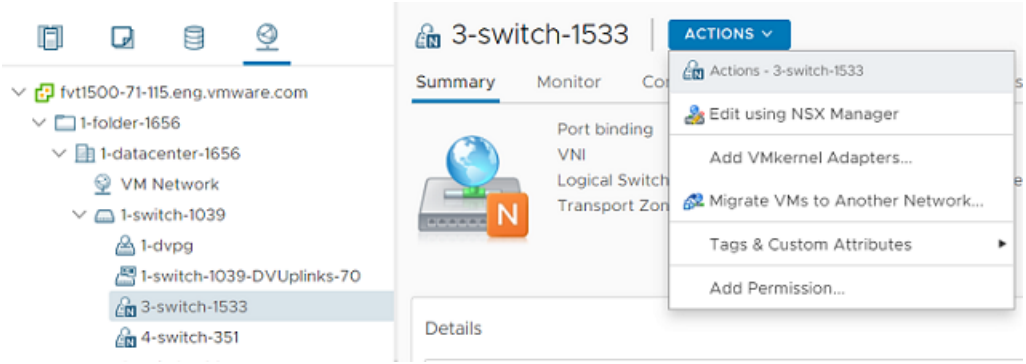
In earlier versions of NSX, a segment created in NSX are represented as an opaque network in vCenter. When running NSX on a VDS switch, a segment is represented as an NSX Distributed Virtual Port Groups.

Any changes to the segments on the NSX network are synchronized in vCenter.

In vCenter, an NSX Distributed Virtual Port Group is represented as



.



Any NSX segment created in NSX is realized in vCenter as an NSX object. A vCenter displays the following details related to NSX segments:

- NSX Manager
- Virtual network identifier of the segment
- Transport zone
- Attached virtual machines

The port binding for the segment is by default set to `Ephemeral`. Switching parameters for the switch that are set in NSX cannot be edited in vCenter and conversely.

> **Important:**
> In a vCenter, an NSX Distributed Virtual port group realized does not require a unique name to differentiate it with other port groups on a VDS switch. So, multiple NSX Distributed Virtual port groups can have the same name. Any vSphere automations that use port group names might result in errors.

In vCenter, you can perform these actions on an NSX Distributed Virtual Port Group:

- Add VMkernel Adapters.
- Migrate VMs to Another Network.

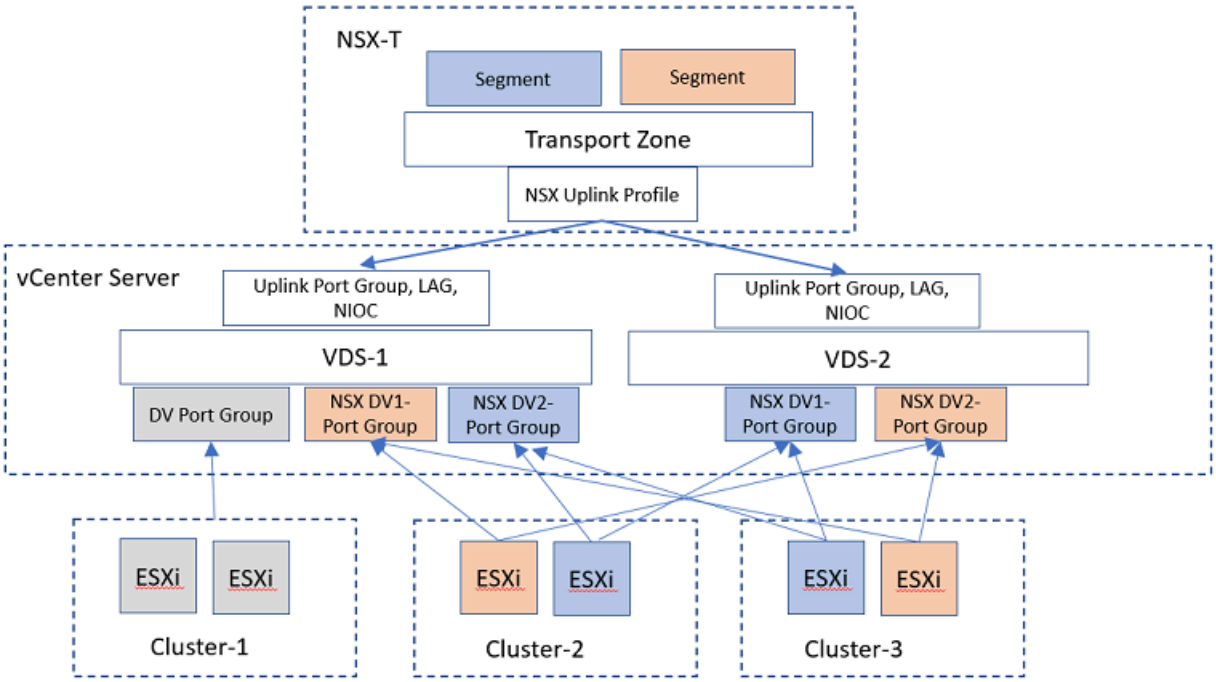
However, NSX objects related to an NSX Distributed Virtual port group can only be edited in NSX Manager. You can edit these segment properties:

- Replication Mode for the segment
- VLAN trunk ID used by the segment
- Switching Profiles (for example, Port Mirroring)
- Ports created on the segment

For details on configuring a vSphere Distributed Virtual port group, refer to the *vSphere Networking Guide*.

NSX Cluster Prepared with VDS

An example of an NSX cluster prepared using VDS as the host switch.



In the sample topology diagram, two VDS switches are configured to manage NSX traffic and vSphere traffic.

VDS-1 and VDS-2 are configured to manage networking for ESX hosts from Cluster-1, Cluster-2, and Cluster-3. Cluster-1 is prepared to run only vSphere traffic, whereas, Cluster-2 and Cluster-3 are prepared as host transport nodes with these VDS switches.

In vCenter, uplink port groups on VDS switches are assigned physical NICs. In the topology, uplinks on `VDS-1` and `VDS-2` are assigned to physical NICs. Depending on the hardware configuration of the ESX host, you might want to plan out how many physical NICs to be assigned to the switch. In addition to assiging uplinks to the VDS switch, MTU, NIOC, LLDP, LAG profiles are configured on the VDS switches.

After VDS switches are configured in NSX, add an uplink profile.

When preparing a cluster by applying a transport node profile (on a VDS switch), the uplinks from the transport node profile is mapped to VDS uplinks.

After preparing the clusters, ESX hosts on cluster-2 and Cluster-3 manage NSXtraffic, while cluster-1 manage vSphere traffic.

APIs to Configure vSphere Distributed Switch on NSX

NSX API commands to support vSphere Distributed Switch on NSX.

API Changes for vSphere Distributed Switch

For detailed information related to API calls, see the *NSX API Guide*.

Note:
Configuration done using API commands is also possible from the vCenter user interface. For more information on creating a NSX transport node using Sphere Distributed Switch as host switch, refer to the *Configure a Managed Host Transport Node* topic in the *NSX Installation Guide*.

API	NSX on vSphere Distributed Switch (VDS)	Changes in API Commands for NSX on vSphere Distributed Switch

Create a Transport node for a Discovered node.	`/api/v1/fabric/discovered-nodes/<external-id/discovered-node-id>?action=create_transport_node`
	`{`
	`"node_id": "d7ef478b-752c-400a-b5f0-207c04567e5d",`
	`"host_switch_spec": {`
	`"host_switches": [`
	`{`
	`"host_switch_name": "vds-1",`
	`"host_switch_id": "50 2b 92 54 e0 80 d8 d1-ee ab 8d a6 7b fd f9 4b",`
	`"host_switch_type": "VDS",`
	`"host_switch_mode": "STANDARD",`
	`"host_switch_profile_ids": [`
	`{`
	`"key": "UplinkHostSwitchProfile",`
	`"value": "159353ae-c572-4aca-9469-9582480a7467"`
	`} ],`
	`"pnics": [],`
	`"uplinks": [`
	`{`
	`"vds_uplink_name": "Uplink 2",`
	`"uplink_name": "nsxuplink1"`
	`} ],`
	`"is_migrate_pnics": false,`
	`"ip_assignment_spec": {`
	`"resource_type": "AssignedByDhcp"`
	`},`
	`"cpu_config": [],`
	`"transport_zone_endpoints": [`
	`{`
	`"transport_zone_id": "06ba5326-67ac-4f2c-9953-a8c5d326b51e",`
	`"transport_zone_profile_ids": [`
	`{`
	`"resource_type": "BfdHealthMonitoringProfile",`
	`"profile_id": "52035bb3-ab02-4a08-9884-18631312e50a"`
	`} ] } ],`
	`"vmk_install_migration": [],`
	`"pnics_uninstall_migration":`
		• `"host_switch_name": "vds-1"`: Is not a administrator entered switch name. The host switch name field is selected from the populated list of vSphere Distributed Switches created in vSphere.
		• `"host_switch_id"`: Is the UUID of the vSphere Distributed Switch object. The corresponding API in vSphere is `vim.DistributedVirtualSwitch.config.uuid`.
		• `"vds_uplink_name"`: An uplink created in vSphere Distributed Switch, mapping uplinks to physical NICs.
		• `"uplink_name"`: An uplink created in NSX that maps uplinks on N-VDS to the uplinks defined in vds_uplink_name.
		• `"is_migrate_pnics": false`: By default, migration of physical NICs when using vSphere Distributed Switch is not supported.
		• `"transport_zone_endpoints"`: Not supported when the host switch type is vSphere Distributed Switch. This field is required when the host switch type is N-VDS. Transport zone endpoint IDs correspond to the host switch it is associated with.
		For more details on `/api/v1/fabric/discovered-nodes/<external-id/discovered-node-id>?action=create_transport_node`, refer to the *NSX API Guide*.

API	NSX on vSphere Distributed Switch (VDS)	Changes in API Commands for NSX on vSphere Distributed Switch
VMkernel NIC	`vim.dvs.DistributedVirtualPort`	• N-VDS: As an NSX administrator, set values to these parameters: – `OpaqueNetworkId` – `OpaqueNetworkType` • NSX on vSphere Distributed Switch: As a vSphere administrator, set values to these parameters: – `SwitchUUID` and `portgroupKey` – `BackingType` of the DVPG must be `NSX`.
Physical NIC to Uplink Mapping	API: `vim.host.NetworkSystem: networkSystem.updateNetworkConfig` Property: `vim.host.NetworkConfig.proxySwitch`	• To map physical NICs to uplinks for vSphere Distributed Switch by calling API command, set the `maxMtu` property.
MTU	API: `vim.dvs.VmwareDistributedVirtualSwitch.reconfigure` Property: `VmwareDistributedVirtualSwitch.ConfigSpec.maxMtu`	• To configure an MTU value for vSphere Distributed Switch by calling API command, set the `maxMtu` property. **Note:** MTU value defined in uplink profiles in NSX are not applied to the host switch.
LAG	API: `vim.dvs.VmwareDistributedVirtualSwitch.updateLacpGroupConfig` Property: `vim.dvs.VmwareDistributedVirtualSwitch.LacpGroupSpec`	• To configure an LAG profile for vSphere Distributed Switch by calling API command, set the `LacpGroupSpec` property.
NIOC	API:`vim.dvs.VmwareDistributedVirtualSwitch.reconfigure` Property: `vim.dvs.VmwareDistributedVirtualSwitch.ConfigSpec.infrastructureTrafficResourceConfig`	• To configure an NIOC profile for vSphere Distributed Switch by calling API command, set the `infrastructureTrafficResourceConfig` property.
LLDP	API: `vim.dvs.VmwareDistributedVirtualSwitch.reconfigure` Property: `vim.dvs.VmwareDistributedVirtualSwitch.ConfigSpec.linkDiscoveryProtocolConfig`	• To configure LLDP profile for vSphere Distributed Switch by calling API command, set the `linkDiscoveryProtocolConfig` property.

Feature Support in a vSphere Distributed Switch Enabled to Support NSX

Comparison of features supported by a VDS switch version earlier to 7.0 and VDS version 7.0 or later (NSX enabled).

IPFIX and Port Mirroring

An NSX transport node prepared with a VDS switch supports IPFIX, port mirroring.

See Port Mirroring on a vSphere Distributed Switch.

See IPFIX Monitoring on a vSphere Distributed Switch.

SR-IOV support

SR-IOV is supported on a vSphere Distributed Switch but not on a NSX Virtual Distributed Switch.

Feature	NSX Virtual Distributed Switch	vSphere Distributed Switch
SR-IOV	No	Yes (vSphere 7.0 and later)

Stateless Cluster Host Profile Support

Feature	NSX Virtual Distributed Switch	vSphere Distributed Switch
Host Profile Stateless	Yes	Yes (vSphere 7.0 and later) No (when VMkernel adapters are connected to NSX Port Group on vSphere Distributed Switch.

Distributed Resource Scheduler Support

Source Host	Destination Host	DRS (NIOC Configured)	vSphere
vSphere Distributed Switch-A	vSphere Distributed Switch-B	No	No
vSphere Distributed Switch-A	vSphere Distributed Switch-A	Yes	7.0

vMotion Support

vMotion between source vSphere Distributed Switch and destination vSphere Distributed Switch. Both VDS switches are enabled to support NSX.

Source / VDS	Destination / VDS	Compute vMotion	Storage vMotion
vSphere Distributed Switch-A (vCenter -A)	vSphere Distributed Switch-A (vCenter-A)	Yes	Yes
vSphere Distributed Switch-A (vCenter -A)	vSphere Distributed Switch-B (vCenter -A)	Yes	Yes
vSphere Distributed Switch-A (vCenter -A)	vSphere Distributed Switch-B (vCenter -B)	Yes	Yes
Segment-A (vCenter -A)	Segment-B (vCenter-A)	No	No
Segment-A (vCenter -A)	Segment-B (vCenter -B)	No	No
Transport Zone-A	Transport Zone-B	No	No
NSX-A	NSX-B	No	No

vMotion between vSphere Distributed Switch (NSX enabled) and NSX Virtual Distributed Switch

Source / VDS	Destination / NSX Virtual Distributed Switch	Compute vMotion	Storage vMotion
vCenter-A	vCenter-A	Yes	Yes
vCenter-A	vCenter-B	Yes	Yes
Segment-A (vCenter-A)	Segment-B (vCenter-A)	No	No
Segment-A (vCenter-A)	Segment-B (vCenter-B)	No	No
Transport Zone-A	Transport Zone-B	No	No
NSX-A	NSX-B	No	No

vMotion between vSphere Distributed Switch (NSX enabled) and vSphere Standard Switch or vSphere Distributed Switch

Source / VDS	Destination / NSX Virtual Distributed Switch	Compute vMotion	Storage vMotion
vCenter-A	vCenter-A	Yes	Yes
vCenter-A	vCenter-B	Yes	Yes
Segment-A (vCenter-A)	Segment-B (vCenter-A)	No	No
Segment-A (vCenter-A)	Segment-B (vCenter-B)	No	No
Transport Zone-A	Transport Zone-B	No	No
NSX-A	NSX-B	No	No

Enhanced Networking Stack

Both VDS and NSX Virtual Distributed Switches support all features of the enhanced networking stack.

Scale Supported in vSphere 7.0

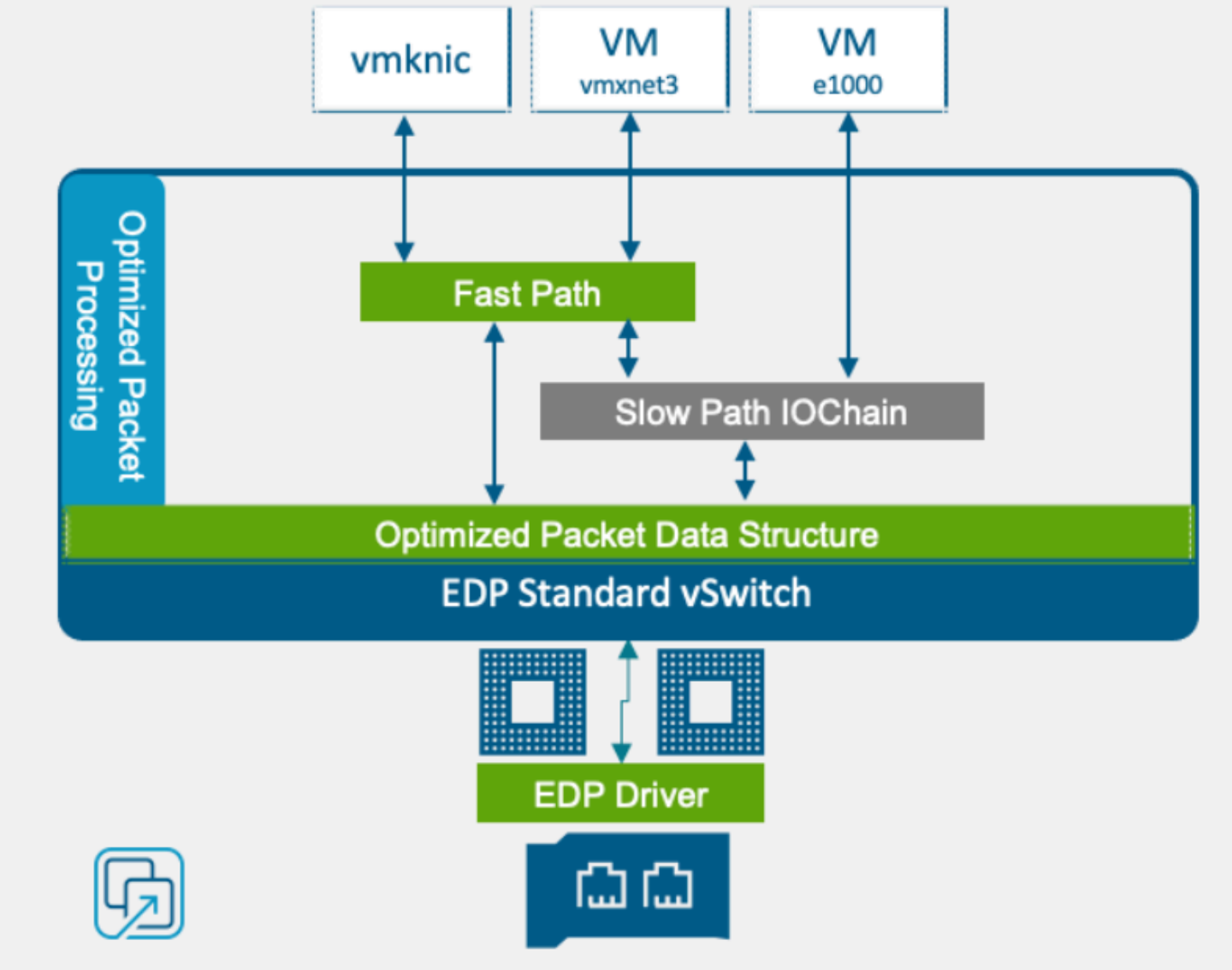
Parameter	NSX Virtual Distributed Switch
Logical Switch	• NSX Distributed Virtual port groups (in vCenter) support 10000 X N, where N is the number of VDS switches in vCenter. • NSX supports 10000 segments.

Relationship between NSX Distributed Virtual port groups and Hostd memory on the host.

NSX Distributed Virtual Port Groups	Minimum Hostd Memory	Supported VMs
5000	600 MB	191
10000	1000 MB	409
15000	1500 MB	682

Enhanced Data Path

Enhanced Data Path (EDP) is a packet forwarding stack designed to provide superior performance in terms of throughput, packet rate, latency, and CPU efficiency.



EDP is suitable for general compute workloads, network workloads and niche workloads like NSX Edge, and telco Network Functions Virtualization (NFV) solutions. It is enabled automatically in new VMware Cloud Foundation (VCF) 9.x workload domains. Environments that were converted, imported, or upgraded to VCF 9.x (and not running the EDP virtual switch mode) can be configured by setting the transport nodes to operate in one of the following modes.

Datapath Modes

Mode	Description	Use Case
EDP Standard	Provides out-of-the-box high-efficiency packet processing through dynamic CPU allocation.	As the default mode, EDP Standard is ideal for general workloads, NSX Edge clusters, and most enterprise workloads.
EDP Dedicated	A prescriptive mode that requires manual reservation of physical CPU cores for the data plane to ensure deterministic performance.	Use for telco Network Functions Virtualization (NFV) and specialized virtual appliances.
Standard (deprecated)	Lacks the performance optimizations of the EDP architecture. This mode is available for legacy configurations.	Use with network adapters that do not support EDP Standard.

Who Should Enable EDP?

The EDP stack is the default networking architecture in VCF 9.x. It serves as a superior alternative to the legacy Standard stack.

- **EDP Standard** is versatile enough for all environments—ranging from general compute and high-performance NVMe-over-Fabrics to specialized NSX Edge and high-performance computing.
- **EDP Dedicated** is designed for telco NFV and specialized virtual appliances.

> **Note:**
>
> Upgrading to VCF 9.x does **not** automatically change your virtual switch from legacy Standard to EDP Standard. While enabling EDP Standard is highly recommended to unlock performance gains, it remains a post-upgrade administrative task.

To enable EDP Standard in environments that were converted, imported, or upgraded to VCF 9.x and are still running the legacy Standard stack, read the next sections in this guide.

When to Use EDP?

EDP Standard is **recommended** for all VCF 9.x environments running compatible network adapters.

EDP Standard is **required** when you enable the following features:

- L4 Equal-Cost Multi-Path in the ESX host switch
- IPFIX (EDP Standard optimizes CPU cost)

Requirements and Restrictions

EDP requires a compatible network interface card (NIC) installed on the ESX host. For a list of compatible NIC devices, refer to the Broadcom compatibility guide.

In the compatibility guide, verify the following requirements for the NIC:

- **ESX Version Compatibility**—Verify that the installed ESX version appears under the Supported Releases list for the NIC.
- **Feature Verification**—Verify that the NIC appears in the results list when filtered by Features > EDP Standard. For telco NFV installations that use poll mode, verify that the NIC appears when filtered by Features > EDP Dedicated.

> **Note:**
>
> In an EDP-enabled environment, changing the NetIOC or MTU configuration causes all uplinks on the vSphere distributed switch (VDS) to restart.

Enabling Enhanced Data Path Standard in a New Environment

EDP Standard can be enabled in a “new environment” as a way to maximize or future-proof a cluster’s packet processing capabilities.

What qualifies as a new environment in this context?

- New VMware Cloud Foundation (VCF) Workload Domains.
- Created VCF environments that will utilize a new NSX Manager.

Overview of enabling EDP Standard in a new environment

1. Understand when you should use EDP, along with the requirements and restrictions described in Enhanced Data Path.
2. In VCF deployments, no manual intervention is required.
   When configuring workload domains, EDP Standard is selected by default as the virtual switch operational mode in the Switch configuration page. If you are running incompatible network adapters, change to legacy (non-EDP) Standard mode.
   See Create a New Workload Domain.

Enabling EDP Standard in Active Environments

Manually enable Enhanced Datapath Standard (without data path impact) on clusters that are **not** configured to use vSphere Configuration Profiles (VCP). To enable EDP Standard in VCP-enabled clusters, simply change the mode using the Transport Node Profile or Workload Domain network settings. VCP-enabled clusters do not require the procedure below.

- Create a Transport Node Profile (TNP) with EDP Standard enabled (required), taking note of the existing Transport Node Profile settings.
  - Create an identical TNP replacement with the mode set to Enhanced Datapath - Standard.
    This new Transport Node Profile will be used to enable EDP Standard in active environments.

    Note:

    Refer to Adding a Host Transport Node Profile for Hosts for general TNP creation steps.
- Configure the vCenter to allow vMotion during mode transitions:
  `config.vpxd.network.allowVmotionBetweenENSModes=true`
  Use this setting in the vCenter where the compute cluster or TNC exists. This configuration ensures VC doesn't block any mode changes during the vMotion of workload VM's from one host to another.
- Reset previous performance configurations (required only if configurations exist). EDP Standard is designed to provide the best performance outcomes with default configurations. Its predecessor (the default Standard mode) may host network optimizations like the following:

  ```
  esxcli system module parameters set -p "DRSS=X, RSS=Y"
  esxcli system settings advanced set -o /Mem/ShareCOSBufSize -i 32
  esxcli system settings advanced set -i 1 -o /Net/NetSchedHClkMQ
  esxcli system settings advanced set -o /Net/NetNetqLoadAvgByteCountShift -i 50
  ```

  Use the following API call after step 1, while each host is in maintenance mode, to reset any host performance settings previously configured.
  PUT https://<NSXMGR-IP>/policy/api/v1/infra/host-switch-profiles/<HPPName>

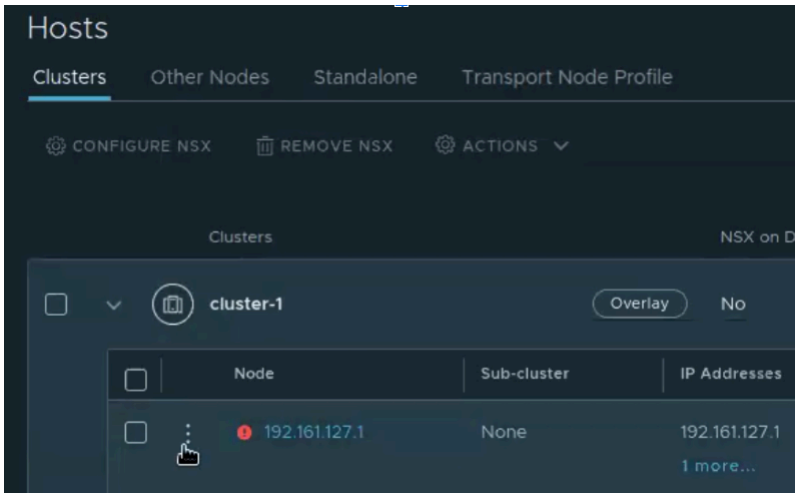
  ```
  {
     "high_performance_configs": [],
     "auto_config": 0,
     "resource_type": "PolicyHighPerformanceHostSwitchProfile"
  }
  ```

  Note:

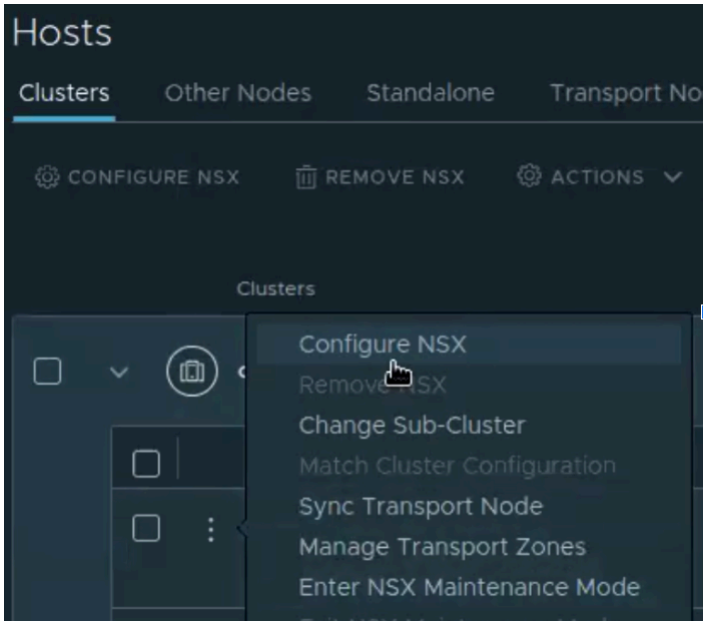
  Replace <NSXMGR-IP> with the NSX Manager IP address, and <HPPName> with a High Performance Profile identified based on your naming convention. Refer to the NSX API Guide for general information using the NSX Rest API.

To enable EDP standard in active environments, use the following procedure. The procedure below uses ESX maintenance mode to transition the mode without any impact to the network forwarding path.

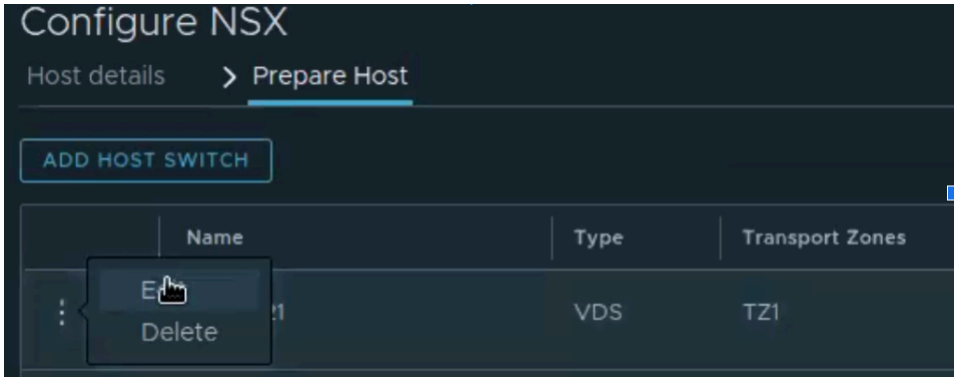
1. Open the vCenter and locate the first Transport Node ESX Host and place the ESX host in Maintenance Mode.
2. Open the NSX Manager, and navigate to **System** > **Configuration - Fabric** > **Hosts** .
3. In the Clusters panel, click the **Node** vertical ellipsis to open the node options.



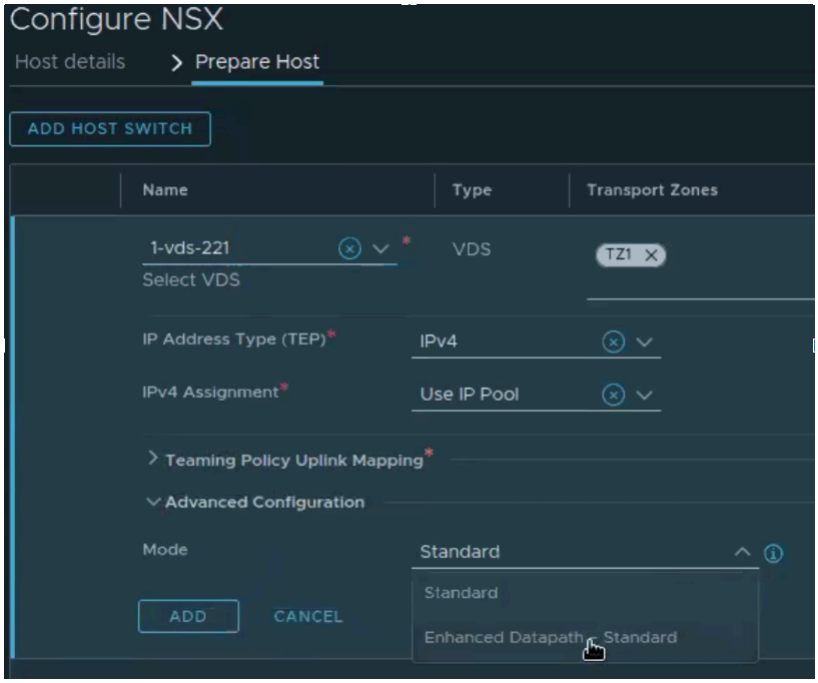
4. In the node options, click Configure NSX. In the Host details panel Click **Next**.



5. In the **Prepare Host** panel, click the vertical ellipsis and **Edit** the host switch.



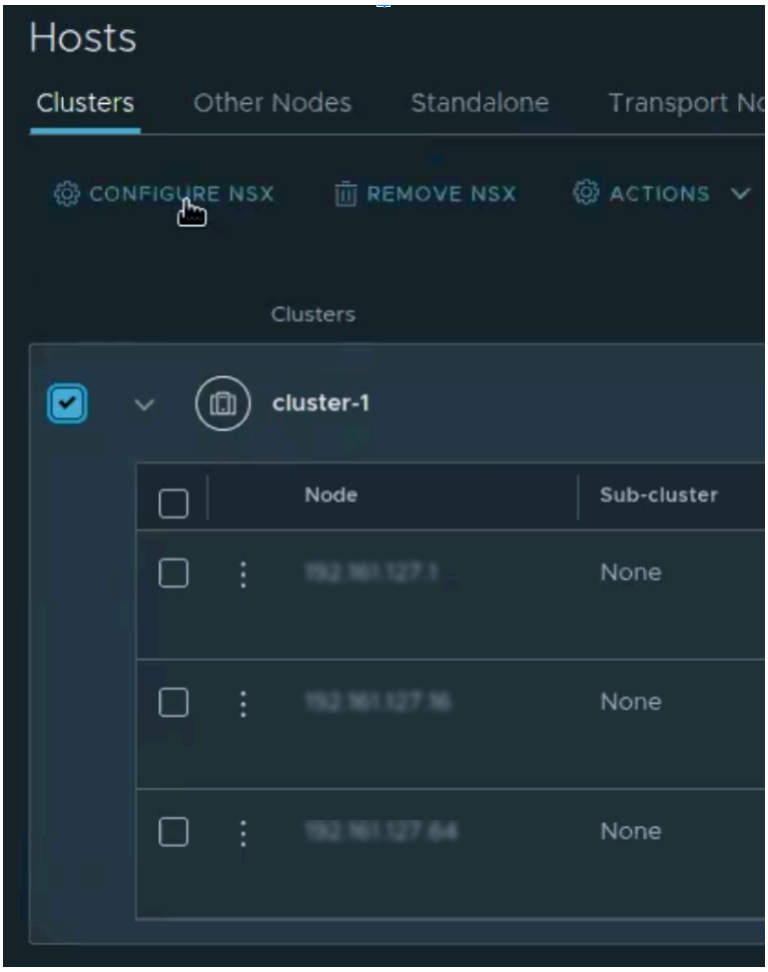
6. Expand Advanced Configuration and change the mode to **Enhanced Datapath - Standard**.



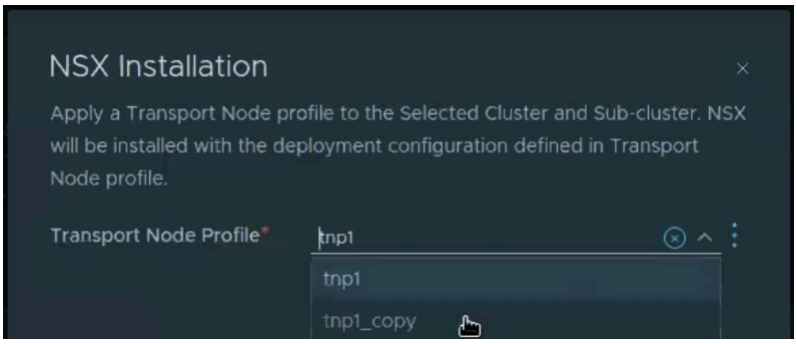
You can verify the mode change was applied on the ESX shell with the command `esxcfg-nics -e`:

```
[root@w1-hs5-c2504:~] esxcfg-nics -e
Name    Driver      ENS Capable  ENS Driven  INTR Capable  INTR Enabled  MAC Address       Description
vmnic0  bnxtnet     True         True        True          True          bc:97:e1:da:be:70 Broadcom … 25Gb …
vmnic1  bnxtnet     True         True        True          True          bc:97:e1:da:be:71 Broadcom … 25Gb …
```

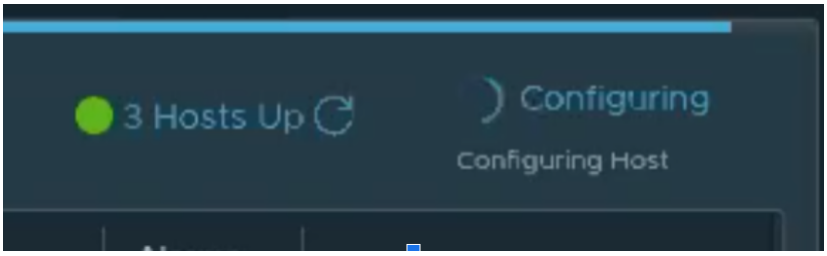
7. In the vCenter, exit **Maintenance Mode**.
8. Repeat the above steps for every host in the Transport Node cluster.
9. Replace the existing Transport Node Profile with the EDP Standard-enabled replacement identified in Prerequisites.
   a) Navigate to **System > Configuration - Fabric > Hosts**
   b) In the Clusters panel, select the Cluster and click **Configure NSX**.



   c) In the NSX Installation screen, select the new EDP Standard TNP and click **Save**.



d) The EDP Standard Transport Nodes become synchronized to the new EDP Standard TNP.



Automated Enablement of EDP Standard

This procedure is a supplement to Enabling EDP Standard in Active Environments. It provides REST API-based steps for the same procedure, which can be used to enable EDP Standard in active environments without impacting the data path.

Configure the vCenter:

```
config.vpxd.network.allowVmotionBetweenENSModes=true
```

Use this setting in the vCenter where the compute cluster or TNC exists. This configuration ensures the vCenter does not block any mode changes during the vMotion of workload VM's from one host to another.

> **Note:**
>
> This procedure is written for technical staff experienced with operational automation. Refer to the NSX REST API Guide for general information.

In the EDP context, an active environment is defined as follows:

- ESX hosts that are already running as NSX Transport Nodes in Standard mode.
- ESX hosts without NSX, but with live workloads that cannot tolerate impact.

The procedure below explains how to enable Enhanced Data Path Standard with the NSX REST API. The procedure is written using a fictional environment with the following components:

- A vLCM or VUM NSX-enabled Transport Node Cluster named “TNC1”.
- A single host or Transport Node named “TN1” within TNC1.
- TNC1 is associated with a Transport Node Profile “TNP1” I.

In the automated procedure, once a host is placed into maintenance mode, the API is used to transition HostSwitchMode=STANDARD to HostSwitchMode-ENS_INTERRUPT.

1. Put TN1 inside the TNC1 in ESX maintenance mode.

   > **Note:**
   >
   > If applicable - apply a High Performance Profile to reset any custom network performance configurations. See the previous topic for more details.

2. Change TN1's HostSwitchMode to ENS_INTERRUPT by calling the TN PUT API. → this will be allowed and will mark the Host as Overridden. Since the Host's are in overridden state TNP1 attached to that TNC1 will not move the HostSwitchMode back to STANDARD on TN1.
   a) API to get all transport-nodes:

   ```
   GET https://<MP_Endpoint>/api/v1/infra/sites/default/enforcement-points/default/
   host-transport-nodes/
   ```

   b) API to update the transport node corresponding to TN1:

   ```
   PUT https://<MP_Endpoint>/api/v1/infra/sites/default/enforcement-points/default/
   host-transport-nodes/<host-transport-node-id>
   ```

   c) GET API to verify the Mode change:

   ```
   GET https://<MP_Endpoint>/api/v1/infra/sites/default/enforcement-points/default/
   host-transport-nodes/<host-transport-node-id>
   ```

3. Turn off the ESX maintenance mode on TN1.
4. Perform steps 1-3 for all TN's in that TNC1 one after the other.
5. Once all TN's within the TNC1 are updated to ENS_INTERRUPT successfully, make a copy of TNP1 and create TNP2 with just one change to make HostSwitchMode=ENS_INTERRUPT for TNP2.
   a) API to GET all transport-node-profiles:

      ```
      GET https://<MP_Endpoint>/policy/api/v1/infra/host-transport-node-profiles/
      ```

   b) Fetch the TNP Id of TNP1 and GET the config:

      ```
      GET https://<MP_Endpoint>/policy/api/v1/infra/host-transport-node-profiles/<TNP1 ID>
      ```

   c) API to create a new TNP (TNP2):

      ```
      PUT https://<MP_Endpoint>/policy/api/v1/infra/host-transport-node-profiles/
      <transport-node-profile-id> to create TNP2.
      ```

6. Update (TNC PUT API) the TNP associated with TNC1 to TNP2.
   a) API to GET all transport-node-collections:

      ```
      GET https://<MP_Endpoint>/api/v1/infra/sites/default/enforcement-points/default/
      transport-node-collections/
      ```

   b) API to update TNC:

      ```
      PUT https://<MP_Endpoint>/api/v1/infra/sites/default/enforcement-points/default/
      transport-node-collections/<transport-node-collections-id>
      ```

      This will try to apply TNP2 with ENS_STANDARD, but since all TN's are already on ENS_STANDARD this is basically a no-op and all overridden TN's will have the override flag removed.

Enhanced Data Path Optional Configurations

EDP is designed to provide optimal services out of the box with its default configuration. If you're enabling EDP for the first time, start with Enhanced Data Path and the enablement procedures provided there.

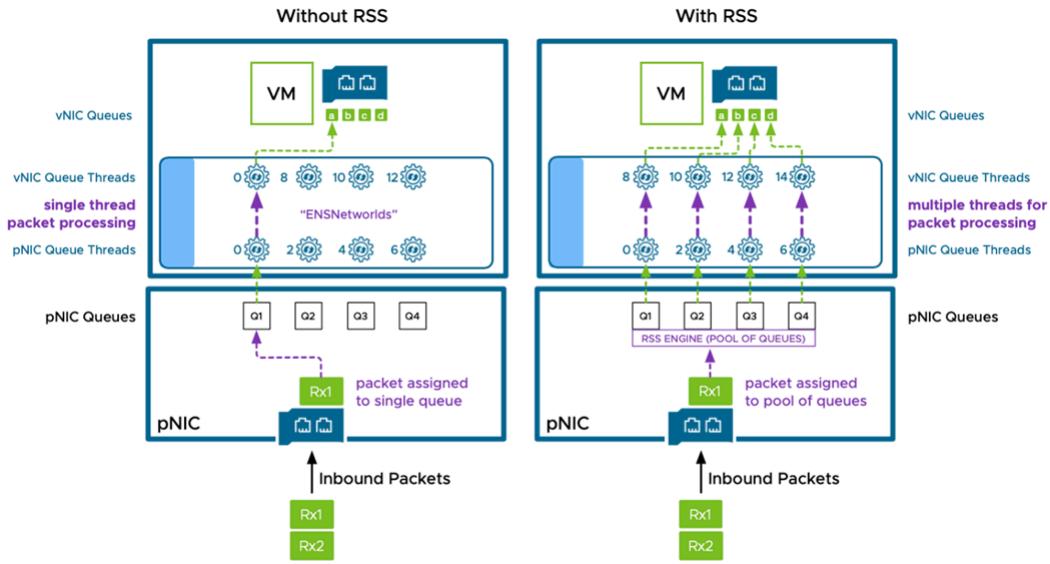
The topics contained within this section provide additional settings that may be helpful under certain conditions with Broadcom technical guidance.

Receive Side Scaling

EDP takes full advantage of RSS to scale out.

Receive Side Scaling allows multiple threads on the receive side for processing incoming traffic for a NIC.

Without RSS, receiving ESX hosts only use one physical queue and hence one lcore (in EDP Dedicated) for packet processing. When the receive side data increases it creates a bottleneck at the single core. The overall throughput performance might decrease. With RSS enabled on the NIC, you can configure multiple hardware queues to process requests to VMs. Before you use a NIC card to leverage the RSS functionality, use the Broadcom Compatibility Guide for I/O to confirm whether the NIC card driver supports RSS.

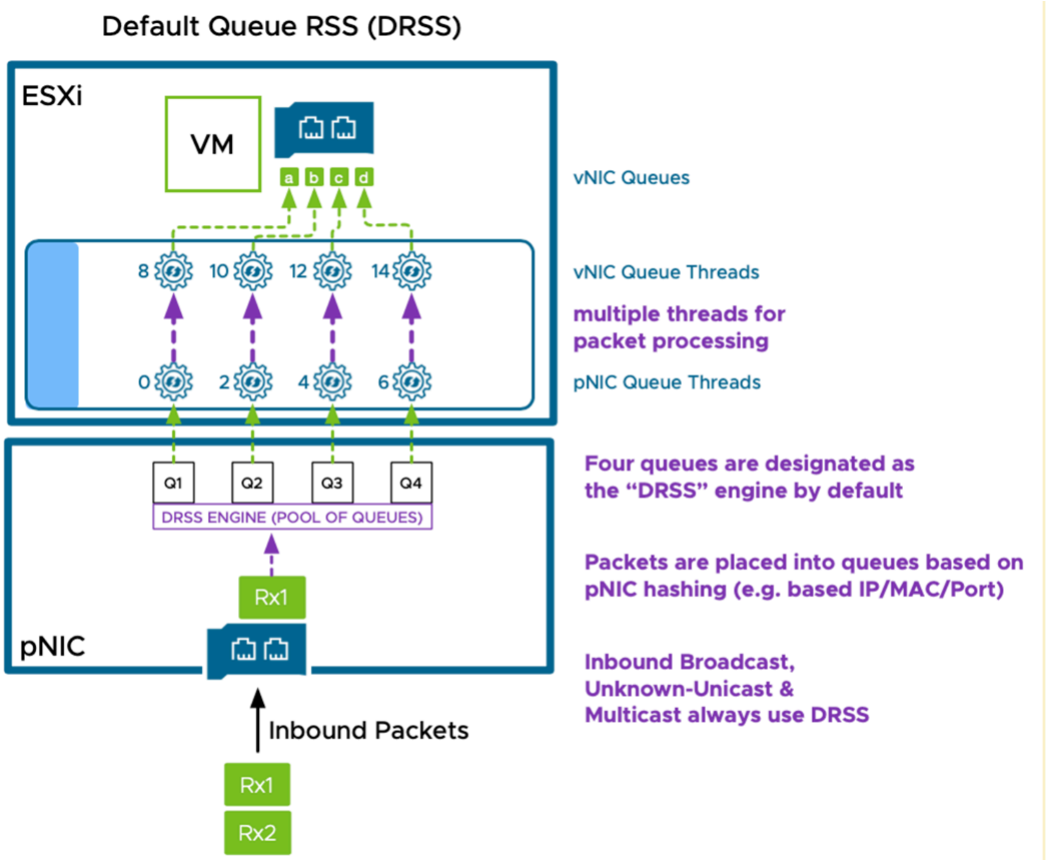


You can choose to enable or disable RSS modes:

RSS Types	EDP Behavior
Default Queue RSS	Enabled by default.
Shared NetQueue RSS	Optional setting. See Configure NetQueue Receive Side Scaling.
Dedicated NetQueue RSS	Optional setting. This setting assigns an entire RSS engine to a single virtual machine. It is not considered a best practice to enable this setting. Engage Broadcom for technical guidance.
Device RSS	Conceptually similar to Default Queue RSS but applies to all pNIC queues. This will be seen on devices that do not support NetQueue. EDP does not interact with Device RSS (this type of RSS is managed fully on the driver).

Configure Default Queue Receive Side Scaling

Enhanced Data Path manages the receive-side data arriving at physical NIC cards. A single port on the physical NIC card makes multiple hardware queues available to receive-side data.



Queues are assigned to packet processing threads from the non-uniform memory access (NUMA) node. When inbound packets - multicast, unknown, or broadcast - arrive at a physical NIC port, they are distributed across several hardware queues. The distribution is based on hashing certain fields (like IP Address, TCP/UDP port) and determining the queue placement based on the hash.

Default Queue RSS reduces bottlenecks from single queue packet processing. DRSS primarily services broadcast, unknown unicast, or multicast (BUM) traffic. Secondarily, DRSS is used as a fallback when single NetQueue and NetQueue RSS placements fail.

Enhanced Data Path configures Default Queue RSS automatically. There are scenarios where Default Queue RSS can be disabled to free up enough physical queues for a NetQueue RSS based configuration. DRSS can be disabled using the module parameter. For example (this command only applies to the intel i40en driver):

```
esxcli system module parameters set -m i40en -p "DRSS=0,0,0 RSS=8,8,8
```

> **Note:**
>
> The command above is a partial example. Exact steps should be obtained from the the vendor NIC driver documentation.

Configure NetQueue Receive Side Scaling

Enable NetQueue Receive Side Scaling to enable VM vNIC requests to be offloaded to a physical NIC. It improves packet performance of the receive-side data.

When a pNIC sends packets to a host, Enhanced Data Path distributes data across different processing threads using a NUMA-aligned RSS engine (a pool of Queues allocated for NetQueue RSS). There are a couple of ways to configure these RSS engines.

These two modes are:

- Shared Netqueue RSS
  RSS engine is shared by multiple vNIC queues. In this mode, multiple hardware queues are made available to vNIC queues. This provides a performance improvement over using a single NetQueue (the default behavior for a VM vNIC).
- Dedicated NetQueue RSS
  RSS engine is dedicated to a single vNIC queue: A dedicated RSS engine completely offloads any request coming from a vNIC to the physical NIC. In this mode, a single RSS engine is dedicated to a single vNIC queue. It improves throughput performance as pNIC manages the receive side data and shares it among the available hardware queues to serve the request. The vNIC queues are co-located on the same logical core or fastpath as pnic queues.

  Note:

  This approach assigns an entire RSS engine to a virtual machine. This option has severe implications (may starve other workloads, or be impossible to use practically in a DRS cluster) .

  While this option is available with EDP Dedicated, it is not considered a best practice.
  This mode is not supported in EDP Standard deployments.

How to Configure Shared NetQueue RSS

NetQueue RSS is configured by the Enhanced Datapath driver module. EDP automatically creates NetQueue RSS engines according to the host pNIC's capabilities. If there was a customized RSS configuration on a server, it should be reset prior to enabling EDP:

Broadcom	esxcli system module parameters set -m bnxtnet -p “”
Mellanox	esxcli system module parameters set -m nmlx5_core -p ""
Intel	esxcli system module parameters set -m i40en -p ""

Configuring VM vNIC for Shared RSS

To configure multiple RSS engines to be available to serve RSS requests from vNICs, configure these parameters in the .vmx file of VM:

`ethernet.pnicfeatures = '4'`, which indicates RSS feature is requested by vNICs.

The VMs connected to the vSphere switch are configured for multiple queues. It means multiple logical cores of a NUMA node can process the Tx and Rx traffic coming from vNICs.

When multiple vNICs request RSS offloading, the Enhanced Network Stack (ENS) does not offload their RSS requests to the pnic, but the shared RSS engine processes their requests. For shared RSS, multiple RSS queues are available but co-location of a vNIC queue or a pNIC queue is not guaranteed.

Configuring VM vNIC for Dedicated RSS

To configure a dedicated RSS engine to process requests from a vNIC, configure these parameters in the .vmx file of the VM: `ethernet.rssoffload=True`.

Note:

This configuration is provided for completeness, and not currently recommended.

Verifying RSS with Queue Counters

Verify that packet flow is distributed on the hardware queues provided by the RSS engine.

Run the following commands. The counters will increment if there is enough activity to distribute traffic across multiple RX queues:

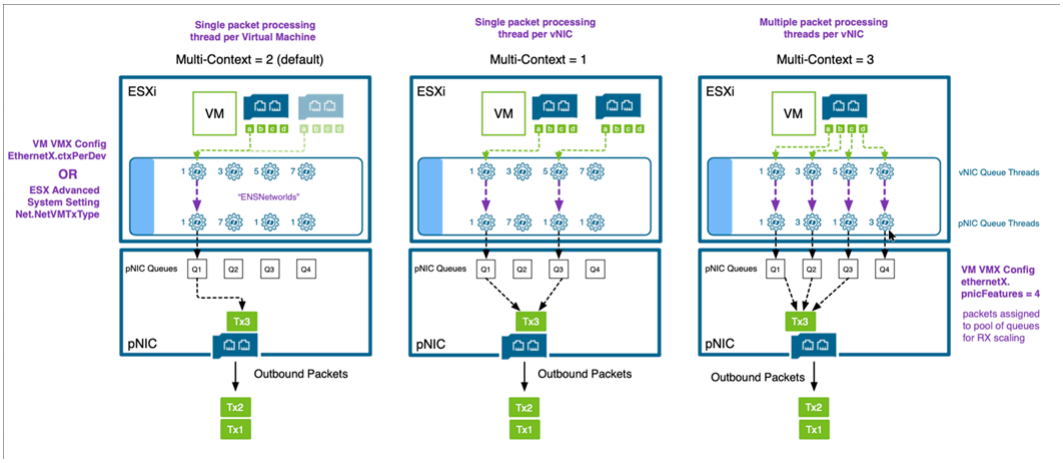
```
vsish
get /net/pNics/vmnicX/stats


Sample output:
rxq0: pkts=0 bytes=0 toFill=2047 toProc=0 noBuf=0 csumErr=0
```

```
rxq1: pkts=0 bytes=0 toFill=2047 toProc=0 noBuf=0 csumErr=0
rxq2: pkts=0 bytes=0 toFill=2047 toProc=0 noBuf=0 csumErr=0
rxq3: pkts=0 bytes=0 toFill=2047 toProc=0 noBuf=0 csumErr=0
rxq4: pkts=0 bytes=0 toFill=2047 toProc=0 noBuf=0 csumErr=0
rxq5: pkts=0 bytes=0 toFill=2047 toProc=0 noBuf=0 csumErr=0
rxq6: pkts=0 bytes=0 toFill=2047 toProc=0 noBuf=0 csumErr=0
rxq7: pkts=0 bytes=0 toFill=2047 toProc=0 noBuf=0 csumErr=0
txq0: pkts=0 bytes=0 toFill=0 toProc=0 dropped=0
txq1: pkts=0 bytes=0 toFill=0 toProc=0 dropped=0
txq2: pkts=0 bytes=0 toFill=0 toProc=0 dropped=0
txq3: pkts=0 bytes=0 toFill=0 toProc=0 dropped=0
txq4: pkts=0 bytes=0 toFill=0 toProc=0 dropped=0
txq5: pkts=0 bytes=0 toFill=0 toProc=0 dropped=0
txq6: pkts=0 bytes=0 toFill=0 toProc=0 dropped=0
txq7: pkts=0 bytes=0 toFill=0 toProc=0 dropped=0
```

Configure VM vNIC RX/TX Scaling

Provide multiple cores to vNICs by configuring the Multiple Context functionality on a host switch running in Enhanced Datapath mode. It helps improve packet performance.



On a host switch configured to run in the Enhanced Datapath mode, you can configure Multiple Context functionality for vNIC traffic. Multiple Context means that multiple threads can serve Tx (transmit) and Rx (receive) vNIC queues, in contrast to the single context, where only one networking thread serves both the Tx queue and Rx queue.

TX = Transmit scaling achieved with the procedure below.

RX = Multi-Context piggybacks on NetQueue RSS without requiring additional configuration on the VM.

1. To verify that host switch is configured to run in **Enhanced Datapath** mode:
   a) Navigate the UI based on the NSX version and select a host:
      - **System** > **Fabric** > **Hosts** and select the **Cluster** tab.
      - **System** > **Fabric** > **Hosts** and select the **Standalone** tab.
   b) Select the transport node.
   c) Select the **Overview** tab and verify `Enhanced Datapath Capable` parameter is set to `Yes`.
2. To configure Multiple Context functionality for vNIC traffic managed through Enhanced Datapath mode, edit configuration options of VMs and set the following parameter value. See the latest *vSphere Virtual Machine Administration guide* for details on how to edit VM configuration options.

   ```
   ethernetX.ctxPerDev = "3"
   ```

   Where, the value `3` indicates that Multiple Context functionality is applied per vNIC queue.

Other supported values for contexts are:

- `ethernetX.ctxPerDev =1` indicates that Multiple Context functionality is applied per vNIC.
- `ethernetX.ctxPerDev =2` indicates that Multiple Context functionality is applied per VM (default). If you set the value to `0` - `ethernetX.ctxPerDev = 0`, the value is configured to `2` (default).

Enhanced Datapath improves packet throughput by using the Multiple Context functionality set for vNIC queues.

Configure High-Performance Host Switch Profiles

You can configure a high-performance host switch profile for hosts to achieve higher network performance.

- Familiarize yourself with the ESX Maintenance Mode when activating high throughput mode on the host and the vSphere Distributed Switch uplinks. See vSphere documentation.
  The high-performance host switch profile is supported on ESX hosts running version 7.0 Update 3 or later.
- Verify that a transport node profile is available. See Add a Transport Node Profile topic in the *NSX Installation Guide*.

You can configure a default high-performance host switch profile or a custom high-performance host switch profile, but not both.

You can apply the high-performance host switch profile to sub-TNP as well.

1. Determine whether you want to configure a default or a custom high-performance host switch profile. Do not configure both.
2. To configure a default high-performance host switch profile, make the following API call.

   ```
   PATCH https://<policy-mgr>/policy/api/v1/infra/host-switch-profiles/HPprofile-default-a
   {
   "high_performance_configs":[],
   "auto_config":0,
   "display_name": "HPprofile-default-a",
   "description": "",
   "resource_type": "HighPerformanceHostSwitchProfile",
   "_system_owned": true,
   "_revision": 0
   }
   ```

3. To configure a custom high-performance host switch profile, make the following API call.

   ```
   PATCH https://<policy-mgr>/policy/api/v1/infra/host-switch-profiles/HPprofile-customized-a
   {
       "high_performance_configs": [
           {
               "high_performance_config_type": "ADV_CONFIG",
               "version": [
                   "8.0.2"
               ],
               "high_performance_config_params": [
                   {
                       "key": "/Net/NetSchedHClkVnicMQ",
                       "value": "1"
                   }
               ]
           },
           {
               "high_performance_config_type": "DRIVER_CONFIG",
               "driver_info": ["nmlx5_core"],
               "version": [
                   "default"
   ```

```
            ],
            "high_performance_config_params": [
                {
                    "key": "netq_rss_ens",
                    "value": "1"
                }
            ]
        },
        {
            "high_performance_config_type": "MISC_CONFIG",
            "version": [
                "default"
            ],
            "uplink_tx_ring_size": 4096,
            "uplink_rx_ring_size": 4096
        }
    ],
    "auto_config": 0,
    "resource_type": "PolicyHighPerformanceHostSwitchProfile"
}
```

4. Identify a host to apply the default or custom high-performance host switch profile.
   a) Log in to the NSX UI.
   b) Select **System** > **Fabric** > **Hosts**.
   c) Select a standalone host or a host within a cluster.
   d) Click the menu icon (three dots) and select **Copy ID to Clipboard**.
5. Identify the transport node profile ID.
   a) Select **System** > **Fabric** > **Hosts** > **Transport Node Profile**.
   b) Identify the transport node profile ID.
6. In the vSphere Client, set the selected host into the maintenance mode.

   When you activate the `pending_host_maintenance_mode` parameter, the host goes into maintenance mode. When in maintenance mode, applying the high-performance configuration on the host starts.

   Note:
   When the host is in maintenance mode, an alarm appears, indicating the state of the host.

   After the host is out of maintenance mode, the alarm resolves automatically.

   Note:
   If you do not put the host in maintenance mode, the high-performance host switch profile is not applied to the host.

   The relevant log messages about the failed operation and its cause are generated if the operation fails.
7. Attach the custom high-performance host switch profile to the transport node profile.

   `PUT https://<nsx-policy-manager>/policy/api/v1/infra/host-transport-node-profiles/<tnp-id>`

   In the `<tnp-id>` parameter, enter the transport node profile ID.

```
...
            {
                "host_switch_name": "DSwitch",
                "host_switch_id": "50 31 7d a4 26 52 3a 8f-71 cf 8c 89 8c 09 d5 89",
                "host_switch_type": "VDS",
                "host_switch_mode": "ENS_INTERRUPT",
                "host_switch_profile_ids": [
                    {
```

```
                        "key": "UplinkHostSwitchProfile",
                        "value": "/infra/host-switch-profiles/fb38b6c9-379b-42cf-b78c-13fc05da2e0d"
                    },
                    {   "key": "HighPerformanceHostSwitchProfile",
                        "value": "/infra/host-switch-profiles/HPprofile-customized-a"  # <--- This is the
  profile created in step 3.
                    }
                ],
...
```

8. Verify that the high-performance host switch profile is applied to transport nodes.

   `GET https://<nsx-manager>/policy/api/v1/infra/host-transport-nodes-profiles/<tnp_id>`

   After the default or custom high-performance host profile is successfully applied, all the required networking sub-configuration is configured, and a success log is generated.

9. Optional: Manually reset all the high-performance host profile configurations to default for emergency recovery.
   a) Locate the DVS switch name on the server.
   b) Run the `hp_dp_script.py` script.

      `python3 <pathname of hp_dp_script.py> -d <DVSSwitchName> -r`

      The pathname of `hp_dp_script.py` is usually `/usr/lib/vmware/high-performance-dp/hp_dp_script.py` on an ESX host.

Logical Core Assignment Modes for EDP Dedicated

Automatically assign logical cores to vNICs such that dedicated logical cores manage the incoming traffic to and outgoing traffic from vNICs.

With a host switch configured in EDP Dedicated mode, if a single logical core is associated to a vNIC, then that logical core processes bidirectional traffic coming into or going out of a vNIC. When multiple logical cores are configured, the host automatically determines which logical core must process a vNIC's traffic.

Logical cores are assigned to vNICs based on one of these parameters.

- CPU-usage: Host predicts the CPU usage to transmit incoming or outgoing traffic at each vNIC direction by using internal statistics. Based on the usage of CPU to transmit traffic, host changes the logical core assignments to balance load among logical cores. The CPU usage mode is more efficient with resources than vNIC-count but vNIC to LCORE assignments may need to be adjusted more frequently. This is the default mode EDP will run in.
- vNIC-count: Host assumes transmission of incoming or outgoing traffic for a vNIC direction requires the same amount of the CPU resource. Each logical core is assigned the same number of vNICs based on the available pool of logical cores. The vNIC-count mode has performance consistency but is less efficient with resources, but is not optimal for asymmetric traffic.

  Note:

  Note: EDP Standard host switches operate in CPU-usage mode by default. vNIC-count mode is not supported on EDP Standard.

Refer to the following procedure.

To switch from one core assignment mode to another mode, run the following command.

`set ens lcore-assignment-mode<hs-name-arg><ens-lc-mode-arg>`

Where, *<ens-lc-mode-arg>* refers to the ENS logical core assignment mode name. This argument can be set to the mode `vNIC-count` or `cpu-usage`.

`vNIC-count` is the logical core assignment based on vNIC/Direction count.

`cpu-usage` is the logical core assignment based on CPU usage.

Where, *<hs-name-arg>* refers to the host switch name.

For example:

```
esx-1> set ens lcore-assignment-mode nsxvswitch cpu-usage
```

Configure an Industrial vSwitch

The Industrial Virtual Switch (IvS) is an NSX virtual switch configuration designed to support real-time PROFINET communication over industrial Ethernet networks, ensuring low latency and bounded jitter.

Prerequisites and Performance Validation

The Industrial vSwitch requires specific hardware and configuration tuning to maintain deterministic timing.

Validated Reference Configuration

Performance benchmarks for the Industrial vSwitch -- specifically the achievement of unidirectional latency within a worst-case jitter bound of 250 microseconds -- were strictly validated using the following reference architecture:

- Processor: Intel Xeon Gold Scalable Processor family.
- Network Interface: NVIDIA Mellanox ConnectX-6 NIC.
- System BIOS: Must be set to a Performance-Tuned Profile. This includes disabling power-saving C-states to ensure low-latency I/O paths.

Processor Compatibility

Performance Consistency --Users can expect relatively consistent performance when utilizing the same Intel Xeon Scalable processor family used in the reference architecture.

Future Generations and Other Architectures -- While validated on 4th Gen Intel Xeon Scalable (Sapphire Rapids), the 5th Gen Intel Xeon Scalable (Emerald Rapids) architecture is expected to provide equivalent or superior performance due to its expanded L3 cache. At this time, CPU architectures from AMD have not been tested with this setup, and the 250-microsecond jitter benchmark cannot be guaranteed for those deployments.

Application Suitability for Operational Technology (OT)

Not all industrial applications require the stringent 250-microsecond jitter threshold. While mission-critical deterministic workloads require this level of precision, many PROFINET applications operate on cycle times that are compatible with higher jitter tolerances. Users should align their hardware choice with the specific timing requirements and synchronization needs of their Operational Technology (OT) environment.

Host Configuration

1. On the ESX hosts to be configured with Industrial vSwitch, configure the hosts in High Performance mode with no Power Saving methods. The examples below show the BIOS settings from a sample server.

Setting Name	Configuration
Enhanced Processor Performance	Enabled
Enhanced Processor Performance Profile	Aggressive
Power Regulator	Static High Performance Mode
Minimum Processor Idle Power Core C-State	No C-States
Minimum Processor Idle Power Package C-State	No Package State
Energy/Performance Bias	Maximum Performance
Collaborative Power Control	Disabled
Intel Turbo Boost Technology	Disabled
Workload Profile	Custom
Time Format	Local Time
Intel Hyper-Threading	Enabled

2. Verify that the NICs used for Industrial vSwitch uplinks support EDP Dedicated/Poll mode. Check the Broadcom Compatibility Guide for the list of supported NICs.

3. Disable Default Queue Receive Side Scaling (DRSS) on the hosts. The IvS prioritizes latency, and aims for minimal and deterministic jitter. To help achieve this, disabling DRSS on the physical NICs in each host is required. Refer to Configure NetQueue Receive Side Scaling for procedures to disable DRSS.

IvS enables the forwarding of PROFINET traffic between virtual Programmable Logic Controllers (vPLCs) running on ESX hosts and I/O devices located outside the vSphere cluster. IvS also supports the Parallel Redundancy Protocol (PRP), providing seamless network redundancy and high availability for traffic across two distinct LANs. Additionally, IvS is natively optimized and tuned for PROFINET performance.

Note:

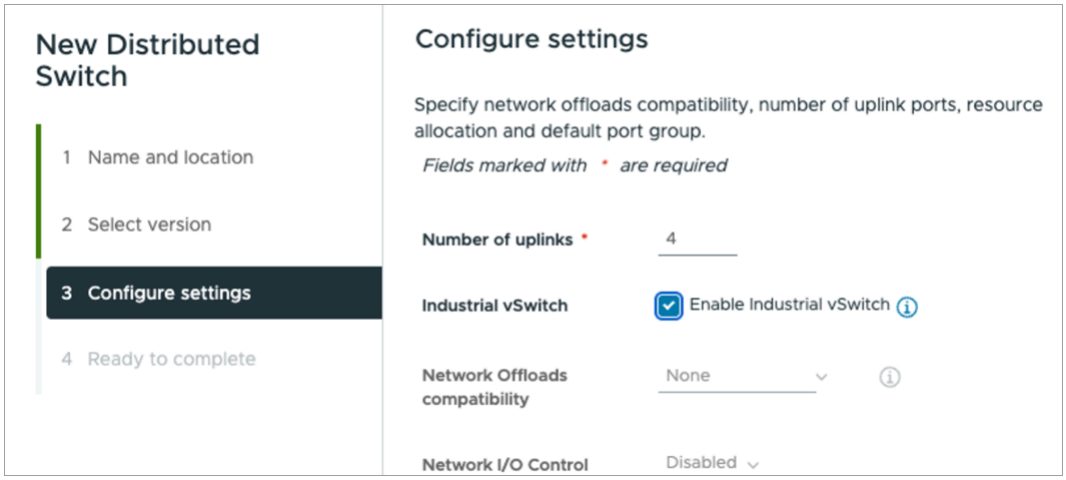
IvS is not compatible with the vDefend suite. If the ports connected to an IvS are configured with DFW, the latency will not meet 500 microseconds and 10% jitter that might be required by PROFINET-RT applications.

The following considerations apply to IvS configuration and use:

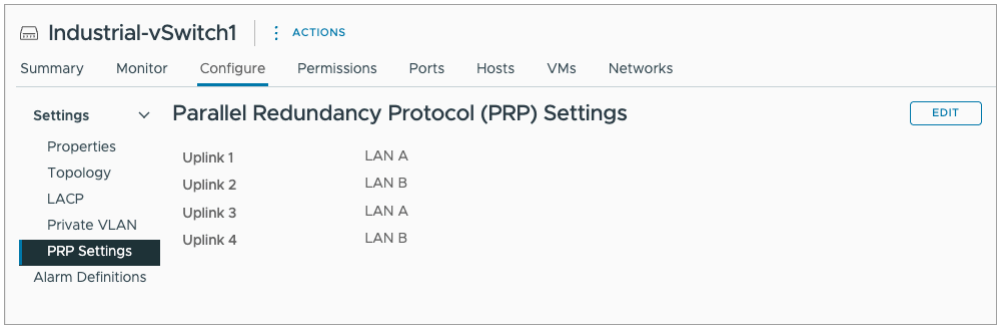
- A maximum of two Industrial vSwitches are supported per Transport Node Profile and per vSphere host, and they can only be configured on VLAN transport zones.
- PRP can only be configured and enabled on one IvS per host.
- Deterministic latency for packets forwarded in Industrial vSwitch is not guaranteed to be bounded during vMotion, software upgrade, or configuration changes.
- For virtual machines connected to the IvS to leverage the supported features, only vmxnet3 interfaces using the latest driver version 1.9 or later are supported. Latency Sensitivity High is also required for the VM settings.

Use the following procedure to configure IvS.

1. In vCenter, create a host switch with the “Industrial vSwitch” option.



2. To leverage Parallel Redundancy Protocol (PRP) in IvS, go to the vSwitch configurations under **Configure** > **Settings** > **PRP Settings** and map the LAN-A and LAN-B networks to the appropriate uplinks connected to those LAN networks. The IvS supports a maximum of 2 uplinks per PRP LAN network.



3. Log in to the NSX Manager or access the NSX interface from vCenter and add the IvS hosts.
4. Navigate to **System** > **Fabric** > **Profiles** > **Uplink Profiles** > **Add Uplink Profile**.
5. Create an Uplink Profile for the IvS created in vCenter by entering the details of a relevant industrial virtual switch.
6. Next to Teaming in the display, click **Set** to enter the Teaming configuration.
7. Click **Add** and then **Apply**.
8. Click **Save**.
9. Navigate to **System** > **Fabric** > **Transport Zones** > **Add Transport Zone**.
10. Enter the Transport Zone name and traffic type **VLAN.**
11. Click **Save**.
12. Navigate to **System** > **Hosts** > **Transport Node Profiles**.
13. Click **Add a Transport Node Profile** (TNP) and enter the profile name.
14. Under Host Switch, click **Set**.
15. Click **Add Host Switch** and complete the configuration:
    a) Enter the vCenter Name.
    b) For Select VDS, select the Industrial vSwitch.
    c) Select the Transport Zone that you created in a previous step.
    d) Select the Uplink Profiles that you created in a previous step.

    The system automatically populates and displays the Teaming Policy Uplink Mapping.
16. Expand Advanced Configuration.

    The system displays prepopulated values for latency and redundancy.
17. For CPU Config, click **Set**.

    The CPU Config screen appears.
18. Set the number of CPUs and number of cores to use for the Enhanced Data Path Dedicated mode. Five CPU cores is the maximum supported.
19. Click **Add** and then **Save**.
20. (Optional) Review and modify any of the Advanced Configuration settings.
21. Click **Add** and then **Save**.
22. Install the Transport Node on the host by applying TNP on the cluster.
    a) Navigate to **System** > **Fabric** > **Hosts** > **Clusters**.
    b) Select the cluster and click **Configure NSX**.
    c) Select the Transport Node Profile created for the IvS, and click **Save**.

    The system applies the Transport Node Profile to the cluster.
23. Navigate to **Networking** > **Segments** > **Profiles** > **Segment Profiles**.
24. Click **Add Segment Profile**, and select **Industrial vSwitch** and enter Segment Profile name.

    For more information, see Understanding Industrial vSwitch Segment Profile.

25. Using the toggles, enable the full **Network Path Latency** and enter the Sampling Rate.
26. Click **Save**.
27. Create a segment for Real-Time networks:
    a) Navigate to **Networking** > **Segments** > **NSX**.
    b) Click **Add Segment** and enter the segment name.
    c) Select the transport zone, and enter the appropriate VLAN number.
    d) Click **Save**.
    e) Expand **Segment Profiles**.
    f) Expand **Industrial Switch** and select the IvS segment profile.
28. Click **Save**.

Configure the vNICs on the vPLC VM and connect them to the respective IvS segment.

L4 Equal-Cost Multi-Path in the ESX Host Switch

You can apply Equal-Cost Multi-Path (ECMP) routing on ESX hosts operating in Enhanced Datapath (EDP) and Standard switch modes.

> **Note:**
>
> This capability is different from northbound ECMP routing. See Understanding ECMP Routing for scaling ECMP paths on edge nodes.

The ECMP has two modes: L3 ECMP and L4 ECMP.

With L4 ECMP, packet distribution is based on 5-tuple hash (IP address source, IP address destination, IP protocol, layer-4 port source, layer-4 port destination). This feature allows better distribution of traffic across all available Service Routers (SR).

- L4 ECMP requires Enhanced Datapath to be. enabled (EDP Standard and EDP Dedicated are both supported).
- Enabling L4 ECMP will significantly increase total flows

With L3 ECMP, you can apply the ECMP mode either directly to a Transport node or through the Transport Node Profile. Note that the ECMP mode configuration is not supported in the sub-TNP level. The TNP level ECMP mode configuration is inherited in the Sub-TNP profile.

To apply the ECMP mode to a Transport node, run the following API.

```
PUT https://<nsx-mgr>/policy/api/v1/infra/sites/<site-id>/enforcement-points/
<enforcementpoint-id>/host-transport-nodes/
```

To apply the ECMP mode through a Transport Node Profile, run the following API.

```
PUT https://<nsx-mgr>/policy/api/v1/infra/host-transport-node-profiles/<transport-node-
profile-id>
```

For more details about the APIs, see the NSX API Guide.

The ECMP 5-tuple feature (hashing on protocol number, source and destination address, and source and destination port) is disabled by default in ESX. You can enable it by setting the parameter `lb_ecmp` to `true` with the following API call.

```
PUT https://<nsx-manager>/policy/api/v1/infra/connectivity-global-config
{
  "lb_ecmp": true,
  ...
}
```

You can see the current `lb_ecmp` value with the API `GET https://<nsx-manager>/policy/api/v1/infra/connectivity-global-config`.

Notes about 5-tuple ECMP:

- When 5-tuple ECMP for ESX is enabled, it will be applied to all ESX hosts. Hosts that do not have ENS enabled might show unexpected behavior.
- Do not enable this feature if you have non-ENS hosts, or a mix of ENS and non-ENS hosts.
- There are appropriate validations to prevent you from enabling L4 ECMP for non-ENS switches.
- When 5-tuple ECMP is enabled, ENS-enabled hosts will consume significantly more flows. When the flow limit is reached, the performance of the host might be impacted.
- Certain topologies, such as a load balancer, where layer 3 does not provide enough path diversity, might benefit more with 5-tuple ECMP enabled.

Enhanced Network Offloads and Advanced NIC Passthrough

VMware Cloud Foundation offloads intensive network datapath tasks from the ESX host CPU to supported hardware NICs. This offloading accelerates networking performance, improves visibility through security flow processing, and reclaims CPU capacity. Network offloads also ensure that host resources remain available for both standard business-critical workloads and performance-dependent AI nodes.

Understanding Network Offload Technologies

VCF provides three different ways to implement network offloads:

- **Uniform Passthrough (UPT) for DPUs:** Maximizes security isolation and x86 resource reclamation by offloading slow-path packet processing to the DPU. This technology ensures that the ESX host CPU is fully dedicated to application workloads while the DPU handles the network control plane.
- **Uniform Passthrough (UPT) for Accelerated NICs:** Enables near line-rate performance using the standard VMXNET3 interface. This technology supports vMotion and removes the need for proprietary guest drivers, simplifying image management.
- **Enhanced DirectPath I/O (EDPIO) for Accelerated NICs:** Delivers the highest possible throughput with lower latency by providing virtual machines with direct hardware access. This technology requires a guest driver but supports GPUDirect RDMA for intensive AI and distributed computing workloads.

The following table provides a summary and comparison of the network offload technologies.

Network Offload Technology	VCF Technology	Hardware Requirement	Minimum VCF Requirement	Best Use Case	Key Advantage
Uniform Passthrough (UPT) for DPUs	vSphere Distributed Services Engine	NVIDIA BlueField 2 and 3 (DPU mode), or AMD Pensando	VCF 5.2 (BlueField 2, AMD) VCF 9.0 (BlueField 3)	Performance networking with minimum x86 usage	x86 isolation
Uniform Passthrough (UPT) for Accelerated NICs	Advanced NIC Passthrough	NVIDIA BlueField 2 and 3 (NIC mode), or NVIDIA ConnectX-7	VCF 9.1	High-performance general workloads	Uses standard VMXNET3 adapter; no guest driver management required
Enhanced DirectPath I/O (EDPIO) for Accelerated NICs	Advanced NIC Passthrough	NVIDIA BlueField 3 (NIC mode), or NVIDIA ConnectX-6 Dx and ConnectX-7	VCF 9.1	AI, machine learning, and distributed computing workloads	Supports GPUDirect RDMA for high-speed GPU communication

To learn more about network offloads, see What is Network Offloads Compatibility.

Advanced NIC Passthrough for Accelerated NICs

VCF 9.1 introduces a high-performance networking tier that uses Advanced NIC Passthrough technology to meet the demands of AI workloads and high-throughput environments. Advanced NIC Passthrough supports the Uniform Passthrough (UPT) and Enhanced DirectPath I/O (EDPIO) offload technologies.

Supported Features

Advanced NIC Passthrough delivers near line-rate performance for up to 100G NICs while retaining support for the following critical features:

- vMotion
- Storage vMotion
- vSphere High Availability (HA)
- vSphere DRS
- Fast suspend/resume

Current Limitations

Advanced NIC Passthrough has the following feature limitations:

- Requires dedicated memory reservation for virtual machines.
- Snapshots of virtual machines are not currently supported.
- Hot add of virtual devices, memory, or vCPU is not supported.

Configure VCF Components for Network Offloads With Advanced NIC Passthrough

To enable network offload capabilities, you must apply configurations across vCenter, the ESX host, and NSX. This section explains how to enable network offloads using Advanced NIC Passthrough.

> **Note:**
>
> For information on network offloads with vSphere Distributed Services Engine, see What is VMware vSphere Distributed Services Engine and Configure Network Offloads with vSphere Distributed Services Engine.

1. Verify that your hardware device model, driver, and firmware can support Advanced NIC Passthrough network offload technologies. For a list of the compatible hardware specifications, search the Broadcom Compatibility Guide for devices compatible with "Enhanced DPIO" or "Uniform Passthrough."
   a. To update the device firmware if needed, see the NVIDIA firmware downloads page.
   b. To update the device driver if needed, see the NVIDIA ConnectX Ethernet Driver for VMware ESX Server page.
2. In vCenter, go to the ESX host settings and navigate to `PCI Devices`. Select **Configure SR-IOV**.
   a. Select **Enable SR-IOV**.
   b. Set the number of virtual functions for the device (for example, 32) , and click **OK**.
3. In vCenter, create a vSphere Distributed Switch (VDS) with the following configurations.
   a. Set the switch version to 9.1.0 or later, compatible with ESX 9.1.0 and later.
   b. Set **Network Offloads Compatibility** to **Advanced NIC Passthrough**.
   c. Add compatible ESX hosts to the VDS.
4. In NSX Manager, edit the host switch configuration for the transport node profile as follows.
   a. Go to the **Advanced Configuration** settings for the host switch.
   b. Set the EDP mode to **Enhanced Datapath - Standard**.
   c. Enable the **Advanced NIC Passthrough** toggle.
5. Proceed to the next section on this page to configure the virtual machines on the hosts that have Advanced NIC Passthrough enabled. These configurations are required for the virtual machines to use Advanced NIC Passthrough technologies.
6. Configure the NSX Edge nodes on the hosts that have Advanced NIC Passthrough enabled. These configurations are required for NSX Edge nodes to use Advanced NIC Passthrough technologies.
   a. To configure NSX Edge nodes to use Uniform Passthrough, go to the **Advanced Resource Reservations** section of the NSX Edge cluster settings and enable the **UPT Mode** toggle.
   b. To configure NSX Edge nodes to use Enhanced DirectPath I/O, complete the NSX Edge configuration steps described in the sections later on this documentation page.

Enable Advanced NIC Passthrough on a Virtual Machine

The following procedures describe how to configure Advanced NIC Passthrough network offload technologies (Uniform Passthrough and Enhanced DirectPath I/O) on a virtual machine (VM).

For more information, see How to Add a Network Interface Controller to a Virtual Machine.

Configure Uniform Passthrough on a VM

You can enable Uniform Passthrough compatibility on a VMXNET3 adapter. Make sure that the virtual machine is of hardware version 20 or later.

1. In vCenter, power off the VM.
2. Right-click the VM and select **Edit Settings**.
3. On the **Virtual Hardware** tab, select **Add New Device** > **Network Adapter**. Expand the section for the new network adapter.
4. Set **Adapter Type** to **VMXNET 3**.
5. Select the **Use UPT Support** checkbox.
6. Power on the VM.

> **Note:**
>
> When you activate Uniform Passthrough, the full memory size of the VM is reserved automatically.

Configure Enhanced DirectPath I/O on a VM

1. In vCenter, power off the VM.
2. Right-click the VM and select **Edit Settings**.
3. On the **Virtual Hardware** tab, select **Add New Device** > **Network Adapter**. Expand the section for the new network adapter.
4. Set **Adapter Type** to **PCI Device passthrough**.
5. For **Device**, click **Change Device** and select the accelerated NIC.
6. Set **Access Type** to **Enhanced DirectPath I/O**.
7. Reserve all guest memory for the VM. This setting is required for powering on the VM.
8. Power on the VM.

Configure a New NSX Edge Node for Enhanced DirectPath I/O

> **Note:**
>
> This release supports Enhanced DirectPath I/O for NSX Edge nodes only through manual deployment of the nodes from an OVA file.

To enable Enhanced DirectPath I/O on a new NSX Edge node, follow the instructions in Install an NSX Edge on ESX Using the vSphere GUI to begin deployment of the OVF. In the OVF deployment wizard, apply the following configurations.

1. On the **Select and name a folder** page of the wizard, select **Customize this virtual machine's hardware** option.
2. On the **Select a computer resource** page of the wizard, deselect the **Automatically power on deployed VM** option.
3. On the **Customize hardware** page of the wizard, complete the following steps to set the Enhanced DirectPath I/O backing for the devices.
   a. Select the first device for management network connectivity and select the appropriate network. Make sure the **Adapter Type** for this network is set to `vmxnet3` as it is the only type that is supported.
   b. Set up the required number of devices planned for use in the uplink profile (For example, two devices are required if you use the default **nsx-edge-multiple-vteps-uplink-profile**). For Enhanced DirectPath I/O backing, select the **Adapter Type** as `PCI Device passthrough` option.
   c. Connect with the required DVPG/segment.
   d. Set up the device with the option **Mellanox NIC/HCA (S/R)**. S/R device supports Suspend/Resume operations of the NSX Edge node for Enhanced DirectPath I/O.

   If you are setting up fewer than four Enhanced DirectPath I/O devices, ensure that the NSX Edge node does not power on automatically after the OVF deployment. Alternatively, you can let the NSX Edge node power on automatically and proceed with the registration.
4. After the OVF is deployed and before powering on the NSX Edge node, remove any extra vmxnet3 devices other than the management network interface. You can remove devices by selecting **Actions** > **Edit** on the NSX Edge node and removing the extra vmxnet3 devices from the end of the network adapters list.

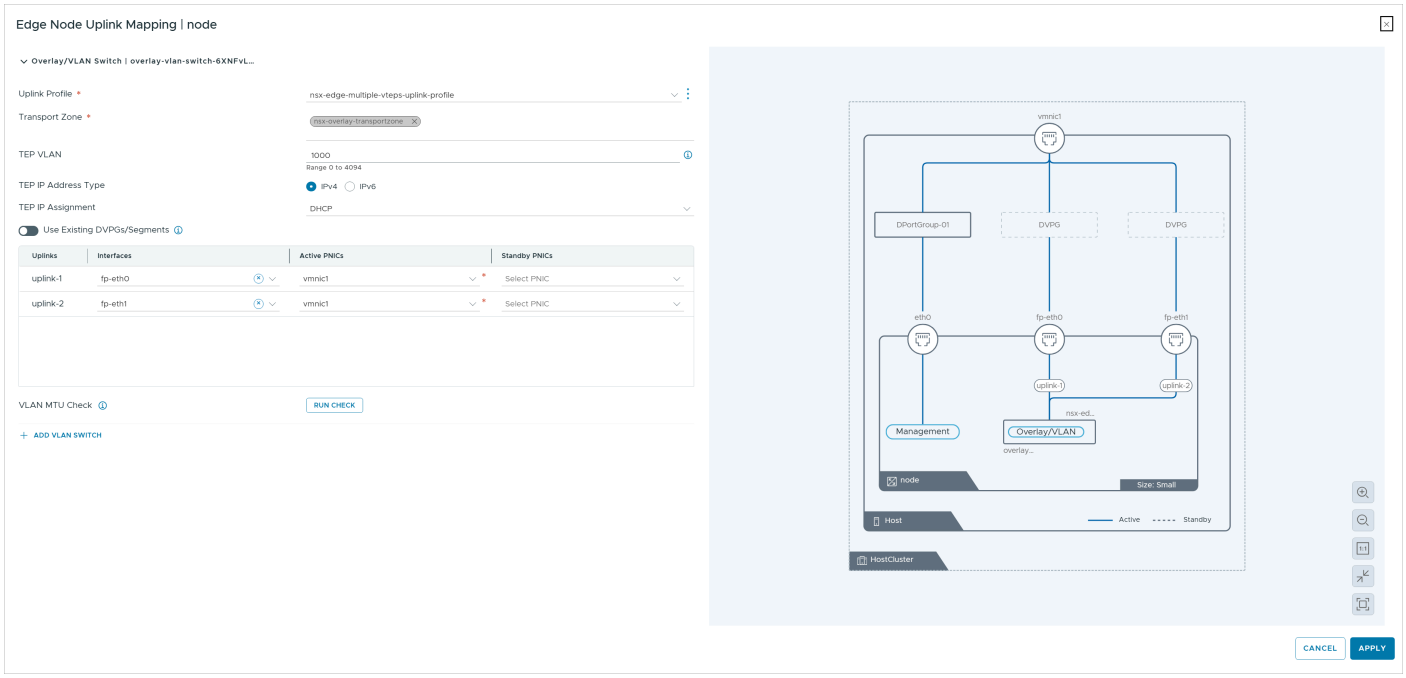
5. When you have only one vmxnet3 interface used for management connectivity and other interfaces that support Enhanced DirectPath I/O, power on the NSX Edge node.
6. Run the CLI command `get interfaces` to display output that resembles the following example. Verify that eth0 is now a management interface that has an assigned IP address for the NSX Edge node to connect to NSX Manager.

```
man-edge2-dvx> get interfaces
Tue Dec 23 2025 UTC 11:35:30.344
Interface: eth0
  IPv4:
    Address: 10.191.0.71/23
    Default gateway: 10.191.0.1
    Broadcast address: 10.191.1.255
  IPv6: not configured
  MAC address: 00:50:56:a7:92:3a
  MTU: 1500
  KNI: False
  Link status: up
  Admin status: up
  RX packets: 436722
  RX bytes: 71403640
  RX errors: 0
  RX dropped: 0
  TX packets: 200620
  TX bytes: 58613310
  TX errors: 0
  TX dropped: 0
  TX collisions: 0
```

7. If you did not configure NSX Manager details during the OVF deployment, use the following command to register the NSX Edge node to NSX Manager.

   ```
   NSX-Edge1> join management-plane <Manager-IP> thumbprint <Manager-thumbprint> username admin
   ```

8. To set up the transport node configuration for the NSX Edge node, configure the host switch on the NSX Edge node in NSX Manager.


Edge Node Uplink Mapping | node
Overlay/VLAN Switch | overlay-vlan-switch-6XNFvL...
Uplink Profile *
nsx-edge-multiple-vteps-uplink-profile
Transport Zone *
nsx-overlay-transportzone
TEP VLAN
1000
Range 0 to 4094
TEP IP Address Type
IPv4
IPv6
TEP IP Assignment
DHCP
Use Existing DVPGs/Segments
Uplinks
Interfaces
Active PNICs
Standby PNICs
uplink-1
fp-eth0
vmnic1
Select PNIC
uplink-2
fp-eth1
vmnic1
Select PNIC
VLAN MTU Check
RUN CHECK
ADD VLAN SWITCH
vmnic1
DPortGroup-01
DVPG
DVPG
eth0
fp-eth0
fp-eth1
uplink-1
uplink-2
nsx-ed...
Management
Overlay/VLAN
overlay...
node
Size: Small
Host
Active
Standby
HostCluster
CANCEL
APPLY

Configure an Existing NSX Edge Node for Enhanced DirectPath I/O

To enable Enhanced DirectPath I/O for an existing NSX Edge node, you must replace the existing node with a new node that is configured for network offloads. Ensure that the following prerequisites are met before you replace an existing NSX Edge node.

- If there are L3 services configured, make sure that the standby NSX Edge node configured for the gateways is healthy and available to serve traffic while you configure the active NSX Edge node for Enhanced DirectPath I/O.
- You have created an NSX Edge node using the OVF workflow that has all the required configurations to use Enhanced DirectPath I/O services.
- Ensure that the replacement operation is performed on one node at a time in an NSX Edge cluster.

Perform the following steps to configure an existing NSX Edge node:

1. Log in to NSX Manager.
2. Go to **System** > **Fabric** > **Edges**.
3. Go to the NSX Edge cluster of the target NSX Edge node and from the three-dot menu select **Replace Edge Cluster Member**.
4. Select the target NSX Edge node to be replaced and select the new NSX Edge node that you created with the required configurations.
5. The target NSX Edge node enters into maintenance mode while the new NSX Edge node takes its place. Once the replacement operation is complete, the Tier-0 gateways from the target NSX Edge node take effect on the replacement NSX Edge node. Ensure that the new NSX Edge node has the same host switch configuration as the target node, so that the new NSX Edge node is homogeneous with the target NSX Edge cluster. If the host switch configuration is misaligned, tunnels to other nodes in the cluster may go down after the replacement operation.

Configure Network Offloads with vSphere Distributed Services Engine

You must configure NSX and enable Enhanced Datapath for vSphere Distributed Services Engine to offload some network operations from your server CPU to the DPU. Network offloads are supported for DPUs on hosts in a vSphere Lifecycle Manager-enabled cluster.

Complete the following prerequisite tasks.

- Verify that Lockdown mode is not enabled on any of the hosts. vSphere Lifecycle Manager might fail to prepare hosts that are enabled to function in Lockdown mode.
- Verify that there is not drift in images between the hosts and cluster. Otherwise, you cannot enable vSphere Lifecycle Manager on the cluster. Remediate hosts in vCenter Server to ensure base image matches on host and cluster.
- Enable vSphere Lifecycle Manager on the host cluster. See the VMware vSphere documentation.
- Create a transport node profile using a vSphere Distributed Switch host switch.

vSphere Distributed Switch (VDS) backed by the DPU on ESX supports the offloading mode after NSX is enabled. Traffic forwarding logic is offloaded from the ESX host to the VDS backed by DPU.

To learn more about network offloads, see What is Network Offloads Compatibility.

1. From a browser, log in with admin privileges to NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.
2. Go to **System** > **Fabric** > **Hosts** > **Clusters**.
3. Select the cluster and click **Configure NSX**.

   Note:

   Identify vSphere Lifecycle Manager-enabled cluster when a cluster is accompanied with vLCM text.

4. To create a transport node profile, click **Add Profile**.
5. Enter details for the transport node profile (TNP) as required.

   Note:

   - For VDS backed by the DPU, set the Host transport node and TNP Host Switch mode to **Enhanced Datapath - Standard** (Recommended) or **Enhanced Datapath - Performance**.
   - In **Add Switch**, select a VDS with offload compatibility. If there is a mismatch, the host will not support network offloads.
   - One cluster allows only one type of host. For example, suppose that you create two distributed switches using vSphere Client: one switch with NVIDIA network offload compatibility and other with AMD Pensando network offload compatibility. Both compatibility options appear on the UI, and you must select the one specific to your requirement. Clusters of different types cannot use the same TNP. Therefore, you need two separate TNPs: one for NVIDIA DPU and one for AMD Pensando DPU.
6. If you have already created transport node profiles, select a transport node profile that uses vSphere Distributed Switch as the host switch and datapath mode as Enhanced Datapath.
7. Click **Apply TNP**.

   If this is the first cluster that is enabled for vSphere Lifecycle Manager, NSX uploads the NSX LCP bundle to the image repository in vCenter Server. vSphere Lifecycle Manager sets NSX as a solution on the cluster. It sets the desired state to the NSX image uploaded to vCenter Server. Then, vSphere Lifecycle Manager begins installation of NSX VIBs on each host, followed by configuration of NSX switch on each transport node.

   As part of the host preparation, vSphere Lifecycle Manager remediates the host, registers the host with NSX Manager, configures NSX switch on the host, and completes the configuration.

   Note:

   - vSphere Lifecycle Manager puts the ESX host backed by AMD Pensando DPU in maintenance mode and reboots it as part of host remediation. If vSphere Lifecycle Manager fails to place the host in maintenance mode, you need to manually power off all VMs and then retry NSX installation.
   - Installing NSX on a vSphere Lifecycle Manager-enabled cluster might take a little more time than when installing on a non-vSphere Lifecycle Manager-enabled cluster. This difference is due to the additional health checks that are included in this combination of products.

Configure Guest Inter-VLAN Routing

On overlay networks, NSX supports routing of inter-VLAN traffic on an L3 domain. During routing, virtual distributed router (VDR) uses VLAN ID to route packets between VLAN subnets.

- Before you associate a VLAN subinterface to a logical switch, ensure that the logical switch does not have any other associations with another VLAN subinterface. If there is a mismatch, inter-VLAN routing on overlay networks might not work.
- Ensure that hosts run ESX v 6.7 U2 or later versions.

Inter-VLAN routing overcomes the limitation of 10 vNICs that can be used per VM. NSX supporting inter-VLAN routing ensures that many VLAN subinterfaces can be created on the vNIC and consumed for different networking services. For example, one vNIC of a VM can be divided into several subinterfaces. Each subinterface belongs to a subnet, which can host a networking service such as SNMP or DHCP. With Inter-VLAN routing, for example, a subinterface on VLAN-10 can reach a subinterface on VLAN-10 or any other VLAN.

Each vNIC on a VM is connected to a switch through the parent logical port, which manages untagged packets.

To create a subinterface, create a child port on a switch using the API with an associated VIF using the API call described in the procedure. The subinterface tagged with a VLAN ID is associated to a new logical switch, for example, VLAN10 is attached to logical switch LS-VLAN-10. All subinterfaces of VLAN10 have to be attached to LS-VLAN-10. This 1–1 mapping between the VLAN ID of the subinterface and its associated logical switch is an important prerequisite. For example, adding a child port with VLAN20 to logical switch LS-VLAN-10 mapped to VLAN-10 makes routing of packets between VLANs non-functional. Such configuration errors make the inter-VLAN routing non-functional.

Starting from NSX 3.2.2, logical port proton APIs are replaced with the corresponding segment port policy APIs.

1. To create subinterfaces for a vNIC, ensure that the vNIC is updated to a parent port. Make the following REST API call:

```
PATCH https://<nsx-mgr-ip>/policy/api/v1/infra/segments/<Segment to which vNIC is connected>/ports/<Seg-
Port-vNIC>
{
   "attachment": {
     "id": "<Attachment UUID of the vNIC>",
     "type": "PARENT"
  },
   "admin_state": "UP",
   "resource_type": "SegmentPort",
   "display_name": "parentport"
}
}
```

   If the logical switch does not have a corresponding segment, you can make the following REST API calls (logical port proton API):

```
PUT https://<nsx-mgr-ip>/api/v1/logical-ports/<Logical-Port UUID-of-the-vNIC>
{
  "resource_type" : "LogicalPort",
  "display_name" : "parentport",
  "attachment" : {
    "attachment_type" : "VIF",
    "context" : {
      "resource_type" : "VifAttachmentContext",
      "vif_type": "PARENT"
    },
    "id" : "<Attachment UUID of the vNIC>"
  },
  "admin_state" : "UP",
  "logical_switch_id" : "UUID of Logical Switch to which the vNIC is connected",
  "_revision" : 0
}
```

2. To create child ports for a parent vNIC port on the N-VDS that is associated to the subinterfaces on a VM, make the following API call:

   Note:
   Before making the API call, verify that a segment exists to connect child ports with the subinterfaces on the VM.

```
PUT https://<nsx-mgr-ip>/policy/api/v1/infra/segments/<Segment to which child port is connected>/ports/
<Child-port>
{
   "attachment": {
     "id": "<Attachment UUID of the CHILD port>",
     "type": "CHILD",
     "context_id": "<Attachment UUID of the PARENT port from Step 1>",
     "traffic_tag": <VLAN ID>,
     "app_id": "<ID of the attachment>", ==> display id(can be any string). Must be unique.
  },
   "address_bindings": [
    {
       "ip_address": "<IP address to the corresponding VLAN>",
       "mac_address": "<vNIC MAC Address>",
       "vlan_id": <VLAN ID>
    }
  ],
   "admin_state": "UP",
   "resource_type": "SegmentPort",
```

```
    "display_name": "<Name of the Child PORT>"
}
```

If the logical switch does not have a corresponding segment, you can make the following REST API calls (logical port proton API):

```
POST https://<nsx-mgr-ip>/api/v1/logical-ports/
{
  "resource_type" : "LogicalPort",
  "display_name" : "<Name of the Child PORT>",
  "attachment" : {
    "attachment_type" : "VIF",
    "context" : {
      "resource_type" : "VifAttachmentContext",
      "parent_vif_id" : "<UUID of the PARENT port from Step 1>",
      "traffic_tag" : <VLAN ID>,
      "app_id" : "<ID of the attachment>",  ==> display id(can give any string). Must be unique.
      "vif_type" : "CHILD"
    },
   "id" : "<ID of the CHILD port>"
  },

  "logical_switch_id" : "<UUID of the Logical switch(not the PARENT PORT's logical switch) to which Child
 port would be connected to>",
  "address_bindings" : [ { "mac_address" : "<vNIC MAC address>", "ip_address" : "<IP address to the
 corresponding VLAN>", "vlan" : <VLAN ID> } ],
  "admin_state" : "UP"
}
```

NSX creates subinterfaces on VMs.

Multi-TEP High Availability

To avoid connectivity loss of workloads (VMs or Containers) associated to a virtual tunnel endpoint (TEP), configure TEP HA host switch profile and apply it to cluster transport nodes or individual transport nodes.

At least two TEPs are configured on a single ESX host. Ensure the uplink profile you use to create a transport node profile (TNP) or individual transport node configuration uses at least two active uplinks (TEPs).

As overlay networks spanning across L2 or L3 networks use encapsulation protocols, packets egressing or ingressing from workloads (VMs or Containers) are encapsulated within the outer packet of a Virtual Tunnel End Point (TEP). If a TEP goes into a faulty state, the VMs associated with it are not reachable. There is a one-to-one mapping between VMs to TEPs.
A TEP is faulty when one of the following points are true:

- All BFD sessions of a TEP are down due to issues in the underlay fabric.
- TEP does not have an IP address.

A healthy TEP has the following characteristics:

- It has an IP address.
- At least one BFD session involving the TEP is Up.

When NSX discovers a TEP is down, vNICs of VMs are migrated from the faulty TEP to a healthy TEP on the same ESX host. If a ESX has two TEPs and both are faulty, then VM vNICs cannot be migrated.

Each VM can have more than one vNIC. Each vNIC is associated to a single TEP. The packets from the vNIC are encapsulated before being sent out on the uplink. There can be multiple vNICs mapped to a single TEP. The mapping is decided based on uplink teaming policies.

Note:
TEP HA only supports IPv4 TEPs.

1. Create a TEP HA host switch profile.

   `PUT https://<nsx-policy-manager>/policy/api/v1/infra/host-switch-profiles/vtephaprofile1`

   ```
   {
   "enabled": "true",
   "failover_timeout":"5",
   "auto_recovery" : "true",
   "auto_recovery_initial_wait" : "300",
   "auto_recovery_max_backoff" : "86400",
   "resource_type": "PolicyVtepHAHostSwitchProfile",
   "display_name": "VtepHAProfile1"
   }
   ```

   where,

   - `enabled`: Indicates TEP HA is enabled or not. The "true" value indicates TEP HA is enabled and "false" indicates it is disabled.
   - `failover_timeout`: Is the TEP High Availability failover timeout (in seconds). This property controls after how much time should VMs vNIC be moved to an alternate healthy TEP once fault is detected at a vmknic. Default timeout is 5 seconds. Maximum value supported is up to 60 seconds.
   - `auto_recovery`: Specifies status of autonomous recovery option for the TEP High Availability feature. When `auto_recovery` is set to true, faulted vmknic is checked internally after every cyclic sequence. If the faulty vmknic recovers, VM is moved back to its initial TEP mapping, provided the VMs vNIC was mapped to that TEP. By default, `auto_recovery` is set to true. During auto recovery VMs face outage if TEP is still in faulty state. The outage time is calculated using the formula: `failover_timeout + bfd timeout + 1`. The default `bfd timeout` is `4` seconds.
   - `auto_recovery_initial_wait`: This property controls after how much time the autonomous recovery should start. The minimum initial wait period is 300 seconds, while the maximum wait period is 3600 seconds. If initial wait period is 300 seconds, then auto recovery starts at 300 seconds.
   - `auto_recovery_max_backoff`: An initial recovery is attempted after `auto_recovery_initial_wait`. If this recovery fails, additional attempts are made after doubling the previous delay time. When the delay reaches `auto_recovery_max_backoff`, the delay stops increasing and all further attempts are made every `auto_recovery_max_backoff`.

2. To attach a TEP HA profile to all ESX hosts, create a TNP with the TEP HA profile key and value properties added to the Host Switch Profiles section or prepare individual transport nodes with the TEP HA profile key and value properties.

   `PUT https://<nsx-policy-manager>/policy/api/v1/infra/host-transport-node-profiles/<tnp-id>`

   In the `<tnp-id>` parameter, enter the transport node profile ID.

   ```
   {
       "host_switch_spec": {
           "host_switches": [
               {
                   "host_switch_name": "vtepha_vds",
                   "host_switch_id": "50 22 ee c4 f6 40 79 8b-0e a4 2b da 6a 4c 36 b3",
                   "host_switch_type": "VDS",
                   "host_switch_mode": "ENS_INTERRUPT",
                   "host_switch_profile_ids": [
                       {
                           "key": "UplinkHostSwitchProfile",
                           "value": "/infra/host-switch-profiles/b32e6ce6-f1ba-4e31-a2c4-33d550505cdd"
                       },
                       {
                           "key": "VtepHAHostSwitchProfile",
                           "value": "/infra/host-switch-profiles/vtephaprofile1"
                       }
   ```

```
                ],
                "uplinks": [
                    {
                        "vds_uplink_name": "Uplink 1",
                        "uplink_name": "uplink-1"
                    },
                    {
                        "vds_uplink_name": "Uplink 2",
                        "uplink_name": "uplink-2"
                    }
                ],
                "is_migrate_pnics": false,
                "ip_assignment_spec": {
                    "ip_pool_id": "/infra/ip-pools/v4",
                    "resource_type": "StaticIpPoolSpec"
                },
                "cpu_config": [],
                "transport_zone_endpoints": [
                    {
                        "transport_zone_id": "/infra/sites/default/enforcement-points/default/transport-
 zones/de47a6b9-fa4c-4bf3-bd75-385859895949",
                        "transport_zone_profile_ids": []
                    }
                ],
                "not_ready": false
            }
        ],
        "resource_type": "StandardHostSwitchSpec"
    },
    "ignore_overridden_hosts": false,
    "resource_type": "PolicyHostTransportNodeProfile"
 }
```

Note:
If you do not configure TNP with TEP HA, NSX creates a default TEP HA profile. This default TEP HA profile is configured with default values. TEP HA is disabled.

3. To know the TEP HA profile ID that you created in Step 1, run the following API:

   ```
   get https://<nsx-manager>/policy/api/v1/infra/host-switch-profiles/vtephaprofile1
   ```

   In the GET response, you can get the value of TEP HA profile from the `id` field.

4. To verify whether TEP HA host switch profile is applied to transport nodes, run the following API.

   ```
   GET https://<nsx-manager>/policy/api/v1/infra/host-transport-nodes-profiles/<tnp_id>
     "host_switch_profile_ids": [
         {
            "key": "VtepHAHostSwitchProfile",
            "value": "/infra/host-switch-profiles/<vtephaprofile1>"
          }
        ],
   ```

   `<vtephaprofile1>` is the ID that is returned when you created TEP HA profile in Step 1.

5. You can also trigger a manual recovery of an faulty TEP and migrate associated VMs to a healthy TEP without waiting for auto recovery to begin.

   ```
   POST https://<nsx-mgr>/policy/api/v1/infra/sites/<site-id>/enforcement-points/<enforcementpoint-
   id>/host-transport-nodes/<host-transport-node-id>/vteps/actions
    {
   ```

```
    "action_type": "TransportNodeVtepRecoveryRequest",
    "device_name": "vmk10"
}
```

where vmk10 is the TEP name.

6. If vmknics of TEPs are faulty, run the alarm API to view details related to the error or view the error on NSX Manager Dashboard.

   NSX might take at least 60 seconds to raise an alarm if a vmknic is faulty. Similarly, when a TEP comes back up again after being in faulty state, NSX can take up to 60 seconds to reflect the new state.

   ```
   GET https://<nsx-manager>/api/v1/alarms?feature_name=tep_health
   ```

7. After fixing either IP or BFD-related errors, verify the `state` parameter to know the vmknics or TEP state.

   ```
   GET https://<nsx-manager>/api/v1/transport-nodes/<node-id>/network/interfaces?source=realtime
   {
           "interfaceId": "vmk10",
           "linkStatus": "UP",
           "adminStatus": "UP",
           "mtu": "1700",
           "interfaceAlias": [{
             "broadcastAddress": "133.117.22.255",
             "ipAddress": {
               "ipv4": 2239043120
             },
             "ipConfiguration": "STATIC",
             "netmask": "255.255.255.0",
             "macAddress": "00:50:56:66:67:a6"
           }],
           "state": "NORMAL"
         },
   ```

If BFD is down, NSX determines whether faulty TEP is back UP using auto recovery or manual recovery mechanism.

If IP is unavailable initially but is reassigned to a faulty TEP, NSX uses an in-built mechanism to determine when TEP is back UP. It does not rely on manual or auto recovery mechanism.

If the faulty TEP does not come back UP:

- During auto recovery or manual recovery process, VM vNIC might face a temporary network outage.
- Network outage might be because during BFD check the VM vnic is still mapped to the faulty TEP.
- Network outage time is calculated using the formula: `failover_timeout + bfd timeout + 1`, which is 10 seconds. The default `bfd timeout` is `4` seconds.

If faulty TEP still does not come back UP, VM vNIC is remapped to a healthy TEP that is available on the host. If no TEPs are healthy, then VM vNIC is mapped to its original TEP.

VTEP Maintenance Mode

You can put a VTEP in maintenance mode provided there is an alternate healthy VTEP on the host.

- Ensure at least one healthy VTEP is present on the host.

1. To put a VTEP in Administrative Down mode, call the following API:

   ```
   POST https://<nsx-mgr>/policy/api/v1/infra/sites/<site-id>/enforcement-points/<enforcementpoint-id>/host-transport-nodes/<host-transport-node-id>/vteps/actions
   {
        "type": "TransportNodeVtepAdminStateMgmtRequest",
        "device_name": "vmk10"
        "admin_state_up": <state>
   }
   ```

where,

`state` is `false` for configuring VTEP in Administrative Down mode.

`state` is `true` for configuring VTEP in non-Administrative Down mode, which is to bring VTEP back up from Administrative Down mode to normal state.

This operation does not persist after restarting the transport node.

2. Move the VM vNIC of the VTEP that is Administrative Down mode to another healthy VTEP.

VM vNIC from VTEP in Administrative Down mode is moved to another healthy VTEP.

Transport Zones and Profiles

Transport zones and profiles are building blocks to prepare hosts for NSX networking.

Create Transport Zones

Transport zones dictate which hosts transport nodes and, therefore, which VMs can participate in the use of a particular network. A transport zone does this by limiting the hosts that can see a segment—and, therefore, which VMs can be attached to the segment. A transport zone can span one or more host clusters. Also, a host transport node can be associated to multiple transport zones.

An NSX environment can contain one or more transport zones based on your requirements. A host can belong to multiple transport zones. A segment can belong to only one transport zone.

NSX does not allow connection of VMs that are in different transport zones in the Layer 2 network. The span of a segment is limited to a transport zone.

Both host transport nodes and NSX Edge nodes use Overlay and VLAN transport zones. Host transport nodes connect to VDS switches while N-VDS switch is configured on NSX Edge transport nodes.

The VLAN transport zone is used by the NSX Edge and host transport nodes for its VLAN uplinks. When an NSX Edge is added to a VLAN transport zone, a VLAN N-VDS is installed on the NSX Edge.

The following system-generated transport zones exist by default:

- A VLAN transport zone with name `nsx-vlan-transportzone`.
- An Overlay transport zone with name `nsx-overlay-transportzone`.
- At least one security transport zone with name `nsx.vlan-tz.security.<UUID>.DSwitch-xx`. The system creates one security transport zone per distributed virtual switch (DVS) specified in the transport node profile. When you activate NSX on DVPGs within a host cluster, the system creates distributed port group segments under these security transport zones.

You can create custom VLAN and Overlay transport zones in addition to the system-generated defaults. Security transport zones are system-generated only and cannot be created by users.

Note:
vMotion is not supported between two segments or logical switches on different VLAN transport zones.

1. From a browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.
2. Select **System** > **Fabric** > **Transport Zones** > **Add Zone**.
3. Enter a name for the transport zone and optionally a description.
4. For the **Traffic Type** drop-down menu, specify whether the transport zone is for overlay or VLAN traffic.

   The options are **Overlay Backed**, **VLAN**, and **Overlay Standard**.

   For a transport zone to forward IPv6 traffic, it must be set for an overlay traffic type.
5. For overlay transport zones, specify the forwarding mode which determines the underlay transport protocol for the traffic encapsulation.

   The forwarding mode can be for **IPv4** or **IPv6**.

Note:
Dual stack (both IPv4 and IPv6) is not supported.

6. Optional: In the **VLANs** field, specify the VLAN IDs or range of VLANs IDs that are allowable for use.

   For example, entering `1-1000, 2000-3000, 4094` would allow VLAN IDs 1 through 1000, 2000 through 3000, and 4094.

   Segments of the transport zone must have VLANs within the specified ranges or match the specified values.

7. Optional: For VLAN transport zones, select the uplink teaming policy from the drop-down list. These named teaming policies can be used by segments attached to the VLAN transport zone which in turn use the named teaming policy specified in uplink profiles to direct traffic. For more information, see Configure Named Teaming Policy.

8. After you add the transport zone, go to the **Transport Zones** page and view the newly added transport zone either from the UI or by running the following API command.

   `GET /policy/api/v1/global-infra/sites/<site-id>/enforcement-points/<enforcementpoint-id>/transport-zones`


```
{
"sort_ascending": true,
"sort_by": "display_name",
"result_count": 1,
"results": [
 {
  "tz_type": "OVERLAY_BACKED",
   "is_default": true,
   "transport_zone_profile_paths": [
     "/infra/transport-zone-profiles/tzp"
 ],
   "nested_nsx": false,
   "resource_type": "PolicyTransportZone",
   "id": "tz",
   "display_name": "tz",
   "path": "/infra/sites/default/enforcement-points/default/transport-zones/tz",
   "relative_path": "tz",
   "parent_path": "/infra/sites/default/enforcement-points/default",
   "unique_id": "8f4a026d-e3f5-4f23-a3ef-46309d573dc1",
   "marked_for_delete": false,
   "overridden": false,
   "_create_user": "admin",
   "_create_time": 1607501697823,
   "_last_modified_user": "admin",
   "_last_modified_time": 1607582307987,
   "_system_owned": false,
   "_protection": "NOT_PROTECTED",
   "_revision": 5
 }
]
}
```


Optionally, create a custom transport-zone profile and bind it to the transport zone. You can create custom transport-zone profiles using the (deprecated) `PATCH /api/v1/infra/transport-zone-profiles` API. There is no UI workflow for creating a transport-zone profile. After the transport-zone profile is created, you can attach it to the transport zone with the `PATCH https://<policy-mgr>/policy/api/v1/infra/sites/default/enforcement-points/nsxt-ep/transport-zones/<transport-zone-id>` API.


```
{
"tz_type": "OVERLAY_BACKED",
"is_default": true,
```

```
"nested_nsx": false,
"transport_zone_profile_paths": [
"/infra/transport-zone-profiles/tzp"
]
}
```

Create an IP Pool for Tunnel Endpoint IP Addresses

NSX Transport Nodes are configured with tunnel endpoint (TEP pool) IP addresses. Tunnel endpoints are the source and destination IP addresses used in the external IP header to identify the hypervisor hosts originating and end the NSX encapsulation of overlay frames. You can use either DHCP or manually configured IP pools for tunnel endpoint IP addresses.

You can also create IP pools for TEP IP addresses during the VCF cluster creation workflow. The VCF workflow or API will create the IP pools if required.

1. From a browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.
2. Navigate to **Networking** > **IP Address Pools**.
3. Click **Add IP Address Pool**.
4. Enter a name and optionally a description.
5. From the **Subnets** column, click **Set** to add the subnets.
6. Click the **Add Subnet** drop-down menu and choose either **IP Block** or **IP Ranges**:
   - Choose **IP Block** to use an IP block of the allowable IP addresses:

Setting	Description
Select IP Blocks	Choose an IP address block.
Size	Specify the size or the number of IP addresses in the IP address block.
Allocation Range	Specify the allocation range for the IP address block. **Note:** For an IPv6 address block, you must specify an allocation range. The allocation range value cannot be more than 1048576.
Auto Assign Gateway	Click the toggle to enable or disable automatic gateway IP assignment.

   - Choose **IP Ranges** to specify the range of IP addresses that can be used:

Setting	Description
Enter IPv4 or IPv6 Ranges	Enter the range of IPv4 or IPv6 addresses. **Note:** Transport nodes and tunnel endpoints (TEPs) do not allow a mixed IP address pool to be used (an IP pool with both IPv4 and IPv6 ranges). Ensure that the IP range consists of only IPv4 or only IPv6 addresses.
CIDR	Enter the IP address range in CIDR format.
Gateway IP	Enter the gateway address.
DNS Server	Enter the list of the DNS (domain name system) server IP addresses separated by commas.
DNS Suffix	Enter the DNS suffix. For example, `corp.local`.

7. Click **Add** and then **Apply**.
8. Click **Save**.

> **Note:**
> IP Pools are configured for only those hosts that participate in overlay traffic (hosts that are members of an overlay transport zone). So, TEP tunnels are not required for host clusters that are member of VLAN transport zones only and/or management host clusters (hosting NSX Appliances / NSX Edge VMs) if management appliances are connected to VLAN backed VDS portgroups.

The IPv4 or IPv6 address pool is listed on the IP pool page.

If you are using API to view the list of IP addresses allocated from an IP pool, run the NSX Manager API. Do not run the Policy API to view IP pool allocations.

For example, the Policy API call does not retrieve IP addresses allocated from an IP pool.

```
GET https://<nsxmanager-IP>/policy/api/v1/infra/ip-pools/TEP-IP-Pool/ip-allocations
```

```
 {
 "results": [],
 "result_count": 0,
 "sort_by": "display_name",
 "sort_ascending": true
 }
```

The NSX Manager API retrieves IP addresses allocated from an IP pool.

```
GET https://<nsxmanager-IP>/api/v1/pools/ip-pools/<ip-pool-UUID>/allocations
```

```
 {
 "results": [
 {
    "allocation_id": "192.85.85.12",
    "_protection": "NOT_PROTECTED"
 },
 {
    "allocation_id": "192.85.85.13",
    "_protection": "NOT_PROTECTED"
 },
 {
    "allocation_id": "192.85.85.14",
    "_protection": "NOT_PROTECTED"
 },
 {
    "allocation_id": "192.85.85.15",
    "_protection": "NOT_PROTECTED"
 },
 {
    "allocation_id": "192.85.85.16",
    "_protection": "NOT_PROTECTED"
 },
 {
    "allocation_id": "192.85.85.17",
    "_protection": "NOT_PROTECTED"
 },
 {
    "allocation_id": "192.85.85.18",
    "_protection": "NOT_PROTECTED"
 },
 {
```

```
    "allocation_id": "192.85.85.19",
    "_protection": "NOT_PROTECTED"
  },
  {
    "allocation_id": "192.85.85.20",
    "_protection": "NOT_PROTECTED"
  },
  {
    "allocation_id": "192.85.85.21",
    "_protection": "NOT_PROTECTED"
  },
  {
    "allocation_id": "192.85.85.11",
    "_protection": "NOT_PROTECTED"
  },
  {
    "allocation_id": "192.85.85.22",
    "_protection": "NOT_PROTECTED"
  }
  ],
  "result_count": 12
  }
```


Create an uplink profile. See Add an Uplink Profile.

Guidance to Set Maximum Transmission Unit

Get guidance on how to set the Maximum Transmission Unit (MTU) value in the different objects or profiles in NSX.

Jumbo Frame Support

The minimum required MTU is 1600 bytes. However, an MTU of 1700 bytes is recommended to address the whole range of functions and future-proof the environment for an expanding Geneve header. For better performance in applications generating large packets and for optimal throughput, increase the MTU to at least 9,000 bytes as long as underlay physical infrastructure supports it and is also set to use jumbo frame MTU of 9000 bytes.

VM MTU

In most of the deployments, the guest VM MTU is set to1500 bytes. So no change is required for the VM MTU if the physical fabric has an MTU of 1700 bytes or higher. To improve the throughput, one can increase the MTU up to 8800 (a estimated number to accommodate bridging and future header expansion) only if underlay physical infrastructure is set to to use 9000 bytes. VM MTU should be set 100 bytes or more (200 preferred) lower than the MTU of the physical fabric.

MTU Configuration

- Global Tunnel Endpoint MTU: To configure the MTU value, go to **System** > **Settings** > **Global Fabric Settings**. The default value for MTU is 1700 bytes. When you set this MTU value, NSX configures the MTU value for all the N-VDS instances used in NSX Transport Nodes.
- Global Logical Interface MTU: To configure the MTU value, go to **Networking** > **Global Networking Config**. The default value for MTU is 1500. When you set this MTU value, NSX configures the MTU value for all the logical router interfaces. If the Global Logical interface MTU value is not specified, the MTU value is taken from the Tier-0 logical router (T-0 Gateway). However, on a specific port, the logical router uplink MTU value can override the Global Logical interface MTU value.
- Uplink Profile MTU: To configure the MTU value, go to **System** > **Profiles** > **Uplink Profiles**. When you set this MTU value, NSX configures the MTU value for NSX Transport nodes that use N-VDS switch. This MTU field is optional in the uplink profile. If you do not configure it, NSX takes the value set in the global Tunnel Endpoint MTU.
- (vSphere) VDS MTU: To configure the MTU value, go to the vCenter and modify the VDS directly. When you set this MTU value, NSX configures the MTU value for NSX Transport Nodes that use vSphere VDS. In this case, MTU value set on the attached uplink profile is not used.

Design Guidance

For optimal throughput, set the Global Tunnel Endpoint MTU , Uplink Profile MTU, and vSphere VDS MTU to at least 9000 bytes as long as:

- The underlying infrastructure supports 9000 bytes.
- The underlying infrastructure is set to use jumbo frame MTU of 9000 bytes.

Otherwise, configure Global TEP MTU, Uplink Profile MTU, and vSphere VDS MTU to minimum 1600 bytes or minimum recommended 1700 bytes.

The Gateway Interface MTU can continue to have default value. If you modified the Gateway Interface MTU, the modified value must be at least 200 bytes less than the Fabric MTU (which refers to Global Tunnel Endpoint MTU or VDS MTU or Uplink Profile MTU).

> **Important:**
> When adjusting the fabric MTU packet size, you must also configure the entire network path (VMkernel ports, virtual switches, physical switches and routers) to support the same MTU packet size. If a device along the path does not support the required frame size and receives a frame larger than its MTU, the device will drop the frame.

Configuring Profiles

Profiles allow you to consistently configure identical capabilities for network adapters across multiple hosts or nodes.

Profiles are containers for the properties or capabilities that you want your network adapters to have. Instead of configuring individual properties or capabilities for each network adapter, you can specify the capabilities in the profiles, which you can then apply across multiple hosts or nodes.

Add an Uplink Profile

An uplink is a link from the NSX Edge nodes or hypervisor nodes to the top-of-rack switches or NSX logical switches. A link is from a physical network interface on an NSX Edge node or hypervisor nodes to a switch.

- See NSX Edge network requirements in NSX Edge Installation Requirements.
- Each uplink in the uplink profile must correspond to an up and available physical link on your hypervisor host or on the NSX Edge node.
  For example, your hypervisor host has two physical links that are up: vmnic0 and vmnic1. Suppose vmnic0 is used for management and storage networks, while vmnic1 is unused. This might mean that vmnic1 can be used as an NSX uplink, but vmnic0 cannot. To do link teaming, you must have two unused physical links available, such as vmnic1 and vmnic2.
  For an NSX Edge, tunnel endpoint and VLAN uplinks can use the same physical link. For example, vmnic0/eth0/em0 might be used for your management network and vmnic1/eth1/em1 might be used for your fp-ethX links.

An uplink profile defines policies for the uplinks. The settings defined by uplink profiles can include teaming policies, active and standby links, transport VLAN ID, and MTU setting.

Consider the following points when configuring Failover Teaming Policy for VM appliance-based NSX Edge nodes and Bare Metal NSX Edge:

- For uplinks used by a teaming policy, you cannot use the same uplinks in a different uplink profile for a given NSX Edge transport node. Standby uplinks are not supported and must not be configured in the failover teaming policy. If the teaming policy uses more than one uplink (active/standby list), you cannot use the same uplinks in the same or a different uplink profile for a given NSX Edge transport node.
- Supported scenarios:
  - Bare Metal NSX Edge supports a single active uplink and a standby uplink. They do not support multiple standby uplinks.
  - NSX Edge VMs do not support any standby uplinks - single or multiple standby uplinks.

Consider the following points when configuring Load Balance Source for VM appliance-based NSX Edge nodes:

- Supports multiple active uplinks.

- You cannot use LAG to configue the teaming policy.
- In the **Active Uplinks** field, enter uplink labels that will be associated to physical NICs when you prepare transport nodes. For example, uplink1, uplink2. When you prepare transport nodes, you will associate uplink1 to pnic1 and uplink2 to pnic2.
- You must use the **Load Balanced Source** teaming policy for traffic load balancing.

Consider the following points when configuring Load Balance Source for Bare Metal NSX Edge:

- Supports multiple active uplinks.
- In the **Active Uplinks** field, you can use LAGs or enter individual uplink lables. For example, LAG1 or uplink1, uplink2.
- A LAG must have two physical NICs on the same N-VDS.
- The number of LAGs that you can actually use depends on the capabilities of the underlying physical environment and the topology of the virtual network. For example, if the physical switch supports up to four ports in an LACP port channel, you can connect up to four physical NICs per host to a LAG.
- In the LACP section, Bare Metal NSX Edge only supports **Source and destination MAC address, IP address and TCP/UDP port**.
- If multiple LAG uplinks are configured on a Bare Metal NSX Edge, enter a unique LAG name for each LAG uplink profile.
- If multi-vtep uplink profile is used for Bare Metal NSX Edge or edge VMs, NSX only supports **Load Balance Source** teaming policy.
- You must use the **Load Balanced Source** teaming policy for traffic load balancing.

Perform the following steps to create an uplink profile:

1. From a browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.
2. Select **System** > **Fabric** > **Profiles** > **Uplink Profiles**.
3. Click **Add Uplink Profile**.
4. Enter an uplink profile name. Add an optional uplink profile description.
5. Teamings: Click Set. Create a default teaming policy and optionally a named teaming policy. In the Set Teamings window, complete the fields.
   a) Click **Add Teaming**.
   b) In the Teaming Policy drop-down field, select from **Failover Order**, **Load Balance Source** or **Load Balance Source MAC address**.
   c) In the Active Uplinks field, enter the active uplink name. For example, `uplink1`, `uplink2` and so on.
   d) In the Standby uplinks field, enter the standby uplink name.
   e) To enable preemptive failover from a standby uplink to an active uplink after the active comes back up, enable Failback. If you do not want the active uplink to carry traffic once it is back up again after a failure (non-preemptive), disable Failback.
   f) Click **Add**.
   g) Click **Apply**.

   For more details on teamings, see Teaming Policy and Configure Named Teaming Policy.
6. LAGs: Click **Set** to configure Link aggregation groups (LAGs) using Link Aggregation Control Protocol (LACP) for the transport network.
   a) On The **Set LAGs** window, click **Add LAG**.
   b) Enter a name for the LAG.
   c) Select the **LACP mode** - Active or Passive.
   d) Select the **LACP Load Balancing mode**.
   e) Enter the number of uplinks.
   f) Select the **LACP time out** type - Slow or Fast.
   g) Click **Add**.
   h) Click **Apply**.

   For more details on LAG, see Link Aggregation Groups.

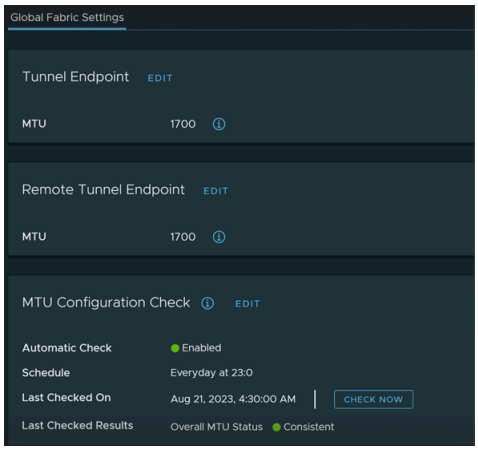
7. Enter a Transport VLAN ID value. The transport VLAN set in the uplink profile tags overlay traffic only and the VLAN ID is used by the Tunnel Endpoint Pools (TEP IP Pools).

   Important:
   While you can choose any of the available pre-created default uplink profiles, note that you can only edit and configure the transport VLAN ID field to a value of your choice. You cannot edit any other field of a pre-created default uplink profile.

8. Enter the MTU value.

   For hosts that use vSphere VDS, configure MTU on the VDS from vCenter. The uplink profile MTU default value is 1700 bytes and is applicable to transport-nodes that use N-VDS.

   Note:
   The MTU field is optional. If you do not configure it, NSX takes the value set in the Tunnel Endpoint MTU field. If both MTU fields are set, uplink profile MTU value ovrerrides the tunnel endpoint MTU value.



   For more information on MTU guidance, see Guidance to Set Maximum Transmission Unit.

9. Configure global tunnel endpoint MTU.

In addition to the UI, you can also view the uplink profiles with the API call `GET /policy/api/v1/infra/host-switch-profiles`.

Link Aggregation Groups

LAG is a mechanism to bundle multiple physical NICs into a single group, which determines how traffic is distributed across physical NICs.

LAG is an optional field. In the LAGs section, click Add for Link Aggregation Groups (LAGs) using Link Aggregation Control Protocol (LACP) for the transport network.

The active and standby uplink names you create can be any text to represent physical links. These uplink names are referenced later when you create transport nodes. The transport node UI/API allows you to specify which physical link corresponds to each named uplink.

Possible LAG hashing mechanism options:

- **Source MAC address**
- **Destination MAC address**
- **Source and destination MAC address**
- **Source and destination IP address and VLAN**
- **Source and destination MAC address, IP address, and TCP/UDP port**

Supported LAG hashing mechanisms on hosts types:

- NSX Edge nodes: Source and destination MAC address, IP address, and TCP/UDP port.

- ESX hosts with VDS in Enhanced Networking Stack (ENS) mode: **Source MAC address**, **Destination MAC address**, and **Source and destination MAC address**.
- ESX hosts with VDS in Standard mode: **Source MAC address**, **Destination MAC address**, **Source and destination MAC address**, and **Source and destination IP address and VLAN**.
- ESX hosts with vSphere Distributed Switch (v 7.0 and later that supports NSX): LACP is not configured in NSX. You need to configure it in vCenter.

Teaming Policy

A teaming policy defines how VDS uses its uplink for redundancy and traffic load balancing.

In the Teaming section, you can either enter a default teaming policy or you can choose to enter a named teaming policy that is only applicable to VLAN networks. Click **Add** to add a naming teaming policy. A teaming policy defines how VDS uses its uplink for redundancy and traffic load balancing. You can configure a teaming policy in the following modes:

- **Failover Order**: Specify an active uplink along with an optional list of standby uplinks. If the active uplink fails, the next uplink in the standby list replaces the active uplink. No actual load balancing is performed with this option. Standby uplinks and multiple active uplinks are not supported for NSX Edge transport nodes. Also, for an NSX Edge transport node, active uplink used in one profile should must not be used in another profile.
- **Load Balance Source**: Maps a virtual interface of a VM to an uplink. Traffic sent by this virtual interface will leave the host through this uplink only, and traffic destined to this virtual interface will necessarily enter the virtual switch through this uplink. Select a list of active uplinks. When you configure a transport node, you can pin each interface of the transport node to one active uplink. This configuration allows use of several active uplinks at the same time. No standby uplink is configured in this case.

  Important:

  To manage VLAN traffic, if you configure a default teaming policy in Load Balance Source mode, then on failure of the first uplink, traffic will not fail over to the second uplink interface.
- **Load Balance Source MAC Address**: Select an uplink based on a hash of the source Ethernet. NSX Edge transport nodes do not support this teaming policy.

  Note:

  - On ESX hosts: For default teaming policy - Load Balance Source MAC, Load Balance Source, and Failover Order teaming policies are supported.
  - On NSX Edge: For default teaming policy, Load Balance Source and Failover Order teaming policies are supported. For named teaming policy, only Failover Order policy is supported.

(ESX hosts and NSX Edge) You can define the following policies for a transport zone:

- A Named teaming policy for every VLAN-based logical switch or segment.
- A Default teaming policy for the entire VDS.

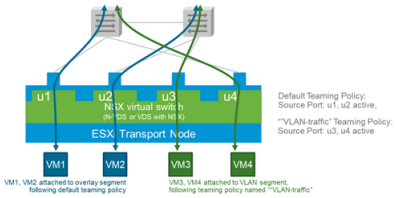
Named teaming policy: A named teaming policy means that for every VLAN-based logical switch or segment, you can define a specific teaming policy mode and uplinks names. This policy type gives you the flexibility to select specific uplinks depending on the traffic steering policy, for example, based on bandwidth requirement.

- If you define a named teaming policy, VDS uses that named teaming policy if it is attached to the VLAN-based transport zone and finally selected for specific VLAN-based logical switch or segment in the host.
- If you do not define any named teaming policies, VDS uses the default teaming policy.

For more details, see Configure Named Teaming Policy.

Configure Named Teaming Policy

On a host running NSX with an N-VDS, you can use the named teaming policies to override the default teaming policy that was configured for some specific VLAN backed segments. This capability is used to steer VLAN traffic to specific uplinks.

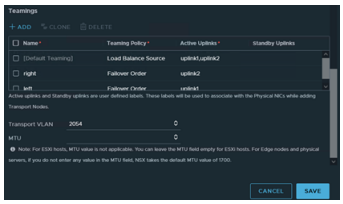


In the figure, VMs (VM1, VM2) connected to overlay networks follow default teaming policy and use uplinks u1 and u2. However, an additional teaming policy `named teaming policy` VLAN-traffic is configured to steer vlan traffic (for VMs 'VM3, VM4' connected to VLAN segment) to uplinks u3 and u4 to separate overlay and VLAN traffic.

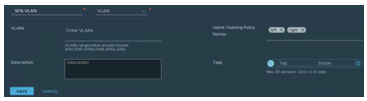
1. Configure an uplink profile, which will be consumed by transport node members of a VLAN Transport zone and define following:

   a) Set a default teaming policy with active uplinks (standby uplinks not supported for NSX Edge Transport nodes).
   b) Set a named teaming policy <p1> with mode 'failover order' and active uplink1.
   c) Set a named teaming policy <p2> with mode 'failover order' and active uplink2 and so on.

      Note:
      For overlay segments, the default teaming is used by default. For VLAN segments, when a named teaming policy is defined, NSX ensures that the named teaming policy overrides the default teaming policy.
   d) Set the Transport VLAN and MTU.
   e) Click **Save**.



2. Configure VLAN Transport zone with teaming policy names specified in the uplink profile.



3. Create VLAN `segment1` on VLAN*X* and associate with teaming policy name <p1>.



4. Create a VLAN `segment2` on VLAN*Y* and associate it with teaming policy name <p2>.
5. Configure a transport node to use uplink profile configured with the named teaming policy.

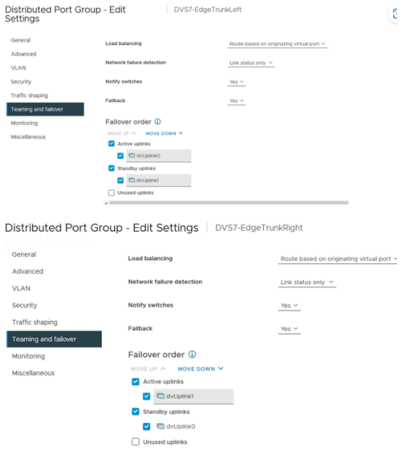
   Use a named teaming policy to pin traffic that egreeses from a specific uplink and ingresses into a ToR switch .

   Note:
   A named teaming policy is only applicable to VLAN segments/logical switches. A named uplink profile only supports the Failover teaming policy. If a VLAN segment does not specify which teaming policy to use, the default teaming policy in the uplink profile will be used.

   If you use (Failover) Active/Standby for a VLAN-based segment, the traffic will be always forwarded through the active uplink. Traffic will not failover to the standby uplink. If you want to configure failover, define it in the distributed port group. This is valid only for Edge VM.

NSX teaming policies apply to traffic entering and existing N-VDS but not the traffic exiting the hypervisor configured with PNICs on VDS. Hosts configured with PNICs on VDS, will follow the teaming policy defined in their respective VDS DVPGs in vCenter.

An example of a VDS portgroup teaming that is defined in vCenter for NSX Edge VM.



Add and attach NSX Edge Bridge Profile to a Segment

The NSX Edge bridge profile specifies the primary NSX Edge node that will be the preferred node for the active bridge and backup node that will be preferred for the backup bridge.

- Verify that the NSX Edge cluster is available with minimum of two edge nodes (BareMetal or VM form factor).
- For the VM form factor NSX Edge, verify NSX Edge uplink VDS Trunk port group in vCenter has the following configuration:
  - Forged transmit
  - MAC learning (recommended) or promiscous mode/sink port configured
  - Active/Standby teaming policy.

  Note:
  Consider dedicating an NSX Edge uplink (vNIC) to bridged traffic so that other kinds of traffic to and from the NSX Edge do not suffer from the performance impact related to promiscuous mode.
- Verify NSX Edge TNs member of NSX Edge Cluster are attached to overlay as well as VLAN Transport Zone.

NSX Edge Bridge connects NSX overlay logical segment with a traditional VLAN at layer 2. The edge bridge leverages DPDK for high performance forwarding. The traffic bridged in or out of the NSX domain is subject to an edge bridge firewall instance. The NSX Edge bridge functionality is mainly for migration scenarios (physical to virtual or virtual to virtual) or for integration of physical, non-virtualized appliances to the virtualized environment.

Note:
Starting with NSX 2.5, the same segment can be attached to several bridges on different Edge Clusters and VLANs.

At the time of the creation of the Bridge Profile, no Bridge is instantiated yet. The Bridge Profile is just a template for the creation of one or several Bridge pairs. Once a Bridge Profile is created, you can attach a segment to it. By doing so, an active Bridge instance is created on the primary Edge, while a standby Bridge is provisioned on the backup Edge. NSX creates a Bridge Endpoint object, which represents this pair of Bridges. The attachment of the segment to the Bridge Endpoint is represented by a dedicated logical port.

1. From a browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.
2. Select **Networking** > **Segments** > **Profiles** > **Edge Bridge Profiles** > **Add Edge Bridge Profile**.
3. Enter the NSX Edge cluster profile details.

Option	Description
Name and Description	Enter a NSX Edge bridge cluster profile name. You can optionally enter the profile details such as, the primary and backup node details.
Edge Cluster	Select the NSX Edge cluster that you can to use.
Primary Node	Designate the preferred NSX Edge node from the cluster.
Backup Node	Designate the back up NSX Edge node if the primary node fails.
Failover Mode	Select either **Preemptive** or **Non-Preemptive** mode. The default HA mode is preemptive, which can slowdown traffic when the preferred NSX Edge node goes back online. The non-preemptive mode does not cause any traffic slowdown. In the preemptive mode, the Bridge on the primary Edge will always become the active bridge forwarding traffic between overlay and VLAN as soon as it is available. In the non-preemptive mode, the Bridge on the primary Edge will remain standby if it becomes available when the Bridge on the backup Edge is already active.

4. After you create a Bridge Profile, associate it to a segment.
5. Select **Networking** > **Segnments** > **NSX** > **Add Segment**.
6. Enter the required details, connect to overlay transport zone and click **Save**.
7. Edit the segment to which you want to add the Bridge Profile.
8. In the Additional Settings section, in the Edge Bridges field, select **Set**.
9. Click **Add Edge Bridge**.
10. Select the Edge Bridge Profile.
11. Select the Transport Zone where the bridged traffic is sent to the N-VDS selected by the transport zone.
12. Select the VLAN ID for the VLAN traffic as well as the physical port you select on the NSX Edge for sending or receiving this VLAN traffic.
13. (Optional) Select the teaming policy to decide how N-VDS balances traffic across its uplinks.
14. Click **Add**.
15. Click **Save**.

The newly created NSX Edge Bridge Profile is associated to a segment to balance VLAN traffic.

- Verify the configuration and state of L2 bridges on the NSX Edge.
  a. SSH to the NSX Edge as an admin.
  b. Run `get bridge`.
  c. Verify that the Device State is `Up`.

NSX Edge bridge state is down

NSX Edge bridge state is down.

NSX Edge bridge state has gone down.

Failure of VLAN uplink or failure of BFD tunnel between two NSX Edge nodes that are part of the bridge.

1. SSH to NSX Edge as admin.
2. Run `get bridge summary` CLI to view existing bridges
3. Run cli `get bridge name <name>` or `get bridge <uuid>` or `get bridge vlan <vlan-id>` to view edge device state status.
4. Run cli `get bridge mac-sync-table` to verify overlay segment mac is learned successfully.
5. run cli `get edge-cluster-status` to verify edge status, admin-state, health-check for all upink interfaces is up with no issues.
6. Verify NSX Edge bridge state is now active.

NSX Edge bridge does not get created

NSX Edge bridge does not get created.

From the NSX Edge node, when you run `get bridge summary`, the output does not show the bridge is created.

VLAN ID or uplink conflict.

1. Verify that VLAN ID used for bridging on a specific uplink is not being used by another VLAN segment for configuration of any other feature.
2. Verify all NSX Edge transport node member of NSX Edge cluster configured with bridging belong to same overlay and VLAN transport zone.
3. Verify the bridge is not configured with multiple VLAN uplinks in a failover teaming policy.

Add a High Availability Profile

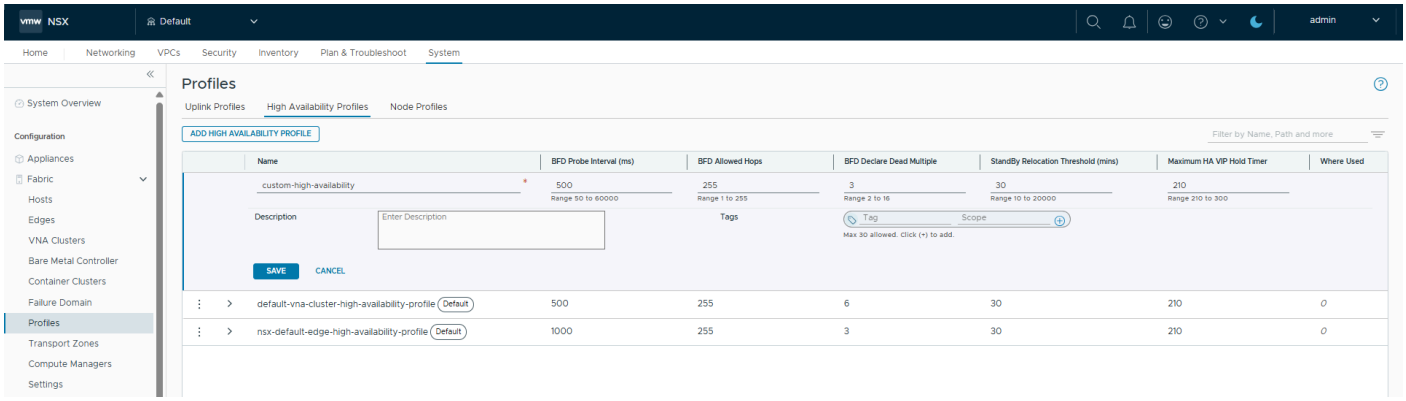
A high availability profile defines the high availability (HA) policies for nodes in an NSX edge cluster or a virtual network appliance (VNA) cluster.

NSX provides the following default high availability profiles:

- nsx-default-edge-high-availability-profile
- default-vna-cluster-high-availability-profile

In addition to the defaults, you can configure custom high availability profiles.

1. From a browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.
2. Select **System** > **Fabric** > **Profiles** > **High Availability Profiles** > **Add High Availability Profile**.



3. Enter the high availability profile details.

Option	Description
Name and Description	Enter a name for the high availability profile. Optionally, enter a description for the profile. For example, `Custom BFD values`.
BFD Probe Interval	Set the time interval, in milliseconds, that Bidirectional Forwarding Detection (BFD) sessions use to detect forwarding path failures. BFD packets are transmitted and received at the configured time interval. The default value is 500 milliseconds.
BFD Allowed Hops	Set the maximum number of hops that a multi-hop BFD session can traverse. The default value is 255 hops.
BFD Declare Dead Multiple	Set the number of BFD packets that can be missed before flagging the BFD session as down. The default value is 3 misses.
Standby Relocation Threshold	**Note:** This setting applies only to active-standby edge clusters and active-standby VNA clusters created through the VPC workflow. Set the threshold period, in minutes, that NSX waits before relocating the standby gateway from a failed standby node to a healthy node. If the failed standby node recovers within the threshold period, standby relocation does not occur. The default and recommended value is 30 minutes. Allowed values can range from 10 to 20000 minutes.
Maximum HA VIP Hold Timer	**Note:** This setting applies only to VNA clusters that were created through the route controller workflow. Set the time interval, in seconds, that NSX waits during BGP connection failures before moving the active VNA node to standby state. The default value is 210 seconds.

4. Click **Save** to add the high availability profile to the list.

Prepare a vSphere Distributed Switch for NSX

Before you configure an NSX transport node using vSphere Distributed Switch (VDS) as a host switch, ensure that the VDS created on a vCenter is configured to manage NSX traffic.

- Verify that ESX hosts have the required number of physical NICs to meet networking requirements. For example, if you plan to configure teaming policies and remote span port mirroring, ensure that a free physical NIC is available to avoid uplink conflicts.
- Ensure that the MTU value of the physical switch port or LACP port is set to 1700 bytes or higher (up to 9000 bytes).

High-level tasks to configure a cluster or a standalone managed host using a VDS switch.

1. In a vCenter, create a VDS. For more information about creating a VDS, see Prepare a vSphere Distributed Switch for NSX.
   - Set the MTU value for the VDS to at least `1600`. A minimum MTU of `9000` is recommended.
   - Add hosts, that you want to prepare for NSX networking, to VDS created in vCenter.
   - Assign Physical NICS of each of the host as uplinks on the VDS.
2. In NSX, add an uplink profile that defines a teaming policy mapping NSX uplinks with VDS uplinks.
3. In NSX, prepare an ESX host using VDS as the host switch.

   At the end of the configuration, the host is prepared as NSX transport node with VDS as the host switch.

Configure the host as a transport node. See Prepare ESX Cluster Hosts as Transport Nodes by Using TNP.

Adding a Host Transport Node Profile for Hosts

A host transport node profile is a template defining a configuration that is applied to a group of hosts that are part of a vCenter cluster. The host transport node profile is not applied to prepare standalone hosts. Prepare vCenter cluster hosts as transport nodes by applying a transport node profile. Transport node profiles define transport zones, member hosts, and the switch configuration which includes the uplink profile, IP assignment, and mapping of physical NICs to uplink virtual interfaces.

- Verify that the hosts are part of a vCenter cluster.
- Verify that cluster hosts are members of VDS version 7.0 or later with at least one uplink on a VDS port group.
- Verify that a transport zone is configured. See Create Transport Zones.
- Verify that NSX Manager cluster nodes are up and available. To verify cluster status, go to **System** > **Appliances** > **Cluster**. See Deploy NSX Manager Nodes to Form a Cluster from the UI.
- Verify that an IP pool is configured, or DHCP must be available in the network deployment. See Create an IP Pool for Tunnel Endpoint IP Addresses.
- Verify that a compute manager is configured. See Add a Compute Manager.
- Verify an uplink profile to use for Host configuration is configured. See Add an Uplink Profile.

> **Note:**
>
> Transport node profiles are only applicable to ESX host members of a vCenter cluster. These profiles cannot be applied to NSX Edge transport nodes.

Transport node creation begins when a transport node profile is applied to a vCenter cluster. NSX Manager prepares the hosts in the cluster and installs the NSX components on all the hosts. Transport nodes for the hosts are created based on the configuration of the transport node profile.

On a cluster prepared with a transport node profile, these outcomes are true:

- When you move an unprepared host into a cluster that has an applied transport node profile, NSX automatically prepares the host as a transport node using the transport node profile.
- When you move a transport node from the cluster to an unprepared cluster or directly as a standalone host under the data center, the following operations ensue. First the transport node configuration applied to the node is removed and then NSX VIBs are removed from the host.

To delete a transport node profile, you must first detach the profile from the associated cluster. The existing transport nodes are not affected. New hosts added to the cluster are no longer automatically converted into transport nodes.

Points to note when you create a Transport Node Profile:

- You can add a maximum of four VDS switches for each configuration: enhanced VDS created for VLAN transport zone, standard VDS created for overlay transport zone, enhanced VDS created for overlay transport zone.
- There is no limit on the number of standard VDS switches created for VLAN transport zones.
- In a single host cluster topology running multiple standard overlay VDS switches and edge VM on the same host, NSX provides traffic isolation such that traffic going through the first VDS is isolated from traffic going through the second VDS and so on. The physical NICs on each VDS must be mapped to the edge VM on the host to allow the north-south traffic connectivity with the external world. Packets moving out of a VM on the first transport zone must be routed through an external router or an external VM to the VM on the second transport zone.
- Each VDS switch name must be unique. NSX does not allow the use of duplicate switch names.
- Each transport zone ID associated with each VDS host in the transport node configuration or transport node profile configuration must be unique.

1. From a browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.
2. Navigate to **System** > **Hosts** > **Transport Node Profile**.
3. Click **Add Transport Node Profile**.
4. Enter a name to identify the transport node profile.

   You can optionally add a description for the transport node profile.
5. Click **Set** under Host Switch. Then click **Add Host Switch** to add the details of the new switch.

   Before you proceed, decide which type of host switch you want to configure on nodes of a cluster.
6. Configure the following fields, and then click **Apply** to save the host switch configuration.

Option	Description
Name	(Hosts managed by a vSphere cluster) Select the vCenter that manages the host switch. Select the VDS that is created in vCenter and attached to your ESX hosts.
Transport Zones	In the Show section, select **Overlay**, **VLAN**, or **All** to view and select the type of transport zones you want for the host switch. These transport zones are realized by associated host switches. Supported transport zone configurations: • You can add multiple VLAN transport zones per host switch. • You must add only one overlay transport zone per host switch. The NSX Manager UI does not allow adding multiple overlay transport zones. If you are creating a TEP-less transport node profile for VLAN-backed VPC subnet, you must select the default transport zone.
Uplink Profile	Select an existing uplink profile from the drop-down menu or create a custom uplink profile. You can also use the default uplink profile. If you keep the MTU value empty, NSX takes the global default MTU value 1700. If you enter an MTU value in the NSX uplink profile, that MTU value overrides the global default MTU value. **Note:** Link Aggregation Groups that are defined in an uplink profile cannot be mapped to VDS uplinks.

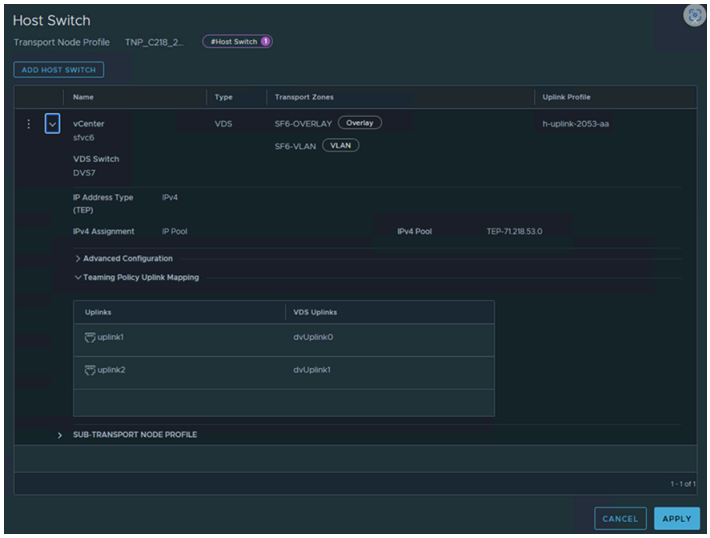
Option	Description
IP Address Type (TEP)	Choose between **IPv4** and **IPv6** to specify the IP version for the tunnel endpoints (TEPs) of the transport node. **Note:** If you are creating a TEP-less transport node profile for a VLAN-backed VPC support, you must select **None**.
IPv4 Assignment	Choose how IPv4 addresses are assigned to the TEPs. The options are: • **Use DHCP**: IPv4 addresses are assigned from a DHCP server. • **Use IPv4 Pool**: IPv4 addresses are assigned from an IP pool. Specify the IPv4 pool name to be used for TEPs.
IPv6 Assignment	Choose how IPv6 addresses are assigned to the TEPs. The options are: **Use DHCPv6**: IPv6 addresses are assigned from a DHCP server. **Use IPv6 Pool**: IPv6 addresses are assigned from an IP pool. Specify the IPv6 pool name to be used for TEPs. **Use AutoConf**: IPv6 addresses are assigned from Router Advertisement (RA).
Teaming Policy Uplink Mapping	Before you map uplinks in NSX with uplinks in VDS, ensure that uplinks are configured on the VDS switch. To configure or view the VDS switch uplinks, go to **vCenter** > **vSphere Distributed Switch**. Click **Actions** > **Settings** > **Edit Settings**. Map uplinks defined in the selected NSX uplink profile with VDS uplinks. The number of NSX uplinks that are presented for mapping depends on the uplink profile configuration. For example, use the following steps to map a VDS link to uplink-1 (active). In the upink-1 (active) row, go to the Physical NICs column, click the edit icon, and type the name of VDS uplink. Likewise, complete mapping for the other uplinks.
Advanced Configuration > Mode	Choose between the following mode options: • **Standard**: This mode is the default datapath mode of operation for all vSphere deployments. This mode is compatible with the broadest set of hardware, but does not offer the latest performance improvements. • **Enhanced Datapath - Standard**: This mode runs an improved packet forwarding stack, recommended for general compute environments and NSX Edge clusters. This mode dynamically and efficiently allocates CPU resources. • **Enhanced Datapath - Dedicated**: This mode is the datapath operation dedicated to telco applications and virtual network functions. This mode requires preallocated and dedicated CPU cores If Mode is set to Enhanced Datapath Dedicated, then an (optional) additional configuration is needed. This configuration takes the number of logical cores per NUMA that the host should dedicate. 1. Click Set. 2. In the CPU Config window, click Add. 3. Enter values for the NUMA Node Index and LCores per NUMA Node fields. 4. To save the values, click Add and Save.
Advanced Configuration > Advanced NIC Passthrough	Turn on this toggle to enable network offloads to physical DPU-based SmartNICs with UPTv2 or Enhanced DirectPath I/O passthrough capabilities. For the Advanced NIC Passthrough feature to take effect, you must configure the selected VDS in vCenter with network offloads compatibility set to **Advanced NIC Passthrough**.

Note:

Uplinks/LAGs, NIOC profile, LLDP profile are defined in vCenter. These configurations are not available in NSX Manager. To manage VMkernel adapters on a VDS switch, go to vCenter to attach VMkernel adapters to Distributed Virtual port groups or NSX port groups.

7. If you have selected multiple transport zones, you can add them to the same switch. To add a new switch, click **Add Host Switch** again to configure a new switch for the other transport zones.

   NSX switches can attach to a single overlay transport zone and multiple VLAN transport zones at the same time.



8. Click **Apply** to complete the configuration.

Apply the transport node profile to an existing vSphere cluster. See Prepare ESX Cluster Hosts as Transport Nodes by Using TNP.

Host Profile integration with NSX

Integrate host profiles extracted from an ESX host with NSX to deploy ESX and NSX VIBs on stateful servers.

Stateful Servers

Integrate host profiles of an ESX host with NSX on stateful servers.

A stateful host is a host that retains all configurations and the installed VIBs even after it is rebooted. While an auto-deploy server is needed for stateless hosts because the boot up files required to bring up a stateless hosts are stored on the auto-deploy server, a stateful host does not need a similar infrastructure. Because the boot up files required to bring up a stateful host is stored on its hard drive.

In this procedure, the reference host is outside of the stateful cluster and the target hosts in the cluster. A target host can be within a cluster or a standalone host outside of the cluster. Prepare a cluster by applying host profile and transport node profile (TN profile) , so that any new target hosts joining the cluster is automatically prepared with NSX VIBs. Configure the target host as a transport node. Similarly, for a standalone host, apply the host profile and configure NSX to install NSX VIBs and when NSX configuration is complete, it becomes a transport node.

Note:
NSX VIBs are installed from TN profile and ESX host configurations are applied by the Host Profiles.
For the supported NSX and ESX versions on stateful servers, see the compatibility matrix.

Prepare a Target Stateful Cluster

Prepare a target stateful cluster so that any new host joining the cluster is automatically deployed with ESX and NSX VIBs.

You can select a host either within the cluster or outside of the cluster to be the reference host. You need to create a reference host because the host profile from the reference host is extracted and applied to a target host. VDS switch type supports migration of VMkernel adapters.

In this procedure, as an example the instructions are to migrate vmk0 (management traffic) and vmk1 (vMotion traffic) to an VDS switch.

1. On the Reference host, deploy a supported ESX build.
   a) In vSphere, add vmk1 adapter. vmk0 is already present to serve management traffic.
2. Configure the reference node as a transport node.
   a) Using vSphere Web Client, ensure a logical switch is created in NSX.
   b) Using vSphere Web Client, ensure that vmk0 and vmk1 are connected to a logical switch on VDS switch.
3. Extract the host profile from the reference host.
4. On a target host that is a standalone host:
   a) Attach the host profile to the target host.
   b) Manually configure NSX on the host. When configuring the host as a transport node because the host profile on the ESX, ensure the following conditions are met.
   c) Host must belong to the same transport zone.
   d) Target host must use the same IP pool that is used by the reference host.
   e) Uplink profile, LLDP, Network mapping for install, VDS configured on the target host must be the same as configured on the reference host.
5. On a target host that is part of a cluster:
   a) Attach the host profile to the stateful target cluster.
   b) Create and apply the TN profile on the cluster.

Scenarios when VMkernel adapters are migrated with and without host profiles applied to NSX.

Host Transport Nodes

You can prepare ESX hosts as NSX transport nodes.

Preparing ESX Cluster Hosts as Transport Nodes through vCenter

To integrate NSX closely with other VCF components, VCF 9.1 now supports overlay VMkernel (vTEP) lifecycle management in vCenter.

Prior to VCF 9.1, vCenter supported transport node profile (TNP) based host cluster preparation through vSphere Configuration Profiles (VCP). This method required a prerequisite to create uplink profiles and transport node profiles in NSX. VCF 9.1 introduces an overlay VMkernel (vTEP) service type, vNetworking, for NSX overlay vTEPs. The lifecycle management of vTEPs is supported through the VCP cluster document. NSX will use the vNetworking-enabled VMK when prepping a cluster for overlay networking.

Note that if vCenter is of version 9.0 or later, then for NSX and VCP to work all ESXi hosts and NSX must also be of version 9.0 or later.

This cluster-level configuration ensures uniformity across member hosts through the Config Manager workflow that includes the following stages:

- Validation
- Compliance check
- Impact
- Remediation

Before you start prepping a VCP cluster, ensure that the following two commands are executed on every host that are included in the cluster.

```
esxcli network firewall ruleset set -e true -r DVFilter
```

```
esxcli system syslog config set --remote-host-max-msg-len 4096
```

The vCenter admin undertakes the following workflow to prepare an ESX cluster as transport nodes.

1. Create a VDS host switch.
2. Create a VCP enabled cluster or enable VCP on an existing cluster.
3. Add hosts to this cluster if required.
4. From the **Configure** tab of this cluster, go to **Desired State** > **Configuration** > **Draft** and create a draft for the cluster by using any of the following options:
   – Create Draft
   – Import From File
   – Import from Host
5. Add the following keys to the VCP doc with the required configuration:
   – esx/network/vmknics
   – esx/network/net_stacks
   – esx/network_vds/switches
   – nsx/fabric/transport_node_config or transport_node_profile
6. Run the pre-check and apply changes.
7. Check the compliance.

For detailed information for each step, see the *vSphere Administration guide*.

Once all the steps are completed, transport nodes are prepared for NSX. A VCP capsule is shown for the VCP-enabled clusters in NSX Manager under **System > Fabric > Hosts > Clusters**. For more information about vSphere Configuration Profiles, see *Enable vSphere Configuration Profiles During Cluster Creation* in the *vSphere Administration guide*.

Note that on a VCP cluster all the CRUD operations including uninstall are blocked from NSX Manager. These operations on a VCP enabled cluster can only be performed from vCenter.

If you try to change a TNP of a VCP cluster from NSX Manager, you will get an error. To attach a new TNP to a VCP cluster, you will have to perform the following steps.

1. Create a new TNP.
2. Go to vCenter.
3. Go to the **Draft** tab of the cluster and create a new draft.
4. Once the configuration is completed, update the NSX Config in Cluster Desired Doc.
5. Run the pre-check and apply changes.
6. Check the compliance.

Once these steps are completed successfully, the transport node cluster and TNs are prepared.

Also, from NSX Manager if you update any property of a TNP that is applied to a VCP cluster (say VDS), you will see a “Preparing hosts is pending message” for that cluster. You will have to go to vCenter and complete the remaining steps from there.

Preparing ESX Hosts as Transport Nodes

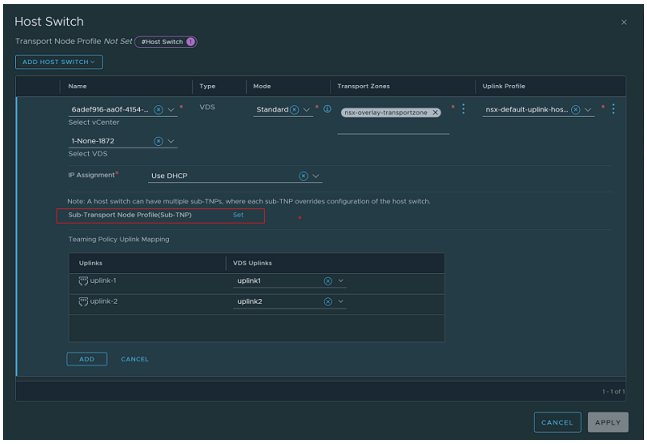
After you create transport zones, IP pools, uplink profiles for transport nodes, prepare an ESX host as a transport node. ESX hosts only support vSphere Distributed Switch (VDS).

Transport Node Profiles

A Transport Node Profile (TNP) is a template to define networking configuration that is applied to a cluster.

Transport Node Profile

To create a TNP, you configure Host Switches with transport zones, uplink profiles, mapping uplinks to VDS uplinks (for a VDS switch) and other configurations. TNP can be created on VDS host switches.



A stretched cluster is a cluster that extends across multiple TEP subnets. A non-stretched cluster is a cluster that is confined to a single TEP subnet. In a non-stretched cluster, a single TNP is sufficient to be applied to a cluster. However, when a cluster is stretched across multiple subnets or L3 domains, you can create sub-clusters consisting of hosts that need the same configuration.

While a TNP represents the global configuration applied to a Host Switch, a sub-TNP represents the local configuration applied to a sub-cluster. When you apply a sub-TNP to a sub-cluster, all the configuration from sub-TNP takes precedence over the host switch configuration.

> **Note:**
> A Sub-TNP can only be created for vSphere Distributed Switch (VDS) switches.

Transport node creation begins when a transport node profile is applied to a vCenter cluster. NSX Manager prepares the hosts in the cluster and installs the NSX components on all the hosts. Transport nodes for the hosts are created based on the configuration specified in the transport node profile.

> **Note:**
> TNP is not used to prepare standalone hosts.

On a cluster prepared with a transport node profile, these outcomes are true:

- When you move an unprepared host into a cluster applied with a transport node profile, NSX automatically prepares the host as a transport node using the transport node profile.
- When you move a transport node from the cluster to an unprepared cluster or directly as a standalone host under the data center, first the transport node configuration applied to the node is removed and then NSX VIBs are removed from the host.

To delete a transport node profile, you must first detach the profile from the associated cluster. The existing transport nodes are not affected. New hosts added to the cluster are no longer automatically converted into transport nodes.

Sub-TNPs and Sub-clusters

A Sub-TNP is a template to define configuration that is applied to hosts that are part of a Sub-cluster. To accommodate different cluster configurations required for sub-clusters, create sub-TNPs.

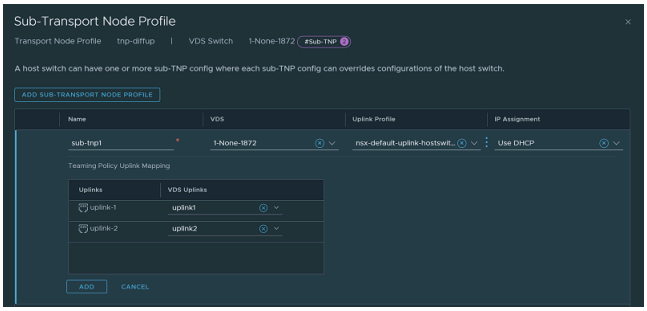
Sub-Transport Node Profile

Starting with NSX 3.2.2, you can configure Sub-TNPs and Sub-clusters.

> **Note:**
> The maximum number of Sub-TNPs under a hostswitch in a TNP is 16.

While a TNP represents the global configuration applied to a Host Switch, a sub-TNP represents the local configuration applied to a sub-cluster. When you apply a sub-TNP to a sub-cluster, the host switch is overriden on the sub-cluster.

You can only apply a sub-TNP to a VDS switch. A sub-TNP can only override the following fields of a host switch: VDS Host Switches ID, uplink profiles and IP assignment.



NSX Manager prepares the hosts in sub-clusters where the sub-TNP is applied and installs the NSX components on all the hosts. Transport nodes for the hosts are created based on the configuration specified in the transport node profile.

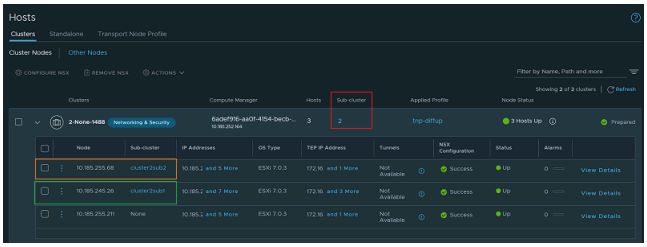
To delete a Sub-TNP, you must first detach the global TNP profile from the associated cluster. The existing transport nodes are not affected. New hosts added to the cluster are no longer automatically converted into transport nodes.

Sub-clusters

Each cluster can have up to 16 sub-clusters.

To manage a cluster that is stretched across different subnets or L3 domains, some hosts will need different configuration, such as a some hosts will need to be in the TEP subnet-1, whereas some other hosts will need to be in TEP subnet-2. So, hosts with similar configuration requirements can be placed in one sub-cluster. And that sub-cluster can be applied with a sub-TNP that provides configuration.

In the following image, the cluster has three hosts: one in each sub-cluster and the last host is not part of any sub-cluster.



Moving Hosts Between Sub-clusters

1. Move host from sub-cluster1 to sub-cluster2.
2. Sub-TNP config of sub-cluster1 is removed from the host.
3. Sub-TNP config of sub-cluster2 is applied to the host.

Limitations of using Sub-clusters

- Sub-clusters can have hosts only from the cluster under which it is created.
- Sub-cluster cannot have hosts which are part of other sub-clusters.
- Maximum number of sub-clusters under a cluster is 16.
- Sub-cluster cannot be deleted if it has some hosts in it. After removing the hosts, sub-cluster can be deleted.
- Minimum supported ESX version for this feature is 7.0.0.
- Stateless hosts cannot be added to a sub-cluster.

Prepare ESX Cluster Hosts as Transport Nodes by Using TNP

If a cluster of ESX hosts is registered to a vCenter, you can apply transport node profiles on the vCenter cluster to prepare all hosts part of the cluster as NSX transport nodes or you can prepare each host individually.

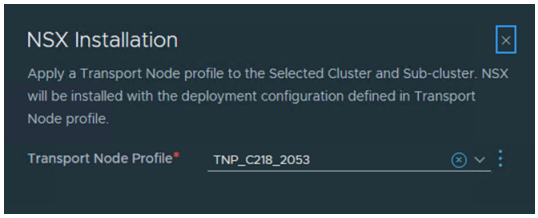
- Verify all hosts that you want to configure as transport nodes are powered on in vCenter.
- Verify all hosts are members of VDS in vCenter with correct uplinks.
- Verify that the system requirements are met.

- Verify vCenter is added as compute manager to NSX Manager.
- Verify NSX Manager cluster is up and stable.
  - UI: Go to **System** > **Appliances** > **NSX Appliances**.
  - CLI: SSH to one of the NSX Manager nodes as an admin and run `get cluster status`.
- The reverse proxy service on all nodes of the NSX Manager cluster must be `Up` and running.
  To verify, run `get service http`. If the service is down, restart the service by running `restart service http` on each NSX Manager node. If the service is still down, contact VMware support.
- Verify that a transport node profile is configured. See Adding a Transport Node Profile.
- Verify that scratch locations of ESX hosts are configured properly and are not shared across them.

Use the following procedure to apply a TNP to a cluster.

> **Note:**
>
> Changing the switch mode when applying a TNP to a cluster can result in data path impact. Refer to Enabling EDP Standard in Active Environments for a non-impactful procedure.

1. From a browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.
2. Select **System** > **Fabric** > **Hosts**.
3. On the **Cluster** tab, select a cluster and click **Configure NSX**.
4. In the **NSX Installation** pop-up window, from the **Transport Node Profile** drop-down menu, select the transport node profile to apply to the cluster. If a transport node is not created, click **Create New Transport Node Profile** to create a new one.



5. Click **Apply** to begin the process of transport node creation of all hosts in the cluster.

   See Adding a Transport Node Profile.

To prepare individual hosts, see Prepare ESX Individual Hosts as Transport Nodes.

Change Sub-cluster

When you want to apply a different networking configuration to a host in a sub-cluster, such as a different uplink profile, move the host to a different Sub-cluster.

To change Sub-cluster a host belongs to, perform these steps:

1. From a browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.
2. Select **System** > **Fabric** > **Hosts**.
3. Select the **Clusters** tab.
4. Select a node and click **Actions** > **Change Sub-cluster**.
5. In the Change Sub-cluster window, select a Sub-cluster that the host must be moved to.
6. Click **Save**.

Detach Cluster TNP

When you want to apply a different cluster TNP, detach the existing one before applying a new TNP.

To detach cluster TNP, perform these steps:

1. From a browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.
2. Select **System** > **Fabric** > **Hosts**.
3. Select the **Clusters** tab.
4. Select a cluster and click **ActionsDetach Transport Node Profile**.

   NSX detached the existing cluster TNP. At the end of the action, the cluster will not have any TNP applied to it.

Migrate VMkernels and Physical NICs to a vSphere Distributed Switch

Manually migrate VMkernel adapters from a vSphere Standard Switch to a vSphere Distributed Switch.

- ESX hosts are prepared as transport nodes using vSphere Distributed Switch.

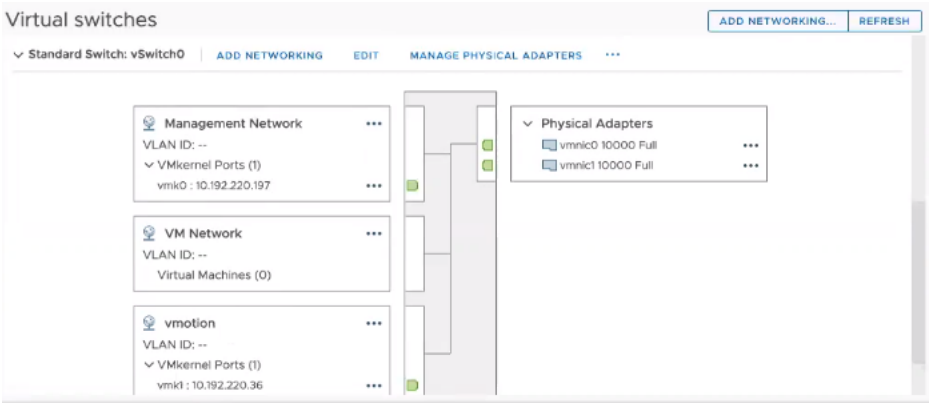
  Note:

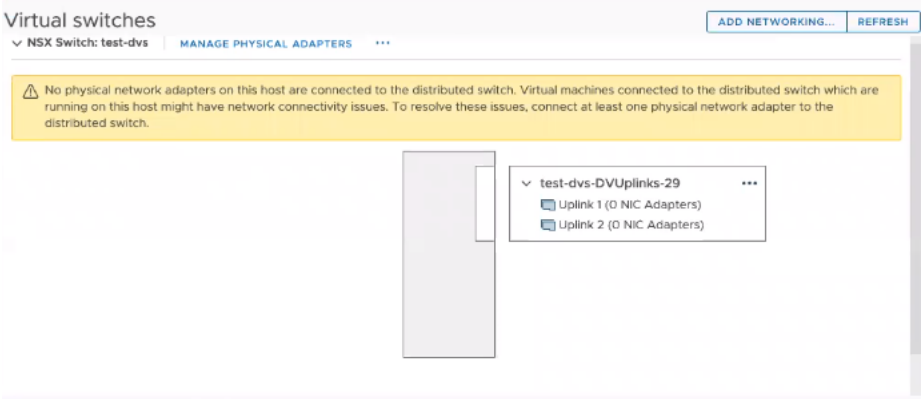
  Transport nodes can be created using vSphere Distributed Switch.

After preparing the transport node with vSphere Distributed Switch host switch type (referred to as an NSX Switch in vCenter), manually migrate VMkernel adapters (vmks) and physical NICs (vmnics) to an NSX Switch on the ESX host.

In the procedure below, consider this switch configuration:

- vmk0, vmk1 are connected to vSwitch0, and vmnic0, vmnic1 are configured as uplink 1 and 2 respectively on the vSwitch0.
- NSX Switch does not have any vmnic or VMkernel adapter configured.





At the end of the procedure, vmnic0, vmnic1 and vmk0, vmk1 are migrated to vSphere Distributed Switch (referred to as an NSX Switch in vCenter).

1. From a browser, log in with admin privileges to a vCenter at https://<vCenterServer-ip-address>.
2. Navigate to **Host** > **Configure** > **Virtual Switches**.
3. View existing vmknics configured on vSwitch0.
4. Make a note of the vmknics to be migrated to the distributed virtual port group of the NSX Switch.
5. Navigate to **Home** > **Networking**, to view all switches configured in the data center.
6. In the Switch page, click **ActionsAdd and Manage Hosts**.
7. Select **Manage Host Networking**.
8. Click **Next**.
9. In the Select Member Hosts window, select `hosts`.
10. Click **Ok**.
11. In the Manage physical adapters window, claim unassigned adapters, as there are available vmnics that can be attached to a switch.
    a) Select an unclaimed uplink and click **Assign uplink**.
    b) Map a vmnic to an uplink on the NSX Switch.
    c) Click **Ok**.
12. In the Manage VMkernel adapters window, assign port groups to NSX Switch.
    a) Select a vmk on vSwitch0 and click **Assign port group**.
    b) Select a NSX port group to assign a vmk to an NSX segment.
    c) Perform steps a and b for the remaining hosts that are managed by the switch.
13. Finish the Add and Manage Hosts wizard.
14. To verify vmk0 and pnics are migrated from vSwitch0 to NSX Switch on the ESX host, navigate to**Host** > **Configure** > **Virtual SwitchesVirtual Switches**. View the updated switch configuration.
15. Alternatively, run the API command, `https://<NSXManager-IP-address>/api/v1/logical-ports`, to verify migration of VMkernel adapters is successful.

    Note:
    All vmk0 ports are set to `Unblocked VLAN` state because management traffic and services are managed by vmk0 ports. These vmk0 ports in `Unblocked VLAN` state allows admins to connect to the vmk0 port if hosts lose connectivity.

Navigate to NSX Manager. In the **System** > **Fabric** > **Hosts** > **Clusters** tab, verify configuration status changed from `Degraded` to `Success`, as vmnics and vmks are migrated to the NSX Switch, the vSphere Distributed Switch.

Prepare ESX Individual Hosts as Transport Nodes

You can configure NSX on individual ESX hosts.

- Verify that the individual host you want to prepare is powered on.
- Verify that the system requirements are met.
- The reverse proxy service on all nodes of the NSX Manager cluster must be `Up` and running.
  To verify, run `get service http`. If the service is down, restart the service by running `restart service http` on each NSX Manager node. If the service is still down, contact Broadcom support.

1. From a browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.
2. Select **System** > **Fabric** > **Hosts**.
3. Select **Standalone** and select a host.
4. On the **Host Details** page, enter details for the following fields.

Option	Description
Name and	Enter the name to identify the standalone host.
IP Addresses	Enter the host IP address.
Description	You can optionally add the description of the operating system used for the host.
Tags	Enter a tag that you want to associate with the host. A tag can be used when you want to group all hosts having a certain OS version, ESX version, and so on.

5. Click **Next**.
6. On the **Prepare Host** tab, click **Add Host Switch**.
7. From the **Select VDS** drop-down menu, select a VDS switch.
8. Configure the following fields:

Option	Description
Name	(Hosts managed by a vSphere cluster) Select the VMware vCenter that manages the host switch. Select the VDS that is created in VMware vCenter and attached to your ESX hosts.
Transport Zone	In the Show section, select Overlay, VLAN or All to view and select the type of transport zones you want for the host switch. These transport zones are realized by associated host switches. Supported transport zone configurations: • You can add multiple VLAN transport zones per host switch. • You must add only one overlay transport zone per host switch. NSX Manager UI does not allow adding multiple overlay transport zones.
Uplink Profile	Select an existing uplink profile from the drop-down menu or create a custom uplink profile. You can also use the default uplink profile. If you keep the MTU value empty, NSX takes the global default MTU value 1700. If you enter an MTU value in the NSX uplink profile, that MTU value will override the global default MTU value. **Note:** Link Aggregation Groups defined in an uplink profile cannot be mapped to VDS uplinks.

Option	Description
IP Address Type (TEP)	Select between IPv4 and IPv6 to specify the IP version for the tunnel endpoints (TEPs) of the transport node.
IPv4 Assignment	Choose how IPv4 addresses are assigned to the TEPs. The options are: • **Use DHCP**: IPv4 addresses are assigned from a DHCP server. • **Use IPv4 Pool**: IPv4 addresses are assigned from an IP pool. Specify the IPv4 pool name to be used for TEPs. • **Use VMkernel Adapter**: Assigns the Management VMkernel interface IP address for the TEP.
IPv6 Assignment	Choose how IPv6 addresses are assigned to the TEPs. The options are: • **Use DHCPv6**: IPv6 addresses are assigned from a DHCP server. • **Use IPv6 Pool**: IPv6 addresses are assigned from an IP pool. Specify the IPv6 pool name to be used for TEPs. • **Use AutoConf**: IPv6 addresses are assigned from Router Advertisement (RA).
Teaming Policy Uplink Mapping	Before you map uplinks in NSX with uplinks in VDS, ensure uplinks are configured on the VDS switch. To configure or view the VDS switch uplinks, go to **VMware Center** > **vSphere Distributed Switch**. Click **Actions** > **Settings** > **Edit Settings**. Map uplinks defined in the selected NSX uplink profile with VDS uplinks. The number of NSX uplinks that are presented for mapping depends on the uplink profile configuration. For example, in the upink-1 (active) row, go to the Physical NICs column, click the edit icon, and type in the name of VDS uplink to complete mapping it with uplink-1 (active). Likewise, complete mapping for the other uplinks.
Advanced Configuration Mode	Choose between the following mode options: • **Standard**: This is the default datapath mode of operation for all vSphere deployments. This mode is compatible with the broadest set of hardware, but does not offer the latest performance improvements. • **Enhanced Datapath - Standard**: This mode runs an improved packet forwarding stack, recommended for general compute environments and NSX Edge clusters. This mode dynamically and efficiently allocates CPU resources. • **Enhanced Datapath - Dedicated**: This is the datapath operation dedicated to telco applications and virtual network functions. This mode requires preallocated and dedicated CPU cores. This is the datapath operation dedicated to telco applications and virtual network functions. This mode requires preallocated and dedicated CPU cores If Mode is set to Enhanced Datapath Dedicated, then an (optional) additional configuration is needed. This configuration takes the number of logical cores per NUMA that the host should dedicate: 1. Click **Set**. 2. In the CPU Config window, click **Add**. 3. Enter values for the NUMA Node Index and LCores per NUMA Node fields. 4. To save the values, click **Add and Save**.

Note:

Uplinks/LAGs, NIOC profile, LLDP profile are defined in VMware vCenter. These configurations are not available in NSX Manager. To manage VMkernel adapters on a VDS switch, go to VMware vCenter to attach VMkernel adapters to Distributed Virtual port groups or NSX port groups.

9. Optional: (Optional) View the ESX connection status.

```
# esxcli network ip connection list | grep 1235
tcp   0   0  192.168.210.53:20514  192.168.110.34:1234  ESTABLISHED1000144459  newreno  nsx-proxy
```

10. On the **Standalone** tab, verify that the NSX Manager connectivity status of hosts in the cluster is Up and NSX configuration state is Success. During the configuration process, each transport node displays the percentage of progress of the installation process. If installation fails at any stage, you can restart the process by clicking the **Resolve** link that is available against the failed stage of the process.

    You can also see that the transport zone is applied to the hosts in the cluster.

    Note:
    If you again configure a host that is part of a cluster that is already prepared by a transport node profile, the configuration state of a node is in `Configuration Mismatch` state.

    Note:
    The **Standalone** tab displays TEP addresses of the host in addition to IP addresses. TEP address is the address assigned to the VMkernel NIC of the host, whereas IP address is the management IP address.

When the hosts are transport nodes, you can create transport zones, logical switches, logical routers, and other network components through the NSX Manager UI or API at any time. When NSX Edge nodes and hosts join the management plane, the NSX logical entities and configuration state are pushed to the NSX Edge nodes and hosts automatically. You can create transport zones, logical switches, logical routers, and other network components through the NSX Manager UI or API at any time. When the hosts are transport nodes, these entities get realized on the host.

Create a logical switch and assign logical ports. See the Advanced Switching section in the *NSX Administration Guide*.

Verify the Transport Node Status

Verify the transport node is created and VDS configured successfully.

1. Log in to the NSX Manager UI.
2. Navigate to **System** > **Fabric** > **Hosts** page.
3. To view the host and VDS status, navigate to **Host Details** > **Overview** tab.
4. Verify host controller connectivity and manager connectivity is UP.
5. Navigate to the Transport Node page and view the VDS status.
6. To view active manager node for this transport node, run `nsxcli -c get managers *`.
7. To view the master controller with 'connected' status and session state up for this transport node, run `nsxcli -c get controllers`.
8. Alternatively, view the VDS on ESX with the `esxcli network ip interface list` command.

    On ESX, verify correct number of vmk interfaces are configured with a VDS name that matches the name you used when you configured the transport zone and the transport node.

```
# esxcli network ip interface list
...

vmk10
   Name: vmk10
   MAC Address: 00:50:56:64:63:4c
```

```
    Enabled: true
    Portset: DvsPortset-1
    Portgroup: N/A
    Netstack Instance: vxlan
    VDS Name: overlay-hostswitch
    VDS UUID: 18 ae 54 04 2c 6f 46 21-b8 ae ef ff 01 0c aa c2
    VDS Port: 10
    VDS Connection: 10
    Opaque Network ID: N/A
    Opaque Network Type: N/A
    External ID: N/A
    MTU: 1700
    TSO MSS: 65535
    Port ID: 67108895

  ...
```

If you are using the vSphere Client, you can view the installed VDS in the UI by selecting host **Configuration** > **Network Adapters**.

9. Check the transport node's assigned tunnel endpoint address.

   The vmk10 interface receives an IP address from the NSX IP pool or DHCP, as shown here:

```
# esxcli network ip interface ipv4 get
Name   IPv4 Address    IPv4 Netmask   IPv4 Broadcast   Address Type  DHCP DNS
-----  --------------  -------------  ---------------  ------------  --------
vmk0   192.168.210.53  255.255.255.0  192.168.210.255  STATIC           false
vmk1   10.20.20.53     255.255.255.0  10.20.20.255     STATIC           false
vmk10  192.168.250.3   255.255.255.0  192.168.250.255  STATIC           false
```

10. Check the API for transport node state information.

    Call the (deprecated API) `GET https://<nsx-mgr>/api/v1/transport-nodes/<transport-node-id>/state` API call.

    Or call the `GET /policy/api/v1/infra/sites/<site-id>/enforcement-points/<enforcement-point-id>/transport-node-status-report`, where the default value for *`<site-id>`* and *`<enforcement-point-id>`* is `default`.

    For example:

```
{ "transport_node_id": "55120a1a-51c6-4c20-b4a3-6f59662c9f6a", "host_switch_states":
 [ { "host_switch_id": "50 21 0c 52 94 22 aa 20-b7 f0 0b da 1c 7c 29 ea", "host_switch_name": "dvs1",
 "endpoints": [ { "device_name": "vmk10", "ip": "172.16.223.175", "default_gateway": "", "subnet_mask":
 "255.255.224.0", "label": 53249 } ],
 "transport_zone_ids": [ "1b3a2f36-bfd1-443e-a0f6-4de01abc963e" ], "host_switch_type": "VDS" } ],
 "maintenance_mode_state": "DISABLED",
 "node_deployment_state": { "state": "success", "details": [] }, "deployment_progress_state":
 { "progress": 100, "current_step_title": "Configuration complete" },
 "state": "success",
 "details": [ { "sub_system_id": "55120a1a-51c6-4c20-b4a3-6f59662c9f6a",
 "sub_system_type": "HostConfig", "state": "success" },
 { "sub_system_id": "55120a1a-51c6-4c20-b4a3-6f59662c9f6a", "sub_system_type": "AppInit", "state":
 "success" },
 { "sub_system_id": "55120a1a-51c6-4c20-b4a3-6f59662c9f6a", "sub_system_type": "LogicalSwitchFullSync",
 "state": "success" } ] }
```

Configure an ESX Host Transport Node with Link Aggregation Group

This procedure describes how to create an uplink profile that has a link aggregation group configured and how to configure an ESX host transport node to use that uplink profile.

- Familiarize yourself with the steps to create an uplink profile. See Create an Uplink Profile.
- Familiarize yourself with the steps to create a host transport node.
- For ESX hosts (connecting to VDS), the LAG configuration is only in vCenter, on the VDS. Therefore configure LAG on the VDS switch in vCenter.

1. From a browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.
2. Select **System** > **Fabric** > **Profiles** > **Uplink Profiles** > **Add Profile**.
3. Enter a name and optionally a description.
   For example, you enter the name `uplink-lag1`.
4. Under **Teamings**, select **Default Teaming**.
5. In the **Active Uplinks** field, do one of the following:
   a) Enter the name of the LAG that you configured on the VDS switch in vCenter.
6. Enter the VLAN ID for **Transport VLAN**.
7. Click **Add** at the bottom of the dialog box.
8. Select **System** > **Fabric** > **Hosts**.
9. Select the **Cluster** tab and select an ESX host.
10. Select **Configure NSX**.
11. In the **Host Details** tab, enter IP address, OS name, admin credentials, and SHA-256 thumbprint of the host.
12. During the switch configuration, depending on the VDS switch, select the uplink profile `uplink-lag1` that was created in step 3.
13. In the **Physical NICs** field, the physical NICs and uplinks dropdown list reflects the new NICs and uplink profile. Specifically, the uplink LAG that is configured on the VDS switch is displayed in the uplink profile drop-down list. Select the VDS uplink for LAG.
14. Enter information for the other fields and complete host preparation.
    The ESX host is prepared as a transport node using the LAG profile.

    Note:
    You can only use LAGs that you define in NSX uplink profile for transport nodes that are prepared using N-VDS. However, for ESX transport nodes prepared using VDS, you can only use LAGs configured on vSphere and it is referred to in the uplink profile.
15. SSH in to ESX host as root and run the following commands:
    a) Run `nsxcli`.
    b) To verify LACP bond is up and active, run `get bonds`.

Quick Start Wizard to Prepare ESX Cluster hosts for Networking and Security

Using the Quick Start wizard, you can configure ESX clusters for Networking and Security.

Starting with NSX 4.2.0, NSX supports only the Networking & Security deployment mode for preparing compute clusters with ESX hosts configured as transport nodes. This change delivers improvements in functionality and flexibility.

Below are the supported deployment modes:

- Pre-4.2.0: NSX supported two deployment modes:
  – Security-Only: Focused solely on security services.
  – Networking & Security: Combined networking and security services.

- NSX 4.2.0 and Later:
  - Only the Networking & Security deployment mode is supported, consolidating the previous options into a deployment mode.

  The single deployment mode allows you to configure the Distributed Firewall (DFW) on Distributed Virtual Portgroups (DVPGs) alongside network virtualization features on the same ESX host.

Default Behavior:

- Fresh NSX 4.2.0 Installation:
  - NSX on DVPG is disabled by default.
  - The cluster is automatically prepared in Networking & Security deployment mode.
  - NSX on DVPG can be manually enabled after cluster preparation.
- Upgrade Scenario (From Pre-4.2.0 Networking & Security to 4.2.0):
  - DFW on DVPG is disabled on the upgraded cluster by default.
  - NSX on DVPG can be manually enabled post-upgrade.
- Upgrade Scenario (From Pre-4.2.0 Security-Only to 4.2.0):
  - The upgraded cluster is automatically prepared for both Networking & Security services.
  - NSX on DVPG is automatically enabled.

If you activate NSX on a cluster that was earlier configured with Security-Only feature, you can now implement the following NSX features on it:

- You can create NSX segments for the clusters you created in vCenter.
- You can edit the auto created Transport Node Profile before applying it to the discovered clusters.
- You can enable NSX Overlay for the discovered clusters.
- You can configure the following NSX features on the discovered clusters using NSX UI/API.
  - Spoof Guard
  - Switch Security
  - Mac Management
  - IP Discovery
  - TraceFlow
  - ENS
  - Advanced Security features, such as DFW and IDS/IPS

    Note:
    Ensure that ESX host version is v7.0.3 or later if you want to use the same VDS to prepare clusters for both the use cases - Security-only or Networking and Security. To avoid issues post cluster nodes preparation, upgrade ESX host to v7.0.3 or later before preparing clusters using the same VDS.

Prepare ESX cluster Hosts for Networking and Security

Use the Quick Start wizard to prepare ESX clusters for networking and security using NSX recommended host configurations.

- Register vCenter as a compute manager in NSX.
- Configure ESX cluster hosts member of VDS in vCenter with correct uplinks.
- (Optional) If you are using VLAN for VTEP pool, configue IP Pool for Host VTEP assignment and Uplink Profile using the TEP VLAN.

The Quick Start wizard helps you finish installation with minimum user input, thus, simplifying the installation process. By default, VLAN networking is the default selection in the wizard.

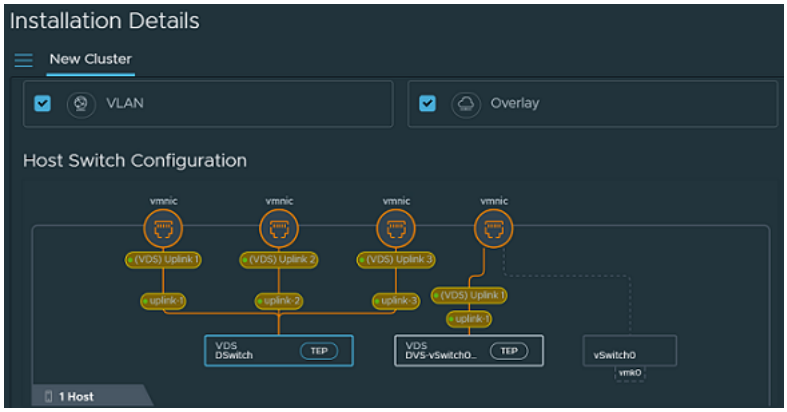
Based on the type of host, the quick start wizard considers the following default configurations:

- ESX hosts running 7.0 and later are prepared on the VDS switch. Configure the desired number of uplinks on the VDS switch in vCenter and set the MTU to 1700.

Each host switch is assigned an auto-created transport zone, uplink profile, and transport node profile.

After preparing ESX clusters for networking and security, you can activate NSX on them. For more information about activating NSX, see Activate NSX on Distributed Switch.

1. From a browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.
2. Navigate to .
3. On the **Prepare Clusters for Networking and Security** card, click **Get Started**.
4. Select the clusters you want to prepare for NSX networking.
5. Click **Install NSX** and then select **Networking and Security**.
6. Depending on your requirement, you can prepare the same cluster for both VLAN and Overlay networking or for one type of networking. With Overlay networking, each host switch is added with a TEP IP address, which is required for overlay networking.



7. View the NSX recommended Host Switch configuration.

   However, you can customize the settings for the cluster, even though it is an optional step.

   Note:
   A dotted line originating from a switch to a physical NIC indicates that it is an existing configuration on the host switch, which will be replaced by a firm line going to the same physical NIC.

8. Even though NSX provides recommendations, you can still customize the configuration. To customize the host switch, select the switch and change the recommended configuration.
   a) **IP Assignment**: Is applicable if overlay is selected for the host switch. Choose IP assignment type to be DHCP or a pre-created IP Pool for the overlay VTEP Pool.
   b) **VDS**: Select the VDS switch as the host switch.
   c) **Transport Zone**: Select a different transport zone that you want the host to be associated with.
   d) **Uplink Profile**: If needed, select a different uplink profile in place of the recommended uplink profile.

      Note:
      If you configure two VDS switches with the same configuration, the wizard recommends the same uplink profile for both the switches.

   e) **Uplink to Physical NIC mapping**: On a VDS switch, all uplinks configured on the VDS switch are mapped to the uplinks in NSX.

      A change to host switch type or uplink to vmnic mapping is reflected in the Host Switch Configuration network representation.

9. Click **Install**.

   View the progress of installation on the **Prepare Clusters for Networking and Security** card. If installation on any of the host fails, retry installation by resolving the error.

10. To view successfully prepared hosts, go to .

The transport nodes are ready for VLAN and Overlay networking.

Activate NSX on Distributed Virtual Port Groups (DVPGs)

NSX allows distributed security across all ports within the DVPGs found in a host cluster. As the host switch is of the type VDS, you can enable DFW capabilities on workload VMs.

The following are the requirements for installing Distributed Security for VDS:

- The vSphere cluster must have at least one VDS configured and ESX cluster hosts must be members of a VDS with uplinks configured.
- A compute manager must be registered in NSX. See Add a Compute Manager.
- Before you deploy and configure Distributed Security on hosts, ensure that NSX is not deployed on such hosts.

DFW supports both DVPGs created in vSphere and segments created in NSX within the same cluster.

Distributed Security provides security-related functionality to your VDS such as:

- Distributed Firewall (DFW)
- Distributed IDS/IPS
- Identity Firewall
- L7 App ID
- Fully Qualified Domain Name (FQDN) Filtering
- Security Intelligence
- NSX Malware Prevention
- NSX Guest Introspection

After you activate security on DVPGs, DVPG ports and DVPG segments are discovered by NSX.

1. From your browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address>.
2. Navigate to **System** > **Quick Start** > **Host Cluster Preparation**.
3. On the **Prepare Clusters for Networking and Security** card, click **Get Started**.
4. Select the clusters where you want to activate distributed security.
5. Click **Install NSX** and under **Networking**, choose either **VLAN** or **Overlay**.

   Note:

   Choosing VLAN enables only NSX VLAN Networking.
6. In the dialog box, click **Install**.
7. Navigate to the **System** > **Fabric** > **Hosts** page to activate security on DVPG within a cluster.
8. On the **Hosts** > **Cluster** page, select the cluster on which you want to activate security on the DVPGs within the cluster.

   Note:

   Ensure you select all clusters where the VDS spans. If you leave out a cluster that VDS spans, security will not be activated on DVPGs of that cluster.
9. On the **Hosts** page, click **Actions** > **Activate NSX on DVPGs**.

   Note:

   Before you activate NSX on DVPGs, protect all the management appliances by adding them to the distributed firewall exclusion list. Sometimes, if "Actions > Activate NSX on DVPGs" is performed immediately after "Actions > Deactivate NSX on DVPGs" hosts may be in a partial success state. To work around this issue, create a dummy DVPG under the Hosts located DVS in the VC. Hosts will retrigger the sync process, and be fully successful.
10. On the **Activate NSX on DVPGs** card, click **Yes** to activate the DVPGs.

    NSX discovers the port groups associated to the VDS switch. To learn more about how to find the discovered ports in NSX, see Distributed Port Groups.

11. To deactivate security on DVPGs within a cluster:
    a) On the **Hosts** > **Cluster** page, select the cluster on which you want to deactivate security on the DVPGs within the cluster.
    b) On the **Hosts** page, click **Actions** > **Deactivate NSX on DVPGs**.

After successfully activating security on a cluster, on the Hosts page, the **NSX on DVPGs** column status is updated to `Yes`. Distributed Security is installed and you can begin using security capabilities such as creating DFW policies and rules for the VDS.

The system creates distributed port group segments under the default security transport zones described in Create Transport Zones.

Managing Transport Nodes

After preparing hosts as transport nodes, you can view status of hosts, switch visualization, and other configuration settings related to transport nodes. You can use it to debug issues if the host transport node is in a degraded or failed state.

Switch Visualization

You get a granular view of a vSphere Distributed Switch (VDS) at an individual host level.

1. From a browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.
2. Select **System** > **Fabric** > **Hosts**.
3. Select **Clusters** or **Unmanaged** tab.
4. For any host, click **View Details**.
5. In the **Host Details** window, select the **Overview** tab and see visualization of VDS switch.

   The VDS switch visualization shows the uplinks and the NICs the switch is connected to.

NSX Maintenance Mode

If you want to avoid vMotion of VMs to a transport node that is not functional, place that transport node in NSX Maintenance Mode.

To put a transport node in NSX Maintenance Mode, select the node, click **Actions** > **NSX Maintenance Mode**.

When you put a host in NSX Maintenance Mode, the transport node cannot participate in networking. Therefore, you must vMotion all VMs to another host before initiating NSX Maintenance Mode. Also, VMs running on other transport nodes that have VDS as the host switch cannot be vMotioned to this transport node. In addition, logical network cannot be configured on ESX hosts as the status of host switch on the ESX host would be shown as `Down`.

Scenarios to put the transport node in NSX Maintenance Mode:

- A transport node is not functional.
- If a host has hardware or software issues that are unrelated to NSX, but you want to retain the node and its configurations in NSX, place the host in NSX Maintenance Mode.
- A transport node is automatically put in NSX Maintenance Mode when an upgrade on that transport node fails.

Any transport node put in the NSX Maintenance Mode is not upgraded.

View Bidirectional Forwarding Detection Status

To monitor the health of NSX overlay fabric, view Bidirectional Forwarding Detection (BFD) status between transport nodes.

Each transport node creates a full mesh of BFD sessions to all the Tunnel Endpoints (TEPs) that are active on one or more logical spans. NSX displays the BFD status among other details related to the transport node.

Both Host Transport nodes (standalone and hosts registered to a vCenter) and Edge nodes display the tunnel status. BFD packets support both GENEVE and STT encapsulation. GENEVE is the default encapsulation.

Note:
For compute transport nodes such as an ESX host, BFD tunnels are formed if ESX has an active port attached to an NSX segment. It means a powered-on VM with a vNIC is connected to an NSX segment.

1. From a browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.
2. Select **System** > **Fabric** > **Hosts**.
3. Select a host and click **View Details**.
4. On the **Host Details** window, select **Monitor** and expand **Transport Node Status**.
5. Select **Tunnels**.
6. On the Tunnel Endpoint, filter tunnels based on the encapsulation protocol they are using. Choose between **GENEVE** or **VXLAN**.
7. In the **Filter by BFD Status** drop-down menu, select ALL to view all BFD statuses or a specific status.

   The Monitor page displays the status of tunnel, BFD diagnostic code, remote node UUID, encapsulation on BFD packets, and tunnel name.

   The tunnel BFD diagnostic code indicates the reason for the change in the session state.

Code	Description	Action
0	No Diagnostic	Code 0: Default diagnostic code seen when the tunnel is Up.
1	Control Detection Time Expired	Code 1: BFD timer has expired. It is seen when the local interface has not received a BFD packet from the remote system resulting in BFD timer expiring. Check if BFD timer is too aggressive with respect to system load and path traffic load. Default BFD timer is 1-sec, up to 3 misses. Change in BFD parameters are not disruptive.
2	Echo Function Failed	Code 2: BFD echo packet loop verification failed. Verify health of transport-node.
3	Neighbor Signalled Session Down	Code 3: Peer node voluntarily brings the session down. Check if peer transport node is in maintenance mode or not healthy. un ICMP ping to verify connectivity to TEPs.
4	Forwarding Plane Reset	Code 4: When Forwarding-Plane is reset and peer does not reply on BFD, so session is marked down

Code	Description	Action

5	Path Down	Code 5: The path to the remote node is down. Validate IP connectivity between the TEPs using ICMP ping. Note that the TEP interfaces on ESX are instantiated on the vxlan netstack and on the tunnel VRF on edges. Be sure to initiate the ping from within the vxlan netstack on ESX or from the tunnel VRF on edges. If you have more than one TEP, be sure to specify the source IP address or interface used for the ping.
		On ESX hosts: `ping ++netstack=vxlan -I <vmk adapter> <remote address>`
		On Edge nodes: `get logical-routers`
		`vrf 0`
		`ping <dst-vtep> source <src-vtep> repeat 3`

6	Concatenated Path Down	`Concat Path Down` represents that the Edge transport node has lost all the BGP/OSPF (northbound) sessions to the northbound router. This error is safe to ignore when not using the Edge cluster for Tier-0 BGP routing purposes, while just the Tier-1 services are used.
7	Administratively Down	Code 7: Session is marked down by the administrator. Verify if local Transport Node is in maintenance mode. Admin CLI to run on TN: `get maintenance-mode`
8	Reverse Concatenated Path Down	Code 8: The path from the remote node to local is down. Test IP connectivity from remote node to local node.

The fabric health BFD sessions are created between the TEP addresses. The tunnel status is a true reflection of the IP connectivity and capability of the network to forward Geneve packets therefore status for all BFD sessions should be `Up`. If the BFD status is down, use the diagnostic code to troubleshoot the issue.

To know the status of BFD sessions on fabric nodes, run the following CLI commands:

- For ESX, run `nsxdp-cli bfd sessions list`.
- For Edge TN, run `get bfd-sessions`.

To verify the fabric health of transport node, call the following API:

`GET /policy/api/v1/infra/sites/<site-id>/enforcement-points/<enforcement-point-id>/transport-node-status-report`

where, <site-id> and <enforcement-point-id> can use the value `default`.

IPv6 Unsupported Features

NSX does not support a few features if you select IPv6 as the addressing scheme while configuring any of these objects or components:

- Host transport node
- NSX Edge transport node
- Transport node profile
- QuickStart card to configure NSX

The unsupported features are:

- EVPN

- Multicast Routing
- NSX Federation

Installing NSX Edge

Install NSX Edge on ESX using the NSX UI, the vSphere web client, or the command-line OVF tool.

NSX Edge VM System Requirements

Before you install an NSX Edge VM, make sure that your environment meets the supported requirements.

Note:
The following conditions apply to the hosts for the NSX Edge nodes:

- NSX Edge nodes are supported only on ESX-based hosts with Intel-based and AMD-based chipsets. Otherwise, vSphere EVC mode may prevent NSX Edge nodes from starting, showing an error message in the console.
- If vSphere EVC mode is enabled for the host for the NSX Edge VM, the CPU must be Haswell or later generation.
- Only VMXNET3 vNIC is supported for the NSX Edge VM.

NSX Edge VM Resource Requirements

Appliance Size	Memory	vCPU	Disk Space	VM Hardware Version	Notes
NSX Edge Small	4 GB	2	200 GB	11 or later (vSphere 7.0 or later)	Ideal for small deployments or environments with low traffic and minimal processing requirements. **Note:** L7 rules for firewall, load balancing and so on are not realized on a Tier-1 gateway if you deploy a small sized NSX Edge VM.
NSX Edge Medium	8 GB	4	200 GB	11 or later (vSphere 7.0 or later)	Ideal for medium-sized environments for L2 through L4 features and when the total throughput requirement is between 2 ~ 10 Gbps: • Routing • NAT • L4 Load Balancer • L4 Gateway Firewall

Appliance Size	Memory	vCPU	Disk Space	VM Hardware Version	Notes
NSX Edge Large	32 GB	8	200 GB	11 or later (vSphere 7.0 or later)	Ideal for larger environments with high traffic. Suggested for these features: • Routing • NAT • L4 Gateway Firewall • L4 Load Balancer • L7 Load Balancer • TLS Inspection See Scaling Load Balancer Resources. For more information about what the different load balance sizes and NSX Edge form factors can support, see VMware Configuration Maximums.
NSX Edge Extra Large	64 GB	16	200 GB	11 or later (vSphere 7.0 or later)	Ideal for environments with the highest throughput requirements. Suggested for these features: • Routing • NAT • L4 Gateway Firewall • L4 Load Balancer • L7 Load Balancer • VPN • TLS Inspection • URL Filtering • Gateway IDS/IPS • Gateway Malware Prevention See Scaling Load Balancer Resources. For more information about what the different load balance sizes and NSX Edge form factors can support, see VMware Configuration Maximums.

NSX Edge VM CPU Requirements

For the DPDK support, the underlaying platform needs to meet the following requirements:

- CPU must have AESNI capability.
- CPU must have 1 GB Huge Page support.

Hardware	Type
CPU	• Intel Xeon Platinum 9xxx, 8xxx • Intel Xeon Gold 6xxx, 5xxx • Intel Xeon Silver 4xxx • Intel Xeon Bronze 3xxx • Intel Xeon E56xx, L56xx, X56xx (Westmere-EP) • Intel Xeon E7-xxxx (Westmere-EX and later CPU generation) • Intel Xeon E5-xxxx (Sandy Bridge and later CPU generation)

Hardware	Type
	• AMD EPYC Series processors

NSX Edge Bare Metal Requirements

Before you configure the NSX Edge bare metal, make sure that your environment meets the supported requirements.

NSX Edge Bare Metal Memory, CPU, and Disk Requirements

Minimum Requirements

Memory	CPU Cores	Disk Space
64 GB	8	200 GB

NSX Edge Bare Metal DPDK CPU Requirements

For DPDK support, the underlying platform needs to meet the following requirements:

- CPU must have AES-NI capability.
- CPU must have 1 GB Huge Page support.
- NSX Edge Bare Metal supports up to 80 cores for the entire system. This means that on a server with a single socket, its CPU can have up to 80 cores. On a server with 2 sockets, each socket cannot have more than 40 cores.

Hardware	Type
CPU	• Intel Xeon Platinum 9xxx, 8xxx • Intel Xeon Gold 6xxx, 5xxx • Intel Xeon Silver 4xxx • Intel Xeon Bronze 3xxx • Intel Xeon E56xx, L56xx, X56xx (Westmere-EP) • Intel Xeon E7-xxxx (Westmere-EX and later CPU generation) • Intel Xeon E5-xxxx (Sandy Bridge and later CPU generation)
	• AMD EPYC Series processors

NSX Edge Bare Metal Hardware Requirements

Verify that the bare metal NSX Edge hardware is listed in this URL. If the hardware is not listed, the storage, video adapter, or motherboard components might not work on the NSX Edge appliance. Bare Metal supports both UEFI and legacy BIOS modes.

NSX Edge Bare Metal NIC Requirements

NIC Type	Description	Vendor ID	PCI Device ID	Firmware Version
NVIDIA ConnectX-4	PCI_DEVICE_ID_MELLANOX_CONNECTX4	15b3	0x1013	12.21.1000 and later versions
NVIDIA ConnectX-4 Lx	PCI_DEVICE_ID_MELLANOX_CONNECTX4LX	15b3	0x1015	14.21.1000 and later versions
NVIDIA ConnectX-5	PCI_DEVICE_ID_MELLANOX_CONNECTX5	15b3	0x1017	16.21.1000 and later versions
NVIDIA ConnectX-5 EX	PCI_DEVICE_ID_MELLANOX_CONNECTX5EX	15b3	0x1019	16.21.1000 and later versions

NIC Type	Description	Vendor ID	PCI Device ID	Firmware Version
NVIDIA ConnectX-6	PCI_DEVICE_ID_MELLANOX_CONNECTX6	15b3	0x101B	20.27.0090 and later versions
NVIDIA ConnectX-6 Dx	PCI_DEVICE_ID_MELLANOX_CONNECTX6DX	15b3	0x101D	22.27.0090 and later versions
NVIDIA ConnectX-6 Lx	PCI_DEVICE_ID_MELLANOX_CONNECTX6LX	15b3	0x101F	26.27.0090 and later versions
Intel X520/Intel 82599	IXGBE_DEV_ID_82599_KX4	8086	0x10F7	n/a
	IXGBE_DEV_ID_82599_KX4_MEZZ	8086	0x1514	n/a
	IXGBE_DEV_ID_82599_KR	8086	0x1517	n/a
	IXGBE_DEV_ID_82599_COMBO_BACKPLANE	8086	0x10F8	n/a
	IXGBE_DEV_ID_82599_CX4	8086	0x10F9	n/a
	IXGBE_DEV_ID_82599_SFP	8086	0x10FB	n/a
	IXGBE_SUBDEV_ID_82599_SFP	8086	0x11A9	n/a
	IXGBE_SUBDEV_ID_82599_RNDC	8086	0x1F72	n/a
	IXGBE_SUBDEV_ID_82599_560FLR	8086	0x17D0	n/a
	IXGBE_SUBDEV_ID_82599_ECNA_DP	8086	0x0470	n/a
	IXGBE_DEV_ID_82599_SFP_EM	8086	0x1507	n/a
	IXGBE_DEV_ID_82599_SFP_SF2	8086	0x154D	n/a
	IXGBE_DEV_ID_82599_SFP_SF_QP	8086	0x154A	n/a
	IXGBE_DEV_ID_82599_QSFP_SF_QP	8086	0x1558	n/a
	IXGBE_DEV_ID_82599EN_SFP	8086	0x1557	n/a
	IXGBE_DEV_ID_82599_XAUI_LOM	8086	0x10FC	n/a
	IXGBE_DEV_ID_82599_T3_LOM	8086	0x151C	n/a
Intel X540	IXGBE_DEV_ID_X540T	8086	0x1528	n/a
	IXGBE_DEV_ID_X540T1	8086	0x1560	n/a
Intel X550	IXGBE_DEV_ID_X550T	8086	0x1563	n/a
	IXGBE_DEV_ID_X550T1	8086	0x15D1	n/a
	IXGBE_DEV_ID_X550EM_A_10G_T	8086	0x15C8	n/a
	IXGBE_DEV_ID_X550EM_A_QSFP	8086	0x15CA	n/a
	IXGBE_DEV_ID_X550EM_A_QSFP_N	8086	0x15CC	n/a

NIC Type	Description	Vendor ID	PCI Device ID	Firmware Version
	IXGBE_DEV_ID_X550EM_X_10G_T	8086	0x15AD	n/a
	IXGBE_DEV_ID_X550EM_X_XFI	8086	0x15B0	n/a
Intel X710	I40E_DEV_ID_SFP_X710	8086	0x1572	9.52 and later versions
	I40E_DEV_ID_KX_C	8086	0x1581	9.52 and later versions
	I40E_DEV_ID_10G_BASE_T	8086	0x1586	9.52 and later versions
	I40E_DEV_ID_10G_BASE_T4	8086	0x1589	9.52 and later versions
	I40E_DEV_ID_10G_SFP	8086	0x104E	9.52 and later versions
	I40E_DEV_ID_10G_B	8086	0x104F	9.52 and later versions
Intel XL710	I40E_DEV_ID_KX_B	8086	0x1580	9.52 and later versions
	I40E_DEV_ID_QSFP_A	8086	0x1583	9.52 and later versions
	I40E_DEV_ID_QSFP_B	8086	0x1584	9.52 and later versions
	I40E_DEV_ID_QSFP_C	8086	0x1585	9.52 and later versions
	I40E_DEV_ID_20G_KR2	8086	0x1587	9.52 and later versions
	I40E_DEV_ID_20G_KR2_A	8086	0x1588	9.52 and later versions
	I40E_DEV_ID_10G_BASE_T_BC	8086	0x15FF	9.52 and later versions
Intel XXV710	I40E_DEV_ID_25G_B	8086	0x158A	9.52 and later versions
	I40E_DEV_ID_25G_SFP28	8086	0x158B	9.52 and later versions
Intel E810-C (100G)	ICE_DEV_ID_E810C_QSFP	8086	0x1592	4.6 and later versions
Intel E810-XXV (25G)	ICE_DEV_ID_E810_XXV_SFP	8086	0x159B	4.6 and later versions
	ICE_DEV_ID_E810_XXV_BACKPLANE	8086	0x1599	4.6 and later versions
	ICE_DEV_ID_E810_XXV_QSFP	8086	0x159A	4.6 and later versions
Intel E810-C	ICE_DEV_ID_E810C_BACKPLANE	8086	0x1591	4.6 and later versions
	ICE_DEV_ID_E810C_SFP	8086	0x1593	4.6 and later versions
Intel E822-C	ICE_DEV_ID_C822N_BACKPLANE	8086	0x1890	4.6 and later versions
	ICE_DEV_ID_C822N_QSFP	8086	0x1891	4.6 and later versions\
	ICE_DEV_ID_C822N_SFP	8086	0x1892	4.6 and later versions
Cisco VIC 1300 series **Important:** See "Additional NIC Requirements."	Cisco UCS Virtual Interface Card 1300	1137	0x0043	n/a

NIC Type	Description	Vendor ID	PCI Device ID	Firmware Version
Cisco VIC 1400 series **Important:** See "Additional NIC Requirements."	Cisco UCS Virtual Interface Card 1400	1137	0x0043	n/a
Broadcom NetXtreme E-Series BCM57508 200G **Important:** See "Additional NIC Requirements."	BROADCOM_DEV_ID_57508	0x14E4	0x1750	226.0.131 and later versions
Broadcom NetXtreme E-Series BCM57504 100G **Important:** See "Additional NIC Requirements."	BROADCOM_DEV_ID_57504	0x14E4	0x1751	226.0.131 and later versions
Broadcom NetXtreme E-Series BCM57414 50G **Important:** See "Additional NIC Requirements."	BROADCOM_DEV_ID_57414	0x14E4	0x16D7	226.0.131 and later versions
Broadcom NetXtreme E-Series BCM57412 25G **Important:** See "Additional NIC Requirements."	BROADCOM_DEV_ID_57412	0x14E4	0x16D6	226.0.131 and later versions

Note:
For all the supported NICs listed above, verify that the media adapters and cables you use follow the vendor's supported media types. Any media adapter or cables not supported by the vendor can result in unpredictable behavior, including the inability to boot up due to an unrecognized media adapter. See the NIC vendor documentation for information about supported media adapters and cables.

Additional NIC Requirements

This section describes additional configurations that are required to support certain NICs on NSX Edge bare metal.

Cisco VICs

To successfully claim Cisco VICs for the NSX Edge datapath, configure multiple RX and TX queues from the Cisco UCS Manager. The number of queues configured must be sufficient for datapath to have one queue per core. For configuration details, refer to the Cisco documentation.

Broadcom NICs

To support strong performance on Broadcom NICs, set the **I/O Snoop Holdoff Response** to 2000 cycles. For example, the following steps describe how to configure this setting on a Dell PowerEdge server.

1. Reboot the server. During the boot process, press F2 to enter System Setup.

2. Select **Device Settings** to display all hardware adapters, and select the adapter for the Broadcom NIC.
3. Navigate to **Device Level Configuration** and locate the **I/O Snoop Holdoff Response** or **Snoop Holdoff** setting.
4. Change the setting to 2000 (2K) cycles.
5. Exit the System Setup and save your configuration changes.
6. Reboot the server for the configuration changes to take effect.

NSX Edge Installation Requirements

The NSX Edge provides routing services and connectivity to networks that are external to the NSX deployment. NSX Edge nodes provide a pool of capacity for running centralized services and is required if you want to deploy a tier-0 router or a tier-1 router with stateful services such as network address translation (NAT), VPN, and so on.

Note:

There can be only one tier-0 router per NSX Edge node. However, multiple tier-1 logical routers can be hosted on one NSX Edge node. NSX Edge VMs of different sizes can be combined in the same cluster; however, it is not recommended.

Table 1032: NSX Edge Deployment, Platforms, and Installation Requirements

Requirements	Description
Supported deployment methods	See NSX Edge Installation Methods.
Supported platforms	NSX Edge is supported as VM (only on ESX) or as physical server (bare metal). Both use the data plane development kit (DPDK) for faster packet processing and high performance.
PXE installation	The Password string must be encrypted with sha-512 algorithm for the root and admin user password.
NSX appliance password	• At least 12 characters • At least one lower-case letter • At least one upper-case letter • At least one digit • At least one special character • At least five different characters • No dictionary words • No palindromes • More than four monotonic character sequence is not allowed
Hostname	When installing NSX Edge, specify a hostname that does not contain invalid characters such as an underscore. If the hostname contains any invalid character, after deployment the hostname will be set to `localhost`. For more information about hostname restrictions, see https://datatracker.ietf.org/doc/html/rfc952 and https://datatracker.ietf.org/doc/html/rfc1123.
VMware Tools	The NSX Edge VM running on ESX has VMTools installed. Do not remove or upgrade VMTools.
System	Verify that the system requirements are met, as described in the following sections. • NSX Edge VM System Requirements • NSX Edge Bare Metal Requirements
Ports	Verify that the required ports are open. See https://ports.broadcom.com/.

Requirements	Description
IP Addresses	Plan your NSX Edge IPv4 and IPv6 IP addressing scheme. However, a NSX Edge configured for IPv4 and IPv6 stack is not supported on an ESX node that is configured only for IPv6 addressing scheme. Also, NSX Edge tunnel endpoints on a dual IPv4 and IPv6 scheme cannot communicate with an ESX node using only IPv6 addressing scheme.
OVF Template	• Verify that you have adequate privileges to deploy an OVF template on the ESX host. • Verify that hostnames do not include underscores. Otherwise, the hostname is set to `localhost`. • A management tool that can deploy OVF templates, such as vCenter or the vSphere Client. The OVF deployment tool must support configuration options to allow for a manual configuration. • The Client Integration Plug-in must be installed.
NTP Server	The same NTP server must be configured on all NSX Edge VMs or Bare Metal Edges in an Edge cluster.

Intel-based Chipsets

NSX Edge nodes are supported on ESX-based hosts with Intel chipsets. If an unsupported chipset type is used, vSphere EVC mode may prevent Edge nodes from starting, showing an error message in the console.

AMD EPYC

NSX Edge nodes are also supported on AMD-based chipsets. NSX Edge nodes can now be deployed on AMD EPYC series chipsets.

NSX Edge Support of vSphere Business Continuity Features

vMotion, DRS, and vSphere HA are supported for NSX Edge nodes. Use vMotion and DRS with caution to avoid traffic disruption, especially, when the BFD timers are set to less than one second.

NSX Edge Installation Scenarios

Important:
When you install NSX Edge from an OVA or OVF file, either from vSphere Web Client or the command line, OVA/OVF property values such as user names, passwords, or IP addresses are not validated before the VM is powered on.

- If you specify a user name for any of the local users, the name must be unique. If you specify the same name, it is ignored and the default names (for example, `admin` or `audit`) are used.
- If the password for the `root` or `admin` user does not meet the complexity requirements, you must log in to NSX Edge through SSH or at the console as `root` with password `vmware` and `admin` with password `default`. You are prompted to change the password.
- If the password for other local users (for example, `audit`) does not meet the complexity requirements, the user account is disabled. To enable the account, log in to NSX Edge through SSH or at the console as the `admin` user and run the command `set user local_user_name` to set the local user's password (the current password is an empty string). You can also reset passwords in the UI using System > User Management > Local Users.

CAUTION:
Changes made to the NSX while logged in with the `root` user credentials might cause system failure and potentially impact your network. You can only make changes using the `root` user credentials with the guidance of VMware Support team.

Note:
The core services on the appliance do not start until a password with sufficient complexity has been set.

After you deploy NSX Edge from an OVA file, you cannot change the VM's IP settings by powering off the VM and modifying the OVA settings from vCenter.

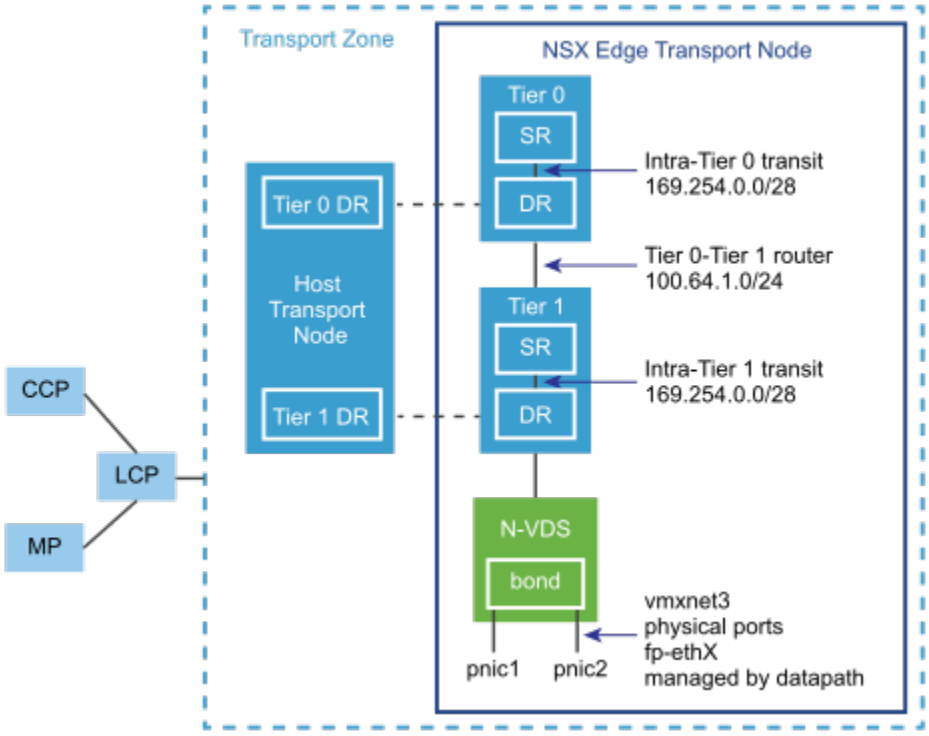
NSX Edge Networking Setup

For all NSX Edge installation methods, make sure that the host networking is prepared before you install NSX Edge.

High-Level View of NSX Edge Within a Transport Zone

The high-level view of NSX shows two transport nodes in a transport zone. One transport node is a host. The other is an NSX Edge.

Figure 405: High-Level Overview of NSX Edge



When you first deploy an NSX Edge, you can think of it as an empty container. The NSX Edge does not do anything until you create logical routers. The NSX Edge provides the compute backing for tier-0 and tier-1 logical routers. Each logical router contains a services router (SR) and a distributed router (DR). When we say that a router is distributed, we mean that it is replicated on all transport nodes that belong to the same transport zone. In the figure, the host transport node contains the same DRs contained on the tier-0 and tier-1 routers. A services router is required if the logical router is going to be configured to perform services, such as NAT. All tier-0 logical routers have a services router. A tier-1 router can have a services router if needed based on your design considerations.

By default, the links between the SR and the DR use the 169.254.0.0/28 subnet. These intra-router transit links are created automatically when you deploy a tier-0 or tier-1 logical router. You do not need to configure or modify the link configuration unless the 169.254.0.0/28 subnet is already in use in your deployment. On a tier-1 logical router, the SR is present only if you select an NSX Edge cluster when creating the tier-1 logical router.

The default address space assigned for the tier-0-to-tier-1 connections is 100.64.0.0/10. Each tier-0-to-tier-1 peer connection is provided a /31 subnet within the 100.64.0.0/10 address space. This link is created automatically when you create a tier-1 router and connect it to a tier-0 router. You do not need to configure or modify the interfaces on this link unless the 100.64.0.0/10 subnet is already in use in your deployment.

Each NSX deployment has a management plane cluster (MP) and a control plane cluster (CCP). The MP and the CCP push configurations to each transport zone's local control plane (LCP). When a host or NSX Edge joins the management plane, the management plane agent (MPA) establishes connectivity with the host or NSX Edge, and the host or NSX Edge

becomes an NSX fabric node. When the fabric node is then added as a transport node, LCP connectivity is established with the host or NSX Edge.

Lastly, the figure shows an example of two physical NICs (pNIC1 and pNIC2) that are bonded to provide high availability. The datapath manages the physical NICs. They can serve as either VLAN uplinks to an external network or as tunnel endpoint links to internal NSX-managed VM networks.

It is a best practice to allocate at least two physical links to each NSX Edge that is deployed as a VM. Optionally, you can overlap the port groups on the same pNIC using different VLAN IDs. The first network link found is used for management. For example, on an NSX Edge VM, the first link found might be vnic1. On a bare-metal installation, the first link found might be eth0 or em0. The remaining links are used for the uplinks and tunnels. For example, one might be for a tunnel endpoint used by NSX-managed VMs. The other might be used for an NSX Edge-to-external TOR uplink.

You can view the physical link information of the NSX Edge, by logging in to the CLI as an administrator and running the `get interfaces` and `get physical-ports` commands. In the API, you can use the `GET /api/v1/transport-nodes/{transport-node-id}/node/network/interfaces` API call. Physical links are discussed in more detail in the next section.

Whether you install NSX Edge as a VM appliance or on bare metal, you have multiple options for the network configuration, depending on your deployment.

Figure 406: NSX Edge Transport Node in VM Form Factor within ESX

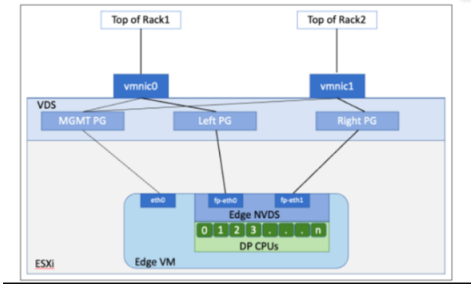
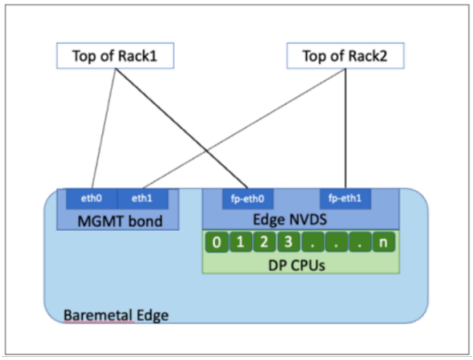


Figure 407: Bare Metal Edge Transport Node



Transport Zones and N-VDS

To understand NSX Edge networking, you must know something about transport zones and N-VDS. Transport zones control the reach of Layer 2 networks in NSX. N-VDS is a software switch that gets created on a transport node. The purpose of N-VDS is to bind logical router uplinks and downlinks to physical NICs. For each transport zone that an NSX Edge belongs to, a single N-VDS gets installed on the NSX Edge.

There are two types of transport zones:

- Overlay for internal NSX tunneling between transport nodes.
- VLAN for uplinks external to NSX.

An NSX Edge can belong to zero VLAN transport zones or many. For zero VLAN transport zones, the NSX Edge can still have uplinks because the NSX Edge uplinks can use the same N-VDS installed for the overlay transport zone. You might do this if you want each NSX Edge to have only one N-VDS. Another design option is for the NSX Edge to belong to multiple VLAN transport zones, one for each uplink.

The most common design choice is three transport zones: One overlay and two VLAN transport zones for redundant uplinks.

To use the same VLAN ID for a transport network for overlay traffic and other for VLAN traffic, such as a VLAN uplink, configure the ID on two different N-VDS, one for VLAN and the other for overlay.

Virtual-Appliance/VM NSX Edge Networking

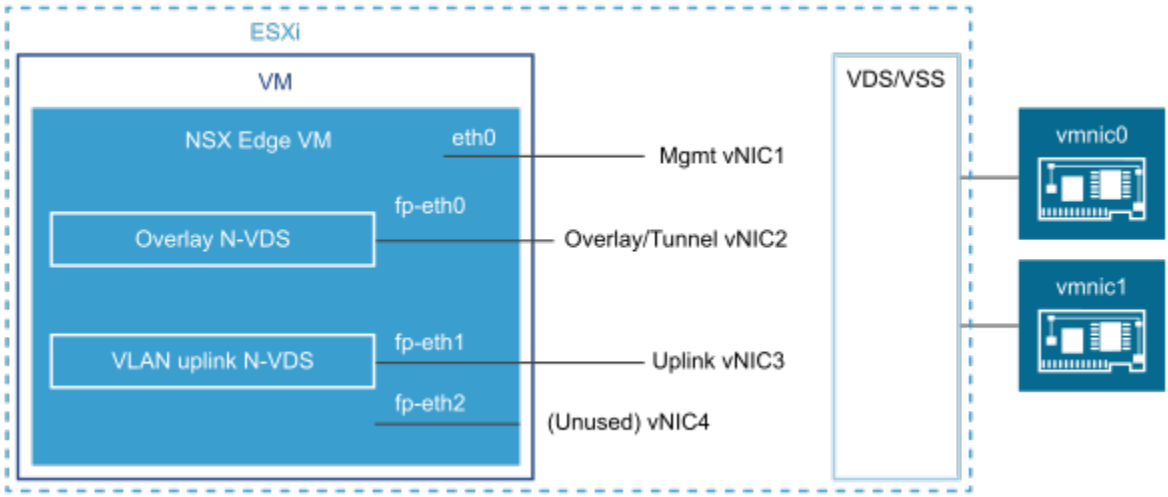
When you install NSX Edge as a virtual appliance or VM, internal interfaces are created, called fp-ethX, where X is 0, 1, 2, and 3. These interfaces are allocated for uplinks to a top-of-rack (ToR) switches and for NSX overlay tunneling.

When you create the NSX Edge transport node, you can select fp-ethX interfaces to associate with the uplinks and the overlay tunnel. You can decide how to use the fp-ethX interfaces.

On the vSphere distributed switch or vSphere Standard switch, you must allocate at least two vmnics to the NSX Edge: One for NSX Edge management and one for uplinks and tunnels.

In the following sample physical topology, fp-eth0 is used for the NSX overlay tunnel. fp-eth1 is used for the VLAN uplink. fp-eth2 and fp-eth3 are not used. vNIC1 is assigned to the management network.

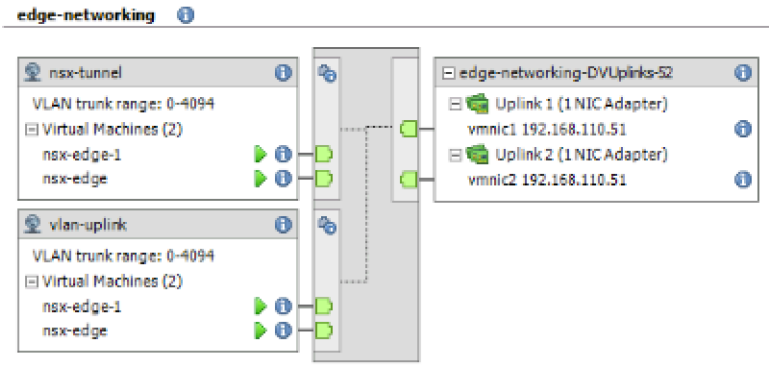
Figure 408: One Suggested Link Setup for NSX Edge VM Networking



The NSX Edge shown in this example belongs to two transport zones (one overlay and one VLAN) and therefore has two N-VDS, one for tunnel and one for uplink traffic.

This screenshot shows the virtual machine port groups, nsx-tunnel, and vlan-uplink.

Figure 409: Edge Networking in vSphere



During deployment, you must specify the network names that match the names configured on your VM port groups. For example, to match the VM port groups in the example, your network ovftool settings can be as follows if you were using the ovftool to deploy NSX Edge:

```
--net:"Network 0-Mgmt" --net:"Network 1-nsx-tunnel" --net:"Network 2=vlan-uplink"
```

The example shown here uses the VM port group names Mgmt, nsx-tunnel, and vlan-uplink. You can use any names for your VM port groups.

The tunnel and uplink VM port groups configured for the NSX Edge do not need to be associated with VMkernel ports or given IP addresses. This is because they are used at Layer 2 only. If your deployment uses DHCP to provide an address to the management interface, make sure that only one NIC is assigned to the management network.

Notice that the VLAN and tunnel port groups are configured as trunk ports. This is required. For example, on a standard vSwitch, you configure trunk ports as follows: . **Host** > **Configuration** > **Networking** > **Add Networking** > **Virtual Machine** > **VLAN ID All (4095)**.

If you are using an appliance-based or VM NSX Edge, you can use standard vSwitches or vSphere distributed switches.

NSX Edge VM can be installed on an NSX prepared host and configured as a transport node. There are two types of deployment:

- NSX Edge VM can be deployed using VSS/VDS port groups where VSS/VDS consume separate pNIC(s) on the host. Host transport node consumes separate pNIC(s) for N-VDS installed on the host. N-VDS of the host transport node co-exists with a VSS or VDS, both consuming separate pNICs. Host TEP (Tunnel End Point) and NSX Edge TEP can be in the same or different subnets.
- NSX Edge VM can be deployed using VLAN-backed logical switches on the N-VDS of the host transport node. Host TEP and NSX Edge TEP can be in the same VLAN or subnet.

Optionally, you can install multiple NSX Edge appliances/VMs on a single host, and the same management, VLAN, and tunnel endpoint port groups can be used by all installed NSX Edges.

With the underlying physical links up and the VM port groups configured, you can install the NSX Edge.

> **Note:**
> If NSX Edge VM uplinks (management, fast-path interfaces) will be connecting to VLAN based portgroups then underlying ESX host does not need to be an NSX transport node. But if NSX Edge VM uplinks will be connecting to NSX VLAN segments (logical switches), then you must configure ESX hosts hosting the edges as transport nodes.

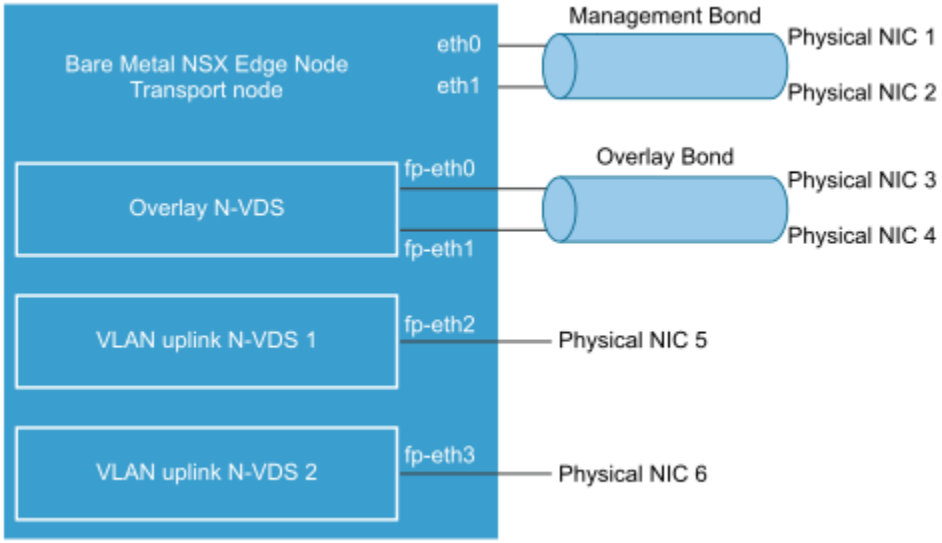
Bare-Metal NSX Edge Networking

The bare-metal NSX Edge contains internal interfaces called fp-ethX, where X is up to 16 interfaces. The number of fp-ethX interfaces created depends on how many physical NICs your bare-metal NSX Edge has. Up to four of these interfaces can be allocated for uplinks to top-of-rack (ToR) switches and NSX overlay tunneling.

When you create the NSX Edge transport node, you can select fp-ethX interfaces to associate with the uplinks and the overlay tunnel.

You can decide how to use the fp-ethX interfaces. In the following sample physical topology, fp-eth0 and fp-eth1 are bonded and used for the NSX overlay tunnel. fp-eth2 and fp-eth3 are used as redundant VLAN uplinks to TORs.

Figure 410: One Suggested Link Setup for Bare-Metal NSX Edge Networking



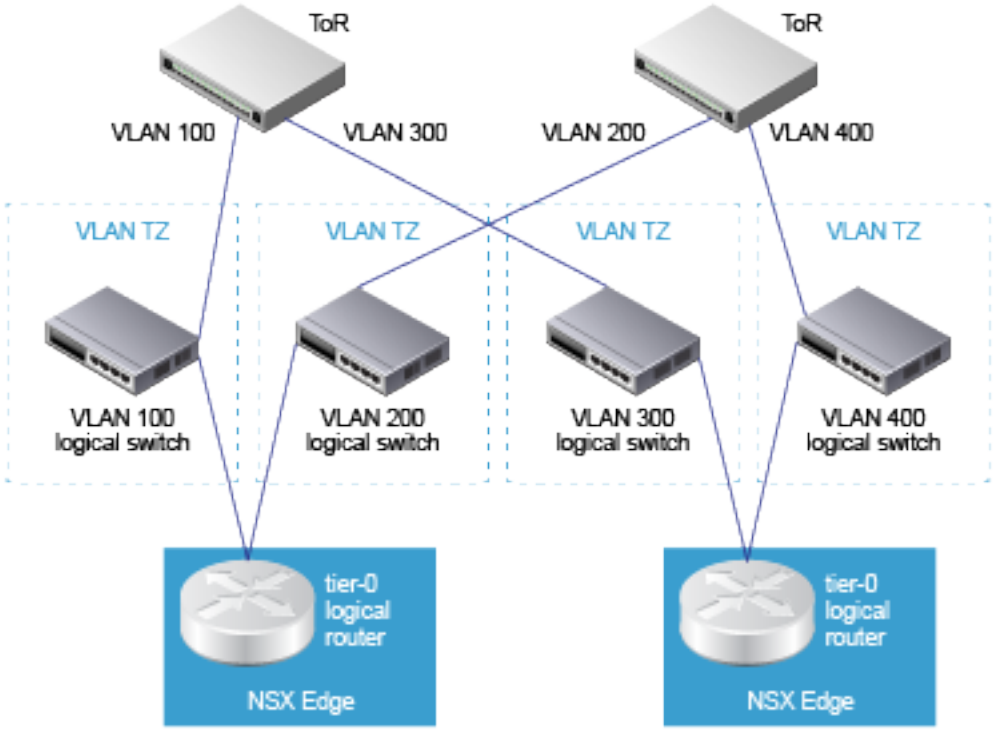
NSX Edge Uplink Redundancy

NSX Edge uplink redundancy allows two VLAN equal-cost multipath (ECMP) uplinks to be used on the NSX Edge-to-external TOR network connection.

When you have two ECMP VLAN uplinks, you must also have two TOR switches for high availability and fully meshed connectivity. Each VLAN logical switch has an associated VLAN ID.

When you add an NSX Edge to a VLAN transport zone, a new N-VDS is installed. For example, if you add an NSX Edge node to four VLAN transport zones, as shown in the figure, four N-VDS get installed on the NSX Edge.

Figure 411: One Suggested ECMP VLAN Setup for NSX Edges to TORs



Note:
For an Edge VM deployed on an ESX host that has the vSphere Distributed Switch (vDS) and not N-VDS, you must do the following:

- On a vDS switch running version prior (<) to 6.6, enable promiscuous mode for the port connected to NSX Edge VM virtual NIC that provides VLAN connectivity. These settings are needed to support bridging or L2VPN and DHCP functionality for VLAN networks.
- On a vDS switch running version equal to or greater than (>=) 6.6, enable mac learning and disable promiscuous mode. These settings ensure that packets are received at the destination where destination mac address does not match the virtual NIC effective MAC address. These settings also ensure packets are received at destinations that are on an NSX Segment. These settings are needed to support bridging or L2VPN and DHCP functionality for VLAN networks.
- Enable forged transmit on the vDS switch. Forged transmit enables sending packets with source mac address not matching the virtual NIC effective MAC addresses. These settings are needed to support bridging or L2VPN and DHCP functionality for VLAN networks.

NSX Edge Installation Methods

Install NSX Edge on an ESX host using NSX Manager UI (recommended method), vSphere Web Client (from UI or vSphere command-line OVF tool) or as physical severs.

NSX Edge Installation Methods

Installation Method	Instructions
NSX Manager (recommended method to install an NSX Edge VM appliance only)	• Ensure NSX Edge network requirements are met. See NSX Edge VM System Requirements. • Create an NSX Edge transport node. See Create an Transport Node.
vSphere web client or vSphere command-line OVF tool	• Ensure NSX Edge network requirements are met. See Installation Requirements. • Choose vSphere web client or vSphere command-line OVF tool to install NSX Edge. – (Web Client) Install NSX Edge on ESX. See Install an on Using the GUI. – (Command-line OVF tool) Install NSX Edge on ESX. See Install on Using the Command-Line OVF Tool. • Join NSX Edge with the Management Plane. See Join with the Management Plane. • Configure an NSX Edge as a transport node. See Edit Transport Node Configuration. • Create an NSX Edge cluster and add the NSX Edge node to the cluster.
Physical server (Automated or Interactive mode using ISO file)	Install NSX Edge using ISO file using PXE on physical servers. • Ensure NSX Edge network requirements are met. • Join NSX Edge with the Management Plane. See Join with the Management Plane. • Configure an NSX Edge as a transport node. See Edit Transport Node Configuration. • Create an NSX Edge cluster and add the NSX Edge node to the cluster.

Create an NSX Edge Cluster and Add Edge Nodes

NSX Edge transport nodes are service appliances that run network and security services. The task of creating an NSX Edge node is intended for administrators who have thorough knowledge of NSX networking concepts and configurations.

To create NSX Edge nodes, complete the following high-level tasks:

1. Add an NSX Edge cluster and configure the NSX Edge cluster settings.
2. Create an NSX Edge node and configure settings.
   a. Configure node placement settings.
   b. Configure management network settings for the NSX Edge node.
   c. Configure the uplink mapping for datapath interfaces of the NSX Edge node.
   d. (Optional) Configure advanced settings for the NSX Edge node.
3. Deploy the NSX Edge node and verify its status.

Each high-level task is described in more detail later on this page.

Essential Prerequisites

Before creating an NSX Edge node, you must meet these essential prerequisites.

1. Identify the vCenter instance where you plan to deploy the NSX Edge node.
2. Identify the vSphere cluster where you plan to deploy the NSX Edge node.
3. Identify the data store to be used for the NSX Edge node. Verify that the data store is accessible to all ESX hosts in the designated vSphere cluster.
4. Prepare a static IP address and gateway for assignment to the NSX Edge management interface. Alternatively, you can use DHCP assignment for the NSX Edge management interface.
5. Identify the distributed virtual port group (DVPG) to which you plan to connect the NSX Edge management interface. Verify that the DVPG is of type VLAN in vCenter. You may already have a DVPG in the VCF management domain.
6. By default, the deployment workflow provides pre-configured uplink profiles (such as **<nsx-edge-multiple-vteps-uplink-profile>**) for the NSX Edge datapath interfaces. If you want to use your own uplink profile instead, configure that uplink profile.
7. Identify the transport VLAN to be used by NSX Edge tunnel endpoints (TEPs) for overlay traffic.
8. Identify the overlay transport zone of the vSphere cluster where you plan to deploy the NSX Edge node. You will apply this overlay transport zone to the NSX Edge node.
9. Prepare the static IP addresses and gateway for assignment to the NSX Edge TEPs. Alternatively, you can use DHCP or IP pool assignment for the TEPs.
10. Identify the active and standby physical network interfaces (pNICs) that you plan to use for the uplink mapping of the NSX Edge datapath interfaces. NSX automatically creates a trunk DVPG in vCenter for the uplink mapping. Alternatively, you can use an existing DVPG as described in the next section, "Optional Prerequisites."

Optional Prerequisites

To use optional features of NSX Edge nodes, prepare the relevant requirements.

- To use NSX Edge nodes in UPT mode, meet the following conditions:
  - Verify that the physical NIC of each ESX host is running on a UPT-compatible driver and firmware.
  - Configure a vSphere Distributed Switch (VDS) to support network offloads, add ESX hosts with compatible physical NICs to that VDS, and enable Advanced NIC Passthrough in the host switch configuration for the transport node profile. For more information, see Enhanced Network Offloads and Advanced NIC Passthrough.
  - Set the memory reservation to 100% on the NSX Edge node.
  - Verify that all NSX Edge datapath interfaces are connected to the VDS that has network offloads (UPT) enabled.
  - Ensure that all NSX Edge nodes in the NSX Edge cluster have homogeneous capabilities to provide equivalent performance in cases of failover.
- To use a resource pool for the NSX Edge node VMs, reserve that resource pool.
- To support the host group affinity feature which uses the best effort restart (BER) policy, ensure that you have created the host group in vCenter according to the network and/or failure boundaries and homogeneity.
- To have the NSX Edge deployment workflow automatically fetch the host overlay network configuration from a vSphere cluster, verify that the vSphere cluster is properly prepared for NSX and enable the **NSX on DVPGs** setting on that vSphere cluster.
- To use existing uplinks for the uplink mapping of the NSX Edge datapath interfaces, you must first prepare these uplinks.

- To connect NSX Edge datapath interfaces to an existing trunk DVPG in vCenter, create the trunk DVPG in vCenter for NSX Edge TEP and uplink traffic. For more information, see the vSphere Networking guide.
- To connect NSX Edge datapath interfaces to NSX VLAN segments, create VLAN trunk segments in NSX.

About NSX Edge Nodes

NSX Edge nodes can host tier-0, tier-1, VPC, and centralized transit gateways. They can be instantiated in virtual machine form factor or as bare metal edges. They are grouped in one or several NSX Edge clusters. Each NSX Edge cluster represents a pool of capacity for networking services.

Note:

The deployment workflow described on this page applies only to NSX Edge nodes in virtual machine form factor. For information about deploying a bare metal edge, see Join NSX Edge with the Management Plane.

An NSX Edge node can belong to only one overlay transport zone, but it can belong to one or more VLAN transport zones depending on your design.

About NSX Edge Clusters

Having an NSX Edge cluster of at least two NSX Edge nodes helps to ensure the availability of nodes. To create a tier-0 gateway or a tier-1 gateway with stateful services (such as NAT and load balancer), you must associate it with an NSX Edge cluster. To associate a tier-0 or tier-1 gateway with a single NSX Edge node, create an NSX Edge cluster that contains the single node.

Each NSX Edge node can belong to only one NSX Edge cluster. An NSX Edge node can be used to back multiple tier-1 gateways in active-standby mode, and multiple VPC and centralized transit gateways. However, there can be only one tier-0 gateway per NSX Edge node.

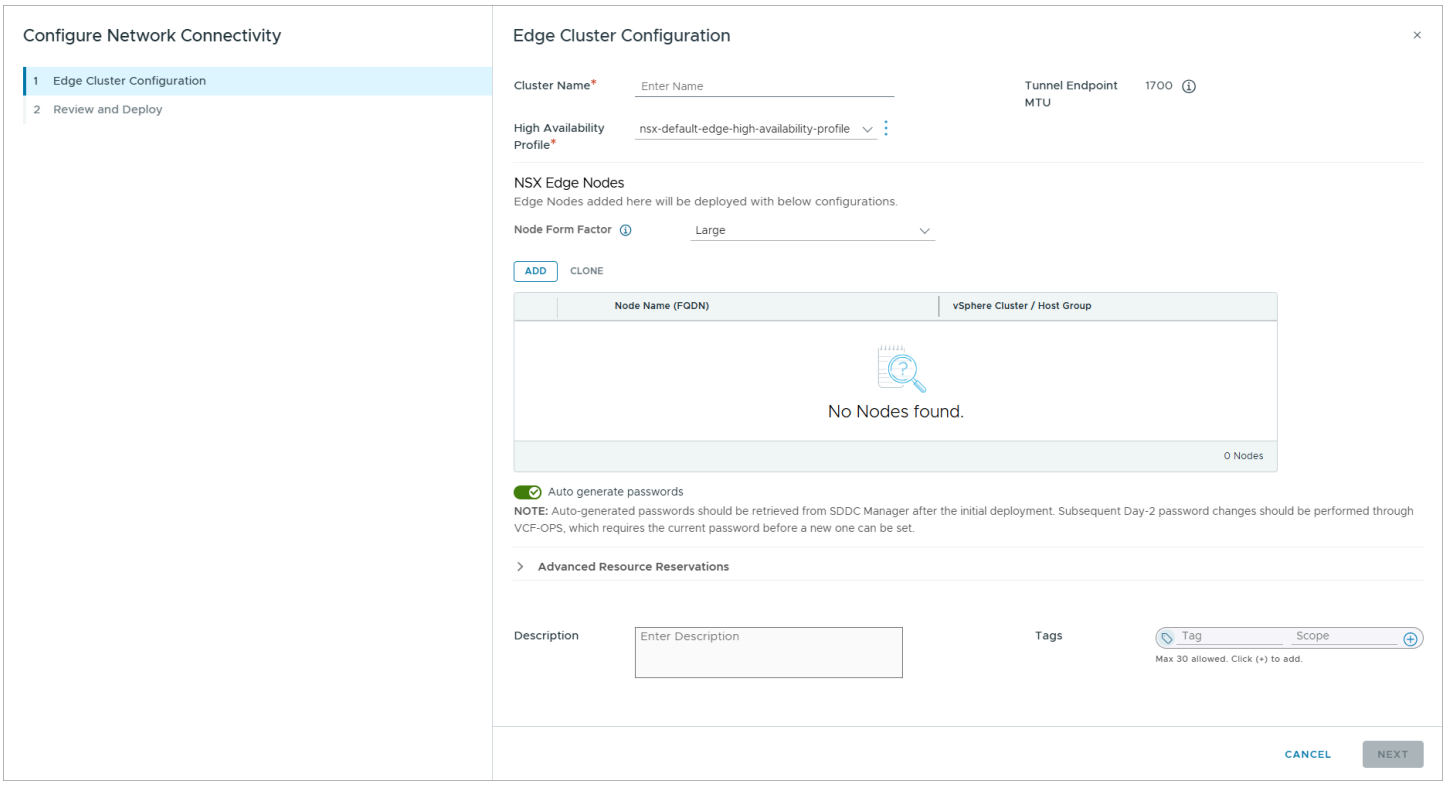
You can edit an existing NSX Edge cluster to add more nodes to it.

Note:

Multiple NSX Edge clusters can be deployed within a single NSX Manager, allowing for the creation of pools of capacity that can be dedicated to specific services: for example, NAT at tier-0 gateways or networking services (NAT, DHCP, VPN, DNS) at tier-1 gateways. Within a given NSX Edge cluster, all NSX Edge nodes must be the same type – either physical servers (bare metal) or VMs. When you deploy an NSX Edge cluster of node VMs using the deployment workflow described on this page, all nodes in the NSX Edge cluster must have the same form factor, or size.

You can also configure a failure domain, which consists of a logical grouping of NSX Edge nodes within an NSX Edge cluster. Failure domains complement auto-placement algorithms and guarantee service availability during a failure affecting multiple NSX Edge nodes. For more information, see Configure Failure Domains.

1. Add an NSX Edge Cluster



1. Select **System** > **Fabric** > **Edges** > **Edge Clusters**.
2. Select **Add Cluster**. The NSX Edge cluster configuration page appears.
3. Enter a unique **Cluster Name**.
4. Select or create a **High Availability Profile**. For more information, see Add a High Availability Profile.
5. For **Edge Form Factor**, designate the size of the NSX Edge nodes to be added to the NSX Edge cluster. All nodes in the NSX Edge cluster must have the same form factor.

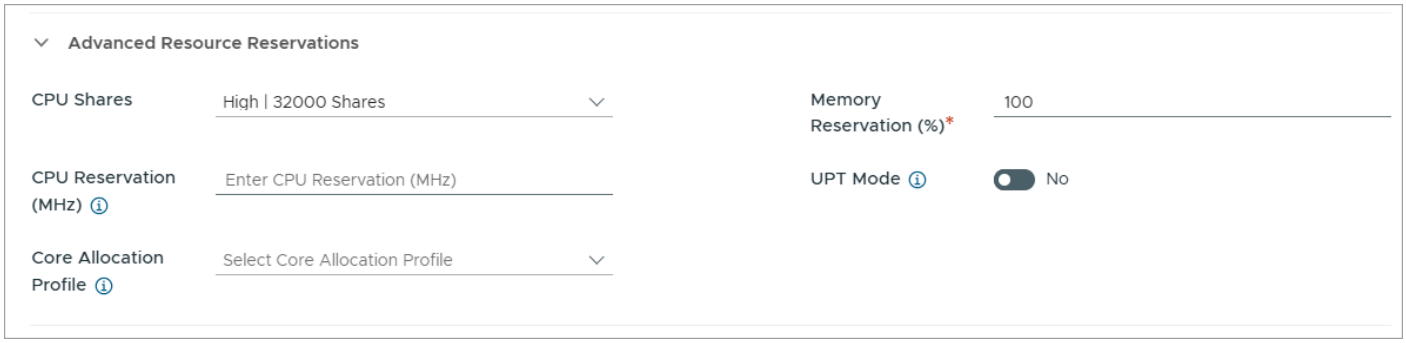
   Ensure that you have met the resource requirements on the vSphere cluster where you plan to deploy the NSX Edge nodes, as described in NSX Edge VM System Requirements.
6. Specify how you want to manage password credentials for the nodes in the NSX Edge cluster.
   – To automatically generate passwords for the NSX Edge nodes and have these passwords managed by SDDC Manager, select the toggle.
   – To manually configure passwords for the NSX Edge nodes, deselect the toggle. Configure the CLI administrator and root administrator credentials, in compliance with the password strength restrictions provided by the wizard. Specify whether to allow root SSH logins to the NSX Edge nodes.

   Note:

   Use the following methods to manage NSX Edge passwords:

   - In the case where you have deployed NSX using SDDC Manager and auto-generated the passwords, use SDDC Manager to retrieve and manage the passwords. Once you have retrieved the passwords, you can also manage them using VCF Operations.
   - In all cases, you can run the **Reset Password** action on the deployed NSX Edge node to manually configure new passwords.
7. (Optional) Under **Advanced Resource Reservations**, select **Configure** to adjust the resource reservation settings for nodes in the NSX Edge cluster.

   For example, the following screenshot shows the settings available in version 9.1.1 and later for nodes with Extra Large form factor.



   a. Select an option for **CPU Shares**.

      CAUTION:

      Configuring CPU shares does not guarantee that CPU cycles will be reserved for the NSX Edge VM. For optimal performance, specify CPU reservations using the **CPU Reservation (MHz)** setting instead.

The NSX Edge form factor determines the options that you can select from this menu, as summarized in the following table.

NSX Edge Form Factor	Options for CPU Shares
Small	• Low | 1000 Shares • Normal | 2000 Shares • High | 4000 Shares (default) • Extra High | 6000 Shares
Medium	• Low | 2000 Shares • Normal | 4000 Shares • High | 8000 Shares (default) • Extra High | 10000 Shares
Large	• Low | 4000 Shares • Normal | 8000 Shares • High | 16000 Shares (default) • Extra High | 18000 Shares
Extra Large	• Low | 8000 Shares • Normal | 16000 Shares • High | 32000 Shares (default) • Extra High | 34000 Shares

b. For **Memory Reservation (%)**, specify the percentage of memory to be allocated to an NSX Edge VM relative to other VMs that are contending for shared memory. The default value is 100%.

CAUTION:

For maximum performance, NSX Edge nodes must be assigned 100% of the available resources. If you do customize resource reservations, restore the reservation to 100% later to get maximum performance.

Reservation percentage is relative to the pre-defined value in the form factor. 100 indicates 100% of memory is reserved for the NSX Edge VM. If you enter 50, it indicates that 50% of the allocated memory is reserved for the NSX Edge node.

Note:

If you plan to enable Uniform Passthrough (UPT) mode, reserve 100% of the allocated memory for the NSX Edge node.

c. Enter a value for **CPU Reservation (MHz)**. This field is empty by default.

The maximum CPU reservation must not exceed the maximum CPU allocation for the NSX Edge VM, or the number of vCPUs multiplied by the physical CPU speed. If the CPU reservation exceeds that value, the NSX Edge VM might fail to start even though the allocation was accepted.

For example, consider a NSX Edge VM with four vCPUs backed by a physical multi-core CPU running at 2.20 GHz base frequency. The maximum CPU allocation for that VM is 4 x 2200 MHz = 8800 MHz. If you set the CPU reservation to 10000 MHz, the ESX host fails to power on the VM because it cannot fulfill the 10000 MHz reservation. If you reconfigure the NSX Edge VM with eight vCPUs, the maximum CPU allocation changes to 8 x 2200 MHz = 17600 MHz. In this case, the 10000 MHz reservation is fulfilled and the VM powers on.

d. In version 9.1.1 and later, for **Core Allocation Profile**, select a profile to configure how NSX Edge allocates available cores between the datapath (fast-path), services, and Linux processes. The menu of available profiles is based on the node form factor specified for the cluster.

If you leave the profile menu unselected or clear the existing selection, NSX Edge uses the default core allocation: 50% for datapath, 25% for services, 25% for Linux.

Note:

After the NSX Edge cluster is created, you can change the core allocation profile if needed by editing the cluster. You must then power off and power on all the NSX Edge nodes in that cluster to make the new profile take effect. For more information, see Broadcom knowledge base article 441953.

Core Allocation Profile	Description	Supported Node Form Factors
L4 FORWARDING	Allocates more fast-path cores to improve forwarding performance of datapath: • 75% for datapath • 0% for services • 25% for Linux	• Bare metal
SECURITY	Allocates more service and Linux cores to support advanced security workloads: • 25% for datapath • 25% for services • 50% for Linux	• Extra Large Edge VM • Bare metal

e. For **UPT Mode**, select the toggle to enable Uniform Passthrough (UPT) mode on NSX Edge datapath interfaces. In UPT mode, NSX Edge nodes can use Uniform Passthrough technology to offload intensive network datapath tasks from the ESX host CPU to supported hardware NICs, accelerating networking performance.

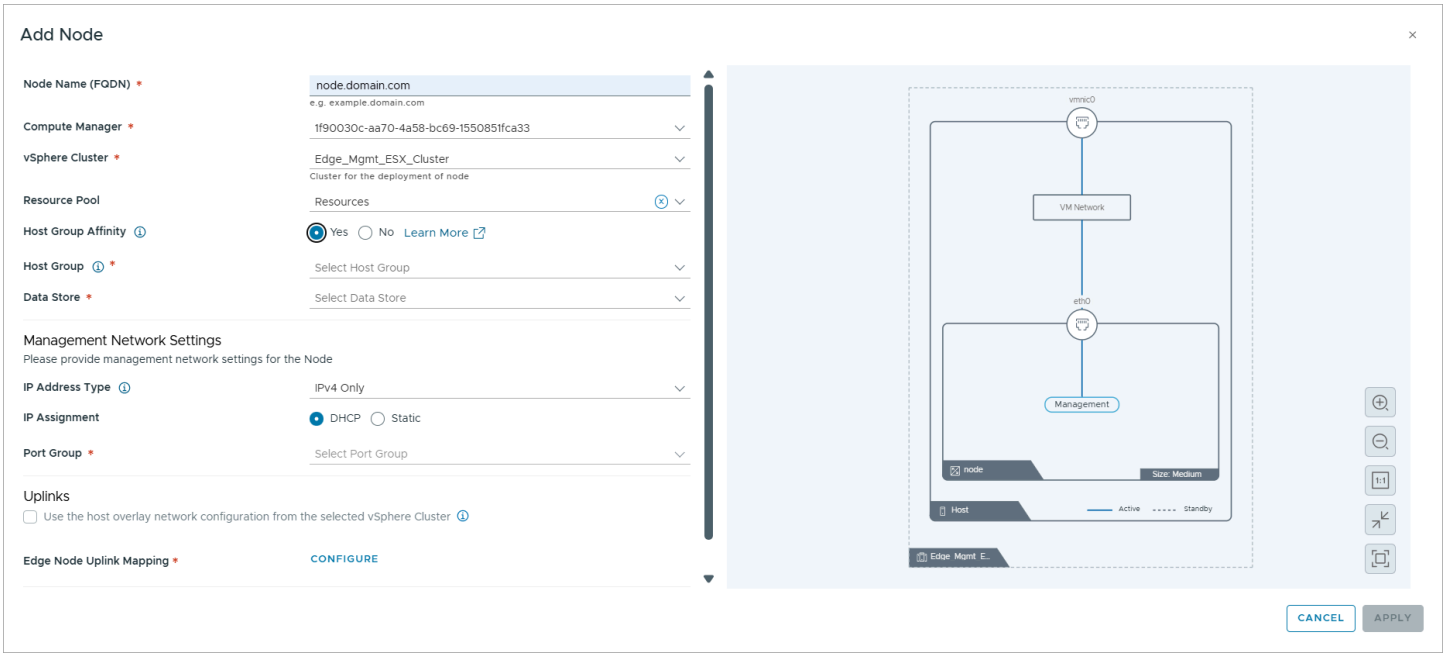
CAUTION:

To make UPT settings effective on NSX Edge VM virtual network adapters, NSX Manager puts the NSX Edge VM into maintenance mode, powers it off and powers it back on again.

8. (Optional) Enter a **Description** and specify **Tags** for the NSX Edge cluster.
9. In the **Add Cluster** page, select **Add** to start the workflow for adding an NSX Edge node. Proceed to the next section for detailed instructions.

2a. Start the NSX Edge Node Workflow and Configure Node Placement Settings

The **Add Node** page contains configurable settings for the new NSX Edge node. As you adjust settings, the preview on the right side of the page updates to reflect the resulting network topology.



1. For **Node Name (FQDN)**, enter a unique FQDN for the NSX Edge node, using the format *subdomain.example.com.*
2. For **Compute Manager**, select the registered vCenter instance where you want to deploy the NSX Edge node.
3. For **vSphere Cluster**, designate the vSphere cluster where you want to deploy the NSX Edge node.
4. (Optional) For **Resource Pool**, assign a resource pool that has been preconfigured for the NSX Edge node.
5. Select a **Data Store** for the NSX Edge node files from the drop-down menu.
6. For **Host Group Affinity**, select **Yes** if you want the NSX Edge node to be pinned to a host or a host group and want to assign a Best Effort Restart (BER) policy to the NSX Edge node. Ensure that you have created the host group in vCenter according to the network and/or failure boundaries and homogeneity. The following considerations apply to this feature.
   – If you deploy an NSX Edge node in a non-homogeneous network, you cannot use vMotion during a host remediation. An example of a non-homogeneous network is a multi-rack horizontal vSphere cluster with a single host per rack and L3 boundaries between the racks. vMotion is not allowed in such cases because migration of the NSX Edge node to another host in another rack will cause connectivity loss with top-of-rack switches.

- When you pin an NSX Edge node to a host group, the node is associated with a host or a host group using an affinity rule.
- During a host upgrade or remediation workflow when the current NSX Edge node has active contexts, NSX also runs prechecks to find if any peer NSX Edge node is healthy and ready for failover before the host automatically enters maintenance mode.

  Note:

  Failover prechecks do not apply when you manually put a host into maintenance mode. For more information, see Broadcom knowledge base article 395307.

  If a healthy peer NSX Edge node is available, a graceful failover is made to this peer node and then the first NSX Edge node is put into maintenance mode and powered off.

  If the precheck detects that no healthy peer NSX Edge node is available for graceful failover, the host upgrade or remediation pauses until you manually resolve the precheck errors.
- If you have enabled DRS for the vSphere cluster, a Best Effort Restart (BER) policy is also applied to the NSX Edge node. The BER policy waits for the guest VM to shut down and ensures that the node gracefully shuts down when its host enters maintenance mode.
  - If the host group contains a single host, the NSX Edge node stays powered off during the host maintenance mode. The node is powered on again when the host exits the maintenance mode. If the NSX Edge node has active contexts and a healthy peer NSX Edge node is available, the peer node takes over the services. Once the host exits maintenance mode, the first NSX Edge node is powered on upon the same host.
  - If the host group contains multiple hosts, a healthy peer NSX Edge node takes over the services of the NSX Edge node that has shut down. Also, that node is powered on upon a compatible host if available. If no compatible host is available due to affinity or availability constraints, the node stays powered off during the host maintenance mode. Once the host exits maintenance mode, the NSX Edge node is powered on again on the same host.

7. If you have set **Host Group Affinity** to **Yes**, select the **Host Group** to which you want to pin the NSX Edge node.

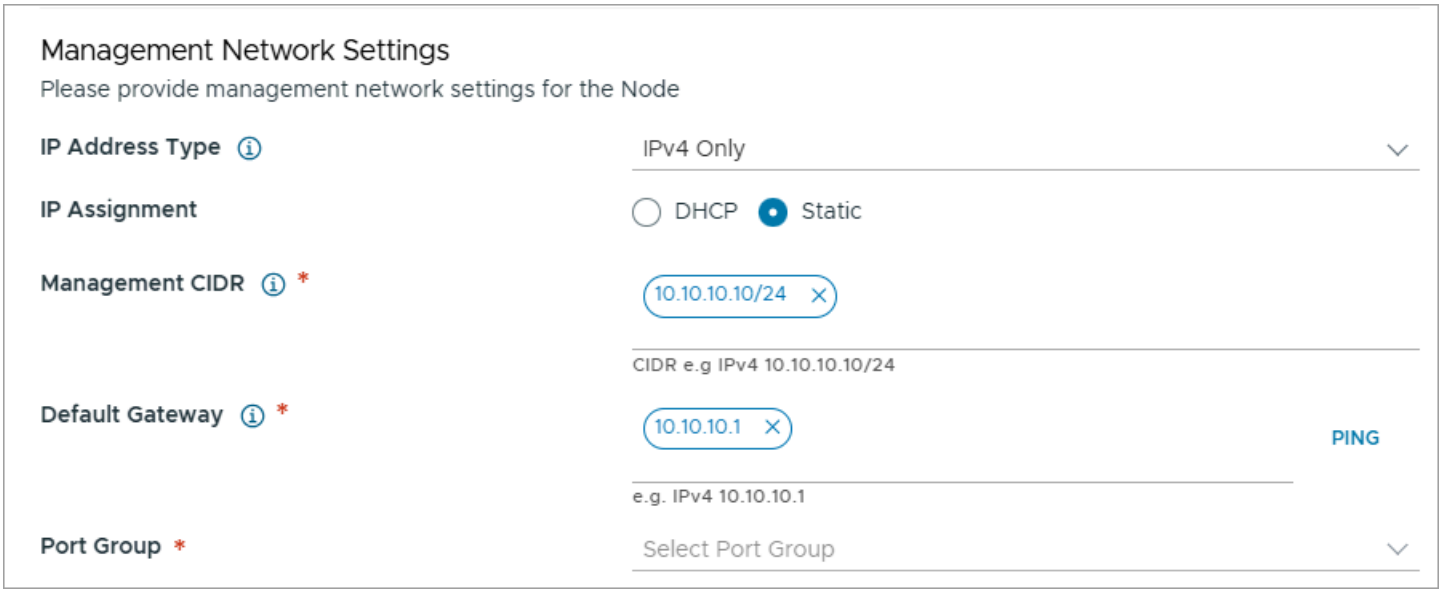
   Note:

   To change the pinned host group after the NSX Edge node is deployed, run the **Edit** action on that node. Once the new host group passes vMotion compatibility checks, the NSX Edge node will undergo vMotion migration to that new host group.
8. Proceed to the next section to configure the NSX Edge management network.

2b. Configure NSX Edge Management Network Settings

These settings specify the IP version and assignment type for the IP address of the NSX Edge node management interface. This IP address is required to communicate with NSX Manager.

The NSX Edge management interface is connected to distributed virtual port groups (DVPGs).



1. In the **Add Node** page, scroll to the **Management Network Settings** section.
2. For **IP Assignment**, specify the IP version of the assignment. You can select from the following options:
   - IPv4 Only
   - IPv4 & IPv6
   - IPv6 Only
3. If you selected **IPv4 Only**, specify whether to use **DHCP** or **Static** assignment type.

   Note:

   If you select DHCP, the IP address of the edge management interface might change upon lease expiration.
4. If you selected static IPv4 Only, IPv4 & IPv6, or IPv6 Only, specify the following:
   - **Management CIDR**: Enter the IP address of the edge node in CIDR notation.
   - **Default Gateway**: Enter the gateway IP address of the NSX Edge node.
5. For **Port Group**, select the distributed virtual port group (DVPG) used to connect to the NSX Edge management interface. This interface must be reachable from NSX Manager.
   The NSX Edge management interface establishes communication with the NSX Manager management interface.
6. Proceed to the next section to configure the uplink mapping for edge datapath interfaces.

2c. Configure the Uplink Mapping for NSX Edge Datapath Interfaces

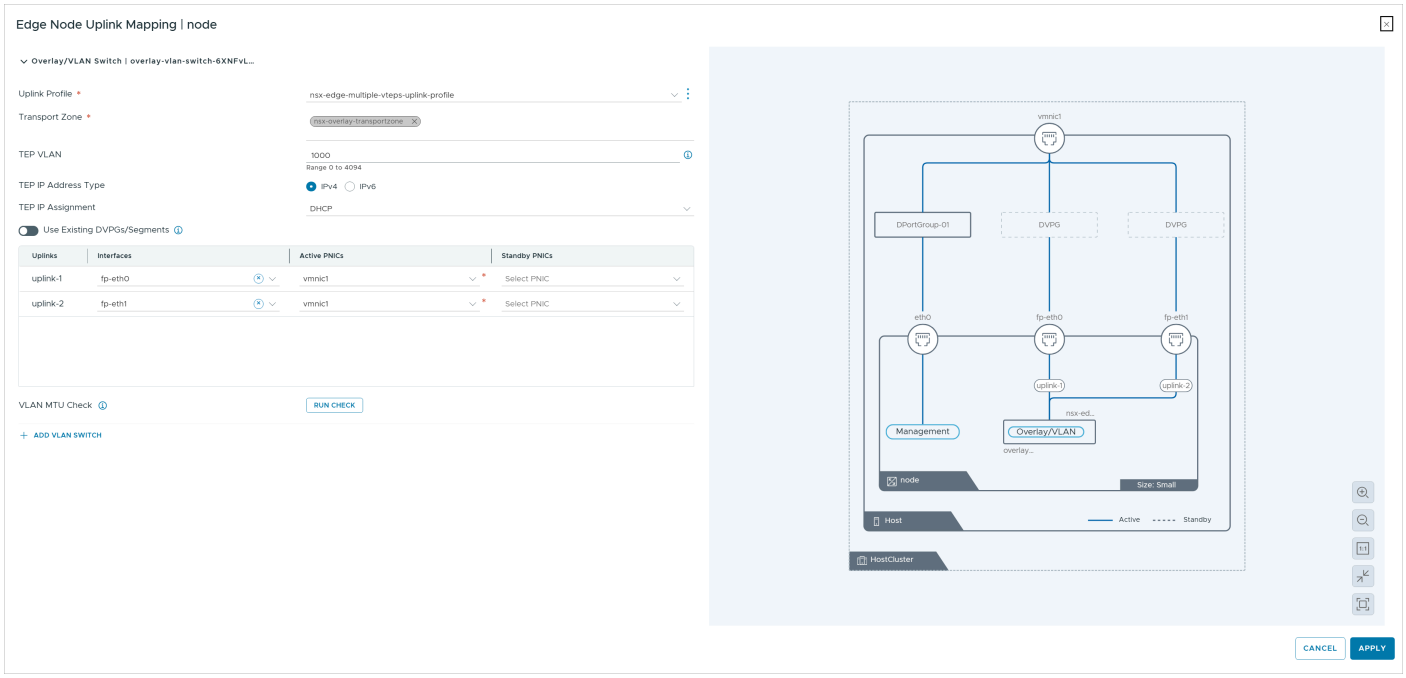
By default, NSX Manager fetches the host TEP configuration from the properly prepared vSphere cluster and uses it for the NSX Edge node uplink mapping. You can accept the fetched configuration, or you can fine-tune datapath settings for the NSX Edge node.

The NSX Edge datapath interfaces are connected to a distributed virtual port group (DVPG). You can either create the DVPG automatically or select an existing DVPG, as described in step 9 of the following procedure.

1. In the **Add Node** page, scroll to the **Uplinks** section.
2. If desired, select the option to use the host overlay configuration from the selected vSphere cluster. The deployment workflow fetches the existing host overlay configuration and applies it to the NSX Edge node TEP VLAN and TEP IP Assignment.

   Note:

   To use this option, you must first prepare the vSphere cluster as described under "Optional Prerequisites" earlier on this documentation page.
3. For **Edge Node Uplink Mapping**, click **Configure**. The NSX Edge node uplink mapping settings are displayed. The selected uplink profile will determine the number of uplinks for the node.


Edge Node Uplink Mapping | node
Overlay/VLAN Switch | overlay-vlan-switch-6XNFvL...
Uplink Profile *
nsx-edge-multiple-vteps-uplink-profile
Transport Zone *
nsx-overlay-transportzone
TEP VLAN
1000
Range 0 to 4094
TEP IP Address Type
IPv4
IPv6
TEP IP Assignment
DHCP
Use Existing DVPGs/Segments
Uplinks
Interfaces
Active PNICs
Standby PNICs
uplink-1
fp-eth0
vmnic1
Select PNIC
uplink-2
fp-eth1
vmnic1
Select PNIC
VLAN MTU Check
RUN CHECK
ADD VLAN SWITCH
vmnic1
DPortGroup-01
DVPG
DVPG
eth0
fp-eth0
fp-eth1
uplink-1
uplink-2
nsx-ed...
Management
Overlay/VLAN
overlay...
node
Size: Small
Host
Active
Standby
HostCluster
CANCEL
APPLY

4. For **Uplink Profile**, select the uplink profile from the drop-down menu. Both default and custom uplink profiles are available for selection. The available number of uplinks depends on the configuration in the selected uplink profile.

   Note:

   NSX Edge nodes support uplink profiles with Failover teaming policy (with a single active uplink) and Loadbalancer Source teaming policy (with multiple active uplinks) only.
5. For **Transport Zone**, select the transport zones that this NSX Edge node belongs to.
   - When assigned to a tier-0 gateway, an NSX Edge node belongs to at least two transport zones, an overlay for NSX connectivity and a VLAN for uplink connectivity.
   - When assigned to a tier-1 gateway, VPC, or centralized transit gateway, an NSX Edge node belongs to an overlay transport zone.

   Note:

   NSX Edge nodes support multiple tunnel endpoints (TEPs) when the following prerequisites are met:

   - TEP configuration must be done on one NSX Edge virtual switch only.
   - All TEPs must use the same transport VLAN for overlay traffic.
   - All TEPs must reside in the same subnet.
   - All TEPs must use the same default gateway.
6. For **TEP VLAN**, specify the unique VLAN ID that identifies the VLAN used for overlay tunnels.
   a. If specified, this value overrides the transport VLAN configured in the selected uplink profile.
   b. If you leave this value unspecified, overlay tunnels use the transport VLAN configured in the selected uplink profile.
7. For **TEP IP Address Type**, select the IP version to be used for the tunnel endpoint (TEP). The options are **IPv4** and **IPv6**.

   Important:

   Ensure that the transport node forwarding mode and TEP IP address type are the same. For example, if the transport node forwarding mode is set to IPv6, set the TEP IP address type to IPv6. If they are different, a loss of traffic may result.
8. For **IP Allocation (TEP)**, choose how IP addresses are assigned to the NSX Edge virtual switch that is configured. The switch is used as the tunnel endpoint of the NSX Edge node. The options are:

   - **IP Pool**: Select the IP pool. See Create an IP Pool for Tunnel Endpoint IP Addresses.
   - **Static IP List**: Specify the following fields:
     - **Static List**: Enter a list of comma-separated IP addresses to be used by the NSX Edge node.
     - **Static Gateway**: Enter the default gateway of the TEP, which is used to route packets to and from another TEP in another network. For example, if ESX TEP is in 20.20.20.0/24 and NSX Edge TEPs are in 10.10.10.0/24, then we use the default gateway to route packets between these networks.
     - (For IPv4) **IP Subnet Mask**: Enter the subnet mask of the TEP network used on the NSX Edge node.
     - (For IPv6) **Prefix Length**: Enter the prefix length to use for IPv6 assignments.
   - **DHCP**: Select to enable DHCP. If you are using IPv6, the underlay network must have an IPv6 router to advertise the default gateway through Router Advertisement (RA).
9. In the mapping table, map each available uplink to a datapath interface, and map the datapath interface to active and standby physical network interfaces (pNICs). For example, for an uplink profile with four uplinks, the datapath interfaces are named fastpath interfaces fp-eth0, fp-eth1, fp-eth2, and fp-eth3.

   You can configure a maximum of four unique datapath interfaces as uplinks on an NSX Edge VM. However, the mapping table only displays the number of interfaces matching the number of uplinks in the uplink profile. For example, it displays two datapath interfaces when the uplink profile has two uplinks.

   - To have the edge deployment workflow automatically create a trunk DVPG on the selected VDS, turn off the **Use Existing DVPG/Segment** toggle.
   - To use an existing DVPG, turn on the **Use Existing DVPG/Segment** toggle. Select the DVPG to use, and click **Apply**.

   A deployment mode with a single NSX Edge virtual switch is recommended for both bare metal edges and NSX Edge VMs. See Configure NSX Edge DPDK Interfaces.

Note:

If the uplink profile applied to the NSX Edge node uses a Named Teaming policy, ensure the following condition: Map all uplinks in the Default Teaming policy to the corresponding physical network interfaces on the NSX Edge VM. This mapping enables traffic to flow through a segment that uses the Named Teaming policies. See Configure Named Teaming Policy.

10. To validate the NSX Edge TEP transport VLAN and MTU configurations, click **Run Check**.
    You can review the results of the health check for any detected configuration issues, such as an MTU mismatch or untrunked VLAN. For more information, see Run a VLAN MTU Check.
11. If needed, use the **Add VLAN Switch** action to add more NSX Edge virtual switches to the configuration.
12. Click **Apply** to save the uplink mapping to the NSX Edge node and return to the **Add Node** page.
13. Optionally configure advanced settings for the NSX Edge node. For more information, see "2d. Advanced Configuration (Optional)."
14. On the **Add Node** page, click **Apply** to save your configurations to the NSX Edge node.

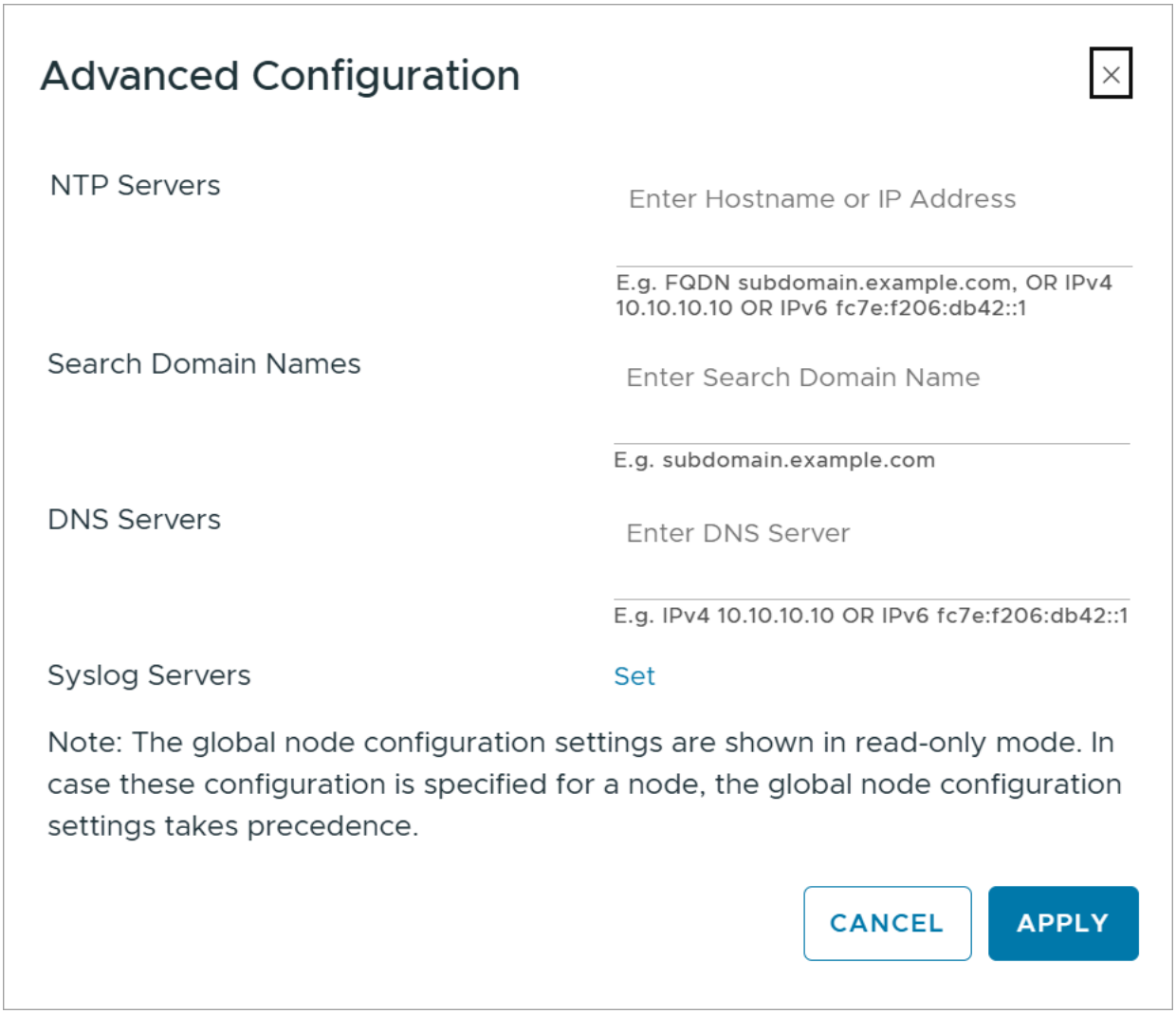
2d. Advanced Configuration (Optional)

You can configure the following advanced settings for an NSX Edge node:

- NTP server
- Search domain
- DNS server
- Syslog server

Note:

Advanced Configuration settings apply specifically to the selected NSX Edge node. As a best practice, configure an NSX Edge global node profile with settings that apply to all NSX Edge nodes, instead of creating a node-specific configuration. For more information, see Configure a Node Profile.



1. In the **Add Node** page, for **Advanced Configuration (Optional)**, select **Configure**.
2. For **NTP Servers**, enter the IP address or FQDN of the NTP server.
3. For **Search Domain Names**, enter search domain names in the format 'example.com', or enter IP addresses.
4. For **DNS Servers**, enter the IP address of the DNS server.
5. For **Syslog Servers**, specify a server for storing the system logs.
6. Click **Apply** to save your configurations to the selected NSX Edge node.

3. Deploy the NSX Edge Nodes

1. To add more nodes to the NSX Edge cluster, repeat the steps beginning from "2a. Start the NSX Edge Node Workflow and Configure Placement Settings."
   Alternatively, you can reuse an existing node's configuration for a new node by selecting **Clone** from the existing node's action menu (three dots). This action copies much of the existing configuration, such as the uplink mapping, to the new node.
2. In the **Edge Cluster Configuration** page, click **Next** to review the NSX Edge cluster configuration.
3. Click **Deploy** to deploy the NSX Edge nodes in the cluster.
4. View the node deployment status on the **Edges** page.
   The deployment workflow takes 20 minutes or longer to reach completion. The workflow deploys and configures the NSX Edge node VMs. Wait for the **Edges** page to show the Configuration status as **Success** and Connectivity Status as **Up** for each NSX Edge node.
5. To verify the NSX Edge node status, use one of the following methods.
   – On the **Edges** page, select the info icon in the Connectivity Status column for the node.

- Open an SSH connection to the NSX Edge node and run the CLI command:
  ```
  get edge-cluster-status | get managers | get controllers | get host-switch
  ```

6. View the NSX Edge node details using one of the following methods.
   - On the **Edges** page, select **View** for the node. Then select the **Monitor** tab in the Edge Node Details screen.
   - Call the API on NSX Manager:
     ```
     GET https://<nsx-mgr>/policy/api/v1/infra/sites/<site-id>/enforcement-points/<enforcementpoint-id>/edge-transport-nodes/<edge-transport-node-id>/state | status
     ```

     The default value for `enforcementpoint-id` and `site-id` is `default`. For more information, see the NSX API Guide.

     Note:

     After an NSX Edge node migrates to a new host using vCenter, the NSX Manager UI may report stale configuration details for the node. These configuration details can include Compute, Datastore, Network, SSH, NTP, DNS, and Search Domains. To refresh the latest NSX Edge node configuration details in NSX Manager, click the refresh icon on the **Edges** page or the Edge Node Details screen.

     Alternatively you can call the API:

     ```
     POST https://<nsx-mgr>/policy/api/v1/infra/sites/<site-id>/enforcement-points/<enforcementpoint-id>/edge-transport-nodes/action/sync-edge-configuration?operation_type=SYNC_ALL
     ```

     Important:

     You can change the IP address of the NSX Edge node from the command-line interface. At the CLI terminal, run the command:

     ```
     set interface eth0 ip <edge-new-ip-address/cidr> gateway <gateway-ip-address> plane mgmt
     ```

Next Steps

To view the status of NSX Edge nodes within a cluster, use one of the following methods.

- UI: On the **Edges** page, select the info icon in the Connectivity Status column for the node. To view more details, select **View** for the node and then select the **Monitor** tab in the Edge Node Details screen.
- CLI:
  ```
  get edge-cluster status
  ```
- API:
  ```
  GET https://<nsx-mgr>/policy/api/v1/infra/sites/<site-id>/enforcement-points/<enforcementpoint-id>/edge-clusters/<edge-cluster-id>/status?source=realtime
  ```

Understand these states of NSX Edge nodes:

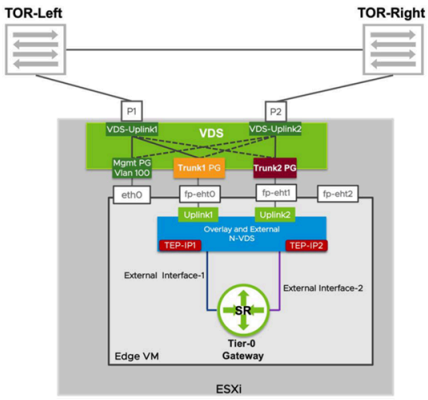
- `Up (routing down)`: The NSX Edge node state is `Up` but tier-0 SR routing daemon is not running on this NSX Edge node as no services are enabled on the it.
- `Admin Down`: The NSX Edge node is in NSX maintenance mode and all services or traffic forwarding is deactivated on this node.
- `Down`: The datapath process is not running, link is down or VTEP tunnels are down.
- `Unreachable`: Between two NSX Edge nodes, one BFD session is run on the management interface, and at least one BFD session is run on each of the VTEP interfaces. An NSX Edge node considers its peer as unreachable only when all BFD sessions to that NSX Edge node (management session and all VTEP sessions) are down.

You can now build logical network topologies and configure services.

Configure NSX Edge DPDK Interfaces

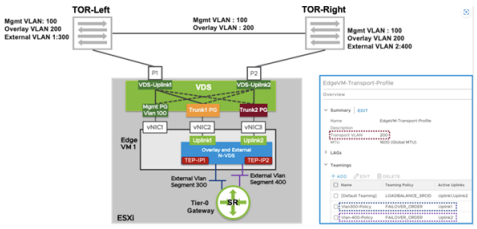
Configure an NSX Edge VM with a node switch to carry both overlay and external traffic.

In this topology, NSX Edge VM is configured with a node switch to carry overlay and external traffic.



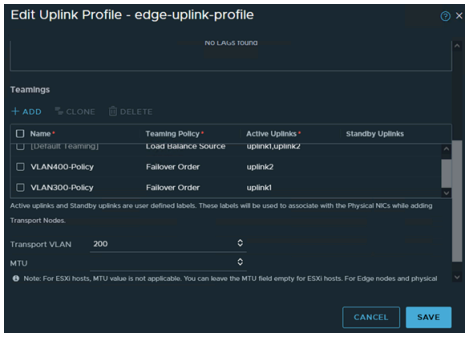
In this topology, the uplink profile attached is configured to use Multi-TEP to provide load balancing for overlay traffic by selecting default teaming to be Load Balancing Source teaming policy with active 'uplink1' and 'uplink2' on transport VLAN 200.

(optional) The uplink profile attached is also configured to use named teaming policy, where 'vlan300-policy' is mapped to uplink1 and 'vlan400-policy' is mapped to uplink2.



To create the topology, follow these steps.

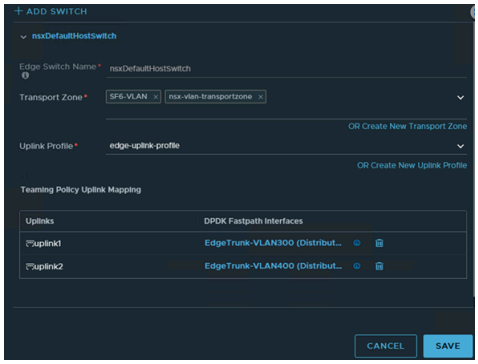
1. Uplink Profile creation with named-teaming policy for VLAN networks and default teaming for overlay networks. If named-teaming does not exist in uplink profile then default teaming is used for all networks.



2. VLAN Transport Zone is created or modified to use named teaming policy VLAN300-Policy and VLAN400-Policy (if using named-teaming policy).



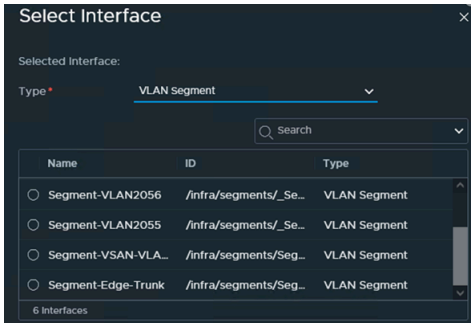
3. In the NSX Edge configuration, “Uplink1” (fp-eth0) is vNIC2 on Edge VM and is mapped to use VLAN 300 Trunk PG and “Uplink2”(fp-eth1) is vNIC3 on Edge VM and is mapped to use VLAN 400 Trunk portgroup.



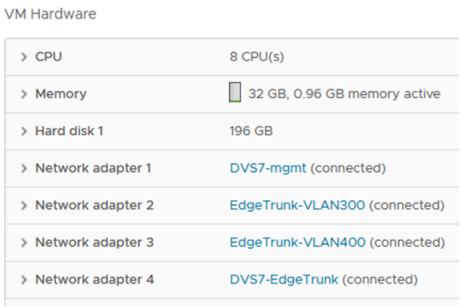
4. NSX Edge VM interfaces can connect to VDS Portgroups in vCenter or NSX VLAN Segments.

   Note:
   If Edge interfaces will be connecting to NSX VLAN Segments, ESX Hosts (hosting the Edge VMs) should be configured as Host Transport Nodes and member of VLAN transport zone.

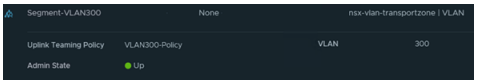
   An example of connecting VM interfaces to NSX VLAN segments.



   An example of an NSX Edge VM with interfaces on vCenter VDS Edge Trunk Portgroups

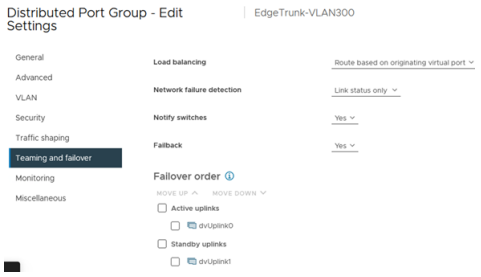


   a) If VM interfaces connect to NSX VLAN segments, named-teaming is enabled on the segments. The diagram shows that External VLAN segment 300 is configured to use a named teaming policy “Vlan300-Policy” that sends traffic from this VLAN on “Uplink1” (vNIC2 of Edge VM). "External VLAN segment 400" is configured to use a named teaming policy “Vlan400-Policy” that sends traffic from this VLAN on “Uplink2” (vNIC3 of Edge VM).



   b) If VM interfaces connect to VDS Portgroups in vCenter, then “Trunk1 PG” is configured to use active uplink as “VDS-Uplink1” and standby uplink as “VDS-Uplink2”. “Trunk2 PG” is configured to use active uplink as “VDS-Uplink2” and standby uplink as “VDS-Uplink1”. This configuration ensures that the traffic sent on “External VLAN

Segment 300”uses vNIC2 of Edge VM to exit the Edge VM and then “VDS-Uplink1” and is sent to the left TOR switch. Similarly, traffic sent on VLAN 400 uses “VDS-Uplink2” and is sent to the TOR switch on the right.



Important:

- ESX TEP and Edge TEP share the same VLAN: Use this configuration only if NSX Edge Trunk 1 portgroup and NSX Edge Trunk 2 portgroup are created from NSX as VLAN segments. Because any traffic between the Edge TEP to its own hypervisor’s TEP does not need to exit the hypervisor.
- ESX TEP and Edge TEP use different VLANs: Use this configuration if NSX EdgeTrunk1 portgroup and NSX EdgeTrunk2 portgroup are created as VDS portgroups from vCenter. Any traffic between the NSX Edge TEP and its hypervisor TEP must exit the ESX. The top of rack switch must then route it back toward the ESX.

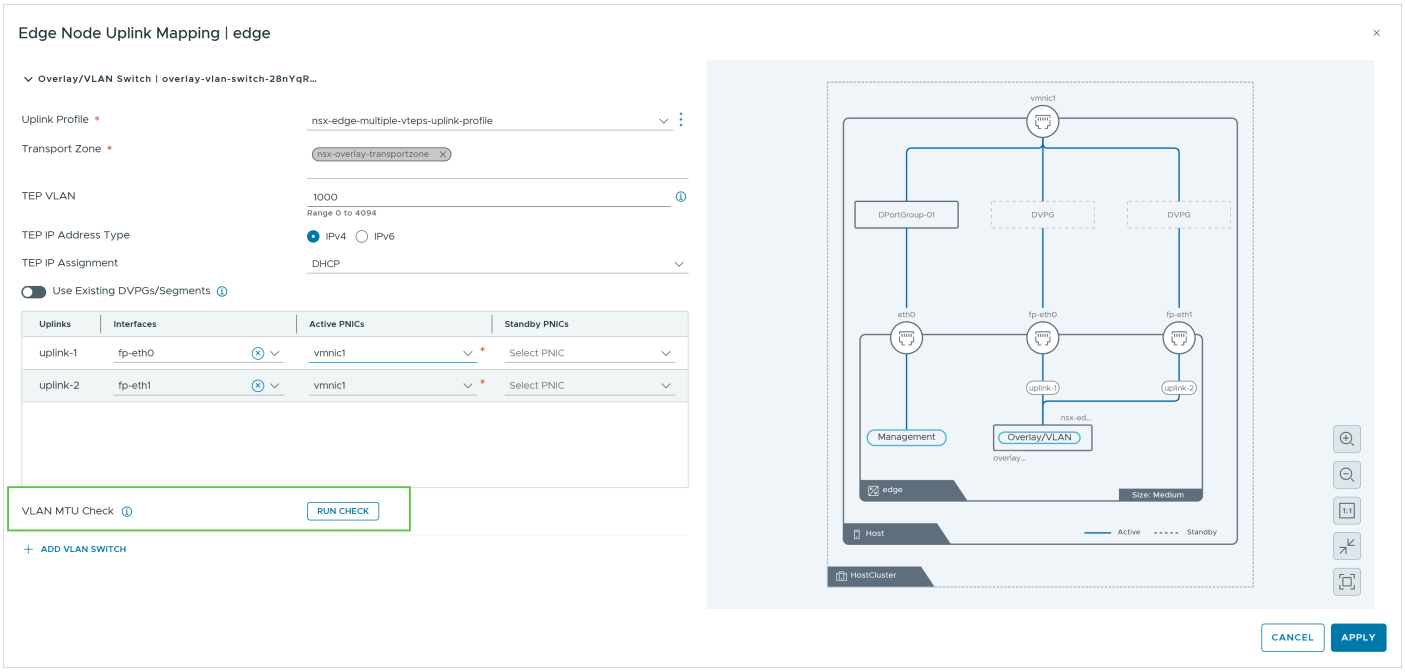
Run a VLAN MTU Check

The VLAN MTU Check validates the configuration of tunnel endpoint (TEP) transport VLANs and maximum transmission unit (MTU) settings during the creation of an edge node. Detecting and correcting VLAN and MTU misconfigurations at this stage helps to ensure that the edge node is deployed successfully without network connectivity problems.

Running the VLAN MTU Check During Edge Creation

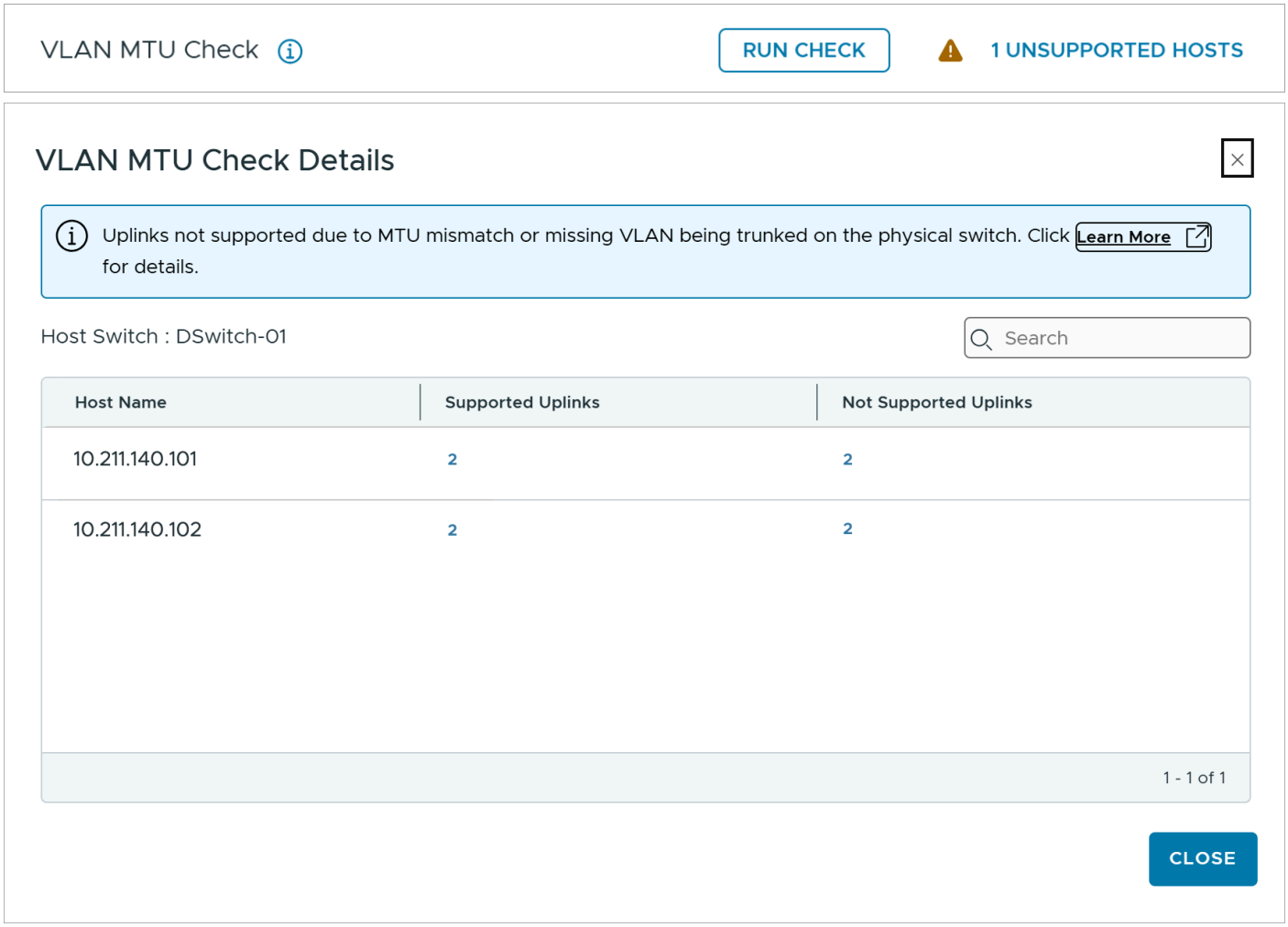
You can run the VLAN MTU Check during the uplinks configuration step of the edge creation workflow.

1. In NSX Manager, start the edge creation workflow.
2. On the **Edge Node Uplink Mapping** page, click **Run Check**.



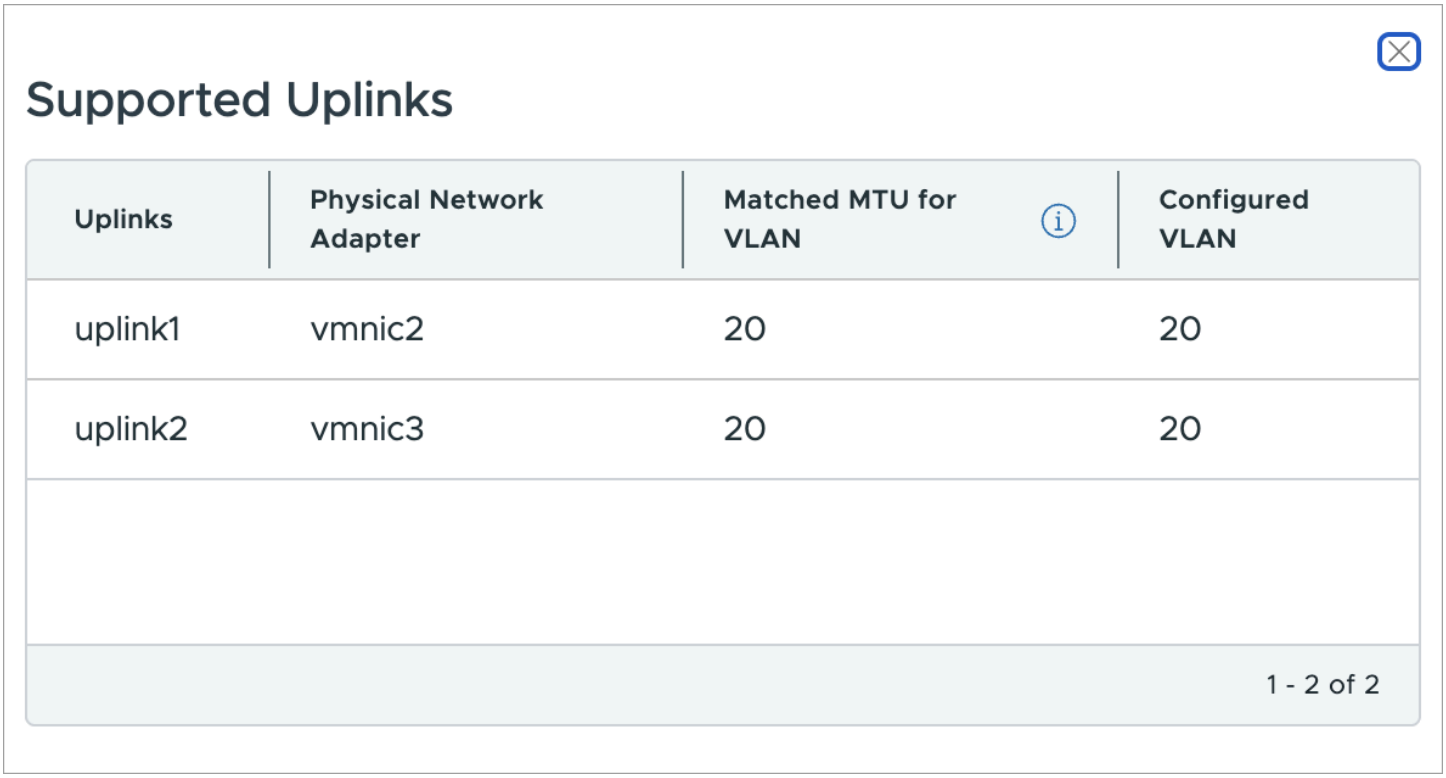
The VLAN MTU Check performs the validation and generates a report showing the number of supported and unsupported uplinks per ESX host in the specified host cluster.

3. Click the report link next to the **Run Check** button to view the report details.



To view report details for a specific uplink, click the number in that uplink column.

Figure 412: Example Report for Matched MTU and Trunked VLAN

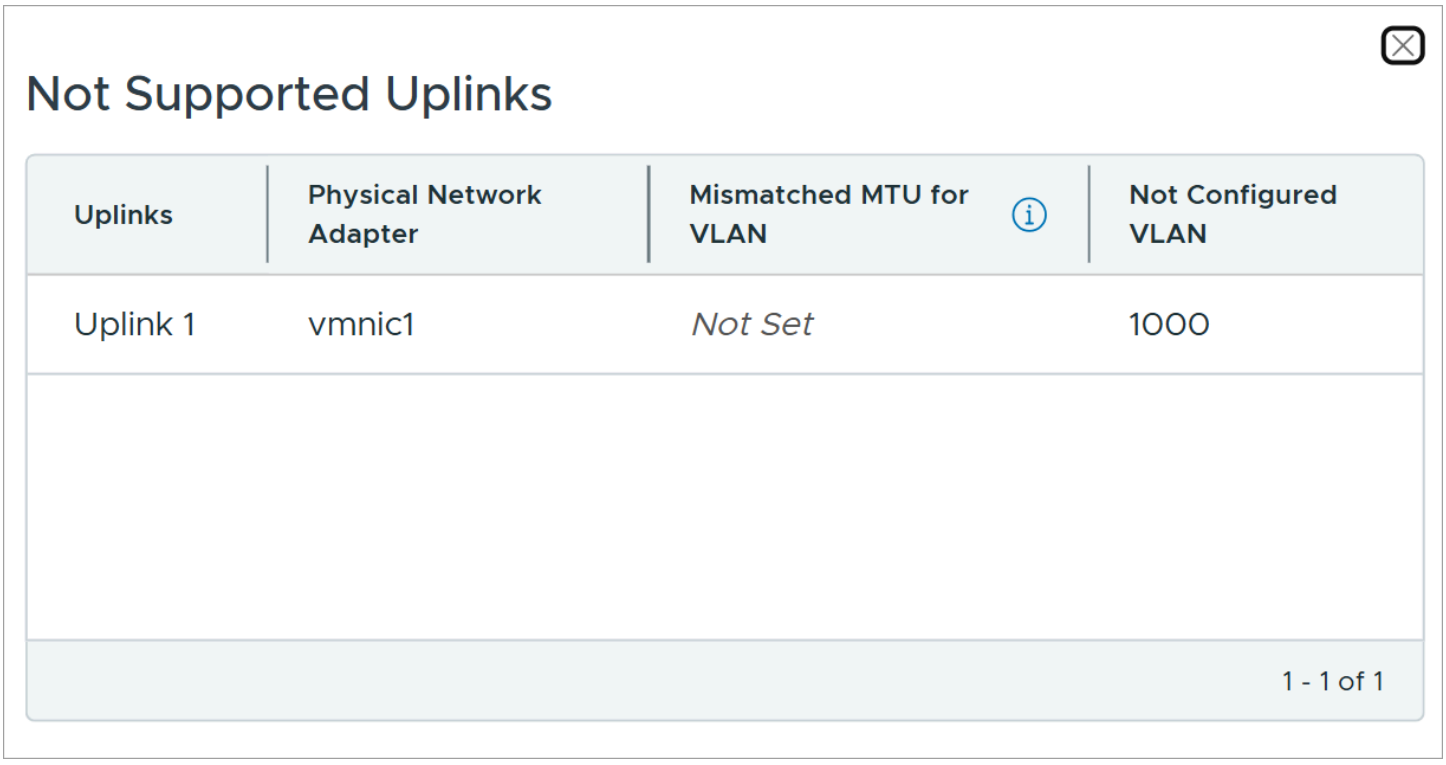


Supported Uplinks

Uplinks	Physical Network Adapter	Matched MTU for VLAN	Configured VLAN
uplink1	vmnic2	20	20
uplink2	vmnic3	20	20

1 - 2 of 2

Figure 413: Example Report for Untrunked VLAN

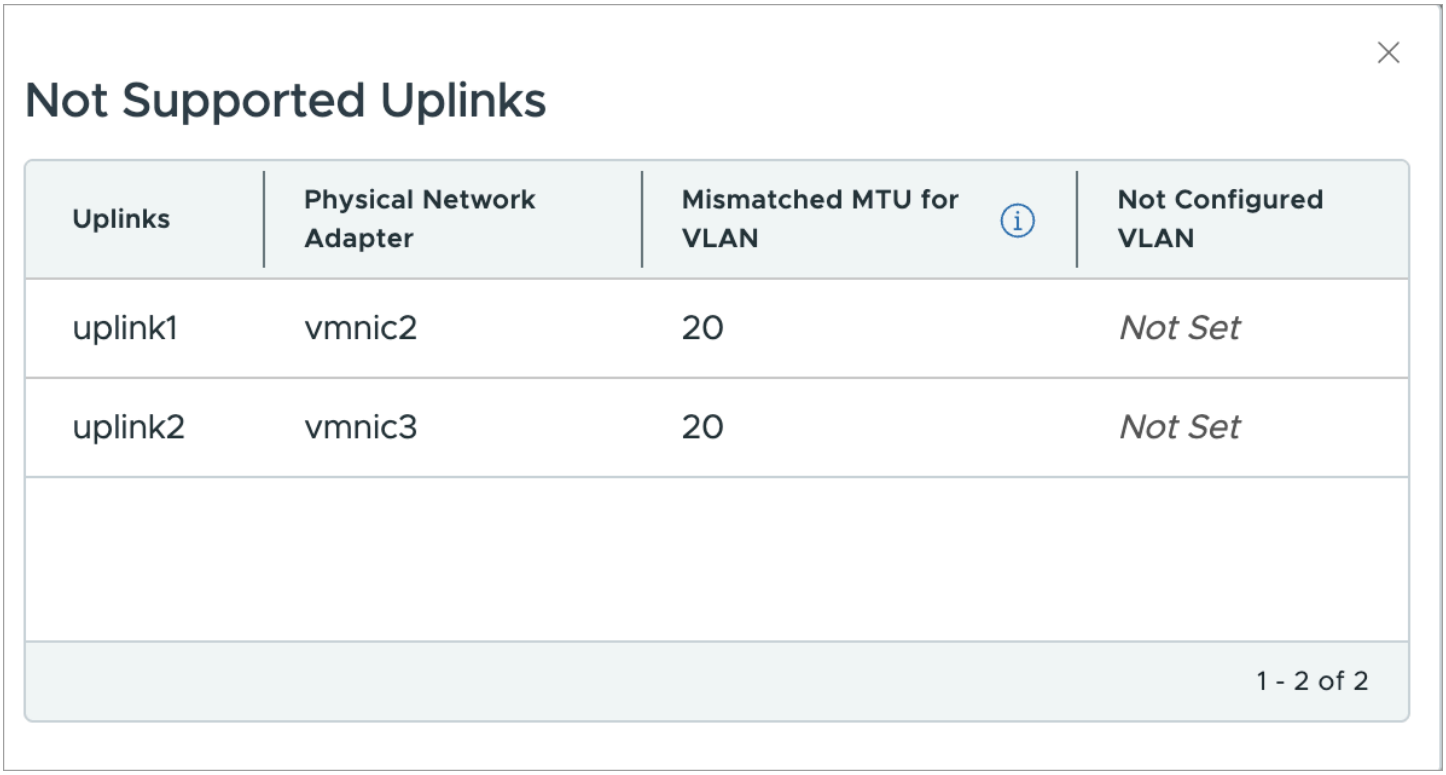


Not Supported Uplinks

Uplinks	Physical Network Adapter	Mismatched MTU for VLAN	Not Configured VLAN
Uplink 1	vmnic1	*Not Set*	1000

1 - 1 of 1

Figure 414: Example Report for Mismatched MTU



An unsupported uplink indicates that a problem exists with the VLAN or MTU configuration for that uplink. For more information, see "Troubleshooting VLAN and MTU Misconfigurations" later on this page.

Running the VLAN MTU Check Using the NSX API

You can also create and run a manual health check of the VLAN and MTU configuration using the following API calls. For detailed information, see the NSX API Guide.

- To create a manual health check:

```
POST https://<nsx-mgr>/policy/api/v1/infra/manual-health-checks/<manual-check-id>

Example payload:
{
    "cluster_scope_filter": {
        "cluster_external_id": "cfb040ff-71c6-4e7a-abd8-cae2a972742f:domain-c20",
        "host_switch_id": "50 33 53 f4 e7 b7 ab f4-f0 6a 37 20 40 16 1a 8a",
        "host_group_name": "test"
    },
    "vlans":{
      "vlan_ranges":[{
         "start": 0,
         "end": 10
       },{
         "start": 20,
         "end": 30
       }]
    }
}
```

- To fetch the results of a manual health check:

```
GET https://<nsx-mgr>/policy/api/v1/infra/manual-health-checks/<manual-check-id>/result
```

Troubleshooting VLAN and MTU Misconfigurations

The following table lists the possible VLAN and MTU misconfigurations and describes how to correct them.

Note:

Because VLAN MTU Check relies on bidirectional active probing between two uplinks, ensure that you have configured at least two uplinks to test in a cluster. You can configure these uplinks as either two uplinks in a single host or at least two hosts in the cluster with a single uplink.

Configuration Issue	Description	Remediation Steps
Misconfigured VLAN trunk	TEP transport VLANs are not properly trunked to the same top-of-rack (ToR) physical switch.	1. Verify the VLAN trunk configuration on the ToR interfaces that the ESX hosts connect to. 2. If the TEP transport VLANs are missing, add the VLANs to the ToR interface configurations.
Mismatched MTU values	MTU of the host switch (vSphere distributed switch) does not match MTU of the ToR physical switch.	1. Check the MTU settings on the host switch and on the ToR physical switch. 2. Adjust the settings so that the MTU of the host switch matches the MTU of the physical switch. For optimal performance, set both MTUs to the higher of the mismatched values.

Manually Deploying NSX Edge Node Outside of NSX Manager

In addition to deploying and configuring NSX Edge from the NSX Manager, you can manually deploy NSX Edge either as a VM in vCenter or a Bare Metal server.

Install an NSX Edge on ESX Using the vSphere GUI

You can use the vSphere Web Client or vSphere Client to interactively install an NSX Edge on ESX. You can also use the new Edge deployment workflow to install an Edge. For more information about using the workflow, see Setting up Network Connectivity. The same workflow is also available in vCenter. For more information, see *Building Your Cloud Infrastructure.* Although you can manually deploy an edge using the following procedure, Broadcom recommends the deployment of Edge through NSX Manager.

See NSX Edge network requirements in Installation Requirements.

1. Locate the NSX Edge node appliance OVA file on the Broadcom Support portal.
   Either copy the download URL or download the OVA file onto your computer.
2. In the vSphere Client, select the host on which to install NSX Edge node appliance.
3. Right-click and select **Deploy OVF template** to start the installation wizard.
4. Enter the download OVA URL or navigate to the saved OVA file, and click **Next**.
5. Enter a name and location for the NSX Edge node , and click **Next**.
   The name you type appears in the vCenter and vSphere inventory.
6. Select a compute resource for the NSX Edge node appliance, and click **Next**.
7. For optimal performance, reserve memory for the NSX Edge node.
   Set the reservation to ensure that the NSX Edge node has sufficient memory to run efficiently. See NSX Edge VM System Requirements.
8. Power on the appliance. Open the console of the NSX Edge node to track the boot process.
   If the console window does not open, make sure that pop-ups are allowed.

9. After the NSX Edge node starts, log in to the CLI with admin credentials.

   Note:
   After NSX Edge node starts, if you do not log in with admin credentials for the first time, the data plane service does not automatically start on the NSX Edge node.

10. Run the `get interface eth0` (without VLAN) to verify that the IP address was applied as expected.

```
nsx-edge-1> get interface eth0

Interface: eth0
  Address: 192.168.110.37/24
  MAC address: 00:50:56:86:62:4d
  MTU: 1500
  Default gateway: 192.168.110.1
  Broadcast address: 192.168.110.255
  ...
```

   Note:
   When bringing up NSX Edge nodes on non-NSX managed host, verify that the minimum MTU setting is set to 1600 (instead of 1500) on the physical host switch for the data NIC.

11. Run the `get managers` command to verify that the NSX Edge node is registered.

```
- 10.173.161.17  Connected (NSX-RPC)
- 10.173.161.140 Connected (NSX-RPC)
- 10.173.160.204 Connected (NSX-RPC)*
```

    The NSX Manager with * next to it is the active manager for the NSX Edge transport node VM.

    If an NSX Edge node is not registered with the management plane, see Join NSX Edge with the Management Plane.

12. Verify that the NSX Edge node has the required connectivity.

    If you enabled SSH, make sure that you can SSH to your NSX Edge node and verify the following:

    - You can ping your NSX Edge node management interface.
    - From the NSX Edge node, you can ping the node's default gateway.
    - From the NSX Edge node, you can ping the hypervisor hosts that are either in the same network or a network reachable through routing.
    - From the NSX Edge node, you can ping the DNS server and NTP Server.

13. Troubleshoot connectivity problems.

    Note:
    If connectivity is not established, make sure the VM network adapter is in the proper network or VLAN.

    By default, the NSX Edge node datapath claims all virtual machine NICs except the management NIC (the one that has an IP address and a default route). If you incorrectly assigned a NIC as the management interface, follow these steps to assign management IP address to the correct NIC.

    a) Log in to the NSX Edge CLI and type the `stop service dataplane` command.
    b) (Static IP) Run the `set interface <interface-name> ip <x.x.x.x/24> gateway <x.x.x.x> plane mgmt` command.
    c) (DHCP) Run the `set interface interface-name dhcp plane mgmt` command.
    d) Type the `start service dataplane` command.

       The datapath fp-ethX ports used for the VLAN uplink and the tunnel overlay are shown in the **get interfaces** and **get physical-port** commands on the NSX Edge node.

Configure NSX Edge as a transport node. See Edit Transport Node Configuration.

Install NSX Edge on ESX Using the Command-Line OVF Tool

If you prefer to automate NSX Edge installation, you can use the VMware OVF Tool, which is a command-line utility.

- Verify that the system requirements are met. See NSX Edge VM System Requirements.
- Verify that the required ports are open. Refer to VMware Ports and Protocols for more details.
- Verify that a datastore is configured and accessible on the ESX host.
- Verify that you have the IP address and gateway, DNS server IP addresses, domain search list, and the NTP Server IP or FQDN for the NSX Edge node to use.
- Create a management VDS and target VM port group in vCenter. Place the NSX appliances onto this management VDS port group network. See Prepare a vSphere Distributed Switch for NSX.
- See NSX Edge network requirements in NSX Edge Installation Requirements.
- Verify that you have adequate privileges to deploy an OVF template on the ESX host.
- Verify that hostnames do not include underscores. Otherwise, the hostname is set to *localhost*.
- OVF Tool version 4.3 or later.
- Know parameters that you can use to deploy a NSX Edge VM and join it to the management plane.

Field Name	OVF Parameter	Field Type
System root password	nsx_passwd_0	Required to install NSX Edge
CLI admin password	nsx_cli_passwd_0	Required to install NSX Edge.
CLI audit password	nsx_cli_audit_passwd_0	Optional
CLI admin username	nsx_cli_username	Optional
CLI audit username	nsx_cli_audit_username	Optional
NSX Manager IP	mpIp	Required to join NSX Edge VM to NSX Manager.
NSX Manager token	mpToken	Required to join NSX Edge VM to NSX Manager. To retrieve token, on the NSX Manager, run POST https://<nsx-manager>/api/v1/aaa/registration-token.
NSX Manager thumbprint	mpThumbprint	Required to join NSX Edge VM to NSX Manager. To retrieve thumbprint, on the NSX Manager node, run get certificate api thumbprint.
Node Id	mpNodeId	Only for internal use.
Hostname	nsx_hostname	Optional
Default IPv4 gateway	nsx_gateway_0	Optional
Management network IP address	nsx_ip_0	Optional
Management network netmask	nsx_netmask_0	Optional
DNS servers	nsx_dns1_0	Optional
Domain Search suffixes	nsx_domain_0	Optional
NTP Servers	nsx_ntp_0	Optional
Is SSH service enabled	nsx_isSSHEnabled	Optional
Is SSH enabled for root login	nsx_allowSSHRootLogin	Optional
Is autonomous Edge	is_autonomous_edge	Optional. Valid values: True, False (default)

- For a standalone host, run the ovftool command with the appropriate parameters.

```
C:\Users\Administrator\Downloads>ovftool
```

```
--name=nsx-edge-1
--deploymentOption=medium
--X:injectOvfEnv
--X:logFile=ovftool.log
--allowExtraConfig
--datastore=ds1
--net:"Network 0=Mgmt"
--net:"Network 1=nsx-tunnel"
--net:"Network 2=vlan-uplink"
--net:"Network 3=vlan-uplink"
--acceptAllEulas
--noSSLVerify
--diskMode=thin
--powerOn
--prop:nsx_ip_0=192.168.110.37
--prop:nsx_netmask_0=255.255.255.0
--prop:nsx_gateway_0=192.168.110.1
--prop:nsx_dns1_0=192.168.110.10
--prop:nsx_domain_0=corp.local
--prop:nsx_ntp_0=192.168.110.10
--prop:nsx_isSSHEnabled=True
--prop:nsx_allowSSHRootLogin=True
--prop:nsx_passwd_0=<password>
--prop:nsx_cli_passwd_0=<password>
--prop:nsx_hostname=nsx-edge
--prop:mpIp=<NSXManager-IP>
--prop:mpToken=<NSXManager-Token>
--prop:mpThumbprint=<NSXManager-Thumbprint>
--prop:is_autonomous_edge=False
<path/url to nsx component ova>
vi://root:<password>@192.168.110.51


Opening OVA source: nsx-<component>.ova
The manifest validates
Source is signed and the certificate validates
Opening VI target: vi://root@192.168.110.24
Deploying to VI: vi://root@192.168.110.24
Transfer Completed
Powering on VM: nsx-edge-1
Task Completed
Completed successfully
```

- For a host managed by vCenter, run the `ovftool` command with the appropriate parameters.

```
C:\Users\Administrator\Downloads>ovftool
--name=nsx-edge-1
--deploymentOption=medium
--X:injectOvfEnv
--X:logFile=ovftool.log
--allowExtraConfig
--datastore=ds1
--net:"Network 0=Mgmt"
--net:"Network 1=nsx-tunnel"
--net:"Network 2=vlan-uplink"
--net:"Network 3=vlan-uplink"
--acceptAllEulas
```

```
--noSSLVerify
--diskMode=thin
--powerOn
--prop:nsx_ip_0=192.168.110.37
--prop:nsx_netmask_0=255.255.255.0
--prop:nsx_gateway_0=192.168.110.1
--prop:nsx_dns1_0=192.168.110.10
--prop:nsx_domain_0=corp.local
--prop:nsx_ntp_0=192.168.110.10
--prop:nsx_isSSHEnabled=True
--prop:nsx_allowSSHRootLogin=True
--prop:nsx_passwd_0=<password>
--prop:nsx_cli_passwd_0=<password>
--prop:nsx_hostname=nsx-edge
--prop:mpIp=<NSXManager-IP>
--prop:mpToken=<NSXManager-Token>
--prop:mpThumbprint=<NSXManager-Thumbprint>
--prop:is_autonomous_edge=False
<path/url to nsx component ova>
vi://administrator@vsphere.local:<password>@192.168.110.24/?ip=192.168.210.53



Opening OVA source: nsx-<component>.ova
The manifest validates
Source is signed and the certificate validates
Opening VI target: vi://administrator@vsphere.local@192.168.110.24:443/
Deploying to VI: vi://administrator@vsphere.local@192.168.110.24:443/
Transfer Completed
Powering on VM: nsx-edge-1
Task Completed
Completed successfully
```

- Set the reservation to ensure that the NSX Edge node has sufficient memory to run efficiently. See NSX Edge VM System Requirements.
- Power on the appliance. Open the console of the NSX Edge node to track the boot process.
- After the NSX Edge node starts, log in to the CLI with admin credentials.
- Run the `get interface eth0` (without VLAN) to verify that the IP address was applied as expected.

```
nsx-edge-1> get interface eth0

Interface: eth0
  Address: 192.168.110.37/24
  MAC address: 00:50:56:86:62:4d
  MTU: 1500
  Default gateway: 192.168.110.1
  Broadcast address: 192.168.110.255
  ...
```

  Note:
  When bringing up NSX Edge nodes on non-NSX managed host, verify that the minimum MTU setting is set to 1600 (instead of 1500) on the physical host switch for the data NIC.
- Verify that the NSX Edge node has the required connectivity.

  If you enabled SSH, make sure that you can SSH to your NSX Edge node and verify the following:

  - You can ping your NSX Edge node management interface.

  - From the NSX Edge node, you can ping the node's default gateway.
  - From the NSX Edge node, you can ping the hypervisor hosts that are either in the same network or a network reachable through routing.
  - From the NSX Edge node, you can ping the DNS server and NTP Server.
- Troubleshoot connectivity problems.

  Note:
  If connectivity is not established, make sure the VM network adapter is in the proper network or VLAN.

  By default, the NSX Edge node datapath claims all virtual machine NICs except the management NIC (the one that has an IP address and a default route). If you incorrectly assigned a NIC as the management interface, follow these steps to assign management IP address to the correct NIC.

  a) Log in to the NSX Edge CLI and type the `stop service dataplane` command.
  b) (Static IP) Run the `set interface <interface-name> ip <x.x.x.x/24> gateway <x.x.x.x> plane mgmt` command.
  c) (DHCP) Run the `set interface interface-name dhcp plane mgmt` command.
  d) Type the `start service dataplane` command.

     The datapath fp-ethX ports used for the VLAN uplink and the tunnel overlay are shown in the **get interfaces** and **get physical-port** commands on the NSX Edge node.

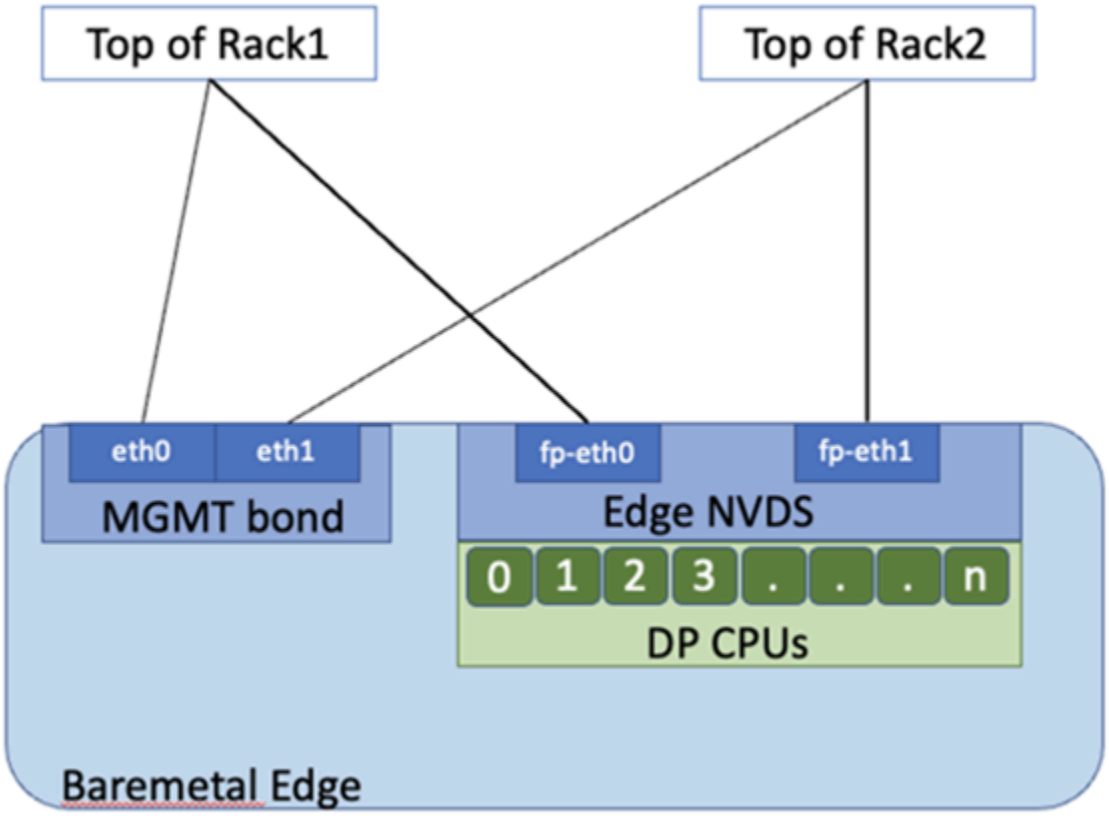
If you did not join the NSX Edge with the management plane, see Join with the Management Plane.

Install NSX Edge on Bare Metal

Use PXE server to automate installation of NSX Edge on a bare metal server or use ISO file to install NSX Edge on a bare metal server.

- Disable sub-NUMA clustering by editing the BIOS settings. NSX does not support sub-NUMA clustering. For more details, see Knowledge Base article 320664: Enabling sub-NUMA clustering on a Bare Metal Edge may result in a shortage of heap memory.
- Starting with NSX v3.1.3, on Bare Metal NSX Edge nodes, you do not need to disable hyperthreading. Hyperthreading is automatically disabled.

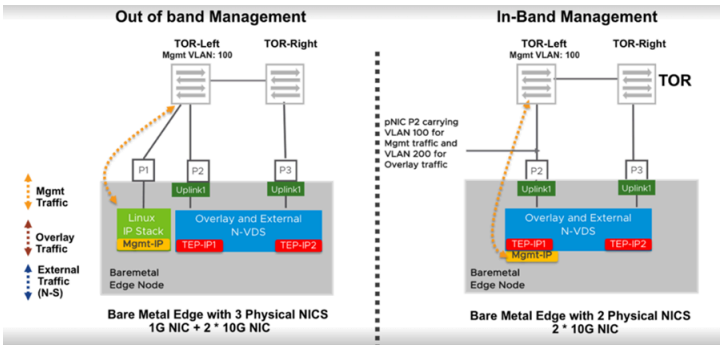
The NSX Edge bare metal node is a dedicated physical server that runs a special version NSX Edge software. The bare metal NSX Edge node requires a NIC supporting the Data Plane Development Kit (DPDK). VMware maintains a list of the compatibility with various vendor NICs. See NSX Edge Bare Metal Requirements.



The NSX Edge nodes in the diagram are illustrated with a single N-VDS switch, configured with two datapath interfaces and two dedicated NICs for high availability of management plane.

Bare Metal NSX Edge nodes can be configured with more than 2 datapath interfaces depending on the number of NICs available on the server. Bare Metal NSX Edge nodes use pNICs as uplinks that directly connect to the top of rack switches. To provide high availability for bare metal edge management, configure two pNICs on the server as an active/standby Linux bond.

The CPUs on the edge are assigned as either datapath (DP) CPUs that provide routing and stateful services or services CPUs that provide load balancing and VPN services.



When a bare metal Edge node is installed, a dedicated interface is retained for management. This configuration is called out-of-band management. If redundancy is desired, two NICs can be used for management plane high availability. Bare metal Edge also supports in-band management where management traffic can leverage an interface being used for overlay or external (N-S) traffic as in the diagram.

For bare metal edge nodes, all the cores from the first node of a multi-NUMA node server will be assigned to NSX datapath. If the bare metal has only one NUMA node, then 50% of the cores will be assigned for the NSX datapath.

Bare Metal NSX Edge nodes support a maximum of two NUMA nodes.

Important:
VMware does not support the sub-NUMA clustering, which is a server hardware feature, on a Bare Metal NSX Edge is not supported as it may result in a shortage of heap memory.

The sub-NUMA clustering functionality changes the socket of heap memory from two NUMA domains to four NUMA domains. This change limits the size of heap memory allocated to each socket and causes a shortage of heap memory for socket 0 that the datapath requires. You must disable the sub-NUMA functionality in the BIOS. Any changes made to the BIOS will require a reboot.

To check if the sub-NUMA functionality is enabled, log in to the bare metal NSX Edge as root and run `lscpu`. The output is also captured in the support bundle. If there are more than two NUMA nodes, it implies that the sub-NUMA functionality is enabled and it must be disabled and must be disabled via the BIOS.

Note:

1. When configuring LACP LAG bonds on Bare Metal NSX Edge nodes, datapath cores (backing NICs) should belong to same NUMA node for load balancing to occur on both devices. If devices forming the bond span multipe NUMA nodes, then bond only uses network device CPU that is local NUMA node (0) to transmit packets. So, not all devices do not get used for balancing traffic that is sent out of the bond device.
   In this case, failover still works as failover is exclusive of load balancing. If the Ethernet device attached to the local NUMA node is down, then the bond sends traffic to the other device even though it is not NUMA local. The load-balancing optimization does not impact failover functionality.
   Run `get dataplane` command to view the NUMA node associated with each datapath interface. To move nics associated with datapath to single NUMA node, physical reconfiguration of the server is required via the BIOS.

Prepare a PXE Server for Bare Metal NSX Edge Installation

PXE is made up of several components: DHCP, HTTP, and TFTP. This procedure demonstrates how to set up a PXE server on Ubuntu.

- A PXE server must be available in your deployment environment. The PXE server can be set up on any Linux distribution.
- Verify that the preseeded configuration file has the parameters net.ifnames=0 and biosdevname=0 set after `--` to persist after reboot.
- See NSX Edge Bare Metal Requirements.
- See NSX Edge network requirements in NSX Edge Installation Requirements.

DHCP dynamically distributes IP settings to NSX components, such as NSX Edge. In a PXE environment, the DHCP server allows NSX Edge to request and receive an IP address automatically.

TFTP is a file-transfer protocol. The TFTP server is always listening for PXE clients on the network. When it detects any network PXE client asking for PXE services, it provides the NSX component ISO file and the installation settings contained in a preseed file.

1. Optional: Use a kickstart file to set up a new TFTP or DHCP services on an Ubuntu server.
   A kickstart file is a text file that contains CLI commands that you run on the appliance after the first boot.

   Name the kickstart file based on the PXE server it is pointing to. For example:
   ```
   nsxcli.install
   ```
   The file must be copied to your Web server, for example at `/var/www/html/nsx-edge/nsxcli.install`.

   In the kickstart file, you can add CLI commands. For example, to configure the IP address of the management interface:

   ```
   stop service dataplane
   set interface eth0 <ip-cidr-format> plane mgmt
   start service dataplane
   ```

   To change the admin user password:

   ```
   set user admin password <new_password> old-password <old-password>
   ```

If you specify a password in the preseed.cfg file, use the same password in the kickstart file. Otherwise, use the default password, which is "default".

To join the NSX Edge with the management plane:

```
join management-plane <manager-ip> thumbprint <manager-thumbprint> username <manager-username> password
 <manager password>
```

2. Create two interfaces, one for management and another for DHCP and TFTP services.

   Make sure that the DHCP/TFTP interface is in the same subnet that the NSX Edge resides in.

   For example, if the NSX Edge management interfaces are going to be in the 192.168.210.0/24 subnet, place eth1 in that same subnet.

```
# The loopback network interface
auto lo
iface lo inet loopback

# PXE server's management interface
auto eth0
iface eth0 inet static
  address 192.168.110.81
  gateway 192.168.110.1
  netmask 255.255.255.0
  dns-nameservers 192.168.110.10

# PXE server's DHCP/TFTP interface
auto eth1
iface eth1 inet static
  address 192.168.210.82
  gateway 192.168.210.1
  netmask 255.255.255.0
  dns-nameservers 192.168.110.10
```

3. Install DHCP server software and configure required settings to set up the PXE server. For more details, see Linux documentation.
4. Install the Apache server and TFTP and other components required to configure the PXE server.
5. Copy or download the NSX Edge installer ISO file to a temporary folder.
6. Mount the ISO file and copy the install components to the TFTP server and the Apache server.

```
            sudo mount -o loop ~/nsx-edge.<build>.iso /mnt
            cd /mnt
sudo cp -fr install/netboot/* /var/lib/tftpboot/
sudo mkdir /var/www/html/nsx-edge
sudo cp -fr /mnt/* /var/www/html/nsx-edge/
```

7. Optional: Edit the `/var/www/html/nsx-edge/preseed.cfg` file to modify the encrypted passwords.

   You can use a Linux tool such as mkpasswd to create a password hash.

```
sudo apt-get install whoissudo mkpasswd -m sha-512

Password:
$6$SUFGqs[...]FcoHLijOuFD
```

   a) Modify the root password, edit `/var/www/html/nsx-edge/preseed.cfg` and search for the following line:

```
d-i passwd/root-password-crypted password $6$tgmLNLMp$9BuAHhN...
```

b) Replace the hash string.

You do not need to escape any special character such as $, ', ", or \.

c) Add the `usermod` command to `preseed.cfg` to set the password for root, admin, or both.

For example, add the following command.

```
usermod --password '\$6\$VS3exId0aKmzW\$U3g0V7BF0DXlmRI.LR0v/VgloxVotEDp00bO2hUF8u/' root; \
usermod --password '\$6\$VS3exId0aKmzW\$U3g0V7BF0DXlmRI.LR0v/VgloxVotEDp00bO2hUF8u/' admin; \
```

The hash string is an example. You must escape all special characters. The root password in the first `usermod` command replaces the password that is set in `d-i passwd/root-password-crypted password $6$tgm....`

If you use the `usermod` command to set the password, the user is not prompted to change the password at the first login. Otherwise, the user must change the password at the first login.

8. Add the following lines to the `/var/lib/tftpboot/pxelinux.cfg/default` file.

Replace 192.168.210.82 with the IP address of your TFTP server.

```
label nsxedge
    kernel ubuntu-installer/amd64/linux
    ipappend 2
    append netcfg/dhcp_timeout=60 auto=true priority=critical vga=normal partman-lvm/
device_remove_lvm=true netcfg/choose_interface=auto debian-installer/allow_unauthenticated=true preseed/
url=http://192.168.210.82/nsx-edge/preseed.cfg mirror/country=manual mirror/http/hostname=192.168.210.82
 nsx-kickstart/url=http://192.168.210.82/nsx-edge/nsxcli.install mirror/http/directory=/nsx-edge
 initrd=ubuntu-installer/amd64/initrd.gz mirror/suite=bionic netcfg/do_not_use_netplan=true --
```

9. Add the following lines to the `/etc/dhcp/dhcpd.conf` file.

Replace 192.168.210.82 with the IP address of your DHCP server.

```
allow booting;
allow bootp;

next-server 192.168.210.82; #Replace this IP address
filename "pxelinux.0";
```

10. Restart the DHCP service.

```
sudo service isc-dhcp-server restart
```

Note:
If an error is returned, for example: "stop: Unknown instance: start: Job failed to start", run `sudo /etc/init.d/isc-dhcp-server stop` and then `sudo /etc/init.d/isc-dhcp-server start`. The `sudo /etc/init.d/isc-dhcp-server start` command returns information about the source of the error.

Install NSX Edge on bare metal using an ISO file. See Install Bare Metal NSX Edge Automatically Using ISO File.

Install Bare Metal NSX Edge Automatically Using ISO File

Use the automated installation of NSX Edge on the bare metal servers because it automatically configures the NSX Edge node according to the hardware configuration and the most frequently used NSX Edge configuration. After the installation is complete, configure the management interface of the nodes as out-of-band or as in-band and provide the final IP configuration of the management interface statically or over DHCP.

- Both BIOS and UEFI boot modes are supported.
- See NSX Edge Bare Metal Requirements.
- If the bare metal servers for NSX Edge nodes have integrated SD card devices, disable those SD cards. For more information, see Knowledge Base article 321176: Booting a Bare Metal server with an NSX Edge ISO fails with the error: "Incorrect CD-ROM detected".
- To prepare a bare metal server as an NSX Edge node, ensure there is a minimum of 200 GB disk space.

For more details about the supported NICs, see Broadcom Compatibility Guide.

1. Locate and download the ISO file for NSX Edge for Bare Metal on the Broadcom Support portal.
2. Log in to the Integrated Lights-Out (ILO) interface of the bare metal server.
3. Click **Launch** in the virtual console preview.
4. Select **Virtual Media** > **Connect Virtual Media**.
   Wait a few seconds for the virtual media to connect.
5. Select **Virtual Media** > **Map CD/DVD** and browse to the ISO file.
6. Select **Next Boot** > **Virtual CD/DVD/ISO**.
7. Select **Power** > **Reset System (warm boot)**.
   The installation duration depends on the bare metal environment.
8. Choose **Automated installation**.
   There might be a pause of 10 seconds after you press Enter.
9. Select the applicable primary network interface. This is for the management network interface.

   During power-on, the installer requests a network configuration. Select static IP settings. If the static IP is not available, use DHCP.

   By default, the root login password is **vmware**, and the admin login password is **default**.

   Note:
   During deployment, the installer automatically selects the largest disk to install NSX on the NSX Edge node.
10. There are three ways to configure a management interface.
    - Untagged interface. This interface type creates an out-of-band management interface.

      (Static) `set interface eth0 ip <CIDR> gateway <gateway-ip> plane mgmt`

      (DHCP) `set interface eth0 dhcp plane mgmt`
    - Tagged interface.

      `set interface eth0 vlan <vlan_ID> plane mgmt`

      (Static) `set interface eth0.<vlan_ID> ip <CIDR> gateway <gateway-ip> plane mgmt`

      (DHCP) `set interface eth0.<vlan_ID> dhcp plane mgmt`
    - In-band interface.

      `set interface mac <mac_address> vlan <vlan_ID> in-band plane mgmt`

      (Static) `set interface eth0.<vlan_ID> ip <CIDR> gateway <gateway-ip> plane mgmt`

      (DHCP) `set interface eth0.<vlan_ID> dhcp plane mgmt`
    - Tagged interface and In-band interface.

      Any existing VLAN management interface must be cleared before creating a new one.

      `clear interface eth0.<vlan_ID>`
    - (Optional) Create a `bond0` interface for management HA interface with multiple interfaces.

      You can configure a bond management interface on an NSX Edge using the following CLI command. Use console to clear existing mangement IP before you create a bond and add an interface to it.

      Note:
      Only active-backup mode is allowed on a bond interface. You can configure VLANs on a bond0 interface.

      Create a bond interface

      ```
      set interface bond0 ip x.x.x.x/mask gateway x.x.x.x plane mgmt mode active-backup members
      eth0,eth1 primary eth0
      ```

      Create vlan inteface on bond0

```
set interface bond0 vlan Y plane mgmt
```

Assign IP address to bond0.yyy

```
set interface bond0.yyy ip x.x.x.x/24 gateway z.z.z.z plane mgmt
```

11. Set physical NICs to be used by NSX dataplane from the list of available PCI devices.
    1. `get dataplane device list`
    2. `reset dataplane device list`
    3. `restart service dataplane`
    4. `get physical-port`

    After selecting physical NICs, restart NSX dataplane services for changes to take effect.

    Note:
    Starting in NSX 3.1.2, you can claim up to 16 physical NICs.

    Note:
    To configure custom NICs for dataplance, run the `set dataplane device list <NIC1>, <NIC2>, <NIC3>` command.
12. To avoid network configuration errors, verify that the physical NICs selected match the NICs configured in the uplink profiles.

Install Bare Metal NSX Edge Interactively Using ISO File

Install NSX Edge devices on bare metal using an ISO file in the interactive mode.

- Starting in NSX 3.2 both UEFI or Legacy BIOS modes are supported.
- See NSX Edge network requirements in NSX Edge Installation Requirements.

1. Locate and download the ISO file for NSX Edge for Bare Metal from the Broadcom Support portal.
2. Log in to the ILO of the bare metal.
3. Click **Launch** in the virtual console preview.
4. Select **Virtual Media** > **Connect Virtual Media**.
   Wait a few seconds for the virtual media to connect.
5. Select **Virtual Media** > **Map CD/DVD** and browse to the ISO file.
6. Select **Next Boot** > **Virtual CD/DVD/ISO**.
7. Select **Power** > **Reset System (warm boot)**.
   The installation duration depends on the bare metal environment.
8. Choose **Interactive Install**.

   This operation takes several seconds to start.
9. In the Configure the keyboard window, select `Yes` if the installer must auto-detect the keyboard or select `No` if the keyboard must not be detected by the console.
10. Select English US as the language.
11. In the Configure the network window, select the applicable primary network interface.
12. Enter the host name that connects to the selected primary interface and click `Ok`.

    During power-on, the installer requests a network configuration mode. Select static IP address and provide one. If static IP address is not available, select DHCP.

    By default, the root login password is **vmware**, and the admin login password is **default**.
13. In the Configure NSX appliance using kickstart window:
    - Enter the URL of the NSX kickstart config file if you want to automate NSX configuration on the bare metal server.

- Leave the field blank if you want to manually configure NSX on the bare metal server.

14. Select a disk size that meets the system requirements.
15. In the Partition disks window, choose one of the following options:
    - Select `Yes` if you want to unmount existing partitions so that new partitions can be created on disks.
    - Select `No` if you want to use existing partitions.

Intel QAT Support for IPSec VPN Bulk Cryptography

Beginning with the NSX 3.0 release, support for the Intel QuickAssist Technology (QAT) is provided on bare metal servers. Intel QAT provides the hardware acceleration for various cryptography operations.

The QAT feature is enabled by default if the NSX Edge is deployed on a bare metal server with an Intel QuickAssist PCIe card that is based on the installed C62x chipset (Intel QuickAssist Adapter 8960 or 8970). The single root I/O virtualization (SR-IOV) interface must be enabled in the BIOS firmware.

To check the status of the QAT feature, enter the following command on the NSX Edge bare metal server CLI.

```
get dataplane qat
```

The possible responses you might receive are listed in the following table.

Status of QAT Feature	Definition
`QAT present, enabled, running`	The QAT feature is enabled and running.
`QAT present, enabled, not running`	The QAT feature has been enabled, but the dataplane service must be restarted for the status change to take effect.
`QAT present, disabled, not running`	The QAT feature is disabled.
`QAT present, disabled, running`	The QAT feature has been disabled, but the dataplane service must be restarted for the status change to take effect.
`QAT not present`	The bare metal server on which you ran the CLI command does not have a QAT device installed.
`QAT not supported in VM`	You ran the CLI command on a VM edge.

To disable or enable the use of an installed QAT device, use the following CLI commands. The expected responses are also shown.

```
set dataplane qat disabled
QAT disabled. Please restart service dataplane to take effect.

set dataplane qat enabled
QAT enabled. Please restart service dataplane to take effect.
```

> **Important:**
> You must enter the `restart service dataplane` command at the CLI prompt for the QAT feature status change to take effect.

Join NSX Edge with the Management Plane

To establish communication between NSX Edges and NSX Manager or NSX Manager cluster, join NSX Edges with NSX Manager. You only need to register NSX Edges with one NSX Manager to ensure communication with the management plane.

Verify that you have admin privileges to log in to the NSX Edges and NSX Manager appliance.

1. Open an SSH session or console session to one of the NSX Manager appliances.
2. Open an SSH session or console session to the NSX Edge node VM.
3. To retrieve the thumbprint of the NSX Manager appliance, at the NSX Manager appliance console, run the `get certificate api thumbprint` command.

   The command output is a string of alphanumeric numbers that is unique to this NSX Manager.

   For example:

   ```
   NSX-Manager1> get certificate api thumbprint
   659442c1435350edbbc0e87ed5a6980d892b9118f851c17a13ec76a8b985f57
   ```

   Alternatively, to retrieve the thumbprint of the cluster, at the NSX Manager appliance console, run `get certificate cluster thumbprint`.

   The output is a string of alphanumeric numbers that is unique to the cluster.
4. To join the NSX Edge node (VM or Bare Metal) to the NSX Manager appliance, run the **join management-plane** command.

   Provide the following information:

   - Hostname or IP address of the NSX Manager with an optional port number
   - User name of the NSX Manager
   - Certificate thumbprint of the NSX Manager
   - Password of the NSX Manager

   ```
   NSX-Edge1> join management-plane <Manager-IP> thumbprint <Manager-thumbprint> username admin
   ```

   Repeat this command on each NSX Edge node VM.
5. If cluster VIP is configured for the NSX Manager, then join NSX Edge Node using the cluster thumbprint. Run the **join management-plane** command.

   Provide the following information:

   - Virtual IP address of the NSX Manager cluster with an optional port number
   - User name of the NSX Manager
   - Certificate thumbprint of the NSX Manager cluster
   - Password of the NSX Manager

   ```
   NSX-Edge1> join  management-plane <Cluster-VIP> username <Manager-username> password <Manager-password> thumbprint <Cluster-tumbprint>
   ```
6. Verify the result by running the `get managers` command on your NSX Edge node VMs.

   ```
   nsx-edge-1> get managers
   - 10.173.161.17  Connected (NSX-RPC)
   - 10.173.161.140 Connected (NSX-RPC)
   - 10.173.160.204 Connected (NSX-RPC)
   ```
7. In the NSX Manager UI, navigate to .

   On the NSX Edge Transport Node page:

   - The **Configuration State** column displays `Configure NSX`. Click `Configure NSX` to begin configuration on the node. If the **NSX Version** column does not display the version number installed on the node, try refreshing the browser window.
   - Before you configure NSX on the NSX Edge node, the **Node Status** and **Tunnel Status** columns display state `Not Available`. The **Transport Zones** and **N-VDS** switches columns display value `0`, indicating there are no transport zones attached or N-VDS switches configured on the NSX Edge node.

When installing NSX Edge using NSX Manager see Create an NSX Edge Cluster and Add Edge Nodes.

When installing NSX Edge manually, see Edit Transport Node Configuration.

Edit an NSX Edge Node Configuration

After manually installing an NSX Edge VM on an ESX host or as a Bare Metal server, you can edit the NSX Edge node configuration.

Before you configure an NSX Edge node, ensure that you have met the "Essential Prerequisites" described in Create an NSX Edge Cluster and Add Edge Nodes.

If you plan to use optional features such as UPT mode or host group affinity, ensure that you have also met the applicable "Optional Prerequisites" described in Create an NSX Edge Cluster and Add Edge Nodes.

To edit the configuration of an NSX Edge node in NSX Manager, navigate to **System** > **Fabric** > **Edges** and select the node in the list. Click the node Actions menu (three dots) and select **Edit**.

The on-screen workflow for editing an NSX Edge node is divided into sequential pages that allow you to configure the following settings:

- Node placement settings.
- Management network settings.
- Uplink mapping for datapath interfaces.
- (Optional) Advanced node settings.

For detailed instructions on how to configure these settings, see Create an NSX Edge Cluster and Add Edge Nodes.

Remove NSX Edge Nodes from an Edge Cluster

Remove NSX Edge nodes with Tier-1 Gateways or Tier-0 Gateway configured with service router (SR), DHCP and metadata proxy.

Before you remove NSX Edge nodes, relocate the gateway configurations to a new standby node:

- To remove Tier-0 gateway configurations on an NSX Edge node, you must manually relocate Tier-0 configurations such as Tier-0 SR, DHCP and metadata proxy configurations to a standby NSX Edge node.
- To remove Tier-1 gateway configurations on an NSX Edge node, do the following in these scenarios:
  - If Tier-1 SR, DHCP and metadata proxy configurations are auto allocated to the NSX Edge node, you can enable the standby relocation functionality to relocate Tier-1 configurations to a new standby NSX Edge node. The procedure describes how to use the standby relocation functionality to relocate the configurations to a new standby node.
  - If Tier-1 SR, DHCP and metadata proxy configurations are manually allocated to the NSX Edge node, you need to manually relocate Tier-1 configurations to a new standby NSX Edge node.

1. From a browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.
2. On NSX Edge nodes configured with Tier-0 configurations, manually move configurations to some other NSX Edgenode.

   Note:
   The standby relocation functionality does not relocate Tier-0 configurations such as Tier-0 SRs, DHCP and metadata proxy configurations to a standby NSX Edge node.

   a) Select **NetworkingTier-0 Gateways**.
   b) To edit a Tier-0 Gateway, select the gateway, click vertical ellipses and click **Edit**.
   c) Navigate to **Interfaces** section and click the **External and Service Interfaces**.
   d) In the **Set Interfaces** window, edit the interface configured for NSX Edge node.
   e) Remove the existing NSX Edge node associated with the interface and select a new NSX Edge node that is configured with the same VLAN connectivity required for the interface and click **Save**.

   The above procedure deletes Tier-0 SRs, DHCP and metadata proxy on Tier-0 Gateways of the existing NSX Edge and moves them to the new NSX Edge node.

f) Delete the NSX Edge node from the edge cluster.

3. On NSX Edge nodes that are auto allocated with Tier-1 SR, DHCP and metadata proxy configurations, follow these steps to trigger the standby relocation functionality to relocate those configurations and remove the NSX Edge node from the NSX Edge cluster:
   a) (Optional) For faster failover, change the BFD timers for NSX Edge cluster by setting it to `500` ms.
   b) Apply the new NSX Edge cluster profile to the transport node profile. It ensures faster failover when the NSX Edge node is powered off.
   c) Select **Networking** > **Tier1 Gateways**.
   d) To edit a Tier-1 Gateway, select the gateway, click vertical ellipses and click **Edit**.
   e) In the Edit view, select the NSX Edge cluster and enable the **Enable Standby Relocation** field.

      Important:
      For standby relocation to function successfully, there must be an additional healthy NSX Edge node in the edge cluster. During the process of removing an NSX Edge node, Tier-1, DHCP or metadata proxy configurations are relocated from an existing NSX Edge to a new standby node.
   f) Select **System** > **Fabric** > **Profiles** > **Edge Cluster Profiles**.
   g) Select the edge cluster profile and click **Edit**.
   h) Set **Standby Relocation Threshold (mins)** that is applied to the edge cluster. The default recommended value is 30 mins and the minimum value is 10 mins.

      Note:
      Only auto allocated Tier-1 SR, DHCP and metadata proxy configurations are relocated to a standby NSX Edge. If the NSX Edge node to be removed contains any manually allocated configurations, such configurations will not be relocated out from existing NSX Edge node to a stanby node. You need to manually change the allocation for those Tier-1 configurations.
   i) Power off the NSX Edge without taking down the node into maintenance mode. If the NSX Edge is running any active service, then all such active service configurations will failover to another NSX Edge because of the HA failover trigger when the node is powered off.
   j) Wait for the duration of standby relocation threshold timeout. After the threshold limit is reached, all Tier-1 service configurations that have standby relocation enabled will be removed from the edge node being powered-off and relocated to some other NSX Edge in the cluster. There can be minor delays in standby relocation to perform the relocation task.
   k) After Tier-1 configurations are relocated to a standby node, remove the NSX Edge that was powered off from the cluster. Select the Edge Cluster, click **Edit** and remove the NSX Edge node from the cluster anc click **Save**.

Relocate and Remove an NSX Edge Node from an NSX Edge Cluster

You can use the NSX relocate and remove API to relocate the service configurations of an NSX Edge node to another standby NSX Edge node in the same NSX Edge cluster and then remove the Edge node from the Edge cluster.

To relocate and remove an Edge node from an Edge cluster, the following conditions are required:

- The Edge node must not have any manually allocated service configurations. Only auto allocated service configurations can be relocated.
- To be available for relocation, standby Edge nodes must not be configured with Layer 2 bridging.
- The Edge cluster must have at least two healthy Edge nodes where the auto allocated service configurations can be relocated to.
- For HA (high availability), the Edge cluster must have more than two Edge nodes that are possible for relocation.

The relocate and remove API relocates the following service configurations:

- Logical routers
- DHCP server
- Metadata proxy
- L2 forwarder

1. Run the API command to get the `member_index` value of the Edge node that you want to relocate and remove from an Edge cluster:

```
GET https://<nsx-manager-IP>/policy/api/v1/edge-clusters/<edge-cluster-id>

{
   "deployment_type": "VIRTUAL_MACHINE",
   "members": [
       {
           "member_index": 11,
           "transport_node_id": "21a19cbf-eaba-4a59-b18d-ff71fe5d76aa",
           "display_name": "edgeVm1New"
       },
       {
           "member_index": 13,
           "transport_node_id": "740cf97d-892b-47bb-97e7-889d92252e80",
           "display_name": "edgeVm2New"
       },
       {
           "member_index": 14,
           "transport_node_id": "cd5ab447-a36a-4bc3-94ff-0a4eea9fb2ad",
           "display_name": "edgeVm3New"
       }
   ],
```

   The `member_index` value is used to specify the Edge node to relocate and remove. Assume that you want to relocate the service configurations for the Edge node named `edgeVm1New`, then its `member_index` value is `11`.

2. Enter the relocate and remove API command and the `member_value` value of the Edge node to relocate and remove:

```
POST https://<nsx-manager-IP>/api/v1/edge-clusters/<edge-cluster-id>?action=relocate_and_remove

{
   "member_index": 11
}
```

3. Run the API command.

   The Edge node enters maintenance mode and its service configurations are transferred to one of the standby Edge nodes in the cluster. After the service configurations are transferred, the Edge node is removed from the Edge cluster and exits maintenance mode.

   Note:
   The API command will not work if:

   - The Edge node has any manually allocated service configurations.
   - The Edge cluster does not have at least two healthy standby Edge nodes.

   CAUTION:

   It is possible for the API command to give a success response, but in the background, the relocation operation fails. If this scenario occurs, then an alarm with the **Event Type** of `Edge Cluster Member Relocate Failure` is raised.

   The recommended action for this scenario is to review the available capacity of the Edge cluster. If more capacity is required, scale your Edge cluster and then retry the API command.

Enable or Disable AutoRefresh on Edge Nodes

Enable or Disable AutoRefresh on Edge nodes.

Starting with 3.2.4, NSX introduces an AutoRefresh flag that will override any mismatch between the state you desire for an NSX Edge node and the state NSX realizes for that NSX Edge server. NSX will not generate mismatch alarms

if desired state and realized state is different. Before NSX 3.2.4, NSX generated mismatch alarms when there was a mismatch between desired and realized state.

If you are upgrading to NSX 3.2.4 or installing NSX 3.2.4, the AutoRefresh feature will be enabled by default.

To modify the AutoRefresh feature, run the following API.

```
PATCH https://{mp}/policy/api/v1/system-config

 Sample Payload:

 {
   "keyValuePairs": [
    {
     "key": "auto_refresh_edge_transport_nodes",
     "value": "true/false"
    }
     ]
 }
```

If you set the value to false, NSX will continue to generate mismatch alarms.

> **Note:**
>
> (NSX Edge nodes deployed from NSX Manager) If you change vSphere parameters from the vSphere Client UI, the desired state will be overriden with the realized state after the next Periodic Refresh task.
>
> If you edit an NSX Edge node from the NSX Edge UI page, mismatch alarm will be generated if there is a mismatch between desired and realized state.

Setting up Network Connectivity

NSX provides a simplified workflow to set up network connectivity and perform network configuration. The workflow improves the operational efficiency of VPC environments by providing a simple way to establish external connectivity and installing Edge nodes and Edge clusters. This simplified workflow for setting up network connectivity is available in both NSX and in vCenter and thus users have a uniform experience of performing network configurations from either of the components.

To simplify the deployment process, the new workflow leverages many defaults settings that are recommended by Broadcom, such as transport zones and uplink profile, that reduces the number of user inputs required to perform network configuration. The workflow also provides a topology visualization that includes user inputs. Contextual visualization helps users to understand the topology better through which they can avoid misconfiguration as they get a view of the entire set up.

If you are using the workflow in a fresh installation, you will need to start with setting up a network connectivity, Centralized or Distributed. You can choose based on the following information.

- Centralized Connectivity: Create a Centralized connection if you have an environment that requires full-scale NSX networking services, such as DHCP, NAT, and Layer 3 services. It connects the Transit Gateway through a Tier-0 router which is provisioned from NSX. The Tier-0 is deployed on NSX Edge nodes and provides the interconnection to the physical fabric. For more information about adding a Transit Gateway with centralized connection, see Add a Transit Gateway with a Centralized Connection.

  Centralized connectivity supports stateful services either at Tier-0 level or at the Transit Gateway level (NAT) and supports dynamic routing, BGP/OSPF, with the physical fabric. The Centralized connectivity requires the setting up of Edge nodes.
- Distributed Connectivity: Distributed Connection allows a distributed connection to the data-center fabric that requires a minimal physical fabric configuration (only VLAN/IP) and without the need to deploy Edge nodes or configure dynamic routing. The Distributed connectivity connects the Transit Gateway directly to a VLAN provisioned in the fabric. This happens directly on the host, which implies that the VLAN must be available on the ESX (on physical uplinks used by NSX vmkernel). For more information about adding a Transit Gateway with distributed connection, see Add a Transit Gateway with a Distributed VLAN Connection.

In addition to on-demand networking and distributed routing, other services available when using distributed connections are 1:1 NAT and distributed DHCP. Stateful services, such as NAT and load balancing, can be provisioned to VPCs using a distributed connectivity by deploying a Virtual Network Appliance (VNA) cluster. For more information about creating a VNA, see Create a Virtual Network Appliance Cluster.

To use this connectivity setup workflow, you must ensure that the ESX cluster must have the ESX host with VDS uplinks configured.

If you have upgraded to VCF 9.0, you can use this workflow to add or delete an edge from edge clusters created in the previous version. Note that the delete option is available only when the Edge does not have any dependent objects. Additionally, you can also use this workflow to set up a Tier-0 gateway for an edge cluster, reset edge passwords, and set VPC external IP blocks. Note that to edit an Edge node, you will have to go to NSX Manager. Navigate to **System** > **Fabric** > **Edges**.

Set up Centralized Connectivity with NSX Edge Clusters

VCF provides a simplified workflow to set up network configuration. The workflow is available in both vCenter and NSX. For more information about setting the connectivity through vCenter, see *Building Your Cloud Infrastructure*. This section describes how to use the workflow from NSX to configure a centralized connectivity along with NSX Edge cluster deployment.

To use the workflow you must have a role of NSX administrator. Also, before you start using the workflow ensure that the following requirements are met:

- Plan for NSX Edge TEP VLAN and IP allocation for TEPs. The same VLAN as the Host TEP can be reused. For more information on creating IP pools for TEPs, see Create an IP Pool for Tunnel Endpoint IP Addresses.
- NSX Edges can be deployed on a host-group within a vSphere cluster or on a vSphere cluster. All hosts in the host-group or the vSphere cluster must have identical access to management network, gateway uplinks, host and Edge TEP networks. If you want to deploy edges on a prepared vSphere cluster, ensure that the following requirements are met:
  - The ESX cluster must have a Transport Node Profile (TNP) attached to it. For more information on applying a TNP, see Prepare Cluster Hosts as Transport Nodes by Using TNP.
  - The **IP Address Type** field should not be set to IPv6 in the Transport Node Profile that is applied to the cluster. Also for IPv4 address type, **IPv4 Assignment Type** should not be set to **Use VMKernel Adapter**. You can check it from **System > Fabric > Hosts > Transport Node Profile** in NSX Manager.
- If you plan to use the NSX Edge Host Affinity feature, ensure that you configure a host group to include a host where the NSX Edge node should be pinned. Select the "Host Group Affinity" option for the host group to enable the NSX Edge Host Affinity feature. The NSX Edge Host Affinity feature automatically enables the Best Effort Restart compute policy for the assigned NSX Edge node. For more information about host pinning, see the KB article Failover pre-checks are not performed when ESX is put into maintenance mode manually.
- NSX Edge node will make a 100% memory reservation. vSphere cluster/resource pool must have enough resources to deploy at least two edge nodes.
- Management network for NSX Edge must have connectivity to NSX Manager.
- If you want to configure dynamic routing with BGP, ensure that you set up two BGP peers per NSX Edge node on ToR switch with an interface IP, local ASN, and optional BGP password. For more information about configuring BGP, see Configure an BGP Neighbors for a Tier-0 Gateway.
- Reserve a local ASN to use for the Tier-0 gateway. For more information about ASN, see BGP Autonomous System Number (ASN) per Tier-0 VRF Gateway and per BGP Neighbor.
- If static routing is desired, it is recommended to configure HA VIP for active/standby Tier-O Gateway and use VIP as the Next Hop IP on ToR switch. You can configure static routing after completing the network connectivity workflow.
- Configure DVPGs for the NSX Edge management. It is recommended that these DVPGs should be different from DVPGs configured for the ESX management.
- Reserve external IP blocks for VPC connectivity. External IP blocks should not overlap with any other used CIDR addresses.
- (Optional) Populate your managed DNS server with DNS entries for NSX Edge components.

To set up centralized gateway connectivity with NSX Edge cluster deployment, perform the following steps:

1. From your browser, log in to the NSX Manager.
2. Go to **Networking** > **VPC Connectivity** > **Transit Gateway**.
3. Click **Setup Network Connectivity**.
4. In the **Span** field, select a span that should be applied to the transit gateway.
5. In the **Connection** field, select **Centralized Connection** and click **Next** .
6. On the Networking Prerequisites dialog box, validate that all the requirements are met. Click **Select all** and click **Continue** only when you are sure the mentioned requirements are fulfilled.
7. On the Edge Cluster page, enter the following details.

Field Name	Description
Cluster Name	Enter a name for the NSX Edge cluster.
Tunnel Endpoint MTU	The maximum packet size allowed over a tunnel is already selected from the Global Fabric Settings. Note that the NSX Edge TEP MTU must be the same across the host MTU within the same overlay Transport Zone, so you must set the Global Fabric MTU accordingly.
NSX Edge Nodes - Define the settings for Edge Nodes	
Edge Form Factor	Select an appropriate size for the Edge cluster. If you do not select any option, the default form factor **Large** is selected.
Search Domain Names	Enter the domain names for search. Example: private.broadcom.com
DNS Servers	Enter the DNS server addresses. Example: 192.161.1.2
Minimum 2 Edge Nodes are required to deploy an Edge Cluster	Configure at least two NSX Edges for high availability. You can add up to 10 edges. For details on configuring an Edge Node, see Adding an Edge Node to a Cluster. Once you have added an NSX Edge, you can also clone it to add another NSX Edge. Cloning is recommended only when the cloned NSX Edge is part of the same vSphere cluster with identical networks (edge management, pNIC mapping, VLAN for NSX Edge TEP, host group affinity).
Edge Node Credentials (optional) - You can define credentials for the NSX admin user to manage log-ins to Edge nodes. Note that if you do not provide CLI and Root admin passwords, they are set by the system that you can reset post the Edge deployment.	
CLI Administrator Credentials	
CLI Username	Enter the name for the CLI user.
CLI Password	Enter the password for the CLI user.
CLI Confirm Password	Enter the password again for confirmation.
Root Administrator Credentials	
System Root Password	Enter the password for the root admin user.
System Root Confirm Password	Enter the password again for confirmation.

8. Click **Next**.
9. On the Workload Domain Connectivity page, enter the following details.

Field Name	Description
Gateway Name	Enter the name for the tier-0 gateway.
High Availability Mode	Select the HA mode for the tier-0 gateway. You can select from the following options: • Active Active • Active Standby
Routing Configuration	
Gateway Routing Type	Select from BGP or Static options. For more details on routing, see the Prerequisite section.

Field Name	Description
Local Autonomous System Number (ASN)	If you selected the BGP gateway routing type, enter Local Autonomous System Number (ASN) that should be assigned to the Tier-0 gateway. For more details on BGP configuration and ASN, see the Prerequisite section.
Two gateway uplinks can be configured for each node	Click Set to set the gateway uplinks for each NSX Edge node. You must configure at least one gateway uplink for an Edge node. For more information about setting gateway uplinks, see Setting Gateway Uplinks for an Edge Node.
VPC Network Configuration	
VPC External IP Blocks	Enter the external IP blocks for VPC that are advertised to allow outside connectivity to VPC either on public subnets or through External IPs
Private -Transit Gateway IP Blocks	Enter the private CIDRs that should be used for inter-VPC communication.

10. Click **Next**.
11. On the Review and Deploy page, review the visuals and check all the configurations.
12. Click **Deploy**.

Add an Edge Node to a Cluster

This section covers the details about adding an NSX Edge node when you are setting up centralized network connectivity from **Networking** > **VPC Connectivity** > **Transit Gateway** in the NSX Manager. You can also use this information when you are adding an NSX Edge node by clicking **Add Edge** in the **Action** column of an existing NSX Edge cluster displayed on the Setup Network Connectivity page once the initial connectivity is set up.

Note that at the time of initial connectivity set up, you will need to add a minimum of two NSX Edge nodes. You can add up to 10 NSX Edges to an NSX Edge cluster. While adding an NSX Edge, you also have the option to clone a previously created NSX Edge and add it to the NSX Edge cluster. Cloning is recommended only when the cloned NSX Edge is part of the same vSphere cluster with identical networks (edge management, pNIC mapping, VLAN for Edge TEP, host group affinity).

The **Add Edge** button for an NSX Edge cluster is enabled only if the NSX Edge cluster is homogenous in terms of configuration, which implies that all member NSX Edge nodes:

- Should have the same form factor
- Should be part of a common Overlay Transport Zone
- Should have the deployment type as Virtual Machine
- Should have a single edge switch
- Should have "nsx-edge-multiple-vteps-uplink-profile" uplink profile

Note that the simplified workflow uses many recommended default settings for edge creation, such as the system-defined multi-TEP uplink profile.

To add an NSX Edge node, perform the following steps:

1. Click **Add** when you are adding an edge node from the Edge Cluster page in the Setup Network Connectivity workflow or click **Add Edge** in the **Action** column of the cluster when you are adding the edge from the Edge Cluster list.

Field Name	Description
Edge Node Name (FQDN)	Enter the name of the NSX Edge node.
Compute Manager	Select the compute manager for the Edge node.
vSphere Cluster	Select the vSphere cluster where the Edge is to be deployed.
Resource Pool	Select the resource pool for the Edge node.

Field Name	Description
Host Group Affinity	If an NSX Edge is deployed in a non-homogeneous network, for example a multi-rack horizontal vSphere cluster with L3 boundaries between the racks, vMotion cannot be used during the host remediation as vMotion of the NSX Edge to another host in another rack can cause connectivity loss with the top-of-rack switches. By selecting the Host Group Affinity option, you can pin an NSX Edge to a host/host group and prevent its vMotion when the host of this NSX Edge goes into the maintenance mode. To pin an NSX Edge with a host, the NSX Edge is associated with a host or a host group using an affinity rule and a Best Effort Restart (BER) policy is applied to the NSX Edge. The BER policy runs validations and ensures that the NSX Edge powers off when its host enters a maintenance mode. While shutting down an NSX Edge, a check is also made to find if any peer NSX Edge is ready for a failover. If a peer Edge is available, a graceful failover is made to this peer Edge and then the edge is powered off. Note that during the precheck if a peer Edge is not available for a graceful failover, host upgrade will not proceed until precheck errors are resolved manually. In case of a single host in a host group, the Edge stays powered off during the host's maintenance mode and is powered on again when the host exits the maintenance mode. If a peer Edge is available, it takes over the services. Once the host exits the maintenance mode, the Edge is powered on upon the same host. In case of multiple hosts in a group, after shutting down the Edge if a peer Edge is available it takes over the services and if a compatible host is available the Edge is powered on upon it. If due to affinity or availability constraints there's no other compatible host that can host the Edge, the Edge will continue to stay powered off during host maintenance mode. Once the host exits the maintenance mode, the Edge is powered on again on the same host. For more information on host groups, see Failover pre-checks are not performed when ESX is put into maintenance mode manually.
Host Group	If you have selected the **Host Group Affinity** as **Yes**, then select a host group to which the Edge needs to be pinned. For more information about host groups, see the prerequisite section about host pinning.
Data Store	Select a data store where the Edge node files are to be saved.
Management Network Settings	
IP Assignment	
IP Assignment Type	Select DHCP or Static.
Port Group	Select a port group that connects to the management interface of the Edge node. The NSX Edge management interface establishes communication with the NSX Manager management interface.
Management CIDR	Enter a CIDR for Management IP if you selected the Static option for IP allocation.
Default Gateway	Enter the default gateway IP address if you selected the Static option for IP allocation.
Uplinks	
Use the host overlay network configuration from the selected vSphere cluster	If this checkbox is selected, the existing overlay network configuration (such as VLAN ID and TEP IP Assignment) is automatically fetched from the hosts in the selected vSphere cluster. This checkbox will get enabled and will be auto-selected if all the prerequisites for a prepared vSphere cluster are met for the Edge deployment. For more information about preparing a vSphere cluster, see the Prerequisites section.
Edge Node Uplink Mapping	ESX is added to at least one VDS by which ESX pNICs get mapped to uplinks on VDS. Map the Edge NIC to these ESX pNICs. The pNICS are also auto-selected when there are two pNICs on a VDS. In this case, if you choose the first pNIC as an active uplink for the first interface (fp-eth0) the other pNIC is automatically selected as a standby uplink. For the second interface (fp-eth1), exactly the reverse pNICs are selected as active and standby uplinks. If there are multiple pNICs or pNICs connected to multiple VDS, pNICs are not auto-selected and you will have to manually map them.
Uplink Profile	The uplink profile is selected by default from the global settings.
TEP VLAN	Select the VLAN ID for Tunnel Endpoints (TEP). This field will be auto populated if the **Use the host overlay network configuration from the selected vSphere cluster** checkbox is selected.
TEP IP Address Type	Select **IPv4** or **IPv6**.

Field Name	Description
IP Allocation (TEP)	Select the IP allocation method for TEP. You can select from the following options. Based on the IP allocation method, provide the other required details. • DHCP - The TEP IP is allocated by the external DHCP server. • IP Pool - Select an IP Pool for allocations. For Creating IP pools for TEP, see the Prerequisites section. • Static IP List - Provide values IPv4 Static List, Static IPv4 Gateway, and IPv4 Subnet Mask. This field will be auto populated if the **Use the host overlay network configuration from the selected vSphere cluster** checkbox is selected.
IP Pool	Select an IP Pool if the option for **IP Allocation (TEP)** is selected as **IP Pool**.
IPv4 Static List	Enter the IPv4 addresses if the option for **IP Allocation (TEP)** is selected as **Static IP List**.
IPv4 Static Gateway	Enter the IPv4 gateway address if the option for **IP Allocation (TEP)** is selected as **Static IP List.**
IPv4 Subnet Mask	Enter the IPv4 subnet gateway if the option for **IP Allocation (TEP)** is selected as **Static IP List.**
VLAN MTU Check	Click Run Check to run a check on the DVS or host switch linked to the prepared transport nodes in the vCenter cluster or host group.

2. Click **Apply**.

Set Gateway Uplinks for an Edge Node

For each edge node, you can set a gateway uplink to establish an uplink connectivity between the gateway and Top-of-the-Rack router. Note that only two Edge nodes can be configured with Gateway uplinks if the High-Availability mode is Active Standby. Enter the following details for the gateway uplink. You can set up to two gateway uplinks for each Edge node.

1. Enter the following details for the gateway uplink.

Field Name	Description
Gateway Interface VLAN	Enter an ID for the VLAN that the Tier-0 gateway will use for the outbound traffic.
Gateway Interface CIDR	Enter the physical gateway address in CIDR notation.
BGP Peer IP	Enter the BGP peer IP to establish a connection between a Tier-0 gateway and a router in your physical infrastructure so that access between Edge and the outside world is enabled.
BFD	Switch on the toggle to enable BFD for backing up BGP and static routes sessions.
MTU	Enter the MTU for connection.
BGP Peer ASN	Enter the ASN for BGP peering.
BGP Peer Password	Enter the BGP peer password.
Confirm Password	Confirm the password.

2. To configure the second uplink, click **Next**.

Set up Distributed Network Connectivity

NSX provides a simplified workflow to set up network connectivity and perform network configuration. This section describes how to use the workflow from NSX to configure a distributed connectivity. The workflow is available in both NSX and vCenter.

You must have a role of NSX administrator to use the workflow. Also, before you start using the workflow ensure that the following requirements are met.

- Dedicated VLANs and subnets must be available for Distributed Transit Gateway connection to Top of Rack (ToR). The same VLAN must be configured on all the ToR switches where the ESX hosts are connected which offers VPCs. All the hosts running VPC workloads must have access to these VLANs.

- Disable 'ICMP redirect' on external gateway for these VLAN(s). External Gateway should allow packets received from ESX over these VLANs forwarded back to the same or different ESX.
- Reserve external IP blocks for VPC connectivity. External IP blocks must be the same as or a subset of dedicated VLAN subnets.

To set up a distributed gateway connectivity on Day 0 when no other connectivity is set up, perform the following steps:

1. From your browser, log in to the NSX Manager.
2. Go to **Networking** > **VPC Connectivity** > **Transit Gateways**.
3. Click **Setup Network Connectivity**.
4. In the **Span** field, select a span that should be applied to the transit gateway.
5. In the **Connection** field, select **Distributed Connection** and click **Next**.
6. On the Networking Prerequisites dialog box, click **Select all** and click **Continue**.
7. On the External Connectivity page, enter the following details.

Field Name	Description
VLAN ID	Enter the ID of the VLAN (for the north side connectivity) that the packets should take when they are sent on the physical network.
Gateway CIDR IPv4 Address	Enter the gateway IP address that is configured on the ToR switches.
VPC Network Configuration - The following configurations are for the default connectivity profile of the Default project and helps for networking within the project. Subnets are allocated IPs from these blocks based on whether they are private, public, or external.	
External IP Blocks	Enter external IP blocks for the VPC.
Private - Transit Gateway IP Blocks	Enter private transit gateway IP Blocks for the VPC

8. Click **Next**.
9. On the VPC Services page, enter details to create a Virtual Network Appliance (VNA) cluster. For more information about creating a VNA cluster, see Create a Virtual Network Appliance Cluster.
10. Click **Next**.
11. On the Review and Deploy page, review all configurations.
12. Click **Deploy**.

Create a Virtual Network Appliance Cluster

A distributed connectivity allows you to set up a VPC network without edge nodes. However, it lacks the support for stateful services such as NAT and load balancer that are available to VPCs using a centralized connectivity. To provision NAT and load balancer services to VPCs using a distributed connectivity, you can deploy a Virtual Network Appliance (VNA) cluster A VNA cluster is a group of Virtual Network Appliances (VNA), which are virtual machines, working collaboratively to provide these stateful networking services.

You can create a VNA cluster by navigating to **System** > **Fabric** > **VNA Clusters** and clicking **Add Cluster**. A minimum of one VNA is required to deploy a VNA cluster, but two VNAs are recommended for high-availability. Once you have created a VNA cluster, you can specify it in a project to provision VPC services for a distributed connectivity.

You can also create a VNA cluster at the time of setting up a distributed network connectivity on Day 0 when no other connectivity is set up by navigating to **Networking** > **VPC Connectivity** > **Transit Gateways**. Click **Setup Network Connectivity** and choose **Distributed VLAN Connection**. Provide the required configuration. On the VPC Services page, you can provide the VNA cluster details. For more information about setting up distributed connectivity, see Set up Distributed Network Connectivity.

Note that with each VNA cluster that you create, the system creates an overlay segment with the default transport zone to which the VNAs in the cluster get connected with. Use the API if you want to specify a different overlay transport zone where the segment is to be created.

If you plan to use the Host Group Affinity feature, ensure that you configure a host group to include a host where the VNA should be pinned. The Host Group Affinity feature automatically enables the Best Effort Restart compute policy for the assigned VNA.

1. Navigate to **System** > **Fabric** > **VNA Clusters** and click **Add Cluster.**
2. In the **Cluster Name** text box, enter the VNA cluster name.
3. In the **Node Form Factor** drop-down, select a form factor that is based on the scaling requirements you have for services that you are planning to use. The form factor is applied at the cluster level and all VNAs in the cluster get deployed with the selected size.
4. Click **Add** to add a VNA to the cluster and provide the following details. At least one VNA is required to deploy a VNA cluster.

Field Name	Description
Node Name (FQDN)	Enter the FQDN for VNA. Each VNA must have a unique FQDN.
Compute Manager	Enter the vCenter where the VNA is to be deployed.
vSphere Cluster	Select a vSphere cluster to host the VNA. You can select a standard vSphere cluster or a stretched vSphere cluster, but all the VNAs in a VNA cluster must be hosted on vSphere clusters of the same type. If the vSphere cluster you select already hosts management virtual machines that are connected to the host Management port group, the **VM Management Port group VLAN** settings are not available.
Resource Pool	Select the resource pool for the VNA.
Host Group Affinity	Slect **Yes** if you want to pin the VNA to a given host or group of hosts. Best Effort Restart (BER) policy is applied to the VNAs. VNAs are gracefully shut down during the host remediation.
Host Group	If you have selected the **Host Group Affinity** as **Yes**, then select a host group where the VNA is to be deployed. Host groups are managed in the vSphere Client under cluster configurations.
Data Store	Select a data store where the VNA is to be deployed.
Management Network Settings	
IP Assignment	Select the IP assignment for the VNA. You can select from the following options: • IPv4 only • IPv4 & IPv6 • IPv6 Only If you select `IPv4 & IPv6` or `IPv6 Only` option, provide values for the management CIDR and the default gateway.
IP Assignment Type	Select an IP assignment type for the VNA. You can select from the following options: • DHCP • Static If you select the `Static` option, provide values for the **Management CIDR** and **Default Gateway**.
Port Group	Select a port group for the management interface of the node. • For management workload, VM management port group is created. If a different port group is required, then create the port group with corresponding VLAN before performing this operation. • For a VI workload, a VM management port group is not created. Create a port group before performing this operation.

5. Click **Apply**. To add another VNA, you can also click **Clone** to copy the same settings.
6. In the **Description** and **Tag** fields enter the description and a tag for the VNA cluster.
7. Click **Save**.

A cluster is created and is shown in the grid. To add more VNAs to the cluster, click **Add Node** for the cluster on the Virtual Network Appliance Clusters page.

Virtual Private Cloud in NSX

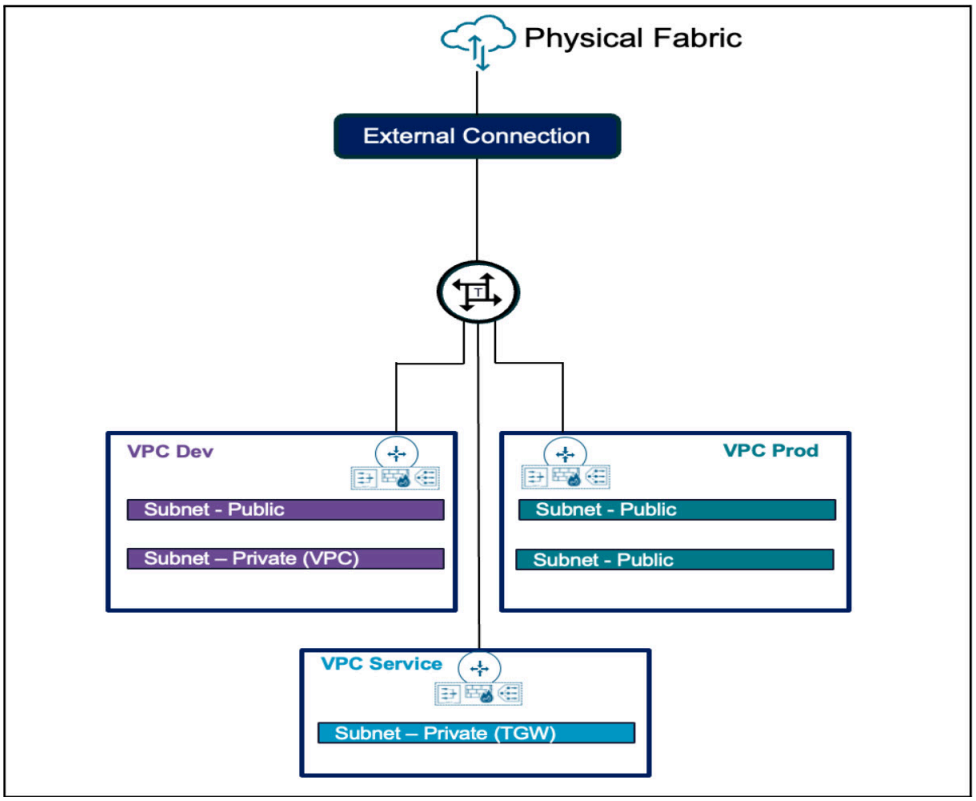
VCF 9.0 and later provides a consistent networking consumption model that is based on Virtual Private Clouds (VPCs) and Transit Gateways (TGW). This model is natively multi-tenant and can be used not only from NSX, but also from VCF (from vCenter, VCF Automation, and monitored from VCF Operations). It aims to provide a standardized, simple, and powerful Cloud networking experience not only to the network administrators but to all cloud users from VCF.

This cloud networking consumption is built using the following objects:

- **External Connections**: Defines outside connectivity for a Transit Gateway to the datacenter. An external connection can be of the following types:
  - Centralized
  - Distributed
  - Distributed VXLAN
- **Transit Gateway (TGW)**: Gateway interconnecting VPCs and connecting to outside datacenter through external connections.
- **Virtual Private Cloud (VPC)**: On-demand logical networks connected together that can be either advertised (public subnets) or exposed through networking services like External IPs or NAT. For simplicity, VPCs in VCF leverage three profiles:
  - VPC Connectivity Profile: Defines outside connectivity configuration for VPCs such as the Transit Gateway they are connected to, the IP Blocks available for consumption, and if a default outbound NAT is configured
  - VPC Service Profile: Defines the common configurations for services available in the VPC such as DHCP, DHCP Relay, or different subnet profiles.
  - VPC Security Profile: VPC Security profiles, which is the default only, activates on a Gateway Firewall for all VPCs in the project. (Stateless Gateway Firewall is part of VCF entitlement. For Stateful Gateway Firewall, you require a vDefend license.)

The following image provides an overview of the new simplified model.

Figure 415: Simplified VPC model



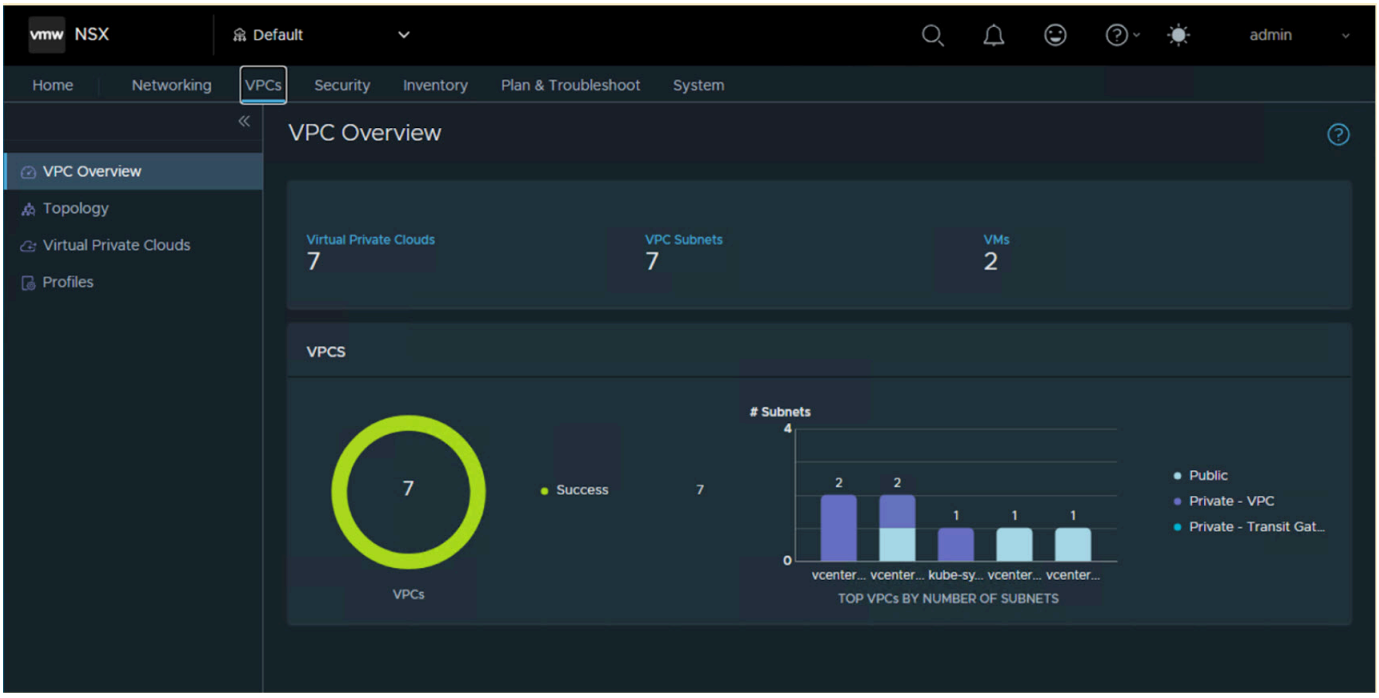
Note:

Note: In addition to the recommended method to create VPCs leveraging the VPC Connectivity Profile, VPC Service Profile and the connection through the Transit Gateways it is supported to have VPCs directly connected to the Tier-0 in user-defined Projects. Those VPCs won't benefit from multiple enhancements provided in VCF 9.0 and later, such as the ability to create Private - Transit Gateway networks for cross- VPC connections, and cannot be configured from vCenter or from VCF Automation.

Overview of VPCs in the Default Project

VPCs are available by default in NSX within the Default project without the need to create tenants (NSX Project). This provision allows administrators to benefit from VPC independently with VPCs in the default context being configurable from vCenter as well.

Figure 416: VPCs in the Default project



Use a multi-tenant platform with the ability to partition networking and security configurations of a single NSX deployment in multiple tenants (NSX Project). Those tenants allow multiple users to configure networking and security and configure Virtual Private Clouds (VPCs). For more information about multi-tenancy, see NSX Projects.

To offer simple on-boarding, a system is created with a Transit Gateway, a default Connectivity Profile, and a default Service Profile in the Default Project. These objects can be configured as part of the workload domain setup deployment and allow consumption of VPCs from NSX or vCenter.

- VPCs created in the Default Project will use the system default overlay transport zone.
- VPCs created in the Default Project will connect to the default Transit Gateway. It is not possible to have VPCs directly connected to Tier-0 in the Default Project.
- VPCs created from vCenter will leverage the default Connectivity and Service profiles but other profiles can be created and used from NSX.

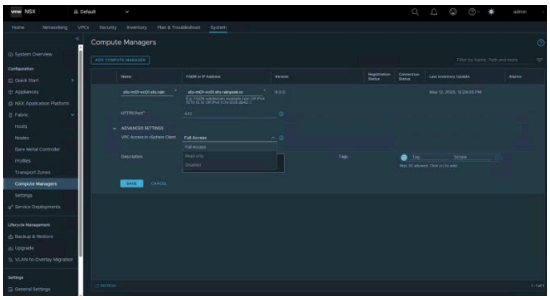
The Default project is the default context for the administrator and as such is not made for multi-tenancy like a user-defined project. You cannot assign a user to the Default project (it is possible to assign a user to a specific set of VPCs only). Additionally, the Default project can access resources such as external connections, external IP blocks, and Edge nodes without having the need to define them as shared resources, which is unlike a user-defined project where you have to define resources that can be shared.

VPC Consumption from vCenter

As mentioned earlier, in VCF 9.0 and later, VPC offers a VCF-wide consumption framework for networking that is simple and also standardized. VPCs are now available in vCenter for consumption so that a vCenter administrator can have a control over compute, networking, and storage from one interface and at the same time also have the visibility so that the objects are consistent between users and interfaces.

In VCF 9.0 and later, VPCs are visible in the vCenter network inventory and the vCenter administrator can also consume the VPCs, subnets, and external IPs for VPCs of the Default project from there. You can also allow or restrict VPC management in vCenter by configuring the required setting in NSX by navigating to **System > Fabric > Compute Manager**. For more information about VPC access, see Add a Compute Manager.

Figure 417: VPC Access Setting



External Connections

An external connection defines how a transit gateway (TGW) connects to networks outside of NSX and which services it supports. Those connections are created from Default (/infra) and define the connection to the outside environment.

Select the connection type that supports the features you require for your network environment.

- **Centralized connection:** Connects the transit gateway through a tier-0 gateway provisioned from NSX. The tier-0 gateway is deployed on NSX Edge nodes and provides interconnection to the physical fabric. This connection type requires you to deploy NSX Edge nodes and manage their lifecycle.

  Centralized connections allow stateful services and dynamic routing with the physical fabric (BGP) at tier-0 level. They also allow NAT configuration at the transit gateway level.
- **Distributed VLAN connection:** Connects the transit gateway directly to VLANs provisioned in the fabric. The connection happens directly on the ESX hosts, which mandates the availability of VLANs on the ESX.

  A distributed VLAN connection allows lightweight connectivity to the data-center fabric without needing to deploy NSX Edge nodes or configure dynamic routing. Stateful services are supported with this connection type using Virtual Network Appliances (VNAs).
- **Distributed VXLAN connection:** Connects the transit gateway directly to the EVPN-VXLAN network fabric to create a unified network fabric that delivers consistent connectivity across virtual and physical domains. Virtual private clouds (VPCs) can establish connectivity to tenant Virtual Routing and Forwarding instances (VRFs) within the EVPN-VXLAN network fabric.

  In version 9.1.1 and later, this connection type supports advanced networking services using Virtual Network Appliances (VNAs).

Figure 418: Transit Gateway with Centralized Connection and Transit Gateway with Distributed Connection (VLAN or VXLAN)

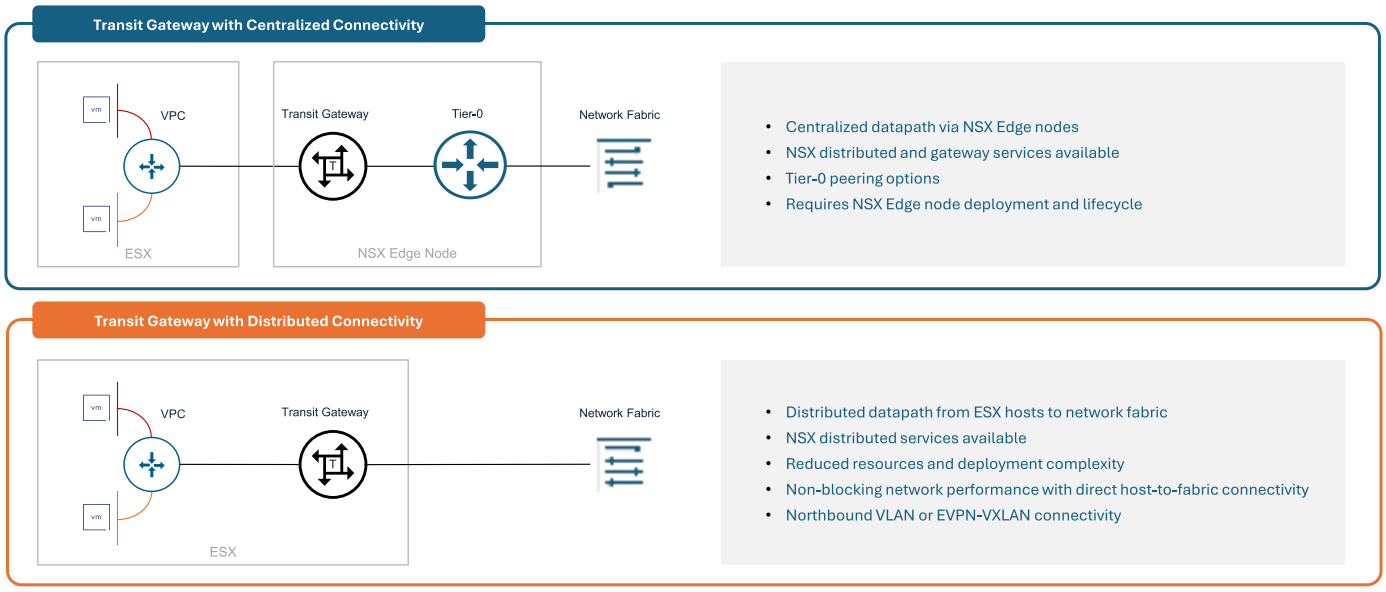
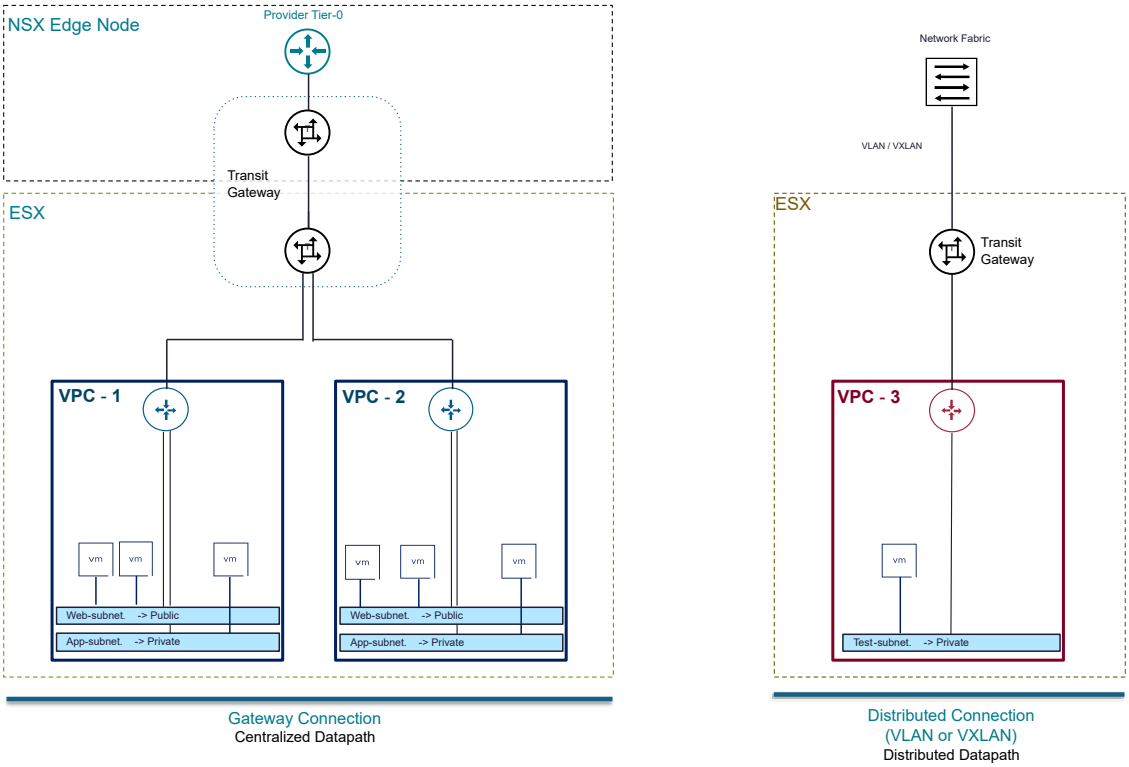


Figure 419: Topology of Transit Gateway with Centralized Connection and Transit Gateway with Distributed Connection (VLAN or VXLAN)



In all cases, an external connection requires that you properly configure the objects to be connected to the transit gateway. For a centralized connection, you must deploy a tier-0 gateway with NSX Edge nodes and properly configure external routing. For dynamic routing such as BGP, select the **Transit Gateway Static** option in the route redistribution settings to enable proper VPC subnet advertisement by upstream tier-0 BGP peers.

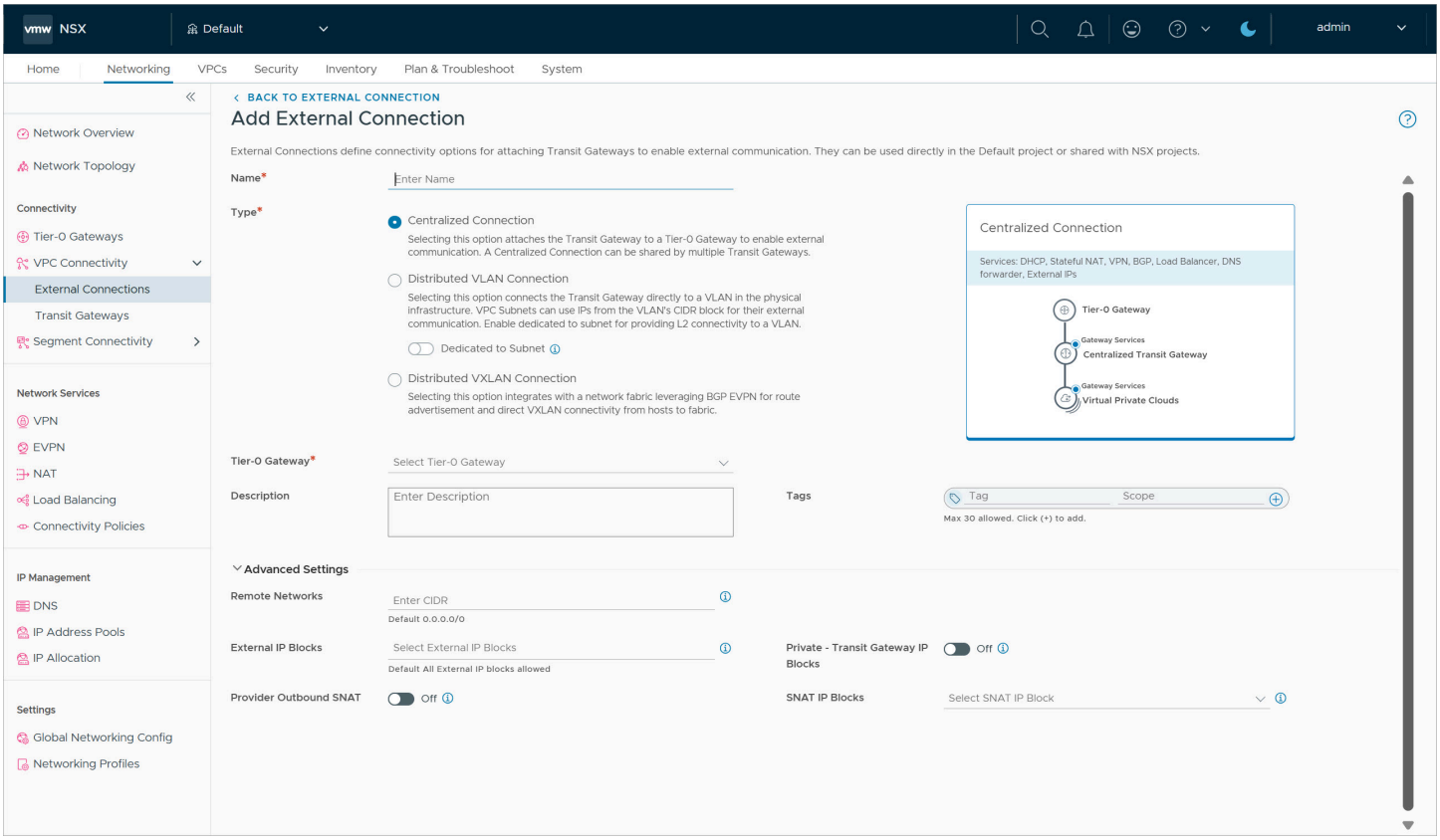
For a distributed VLAN connection, ensure that the VLANs used for outside connections are available to ESX hosts on the same uplinks (physical NICs) as the NSX overlay vmkernel adapter. For a distributed VXLAN connection, ensure that the ESX host (NSX overlay vmkernel adapter) has reachability to the TEP addresses of the EVPN gateway.

You can share an external connection with a custom project and attach it to a transit gateway.

Add a Centralized Connection

Perform the following steps to create a centralized connection.

Figure 420: Centralized Connection



Note:

For a centralized connection, you must deploy a tier-0 gateway with NSX Edge nodes and properly configure external routing. For dynamic routing such as BGP, enable the **Transit Gateway Static** option in the route redistribution settings to enable proper VPC subnet advertisement by upstream tier-0 BGP peers.

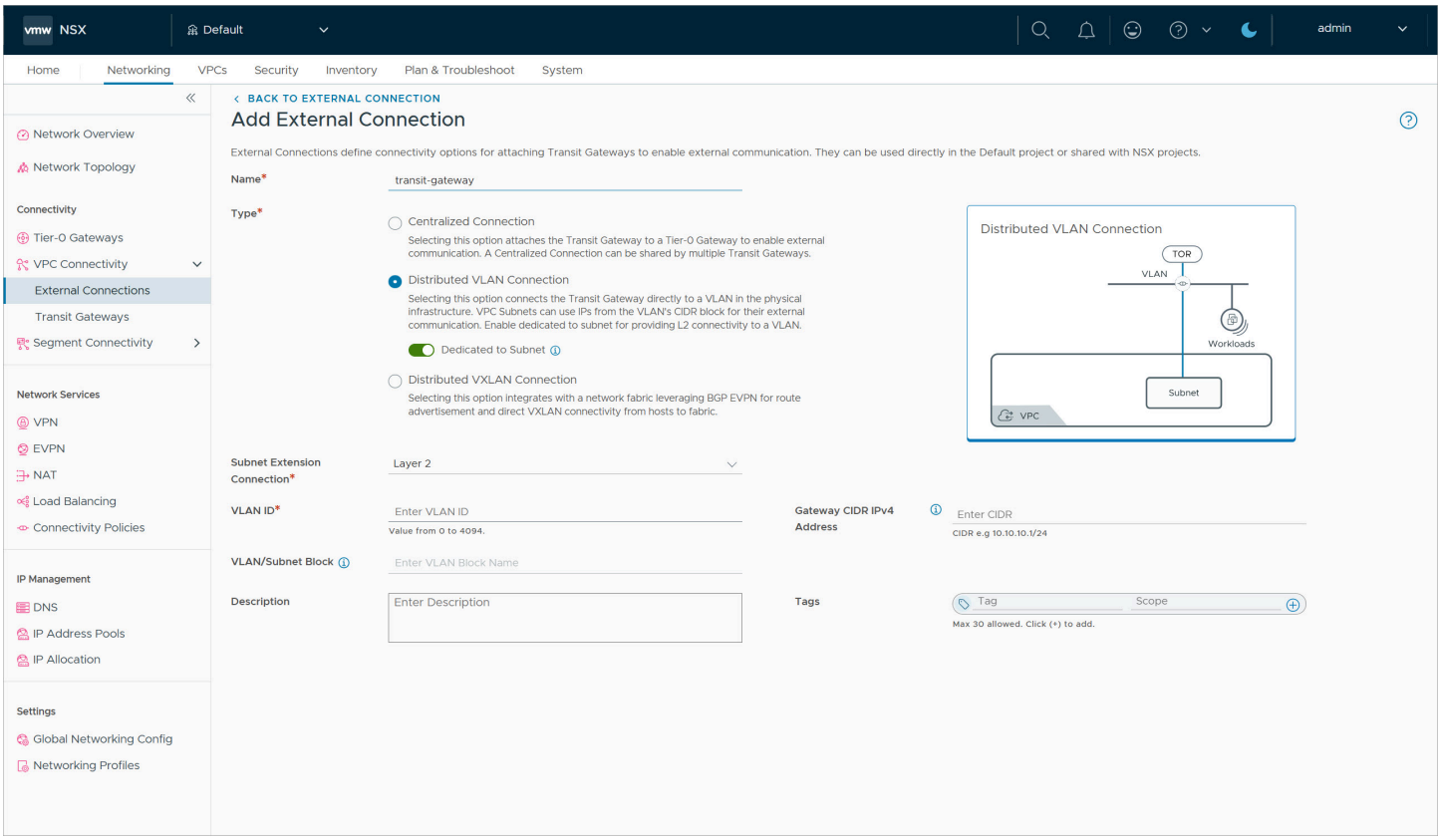
1. From your browser, log in to the NSX Manager.
2. Navigate to **Networking** > **VPC Connectivity** > **External Connection**.
3. Click **Add External Connection**.
4. Enter the following fields:

Field Name	Description
Name	Enter a name for the external connection.
Type	Click to select the type of connection you want to create. The **Centralized Connection** option is selected by default.
Tier-0 Gateway	Select the Tier-0 or Tier-0 VRF gateway through which you want to route the traffic to external networks.
Description	Enter a description for this connection.
Tags	Enter a tag and scope for this connection.
Advanced Settings	
Inbound Remote Networks	Provide a CIDR for inbound remote networks. Inbound remote networks are used to control north-bound traffic when a transit gateway has multiple connections. As the default route can be supported on only one connection, inbound remote networks must be specified for all the other connections to route the traffic. Hence, with a transit gateway with multiple connections there can be one connection using the default route to route the traffic to the Internet and another connection using the inbound remote network to route the traffic to some dedicated networks meant for shared services like DNS. When you are specifying an inbound remote network, consider the following criteria: • Remote networks to be reachable through this connection for the VPCs. The default gateway is 0.0.0.0/0. • Inbound remote networks must be unique per gateway connection and ECMP networks are not supported. • Inbound networks are supported even with a single attachment. If it is not specified, the default route is plumbed.
Authorize External IP Blocks	You can control which external IP blocks should be advertised through the external connection. By default, all external IP blocks are allowed to be advertised if you do not specify any value in this field as this field is optional. However, this field becomes mandatory if you turn on the **Provider Outbound SNAT** toggle. In that case, only external IP blocks that are specified in this field are advertised. If a transit gateway has multiple connections, for each connection you can authorize external IP blocks that the connection can advertise. Traffic from these blocks is routed to the external network without applying any NAT rules.
Advertised Private IP Blocks	If you turn on this toggle, then the private transit gateway IP blocks are advertised through the external connection. The default behavior is to advertise only external IP blocks (all external IP blocks unless you mention the specific ones to be advertised in the**Authorize External IP Blocks** field). If you turn on this toggle, you must also add the transit gateway route advertisement rule, "TGW_PRIVATE", to the transit gateway attachment to get the private transit gateway IP blocks advertised.
Provider Outbound SNAT	If this toggle is on, a provider can control the outgoing traffic irrespective of the subnet type. Any other NAT rule that is created on a transit gateway uplink has a higher priority than the provider outbound SNAT rule. You can control all outbound traffic by turning on this toggle that works in conjunction with **Authorize External IP Blocks** and **SNAT IP Block** fields.
SNAT IP Block	Specify the IP ranges or IP CIDRs to be used for the provider outbound SNAT translated IP. You must also specify this IP block in the **Authorize External IP Blocks** field if the **Provider Outbound SNAT** toggle is on.

5. Click **Save**.

Add a Distributed VLAN Connection

Distributed VLAN Connection



Perform the following steps to create a distributed VLAN connection.

1. From your browser, log in to the NSX Manager.
2. Navigate to **Networking** > **VPC Connectivity** > **External Connection**.
3. Click **Distributed VLAN Connection**.
4. If you want to create a subnet-exclusive distributed VLAN connection for VLAN extension, turn on the toggle **Dedicated to Subnet**. Once you have created a distributed VLAN connection, you can share it with a project whose VPC subnet will be extended to a VLAN. For more information about extending a VPC subnet to VLAN, see Extending VPC Subnet to VLAN and Onboarding DVPG VMs to VPC Subnets.
5. Enter the following fields:

Field Name	Description
Name	Enter a name for the external connection.
Type	Click the Distributed Connection option.
Subnet Extension Connection	If you are creating this connection for VLAN extension by turning the **Dedicated to Subnet** toggle on, select whether you want only the Layer 2 connectivity or both the Layer 2 and Layer 3 connectivity. In the Layer 2 connectivity, the overlay VM traffic cannot go through VPC and Distributed Transit Gateway (DTGW) to the external network. In the Layer 3 connectivity, the extended subnet is connected to the VPC gateway and traffic can go through VPC and Distributed Transit Gateway (DTGW) to the external network.
VLAN ID	Enter the VLAN ID to which the traffic is to be routed. This VLAN must be dedicated for external connectivity of workloads deployment in VPCs and must be stretched across all clusters where VPCs are deployed. Also, note that the External IP block must be a subset of VLAN CIDRs and any in-use IP address on the physical network must be pre-allocated on NSX to avoid duplicate IP assignment. If you are creating this connection for VLAN extension, provide the ID of the VLAN that is to be extended.
Gateway CIDR IPv4 Address	Enter the gateway address and netmask through which the traffic should be routed to the external network. If you are creating this connection for VLAN extension, provide the CIDR of VLAN's gateway.
VLAN/Subnet Block	If you have turned on the **Dedicated to Subnet** toggle, provide the IP address block that you created to be exclusively used for VLAN extension.
Description	Enter a description for this connection.
Tags	Enter a tag and scope for this connection.

6. Click **Save**.

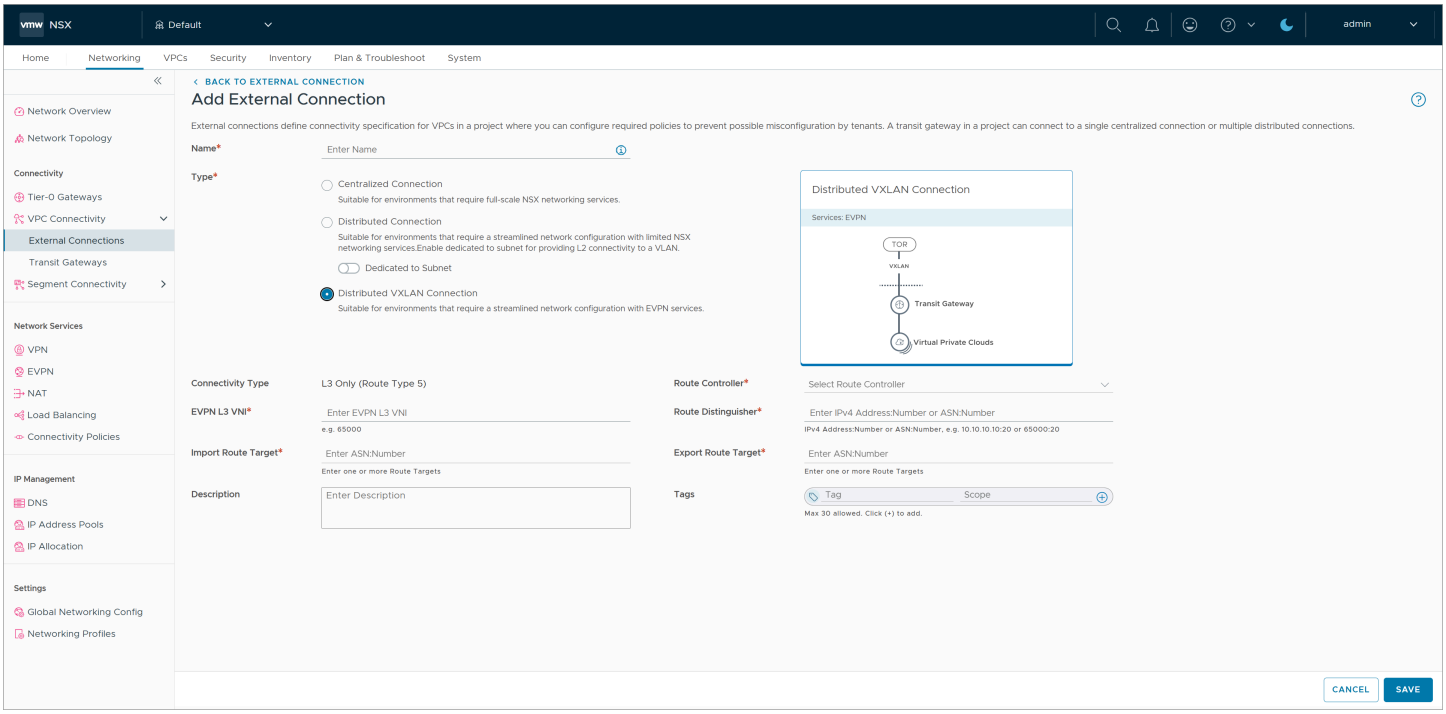
Add a Distributed VXLAN Connection

With a distributed VXLAN connection, you can specify a Layer 3 VNI mapping for distributed VXLAN connectivity. Distributed VXLAN connectivity integrates the virtual NSX fabric with a physical EVPN-VXLAN network fabric to create a unified networking environment.

Create a Route Controller.

> **Note:**
>
> Starting with the VCF version 9.1.1, the route-controller no longer advertises a default route to the ESX. If a default route must be advertised to the ESX, it needs to be originated by the BGW.

For more information about how a distributed VXLAN connection works in the context of a unified networking environment, see Distributed VXLAN Connectivity.


NSX
Default
admin
Home
Networking
VPCs
Security
Inventory
Plan & Troubleshoot
System
Network Overview
Network Topology
Connectivity
Tier-0 Gateways
VPC Connectivity
External Connections
Transit Gateways
Segment Connectivity
Network Services
VPN
EVPN
NAT
Load Balancing
Connectivity Policies
IP Management
DNS
IP Address Pools
IP Allocation
Settings
Global Networking Config
Networking Profiles
BACK TO EXTERNAL CONNECTION
Add External Connection
External connections define connectivity specification for VPCs in a project where you can configure required policies to prevent possible misconfiguration by tenants. A transit gateway in a project can connect to a single centralized connection or multiple distributed connections.
Name*
Enter Name
Type*
Centralized Connection
Suitable for environments that require full-scale NSX networking services.
Distributed Connection
Suitable for environments that require a streamlined network configuration with limited NSX networking services.Enable dedicated to subnet for providing L2 connectivity to a VLAN.
Dedicated to Subnet
Distributed VXLAN Connection
Suitable for environments that require a streamlined network configuration with EVPN services.
Distributed VXLAN Connection
Services: EVPN
TOR
VXLAN
Transit Gateway
Virtual Private Clouds
Connectivity Type
L3 Only (Route Type 5)
EVPN L3 VNI*
Enter EVPN L3 VNI
e.g. 65000
Import Route Target*
Enter ASN:Number
Enter one or more Route Targets
Description
Enter Description
Route Controller*
Select Route Controller
Route Distinguisher*
Enter IPv4 Address:Number or ASN:Number
IPv4 Address:Number or ASN:Number, e.g. 10.10.10.10:20 or 65000:20
Export Route Target*
Enter ASN:Number
Enter one or more Route Targets
Tags
Tag
Scope
Max 30 allowed. Click (+) to add.
CANCEL
SAVE

1. From your browser, log in to the NSX Manager.
2. Navigate to **Networking** > **VPC Connectivity** > **External Connections**.
3. Select **Add External Connection** and select **Distributed VXLAN Connection** as the Type.
4. Specify the following fields:

Field Name	Description
Name	Enter a name for the external connection.
Type	Select **Distributed VXLAN Connection**.
Connectivity Type	This read-only field is set by default to **L3 Only (Route Type 5)**.
Route Controller	Select the Route Controller to use for this distributed VXLAN connection.
EVPN L3 VNI	Enter the unique VXLAN Network Identifier (VNI) that identifies the L3 routing instance in the physical network fabric.
Direct Prefix Routing	Starting with the VCF version 9.1.1, you can allow specific prefixes (such as storage services) that require lower latency to bypass the Border Gateway (BGW). You can turn on this toggle to route the latency sensitive traffic directly from ESX TEP to the destination fabric VTEP (non-BGW ToR). To achieve the direct communication, specific hosts or aggregate prefixes are installed directly on the ESX TEP. The traffic is routed straight to the destination fabric VTEP without going through the BGW. Even with the toggle on, the default route (0.0.0.0/0) through the BGW functions as is for the standard Internet and external traffic provided the BGW had advertised the default route (0.0.0.0/0) to the Route Controller.
Route Distinguisher	Enter the 64-bit prefix to attach to every IP address, ensuring that routes remain unique across overlapping IP addresses.
Import Route Target	Specify the route target label to attach to every incoming route received from the EVPN gateway. The **Import Route Target** can take the same value as the **Route Distinguisher**.
Export Route Target	Specify the route target label to attach to every outgoing route sent to the EVPN gateway. The **Export Route Target** must match the **Import Route Target**.
Description	Enter a description for this connection.
Tags	Enter a tag and scope for this connection.

5. Click **Save**.

Add a Transit Gateway of distributed VXLAN Connection type, and attach the distributed VXLAN connection to the Transit Gateway.

Transit Gateways

A Transit Gateway (TGW) provides both inter-VPC communication and connection to the outside network through External Connections. This model keeps an inter-VPC communication inside a tenant. Note that with a TGW, there is no need for sharing a Tier-0 directly with an NSX Project. Instead, an External Connection is shared to the project to which the TGW is connected.

A Transit Gateway can be created with a centralized connectivity or it can be created with a distributed connectivity. NSX automatically creates a default TGW for every tenant (NSX Project), including Default. Based on your network requirements, you can also create additional TGWs (both CTGWs and DTGW) for a project. With multiple TGWs in a project, a VPC can be re-attached or made to point to another TGW as and when required. You can move a VPC to another TGW by changing the VPC's connectivity profile. When you move a VPC to another TGW, the system runs

validation checks at different levels, such as IPAM validation and VPC span validation. Ensure that when you move a VPC to another TGW, the dependent configurations are up to the mark.

A TGW can also have multiple External Connections to support complex routing, such as separating Internet and Intranet paths. All external connections for a TGW must be of the same type, that is, either all centralized or all distributed. A TGW cannot have a mixed type of connections. Note that a VPC in a project can connect to only one TGW.

When you create a TGW, the system also creates a corresponding connectivity profile for this TGW. To connect a VPC with the required TGW, you must link that VPC with the connectivity profile of that TGW.

Centralized Transit Gateway

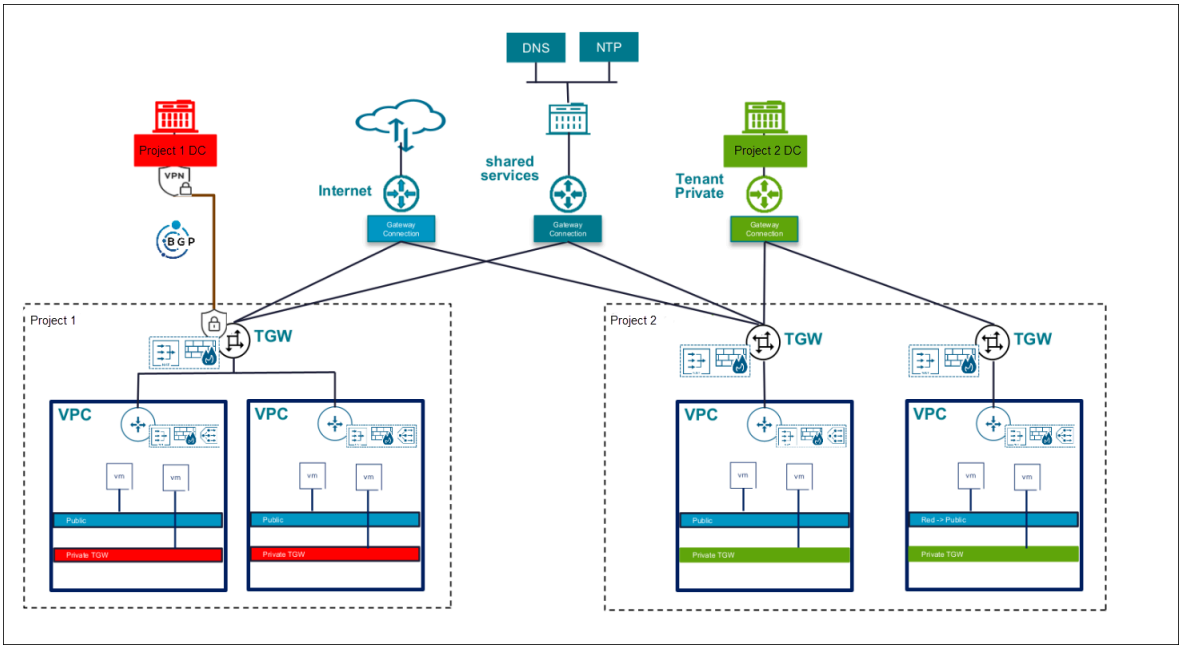
A centralized transit gateway (CTGW) is created by attaching a TGW with a centralized external connection. Previously, a CTGW shared the same High Availability (HA) mode and placement on the NSX Edge cluster as the Tier-0/VRF gateway to which it connected. In this case, if the Tier-0/VRF had an active-active HA, CTGW was not able to support stateful services (auto SNAT).

Now, a CTGW can be deployed on an Edge cluster which can be different from the Edge cluster of the Tier-0/VRF gateway to which it connects. Also, the CTGW can now have an independent HA mode that supports configuring multiple topologies and assigning different HA modes per tenant. Note that CTGW sharing its HA mode with the Tier-0 gateway to which it connects is deprecated now and post upgrade any new CTGW that you create will have indepedent an HA mode.

A CTGW can have the following HA modes:

- Active active - If you need a better traffic throughput, define the HA mode as active-active.
- Active standby - If you require services such as NAT, define the HA mode as active-standby

If there are multiple CTGWs in a project, they can have different HA modes. The following image shows multiple centralized transit gateways in a project.



You can also add NAT rules for a CTGW for external address binding.

Distributed Transit Gateway

A distributed transit gateway (DTGW) is created by attaching a TGW with a distributed external connection. If there are multiple DTGWs in a project, each DTGW can use the same or a different VLAN but public subnets or external IPs must not overlap within different DTGWs.

VPCs can be migrated between DTGWs by changing the connectivity profile of VPCs. NSX triggers validations at the policy layer when a VPC is moved across DTGWs. You must take care of the prerequisites before moving a VPC across DTGWs. You can also move a VPC with a VLAN extension subnet to a different DTGW. To move a VPC with a VLAN

extension subnet, you must attach the correlated distributed VLAN connection to the destination DTGW and then detach the distributed VLAN connection from the source DTGW.

You can also add NAT rules for a DTGW for external address binding by creating a virtual network appliance (VNA).

Network Span

Network span is a logical construct that defines a collection of both VCP and non-VCP vSphere clusters. A span is applied to a TGW (Central TGW or Distributed TGW) that makes VPC subnets connected to the TGW available only across vSphere clusters of the span instead of all vSphere clusters in the environment. Thus, with a network span you can limit the availability of VPC subnets to a specific set of vSphere clusters.

Spans can contain overlapping clusters, that is, a cluster can belong to multiple spans. Two special types of spans are:

- Default span: By default, all vCenter clusters in the environment belong to a default span.
- Exclusive span: When you want to specify a set of vCenter clusters that are dedicated for a workload and should not be shared across the environment, you can create a span of those clusters and mark the span as an Exclusive Span. Exclusive span prevents overlapping of clusters between spans and also removes those clusters from the default span.

For more information about adding a network span, see Add a Network Span.

Add a Transit Gateway with a Centralized Connection

You can create a Transit Gateway (TGW) with one or more centralized external connections.

> **Note:**
>
> For a Transit Gateway with a centralized connection, you must deploy a tier-0 gateway with NSX Edge nodes and properly configure external routing. For dynamic routing such as BGP, enable the **Transit Gateway Static** option in the route redistribution settings to enable proper VPC subnet advertisement by upstream tier-0 BGP peers.

Perform the following steps to create a Centralized Transit Gateway (CTGW) for a project.

1. From your browser, log in to NSX Manager.
2. From the **Project** drop-down, select the project for which you want to create the Transit Gateway.
3. Go to **Networking** > **VPC Connectivity** > **Transit Gateways**.
4. Click **Add Transit Gateway**.
5. Enter the following information:

Field Name	Description
Gateway Name	Enter a gateway name.
Span	Select a span with which you want to associate this transit gateway. The subnets connected to this transit gateway will be available to all the vSphere clusters of the associated span. Note that if you do specify a span, the default span is associated with the transit gateway. In this scenario, all VPC subnets that are connected to this transit gateway will be available to all vSphere clusters except to the vSphere clusters of exclusive spans.
Connection	Select the **Centralized Connection** option.
Additional Settings	

Field Name	Description
High Availability Mode	The high-availability (HA) configuration allows the transit gateway to have an independent HA mode from the attached Tier-0/VRF gateway HA mode. Select one of the following High-Availability modes: • Active Active: In the Active-Active mode, traffic is load-balanced across all NSX Edge nodes where the transit gateway is deployed. Stateful services like NAT are not supported in Active-Active mode. • Active Standby: In Active-Standby mode, all traffic is processed by an elected active NSX Edge node. If the active member fails, a new member is elected to be active. Stateful services like NAT are not supported in Active-Active mode. When a Centralized Transit Gateway (CTGW) is created, irrespective of the HA mode that you select the system auto allocates two NSX Edge nodes from the designated edge cluster for the transit gateway realization. The NSX Edge nodes selection is based on the least utilized or least allocated principle that prioritizes nodes with the lowest number of existing transit gateways and Tier-1 gateways.
TGW Edge Cluster	Select an NSX Edge cluster where this transit gateway should be realized. If you are creating a transit gateway for a custom project, you can select from NSX Edge clusters that are shared with the custom project. For the default project, all NSX Edge clusters are available.

Note:

If you do not set the **TGW Edge Cluster**, the system selects an NSX edge cluster as follows:

- If the NSX Edge cluster of the connected tier-0 gateway is shared with the custom Project, the system selects that edge cluster.
- If the NSX Edge cluster of the connected tier-0 gateway is not shared with the custom Project, the system selects any edge cluster (preferably the first in the shared list) from the edge clusters shared with the custom Project.

6. Click Next and on the Workload Domain Connectivity page, enter the following information.

Field Name	Description
External Connection	Select one of the following options: • Create New - Click to create a new centralized external connection by providing a name and selecting a Tier-0 gateway. • Use Existing - Click to select from the existing centralized external connections.
VPC Network Configuration	
VPC External IP Blocks	Enter external IP blocks to be used for creating public subnets, external IPs, or translated IP for the system and user-defined NAT rules. The entered IP blocks allow outside connectivity to VPC workloads either on VPC public subnets or through external IPs.
Private - Transit Gateway IP Blocks	Enter private IP blocks to create private subnets in the VPC. The entered IP blocks are available for inter-VPC communication and the transit gateway, by default, does not advertise them to the outside datacenter.
Default Outbound NAT	Turn on the toggle to ensure that workloads on VPC private subnets have outbound access outside of VPC. One SNAT rule per VPC is auto-plumbed and the translated IP is taken from the external IP block for this VPC. Note that if you enable this toggle, the connectivity profile that the system creates for this CTGW by default will have **N-S Service** and **Default Outbound NAT** toggles as enabled.

7. Click **Save**.

Add NAT Rules for a Centralized Transit Gateway

You can add NAT rules for a centralized transit gateway (CTGW) for external address binding.

For creating a NAT rule for CTGW, consider the following conditions:

- DNAT rule should be configured at the nearest point to the translated IP, that is, the VPC.
- SNAT rule should be configured at the furthest point till the Match IP is visible - VPC for Private IP and CTGW for Project IP.

Perform the following steps to add a NAT rule for VPC.

1. From your browser, log in to NSX Manager.
2. From the **Project** drop-down, select the project for which you want to configure the transit gateway.
3. Go to **Netwoking** > **VPC Connectivity** > **Transit Gateways**.
4. Click the down arrow to expand the configurations of the required transit gateway.
5. Under the Services section, click Set in the **NAT** field.
6. Click **Add NAT Rule** .
7. Enter the following information.

Field Name	Description
Name	Enter a NAT name.
Action	Select an action. The following actions are available: • SNAT • DNAT • Reflexive • NO SNAT • NO DNAT
Match Source IP | Port	Enter the source network address and source port. Specify an IP address, or an IP address range in CIDR format. For SNAT, NO_SNAT and Reflexive rules, this is a required field and represents the source network of the packets leaving the network. For DNAT and NO_DNAT rules, optionally it can contain a source network of incoming packets. For a NULL value, this field represents any network.
Match Destination IP | Port	Enter a destination IP address or an IP address range in CIDR format and destination port. For DNAT and NO_DNAT rules, this is a required field and represents the source network of the packets leaving the network. This field is not applicable for Reflexive.
Select Protocol	Select the required protocol.
Translated IP | Port	Enter a value for translated IP. Specify an IPv4 address or an IP address range in CIDR format. If translated IP is less than the match IP for SNAT it will work as PAT.
Apply To	Select the required connection to apply the rule.
Enabled	Turn on the toggle to enable the rule.
Logging	Turn on the toggle to enable logging.
Priority	Enter a priority value. A lower value means a higher priority. The default is 0. A No SNAT or No DNAT rule should have a higher priority than other rules.
Firewall	The available settings are: • Match External Address - The firewall will be applied to the external address of a NAT rule. – For SNAT, the external address is the translated source address after NAT is done. – For DNAT, the external address is the original destination address before NAT is done. – For REFLEXIVE, to egress traffic, the firewall is applied to the translated source address after NAT is done. For ingress traffic, the firewall is applied to the original destination address before NAT is done. • Match Internal Address - Indicates the firewall will be applied to the internal address of a NAT rule. – For SNAT, the internal address is the original source address before NAT is done.

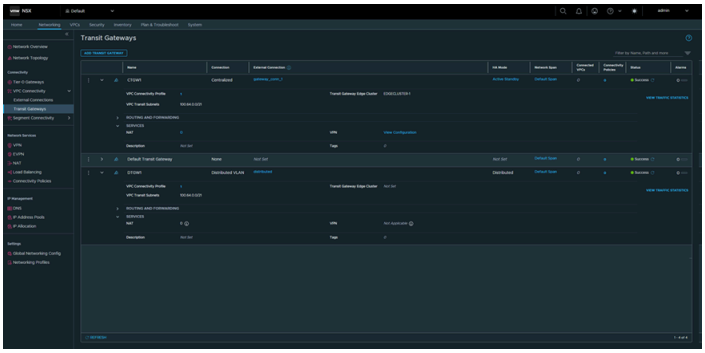
Field Name	Description
	– For DNAT, the internal address is the translated destination address after NAT is done. – For REFLEXIVE, for egress traffic, the firewall is applied to the original source address before NAT is done. For ingress traffic, the firewall is applied to the translated destination address after NAT is done. • Bypass - The packet bypasses firewall rules.
Description	Enter a description for NAT.
Tag	Enter tag and scope for NAT

8. Click **Save**.

Enabling an IPSec VPN Service on a Centralized Transit Gateway

In a multi-tenancy environment, a tenant might want to securely connect multiple data centers that are hosted across multiple provider infrastructures like public clouds (AWS and Azure) or a private cloud. For a secure connection, a tenant can enable an IPSec VPN service on a centralized transit gateway (CTGW) that is provisioned by the tenant's provider.

The following image shows that a IPSec VPN service can be configured only for a CTGW.



An IPSec VPN service can be enabled only on a CTGW that is configured in the active-standby mode. Per CTGW, you can configure only one IPsec VPN service. The number of VPN sessions and tunnels that are supported for CTGW VPN are the same as supported for tier-0 and tier-1 VPN.

You have to perform the following tasks to enable IPSec VPN service on a CTGW:

- Add an IPSec VPN Service for a Centralized Transit Gateway
- Add Local Endpoints for an IPSec VPN Service with unique IP addresses identified from the external IP blocks.
- Add IPSec VPN Sessions for a Centralized Transit Gateway

The IPSec VPN service on CTGW is supported for the following topologies:

- CTGW with a shared tier-0 across different tenants
- CTGW with a dedicated tier-0 for a tenant
- CTGW with multiple tier-0

Add an IPSec VPN Service for a Centralized Transit Gateway

If you are configuring an IPSec VPN service for a centralized transit gateway (CTGW), ensure that the CTGW is configured in the active-standby mode.

To configure an IPSec VPN service on CTGW, perform the following steps:

1. From your browser, log in to NSX Manager.
2. Go to **Networking** > **VPC Connectivity** > **Transit Gateways**.
3. Click the arrow to expand the configurations of the required transit gateway (CTGW with an active-standby HA).
4. Expand the **Services** section and click **View Configuration** in the **VPN** field.
5. On the **VPN Service** tab, click **Add IPSec Service**.
6. Enter the following information:

Field Name	Description
Name	Enter a name for the IPSec VPN service.
Service Type	Displays the service name as **IPSec**.
Sessions	Click **Set** after creating the VPN service to add sessions for this service.
Admin Status	By default the toggle is tuned on which implies that the IPSec VPN service is enabled on the CTGW after the new IPSec VPN service is configured.
IKE Log Level	Select a log level for internet key exchange (IKE).
Session Sync	By default the toggle is on which implies that a stateful synchronization of IPSec VPN sessions on the standby HA node is activated.
Description	Enter a description for the service.
Tags	Enter tags for the service.
Global Bypass Rules - You can allow data packets exchange between the specified local and remote IP addresses without any IPSec protection. Bypass rules always have a higher priority over protection rules and these rules apply for all IPSec VPN sessions that you create within this IPSec VPN service. If you enable these rules, data packets are exchanged between the specified local and remote IP sites even if their IP addresses are specified in the IPSec session rules. In the **Local Networks** and **Remote Networks** text boxes, enter the list of local and remote subnets between which the bypass rules are applied.	
Local Networks	Enter the list of local subnets for the bypass rules.
Remote Networks	Enter the list of remote subnets for the bypass rules.

7. Click **Save**.

   After the new IPSec VPN service is created successfully, the system prompts whether you want to continue with the rest of the IPSec VPN configuration. If you click **Yes**, the system takes you back to the Add IPSec VPN Service panel. The **Set** link in the **Sessions** field link gets activated and you can click it to add sessions for the IPSec VPN service.

Add Local Endpoints for an IPSec VPN Service

You must configure a local endpoint to use with the IPSec VPN service that you have configured for a centralized transit gateway (CTGW).

- Ensure that you have configured an IPSec VPN service to which this local endpoint is to be associated.
- Ensure that you have obtained the information about the certificate that the local endpoint must use if you are adding the local endpoint for an IPSec VPN session that uses a certificate-based authentication mode.
- Ensure that you have requested a unique IP address from the external IP block to assign to this local endpoint. You can also request one at the time of configuring the local endpoint.

To configure certificates for a local endpoint, you can import certificates to the project or you can use the certificates that are shared with the project by the provider.

To add a local endpoint for an IPSec VPN session, perform the following steps:

1. From your browser, log in to NSX Manager.
2. Go to **Networking** > **VPC Connectivity** > **Transit Gateways**.
3. Click the arrow to expand the configurations of the required transit gateway (CTGW with an active-standby HA).
4. Expand the **Services** section and click **View Configuration** in the **VPN** field.
5. On the **Local Endpoints** tab, click **Add Local Endpoint** and enter the following information.

Field Name	Description
Name	Enter a name of the local endpoint.
IP Address	Select an IP address that is already allocated from an external IP block for the local endpoint. You can also click **Request IP** to request an IP address allocation from an external IP block.
Site Certificate	Select a site certificate that the local endpoint must use. The certificate is used only if the IPSec VPN session to with which this endpoint is associated uses a certificate-based authentication mode.
Trusted CA / Self Signed Certificates	Select trusted CA or self-signed certificates that the local endpoint will use to verify certificates of a remote peer.
Certificate Revocation List	Select a certificate revocation list for the local endpoint to block a remote peer if the certificate presented by it is compromised.
Local ID	Enter a local ID for the local endpoint. This local ID is configured as remote ID on the remote site. For IPSec VPN sessions with certificate-based authentication, the local ID is derived from the certificate that is associated with the local endpoint.
Description	Enter a description for the local endpoint.
Tags	Enter tags for the local endpoint.

6. Click **Save**.

Add IPSec VPN Sessions for a Centralized Transit Gateway

You can configure both policy-based and route-based IPSec VPN sessions for a centralized transit gateway (CTGW). With a route-based IPSec VPN session, only static routes are supported. Dynamic routing using BGP is not yet supported on CTGW.

You can add a session only after creating an IPSec VPN service first. If you have not created an IPSec VPN service before adding a session, the **IPSec Sessions** tab remains deactivated.

When you configure profiles (IPSec, IKE, and DPD) for a policy-based or route-based session, you can use profiles that are added to the project or use the profiles that are shared with the project by the provider.

To configure a session for an IPSec VPN service, perform the following steps:

1. From your browser, log in to NSX Manager.
2. Go to **Netwoking** > **VPC Connectivity** > **Transit Gateways**.
3. Click the arrow to expand the configurations of the required transit gateway (CTGW with an active-standby HA).
4. Expand the **Services** section and click **View Configuration** in the **VPN** field.
5. On the **IPSec Sessions** tab, click **Add IPSec Session** and perform one of the following actions:
   - Click **Policy Based** to add a policy-based session and enter the following information.

Field Name	Description
Name	Enter a name for the session.
Local Endpoint	Select or create a new local endpoint for the session.
Remote IP	Enter the local endpoint IP address of the remote site.

Field Name	Description
Compliance Suite	Select a security compliance suite for the session. If you select a compliance suite other than **None**, the **Authentication Mode** is set to **Certificate** and the values for IKE and IPSec profiles are set to the system-defined profiles for the selected security compliance suite. You cannot edit these system-defined profiles.
Admin Status	By default the toggle is tuned on which implies that the session is enabled.
Authentication Mode	If **Compliance Suite** is set to **None**, select one of the following authentication modes. • PSK - A secret key that is shared between the CTGW and the remote site is used for the IPSec VPN session. • Certificate - The site certificate that is used to configure the local endpoint is used for authentication.
Pre-shared Key	If the **Authentication Mode** is set to **PSK**, enter a pre-shared key. The maximum length of this key is 128 characters.
Enter a subnet IP address of least one network, either local or remote.	
Local Networks	Enter at least one CTGW private subnet address or public subnet address for this policy-based IPSec VPN session. For now, VPN is not supported for private subnets.
Remote Networks	Enter at least one subnet address of the remote site.

Remote ID	Enter an ID to identify the peer site.
	For peer sites using PSK authentication, this ID value must be the IP address or the FQDN of the peer site. For peer sites using certificate authentication, this ID value must be the common name (CN) or distinguished name (DN) used in the peer site's certificate.
	If the peer site's certificate contains an email address in the DN string, for example,
	`C=US, ST=California, O=MyCompany, OU=MyOrg, CN=Site123/ emailAddress=user1@mycompany.com`
	then enter the **Remote ID** value using the following format as an example.
	`C=US, ST=California, O=MyCompany, OU=MyOrg, CN=Site123, MAILTO=user1@mycompany.com"`
	If the local site's certificate contains an email address in the DN string and the peer site uses the strongSwan IPsec implementation, enter the local site's ID value in that peer site. The following is an example.
	`C=US, ST=California, O=MyCompany, OU=MyOrg, CN=Site123, E=user1@mycompany.com"`

Description	Enter a description for the session.
Tags	Enter tags for the session.
Advanced Properties - By default, the system-generated profiles are used. Select another available profile if you do not want to use the default.	
IKE Profiles	Use the system-generated IKE profile. You can also select another IKE profile or create one.
IPSec Profiles	Use the system-generated IPSec profile. You can also select another IPSec profile or create one.
DPD Profiles	Use the system-generated DPD profile. You can also select another DPD profile or create one.
Connection Initiation Mode	Select the preferred connection initiation mode. Connection initiation mode defines the policy used by the local endpoint in the process of tunnel creation. The default value is Initiator. You can select from the following options: • Initiator - In this mode, the local endpoint initiates the IPSec VPN tunnel creation and responds to incoming tunnel setup requests from the peer gateway. • On Demand - In this mode, the local endpoint initiates the IPSec VPN tunnel creation after the first packet matching the policy rule is received. It also responds to the incoming initiation request. • Respond Only - The IPSec VPN never initiates a connection. The peer site always initiates the connection request and the local endpoint responds to that connection request.

Field Name	Description
TCP MSS Clamping	If you want to reduce the maximum segment size (MSS) payload of the TCP session during the IPSec connection, enable the toggle. Select the **TCP MSS Direction** value and optionally set the **TCP MSS Value**.

- Click **Route Based** to add a route-based session and enter the following information.

Field Name	Description
Name	Enter a name for the session.
Local Endpoint	Select or create a new local endpoint for the session.
Remote IP	Enter the local endpoint IP address of the remote site.
Compliance Suite	Select a security compliance suite for the session. If you select a compliance suite other than **None**, the **Authentication Mode** is set to **Certific ate**. Also, the values for IKE and IPSec profiles are set to the system-defined profiles for the selected security compliance suite. You cannot edit these system-defined profiles.
Admin Status	By default the toggle is tuned on which implies that the session is enabled.
Authentication Mode	If **Compliance Suite** is set to **None**, select one of the following authentication modes. • PSK - A secret key that is shared between the CTGW and the remote site is used for the IPSec VPN session. • Certificate - The site certificate that was is used to configure the local endpoint is used for authentication.
Pre-shared Key	If the **Authentication Mode** is set to **PSK**, enter a pre-shared key. The maximum length of this key is 128 characters.
Tunnel Interface	Enter the virtual tunnel interface (VTI) IP address. Ensure that this address is not from any of the subnet IP blocks defined and it does not conflict with the following subnets: • Internal Transit Subnet • Inter VRF Transit Subnet • CTGW Transit Subnet

Remote ID	Enter an ID to identify the peer site.
	For peer sites using PSK authentication, this ID value must be the IP address or the FQDN of the peer site. For peer sites using certificate authentication, this ID value must be the common name (CN) or distinguished name (DN) used in the peer site's certificate.
	If the peer site's certificate contains an email address in the DN string, for example,
	`C=US, ST=California, O=MyCompany, OU=MyOrg, CN=Site123/ emailAddress=user1@mycompany.com`
	then enter the **Remote ID** value using the following format as an example.
	`C=US, ST=California, O=MyCompany, OU=MyOrg, CN=Site123, MAILTO=user1@mycompany.com"`
	If the local site's certificate contains an email address in the DN string and the peer site uses the strongSwan IPsec implementation, enter the local site's ID value in that peer site. The following is an example.
	`C=US, ST=California, O=MyCompany, OU=MyOrg, CN=Site123, E=user1@mycompany.com"`

Description	Enter a description for the session.
Tags	Enter a description for the session.
Advanced Properties - By default, the system-generated profiles are used. Select another available profile if you do not want to use the default.	
IKE Profiles	Use the system-generated IKE profile. You can also select another IKE profile or create one.
IPSec Profiles	Use the system-generated IPSec profile. You can also select another IPSec profile or create one.
DPD Profiles	Use the system-generated DPD profile. You can also select another DPD profile or create one.

Field Name	Description
Connection Initiation Mode	Select the preferred connection initiation mode. Connection initiation mode defines the policy used by the local endpoint in the process of tunnel creation. The default value is Initiator. You can select from the following options: • Initiator - The default value. In this mode, the local endpoint initiates the IPSec VPN tunnel creation and responds to incoming tunnel setup requests from the peer gateway. • On Demand - In this mode, the local endpoint initiates the IPSec VPN tunnel creation after the first packet matching the policy rule is received. It also responds to the incoming initiation request. • Respond Only - The IPSec VPN never initiates a connection. The peer site always initiates the connection request and the local endpoint responds to that connection request.
TCP MSS Clamping	If you want to reduce the maximum segment size (MSS) payload of the TCP session during the IPSec connection, enable the toggle. Select the **TCP MSS Direction** value and optionally set the **TCP MSS Value**. For more information about TCP MSS clamping see, Understanding TCP MSS Clamping.

6. Click **Save**.

After you have created the session, you can view the session status and session-specific alarms in the **Status** and **Alarms** fields.

If you are creating a route-based session, add a static route with a prefix whose scope is set to the path of the route-based VPN session, that is, the next hop IP should point to the remote VTI you provided.

Add Static Routes for a Centralized Transit Gateway

A centralized gateway (CTGW) with multiple connections now also supports a route-based VPN with only static routes. You can define north-bound static routes for a CTGW that points to an external gateway connection or IPSec VPN.

To define a static route, you have to provide one particular prefix network and a gateway connection to which the traffic is to be routed. For an IPSec VPN, it shall be the route-based VPN path. Static routes are also available for transit gateways with a single connection.

Perform the following steps to add a static route.

1. From your browser, log in to NSX Manager.
2. From the **Project** drop-down, select the project for which you want to configure the transit gateway.
3. Go to **Networking** > **VPC Connectivity** > **Transit Gateways**.
4. Click the arrow to expand the configurations of the required transit gateway.
5. Expand the **Routing And Forwarding** section and click 0 in the **Static Routes** field.
6. Click **Add Static Route**.
7. In the **Name** field, enter a name for the static route.
8. In the **Network** field, enter a CIDR for the network.
9. Click Set in the **Next Hop** field and click **Add Next Hop**.
10. In the **Scope** drop-down, select a scope and in the **Admin Distance** field provide the admin distance value.
11. Click **Add** to save the next hop.
12. Click **Apply** and then click **Save** to save the static route.
13. Click **Close** to exit the configuration workflow.

Add a Transit Gateway with a Distributed VLAN Connection

You can create a transit gateway with one or more distributed VLAN connections.

Perform the following steps to create a transit gateway for a project.

1. From your browser, log in to NSX Manager.
2. From the **Project** drop-down, select the project for which you want to create the transit gateway.
3. Go to **Networking** > **VPC Connectivity** > **Transit Gateways**.
4. Click **Add Transit Gateway**.
5. Enter the following information:

Field Name	Description
Gateway Name	Enter a gateway name.
Span	Select a span with which you want to associate this transit gateway. The subnets connected to this transit gateway will be available to all the vSphere clusters of the associated span. Note that if you do specify a span, the default span is associated with the transit gateway. In this scenario, all subnets that are connected to this transit gateway will be available to all vSphere clusters except to the clusters of exclusive spans.
Connection	Select the **Distributed VLAN Connection** option.

6. Click Next and on the External Network Connectivity page, enter the following information.

Field Name	Description
External Connection	Select one of the following options: • Create New - Click to create a new distributed VLAN connection by providing a name and other details. • Use Existing - Click to select from the existing distributed VLAN connections.
VLAN ID	Provide the VLAN ID to which the traffic is to be routed.
Gateway CIDR IPv4 Address	Provide the gateway address and netmask through which the traffic should be routed to the external network.
VPC Network Configuration	

Field Name	Description
VPC External IP Blocks	Enter external IP blocks that can be used to create public subnets in the VPC. The entered IP blocks allow outside connectivity to VPC workloads either on public subnets or through external IPs.
Private - Transit Gateway IP Blocks	Enter private transit gateway IP blocks to create private subnets in the VPC. The entered IP blocks are available for inter-VPC communication and are not advertised by the transit gateway to the outside datacenter.

7. Click Next and on the VPC services page provide the following information.

Field Name	Description
Virtual Network Appliance Cluster	Select a virtual appliance cluster for the transit gateway.
Default Outbound NAT	Turn on the toggle to ensure that workloads on private subnets have outbound access outside of the VPC. One SNAT rule per VPC will be auto-plumbed with IP used for translation taken from an external IP block for this VPC.

8. In the **NAT** field, click **Set** and add a NAT rule for the IP taken from an External IP Block. Note that NAT is supported for a TGW that has a centralized external connection. For more information about configuring NAT for a TGW, see Add a NAT Rule for Transit Gateway.
9. Click **Save**.

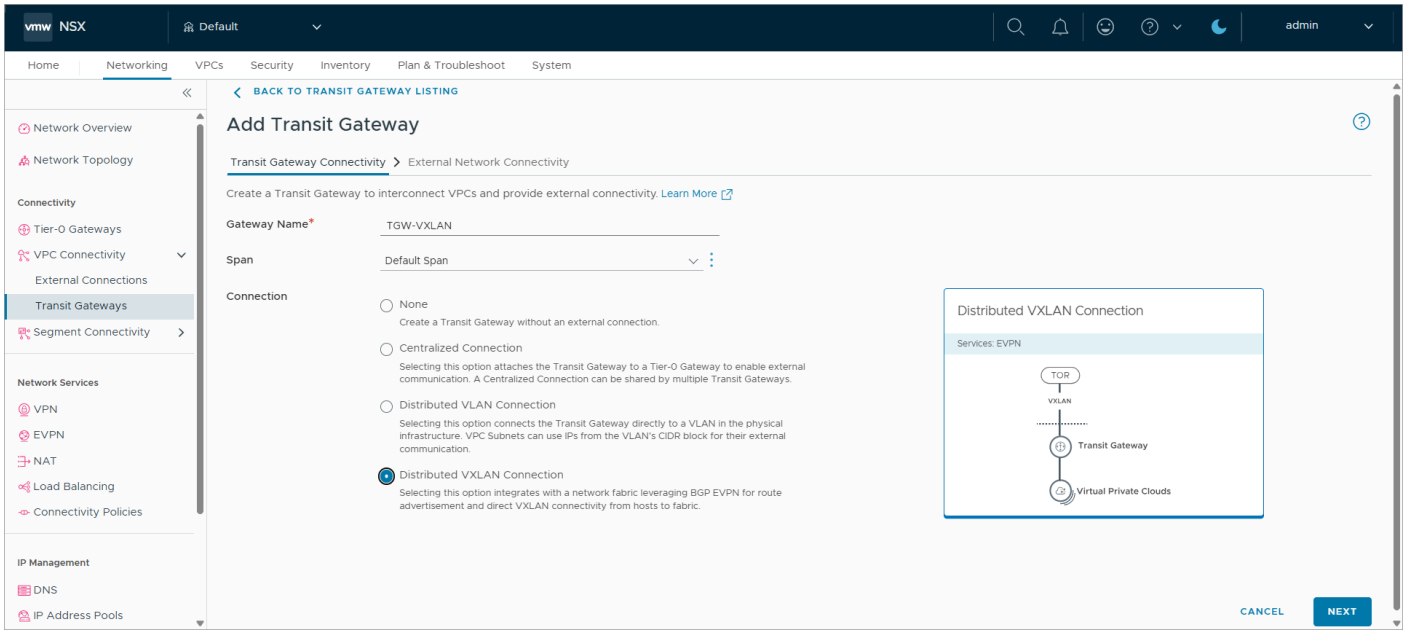
Add a Transit Gateway with a Distributed VXLAN Connection

You can create a transit gateway with one or more distributed VXLAN connections.

Create a Route Controller.

Perform the following steps to create a transit gateway for a project.

1. From your browser, log in to NSX Manager.
2. From the **Project** drop-down, select the project for which you want to create the transit gateway.
3. Go to **Networking** > **VPC Connectivity** > **Transit Gateways**.
4. Click **Add Transit Gateway**.

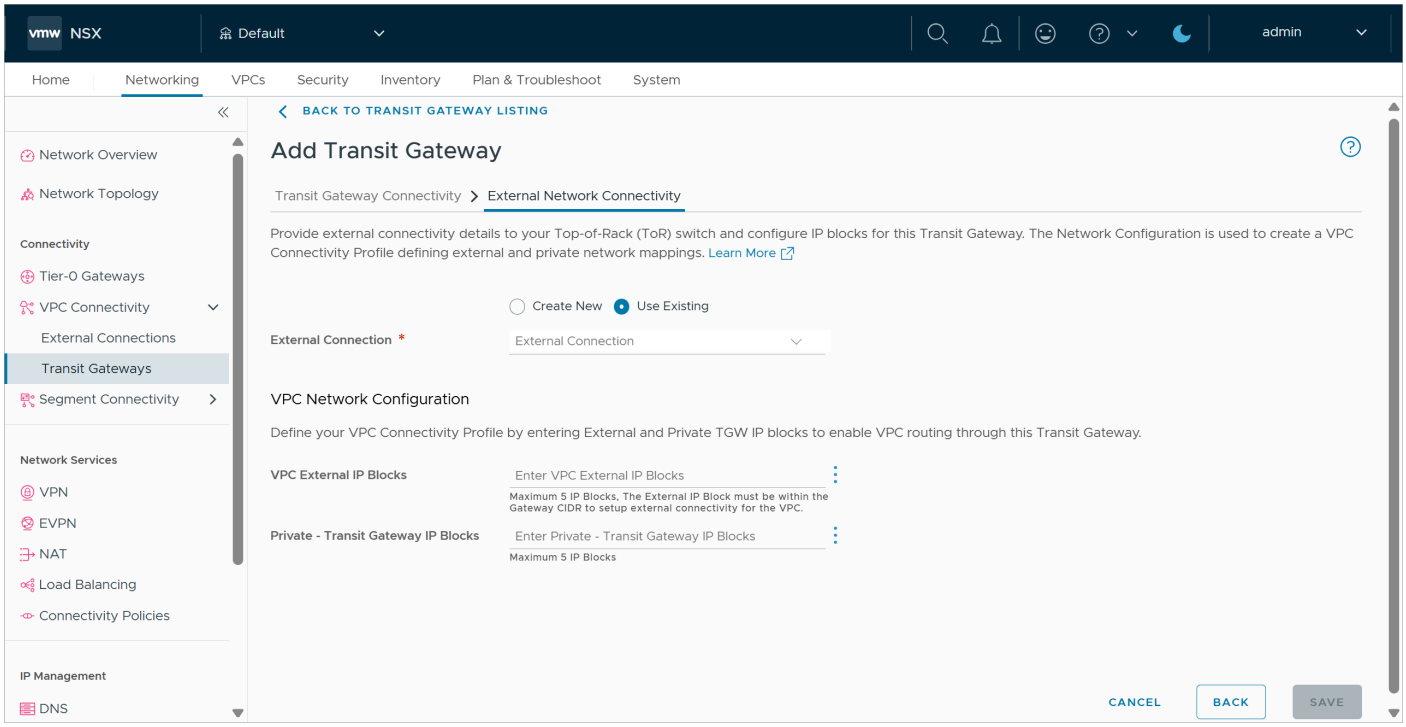


5. Enter the following information:

Field Name	Description
Gateway Name	Enter a gateway name.
Span	Select a span with which you want to associate this transit gateway. The subnets connected to this transit gateway will be available to all the vSphere clusters of the associated span. If you do specify a span, the default span is associated with the transit gateway. In this scenario, all subnets connected to this transit gateway will be available to all vSphere clusters except the vSphere clusters of exclusive spans.
Connection	Select the **Distributed VXLAN Connection** option. Select this type to align the VPC tenant with a Virtual Routing and Forwarding (VRF) tenant in an external EVPN-VXLAN network fabric.

6. Click **Next** and on the External Network Connectivity page, enter the following information.

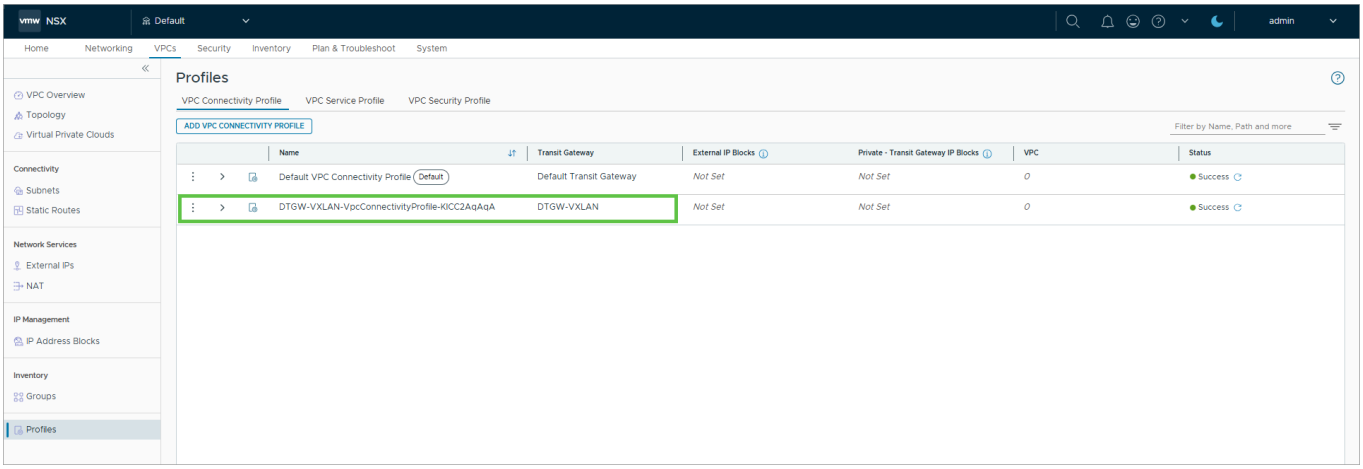
Field Name	Description
External Connection	Select one of the following options: • **Create New** - Click to create a new distributed VXLAN connection by providing a name and other settings. For detailed information on each setting, see Add a Distributed VXLAN Connection. • **Use Existing** - Click to select from existing distributed VXLAN connections.
VPC Network Configuration	
VPC External IP Blocks	Enter external IP blocks that can be used to create public subnets in the VPC. The entered IP blocks allow outside connectivity to VPC workloads either on public subnets or through external IPs.
Private - Transit Gateway IP Blocks	Enter private IP blocks to create private subnets in the VPC. The entered IP blocks are available for inter-VPC communication and are not advertised by the Transit Gateway to the outside datacenter.



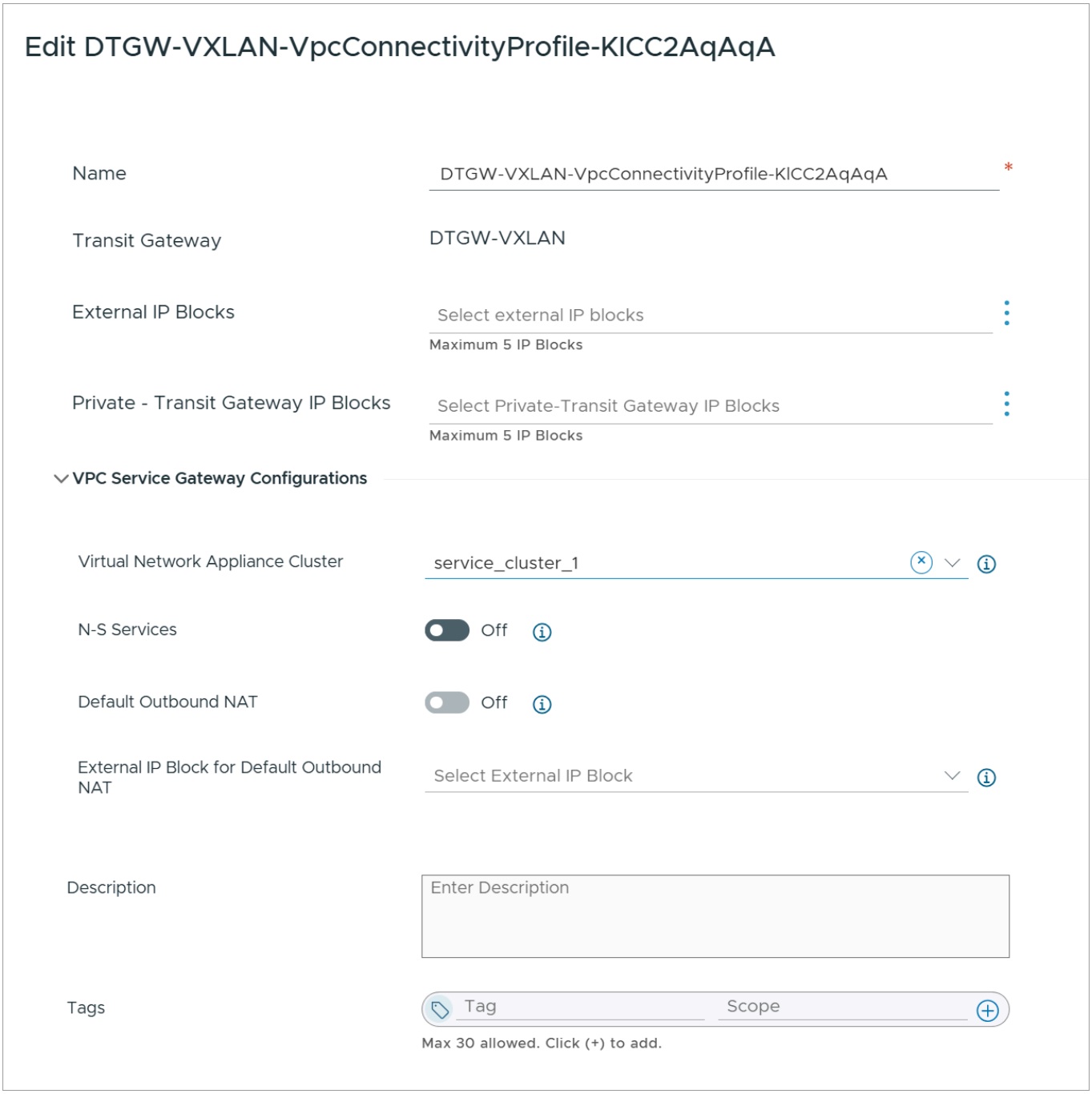
7. Click **Save**.

In version 9.1.1 and later, complete the following steps to configure support for advanced networking services such as NAT, load balancing, and DHCP relay.

1. Enable Advanced Networking Services for Distributed VXLAN Connections.
2. Create a Virtual Network Appliance Cluster that will provision the advanced networking services.
3. Configure the VPC connectivity profile assigned to the distributed VXLAN transit gateway. NSX automatically created and assigned this connectivity profile when you created the transit gateway.
   a. Click the **Project** drop-down menu, and select the project containing the distributed VXLAN transit gateway that you created.
   b. Go to **VPCs** > **Profiles** > **VPC Connectivity Profile** and find the profile assigned to the distributed VXLAN transit gateway.



   c. Click the actions menu (three dots) for the connectivity profile and select **Edit**. The profile settings are displayed.



d. Configure all profile settings as instructed in Add a VPC Connectivity Profile. In particular, for **VPC Service Gateway Configurations** > **Edge / Virtual Network Appliance Cluster**, select the virtual network appliance (VNA) cluster that you created.

Edit a Transit Gateway

You can edit the default Transit Gateway (TGW) of a custom project (including Default) to configure additional connections or VPC transit subnets.

To edit the default TGW of a project, perform the following steps:

1. Select the project and go to**Networking** > **VPC Connectivity** > **Transit Gateways**.
2. Click the three-dot menu next to the required TGW and select **Edit.**
3. Make the required updates and click **Save**.

Configure Connectivity Policies

You can create connectivity policies to specify which VPCs under the same Transit Gateway (TGW) should communicate with each other. With connectivity policies, there is no need to create multiple Transit Gateways or complex firewall rules to restrict or establish communication rules between the required VPCs.

The scope of connectivity policy determines whether subnets in VPCs with the appropriate access mode can communicate with each other or not. Note that connectivity policy works in conjunction to the subnet access mode. If the subnet for a VPC based on subnet access mode does not have a reachability to a subnet of a different VPC, it will continue to be the same even with connectivity policy configuration. In case the subnet access mode allows reachability then the connectivity policy (if configured) will come into picture to allow or deny the reachability.

You can create a connectivity policy by first grouping the required VPCs that are under the same Transit Gateway and assigning one the following connectivity scopes to the connectivity policy.

- Isolated: If the Isolated scope is assigned to a VPC group, the VPCs in that group cannot communicate with each other and with VPCs of other Community or other Isolated groups. They can communicate with Promiscuous VPCs only.
- Community: If the Community scope is assigned to a VPC group, the VPCs in that group can communicate with each other and with Promiscuous VPCs. They cannot communicate with VPCs of other Community or Isolated VPCs.
- Promiscuous: If the Promiscuous scope is assigned to a VPC group, all member VPCs in the group can communicate with each other and with all the connected VPCs. Generally, these are VPC groups that host services such as NAT and DNS.

Note that VPCs that are not added to any Connectivity Policy behave as if they are in a Community. They can communicate with each other and all the Promiscuous VPCs. Also, connectivity policies do not impact the North-South traffic through the TGW external attachment.

1. Navigate to **Networking** > **Connectivity Policies**.
2. Click **Add Policy**.
3. Enter values for the editable fields.

Field Name	Description
Name	Enter the name of the connectivity policy.
Type	Select the type of connectivity policy you want to create. You can select from the following option: • Community • Isolated • Promiscuous
Group	Select a VPC group to which you want to apply this policy. You can also click the three dots menu and create a new VPC group by adding the required VPC members. Only static VPC member groups are supported. Also, a VPC can be a part of only one connectivity policy.
VPC Members	This non-editable field displays the number of VPCs in the policy.
Status	This non-editable field displays the status of the policy.

4. Click **Save**.

Virtual Private Clouds Overview

Virtual Private Clouds (VPCs) are logically isolated networking subnets routed together by a VPC Gateway. VPCs are defined under their NSX Project by either an Enterprise Admin or a Project Admin.

A VPC connects to a Transit Gateway (recommended) and can have the following types of subnets:

- **Private - VPC**: A subnet scoped to the VPC and routed with other subnets inside the same VPC, but not to the outside network. NAT is needed if outside communication is required.

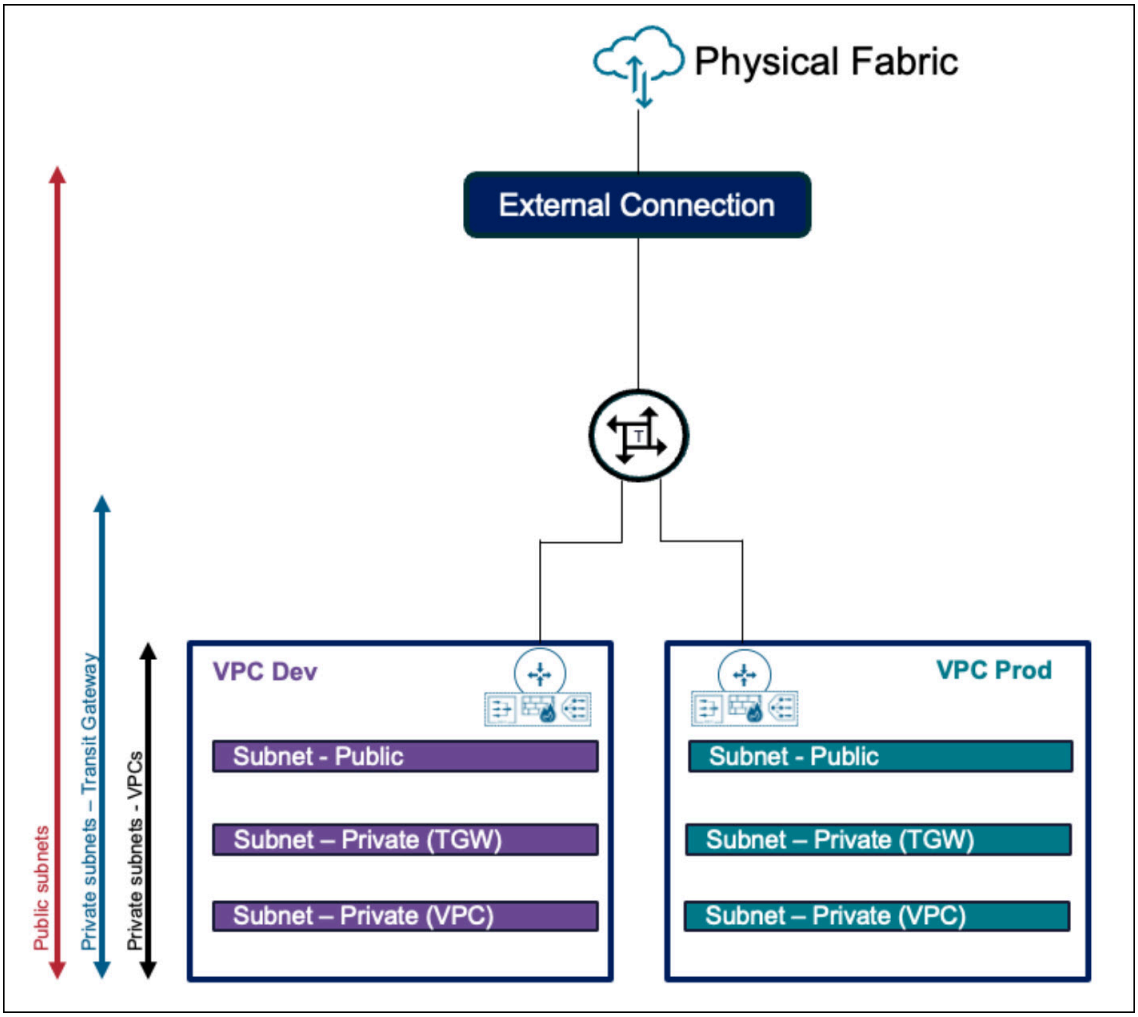
- **Private - Transit Gateway**: Subnets allowing inter-VPC connectivity below the Transit Gateway without NAT. IP translation is needed if workloads are required to be reachable to an outside network.
- **Public**: A subnet that is accessible from outside the environment, from other VPCs but also from customers and workload above the Transit Gateway.

In case you have to grant access to specific workloads on private networks from the outside environment, you can use External IPs.

External IPs – IPs that allow outside connectivity for a VM on a private network by performing IP translation. It is implemented with 1:1 NAT on the workload picking an IP from External IP block and assigning it to the VM.

Regardless of External Connection type (Centralized or Distributed), it is possible to configure static routes, groups, and DHCP server in a VPC.

Figure 421: VPC in NSX



Services in VPC leveraging Edge Nodes

In case of environments with centralized connection and edge nodes made available through the VPC connectivity profile, the VPC can have the following services as part of the VCF entitlement:

- NAT: full NAT configuration
- Stateless North-South Firewall (Centralized)
- Load Balancer (only for Supervisor Cluster)

A VPC can also benefit from the following additional features (with proper licenses):

- Integration with AVI Load Balancer

- Stateful North-South Firewall (Centralized)

Services in VPC leveraging Virtual Network Appliances

In case of environments with distributed connection and virtual network appliances made available through the VPC connectivity profile, the VPC can have the following services as part of the VCF entitlement:

- NAT
- Load Balancer (only for Supervisor Cluster)

A VPC can also benefit from the following additional feature (with proper licenses):

- Integration with AVI Load Balancer

Add a VPC Connectivity Profile

A VPC connectivity profile defines a common northbound connectivity configuration for VPCs.

When a project is created, a default connectivity profile is also created for it. Based on your requirements, you can add more connectivity profiles for a project.

The system also, by default, creates a connectivity profile whenever a new transit gateway (TGW) is created in a project. With multiple TGWs in a project, there are multiple connectivity profiles. A VPC can be re-attached or made to point to another TGW as and when required by changing the VPC connectivity profile of the VPC.

1. From your browser, log in to the NSX Manager.
2. Click the **Project** drop-down menu, and select the project for which you want to add a connectivity profile. By default, the Default project is selected.
3. Go to **VPCs** > **Profiles** > **VPC Connectivity Profile**.
4. Click **Add VPC Connectivity Profile**.
5. Enter the following fields.

Field Name	Description
Name	Enter a name for the VPC connectivity profile.
Transit Gateway	By default, the system-created transit gateway is already selected. You can assign any other Transit Gateway (TGW) based on the requirement.
External IP Blocks	Enter external IP blocks to be used for creating public subnets, external IPs, and translated IP for system and user-defined NAT rules. The entered IP blocks allow outside connectivity to VPC workloads either on public subnets or through external IPs.
Private-Transit Gateway IP Blocks	Enter private IP blocks to create private subnets in the VPC. The entered IP blocks are available for inter-VPC communication. Networks created from the Private - Transit Gateway IP blocks by default are not advertised.
VPC	Shows the number of VPCs to which this VPC connectivity profile is applied.
Status	Shows if the connectivity profile is successfully created.
VPC Service Gateway Configurations	
Edge / Virtual Network Appliance Cluster	Select an edge or a virtual network appliance cluster for this connectivity profile based on whether you want to connect the associated VPC with a centralized or a distributed transit gateway. For more information about adding edge clusters, see Create an NSX Edge Cluster and Add Edge Nodes For more information about adding a virtual network appliance cluster, see Create a Virtual Network Appliance Cluster.
N-S Services	Turn on the toggle to enable service gateway for the connected subnets and to support centralized services such as N-S firewall, NAT, or gateway QoS profile in the VPC.

Field Name	Description
Default Outbound NAT	Turn on the toggle if you want the workloads on private subnets to have an outbound access outside of VPC. One SNAT rule per VPC is auto-plumbed and the translated IP for auto-plumbed NAT rule is taken from any of the IP blocks from the **External IP Blocks** field unless the user has specified a block in the **External IP Block for Default Outbound NAT** field. If you have selected a transit gateway with a distributed connection (DGTW) for this connectivity profile and you have turned on this toggle, then you must create a Virtual Network Appliance (VNA) to enable the NAT service for private subnets. This toggle is available only if the **N-S Services** toggle is turned on.
External IP Block for Default Outbound NAT	The option list of this field is populated with all the IP blocks provided in the **External IP Blocks** field. You can select IP blocks that are to be used for the translated IP for auto-plumbed outbound NAT.
N-S Ingress QoS Profile	Select an appropriate ingress profile. For more information, see Segment Profiles.
N-S Egress QoS Profile	Select an appropriate egress profile. For more information, see Segment Profiles.
Description	Enter a description for the connectivity profile.
Tags	Enter tags and scope for this connectivity profile.

6. Click **Save**.

Add a VPC Service Profile

A VPC service profile is a collection of services (DHCP, DNS Forwarder, and subnet profiles). A service profile simplifies the configuration of common services on multiple VPCs.

When a project is created, a default service profile is also created for it. Based on your requirements, you can add more service profiles for a project.

1. From your browser, log in to an NSX Manager.
2. Click the **Project** drop-down menu, and select the project for which you want to add a service profile. By default, the Default project is selected.
3. Go to **VPCs** > **Profiles** > **VPC Service Profile** .
4. Click **Add VPC Service Profile**.
5. Enter the following fields.

Field Name	Description
Name	Enter a name for the VPC service profile.
VPC	Shows where the service profile is used.
Status	Shows if the service profile is successfully created.
DHCP Configuration - Configure the following settings.	
Enable DHCP Server Config	Turn on the toggle to enable the DHCP server. If you create a subnet for a VPC and select the DHCP server option for the subnet, the subnet inherits DHCP settings configured here and are applied to the subnet. A DHCP server can either be a Centralized or a Distributed. By default, the DHCP server mode is Distributed. In case the VPC has a Distributed connectivity, the DHCP server mode can only be Distributed DHCP. If the VPC has a Centralized connectivity, the DHCP server mode can either be a Centralized DHCP server or a Distributed server with the default mode being a Distributed server. You can use the NSX API to change the DHCP server mode. For more information about the API, see the *NSX API Guide*.
DNS Server IPs	Enter the IPs of DNS servers provided to clients by the DHCP server. Note that if you require to configure DNS servers, the **Enable DHCP Server Config** toggle must be on.
Lease Time (Seconds)	Enter the time for which clients can lease IP addresses from the DHCP server.

Field Name	Description
NTP Server IPs	Enter the IPs of NTP servers provided to clients by the DHCP server.
Enable DHCP Relay Config	Turn on the toggle to enable the relay-type DHCP configuration for relaying DHCP client requests to external DHCP servers deployed outside the VPC. If you create a public subnet and select the DHCP relay option for the subnet, the settings configured here are applied to the subnet's DHCP relay. **Note:** DHCP relay has the following support limitations: • DHCP client requests can only be relayed to external DHCP servers with IP addresses located outside the NSX environment. • DHCP relay is supported for public subnets only. It is not supported for private subnets. • In version 9.1.1 and later, DHCP relay is supported for transit gateways with centralized, distributed VLAN, and distributed VXLAN connections. • In version 9.1.0, DHCP relay is supported only for transit gateways with centralized connections.
External DHCP Server Addresses	Enter addresses of external DHCP servers. Note that if you require to configure external DHCP servers, the **Enable DHCP Relay Config** toggle must be on.
Subnet Profiles - Attach the following profiles to the VPC. For more information about these profiles, see Segment Profiles	
IP Discovery	Select an IP discovery profile or create a new profile to use. For more information, see Create an IP Discovery Segment Profile.
Spoofguard	Select a spoofguard profile or create a new profile to use. For more information, see Create an Spoof Guard Segment Profile.
MAC Discovery	Select a MAC discovery profile or create a new profile to use. For more information, see Create an MAC Discovery Segment Profile.
Segment Security	Select a segment security profile or create a new profile to use. For more information, see Create a Segment Security Segment Profile.
QoS	Select a QoS profile or create a new profile to use. For more information, see Create an QoS Segment Profile.
Description	Enter a description for the service profile.
Tags	Enter tags and scope for this connectivity profile.

6. Click **Save**.

Create a VPC

Prerequisites

- You must have a role of a Project Admin or an Enterprise Admin.
- You must have a VPC Connectivity Profile and VPC Service Profile configured.
- If you want to configure stateful rules, ensure you have the VMware Firewall with Advanced Threat Prevention license.

1. From your browser, log in to the NSX Manager.
2. Click the **Project** drop-down menu, and select the project where you want to add an NSX VPC. By default, the Default project is selected.
3. Go to **VPCs** > **Virtual Private Cloud**.
4. Click **Add VPC**.
5. On the Basic Info page, enter the following information.

Field Name	Description
Name	Enter a name for the VPC.
Private- VPC IP CIDRs	Enter a block of IP CIDRs that can be assigned to private subnets of this VPC. These CIDRs are local to VPCs and can overlap between the VPCs.
Short Log Identifier	Enter a short log Identifier that should be used to identify the VPC context in log.
Description	Enter a description for the VPC.
Tags	Enter a tag and scope for the VPC.
Advanced Settings	
Connectivity Profile	Select a connectivity profile for this VPC. For more information about the VPC Connectivity Profile, see Add a VPC Connectivity Profile. The connectivity profile is used to connect VPCs through a transit gateway and does not require you to share a Tier-0 gateway with the project. Note that if your connectivity profile has some missing configurations, you can configure them from the links in the warning that are displayed on the topology graphic. Also, if you want to move a VPC to another transit gateway (TGW), you can update the connectivity profile and make it point to another TGW.
Connectivity Policy	Select an appropriate policy to define the reachability of this VPC with respect to other VPCs under the same Transit Gateway. For more information about connectivity policy, see Configure Connectivity Policies.
Service Profile	Select a service profile for this VPC. For more information about the VPC Service Profile, see Add a VPC Service Profile.
Enable for Avi Load Balancer	Turn on this toggle to enable the discovery of this VPC in the AVI load balancer. This feature provides multi-cloud load balancing, web application firewall, application analytics, and container ingress services from the data center to the cloud. For the toggle to be available, the VPC must have one private IP CIDR.

6. Click **Save and Go Next** .
7. On the VPC created confirmation window, click **Yes, Continue**.
8. On the Additional Configurations page, enter the following information.

Field Name	Description
Users	Click Set and set role assignments to user or user groups. Note that you can add roles to users, only if your role is permitted to assign roles of other users.
Connectivity	
Subnets	Click Set and add subnets for the VPC. For more information, see Add a Subnet for the VPC.
Static Routes	Click Set to set the static routes.
Network Services	
External IPs	Click Set to set external IPs for the VMs attached to subnets. You will need to select the VM, the NIC, and whether to auto-assign the IPs. If you want to provide an IP from an IP block, use the NSX API.
NAT	Note that NAT is not available for a DTGW.

Field Name	Description
Security - Note that Groups and N-S Firewall are available only if you have the VMware Firewall with Advanced Threat Prevention license.	
E-W Firewall Rules	Click Set to add E-W firewall rules. For more information, see the vDefend documentation.
Groups	Click Set to add groups on which security rules are to be applied. For more information, see the vDefend documentation.
N-S Firewall Rules	Click Set to add N-S firewall rules. For more information, see the vDefend documentation.

9. Click **Finish**.

Add a Subnet to a VPC

Perform the following steps to add a subnet to a VPC. You can also create a subnet that can be extended to a VLAN by turning on the **VLAN Extension** toggle.

1. From your browser, log in to the NSX Manager.
2. Go to **VPCs** > **Virtual Private Cloud**.
3. Select the VPC to which you want to add the subnet.
4. In the **Subnets** column, click the link.
5. Click **Add Subnet**.
6. Enter the following fields.

Field Name	Description
Name	Enter the name of the subnet.
VLAN Extension	Turn on the toggle if you want to extend the subnet to a VLAN Network. For more information about extending a VPC subnet to VLAN, see Extending VPC Subnet to VLAN and Onboarding DVPG VMs to VPC Subnets.
Access Mode	Select an access mode for the subnet to specify the subnet's connectivity scope or accessibility. • Private VPC: The subnet will be reachable only from inside the VPC, that is, virtual machines (VMs) on a subnet with this access mode can communicate freely with other VMs on any other subnet within the same VPCs. This mode is ideal for the subnet workload hosting backend tiers of a standard three-tier application, such as database and application servers, which should only be accessed by web servers within the same VPC. • Private Transit Gateway: The subnet will be reachable from VPCs under the Project, but not advertised beyond the transit gateway that connects to the outside datacenter network. This mode enables communication for workloads both within their own VPC and between other VPCs that are connected to the same transit gateway. This mode is ideal for subnet workloads hosting shared services, such as Active Directory and DNS servers. • Public: The subnet will be reachable from within the VPC and from the outside datacenter network. This mode is ideal for subnet workloads hosting front-end web servers that need to be directly reached from the corporate network for management or user access. • L2 Only: The subnet will provide a Layer 2 connectivity for VLAN extended networks with optional Layer 3 gateway access. Note that this option is available only when the **VLAN Extension** toggle is turned on. For more information about extending a VPC subnet to VLAN, see Extending VPC Subnet to VLAN and Onboarding DVPG VMs to VPC Subnets.
IP Block	Select an IP Block for the subnet. Turn on the toggle **Auto allocate Subnet CIDR from IP Blocks** to auto allocate the IPs and select a subnet size. Based on the type of subnet you create, IPs are allocated from the private, public, or private TGW CIDRs you defined in the connectivity profile of the VPC. If you choose to turn off the auto allocation of subnet CIDR, then enter the IP CIDR.
Network Span	Displays the span of the subnet, that is, the clusters where it will be deployed.
Virtual Machines	Shows the virtual machines attached to this subnet.
Subnet Ports	Shows the subnet ports.

Field Name	Description
Status	Shows whether the subnet is successfully created.
Alarms	Shows number of alarms generated from the subnet.
VLAN Connection	If you have turned the **VLAN Extension** toggle on for creating this subnet for VLAN extension, enter the VLAN connection.
VPC Gateway Connectivity	Turn on the toggle if you want the subnet to connect to other VPCs. If you are creating this subnet for VLAN extension and you turn this toggle on, then you must connect the DLC with the Transit Gateway of VPC subnet. When this toggle is on, overlay VMs will use the VPC gateway for routing and the VLAN VMs will use the external gateway of the VLAN for routing.
DHCP Configuration - Note that DHCP Configuration options are available only if the DHCP settings are configured in the Service Profile linked to the VPC. The section is not displayed if you are creating the subnet for VLAN extension. DHCP is not applicable if the **VLAN Extension** toggle is turned on.	
DHCP Config	Select one of the following options: • **None** : DHCP will not be enabled in the subnet and all IP addresses will be set statically. • **DHCP Server**: The DHCP server specified in the service profile will automatically assign IP addresses and network settings to VMs connected to the subnet. You can set **Classless Static Routes** or **Generic DHCP Options** if you want such DHCP options to be configured to the subnet VMs. If you don't want the DHCP server to dynamically allocate the whole range of this subnet IP CIDR to the subnet VMs, you can specify the IP ranges that you want to exclude through the **Reserved IP Ranges** field. The **Static Bindings** should be a part of the **Reserved IP Ranges** . For more information, see NSX DHCP Configuration Settings: Reference. • **DHCP Relay** : The DHCP relay specified in the service profile will forward DHCP messages from VMs connected to the subnet to an external DHCP. **Note:** DHCP relay is supported only for public subnets that are connected to a transit gateway with the following connection types: • In version 9.1.1 and later, centralized, distributed VLAN, and distributed VXLAN connections are supported. • In version 9.1.0, only centralized connections are supported.
Description	Enter a description for this subnet.
Tags	Enter a tag and scope for this subnet.

7. Click **Save**.

Assign External IPs to a VM in a VPC

You can assign external IPs to virtual machines (VM) that are connected to private subnets (Private - VPC or Private Transit Gateway) of a VPC.

1. From your browser, log in to the NSX Manager.
2. Go to **VPCs** > **Network Services** > **External IPs**.
3. In the **VPC** field, select a VPC for assigning external IPs to its attached VMs.

   The page displays a list of external IPs allocated to this VPC along with other specific details of allocation, such as virtual machine, network adapter, and VPC subnet.
4. Click **Assign External IP**.
5. Select one of the following options:
   - **Assign IP automatically from IP Block** - Select this option if you want to auto assign IP from the IP address block.
   - **Specify IP to assign**- Select this option if you want to assign a specific IP from the allocated IPs. You can also make a request for a specific IP from the IP address block.

6. In the **Virtual Machine** field, select the required VM to which the external IP is to be allocated.
7. In the **Network Adapter** field, select the required network adapter. Note that only those network adapters are displayed which do not have an external IP already assigned.
8. Click **Save**.

Add a NAT Rule for a VPC

You can add a NAT rule at the VPC level to provide external connectivity to the internal or VPC private networks.

To create a NAT rule for a VPC, you must ensure that the following conditions are met:

- You have shared external IP blocks with the VPC from which the translated IP is to be selected.
- You have created a Virtual Network Appliance (VNA) if the VPC is connected transit gateway with distributed connection. For more information about creating a VNA, see Create a Virtual Network Appliance Cluster.
- You have selected an edge cluster or the VNA cluster with the project.
- You have activated north-south services and default outbound NAT in the connectivity profile that is linked to the VPC. For more information about connectivity profile, see Add a VPC Connectivity Profile.

You can create 1:1 NAT or 1:N NAT rules based on your requirements and number of public IP addresses available to you.

A 1:1 NAT rule creates a direct, dedicated relationship between a single internal (private) IP address and a single external (public) IP address. You can create a 1:1 NAT rule for servers, such as a web server or mail server, that need to be reachable from external entities requiring a consistent IP to connect with. To create a 1:1 NAT rule, you can use the following procedure or use the external IP binding by assigning a specific IP to a VM. For more information about external IP binding, see Assign External IPs to a VM in a VPC.

A 1:N NAT rule maps multiple internal (private) IP addresses to a single external (public) IP address by using different source ports to track the sessions. You can create a 1:N rule if you have limited public IP addresses that you want to conserve while providing connectivity. If you want to create a 1:N rule for a VPC that is connected transit gateway with distributed connection, you must create a VNA cluster first.

Consider the following conditions when you are creating a NAT rule for VPCs attached to a centralized transit gateway (CTGW) running on an edge node:

- DNAT rule should be configured at the nearest point to the translated IP, that is, the VPC.
- SNAT rule should be configured at the furthest point till the Match IP is visible - VPC for Private IP and centralized transit gateway for Project IP.

Consider the following conditions when you are creating a NAT rule for VPCs attached to a distributed transit gateway (DTGW):

- For SNAT, the source network must not overlap with external IP blocks.
- For DNAT, the translated network must not overlap with external IP blocks.
- Reflexive NAT is not supported. Instead, External IP binding (EIP) can be used for creating a 1:1 NAT rule.
- Default SNAT is allowed for both private and TGW private subnets and is performed on the single centralized service port.

Perform the following steps to add a NAT rule for VPC.

1. From your browser, log in to the NSX Manager.
2. Click the **Project** drop-down menu, and select the required project. By default, the Default project is selected.
3. Go to **VPCs** > **Network Services** > **NAT**.
4. Click **Add NAT Rule**.
5. Enter the following information.

Field Name	Description
Name	Enter a NAT rule name.
Action	Select an action. The following actions are available:

Field Name	Description
	• SNAT • DNAT • Reflexive (This option is not available for VPC with distributed connectivity) • NO SNAT • NO DNAT
Match Source IP | Port	Enter the source IP address and source port. Specify an IP address, or an IP address range in CIDR format. For SNAT, NO_SNAT, and Reflexive rules, this is a required field and represents the source network of the packets leaving the network. For DNAT and NO_DNAT rules, optionally it can contain a source network of incoming packets. For a NULL value, this field represents any network.
Match Destination IP | Port	Enter a destination IP address or an IP address range in CIDR format and destination port. This field is required for DNAT and NO_DNAT rules and represents the network from where the inbound packets originate. This field is not applicable for Reflexive.
Select Protocol	Select the required protocol.
Translated IP | Port	Enter a value for translated IP. Specify an IPv4 address, or an IP address range in CIDR format. If translated IP is less than the match IP for SNAT, it will work as 1:N NAT. If you select SNAT or Reflexive rule, you can request allocation of IP/CIDR from the required external IP block.
Enabled	Turn on the toggle to enable the rule.
Logging	Turn on the toggle to enable logging.
Priority	A lower value means a higher priority. The default is 0. A No SNAT or No DNAT rule should have a higher priority than other rules for NO-NAT to be executed.
Firewall	The available settings are: • Match External Address - The firewall will be applied to the external address of a NAT rule. – For SNAT, the external address is the translated source address after NAT is done. – For DNAT, the external address is the original destination address before NAT is done. – For REFLEXIVE, to egress traffic, the firewall is applied to the translated source address after NAT is done. For ingress traffic, the firewall is applied to the original destination address before NAT is done. • Match Internal Address - Indicates the firewall will be applied to the internal address of a NAT rule. – For SNAT, the internal address is the original source address before NAT is done. – For DNAT, the internal address is the translated destination address after NAT is done. – For REFLEXIVE, for egress traffic, the firewall is applied to the original source address before NAT is done. For ingress traffic, the firewall is applied to the translated destination address after NAT is done. • Bypass - The packet bypasses firewall rules.
Description	Enter a description for NAT.
Tag	Enter tag and scope for NAT.

6. Click **Save**.

Extending VPC Subnet to VLAN and Onboarding DVPG VMs to VPC Subnets

You can use the VLAN Extension feature to create a subnet that can connect VLAN workloads to the overlay. You can also use VLAN Extension to migrate workloads from a DVPG VLAN to an overlay network so that you can avail benefits of VPC resources by gaining full access to them. With VLAN extension, a subnet is bridged to a VLAN that makes the VLAN connection and its associated CIDR block reachable by that single VPC subnet. The subnet is bridged to a VLAN by creating a Distributed VLAN Connection (DLC). The VLAN can also be attached to the subnet's Transit Gateway (TGW) to provide routing through VPC. If you have physical workloads on the VLAN side, you can also provide connectivity between these physical workloads and VPC through VLAN extension.

To extend a subnet to a VLAN, implement the following workflow.

1. Create an IP address block with the same CIDR as that of the VLAN to which the VPC subnet has to be extended. Note that the visibility of this IP address block must be external and the **Reserved for Specific Subnet** toggle must be on. Once you reserve this IP address block, it can only be assigned to a single subnet that will be extended to the VLAN and no other subnet can share this block. For more information about creating an IP address block, see Add an NSX IP Address Block.
2. Configure an external connection of type "distributed" and mark it exclusively dedicated to a subnet. This distributed VLAN connection should be created with the same IP address block that you created earlier and with the gateway IP set to match the gateway on the VLAN side. For more information about creating a subnet-exclusive distributed VLAN connection, see Add a Distributed VLAN Connection.
3. If you want a Layer 3 connectivity, create an attachment of the distributed VLAN connection with the Transit Gateway of the extended VPC subnet by specifying this connection as an external connection of the Transit Gateway.
4. Now create a VPC subnet with **Access Mode** as public and **VLAN Extension** as enabled to extend the VLAN. VMs can be attached to this VPC subnet or this subnet can be a bridge for VMs on the VLAN side. For more information about creating a subnet, see Add a Subnet to a VPC

You can also share the distributed VLAN connection as a normal external connection with the custom project whose subnet you want to extend to the VLAN and then in the custom project create a subnet with the VLAN extension enabled by using the subnet exclusive IP block.

Once the subnet is created, you can move the VMs to this subnet vSphere.

This VLAN extended subnet is directly mapped, deriving the gateway IPs and knowing the VLAN to which it needs to extend (L2 extension).

- The native subnet connects to the VPC based on the configuration of the distributed VLAN connection.
- VMs can connect to this public subnet and can bridge to VMs connected to the DVPG.

Configure VLAN-backed VPC Subnets

Starting with VCF 9.1, subnets with layer 2 support can be realized with the VLAN-backed VPC (TEP-less Mode). This enables easier adoption of VPC in VCF due to elimination of TEP infra requirements.

Perform the following steps to configure a VLAN-backed VPC subnet.

1. Prepare a cluster with TEP-less mode and add the cluster to the TEP-less span.
2. Create a transit gateway with the TEP-less span.
3. Create a VPC.
4. Create a distributed VLAN external connection.
5. Add a subnet with L2_Only mode to the VPC.
6. Attach the required VMs to the subnet.

1. Prepare a cluster with TEP-less mode.
   a) Create a TEP-less Transport Node Profile (TNP): Create a TNP by following the usual steps. When you set the switch, ensure that you select the default transport zone and you select **None** for the **IP Address Type (TEP)** option. For more information about creating a TNP, see Adding a Host Transport Node Profile for Hosts.
   b) Create a Span: A span must be created for the TEP-less mode. For more information about creating a span see Add a Network Span.
   c) Prepare the cluster: Go to **System** > **Fabric** > **Hosts** and click the three-dot menu of the required cluster. Select **Configure NSX (Advanced)** and apply the TEP-less TNP you created and the span to the cluster. The cluster inherits properties from the TNP and becomes a VLAN-backed networking stack.
2. Create a transit gateway.
   a) Go to **Networking** > **VPC Connectivity** > **Transit Gateways**.
   b) Click **Add Transit Gateway**.
   c) Set the **Span** field to the span that you created in the previous step.
   d) Set **Connection** to **None**.
3. Create a VPC.
   a) Create the VPC by following the usual steps. For more information about creating a VPC, see Create a VPC.
   b) Set the connectivity profile to the profile that the system created for the transit gateway that you created in the previous step.
4. Create a distributed VLAN external connection.
   a) Create a distributed VLAN external connection by following the usual steps. For more information about creating a distributed VLAN external connection, see Add a Distributed VLAN Connection.
   b) Turn on the **Dedicated to Subnet** toggle.
   c) Set the **Subnet Extension Connection** to **Layer 2**.
5. Add a subnet to the VPC.
   a) Create a subnet by following the usual steps. For more information about creating a subnet, see Add a Subnet to a VPC.
   b) Set the **Access Mode** to **L2 only**.
   c) Set **VLAN Connection** to the distributed VLAN external connection that you created in the previous step.
6. Attach the required VMs to the subnet.
   a) Go to the vSphere Client and attach the required VMs to the subnet that you created in the previous step. For more information about attaching a VM to the subnet, see *Connect Virtual Machines to a Subnet* under *Managing Virtual Private Clouds in vCenter* in the *Building Cloud Infrastructure* section.

Virtual Private Network (VPN)

NSX supports IPSec Virtual Private Network (IPSec VPN) and Layer 2 VPN (L2 VPN) on an NSX Edge node.

- IPSec VPN offers site-to-site connectivity between an NSX Edge node and remote sites.
- L2 VPN allows you to extend L2 networks across data centers securely by enabling virtual machines to keep their network connectivity across geographical boundaries while using the same IP address

You must have a working NSX Edge node with at least one configured Tier-0 or Tier-1 gateway before you can configure a VPN service.

You can configure VPN services through NSX Manager or the REST API.

> **Important:**
> To configure VPN services, you must use new objects, such as Tier-0 gateways, that were created using the NSX Manager UI or Policy APIs that are included with NSX.

System-default configuration profiles with predefined values and settings are made available for your use during a VPN service configuration. You can also define new profiles with different settings and select them during the VPN service configuration.

Understanding IPSec VPN

Internet Protocol Security (IPSec) VPN secures traffic flowing between two networks connected over a public network through IPSec gateways called endpoints. NSX Edge only supports a tunnel mode that uses IP tunneling with Encapsulating Security Payload (ESP). ESP operates directly on top of IP, using IP protocol number 50.

IPSec VPN uses the IKE protocol to negotiate security parameters. The default UDP port is set to 500. If NAT is detected in the gateway, the port is set to UDP 4500.

NSX Edge supports a policy-based or a route-based IPSec VPN.

IPSec VPN service is supported on Tier-0, Tier-0 VRF, and Tier-1 gateways. See Add an NSX Tier-0 Gateway or Add an NSX Tier-1 Gateway for more information. The Tier-0 or Tier-1 gateway must be in `Active-Standby` high-availability mode when used for an IPSec VPN service. You can use segments that are connected either to Tier-0, Tier-0 VRF, or Tier-1 gateways when configuring an IPSec VPN service.

> **Note:**
> IPSec VPN session is not supported between a parent Tier-0 gateway and a Tier-0 VRF gateway that is attached to this parent Tier-0 gateway.

An IPsec VPN service in NSX uses the gateway-level failover functionality to support a high-availability service at the VPN service level. Tunnels are re-established on failover and VPN configuration data is synchronized. . The IPSec VPN state is synchronized to the standby NSX Edge node when the current active NSX Edge node fails and the original standby NSX Edge node becomes the new active NSX Edge node without renegotiating the tunnels. This feature is supported for both policy-based and route-based IPSec VPN services.

Pre-shared key mode authentication and IP unicast traffic are supported between the NSX Edge node and remote VPN sites. In addition, certificate authentication is supported. Only certificate types signed by one of the following signature hash algorithms are supported.

- SHA256withRSA
- SHA384withRSA
- SHA512withRSA

Using Policy-Based IPSec VPN

Policy-based IPSec VPN requires a VPN policy to be applied to packets to determine which traffic is to be protected by IPSec before being passed through the VPN tunnel.

This type of VPN is considered static because when a local network topology and configuration change, the VPN policy settings must also be updated to accommodate the changes.

When using a policy-based IPSec VPN with NSX, you use IPSec tunnels to connect one or more local subnets behind the NSX Edge node with the peer subnets on the remote VPN site.

You can deploy an NSX Edge node behind a NAT device. In this deployment, the NAT device translates the VPN address of an NSX Edge node to a publicly accessible address facing the Internet. Remote VPN sites use this public address to access the NSX Edge node.

You can place remote VPN sites behind a NAT device as well. You must provide the remote VPN site's public IP address and its ID (either FQDN or IP address) to set up the IPSec tunnel. On both ends, static one-to-one NAT is required for the VPN address.

> **Note:**
> DNAT is not supported on tier-0 or tier-1 gateways where policy-based IPSec VPN are configured.

IPSec VPN can provide a secure communications tunnel between an on-premises network and a network in your cloud software-defined data center (SDDC). For policy-based IPSec VPN, the local and peer networks provided in the session must be configured symmetrically at both endpoints. For example, if the cloud-SDDC has the *`local`* networks configured as `X, Y, Z` subnets and the *`peer`* network is `A`, then the on-premises VPN configuration must have `A` as the *`local`* network and `X, Y, Z` as the *`peer`* network. This case is true even when `A` is set to `ANY (0.0.0.0/0)`. For example, if the cloud-SDDC policy-based VPN session has the *`local`* network configured as `10.1.1.0/24` and the *`peer`* network

as `0.0.0.0/0`, at the on-premises VPN endpoint, the VPN configuration must have `0.0.0.0/0` as the local network and `10.1.1.0/24` as the *`peer`* network. If misconfigured, the IPSec VPN tunnel negotiation might fail.

The size of the NSX Edge node determines the maximum number of supported tunnels, as shown in the following table.

Table 1033: Number of IPSec Tunnels Supported

Edge Node Size	# of IPSec Tunnels Per VPN Session (Policy-Based)	# of Sessions Per VPN Service	# of IPSec Tunnels Per VPN Service (16 tunnels per session)
Small	N/A (POC/Lab Only)	N/A (POC/Lab Only)	N/A (POC/Lab Only)
Medium	128	128	2048
Large	128 (soft limit)	256	4096

Restriction:
The inherent architecture of policy-based IPSec VPN restricts you from setting up a VPN tunnel redundancy.

For information about configuring a policy-based IPSec VPN, see Add an NSX IPSec VPN Service.

Using Route-Based IPSec VPN

Route-based IPSec VPN provides tunneling on traffic based on the static routes or routes learned dynamically over a special interface called virtual tunnel interface (VTI) using, for example, BGP as the protocol. IPSec secures all the traffic flowing through the VTI.

Note:

- OSPF dynamic routing is not supported for routing through IPSec VPN tunnels.
- Dynamic routing for VTI is not supported on VPN that is based on Tier-1 gateways.
- Load balancer over IPSec VPN is not supported for route-based VPN terminated on Tier-1 gateways.

Route-based IPSec VPN is similar to Generic Routing Encapsulation (GRE) over IPSec, with the exception that no additional encapsulation is added to the packet before applying IPSec processing.

In this VPN tunneling approach, VTIs are created on the NSX Edge node. Each VTI is associated with an IPSec tunnel. The encrypted traffic is routed from one site to another site through the VTI interfaces. IPSec processing happens only at the VTI.

VPN Tunnel Redundancy

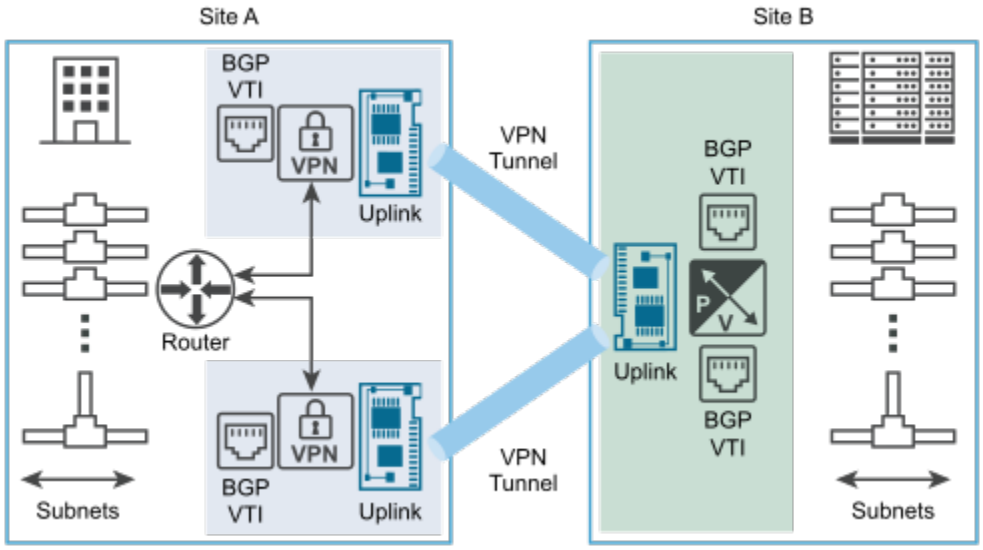
You can configure VPN tunnel redundancy with a route-based IPSec VPN session that is configured on a Tier-0 gateway. With tunnel redundancy, multiple tunnels can be set up between two sites, with one tunnel being used as the primary with failover to the other tunnels when the primary tunnel becomes unavailable. This feature is most useful when a site has multiple connectivity options, such as with different ISPs for link redundancy.

Important:

- In NSX, IPSec VPN tunnel redundancy is supported using BGP only.
- Do not use static routing for route-based IPSec VPN tunnels to achieve VPN tunnel redundancy.

The following figure shows a logical representation of IPSec VPN tunnel redundancy between two sites. In this figure, Site A and Site B represent two data centers. For this example, assume that NSX is not managing the Edge VPN Gateways in Site A, and that NSX is managing an Edge Gateway virtual appliance in Site B.

Figure 422: Tunnel Redundancy in Route-Based IPSec VPN



As shown in the figure, you can configure two independent IPSec VPN tunnels by using VTIs. Dynamic routing is configured using BGP protocol to achieve tunnel redundancy. If both IPSec VPN tunnels are available, they remain in service. All the traffic destined from Site A to Site B through the NSX Edge node is routed through the VTI. The data traffic undergoes IPSec processing and goes out of its associated NSX Edge node uplink interface. All the incoming IPSec traffic received from Site B VPN Gateway on the NSX Edge node uplink interface is forwarded to the VTI after decryption, and then usual routing takes place.

You must configure BGP HoldDown timer and KeepAlive timer values to detect loss of connectivity with peer within the required failover time. See Configure BGP.

Understanding Layer 2 VPN

With Layer 2 VPN (L2 VPN), you can extend Layer 2 networks (VNIs or VLANs) across multiple sites on the same broadcast domain. This connection is secured with a route-based IPSec tunnel between the L2 VPN server and the L2 VPN client.

> **Note:**
> This L2 VPN feature is available only for NSX and does not have any third-party interoperability.

The extended network is a single subnet with a single broadcast domain, which means the VMs remain on the same subnet when they are moved between sites. The VMs' IP addresses do not change when they are moved. So, enterprises can seamlessly migrate VMs between network sites. The VMs can run on either VNI-based networks or VLAN-based networks. For cloud providers, L2 VPN provides a mechanism to onboard tenants without modifying existing IP addresses used by their workloads and applications.

In addition to supporting data center migration, an on-premises network extended with an L2 VPN is useful for a disaster recovery plan and dynamically engaging off-premise compute resources to meet the increased demand.

L2 VPN services are supported on both Tier-0 and Tier-1 gateways. Only one L2 VPN service (either client or server) can be configured for either Tier-0 or Tier-1 gateway.

Each L2 VPN session has one Generic Routing Encapsulation (GRE) tunnel. Tunnel redundancy is not supported. An L2 VPN session can extend up to 4094 L2 segments.

VLAN-based and VNI-based segments can be extended using L2 VPN service on an NSX Edge node that is managed in an NSX environment. You can extend L2 networks from VLAN to VNI, VLAN to VLAN, and VNI to VNI.

Segments can be connected to either Tier-0 or Tier-1 gateways and use L2 VPN services.

Also supported is VLAN trunking using virtual distributed switching (VDS) 7.0 or later running NSX. If there are sufficient compute and I/O resources, an NSX Edge cluster can extend multiple VLAN networks over a single interface using VLAN trunking.

Beginning with NSX 3.0, the L2 VPN path MTU discovery (PMTUD) feature is enabled by default. With the PMTUD enabled, the source host learns the path MTU value for the destination host through the L2 VPN tunnel and limits the length of the outgoing IP packet to the learned value. This feature helps avoid IP fragmentation and reassembly within the tunnel, as a result improving the L2 VPN performance.

The L2 VPN PMTUD feature is not applicable for non-IP, non-unicast, and unicast packets with the DF (Don't Fragment) flag cleared. The global PMTU cache timer expires every 10 minutes. To disable or enable L2 VPN PMTUD feature, see Enable and Disable L2 VPN Path MTU Discovery.

The L2 VPN service support is provided in the following deployment scenarios.

- Between an NSX L2 VPN server and an L2 VPN client hosted on an NSX Edge that is managed in an NSX Data Center for vSphere environment. A managed L2 VPN client supports both VLANs and VNIs.
- Between an NSX L2 VPN server and an L2 VPN client hosted on a standalone or unmanaged NSX Edge. An unmanaged L2 VPN client supports VLANs only.
- Between an NSX L2 VPN server and an L2 VPN client hosted on an autonomous NSX Edge. An autonomous L2 VPN client supports VLANs only.
- Beginning with NSX 2.4 release, L2 VPN service support is available between an NSX L2 VPN server and NSX L2 VPN clients. In this scenario, you can extend the logical L2 segments between two on-premises software-defined data centers (SDDCs).

The following table lists the compatible NSX versions that can be used for the L2 VPN server and client.

Table 1034: NSX L2 VPN Client

L2 VPN Server Version (NSX)	L2 VPN Client Version (NSX) Validated	L2 VPN Client Version (NSX) Supported Not Validated
4.1.0	4.1.0, 4.0.0.1, 4.0.1,3.2.2	3.1.x and later 3.1.x, 3.2.x, and 4.x versions that are not listed under the Validated column
4.0.1.1	4.0.1.1, 4.0.0.1, 3.2.1.2	3.1.x and later
3.2.0	3.2.0, 3.1.3, 3.1.2	3.1.x and later
3.1.3	3.1.3, 3.1.2, 3.1.1	3.0.x and later
3.1.2	3.1.2, 3.1.1, 2.5.3	3.0.x and later
3.1.1	3.1.1, 3.1.0, 3.0.1	3.0.x and later
3.1.0	3.1.0, 3.0.1, 3.0.0	3.0.x and later
3.0.3	3.0.3, 3.0.2, 3.0.1	2.5.x and later
3.0.2	3.0.2, 3.0.1, 2.5.2	2.5.x and later
3.0.0	3.0.0, 2.5.0, 2.5.1	2.5.x and later

The following table lists the compatible NSX and NSX-v versions that can be used for the L2 VPN server and client.

Table 1035: NSX for vSphere L2VPN Client

L2 VPN Server Version (NSX)	L2 VPN Client Version (NSX-v) Validated	L2 VPN Client Version (NSX-v) Supported Not Validated
3.2.x	6.4.12	6.4.x and later
3.1.x	6.4.8	6.4.x and later

Enable and Disable L2 VPN Path MTU Discovery

You can enable or disable the L2 VPN path MTU (PMTU) discovery feature using CLI commands. By default L2 VPN PMTU discovery is enabled.

You must have the user name and password for the admin account to log in to the NSX Edge node.

1. Log in with admin privileges to the CLI of the NSX Edge node .
2. To check the status of the L2 VPN PMTU discovery feature, use the following command.

   ```
   Nsxedge> get dataplane l2vpn-pmtu config
   ```

   If the feature is enabled, you see the following output: `l2vpn_pmtu_enabled : True`.

   If the feature is disabled, you see the following output: `l2vpn_pmtu_enabled : False`.
3. To disable the L2 VPN PMTU discovery feature, use the following command.

   ```
   nsxedge> set dataplane l2vpn-pmtu disabled
   ```

Adding NSX VPN Services

You can add either an IPSec VPN (policy-based or route-based) or an L2 VPN service using the NSX Manager user interface (UI).

The following sections provide information about the workflows required to set up the VPN service that you need. The topics that follow these sections provide details on how to add either an IPSec VPN or an L2 VPN service using the NSX Manager UI.

Policy-Based IPSec VPN Configuration Workflow

Configuring a policy-based IPSec VPN service workflow requires the following high-level steps.

1. Create and enable an IPSec VPN service using an existing Tier-0 or Tier-1 gateway. Refer to Add an NSX IPSec VPN Service.
2. Create a DPD (dead peer detection) profile, if you prefer not to use the system default. Refer to Add DPD Profiles.
3. To use a non-system default IKE profile, define an IKE (Internet Key Exchange) profile. Refer to Add IKE Profiles.
4. Configure an IPSec profile using Add IPSec Profiles.
5. Use Add Local Endpoints to create a VPN server hosted on the NSX Edge.
6. Configure a policy-based IPSec VPN session, apply the profiles, and attach the local endpoint to it. Refer to Add a Policy-Based IPSec Session. Specify the local and peer subnets to be used for the tunnel. Traffic from a local subnet destined to the peer subnet is protected using the tunnel defined in the session.
7. To get a representative configuration of VPN on the remote VPN endpoint, use the **Download Configuration** feature. This file contains parameters that come from the IPSec VPN session configured in step 6 and can be used to configure the remote endpoint of the VPN session. Refer to Download the Remote Side IPSec VPN Configuration File for details.

Route-Based IPSec VPN Configuration Workflow

A route-based IPSec VPN configuration workflow requires the following high-level steps.

1. Configure and enable an IPSec VPN service using an existing Tier-0 or Tier-1 gateway. Refer to Add an NSX IPSec VPN Service.
2. Define an IKE profile if you prefer not to use the default IKE profile. Refer to Add IKE Profiles.
3. If you decide not to use the system default IPSec profile, create one using Add IPSec Profiles.
4. Create a DPD profile if you want to do not want to use the default DPD profile. Refer to Add DPD Profiles.
5. Add a local endpoint using Add Local Endpoints.
6. Configure a route-based IPSec VPN session, apply the profiles, and attach the local endpoint to the session. Provide a VTI IP in the configuration and use the same IP to configure routing. The routes can be static or dynamic (using BGP). Refer to Add a Route-Based IPSec Session.

7. To get a representative configuration of VPN on the remote VPN endpoint, use the **Download Configuration** feature. This file contains parameters that come from the IPSec VPN session configured in step 6 and can be used to configure the remote endpoint of the VPN session. Refer to Download the Remote Side IPSec VPN Configuration File for details.

L2 VPN Configuration Workflow

Configuring an L2 VPN requires that you configure an L2 VPN service in Server mode and then another L2 VPN service in Client mode. You also must configure the sessions for the L2 VPN server and L2 VPN client using the peer code generated by the L2 VPN Server. Following is a high-level workflow for configuring an L2 VPN service.

1. Create an L2 VPN Service in Server mode.
   a. Configure a route-based IPSec VPN tunnel with a Tier-0 or Tier-1 gateway and an L2 VPN Server service using that route-based IPSec tunnel. Refer to Add an L2 VPN Server Service.
   b. Configure an L2 VPN server session, which binds the newly created route-based IPSec VPN service and the L2 VPN server service, and automatically allocates the GRE IP addresses. Refer to Add an L2 VPN Server Session.
   c. Add segments to the L2 VPN Server sessions. This step is also described in Add an L2 VPN Server Session.
   d. Use Download the Remote Side L2 VPN Configuration File to obtain the peer code for the L2 VPN Server service session, which must be applied on the remote site and used to configure the L2 VPN Client session automatically.
2. Create an L2 VPN Service in Client mode.
   a. Configure another route-based IPSec VPN service using a different Tier-0 or Tier-1 gateway and configure an L2 VPN Client service using that Tier-0 or Tier-1 gateway that you just configured. Refer to Add an L2 VPN Client Service for information.
   b. Define the L2 VPN Client sessions by importing the peer code generated by the L2 VPN Server service. Refer to Add an L2 VPN Client Session.
   c. Add segments to the L2 VPN Client sessions defined in the previous step. This step is described in Add an L2 VPN Client Session.

Add an NSX IPSec VPN Service

NSX supports a site-to-site IPSec VPN service between a Tier-0 or Tier-1 gateway and remote sites. You can create a policy-based or a route-based IPSec VPN service. You must create the IPSec VPN service first before you can configure either a policy-based or a route-based IPSec VPN session.

- Familiarize yourself with the IPSec VPN. See Understanding IPSec VPN.
- You must have at least one tier-0 or tier-1 gateway configured and available for use. See Add an NSX Tier-0 Gateway or Add an NSX Tier-1 Gateway for more information.
- When configuring NSX with both NAT and IPSec, it is important to follow the correct sequence of steps to ensure proper functionality. Specifically, configure NAT before setting up the VPN connection. If you inadvertently configure the VPN before NAT, for instance, by adding a NAT rule after your VPN session is configured, the VPN tunnel status will remain down. You must reenable or restart the VPN configuration to reestablish the VPN tunnel. To avoid this issue, always configure NAT before setting up the VPN connection in NSX or perform the workaround to resolve the issue.

  Note:
  IPSec VPN is not supported in the NSX limited export release.

IPSec VPN is not supported when the local endpoint IP address goes through NAT in the same logical router that the IPSec VPN session is configured.

1. With admin privileges, log in to NSX Manager.
2. Navigate to **Networking** > **VPN** > **VPN Services**.
3. Select **Add Service** > **IP Sec**.
4. Enter a name for the IPSec service.
   This name is required.

5. From the **Tier-0/Tier-1 Gateway** drop-down menu, select the Tier-0 or Tier-1 gateway to associate with this IPSec VPN service.
6. Enable or disable **Admin Status**.

   By default, the value is set to `Enabled`, which means the IPSec VPN service is enabled on the Tier-0 or Tier-1 gateway after the new IPSec VPN service is configured.
7. Set the value for **IKE Log Level**.

   The default is set to the `Info` level.
8. Enter a value for **Tags** if you want to include this service in a tag group.
9. To enable or disable the stateful synchronization of VPN sessions, toggle **Session sync**.

   By default, the value is set to `Enabled`.
10. Click **Global Bypass Rules** if you want to allow data packets to be exchanged between the specified local and remote IP addresses without any IPSec protection. In the **Local Networks** and **Remote Networks** text boxes, enter the list of local and remote subnets between which the bypass rules are applied.

    If you enable these rules, data packets are exchanged between the specified local and remote IP sites even if their IP addresses are specified in the IPSec session rules. The default is to use the IPSec protection when data is exchanged between local and remote sites. These rules apply for all IPSec VPN sessions created within this IPSec VPN service.
11. Click **Global Bypass Rules** if you want to allow data packets to be exchanged between the specified local and remote IP addresses without any IPSec protection. In the **Local Networks** and **Remote Networks** text boxes, enter the list of local and remote subnets between which the bypass rules are applied.

    If you enable these rules, data packets are exchanged between the specified local and remote IP sites even if their IP addresses are specified in the IPSec session rules. The default is to use the IPSec protection when data is exchanged between local and remote sites. These rules apply for all IPSec VPN sessions created within this IPSec VPN service.
12. Click **Save**.

    After the new IPSec VPN service is created successfully, you are asked whether you want to continue with the rest of the IPSec VPN configuration. If you click **Yes**, you are taken back to the Add IPSec VPN Service panel. The **Sessions** link is now enabled and you can click it to add an IPSec VPN session.

After one or more IPSec VPN sessions are added, the number of sessions for each VPN service will appear in the **VPN Services** tab. You can reconfigure or add sessions by clicking the number in the **Sessions** column. You do not need to edit the service. If the number is zero, it is not clickable and you must edit the service to add sessions.
Use information in Adding IPSec VPN Sessions to guide you in adding an IPSec VPN session. You also provide information for the profiles and local endpoint that are required to finish the IPSec VPN configuration.

Add an L2 VPN Service

You configure an L2 VPN service on a Tier-0 or Tier-1 gateway. To enable the L2 VPN service, you must first create an IPSec VPN service on the Tier-0 or Tier-1 gateway, if it does not exist yet. You then configure an L2 VPN tunnel between an L2 VPN server (destination gateway) and an L2 VPN client (source gateway).

- Familiarize yourself with IPsec VPN and L2 VPN. See Understanding IPSec VPN and Understanding Layer 2 VPN.
- You must have at least one Tier-0 or Tier-1 gateway configured and available for use. See Add an NSX Tier-0 Gateway or Add an NSX Tier-1 Gateway.

To configure an L2 VPN service, use the information in the topics that follow in this section.

Add an L2 VPN Server Service

To configure an L2 VPN Server service, you must configure the L2 VPN service in server mode on the destination NSX Edge to which the L2 VPN client is to be connected.

1. With admin privileges, log in to NSX Manager.
2. Optional: If an IPSec VPN service does not exist yet on either a Tier-0 or Tier-1 gateway that you want to configure as the L2 VPN server, create the service using the following steps.
   a) Navigate to the **Networking** > **VPN** > **VPN Services** tab and select **Add Service** > **IP Sec**.
   b) Enter a name for the IPSec VPN service.
   c) From the **Tier-0/Tier-1 Gateway** drop-down menu, select the gateway to use with the L2 VPN server.
   d) If you want to use values different from the system defaults, set the rest of the properties on the Add IPSec Service pane, as needed.
   e) Click **Save** and when prompted if you want to continue configuring the IPSec VPN service, select **No**.
3. Navigate to the **Networking** > **VPN** > **VPN Services** tab and select **Add Service** > **L2 VPN Server** to create an L2 VPN server.
4. Enter a name for the L2 VPN server.
5. From the **Tier-0/Tier-1 Gateway** drop-down menu, select the same Tier-0 or Tier-1 gateway that you used with the IPSec service you created a moment ago.
6. Enter an optional description for this L2 VPN server.
7. Enter a value for **Tags** if you want to include this service in a tag group.
8. Enable or disable the **Hub & Spoke** property.

   By default, the value is set to `Disabled`, which means the traffic received from the L2 VPN clients is only replicated to the segments connected to the L2 VPN server. If this property is set to `Enabled`, the traffic from any L2 VPN client is replicated to all other L2 VPN clients.
9. Click **Save**.

   After the new L2 VPN server is created successfully, you are asked whether you want to continue with the rest of the L2 VPN service configuration. If you click **Yes**, you are taken back to the Add L2 VPN Server pane and the **Session** link is enabled. You can use that link to create an L2 VPN server session or use the tab.

After one or more L2 VPN sessions are added, the number of sessions for each VPN service will appear in the **VPN Services** tab. You can reconfigure or add sessions by clicking the number in the **Sessions** column. You do not need to edit the service. Note that if the number is zero, it is not clickable and you must edit the service to add sessions. Configure an L2 VPN server session for the L2 VPN server that you configured using information in Add an L2 VPN Server Session as a guide.

Add an L2 VPN Client Service

After configuring the L2 VPN Server service, configure the L2 VPN service in the client mode on another NSX Edge instance.

1. With admin privileges, log in to NSX Manager.
2. Optional: If an IPSec VPN service does not exist yet on either a Tier-0 or Tier-1 gateway that you want to configure as the L2 VPN client, create the service using the following steps.
   a) Navigate to the **Networking** > **VPN** > **VPN Services** tab and select **Add Service** > **IP Sec**.
   b) Enter a name for the IPSec VPN service.
   c) From the **Tier-0/Tier-1 Gateway** drop-down menu, select a Tier-0 or Tier-1 gateway to use with the L2 VPN client.
   d) If you want to use values different from the system defaults, set the rest of the properties on the Add IPSec Service pane, as needed.
   e) Click **Save** and when prompted if you want to continue configuring the IPSec VPN service, select **No**.
3. Navigate to the **Networking** > **VPN** > **VPN Services** tab and select **Add Service** > **L2 VPN Client**.
4. Enter a name for the L2 VPN Client service.
5. From the **Tier-0/Tier-1 Gateway** drop-down menu, select the same Tier-0 or Tier-1 gateway that you used with the route-based IPSec tunnel you created a moment ago.
6. Optionally set the values for **Description** and **Tags**.
7. Click **Save**.

   After the new L2 VPN client service is created successfully, you are asked whether you want to continue with the rest of the L2 VPN client configuration. If you click **Yes**, you are taken back to the Add L2 VPN Client pane and the **Session** link is enabled. You can use that link to create an L2 VPN client session or use the **Networking** > **VPN** > **L2 VPN Sessions** tab.

After one or more L2 VPN sessions are added, the number of sessions for each VPN service will appear in the **VPN Services** tab. You can reconfigure or add sessions by clicking the number in the **Sessions** column. You do not need to edit the service. If the number is zero, it is not clickable and you must edit the service to add sessions.
Configure an L2 VPN client session for the L2 VPN Client service that you configured. Use the information in Add an L2 VPN Client Session as a guide.

Adding IPSec VPN Sessions

After you have configured an IPSec VPN service, you must add either a policy-based IPSec VPN session or a route-based IPSec VPN session, depending on the type of IPSec VPN you want to configure. You also provide the information for the local endpoint and profiles to use to finish the IPSec VPN service configuration.

Using Certificate-Based Authentication for IPSec VPN Sessions

When you use certificate-based authentication for an IPSec VPN session, you must configure the certificate details for the IPSec session in the associated local endpoint.

> **Note:**
>
> Wildcard certificates are not supported for IPSec VPN.

Refer to the following workflow for details on how to configure the certificate details for a IPSec VPN session.

Configure Certificate-Based Authentication for an IPSec VPN Session

1. Create and enable an IPSec VPN service using an existing Tier-0 or Tier-1 gateway. See Add an IPSec VPN Service.
2. If you do not have the necessary server certificates or CA certificates in NSX Manager, import the certificates. See Import a Self-signed Certificate for NSX and Import a CA Certificate for NSX.
3. Use Add Local Endpoints to create a VPN server hosted on the logical router and select the certificates for it.
   The local ID is derived from the certificate associated with the local endpoint and depends on the X509v3 extensions present in the certificate. The local ID can be either the X509v3 extension Subject Alternative Name (SAN) or Distinguished Name (DN). The **Local ID** is not required and the ID specified there is ignored. However, for the remote VPN gateway, you need to configure the local ID as remote ID in the peer VPN gateway.

– If `X509v3 Subject Alternative Name` is found in the certificate, then one of the SAN strings is taken as the local ID value.
  If the certificate has multiple SAN fields, then following order is used to select the local ID.

Order	SAN Field
1	IP Address
2	DNS
3	Email Address

  For example, if the configured site certificate has the following SAN fields,

```
X509v3 Subject Alternative Name:
DNS:Site123.vmware.com, email:user1@company.com, IP Address:1.1.1.1
```

  then the IP address `1.1.1.1` is used as the local ID. If the IP address is not available, then the DNS string is used. And if the IP address and the DNS are not available, then the email address is used.
– If `X509v3 Subject Alternative Name` is not present in the certificate, then the Distinguished Name (DN) is used as the local ID value.
  For example, if the certificate does not have any SAN fields, and its DN string is

```
C=US, ST=California, O=MyCompany, OU=MyOrg, CN=Site123
```

  then the DN string automatically becomes the local ID. The local ID is the peer ID on the remote site.

  Note:
  If the certificate details are not properly configured, it might cause the VPN session to go down with the `Down` alarm of `Authentication failed`.

4. Configure either a policy-based or route-based IPSec VPN session. See Add a Policy-Based IPSec Session or Add a Route-Based IPSec Session.
   Make sure to configure the following settings.
   a. From the **Authentication Mode** drop-down menu, select **Certificate**.
   b. In the **Remote ID** textbox, enter a value to identify the peer site.
      The remote ID must be a distinguished name (DN), IP address, DNS, or an email address used in the peer site's certificate.

      Note:

      If the peer site's certificate contains an email address in the DN string, for example,

```
C=US, ST=California, O=MyCompany, OU=MyOrg, CN=Site123/emailAddress=user1@mycompany.com
```

      then enter the **Remote ID** value using the following format as an example.

```
C=US, ST=California, O=MyCompany, OU=MyOrg, CN=Site123, MAILTO=user1@mycompany.com
```

Add a Policy-Based IPSec Session

When you add a policy-based IPSec VPN, IPSec tunnels are used to connect multiple local subnets that are behind the NSX Edge node with peer subnets on the remote VPN site.

- You must have configured an IPSec VPN service before proceeding. See Add an NSX IPSec VPN Service.
- Obtain the information for the local endpoint, IP address for the peer site, local network subnet, and remote network subnet to use with the policy-based IPSec VPN session you are adding. To create a local endpoint, see Add Local Endpoints.
- If you are using a Pre-Shared Key (PSK) for authentication, obtain the PSK value.
- If you are using a certificate for authentication, ensure that the necessary server certificates and corresponding CA-signed certificates are already imported.
  See Certificates in NSX.

- If you do not want to use the defaults for the IPSec tunnel, IKE, or dead peer detection (DPD) profiles provided by NSX, configure the profiles you want to use instead. See Adding Profiles for information.

The following steps use the **IPSec Sessions** tab on the NSX Manager UI to create a policy-based IPSec session. You also add information for the tunnel, IKE, and DPD profiles, and select an existing local endpoint to use with the policy-based IPSec VPN.

> **Note:**
>
> You can also add the IPSec VPN sessions immediately after you have successfully configured the IPSec VPN service. You click **Yes** when prompted to continue with the IPSec VPN service configuration and select **Sessions** > **Add Sessions** on the Add IPsec Service panel. The first few steps in the following procedure assume you selected **No** to the prompt to continue with the IPSec VPN service configuration. If you selected **Yes**, proceed to step 3 in the following steps to guide you with the rest of the policy-based IPSec VPN session configuration.

1. With admin privileges, log in to NSX Manager.
2. Navigate to the **Networking** > **VPN** > **IP Sec Sessions** tab.
3. Select **Add IPSec Session** > **Policy Based**.
4. Enter a name for the policy-based IPSec VPN session.
5. From the **VPN Service** drop-down menu, select the IPSec VPN service to which you want to add this new IPSec session.

   > **Note:**
   > If you are adding this IPSec session from the **Add IPSec Sessions** dialog box, the VPN Service name is already indicated above the **Add IPSec Session** button.

6. Select an existing local endpoint from the drop-down menu.

   This local endpoint value is required and identifies the local NSX Edge node. If you want to create a different local endpoint, click the three-dot menu (

) and select **Add Local Endpoint**.
7. In the **Remote IP** text box, enter the required IP address of the remote site.

   This value is required.
8. Enter an optional description for this policy-based IPSec VPN session.

   The maximum length is 1024 characters.
9. To enable or disable the IPSec VPN session, click **Admin Status**.

   By default, the value is set to `Enabled`, which means the IPSec VPN session is to be configured down to the NSX Edge node.
10. Optional: From the **Compliance suite** drop-down menu, select a security compliance suite.

    > **Note:**
    > Compliance suite support is provided beginning with NSX 2.5. See About Supported Compliance Suites for more information.

    The default value selected is `None`. If you select a compliance suite, the **Authentication Mode** is set to `Certificate` and in the **Advanced Properties** section, the values for **IKE profile** and **IPSec profile** are set to the system-defined profiles for the selected security compliance suite. You cannot edit these system-defined profiles.
11. If the **Compliance Suite** is set to `None`, select a mode from the **Authentication Mode** drop-down menu.

    The default authentication mode used is `PSK`, which means a secret key shared between NSX Edge and the remote site is used for the IPSec VPN session. If you select `Certificate`, the site certificate that was used to configure the local endpoint is used for authentication.

    For more information about certificate-based authentication, see Using Certificate-Based Authentication for IPSec VPN Sessions.

12. In the Local Networks and Remote Networks text boxes, enter at least one IP subnet address to use for this policy-based IPSec VPN session.

    These subnets must be in a CIDR format.

13. If **Authentication Mode** is set to `PSK`, enter the key value in the **Pre-shared Key** text box.

    This secret key can be a string with a maximum length of 128 characters.

    CAUTION:
    Be careful when sharing and storing a PSK value because it contains some sensitive information.

14. To identify the peer site, enter a value in **Remote ID**.

    For peer sites using PSK authentication, this ID value must be the IP address or the FQDN of the peer site. For peer sites using certificate authentication, this ID value must be the common name (CN) or distinguished name (DN) used in the peer site's certificate.

    Note:
    If the peer site's certificate contains an email address in the DN string, for example,

    ```
    C=US, ST=California, O=MyCompany, OU=MyOrg, CN=Site123/emailAddress=user1@mycompany.com
    ```

    then enter the **Remote ID** value using the following format as an example.

    ```
    C=US, ST=California, O=MyCompany, OU=MyOrg, CN=Site123, MAILTO=user1@mycompany.com"
    ```

    If the local site's certificate contains an email address in the DN string and the peer site uses the strongSwan IPsec implementation, enter the local site's ID value in that peer site. The following is an example.

    ```
    C=US, ST=California, O=MyCompany, OU=MyOrg, CN=Site123, E=user1@mycompany.com"
    ```

15. To change the profiles, initiation mode, TCP MSS clamping mode, and tags used by the policy-based IPSec VPN session, click **Advanced Properties**.

    By default, the system-generated profiles are used. Select another available profile if you do not want to use the default. If you want to use a profile that is not configured yet, click the three-dot menu () to create another profile. See Adding Profiles.

    a) If the **IKE Profiles** drop-down menu is enabled, select the IKE profile.
    b) Select the IPsec tunnel profile, if the **IPSec Profiles** drop-down menu is not disabled.
    c) Select the preferred DPD profile if the **DPD Profiles** drop-down menu is enabled.
    d) Select the preferred mode from the **Connection Initiation Mode** drop-down menu.

       Connection initiation mode defines the policy used by the local endpoint in the process of tunnel creation. The default value is `Initiator`. The following table describes the different connection initiation modes available.

Table 1036: Connection Initiation Modes

Connection Initiation Mode	Description
Initiator	The default value. In this mode, the local endpoint initiates the IPSec VPN tunnel creation and responds to incoming tunnel setup requests from the peer gateway.
On Demand	In this mode, the local endpoint initiates the IPSec VPN tunnel creation after the first packet matching the policy rule is received. It also responds to the incoming initiation request.
Respond Only	The IPSec VPN never initiates a connection. The peer site always initiates the connection request and the local endpoint responds to that connection request.

    e) If you want to reduce the maximum segment size (MSS) payload of the TCP session during the IPSec connection, enable **TCP MSS Clamping**, select the **TCP MSS direction** value, and optionally set the **TCP MSS Value**.

       See Understanding TCP MSS Clamping for more information.

f) If you want to include this session as part of a specific group, enter the tag name in **Tags**.

16. Click **Save**.

When the new policy-based IPSec VPN session is configured successfully, it is added to the list of available IPsec VPN sessions. It is in read-only mode.

- Verify that the IPSec VPN tunnel status is Up. See Monitor and Troubleshoot VPN Sessions for information.
- If necessary, manage the IPSec VPN session information by clicking the three-dot menu (

) on the left-side of the session's row. Select one of the actions you are allowed to perform.
- To configure the peer VPN device, see Download the Remote Side IPSec VPN Configuration File.

Add a Route-Based IPSec Session

When you add a route-based IPSec VPN, tunneling is provided on traffic that is based on routes that were learned dynamically over a virtual tunnel interface (VTI) using a preferred protocol, such as BGP. IPSec secures all the traffic flowing through the VTI.

- You must have configured an IPSec VPN service before proceeding. See Add an NSX IPSec VPN Service.
- Obtain the information for the local endpoint, IP address for the peer site, and tunnel service IP subnet address to use with the route-based IPSec session you are adding. To create a local endpoint, see Add Local Endpoints.
- If you are using a Pre-Shared Key (PSK) for authentication, obtain the PSK value.
- If you are using a certificate for authentication, ensure that the necessary server certificates and corresponding CA-signed certificates are already imported.
  See Certificates in NSX.
- If you do not want to use the default values for the IPSec tunnel, IKE, or dead peer detection (DPD) profiles provided by NSX, configure the profiles you want to use instead. See Adding Profiles for information.

The steps described in this topic use the **IPSec Sessions** tab to create a route-based IPSec session. You also add information for the tunnel, IKE, and DPD profiles, and select an existing local endpoint to use with the route-based IPSec VPN.

> **Note:**
>
> You can also add the IPSec VPN sessions immediately after the IPSec VPN service is successfully configured. Click **Yes** when prompted to continue with the IPSec VPN service configuration and select **Sessions** > **Add Sessions** on the Add IPSec Service panel. The first few steps in the following procedure assume you selected **No** to the prompt to continue with the IPSec VPN service configuration. If you selected **Yes**, proceed to step 3 to guide you with the rest of the route-based IPSec VPN session configuration.

1. With admin privileges, log in to NSX Manager.
2. Navigate to **Networking** > **VPN** > **IP Sec Sessions**.
3. Select **Add IPSec Session** > **Route Based**.
4. Enter a name for the route-based IPSec session.
5. From the **VPN Service** drop-down menu, select the IPSec VPN service to which you want to add this new IPSec session.

   > **Note:**
   > If you are adding this IPSec session from the **Add IPSec Sessions** dialog box, the VPN Service name is already indicated above the **Add IPSec Session** button.

6. Select an existing local endpoint from the drop-down menu.
   The local endpoint value is required and identifies the local NSX Edge node. If you want to create a different local endpoint, click the three-dot menu (

) and select **Add Local Endpoint**.

7. In the **Remote IP** text box, enter the IP address of the remote site.
   This is a required value.
8. Enter an optional description for this route-based IPSec VPN session.
   The maximum length is 1024 characters.
9. To enable or disable the IPSec session, click **Admin Status**.
   By default, the value is set to `Enabled`, which means the IPSec session is to be configured down to the NSX Edge node.
10. Optional: From the **Compliance suite** drop-down menu, select a security compliance suite.

    Note:
    Compliance suite support is provided beginning with NSX 2.5. See About Supported Compliance Suites for more information.

    The default value is set to `None`. If you select a compliance suite, the **Authentication Mode** is set to `Certificate` and in the **Advanced Properties** section, the values for **IKE profile** and **IPSec profile** are set to the system-defined profiles for the selected compliance suite. You cannot edit these system-defined profiles.
11. Enter an IP subnet address in **Tunnel Interface** in the CIDR notation.
    This address is required.
12. If the **Compliance Suite** is set to `None`, select a mode from the **Authentication Mode** drop-down menu.
    The default authentication mode used is `PSK`, which means a secret key shared between NSX Edge and the remote site is used for the IPSec VPN session. If you select `Certificate`, the site certificate that was used to configure the local endpoint is used for authentication.

    For more information about certificate-based authentication, see Using Certificate-Based Authentication for IPSec VPN Sessions.
13. If you selected `PSK` for the authentication mode, enter the key value in the **Pre-shared Key** text box.
    This secret key can be a string with a maximum length of 128 characters.

    CAUTION:
    Be careful when sharing and storing a PSK value because it contains some sensitive information.
14. Enter a value in **Remote ID**.
    For peer sites using PSK authentication, this ID value must be the IP address or the FQDN of the peer site. For peer sites using certificate authentication, this ID value must be the common name (CN) or distinguished name (DN) used in the peer site's certificate.

    Note:
    If the peer site's certificate contains an email address in the DN string, for example,
    ```
    C=US, ST=California, O=MyCompany, OU=MyOrg, CN=Site123/emailAddress=user1@mycompany.com
    ```
    then enter the **Remote ID** value using the following format as an example.
    ```
    C=US, ST=California, O=MyCompany, OU=MyOrg, CN=Site123, MAILTO=user1@mycompany.com"
    ```
    If the local site's certificate contains an email address in the DN string and the peer site uses the strongSwan IPsec implementation, enter the local site's ID value in that peer site. The following is an example.
    ```
    C=US, ST=California, O=MyCompany, OU=MyOrg, CN=Site123, E=user1@mycompany.com"
    ```
15. If you want to include this IPSec session as part of a specific group tag, enter the tag name in **Tags**.
16. To change the profiles, initiation mode, TCP MSS clamping mode, and tags used by the route-based IPSec VPN session, click **Advanced Properties**.
    By default, the system-generated profiles are used. Select another available profile if you do not want to use the default. If you want to use a profile that is not configured yet, click the three-dot menu (
    
) to create another profile. See Adding Profiles.

a) If the **IKE Profiles** drop-down menu is enabled, select the IKE profile.
b) Select the IPsec tunnel profile, if the **IPSec Profiles** drop-down menu is not disabled.
c) Select the preferred DPD profile if the **DPD Profiles** drop-down menu is enabled.
d) Select the preferred mode from the **Connection Initiation Mode** drop-down menu.

Connection initiation mode defines the policy used by the local endpoint in the process of tunnel creation. The default value is `Initiator`. The following table describes the different available connection initiation modes.

Table 1037: Connection Initiation Modes

Connection Initiation Mode	Description
Initiator	The default value. In this mode, the local endpoint initiates the IPSec VPN tunnel creation and responds to incoming tunnel setup requests from the peer gateway.
On Demand	Do not use with the route-based VPN. This mode applies to policy-based VPN only.
Respond Only	The IPSec VPN never initiates a connection. The peer site always initiates the connection request and the local endpoint responds to that connection request.

17. If you want to reduce the maximum segment size (MSS) payload of the TCP session during the IPSec connection, enable **TCP MSS Clamping**, select the **TCP MSS** direction value, and optionally set the **TCP MSS Value**.

    See Understanding TCP MSS Clamping for more information.

18. If you want to include this IPSec session as part of a specific group tag, enter the tag name in **Tags**.
19. Click **Save**.

When the new route-based IPSec VPN session is configured successfully, it is added to the list of available IPsec VPN sessions. It is in read-only mode.

- Verify that the IPSec VPN tunnel status is Up. See Monitor and Troubleshoot VPN Sessions for information.
- Configure routing using either a static route or BGP. See Configure a Static Route or Configure BGP.
- If necessary, manage the IPSec VPN session information by clicking the three-dot menu () on the left-side of the session's row. Select one of the actions you can perform.
- To configure the peer VPN device, see Download the Remote Side IPSec VPN Configuration File.

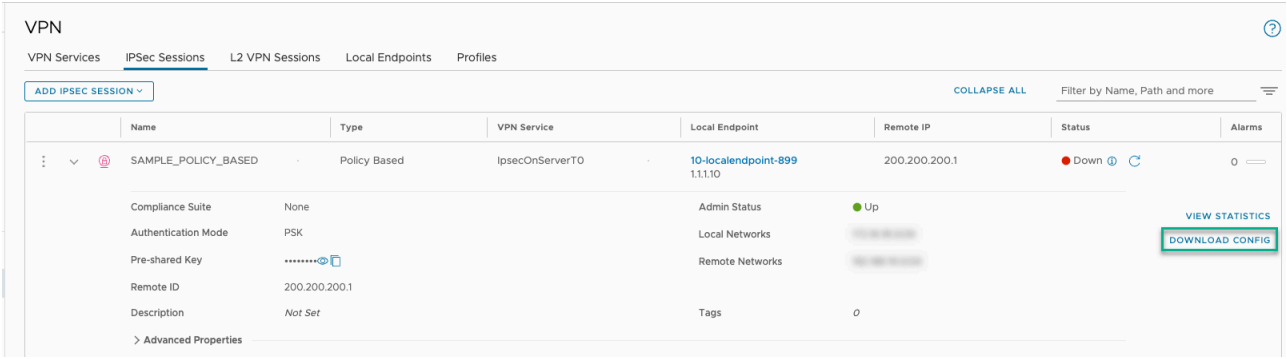
Download the Remote Side IPSec VPN Configuration File

To set up IPSec between two sites you must configure the two VPN endpoints with matching attributes. This topic helps you to understand the requirements to ensure that your IPSec VPN device vendor attributes match the VPN-related attributes of your local IPSec VPN session.

Each IPSec VPN vendor has their own format of accepting the configurations. In certain cases there are default values used for a few parameters. You can use the **Download Config** feature in the NSX IPSec VPN Sessions UI which provides all the VPN-related configurations that an administrator can use to configure a peer VPN vendor device. It is based on the IPSec VPN session configured at NSX. The feature presents all hidden/default attributes for the IPSec VPN session to allow the administrator to configure the peer VPN device, which may have different default values. Any configuration mismatch can lead to the IPsec VPN tunnel not coming up properly.

Clicking **DOWNLOAD CONFIG**, as shown in the IPSec VPN Sessions Download Config Button image, downloads a text file that contains relevant attributes that might be required to configure the IPSec VPN session counterpart at the peer VPN device.

Figure 423: IPSec VPN Sessions Download Config Button



Procedure

1. Ensure you have configured an IPSec VPN service and a session successfully before proceeding.
2. Go to the **Networking** > **VPN** > **IPSec Sessions** tab to access the **Download Config** button.
3. In the table of IPSec VPN Sessions, expand the row for the session you plan to use for the IPSec VPN session configuration. For example, the *Sample_Policy_Based* row is expanded in the IPSec VPN Sessions Download Config Button image.
4. Click **Download Config** and click **Yes** on the Warning dialog box to download a text file.
5. Use the downloaded config file to configure the policy or route-based IPSec VPN session attributes at the peer VPN endpoint to ensure it contains the required matching values.

The following sample text file is similar to the file that gets downloaded. The file name, *Sample_Policy_Based.txt* is a policy-based IPSec VPN session configured in NSX. The name of the file downloaded is based on the name of the session. For example <session-name>.txt.

```
 # Suggestive peer configuration for Policy IPSec Vpn Session
#
# IPSec VPN session path          : /infra/tier-0s/ServerT0_AS/ipsec-vpn-services/IpsecOnServerT0/sessions/
SAMPLE_POLICY_BASED
# IPSec VPN session name          : SAMPLE_POLICY_BASED
# IPSec VPN session description   :
# Tier 0 path                     : /infra/tier-0s/ServerT0_AS
#
# Enforcement point path    : /infra/sites/default/enforcement-points/default
# Enforcement point type    : NSX
#
# Suggestive peer configuration for IPSec VPN Connection
#
# IPSecVPNSession Id         : e7f34d43-c894-4dbb-b7d2-c899f81b1812
# IPSecVPNSession name       : SAMPLE_POLICY_BASED
# IPSecVPNSession description:
# IPSecVPNSession enabled    : true
# IPSecVPNSession type       : Policy based VPN
# Logical router Id          : 258b91be-b4cb-448a-856e-501d03128877
# Generated Time             : Mon Apr 29 07:07:43 GMT 2024
#
# Internet Key Exchange Configuration [Phase 1]
# Configure the IKE SA as outlined below
IKE version                  : IKE_V2
Connection initiation mode   : INITIATOR
Authentication method        : PSK
Pre shared key               : nsxtVPN!234
Authentication algorithm     : [SHA2_256]
Encryption algorithm         : [AES_128]
SA life time                 : 86400
```

```
Negotiation mode             : Not applicable for ikev2
DH group                     : [GROUP14]
Prf Algorithm                : [SHA2_256]
#
# IPsec_configuration [Phase 2]
# Configure the IPsec SA as outlined below
Transform Protocol              : ESP
Authentication algorithm        :
Sa life time                    : 3600
Encryption algorithm            : [AES_GCM_128]
Encapsulation mode              : TUNNEL_MODE
Enable perfect forward secrecy  : true
Perfect forward secrecy DH group: [GROUP14]
#
# IPsec Dead Peer Detection (DPD) settings
DPD enabled         : true
DPD probe interval  : 60
#
# IPSec VPN Session Configuration
Peer address    : 1.1.1.10 # Peer gateway public IP.
Peer id         : 1.1.1.10
#
Local address   : 200.200.200.1 # Local gateway public IP.
Local id        : 200.200.200.1
#
# Policy Rules
#Rule1
Sources: [192.168.19.0/24]
Destinations: [172.16.18.0/24]
```

The IPSec VPN Sessions Configuration File Attributes table contains the attributes in the *Sample_Policy_Based.txt* VPN session config file to use when configuring IPSec VPN at the peer VPN device.

Table 1038: IPSec VPN Sessions Configuration File Attributes

Category	Attribute Name	Meaning and Value of Attribute to be Configured at the Peer VPN Device	Peer Device Configurability
ISAKMP Phase 1Parameters	IKE version	IKE protocol version	Mandatory
	Connection initiation mode	Whether the device initiates IKE connection	Optional. Mandatory if NSX IPSec is configured with **Connection Initiation Mode** = "Respond Only."
	Authentication method	Authentication mode for IKE - Pre-Shared Key or Certificate	Mandatory
	Pre shared key	Value of Shared Key if the Authentication Mode is PSK	Mandatory
	Authentication algorithm	Authentication algorithm to be used for IKE	Mandatory
	Encryption algorithm	Encryption algorithm to be used for IKE	Mandatory
	SA life time	Lifetime of IKE Security Association (SA) in seconds	Optional

Category	Attribute Name	Meaning and Value of Attribute to be Configured at the Peer VPN Device	Peer Device Configurability
	Negotiation mode	Mode of IKEv1 protocol - Only Main mode is supported. Not relevant for IKEv2	Mandatory
	DH group	Diffie Hellman Group to be used for IKE SA negotiation	Mandatory
	Prf Algorithm	Pseudo Random function to be used for IKE SA negotiation	Mandatory
	Peer address	IP address of the VPN endpoint at the NSX side	Mandatory
	Peer id	Identity of the VPN endpoint at the NSX side	Mandatory
	Local Address	Address of the VPN endpoint at the peer endpoint side (in configuration done in NSX side)	Mandatory
	Local ID	Identity of the VPN endpoint to be configured at the peer endpoint side	Mandatory
ISAKMP Phase 2 Parameters	Transform Protocol	Transform Protocol	Transform Protocol
	Authentication algorithm	Integrity protection algorithm for IPSec packets	Mandatory
	SA Lifetime	Lifetime of IPSec SA, in seconds. Keys are refreshed as the SA lifetime approaches.	Optional
	Encryption algorithm	Encryption protection for IPSec packets	Mandatory
	Encapsulation mode	Mode for IPSec Tunnel (Tunnel or Transport)	Mandatory. Only Tunnel Mode is supported.
	Enable perfect forward secrecy	PFS (enabled or inactive)	Mandatory
	Perfect forward secrecy DH group	DH group to be used for PFS	Mandatory
	Sources	Applies to policy-based VPN. This is the subnet or subnets behind the peer VPN endpoint.	Mandatory for policy-based VPN
	Destinations	Applies to policy-based VPN. This is the subnet or subnets behind the NSX VPN endpoint for which traffic needs to be tunneled over IPSec.	Mandatory for policy-based VPN
Other Parameters	DPD enabled	Whether Dead Peer Detection is enabled	Optional
		Frequency at which DPD is performed (in seconds)	Optional

About Supported Compliance Suites

You can specify a security compliance suite to use to configure the security profiles used for an IPSec VPN session.

A security compliance suite has predefined values that are used for different security parameters and that cannot be modified. When you select a compliance suite, the predefined values are automatically used for the security profile of the IPSec VPN session you are configuring.

The following table lists the compliance suites that are supported for IKE profiles in NSX and the values that are predefined for each.

Compliance Suite Name	IKE Version	Encryption Algorithm	Digest Algorithm	Diffie Hellman Group
CNSA	IKE V2	AES 256	SHA2 384	Group 15, Group 20
FIPS	IKE FLEX	AES 128	SHA2 256	Group 20
Foundation	IKE V1	AES 128	SHA2 256	Group 14
PRIME	IKE V2	AES GCM 128	Not Set	Group 19
Suite-B-GCM-128	IKE V2	AES 128	SHA2 256	Group 19
Suite-B-GCM-256	IKE V2	AES 256	SHA2 384	Group 20

Note:
The AES 128 and AES 256 algorithms use the CBC mode of operation.

The following table lists the compliance suites that are supported for IPSec profiles in NSX and the values that are predefined for each.

Compliance Suite Name	Encryption Algorithm	Digest Algorithm	PFS Group	Diffie-Hellman Group
CNSA	AES 256	SHA2 384	Enabled	Group 15, Group 20
FIPS	AES GCM 128	Not Set	Enabled	Group 20
Foundation	AES 128	SHA2 256	Enabled	Group 14
PRIME	AES GCM 128	Not Set	Enabled	Group 19
Suite-B-GCM-128	AES GCM 128	Not Set	Enabled	Group 19
Suite-B-GCM-256	AES GCM 256	Not Set	Enabled	Group 20

Note:
The AES 128 and AES 256 algorithms use the CBC mode of operation.

Adding L2 VPN Sessions

After you have configured an L2 VPN server and an L2 VPN client, you must add L2 VPN sessions for both to complete the L2 VPN service configuration.

Add an L2 VPN Server Session

After creating an L2 VPN Server service, you must add an L2 VPN session and attach it to an existing segment.

- You must have configured an L2 VPN Server service before proceeding. See Add an L2 VPN Server Service.
- Obtain the information for the local endpoint and remote IP to use with the L2 VPN Server session you are adding. To create a local endpoint, see Add Local Endpoints.
- Obtain the values for the pre-shared key (PSK) and the tunnel interface subnet to use with the L2 VPN Server session.
- Obtain the name of the existing segment you want to attach to the L2 VPN Server session you are creating. See Create an NSX Segment for information.

The following steps use the **L2 VPN Sessions** tab on the NSX Manager UI to create an L2 VPN Server session. You also select an existing local endpoint and segment to attach to the L2 VPN Server session.

Note:
You can also add an L2 VPN Server session immediately after you have successfully configured the L2 VPN Server service. You click **Yes** when prompted to continue with the L2 VPN Server configuration and select **Sessions** > **Add Sessions** on the Add L2 VPN Server panel. The first few steps in the following procedure assume you selected **No** to the prompt to continue with the L2 VPN Server configuration. If you selected **Yes**, proceed to step 3 in the following steps to guide you with the rest of the L2 VPN Server session configuration.

1. With admin privileges, log in to NSX Manager.
2. Navigate to the **Networking** > **VPN** > **L2 VPN Sessions** tab.
3. Select **Add L2 VPN** > **L2 VPN Server**.
4. Enter a name for the L2 VPN Server session.
5. From the **VPN Service** drop-down menu, select the IPsec service on the same Tier-0 gateway for which the L2 VPN session is being created.

   Note:
   If you are adding this L2 VPN Server session from the Set L2VPN Server Sessions dialog box, the L2 VPN Server service is already indicated above the **Add L2 Session** button.
6. Select an existing local endpoint from the drop-down menu.

   If you want to create a different local endpoint, click the three-dot menu (
   
) and select **Add Local Endpoint**.
7. Enter the IP address of the remote site under Remote IP.
8. To enable or disable the L2 VPN Server session, click **Admin Status**.

   By default, the value is set to `Enabled`, which means the L2 VPN Server session is to be configured down to the NSX Edge node.
9. Enter the secret key value in **Pre-shared Key**.

   CAUTION:
   Be careful when sharing and storing a PSK value because it is considered sensitive information.
10. Enter an IP subnet address in the **Tunnel Interface** using the CIDR notation.

    For example, 4.5.6.6/24. This subnet address is required.
11. Enter a value in **Remote ID**.

    For peer sites using certificate authentication, this ID must be the common name in the peer site's certificate. For PSK peers, this ID can be any string. Preferably, use the IP address of the VPN or an FQDN for the VPN services as the remote ID.
12. If you want to include this session as part of a specific group, enter the tag name in **Tags**.
13. Click **Advanced Properties.** if you want to reduce the maximum segment size (MSS) payload of the TCP session during the L2 VPN connection.

    By default, **TCP MSS Clamping** is enabled and the **TCP MSS Direction** is set to `Both`. See Understanding TCP MSS Clamping for more information.
    a) Enable or disable **TCP MSS Clamping**.
    b) Set the **TCP MSS Value**, if necessary. If the field is left blank, the value is automatically assigned.
14. Click **Save** and click **Yes** when prompted if you want to continue with the VPN service configuration.

    You are returned to the Add L2VPN Sessions panel and the **Segments** link is now enabled.

15. Attach an existing segment to the L2 VPN Server session.
    a) Click **Segments** > **Set Segments**.
    b) In the **Set Segments** dialog box, click **Set Segment** to attach an existing segment to the L2 VPN Server session.
    c) From the **Segment** drop-down menu, select the VNI-based or VLAN-based segment that you want to attach to the session.
    d) Enter a unique value in the **VPN Tunnel ID** that is used to identify the segment that you selected.
    e) In the **Local Egress Gateway IP** text box, enter the IP address of the local gateway that your workload VMs on the segment use as their default gateway. The same IP address can be configured in the remote site on the extended segment.
    f) Click **Save** and then **Close**.

    In the Set L2VPN Sessions pane or dialog box, the system has incremented the **Segments** count for the L2 VPN Server session.
16. To finish the L2 VPN Server session configuration, click **Close Editing**.

In the **VPN Services** tab, the system incremented the **Sessions** count for the L2 VPN Server service that you configured.

If you have attached one or more segments to the session, you see the number of segments for each session in the **L2 VPN Sessions** tab. You can reconfigure or add segments by clicking the number in the **Segments** column. You do not need to edit the session. If the number is zero, it is not clickable and you must edit the session to add segments.

To complete the L2 VPN service configuration, you must also create an L2 VPN service in Client mode and an L2 VPN client session. See Add an L2 VPN Client Service and Add an L2 VPN Client Session.

Add an L2 VPN Client Session

You must add an L2 VPN Client session after creating an L2 VPN Client service, and attach it to an existing segment.

- You must have configured an L2 VPN Client service before proceeding. See Add an L2 VPN Client Service.
- Obtain the IP addresses information for the local IP and remote IP to use with the L2 VPN Client session you are adding.
- Obtain the peer code that was generated during the L2 VPN server configuration. See Download the Remote Side L2 VPN Configuration File.
- Obtain the name of the existing segment you want to attach to the L2 VPN Client session you are creating. See Create an NSX Segment.

The following steps use the **L2 VPN Sessions** tab on the NSX Manager UI to create an L2 VPN Client session. You also select an existing local endpoint and segment to attach to the L2 VPN Client session.

> **Note:**
>
> You can also add an L2 VPN Client session immediately after you have successfully configured the L2 VPN Client service. Click **Yes** when prompted to continue with the L2 VPN Client configuration and select **Sessions** > **Add Sessions** on the Add L2 VPN Client panel. The first few steps in the following procedure assume you selected **No** to the prompt to continue with the L2 VPN Client configuration. If you selected **Yes**, proceed to step 3 in the following steps to guide you with the rest of the L2 VPN Client session configuration.

1. With admin privileges, log in to NSX Manager.
2. Select the **Networking** > **VPN** > **L2 VPN Sessions**.
3. Select **Add L2 VPN Session** > **L2 VPN Client**.
4. Enter a name for the L2 VPN Client session.
5. From the **VPN Service** drop-down menu, select the L2 VPN Client service with which the L2 VPN session is to be associated.

   Note:
   If you are adding this L2 VPN Client session from the Set L2VPN Client Sessions dialog box, the L2 VPN Client service is already indicated above the **Add L2 Session** button.

6. In the **Local IP address** text box, enter the IP address of the L2 VPN Client session.
7. Enter the remote IP address of the IPSec tunnel to be used for the L2 VPN Client session.
8. In the **Peer Configuration** text box, enter the peer code generated when you configured the L2 VPN Server service.
9. Enable or disable **Admin Status**.

   By default, the value is set to `Enabled`, which means the L2 VPN Server session is to be configured down to the NSX Edge node.
10. Click **Save** and click **Yes** when prompted if you want to continue with the VPN service configuration.
11. Attach an existing segment to the L2 VPN Client session.
    a) Select **Segments** > **Add Segments**.
    b) In the **Set Segments** dialog box, click **Add Segment**.
    c) From the **Segment** drop-down menu, select the VNI-based or VLAN-based segment you want to attach to the L2 VPN Client session.
    d) Enter a unique value in the **VPN Tunnel ID** that is used to identify the segment that you selected.
    e) Click **Close**.
12. To finish the L2 VPN Client session configuration, click **Close Editing**.

In the **VPN Services** tab, the sessions count is updated for the L2 VPN Client service that you configured.

If you have attached one or more segments to the session, you see the number of segments for each session in the **L2 VPN Sessions** tab. You can reconfigure or add segments by clicking the number in the **Segments** column. You do not need to edit the session. If the number is zero, it is not clickable and you must edit the session to add segments.

Download the Remote Side L2 VPN Configuration File

To configure the L2 VPN client session, you must obtain the peer code that was generated when you configured the L2 VPN server session.

- You must have configured an L2 VPN server service and a session successfully before proceeding. See Add an L2 VPN Server Service.

1. With admin privileges, log in to NSX Manager.
2. Navigate to the **Networking** > **VPN** > **L2 VPN Sessions** tab.
3. In the table of L2 VPN sessions, expand the row for the L2 VPN server session you plan to use for the L2 VPN client session configuration.
4. Click **Download Config** and click **Yes** on the Warning dialog box.

   A text file with the name `L2VPNSession_<name-of-L2-VPN-server-session>_config.txt` is downloaded. It contains the peer code for the remote side L2 VPN configuration.

   CAUTION:
   Be careful when storing and sharing the peer code because it contains a PSK value, which is considered sensitive information.

   For example, `L2VPNSession_L2VPNServer_config.txt` contains the following configuration.

```
[
  {
    "transport_tunnel_path": "/infra/tier-0s/ServerT0_AS/locale-services/1-policyconnectivity-693/ipsec-
vpn-services/IpsecService1/sessions/Routebase1",
    "peer_code":
"MCw3ZjBjYzdjLHsic2l0ZU5hbWUiOiJSb3V0ZWJhc2UxIiwic3JjVGFwSXAiOiIxNjkuMjU0LjY0LjIiLCJkc3RUYXBJcCI6IjE2OS4yNTQuNjQuMSI
vbiI6ImlrZXYyIiwiZW5jYXBQcm90byI6ImdyZS9pcHNlYyIsImRoR3JvdXAiOiJkaDE0IiwiZW5jcnlwdEFuZERpZ2VzdCI6ImFlcy1nY20vc2hhLTI
6IlZNd2FyZTEyMyIsInR1bm5lbHMiOlt7ImxvY2FsSWQiOiI2MC42MC42MC4xIiwicGVlcklkIjoiNTAuNTAuNTAuMSIsImxvY2FsVnRpSXAiOiIxNjk
  }
]
```

5. Copy the peer code, which you use to configure the L2 VPN client service and session.

   Using the preceding configuration file example, the following peer code is what you copy to use with the L2 VPN client configuration.


```
MCw3ZjBjYzdjLHsic2l0ZU5hbWUiOiJSb3V0ZWJhc2UxIiwic3JjVGFwSXAiOiIxNjkuMjU0LjY0LjIiLCJkc3RUYXBJcCI6IjE2OS4yNTQuNjQuMSIs
vbiI6ImlrZXYyIiwiZW5jYXBQcm90byI6ImdyZS9pcHNlYyIsImRoR3JvdXAiOiJkaDE0IiwiZW5jcnlwdEFuZERpZ2VzdCI6ImFlcy1nY20vc2hhLTI
6IlZNd2FyZTEyMyIsInR1bm5lbHMiOlt7ImxvY2FsSWQiOiI2MC42MC42MC4xIiwicGVlcklkIjoiNTAuNTAuNTAuMSIsImxvY2FsVnRpSXAiOiIxNjk
```


Configure the L2 VPN Client service and session. See Add an L2 VPN Client Service and Add an L2 VPN Client Session.

Add Local Endpoints

You must configure a local endpoint to use with the IPSec VPN that you are configuring.

- If you are using a certificate-based authentication mode for the IPSec VPN session that is to use the local endpoint you are configuring, obtain the information about the certificate that the local endpoint must use.
- Ensure that you have configured an IPSec VPN service to which this local endpoint is to be associated.

The following steps use the **Local Endpoints** tab on the NSX Manager UI. You can also create a local endpoint while in the process of adding an IPSec VPN session by clicking the three-dot menu (



) and selecting **Add Local Endpoint**. If you are in the middle of configuring an IPSec VPN session, proceed to step 3 in the following steps to guide you with creating a new local endpoint.

1. With admin privileges, log in to NSX Manager.
2. Navigate to and click **Add Local Endpoint**.
3. Enter a name for the local endpoint.
4. From the **VPN Service** drop-down menu, select the IPSec VPN service to which this local endpoint is to be associated.
5. Enter an IP address for the local endpoint.

   For an IPSec VPN service running on a Tier-0 gateway, the local endpoint IP address must be different from the Tier-0 gateway's uplink interface IP address. The local endpoint IP address you provide is associated with the loopback interface for the Tier-0 gateway and is also published as a routable IP address over the uplink interface.

   For an IPSec VPN service running on a Tier-1 gateway, the local endpoint IP address must be different from the Tier-1 gateway's uplink interface IP address. For the local endpoint IP address to be routable, the route advertisement for IPSec local endpoints must be enabled in the Tier-1 gateway configuration. See Add an NSX Tier-1 Gateway for more information.
6. If you are using a certificate-based authentication mode for the IPSec VPN session, from the **Site Certificate** drop-down menu, select the certificate that is to be used by the local endpoint.
7. Optional: Optionally add a description in **Description**.
8. Enter the **Local ID** value that is used for identifying the local NSX Edge instance.

   This local ID is configured as remote ID on the remote site. The local ID must either be the IP address or FQDN of the local site. For IPSec VPN sessions with certificate-based authentication and are associated with the local endpoint, the **Local ID** is derived from the certificate associated with the local endpoint. The ID specified in the **Local ID** text box is ignored. The local ID derived from the certificate for a VPN session depends on the extensions present in the certificate.

   - If the X509v3 extension `X509v3 Subject Alternative Name` is not present in the certificate, then the Distinguished Name (DN) is used as the local ID value.

     For example, if the certificate does not have any Subject Alternative Name (SAN) fields and its DN string is:

     ```
     C=US, ST=California, O=MyCompany, OU=MyOrg, CN=Site123
     ```

     then the DN string is used as the local ID. This local ID is the peer ID on the remote site.
   - If the X509v3 extension `X509v3 Subject Alternative Name` is found in the certificate, then one of the SAN fields is taken as the local ID value.

If the certificate has multiple SAN fields, then the following order is used to select the local ID.

Order	SAN Field
1	IP Address
2	DNS
3	Email Address

For example, if the configured site certificate has the following SAN fields:

```
x509v3 Subject Alternative Name:
DNS:Site123.vmware.com, email:user@company.com, IP Address:1.1.1.1
```

then the IP address `1.1.1.1` is used as the local ID. If the IP address is not available, then the DNS string is used. And if the IP address and DNS are not available, then the email address is used.

To see the local ID that is used for an IPSec VPN session, do the following:

1. Navigate to **Networking** > **VPN** and then click the **IPSec Sessions** tab.
2. Expand the IPSec VPN session.
3. Click **Download Config** to download the configuration file which contains the local ID as the remote ID to be configured at the remote VPN endpoint.

9. From the **Trusted CA Certificates** and **Certificate Revocation List** drop-down menus, select the appropriate certificates that are required for the local endpoint.
10. Optional: Specify a tag.
11. Click **Save**.

Adding Profiles

NSX provides the system-generated IPSec tunnel profile and an IKE profile that are assigned by default when you configure either an IPSec VPN or L2 VPN service. A system-generated DPD profile is created for an IPSec VPN configuration.

The IKE and IPSec profiles provide information about the algorithms that are used to authenticate, encrypt, and establish a shared secret between network sites. The DPD profile provides information about the number of seconds to wait in between probes to detect if an IPSec peer site is alive or not.

If you decide not to use the default profiles provided by NSX, you can configure your own profile using the information in the topics that follow in this section.

Add IKE Profiles

The Internet Key Exchange (IKE) profiles provide information about the algorithms that are used to authenticate, encrypt, and establish a shared secret between network sites when you establish an IKE tunnel.

NSX provides system-generated IKE profiles that are assigned by default when you configure an IPSec VPN or L2 VPN service. The following table lists the default profiles provided.

Table 1039: Default IKE Profiles Used for IPSec VPN or L2 VPN Services

Default IKE Profile Name	Description
nsx-default-l2vpn-ike-profile	• Used for an L2 VPN service configuration. • Configured with IKE V2, AES CBC 128 encryption algorithm, SHA2 256 algorithm, and Diffie-Hellman group14 key exchange algorithm.
nsx-default-l3vpn-ike-profile	• Used for an IPSec VPN service configuration. • Configured with IKE V2, AES CBC 128 encryption algorithm, SHA2 256 algorithm, and Diffie-Hellman group 14 key exchange algorithm.

Instead of the default IKE profiles used, you can also select one of the compliance suites supported starting with NSX 2.5. See About Supported Compliance Suites for more information.

If you decide not to use the default IKE profiles or compliance suites provided, you can configure your own IKE profile using the following steps.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **VPN** and then click the **Profiles** tab.
3. Select the **IKE Profiles** profile type, and click **Add IKE Profile**.
4. Enter a name for the IKE profile.
5. From the **IKE Version** drop-down menu, select the IKE version to use to set up a security association (SA) in the IPSec protocol suite.

Table 1040: IKE Versions

IKE Version	Description
IKEv1	When selected, the IPSec VPN initiates and responds to an IKEv1 protocol only.
IKEv2	This version is the default. When selected, the IPSec VPN initiates and responds to an IKEv2 protocol only.
IKE-Flex	If this version is selected and if the tunnel establishment fails with the IKEv2 protocol, the source site does not fall back and initiate a connection with the IKEv1 protocol. Instead, if the remote site initiates a connection with the IKEv1 protocol, then the connection is accepted.

6. Select the encryption, digest, and Diffie-Hellman group algorithms from the drop-down menus. You can select multiple algorithms to apply or deselect any selected algorithms you do not want to be applied.

Table 1041: Algorithms Used

Type of Algorithm	Valid Values	Description
Encryption	• AES 128 (default) • AES 256 • AES GCM 128 • AES GCM 192 • AES GCM 256	The encryption algorithm used during the Internet Key Exchange (IKE) negotiation. The AES 128 and AES 256 algorithms use the CBC mode of operation. The AES-GCM algorithms are supported when used with IKEv2. They are not supported when used with IKEv1.
Digest	• SHA2 256 (default) • SHA1 • SHA2 384 • SHA2 512	The secure hashing algorithm used during the IKE negotiation. If AES-GCM is the only encryption algorithm selected in the **Encryption Algorithm** text box, then no hash algorithms can be specified in the **Digest Algorithm** text box, per section 8 in RFC 5282. In addition, the Psuedo-Random Function (PRF) algorithm PRF-HMAC-SHA2-256 is implicitly selected and used in the IKE security association (SA) negotiation. The PRF-HMAC-SHA2-256 algorithm must also be configured on the peer gateway in order for the phase 1 of the IKE SA negotiation to succeed. If more algorithms are specified in the **Encryption Algorithm** text box, in addition to the AES-GCM algorithm, then one or more hash algorithms can be selected in the **Digest Algorithm** text box. In addition, the PRF algorithm used in the IKE SA negotiation is implicitly determined based on the hash algorithms configured. At least one of the matching PRF algorithms must also be configured on the peer gateway in order for the phase 1 of the IKE SA negotiation to succeed. For example, if the **Encryption Algorithm** text box contains AES 128 and AES GCM 128 and SHA1 is specified in the **Digest Algorithm** text box, then the PRF-HMAC-SHA1 algorithm is used during the IKE SA negotiation. It must also be configured in the peer gateway.

Type of Algorithm	Valid Values	Description
Diffie-Hellman Group	• Group 14 (default) • Group 2 • Group 5 • Group 15 • Group 16 • Group 19 • Group 20 • Group 21	The cryptography schemes that the peer site and the NSX Edge use to establish a shared secret over an insecure communications channel.

Note:
When you attempt to establish an IPSec VPN tunnel with a GUARD VPN Client (previously QuickSec VPN Client) using two encryption algorithms or two digest algorithms, the GUARD VPN Client adds additional algorithms in the proposed negotiation list. For example, if you specified AES 128 and AES 256 as the encryption algorithms and SHA2 256 and SHA2 512 as the digest algorithms to use in the IKE profile you are using to establish the IPSec VPN tunnel, the GUARD VPN Client also proposes AES 192 (using CBC mode) and SHA2 384 in the negotiation list. In this case, NSX uses the first encryption algorithm you selected when establishing the IPSec VPN tunnel.

7. Enter a security association (SA) lifetime value, in seconds, if you want it different from the default value of 86400 seconds (24 hours).
8. Provide a description and add a tag, as needed.
9. Click **Save**.

A new row is added to the table of available IKE profiles. To edit or delete a non-system created profile, click the three-dot menu (



) and select from the list of actions available.

Add IPSec Profiles

The Internet Protocol Security (IPSec) profiles provide information about the algorithms that are used to authenticate, encrypt, and establish a shared secret between network sites when you establish an IPSec tunnel.

NSX provides system-generated IPSec profiles that are assigned by default when you configure an IPSec VPN or L2 VPN service. The following table lists the default IPSec profiles provided.

Table 1042: Default IPSec Profiles Used for IPSec VPN or L2 VPN Services

Name of Default IPSec Profile	Description
nsx-default-l2vpn-tunnel-profile	• Used for L2 VPN. • Configured with AES GCM 128 encryption algorithm and Diffie-Hellman group 14 key exchange algorithm.
nsx-default-l3vpn-tunnel-profile	• Used for IPSec VPN. • Configured with AES GCM 128 encryption algorithm and Diffie-Hellman group 14 key exchange algorithm.

Instead of the default IPSec profile, you can also select one of the supported compliance suites. See About Supported Compliance Suites for more information.

If you decide not to use the default IPSec profiles or compliance suites provided, you can configure your own using the following steps.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **VPN** and then click the **Profiles** tab.
3. Select the **IPSec Profiles** profile type, and click **Add IPSec Profile**.
4. Enter a name for the IPSec profile.
5. From the drop-down menus, select the encryption, digest, and Diffie-Hellman algorithms. You can select multiple algorithms to apply.

   Deselect the ones you do not want used.

Table 1043: Algorithms Used

Type of Algorithm	Valid Values	Description
Encryption	• AES GCM 128 (default) • AES 128 • AES 256 • AES GCM 192 • AES GCM 256 • No Encryption Auth AES GMAC 128 • No Encryption Auth AES GMAC 192 • No Encryption Auth AES GMAC 256 • No Encryption	The encryption algorithm used during the Internet Protocol Security (IPSec) negotiation. The AES 128 and AES 256 algorithms use the CBC mode of operation.
Digest	• SHA1 • SHA2 256 • SHA2 384 • SHA2 512	The secure hashing algorithm used during the IPSec negotiation.
Diffie-Hellman Group	• Group 14 (default) • Group 2 • Group 5 • Group 15 • Group 16 • Group 19 • Group 20 • Group 21	The cryptography schemes that the peer site and NSX Edge use to establish a shared secret over an insecure communications channel.

6. Deselect **PFS Group** if you decide not to use the PFS Group protocol on your VPN service.

   It is selected by default.
7. In the **SA Lifetime** text box, modify the default number of seconds before the IPSec tunnel must be re-established.

   By default, an SA lifetime of 24 hours (86400 seconds) is used.
8. Select the value for **DF Bit** to use with the IPSec tunnel.

   The value determines how to handle the "Don't Fragment" (DF) bit included in the data packet received. The acceptable values are described in the following table.

Table 1044: DF Bit Values

DF Bit Value	Description
COPY	The default value. When this value is selected, NSX copies the value of the DF bit from the received packet into the packet which is forwarded. This value implies that if the data packet received has the DF bit set, after encryption, the packet also has the DF bit set.
CLEAR	When this value is selected, NSX ignores the value of the DF bit in the data packet received, and the DF bit is always 0 in the encrypted packet.

9. Provide a description and add a tag, if necessary.
10. Click **Save**.

A new row is added to the table of available IPSec profiles. To edit or delete a non-system created profile, click the three-dot menu (



) and select from the list of actions available.

Add DPD Profiles

A DPD (Dead Peer Detection) profile provides information about the number of seconds to wait in between probes to detect if an IPSec peer site is alive or not.

NSX provides a system-generated DPD profile, named `nsx-default-l3vpn-dpd-profile`, that is assigned by default when you configure an IPSec VPN service. This default DPD profile is a periodic DPD probe mode.

If you decide not to use the default DPD profile provided, you can configure your own using the following steps.

1. With admin privileges, log in to NSX Manager.
2. Navigate to **Networking** > **VPN** > **Profiles**.
3. Select **DPD Profiles** from the **Select Profile Type** drop-down menu, and click **Add DPD Profile**.
4. Enter a name for the DPD profile.
5. From the **DPD Probe Mode** drop-down menu, select **Periodic** or **On Demand** mode.

   For a periodic DPD probe mode, a DPD probe is sent every time the specified DPD probe interval time is reached.

   For an on-demand DPD probe mode, a DPD probe is sent if no IPSec packet is received from the peer site after an idle period. The value in **DPD Probe Interval** determines the idle period used.
6. In the **DPD Probe Interval** text box, enter the number of seconds you want the NSX Edge node to wait before sending the next DPD probe.

   For a periodic DPD probe mode, the valid values are between 3 and 360 seconds. The default value is 60 seconds.

   For an on-demand probe mode, the valid values are between 1 and 10 seconds. The default value is 3 seconds.

   When the periodic DPD probe mode is set, the IKE daemon running on the NSX Edge sends a DPD probe periodically. If the peer site responds within half a second, the next DPD probe is sent after the configured DPD probe interval time has been reached. If the peer site does not respond, then the DPD probe is sent again after waiting for half a second. If the remote peer site continues not to respond, the IKE daemon resends the DPD probe again, until a response is received or the retry count has been reached. Before the peer site is declared to be dead, the IKE daemon resends the DPD probe up to a maximum of times specified in the **Retry Count** property. After the peer site is declared dead, the NSX Edge node then tears down the security association (SA) on the dead peer's link.

   When the on-demand DPD mode is set, the DPD probe is sent only if no IPSec traffic is received from the peer site after the configured DPD probe interval time has been reached.
7. In the **Retry Count** text box, enter the number of retries allowed.

   The valid values are between 1 and 100. The default retry count is 5.

8. Provide a description and add a tag, as needed.
9. To enable or disable the DPD profile, click the **Admin Status** toggle.

   By default, the value is set to **Enabled**. When the DPD profile is enabled, the DPD profile is used for all IPSec sessions in the IPSec VPN service that uses the DPD profile.
10. Click **Save**.

A new row is added to the table of available DPD profiles. To edit or delete a non-system created profile, click the three-dot menu (



) and select from the list of actions available.

Add an Autonomous Edge as an L2 VPN Client

You can use L2 VPN to extend your Layer 2 networks to a site that is not managed by NSX. An autonomous NSX Edge, also referred to as NSX Edge for VMware ESX, can be deployed on the site, as an L2 VPN client. The autonomous for VMware NSX Edge is simple to deploy, easily programmable, and provides high-performance VPN. The autonomous NSX Edge is deployed using an OVF file on a host that is not managed by NSX. You can also enable high availability (HA) for VPN redundancy by deploying primary and secondary autonomous Edge L2 VPN clients.

- Create a port group and bind it to the vSwitch on your host. Ensure that this port group accepts promiscuous mode and forged transmits from the port group's security settings. For instructions, see Configure an NSX Edge Uplink Port in ESX.
- Create a port group for your internal L2 extension port.
- Obtain the IP addresses for the local IP and remote IP to use with the L2 VPN client session you are adding.
- Obtain the peer code that was generated during the L2 VPN server configuration.

1. Using vSphere Client, log in to the vCenter that manages the non-NSX environment.
2. Select **Hosts and Clusters** and expand clusters to show the available hosts.
3. Right-click the host where you want to install the autonomous NSX Edge and select **Deploy OVF Template**.
4. Enter the URL to download, https://support.broadcom.com/group/ecx/downloads, select the version, and click Download Now to install the NSX Edge for VMware ESX OVA file from the Internet or click **Browse** to locate the folder on your computer that contains the autonomous NSX Edge for VMware ESX file and click **Next**.

   This appliance can be used for both autonomous and managed Edges.
5. On the **Select name and folder** page, enter a name for the autonomous NSX Edge and select the folder or data center where you want to deploy. Then click **Next**.
6. On the **Select a compute resource** page, select the destination of the compute resource.
7. On the OVF Template Details page, review the template details and click **Next**.
8. On the **Configuration** page, select a deployment configuration option.
9. On the **Select storage** page, select the location to store the files for the configuration and disk files.
10. On the **Select networks** page, configure the networks that the deployed template must use. Select the port group you created for the uplink interface, the port group that you created for the L2 extension port, and enter an HA interface. Click **Next**.
11. On the **Customize Template** page, enter the following values and click **Next**.
    1. Type and retype the CLI admin password.
    2. Type and retype the CLI enable password.
    3. Type and retype the CLI root password.
    4. Enter the IPv4 address for the Management Network.
    5. Enable the option to deploy an autonomous Edge.

6. Enter the **External Port** details for VLAN ID, exit interface, IP address, and IP prefix length such that the exit interface maps to the Network with the port group of your uplink interface.
   If the exit interface is connected to a trunk port group, specify a VLAN ID. For example, `20,eth2,192.168.5.1,24`. You can also configure your port group with a VLAN ID and use VLAN 0 for the **External Port**.
7. To configure High Availability, enter the **HA Port** details where the exit interface maps to the appropriate HA Network.
8. When deploying an autonomous NSX Edge as a secondary node for HA, select **Deploy this autonomous-edge as a secondary node**.
   Use the same OVF file as the primary node and enter the primary node's IP address, user name, password, and thumbprint.
   To retrieve the thumbprint of the primary node, log in to the primary node and run the following command:
   ```
   get certificate api thumbprint
   ```
   Ensure that the VTEP IP addresses of the primary and secondary nodes are in the same subnet and that they connect to the same port group. When you complete the deployment and start the secondary-edge, it connects to the primary node to form an edge-cluster .

12. On the **Ready to complete** page, review the autonomous Edge settings and click **Finish**.

    Note:
    If there are errors during the deployment, a message of the day is displayed on the CLI. You can also use an API call to check for errors:
    ```
    GET https://<nsx-mgr>/api/v1/node/status
    ```
    The errors are categorized as soft errors and hard errors. Use API calls to resolve the soft errors as required. You can clear the message of day using an API call:
    ```
    POST /api/v1/node/status?action=clear_bootup_error
    ```

13. Power on the autonomous NSX Edge appliance using the vSphere Client. Open the console of the NSX Edge node to track the boot process using **Launch Remote Console**.
14. After the NSX Edge starts, log in to the Edge node using the console or SSH (provided SSH is enabled at the time of install) with admin credentials.

    Note:
    After the NSX Edge node starts, if you do not log in with admin credentials for the first time, the data plane service does not automatically start on the NSX Edge node.

15. Select **L2VPN** > **Add Session** and enter the following values:
    1. Enter a session name.
    2. Enter the local IP address and the remote IP address.
    3. Enter the peer code from the L2VPN server. See Download the Remote Side L2 VPN Configuration File for details on obtaining the peer code.
16. Click **Save**.
17. Select **Port** > **Add Port** to create an L2 extension port.
18. Enter a name, a VLAN, and select an exit interface.
19. Click **Save**.
20. Select **L2VPN** > **Attach Port** and enter the following values:
    1. Select the L2 VPN session that you created.
    2. Select the L2 extension port that you created.
    3. Enter a tunnel ID.
21. Click **Attach**.
    You can create additional L2 extension ports and attach them to the session if you need to extend multiple L2 networks.

22. Use the browser to log in to the autonomous NSX Edge or use API calls to view the status of the L2VPN session.

    Note:
    If the L2VPN server configuration changes, ensure that you download the peer code again and update the session with the new peer code.

Configure an NSX Edge Uplink Port in ESX

There are three options to configure uplink port changes on the ESX host that hosts the NSX autonomous Edge.

Consider performing this task if:

- Your east/west traffic between servers is timing out or works intermittently before timing out.
- The host kernel entry for migrated VMs are overwritten with the NSX Edge VM TEP address, instead of pointing to the correct TEP address of the ESX host hosting the VM.
- You are migrating from NSX-V to NSX
- Promiscuous mode is set to Accept on the virtual switch portgroup and the virtual machine guest OS places its vNIC in promiscuous mode.
- When running a packet capture within the VM, multicast and broadcast packets are received multiple times.
- The vSwitch is configured with NIC teaming and the load balancing policy is route based on originating port ID, route based on source MAC hash, use explicit failover order, or route based on physical NIC load.
- Multicast applications and protocols (such as CARP) running in virtual machines in promiscuous mode experience problems due to duplicated receive packets.

Option 1: From the ESX host, configure the NSX Edges to use sink port mode and enable promiscuous mode on the trunk vNic using the CLI

This option configures the NSX Edge from the ESX host using the CLI.

1. SSH to the ESX host that hosts the autonomous NSX Edge.
2. To enable promiscuous mode when using a virtual switch to configure trunk interface and prevent the issues above, run the command:

   ```
   esxcli system settings advanced set -o
           /Net/ReversePathFwdCheckPromisc -i 1
   ```

3. To check that the setting is enabled, run the following command:

   ```
   esxcli system settings advanced list -o
           /Net/ReversePathFwdCheckPromisc
   ```

   ```
   Path: /Net/ReversePathFwdCheckPromisc
      Type: integer
      Int Value: 1
      Default Int Value: 0
      Min Value: 0
      Max Value: 1
      String Val ue:
      Default String Value:
      Valid Characters:
      Description: Block duplicate packet in a teamed environment when
      the virtual switch is set to Promiscuous mode.
   ```

   This setting will discard packets coming from uplinks that are not associated with the particular client when promiscuous mode is enabled and will prevent duplicate packets from being received by the guest operating system. This will affect all promiscuous mode virtual machine and vmkernel interfaces on the ESX host.
4. In order for the setting to take effect, in the PortGroup security policy, set **Promiscuous Mode** from **Accept** to **Reject** and back to **Accept** to activate the configured change.

Option 2: From the NSX Edge UI, change advanced settings for the NSX Edge host

This option configures the NSX Edge using the UI.

1. Log in to the ESX host UI.

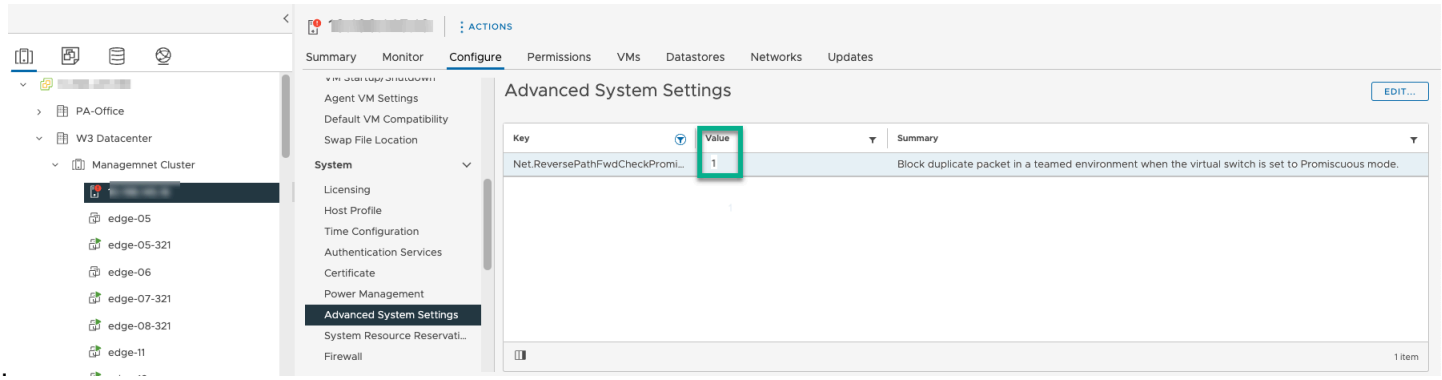
2. Go to the ESX host advanced settings and change the Net.ReversePathFwdCheckPromisc value.
   a. Select **Manage** > **Advanced Settings**.
   b. Enter **ReversePathFwdCheckPromisc** in the search field in the upper right hand window corner.
   c. Select the key value in the table and click **ActionsEdit option** and enter **1** in the new window.

Option 3: From the VMware vCenter server UI, change advanced settings for the ESX host

This option configures the ESX host using the vCenter UI.

1. Log in to the vCenter UI.
2. Go to the ESX host advanced settings and change the Net.ReversePathFwdCheckPromisc value.
   a. Select the ESX host from the management cluster, then **Configure** > **Advanced System Settings**.
   b. Select `Net.ReversePathFwdCheckPromisc` and click **Edit** and update the value to **1**.

For example, the following image shows the vCenter interface with the Net.ReversePathFwdCheckPromisc value set to

1. 

Check the Realized State of an IPSec VPN Session

After you send a configuration update request for an IPSec VPN session, you can check to see if the requested state has been successfully processed in the NSX local control plane on the transport nodes.

- Familiarize yourself with IPSec VPN. See Understanding IPSec VPN.
- Verify the IPSec VPN is configured successfully. See Add an NSX IPSec VPN Service.
- You must have access to the NSX Manager API.

When you create an IPSec VPN session, multiple entities are created: IKE profile, DPD profile, tunnel profile, local endpoint, IPSec VPN service, and IPSec VPN session. These entities all share the same `IPSecVPNSession` span, so you can obtain the realization state of all the entities of the IPSec VPN session by using the same `GET` API call. You can check the realization state using only the API.

1. Send a `POST`, `PUT`, or `DELETE` request API call.

   For example:

```
PUT https://<nsx-mgr>/api/v1/vpn/ipsec/sessions/8dd1c386-9b2c-4448-85b8-51ff649fae4f
{
   "resource_type": "PolicyBasedIPSecVPNSession",
   "id": "8dd1c386-9b2c-4448-85b8-51ff649fae4f",
   "display_name": "Test RZ_UPDATED",
   "ipsec_vpn_service_id": "7adfa455-a6fc-4934-a919-f5728957364c",
   "peer_endpoint_id":  "17263ca6-dce4-4c29-bd8a-e7d12bd1a82d",
   "local_endpoint_id": "91ebfa0a-820f-41ab-bd87-f0fb1f24e7c8",
   "enabled": true,
   "policy_rules": [
       {
           "id": "1026",
           "sources": [
```

```
                {
                    "subnet": "1.1.1.0/24"
                }
            ],
            "logged": true,
            "destinations": [
                {
                    "subnet": "2.1.4..0/24"
                }
            ],
            "action": "PROTECT",
            "enabled": true,
            "_revision": 1
        }
    ]
}
```

2. Locate and copy the value of `x-nsx-requestid` from the response header returned.
   For example:
   ```
   x-nsx-requestid   e550100d-f722-40cc-9de6-cf84d3da3ccb
   ```
3. Request the realization state of the IPSec VPN session using the following `GET` call.

   ```
   GET https://<nsx-mgr>/api/v1/vpn/ipsec/sessions/<ipsec-vpn-session-id>/state?request_id=<request-id>
   ```

   The following API call uses the `id` and `x-nsx-requestid` values in the examples used in the previous steps.
   ```
   GET https://<nsx-mgr>/api/v1/vpn/ipsec/sessions/8dd1c386-9b2c-4448-85b8-51ff649fae4f/state?request_id=e550100d-f722-40cc-9de6-cf84d3da3ccb
   ```

   Following is an example of a response you receive when the realization state is `in_progress`.
   ```
   {
     "details": [
       {
         "sub_system_type": "TransportNode",
         "sub_system_id": "fe651e63-04bd-43a4-a8ec-45381a3b71b9",
         "state": "in_progress",
         "failure_message": "CCP Id:ab5958df-d98a-468e-a72b-d89dcdae5346, Message:State realization is in progress at the node."
       },
       {
         "sub_system_type": "TransportNode",
         "sub_system_id": "ebe174ac-e4f1-4135-ba72-3dd2eb7099e3",
         "state": "in_sync"
       }
     ],
     "state": "in_progress",
     "failure_message": "The state realization is in progress at transport nodes."
   }
   ```

   Following is an example of a response you receive when the realization state is `in_sync`.
   ```
   {
       "details": [
           {
               "sub_system_type": "TransportNode",
               "sub_system_id":  "7046e8f4-a680-11e8-9bc3-020020593f59",
               "state": "in_sync"
   ```

```
        }
    ],
    "state": "in_sync"
}
```

The following are examples of possible responses you receive when the realization state is `unknown`.

```
{
    "state": "unknown",
    "failure_message": "Unable to get response from any CCP node. Please retry operation after some time."
}
{
    "details": [
       {
          "sub_system_type": "TransportNode",
          "sub_system_id": "3e643776-5def-11e8-94ae-020022e7749b",
          "state": "unknown",
          "failure_message": "CCP Id:ab5958df-d98a-468e-a72b-d89dcdae5346, Message: Unable to get response
 from the node. Please retry operation after some time."
       },
       {
          "sub_system_type": "TransportNode",
          "sub_system_id": "4784ca0a-5def-11e8-93be-020022f94b73",
          "state": "in_sync"
       }
    ],
    "state": "unknown",
    "failure_message": "The state realization is unknown at transport nodes"
}
```

After you perform an entity `DELETE` operation, you might receive the status of `NOT_FOUND`, as shown in the following example.

```
{
   "http_status": "NOT_FOUND",
   "error_code": 600,
   "module_name": "common-services",
   "error_message": "The operation failed because object identifier
 LogicalRouter/61746f54-7ab8-4702-93fe-6ddeb804 is missing: Object identifiers are case sensitive.."
}
```

If the IPSec VPN service associated with the session is disabled, you receive the `BAD_REQUEST` response, as shown in the following example.

```
{
    "httpStatus": "BAD_REQUEST",
    "error_code": 110199,
    "module_name": "VPN",
    "error_message": "VPN service f9cfe508-05e3-4e1d-b253-fed096bb2b63 associated with the session
 8dd1c386-9b2c-4448-85b8-51ff649fae4f is disabled. Can not get the realization status."
}
```

Understanding TCP MSS Clamping

TCP MSS clamping enables you to reduce the maximum segment size (MSS) value used by a TCP session during connection establishment through an IPSec tunnel. This feature is supported starting with NSX 2.5.

TCP MSS is the maximum amount of data in bytes that a host is willing to accept in a single TCP segment. Each end of a TCP connection sends its desired MSS value to its peer-end during a three-way handshake, where MSS is one of the TCP header options used in a TCP SYN packet. TCP MSS is calculated based on the maximum transmission unit (MTU) of the egress interface of the sender host.

When a TCP traffic goes through an IPSec VPN or any kind of VPN tunnel, additional headers are added to the original packet to keep it secure. For IPSec tunnel mode, additional headers used are IP, ESP, and optionally UDP (if port translation is present in the network). Because of these additional headers, the size of the encapsulated packet goes beyond the MTU of the VPN interface. The packet can get fragmented or dropped based on the DF policy.

To avoid packet fragmentation or drop, you can adjust the MSS value for the IPSec session by enabling the TCP MSS clamping feature. Navigate to **Networking** > **VPN** > **IPSec Sessions**. When you are adding an IPSec session or editing an existing one, expand the **Advance Properties** section, and enable **TCP MSS Clamping**.

You can configure the pre-calculated MSS value suitable for the IPSec session by setting both **TCP MSS Direction** and **TCP MSS Value**. The configured MSS value is used for MSS clamping. You can opt to use the dynamic MSS calculation by setting the **TCP MSS Direction** and leaving **TCP MSS Value** blank. The MSS value is auto-calculated based on the VPN interface MTU, VPN overhead, and the path MTU (PMTU) when it is already determined. The effective MSS is recalculated during each TCP handshake to handle the MTU or PMTU changes dynamically.

Troubleshooting VPN Problems

This section provides information to help you resolve problems that you might encounter while using the VPN features in NSX.

Monitor and Troubleshoot VPN Sessions

After you configure an IPSec or L2 VPN session, you can monitor the VPN tunnel status and troubleshoot any reported tunnel issues using the NSX Manager user interface.

1. With admin privileges, log in to NSX Manager.
2. Navigate to the or tab.
3. Expand the row for the VPN session that you want to monitor or troubleshoot.
4. To view the status of the VPN tunnel status, click the info icon.
   The Status dialog box appears and displays the available statuses.
5. To view the VPN tunnel traffic statistics, click **View Statistics** in the Status column.
   The Statistics dialog box displays the traffic statistics for the VPN tunnel.
6. To view the error statistics, click the **View More** link in the Statistics dialog box.
7. To close the **Statistics** dialog box, click **Close**.

Alarms When an IPsec VPN Session or Tunnel Is Down

When an IPsec VPN session or tunnel is down, an alarm is raised and the reason for the `Down` alarm is displayed on the Alarms dashboard or the VPN page on the NSX Manager user interface.

Use the following tables to locate the `Reason` message that you see on the NSX Manager user interface and review the possible cause for the `Down` alarm. To resolve the alarm, perform the necessary actions listed for the specific `Reason` message and possible cause for the `Down` alarm.

Table 1045: Causes and Solutions for an IPsec VPN Session Is Down Alarm

Reason for the IPsec VPN Session Down Alarm	Possible Cause	Necessary Actions to Resolve the Alarm Message
`Authentication failed`	The IKE SA establishment between the VPN gateways failed due to a failure in authentication. Authentication of the IKE SA depends on the pre-shared key, Local ID, and Remote ID values.	• Verify the `Local ID` and `Remote ID` values. The Local ID value must be set as the Remote ID value in the peer VPN gateway. • Verify the `Pre-shared Key` value. It must match exactly in both the VPN gateways.
`No proposal chosen`	The IKE transform configuration in both the local and peer configuration file are inconsistent.	Ensure that the following properties are configured the same for both gateways. • `DH groups` • `Digest and encryption algorithms`
`Peer sent delete`	The peer gateway initiated a delete case. A `DELETE` payload is received for IKE SA.	To determine why the peer gateway sent a `DELETE` payload, examine the syslogs on the NSX Edge and on the peer gateway side.
`Peer not responding`	The IKE SA negotiation timed out.	• Verify that the remote gateway is up. • Verify the connectivity to the remote gateway.
`Invalid syntax`	• IKE proposals or transforms are not formed correctly. • There are malformed IKE payloads.	To debug the invalid syntax, analyze the edge syslogs.
`Invalid spi`	An invalid SPI value was received in the IKE payload.	To debug the invalid SPI value, analyze the edge syslogs.
`Configuration failed`	The session configuration realization failed in NSX Edge due to some constraints or certain criteria. The reason is listed in the session dump under `Session_Config_Fail_Reason`.	Resolve the error using the reason displayed in the session dump under `Session_Config_Fail_Reason`.
`Negotiation not started`	The IKE SA negotiation has not started.	• Verify that the `Connection Initiation Mode` property in the session configuration is set to `Respond Only`. • Verify the VPN configuration of both gateways. At least one of the gateways must be set to `Initiator`.
`IPsec service not enabled`	Status of the VPN service used for the session is not active.	Verify if the Admin Status in the IPsec VPN service configuration is not enabled.
`Session not enabled`	Admin has not enabled the session.	Enable the session to resolve this error.
`SR state is not Active`	SR is in a standby state.	Verify VPN session status on the NSX Edge node where HA peer SR is in the Active state.

Table 1046: Causes and Solutions for an IPsec VPN Tunnel Is Down Alarm

Reason for the IPsec VPN Tunnel Down Alarm	Possible Cause	Necessary Actions to Resolve the Alarm Message
`Peer sent delete`	The peer gateway sent a `DELETE` payload for the IPSEC SA.	To understand why the peer gateway sent a `DELETE` payload, you must check the syslogs on the NSX Edge and on the peer gateway side.
`No proposal chosen`	The ESP transform configuration is not consistent in the configurations for both the local and peer gateways.	Ensure that the following properties are configured the same for both gateways. • `DH groups` • `Digest and encryption algorithms` • The `PFS` is activated or not.
`TS unacceptable`	The IPsec SA setup has failed due to a mismatch in the policy rule definition between the gateways for the tunnel configuration.	Check the local and remote network configuration on both gateways.
`IKE SA down`	The IKE SA session is down.	First, check the session down reason in the edge syslogs. Refer to the Necessary Actions column in the previous table to resolve the session down errors, and then check whether the tunnel down reason still persists.
`Invalid syntax`	• The proposals or transforms are not formed correctly. • There are malformed payloads.	To debug the invalid syntax, analyze the edge syslogs.
`Invalid spi`	An invalid SPI value was received in the ESP payload.	To debug the invalid SPI value, analyze the edge syslogs.
`No IKE peers`	All IKE peers are dead. There are no peer gateways left with whom to try to establish a connection.	• Check if the remote gateway is up. • Check the connectivity to the configured peer gateways.
`IPsec negotiation not started`	The IPsec SA negotiation has not started.	The IKE SA is not up yet. Check the session down reason in the edge syslogs and resolve the errors.

Network Address Translation (NAT)

Network address translation (NAT) maps one IP address space to another. You can configure NAT on tier-0 and tier-1 gateways.

- Source NAT (SNAT) - Translates a source IP address of outbound packets so that packets appear as originating from a different network. Supported on tier-0/tier-1 gateways running in active-standby mode. For one-to-one SNAT, the SNAT translated IP address is not programmed on the loopback port, and there is no forwarding entry with an SNAT translated IP as the prefix. For n-to-one SNAT, the SNAT translated IP address is programmed on the loopback port, and users will see a forwarding entry with an SNAT-translated IP address prefix. NSX SNAT is designed to be applied to traffic that egresses the NSX environment.
- Destination NAT (DNAT) - Translates the destination IP address of inbound packets so that packets are delivered to a target address into another network. Supported on tier-0/tier-1 gateways running in active-standby mode. NSX DNAT is designed to be applied to traffic that ingresses the NSX environment.
- Reflexive NAT - (sometimes called stateless NAT) Translates addresses passing through a routing device. Inbound packets undergo destination address rewriting, and outbound packets undergo source address rewriting. It is not keeping a session as it is stateless. Supported on tier-0 gateways running in active-active or active-standby mode, and on tier-1 gateways.
- Stateful NAT services are supported when using Stateful Active-Active HA mode on the tier-0 or tier-1 gateways.

You can also disable SNAT or DNAT for an IP address or a range of addresses (No-SNAT/No-DNAT). If an address has multiple NAT rules, the rule with the highest priority is applied.

Note:

If there is a service configured in a DNAT rule, `translated_port` will be realized on NSX Manager as `destination_port`. This means that the service will be the translated port while the translated port is used to match the traffic as destination port. If there is no service configured, the port will be ignored.

If you are creating a NAT rule from a Global Manager in an NSX Federation environment, you can select site-specific IP addresses for NAT. Note the following:

- Do not click **Set** under **Apply To** if you want the default option of applying the NAT rule to all locations.
- Under **Apply To**, click **Set** and select the locations whose entities you want to apply the rule to and then select **Apply NAT rule to all entities**.
- Under **Apply To**, click **Set**, select a location and then select **Interfaces** from the **Categories** drop-down menu. You can select specific interfaces to which you want to apply the NAT rule.
- SNAT configured on a tier-0 gateway's external interface processes traffic from a tier-1 gateway, and from another external interface on the tier-0 gateway.
- NAT is configured on the uplinks of the tier-0/tier-1 gateways and processes traffic going through this interface. This means that tier-0 gateway NAT rules will not apply between two tier-1 gateways connected to the tier-0.

NAT Support Matrices

Configuration Fields

Rule Type	Match			Translated IP | Port			
	Source IP	**Destination IP**	**Port**	**Translated IP**	**Service protocol**	**Service source port**	**Service destination port**
SNAT	Optional	Optional	Optional (is used to match destination port)	Required	Optional	Optional (match as source port)	Optional (setting is ignored)
DNAT	Optional	Required	Optional (is used to match destination port)	Required	Optional	Optional (match as source port)	Optional (use as translated port)
NO_SNAT	Required	Optional	Optional (is used to match destination port)	N/A	Optional	Optional (match as source port)	Optional (setting is ignored)
NO_DNAT	Optional	Required	Optional (is used to match destination port)	N/A	Optional	Optional (match as source port)	Optional (setting is ignored)
REFLEXIVE	Required	N/A	N/A	Required	N/A	N/A	N/A
NAT64	Optional (IPv6)	Required (IPv6)	Optional (is used to match destination port)	Required (IPv6)	Required	Optional (match as source port)	Optional (setting is ignored)

Configuration Use Cases

Types	**1:1**	**n:n**	**n:m**	**n:1**	**1:m**
SNAT	Yes	Yes	Yes	Yes	No
DNAT	Yes	Yes	N/A	Yes	N/A

Types	1:1	n:n	n:m	n:1	1:m
NO_SNAT	-	-	-	-	-
NO_DNAT	-	-	-	-	-
REFLEXIVE	Yes	Yes	N/A	No	No
NAT64	Yes	Yes	No	Yes	No

NAT Traffic Flow Support on Interfaces

Note:

The following support matrix refers to rules applied to any object type.

Traffic Flow Support on Interfaces	DNAT	SNAT	NO_DNAT	NO_SNAT	REFLEXIVE	NAT64
Inter-VRF → Uplink	Yes	Yes	Yes	Yes	Yes	Yes
Inter-VRF → Downlink	Yes	No	Yes	No	Yes	Yes
Downlink → Inter-VRF	No	Yes	No	Yes	Yes	No
Uplink → Inter-VRF	Yes	Yes	Yes	Yes	Yes	Yes
Uplink/External → Uplink/External	Yes	Yes	Yes	Yes	Yes	Yes
External (Uplink) → Connected Segment (Downlink)	Yes	No	Yes	No	Yes	Yes
Uplink → Service Interface	Yes	Yes	Yes	Yes	Yes	Yes
Backplane → Backplane	No	No	No	No	No	No
Downlink → Uplink	No	Yes	No	Yes	Yes	No
Downlink → Service Interface	No	Yes	No	Yes	Yes	No
Service Interface → Service Interface	Yes	Yes	Yes	Yes	Yes	Yes
Service Interface → Uplink	Yes	Yes	Yes	Yes	Yes	Yes
Service Interface → Downlink	Yes	No	Yes	No	Yes	Yes

Configure an NSX NAT/DNAT/No SNAT/No DNAT/Reflexive NAT

You can configure different types of NAT for IPv4 on a tier-0 or tier-1 gateway.

Note:
If there is a service configured in this NAT rule, the translated_port will be realized on NSX Manager as the destination_port. This means the service will be the translated port while the translated port is used to match the traffic as destination port. If there is no service configured, the port will be ignored.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **NAT**.
3. From the **View** drop-down menu, select **NAT**.
4. Click **Add NAT Rule**.
5. Enter a **Name**.
6. Select an action.

Gateway	Available Actions
Tier-1 gateway	Available actions are **SNAT**, **DNAT**, **Reflexive**, **NO SNAT**, and **NO DNAT**.
Tier-0 gateway in active-standby mode	Available actions are **SNAT**, **DNAT**, **NO SNAT**, and **NO DNAT**.
Tier-0 gateway in active-active mode	The available action is **Reflexive**.

7. In the **Source IP** field, enter the source network address.

   Specify an IP address, or an IP address range in CIDR format. For **SNAT**, **NO_SNAT** and **Reflexive** rules, this is a required field and represents the source network of the packets leaving the network.

   For **DNAT** and **NO_DNAT** rules, optionally it can contain source network of incoming packets.

   For a NULL value, this field represents any network.
8. Enter a **Destination**.

   Specify an IP address, or an IP address range in CIDR format. For **DNAT** and **NO_DNAT** rules, this is a required field and represents the source network of the packets leaving the network. This field is not applicable for **Reflexive**.
9. Enter a value for **Destination Port**.
10. Enter a value for **Translated IP**.

    Specify an IPv4 address, or an IP address range in CIDR format. If translated IP is less than the match IP for SNAT it will work as PAT.
11. From the **Select Service** drop-down menu, select a service, which has an associated port number.

    If there is a service interface configured in a NAT rule, `translated_port` will be realized on NSX Manager as `destination_port`. This means that the service will be the translated port while the translated port is used to match the traffic as destination port. If there is no service configured, the port will be ignored.
12. For **Apply To**, click **Set** and select objects that this rule applies to.

    The available objects are **Tier-0 Gateways**, **Interfaces**, **Labels**, **Service Instance Endpoints**, and **Virtual Endpoints**.

    Note:
    If you are using NSX Federation and creating a NAT rule from a Global Manager appliance, you can select site-specific IP addresses for NAT. You can apply the NAT rule to any of the following location spans:

    - Do not click **Set** if you want to use the default option of applying the NAT rule to all locations.
    - Click **Set**. In the **Applied To | New Rule** dialog box, select the locations whose entities you want to apply the rule to and then click **Apply**.
    - Click **Set**. In the **Applied To | New Rule** dialog box, select a location and then select **Interfaces** from the **Categories** drop-down menu. You can select specific interfaces to which you want to apply the NAT rule.
    - Click **Set**. In the **Applied To | New Rule** dialog box, select a location and then select **VTI** from the **Categories** drop-down menu. You can select specific VTIs to which you want to apply the NAT rule.
13. Toggle the **Enabled** field to **Yes**.
14. Optional: Select a firewall setting.

    The available settings are:

    - **Match External Address** - The firewall will be applied to external address of a NAT rule.
      – For SNAT, the external address is the translated source address after NAT is done.

  - For DNAT, the external address is the original destination address before NAT is done.
  - For REFLEXIVE, to egress traffic, the firewall is applied to the translated source address after NAT is done. For ingress traffic, the firewall is applied to the original destination address before NAT is done.
- **Match Internal Address** - Indicates the firewall will be applied to internal address of a NAT rule.
  - For SNAT, the internal address is the original source address before NAT is done.
  - For DNAT, the internal address is the translated destination address after NAT is done.
  - For REFLEXIVE, for egress traffic, the firewall is applied to the original source address before NAT is done. For ingress traffic, the firewall is applied to the translated destination address after NAT is done.
- **Bypass** - The packet bypasses firewall rules.

15. Optional: Toggle the **Logging** button to enable logging.
16. Specify a priority value.

    A lower value means a higher priority. The default is 0. A **No SNAT** or **No DNAT** rule should have a higher priority than other rules.
17. Optional: **Apply to Policy Based VPN**: Applies only for the **DNAT** or **No DNAT** rule category. The rule is applied based on the priority value. Despite the **Bypass** or **Match** settings, the settings applied for the **Apply To** parameter of a NAT policy are still honored.

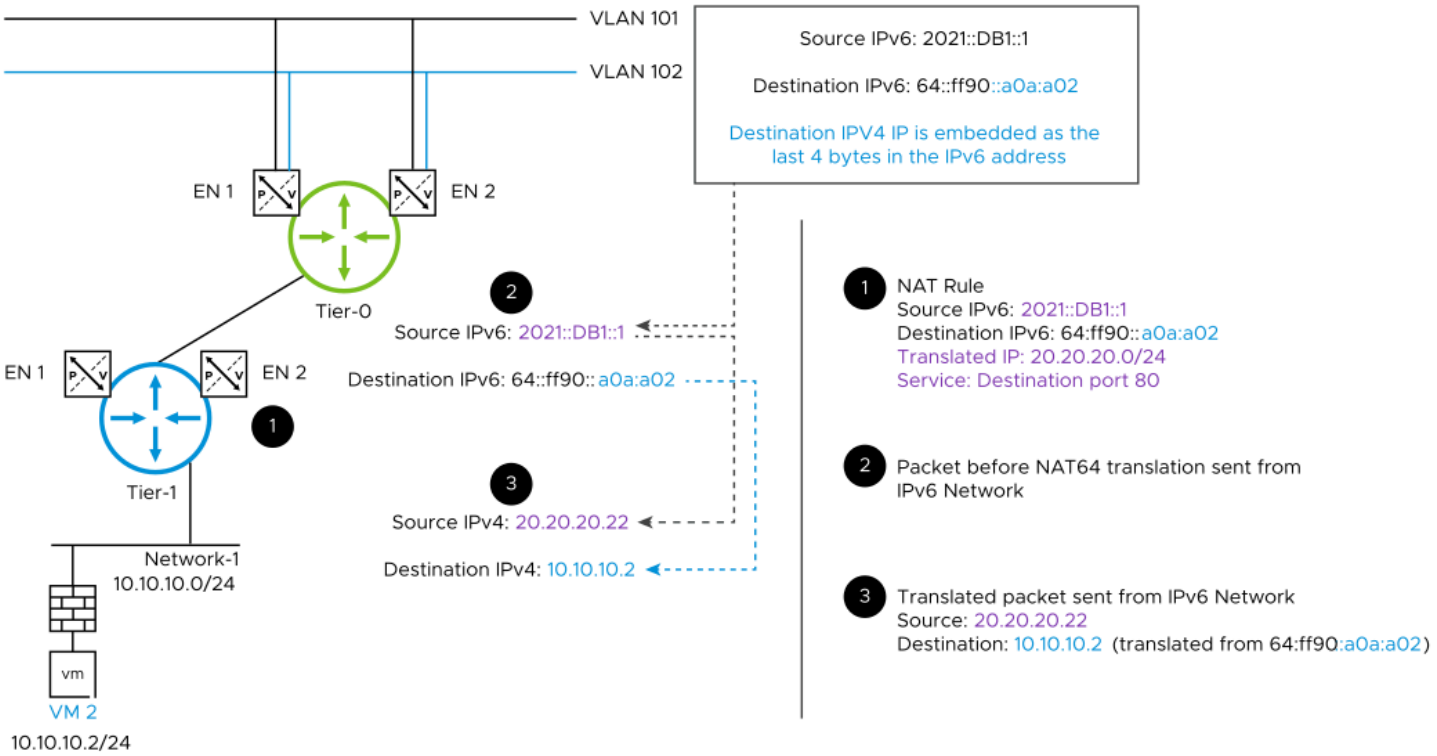
    - **Bypass:** NAT Rule is not applied to the traffic decrypted from a policy-based IPSec VPN tunnel. This is the default setting.
    - **Match**: If the traffic is decrypted from a policy-based IPSec VPN tunnel, the NAT policy is evaluated and matched. NAT policy is NOT evaluated on the traffic that is not decrypted from a policy-based IPSec VPN tunnel.

    For a NAT policy to hit the decrypted traffic, the policy must be set to **Match** and the interface where encrypted traffic is sent/received must be set in the **Apply To** parameter of the NAT policy. For more information on the **Apply To** parameter, see Network Address Translation (NAT).
18. Click **Save**.

Add an NSX NAT64

NAT64 is a mechanism for translating IPv6 packets to IPv4 packets, and vice versa. NAT64 allows IPv6-only clients to contact IPv4 servers using unicast UDP or TCP. NAT64 only allows an IPv6-only client to initiate communications to an IPv4-only server. To perform IPv6-to-IPv4 translation, binding and session information is saved. NAT64 is stateful.

The following diagram shows details of NAT64 translation.



Note:

- NAT64 is only supported for external IPv6 traffic coming in through the NSX Edge uplink to the IPv4 server in the overlay.
- NAT64 supports TCP and UDP. Packets of all other protocol types are discarded. NAT64 does not support ICMP, fragmentation, or IPV6 packets that have extension headers.
- When a NAT64 rule and an NSX load balancer are configured on the same Edge node, using the NAT64 rule to direct IPv6 packets to the IPv4 load balancer is not supported.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **NAT**.
3. Select **NAT64** as the view.
4. Click **Add NAT 64 Rule**.
5. Enter a **Name**.
6. Enter a **Source**.

   Specify an IPv6 address, or an IPv6 address range in CIDR format. For example, 2001:DB7:1::1 or 2001:DB7:1::/64.

   If this text box is left blank, the NAT rule applies to all sources outside of the local subnet.
7. Enter a **Destination**.

   Specify an IPv6 address, or an IPv6 address range in CIDR format with subnet size 96. For example, 64:ff9b::0B01:0101 or 2001:DB8::/96.
8. Enter a value for **Translated IP**.

   Specify an IPv4 address, an IPv4 address range, or a comma-separated list of IPv4 addresses. For example, 10.1.1.1, 10.1.1.1-10.1.1.2, or 10.1.1.1,10.1.1.2.

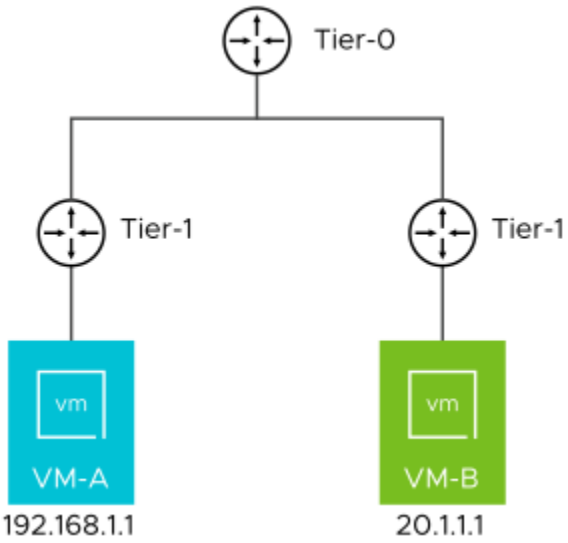
9. Toggle **Enable** to enable the rule.
10. Optional: In the **Service** column, click **Set** to select services.
11. Optional: Enter a value for **Translated Port**.
12. Optional: For **Apply To**, click **Set** and select objects that this rule applies to.
    The only option available is **Interfaces**.
13. Optional: Toggle the **logging** button to enable logging.
14. Optional: Specify a priority value.
    A lower value means a higher priority. The default is 0.
15. The **Firewall** setting is set to **Bypass** and cannot be changed.
16. Click **Save**.

NAT and Gateway Firewall

A NAT firewall allows internet traffic to pass through the gateway if a device on the private network requested it. Any unsolicited requests or data packets are discarded, preventing communication with potentially dangerous devices.

If a tier-1 gateway has both SNAT and gateway firewall (GWFW) configured, and if the GWFW is not configured to be stateful, you must configure NO SNAT for the tier-1 gateway's advertised subnets. Otherwise, traffic to IP addresses in these subnets will fail.

In the example below, T1-A is the gateway and there is a SNAT rule configured that translates any traffic from its attached subnet 192.168.1.0/0 to 10.1.1.1.



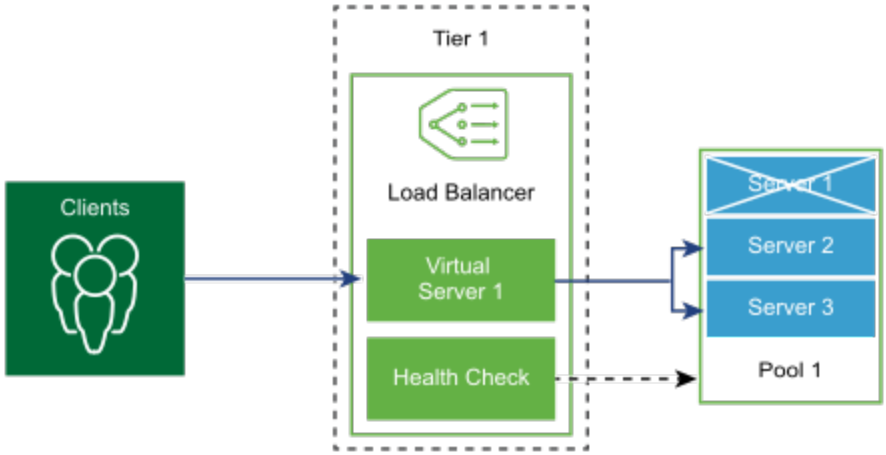
Here are some traffic scenarios:

1. Any traffic stream that is initiated from VM-A/192.168.1.1 will get translated to 10.1.1.1 as the source IP, regardless if gateway firewall is stateful, stateless, or disabled. When the traffic from VM-C or VM-B returns for that flow, they will have a destination IP of 10.1.1.1; T1-A will match it up with the SNAT flow and translate it correctly so that it flows back to VM-A. The SNAT rule works as expected, and there are no issues.
2. VM-B/20.1.1.1 initiates a traffic flow to VM-A/192.168.1.1. Here, there's a difference in behavior when T1-A has a stateful firewall versus when it has no firewall or stateless firewall. The firewall rules permit the traffic between VM-B and VM-A. To have this scenario, configure a NO-NAT rule for traffic matching 192.168.1.0/24 to 20.1.1.0/24. When this NO-NAT rule exists, then there will be no difference in behavior.
3. If T1-A has stateful firewall, the T1-A firewall will create a firewall connection entry for the TCP SYN packet from VM-B/20.1.1.1 to VM-A/192.168.1.1. When VM-A replies, T1-A will match the reply packet with the stateful connection entry, and forward the traffic from VM-A/192.168.1.1 to VM-B/ 20.1.1.1 with no SNAT translation. This is because the firewall will skip the SNAT lookup when the return traffic matches up with a firewall connection entry.

4. If T1-A has firewall disabled or stateless, the T1-A firewall will forward the TCP SYN packet from VM-B/20.1.1.1 to VM-A/192.168.1.1 without creating a firewall connection entry, because it's either stateless or no firewall. When VM-A/192.168.1.1 replies back to VM-B/20.1.1.1, T1-A sees that there's no firewall connection entry, performs SNAT on it, and translates the source IP from VM-A/192.168.1.1 to 10.1.1.1. When that reply gets back to VM-B, VM-B will drop the traffic because the source IP address is 10.1.1.1 instead of VM-A/192.168.1.1.

NSX Native Load Balancer

NSX Native Load Balancer is a logical load balancer that offers high-availability service for applications and distributes the network traffic load among multiple servers.



The load balancer distributes incoming service requests evenly among multiple servers in such a way that the load distribution is transparent to users. Load balancing helps in achieving optimal resource utilization, maximizing throughput, minimizing response time, and avoiding overload.

You can map a virtual IP address to a set of pool servers for load balancing. The load balancer accepts TCP and UDP requests on the virtual IP address and decides which pool server to use.

Depending on your environment needs, you can scale the load balancer performance by increasing the existing virtual servers and pool members to handle heavy network traffic load.

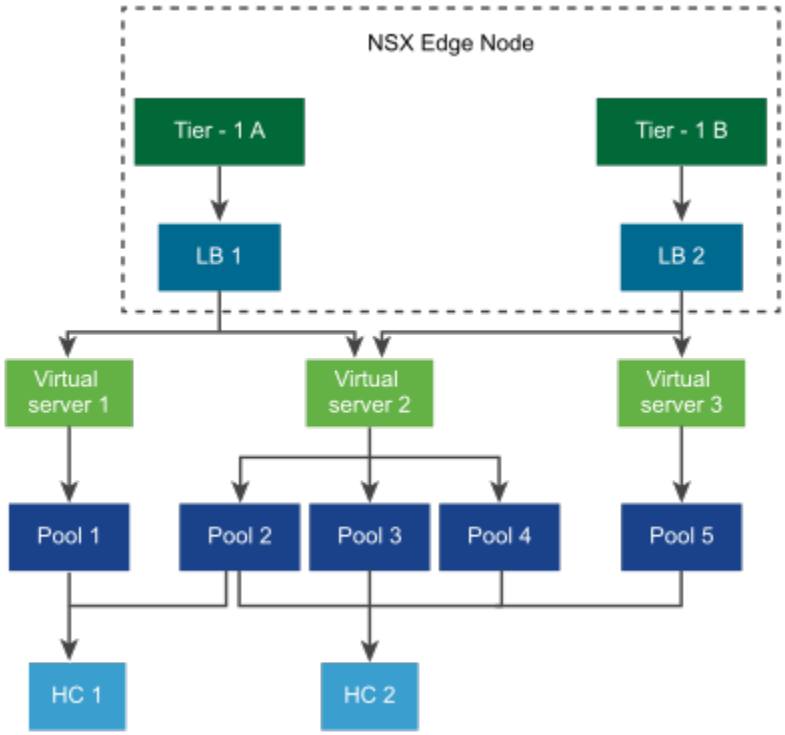
> **Note:**
> Logical load balancer is supported only on the tier-1 gateway. One load balancer can be attached only to a tier-1 gateway.

Key Load Balancer Concepts

Load balancer includes virtual servers, server pools, and health checks monitors.

A load balancer is connected to a Tier-1 logical router. The load balancer hosts single or multiple virtual servers. A virtual server is an abstract of an application service, represented by a unique combination of IP, port, and protocol. The virtual server is associated to single to multiple server pools. A server pool consists of a group of servers. The server pools include individual server pool members.

To test whether each server is correctly running the application, you can add health check monitors that check the health status of a server.



Scaling Load Balancer Resources

When you configure a load balancer, you can specify a size (small, medium, large, or extra large). The size determines the number of virtual servers, server pools, and pool members the load balancer can support.

A load balancer runs on a tier-1 gateway, which must be in active-standby mode. The gateway runs on NSX Edge nodes. The form factor of the NSX Edge node (small, medium, large, or extra large) determines the number of load balancers that the NSX Edge node can support.

For more information about what the different load balance sizes and NSX Edge form factors can support, see https://configmax.broadcom.com.

Note that using a small NSX Edge node to run a small load balancer is not recommended in a production environment.

You can call an API to get the load balancer usage information of an NSX Edge node:

```
GET /policy/api/v1/infra/lb-node-usage?node_path=<node-path>
```

The usage information includes the number of load balancer objects (such as load balancer services, virtual servers, server pools, and pool members) that are configured on the node. For more information, see the *NSX API Guide*.

Supported Load Balancer Features

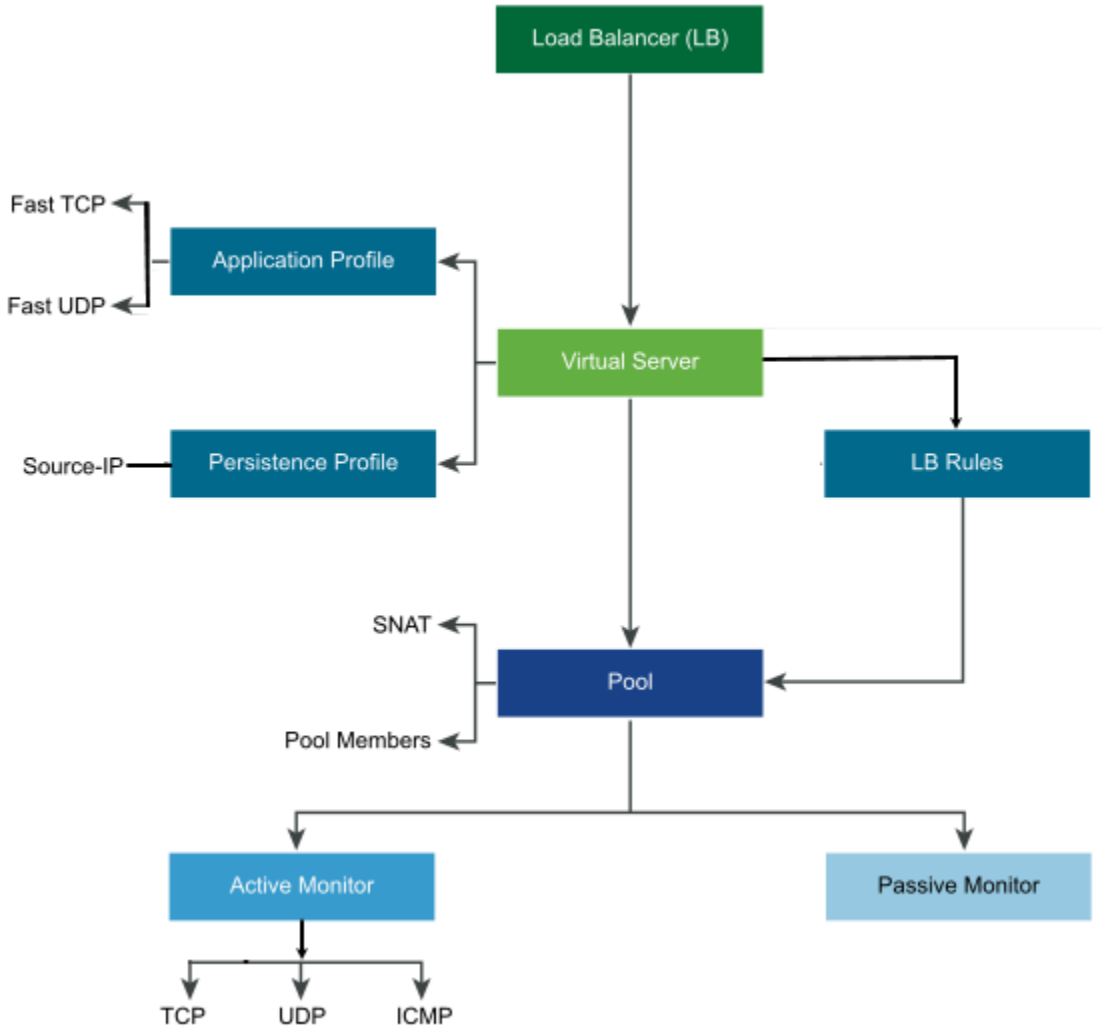
You can use an NSX load balancer with the management appliance and with VCF Automation All Apps. A load balancer supports the following features.

- Layer 4 (TCP and UDP) - The NSX load balancer can make traffic distribution decisions based on information at the transport layer of the network. Decisions are based on IP addresses and TCP/UDP socket information (source/destination IP and port) only. It does not inspect or make load balancer descisions based on Layer 7 (application layer) data, such as HTTP or HTTPs requests. This means no content-based routing URL-based redirection, or advanced apllication-aware features.
- Server pools - Static and dynamic with NSGroup
- Persistence - Source-IP
- Health check monitors - Active monitor which includes TCP, UDP, and ICMP, and passive monitor
- SNAT - Transparent, Automap, and IP List

- The Load Balancing runbook lists the topology for the given load balancer instance.

Note: SSL -Terminate-mode and proxy-mode is not supported in NSX limited export release.

Figure 424:

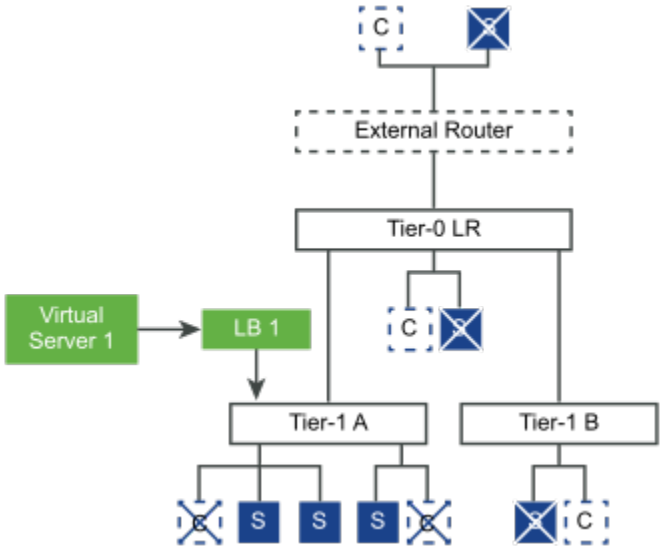


Load Balancer Topologies

Load balancers are typically deployed in either inline or one-arm mode. One-arm mode requires virtual server Source NAT (SNAT) configuration, and inline mode does not.

Inline Topology

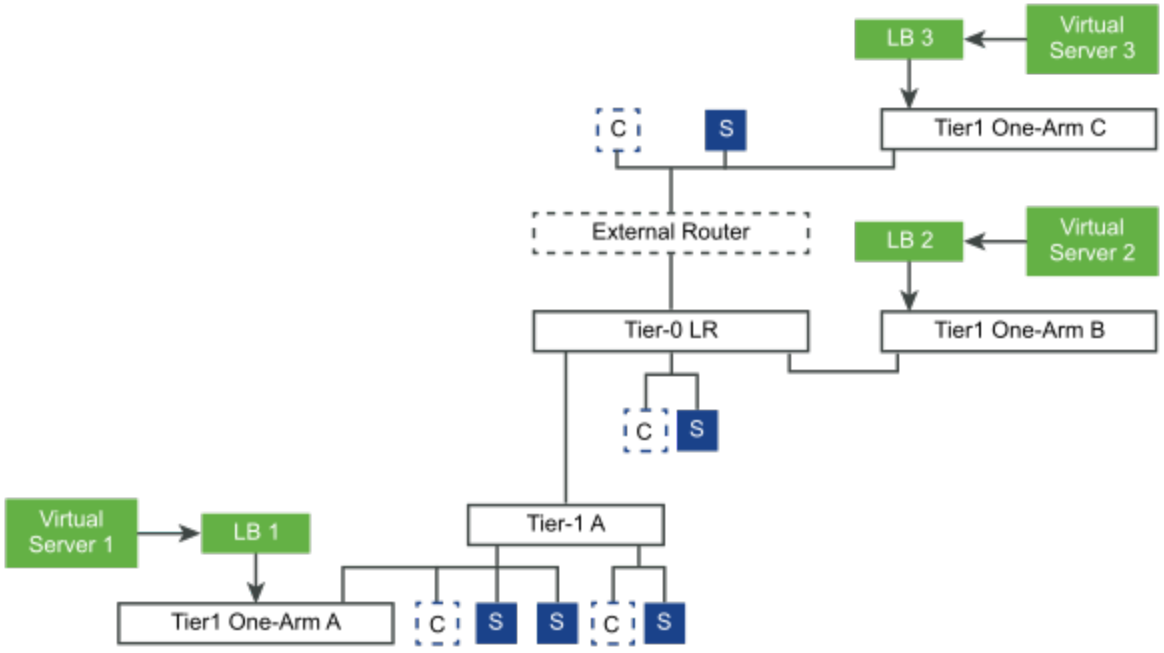
In the inline mode, the load balancer is in the traffic path between the client and the server. Clients and servers should not be connected to overlay segments on the same tier-1 logical router if SNAT on the load balancer is not desired. If clients and servers are connected to overlay segments on the same tier-1 logical router, SNAT is required.



One-Arm Topology

In one-arm mode, the load balancer is not in the traffic path between the client and the server. In this mode, the client and the server can be anywhere. The load balancer performs Source NAT (SNAT) to force return traffic from the server destined to the client to go through the load balancer. This topology requires virtual server SNAT to be enabled.

When the load balancer receives the client traffic to the virtual IP address, the load balancer selects a server pool member and forwards the client traffic to it. In the one-arm mode, the load balancer replaces the client IP address with the load balancer IP address so that the server response is always sent to the load balancer. The load balancer forwards the response to the client.



Special Use Case When no overlay is used and everything is VLAN, overlay still must be configured on the edge nodes hosting the tier-1 one-arm load balancer. This is because edge nodes must have at least one tunnel end point UP for high availability between edge nodes. When the tunnels are UP they will agree on which edge node will run the active or standby role of each tier-0 and tier-1 gateway.

Tier-1 Service Chaining

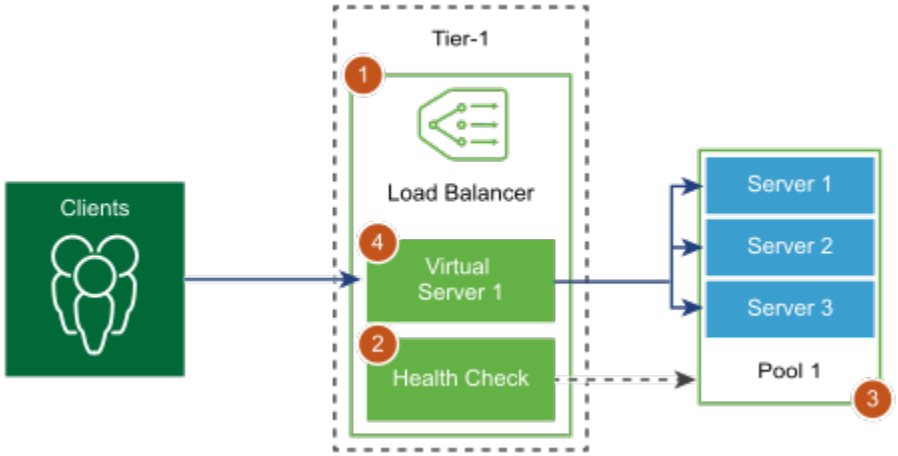
If a tier-1 gateway or logical router hosts different services, such as NAT, firewall, and load balancer, the services are applied in the following order:

- Ingress
  DNAT - Firewall - Load Balancer
  Note: If DNAT is configured with Firewall Bypass, firewall is skipped but not load balancer.
- Egress
  Load Balancer - Firewall - SNAT

Setting Up Load Balancer Components

To use logical load balancers, you must start by configuring a load balancer and attaching it to a tier-1 gateway.

Next, you set up health check monitoring for your servers. You must then configure server pools for your load balancer. Finally, you must create a layer 4 or layer 7 virtual server for your load balancer and attach the newly created virtual server to the load balancer.



Add Load Balancers

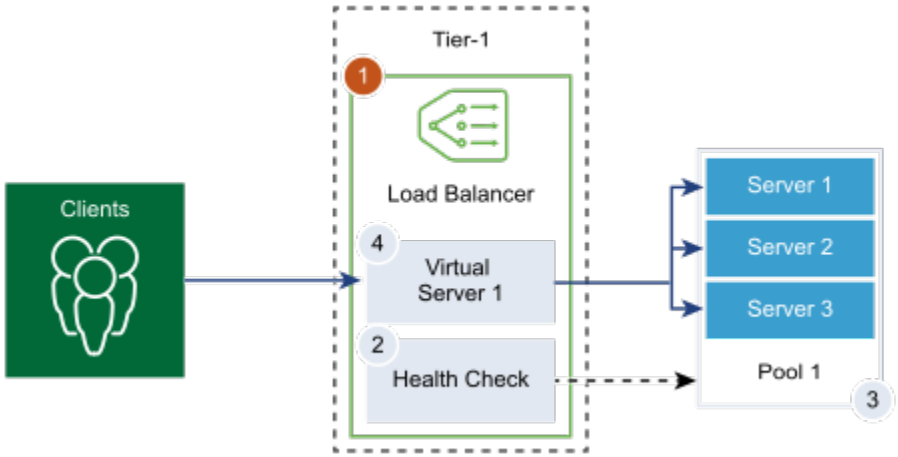
Load balancer is created and attached to the tier-1 gateway.

Verify that a tier-1 gateway is configured. See Tier-1 Gateway.

You can configure the level of error messages you want the load balancer to add to the error log.

Note:

- Avoid setting the log level to DEBUG on load balancers with a significant traffic due to the number of messages printed to the log that affect performance.
- Load balancer over IPSec VPN is not supported for route-based VPN terminated on Tier-1 gateways.



1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Load Balancing** > **Add Load Balancer**.
3. Enter a name and a description for the load balancer.
4. Select the load balancer size based on your needs of virtual servers and pool members and available resources.
5. Select the already configured tier-1 gateway to attach to this load balancer from the drop-down menu.
   The tier-1 gateway must be in the Active-Standby mode.
6. Define the severity level of the error log from the drop-down menu.
   Load balancer collects information about encountered issues of different severity levels to the error log.
7. Optional: Enter tags to make searching easier.
   You can specify a tag to set a scope of the tag.
8. Click **Save**.
   The load balancer creation and attaching the load balancer to the tier-1 gateway takes about three minutes and the configuration status to appear green and Up.

   If the status is Down, click the information icon and resolve the error before you proceed.
9. Optional: Delete the load balancer.
   a) Detach the load balancer from the virtual server and tier-1 gateway.
   b) Select the load balancer.
   c) Click the vertical ellipses button.
   d) Select **Delete**.

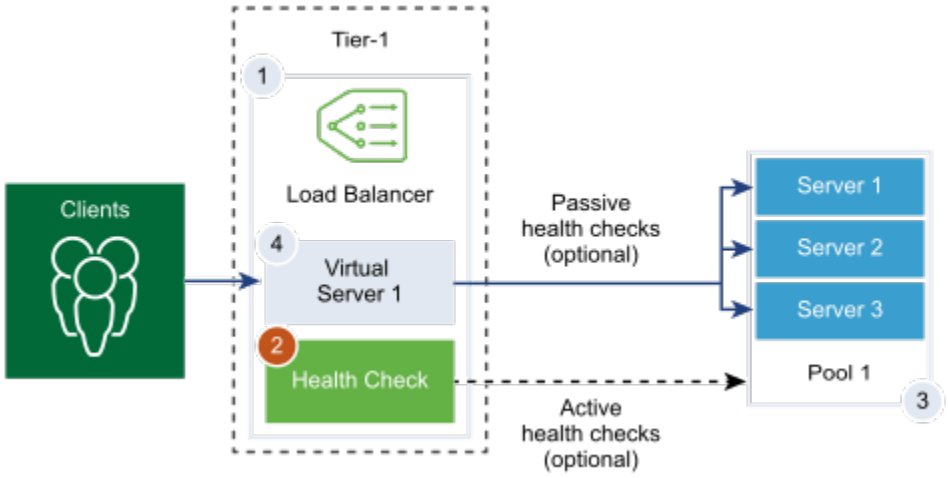
Add an Active Monitor

The active health monitor is used to test whether a server is available. The active health monitor uses several types of tests such as sending a basic ping to servers.

Servers that fail to respond within a certain time period or respond with errors are excluded from future connection handling until a subsequent periodic health check finds these servers to be healthy.

Active health checks are performed on server pool members after the pool member is attached to a virtual server and that virtual server is attached to a tier-1 gateway. The tier-1 uplink IP address is used for the health check.

> **Note:**
> More than one active health monitor can be configured per server pool.



1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Load Balancing** > **Monitors** > **Active** > **Add Active Monitor**.
3. Select a protocol for the server from the drop-down menu.
   You can also use predefined protocols: ICMP, TCP, and UDP for NSX Manager.
4. Select the **ICMP** protocol.
5. Complete step 5 and assign the data size in byte of the ICMP health check packet.
6. Select the **TCP** protocol.
7. Complete step 5 and you can leave the TCP data parameters empty.
   If both the data sent and expected are not listed, then a three-way handshake TCP connection is established to validate the server health. No data is sent.

   Expected data if listed has to be a string. Regular expressions are not supported.
8. Select the **UDP** protocol.
9. Complete step 5 and configure the UDP data.

Required Option	Description
UDP Data Sent	Enter the string to be sent to a server after a connection is established.
UDP Data Expected	Enter the string expected to receive from the server. Only when the received string matches this definition, is the server is considered as UP.

Associate the active health monitor with a server pool. See Add a Server Pool.

Add a Passive Monitor

Load balancers perform passive health checks to monitor failures during client connections and mark servers causing consistent failures as DOWN.

Passive health check monitors client traffic going through the load balancer for failures. For example, if a pool member sends a TCP Reset (RST) in response to a client connection, the load balancer detects that failure. If there are multiple consecutive failures, then the load balancer considers that server pool member to be temporarily unavailable and stops sending connection requests to that pool member for some time. After some time, the load balancer sends a connection request to verify that the pool member has recovered. If that connection is successful, then the pool member is considered healthy. Otherwise, the load balancer waits for some time and tries again.

Passive health check considers the following scenarios to be failures in the client traffic.

- For server pools associated with Layer 7 virtual servers, if the connection to the pool member fails. For example, if the pool member sends a TCP RST when the load balancer tries to connect or perform an SSL handshake between the load balancer and the pool member fails.
- For server pools associated with Layer 4 TCP virtual servers, if the pool member sends a TCP RST in response to client TCP SYN or does not respond at all.
- For server pools associated with Layer 4 UDP virtual servers, if a port is unreachable or a destination unreachable ICMP error message is received in response to a client UDP packet.

Server pools associated to Layer 7 virtual servers, the failed connection count is incremented when any TCP connection errors, for example, TCP RST failure to send data or SSL handshake failures occur.

Server pools associated to Layer 4 virtual servers, if no response is received to a TCP SYN sent to the server pool member or if a TCP RST is received in response to a TCP SYN, then the server pool member is considered as DOWN. The failed count is incremented.

For Layer 4 UDP virtual servers, if an ICMP error such as, port or destination unreachable message is received in response to the client traffic, then it is considered as DOWN.

Note:
One passive health monitor can be configured per server pool.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Load Balancing** > **Monitors** > **Passive** > **Add Passive Monitor**.
3. Enter a name and description for the passive health monitor.
4. Configure the values to monitor a service pool.
   You can also accept the default active health monitor values.

Option	Description
Fail Count	Set a value when the consecutive failures reach this value, the server is considered temporarily unavailable.
Timeout Period	Set the number of times the server is tested before it is considered as DOWN.
Tags	Enter tags to make searching easier. You can specify a tag to set a scope of the tag.

   For example, when the consecutive failures reach the configured value 5, that member is considered temporarily unavailable for 5 seconds. After this period, that member is tried again for a new connection to see if it is available. If that connection is successful, then the member is considered available and the failed count is set to zero. However, if that connection fails, then it is not used for another timeout interval of 5 seconds.

Associate the passive health monitor with a server pool. See Add a Server Pool.

Add a Server Pool

A server pool consists of one or more servers that are configured and running the same application. A single pool can be associated to both Layer 4 and Layer 7 virtual servers.

- If you use dynamic pool members, a NSGroup must be configured. See Add a Group.

- Verify that a passive health monitor is configured. See Add a Passive Monitor.

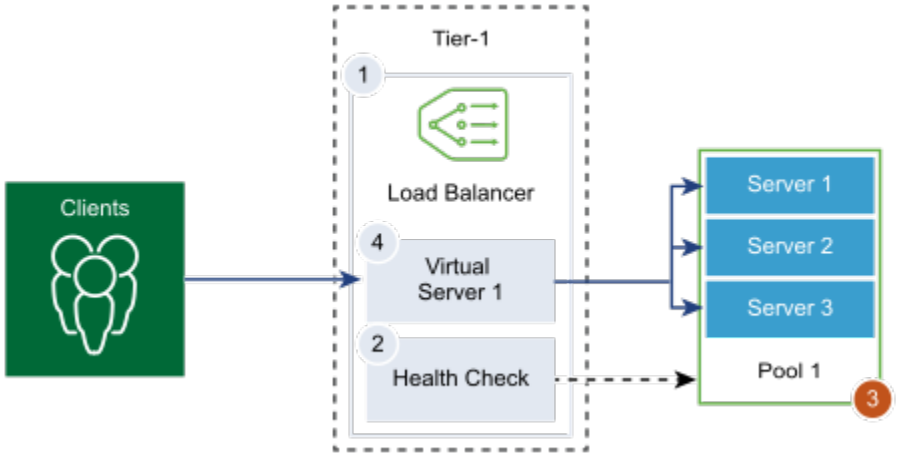
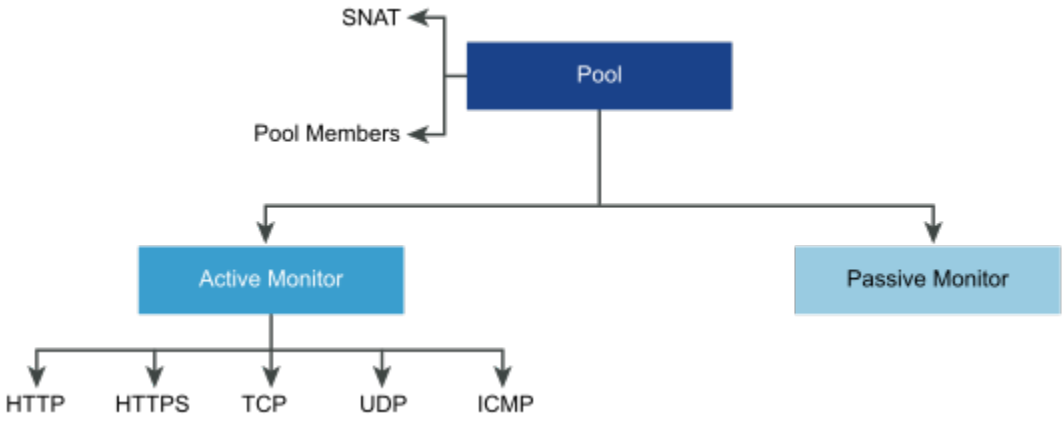


Figure 425: Server Pool Parameter Configuration



1. With admin privileges, log in to NSX Manager.
2. Select .
3. Enter a name and description for the load balancer server pool.
   You can optionally describe the connections managed by the server pool.
4. Select the algorithm balancing method for the server pool.
   Load balancing algorithm controls how the incoming connections are distributed among the members. The algorithm can be used on a server pool or a server directly.

   All load balancing algorithms skip servers that meet any of the following conditions:

   - Admin state is set to DISABLED.
   - Admin state is set to GRACEFUL_DISABLED and no matching persistence entry.
   - Active or passive health check state is DOWN.
   - Connection limit for the maximum server pool concurrent connections is reached.

Option	Description
ROUND_ROBIN	Incoming client requests are cycled through a list of available servers capable of handling the request. Ignores the server pool member weights even if they are configured.
WEIGHTED_ROUND_ROBIN	Each server is assigned a weight value that signifies how that server performs relative to other servers in the pool. The value

Option	Description
	determines how many client requests are sent to a server compared to other servers in the pool. This load balancing algorithm focuses on fairly distributing the load among the available server resources.
LEAST_CONNECTION	Distributes client requests to multiple servers based on the number of connections already on the server. New connections are sent to the server with the fewest connections. Ignores the server pool member weights even if they are configured.
WEIGHTED_LEAST_CONNECTION	Each server is assigned a weight value that signifies how that server performs relative to other servers in the pool. The value determines how many client requests are sent to a server compared to other servers in the pool. This load balancing algorithm focuses on using the weight value to distribute the load among the available server resources. By default, the weight value is 1 if the value is not configured and slow start is enabled.
IP-HASH	Selects a server based on a hash of the source IP address and the total weight of all the running servers.

5. Click **Select Members** and elect the server pool members.

   A server pool consists of single or multiple pool members.

Option	Description
Enter individual members	Enter a pool member name, IPv4 or IPv6 address, and a port. IP addresses can be either IPv4 or IPv6. Mixed addressing is not supported. Note that the pool members IP version must match the VIP IP version. For example, VIP-IPv4 with Pool-IPv4, and IPv6 with Pool-IPv6. Each server pool member can be configured with a weight for use in the load balancing algorithm. The weight indicates how much more or less load a given pool member can handle relative to other members in the same pool. You can set the server pool admin state. By default, the option is enable when a server pool member is added. If the option is disabled, active connections are processed, and the server pool member is not selected for new connections. New connections are assigned to other members of the pool. If gracefully disabled, it allows you to remove servers for maintenance. The existing connections to a member in the server pool in this state continue to be processed. Toggle the button to designate a pool member as a backup member to work with the health monitor to provide an Active-Standby state. Traffic failover occurs for backup members if active members fail a health check. Backup members are skipped during the server selection. When the server pool is inactive, the incoming connections are sent to only the backup members that are configured with a sorry page indicating an application is unavailable. **Max Concurrent Connection** value assigns a connection maximum so that the server pool members are not overloaded and skipped during server selection. If a value is not specified, then the connection is unlimited.
Select a group	Select a pre-configured group of server pool members.

Option	Description
	Enter a group name and an optional description. Set the compute member from existing list or create one. You can specify membership criteria, select members of the group, add IP addresses, and MAC addresses as group members, and add Active Directory groups. IP addresses can be either IPv4 or IPv6. Mixed addressing is not supported. The identity members intersect with the compute member to define membership of the group. Select a tag from the drop-down menu. You can optionally define the maximum group IP address list.

6. Click **Set Monitors** and select one or more active health check monitors for the server. Click **Apply**.

   The load balancer periodically sends an ICMP ping to the servers to verify health independent of data traffic. You can configure more than one active health check monitor per server pool.

7. Select the Source NAT (SNAT) translation mode.

   Depending on the topology, SNAT might be required so that the load balancer receives the traffic from the server destined to the client. SNAT can be enabled per server pool. If the client and pool member are in the same segment, SNAT must be enabled.

SNAT Translation Mode	Description
Automap Mode	Load Balancer uses the interface IP address and ephemeral port to continue the communication with a client initially connected to one of the server's established listening ports. The interface is the uplink or service link of service router which Load Balancer is attaching to. It may be a private IP address. Make sure that this SNAT IP is reachable from the pool member.
Deactivated	Deactivate SNAT translation mode.
IP Pool	Specify a single IPv4 or IPv6 address range. For example, 1.1.1.1-1.1.1.10 to be used for SNAT while connecting to any of the servers in the pool. IP addresses can be either IPv4 or IPv6. Mixed addressing is not supported. By default, the port range 4096 through 65535 is used for all configured SNAT IP addresses. The port range 1000 through 4095 is reserved for purposes such as health checks, and connections initiated from Linux applications. If multiple IP addresses are present, then they are selected in a Round Robin manner. If a virtual server IP port is in the SNAT default port range 4096 through 65535, make sure that the virtual server IP is **not** in the SNAT IP pool. Enable port overloading to allow the same SNAT IP and port to be used for multiple connections if the tuple (source IP, source port, destination IP, destination port, and IP protocol) is unique after the SNAT process is performed. You can also set the port overload factor to allow the maximum number of times a port can be used simultaneously for multiple connections.

8. Click **Additional Properties**, and toggle the button to enable TCP Multiplexing.

   With TCP multiplexing, you can use the same TCP connection between a load balancer and the server for sending multiple client requests from different client TCP connections.

9. Set the **Max Multiplexing Connections** per server that are kept alive to send future client requests.
10. Enter the **Min Active Members** the server pool must always maintain.
11. Select a passive health monitor for the server pool from the drop-down menu.
12. Select a tag from the drop-down menu.

Setting Up Virtual Server Components

You can set up the Layer 4 virtual servers and configure several virtual server components such as, application profiles, persistent profiles, and load balancer rules.

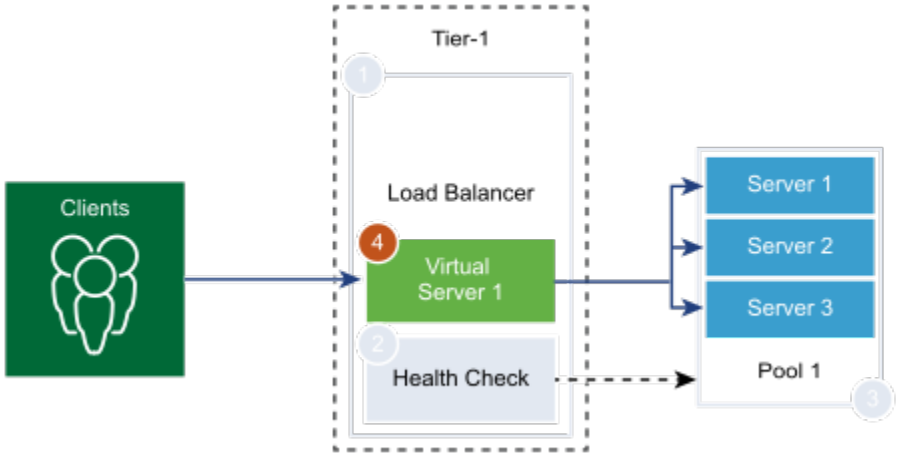
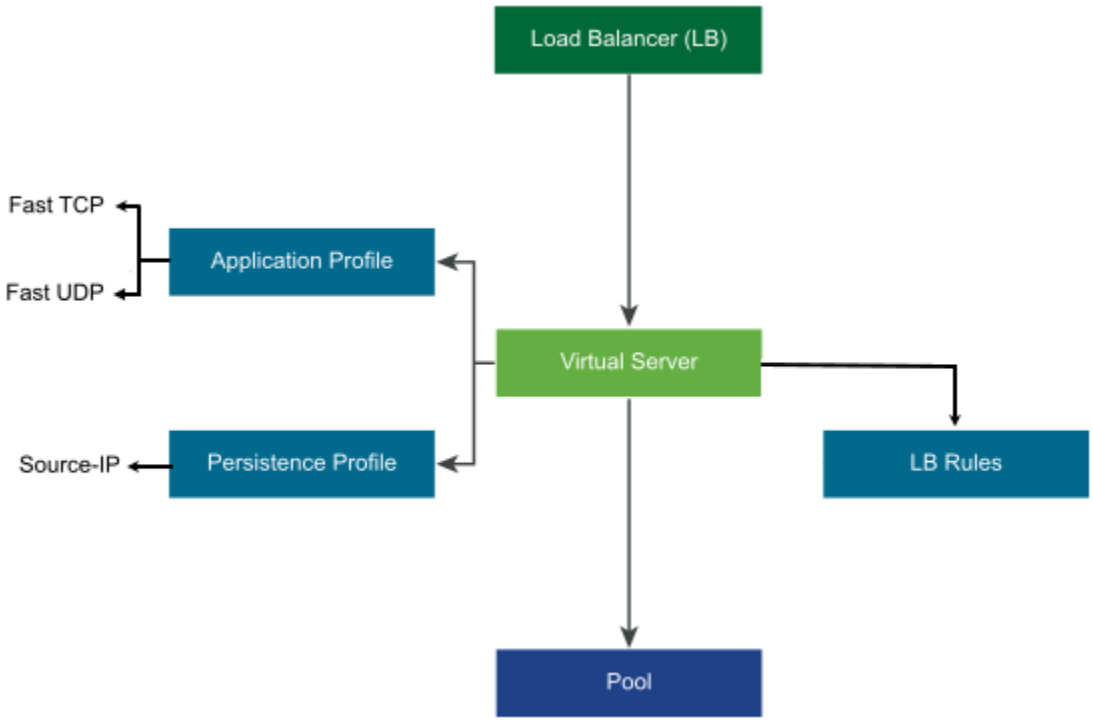


Figure 426: Virtual Server Components



Add a Persistence Profile

To ensure stability of stateful applications, load balancers implement persistence which directs all related connections to the same server. Different types of persistence are supported to address different types of application needs.

Some applications maintain the server state such as, shopping carts. Such state can be per client and identified by the client IP address or per HTTP session. Applications can access or modify this state while processing subsequent related connections from the same client or HTTP session.

The source IP persistence profile tracks sessions based on the source IP address. When a client requests a connection to a virtual server that enables the source address persistence, the load balancer checks if that client was previously connected, and if so, returns the client to the same server. If not, the load balancer selects the server pool member based on the pool load balancing algorithm. Source IP persistence profile is used by Layer 4 and Layer 7 virtual servers.

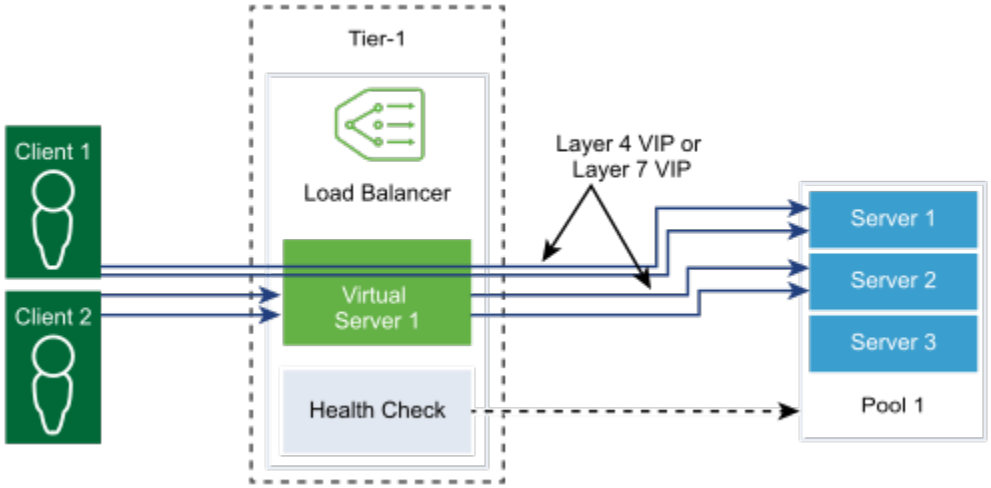
If rule persistence, cookie persistence, and server keep-alive are all configured, the load balancer follows the priority of rule persistence > cookie persistence > server keep-alive.

The Cookie persistence profile offers 3 modes:

- Cookie Insert - the load balance inserts its own cookie with the pool member information (encoded or not) in the server response to the client. The client then forwards the received cookies in subsequent requests (NSX cookie included), and the load balancer uses that information to provide the pool member persistence. The NSX cookie is trimmed from the client request when sent to the pool member.
- Cookie Prefix - the load balancer appends the pool member information (encoded or not) in the server response to the client. The client then forwards the received HTTP cookie in subsequent requests (with the NSX prepended information), and the load balancer uses that information to provide the pool member persistence. The NSX cookie prefix is trimmed from the client request when sent to the pool member.
- Cookie Rewrite - the load balancer replace server cookie value with the pool member information (encoded or not) in the server response to the client. The client then forwards the received HTTP cookie in subsequent requests (with the NSX prepended information), and the load balancer uses that information to provide the pool member persistence. The original server cookie is replaced in the client request when sent to the pool member.

Cookie persistence is available only on L7 virtual servers. Note that a blank space in a cookie name is **not** supported.

The generic persistence profile supports persistence based on the HTTP header, cookie, or URL in the HTTP request. Therefore, it supports application session persistence when the session ID is part of the URL. This profile is not associated with a virtual server directly. Specify this profile when you configure a load balancer rule for request forwarding and response rewrite.



1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Load Balancing** > **Profiles** > **Persistence** > **Add Persistence Profiles**.
3. Select **Source IP** to add a source IP persistence profile and enter the profile details.
   You can also accept the default Source IP profile settings.

Option	Description
Name and Description	Enter a name and a description for the Source IP persistence profile.
Share Persistence	Toggle the button to share the persistence so that all virtual servers this profile is associated with can share the persistence table. If the persistence sharing is not enabled in the Source IP persistence profile associated to a virtual server, each virtual server that the profile is associated to maintains a private persistence table.
Persistence Entry Timeout	Enter the persistence expiration time in seconds. The load balancer persistence table maintains entries to record that client requests are directed to the same server. The very first connection from new client IP is load balanced to a pool member based on the load balancing algorithm. NSX will store that persistence entry on the LB persistence-table which is viewable on the Edge Node hosting the T1-LB active via the CLI command:`get load-balancer <LB-UUID> persistence-tables`. • When there are connections from that client to the VIP, the persistence entry is kept. • When there are no more connections from that client to the VIP, the persistence entry begins the timer count down specified in the "Persistence Entry Timeout" value. If no new connection from that client to the VIP is made before the timer expires, the persistence entry for that client IP is deleted. If that client comes back after the entry is deleted, it will be load balanced again to a pool member based on the load balancing algorithm.
Purge Entries When Full	A large timeout value can lead to the persistence table quickly filling up when the traffic is heavy. When this option is enabled, the oldest entry is deleted to accept the newest entry. When this option is disabled, if the source IP persistence table is full, new client connections are rejected.
HA Persistence Mirroring	Toggle the button to synchronize persistence entries to the HA peer. When HA persistence mirroring is enabled, the client IP persistence remains in the case of load balancer failover.
Tags	Enter tags to make searching easier. You can specify a tag to set a scope of the tag.

4. Select a **Cookie** persistence profile, and enter the profile details. Cookie persistence is available only on L7 virtual servers. Note that a blank space in a cookie name is **not** supported.

Option	Description
Name and Description	Enter a name and a description for the Cookie persistence profile.
Share Persistence	Toggle the button to share persistence across multiple virtual servers that are associated to the same pool members. The Cookie persistence profile inserts a cookie with the format, *<name>.<profile-id>.<pool-id>*. If the persistence shared is not enabled in the Cookie persistence profile associated with a virtual server, the private Cookie persistence for each virtual server is used and is qualified by the pool member.

Option	Description
	The load balancer inserts a cookie with the format, *<name>.<virtual_server_id>.<pool_id>*.
Cookie Mode	Select a mode from the drop-down menu. • INSERT - Adds a unique cookie to identify the session. • PREFIX - Appends to the existing HTTP cookie information. • REWRITE - Rewrites the existing HTTP cookie information.
Cookie Name	Enter the cookie name. A blank space in a cookie name is **not** supported.
Cookie Domain	Enter the domain name. HTTP cookie domain can be configured only in the **INSERT** mode.
Cookie Fallback	Toggle the button so that the client request is rejected if cookie points to a server that is in a DISABLED or is in a DOWN state. Selects a new server to handle a client request if the cookie points to a server that is in a DISABLED or is in a DOWN state.
Cookie Path	Enter the cookie URL path. HTTP cookie path can be set only in the **INSERT** mode.
Cookie Garbling	Toggle the button to disable encryption. When garbling is disabled, the cookie server IP address and port information is in a plain text. Encrypt the cookie server IP address and port information.
Cookie Type	Select a cookie type from the drop-down menu. **Session Cookie** - Not stored. Will be lost when the browser is closed. **Persistence Cookie** - Stored by the browser. Not lost when the browser is closed.
HttpOnly Flag	When enabled, this option prevents a script running in the browser from accessing cookies. HttpOnly Flag is only available in the **INSERT** mode.
Secure Flag	When enabled, this option causes web browsers to send cookies over https only. Secure Flag is only available in the **INSERT** mode.
Max Idle Time	Enter the time in seconds that the cookie type can be idle before a cookie expires.
Max Cookie Age	For the session cookie type, enter the time in seconds a cookie is available.
Tags	Enter tags to make searching easier. You can specify a tag to set a scope of the tag.

5. Select **Generic** to add a generic persistence profile and enter the profile details.

Option	Description
Name and Description	Enter a name and a description for the Source IP persistence profile.
Share Persistence	Toggle the button to share the profile among virtual servers.
Persistence Entry Timeout	Enter the persistence expiration time in seconds. The load balancer persistence table maintains entries to record that client requests are directed to the same server.

Option	Description
	The very first connection from new client IP is load balanced to a pool member based on the load balancing algorithm. NSX will store that persistence entry on the LB persistence-table which is viewable on the Edge Node hosting the T1-LB active via the CLI command:`get load-balancer <LB-UUID> persistence-tables`. • When there are connections from that client to the VIP, the persistence entry is kept. • When there are no more connections from that client to the VIP, the persistence entry begins the timer count down specified in the "Persistence Entry Timeout" value. If no new connection from that client to the VIP is made before the timer expires, the persistence entry for that client IP is deleted. If that client comes back after the entry is deleted, it will be load balanced again to a pool member based on the load balancing algorithm.
HA Persistence Mirroring	Toggle the button to synchronize persistence entries to the HA peer.
Tags	Enter tags to make searching easier. You can specify a tag to set a scope of the tag.

Add Layer 4 Virtual Servers

Virtual servers receive all the client connections and distribute them among the servers. A virtual server has an IP address, a port, and a protocol. For Layer 4 virtual servers, lists of ports ranges can be specified instead of a single TCP or UDP port to support complex protocols with dynamic ports.

- Verify that application profiles are available.
- Verify that persistent profiles are available. See Add a Persistence Profile.
- Verify that SSL profiles for the client and server are available. See Add an SSL Profile.
- Verify that server pools are available. See Add a Server Pool.
- Verify that load balancer is available. See Add Load Balancers.

A Layer 4 virtual server must be associated to a primary server pool, also called a default pool.

If a virtual server status is disabled, any new connection attempts to the virtual server are rejected by sending either a TCP RST for the TCP connection or ICMP error message for UDP. New connections are rejected even if there are matching persistence entries for them. Active connections continue to be processed. If a virtual server is deleted or disassociated from a load balancer, then active connections to that virtual server fail.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Load Balancing** > **Virtual Servers** > **Add Virtual Server**.
3. Select a **L4 TCP** or a **L4 UDP** protocol and enter the protocol details.

   Layer 4 virtual servers support either the Fast TCP or Fast UDP protocol, but not both.

   For Fast TCP or Fast UDP protocol support on the same IP address and port, for example DNS, a virtual server must be created for each protocol.

L4 TCP Option	L4 TCP Description
Name and Description	Enter a name and a description for the Layer 4 virtual server.
IP Address	Enter the virtual server IP address. Both IPv4 and IPv6 addresses are supported. Note that the pool members IP version must match the VIP IP version. For example, VIP-IPv4 with Pool-IPv4, and IPv6 with Pool-IPv6.
Ports	Enter the virtual server port number.

L4 TCP Option	L4 TCP Description
Load Balancer	Select an existing load balancer to attach to this Layer 4 virtual server from the drop-down menu.
Server Pool	Select an existing server pool from the drop-down menu. The server pool consists of one or more servers, also called pool members that are similarly configured and running the same application. You can click the vertical ellipses to create a server pool.
Application Profile	Based on the protocol type, the existing application profile is automatically populated. Click the vertical ellipses to create an application profile.
Persistence	Select an existing persistence profile from the drop-down menu. Persistence profile can be enabled on a virtual server to allow Source IP related client connections to be sent to the same server.
Access List Control	When you enable Access List Control (ALC), all traffic flowing through the load balancer is compared with the ACL statement, which either drops or allows the traffic. ACL is disabled by default. To enable, click **Configure**, and select **Enabled**. Select an Action: • Allow - Allows connections matching the selected group. All other connections are dropped. • Drop - Allows connections not matching the selected group. A dropped connection generates a log entry is access log is enabled. Select a **Group**. The IP addresses included in this group are either dropped or allowed by the ACL.
Max Concurrent Connection	Set the maximum concurrent connection allowed to a virtual server so that the virtual server does not deplete resources of other applications hosted on the same load balancer.
Max New Connection Rate	Set the maximum new connection to a server pool member so that a virtual server does not deplete resources.
Sorry Server Pool	Select an existing sorry server pool from the drop-down menu. The sorry server pool serves the request when a load balancer cannot select a backend server to the serve the request from the default pool. You can click the vertical ellipses to create a server pool.
Default Pool Member Port	Enter a default pool member port if the pool member port for a virtual server is not defined. For example, if a virtual server is defined with a port range of 2000–2999 and the default pool member port range is set as 8000-8999, then an incoming client connection to the virtual server port 2500 is sent to a pool member with a destination port set to 8500.
Admin State	Toggle the button to disable the admin state of the Layer 4 virtual server.
Access Log	Toggle the button to enable logging for the Layer 4 virtual server.
Tags	Enter tags to make searching easier. You can specify a tag to set a scope of the tag.

L4 UDP Option	L4 UDP Description
Name and Description	Enter a name and a description for the Layer 4 virtual server.
IP Address	Enter the virtual server IP address. Both IPv4 and IPv6 addresses are supported. Note that the pool members IP version must match the VIP IP version. For example, VIP-IPv4 with Pool-IPv4, and IPv6 with Pool-IPv6.
Ports	Enter the virtual server port number.
Load Balancer	Select an existing load balancer to attach to this Layer 4 virtual server from the drop-down menu.
Server Pool	Select an existing server pool from the drop-down menu. The server pool consists of one or more servers, also called pool members that are similarly configured and running the same application. You can click the vertical ellipses to create a server pool.
Application Profile	Based on the protocol type, the existing application profile is automatically populated. You can click the vertical ellipses to create an application profile.
Persistence	Select an existing persistence profile from the drop-down menu. Persistence profile can be enabled on a virtual server to allow Source IP related client connections to be sent to the same server.
Max Concurrent Connection	Set the maximum concurrent connection allowed to a virtual server so that the virtual server does not deplete resources of other applications hosted on the same load balancer.
Access List Control	When you enable Access List Control (ALC) all traffic flowing through the load balancer will be compared with the ACL statement, which will either drop it or allow it. ACL is disabled by default. To enable, click **Configure**, and check **Enabled**. Select an Action: • Allow - Allows connections matching the selected group. All other connections are dropped • Drop - Allows connections not matching the selected group. A dropped connection generates a log entry is access log is enabled. Select a **Group**. The IP addresses included in this group are either dropped or allowed by the ACL.
Max New Connection Rate	Set the maximum new connection to a server pool member so that a virtual server does not deplete resources.
Sorry Server Pool	Select an existing sorry server pool from the drop-down menu. The sorry server pool serves the request when a load balancer cannot select a backend server to the serve the request from the default pool. You can click the vertical ellipses to create a server pool.
Default Pool Member Port	Enter a default pool member port if the pool member port for a virtual server is not defined. For example, if a virtual server is defined with port range 2000–2999 and the default pool member port range is set as 8000-8999, then an incoming client connection to the virtual

L4 UDP Option	L4 UDP Description
	server port 2500 is sent to a pool member with a destination port set to 8500.
Admin State	Toggle the button to disable the admin state of the Layer 4 virtual server.
Access Log	Toggle the button to enable logging for the Layer 4 virtual server.
Log Significant Event Only	This field can only be configured if access logs are enabled. Connections that cannot be sent to a pool member are treated as a significant event such as "max connection limit," or "Access Control drop."
Tags	Enter tags to make searching easier. You can specify a tag to set a scope of the tag.

Add an SSL Profile

SSL profiles configure application-independent SSL properties such as, cipher lists and reuse these lists across multiple applications. SSL properties are different when the load balancer is acting as a client and as a server, as a result separate SSL profiles for client-side and server-side are supported.

> **Note:**
> SSL profile is not supported in the NSX limited export release.

Client-side SSL profile refers to the load balancer acting as an SSL server and stopping the client SSL connection. Server-side SSL profile refers to the load balancer acting as a client and establishing a connection to the server.

You can specify a cipher list on both the client-side and server-side SSL profiles. NSX Managers come with the following default SSL Profiles:

- default-balanced-client-ssl-profile
- default-balanced-server-ssl-profile
- default-high-compatibility-client-ssl-profile
- default-high-compatibility-server-ssl-profile
- default-high-security-client-ssl-profile,
- default-high-security-server-ssl-profile.

The "balanced" SSL Profile supports a mix of SSL protocols and ciphers to offer a perfect mix of performance and security to clients/servers. The "high-compatibility" SSL Profile supports a large range of SSL protocols and ciphers to offer access to the widest range of clients/servers. The "high-security" SSL Profile supports the highest-secured SSL protocols and ciphers to offer the most secured access to clients/servers. Additionally, custom SSL profiles can be created.

The cipher TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 is also added to the balanced and high compatibility client/server profiles to support the communication between the NSX Manager and a load balancer virtual server for vIDM. With this new addition, you no longer need to create a custom profile and add the cipher TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 as a workaround for supporting the communication between the NSX Manager and a vIDM LB VIP.

SSL session caching allows the SSL client and server to reuse previously negotiated security parameters avoiding the expensive public key operation during the SSL handshake. SSL session caching is disabled by default on both the client-side and server-side.

SSL session tickets are an alternate mechanism that allows the SSL client and server to reuse previously negotiated session parameters. In SSL session tickets, the client and server negotiate whether they support SSL session tickets during the handshake exchange. If supported by both, server can send an SSL ticket, which includes encrypted SSL session parameters to the client. The client can use that ticket in subsequent connections to reuse the session. SSL session tickets are enabled on the client-side and disabled on the server-side.

Figure 427: SSL Offloading

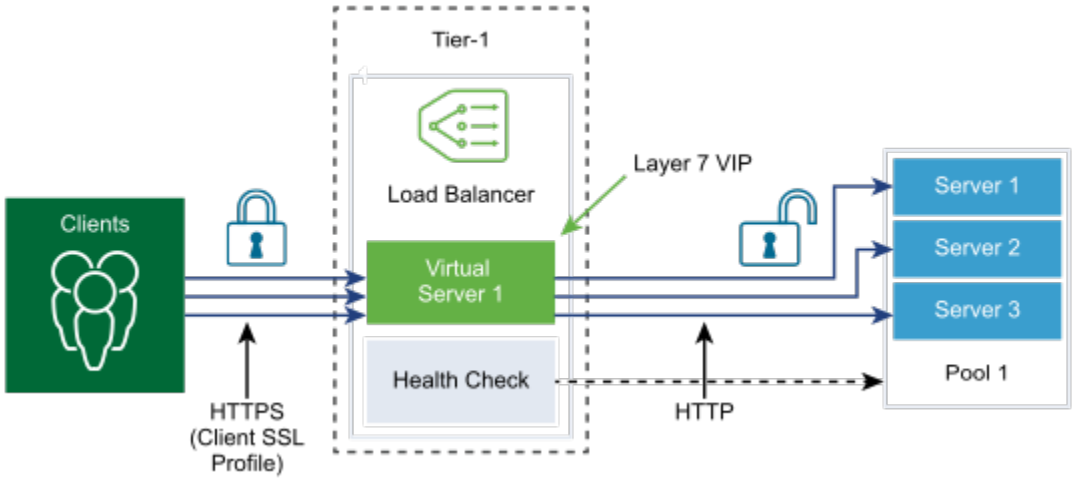
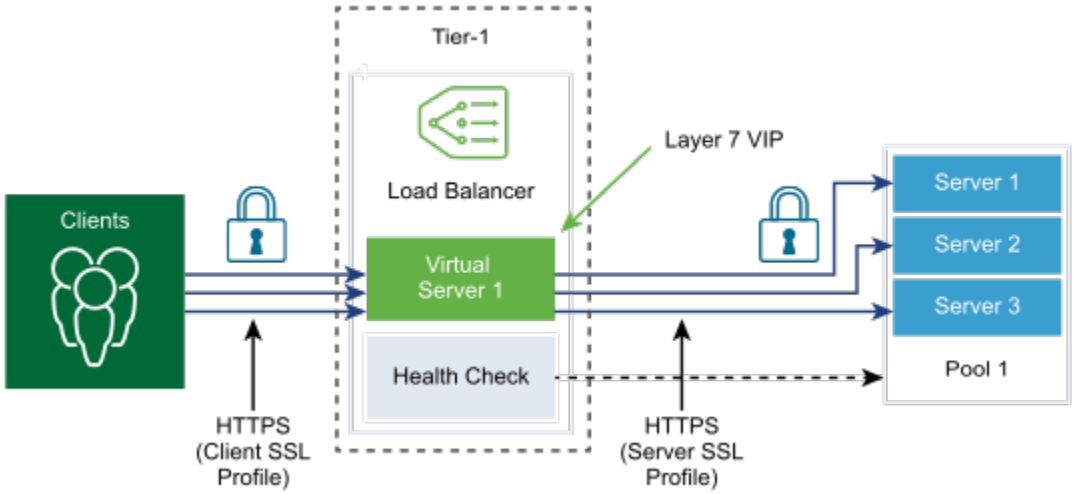


Figure 428: End-to-End SSL



1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Load Balancing** > **Profiles** > **SSL Profile**.
3. Select a **Client SSL Profile** and enter the profile details.

Option	Description
Name and Description	Enter a name and a description for the Client SSL profile.
SSL Suite	Select the SSL Cipher group from the drop-down menu and available SSL Ciphers and SSL protocols to be included in the Client SSL profile are populated. Balanced SSL Cipher group is the default.
Session Caching	Toggle the button to allow the SSL client and server to reuse previously negotiated security parameters avoiding the expensive public key operation during an SSL handshake.
Tags	Enter tags to make searching easier. You can specify a tag to set a scope of the tag.
Supported SSL Ciphers	Depending on the SSL suite, you assigned the supported SSL Ciphers are populated here. Click **View More** to view the entire list.

Option	Description
	If you selected **Custom**, you must select the SSL Ciphers from the drop-down menu.
Supported SSL Protocols	Depending on the SSL suite, you assigned the supported SSL protocols are populated here. Click **View More** to view the entire list. If you selected **Custom**, you must select the SSL Ciphers from the drop-down menu.
Session Cache Entry Timeout	Enter the cache timeout in seconds to specify how long the SSL session parameters must be kept and can be reused.
Prefer Server Cipher	Toggle the button so that the server can select the first supported cipher from the list it can support. During an SSL handshake, the client sends an ordered list of supported ciphers to the server.

4. Select a **Server SSL Profile** and enter the profile details.

Option	Description
Name and Description	Enter a name and a description for the Server SSL profile.
SSL Suite	Select the SSL Cipher group from the drop-down menu and available SSL Ciphers and SSL protocols to be included in the Server SSL profile are populated. Balanced SSL Cipher group is the default.
Session Caching	Toggle the button to allow the SSL client and server to reuse previously negotiated security parameters avoiding the expensive public key operation during an SSL handshake.
Tags	Enter tags to make searching easier. You can specify a tag to set a scope of the tag.
Supported SSL Ciphers	Depending on the SSL suite, you assigned the supported SSL Ciphers are populated here. Click **View More** to view the entire list. If you selected **Custom**, you must select the SSL Ciphers from the drop-down menu.
Supported SSL Protocols	Depending on the SSL suite, you assigned the supported SSL protocols are populated here. Click **View More** to view the entire list. If you selected **Custom**, you must select the SSL Ciphers from the drop-down menu.
Session Cache Entry Timeout	Enter the cache timeout in seconds to specify how long the SSL session parameters must be kept and can be reused.
Prefer Server Cipher	Toggle the button so that the server can select the first supported cipher from the list it can support. During an SSL handshake, the client sends an ordered list of supported ciphers to the server.

Groups Created for Server Pools and Virtual Servers

NSX Manager automatically creates groups for load balancer server pools and VIP ports.

Load Balancer created groups are visible under **Inventory** > **Groups**.

Server pool groups are created with the name NLB.PoolLB.*Pool_NameLB_Name* with group member IP addresses assigned:

- Pool configured with no LB-SNAT (transparent): 0.0.0.0/0
- Pool configured with LB-SNAT Automap: T1-Uplink IP 100.64.x.y and T1-ServiceInterface IP
- Pool configured with LB-SNAT IP-Pool: LB-SNAT IP-Pool

VIP Groups are created with the name NLB.VIP.*virtual server name* and the VIP group member IP addresses are VIP IP@.

For server pool groups, you can create an allow traffic distributed firewall rule from the load balancer (NLB.PoolLB.*Pool_NameLB_Name*). For Tier-1 gateway firewall, you can create an allow traffic from clients to LB VIP NLB.VIP.*virtual server name*.

Distributed Load Balancer

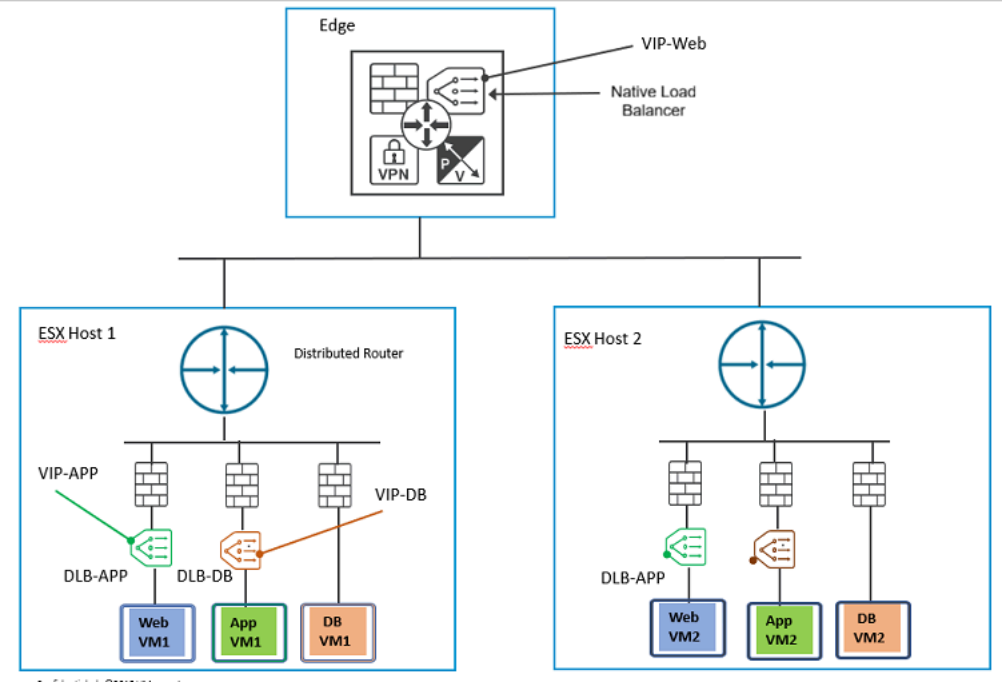
A Distributed Load Balancer configured in NSX can help you effectively load balance East-West traffic and scale traffic because it runs on each ESX host.

> **Important:**
> Distributed Load Balancer is supported only for Kubernetes (K8s) cluster IPs managed by vSphere with Kubernetes. Distributed Load Balancer is not supported for any other workload types. As an administrator, you cannot use NSX Manager GUI to create or modify Distributed Load Balancer objects. These objects are pushed by vCenter through NSX API when K8 cluster IPs are created in vCenter.

A distributed load balancer is realized on each hypervisor where load balancing workloads, such as clients and servers, are deployed, ensuring traffic is load balanced on each hypervisor in a distributed way.

A distributed load balancer can be configured on the NSX network along with a native load balancer.

> **Note:**
>
> Starting with VCF 9.1, Distributed Load Balancer functions independently of VMware vDefend Firewall. You can use Distributed Load Balancer without having to enable vDefend Firewall.



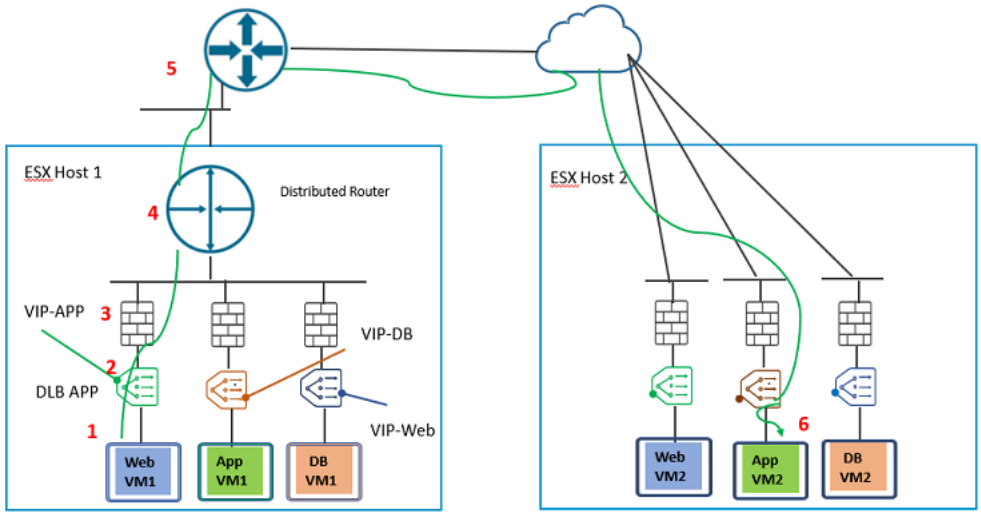
In the diagram, an instance of the Distributed Load Balancer is attached to a VM group. As the VMs are downlinks to the distributed logical router, Distributed Load Balancer only load balances east-west traffic.

To cater load balancing requirements of each component or module of an application, a distributed load balancer can be attached to each tier of an application. For example, to serve a user request, a frontend of the application needs to reach out to the middle module to get data. However, the middle layer might not be deployed to serve the final data to the

user, so it needs to reach out the backend layer to get additional data. For a complex application, many modules might need to interact with each other to get information. Along with complexity, when the number of user request increase exponentially, a distributed load balancer can efficiently meet the user needs without taking a performance hit. Configuring a Distributed Load Balancer on every host achieves issues of scale and packet transmission efficiency.

Understanding Traffic Flow with a Distributed Load Balancer

Understand how traffic flows between VMs that are connected to an instance of a distributed load balancer (DLB).



As an administrator, ensure:

- Virtual IP addresses and pool members connected to a DLB instance must have unique IP address for traffic to be routed correctly.

Traffic flow between Web VM1 and APP VM2.

1. When Web VM1 sends out a packet to APP VM2 it is received by the VIP-APP.
   The DLB APP is attached to the policy group consisting of Web tier VMs. Similarly, DLB-APP hosting VIP-DB must be attached to the policy group consisting of App tier VMs.
2. The VIP-APP hosted on DLB APP receives the request from Web VM1.
3. Before reaching the destination VM group, the packet is filtered by distributed firewall rules.
4. After the packets are filtered based on the firewall rules, it is sent to the Tier-1 router.
5. It is further routed to the the physical router.
6. The route is completed when the packet is delivered to the destination App VM2 group.

As DLB VIPs can only be accessed from VMs connected to downlinks of Tier-0 or Tier-1 logical routers, DLB provides load balancing services to east-west traffic.

A DLB instance can co-exist with an instance of VMware vDefend Distributed Firewall. With DLB and vDefend Distributed Firewall enabled on a virtual interface of a hypervisor, first the traffic is load balanced based on the configuration in DLB and then vDefend Distributed Firewall rules are applied on traffic flowing from a VM to the hypervisor. DLB rules are applied on traffic originating from downlinks of a Tier-0 or Tier-1 logical routers going to the destination hypervisor. DLB rules cannot be applied on traffic flowing in the reverse direction - originating from outside the host going to a destination VM.

For example, if the DLB instance is load balancing traffic from Web-VMs to App-VMs, then to allow such traffic to pass through vDefend Distributed Firewall, ensure that the vDefend Distributed Firewall rule is set to value "Source=Web-VMs, Destination=App-VMs, Action=Allow".

Create and Attach a Distributed Load Balancer Instance

Unlike NSX Native Load Balancer, a Distributed Load Balancer (DLB) instance is attached to virtual interfaces of a VM group.

- Add a policy group consisting of VMs. For example, such a VM group can be related to the App tier that receives requests from a VM on the Web-tier.

At the end of the procedure a DLB instance is attached to the virtual interfaces of a VM group. It is only possible to create and attach a DLB instance through API commands.

Run `Put /policy/api/v1/infra/lb-services/<mydlb>`.

{

"connectivity_path" : "/infra/domains/default/groups/<clientVMGroup>",

"enabled" : true,

"size" : "DLB",

"error_log_level" : "INFO",

"access_log_enabled" : false,

"resource_type" : "LBService",

"display_name" : "mydlb"

}

Where,

- `connectivity_path`: If the connectivity path is set to `Null` or `Empty`, the DLB instance is not applied to any transport nodes.
- `size`: Set to value `DLB`. As each application or virtual interface gets an instance of DLB, there is just a single size form factor of the DLB instance.
- `enabled`: By default, the created DLB instance is enabled. You cannot disable the DLB instance.
- `error_log_level`: The only supported level is `INFO`.

A DLB instance is created and attached to the VM group. The DLB instance created on the Web-tier is attached to all the virtual interfaces of the Web-tier VM group.

After creating a DLB instance, log in to the NSX Manager, go to **Networking -> Load Balancing -> Load Balancers** to monitor the DLB instance and view its configuration.

To edit the DLB configuration, use the NSX API.

Next, Create a Server Pool for Distributed Load Balancer.

Create a Server Pool for Distributed Load Balancer

Create a load balancer pool to include virtual machines that consume DLB services.

- Create a VM group that consumes DLB service.
- Create and attach a DLB instance to a VM group.

This task can be done both from the NSX UI and NSX API.

The API command to create a DLB pool is `PUT https://<NSXManager_IPaddress>/policy/api/v1/infra/lb-pools/<lb-pool-id>`

1. With admin privileges, log in to NSX Manager.
2. Go to **Networking** > **Load Balancing** > **Server Pools**.
3. Click **Add Server Pool**.
4. Enter values in these fields.

Field	Description
Name	Enter name of the DLB pool.
Algorithm	**Weighted Round Robin**, **Round Robin**, **Weighted Least Connection**, **Least Connection** and **IP Hash** are the supported algorithms. Since Distributed Load Balancer runs locally on each ESX server, these algorithms are local to each ESX server. There is no synchronization of load balancing connection information between different ESX servers of a cluster. • **Weighted Round Robin**: Use this algorithm to send connections to pool members based on the weights assigned to each pool member. For example, if you assign pool member A with weight 3, pool member B with weight 2 and pool member C with weight 1, then out of a total of 6 client connections, pool member A receives 3 connections, pool member B receives 2 connections and pool member C receives 1 connection. • **Round Robin**: Use this algorithm to send equal number of connections to each pool member. • **Least Connection**: Use this algorithm so that a pool member with the least number of active connections. Each pool member is configured to a slow start (Slow Start is set to True). When it receives connections, the status of the pool member is set to Slow Start is False. • **Weighted Least Connection**: Use this algorithm, to send connections to pool members based on the weights assigned to each pool member. • **IP Hash**: Use this algorithm to send connections based on the hash of IP addresses. **Note:** Do not use **IP Hash** if you want to persist connections to the same pool member even after the number of pool members change.
Members/Group	Click **Select Members** and on the **Configure Server Pool Members** window, do one of the following: • Select **Enter individual members**. To add a new member, click **Add Member** and enter values in the mandatory fields. • Select **Select a group** and **Add Group** or select an existing group. To add a new group, enter values in these fields. – **Name** – **Compute Members**: Click **Set Members** to add a group that includes all the pool members. – **IP Revision Filter**: Both IPv4 and IPv6 are supported. – **Port**: Default port for all the dynamic pool members.
SNAT Translation Mode	Set this field to **Disabled** state. SNAT translation is not supported in a Distributed Load Balancer.

5. Click **Save**.

Server pool members are added for the Distributed Load Balancer.
See Create a Virtual Server with a Fast TCP or UDP Profile.

Create a Virtual Server with a Fast TCP or UDP Profile

Create a virtual server and bind it to a Distributed Load Balancer service.

- Create a server pool for the Distributed Load Balancer.
- To use IPv6 addresses as the virtual IP of Distributed Load Balancer, on the Global Networking Config page (**Networking** > **Global Networking Config**), ensure **L3 Forwarding Mode** to **IPv4 and IPv6**.

This task can be performed both from the NSX UI and NSX APIs.

The API command to create a virtual server is `PUT https://<NSXManager_IPAddress>/policy/api/v1/infra/lb-virtual-servers/<lb-virtual-server-id>`.

1. With admin privileges, log in to NSX Manager.
2. Go to **Networking** > **Load Balancing** > **Virtual Servers**.
3. Click **Add Virtual Server -> L4 TCP**.
4. To configure a virtual server for a Distributed Load Balancer, only the following fields are supported.

Field	Description
Name	Enter a name for the virtual server.
IP Address	Supports both IPv4 and IPv6 addresses. Enter the IP address of the Distributed Load Balancer virtual server. All client connections arrive at this IP address of the Distributed Load Balancer virtual server.
Ports	Virtual server port number. Multiple ports or port ranges are not supported in the virtual server of a Distributed Load Balancer.
Load Balancer	Attach the Distributed Load Balancer instance that is associated to the virtual server. The virtual server then knows which policy group the load balancer is servicing.
Server Pool	Select the server pool. The server pool contains backend servers. Server pool consists of one or more servers that are similarly configured and are running the same application. It is also referred to as pool members. **Note:** If the virtual IP address of the Distributed Load Balancer is IPv4, the server pool members must be of the same versions. Likewise if you use IPv6 version of virtual IP address.
Application Profile	Select the application profile for the virtual server. The application profile defines the application protocol characteristics. It is used to influence how load balancing is performed. The supported application profiles are: • Load Balancer Fast TCP Profile **Note:** The close timeout setting for the Fast TCP Profile is not supported. The setting does not take effect if you configure it. • Load Balancer Fast UDP Profile

The Distributed Load Balancer configuration is complete.

Verify whether the DLB is distributing traffic to all the servers in the pool based on the algorithm defined in the configuration. If you choose the Round_Robin algorithm, then DLB must be able to choose servers from the pool in a round robin fashion.

In the ESX host, verify whether the DLB configuration is complete.

See Verifying Distributed Load Balancer Configuration on ESX Hosts.

Verifying Distributed Load Balancer Configuration on ESX Hosts

Verify whether the Distributed Load Balancer was configured completely on ESX hosts.

After you securely connect to the ESX host, run `/opt/vmware/nsx-nestdb/bin/nestdb-cli`. From the `nestdb-cli` prompt, run the following commands.

Command	Sample Response
To view the configured DLB service, run `get LbServiceMsg`.	`{'id': {'left': 13946864992859343551, 'right': 10845263561610880178}, 'virtual_server_id': [{'left': 13384746951958284821, 'right': 11316502527836868364}], 'display_name': 'mydlb', 'size': 'DLB', 'enabled': True, 'access_log_enabled': False, 'log_level': 'LB_LOG_LEVEL_INFO', 'applied_to': {'type': 'CONTAINER', 'attachment_id': {'left': 2826732686997341216, 'right': 10792930437485655035}}}`
To view the virtual server configured for DLB, run `get LbVirtualServerMsg`.	`{'port': '80', 'revision': 0, 'display_name': 'mytcpvip', 'pool_id': {'left': 4370937730160476541, 'right': 13181758910457427118}, 'enabled': True, 'access_log_enabled': False, 'id': {'left': 13384746951958284821, 'right': 11316502527836868364}, 'ip_protocol': 'TCP', 'ip_address': {'ipv4': 2071690107}, 'application_profile_id': {'left': 152703408922455365 7, 'right': 10785436903467108397}}`
To view configuration of the DLB pool members, run `get LbPoolMsg`.	`{'tcp_multiplexing_number': 6, 'display_name': 'mylbpool', 'tcp_multiplexing_enabled': False, 'member': [{'port': '80', 'weight': 1, 'display_name': 'Member_VM30', 'admin_state': 'ENABLED', 'ip_address': {'ipv4': 3232261280}, 'backup_member': False}, {'port': '80', 'weight': 1, 'display_name': 'Member_VM31', 'admin_state': 'ENABLED', 'ip_address': {'ipv4': 3232261281}, 'backup_member': False}, {'port': '80', 'weight': 1, 'display_name': 'Member_VM32', 'admin_state': 'ENABLED', 'ip_address': {'ipv4': 3232261282}, 'backup_member': False}], 'id': {'left': 4370937730160476541, 'right': 13181758910457427118}, 'min_active_members': 1, 'algorithm': 'ROUND_ROBIN'}`

Command	Sample Response
To view NSX controller configuration pushed to the ESX host, run `get ContainerMsg`.	`{'container_type': 'CONTAINER', 'id': {'left': 2826732686997341216, 'right': 10792930437485655035}, 'vif': ['cd2e482b-2998-480f-beba-65fbd7ab1e62', 'f8aa2a58-5662-4c6b-8090-d1bd19174205', '83a1f709-e675-4e42-b677-ff501fd0f4ec', 'b8366b39-4c81-41fc-b89e-de7716462b2f'], 'name': 'default.clientVMGroup', 'mac_address': [{'mac': 52237218275}, {'mac': 52243694681}, {'mac': 52233233291}, {'mac': 52239463383}], 'ip_address': [{'ipv4': 16844388}, {'ipv4': 16844644}, {'ipv4': 16844132}, {'ipv4': 3232261283}, {'ipv4': 16844298}, {'ipv4': 16844554}, {'ipv4': 16844042}]}`
To view application profile configuration on the ESX host, run `get LbApplicationProfileMsg`.	`{'display_name': 'default-tcp-lb-app-profile', 'id': {'left': 1527034089224553657, 'right': 10785436903467108397}, 'application_type': 'FAST_TCP', 'fast_tcp_profile': {'close_timeout': 8, 'flow_mirroring_enabled': False, 'idle_timeout': 1800}}`

Distributed Load Balancer Statistics and Diagnostics

NSX API and CLI commands to monitor statistics for Distributed Load Balancer instances.

CLI Commands for Distributed Load Balancer

Action	Command
Display all load balancers.	`get load-balancers`
Display a specific load balancer.	`get load-balancer <UUID_LoadBalancer>`
Show statistics of all pools of the specified load balancer	`get load-balancer <UUID_LoadBalancer> pools stats`
Show load balancer virtual-server configuration.	`get load-balancer <UUID_LoadBalancer> virtual-servers`
Show statistics of the specified load balancer and pool	`get load-balancer <UUID_LoadBalancer> pool <UUID_Pool> stats`
Show load balancer pools configuration	`get load-balancer <UUID_LoadBalancer> pools`
Show statistics of all virtual servers of the specified load balancer	`get load-balancer <UUID_LoadBalancer> virtual-servers stats`
Show statistics of the specified load balancer and virtual server	`get load-balancer <UUID_LoadBalancer> virtual-server <UUID_VirtualSerever> stat`
Clear statistics of the specified load balancer and pool	`clear load-balancer <UUID_LoadBalancer> pool <UUID_Pool> stats`
Clear statistics of all pools of the specified load balancer	`clear load-balancer <UUID_LoadBalancer> pools stats`
Clear statistics of the specified load balancer	`clear load-balancer <UUID_LoadBalancer> stats`
Clear statistics of the specified load balancer and virtual server	`clear load-balancer <UUID_LoadBalancer> virtual-server <UUID_VirtualServer> stats`
Clear statistics of all virtual servers of the specified load balancer	`clear load-balancer <UUID_LoadBalancer> virtual-servers stats`
Display distributed load balancer statistics	`get load-balancer <UUID_LoadBalancer> stats`

API commands for Distributed Load Balancer

Action	Command
Get Distributed Load Balancer statistics **Note:** If you do not specify an enforcement point path, API fetches information from all enforcement paths and displays aggregated information. If you specify an enforcement point path, only information for that path is retrieved and displayed in the response body of the API call.	`GET /policy/api/v1/infra/lb-services/LB_Service/statistics?source=realtime&enforcement_point_path=/infra/sites/default/enforcement-points/default`
Get Distributed Load Balancer virtual server statistics	`GET /infra/lb-services/<lb-service-id>/lb-virtual-servers/<lb-virtual-server-id>/statistics?source=realtime&enforcement_point_path=/infra/sites/default/enforcement-points/default`
Get Distributed Load Balancer pool statistics	`GET /infra/lb-services/<lb-service-id>/lb-pools/<lb-pool-id>/statistics?source=realtime&enforcement_point_path=/infra/sites/default/enforcement-points/default`

Distributed Load Balancer Operational Status

Know the operational status of the distributed load balancer service in NSX Manager UI and on ESX hosts.

As distributed load balancer service scales linearly as the number of hosts increases ESX, a single distributed load balancer service can support several ESX hosts. In turn, each ESX host can support multiple virtual interfaces (VIFs), across many ESX hosts. The consolidated status of distributed load balancer at NSX Manager level is calculated using the consolidated status of all the associated ESX hosts. The consolidated status of distributed load balancer at ESX host level is calculated using the individual status of all associated VIFs on that ESX host.

Status at NSX Manager

Status	Description
`Up`	Status is `Up` when all the related transport nodes return status for the distributed load balancer service as `ready`.
`Degraded`	Status is `Degraded` when all the following conditions are true: • At least one transport node returns status for the distributed load balancer service as `ready` or `partially ready` • Not all the related transport nodes return status for the load balancer service as `ready`.
`Down`	Status is `Down` when one of the following conditions is true: • All the related transport nodes return `not ready`. • At least one transport node returns `not ready` and no transport node returns `ready`.
`Unknown`	Status is `Unknown` when all the related transport nodes return status for the distributed load balancer service as `Unknown`.
`Disabled`	Status is `Disabled` when the distributed load balancer service is enabled but the connectivity path is not specified.

Status at ESX Host

Status	Description
`ready`	The consolidated status for the distributed load balancer service on the ESX Host is `ready` when the status of all associated VIFs on this ESX Host are `ready`. **Note:** • `ready` status on VIF means that the distributed load balancer instance is the oldest and applied.
`not ready`	The consolidated status for the distributed load balancer service on the ESX Host is `not ready` when no associated VIF is `ready`.
`partially ready`	The consolidated status for the distributed load balancer service on the ESX Host is `partially ready` when both of the following conditions are true: • At least one associated VIF is `ready`. • At least one associated VIF is `not ready` or `conflict`. **Note:** • `not ready` status on VIF means that the distributed load balancer service instance is the oldest, should be applied, but not applied. • `conflict` status on VIF means that the distributed load balancer service instance is not the oldest and not applied.

Detailed Status Through API

Run the following API to get detailed status of distributed load balancer instance running at a transport node.

```
GET https://<manager IP>/policy/api/v1/infra/lb-services/<DLBname>/detailed-status?
source=realtime&include_instance_details=true&transport_node_ids=node1_uuid
```

Sampled response:

```
{
  "results":
    {
      "service_path": "/infra/lb-services/mydlb",
      "service_status": "UP",
      "virtual_servers": [
        {
          "virtual_server_path": "/infra/lb-virtual-servers/mytcpvip",
          "status": "UP",
          "last_update_timestamp": 1591344963509,
          "resource_type": "LBVirtualServerStatus"
        }
      ],
      "pools": [
        {
          "pool_path": "/infra/lb-pools/mylbpool",
          "status": "UP",
          "last_update_timestamp": 1591344963509,
          "resource_type": "LBPoolStatus"
        }
      ],
      "last_update_timestamp": 1591344963509,
```

```
      "instance_detail_per_tn": [
        {
          "transport_node_id": "b09b7b6c-a60d-11ea-835e-d95476fe6438",
          "instance_detail_per_status": [
            {
              "status": "READY",
              "instance_number": 3,
              "instance_details": [
                {
                  "attachment_display_name": "12-vm_Client_VM_Ubuntu_1404-local-1762/12-
vm_Client_VM_Ubuntu_1404-local-1762.vm@b09b7b6c-a60d-11ea-835e-d95476fe6438"
                },
                {
                  "attachment_display_name": "10-vm_Client_VM_Ubuntu_1404-local-1762/10-
vm_Client_VM_Ubuntu_1404-local-1762.vm@b09b7b6c-a60d-11ea-835e-d95476fe6438"
                },
                {
                  "attachment_display_name": "11-vm_Client_VM_Ubuntu_1404-local-1762/11-
vm_Client_VM_Ubuntu_1404-local-1762.vm@b09b7b6c-a60d-11ea-835e-d95476fe6438"
                }
              ]
            },
            {
              "status": "NOT_READY",
              "instance_number": 0
            },
            {
              "status": "CONFLICT",
              "instance_number": 0
            }
          ]
        }
      ],
      "enforcement_point_path": "/infra/sites/default/enforcement-points/default",
      "resource_type": "LBServiceStatus"
    }
  ],
  "intent_path": "/infra/lb-services/mydlb"
}
```


Status Through CLI

Run the following CLI command to get status of the distributed load balancer.

```
get load-balancer <UUID_LoadBalancer> status
 Load Balancer
 UUID : 8721fb3e-dbef-4d9a-8f48-432e893883f1
 Display-Name : DLB_Service21
 Status : ready
 Ready LSP Count : 4
 Not Ready LSP Count: 0
 Conflict LSP Count : 0
 Ready LSP          : 95cf00d0-d203-46d0-8215-6cd95ec51757
 Not Ready LSP      :
 Conflict LSP       :
```

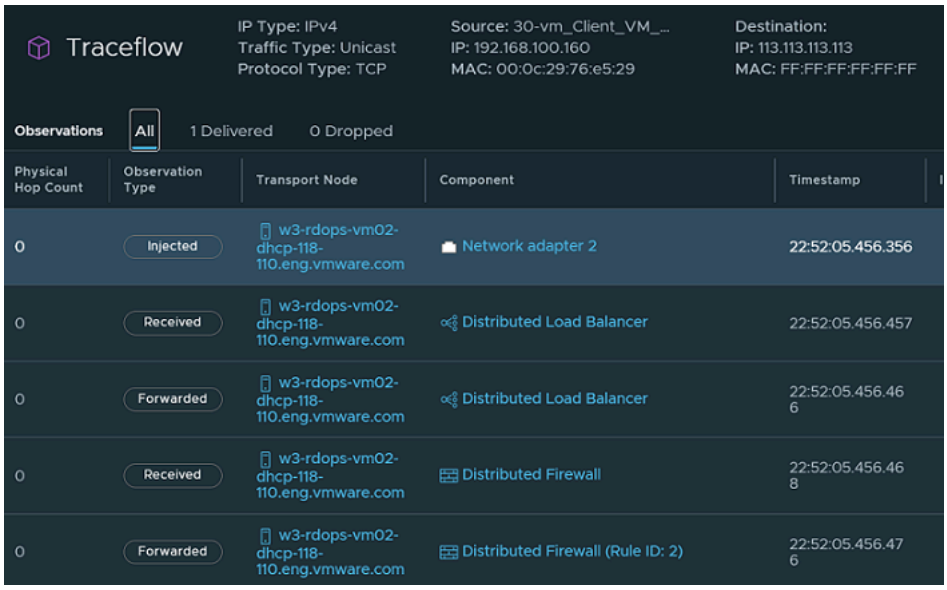
Run Traceflow on Distributed Load Balancer

Run Traceflow between virtual machines or interfaces where Distributed Load Balancer is enforced and the Distributed Load Balancer virtual IP address (VIP).

Use Traceflow for debugging purposes, when:

- On a client guest VM, where a Distributed Load Balancer service is applied, if the communication is lost between the guest VMs to Distributed Load Balancer.
- Or when guest VMs sending east-west traffic to another VM in the network drops.

1. With admin privileges, log in to NSX Manager.
2. Navigate to the **Plan & Troubleshoot** tab, and click **Traceflow**.
3. Enter source (for example, virtual machine where Distributed Load Balancer is enforced) and destination (Distributed Load Balancer VIP).
4. Click **Trace**.



5. View results.
   - As DLB is not based on rules, if Distributed Load Balancer drops a packet, there are not rule IDs that show up in the traceflow output details.
   - If Distributed Load Balancer and VMware vDefend Distributed Firewall services are applied to a client virtual machine, a member of a group, first Distributed Load Balancer service is applied followed by vDefend Distributed Firewall rules service.

Feature Limitations

The following features are not supported by Distributed Load Balancer (DLB) in this release.

- Persistence profiles configured on the virtual server
- Default Distributed Load Balancer (DLB) instance, where `connectivity_path` is set to ALL instead of a specific VM group
- Log-related configurations for the DLB instance and virtual server
- TCP close timeout setting in the DLB application profile

Configuring an External Load Balancer

You can configure an external load balancer to distribute traffic to the NSX Managers in a manager cluster.

An NSX Manager cluster does not require an external load balancer. The NSX Manager virtual IP (VIP) provides resiliency in the event of a Manager node failure but has the following limitations:

- VIP does not perform load balancing across the NSX Managers.
- VIP requires all the NSX Managers to be in the same subnet.
- VIP recovery takes about 1 - 3 minutes in the event of a Manager node failure.

An external load balancer can provide the following benefits:

- Load balance across the NSX Managers.
- The NSX Managers can be in different subnets.
- Fast recovery time in the event of a Manager node failure.

Authentication Methods When Accessing NSX Manager

The following authentication methods are supported by NSX Manager. For more information about the authentication methods, see the *NSX API Guide*.

- HTTP Basic Authentication
- Session-Based Authentication
- Authentication using an X.509 certificate and a Principal Identity
- Authentication in VMware Cloud on AWS

The session-based authentication method (used when you access NSX Manager from a browser) requires source-IP persistence (all requests from the client must go to the same NSX Manager). The other methods do not require source-IP persistence (requests from the client can go to different NSX Managers).

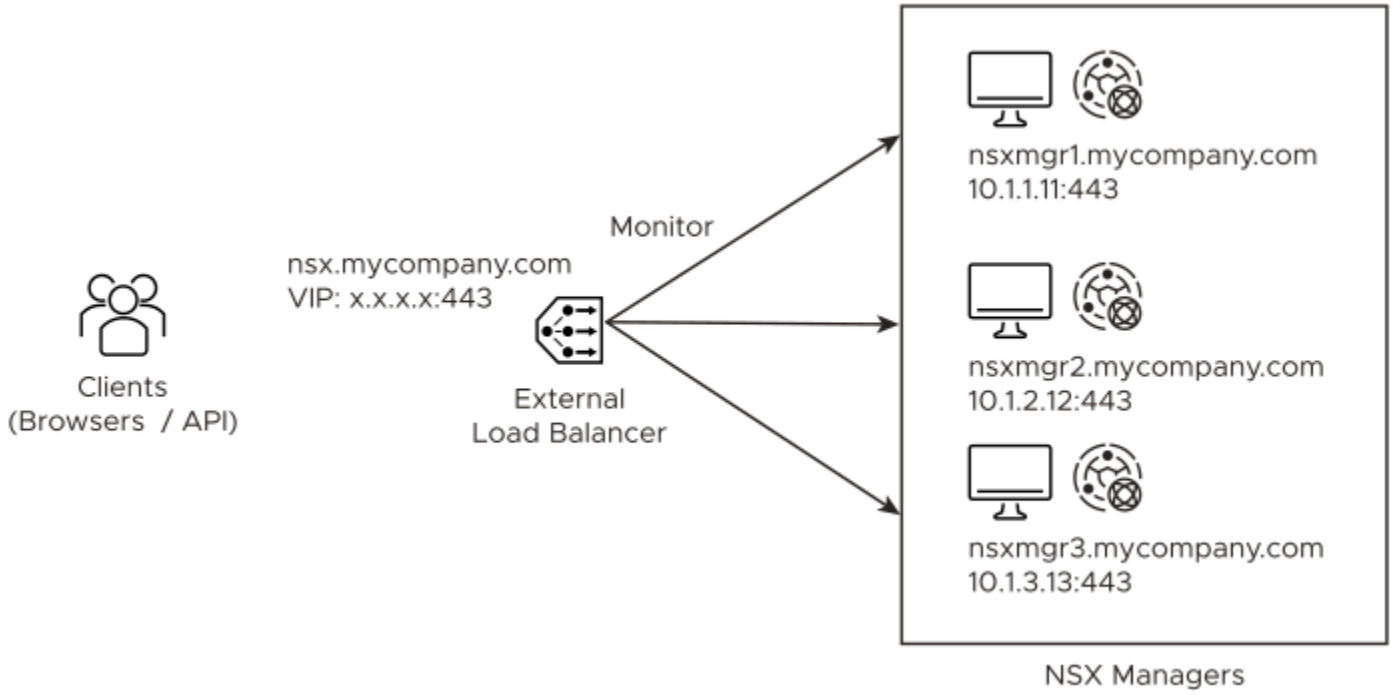
Recommendations

- Create a single VIP on the load balancer with source-IP persistence configured to handle all the authentication methods.
- If you have applications or scripts that might generate a lot of requests to NSX Manager, create a second VIP without source-IP persistence for these applications or scripts. Use the first VIP for browser access to NSX Manager only.

The VIP must have the following configurations:

- Type: Layer4-TCP
- Port: 443
- Pool: NSX Manager Pool
- Persistence: Source-IP persistence for the first VIP. None for the second VIP (if present).

Example of an external load balancer configuration:



NSX Manager's Certificate

The clients access NSX Manager using FQDN name (for example, nsx.mycompany.com). This FQDN is resolved to the load balancer's VIP. To avoid any certificate mismatch, each NSX Manager must have a certificate that is valid for the VIP's FQDN name. Therefore, you must configure each NSX Manager with a SAN certificate that is valid for its own name (for example, nsxmgr1.mycompany.com) and the VIP's FQDN.

Monitoring the Health of NSX Managers

The load balancer can check that each NSX Manager is running with the following API:

```
GET /api/v1/reverse-proxy/node/health
```

The request headers are:

- Header 1
  - Name: Content-Type
  - Value: application/json
- Header 2
  - Name: Accept
  - Value: application/json

A response indicating that the NSX Manager is running will be:

```
"healthy" : true
```

Note that the format of the response is `"healthy"<space>:<space>true`.

Ethernet VPN (EVPN)

EVPN (Ethernet VPN) is a standards-based BGP control plane that allows you to extend Layer 2 and Layer 3 connectivity between different data centers.

Overview of BGP EVPN

Ethernet VPN (EVPN) is a standards-based BGP distributed control plane for Network Virtualization Overlay (NVO), that provides Layer 2 (bridging) and Layer 3 (routing) connectivity over IP or IP/MPLS underlay networks. BGP EVPN was

initially designed to be used with MPLS data plane to address limitations of VPLS in service provider networks. However, EVPN has been widely adopted in data centers as a control plane mechanism for VXLAN overlay networking due to advantages in BGP scalability and flexibility.

Some of the key characteristics and benefits for BGP EVPN are:

- BGP-based control plane learning for Layer 2 and Layer 3 end host reachability information. This replaces flood-and-learn behavior of legacy L2VPN solutions, such as VXLAN, SPB, and Trill.
- ARP suppression to minimize unnecessary ARP and ND message flooding.
- Support for bridging-only and/or integrated routing and bridging.
- Support for MAC and IP mobility and multihoming.

MP-BGP EVPN Address Family

A new MP Address Family Indicator/Subsequent Address Family Indicator (AFI/SAFI) is defined for EVPN: l2vpn (25) / evpn (70). For two BGP speakers to exchange EVPN Network Layer Reachability Information (NLRI), they must negotiate the EVPN BGP capability at the start of the BGP session to ensure that both peers can support such NLRI.

Route Distinguisher and Route Targets

BGP EVPN uses the same mechanisms of other BGP VPN technologies to ensure uniqueness and multi-tenancy:

Mechanism	Description
Route Distinguisher (RD)	• Used in EVPN to make addresses globally unique. • The same encoding defined in RFC 4364 is applicable for BGP EVPN. • RD type 0 has an Administrator subfield of 2 bytes and Assigned Number subfield of 4 bytes. The Administrator subfield must contain an Autonomous System number. • RD type 1 has an Administrator subfield of 4 bytes and Assigned Number subfield of 2 bytes. The Administrator subfield must contain an IP address.
Route Target (RT)	• Used in EVPN to indicate virtual network membership by importing and exporting RTs as required.

EVPN Route Types

EVPN NLRI is further classified by the following route types:

Route Type	Description	Purpose	NSX Inline Mode	NSX Route Server Mode
Type 1 (RT-1)	Ethernet Auto-Discovery (A-D) route	Used in data centers to support EVPN active-active of multihoming.	No	Yes (receive only for DC-GW mutlihoming)
Type 2 (RT-2)	MAC/IP Advertisement route	Advertises reachability of a specific MAC address, and optionally MAC and IP address binding.	No	Yes
Type 3 (RT-3)	Inclusive Multicast Ethernet Tag route	Advertises reachability of a VNI associated to a particular VTEP in a virtual network. Type 3 routes are required for BUM traffic delivery across EVPN networks.	No	Yes
Type 4 (RT-4)	Ethernet Segment route	Used in data centers with multihomed endpoints, and used for Designated Forwarder Election to ensure that only one of the VTEPs forwards BUM traffic.	No	No
Type 5 (RT-5)	IP Prefix route	Advertises IPv4 and IPv6 prefixes reachability. This advertisement of prefixes into the EVPN domain provides the ability to build L3VPN similar services.	Yes	Yes

Route Type	Description	Purpose	NSX Inline Mode	NSX Route Server Mode
Type 6 (RT-6)	Selective Multicast Ethernet Tag route	Used to advertise the intent of the host or VM to receive multicast traffic for a certain Multicast Group (*,G) or Source-Group combination (S,G).	No	No

Virtual Routing and Forwarding

Virtual routing and forwarding (VRF) makes it possible to instantiate isolated routing and forwarding tables within a router. These routing and forwarding table instances isolate Layer 3 domains and segments from each other creating a multi-tenant network, either locally within the router or across multiple routers.

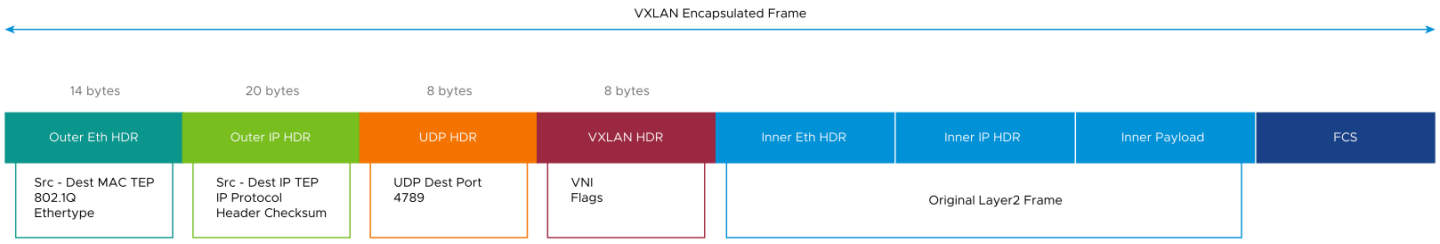
VRFs are supported in NSX by deploying tier-0 VRF gateways. A tier-0 VRF gateway must be linked to a parent tier-0 gateway and inherits some of the tier-0 settings, such as the HA mode, edge cluster, internal transit subnet, T0-T1 transit subnets, and BGP local ASN.

Multiple tier-0 VRF gateways can be created under the same parent tier-0, allowing the separation of segments and tier-1 gateways into multiple isolated tenants. With tier-0 VRF gateways, tenants can use overlapping IP addresses without any interference or communication with each other.

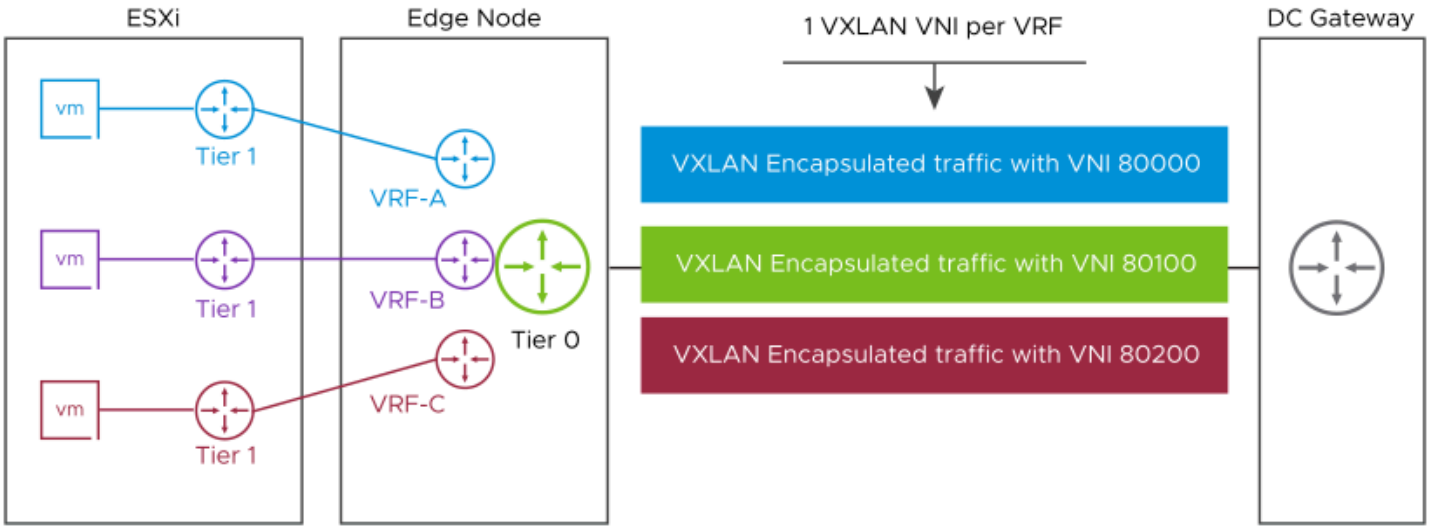
In the context of EVPN, each Layer 3 VRF is identified by a global unique Virtual Network Identifier (VNI). The VNI for each VRF must match in NSX Edge nodes and data center gateways.

VXLAN Encapsulation and VNI

VXLAN encapsulation as defined in RFC7348, is used between NSX tunnel endpoints (edge nodes for Inline mode and hypervisors for Route Server mode) and external routers in order to ensure data plane compatibility with other vendors. Inside the NSX domain, GENEVE encapsulation is still used.



The VNI is a 24-bit identifier used to identify a particular virtual network segment. When EVPN is used to advertise IP prefixes reachability by using Route Type 5 and the encapsulation type as VXLAN then the VNI identifies the tenant VRF instance. As defined in RFC9135, this VNI is advertised in the BGP control plane along with the prefix routes as well as used in the data plane encapsulation to differentiate the traffic between VRFs. The VNI for each VRF must match in NSX Edge nodes and data center gateways.



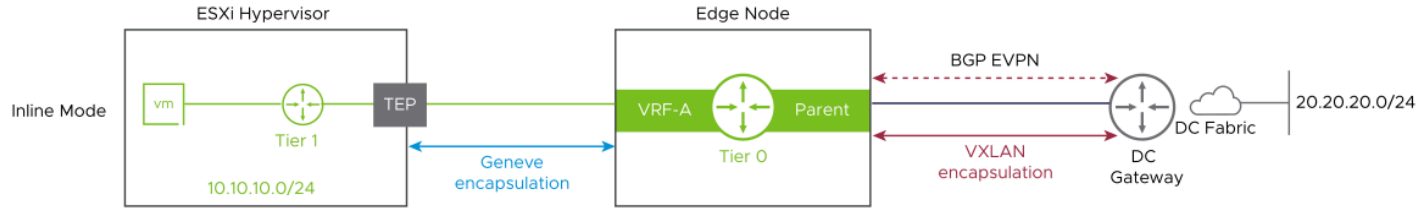
EVPN Support in NSX

NSX leverages BGP EVPN technology to interconnect and extend NSX-managed overlay networks to other data center environments not managed by NSX, VXLAN encapsulation is used between NSX TEPs (edge nodes and hypervisors) and external network devices to ensure data plane compatibility.

Two connectivity modes are supported for EVPN implementation in NSX:

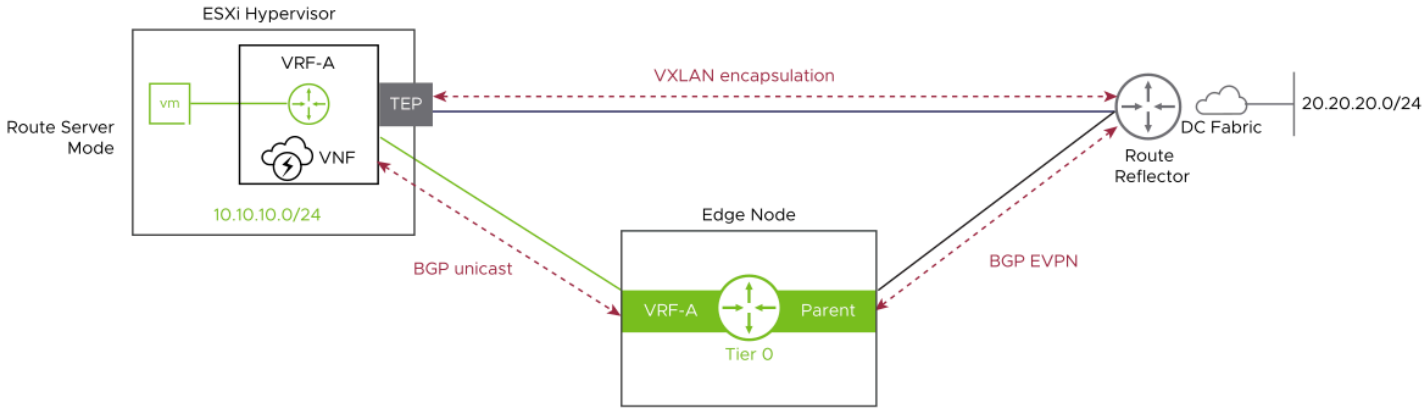
Inline Mode

In this mode, the tier-0 gateway establishes MP-BGP EVPN control plane sessions with external routers to exchange routing information. In the data plane, edge nodes forwards all the traffic exiting the local data center to the data center gateways and incoming traffic from the remote data center to the hypervisors in the local data center. Since the edge nodes are in the data forwarding path, this model is called the Inline model.



Route Server Mode

In this mode, the tier-0 gateway establishes MP-BGP EVPN control plane to exchange routing information with the external router or route reflectors. In the data plane, ESX hypervisors forward the traffic to external networks either to the data center gateways or remote ToR switches over VXLAN tunnels. TEPs used for the data plane VXLAN encapsulation are the same than the ones used for GENEVE encapsulation.



Route Distinguishers and Route Targets in NSX

With NSX BGP implementation, route distinguishers (RD) can be either set automatically or manually. The following table details the supported RD modes in the Inline and Route Server modes.

Mode	Auto RD	Manual RD
Inline	• Supported. • Only type-1 is supported. • You must configure the RD Admin field. The RD Admin field must be in the format of an IP address. • The RD admin field is used to fill the Administrator subfield in the RD. • The 2-byte Assigned Number subfield will be allocated a random number in the range for each RD generation. • Generated auto RD is checked against other manually configured RDs to avoid any duplicates.	• Supported. • Both type-0 and type-1 are allowed, but type-1 is recommended. • No RD Admin field is required to be configured. • Configure manual RD is checked against other auto RDs to avoid any duplicates.
Route Server	• Not supported.	• Supported. • Both type-0 and type-1 are allowed, but type-1 is recommended. • No RD Admin field is required to be configured. • Configured manual RD is checked against other auto RDs to avoid any duplicates.

Limitations and Caveats

- NSX supports L3 EVPN by advertising and receiving IP prefixes as EVPN Route Type-5.
- NSX generates a unique router MAC for every NSX Edge VTEP in the EVPN domain. However, there may be other nodes in the network that are not managed by NSX, for example, physical routers. You must make sure that the router MACs are unique across all the VTEPs in the EVPN domain.
- The EVPN feature supports NSX Edge nodes to be either the ingress or the egress of the EVPN virtual tunnel endpoint. If an NSX Edge node receives EVPN Route Type-5 prefixes from its eBGP peer that needs to be redistributed to another eBGP peer, the routes are re-advertised without any change to the next hop.
- In multi-path network topologies, it is recommended that ECMP is enabled for the NSX BGP EVPN control plane, so that all the possible paths can be advertised by the tier-0 gateway. This will avoid any potential traffic blackhole due to asymmetric data path forwarding.
- Route maps are not supported for EVPN address family.
- Recursive route resolution for gateway IP via default static route is not supported.

Limitations and caveats for Inline mode:

- Only BGP Graceful Restart in Helper Mode is supported.
- Only eBGP is supported between tier-0 SRs and external routers.
- Only one TEP is supported per edge node. The use of loopback interfaces for TEP is highly recommended.

Limitations and caveats for Route Server mode:

- The High Availability mode on the tier-0 must be set to active-active.
- The manual Route Distinguisher and manual Route Targets are supported.
- BGP Graceful Restart, Helper Mode, and Restarted Mode are not supported.
- Only eBGP is supported between hosted VNFs and tier-0 VRF gateways.
- eBGP multihop using loopbacks is required between tier-0 SRs and external routers. Using uplinks for eBGP neighbor session is not supported for EVPN Router Server mode operation.
- The VNF uplink towards the tier-0 SR VRF must be in the same subnet as the Integrated Routing and Bridging (IRB) on the data center gateways.

EVPN Inline Mode

In the EVPN Inline mode, MP-BGP sessions with the L2VPN EVPN address family are configured between tier-0 gateways and the external routers. The tier-0 gateway will negotiate the L2VPN EVPN AFI/SAFI with the external BGP peer (data center gateway) and start exchanging EVPN information. In this mode, the edge nodes are in the datapath between internal workloads and external networks.

The NSX EVPN Inline mode is based on the "Interface-less IP-VRF-to-IP-VRF Model" as defined in the IETF RFC9136. In this mode, the tier-0 gateway advertises to the data center gateway only Route Type 5 (RT-5) routes, that includes:

- Prefixes of segments connected directly to the tier-0 VRF gateway.
- Tier-1 segment prefixes redistributed to the tier-0 VRF gateway.
- Other redistributed sources, such as static routes and connected interfaces.
- BGP prefixes learned via southbound BGP sessions inside the VRF.

All routes received from and advertised to the data center gateways as RT-5 has an EVPN Router's MAC Extended Community with the MAC address of the corresponding peer uplink. Since "Interface-less IP-VRF-to-IP-VRF Model" is used, there is no recursive route lookup to resolve the RT-5 route. The packet is encapsulated in VXLAN using the RT-5's next hop as destination IP address and EVPN Router's MAC Extended Community as MAC address.

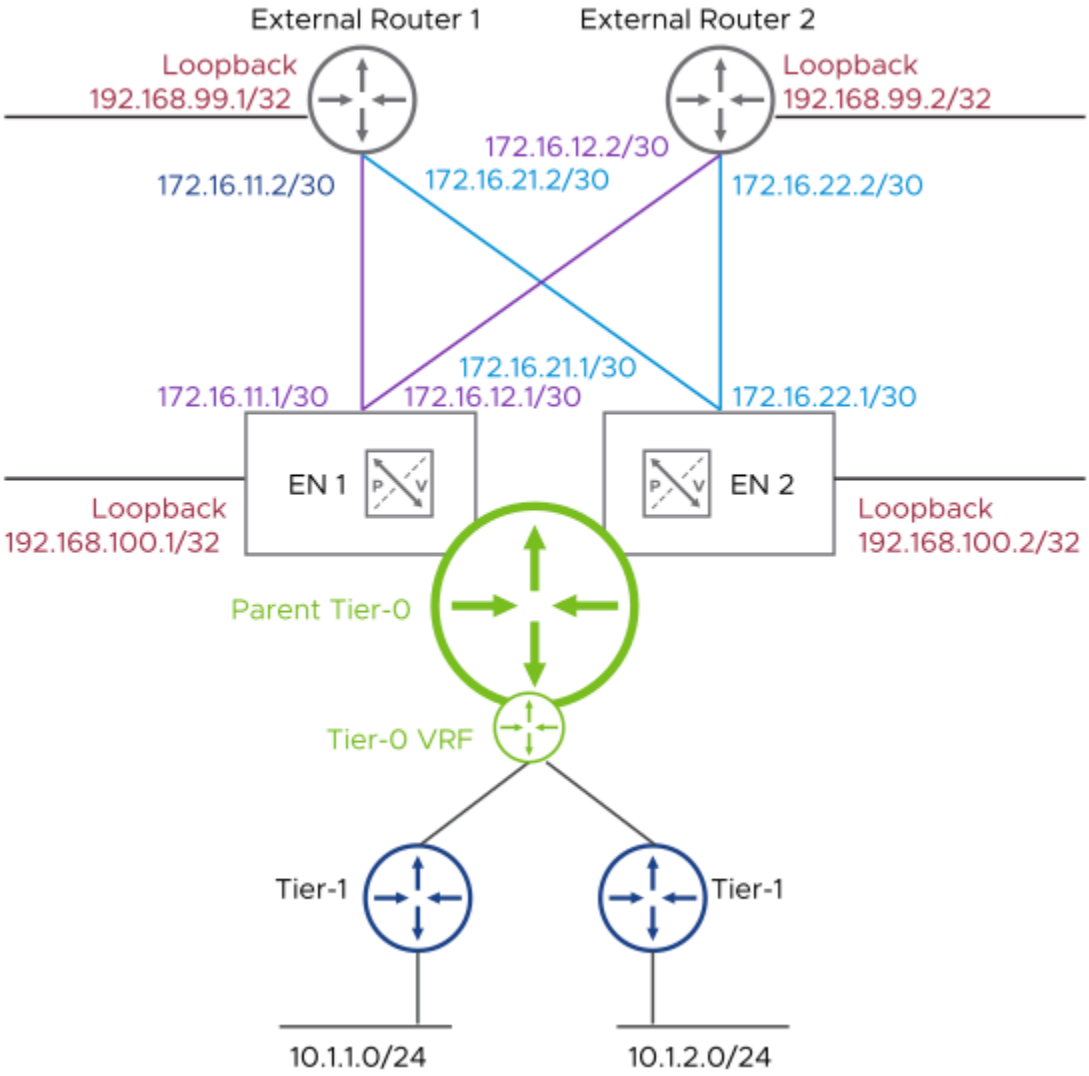
Inside the NSX domain, the control plane is still handled by the central control plane (CCP) and the encapsulation protocol among internal TEPs is still GENEVE.

EVPN Inline Mode Configuration Workflow

A typical BGP EVPN Inline mode deployment topology has the following characteristics:

- There are point-to-point uplinks between edge nodes and external routers over individual VLAN segments.
- There are BGP peering sessions between edge nodes and external routers using loopback interfaces.
- Loopback reachability can be achieved using either static routing or OSPF protocol.

The following diagram depicts a typical BGP EVPN Inline mode deployment topology:



Follow this workflow to configure EVPN with Inline mode.

1. Create a VNI pool. See Add an NSX EVPN/VXLAN VNI Pool.
2. Configure a tier-0 gateway and enable EVPN. See Configure a Tier-0 Gateway for EVPN Inline Mode.
3. Configure BGP neighbors. See Configure an NSX BGP Neighbors for a Tier-0 Gateway.
4. Configure a tier-0 VRF gateway. See Configure a Tier-0 VRF Gateway for EVPN Inline Mode.
5. Verify the BGP neighbor session status.
   a) Select **Networking** > **Tier-0 Gateways**.
   b) Click the menu icon (three dots) of the tier-0 gateway and select **Generate BGP Summary**.
   c) Verify that **Connection Status** for the neighbor is **Established** and that **Address Families** displays **L2VPN EVPN**.
6. Verify the tier-0 VRF gateway forwarding table.
   a) Select **Networking** > **Tier-0 Gateways**.
   b) Click the menu icon (three dots) of the tier-0 VRF gateway and select **Download Forwarding Table**.
   c) Verify that the remote routes received from the external router are installed in the tier-0 VRF gateway forwarding table.

Configure a Tier-0 Gateway for EVPN Inline Mode

Use this procedure to configure a tier-0 gateway for EVPN Inline mode.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Tier-0 Gateways**.
3. Create a tier-0 gateway and configure uplink interfaces between the tier-0 gateway and data center gateways. See Add an NSX Tier-0 Gateway.
4. Optional: Configure loopback interfaces for each edge node to be used as the BGP source address and VXLAN VTEPs.
5. Optional: If connecting to BGP peers using loopbacks, make sure that the data center gateway external loopbacks are reachable from the parent tier-0 gateway.

   This connectivity can be achieved either by using OSPF or static routes.
6. Enable EVPN.
   a) Expand the **EVPN Settings** section.
   b) For **EVPN Mode**, select **Inline**.
   c) For **EVPN/VXLAN VNI Pool**, select a VNI pool.
   d) For **EVPN Tunnel Endpoint**, click **Set** > **Add EVPN Local Tunnel Endpoint**.
      1. Enter a name for the endpoint.
      2. Select the edge node.
      3. Enter the IP address for the VXLAN TEP.
      4. Click **Save** and then **Close**.
7. Optional: If the VXLAN TEP IP address is different than the loopback interface IP address used for the BGP neighbor configuration, make sure that the local tunnel endpoints is redistributed into the tier-0 BGP routing table.
   a) Expand the **Route Re-distribution** section.
   b) Click **Set** > **Add Route Re-distribution**.
      1. Enter a name for the re-distribution policy.
      2. For **Destination Protocol**, select **BGP**.
      3. Click **Set** and then select **EVPN TEP IP**.
      4. Click **Apply**.
      5. Click **Add** and then **Apply**.

Configure an NSX BGP Neighbors for a Tier-0 Gateway

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Tier-0 Gateways**.
3. Click the menu icon (three dots) for the tier-0 gateway and select **Edit**.
4. Expand the **BGP** section.
5. Click the **BGP** toggle to enable BGP.
6. Enter the **Local AS** number.
7. Configure BGP neighbors.
   a) For **BGP Neighbors**, click **Set** > **Add BGP Neighbor**.
   b) Enter the IP address of the neighbor.
   c) Enable BFD if required.
   d) Enter the **Remote AS number** of the neighbor.
   e) For **Source Addresses**, enter the source IP address.

      There should be one or more addresses of created external interfaces or loopback.
   f) For **Route Filter**, click **Set** > **Add Route Filter**.
      1. For **IP Address Family**, select **L2VPN EVPN**.
      2. Specify the desired maximum routes.

3. Click **Add** and then **Apply**.

8. Click **Save** and then **Close**.

Configure a Tier-0 VRF Gateway for EVPN Inline Mode

Use this procedure to configure a tier-0 VRF gateway for EVPN Inline mode.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Tier-0 Gateways**.
3. Add a VRF gateway.
   a) Click **Add Gateway** > **VRF**.
   b) Enter a name for the gateway.
   c) Select a tier-0 gateway with EVPN Inline mode enabled to connect to.

      Some advanced configurations are inherited from the parent tier-0 gateway, such as HA mode, edge cluster, internal transit subnet, T0-T1 transit subnets, and BGP Local ASN.
4. Expand the **VRF Settings** section.
   a) Enter the route distinguisher.
      - If the **Enable the RD per Edge node** flag is turned on in the parent tier-0 Gateway, the VRF settings section of the tier-0 VRF gateway UI will show a banner indicating “The Edge node Router distinguisher pool” is in use. If **RD Admin Address** is configured, then a pool of 10 route distinguishers is auto-assigned. To override the auto-assigned RD values, configure **Edge Node Route Distinguisher Pool**.
      - If the **Enable the RD per Edge node** flag is turned off in the parent tier-0 Gateway and if the connected parent tier-0 gateway has **RD Admin Address** configured, then the **Route Distinguisher** field is automatically populated. Enter a new value if you want to override the assigned route distinguisher.

   b) For **Route Targets**, click **Set** > **Add Route Target** to add route targets.
      - For EVPN, only the **Manual** mode is supported.
      - Specify one or more **Import Route Targets** to install the desired received routes from BGP peers to the VRF routing table.
      - Specify one or more **Export Route Targets** to label advertised VRF routes.

   c) Click **Add** and then **Apply**.
   d) For **EVPN Transit VNI**, enter the L3 transit VNI value for the VRF.

      This VNI must be unique per VRF and belong to the configured EVPN/VXLAN VNI pool. Make sure that the same VNI value for the VRF is configured on the external router.
5. Click **Save** and then **Yes** to continue configuring the VRF gateway.
6. Expand the **Route Re-distribution** section.
   a) Click **Set** > **Add Route Re-Distribution**.
   b) Enter a name for the re-distribution policy.
   c) Click **Set** to select available sources, such as tier-0 connected interfaces and segments.
   d) Click **Apply**.
   e) Click **Add** and then **Apply**.
7. Make sure that the created segments or tier-1 gateways are connected to the new tier-0 VRF gateway.

EVPN Route Server Mode

In the EVPN Route Server mode, the tier-0 service router (SR) hosted on the edge node acts as a BGP route server, establishing BGP control plane sessions with southbound VNFs and external data center routers. ESX hypervisors exchange the user plane traffic directly with the data center fabric routers using VXLAN encapsulation, bypassing the edge node in the data path.

From the BGP control plane perspective, there are two types of sessions:

Session Type	Description
Between hosted VNFs and tier-0 VRF gateway.	• BGP IPv4 unicast and IPv6 unicast sessions from the VNF to the tier-0 VRF service ports. • IP prefixes learned from the VNF via the BGP IPv4/IPv6 unicast sessions are advertised as EVPN Route-Type 5 towards the external route with the corresponding VRF route distinguisher and route targets. • IP prefixes (RT-5) learned from the external router via the BGP EVPN session are injected in the tier-0 VRF routing table based on the route target policies and are advertised as IPv4/IPv6 unicast routes to the VNF.
Between tier-0 SR and DC gateways.	• BGP IPv4 session with L2VPN EVPN address family from tier-0 SR to the loopback of DC gateways. • The IP prefixes learned from the VNF via the BGP IPv4/IPv6 unicast sessions are advertised as EVPN Route-Type 5 towards the external router with the corresponding VRF route distinguisher and route targets. • IP prefixes (RT-5) learned from the external router via the BGP EVPN session are injected in the tier-0 VRF routing table based on the route target policies and are advertised as IPv4/IPv6 unicast routes to the VNF. • EVPN routes of type 2 (RT-2) are exchanged between tier-0 SR and DC gateways to advertise MAC and IP/MAC bindings for RT-5 routes next hops. • EVPN routes of type 1 (RT-1) are sent by the DC gateway to the tier-0 SRs when multiple data center gateways are configured as L2ECMP EVPN multihoming. RT-1 routes are used to create a list of data center gateway VTEPs to be used as next hop by ESX nodes.

The NSX EVPN Route Server mode is based on the "Interface-ful IP-VRF-to-IP-VRF with SBD IRB" as defined in the IETF RFC 9136. The RFC 9136 introduces a concept called overlay index in EVPN. A key concept of EVPN RT-5 is the overlay index, which can be a gateway IP address, a MAC, or an ESI. When a node receives an EVPN RT-5 with an overlay index specified, the receiving node performs a recursive route resolution to find the appropriate node to forward the data packets for the corresponding IP prefix.

NSX EVPN Route Server mode implements the gateway IP address as the overlay index. The tier-0 SR also advertises to the external router an additional EVPN type-2 route with the appropriate MAC/IP (gateway IP) binding and the corresponding VXLAN TEP address.

The gateway IP address in this case will be the IPv4 BGP next hop for a given prefix as advertised by the VNF to the tier-0 VRF gateway. The recursive route resolution uses respective RT-2 to learn the ESX TEP address where the VNF is hosted.

Data Center Gateway Requirements

The data center gateway router connected to the edge node tier-0 SR must support the "Interface-ful IP-VRF-to-IP-VRF with SBD IRB Mode" described in the IETF RFC 9136, section 4.4.2.

Virtual Network Function (VNF) Requirements

A VNF is typically a virtual machine used for some networking function such as a virtual router, firewall, or a Telco 5G core application. In the context of EVPN Route Server mode, the VNF is hosted by an ESX hypervisor and should support 802.1Q-tagged interfaces and regular BGP protocol with IPv4 and IPv6 unicast address families.

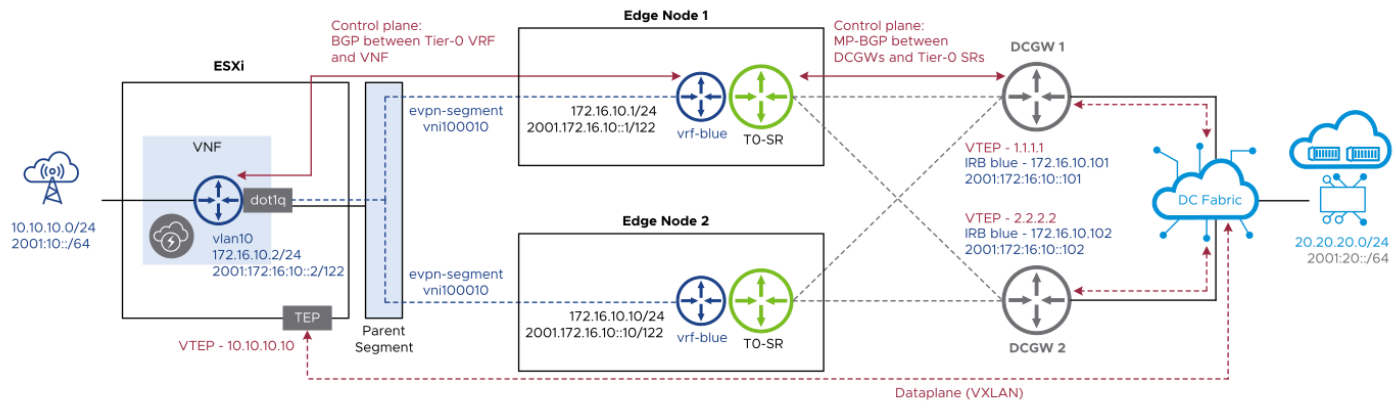
EVPN Route Server Mode Configuration Workflow

A typical BGP EVPN Route Server mode deployment topology has the following characteristics:

- The tier-0 gateway must be in active-active mode.
- There are least two data center gateways connected to the edge nodes.
- There are point-to-point uplinks between edge nodes and data center gateways over VLAN segments.
- There are eBGP peering sessions between edge nodes and data center gateways using loopback interfaces.

- The ESX node TEP network must have connectivity to the data center gateway VTEP IP addresses.
- There are southbound VMs and workloads connected to the VNF southbound interfaces using regular NSX segments.
- There are eBGP peering sessions between the VNF and service ports of tier-0 VRF gateways.

The following diagram depicts a typical BGP EVPN Route Server mode deployment topology:



Follow this workflow to configure EVPN with Route Server mode.

1. Configure a VNI pool. See Add an NSX EVPN/VXLAN VNI Pool.
2. Configure an EVPN tenant. See Configure an NSX EVPN Tenant.
3. Configure EVPN BFD between the ESX nodes and data center gateways. See Configure an NSX EVPN BFD.
4. Configure a tier-0 gateway and enable EVPN. See Configure an NSX Tier-0 Gateway for EVPN Route Server Mode.
5. Configure BGP neighbors. See Configure an NSX BGP Neighbors for a Tier-0 Gateway.
6. Configure a tier-0 VRF gateway. See Configure an NSX Tier-0 VRF Gateway for EVPN Route Server Mode.
7. Configure networking for onboarding the tenant VNF. See Onboard a Tenant VNF for an NSX EVPN Route Server Mode.
8. In NSX, verify the following:

Verification	Steps
Verify the tier-0 SR BGP neighbor session status.	1. Select **Networking** > **Tier-0 Gateways**. 2. Click the menu icon (three dots) for the tier-0 VRF gateway and select **Generate BGP Summary**. 3. Verify that **Connection Status** displays **Established**. 4. Verify that **Address Families** displays **L2VPN EVPN**.
Verify the tier-0 VRF BGP neighbor session status.	1. Select **Networking** > **Tier-0 Gateways**. 2. Click the menu icon (three dots) for the tier-0 VRF gateway and select **Generate BGP Summary**. 3. Verify that the neighbor **Connection Status** displays **Established**.
Verify the tier-0 VRF gateway routing table.	1. Select **Networking** > **Tier-0 Gateways**. 2. Click the menu icon (three dots) for the tier-0 VRF gateway and select **Download Routing Table**. 3. Select the transport node (edge node) and for **Source**, select **BGP**. 4. Click **Download**. 5. Verify that the remote nodes received from the external router are installed in the tier-0 VRF gateway routing table.

9. In the ESXNSX CLI, verify the following.
   - Verify the status of the VTEP group for the VNI.
     `get logical-switch <vni> vtep-group`

- Verify the MAC address for the L2 VNI.
  ```
  get logical-switch <vni> mac-table
  ```
- Verify the ARP table for the L2 VNI.
  ```
  get logical-switch <vni> arp-table
  ```

Configure an NSX EVPN Tenant

Configuring an EVPN tenant is required for EVPN Route Server mode.

To configure an EVPN tenant, you must specify one or more VLAN-VNI mappings.

1. With admin privileges, log in to NSX Manager.
2. Click **Networking** > **EVPN Tenant** > **Add EVPN Tenant**.
3. Enter a name.
4. For **Overlay Transport Zone**, select one from the drop-down list.
5. Select an existing EVPN/VXLAN VNI pool or create a new pool by clicking the menu icon (three dots).
6. For **VLAN-VNI Mapping**, click **Set** to create one or more mappings.
   The VNIs must be within the range specified in the EVPN/VXLAN VNI pool. You can specify a single value for VLAN and VNI, or a range of values such as 10-20 for VLAN and 77010-77020 for VNI. For each VLAN-VNI mapping specified in the EVPN tenant, a special segment for the VNI will be created automatically. You can see the VRF segments by navigating to **Networking** > **Segments**.
7. Click **Save**.

Configure an NSX EVPN BFD

In EVPN Server Route mode, BFD (Bidirectional Flow Detection) between the ESX nodes and data center gateways can be "optionally" enabled for TEP fast failure detection.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Global Networking Config**.
3. Click **Edit**.
4. For **EVPN BFD Profile**, select a BFD profile.
   If you need to create a new profile with required BFD timers, click the menu icon (three dots) and select **Create New**.
5. For **Enable EVPN BFD**, make sure that the toggle is on.
6. Click **Save**.

Configure an NSX Tier-0 Gateway for EVPN Route Server Mode

Use this procedure to configure a tier-0 gateway for EVPN Route Server mode.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Tier-0 Gateways**.
3. Create a tier-0 gateway and configure uplink interfaces between the tier-0 gateway and data center gateways. See Add an NSX Tier-0 Gateway.

   Note:
   For EVPN Route Server mode, only active-active HA mode is supported.
4. Optional: Configure loopback interfaces for each edge node to be used as the BGP source address and VXLAN VTEPs.
5. Make sure that the data center gateway external loopbacks are reachable from the parent tier-0 gateway.
   This connectivity can be achieved either by using OSPF or static routes.

6. Enable EVPN.
   a) Expand the **EVPN Settings** section.
   b) For **EVPN Mode**, select **Route Server**.
   c) Select the EVPN tenant.
   d) Click **Save** and then **Close**.

Configure an NSX BGP Neighbors for a Tier-0 Gateway

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Tier-0 Gateways**.
3. Click the menu icon (three dots) for the tier-0 gateway and select **Edit**.
4. Expand the **BGP** section.
5. Click the **BGP** toggle to enable BGP.
6. Enter the **Local AS** number.
7. Configure BGP neighbors.
   a) For **BGP Neighbors**, click **Set** > **Add BGP Neighbor**.
   b) Enter the IP address of the neighbor.
   c) Enable BFD if required.
   d) Enter the **Remote AS number** of the neighbor.
   e) For **Source Addresses**, enter the source IP address.
      There should be one or more addresses of created external interfaces or loopback.
   f) For **Route Filter**, click **Set** > **Add Route Filter**.
      1. For **IP Address Family**, select **L2VPN EVPN**.
      2. Specify the desired maximum routes.
      3. Click **Add** and then **Apply**.
8. Click **Save** and then **Close**.

Configure an NSX Tier-0 VRF Gateway for EVPN Route Server Mode

Use this procedure to configure a tier-0 VRF gateway for EVPN Route Server mode.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Tier-0 Gateways**.
3. Add a VRF gateway.
   a) Click **Add Gateway** > **VRF**.
   b) Enter a name for the gateway.
   c) Select a tier-0 gateway with EVPN Route Server mode enabled to connect to.
      Some advanced configurations are inherited from the parent tier-0 gateway, such as HA mode, edge cluster, internal transit subnet, T0-T1 transit subnets, and BGP Local ASN.
4. Expand the **VRF Settings** section and configure the L3 VRF settings.

   Note:
   The L3 VRF settings correspond with the EVPN RT-5 routes advertised and received from the data center gateways. The imported/exported RTs needs to match with the corresponding L3 VRF configuration in the data center gateway side.

a) Enter the route distinguisher.
   - If the **Enable the RD per Edge node** flag is turned on in the parent tier-0 gateway, the VRF settings section of the tier-0 VRF gateway UI will show a banner indicating “The **Edge node Router distinguisher pool** is in use. Configure **Edge Node Route Distinguisher Pool**".
   - If the **Enable the RD per Edge node** flag is turned off in the parent tier-0 gateway, enter a new value in the **Route distinguisher** field.

b) For **Route Targets**, click **Set** > **Add Route Target** to add route targets.
   - For EVPN, only the **Manual** mode is supported.
   - Specify one or more **Import Route Targets** to install the desired routes from BGP peers to the VRF routing table.
   - Specify one or more **Export Route Targets** to label advertised VRF routes.

c) Click **Add** and then **Apply**.

5. Configure the L2 VNI settings.

   Note:
   The L2 VNI settings correspond with the EVPN RT-3, RT-2, and RT-1 routes advertised and received from the data center gateways. The imported/exported RTs needs to match with the corresponding L2 VNI configuration in the data center gateway side.

   a) For **L2 VNI**, click **Set** > **Add L2 VNI**.
   b) Select an L2 VNI.

      The VNI must be unique and belong to the VLAN-VNI mappings from an EVPN tenant.
   c) Enter the route distinguisher.
   d) For **Route Targets**, click **Set** > **Add**.
      - For EVPN, only the **Manual** mode is supported.
      - Specify one or more **Import Route Targets** to install the desired routes from BGP peers to the VRF routing table.
      - Specify one or more **Export Route Targets** to label advertised VRF routes.

   e) Click **Add** and then **Apply**.
   f) For **VTEP Groups**, click the toggle to enable or disable the creation of VTEP groups with data center gateways for the L2 VNI.

      Note:
      The **VTEP Groups** option will enable northbound L2ECMP EVPN multihoming between the VNF and DC gateways.

6. Click **Save** and then **Yes** to continue configuring the VRF gateway.
7. Expand the **Route Re-distribution** section.
   a) Click **Set** > **Add Route Re-Distribution**.
   b) Enter a name for the re-distribution policy.
   c) Click **Set** to select available sources, such as tier-0 connected interfaces and segments.
   d) Click **Apply**.
   e) Click **Add** and then **Apply**.

Onboard a Tenant VNF for an NSX EVPN Route Server Mode

For EVPN Route Server mode, you need to onboard a virtual network function (VNF). A VNF is typically a virtual machine used for some networking function such as a virtual router, firewall, or a Telco 5G core application. In the context of EVPN Route Server mode, the VNF is hosted by an ESX hypervisor and should support 802.1Q-tagged interfaces and regular BGP protocol with IPv4 and IPv6 unicast address families.

1. With admin privileges, log in to NSX Manager.
2. Create a parent overlay segment to connect the VNF virtual machine.
   a) Select **Networking** > **Segments**.
   b) Click **NSX** > **Add Segment**.
   c) Enter a name for the segment.
   d) For **Connected Gateway**, select **None**.
   e) Select an overlay transport zone.
   f) For the EVPN configuration section, select the EVPN tenant.
   g) Click **Save**.
3. Create a service interface to establish a BGP IPv4 session between the Tier-0 VRF gateway and the VNF.
   a) Select **Networking** > **Tier-0 Gateways**.
   b) Click
   
   next to the object you want to edit and click **Edit**.
   c) Expand the **Interfaces** section.
   d) Click **Set** > **Add Interface**.
   e) Enter a name for the interface.
   f) For **Type**, select **External**.
   g) Enter an IP address.
   h) Select a segment to connect to.
   The segment should be one of the EVPN automatically created segments with the appropriate VLAN to communicate with the VNF.
   i) Select an edge node.
   j) Click **Save** and then **Close**.
4. In vCenter, connect the VNF virtual machine uplink interface to the NSX parent segment created from the previous step.
5. Link the VNF segment port to the corresponding EVPN VLAN.
   a) In NSX, select **Networking** > **Segments**.
   b) Click the menu icon (three dots) for the parent segment and select **Edit**.
   c) For **Ports / Interfaces**, click **Set**.
   For each VNF interface attached to the parent segment, you should see the corresponding segment port.
   d) Click the menu icon (three dots) of the segment port and select **Edit**.
   e) For **EVPN VLAN**, add the corresponding VLAN.
   The VLAN should match the VLAN/VNI mapping for the VRF.
   f) Click **Save** and then **Close**.
6. Configure the BGP IPv4 session between the tier-0 VRF gateway and the VNF.
   a) Select **Networking** > **Tier-0 Gateways**.
   b) Click the menu icon (three dots) of the tier-0 VRF gateway and select **Edit**.
   c) Expand the **BGP** section.
   d) For **BGP**, click the toggle to enable BGP.
   e) You can configure advanced BGP settings, such as ECMP.
   f) For **BGP Neighbors**, click **Set** > **Add BGP Neighbor**.
      1. Enter the neighbor IP address.
      2. For **BFD**, click the toggle to enable or disable the BFD session with the VNF.
      3. Enter the remote AS number of the neighbor.
      4. For **Source IP Address**, it is not required. The system automatically uses the service port interface IP address previously created.
      5. For **Route Filter**, click **Set** > **Add Route Filter** to enable IP address families and the desired maximum routes.

6. For **IP Address Family**, select **IPv4** or **IPv6**.
7. Click **Add** and then **Apply**.

g) Expand the **Timers & Password** section.
h) Configure the BFD timers and BGP password.
i) Click **Save** and then **Close**.

Distributed VXLAN Connectivity

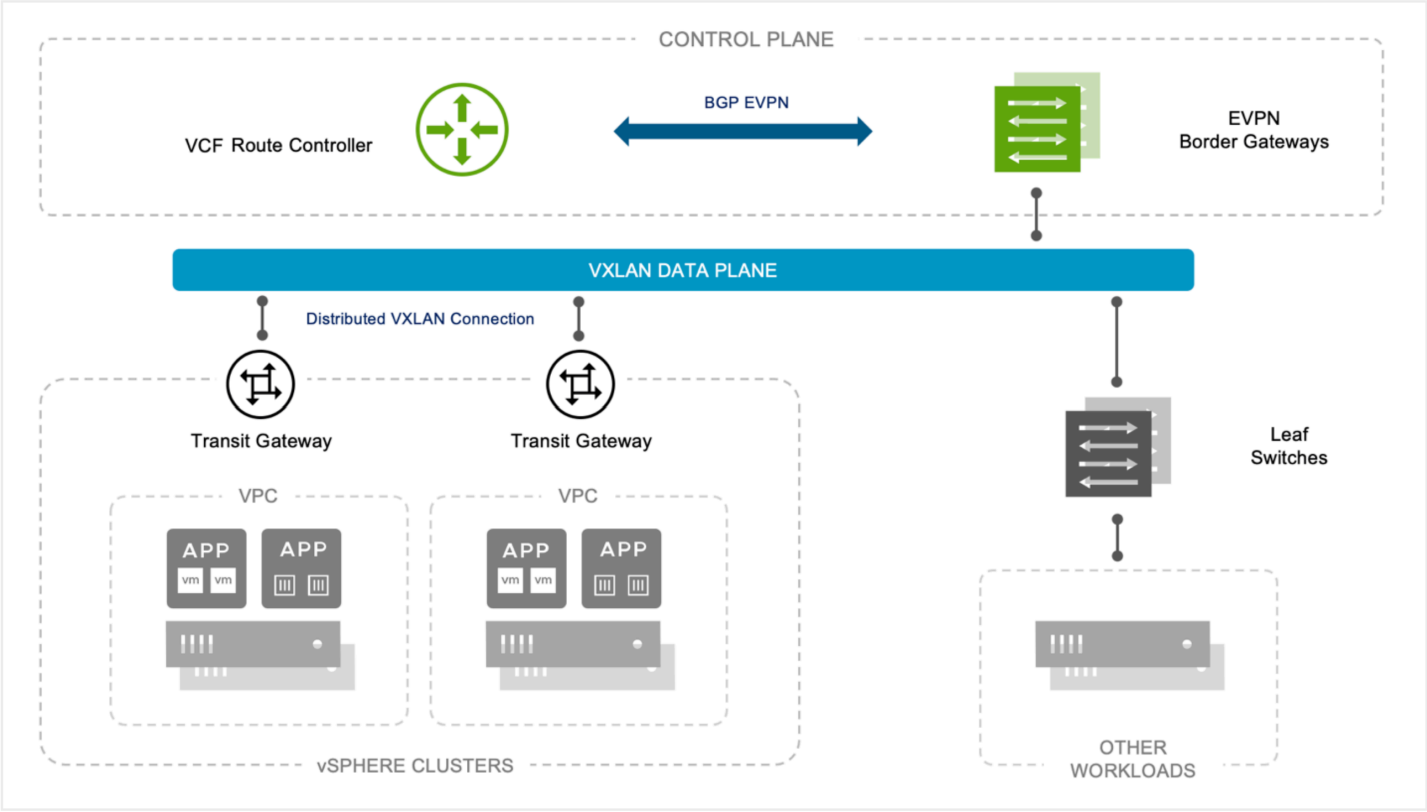
Distributed VXLAN connectivity integrates NSX and EVPN-VXLAN environments to create a unified network fabric that delivers consistent connectivity across virtual and physical domains. Virtual private clouds (VPCs) connect to the physical EVPN-VXLAN network fabric, using distributed VXLAN Transit Gateways to connect to tenant Virtual Routing and Forwarding instances (VRFs) within the EVPN-VXLAN fabric.

How Distributed VXLAN Connectivity Works

Ethernet VPN (EVPN) is widely used in data centers as a distributed control plane for the physical network fabric. The EVPN control plane uses Border Gateway Protocol (BGP) to advertise both Layer 2 MAC addresses and Layer 3 IP addresses across the physical network.

The EVPN control plane uses Virtual eXtensible Local Area Network (VXLAN) as a tunneling protocol for the data plane to encapsulate Layer 2 packets for transmission over the underlay Layer 3 network. The VXLAN protocol is the physical counterpart to the Geneve overlay protocol, which handles packet encapsulation and tunneling within the virtual NSX environment.

Figure 429: Unified Network Fabric with Distributed VXLAN Connectivity



With distributed VXLAN connectivity, the NSX data plane continues to use Geneve for east-west traffic between ESX hosts while it maintains connectivity with the external EVPN-VXLAN network fabric. The required control plane connectivity is established first by configuring a Route Controller to serve as the common control plane aligning the virtual and physical environments. Using Multiprotocol Border Gateway Protocol (MP-BGP), the Route Controller peers directly with EVPN gateways in the physical EVPN-VXLAN network fabric.

Next, you align the virtual private cloud (VPC) tenants in NSX with VRF tenants in the EVPN-VXLAN fabric. The alignment involves mapping the distributed VXLAN Transit Gateway through the distributed VXLAN connection to a corresponding Layer 3 VNI in the physical EVPN-VXLAN network fabric.

Once control plane and tenant associations are in place, the two environments can advertise and learn workload routes dynamically. As a result, NSX virtual workloads and physical endpoints coexist in one cohesive routing domain. In the data plane, traffic from the ESX host flows seamlessly over VXLAN encapsulation, extending Layer 3 reachability across the entire network topology.

Note:

The ESX TEP VMkernel interface can handle both Geneve and VXLAN traffic simultaneously.

For a summary of the configuration steps and more details about how distributed VXLAN connectivity works, see the next sections on this page.

Configuration Summary

To configure distributed VXLAN connectivity, complete the following high-level steps:

1. In version 9.1.1 and later, enable support for advanced networking services.
2. Establish a common control plane by creating a Route Controller.
   a. Create the Route Controller cluster.
   b. Configure the BGP interface of the Route Controller.
   c. Configure BGP sessions between the Route Controller and BGP peers in the physical EVPN-VXLAN network fabric.
3. Map VPC tenants in NSX to VRF tenants in EVPN-VXLAN.
   a. Add an external connection of distributed VXLAN type.
   b. Add a Transit Gateway, and attach the distributed VXLAN connection to the Transit Gateway.
4. In version 9.1.1 and later, configure the following to use advanced networking services.
   a. Create a virtual network appliance (VNA) cluster to provision the services.
   b. Configure the connectivity profile associated with the transit gateway.

Route Controller: The Common Control Plane

The Route Controller functions as the common control plane that unites the virtual NSX and physical EVPN-VXLAN network fabrics. Using the industry-standard MP-BGP protocol with EVPN, the Route Controller establishes BGP sessions with the EVPN gateways in the external physical network.

This setup enables robust and scalable routing information exchange, ensuring dynamic and efficient path selection across the entire infrastructure. Network administrators gain a continuously updated, synchronized view of the unified network topology.

Tenant Alignment Across Virtual and Physical Fabrics

In an EVPN-VXLAN network fabric, every VRF tenant operates within its own Layer 3 VNI (L3VNI) — the logical VXLAN segment that carries inter-subnet routing information (EVPN Type 5 routes).

In NSX, the distributed VXLAN Transit Gateway provides routed connectivity between VPCs and external networks.

Distributed VXLAN connectivity relies on a one-to-one mapping between each distributed VXLAN Transit Gateway and a corresponding L3VNI in the physical network fabric. To specify this mapping, you first configure an external connection of Distributed VXLAN type that defines the target L3VNI. Then you attach this external connection to the Transit Gateway for the corresponding VPCs.

The result is a unified network with seamless connectivity across both NSX and EVPN-VXLAN network fabrics.

Dynamic Route Exchange

Once configured, distributed VXLAN connectivity allows for the automatic exchange of EVPN Route Type 5 routes between the virtual network and physical EVPN-VXLAN network fabric.

From NSX to EVPN-VXLAN network fabric:

- The Route Controller advertises /32 prefixes for each virtualized workload, identifying each workload and its virtual tunnel endpoint (VTEP) on the ESX host.

- You can summarize these routes at the EVPN gateway of the EVPN-VXLAN fabric to optimize the fabric's routing tables.

From EVPN-VXLAN network fabric to NSX:

- The EVPN gateway advertises prefixes representing external workloads, providing visibility and reachability to NSX. Virtual NSX workloads and other endpoints in the physical network fabric coexist in one cohesive routing domain.

This dynamic, standards-based route exchange removes the need for static and manual configurations and enables cloud-scale mobility across both fabrics.

End-to-End VXLAN Data Path

In the data plane, distributed VXLAN connectivity uses a symmetric model to provide consistent routing behavior no matter where workloads are located, delivering line-rate forwarding between NSX, the EVPN-VXLAN network fabric, and external resources.

1. NSX forwards workload traffic through the VTEP on the ESX host to the EVPN gateway. The ESX VMkernel interface can handle both Geneve and VXLAN encapsulation simultaneously.
2. The EVPN gateway routes packets through the appropriate Layer 3 VNI to the destination VTEP (such as a leaf switch) in the EVPN-VXLAN network fabric.

Advanced Networking Services

In version 9.1.1 and later, distributed VXLAN connectivity supports the following advanced networking services:

- Network address translation (NAT), including VPC Default Outbound NAT (SNAT)
- NSX Native Load Balancer for VCF infrastructure
- AVI Load Balancer in active/standby and active/active L2 Scale-Out modes
- Distributed DHCP Server
- DHCP Relay to external servers

To use advanced networking services, first enable the feature by configuring the minimum OS version for all host transport nodes in your NSX deployment. Then create a virtual network appliance (VNA) cluster that will provision services to the VPC. Finally, create a transit gateway with distributed VXLAN connectivity and configure the associated connectivity profile that defines the northbound connectivity configuration for VPCs.

Create a Route Controller

The Route Controller is a two-node cluster that serves as the common control plane for distributed VXLAN connectivity. Distributed VXLAN connectivity integrates the virtual NSX fabric with the external EVPN-VXLAN fabric to create a single, unified networking environment.

Nodes in the NSX environment use Multiprotocol Border Gateway Protocol (MP-BGP) to exchange routing information with devices in the physical fabric, such as an EVPN gateway.

You configure the BGP interface of the Route Controller and settings for the communication sessions between the Route Controller and BGP peers in the physical fabric. For more information about Route Controllers in the context of a unified networking environment, see Distributed VXLAN Connectivity.

High Availability Mode for Route Controller

- You can deploy one or two nodes per Route Controller cluster. Unless you are configuring a lab environment, deploy two nodes to realize the benefits of high availability.
- By default, the Route Controller uses the settings in the **default-vna-cluster-high-availability-profile**. You can edit the Route Controller to use a different high availability profile.
- The Route Controller uses the **Maximum HA VIP Hold Timer** setting in the high availability profile to determine the availability status of nodes in the cluster.
- Starting with the VCF version 9.1.1, the Active Active mode is also available for a route controller. The Active Active mode prevents routing convergence delays and traffic disruptions that can occur in the Active Standby mode when an active route controller fails. If you select the Active Standby mode, you require a minimum of one node for the cluster. If you select the Active Active mode, you require a minimum of two nodes for the cluster and can have up to eight

active Route Controller nodes for this mode. When you are configuring a Route Controller node on the first day, you are allowed to add only the minimum number of required nodes . You can add more nodes later by editing the Route Controller node.

For the Active Active mode, BGW must be manually configured with separate BGP neighbors for each Route Controller node.

- In the Active Standby mode, both Route Controller nodes share the same VLAN and only the active node establishes BGP sessions. In the Active-Active mode, each Route Controller node acts as an independent router establishing its own BGP sessions with the border gateways and having no restriction of sharing the same VLAN.

  The following table shows the difference between the two modes:

Feature	Active Standby	Active Active
VLAN restriction	Both nodes must use the same VLAN/subnet.	Each node can use the same VLAN or a different VLAN/subnet. Each interface can be placed in a different failure domain based on the VLAN availability.
Source of BGP sessions	Uses a BGP floating IP (VIP). The active node owns the VIP and sources all BGP sessions from it.	No floating IP is used. Each node sources BGP from its own interface IP.
Number of interfaces	One shared interface (through VIP).	One interface per Route Controller node, each with its own IP.
Node behavior	Only the active node peers. The standby does nothing until failover.	Each node peers independently and acts like a standalone router for BGP. Both Route Controller nodes advertise the same VM IP prefixes to the BGW simultaneously.

- When creating BGP peers for the Active Active mode, you must specify:
  - Neighbor IP - the border gateway address.
  - Source IP - the specific Route Controller node's interface IP.
  - Gateway IP - optional; required for multihop BGP reachability to the directly connected gateway.

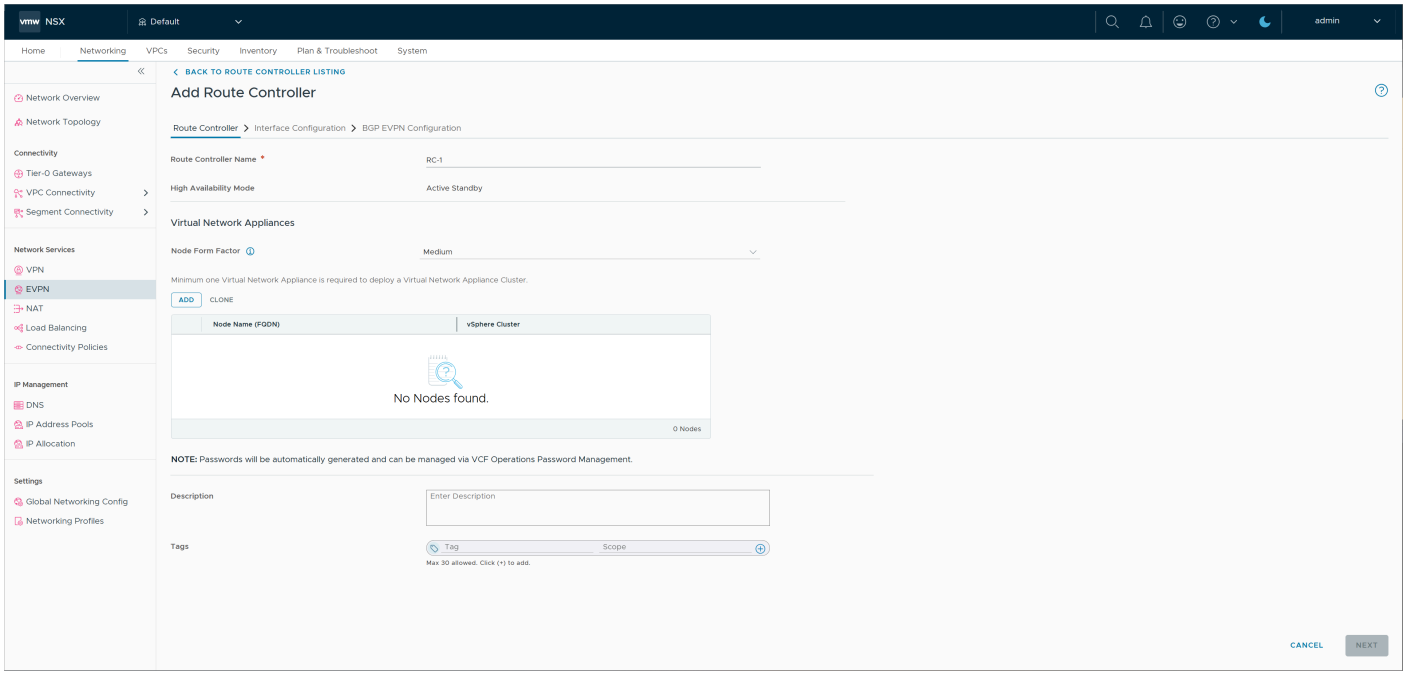
    If you want the same neighbor on both Route Controllers, you must configure the neighbor twice (once per source IP) because multihop scenarios might need different gateway IPs per node.

    Note:

    Starting from the VCF version 9.1.1, the route-controller no longer advertises a default route to the ESX. If a default route must be advertised to the ESX, it needs to be originated by the BGW.

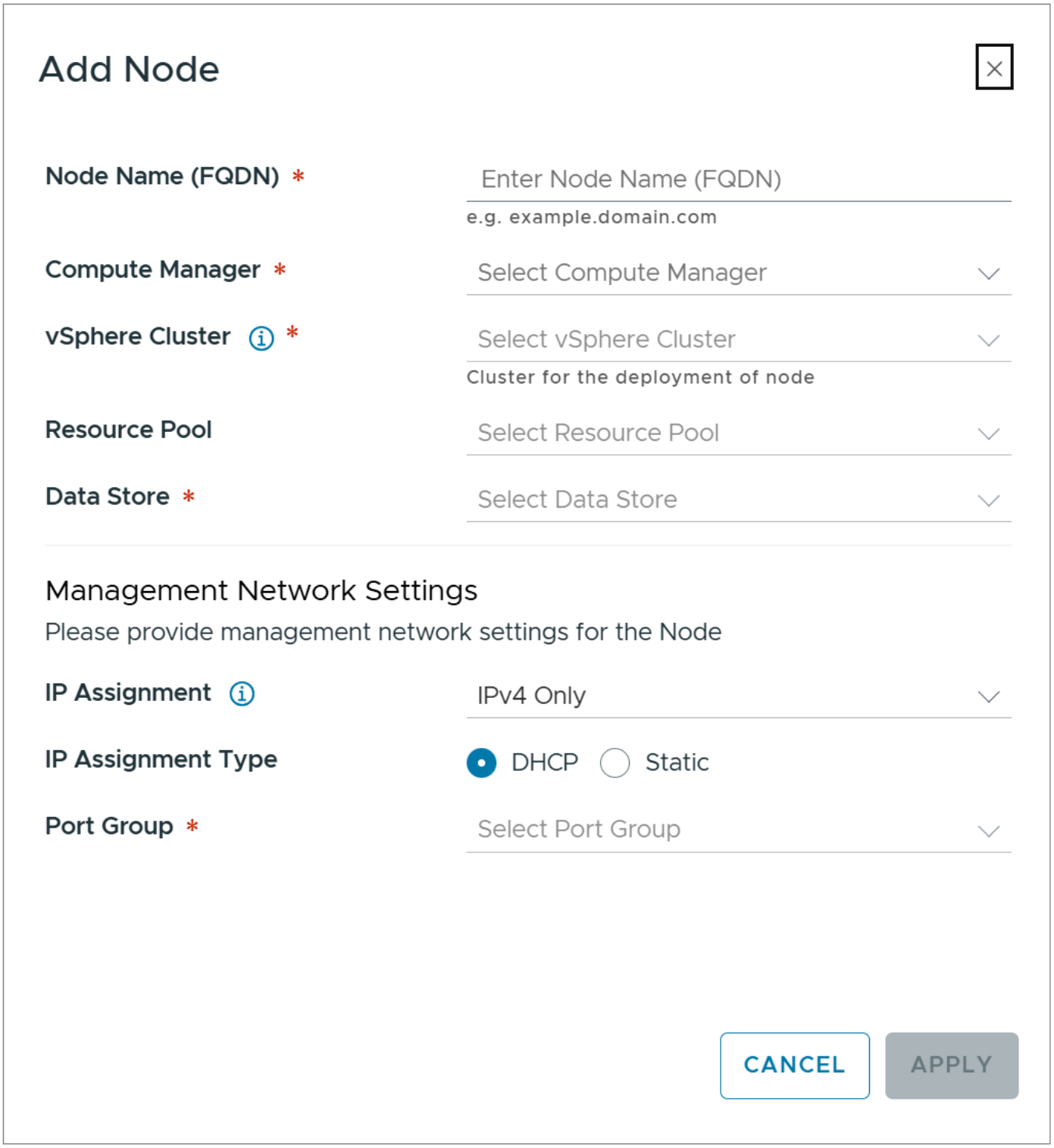
1. From a browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.
2. Navigate to **Networking** > **EVPN** > **Route Controller**.
3. Click **Add Route Controller**.

   The first page of the Add Route Controller workflow appears.


NSX
Default
admin
Home
Networking
VPCs
Security
Inventory
Plan & Troubleshoot
System
BACK TO ROUTE CONTROLLER LISTING
Add Route Controller
Network Overview
Network Topology
Connectivity
Tier-0 Gateways
VPC Connectivity
Segment Connectivity
Network Services
VPN
EVPN
NAT
Load Balancing
Connectivity Policies
IP Management
DNS
IP Address Pools
IP Allocation
Settings
Global Networking Config
Networking Profiles
Route Controller
Interface Configuration
BGP EVPN Configuration
Route Controller Name *
RC-1
High Availability Mode
Active Standby
Virtual Network Appliances
Node Form Factor
Medium
Minimum one Virtual Network Appliance is required to deploy a Virtual Network Appliance Cluster.
ADD
CLONE
Node Name (FQDN)
vSphere Cluster
No Nodes found.
0 Nodes
NOTE: Passwords will be automatically generated and can be managed via VCF Operations Password Management.
Description
Enter Description
Tags
Tag
Scope
Max 30 allowed. Click (+) to add.
CANCEL
NEXT

4. Enter the **Route Controller Name**.
5. Enter the **High Availability Mode**.
6. For **Form Factor**, specify the size of nodes in the Route Controller cluster.
   - **Small** - Suitable for proof-of-concept (PoC) and lab environments only.
   - **Medium** - Suitable for production environments requiring enterprise-grade prefix scale.

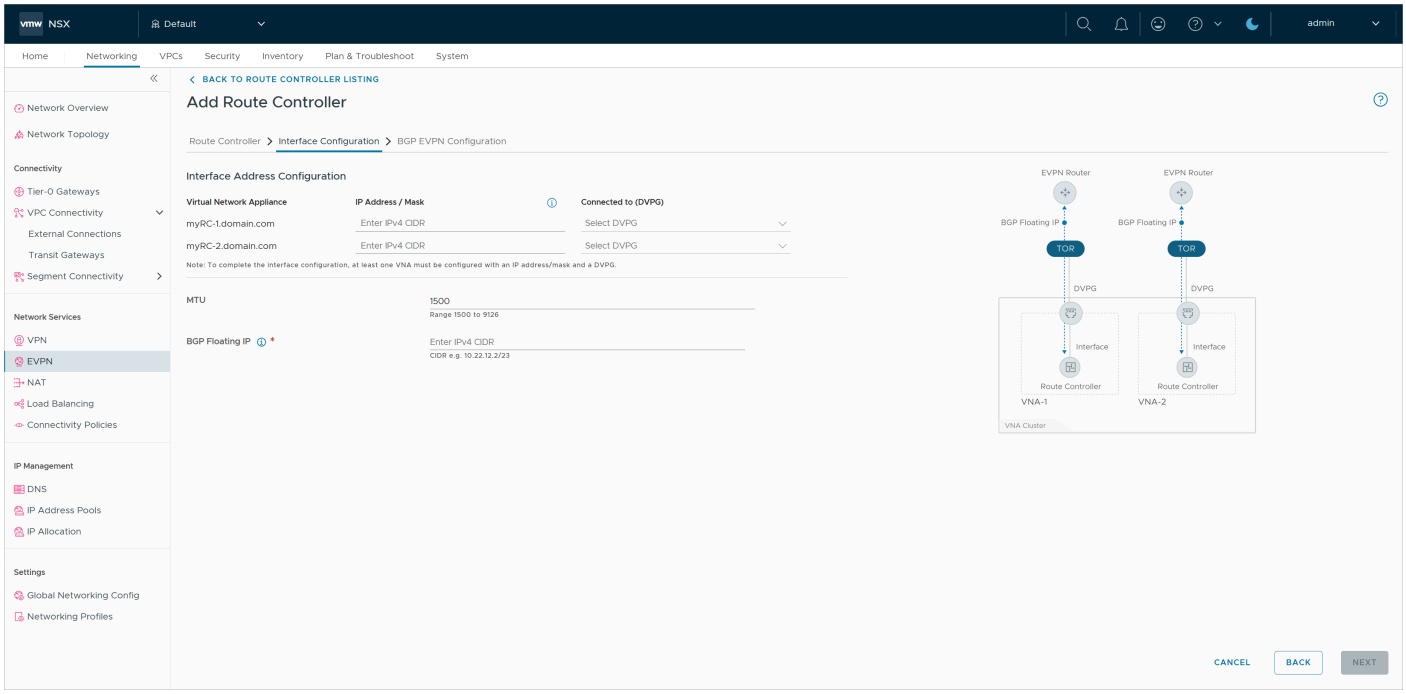
7. Click **Add** to add a new node to the Route Controller cluster and configure node settings.



Setting	Description
vSphere Cluster	Designate the vSphere cluster that the node will join.
Resource Pool	Assign a resource pool for the node.
Host Group Affinity	Select **Yes** if you want to pin the node to a given host or group of hosts. Best Effort Restart (BER) policy is applied to the node. Nodes are gracefully shut down during the host remediation.
Host Group	If you have selected the **Host Group Affinity** as **Yes**, then select a host group where the node is to be deployed. Host groups are managed in the vSphere Client under cluster configurations.
Data Store	Select the data store from which to install the Route Controller.
Management Network Settings	The following settings define the Route Controller management interface, which enables SSH management access to the Route Controller and internal communication with NSX Manager.
IP Assignment	Specify the IP version used to assign IP addresses to the Route Controller node.
IP Assignment Type	If you selected static IPv4 Only, IPv4 & IPv6, or IPv6 Only, specify the following: • **Management CIDR**: Enter the IP address of the node in CIDR notation. • **Default Gateway**: Enter the gateway IP address of the node.
Port Group	Select the vSphere distributed virtual port group (DVPG) used to connect to the management interface of the Route Controller node. This interface must be reachable from NSX Manager. The Route Controller node establishes communication with the NSX Manager management interface.

Click **Apply** to save your node configuration.

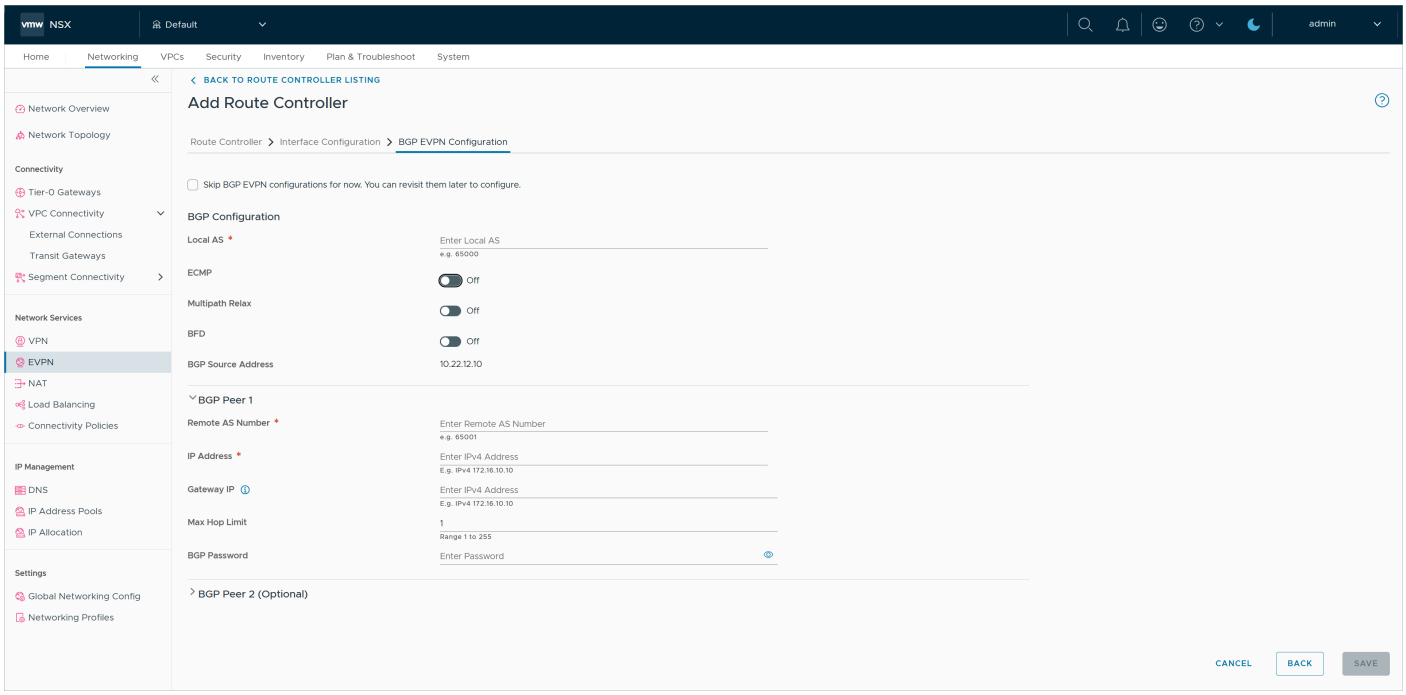
8. Repeat the previous step to add a second node to the cluster. You can select the first node in the list and click **Clone** to quickly create the second node based on existing settings.
9. Specify a **Description** and any **Tags** for the Route Controller. Then click **Next** to proceed to the next stage of the workflow.
10. Configure the BGP interface of the Route Controller to establish BGP peering with EVPN gateways in the external EVPN-VXLAN network fabric.



Setting	Description
MTU	Specify the Maximum Transmission Unit (MTU), in bytes, that is allowed for packets transmitted through the BGP interface.
BGP Floating IP	Specify the BGP source address in CIDR notation in case you selecetd the Active Standby mode. This IP address acts as a virtual IP (VIP) and is owned by the active Route Controller node. This IP address must belong to the same subnet as the interfaces that you specified earlier.

Click **Next** to proceed to the next stage of the workflow.

11. Specify the BGP EVPN configuration for the BGP session between the Route Controller and the BGP peers on the EVPN gateways in the physical network fabric. Complete the configuration for every BGP peer in the physical network fabric.



Alternatively, you can opt to skip the BGP EVPN configuration for now and finish creating the Route Controller. You can return to the BGP EVPN settings later to complete the configuration.

Setting	Description
BGP Configuration	
Local AS	Enter the unique Autonomous System (AS) number that identifies the network containing the Route Controller.
ECMP	Specify whether to enable Equal-Cost Multi-Path (ECMP) routing for the BGP session.
Multipath Relax	Specify whether to allow load sharing across multiple paths that differ only in AS-path attribute values but have the same AS-path length.
BFD	Specify whether to enable Bidirectional Forwarding Detection (BFD) for the BGP session.
BGP Source Address	This read-only field is automatically set to the **BGP Floating IP** configured in the previous step in case you selected the Active Standby mode.
BGP Peer	
Remote AS Number	Enter the unique AS number that identifies the physical network fabric containing the BGP peer on the EVPN gateway.
IP Address	Enter the IP address of the BGP neighbor with which the Route Controller node will establish its peering session.
Source Address	Enter the IP address of the node fom which the BGP peering session needs to originate from in case you selected the Active Active mode.
Gateway IP	Enter the gateway IP address of the BGP peer. For example, the gateway of the BGP peer might be the Layer 3 switch that sits between the Route Controller and the BGP peer on the EVPN gateway.
Max Hop Limit	Specify the maximum number of hops allowed between the Route Controller and the BGP peer on the EVPN gateway.
BGP Password	Optionally specify a password to protect the BGP session.

12. Click **Save** to apply your configurations and create the Route Controller.

    After completion, the new Route Controller appears on the following pages:

    - **Networking** > **EVPN** > **Route Controller**
    - **System** > **Fabric** > **VNA Clusters**

13. (Optional) Customize the high availability profile for the Route Controller cluster.

    Note:

    You must create the Route Controller first before you can customize its high availability profile.

    By default, a new Route Controller uses the high availability settings in the **default-vna-cluster-high-availability-profile**. The **Maximum HA VIP Hold Timer** setting in the profile defines how long the Route Controller can wait during BGP connection failures before it moves the virtual IP (VIP) to the standby node. The default value of this setting is 210 seconds.

    To modify the **Maximum HA VIP Hold Timer** setting and other high availability settings, you must create a custom high availability profile and apply it to the Route Controller.

a) Go to **System** > **Fabric** > **Profiles** > **High Availability Profiles** and select **Add High Availability Profile**. Customize the high availability settings and save the new profile, as described in Add a High Availability Profile.
b) Go to **System** > **Fabric** > **VNA Clusters** and locate the Route Controller cluster in the list. Select the **Edit** action and specify the custom high availability profile that you created in the previous step.

Map the virtual private cloud (VPC) tenants in NSX to Virtual Routing and Forwarding (VRF) tenants in the EVPN-VXLAN fabric:

1. Add an external connection of distributed VXLAN type.
2. Add a Transit Gateway with a distributed VXLAN connection.

Enable Advanced Networking Services for Distributed VXLAN Connections

In version 9.1.1 and later, distributed VXLAN connectivity supports advanced networking services such as network address translation (NAT), load balancing, and DHCP relay. To enable support for this feature, verify that all host transport nodes meet the minimum ESX requirements and then use the NSX API to configure the `host_tn_min_os_version` value for your NSX deployment.

Verify that each host transport node in your NSX environment is running ESX 9.1.1.0 or later.

For a detailed list of supported services, see the "Advanced Networking Services" section in Distributed VXLAN Connectivity.

1. Use the following API call and request payload to set the `host_tn_min_os_version` value.

```
PUT https://<nsx-mgr>/policy/api/v1/infra/sites/default/enforcement-points/default/host-transport-nodes/
feature-opt-in

{
   "host_tn_min_os_version": "9.1.1.0"
}
```

2. To retrieve the current `host_tn_min_os_version` value, you can use the following API call.

```
GET https://<nsx-mgr>/policy/api/v1/infra/sites/default/enforcement-points/default/host-transport-nodes/
feature-opt-in
```

Example response:

```
{
  "host_tn_min_os_version": "9.1.1.0",
  "resource_type": "FeatureOptIn",
  "id": <id>,
  "display_name": <display_name>,
  "_create_time": 1780982377053,
  "_create_user": "admin",
  "_last_modified_time": 1780982377053,
  "_last_modified_user": "admin",
  "_system_owned": false,
  "_protection": "NOT_PROTECTED",
  "_revision": 0
}
```

IP Address Management (IPAM)

To manage IP addresses, you can configure DNS (Domain Name System), DHCP (Dynamic Host Configuration Protocol), IP address pools, and IP address blocks.

> **Note:**
> IP blocks are used by NSX Container Plugin (NCP). For more info about NCP, see the *NSX Container Plug-in for Kubernetes and Cloud Foundry - Installation and Administration Guide*.

Add an NSX DNS Zone

You can configure DNS zones for your DNS service. A DNS zone is a distinct portion of the domain name space in DNS.

When you configure a DNS zone, you can specify a source IP for a DNS forwarder to use when forwarding DNS queries to an upstream DNS server. If you do not specify a source IP, the DNS query packet's source IP will be the DNS forwarder's listener IP. Specifying a source IP is needed if the listener IP is an internal address that is not reachable from the external upstream DNS server. To ensure that the DNS response packets are routed back to the forwarder, a dedicated source IP is needed. Alternatively, you can configure SNAT on the logical router to translate the listener IP to a public IP. In this case, you do not need to specify a source IP.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **IP Management** > **DNS**.
3. Click the **DNS Zones** tab.
4. To edit a default zone, click
   
   next to the object you want to edit and click **Edit**.
   a) Enter a name and optionally a description.
   b) Enter the IP address of up to three DNS servers.
   c) Optional: Enter an IP address in the **Source IP** field.
5. To edit an FQDN zone, click
   
   next to the object you want to edit and click **Edit**.
   a) Enter a name and optionally a description.
   b) Enter a FQDN for the domain.
   c) Enter the IP address of up to three DNS servers.
   d) Optional: Enter an IP address in the **Source IP** field.
6. Click **Save**.

Add an NSX DNS Forwarder Service

You can configure a DNS forwarder to forward DNS queries to external DNS servers.

Before you configure a DNS forwarder, you must configure a default DNS zone. Optionally, you can configure one or more FQDN DNS zones. Each DNS zone is associated with up to 3 DNS servers. When you configure a FQDN DNS zone, you specify one or more domain names. A DNS forwarder is associated with a default DNS zone and up to 5 FQDN DNS zones. When a DNS query is received, the DNS forwarder compares the domain name in the query with the domain names in the FQDN DNS zones. If a match is found, the query is forwarded to the DNS servers specified in the FQDN DNS zone. If a match is not found, the query is forwarded to the DNS servers specified in the default DNS zone.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **IP Management** > **DNS**.
3. Click **DNS Services**.
4. Click **Add DNS Service**.
5. Enter a name.
6. Select a tier-0 or tier-1 gateway.
7. Enter the IP address of the DNS service.
   Clients send DNS queries to this IP address, which is also known as the DNS forwarder's listener IP.

8. Select a default DNS zone.
9. Select up to five FQDN zones.
10. Select a log level.
11. Enter a description.
12. Click the **Admin Status** toggle to enable or disable the DNS service.
13. Optional: Change the cache size.
14. Click **Save**.

Add an IP Address Pool in NSX

You can configure IP address pools for use by components such as DHCP.

1. With admin privileges, log in to NSX Manager.
2. Navigate to **Networking** > **IP Address Pools**.
3. Click **Add IP Address Pool**.
4. Enter a name and optionally a description.
5. From the **Subnets** column, click **Set** to add the subnets.
6. Click the **Add Subnet** drop-down menu and choose either **IP Block** or **IP Ranges**:
   - Choose **IP Block** to use an IP block of the allowable IP addresses:

Setting	Description
Select IP Blocks	Choose an IP address block.
Size	Specify the size or the number of IP addresses in the IP address block.
Allocation Range	Specify the allocation range for the IP address block. **Note:** For an IPv6 address block, you must specify an allocation range. The allocation range value cannot be more than 1048576.
Auto Assign Gateway	Click the toggle to enable or disable automatic gateway IP assignment.

   - Choose **IP Ranges** to specify the range of IP addresses that can be used:

Setting	Description
Enter IPv4 or IPv6 Ranges	Enter the range of IPv4 or IPv6 addresses. **Note:** Transport nodes and tunnel endpoints (TEPs) do not allow a mixed IP address pool to be used (an IP pool with both IPv4 and IPv6 ranges). Ensure that the IP range consists of only IPv4 or only IPv6 addresses.
CIDR	Enter the IP address range in CIDR format.
Gateway IP	Enter the gateway address.
DNS Server	Enter the list of the DNS (domain name system) server IP addresses separated by commas.
DNS Suffix	Enter the DNS suffix. For example, `corp.local`.

7. Click **Add** and then **Apply**.
8. Click **Save**.

Note:
After you add an IP address pool and IP addresses have been allocated from the pool, you will not be able to delete the pool. If you want to expand the pool, you must add new IP ranges. For example, if the existing range is 192.168.1.11 - 192.168.1.20 and you want to expand it to 192.168.1.10 - 192.168.1.30, add the following two ranges:

- 192.168.1.10 - 192.168.1.10
- 192.168.1.21 - 192.168.1.30

Add an NSX IP Address Block

You can configure IP address blocks for use by other resources and view detailed statistics about IP block usage by clicking the View Statistics link of the respective IP address block from the list. You can also view IPs allocated to a tenant and the service using the allocated IP.

Once an IP address block with external visibility is created, it can be shared with tenants and the tenants can use the block to request for an allocation. These allocations are then used by services like NAT or static-routes.

You can also run the available_subnets API on an IP address block to check the number of free subnets of a particular size that the IP block has.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **IP Management** > **IP Address Pools**.
3. Click the **IP Address Blocks** tab.
4. Click **Add IP Address Block**.
5. Enter the following details.

Field Name	Description
Name	Enter a name for this IP address block.
Visibility	If you are creating this IP address block specifically for the consumption by a VPC, define whether it should be used for public subnets (external) or private subnets (private).
CIDRs/Ranges	Define the CIDRs and IP ranges for this IP address block. These CIDRs or IP ranges can be shared with tenants or requested by them to carve the subnets. Note that CIDRs must not overlap with the IP ranges. Also, you cannot specify a mixed address type. They should all be either IPv4 or IPv6 type. If you are creating this IP address block for VLAN extension, then enter the CIDR of that VLAN.
Excluded IPs	Specify the IPs from the mentioned CIDRs/Ranges field that you want to reserve and keep them away from getting allocated.
Overall Usage	Displays the block's consumption information.
Status	Displays if the IP address block is successfully created.
Reserved for Specific Subnet	Turn on the toggle if you want to reserve the IP address block for VLAN extension. In this case the IP address block can be assigned only to a single VLAN-extended subnet. Note that the visibility is external in this case and the address block contains only a single CIDR.
Description	Provide a description for the IP address block.
Tags	Specify tags for the IP address block.

6. Click **Save**.

Allocate IP or CIDR from an IP Address Block

You can view details of each IP address block and where it is being consumed. You can also request for an IP or CIDR from the IP address block for allocating it to a project or VPC.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **IP Management** > **IP Allocation**.
3. Click **Request IP/CIDR** .
4. Enter the following details.

Field Name	Description
Allocate To	Specify whether you want to request the allocation for a project or VPC.
Request Type	Specify the type of allocation, IP or CIDR, that you require.
VPC	If you selected `VPC` in the **Allocate To** field, select the VPC for which the allocation is required.
IP Address Block	Select the IP address block from which an IP or CIDR should be allocated.
Auto-Allocate from IPAM	Turn off the toggle when you do not want the system to automatically allocate the IP/CIDR to a project or VPC, but when you want to make specific allocations.
IP Address	Enter the specific IP address that you want to allocate. This field is displayed when **Request Type** is IP Address and the **Auto-Allocate from IPAM** toggle is turned off.
CIDR	Enter the specific CIDR that you want to allocate. This field is displayed when **Request Type** is CIDR and the **Auto-Allocate from IPAM** toggle is turned off.

5. Click **Save**.

Integrating NSX with Infoblox

You can now integrate Infoblox NIOS DDI, a third-party IP address management (IPAM) system, with NSX.

As many enterprises use a third-party IPAM (such as Infoblox) to centrally manage their IP addresses, they would want to continue using the same IPAM even for VCF so that they do not have to reserve separate CIDRs for VCF.

In Infoblox NIOS DDI, a network container is equivalent to an NSX IP address block. Both represent a top-level IP CIDR range. With the integration of NSX and Infoblox, NSX imports these network containers as NSX External IP address blocks thus providing enterprises the flexibility to continue using Infoblox instead of NSX IPAM. The imported network containers can then be consumed like native NSX IP blocks to create subnets and to allocate IP addresses to workloads.

Requirements and Limitations

The Infoblox Grid Master must present a valid TLS certificate to NSX for successful registration. The following certificate configurations are supported according to standard HTTPS client requirements:

- CA-signed certificates: Infoblox can present a complete chain, a chain without the root CA, or only the leaf certificate. However, chains with missing intermediate certificates are not supported.
- Self-signed certificates: The certificate chain must contain only the self-signed certificate without any CA certificates.

The following features are not supported with Infoblox integration:

- IPv6
- Subnet exclusive IP blocks (Direct VLAN feature)
- Excluded IPs
- Allocation of a subnet of size equal to its block size

To use Infoblox network containers for VPC subnets that are connected to a distributed Transit Gateway (DTGW), a container matching the entire CIDR of the physical VLAN network must be dedicated to VCF.

If the network container overlaps with an existing network on the Network View, create a network container in another Network View and dedicate it to VCF.

Guest Customization for VM Launch

When launching a VM in a VCF Automation which is integrated with Infoblox NIOS, the following Cloud-Init Raw Configuration is needed to ensure that a DNS Host Record is added:

1. In the VM deployment wizard, navigate to Guest Customization.
2. Select the **Cloud-Init** radio button.
3. Click the **Raw Configuration** tab.
4. Enter the following YAML block:

```
#cloud-config
fqdn: <FQDN>
hostname: <hostname>
prefer_fqdn_over_hostname: true
create_hostname_file: true
```

De-register Infoblox

To de-register Infoblox, first remove all imported objects (containers as IP blocks) and then de-register from UI or by using API.

Register Infoblox NIOS DDI with NSX

- Before you register Infoblox with NSX, define the following Extensible Attributes (EAs) in the Infoblox Grid Manager by navigating to **Administration** > **Extensible Attributes**. Each attribute must be of the String data type.
  - VCFNetworkingAllocatedPermanently
  - VCFNetworkingCreationTimestamp
  - VCFNetworkingNsxClusterId
  - VCFNetworkingReservedIp
  - VCFNetworkingObjectID
  - VM ID
  - Created-By-VCF
  - IpObjectID
- Verify that you have the required user credentials to access and integrate your Infoblox NIOS DDI on NSX. Open the **Administration** tab in the Infoblox Grid Manager and configure administrator, groups, and roles entries. The user account must be a member of a group that meets one of the following criteria:
  - Must have administrator or superuser permissions
  - Custom group with IPAM, DNS, Grid, and Extensible Attributes permissions

  These permissions enable NSX to read network containers, create and delete networks, allocate and release IP addresses, manage DNS host records, and add extensible attributes to these objects. For more information about configuring user permissions, see the Infoblox product documentation.

Perform the following steps to register Infoblox.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **IP Management** > **IP Address Pools**.
3. Click the **IP Service Registration** tab.
4. Click **Register** to register Infoblox and enter the following details.

Field	Description
Server FQDN/IP	**Enter the FQDN or IP of Infoblox that you want to integrate with NSX.**
Port Number	Enter the port number of Infoblox.
Username	Enter the user name to access Infoblox.
Password	Enter the password for Infoblox.

5. Click **Register**.

   The TLS certificate of the Infoblox Grid Master is displayed. Review the certificate details.

6. Click **Accept** to accept the trust certificate.

   The Infoblox gets registered.

7. Edit the configuration to provide Network and DNS views by clicking **Actions** > **Edit Registration**.

8. To use network containers from Infoblox as IP address blocks, import network containers by clicking **Actions** > **Import Network Containers**.

   The containers get imported and you can view them by going to the **IP Address Blocks** tab.

Networking Settings

You can configure networking settings for IPv6, VNI (Virtual Network Identifier) pools, gateways, multicast, and BFD (Bidirectional Forwarding Detection).

Configuring Multicast on an NSX Tier-0 or Tier-1

You can configure multicast on a tier-0 gateway and optionally on a tier-1 gateway for an IPv4 network to send the same multicast data to a group of recipients. In a multicast environment, any host, regardless of whether it is a member of a group, can send to a group. However, only the members of a group will receive packets sent to that group.

The multicast feature has the following capabilities and limitations:

- PIM Sparse Mode with IGMPv2.
- No Rendezvous Point (RP) or Bootstrap Router (BSR) functionality on NSX. However, RP information can be learned via PIM Bootstrap Messages (BSMs). In addition, multiple Static RPs can be configured.
  When a Static RP is configured, it serves as the RP for all multicast groups (224/4). If candidate RPs learned from BSMs advertise candidacy for the same group range, the Static RP is preferred. However, if candidate RPs advertise candidacy for a specific group or range of groups, they are preferred as the RP for those groups.
- The Reverse Path Forwarding (RPF) check for all multicast-specific IPs (senders of data traffic, BSRs, RPs) requires that a route to each of them exists.
- The RPF check requires a route to each multicast-specific IP with an IP address as the next hop. Reachability via device routes, where the next hop is an interface index, is not supported.
- Both tier-0 and tier-1 gateways are supported. To enable multicast on a tier-1 gateway, an Edge cluster must be selected and the tier-1 gateway must be linked to a tier-0 gateway that also has multicast enabled.
- All uplinks on a tier-0 gateway are supported.
- Multiple Static RPs with discontinuous group ranges are supported.
- IGMP local groups on uplink interfaces are supported.
- PIM Hello Interval and Hold Time are supported.
- The NSX Edge cluster can be in active-active or active-standby mode. Note that since unicast reachability to NSX in an active-active cluster is via ECMP, it is imperative that the northbound PIM router selects the ECMP path that matches a PIM neighbor to send PIM Join/Prune messages to NSX. In this way it will select the active Edge node which is running PIM.
- For both active-active and active-standby modes, PIM must be enabled on at least one of the tier-0 uplinks for multicast to work.
- East-west multicast replication: up to 4 VTEP segments for maximum replication efficiency.
- ESX host and NSX Edge only.
- Layer 2 bridge attached to a downlink segment not supported.
- Gateway Firewall services are not supported for multicast. Distributed Firewall and Bridge Firewall are supported.
- Multi-site (NSX Federation) not supported.
- Multi-VRF not supported.

Multicast Configuration Prerequisites

Underlay network configurations:

- Acquire a multicast address range from your network administrator. This will be used to configure the Multicast Replication Range when you configure multicast on a tier-0 gateway (see Configure Multicast Routing on a Tier-0 Gateway).

- Enable IGMP snooping on the layer 2 switches to which GENEVE participating transport nodes are attached. If IGMP snooping is enabled on layer 2, IGMP querier must be enabled on the router or layer 3 switch with connectivity to multicast enabled networks.

Multicast Configuration Steps

1. Create an IGMP profile. See Create an IGMP Profile.
2. Optionally create a PIM profile to configure a Static Rendezvous Point (RP). See Add an NSX PIM Profile.
3. Configure a tier-0 gateway to support multicast. See Add an NSX Tier-0 Gateway and Configure Multicast Routing on a Tier-0 Gateway.
4. Optionally configure tier-1 gateways to support multicast. See Add an NSX Tier-1 Gateway.

Create an IGMP Profile

Internet Group Management Protocol (IGMP) is a multicast protocol used in IPv4 networks.

Note that the IGMP snooping timeout for reports is 2 times the general query timeout. By default, the IGMP snooping timeout value is 120 seconds. On ESX, the default IGMP snooping timeout value is 60 seconds.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Networking Profiles**.
3. Click the **Multicast** tab.
4. Click **Add IGMP Profile**.
5. Enter a profile name and the following profile details.

Option	Description
Query Interval (seconds)	Interval between general query messages. A larger value causes IGMP queries to be sent less often. Default: 30. Range: 1 - 1800.
Query Max Response Time (seconds)	Maximum allowed time before sending a response to a membership query message. Default: 10. Range: 1 - 25.
Last Member Query Interval (seconds)	Maximum amount of time between group-specific query messages, including those sent in response to leave-group messages. Default: 10. Range: 1 - 25.
Robustness Variable	Number of IGMP query messages sent. This helps alleviate the risk of loss of packets in a busy network. A larger number is recommended in a network with high traffic. Default: 2. Range: 1 - 7.

Add an NSX PIM Profile

Protocol Independent Multicast (PIM) is a collection of multicast routing protocols for IP networks. It is not dependent on a specific unicast routing protocol and can leverage whichever unicast routing protocols are used to populate the unicast routing table.

This step is optional. It is needed only if you want to configure a Static Rendezvous Point (RP). A Rendezvous Point is a router in a multicast network domain that acts as a shared root for a multicast shared tree. If a Static RP is configured, it is preferred over the RPs that are learned from the elected Bootstrap Router (BSR).

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Networking Profiles** > **Multicast**.
3. In the **Select Profile type** drop-down menu, select **PIM Profiles**.
4. Click **Add PIM Profile**.
5. Enter a profile name.
6. Enable or disable Bootstrap Message (BSM) processing.
7. Add one or more Static Rendezvous Point (RP) addresses.
8. Click **Save**.

About IGMP Join

When local IGMP join command for a group is configured on the uplink of a tier-0 service router (SR), IGMP group is joined on the uplink interface on the Edge Linux.

The designated router (DR) on the uplink VLAN interface will issue a PIM (*,g) join towards the Rendezvous Point (RP) upon receiving an IGMP report for the group g.

Traffic flow:

1. IGMP join is configured on the uplink of the non-active multicast node of the tier-0 SR and multicast traffic is forwarded to the non-active node from the RP. Since north-south traffic on non-active multicast node is not processed regardless of it is DR or non-DR for the uplink interface, the packet will be dropped on the non-active node. (s,g) mroute will not be present.
2. IGMP join is configured on the uplink of the active multicast node of the tier-0 SR and multicast traffic is forwarded to the active node from the RP.
   - If the node is non-DR for the uplink interface, the traffic will be dropped.
   - If the node is DR for the uplink interface:
     - If IGMP join is given on the uplink interface which is also the interface to reach Source, then the incoming interface will be same as the outgoing interface for the (s,g) and the mroute will not be installed.
     - If IGMP join is given on the uplink interface which is different from the incoming interface for a given (s,g) then the (s,g) mroute will be installed.

Add an NSX EVPN/VXLAN VNI Pool

You can create a VNI pool to be used when you configure EVPN for a tier-0 gateway. VNI pools cannot have values that overlap.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Global Networking Config** > **EVPN/VXLAN VNI Pool**.
3. To edit an EVPN/VXLAN VNI Pool , click the three-dots and click **Edit in Local Manager**.
4. Enter a name for the pool.
5. Enter a start value.
   The value must be from 75001 to 16777215.
6. Enter an end value.
   The value must be from 75001 to 16777215.

7. Click **Save**.

Configure Global Gateway Settings

You can configure some settings that apply to all gateways.

The settings **EVPN BFD Profile** and **Enable EVPN BFD** are required by the EVPN feature to support layer-2 ECMP and BFD. For more information, see Ethernet VPN (EVPN). For information about adding or editing a BFD profile, see Add an NSX BFD Profile.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Global Networking Config** > **Global Networking Config**.
3. In the **Global Networking Config** tab, click **Edit**.
4. Optional: Update any of the following settings.

Option	Description
Location	Select the sites to apply the global networking configuration.
Gateway Interface MTU	The default is 1500.
Layer-3 Forwarding Mode	The options are **IPv4 Only** and **IPv4 and IPv6**. The default is **IPv4 Only**.
EVPN BFD Profile	The default is **default-external-gw-bfd-profile**.
Enable EVPN BFD	The default is **On**.

5. Click **Save**.

Add a Gateway QoS Profile

Create a QoS (quality of service) profile for your tier-1 gateways to define limits on the traffic rates. You can specify the permitted information rate and the burst size to set the limitations. Any traffic that does not conform to the QoS policy, is dropped. QoS profiles can be set for both ingress and egress traffic, for all traffic types (unicast, BUM, IPv4/IPv6). You can choose to create a different profile for each tier-1 gateway.

Limitations

- Gateway QoS profile is supported only on tier-1 gateways.
- QoS policies on tier-1 gateways apply only to north-south traffic and not to tier-1 gateway overlay segments or service interfaces.
- The tier-1 gateways must be in active-standby mode with an NSX Edge cluster.
- QoS profile is not supported on tier-1 gateways that are configured for distributed routing only.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Networking Profiles**.
3. Click the **Gateway QoS** tab.
4. Click **Add Gateway QoS Profiles**.
5. Enter a name for the profile.
6. Enter the commited bandwidth limit that you want to set for the traffic.
7. Enter the burst size. Use the following guidelines for burst size.
   - *B* is the burst size in bytes.
   - *R* is the committed rate (or bandwidth) in Mbps.
   - *I* is the time interval in milliseconds, to refill or withdraw tokens(in bytes) from the token bucket. Use the `get dataplane` command from the NSX Edge CLI to retrieve the time interval, *Qos_wakeup_interval_ms*. The default value for *Qos_wakeup_interval_ms* is 50ms. However, this value is automatically adjusted by the dataplane based on the QoS configuration.

   The constraints for burst size are:

   - `B >= R * 1000,000 * I / 1000 / 8` because burst size is the maximum amount of tokens that can be refilled in each interval.

- `B >= R * 1000,000 * 1 / 1000 / 8` because the minimum value for *`I`* is 1 ms, taking into account dataplane CPU usage among other constraints.
- `B >= MTU of SR port` because at least the MTU-size amount of tokens need to be present in the token bucket for an MTU-size packet to pass rate-limiting check.

Since the burst size needs to satisfy all three constraints, the configured value of burst size would be:

```
Max (R * 1000,000 * I / 1000 / 8, R * 1000,000 * 1 / 1000 / 8, MTU)
```

For example, if *`R`* = 100 Mbps, *`I`* = 50 ms, and *`MTU`* = 1500, then

```
B >= max (100 * 1000,000 * 50 / 1000/ 8, 100 * 1000,000 * 1 / 1000/ 8, 1500) = 625000 in bytes
```

8. Click **Save**.

Add an NSX BFD Profile

BFD (Bidirectional Forwarding Detection) is a protocol that can detect forwarding path failures. You can create a BFD profile for your Tier-0 static routes.

1. With admin privileges, log in to NSX Manager.
2. Select **Networking** > **Networking Profiles** > **BFD**.
3. Click **Add BFD Profile**.
4. Enter a name for the profile.
5. Enter values for the **Interval(ms)** and **Declare Dead Multiple**. The minimum value of Interval is 500 ms for a VM Edge. The default value is 500 ms.
6. Click **Save**.

Add an NSX DHCP Profile

Before you can configure DHCP on a segment, you must add a DHCP profile in your network. You can create two types of DHCP profiles: DHCP server profile and DHCP relay profile.

A DHCP profile can be used simultaneously by multiple segments and gateways in your network.

- On an overlay segment that is connected to the downlink interface of a tier-0 or a tier-1 gateway, you can attach either a DHCP server profile or a DHCP relay profile.
- On an overlay or VLAN segment that is connected to the service interface of a tier-0 or a tier-1 gateway, you can attach either a DHCP server profile or a DHCP relay profile.
- On an isolated segment (overlay or VLAN) that is not connected to a gateway, you can attach only a DHCP server profile. Isolated segment supports only a Segment DHCP server.

Add an NSX DHCP Server Profile

You can add multiple DHCP server profiles in your network. Further, you can attach a single DHCP server profile to multiple DHCP servers.

- Edge nodes are deployed in the network.
- Edge cluster is added in the network.

1. From your browser, log in with **admin** privileges to an NSX Manager at `https://`*`nsx-manager-ip-address`*.
2. Select **Networking** > **Networking Profiles**.
3. Click the **DHCP** tab, and then click **Add DHCP Profile**.
4. Enter a unique name to identify the DHCP server profile.
5. In the **Profile Type** drop-down menu, ensure that **DHCP Server** is selected.
6. Optional: Enter the IP address of the DHCP server in a CIDR format.

   Note:
   A maximum of two DHCP server IP addresses are supported. You can enter one IPv4 address and one IPv6 address. For an IPv4 address, the prefix length must be <= 30, and for an IPv6 address, the prefix length must

be <= 126. The DHCP server IP address must not overlap with the addresses used in DHCP ranges and DHCP static binding.

If no server IP address is specified, 100.96.0.1/30 is autoassigned to the DHCP server.

The server IP address cannot be any of the following:

- Multicast IP address
- Broadcast IP address
- Loopback IP address
- Unspecified IP address (address with all zeroes)

CAUTION:
After a DHCP server profile is used in your network, preferably avoid editing the server IP addresses in the DHCP server profile. It might cause a failure while renewing or releasing the IP addresses that are leased to the DHCP clients.

7. Select an edge cluster.

   Follow these guidelines:

   - If you are using a Segment DHCP server on a segment, you must select an edge cluster in the DHCP server profile. If an edge cluster is unavailable in the profile, an error message is displayed when you save the segment.
   - If you are using a Gateway DHCP server on the segment, select an edge cluster either in the gateway, or DHCP server profile, or both. If an edge cluster is unavailable in either the profile or the gateway, an error message is displayed when you save the segment.

   CAUTION:
   After a DHCP server profile is used in your network, preferably avoid changing the edge cluster in the DHCP server profile. It might lead to a loss of existing DHCP leases that are assigned to the DHCP clients.

   When a DHCP server profile is attached to a segment that uses a Segment DHCP server, the DHCP service is created in the edge cluster that you specified in the DHCP profile. However, if the segment uses a Gateway DHCP server, the edge cluster in which the DHCP service is created depends on a combination of several factors. For a detailed information about how an edge cluster is selected for DHCP service, see Scenarios: Selection of Edge Cluster for NSX DHCP Service.

8. By default, the edges for the DHCP server are autoallocated from the edge cluster and DHCP HA is configured.

   The system selects a pair of active and standby edge nodes automatically from the available nodes in the edge cluster. If you want to allocate the edges manually from the edge cluster, go to the next step.

9. Optional: Manually allocate the edges from the cluster.
   a) Turn off the **Auto Allocate Edges** toggle button.
   b) Required: From the first drop-down menu, select an edge node.
   c) Optional: From the second drop-down menu, select another edge node.

      The first edge node becomes the active edge, and the second edge node becomes the standby edge. If second edge node is not selected, DHCP HA is not configured.

10. Optional: Click the **Standby Relocation** toggle button to enable standby relocation. By default, this option is set to No.

    Standby relocation means that if the edge node where the active or standby DHCP server is running fails, a new active or standby DHCP server is created on another edge node to maintain high availability. That is, if the edge node where the active DHCP server is running fails, a new active DHCP server is created on another edge node in the same edge cluster. On similar lines, if the edge node where the standby DHCP server is running fails, a new standby DHCP server replaces it on another edge node in the same edge cluster.

    To enable standby relocation, **Auto Allocate Edges** must be set to Yes.

11. Optional: In the **Tag** drop-down menu, enter a tag name. When you are done, click **Add Item(s)**.

    The maximum length of the tag name is 256 characters.

    If tags exist in the inventory, the **Tag** drop-down menu displays a list of all the available tags and their scope. You can select an existing tag from the drop-down menu and add it to the DHCP profile.

12. Click **Save**.

Add an NSX DHCP Relay Profile

You can add a DHCP relay profile to relay the DHCP traffic to remote DHCP servers. The remote or external DHCP servers can be in any subnet, outside the SDDC, or in the physical network.

1. From your browser, log in with **admin** privileges to an NSX Manager at `https://nsx-manager-ip-address`.
2. Select **Networking** > **Networking Profiles**.
3. Click the **DHCP** tab, and then click **Add DHCP Profile**.
4. Enter a unique name to identify the relay profile.
5. In the **Profile Type** drop-down menu, select **DHCP Relay**.
6. Required: Enter the IP addresses of the remote DHCP servers.

   Both DHCPv4 and DHCPv6 servers are supported. You can enter multiple IP addresses. The server IP addresses of the remote DHCP servers must not overlap with the addresses that are used in DHCP ranges and DHCP static binding.

   The server IP address cannot be any of the following:

   - Multicast IP address
   - Broadcast IP address
   - Loopback IP address
   - Unspecified IP address (address with all zeroes)
7. Optional: In the **Tag** drop-down menu, enter a tag name. When you are done, click **Add Item(s)**.

   The maximum length of the tag name is 256 characters.

   If tags exist in the inventory, the **Tag** drop-down menu displays a list of all the available tags and their scope. You can select an existing tag from the drop-down menu and add it to the DHCP profile.
8. Click **Save**.

vDefend Firewall with Advanced Threat Prevention

VMware vDefend Firewall with Advanced Threat Prevention (ATP) is a software-defined Layer 2–7 firewall purpose-built to secure virtualized workloads in a private cloud. It provides stateful firewalling with threat prevention capabilities that protect organizations against advanced threats. VMware vDefend Firewall is available in two form factors: a Distributed Firewall that can be deployed at each vSphere workload and a Gateway Firewall that can be deployed on a vSphere host, either as a Virtual Machine (VM) or as an ISO image on a physical server.

Advanced Threat Prevention (ATP) combines multiple detection technologies—Intrusion Detection/Prevention System (IDS/IPS), Network Sandboxing (Malware Prevention), and Network Traffic Analysis (NTA)—with aggregation, correlation, and context engines from Network Detection and Response (NDR).

For more details about VMware vDefend Firewall, see the VMware vDefend Firewall documentation.

For more details about VMware vDefend Advanced Threat Prevention, see the VMware vDefend ATP documentation.

Inventory

You can configure services, groups, context profiles, and virtual machines for the NSX inventory.

When you click the **Inventory** tab, an overview of the inventory objects is displayed, showing the number of groups, services, virtual machines, context profiles, and L7 access profiles that are in the inventory. In addition, the following information about groups is shown:

- the number of groups used in policies

- the number of groups not used in policies
- the number of groups with members
- the number of groups without members
- the number of identity groups
- the number of identity groups used in policies
- the number of identity groups not used in policies

Add a Network Span

A network span provides the ability for the administrative user to limit the availability of VPC networks on a specific set of vSphere clusters. For more information about network spans, see Transit Gateways.

Perform the following steps to add a network span.

1. With the admin privileges, log in to NSX Manager.
2. Navigate to **Inventory** > **Network Spans.**
3. Click **Add Network Span**.
4. Enter a name for the new network span.
5. If you want to create an exclusive span, in the **Exclusive Span** field turn on the toggle.
6. In the **Members** field, click set to select the clusters for the span.
7. In the **Description** and **Tags** fields, enter a description and set tags for the span.
8. Click **Save**.

Add a Service

You can configure a service, and specify parameters for matching network traffic such as a port and protocol pairing.

You can also use a service to allow or block certain types of traffic in firewall rules. You cannot change the type after you create a service. Some services are predefined and cannot be modified or deleted.

1. With admin privileges, log in to NSX Manager.
2. Select **Inventory** > **Services**.
3. Click **Add Service**.
4. Enter a name.
5. Click **Set**.
6. Select a type.
   The choices are **Layer 2** and **Layer 3 and above**.
7. Under the **Port-Protocol** tab, click **Add Service Entry** to add one or more service entries.
   For layer 2, the only available service type is **Ether**.

   For layer 3 and above, the available service types are **IP**, **IGMP**, **ICMPv4**, **ICMPv6**,**ALG**, **TCP**, and **UDP**.

   Note:

   The following built-in ALGs for DFW and GFW are supported: FTP and TFTP.
8. Click the **Services** tab to add one or more services.
   Any service that you add is considered a nested service because it is included in the service that you are creating. The recommended maximum level of nesting is 3. An example of three levels of nesting: service A includes service B, service B includes service C, and service C includes service D. In addition, cyclic nesting is not allowed. In the previous example, service C cannot include service A or B.

9. Click **Apply**.
10. Optional: Add one or more tags.
11. Optional: Enter a description.
12. Click **Save**.

Add a Group

Groups include different objects that are added both statically and dynamically, and can be used as the source and destination of a firewall rule.

Groups can be configured to contain a combination of virtual machines, IP sets, MAC sets, segment ports, segments, AD user groups, and other groups. Dynamic inclusion of groups can be based on tag, machine name, OS name, or computer name.

> **Note:**
> If you create a group in the API using LogicalPort based criteria, you cannot edit the group in the UI using the AND operator between SegmentPort criteria. If you create a group using Segment, Segment Port, Distributed Port Groups, or Distributed Ports based criteria, disable the "Trust on First Use" option in the group's IP Discovery Profile. Otherwise, the original IP Address of the interface will remain in your group even if its IP Address changes.

If Malicious IP Feed is enabled, a list of known malicious IPs are downloaded from NTICS cloud service. You can create groups to include these downloaded IPs and configure firewall rules to block access to them. A Generic or IP Addresses Only group cannot be converted to IP Addresses Only group with malicious IPs, and the reverse. However, a Generic group can be converted to an IP Addresses Only group without malicious IPs.

Groups can also be excluded from firewall rules, and there are a maximum of 100 groups that can be on the list. IP sets, MAC sets, and AD groups cannot be included as members in a group that is used in a firewall exclusion list. See the VMware vDefend documentation for more information.

If you use Active Directory groups as the source, a single Active Directory group can be used. If both IP and Active Directory groups are needed at the source, create two separate firewall rules.

Groups consisting of only IP addresses, MAC addresses, or Active Directory groups cannot be used in the **Applied to** text box.

> **Note:**
> When a host is added to or removed from a vCenter, the external ID of the VMs on the host changes. If a VM is a static member of a group and the VM's external ID changes, the NSX Manager UI will no longer show the VM as a member of the group. However, the API that lists the groups will still show that the group contains the VM with its original external ID. If you add a VM as a static member of a group and the VM's external ID changes, you must add the VM again using its new external ID. You can also use dynamic membership criteria to avoid this issue.

For Policy Groups containing IPs or MAC addresses, the NSGroup listing API will **NOT** display the ‘members’ attribute. This applies to Groups containing a combination of static members also. For example, if a Policy Group contains IP and DVPG, the NSGroup listing API will not display the members attribute.

For Policy Groups not containing IPs, MAC addresses, or Identity Groups, the member attribute will be displayed in the NSGroup response. However new members and criteria introduced in NSX (such as DVPort and DVPG) will not be included in the MP group definition. Users can view the definition in Policy.

Tags in NSX are case-sensitive, but a group that is based on tags is "case- insensitive." For example, if the dynamic grouping membership criterion is `VM Tag Equals 'quarantine'`, the group includes all VMs that contain either the tags 'quarantine' or 'QUARANTINE'.

1. With admin privileges, log in to NSX Manager.
2. Select **Inventory** > **Groups** from the navigation panel.
3. Click **Add Group**, then enter a group name.
4. If you are adding a group from a Global Manager for NSX Federation, either accept the default region selection, or select a region from the drop-down menu. Once you create a group with a region, you cannot edit the region selection. However, you can change the span of the region itself by adding or removing locations from it. You can create customized regions before you create the group.

   For groups added from a Global Manager in an NSX Federation environment, selecting a region is mandatory. This text box is not available if you are not using the Global Manager.
5. Click **Set**.
6. In the **Set Members** window, select the **Group Type**.

Group Type	Description
Generic	This group type is the default selection. A Generic group definition can consist of a combination of membership criteria, manually added members, IP addresses, MAC addresses, and Active Directory groups. Generic groups with only manually added IP address members are not supported for use in the **Applied To** field in DFW rules. It is possible to create the rule, but it will not be enforced. When you define membership criteria in the group, the members are dynamically added in the group based on one or more criteria. Manually added members include objects, such as segment ports, distributed ports, distributed port groups, VIFs, virtual machines, and so on.
IP Addresses Only	This group type contains only IP addresses (IPv4 or IPv6). **IP Addresses Only** groups with only manually added IP address members are not supported for use in the **Applied To** in DFW rules. It is possible to create the rule, but it will not be enforced. If the group type is **Generic**, you can edit its type to **IP Addresses Only** group but not to **IP Addresses Only** with malicious IPs group. In this case, only the IP addresses are retained in the group. All the membership criteria and other group definitions are lost. After a group of type **IP Addresses Only** or **IP Addresses Only** with malicious IPs is realized in NSX, you cannot edit the group type to **Generic**.
IP Addresses Only with malicious IPs	If you have enabled Malicious IP Feeds, you can create an **IP Addresses Only** group with malicious IPs by switching on **Add Pre-Defined Malicious IPs**. You can also specify IPs and IP addresses only groups that should be treated as exceptions and must not be blocked. Note that once you have switched on the toggle **Add Pre-Defined Malicious IPs**, you cannot turn it off while editing the group.
Antrea	This group type is available only when your NSX environment has one or more Antrea Kubernetes clusters registered to it.

7. Optional: On the **Membership Criteria** page, click **Add Criterion** to add members in the generic group dynamically based on one or more membership criteria.

   If you have registered Kubernetes clusters with Antrea CNI in your NSX deployment, you can create generic groups with Kubernetes member types in dynamic membership criteria to match traffic entering into or leaving from these Antrea Kubernetes clusters.

   A membership criterion can have one or more conditions. The conditions can use the same member type or a mix of different member types. In a single membership criterion, conditions based on NSX member types cannot be mixed

with conditions based on Kubernetes member types. However, you can have one criterion based on only NSX member types, and another criterion based on only Kubernetes member types, and then join both criteria with an OR operator.

Some restrictions apply to adding multiple conditions with mixed NSX member types or mixed Kubernetes member types in a membership criterion. To learn more about membership criteria, see Overview of NSX Group Membership Criteria.

Note:
In a multi-tenant NSX environment, Kubernetes cluster resources are not exposed to the project inventory. Therefore, inside a project, you cannot create generic groups with Kubernetes member types in dynamic membership criteria.

8. Optional: Click **Members** to add static members in the group.

   The available member types are:

   - **Groups** - If you are using NSX Federation, you can add a group as a member that has an equal or smaller span than the region you selected for the group you are creating from the Global Manager.
   - **NSX Segments** - IP addresses assigned to a gateway interface, and NSX load balancer virtual IP addresses are not included as segment group members.
   - **Segment Ports**
   - **Distibuted Port Groups**
   - **Distributed Ports**
   - **VIFs**
   - **Virtual Machines**
   - **Physical Servers**

9. Optional: Click **IP Addresses** or **MAC Addresses** to add IP and MAC addresses as group members. IPv4 addresses, IPv6 addresses, and multicast addresses are supported.

   Click **Action** > **Import** to import IP/MAC Addresses from a TXT file or a CSV file containing comma-separated IP/MAC values.

10. Optional: Click **AD Groups** to add Active Directory Groups. Groups with Active Directory members can be used in the source text box of a distributed firewall rule for Identity Firewall. Groups can contain both AD and compute members.

    Note:
    If you are using NSX Federation, you cannot create groups from the Global Manager to include AD user groups.

11. Optional: Enter a description and tag.

12. Click **Apply**

    Groups are listed, with an option to view the members and where the group is used.

Overview of NSX Group Membership Criteria

You can define membership criteria to add members dynamically in an NSX group based on one or more criteria.

A criterion can have one or more conditions. The conditions can use the same member type or a mix of different member types.

Some restrictions apply to adding multiple conditions with mixed NSX member types or mixed Kubernetes member types in a membership criterion. See the *Restrictions for Criteria with Mixed NSX Member Types* section and the *Restrictions for Criteria with Mixed Kubernetes Member Types* section later in this documentation.

By default, NSX uses the logical AND operator after each condition in a membership criterion. Other logical operators are not supported to join the conditions in a membership criterion.

To join criteria, OR and AND operators are available. By default, system selects the OR operator to join two criteria. AND operator is supported between two criteria only when:

- Both criteria use the same member type.
- Both criteria use a single condition.

The following restrictions apply to adding multiple conditions:

- A maximum of five conditions with the same member type is supported in a single membership criterion. For example, in a criterion, you can add a maximum of five conditions with the Virtual Machine member type.
- A maximum of 15 conditions with mixed member types are supported in a single membership criterion. For example, in a criterion, you can add a maximum of 15 conditions with a mix of Segment and Segment Port member types.
- A maximum of 35 conditions with mixed member types are supported in a generic group.

A group can have a maximum of five membership criteria. However, the total number of criteria that you can add in a group is determined by the number of conditions in each criterion. See the following examples.

Example 1

A group with three membership criteria and a total of 35 conditions:

- Criterion 1 has 15 conditions with mixed member types.
- Criterion 2 has 15 conditions with mixed member types.
- Criterion 3 has 5 conditions with the same member type.

Example 2

A group with four membership criteria and a total of 35 conditions:

- Criterion 1 has 15 conditions with mixed member types.
- Criterion 2 has 14 conditions with mixed member types.
- Criterion 3 has four conditions with the same member type.
- Criterion 4 has two conditions with the same member type.

Example 3

A group with five membership criteria and a total of 22 conditions:

- Criterion 1 has 10 conditions with mixed member types.
- Criterion 2 has three conditions with the same member type.
- Criterion 3 has four conditions with the same member type.
- Criterion 4 has three conditions with the same member type.
- Criterion 5 has two conditions with mixed member types.

Because this group has reached the limit of five criteria, you cannot add another membership criterion. However, you can add more conditions, if required, in any of the five criteria until you don't exceed the following upper limits mentioned earlier:

- A maximum of five conditions with the same member type in a single criterion.
- A maximum of 15 conditions with mixed member types in a single criterion.
- A total of 35 conditions in the generic group.

Restrictions for Criteria with Mixed NSX Member Types

Member Type	Criterion With Mixed Member Types	Tag Operator	Scope Operator
Virtual Machine	Not Supported	• Equals - one tag can be selected. • Contains • Starts with • Ends with	• Equals
Segment	Supported Conditions based on Segment can be mixed with conditions based on Segment Port	• Equals - one tag can be selected. • Not Equals - one tag can be selected.	• Equals • Not Equals - if selected, the tag operator is removed.
Segment Port	Supported Conditions based on Segment Port can be mixed with conditions based on NSX Segment	• Equals - one tag can be selected. • Not Equals - one tag can be selected. • Not In - a maximum of five tags can be selected.	• Equals • Not Equals - if selected, the tag operator is removed.

Member Type	Criterion With Mixed Member Types	Tag Operator	Scope Operator
Distributed Port Groups	Supported Conditions based on Distributed Port Group can be mixed with conditions based on Distributed Port	• Equals - one tag can be selected. • Not Equals - one tag can be selected.	• Equals • Not Equals - if selected, the tag operator is removed.
Distributed Ports	Supported Conditions based on Distributed Port can be mixed with conditions based on Distributed Port Group	• Equals - one tag can be selected. • Not Equals - one tag can be selected. • Not In - a maximum of five tags can be selected.	• Equals • Not Equals - if selected, the tag operator is removed.
IP Set - This member type will be deprecated in the future. It is currently available to achieve backward compatibility with preexisting NSGroups or Groups based on IP Set tag-based criterion. We recommend you to use **Group** as the member type and add tag-based groups of type 'IP Addresses Only' in a membership criterion.	Not Supported	• Equals - one tag can be selected.	• Equals
Group - Use this member type to add tag-based groups of type 'IP Addresses Only' in a membership criterion.	Not Supported	Equals - one tag can be selected.	Equals

Overview of Membership Criteria with Kubernetes Member Types

Starting in NSX 4.1, you can create generic groups with Kubernetes member types in dynamic membership criteria to match traffic entering into or leaving from Antrea Kubernetes clusters.

You can then use these generic groups in distributed firewall rules or gateway firewall rules to secure traffic between VMs in the NSX environment and pods in Antrea Kubernetes clusters.

This feature requires Antrea- NSX Interworking version that is available with VMware Container Networking™ with Antrea™ 1.6.0 or later. See the 1.6.0 release notes.

In a generic group, Kubernetes member types are available on the **Membership Criteria** page only when at least one Antrea Kubernetes cluster is registered to your NSX environment.

> **Note:**
> In a multi-tenant NSX environment, Kubernetes cluster resources are not exposed to the project inventory. Therefore, inside a project, you cannot create generic groups with Kubernetes member types in dynamic membership criteria.

The following table lists the Kubernetes member types that are available in the **Default** view of NSX 4.1 onwards to add dynamic membership criteria in a generic group. Each Kubernetes member type belongs to either cluster scope or namespace scope.

Member Type	Scope
Kubernetes Cluster	Cluster
Kubernetes Namespace	Namespace
Kubernetes Node	Cluster

Member Type	Scope
Kubernetes Service	Namespace
Kubernetes Ingress	Namespace
Kubernetes Gateway	Namespace
Antrea Egress	Cluster
Antrea IP Pool	Cluster

Naming Conventions Used for Conditions With Kubernetes Member Types

The following table explains the naming conventions that are used in this documentation to represent the different conditions that you can add in dynamic membership criteria, which are based on Kubernetes member types.

Naming Convention for Condition	Meaning
Kubernetes Cluster condition	Conditions in the dynamic membership criteria are based on the Kubernetes Cluster member type.
Kubernetes Namespace condition	Conditions in the dynamic membership criteria are based on the Kubernetes Namespace member type.
Resource condition	Conditions in the dynamic membership criterion are based on any of these Kubernetes member types: • Kubernetes Service • Kubernetes Ingress • Kubernetes Gateway • Antrea Egress • Antrea IP Pool • Kubernetes Node
Cluster-scoped resource condition	Conditions in the dynamic membership criteria are based on any of these Kubernetes member types: • Antrea Egress • Antrea IP Pool • Kubernetes Node
Namespace-scoped resource condition	Conditions in the dynamic membership criteria are based on any of these Kubernetes member types: • Kubernetes Service • Kubernetes Ingress • Kubernetes Gateway

Restrictions for Criteria with Mixed Kubernetes Member Types

The following table provides a high-level summary of the restrictions or validations that apply to using Kubernetes member types in a single membership criterion. For examples of validations, see the *Validations for Dynamic Grouping with Kubernetes Member Types* section later in this documentation.

Member Type	Restrictions for Using the Member Type in a Criterion	Supported Attributes	Tag Operator	Scope Operator
Kubernetes Cluster	Cannot be used alone in a criterion. Only one cluster condition is allowed in a criterion. Must be mixed with at least one Kubernetes resource condition. Optionally, can be mixed with Kubernetes Namespace conditions and Kubernetes resource conditions.	Name	Not Supported	Not Supported
Kubernetes Namespace	Cannot be used alone in a criterion. Cannot be mixed with cluster-scoped resource conditions. Must be mixed with namespace-scoped resource conditions. Optionally, can be mixed with a Kubernetes Cluster condition.	Name Tag	Equals - one tag can be selected	Equals
Antrea Egress	Can be used alone in a criterion. Optionally, can be mixed with a Kubernetes Cluster condition.	Name Tag	Equals - one tag can be selected	Equals
Antrea IP Pool	Can be used alone in a criterion. Optionally, can be mixed with a Kubernetes Cluster condition.	Name Tag	Equals - one tag can be selected	Equals
Kubernetes Ingress	Can be used alone in a criterion. Optionally, can be mixed with only Kubernetes Namespace conditions or only Kubernetes Cluster condition, or both.	Name Tag	Equals - one tag can be selected	Equals
Kubernetes Gateway	Can be used alone in a criterion. Optionally, can be mixed with only Kubernetes Namespace conditions or only Kubernetes Cluster condition, or both.	Name Tag	Equals - one tag can be selected	Equals

Member Type	Restrictions for Using the Member Type in a Criterion	Supported Attributes	Tag Operator	Scope Operator
Kubernetes Service	Can be used alone in a criterion. Optionally, can be mixed with only Kubernetes Namespace conditions or only Kubernetes Cluster condition, or both.	Name Tag	Equals - one tag can be selected	Equals
Kubernetes Node	Can be used alone in a criterion. Only a single node condition is allowed. Optionally, can be mixed with a Kubernetes Cluster condition. Cannot be mixed with Kubernetes Namespace conditions or Kubernetes resource conditions.	IP Address Pod CIDR	Not supported See Note after this table.	Not supported

Note:
When you create a generic group with Kubernetes Node member type by using the API, only the Matches operator is allowed. This operator can take only the * value. The * wildcard character will match all the nodes in the K8s cluster (if the Kubernetes Node condition is mixed with a Kubernetes Cluster condition), or it will match nodes across all clusters (if the Kubernetes Node condition is used alone).

Validations for Dynamic Grouping with Kubernetes Member Types

Validation 1

A membership criterion can have a maximum of one Kubernetes Cluster condition. To match a single Kubernetes cluster by name, use the Equals operator and enter the cluster name.

Note:
Kubernetes cluster name must be unique.

If you want to match multiple clusters in a single Kubernetes Cluster condition, you can use one of these operators:

- In
- Starts With
- Ends With

Example:

Matches Single K8s Cluster	Matches Multiple K8s Clusters
Criterion: Kubernetes Cluster Name Equals ClusterA	Criterion: Kubernetes Cluster Name In ClusterA,ClusterB,ClusterC A maximum of five comma-separated values are allowed. The values must not be separated with spaces.

Validation 2

A membership criterion with a Kubernetes Cluster condition can be mixed with any one of the Kubernetes resource conditions. If you add Kubernetes Namespace conditions too in the same criterion, then resource conditions must be limited only to namespace-scoped resource conditions.

Examples:

- A membership criterion with a Kubernetes Cluster condition and three Kubernetes Ingress conditions is valid.

- A membership criterion with a Kubernetes Cluster condition and two Antrea Egress conditions is valid.
- A membership criterion with a Kubernetes Cluster condition and three Antrea IP Pool conditions is valid.
- A membership criterion with a Kubernetes Cluster condition, one Kubernetes Namespace condition, and three Kubernetes Gateway conditions is valid.

Validation 3

A membership criterion must include at least one Kubernetes resource condition. A membership criterion is invalid if it contains:

- Only Kubernetes Cluster condition
- Only Kubernetes Namespace condition
- Only Kubernetes Cluster condition and Kubernetes Namespace condition

Examples:

- A membership criterion with one Kubernetes Cluster condition, one Kubernetes Namespace condition, and one Kubernetes Ingress condition is valid.
- A membership criterion with one Kubernetes Cluster condition, two Kubernetes Namespace conditions, and three Kubernetes Ingress conditions is valid.

Validation 4

A membership criterion can have only a single Kubernetes Node condition. Optionally, a Kubernetes Node condition can be mixed with only a Kubernetes Cluster condition.

A Kubernetes Node condition cannot be mixed with Kubernetes Namespace conditions or Kubernetes resource conditions.

A membership criterion with only a Kubernetes Node condition can stand alone. However, a group with only a Kubernetes Node condition will match nodes of all Antrea Kubernetes clusters that are registered to NSX.

Tag operator and Scope operator are currently not supported for the Kubernetes Node condition.

Kubernetes Node condition supports the following two properties.

Property	Description
IP Address	Matches the internal IP address of all nodes of specified Antrea Kubernetes clusters.
Pod CIDR	Matches the Pod CIDR of all nodes of specified Antrea Kubernetes clusters.

Validation 5

If a membership criterion includes Kubernetes Cluster condition and Kubernetes Namespace conditions, then it must include at least one namespace-scoped Kubernetes resource condition. You cannot mix any of the following cluster-scoped Kubernetes resource conditions in the same criterion:

- Antrea Egress
- Antrea IP Pool
- Kubernetes Node

Examples:

- A membership criterion with one Kubernetes Cluster condition, two Kubernetes Namespace conditions, and Kubernetes Gateway conditions is valid.
- A membership criterion with one Kubernetes Cluster condition, four Kubernetes Namespace conditions, and three Kubernetes Service conditions is valid.
- A membership criterion with one Kubernetes Cluster condition, one Kubernetes Namespace condition, and one Antrea Egress condition is invalid because Antrea Egress is a cluster-scoped resource.

Validation 6

A membership criterion must include at least one Kubernetes resource condition. A resource condition can stand alone in a criterion. However, if you add multiple resource conditions in a criterion, then all resource conditions must be of the same member type.

> **Note:**
> Kubernetes Cluster condition and Kubernetes Namespace conditions are used for defining the scope of a criterion. They are not Kubernetes resource conditions, and hence they are not limited by this validation rule.

Examples:

- A membership criterion with five Kubernetes Service conditions is valid.

- A membership criterion with one Kubernetes Cluster condition, three Kubernetes Namespace conditions, and four Kubernetes Service conditions is valid.
- A membership criterion with one Kubernetes Cluster condition, three Kubernetes Namespace conditions, four Kubernetes Service conditions, and three Kubernetes Ingress conditions is invalid. The reason is that you mixed resource conditions of two different member types (Kubernetes Service and Kubernetes Ingress) in the same criterion.
  However, you can create separate criteria with a resource condition based on a single member type and then join both criteria with an OR operator, as shown below:
  Criterion 1:
  One Kubernetes Cluster condition + three Kubernetes Namespace conditions + four Kubernetes Service conditions
  OR
  Criterion 2:
  One Kubernetes Cluster condition + three Kubernetes Namespace conditions + three Kubernetes Ingress conditions

Validation 7

In a single membership criterion, conditions based on NSX member types cannot be mixed with conditions based on Kubernetes member types. However, you can have a group with one criterion based on only NSX members types, and other criterion based on only Kubernetes member types, and then join both criteria with an OR operator.

Example:

Valid	Invalid
Criterion 1: Virtual Machine conditions OR Criterion 2: Kubernetes Cluster condition + Kubernetes Gateway conditions	Criterion: NSX Segment condition + Segment Port condition AND Kubernetes Cluster condition + Kubernetes Gateway conditions

Profiles

There are two types of profiles: context profiles and layer 7 access profiles.

Profiles enable creating attributes key value pairs such as layer 7 App ID, and domain names. After a profile has been defined, it can be used in one or more distributed firewall rules, or gateway firewall rules.

Context Profiles

Context Profiles are used in firewall rules.

There are five attributes for use in context profiles: App ID, Custom URL, Domain (FQDN) Name, URL Category, and URL Reputation.

Select App IDs can have one or more sub attributes, such as TLS_Version and CIPHER_SUITE. Both App ID and FQDN can be used in a single context profile. Multiple App IDs can be used in the same profile. One App ID with sub attributes can be used - sub attributes are cleared when multiple App ID attributes are used in a single profile.

Both system defined and user defined Fully Qualified Domain Names (FQDNs) are supported. You can see the list of FQDNs when you add a new context profile of attribute type *Domain (FQDN) Name*.You can also see a list of FQDNs, and where they are used by navigating to **Inventory** > **Profiles** > **FQDNs**.

1. Select **Inventory** > **Profiles**.
2. Select **Security** > **Profiles**.
3. Select the **Context Profile** tab and click **Add Context Profile**.
4. Enter a **Profile Name**, and optional **Description**.
5. In the Attributes column, click **Set**.
6. Click **Add Attribute**, and select one or more attributes from the drop-down menu: **App ID**, **Custom URL**, **Domain (FQDN) Name**, **URL Category**, or **URL Reputation**.

Attribute	Procedure
App ID - Advanced App IDs require a custom profile.	1. Enter the name of the advanced App ID you'd like to use in firewall rules. 2. Click **Add**. 3. Click **Apply**.
Custom URL	To create a custom URL: 1. Click the three dot menu  and select **Add Custom URL**. 2. Enter the URL. 3. Click **Save**.
Domain (FQDN) Name	Select a system FQDN by scrolling down the list. To create a user-defined FQDN: 1. Click the three dot menu  and select **Add FQDN**. 2. Enter the domain name in the form `*.[hostname].[domain]`. For example, *.abracadabra.com. Do not include http:// or any other header. 3. Click **Save**. The newly created FQDN appears in the attribute value column. 4. Search and add additional FQDNs. 5. Click **Apply**.
URL Category	1. Select one or more URL categories by scrolling down the list. 2. Click **Add**. 3. Click **Apply**.
URL Reputation	1. Select one or more of the attributes by clicking in the box. 2. Click **Add**. 3. Click **Apply**. See the "FQDN Analysis Dashboard" page in the VMware vDefend documentation for more information about URL reputation.

7. Optional: If you have selected an attribute with sub attributes such as SSL or CIFS, click **Set** in the Sub Attributes/ Values column.
   a) Click **Add Sub Attribute** and select TLS_VERSION, TLS_CIPHER_SUITE, or CIFS_SMB_VERSION.
   b) Select one or more sub attributes.
   c) Click **Add**. Another sub attribute can be added by clicking **Add Sub Attribute**.
   d) Click **Apply**.
8. Optional: Enter a tag or scope. See NSX Tags for more information.
9. Click **Save**.

Apply this context profile to a layer 7 distributed firewall rule (for layer 7 or domain name), or gateway firewall rule (for layer 7).

L7 Access Profiles

An L7 access profile can contain multiple entries with different attribute types. L7 access profiles can be used in both Tier-0 and Tier-1 gateway firewall rules.

L7 access profiles have a default entry, and more entries can be added above the default entry. Entries in the profile are evaluated in the listing order, and action is taken upon the first match.

Before creating application firewall rules on the Tier-0 gateway firewall, it is important to manually add gateway firewall rules to allow routing protocols such as BGP, OSPF, and the failure detection protocol BFD. These rules should be added before any application rules to ensure that routing peering respective failure detection protocol is not impacted when changing application firewall rules.

1. Select **Inventory** > **Profiles**.
2. Select the **L7 Access Profile** tab, and click **Add L7 Access Profile**.
3. Enter a **Profile Name**, and optional **Description**.
4. In the Attribute Entities column, click **Set**.
5. Click **Add Attribute Type**, and select one or more attributes from the drop-down menu. For Tier-0 gateway firewall rules, App ID is the only supported attribute type.

Attribute Type	Attribute Value
App ID - over 750	To view available App IDs, scroll down the list, or select App IDs.
URL Category - 80+ categories including social media, banking, phishing, etc.	Select one or more URL categories by scrolling down the list.
Custom URL - with regular expression	For more details see Custom URLs.
URL Reputation	Choose one or more of these reputations: High Risk, Suspicious, Moderate Risk, Low Risk, Trustworthy, Unknown

6. Select the rule action:
   - Allow - allows matched traffic.
   - Reject - rejects matched traffic.
   - Reject with Response - rejects and sends the client the response page. This option is not available for the App ID attribute type. Navigate to **Security** > **Gateway Firewall** > **Settings** > **URL Filtering** to view and customize the **Reject with Response** message.
     The **Reject with Response** page is sent only for http traffic. The response page will contain the URL (first 10 bytes show), Category, Source IP and message-text. Enter the message for the **Reject with Response** page. Click **Preview Page** to view the page that will be sent when access to a URL is blocked by a policy.

7. By default, logging is turned off. Toggle the button to activate logging.
8. By default the newly added entry is turned on. Toggle the button to deactivate the entry.
9. Click **Add**.
10. To edit or delete an entry, click the menu icon () and select **Edit** or **Delete**.
11. To change the settings of the default entry, click the menu icon () and select **Edit**. Note that the default entry cannot be turned off. By default, logging is turned off for the default entry. After the final processing of the default entry within the L7 Access Profile, the gateway firewalling evaluation process is stopped.
12. Click **Add** to activate the changed setting for the default entry or click **Cancel.**
13. Click **Apply**.
14. Click **Save**.

Attribute Types

App IDs

The layer 7 attribute App ID (Application Identification) identifies network traffic based on the application or protocol instead of port numbers and IP addresses.

Enforcement based on App IDs lets you allow or deny applications to run on any port, or to force applications to run on their standard port. VMware Deep Packet Inspection enables matching packet payload against defined patterns, commonly referred to as signatures. Signature-based identification and enforcement lets you match a particular application or protocol with the version of that protocol, for example TLS version 1.2, TLS version 1.3, or different versions of CIFS traffic. This allows you to have visibility into or restrict the use of protocols that have known vulnerabilities.

Layer 7 App IDs are used in Context Profiles and L7 Access Profiles. Context Profile is applicable to both distributed firewall and gateway firewall. L7 Access Profile is only applicable to gateway firewall.

To see the complete list, go to vDefend Threat Intelligence Service.

> **Note:**
> NFS version 4 is not a supported attribute.

- Gateway firewall rules do not support the use of FQDN attributes or other sub attributes in context profiles.
- Context profiles are not supported on tier-0 gateway firewall policy.

Supported App IDs and FQDNs:

- For FQDN, users need to configure a high priority rule with a DNS App ID for the specified DNS servers on port 53.
- SYSLOG App ID is detected only on standard ports.

FQDNs

A fully qualified domain name (FQDN) is the complete domain name for a specific host on the Internet. FQDNs are used in firewall rules to allow or reject traffic going to specific domains.

The FQDN attribute type is used in distributed firewall FQDN Filtering policies, as described in the VMware vDefend documentation. NSX supports custom FQDNs that are defined by an administrator in addition to the pre-defined list of FQDNs. Custom FQDN supports the following:

- FQDN supports processing of DNS response record packets containing canonical names (CNAMEs).

- Full FQDN names such as maps.google.com or myapp.corp.com
- Partial REGEX with * at the beginning only such as *eng.northpole.com or *yahoo.com
- FQDN name length up to 64 characters
- FQDN names must end with the registered top level domain (TLD) such as .com, .org, or .net

When creating a custom FQDN, using a wildcard domain is best practice. For example, using `*.example.com`, would include sub domains such as `americas.example.com` and `emea.example.com`. Using `example.com`, would not include any sub domains.

Note:
Custom FQDNs do not support custom top level domain names, such as www.example.john.

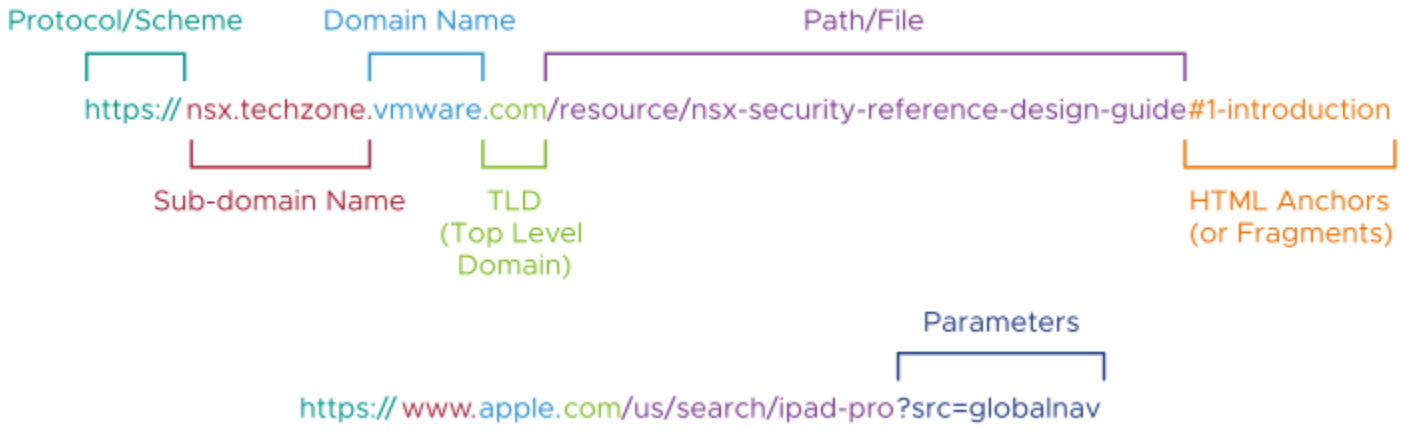
1. From your browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address>.
2. Select **Inventory** > **Profiles**.
3. Select **Security** > **Profiles**.
4. Select the **Attribute Types** tab, and **FQDNs**.
   A table of system-generated FQDNs appears.
5. Select **Actions** > **Add FQDN**.
6. Enter the domain name in form `*[hostname].[domain]`. For example, *abracadabra.com
   Do not include http:// or any other header.
7. Click **Save**.
   The user-defined FQDN is shown in the table of available FQDNs, with User in the **Created By** column.
8. Optional: To display a subset of FQDNs, click **Filter by Name, Path and more** and select **Created by** or **Domain**.

FQDNs can be used in context profiles for distributed firewall rules.

Custom URLs

Custom URLs are used in L7 access profiles as an attribute type for URL filtering, and context profiles.

NSX allows users to configure full or partial domain or URL with special characters. The following image depicts different elements of a URL (Uniform Resource Locator).



NSX supports the following custom URL patterns:

- Only IANA registered Top Level Domain (TLD), and wildcard characters * and ^ are supported.
- Entered URL should be a valid URL. For example, www...google+com will be rejected.
- Do not include http://, https://, ftp://, etc.
- Users must enter a domain name, but may optionally include URI as well. For example, google.com and google.com/news are valid, but /news is invalid input.
- URI must not include query parameters. For example, www./google.com/query?keyboard=news will be rejected.

- Special characters * and ^ can be used for wildcard matching both in the domain name and the URI path. * will match one or more words where as ^ will match exactly one word. For example, *.google.com will match news.google.com and also local.news.google.com. However, ^ will only match news.google.com, but will not match local.news.google.com. Similarly, google.com/* will match google.com/news and also google.com/news/sanjose. However, google.com/^ will only match google.com/news, but will not match google.com/news/sanjose.
- * and ^ match only full words, they cannot be used to match partial words. For example, *google.com cannot be used to match mygoogle.com. However, *.google.com can be used to match my.google.com. Similarly, google.com/*news and google.com/n* are invalid and will be rejected.
- * and ^ can appear more than once in the URL. For example, *.google.* and *.google.com/^/news/* are valid.
- As ^ matches only a single word, two ^s back to back is allowed. For example, ^.^.google.com is valid and matches local.news.google.com, but does not match my.local.news.google.com or news.google.com. Similarly, google.com/^/^ is valid and matches google.com/news/sanjose, but does not match google.com/news or google.com/news/sanjose/today.
- When entering just the domain name (without URI), user may enter the domain name with a / at the end or without it and they have different behavior. If the user enters without an ending /, then it is treated as a partial match. For example, *.google.com will match news.google.com, news.google.com.us, news.google.com.us/, news.google.com.us/local, etc. If the user enters the domain name with a / at the end, then it is treated as an exact match. For example, *.google.com/ will match news.google.com and news.google.com/, but will not match news.google.com.us or news.google.com/local.
- When the entered URL has a path, then the presence of / at the end is not treated in any special way and the URL is treated as an exact match. For example, google.com/news/ will only match google.com/news/, but will not match google.com/news or google.com/newslatest or google.com/news/latest.
- URL matching can be used to match HTTP URL (Host header + URI) or TLS SNI. If the user specified URL has a path, it will not match the TLS SNI. However, if TLS Inspection is enabled and the traffic is decrypted, then the internal HTTP URL can be used for matching URL with path. For example, *.google.com and *.google.com/ can match HTTP URL or TLS SNI (without TLS Inspection), but *.google.com/news will not match TLS SNI (without TLS Inspection).

Examples

User Input	Example
espn.com	Matches: espn.com, espn.com/, espn.com.us, espn.com.us/, espn.com/sports, espn.com.us/sports/p Does not match: premium.espn.com
espn.com/	Matches: espn.com, espn.com/ Does not match: premium.espn.com, espn.com.us, espn.com.us/, espn.com/sports, espn.com.us/sports/
espn.com/sports	Matches: espn.com/sports Does not match: espn.com/sportsnba, esp.com/sports/, espn.com/sports/nba
espn.com/sports/	Matches: espn.com/sports Does not match: esp.com/sports, espn.com/sports/nba
*espn.com	Matches: premium.espn.com, replay.preimum.espn.com, instant.replay.premium.espn.com, premium.espn.com/, premium.espn.com.us, premium.espn.com.us/, premium.espn.com/sports, premium.espn.com.us/sports Does not match: espn.com, .espn.com
.espn.	Matches: www.espn.com, premium.espn.com, www.espn.us, premium.espn.com.us/, latest.espn.com/sports, www.espn.com.us/sports/ replay.premium.espn.com, instant.replay.premium.espn.com instant.replay.premium.espn.com.us/ Does not match: espn.com, www.espn

User Input	Example
espn.*.us/	Matches: espn.news.us, espn.news.us/, espn.local.news.us Does not match: espn.us, www.espn.us, espn.news.us/sports
.espn./*	Matches: www.espn.com/sports, replay.premium.espn.com.us/sports/nba/ Does not match: www.espn.com, www.espn.com/
^.espn.com	Matches: www.espn.com, www.espn.com/, www.espn.com/sports, www.espn.com.us/ Does not match: espn.com, news.local.espn.com
^.espn.^	Matches: www.espn.com, www.espn.com/,www.espn.com.us, www.espn.com.us/sports Does not match: news.local.espn.com, www.espn.com.us
espn.^/	Matches: espn.com, espn.com/ Does not match: espn.com.us, espn.com/sports
www.^.^.com/^/^	Matches www.local.espn.com/sports/nba Does not match www.personal.local.espn.com/sports/nba, www.local.espn.com/sports/nba/

URL Categories

URL categories are used by in context profiles and L7 access profiles.

To view available URL categories and where they are used, navigate to **Inventory** > **Profiles** > **Attribute Types** and select the **URL Categories** tab. Click **Where Used** to view the attribute entries that contain the URL category.

Containers

You can view container objects in the NSX inventory and do basic diagnostic or troubleshooting tasks on container clusters in the NSX Manager UI. High level details about container objects such as, Namespaces, Pods, Kubernetes Services, Network Policies, and so on, are displayed in the UI.

If the **Containers** page is active in your browser, and some changes happened in the container cluster inventory in the background, the inventory details in the UI are not auto-refreshed. You must manually click the **Refresh** icon at the bottom of the page a few times to ensure that inventory updates are pushed to the NSX Manager UI.

View Details of Namespaces

You can filter the list of namespaces in the table using several criteria, such as ID, Label, Cluster Name, CNI Type, and many other criteria. At a high level, for each namespace, the table shows the number of Kubernetes Services, Pods, the name of container cluster in which the namespace is created, CNI type of container cluster, and so on.

Type of container cluster includes: Kubernetes, OpenShift, AKS, EKS, GKE, TKGm, to name a few. CNI Type includes: NSX Container Plugin (NCP), Antrea.

For details about Kubernetes Services and Pods in a specific namespace, click the hyperlinked number in the respective columns.

For example, if you want to view the list of all Pods that are associated with a specific Kubernetes Service, click the number in the **Pods** column. The **Pods Details** window opens. This pop-up window shows the status of Pods, Pod IP address, Labels associated with Pod, and other details. For container clusters with NCP as the CNI, only IP addresses of the Pod's corresponding segment ports are shown. If the network interfaces of the Pods are not attached to NSX segments, the Pod IP addresses are not shown.

Expand a row in the table to view more details about the namespace. For example:

- Number of Ingress Rules and the details of each Ingress Rule.

- Number of Network Policies.
- Number of Labels and the details of each Label.

This list does not mention all the inventory details that you can view for a namespace. Check the **Namespaces** page in the NSX Manager UI to know more.

Note:

- For Antrea container clusters, the **Namespaces** page shows information about Kubernetes Network Policies and Antrea Network Policies, which have their scope limited to a namespace.
- The **Networking** and **Networking Status** columns show information only for container clusters that use NCP as the CNI. For container clusters with Antrea as the CNI, these two columns are not applicable.

View Details of Container Clusters

You can filter the list of container clusters in the table using several criteria, such as External ID, Cluster Name, CNI Type, and many other criteria. At a high level, for each container cluster, the table shows the number of Kubernetes Services, Pods, and Nodes, CNI Type, and Networking information.

For details about Kubernetes Services, Pods, Nodes, and Networking in a specific container cluster, click the hyperlinked number in the respective columns. The **Networking Status** column shows information only for container clusters that use NCP as the CNI. For container clusters with Antrea as the CNI, this column is not applicable.

Expand a row in the table to view more details about the container cluster. For example:

- Infrastructure type of the container cluster (example values: vSphere, AWS, Azure, Google, VMware Cloud, and so on).
- Antrea version (for Antrea container clusters).
- Number of Network Policies in the container cluster (for Antrea container clusters, this number refers to Antrea Cluster Network Policies).
- Number of namespaces in the cluster and the details of each namespace.

This list does not mention all the inventory details that you can view for a container cluster. Check the **Clusters** page in the NSX Manager UI to know more.

Related Documentation

To learn about NSX Container Plugin, see the NCP documentation.

To learn about integrating Antrea container clusters to NSX, go to Integration of Kubernetes Clusters with Antrea CNI.

NSX Tags

Tags help you to label NSX objects so that you can quickly search or filter objects, troubleshoot and trace, and do other related tasks.

You can create tags using both the UI and APIs. Each tag has the following two attributes:

- Tag (refers to the tag name. It is required, must be unique and case-sensitive.)
- Scope (optional)

Tag scope is analogous to a key and tag name is analogous to a value. For example, let us say, you want to label all virtual machines based on their operating system (Windows, Mac, Linux). You can create three tags, such as Windows, Linux, and Mac, and set the scope of each tag to OS. Other examples of tag scope can be tenant, owner, name, and so on.

After you save a tag, you cannot update the name and scope. However, you can unassign or remove tags from objects.

For information about the maximum number of tags supported in NSX objects, see the VMware Configuration Maximums tool at https://configmax.vmware.com/home.

Following are some of the operations that you can do with tags:

- Assign or unassign tags to an object.

- Assign or unassign a single tag to multiple objects simultaneously (supported only for VMs).
- View a list of all tags in the inventory.
- Filter the list of tags by tag name, tag source, and tag scope.
- View a list of objects that are assigned a specific tag.

Note:
When a VM disappears from the vCenter inventory for more than 30 minutes, tags on the VM are lost. If the same VM reappears in the vCenter inventory after 30 minutes, NSX treats it as a new VM, and you must add the tags again on the VM. This behavior is expected and as per design. For example, this behavior is seen when using array-based replication with VMware Live Site Recovery™.

Use Cases of Tags

The following table describes some use cases of using tags.

Use Case	Description
Manageability	• Simplify searching of objects in a large-scale inventory management. • Provide more information to differentiate objects that share similar or unclear names.
Third-party sharing and context sharing	• Annotate objects with custom information. • Allow third-party non-NSX systems to add metadata information in an automated fashion. For example, metadata from partners, container platforms, and so on. • Capture attributes or properties and relationships that are learned using NSX discovery agent, inventory collection, Guest Introspection, VM Tools, and so on.
Security	• Create grouping membership criteria. • Specify the firewall source and destination.
Troubleshooting (Traceability)	• Trace a firewall rule into the logs (Rule tags) • Trace and correlate objects back to an OpenStack network.

System Tags

System tags are tags that are system-defined, and you cannot add, edit, or delete them.

Table 1047: System Tags in Other NSX Objects

Objects	System Tags
Group	• autoPlumbing • abstractionPath • NLB-VIP_ID • NLB-Lb-ID • NLB-Pool_ID
Segment	• subnet-cidr
IP Address Pool IP Address Block	• abstractionPath

Discovered Tags

NSX can discover and synchronize tags from the following:

- Amazon Web Services
- Microsoft Azure
- Kubernetes container clusters with NSX Container Plugin (NCP)

- Kubernetes container clusters with Antrea network plug-in
- OpenShift container clusters with NCP

Discovered tags are displayed for workload VMs and container cluster objects. You cannot edit discovered tags in the NSX UI.

For example:

- Tags that are added to container cluster objects in the Kubernetes container clusters with NCP CNI or Antrea CNI, and OpenShift container clusters with NCP CNI.

Tag Prefix	Meaning
dis:aws	Tags discovered from Amazon Web Services (AWS).
dis:azure	Tags discovered from Microsoft Azure.
dis:k8s	Tags discovered from Kubernetes container clusters with NCP CNI, Antrea CNI, and OpenShift container clusters with NCP CNI.

You can enable or disable the discovery of AWS tags at the time of adding the AWS account. Similarly, you can enable or disable Microsoft Azure tags at the time of adding the Microsoft Azure subscription.

Add Tags to an NSX Object

You can select existing tags that are available in the NSXinventory or create new tags to add to an object.

The following procedure explains the steps for adding tags to a single object. For this documentation, the segment object is considered. The steps for adding tags to other objects remain the same. You can navigate to the specific object page, and follow similar steps to add tags to that object.

For information about the maximum number of tags supported for objects in NSX, see the VMware Configuration Maximums tool at https://configmax.vmware.com/home.

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Ensure that you are in the edit mode of an object to assign tags to it. Objects can be segments, groups, tier-0 gateways, tier-1 gateways, and so on.
   For example, to tag segments, click **Networking** > **Segments**. Next to the segment that you want to edit, click 
   and then click **Edit**.
3. In the **Tag** drop-down menu, enter a tag name. When you are done, click **Add Item(s)**.
   The maximum length of the tag name is 256 characters.

   If tags exist in the inventory, the **Tag** drop-down menu displays a list of the available tags and their scope. You can select an existing tag from the drop-down menu and assign it to the segment.

   Note:
   If you are assigning a tag to the virtual machine, do not assign the Edge_NSGroup tag to the VM. The system automatically assigns this tag to edge VMs for including them in the DFW exclusion list.
4. Optional: Enter a tag scope.
   For example, let us say, you want to tag segments based on department names, such as sales, marketing, finance, and so on, in your organization. Create tags such as sales, marketing, and finance, and set the scope of each tag to department.

   The maximum length of the scope is 128 characters.

   If you selected an existing tag from the inventory, the scope of the selected tag is applied automatically. Otherwise, you can enter a scope for the new tag that you are creating.
5. Click the **+** icon or press Enter.
   The tag is added to the segment.

6. Add more tags to the segment, if required.
7. Click **Save**.

Add a Tag to Multiple NSX Objects

You can add a tag to multiple objects simultaneously. However, this feature is supported only for the virtual machine object.

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Click **Inventory** > **Tags**.
3. Click **Add Tag**.
4. Enter a tag name.
   The maximum length of the tag name is 256 characters.
5. Optional: Enter a tag scope.
   For example, let us say, you want to tag virtual machines based on their operating system (Windows, Mac, Linux). Create three tags, such as Windows, Linux, and Mac, and set the scope of each tag to OS.
   The maximum length of the scope is 128 characters.
6. Under the **Assigned To** column, click **Set Virtual Machines**.
7. Required: Select the virtual machines to which you want to assign the tag, and click **Apply**.
   You must assign a tag to at least one virtual machine before you can save the tag.
8. Click **Save**.

- If the tag is assigned to many virtual machines, the assignment might take some time. When the assignment is in progress, the **Last Assignment Status** shows `Running`. After the tag is assigned successfully to all the selected virtual machines, the **Last Assignment Status** column changes to `Successful`.
- If a partial assignment occurs, the system does not roll back the tag assignment from the VMs on which the tag is applied. For example, assume that you selected 100 VMs for a bulk tag assignment, and the assignment fails for 10 VMs. The tag that is assigned on the remaining 90 VMs is not rolled back.
  In such partial assignment situations, run the following API to retrieve the status of the tagging operation:
  ```
  GET /api/v1/infra/tags/tag-operations/<operation-id>/status
  ```
  You can also retrieve the realized status of the tagging operation with the following API:
  ```
  GET /api/v1/infra/realized-state/realized-entities?intent_path=/infra/tags/tag-operations/<operation-id>
  ```
  For more details about these APIs, see the *NSX API Guide*.

If you have a long list of tags in the inventory, you can filter or search tags to find the tags of your interest quickly. You can filter on source, scope, and tag (name of the tag). You can also sort tags in the UI. However, due to the case-sensitive nature of tags, tags are sorted only in a lexical order.

The following limitations apply to searching or filtering tags:

- You cannot filter tags on the source and scope attributes simultaneously because both work on the scope attribute of the tag.
- The API does not support filtering tags with special characters, such as *, &, /, \, and so on. However, you can use special characters to filter tags in the UI.

Unassign Tags from an NSX Object

You can remove tags that you had assigned previously to an object.

The following procedure explains the steps for unassigning or removing tags from a single NSX object. For this procedure, the segment object is considered. The steps for unassigning tags from other objects remain the same. You can navigate to the specific feature page, and follow similar steps to unassign tags from the object.

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Edit an object.
   For example, click **Networking** > **Segments**. Next to the segment that you want to edit, click
   
   ,
   and then click **Edit**.
3. Click the **X** icon for each tag that you want to unassign from the segment.
4. Click **Save**.

Unassign a Tag from Multiple NSX Objects

You can unassign a tag from multiple objects simultaneously. However, this feature is supported only for virtual machines.

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Click **Inventory** > **Tags**.
3. Next to the tag name, click
   
   ,
   and then click **Edit**.
4. In the **Assigned To** column, click the number of virtual machines that are assigned this tag.
5. Click the **X** icon for each virtual machine that you want to unassign this tag.

   Note:

   - You can do a bulk unassignment of a tag on a maximum of 1000 virtual machines at one go.
   - When a tag is unassigned from all objects, the tag is deleted automatically from the inventory after five days.
6. Click **Apply**, and then click **Save**.

Multisite and NSX Federation

There are two options for managing NSX across multiple locations.

Table 1048: Comparison of Multisite and NSX Federation

	Multisite	NSX Federation
Availability	NSX 2.3	NSX 3.0
Environments	Two locations in metropolitan regions (<10 ms across locations) with stretched VLAN	Other use cases
Number of NSX Manager clusters	1	1 per location
Network Services	All features supported: • Switching (Overlay and VLAN) • IPAM (DHCP and DNS) • Routing (VRF, EVPN, NAT and route redistribution) • Layer4+ services (Load Balancing, VPN)	All features supported from GM: • Switching (Overlay and VLAN) • IPAM (DHCP Relay and static binding, and DNS) • L2-Bridge • Routing (NAT and route redistribution) • Routing protocols (BGP, Static) Exceptions: • T0-VRF • DHCP dynamic binding • Routing protocols (OSPF) • Routing VPN and EVPN • Load Balancing

	Multisite	NSX Federation
Security Services	All features supported: • Distributed Firewall • Gateway Firewall • FQDN Filtering • L7 App ID context support • Identity Firewall • Distributed IDS • Malware Prevention • Network Introspection • Endpoint Protection • Time-Based Firewall	All features supported from GM: • Distributed Firewall • Gateway Firewall • FQDN Filtering • L7 App ID context support • Time-Based Firewall Exceptions: • Identity Firewall • Distributed IDS • Malware Prevention • Network Introspection and Endpoint Protection
High-Availability for Management Plane	NSX Manager VMs recovery • With SRM from NSX. See Working with VMware Live Site Recovery and Multisite Environments. • With vSphere HA.	NSX GM and LM VMs recovery with SRM.
High-Availability for Compute VMs	Compute VMs recovery • With SRM from NSX. See Working with VMware Live Site Recovery and Multisite Environments . • With vSphere HA.	Compute VMs recovery • With SRM from NSX with one limitation: Distributed Firewall on Workload VMs cannot be based on NSX tags. • With SRM from NSX without any limitations.

Also see the NSX Multi-location Design Guide and VMware Site Recovery Manager.

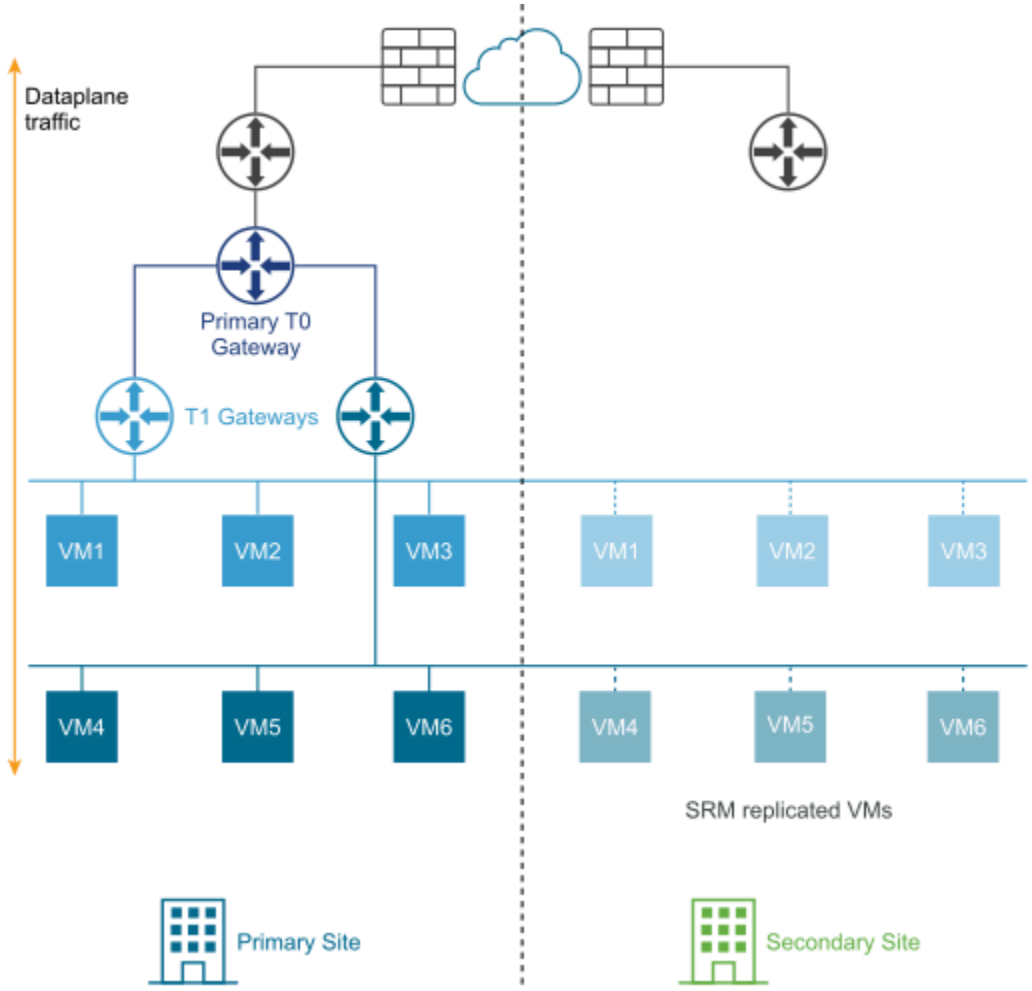
NSX Multisite

NSX supports multisite deployments where you can manage all the sites from one NSX Manager cluster.

Two types of multisite deployments are supported:

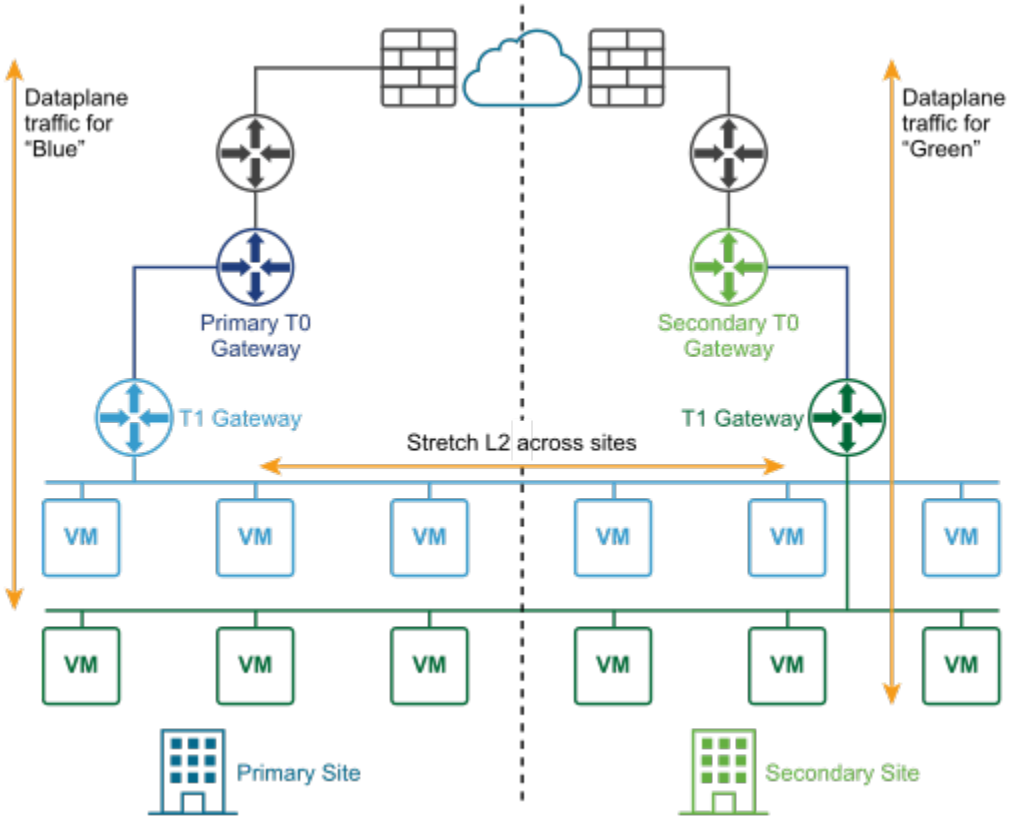
- Disaster recovery
- Active-active

The following diagram illustrates a disaster recovery deployment.



In a disaster recovery deployment, NSX at the primary site handles networking for the enterprise. The secondary site stands by to take over if a catastrophic failure occurs at the primary site.

The following diagram illustrates an active-active deployment.



You can deploy two sites for automatic recovery of the management plane and the data plane.

Automatic Recovery of the Management Plane

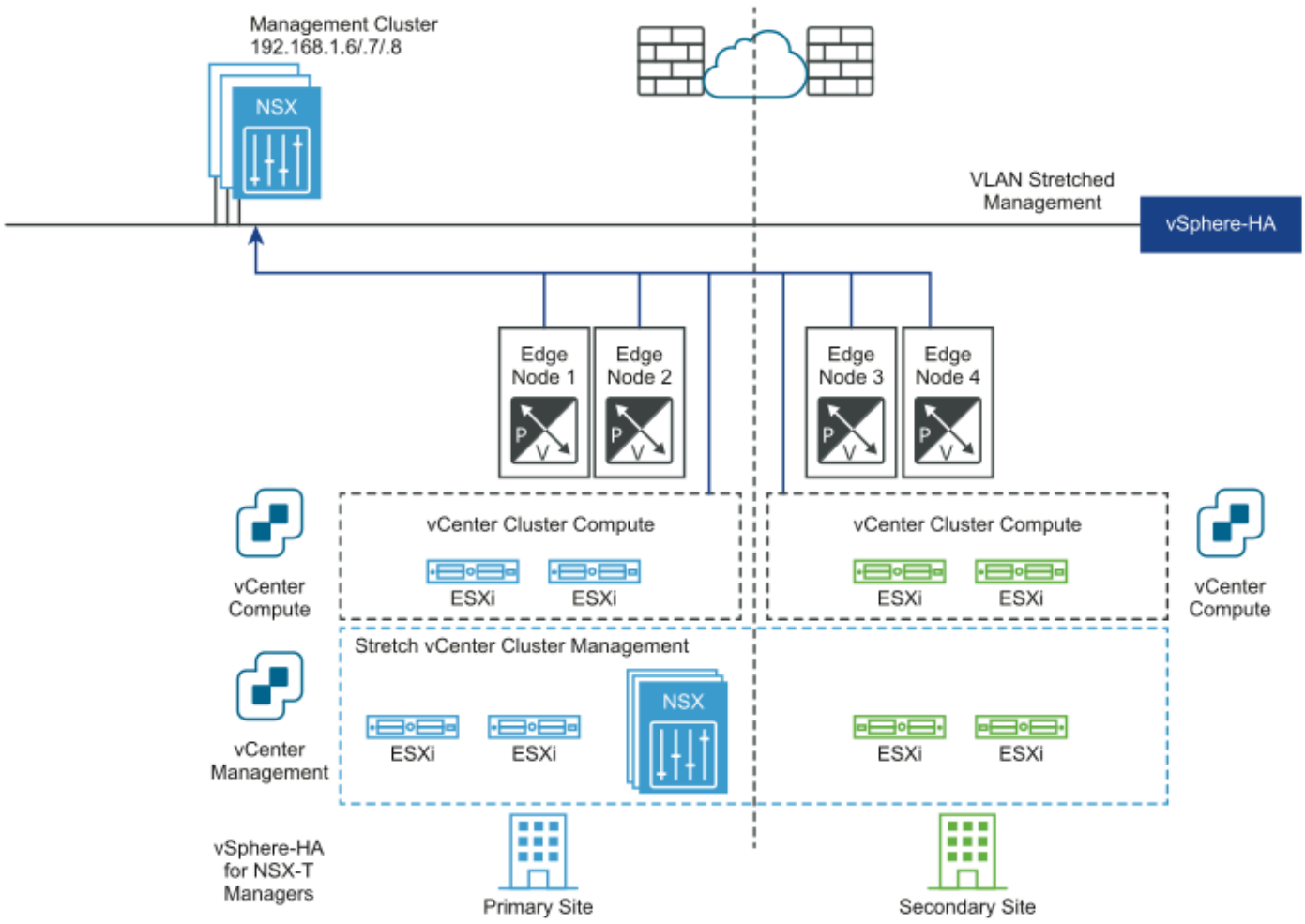
Requirements:

- A stretched vCenter cluster with HA across sites configured.
- A stretched management VLAN.

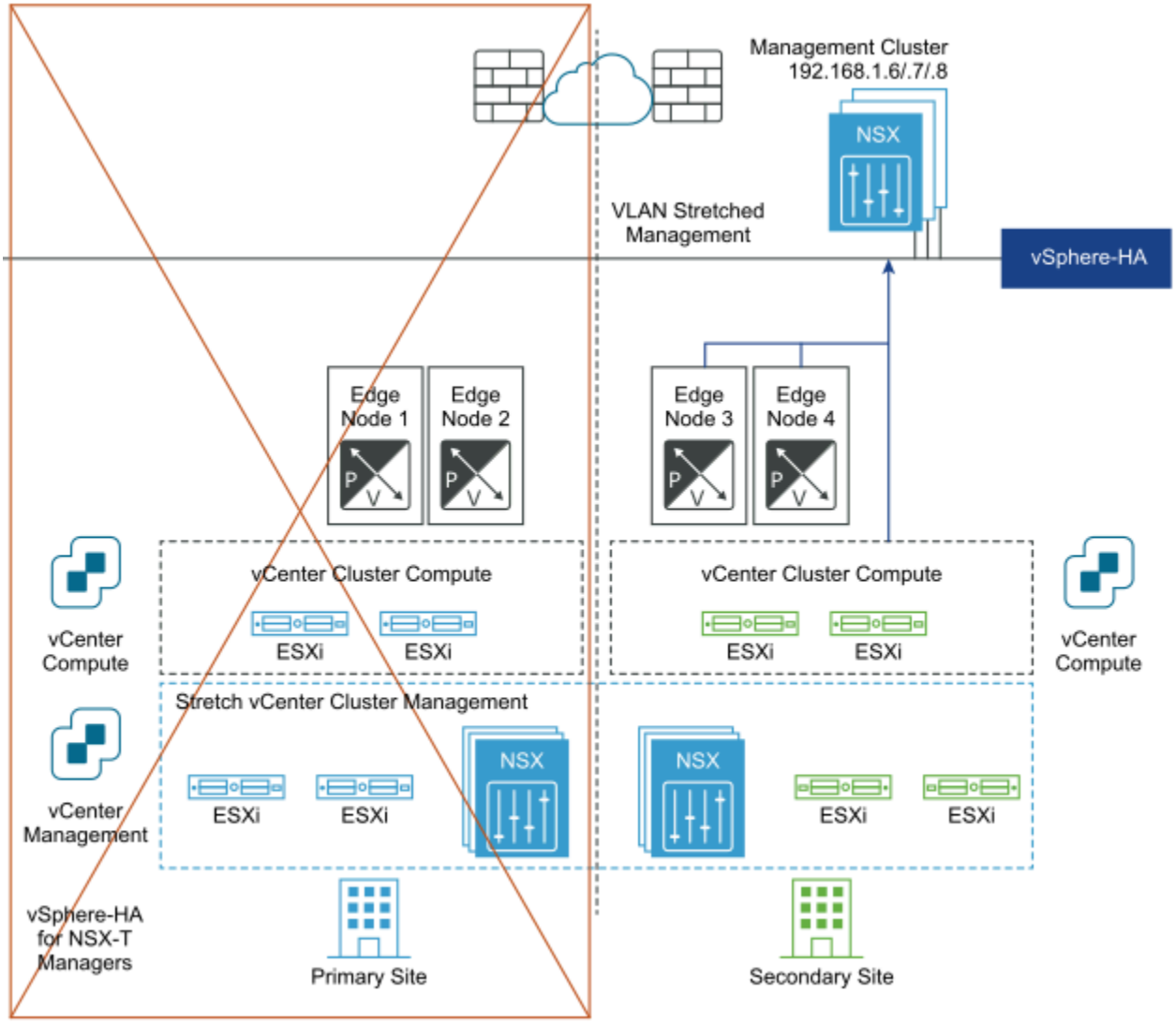
The NSX Manager cluster is deployed on the management VLAN and is physically in the primary site. If there is a primary site failure, vSphere HA will restart the NSX Managers in the secondary site. All the transport nodes will reconnect to the restarted NSX Managers automatically. This process takes about 10 minutes. During this time, the management plane is not available but the data plane is not impacted.

The following diagrams illustrate automatic recovery of the management plane.

Before the disaster:



After disaster recovery:



Automatic Recovery of the Data Plane

You can configure failure domains for Edge nodes to achieve automatic recovery of the data plane. You can group Edge nodes within an Edge cluster in different failure domains. NSX Manager will automatically place any new active tier-1 gateway in the preferred failure domain, and the standby tier-1 gateway in the other domain. Note that a T1 deployed prior to the failure domain creations will keep their original Edge Node placement and might not be running where you want. If you want to fix their placement, edit the T1 and manually select the Edge Nodes for T1-Active and T1-Standby.

Requirements:

- The maximum latency between Edge nodes is 10 ms.
- If asymmetric north-south routing is not achievable, for example a physical firewall is used northbound to the NSX Edge node, then the HA mode for the tier-0 gateway must be active-standby, and the failover mode must be preemptive.
- If asymmetric north-south routing is possible, for example the two locations are two buildings without any physical firewall between them, then the HA mode for the tier-0 gateway can be active-active.

The failover mode of the tier-1 gateway can be preemptive or non-preemptive, but preemptive is recommended to guarantee that the tier-0 and tier-1 gateways are in the same location.

Configuration steps:

- Using the API, create failure domains for the two sites, for example, `FD1A-Preferred_Site1` and `FD2A-Preferred_Site1`.

Note:
If you want all your tier-1 with the T1-Active in the Edge nodes and the T1-Standby in other Edge nodes to be part of the same failure domain (FD1A-Preferred_Site1 and FD2A-Preferred_Site1); then you must first create your tier-1 with the option preemptive and then create your failure domain primary (FD1A-Preferred_Site1) set to `preferred_active_edge_services = true`. For example,

```
POST /api/v1/failure-domains
{
"display_name": "FD1A-Preferred_Site1",
"preferred_active_edge_services": "true"
}

POST /api/v1/failure-domains
{
"display_name": "FD2A-Preferred_Site1",
"preferred_active_edge_services": "false"
}
```

- Using the API, configure an Edge cluster spanning the two sites. For example, the cluster has Edge nodes `EdgeNode1A` and `EdgeNode1B` in the primary site, and Edge nodes `EdgeNode2A` and `EdgeNode2B` in the secondary site. The active tier-0 and tier-1 gateways will run on `EdgeNode1A` and `EdgeNode1B`. The standby tier-0 and tier-1 gateways will run on `EdgeNode2A` and `EdgeNode2B`.
- Using the API, associate each Edge node with the failure domain for the site. First call the `GET /api/v1/transport-nodes/<transport-node-id>` API to get the data about the Edge node. Use the result of the GET API as the input for the `PUT /api/v1/transport-nodes/<transport-node-id>` API, with the additional property, `failure_domain_id`, set appropriately. For example,

```
GET /api/v1/transport-nodes/<transport-node-id>
Response:
{
    "resource_type": "TransportNode",
    "_revision": 15"
    "description": "Updated NSX configured Test Transport Node",
    "id": "77816de2-39c3-436c-b891-54d31f580961",
    ...
}

PUT /api/v1/transport-nodes/<transport-node-id>
{
    "resource_type": "TransportNode",
    "_revision": 15"
    "description": "Updated NSX configured Test Transport Node",
    "id": "77816de2-39c3-436c-b891-54d31f580961",
    ...
    "failure_domain_id": "<UUID>",
}
```

- Using the API, configure the Edge cluster to allocate nodes based on failure domain. First call the `GET /api/v1/edge-clusters/<edge-cluster-id>` API to get the data about the Edge cluster. Use the result of the GET API as the input for the `PUT /api/v1/edge-clusters/<edge-cluster-id>` API, with the additional property, `allocation_rules`, set appropriately. For example,

```
GET /api/v1/edge-clusters/<edge-cluster-id>
Response:
{
    "_revision": 0,
    "id": "bf8d4daf-93f6-4c23-af38-63f6d372e14e",
    "resource_type": "EdgeCluster",
    ...
}
```

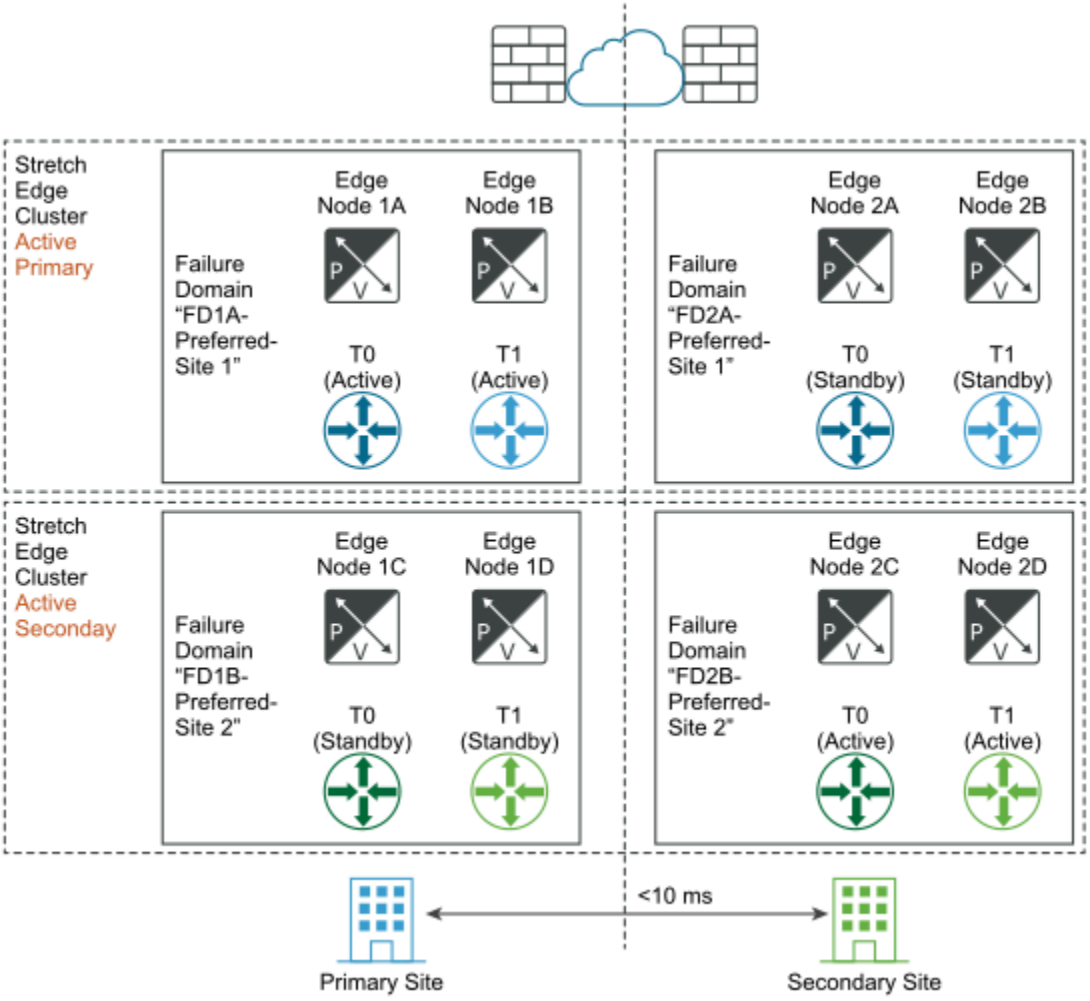
```
PUT /api/v1/edge-clusters/<edge-cluster-id>
{
    "_revision": 0,
    "id": "bf8d4daf-93f6-4c23-af38-63f6d372e14e",
    "resource_type": "EdgeCluster",
    ...
    "allocation_rules": [
        {
            "action": {
                      "enabled": true,
                      "action_type": "AllocationBasedOnFailureDomain"
                      }
        }
    ],
}
```

- Create tier-0 and tier-1 gateways using the API or NSX Manager UI.

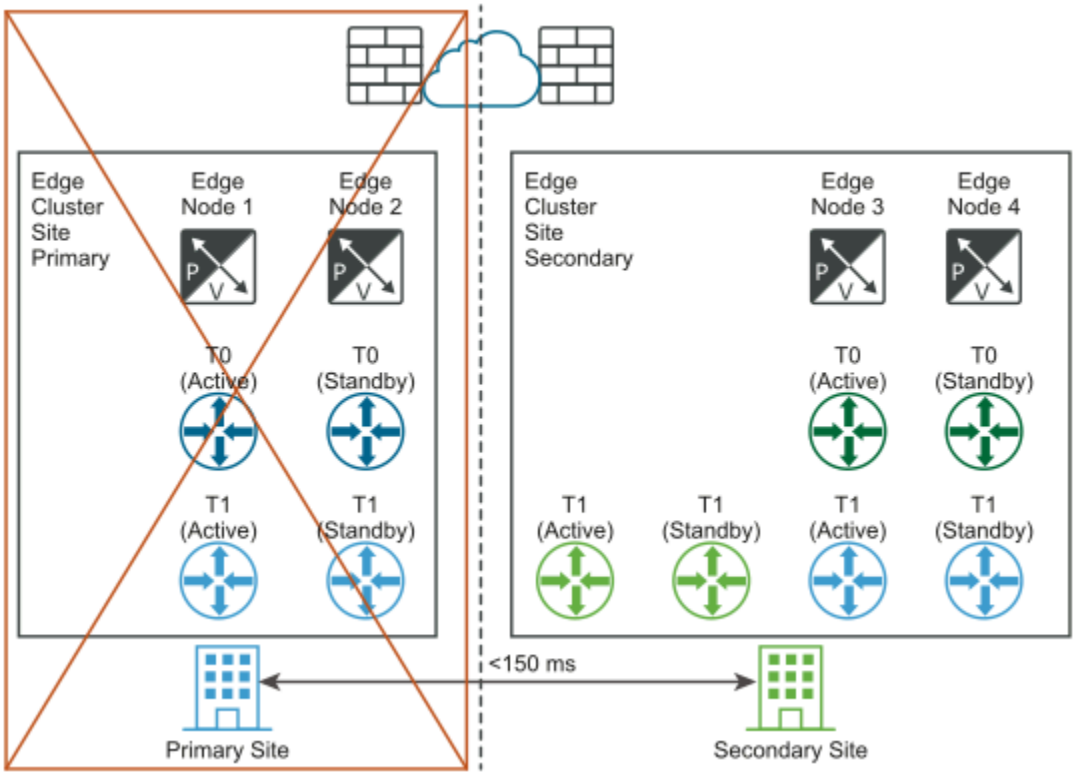
In case of a full primary site failure, the tier-0 standby and tier-1 standby in the secondary site automatically take over and become the new active gateways.

The following diagrams illustrates automatic recovery of the data plane.

Before the disaster:



After disaster recovery:



In case of a failure of one of the Edge nodes in the primary site and not full site failure, it is important to note that the same principle applies. For example, in the diagram, "Before the disaster", assume that Edge node 1B hosts tier-1-blue active and that Edge node 2B hosts the tier-1-blue standby. If Edge node 1B fails, the standby tier-1-blue on Edge node 2B takes over and become the new tier-1-blue active gateway.

Requirements for Multisite Deployments

Inter-site Communication

- The bandwidth must be at least 1 Gbps and the latency (RTT) must be less than 10 ms.
- MTU must be at least 1700. 9000 is recommended.

NSX Manager

- With automatic recovery of management plane with VLAN management stretched between sites. vSphere HA across sites for NSX Manager VMs.

Data Plane

- The same internet provider must be used if public IP addresses are exposed through services such as NAT or load balancer.
- With automatic recovery of management plane
  - Maximum latency between locations is 10 ms.
  - The HA mode for the tier-0 gateway must be active-standby and the failover mode must be preemptive to guarantee no asymmetric routing.
  - The HA mode for the tier-0 gateway can be active-active if asymmetric routing is acceptable (such as different buildings in a metropolitan region).
- Cloud Management System (CMS)
  - The CMS must support an NSX plug-in. In this release, VMware Integrated OpenStack (VIO) and vRealize Automation (vRA) satisfy this requirement.

Limitations

- No local-egress capabilities. All north-south traffic must occur within one site.

- The compute disaster recovery software must support NSX, for example, VMware SRM 8.1.2 or later.
- When restoring the NSX Manager in a multi-site environment do the following on the secondary/primary site:
  - After the restore process pauses at the **AddNodeToCluster** step, you must first remove the existing VIP and set the new virtual IP from the **System** > **Appliances** UI page before you add additional manager nodes.
  - Add new nodes to a restored one-node cluster after the VIP is updated.

Working with VMware Live Site Recovery and Multisite Environments

You can use VMware Live Site Recovery™ with NSX Multisite for disaster recovery use cases.

For detailed instructions on using VMware Live Site Recovery, see the Documentation.

VMware Live Site Recovery supports the following workflows with NSX Multisite:

- NSX Management VMs support full and test recovery of Management VMs (supported with or without NSX management cluster VIP).
- NSX Compute VMs support full and test recovery of compute VMs. Recovered VMs in the disaster recovery site have their NSX tags and any firewall rules based on these NSX tags or other VM attributes, such as VM name. If a VM gets deleted from ESX, NSX removes the VM from its inventory but saves its VM tag information internally for 30 minutes. If that VM, with same instanceUuid, gets recovered in that time frame it gets its original VM tags. After 30 minutes, the NSX admin must reconfigure its VM tags. For VMs on ESX, instanceUUid and external Id values are the same.

  Note:

  For VMware Live Site Recovery planned migration:

  - VMware Live Site Recovery with array-based replication removes the VMs from the protected ESX site and recovers those VMs to the recovery ESX site. If that process takes more than 30 minutes, the VMs lose their NSX tags. As a result, the NSX admin must reconfigure those NSX tags.
  - VMware Live Site Recovery with vSphere Replication does not remove the VMs from the protected ESX site. It powers them off and recover those VMs to the recovery ESX site. As a result, the VMs never lose their VM tags.

For NSX Federation disaster recovery support, see Disaster Recovery for Global Manager.

NSX Federation

With NSX Federation, you can manage multiple NSX environments with a single pane of glass view, create gateways and segments that span one or more locations, and configure and enforce firewall rules consistently across locations.

Once you have installed the Global Manager and have added locations, you can configure networking and security from Global Manager.

For information about the initial NSX Federation configuration, including installing Global Manager and adding locations, see Getting Started with.

Overview of NSX Federation

Before you configure your NSX Federation environment, understand what feaures are supported, how NSX Federation shares information across locations, and how the user interface works.

See NSX Federation Key Concepts for definitions of terminology specific to NSX Federation.

NSX Federation Key Concepts

NSX Federation introduces some new terms and concepts, such as remote tunnel endpoint (RTEP), span, and region.

NSX Federation Systems: Global Manager and Local Manager

An NSX Federation environment includes two types of management systems:

- Global Manager: A system similar to NSX Manager that federates multiple Local Managers.
- Local Manager: An NSX Manager system in charge of network and security services for a location.

NSX Federation Span: Local and Stretched

When you create a networking object from Global Manager, it can span one or more locations.

- Local: The object spans only one location.
- Stretched: The object spans more than one location.

You do not directly configure the span of a segment. A segment has the same span as the gateway it is attached to.

NSX Federation Regions

Security objects have a region. The region can be one of the following:

- Location: Each location automatically creates a region. This region has the span of that location.
- Global: A region that has the span of all available locations.
- Custom Region: You can create regions that include a subset of the available locations.

NSX Federation Tunnel Endpoints

In an NSX Federation environment, there are two types of tunnel endpoints.

- Tunnel End Point (TEP): The IP address of a transport node (Edge node or Host) used for Geneve encapsulation within a location.
- Remote Tunnel End Points (RTEP): The IP address of a transport node (Edge node only) used for Geneve encapsulation across locations.

Features and Configurations Supported in NSX Federation

This page describes the features and configurations supported in NSX Federation.

Configuration Maximums

An NSX Federation environment has the following configuration maximums:

- For most configurations, the Local Manager cluster has the same configuration maximums as an NSX Manager cluster. Go to VMware Configuration Maximums tool and select NSX.
  Select the NSX Federation category for NSX in the VMware Configuration Maximums tool for exceptions and other NSX Federation-specific values.
- For a given location, the following configurations contribute to the configuration maximum:
  - Objects that were created on the Local Manager.
  - Objects that were created on the Global Manager and include the location in its span.

  You can view the capacity and usage on each Local Manager. See View the Usage and Capacity of Categories of Objects.

Feature Support

Note that in NSX Federation, Service insertion (Network Introspection) support only occurs when an NSX Federation environment has a Global Manager (GM) deployed under the following conditions:

- All service-insertion related configuration such as partner service registration, deployment and consumption, is done from a Local Manager (LM).
- Only objects configured on the LM are used with service insertion. This includes groups, segments, and any other constructs. Service insertion cannot be applied to workloads connected to a stretched/global segment defined from the GM, or any segment connected to a logical router created from the GM. Groups created from the Global Manager should not be used within service insertion redirection polices.

Important:

- NSX Federation locations must run on environments where administrators have full control of the underlay fabric.
- NSX Federation does not currently support Global Manager and Local Manager hosted on VMware Cloud on AWS, VMware Cloud on Dell, Azure VMware Solution, Google Cloud VMware Engine, Oracle Cloud VMware Solution, or Alibaba VMware Cloud Service.

After Local Manager is registered to Global Manager:

- You can continue to configure on Local Manager. For more details, refer to Understanding NSX Federation.
- Some objects configured from Local Manager can have options/settings that are Global Manager objects. For example, you can connect an LM_created-t1 to a GM_created-t0.
- Some objects configured from Local Manager cannot be configured with options/settings that are Global Manager objects. For example, you cannot connect an LM_created-segment to a GM_created-t0. Note that you can only edit those LM-objects from Local Manager; Global Manager does not see those object. For more information, refer to "Logical Configuration Ownership" in the *NSX-T Multi-Location Design Guide* for your release.

The Features Supported in NSX Federation table describe the features available in Global Manager.

Table 1049: Features Supported in NSX Federation

Feature	Details	Related Links
Tier-0 Gateway	• Active-active and active-standby. • Active-active only	Add a Tier-0 Gateway from Global Manager
Tier-1 Gateway		Add a Tier-1 Gateway from Global Manager
Segments	You can include Layer 2 bridge configuration from Global Manager.	Add a Segment from Global Manager and Configure Bridging on Global Manager
Groups	Some limitations. See Security in NSX Federation.	See the section on creating groups from Global Manager in the VMware vDefend documentation.
Distributed Firewall	Drafts of the security policies are now available on Global Manager. This includes support for auto and manual drafts.	See the section on creating drafts in Global Manager in the VMware vDefend documentation.
Firewall Exclusion List	Available.	See the VMware vDefend documentation.
Time Based Firewall Rules	Available.	See the VMware vDefend documentation.
Gateway Firewall	Only Layer 3 and 4 rules are supported.	See the section on creating gateway policies and rules from Global Manager in the VMware vDefend documentation.

Feature	Details	Related Links
Network Address Translation (NAT)	1. Tier-0 Gateway: – Active-active: You can configure stateless NAT only, that is, with action type Reflexive. – Active-standby: You can create stateful or stateless NAT rules. 2. Tier-1 Gateway: – You can create stateful or stateless NAT rules. – Stateless NAT rules are pushed to all locations in the gateway's span unless scoped to one or more locations specifically. – Stateful NAT rules are also pushed to all locations in the gateway's span or to the specific location selected. However, stateful NAT rules are realized and enforced only on the primary location.	Configure an NSX NAT/DNAT/No SNAT/No DNAT/Reflexive NAT
DNS		See Add an NSX DNS Forwarder Service
DHCP and SLAAC	• DHCP Relay is supported on segments and gateways. • DHCPv4 server is supported on gateways with DHCP static bindings configured on segments. • You can assign IPv6 addresses using SLAAC with DNS Through RA (DAD detects duplicates within a location only).	• DHCP Relay: Add an NSX DHCP Relay Profile • DHCP Server (supported on gateway only): – Add an NSX DHCP Server Profile – Attach an NSX DHCP Profile to a Tier-0 or Tier-1 Gateway – NSX DHCP Configuration Settings: Reference • IPv6 address assignment: Create NSX SLAAC and DAD Profiles for IPv6 Address Assignment
Distributed Malware Detection and Prevention	Starting with NSX 4.1.2: • Deploy NSX Application Platform (NAPP) on each federated Local Manager. • Deploy and manage Malware Prevention from the Local Manager UI. • You can use stretched and non-stretched segments for the virtual machine endpoints.	See the section on Malware Prevention for vDefend in the VMware vDefend documentation.
Network Detection and Response	Starting with NSX 4.1.2: • Deploy NAPP on each federated Local Manager. • Deploy and manage Network Detection and Response (NDR) from the Local Manager UI. • You can use stretched and non-stretched segments to collect NDR events.	For more details about Network Detection and Response, see the VMware vDefend ATP documentation.

Feature	Details	Related Links
Using objects created on Global Manager in a Local Manager configuration	1. Most configurations are supported. For example: – Connecting a Local Manager tier-1 gateway to a Global Manager tier-0 gateway. – You can use a Global Manager group in a Local Manager distributed firewall rule. 2. These configurations are **not** supported: – Connecting a Local Manager segment to a Global Manager tier-0 or tier-1 gateway. – Connecting a load balancer to a Global Manager tier-1 gateway.	
Network Monitoring	• Expanded communication monitoring between Local Manager and Global Manager. • Traceflow across NSX instances in the same Federation.	
Certificates	Local Manager self-signed certificates generate only when the Local Manager is in the NSX Federation environment. That same certificate gets deleted if Local Manager moves out of the NSX Federation environment.	Certificates for NSX and NSX Federation
LDAP	Authenticate Global Manager users using a directory service such as Active Directory over LDAP or OpenLDAP.	Integration with LDAP
Backup and Restore	Backup with FQDN or IP is supported.	Backup and Restore in NSX Federation
Users	You can add new local users and remove audit and guest users.	Manage Local User Accounts
vMotion between locations	Tag replication across locations is supported.	

Understanding NSX Federation

In NSX Federation, you make configuration changes on the active Global Manager. The active Global Manager then synchronizes the changes with the relevant Local Managers and the standby Global Manager, if you have one. Local Managers also sync some information with each other and to the Global Manager.

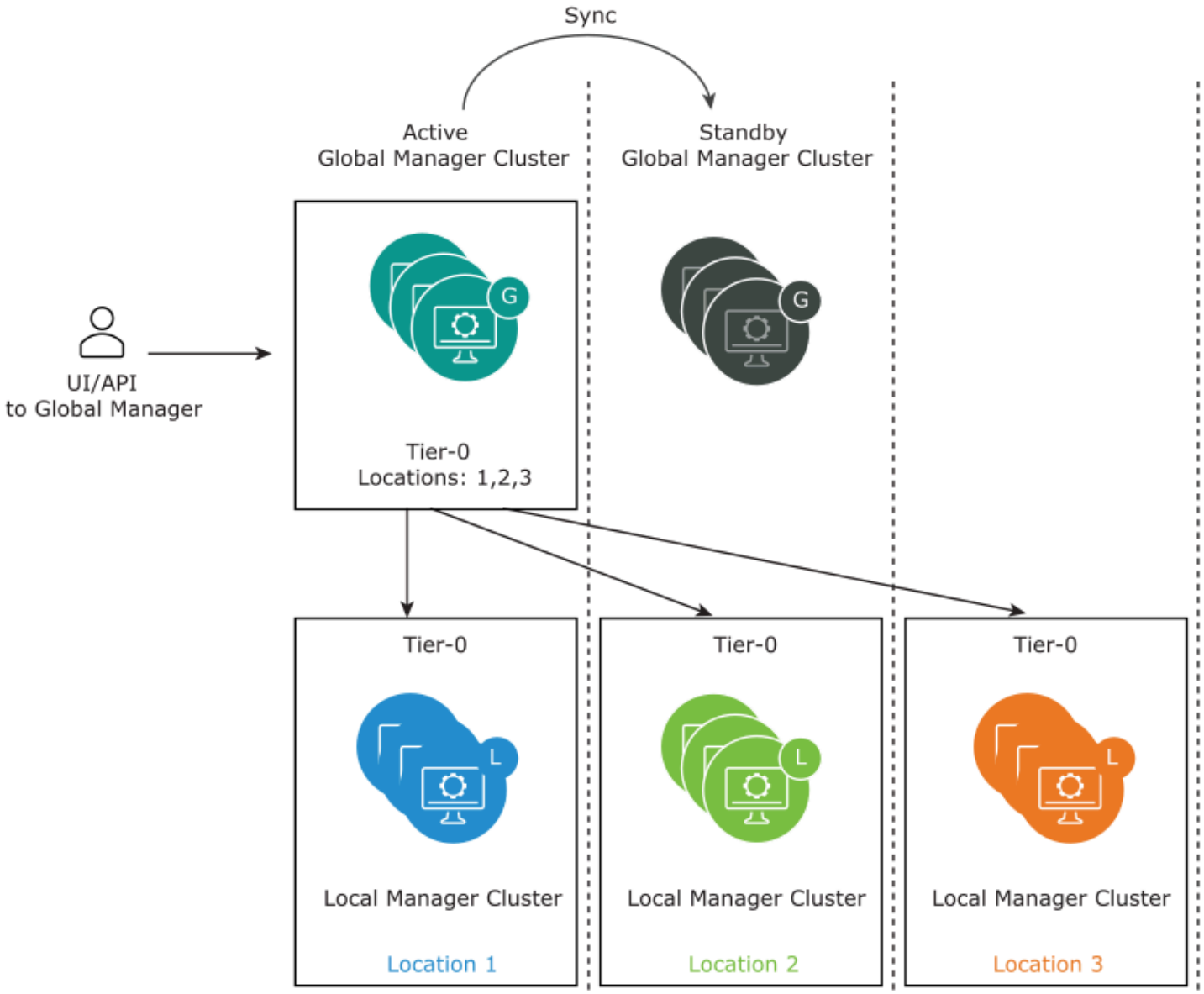
Making Changes on Global Manager

The Global Manager provides a user interface similar to the NSX Manager interface.

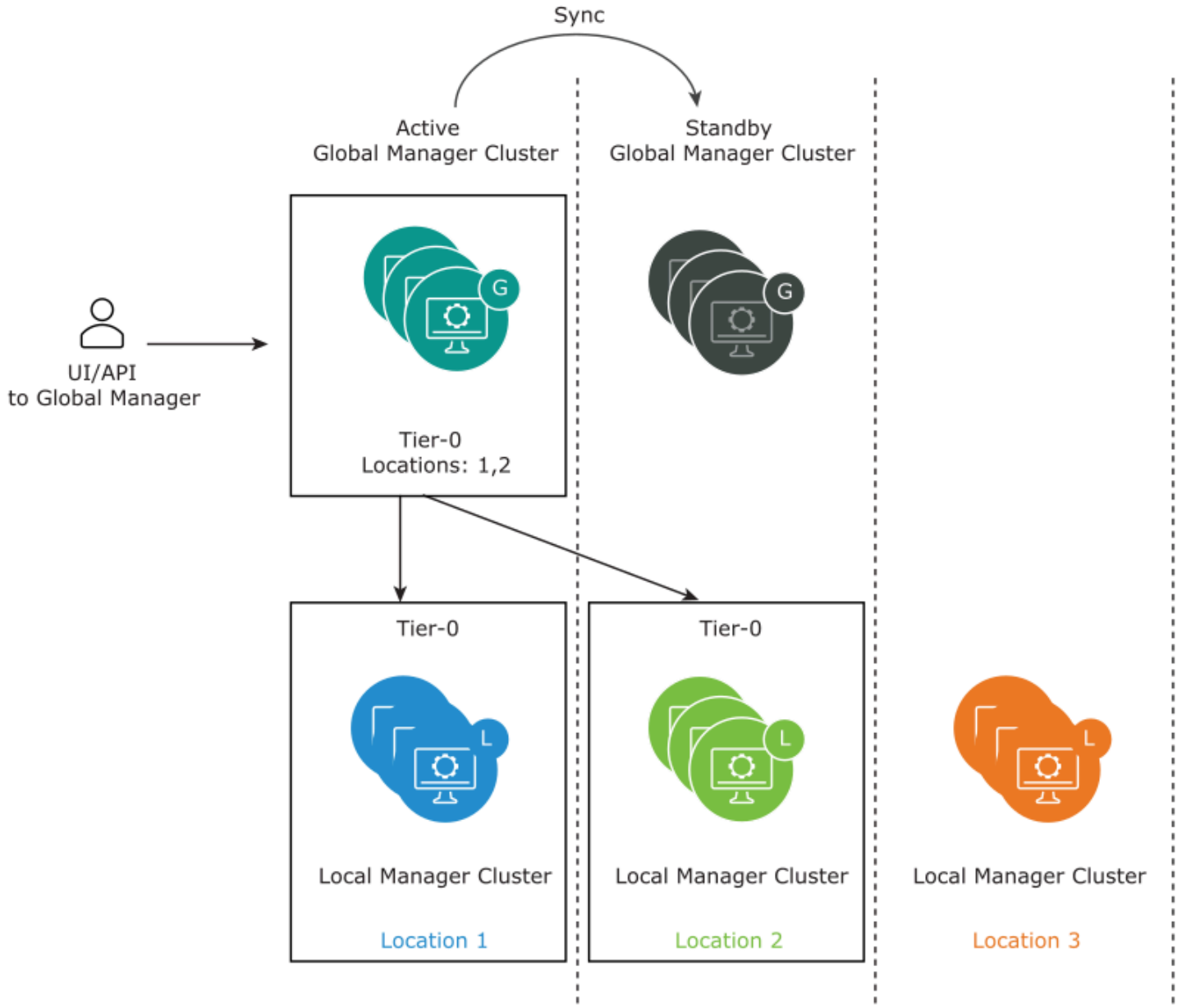
If you choose, you can configure all objects on the Global Manager, independent of span.

The Global Manager syncs a configuration with a Local Manager only if the configuration is relevant to that location. For example, if you create a tier-0 gateway and add it to Location 1, Location 2, and Location 3, the configuration gets synchronized with all three Local Managers. Local Managers can only synchronize once with Global Manager during a configuration import.

If you have a standby Global Manager, the configurations synchronize between the active Global Manager and the standby Global Manager.

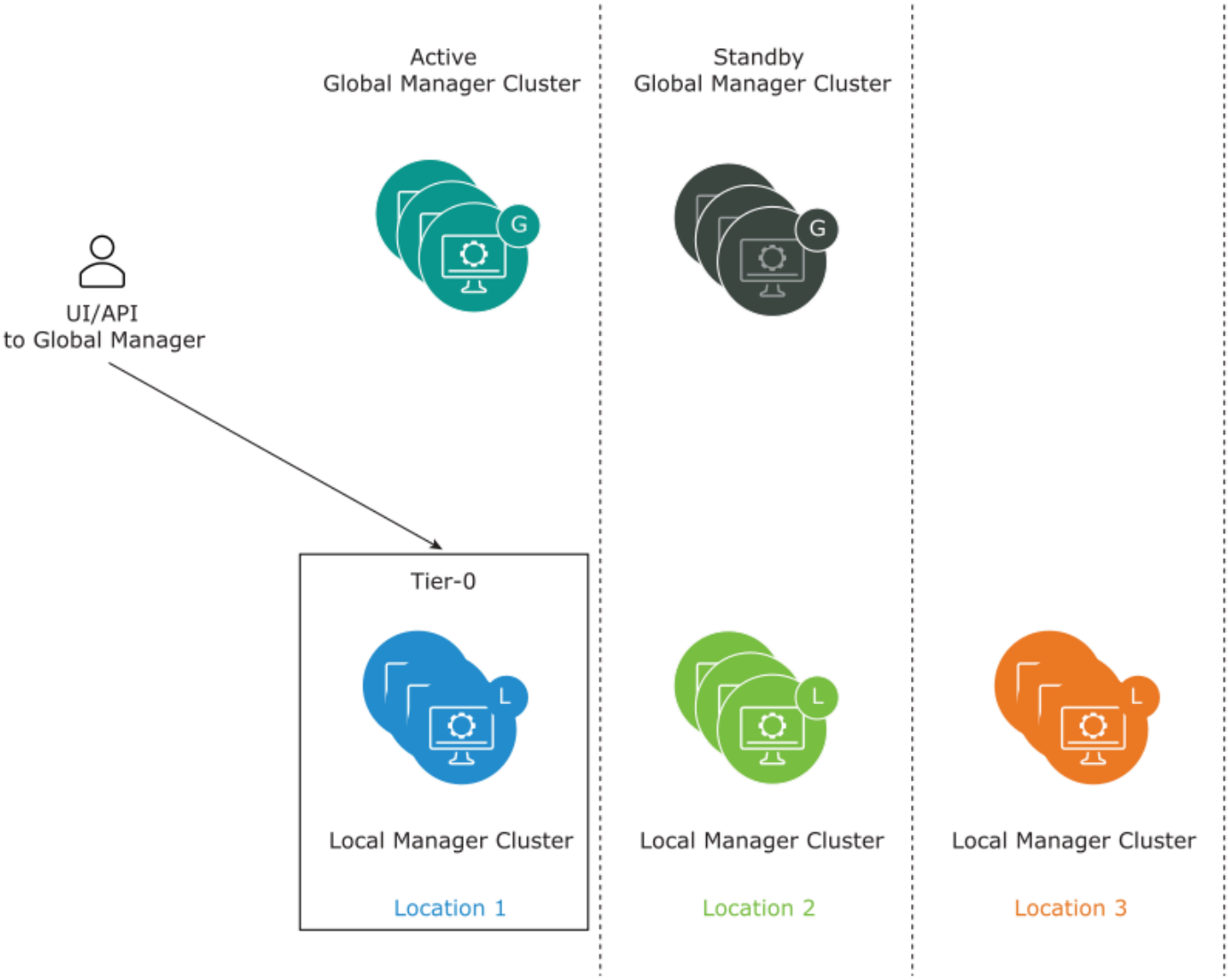


If the tier-0 gateway is added only to Location 1 and Location 2, the configuration is not synced with Location 3.



Making Changes on Local Managers

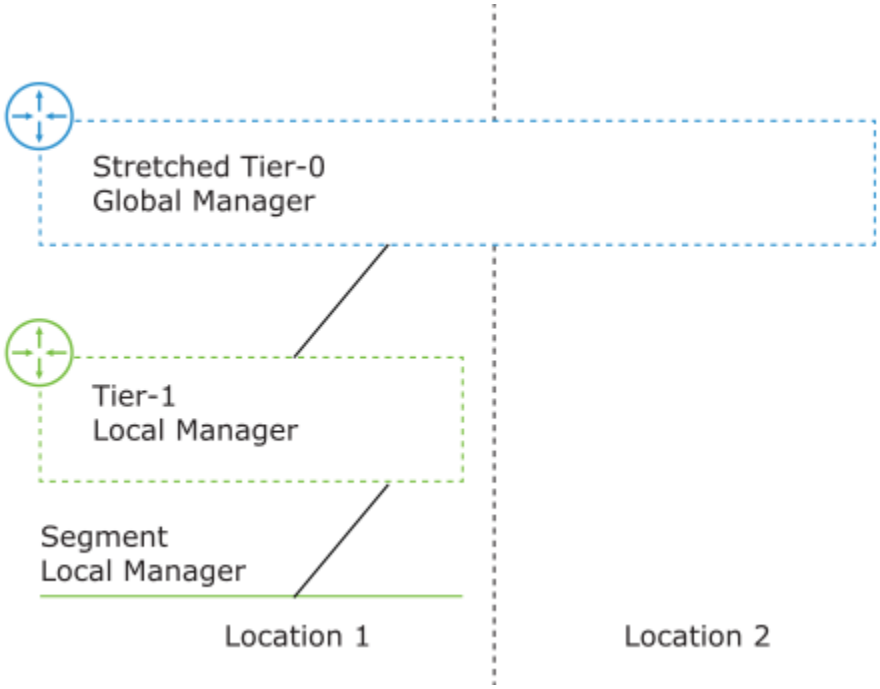
To create objects on a specific Local Manager, you use that Local Manager. These objects do not sync with the active Global Manager or any other Local Managers.



Realizing Global Manager Changes on Local Managers

The Global Manager validates change against the Global Manager and the Local Manager configurations. When a Local Manager receives a configuration from the Global Manager, it realizes the configuration in the fabric nodes of that Local Manager. During this realization, errors or conflicts might get detected. To monitor configuration flow, use the NSX Federation monitoring dashboard. See Monitoring NSX Federation Locations for details.

For example, you can create a tier-0 gateway from Global Manager, and then from a Local Manager you can create and attach a tier-1 gateway to the tier-0 gateway.



Because Local Managers now sync their configurations to the Global Manager, the Global Manager context the tier-0 gateway now appears to be connected. You can delete the tier-0 gateway from the Global Manager, and this change gets synchronized to the Local Managers. When the changes in each location get realized, the following occurs:

- The tier-0 gateway might get deleted from the Local Manager in Location 2.
- The tier-0 gateway might get deleted from the Local Manager in Location 1.
- The tier-0 gateway gets marked for deletion on the Global Manager.

When the tier-0 disconnects from the tier-1 in Location 1, the tier-0 gets deleted from Global Manager.

Most problems are displayed on the user interface. You can also display problems using these API calls.

- On Global Manager:
  ```
  GET /global-manager/api/v1/global-infra/realized-state/alarms
  ```
- On Local Manager:
  ```
  GET /policy/api/v1/infra/realized-state/alarms
  ```

Getting Started with NSX Federation

To get started with NSX Federation, you install the Global Manager, configure the Global Manager as active, and add locations.

Task	Details
Check the requirements for Federation.	See NSX Federation Requirements.
Install the Global Manager.	See Install the Active and Standby Global Manager.
Make the Global Manager cluster active.	See Make the Global Manager Active and Add Standby Global Manager.
Add Locations to the active Global Manager.	See Add a Location .

For further configuration tasks, such as preparing Edge clusters for stretched networking, and creating objects from the Global Manager, see *Federation* in the *NSX Administration Guide*.

NSX Federation Requirements

To support NSX Federation, your environment must meet various requirements, including round-trip time, software versions, and ports.

- There must be a maximum round-trip time of 500 ms between the following nodes:
  - Active Global Manager and standby Global Manager.
  - Global Manager and Local Manager.
  - Local Manager and remote Local Manager if you have cross-location security configuration only and VMware NSX® Edge™ Nodes RTEP and remote Edge Nodes RTEP if you have cross-network configurations.
- The Global Manager and Local Manager appliances must all have NSX 3.1.0 or later installed. All appliances in an NSX Federation environment must have the same version installed.
- The required ports must be open to allow communication between the Global Manager and Local Manager. See VMware Ports and Protocols.
- There must be connectivity without NAT between the following nodes:
  - Global Manager and Local Manager.
  - Local Manager and remote Local Manager.
  - Edge node RTEP and remote Edge node RTEP.
- Verify that each location has a default overlay transport zone configured. From each Local Manager, select **System > Fabric > Transport Zones**. Select an overlay transport zone, and click **Actions > Set as Default Transport Zone**.
- Global Manager supports only Policy Mode. NSX Federation does not support Manager Mode.

An NSX Federation environment has the following configuration maximums:

- For most configurations, the Local Manager cluster has the same configuration maximums as an NSX Manager cluster. Go to the VMware Configuration Maximums tool and select NSX. Select the NSX Federation category for NSX in the VMware Configuration Maximums tool for exceptions and other NSX Federation-specific values.
- For a given location, the following configurations contribute to the configuration maximum:
  - Objects that were created on the Local Manager.
  - Objects that were created on the Global Manager and include the location in its span.

You can view the capacity and usage on each Local Manager. See *View the Usage and Capacity of Categories of Objects* in the *NSX Administration Guide*.

Configuring the Global Manager and Local Managers

An NSX Federation environment contains an active and a standby Global Manager cluster and one or more Local Manager clusters.



Install the Active and Standby Global Manager
To use NSX Federation, you must install the Global Manager.

- Verify that your environment meets the requirements for NSX Manager.
- Decide which locations will contain the active and standby Global Manager appliances.

  Important:
  To consume the latest features of a given NSX version, all Global Manager and Local Manager appliances in an NSX Federation environment must have the same version of NSX installed.

Installing a Global Manager appliance is similar to installing an NSX Manager appliance. The only difference is that when you deploy the appliance, you select *`NSX Global Manager`* for the role.

Install a standby Global Manager appliance for high availability and disaster recovery. The standby Global Manager appliance must be installed in a different location with a latency of 500ms or less.

1. To install the first Global Manager appliance on vSphere: Install NSX Manager and Available Appliances.
   - Select `Medium`, `Large`, or `Extra Large` for the deployment configuration size.
   - Select `NSX Federation Global Manager` for the **Rolename**.
2. Log in to the NSX Manager appliance.
   See Log In to the Newly Installed NSX Manager.
3. Configure a compute manager.
   See Add a Compute Manager.

   Note:
   If you are at this step while installing the standby Global Manager, you must configure a separate compute manager. Do not use the same compute manager that you configured for the active Global Manager.
4. Create a Global Manager cluster.
   - On vSphere with a compute manager configured: See Deploy NSX Manager Nodes to Form a Cluster from the UI.
   - On vSphere without a compute manager configured: Repeat the NSX Manager install on vSphere steps to install three appliances, then form the cluster. See Form an NSX Manager Cluster Using the CLI.
5. Configure a VIP for the Global Manager cluster.
   See Configure a Virtual IP Address for a Cluster.
6. In a different location, install a standby Global Manager appliance and form a cluster by repeating these steps.

Select the designated Global Manager appliance as active and connect it with the standby Global Manager.

Make the Global Manager Active and Add Standby Global Manager

After you have deployed a Global Manager appliance, you can make the Global Manager active.

Adding a standby Global Manager is optional but recommended for high availability of the Global Manager.

1. Log in to the appliance at https://global-manager-ip-or-fqdn/.
2. Select **System** > **Location Manager**. In the **Global Manager** tile, click **Make Active**. Provide a descriptive name for the active Global Manager and click **Save**.
3. Optional: Add a standby Global Manager cluster.
   a) Install a new Global Manager appliance in a secondary location and start it. Follow the same instructions as for installing the primary Global Manager, see Install the Active and Standby Global Manager.
   b) From the active Global Manager, add this newly installed Global Manager appliance as standby.
      Navigate back to your active Global Manager and click **Add Standby** and provide the following information:

Option	Description
Global Manager Name	Provide a name for the standby Global Manager.
FQDN/IP	Enter the FQDN or IP address of the Global Manager cluster VIP at the secondary location. Do not enter an individual Global Manager FQDN or IP.
Username and **Password**	Provide the admin user's credentials for the Global Manager at the secondary location.

SHA-256 Thumbprint	Log in to any Global Manager node at the secondary location and run this command:
	`get certificate cluster thumbprint`
	The result is the cluster VIP certificate thumbprint:
	`bfae1a0a...`
	If the GM-Standby is a single Manager VM, use the same command.

Check Compatibility	Click **Check Compatibility** to ensure that the Global Manager can be added as standby. This checks that the NSX version is compatible.

c) Click **Save**.
d) Click **Make Standby**.

Add a Location
After you add a location to Global Manager, you can create objects from Global Manager that span that location.

- Verify that the NSX environment you are adding has the latest version of NSX installed.
  This new NSX location can be a new NSX environment or an NSX environment with an existing Network and Security configuration.
- The NSX environment in the new location must have three NSX Manager nodes deployed and a cluster VIP configured. See Configure a Virtual IP Address for a Cluster
- .
  For a proof-of-concept environment, you can add a location that has only one NSX Manager node, but you must still configure a cluster VIP.
- Verify that the latency between the Global Manager and the location is 500 ms or less for non-stretched networks or 150 ms or less for stretched networks.

You can find the number of supported locations in the VMware Configuration Maximums tool. Select the appropriate version of NSX, select the NSX Federation category, and click **View Limits**.

Only use the admin account credentials to register the Local Manager with the Global Manager.

After you add a location to the Global Manager, the NSX Manager is called a Local Manager (LM).

1. Log in to the Global Manager at https://global-manager-ip-or-fqdn/.
2. Select **System** > **Location Manager** and click **Add On-Prem Location**.
3. In the **Add New Location** dialog box, enter the Location details.

Option	Description
Location Name	Provide a name for the location.
FQDN/IP	Enter the FQDN or IP address of the NSX Manager cluster VIP. Do not enter an individual NSX Manager FQDN or IP.
Username and Password	Provide the admin user's credentials for the NSX Manager at the location. Do not use any other account to register the Local Manager with the Global Manager.

SHA-256 Thumbprint	Log in to any NSX Manager node in the cluster and run this command:
	`get certificate cluster thumbprint`
	The result is the cluster VIP certificate thumbprint:
	`bfae1a0a...`

Check Compatibility	Click **Check Compatibility** to ensure that the location can be added. This checks that the NSX version is compatible.

If you want to create gateways and segments that span more than one location, you must configure a remote tunnel endpoint (RTEP) on Edge nodes in each location to handle the cross-location traffic. See Configure Edge Nodes for Stretched Networking. After you add a location to your Global Manager, you can import your configurations from that location's Local Manager appliance into the Global Manager. See Importing Configurations from Local Manager.

Importing Configurations from Local Manager

After you successfully add a Local Manager location to the Global Manager, you can import all network and security Local Manager configurations to the Global Manager.

- The Local Manager appliance must register with the Global Manager.
- The Local Manager appliance must have a backup that you can restore in case the importing procedure fails.
- You must remove configurations for unsupported features from your Local Manager appliance. You are provided guidance in the NSX UI on how to resolve any importing conflicts.

- The Global Manager and the Local Manager must be on same version.

You can only import the entire Local Manager configuration into the Global Manager. There is no support for partial configuration import. You can only import the configurations once.

Local Manager Configurations Supported for Importing into Global Manager

- Context Profiles
- DHCP
- DNS
- Firewall Security Policies
- Gateway Profiles
- Groups
- NAT
- Security Profiles
- Services
- T0 Gateway
- T1 Gateway
- Time-based firewall (import/onboard now supported)

Local Manager Configurations Not Supported for Importing into Global Manager

The following features are not supported with NSX Federation. Import of configurations into the Global Manager is blocked if you have any of these configurations in your Local Manager. You must remove unsupported configurations to proceed with importing. After your supported Local Manager configurations are successfully imported into Global Manager, you can add the configurations for any of the unsupported features back into your Local Manager.

- DHCP dynamic binding
- Distributed IDS
- Distributed security for vCenter VDS Port Group only (Global Manager does not see the vCenter VDS port groups to assign them in security groups. However, Global Manager can use dynamic membership in groups based on vCenter VDS port groups tags added by Local Managers.)
- Endpoint protection
- Forwarding policies
- Guest introspection
- Identity firewall
- IDS/IPS
- L2 Bridge
- Load balancer
- Malware prevention
- Metadata proxy
- Multicast
- Network detection and response
- Network introspection
- Routing protocols (OSPF)
- Routing VPN and EVPN
- Service insertion
- T0 VRF
- TLS inspection
- URL filtering

> **Note:**
> If you use a load balancer in the Local Manager, you cannot import the load balancer, but you can still import other configurations if you meet certain conditions as described in the section "Importing Configurations if you have a Load Balancer service on the Local Manager".

Importing Configurations if you have a Load Balancer service on the Local Manager

If your Local Manager has a load balancer service, you can import configurations except the load balancer, if you meet the following conditions:

- The load balancer service must be in one-arm mode on a standalone tier-1 gateway.
- The standalone tier-1 gateway that the one-arm load balancer is attached to:
  - must have only the load balancer service and no other services
  - must not have any downlink segments
  - must not share Gateway Firewall rules with any other tier-0 or tier-1 gateways.
- Groups used in load balancer service must not be used in any firewall rules. If you have groups common to both load balancer and firewall rules, you must either remove the group from the firewall rule or create an identical group to use with the load balancer.

Configurations Created by a Principal Identity User in Local Manager

If you have configurations in the Local Manager that are created by the Principal Identity user and the same Principal Identity user is not present in the Global Manager, import is blocked.

You have the following options for importing these entities:

- The system displays a list of Principal Identity usernames that are being used on the Local Managerr to create configurations. Create each of these Principal Identity users in the Global Manager before proceeding to import.
- If you do not want to create Principal Identity usernames in Global Manager, remove all configurations in the Local Manager that are created using the Principal Identity username. You can then proceed with importing other configurations from the Local Manager.

1. Log in to the Global Manager and navigate to **System** > **Location Manager**.
2. A system message appears for each location that has been successfully added into the Global Manager and has objects that can be imported.
3. Click **Import Now** from the system message. You can also import objects by clicking **Actions** > **Import** from the location tile.
4. You see a list of objects that can be imported into the Global Manager.
   a) If there are naming conflicts, you can provide a prefix or suffix for configurations. The total length of the object including the prefix and suffix must not exceed 256 characters.

      The prefix or suffix gets applied to the following objects being imported:

      - Tier-0 gateway
      - Tier-1 gateway
      - Segments
      - DNS zones
      - DHCP profiles
      - Switching profiles: IPv6, VNI Pool, Gateway QoS, BFD, IPFIX
      - Security profiles: IPFIX, Flood-Protection, DNS Security, Session Timer, Context Profiles
      - L4-L7 services (all services listed under **Inventory** > **Services**).

      The prefix or suffix does not get applied to the inventory and the firewall objects, that is: groups, firewall policies and firewall rules, and to system-created profiles and services.
   b) For other conflicts, follow the guidance provided in the UI.

Global Manager owns any Local Manager objects imported into the Global Manager. These objects appear in the Local Manager with this icon:



You can only modify these objects from the Global Manager now.

Configure Edge Nodes for Stretched Networking

If you want to create gateways and segments that span more than one location, you must configure a remote tunnel endpoint (RTEP) on Edge nodes in each location.

- Verify that each location participating in the stretched network has at least one Edge cluster.

- For RTEP networks, determine which layer 3 networks and VLANs to use.
  - Intra-location tunnel endpoints (TEP) and inter-location tunnel endpoints (RTEP) must use separate VLANs and layer 3 subnets.
- Verify that all RTEP networks used in a given NSX Federation environment have IP connectivity to each other.
- Verify that external firewalls allow cross-location RTEP tunnels. See VMware Ports and Protocols at https://ports.broadcom.com/home/NSX.
- Configure the MTU for RTEP on each Local Manager. The default is 1500. Set the RTEP MTU to be as high as your physical network supports. On each Local Manager, select **System** > **Fabric** > **Settings**. Click **Edit** next to **Remote Tunnel Endpoint**.
- Optionally, if you do not use DHCP for RTEP, configure the RTEP IP pool for your site to configure the RTEPs for the Edge cluster. For details, go to "Add an IP Address Pool" in the *NSX Administration Guide*.

When you configure an RTEP, do it on an Edge cluster basis. All Edge nodes in the cluster must have an RTEP configured. You do not need to configure all Edge clusters with RTEP. RTEPs are required only if the Edge cluster is used to configure a gateway that spans more than one location.

You can configure the TEP and RTEP to use the same physical NIC on the Edge node or use separate physical NICs.

This procedure describes this task starting from your Local Manager. You can also configure RTEPs from your Global Manager by using the Location Manager site selection drop-down to choose the Local Manager.

1. From your browser, log in with admin privileges to the Local Manager at https://<local-manager-ip-address>.
2. To configure a new RTEP, select **System** > **Quick Start**.
3. Click **Configure Remote Tunnel Endpoint** > **Getting Started**.
4. You can select all Edge Nodes in this cluster or one node at a time. Provide the following details for the RTEP configuration:

Option	Description
Edge Switch	Select an edge switch from the drop-down menu.
Teaming Policy Name	(Optional) Select a teaming policy if you have one configured.
RTEP VLAN	Enter the VLAN ID for the RTEP network. Valid values are between 1 and 4094.
IP Assignment	Select an option from the drop-down. For example, select Use IP Pool and choose an option from the drop-down list.
IP Pool for all nodes	Select an IP pool for all nodes in this Edge Cluster. If you want to assign an IP address to an individual node, you can edit the RTEP configuration later.
Inter Location MTU	The default is 1500.

5. Click **Save**.

   The green notification banner shows that all the Edge nodes in this edge cluster are configured successfully.
6. To add the RTEPs to the edge cluster for your other site locations, repeat steps 2 through 5.
7. To view or edit an existing Edge transport node:
   a) Select **System** > **Fabric** > **Nodes** > **Edge Transport Nodes**.
   b) Select an Edge node, then click **Tunnels**. If an RTEP is configured, it is displayed in the **Remote Tunnel Endpoint** section.
   c) Click **Edit** to modify the RTEP configuration.

   The Configure Edge Nodes for Stretched Networking screen opens in the Local Manager with that Edge cluster selected.

Remove a Location

Removing a location from the Global Manager removes all objects created from the Global Manager that have a Global scope and are not specific to this location.

Before you remove a location, you must delete all objects created from the Global Manager that are specific to this location, such as tier-0 gateway, or firewall policies.

Removing a location is disruptive.

All configurations created by the Global Manager that are not specific to this location, such as groups and firewall sections with a Global scope, will be removed from the NSX Manager at this location.

1. Log in to the Global Manager at https://global-manager-ip-fqdn.
2. Navigate to **System** > **Location Manager.**
3. From the location that you want to remove, click **Actions** > **Remove**.

   You see a message describing the effect of removing a location. If you have not previously removed objects created from the Global Manager that are specific to this location, such as tier-0 gateways or firewall policies, you cannot remove the location. The system automatically removes all other configurations from the NSX Manager at this location, that have a Global scope in your NSX Federation deployment.

Remove a Location When the Global Manager is Unavailable

This procedure is used only if the Global Manager has been deleted **BEFORE** the Local Managers (LM) are aware of its deletion.

This is a possible use case at the completion of a Federation test. In this case, the LM will constantly try to connect to its configured Global Manager (which has been deleted), and share with other Local Managers its stretch GM group members. The API calls will remove the Global Manager constructs on each LM, even when the GM has been deleted.

The below API calls on the Local Manager, will remove the Local Manager, its registration to the Global Manager, and its registration to other Local Managers.

> **Note:**
> The following API calls done on the Local Manager are disruptive.

All configurations created by the Global Manager that are not specific to this location, such as groups and firewall sections with a Global scope, will be removed from the NSX Manager at this location.

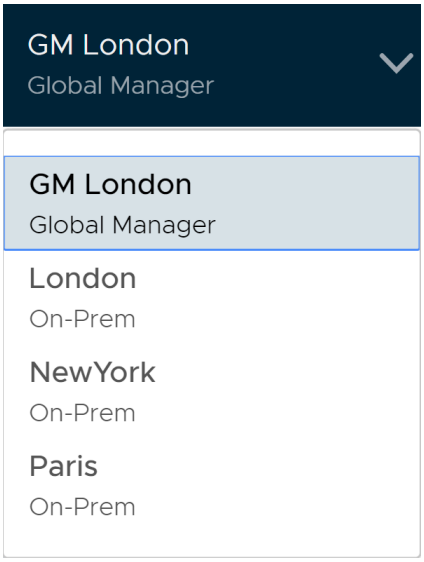
1. To remove a Global Manager that is active, a Global Manager that is standby, and other Local Managers registrations from the Local Manager use the site manager API (at the Local Manager) `POST https://<LM>/api/v1/sites?action=offboard_local..`
2. To remove Global Manager objects from a Local Manager, run the Local Manager API (at the Local Manager) `POST https://<LM>/policy/api/v1/infra/site?action=offboard`.
3. Optional: The offboarding progress can be monitored by using the API call `GET https://<LM>/policy/api/v1/infra/site/offboarding-status`.

Using the Global and Local Manager Web Interfaces

You can use the Global Manager to create objects that are limited to one location, or span multiple locations. You can also monitor communication details between Local Manager and Global Manager using the Location Manager.

Location Drop-Down Menu on Global Manager

When you log into the active Global Manager web interface, you see a Location drop-down menu in the top navigation bar. Using this menu, you can switch between the active Global Manager and any associated Local Managers. You can also select **System > Location Manager** to view communication activity.



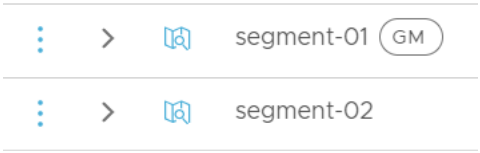
Local and Global Objects

Objects created on a Local Manager are local objects. Objects created from the Global Manager are global objects, though their span might not include all available locations. You can import objects to the Global Manager using the Location Manager.

On a Local Manager, you can see local objects, and any global objects that apply to that location. The global objects have the following Global icon next to them:



This screenshot from the Local Manager web interface shows two segments. The segment that has the Global icon next to it indicates that it was created on the Global Manager. The segment that has no Global icon indicates that it was created on the Local Manager.

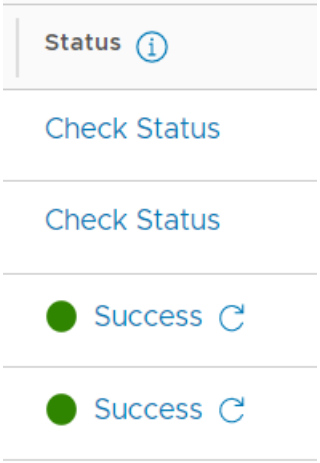


Because all objects on the Global Manager are global, there is no icon displayed when you are logged into the active Global Manager.

Status of Local and Global Objects

Local Managers display the status of both global and local objects. The active Global Manager displays any objects imported from the Local Managers. For details on importing objects from Local to Global Managers using Location Manager, see Monitoring NSX Federation Locations.

To retrieve the latest status from the Local Managers, click **Check Status** for the object. To refresh the status, click the **Refresh** icon.



Monitoring NSX Federation Locations

To monitor the configuration flow between all Global Managers (GM) and Local Managers (LM) across a single NSX Federation federated domain for on-premises solutions you can use the **System > Location Manager** menu or the Location drop-down menu on the GM page. After you configure Locations, alarms display any communication issues between the GM and the LM on the Location Manager page.

You must already have a Location added. See "Add a Location" in the *NSX Installation Guide*.

Use the Location Manager to view latency data, monitor communication synchronizations, check policies, and get better visibility into the health between the different components of the federated domain. The Global Manager View of Location Manager figure shows the Global and Local Manager details. For callout details, see the Callouts for Global Manager View of Location Manager table.

Figure 430: Global Manager View of Location Manager

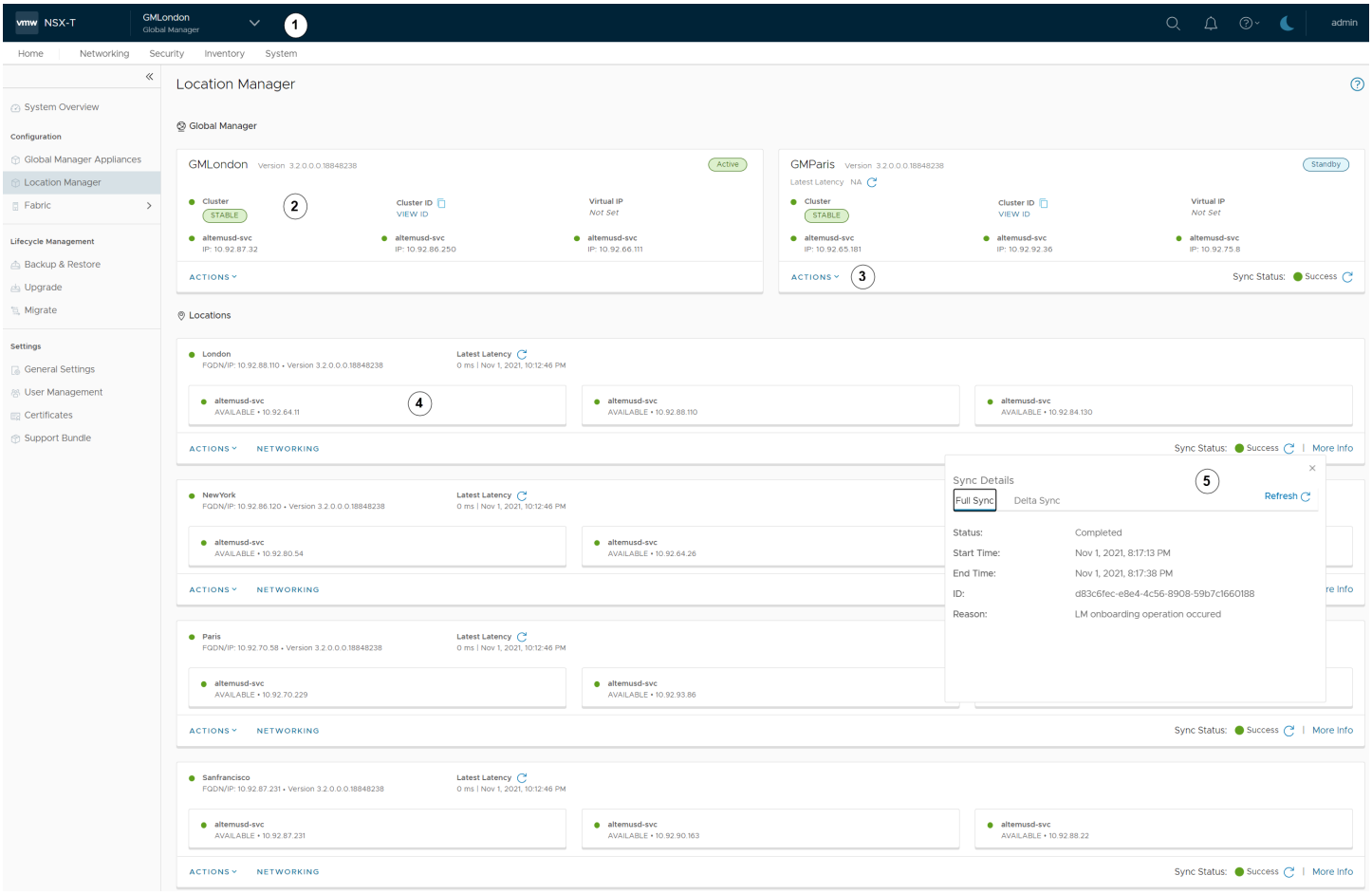


Table 1050: Callouts for Global Manager View of Location Manager

ID	Options	Description
1	Set location view.	Switch to different GM-active and LM views by selecting dropdown. After logging into the active GM, this drop-down indicates what instance you are logged into and lets you switch to other LM locations views. You can also switch to the standby GM, not using this dropdown, but by selecting the **Actions** menu in the Standby pane and clicking **Access Standby GM**.
2	Active GM information.	• GM-Active details. • Tasks such as GM Active Name.
3	GM Standby information.	• GM Standby details such as Sync Status. • Other details such as GM Standby Name, Access Standby GM View, and Remove GM Standby.
4	LM pane information.	• LM details such as Latency Data, Sync Status, and Networking RTEP configuration. • Tasks such as edit settings, remove, evacuate location, and import to GM. Evacuate location allows network recovery for sites that have not lost communication. For example, if you want to migrate a data center or perform a disaster recovery test. Import to GM allows the former LM configuration to be pushed up to GM (see "Importing Configurations from Local Manager" in the *NSX Installation Guide*.

ID	Options	Description
5	LM More Info pane.	• View detailed Full Sync from GM Active to LM information such as Start Time, End Time, and Reason. • View detailed Delta Sync from GM Active to LM information such as Message Queues on GM and LM.

The Local Manager View of Location Manager figure shows an example of the window details. For callout details, see the Callouts for Local Manager View table.

Figure 431: Local Manager View of Location Manager

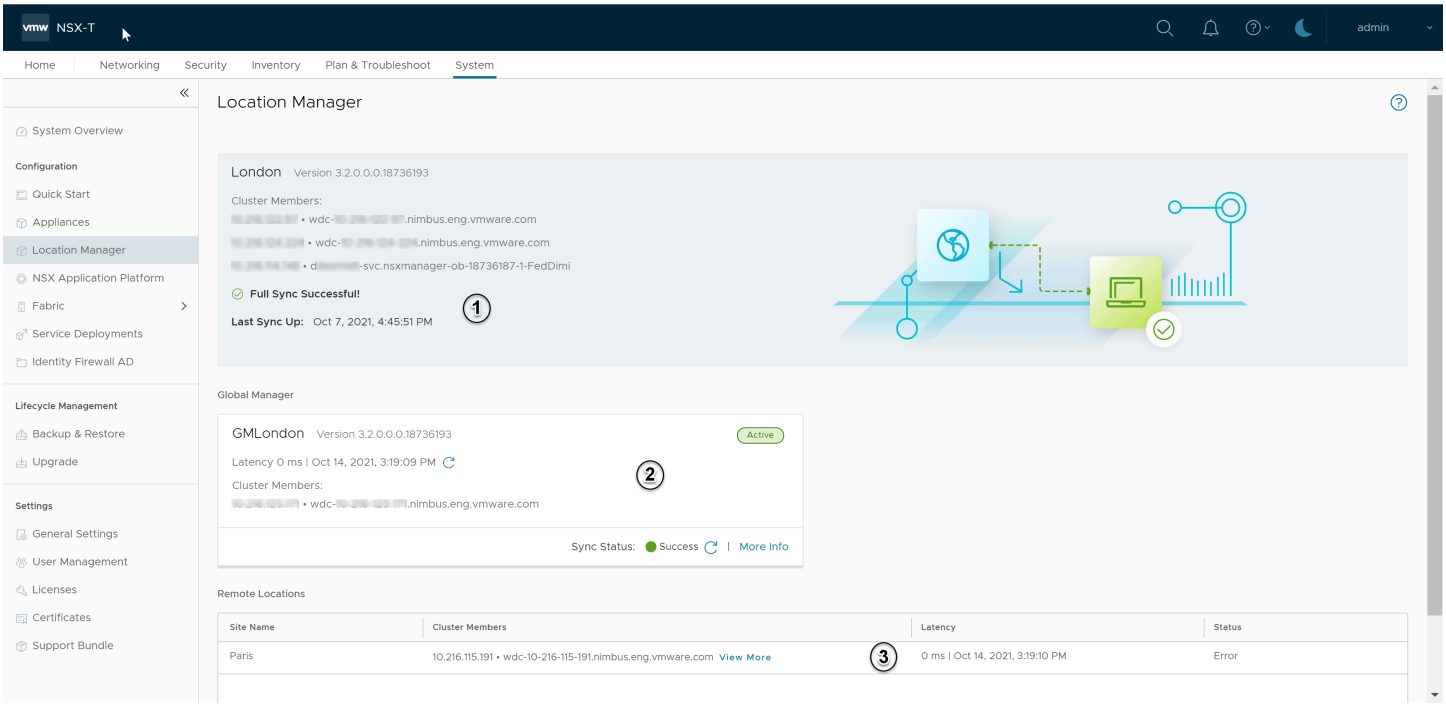


Table 1051: Callouts for Local Manager View of Location Manager

ID	Task	Description
1	View LM details.	NSX version, cluster IPs and names, status of full sync and when it completed.
2	View related active GM details.	• Software versions, latency details between GM and LM, cluster details, and color-coded sync status. • Green is communication is OK. Orange is some backlog in the queue based on threshold. Red is sync is not occurring. • To see the full sync details including status, start and end times, ID, and status reasons, click More Info.
3	View remote location details.	Cluster details, including View More information, latency data between LM and GM, and current status.

The Supported Communication Channel Details table provides details about the Location Manager displays.

Table 1052: Supported Communication Channel Details

Supported Communication Channels	Display Details
All channels	• Status (color-coded) — Green - OK. — Orange - Sync between active GM and Standby GM is in progress. — Red - Communication is lost. • When latency is over 500 ms, an alarm gets raised. For NSX Federation events, see the Event Catalog.
On a GM and LM UI for each remote LM	• Current queue levels contain the number of queued messages compared to the queue size. • Last measured latency - Uses ICMP or TCP traffic between the local LM and the remote LM to measure the network round trip in milliseconds (ms).
On a LM UI for a GM active cluster and a GM standby cluster	• Current queue levels contain the number of queued messages compared to the queue size for GM active only. • Last measured latency - Measures the network round trip in ms using ICMP or TCP traffic between the local LM and the active and standby GM. • When latency is over 500 ms, an alarm gets raised.
On an active GM UI for a standby GM cluster	• When Active-Standby GM replication is not working, an alarm gets raised.

Overriding Global Manager Configurations on Local Manager

When you create an object from Global Manager, the same configuration is propagated to all relevant locations. You can override some Global Manager configurations on a Local Manager.

To override a configuration, click the three dots menu
()
next to the configuration, and click **Edit**. If the **Edit** menu item is dimmed, you cannot override this configuration.

If a configuration is overridden, you see this icon in the status column on both Global Manager and Local Manager:

.

To remove an override, click the three dots menu
()
next to the configuration, and click **Revert**. The configuration from Global Manager is restored.

If you override a configuration from Global Manager on a Local Manager, and then you delete the configuration from the Global Manager, the configuration persists on the Local Manager. When you revert the configuration, the configuration is deleted from Local Manager.

You can get a list of all configurations that have been overridden. Make this API request to the Global Manager: `GET https://<global-mgr>/global-manager/api/v1/global-infra/overridden-resources`.

Gateway Configurations

Gateway configurations are found in **Networking** > **Tier-0 Gateways**.

You can modify the following gateway configurations:

- Tier-0 Gateway BGP Configuration
- Tier-0 Gateway Interfaces

Profile Configurations

Profile configurations on Global Manager are used in all Local Managers. There is no span setting for a profile configuration.

You can override the following global profile configurations from Local Manager:

- Segment Profiles: **Networking** > **Segments** > **Segment Profiles**
  - IP Discovery Profiles
  - MAC Discovery Profiles
  - QoS Profiles
  - Segment Security Profiles
  - SpoofGuard Profiles
- Networking Profiles: **Networking** > **Networking Profiles**
  - IPv6 DAD Profiles
  - IPv6 ND Profiles
  - Gateway QoS Profiles
  - BFD Profiles
- Context Profiles: **Inventory** > **Profiles**
- Security Profiles: **Security** > **General Settings** > **Firewall**
  - Firewall Session Timer Profile
  - Edge Gateway Flood Protection Profiles
  - Firewall Flood Protection Profiles
  - DNS Security Profiles
  - CPU and Memory Threshold Profiles are API only:
    - Override with `PUT/PATCH https://<local-manager>/policy/api/v1/global-infra/settings/firewall/cpu-mem-thresholds-profiles/<id>?override=true`.
    - Revert with `DELETE https://<local-manager>/policy/api/v1/global-infra/settings/firewall/cpu-mem-thresholds-profiles/<id>?override=true`.

Networking in NSX Federation

Tier-0 gateways, tier-1 gateways, and segments can span one or more locations in the NSX Federation environment.

When you plan your network topology, keep these requirements in mind:

- Tier-0 and tier-1 gateways can have a span of one or more locations.
- The span of a tier-1 gateway must be equal to, or a subset of, the span of the tier-0 gateway it is attached to.
- A segment has the same span as the tier-0 or tier-1 gateway it is attached to. Isolated segments are not realized until they are connected to a gateway.
- NSX Edge nodes in the Edge Cluster selected on the Global Manager for tier-0 and tier-1 gateways must be configured with the Default TZ Overlay.

You can create different topologies to achieve different goals.

- You can create segments and gateways that are specific to a given location. Each site has its own configuration, but you can manage everything from the Global Manager interface.
- You can create segments and gateways that span locations. These stretched networks provide consistent networking across sites.

Tier-0 Gateway Configurations in NSX Federation

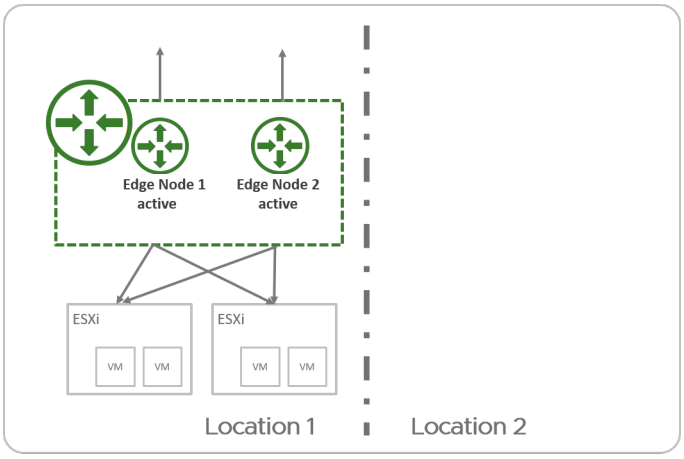
With NSX Federation, you can deploy a tier-0 gateway that is limited to a single location, or you can stretch it across multiple locations.

Tier-0 gateways can have one of the following configurations:

- Non-stretched tier-0 gateway.
- Stretched active-active with primary and secondary locations.
- Stretched active-active with all primary locations.
- Stretched active-standby with primary and secondary locations.

Non-Stretched Tier-0 Gateway

You can create a tier-0 gateway from Global Manager that spans only one location. This is similar to creating the tier-0 gateway on the Local Manager directly, but has the advantage that you can manage it from Global Manager.



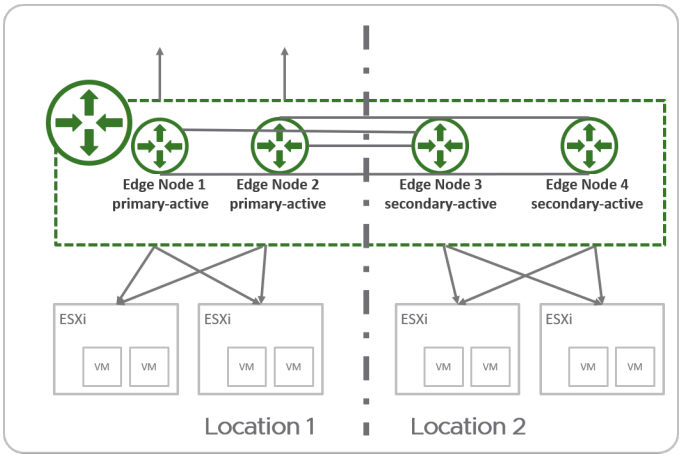
Stretched Active-Active Tier-0 Gateway with Primary and Secondary Locations

In an active-active tier-0 gateway with primary and secondary locations, the following applies:

- All Edge nodes are active at the same time, therefore the tier-0 cannot run stateful services.
- All traffic enters and leaves through the Edge nodes in the primary location.

If both the tier-0 gateway and the linked tier-1 gateway have primary and secondary locations, configure the same location to be primary for both gateways to reduce cross-location traffic.

> **Important:**
> In this topology, NSX ensures that all egress traffic leaves from each location.
>
> If your environment has stateful services, such as external firewall, on the physical network, you must ensure that the return traffic enters through the primary location. For example, you can add AS path prepending on the BGP peers in your secondary locations.
>
> If you do not have stateful services on your physical network, and you choose to have asymmetric routing, you must deactivate Unicast Reverse Path Forwarding (uRPF) on all externally tier-0 interfaces.

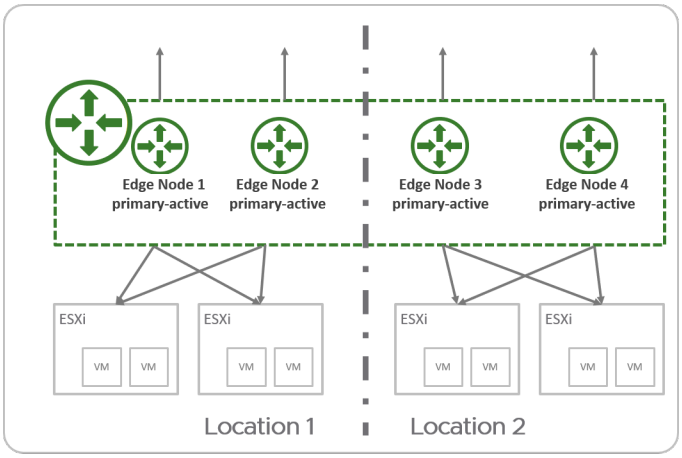


Stretched Active-Active Tier-0 Gateway with All Primary Locations

In an active-active tier-0 gateway with all primary locations, the following applies:

- All Edge nodes are active at the same time, therefore the tier-0 cannot run stateful services.
- All traffic enters and leaves through Edge nodes in the same location as the workloads.

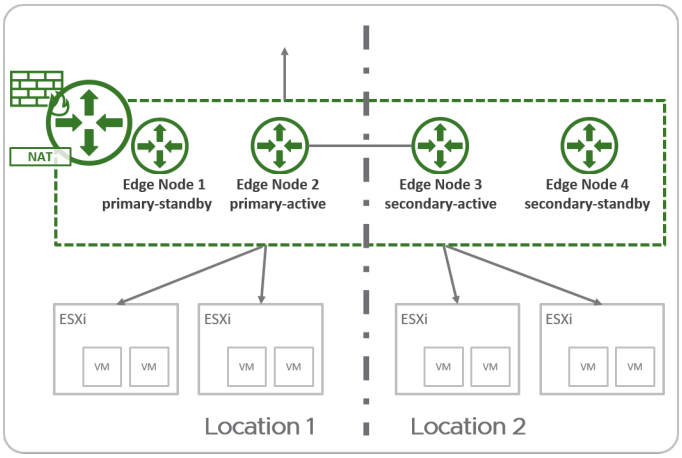
> **Important:**
> This topology allows traffic to egress locally from each location. You must ensure that return traffic enters the same location to allow external stateful services (such as an external firewall). For example, you can configure an external location-specific NAT IP (such as on the external firewall) so that return traffic is always routed back to the same location that it left.



Stretched Active-Standby Tier-0 Gateway with Primary and Secondary Locations

In an active-standby tier-0 gateway with primary and secondary locations, the following applies:

- Only one Edge node is active at a time, therefore the tier-0 can run stateful services.
- All traffic enters and leaves through the active Edge node in the primary location.



For active standby tier-0 gateways, the following services are supported:

- Network Address Translation (NAT)
- Gateway Firewall
- DNS
- DHCP

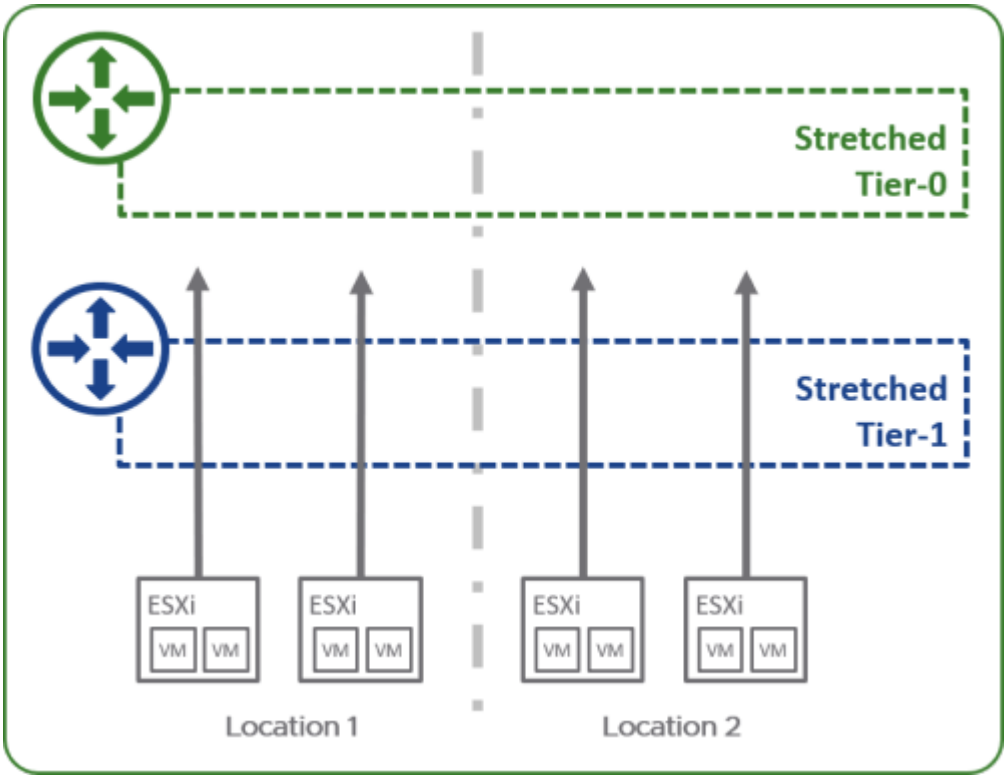
See Features and Configurations Supported in NSX Federation for more information.

Tier-1 Gateway Configurations in NSX Federation

With NSX Federation, you can deploy a tier-1 gateway to provide distributed routing only, or you can configure services on it.

Tier-1 Gateway for Distributed Routing Only

You can create a tier-1 gateway in NSX Federation for distributed routing only. This gateway has the same span as the tier-0 gateway it is linked to. The tier-1 does not use Edge nodes for routing. All traffic is routed from host transport nodes to the tier-0 gateway. However, to enable cross-location forwarding, the tier-1 allocates two Edge nodes from the Edge cluster configured on the linked tier-0 to use for that traffic.



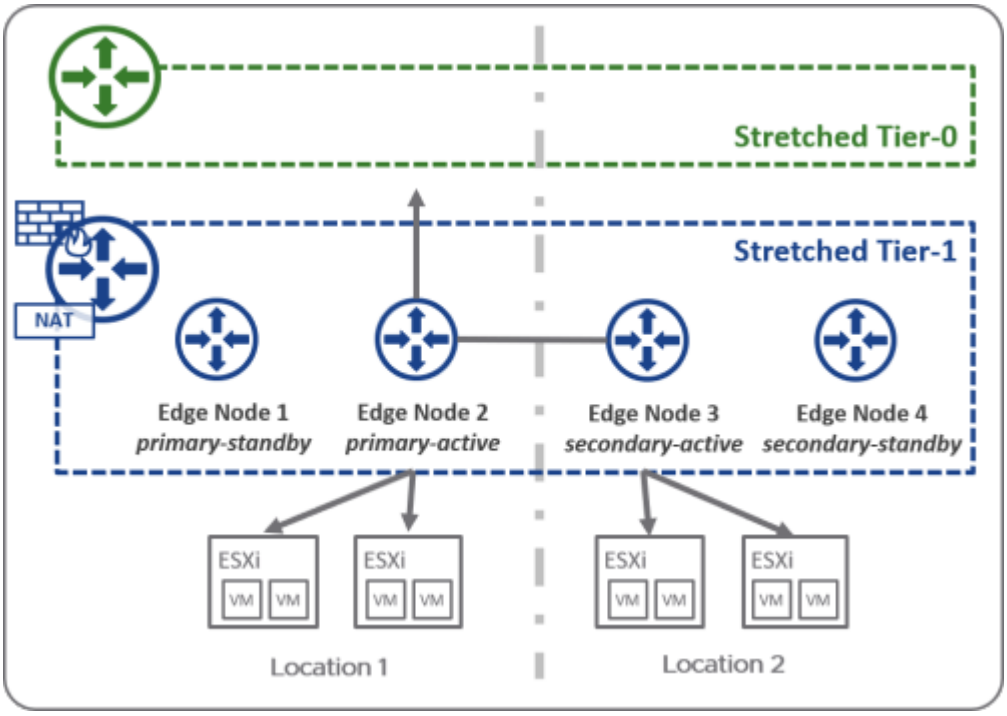
Tier-1 Gateway with Services or Custom Span

You configure the tier-1 gateway with Edge clusters if you need one of the following configurations:

- You want to run services on the tier-1 gateway.
- You want to deploy a tier-1 gateway that has a different span than the linked tier-0 gateway.
  You can remove locations, but you cannot add locations that are not already included the span of the tier-0 gateway.

You select one of the locations to be the primary location. All other locations are secondary. The HA mode for the tier-1 gateway is Active Standby. All traffic passing through this tier-1 gateway passes through the active edge node in the primary location.

If both the tier-1 gateway and the linked tier-0 gateway have primary and secondary locations, configure the same location to be primary for both gateways to reduce cross-location traffic.



Configure Edge Nodes for Stretched Networking

If you want to create gateways and segments that span more than one location, you must configure a remote tunnel endpoint (RTEP) on Edge nodes in each location.

- Verify that each location participating in the stretched network has at least one Edge cluster.
- For RTEP networks, determine which layer 3 networks and VLANs to use.
  - Intra-location tunnel endpoints (TEP) and inter-location tunnel endpoints (RTEP) must use separate VLANs and layer 3 subnets.
- Verify that all RTEP networks used in a given NSX Federation environment have IP connectivity to each other.
- Verify that external firewalls allow cross-location RTEP tunnels. See VMware Ports and Protocols at https://ports.broadcom.com/home/NSX.
- Configure the MTU for RTEP on each Local Manager. The default is 1500. Set the RTEP MTU to be as high as your physical network supports. On each Local Manager, select **System** > **Fabric** > **Settings**. Click **Edit** next to **Remote Tunnel Endpoint**.
- Optionally, if you do not use DHCP for RTEP, configure the RTEP IP pool for your site to configure the RTEPs for the Edge cluster. For details, go to "Add an IP Address Pool" in the *NSX Administration Guide*.

When you configure an RTEP, do it on an Edge cluster basis. All Edge nodes in the cluster must have an RTEP configured. You do not need to configure all Edge clusters with RTEP. RTEPs are required only if the Edge cluster is used to configure a gateway that spans more than one location.

You can configure the TEP and RTEP to use the same physical NIC on the Edge node or use separate physical NICs.

This procedure describes this task starting from your Local Manager. You can also configure RTEPs from your Global Manager by using the Location Manager site selection drop-down to choose the Local Manager.

1. From your browser, log in with admin privileges to the Local Manager at https://<local-manager-ip-address>.
2. To configure a new RTEP, select **System** > **Quick Start**.
3. Click **Configure Remote Tunnel Endpoint** > **Getting Started**.
4. You can select all Edge Nodes in this cluster or one node at a time. Provide the following details for the RTEP configuration:

Option	Description
Edge Switch	Select an edge switch from the drop-down menu.
Teaming Policy Name	(Optional) Select a teaming policy if you have one configured.
RTEP VLAN	Enter the VLAN ID for the RTEP network. Valid values are between 1 and 4094.
IP Assignment	Select an option from the drop-down. For example, select Use IP Pool and choose an option from the drop-down list.
IP Pool for all nodes	Select an IP pool for all nodes in this Edge Cluster. If you want to assign an IP address to an individual node, you can edit the RTEP configuration later.
Inter Location MTU	The default is 1500.

5. Click **Save**.

   The green notification banner shows that all the Edge nodes in this edge cluster are configured successfully.
6. To add the RTEPs to the edge cluster for your other site locations, repeat steps 2 through 5.
7. To view or edit an existing Edge transport node:
   a) Select **System** > **Fabric** > **Nodes** > **Edge Transport Nodes**.
   b) Select an Edge node, then click **Tunnels**. If an RTEP is configured, it is displayed in the **Remote Tunnel Endpoint** section.
   c) Click **Edit** to modify the RTEP configuration.

   The Configure Edge Nodes for Stretched Networking screen opens in the Local Manager with that Edge cluster selected.

Add a Tier-0 Gateway from Global Manager

You can add a tier-0 gateway from the Global Manager. This gateway can have a span of one or more locations. This span affects the span of the tier-1 gateways and sengments attached to it.

- If you are creating a tier-0 gateway that spans more than one location, verify that each location has Edge nodes configured with RTEPs for stretched networking. Refer to Configure Edge Nodes for Stretched Networking.
- If you plan to configure the gateway DHCP server, refer to Attach an NSX DHCP Profile to a Tier-0 or Tier-1 Gateway.

For details about tier-0 gateway configurations in NSX Federation, refer to Tier-0 Gateway Configurations in NSX Federation.

The following settings must be kept consistent across locations. If you change these settings from the Global Manager web interface, those changes are automatically applied on all locations. However, if you change these settings using the API, you must manually make the same changes in each location.

- Local AS
- ECMP settings
- Multipath Relax settings
- Graceful Restart

> **Important:**
>
> When you create a tier-0 gateway from Global Manager, you must configure an external interface in each location that the tier-0 is stretched to. Each external interface must be connected to a segment that was created from Global Manager, with the **Connectivity** set to None and the **Traffic type** set to VLAN. Refer to Add a Segment from Global Manager. The Edge nodes configured with those external interfaces are used for inter-location communication, even if northbound communication is not needed.

1. From your browser, log in with admin privileges to the active Global Manager at `https://<global-manager-ip-address>`.
2. Select **Networking** > **Tier-0 Gateways**.
3. Enter a name for the gateway.
4. Select an HA (high availability) mode to configure within each location.

   The default mode is active-active. In the active-active mode, traffic is load balanced across edge nodes in all locations. In the active-standby mode, an elected Edge node processes traffic in each location. If the active node fails, the standby node becomes active.
5. If the HA mode is active-standby, select a failover mode.

Option	Description
Preemptive	If the preferred node fails and recovers, it will preempt its peer and become the active node. The peer will change its state to standby.
Non-preemptive	If the preferred node fails and recovers, it will check if its peer is the active node. If so, the preferred node will not preempt its peer and will be the standby node.

6. Optional: Add **DHCP Config** on the gateway.
7. Specify the span of this tier-0 gateway by providing the following details for each location. To add additional locations, click **Add Location**.

Option	Description
Location	Select the location from the drop-down menu.
Edge Cluster	Select an Edge cluster from this location. If you are configuring a stretched tier-0, you must select an Edge cluster that contains Edge nodes that are configured with an RTEP.
Mode	Each location of the tier-0 gateway can have a mode of **Primary** or **Secondary**. • If the HA mode is **Active Active**, you can configure the tier-0 or VRF gateway with all locations mode set to primary. 1. Select the **Mark all locations as Primary** toggle to mark all locations as primary. • If the HA mode is **Active Active** or **Active Standby**, you can configure the tier-0 gateway with one location set to **Primary**, and all others set to **Secondary**. 1. Select **Primary** mode for one location. In all other locations, set mode to **Secondary**. 2. For secondary locations, you must select a fallback preference.

8. Click **Additional Settings**.
   a) Turn on the **Multi-VRF Inter SR** toggle to run MP BGP between edges for inter-node route sync for teir-0 and VRF gateways. Once the toggle is enabled, you can create and stretch a VRF gateway connected to this tier-0 gateway. To enable inter SR routing on a VRF gateway connected to this tier-0 gateway, you must also turn on the **Inter SR iBGP** toggle in the BGP section while configuring a VRF gateway that must be in active-active mode.
   b) In the **Internal Transit Subnet** field, enter a subnet.

      This is the subnet used for communication between components within this gateway. The default is 169.254.0.0/24.
   c) In the **T0-T1 Transit Subnets** field, enter one or more subnets.

      These subnets are used for communication between this gateway and all tier-1 gateways that are linked to it. After you create this gateway and link a tier-1 gateway to it, you will see the actual IP address assigned to the link on the tier-0 gateway side and on the tier-1 gateway side. The address is displayed in **Additional Settings** > **Router Links** on the tier-0 gateway page and the tier-1 gateway page. The default is 100.64.0.0/16.

After the tier-0 gateway is created, you can change the **T0-T1 Transit Subnets** by editing the gateway. Note that this will cause a brief disruption in traffic.

d) In the **Inter-site Transit Subnet** field, enter a subnet. This subnet is used for cross-location communication between gateway components. The default is 169.254.32.0/20.

9. Click **Save**.
10. To configure interfaces, expand the **Interfaces and GRE Tunnels** category and click **Set** or the hyperlinked number for **External and Service Interfaces**. Configure an external interface for each location that the tier-0 gateway spans.
    a) Click **Add Interface**.
    b) Enter a name.
    c) Select a location.
    d) Select a type.

    If the HA mode is active-standby, the choices are **External**, **Service**, and **Loopback**. If the HA mode is active-active, the choices are **External** and **Loopback**.

    Service interfaces are supported only on gateways that span one location. If the gateway is stretched, service interfaces are not supported.
    e) Enter an IP address in CIDR format.
    f) Select a segment.

    The segment must be created from the Global Manager, with the **Connectivity** set to None and the **Traffic type** set to VLAN. See Add a Segment from Global Manager.
    g) If the interface type is not **Service**, select an NSX Edge node.
    h) Optional: If the interface type is not **Loopback**, enter an MTU value.
    i) Skip **PIM** configuration.

    Multicast is not supported in NSX Federation.
    j) Optional: Add tags and select an ND profile.
    k) Optional: If the interface type is **External**, for **URPF Mode**, you can select **Strict** or **None**.

    Asymmetric routing or forwarding is not recommended. Therefore, **URPF Mode** is set to **Strict** by default.

    For symmetric routing, ensure that the configuration between tier-0 gateway and northbound routers is such that the same set of prefixes are advertised from each edge node within a tier-0 gateway on a given site. Also, the same set of prefixes must be learned from TORs on all edge nodes of a tier-0 gateway on a given site.

    If you have BGP between the tier-0 gateway and northbound routers and you are using route-maps or filters, ensure that this configuration is consistent for inbound and outbound directions for all edge nodes.

    For a federation environment with primary and secondary sites, you should advertise longer AS paths for BGP advertisements on secondary site BGP neighbors to resolve asymmetric forwarding.

    If there is a need for asymmetric routing, set **URPF Mode** to **None**.
    l) Optional: After you create an interface, you can download the aggregate of ARP proxies for the gateway by clicking the menu icon (three dots) for the interface and selecting **Download ARP Proxies**.

    You can also download the ARP proxy for a specific interface by expanding a gateway and then expanding **Interfaces**. Click an interface and click the menu icon (three dots) and select **Download ARP Proxy**.

    Note:
    You cannot download the ARP proxy for loopback interfaces.
11. Click **Routing** to add IP prefix lists, community lists, static routes, and route maps.

    When you add a static route on a tier-0 gateway, the default behavior is that the static routes are pushed to all locations configured on the gateway. However, the routes are enabled only on the primary locations. This ensures that on the secondary locations, the routes that are learned from the primary location are preferred.

    If you want to change this behavior, you can use the **Enabled on Secondary** setting and the **Scope** setting.

    If you select **Enabled on Secondary**, the static route is also enabled on the secondary locations.

When you add a next hop for a static route, you can set the **Scope**. The scope can be an interface, a gateway, or a segment. On a tier-0 gateway created from Global Manager, the scope can also be a location. You can use the scope setting to configure different next hops for each location.

12. Click **BGP** to configure BGP.

    When you configure BGP on a tier-0 gateway from the Global Manager, most settings apply to all locations.

    Some of the settings within the BGP configuration, such as **Route Aggregation** and **BGP Neighbors** prompt you to provide separate values for each location.

    See Configure BGP for more information about configuring BGP.

13. To configure route redistribution, click **Route Redistribution**, and for each location, click **Set**.

    Select one or more of the sources:

    - Tier-0 subnets: **Static Routes**, **NAT IP**, **IPsec Local IP**, **DNS Forwarder IP**, **Router Link**, **Connected Interfaces & Segments**.
      Under **Connected Interfaces & Segments**, you can select one or more of the following: **Service Interface Subnet**, **External Interface Subnet**, **Loopback Interface Subnet**, **Connected Segment**.
    - Tier-0 subnets: **Static Routes**, **NAT IP**, **IPsec Local IP**, **DNS Forwarder IP**, **EVPN TEP IP**, **Connected Interfaces & Segments**.
      Under **Connected Interfaces & Segments**, you can select one or more of the following: **Service Interface Subnet**, **External Interface Subnet**, **Loopback Interface Subnet**, **Connected Segment**.
    - Advertised tier-1 subnets: **DNS Forwarder IP**, **Static Routes**, **LB VIP**, **NAT IP**, **LB SNAT IP**, **IPsec Local Endpoint**, **Connected Interfaces & Segments**.
      Under **Connected Interfaces & Segments**, you can select **Service Interface Subnet** and/or **Connected Segment**.

Set up a tier-1 gateway from Global Manager.

Add a Tier-1 Gateway from Global Manager

A gateway can be configured in one or more locations. These locations are the span of the gateway. A tier-1 gateway cannot have a greater span than the tier-0 gateway it is connected to.

- Verify you have a tier-0 gateway configured.
- If you plan to configure the gateway DHCP server, refer to Attach an NSX DHCP Profile to a Tier-0 or Tier-1 Gateway.

Refer to Tier-1 Gateway Configurations in NSX Federation for details about tier-1 gateway configuration options in NSX Federation.

1. From your browser, log in with admin privileges to an NSX Manager at https://<global-manager-ip-address>.
2. Select **Networking** > **Tier-1 Gateways**.
3. Click **Add Tier-1 Gateway**.
4. Enter a name for the gateway.
5. Select a tier-0 gateway to connect to this tier-1 gateway to create a multi-tier topology.
   - If you select a tier-0 gateway, the Locations configuration is populated with the same locations that are configured on the tier-0. If needed, you can modify the locations configuration in the Locations section.
   - If you do not select a tier-0 gateway, you can select locations. However, if you later connect the tier-1 gateway to a tier-0 gateway, you might need to update the locations to create a valid configuration.
6. In **Locations**, you can change the **Enable Edge Clusters for Services or Custom Span** setting. It is disabled by default.
   - Leave **Enable Edge Clusters for Services or Custom Span** disabled if you want the tier-1 gateway to have the same span as the tier-0 gateway, and you do not need to enable services on the tier-1 gateway. The tier-1 gateway will perform distributed routing only.

- Enable **Enable Edge Clusters for Services or Custom Span** if you want to choose a subset of locations for the tier-1 gateway, or if you want to enable services on the tier-1 gateway.

If you enable **Enable Edge Clusters for Services or Custom Span**, enter the location, cluster, and mode information.

a) Select a location from the drop-down menu. If you linked this tier-1 gateway to a tier-0 gateway, the locations of that tier-0 gateway are automatically listed. If needed, you can delete a location.
b) Select an NSX Edge cluster for each location. If the tier-1 gateway spans more than one location, the Edge clusters must already be configured with an RTEP for each of its Edge Nodes.
c) Optional: To select specific Edge nodes, click **Set** next the Edge cluster.

   Edge nodes are automatically allocated if you do not select Edge nodes.
d) Select a mode for each location. Mode can be Primary or Secondary.

   Only one location can be configured with Primary mode. All northbound traffic from this tier-1 gateway is sent through this location.

7. If you have enabled Edge clusters, select a failover mode.

Option	Description
Preemptive	If the preferred NSX Edge node fails and recovers, it will preempt its peer and become the active node. The peer will change its state to standby.
Non-preemptive	If the preferred NSX Edge node fails and recovers, it will check if its peer is the active node. If so, the preferred node will not preempt its peer and will be the standby node. This is the default option.

8. Optional: Add **DHCP Config** on the gateway.
9. Skip selecting a size from the **Edge Pool Allocation Size** drop-down menu.
10. If you have enabled Edge clusters, select a setting for **Enable StandBy Relocation**.

    Standby relocation means that if the Edge node where the active or standby logical router is running fails, a new standby logical router is created on another Edge node to maintain high availability. If the Edge node that fails is running the active logical router, the original standby logical router becomes the active logical router and a new standby logical router is created. If the Edge node that fails is running the standby logical router, the new standby logical router replaces it.
11. Optional: Click **Route Advertisement**.

    Select one or more of the following:

    - **All Static Routes**
    - **All NAT IP's**
    - **All DNS Forwarder Routes**
    - **All LB VIP Routes**
    - **All Connected Segments and Service Ports**
    - **All LB SNAT IP Routes**
    - **All IPSec Local Endpoints**
12. Click **Save**.
13. Optional: Click **Route Advertisement**.
    a) In the **Set Route Advertisement Rules** field, click **Set** to add route advertisement rules.
14. Optional: Click **Additional Settings**.
    a) For IPv6, you can select or create an **ND Profile** and a **DAD Profile**.

       These profiles are used to configure Stateless Address Autoconfiguration (SLAAC) and Duplicate Address Detection (DAD) for IPv6 addresses.
    b) Select an **Ingress QoS Profile** and an **Egress QoS Profile** for traffic limitations.

       These profiles are used to set information rate and burst size for permitted traffic. See Add a Gateway QoS Profile for more information on creating QoS profiles.

    If this gateway is linked to a tier-0 gateway, the **Router Links** field shows the link addresses.

15. Optional: Click **Service Interfaces** and **Set** to configure connections to segments. Required in some topologies such as VLAN-backed segments or one-arm load balancing.

    Service interfaces are supported only on gateways that span one location. If the gateway is stretched, service interfaces are not supported.

    a) Click **Add Interface**.
    b) Enter a name and IP address in CIDR format.

       If you configure multicast on this gateway, you must not configure tier-1 addresses as static RP address in the PIM profile.
    c) Select a segment.
    d) In the **MTU** field, enter a value between 64 and 9000.
    e) For **URPF Mode**, you can select **Strict** or **None**.

       URPF (Unicast Reverse Path Forwarding) is a security feature.
    f) Add one or more tags.
    g) In the **ND Profile** field, select or create a profile.
    h) Click **Save**.
    i) Optional: After you create an interface, you can download the ARP proxies for the gateway by clicking the menu icon (three dots) for the interface and selecting **Download ARP Proxies**.

       You can also download the ARP proxy for a specific interface by expanding a gateway and then expanding **Service Interfaces**. Click an interface and click the menu icon (three dots) and select **Download ARP Proxy**.

16. Optional: Click **Static Routes** and **Set** to configure static routes.

    a) Click **Add Static Route**.
    b) Enter a name and a network address in the CIDR or IPv6 CIDR format.
    c) Click **Set Next Hops** to add next hop information.
    d) Click **Save**.

Add a Segment from Global Manager

You can add two kinds of segments: overlay-backed segments and VLAN-backed segments. When you create segments from Global Manager, only overlay-backed segments can span multiple locations.

Verify that each location has a default overlay transport zone configured. The default overlay transport zone is used to create global overlay segments. From each Local Manager, select **System** > **Fabric** > **Transport Zones**. Select an overlay transport zone, and click **Actions** > **Set as Default Transport Zone**.

You can view segments ports from Global Manager, but you cannot create or modify them. If you need to create or modify a segment port, you must do it from the Local Manager.

> **Important:**
>
> Do not change the gateway connectivity of a segment in NSX Federation. Changing the gateway affects the span of the segment. If the span changes in such a way that it excludes a location, the segment is deleted on the excluded location. You must disconnect all VMs before you shrink the span of a segment.

1. From your browser, log in with admin privileges to a Global Manager at https://<global-manager-ip-address>.
2. Select **Networking** > **Segments**.
3. Click **Add Segment**.
4. Enter a name for the segment.
5. Select the Connectivity, Traffic Type, and Locations for this segment.

Table 1053: Segment Configurations

Connectivity	Traffic Type	Location and Transport Zone	Details
A global tier-0 or tier-1 gateway	Overlay	The **Location** section is populated with the following configurations: • the same locations that are configured on the attached gateway. • the default overlay transport zone for each location.	Use this configuration to create a global overlay-backed segment connected to the selected global gateway.
None	VLAN	You must select one location for this segment. You must also select a transport zone from that location.	Use this configuration to create a global VLAN-backed segment to use for a tier-0 external interface.
None	Overlay	No locations or transport zones can be selected.	This segment is created on the Global Manager but is not realized in any Local Managers. You can attach it to a gateway later.

Creating a VLAN-backed segment that is attached to a gateway is not supported.

6. Enter the Gateway IP address of the subnet in a CIDR format. A segment can contain an IPv4 subnet, or an IPv6 subnet, or both.
   - If a segment is not connected to a gateway, subnet is optional.
   - If a segment is connected either to a tier-1 or tier-0 gateway, subnet is required.

   Subnets of one segment must not overlap with the subnets of other segments in your network. A segment is always associated with a single virtual network identifier (VNI) regardless of whether it is configured with one subnet, two subnets, or no subnet.
7. Skip **Set DHCP Config**.

   Only static bindings are supported on a segment created from Global Manager. See Features and Configurations Supported in NSX Federation.
8. In the **VLAN** field, specify the VLAN IDs.
   - If the transport zone is of type VLAN, you must configure a VLAN ID or a range of VLAN IDs.
     – The configuration of a single VLAN ID defines the VLAN ID of the VLAN backing the segment in the physical infrastructure.
     – The configuration of a range of VLAN IDs enables guest VLAN tagging for the segment and defines the VLAN IDs for the VLAN tags allowed on the segment.
   - If the transport zone is of type overlay, you can optionally enter a range of VLAN IDs, which enables guest VLAN tagging on the segment and defines the VLAN IDs for the VLAN tags allowed on the segment.

     Note:
     The VLAN IDs must be within the allowable range specified by the transport zone.

9. Optional: Select an uplink teaming policy for the segment.

   This drop-down menu displays the named teaming policies, if you have added them in the VLAN transport zone. If no uplink teaming policy is selected, the default teaming policy is used.

   - Named teaming policies are not applicable to overlay segments. Overlay segments always follow the default teaming policy.
   - For VLAN-backed segments, you have the flexibility to override the default teaming policy with a selected named teaming policy. This capability is provided so that you can steer the infrastructure traffic from the host to specific VLAN segments in the VLAN transport zone. Before adding the VLAN segment, ensure that the named teaming policy names are added in the VLAN transport zone.

10. Click **Save**.
11. To continue configuring the segment, click **Yes** when prompted.
12. To select segment profiles, click **Segment Profiles**.
13. Click **Save**.

Configure Bridging on Global Manager

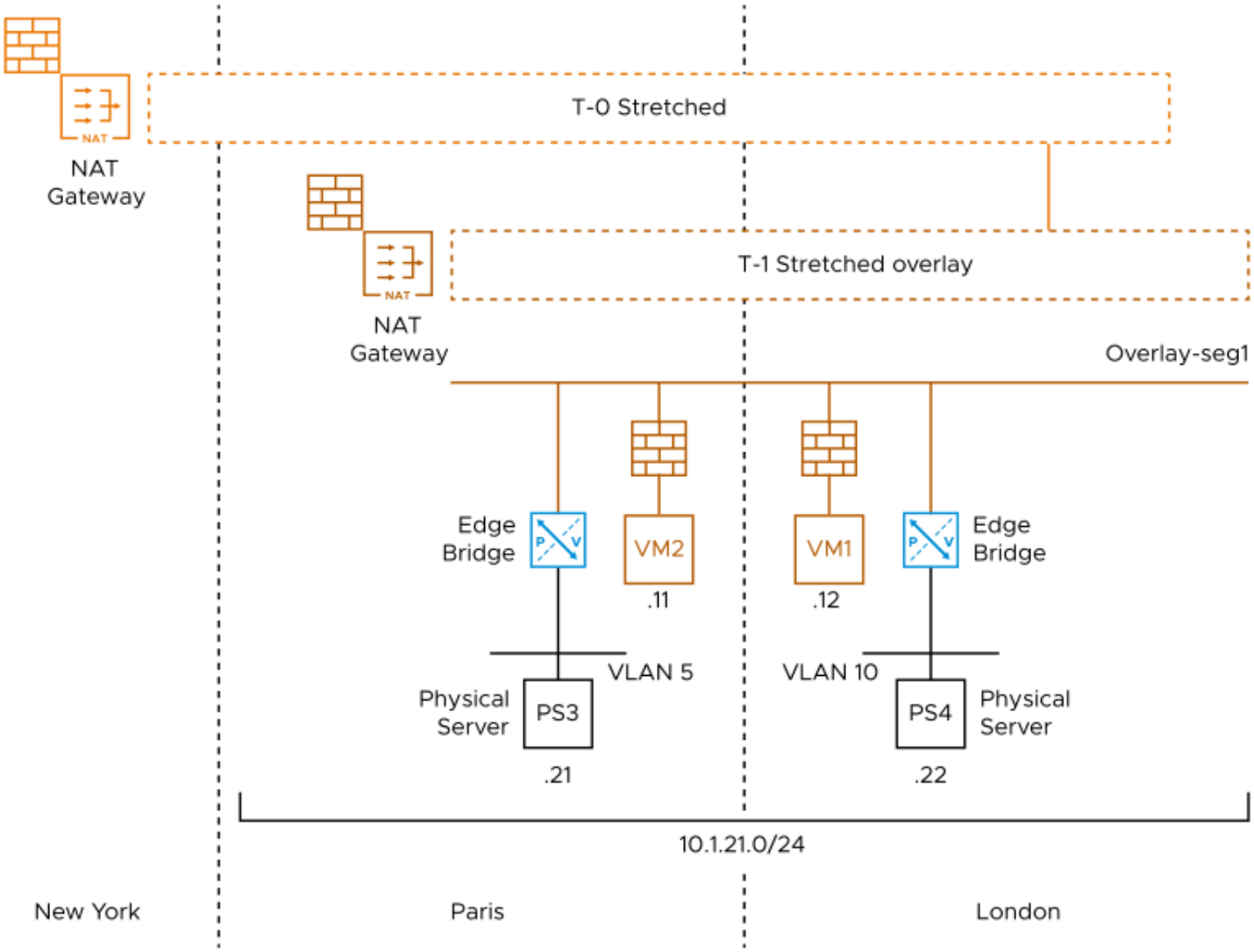
Bridging allows the communication between overlay segments and physical VLANs. Starting with NSX 3.2.2, you can configure edge bridges from the Global Manager and apply them to stretched and non-stretched segments.

For example, in the logical diagram, Edge Bridging in NSX:

- VM2 connects to an overlay segment (overlay-seg1) and a physical server (PS3) on physical VLAN5. All are in the subnet 10.1.21.0/24.
- VM1 connects to an overlay segment (overlay-seg1) and physical server (PS4) on physical VLAN10. All are on the subnet 10.1.21.0/24.

The edge bridge allows the communication between overlay and VLAN.

Figure 432: Edge Bridging in NSX



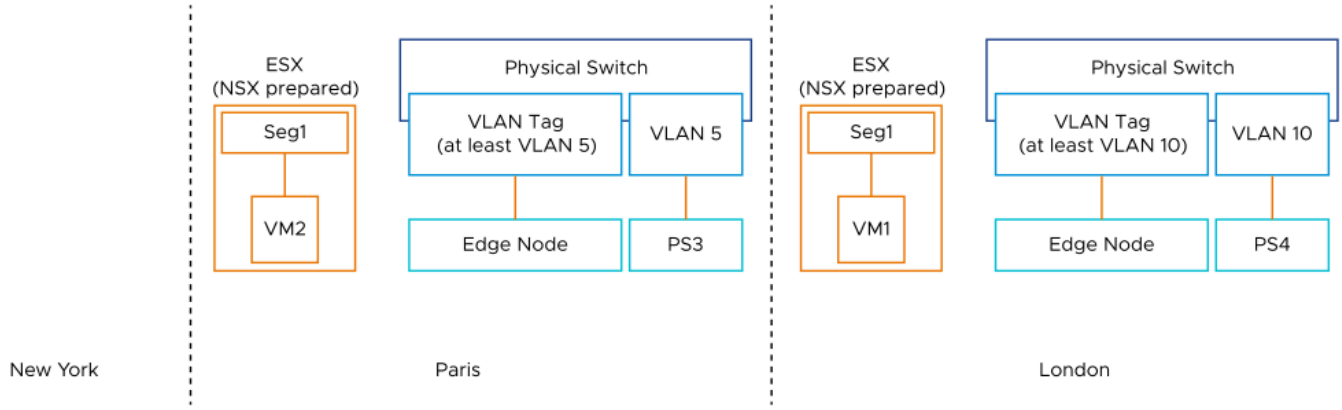
This functionality includes the following support as depicted in the edge bridging in NSX Federation diagram:

- Bridge creation on a segment on a given location.
- The segment can span one site or multiple sites. For instance, if you had three sites (New York, Paris, and London), you have a bridge segment, overlay-seg1, that spans Paris and London.
- The bridge is present on one site in one edge cluster with an edge node on the same site. For instance, the bridging profile bridge-paris defines the bridge in Paris.
- Multiple bridges can be assigned to the same segment. For example, you can bridge the segment overlay-seg1 in Paris with bridge-Paris to a given VLAN, such as VLAN 5.

You can use overlay/VLAN bridging on edge nodes which you configure using edge bridge profiles. The edge bridge profile contains the edge node primary and edge node backup of an edge cluster. You then associate that edge bridge profile in your overlay-segment.

> **Note:**
> As shown in the NSX Edge Bridge Physical View diagram, the edge nodes must have connectivity to those physical VLANs (VLAN5 and VLAN10) to offer the edge bridge.

Figure 433: Edge Bridge Physical View



For detailed procedures on edge bridging configuration, see Edge Bridging: Extending Overlay Segments to VLAN. Use the same procedure for Global Manager and Local Manager.

Security in NSX Federation

You can create distributed and gateway firewall rules from the Global Manager with global, regional or local spans. For more details about implementing security in NSX Federation, see VMware vDefend Firewall documentation.

Role-Based Access Control in NSX Federation

You can configure NSX Federation role-based access control (RBAC) to restrict system access to authorized users. NSX Federation RBAC works similarly to NSX RBAC for authorized users. This topic provides some optional configuration information for RBAC on NSX Federation when it is used with specific authentication providers.

Most authentication and authorization tasks use the same procedures as described under the Authentication and Authorization section of the *NSX Administration Guide*. One exception is that the VMware Workspace ONE Access (vIDM) and LDAP configuration is not synchronized from the active or the standby Global Managers (GM) to the Local Managers (LM). This requires that you configure each GM or LM (NSX cluster) separately for vIDM and LDAP. It also requires that users have the same role bindings on each NSX Federation server for seamless access.

For example, if you use NSX Federation and vIDM or LDAP authentication and want to switch between the GM and the LM server using the Location drop-down menu from the GM page, ensure you complete the following high-level tasks, so your configuration is set up properly. These configuration tasks help users that use vIDM and LDAP authentication providers avoid user permission error messages.

Task	Go To
Configure VIDM or LDAP on both the active and the standby Global Manager servers separately.	• Integration with VMware Workspace ONE Access • Integration with LDAP
Configure VIDM or LDAP on each Local Manager server.	• Integration with VMware Workspace ONE Access • Integration with LDAP
Ensure that users that want to switch between the GM and LM servers using the Location drop-down menu have the same user roles on both GM and LM servers. If the user has a role on GM, but no role on LM, users might see a permission error such as "The user does not have permission on any feature."	• Using the Global and Local Manager Web Interfaces • Manage Local User Accounts • Create or Manage Custom Roles • Add a Role Assignment or Principal Identity

To ensure that the Location drop-down menu allows your user to switch between the GM and the LM servers, after you update the user roles on the LM server from read only to write or mirror the GM roles, verify that the task completes. For details, go to Using the Global and Local Manager Web Interfaces and Monitoring NSX Federation Locations.

Traceflow in Federation

Traceflow is a troubleshooting tool used to check connectivity between two NSX entities.

Traceflow in Federation is supported starting with NSX 3.2.1. The Global Manager (GM) serves as a single point to trigger traceflow all the sites. The GM gets VM inventory information from each of the sites. Only VMs (and their IP addresses that are known to the Global Manager are supported, VMs attached to ports not known to the GM are not supported. Segment ports on segments that are stretched are supported on GM. Only such ports can be sources/destination in Federation traceflow.

1. From your browser, log in with admin privileges to the active Global Manager at https://<global-manager-ip-address>.
2. Select **Plan &Troubleshoot** > **Traceflow**.
3. In the Packet Information box perform the following tasks:
   a) Select an IPv4 or IPv6 address type.
   b) Select a traffic type. For IPv4 addresses the traffic type choices are: Unicast, Multicast, and Broadcast. For IPv6 address the traffic type choices are: Unicast or Multicast.
   c) Select a **Protocol Type**, and provide related information.

Protocol	Parameters
DHCP	Select a DHCP OP code: **Boot Request** or **Boot Reply**.
DHCPv6 (IPv6 only)	Select a DHCP message type: **Solicit**, **Advertise**, **Request**, or **Reply**.
DNS	Specify an address and select a message type: **Query** or **Response**.
ICMP	Specify an ICMP ID and a sequence.
ICMPv6 (IPv6 only)	Specify an ICMP ID and a sequence.
TCP	Specify a source port, a destination port, and TCP flags. • The default flag is SYN. • SYN cannot be combined with RST or FIN. • If SYN is not selected, you must select ACK or RST. • ACK cannot be combined with FIN, PSH, or URG.
UDP	Specify a source port and a destination port.
ARP	Select an ARP OP Code: **ARP Request** or **ARP Reply** .
NDP (IPv6 only)	For Unicast traffic, specify the **Destination IP** address. For Multicast traffic, specify the **Destination IP** and **Destination MAC** address.

4. Specify the source and destination information according to the traffic type.

Traffic Type	Source	Destination
Unicast	Select a VM or a logical port/interface. For a VM: • Select a VM from the drop-down list. • Select a virtual interface. • The IP address and MAC address are displayed if VMtools is installed in the VM, or if the VM is deployed using OpenStack plug-in (address bindings will be used in this case). If the VM has more than one IP address, select one from the drop-down list. • If the IP address and MAC address are not displayed, enter the IP address and MAC address in the text boxes. For a logical port/interface: • Select an attachment type: **VIF**, **DHCP**, **Edge Uplink**, or **Edge Centralized Service**. • Select a port.	Select a VM, a logical port/interface, or IP-MAC. For a VM: • Select a VM from the drop-down list. • Select a virtual interface. • The IP address and MAC address are displayed if VMtools is installed in the VM or if the VM is deployed using OpenStack plug-in (address bindings will be used in this case). If the VM has more than one IP address, select one from the drop-down list. • If the IP address and MAC address are not displayed, enter the IP address and MAC address in the text boxes. For a logical port/interface: • Select an attachment type: **VIF**, **DHCP**, **Edge Uplink**, or **Edge Centralized Service**. • Select a port. For IP-MAC: • Select the trace type (layer 2 or layer 3). For layer 2, enter an IP address and a MAC address. For layer 3, enter an IP address.
Multicast	Same as above.	Enter an IP Address. It must be a multicast address from 224.0.0.0 - 239.255.255.255.
Broadcast	Same as above.	Enter a subnet prefix length.

5. Optional: Click **Advanced Settings** to see the advanced options.
   Enter the desired values or input for the following fields, and click **Save**:

Option	Description
Hop Limit	This value should be between 2-255.
Next Header	This value should be between 2-255.
Ethertype	The default is 2048.
Timeout	The default is 30 seconds, and the range is 15-60 seconds.
Frame Size	The default is 128.
Payload Type	Select **Base64**, **Hex**, **Plaintext**, **Binary**, or **Decimal**.
Payload Data	Payload formatted based on selected type.

6. Click **Trace**.

   The output includes a graphical map of the topology and a table listing the observed packets. The first packet listed has the observation type `Injected` and shows the packet that is injected at the injection point.

   You can apply a filter (**All**, **Delivered**, **Dropped**) on the observations that are displayed. If there are dropped observations, the **Dropped** filter is applied by default. Otherwise, the **All** filter is applied.

   The graphical map shows the backplane and router links. Note that bridging information is not displayed.

Prevent Password Lockout on Local Manager Nodes

Password lockout can occur after a Local Manager is imported to a Global Manager.

If you reset the admin password of the Local Manager, the admin account is locked out. As a result, the connectivity between the Global Manager cluster and the Local Manager cluster fails with a general system error. The Global Manager nodes in both regions must be added to an allowlist of trusted nodes for the Local Manager cluster to avoid lockdown. This can be prevented by adjusting the `lockout_immune_addresses` parameter on the Local Manager.

1. Log in to the host that has access to your data center.
2. Update the allowlist of trusted sources on the Local Manager appliance by using the Postman PUT method.

   You add the IP addresses of the Global Manager cluster to the `lockout_immune_addresses` list of the Local Manager cluster.

   a) Start the Postman application in your web browser and log in.
   b) On the **Authorization** tab, enter the following settings and click **Update request**.

Setting	Value
Type	Basic Auth
User name	admin
Password	*nsx_admin_password*

   c) On the **Headers** tab, add a key by using the following details.

Setting	Value
Key	Content-Type
Key Value	application/json

   d) In the request pane at the top, send the following HTTP request.

Setting	Value
HTTP request method	`GET`
URL	`https://<nsx_manager_FQDN>/api/v1/cluster/api-service` (change the FQDN to your local NSX Manager FQDN)

   e) After a successful response (“`status: 200 OK`”), copy the JSON-formatted body response from the **Body** tab to a text-editor.
3. Add the `lockout_immune_addresses` information to the JSON response:
   a) Search for the `lockout_immune_addresses` line in the JSON response. If the line cannot be found in the JSON response, add a new line with `lockout_immune_addresses,`. Note that a “,” must be added to end of the previous line.
   b) Add the IP addresses of all global NSX Managers (including the VIP addresses) between the brackets in the following format, leaving the quotes intact: `"lockout_immune_addresses”:[ “172.16.11.95”, “172.16.11.96”, “172.16.11.97”, “172.16.11.98” ]`
4. Send the new security configuration to the local manager using the Postman PUT method:
   a) Take the previous Postman HTTP request and change the HTTP request method from `GET` to `PUT`.
   b) On the Body tab, paste the new JSON formatted security configuration from your code-/text editor.
   c) Send the new HTTP request and confirm a successful response (“`status: 200 OK`”).
5. Repeat steps 2 through 4 for all NSX Managers.

Backup and Restore in NSX Federation

You can configure and start backups for Global Manager (GM) and each Local Manager (LM) from within the Global Manager. NSX Managers are called Local Managers if they are federated with a Global Manager.

- Log in to the active Global Manager and select **System** > **Backup & Restore**. Each Global Manager and Local Manager in the environment is listed. See Configure Backups for instructions.

- Backup and restore on a standby Global Manager is not supported. Backup configuration for the standby Global Manager is turned off, as this is not required. If there is an issue with a standby Global Manager, delete the standby Global Manager. Then deploy a new Global Manager and onboard it to the active Global Manager.
- You cannot restore a Local Manager from within the Global Manager. To restore a Local Manager backup, log in to the Local Manager to restore. If you create the Local Manager backup after the Local Manager is onboarded to the Global Manager, then to recover the Local Manager, you only have to restore Local Manager. After restore completes, Local Manager is automatically connected to Global Manager. See Restore a Backup for instructions.
- The system treats backup and restore operations as specific to each appliance, whether it is the Global Manager or the Local Manager you are backing up or restoring. The Global Manager's backup contains a backup of the database of that appliance only. The Local Manager contains a backup of the database and inventory of that appliance only.
- If you are restoring a Global Manager and a Local Manager, select backup timestamps of each appliance as close to each other as possible.
- After each appliance is restored, the `async replicator` service restores communication between the Global Manager and each Local Manager.
- Storing libraries during backup is not supported in NSX backup. During restore NSX will always have default version libraries. If you are using VMware NSX® Application Platform, you must upload Kubernetes tools if requested.

Backup Scenarios in NSX Federation

Scenario	Backup Workflow
Global Manager has any of the following changes: • Registering of a new LM • Networking configuration • Security configuration	Back up only the active Global Manager.
A Local Manager has any of the following changes: • Registering to GM • Networking configuration • Security configuration • Fabric changes, such as: – Host transport node added or removed (ESX) – Edge transport nodes added or removed	Back up the Local Manager. You can perform this task from the Global Manager or the Local Manager.

Restore Scenarios in NSX Federation

Note, if you had a cluster of Global Manager appliances, you can only restore one node using the restore process. You must create the cluster after the restore of the first node completes.

This table uses example names for active and standby to illustrate the workflows. For detailed steps, see Backing Up and Restoring NSX Manager or Global Manager. For workflows about planned and unplanned Federation NSX disaster recovery scenarios, see Disaster Recovery for Global Manager.

Scenario	Restore Workflow
Your active Global Manager, for example, GM-Loc1-Active, is lost, but you have a standby Global Manager, (GM-Loc2-Standby).	This workflow performs a switch-over to make the standby server active. 1. Go to the standby server, GM-Loc2-Standby, and make it active (resulting in GM-Loc2-Active). 2. Recreate a standby GM-Loc1 cluster by installing the NSX Manager OVA onto a new server and name it GM-Loc1-Standby. Then make GM-Loc1-Standby active (resulting in GM-Loc1-Active). This automatically switches GM-Loc2-Active to become GM-Loc2-Standby.
Your active Global Manager, GM-Loc1-Active, is lost and you do not have a standby. For production environments, we recommend having a standby GM server.	1. Install a new GM-VM with the same IP address as the lost GM-Loc1-Active server. Use the NSX Manager OVA (resulting in GM-Loc1). Ensure you delete the active server before proceeding with this install. For instructions, see the *NSX Installation Guide*. 2. After installation, access the UI, select **System** > **Backup & Restore** then: a. Enter the SFTP server information where the backup is stored. b. Select the latest backup file. c. Click on the Restore.
Your standby Global Manager, GM-Loc2-Standby, is lost.	1. On your active GM, GM-Loc1-Active, remove the current GM standby, GM-Loc2-Standby, using **System** > **Location Manager** . 2. Install a new GM-VM using the NSX Manager OVA (resulting in GM-Loc2). You can optionally use the same IP address as the old GM-Loc2-Standby. 3. On the active GM, GM-Loc1-Active, add the newly installed GM-VM, GM-Loc2, as standby. To add the standby after installation completes, on the active GM, select **System** > **Location Manager** (resulting in GM-Loc2-Standby).
A Local Manager, LM-Loc1, is lost.	Restore the Local Manager, LM-Loc1, from the Local Manager. See Backing Up and Restoring NSX Manager or Global Manager. When restored, configurations from the Global Manager are synchronized with the Local Manager.
Both the Global Manager and the Local Manager are lost.	If you are restoring both the Global Manager, GM-Loc1-Active, and the Local Manager, LM-Loc1, use the latest backups of each appliance. When the Global Manager and the Local Manager are restored, the Global Manager, GM-Loc1-Active, pushes the configurations to the Local Manager, LM-Loc1. You must manually resolve any discrepancies in inventory and fabric related changes between the Local Manager and the Global Manager.

Disaster Recovery for Global Manager

In an NSX Federation environment, if you lose your active Global Manager, you can switch to the standby Global Manager.

These workflows describe the following scenario where GM denotes the Global Manager appliance.

You have a GM service to federate the network and security services across your locations. This includes:

- One GM cluster in location `Loc1` set **Active** (`GM-Loc1`).
- One GM cluster in location `Loc2` set **Standby** (`GM-Loc2`).

Planned Switchover to Standby Location

The active GM service is in Loc1 and the stretched tier-0/tier-1 primary routing services are also in Loc1.

1. Switch your GM service in `Loc2`.
   a. Log in to the standby GM – `GM-Loc2`.
   b. Select the **Active** drop-down menu and click **Make Active**.

The system starts the process of making GM-Loc2 active. After the process completes, GM-Loc2 changes status to Active and GM-Loc1 changes status to Standby.

If the planned failover is unsuccessful use the Force Option.

Note:
If the synchronization between GM-Loc1 and GM-Loc2 is not successful for some time, you might have possible configuration data loss when GM-Loc2 is forced to be GM Active.

c. To check if GM-Loc1 and GM-Loc2 are synchronizing, run `/etc/init.d/corfu-log-replication-server status` in an SSH root shell command on both GM-Loc1 and GM-Loc2. If synchronization is not running, contact technical support.

2. (Optional) Switch primary stretched tier-0/tier-1 `Loc1` to `Loc2`.
   a. To move the stretched tier-0 and tier-1 gateways to the secondary site, follow the network recovery workflow. See Network Recovery for Local Managers.
   b. Recover the compute VMs using your preferred method. For example, use VMware Site Recovery Manager from `Loc1` to `Loc2`.

Unplanned Switchover to the Standby Location

If the site `Loc1` becomes unavailable, `GM-Loc1`, `LM-Loc1`, as well as `Loc1` Edge Nodes stop responding. Proceed with a forced failover.

1. Recover your GM Service in `Loc2` .
   a. Log in to the standby GM – `GM-Loc2`.
   b. Select the **Actions** drop-down menu and click **Make Active**.
   The system starts the process of making GM-Loc2 active. After the process completes, GM-Loc2 changes status to **Active** and all LMs (other than LM-Loc1 unavailable) synchronizes with GM-Loc2 from the last synchronized configuration received from GM-Loc1.
   If `GM-Loc1` is online after `GM-Loc2` changes to active, the status of `GM-Loc1` is NONE. To make `GM-Loc1` standby:
      a. Log in to the active GM – `GM-Loc2`.
      b. From the tile where GM-Loc1 shows the status of NONE, select the **Actions** drop-down menu and click **Make Standby**.
2. Move your primary stretched tier-0/tier-1 **Loc1** to **Loc2**.
   a. Follow the network recovery workflow to move stretched tier-0 and tier-1 gateways to the secondary site. See instructions at Network Recovery for Local Managers.
   b. Recover the compute VMs using your preferred method. For example, use VMware Site Recovery Manager from to `Loc2`.

See the *Disaster Recovery* section in the NSX Multi-location Design Guide for more details. For details about using Site Recovery Manager, see Working with VMware Live Site Recovery and Multisite Environments.

Working with Site Recovery Manager and NSX Federation

You can use VMware Live Site Recovery™ with NSX Federation for disaster recovery use cases.

Site Recovery Manager supports the following workflows with NSX Federation:

- NSX Federation Global Manager (GM) VMs support full and test recovery of GM VMs (supported with or without NSX Federation management cluster VIP).
- Compute VMs support full and test recovery of compute VMs. Recovered VMs in the disaster recovery site have their NSX tags and firewall rules based on these NSX tags or not such as IP addresses and VM names.

To ensure that groups and firewall rules replicate at the disaster recovery location during recovery, the NSXLocal Manager managing the disaster recovery location must have the NSX tags present at recovery time.

In NSX Federation, two fields support storage array replication: `replication _type` and `tag_delay_delete_time`. If you use storage array replication from your primary site to your recovery site in SRM, then create or update the tag replication policy `replication type` as STORAGE_ARRAY_REPLICATION.

Running a protection plan on SRM to migrate virtual machines using disaster recovery or planned migration options from your primary site to your recovery site will move the virtual machines to the recovery site and delete the virtual machines from primary site. When the concerned virtual machines are deleted from primary site and appear on recovery site within the tag_delay_delete_time setting, then tags from primary site get replicated on the virtual machines on recovery site. For storage array replication, the time for VMs to appear on the recovery site after they are deleted from the primary site depends on the VM count configured to failover, storage array performance, and ESX host.

If vSphere replication is being used to replicate VMs from primary site to recovery site in SRM, then the virtual machine is not deleted from primary site. Therefore, there is no need to specify replication_type or tag_delay_delete_time in the tag replication policy.

> **Note:**
> For Global Manager, if replication types VSPHERE_REPLICATION or OTHER options are used, then the tag_delay_time value gets set to 0. The replication_type is set to default value of enum (invalid) and the tag_delay_delete_time is set to 0 minutes.

Configure VM Tag Replication across LMs using GM API

In NSX Federation, to configure VM tag replication across Local Managers, run the following Global Manager API:

```
PUT https://{{gm}}/global-manager/api/v1/global-infra/vm-tag-replication-policies/policy1
{
   "display_name":"vm tag replication policy Paris to London",
   "description":"vm tag replication policy1",
   "protected_site": "/global-infra/sites/LM_Paris",
   "replication_type": "STORAGE_ARRAY_REPLICATION",
   "tag_delay_delete_time": 30,
   "recovery_sites": [
       "/global-infra/sites/LM_London"
   ],
   "groups":[
       "/global-infra/domains/default/groups/Web-VM-Group",
       "/global-infra/domains/default/groups/DB-VM-Group"
   ],
   "vm_match_criteria": "MATCH_BIOS_UUID_NAME"
```

For NSX Federation, to configure VM tag replication across Local Managers, run the following Global Manager API:

```
PUT https://{{gm}}/global-manager/api/v1/global-infra/vm-tag-replication-policies/policy1
{
   "display_name":"vm tag replication policy Paris to London",
   "description":"vm tag replication policy1",
   "protected_site": "/global-infra/sites/LM_Paris",
   "recovery_sites": [
       "/global-infra/sites/LM_London"
   ],
   "groups":[
       "/global-infra/domains/default/groups/Web-VM-Group",
       "/global-infra/domains/default/groups/DB-VM-Group"
   ],
   "vm_match_criteria": "MATCH_BIOS_UUID_NAME"
```

LM_Paris sends the tag information of the VMs for the BIOS UUID of the VMs in the groups Web-VM-Group + DB-VM-Group to LM_London. Before the recovery of the London VMs by SRM, LM_London does not have the VMs with the BIOS UUID and the VMs are not visible in LM_London yet. However, when SRM recovers the VMs in London, LM_London sees those VMs with the BIOS UUID and applies their NSX tags on them. The VMs get their security based on NSX tags.

> **Note:**
> `vm_match_criteria` has two possible values MATCH_BIOS_UUID_NAME or MATCH_NSX_ATTACHMENT_ID. At the recovery, SRM copies both so any configuration is valid with SRM.

However, if another product completes VM replication and copies one, but not the other value, then configure GM with the appropriate vm_match_criteria value.

To estimate how much time it might take for the VM to disappear from the primary site and appear on the protected site, you can perform a test on the protection plan and measure the time interval between the end of the **Synchronize storage** and completion of the **Power on priority X VMs** steps. Set your `tag_delay_delete_time` to be more than this estimated time. Another tip is to run the protection plan in VMware Live Site Recovery and set the `tag_delay_delete_time` to always be more than time it takes between the step **Shut Down VMs at the protected site** and to complete the step to **Power on priority X VMs** while doing run on protection plan in SRM.

Check VM Tag Replication across LMs using GM API

To get details on VM tag replication across Local Managers run the following Global Manager API :

```
GET https://{{gm}}/global-manager/api/v1/global-infra/vm-tag-replication-policies
```

The output returns something similar to:

```
{
 "protected_site": "/global-infra/sites/LM_Paris",
 "recovery_sites": [
   "/global-infra/sites/LM_London"
 ],
 "vm_match_criteria": "MATCH_BIOS_UUID_NAME",
 "groups": [
   "/global-infra/domains/default/groups/Web-VM-Group",
   "/global-infra/domains/default/groups/DB-VM-Group"
 ],
 "resource_type": "VMTagReplicationPolicy",
 "id": "policy1",
 "display_name": "vm tag replication policy Paris to London",
 "description": "vm tag replication policy1",
 "path": "/global-infra/vm-tag-replication-policies/policy1",
 "relative_path": "policy1",
 "parent_path": "/global-infra",
 "unique_id": "9ee18586-5480-41d9-8223-690c9226d763",
 "marked_for_delete": false,
 "overridden": false,
 "_create_time": 1638413861377,
 "_create_user": "admin",
 "_last_modified_time": 1638413861377,
 "_last_modified_user": "admin",
 "_system_owned": false,
 "_protection": "NOT_PROTECTED",
 "_revision": 0
}
```

NSX supports only one entry from recovery sites. For details, see the vm-tag-replication-policies/policy-name API in the *NSX Global Manager REST API Guide*.

Network Recovery for Local Managers

If a Local Manager is lost, you can recover networking configurations from it using the auto-detected Network Recovery option in the Global Manager.

You must have at least one stretched tier-0 or tier-1 gateway set up designating a Location Manager as primary. The loss of this primary Location Manager for the tier-0 or tier-1 gateway triggers the option of network recovery in the Global Manager.

- The Global Manager detects the loss of connection and prompts you to perform **Network Recovery**.

- In the first step of recovery, you recover the tier-0 gateway. You can change the preferred primary location if you want it to be different from the one you set in the fallback preference.
- In the second step, you select a preferred primary location for tier-1 gateways that have a subset of the span of the locations covered by the tier-0 network. The preferred primary location for such tier-1 gateways would be different from tier-0 gateways and you must either accept the fallback preference established by the tier-0 gateway, or elect not to move the gateway.
- In the final step, you can view the list of networking constructs that cannot be recovered because they do not have a secondary location configured.

  Note:

  If you have a tier-0 and tier-1 gateway set up using a Location Manager as primary, but the tier-0 and tier-1 gateway do not have any services attached to them, for example, tier-0 and tier-1 without NAT and firewall, then the data plane traffic still works after the loss of the primary Location Manager. For tier-0/tier-1 configuration without service, Network Recovery is not mandatory for the recovery of data plane, even though the Network Recovery option appears in the Global Manager.

1. From your browser, log in with admin privileges to the active Global Manager at https://<global-manager-ip-address>.
2. Select **System** > **Location Manager**.
3. A banner appears on this page noting the location that is down. Click **Network Recovery** on the banner and start the workflow for **Location Disaster Recovery** in the following steps.
4. **Tier-0 Gateways**: For each tier-0 gateway that has the failed location set as primary, you have the option to select a new primary location. This new primary location can be different from the fallback preference you elected when creating the tier-0 gateway. You can also elect to not move the tier-0 gateway. Click **Apply Configuration** for each tier-0 gateway after selecting a new primary location or retaining the priority set earlier.
5. Click **Next**.
6. Tier-1 A/S gateways are listed for recovery only if their span differs from the span of the tier-0 gateway. If tier-1 A/S gateways follow the same span as the tier-0 gateway, the same locations are selected to be primary as for tier-0 gateways. For a different span, you can either select a different location as primary or elect to not move the tier-1 gateway at all.
7. After you make your selections for each tier-1 gateway, click **Accept** and **Next** to proceed.
8. Under **Single Location Entities** you can see a list of tier-0 and tier-1 gateways that cannot be moved to a new primary location because they exist only in the failed location. Click **Next** to proceed.

The stretched tier-0 and tier-1 gateways are moved to the new location which that you designated as primary.

For more details, go to the Broadcom Communities discussion forum to review the *NSX Multi-Location Design Guide*, section 4.4.2, *Data Plane Recovery*.

NSX Multi-tenancy

NSX supports multi-tenancy where you can configure multiple tenants on a single NSX deployment. Multi-tenancy enables you to isolate security and networking configurations across tenants.

NSX users across tenants can access the same NSX Manager to provision networking and security objects on shared hosts.

NSX Projects

A project in NSX is analogous to a tenant. By creating projects, you can isolate security and networking objects across tenants in a single NSX deployment.

Let us assume that an organization has NSX deployed at its site. This organization currently has all its infrastructure, networking, and security configurations in the default space, which is owned by the Enterprise Admin. You will learn more about the default space later in this documentation.

This organization has the following objectives:

- Isolate networking and security configurations for its three departments: Sales, Marketing, and Operations.
- Delegate the tasks of creating and managing networking and security configurations for each department to a specific set of NSX users, and avoid giving these users visibility to all the objects in the system.
- By default, allow workload VMs within a department to communicate only with other workload VMs (including the DHCP server) in the same department.
- By default, block communication with workloads outside the department. If such a communication is required, the system must allow adding new rules or modify existing rules in the default security policy.

To achieve these objectives, the organization can implement multi-tenancy in its NSX deployment by using the Project feature. For example, it can create three projects with these names:

- Sales
- Marketing
- Operations

In a multi-tenancy deployment, users in each project have access to objects that they create in their project and can consume objects (in a read-only mode) that the Enterprise Admin has shared with their project from the default space.

Note:

- Setting up multi-tenancy in your NSX deployment is optional and its implementation has no impact on your existing NSX configuration.
- Multi-tenancy is currently not supported in an NSX Federation environment.

Multi-tenancy Policy Data Model

The NSX Policy data model is hierarchical and has two system-created branches:

- `/infra` branch is managed by the Enterprise Admin. The objects under this branch are displayed in the **Default** view on the UI.
  User roles other than the Enterprise Admin also exist in the `/infra` branch. The users in this branch are not tied to any specific projects. This documentation refers to such users as "system-wide" users. These users can configure a subset of objects under the `/infra` branch.
  System-wide users have access to all the objects in the system. That is, they have access to objects inside the **Default** view (`/infra` branch) and inside the projects.
  At times, this documentation uses the term "default space" to refer to objects under the **Default** view. In other words, the terms "default space" and "**Default** view" are used interchangeably. They both mean the same. To learn more about the **Default** view, see the *Overview of the Default View (Default Space)* subsection later in this documentation.
- `/orgs/default` branch holds the multi-tenancy objects. Every project has its own space to host objects that it owns.

Projects are created under `/orgs/default` to support independent sets of networking and security configurations for each tenant.

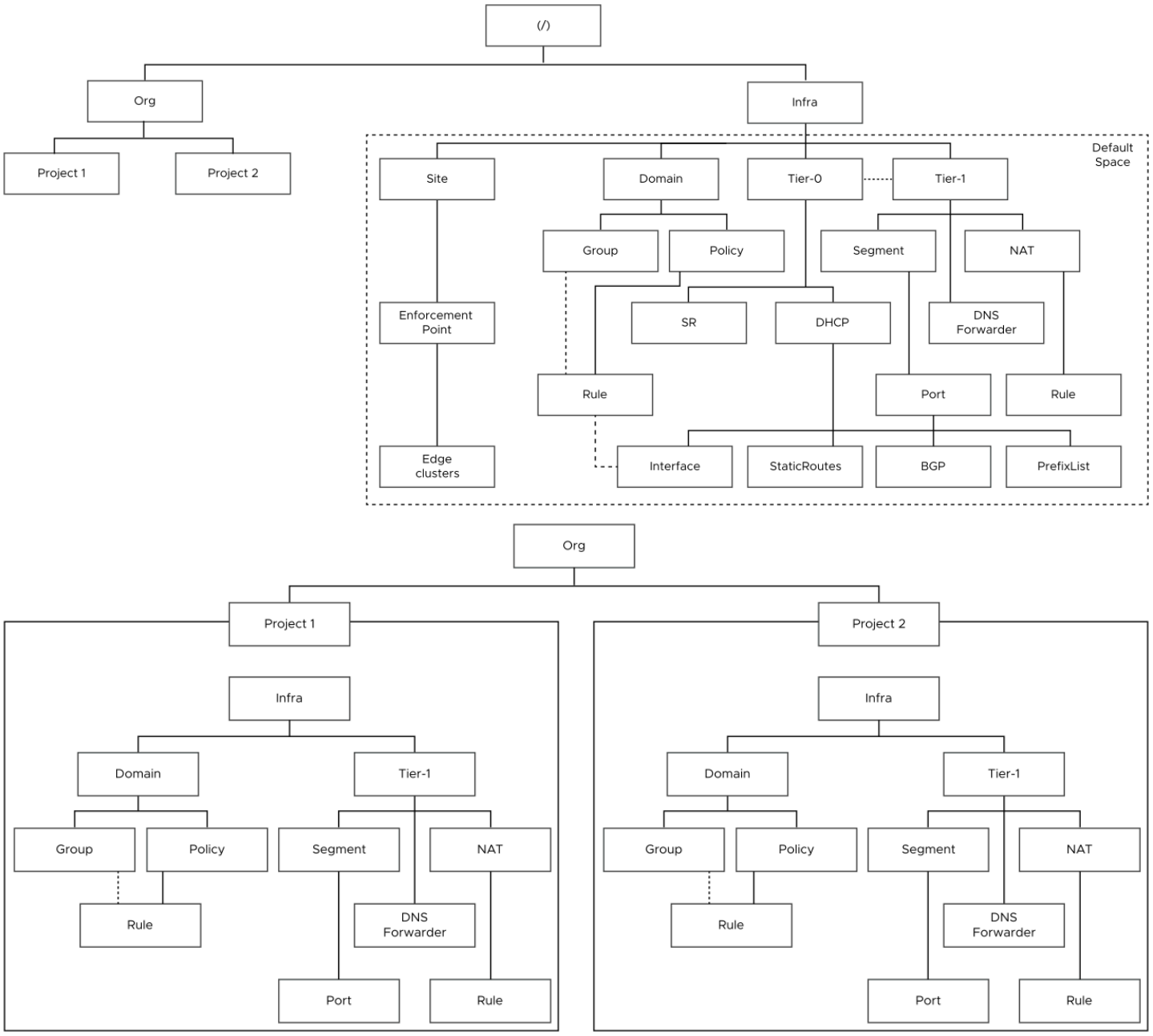
Project configurations are set up under `/orgs/default/projects/<project-id>/infra`

The following diagrams illustrate the data model for multi-tenancy. These diagrams represent a partial view of the data model only to understand the concept. The Policy data model has several objects, which are not shown.

The first figure shows the default space and two projects under the org. The next figure shows the hierarchy of objects in both the projects. Under the org, projects 1 and 2 have their own hierarchy of NSX networking and security objects that are created inside the project. Objects created inside a project are owned by that project.

Tier-0 gateways and edge clusters are owned by the default space, and they can be allocated to projects under the org. You cannot create tier-0 gateways and edge clusters inside a project.

Each project can optionally have its own tier-1 gateways, which must be configured in the project. In other words, the tier-1 gateways must be owned by the project. A project cannot use the tier-1 gateways that are configured in the default space.



Default Org

An NSX deployment has one default org. You cannot create, modify, or delete the default org. The org object is created by the system at startup. The tier-0 gateways and edge clusters in the system can be allocated to projects under the org.

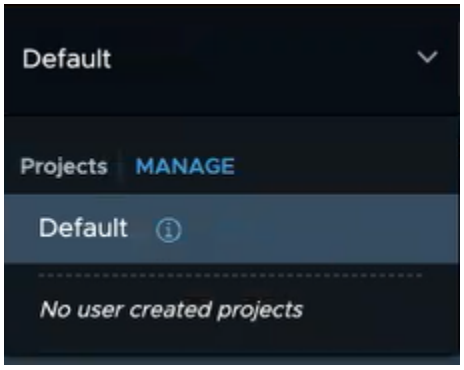
The org object is created by the system with the following identifier:

`/orgs/default`

The org object is not visible in the UI.

Understanding the Project Drop-Down Menu

The **Project** drop-down menu is available on the application bar, which is at the top of the NSX Manager UI. To view this menu, click **Default**, as shown in the following screen capture.



This menu displays the list of projects that you are are provided access to. You can use this menu to switch between projects and manage the objects in your assigned projects.

Overview of the Default View (Default Space)

When you log in to NSX Manager for the first time, the **Project** drop-down menu shows only the **Default** view. No user-created projects exists in the system, as shown in the preceding screen capture.

The **Default** view is visible to the Enterprise Admin and to the other system-wide user roles that are not assigned to any specific projects. This view contains:

- All the objects in your NSX fabric, such as hosts, tier-0 gateways, edge clusters, transport zones, and so on.
- Global user management objects.
- Objects under the `/infra` space of the hierarchical Policy data model, such as tier-1 gateways, segments, groups, firewall policies, and so on.
- Resource shares

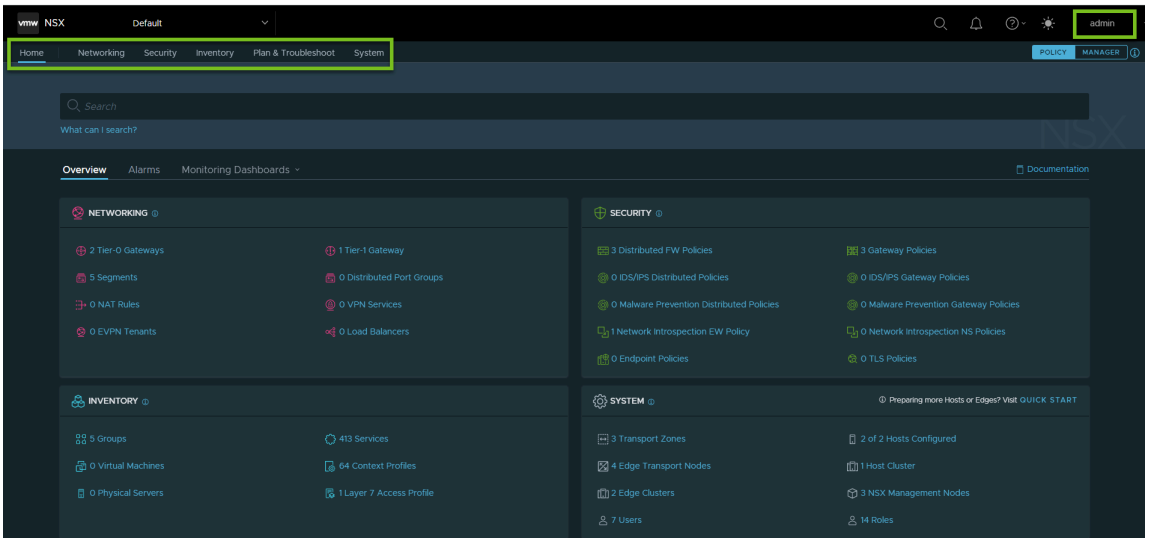
In short, the **Default** view contains NSX objects that do not belong to any project. The only exception is that on the **Virtual Machines** page of the **Default** view, all the VMs in the system are displayed. That is, VMs that are connected to:

- Segments in the default space.
- Segments in the projects.
- Subnets in the NSX VPCs within the project.

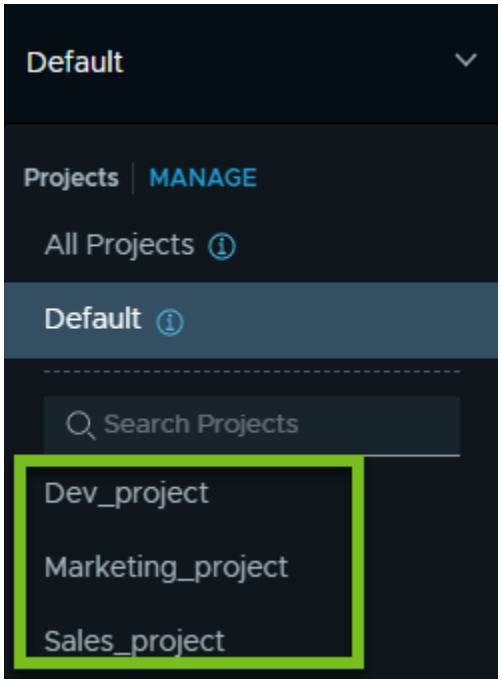
VMs that are not connected to any segment in NSX are also displayed in the default space. Such VMs are shown as `Not Connected`.

This exception allows an Enterprise Admin to view all the VMs that are running in the system from the default space itself. The Enterprise Admin can assign tags to any VM in the system and apply security policies on them.

When an Enterprise Admin logs into NSX Manager, the **Default** view is displayed, as shown in the following screen capture. Observe that all the tabs are displayed in the UI. The **Overview** page shows a high-level summary of the objects in the default space.

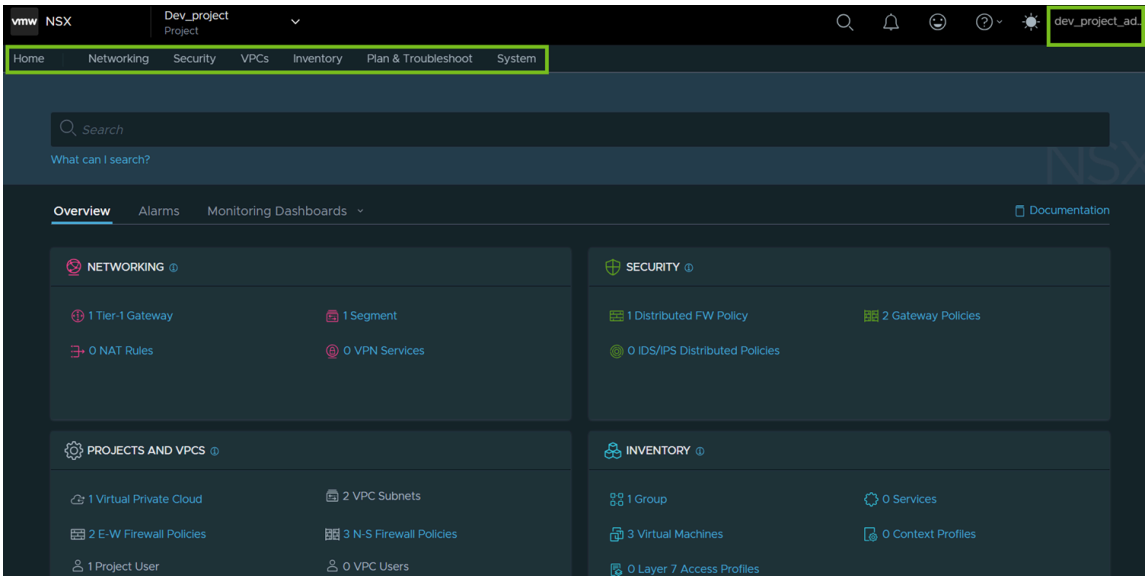


After one or multiple projects are created in your NSX deployment, these projects are displayed in the **Project** drop-down menu, as shown in the following screen capture.



An Enterprise Admin can view all the projects in the system. Other system-wide user roles, such as an Auditor, can also view all the projects in the system. Users that are assigned to specific projects with roles, such as Project Admin, Security Admin, Network Admin, Security Operator, and Network Operator can view the projects that they have access to.

For example, when a Project Admin logs into NSX Manager, the project-specific view is displayed, as shown in the following screen capture. The **Overview** page displays a high-level summary of the objects that are created in the project.



When you upgrade from an earlier version of NSX to NSX 4.1 or later, the default space will host all the existing infrastructure, networking and security configurations. You can continue to use the default space for all the networking and security requirements in your organization. The properties of the existing networking and security objects or the path of those objects are not modified. Creating projects is optional.

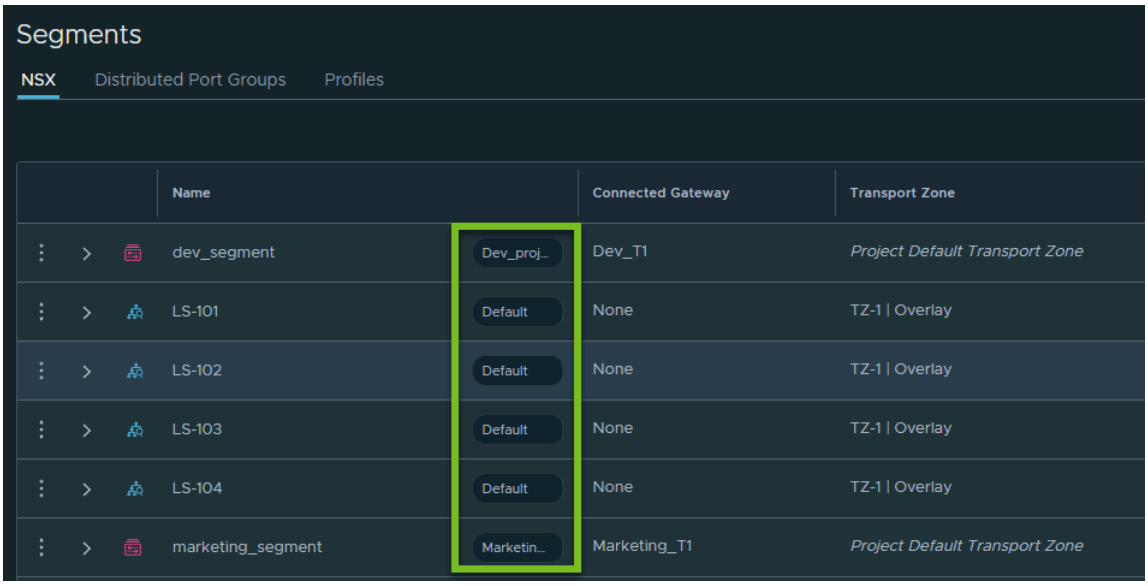
YYou can switch between project views and the **Default** view to view the objects under each of them. In addition, the **Project** menu also shows the **All Projects** view, which is described in the following section.

Overview of the All Projects View

- The **All Projects** view is available to all the system-wide user roles that are not assigned to any specific project. That is, this view is available to all the users that have access to the default space. For example, Enterprise Admin, Auditor, and so on.
- This view is available in the **Project** drop-down menu only after at least one project is added in your NSX deployment.
- This view displays networking and security configurations across all projects, including the default space.
- The objects in this view are displayed in a **read-only** mode.

In the **All Projects** view, the networking and security objects display a pill-shaped icon next to the object name to indicate whether the object is owned by the default space or a project.

For example, the following screen capture shows the list of segments on the **Segments** page. The pill-shaped icons that are highlighted in the green box indicate who owns each segment – default space or the project.



NSX Virtual Private Clouds

A project can optionally contain one or more NSX Virtual Private Clouds (VPC).

A VPC represents a self-contained private network within an NSX project that application developers or DevOps engineers can use to host their applications and consume networking and security objects by using a self-service consumption model.

NSX VPCs represent an additional layer of multi-tenancy within a project. It provides a simplified consumption model of networking and security services, which is aligned to the experience that you would have in a public cloud environment.

NSX VPCs can be created in projects and also in the default space.

VPC configurations are set up under the following path of the NSX Policy data model:

```
/orgs/default/projects/<project-id>/vpcs/<vpc-id>
```

To learn more about NSX VPC, see NSX Virtual Private Clouds.

Add an NSX Project

Projects help you to isolate networking and security configurations across tenants in a single NSX deployment.

You must be assigned the Enterprise Admin role.

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Click **Default**, and then click **Manage**.
3. Click **Add Project**.
4. Required: In the **Name** field, enter a name for the project.
5. In the **External Connections** field, click the Set link and enter the external connections for the project VPC's and also for the default transit gateway of the project.
6. In the **External IP Blocks** field, click the Set link and select the external IP blocks for the project and also for the default connectivity profile of the project.

   The selected IP blocks will become available to you when you add public subnets in the NSX VPCs within the project. The system will assign CIDR blocks to the public subnets in the NSX VPCs from these external IPv4 blocks. VPC users can also use the external IP blocks for adding NAT rules in the NSX VPCs.

   If no IP blocks are available for selection, click

   ,

   and then click **Create New** to add an IP address block.

   The external IP blocks must not overlap each other within a project, and they must not overlap on the same tier-0 gateway.

   For example, let us assume that Project A is connected to tier-0 gateway A, Project B is connected to tier-0 gateway B. The tier-0 gateways of these two projects are isolated. In this case, projects A and B can use the same or overlapping external IP blocks because they are connected to separate tier-0 gateways.
7. In the **Edge/Virtual Network Appliance Clusters** field, click the Set link and select an edge cluster or a virtual network appliance (VNA) cluster to associate with this project. Also select an edge cluster or a VNA cluster for the default connectivity profile.

   The selected edge clusters can be consumed inside the project in the future. For example, the edge clusters can be consumed for running centralized services such as NAT, Gateway Firewall, DHCP, and so on, that you configure on the tier-1 gateways inside the project. The edge clusters that are associated with the project need not necessarily run the tier-0 gateways that are associated with the project.

   Edge Clusters specified at the Project level are required for centralized services (NAT, Gateway firewall, VPN, DHCP). If no services are required, Edge Cluster can be skipped.

   An Edge Cluster can be assigned to multiple projects. That is, allocating an edge cluster to one project (say project 1) does not prevent you from allocating it to other projects (say project 2 and project 3).

   Note:

   The edge clusters that are associated with the default overlay transport zone are displayed in the drop-down menu.
8. In the **Short log identifier** text box, enter a string that the system can use to identify the logs that are generated in the context of this project.

   The short log identifier is applied to the security logs and audit logs.

   If you have dedicated a tier-0/VRF gateway to a project by configuring the `dedicated_resources` parameter in the `project` API, the short log identifier is appended to the log messages that are generated in the edge syslog for the centralized services, which are running on the tier-0/VRF gateway. To learn more, see Enabling Project Context in NSX Edge Syslog.

   The identifier must be unique across all the projects in your NSX environment.

   The identifier must not exceed eight alphanumeric characters. If it is not specified, the system autogenerates it when you save the project. After the identifier is set, you cannot modify it.

9. Expand Advanced Configuration and enter the remaining information.
10. In the **Tier-0 Gateways** field, select a tier-0 or a tier-0 VRF gateway that the workloads in this project can use for north-south connectivity with the physical network outside NSX.

    You can select multiple gateways if required. If no gateway is selected, the workloads in the project will not have north-south connectivity.

    Note:
    By default, the project is created in the default overlay transport zone of the system. Therefore, the tier-0/VRF gateways that are associated with the default transport zone are displayed in the drop-down menu.

    A tier-0 or a tier-0 VRF gateway can be assigned to multiple projects. That is, allocating a tier-0/VRF gateway to one project (say project 1) does not prevent you from allocating it to other projects (say project 2 and project 3).
11. Turn on the **Organize NSX Portgroups in vCenter folders** toggle if you want the system to create folders for this project on the VMware vCenter.

    By default, this toggle is turned off.

    Note:

    This toggle is applicable only when vCenter servers that are registered to your NSX deployment are at version 8.0 Update 3 or later.

    If this toggle is turned on, and the system detects that vCenter is at a version lower than 8.0 Update 3, a warning message is displayed, and this setting is not enforced in the project. That is, folders are not created for the project in vCenter.

    Important:

    - If **Multi NSX** is activated for any of the vCenter servers that are registered with your NSX deployment, folders cannot be created for the project in vCenter.
    - Conversely, if you have created a project in NSX with the **Organize NSX Portgroups in vCenter folders** toggle turned on, you cannot activate **Multi NSX** for the vCenter servers that are registered as compute managers in your NSX deployment.

    When you turn on this toggle, the setting is propagated to all the NSX VPCs that are added in this project. However, this setting is read-only in an NSX VPC, and cannot be overridden in the VPC.

    To understand the folder hierarchy that is created in vCenter when this toggle is turned on for the project, see Example: Organizing NSX Port Groups in vCenter Folders.

    The hierarchical organization of Project and VPC folders helps the vSphere administrator easily manage the NSX port groups in vCenter. For example, let us assume that the vSphere admin wants to assign a **read-only** permission to a vSphere user named "Tom" for all the 10 NSX port groups in VPC X. The vSphere admin can assign the **read-only** permission to Tom on the `VPC X` folder, and choose to propagate this permission to all the children in this folder.

    Another advantage of the folder hierarchy is that if NSX segments are created with the same name in the default space, project, and VPC, these segments are easier to find in the **Inventory** pane of the vSphere Client UI. The reason is that the NSX port groups corresponding to segments in Project and subnets in the VPC are grouped under their respective project and VPC folders.

    NSX port groups corresponding to segments in the default space are not organized in vCenter folders. These NSX port groups are placed under the vSphere Distributed Switch (VDS) object.

    The lifecycle of the project and VPC folders is managed by NSX. vSphere users cannot rename, move, or delete the project or VPC folders because these folders are owned by NSX. When a project or VPC is deleted in NSX, the corresponding folder is deleted from vCenter. On similar lines, when projects or VPCs are renamed in NSX, the corresponding folder names change in vCenter.

    In vCenter, the following actions are permitted on folders and port groups that are created by NSX:

    - Assign permissions to vSphere users or groups.
      For example, by associating the **Assign Network** privilege to users or groups for a folder, a vSphere Admin can restrict or limit the users that can connect workload VMs to the port groups, which belong to NSX projects or VPCs.

- Add/enable/disable alarms.
- Assign or remove vSphere tags.
- Edit NSX port groups by using the NSX Manager UI.

After the project is saved, you can turn off this toggle, if required. This action does not impact the workloads that are connected to the project segments or NSX VPC subnets. Only the placement of the NSX port groups change in vCenter. That is, when this toggle is turned off, the project and VPC folders are deleted, and the NSX port groups are moved under the **VDS** object in vCenter.

12. In the **Network Spans** field, click the Set link and select the spans for this project where project's VPC networks will be deployed.
13. Use the **Activate Default Distributed Firewall Rules** toggle to turn on or turn off the default distributed firewall rules for this project.

    The default DFW rules allow communication between workload VMs within the project, including communication with the DHCP server. All other communication is blocked.

    This toggle is editable only when you apply an appropriate security license in your NSX deployment that entitles the system to the distributed firewall security feature.. This setting only turns on/off the default distributed firewall rules for the project. It does not turn off the distributed firewall in the project.

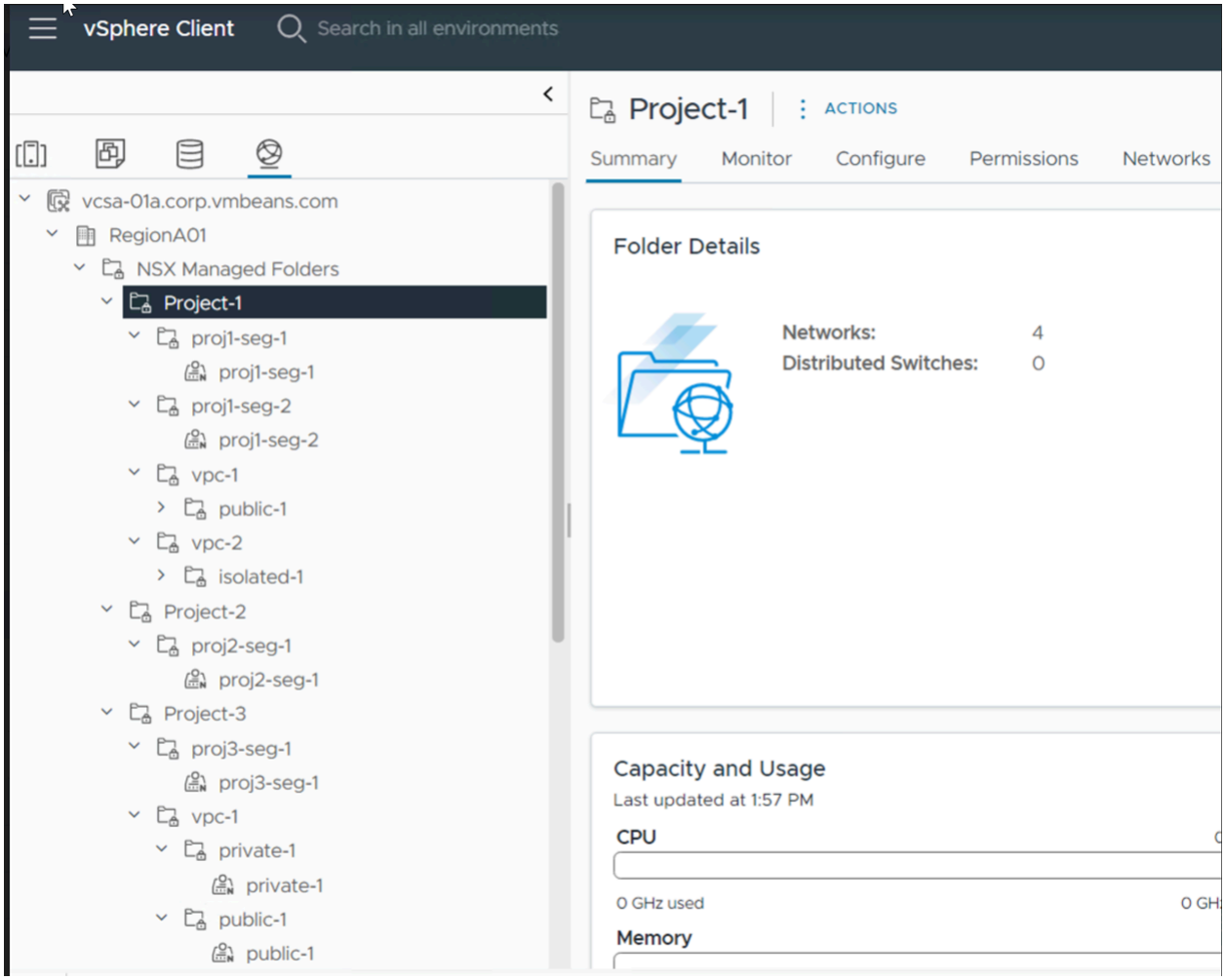
    For instance, if a distributed firewall service is activated in your NSX platform, you can still deactivate the default firewall rules of the project. In this case, the system-wide default distributed firewall rules that are configured in the default space will be applied to the project.

    For an explanation of the default state of the **Activate Default Distributed Firewall Rules** toggle in the project under various scenarios, see NSX Project Default Firewall Rules: Licensing Considerations.
14. In the **Default ID Suffix** field, specify a suffix if you want to suffix the project's default object IDs.
15. In the **Description** field, optionally enter a description for the project.
16. In the **Tags** field, optionally enter any tags for this project.
17. Click **Save**.

If you have turned on the **Organize NSX Portgroups in vCenter folders** toggle for the project, do these steps:

1. Check the status of the project. When folders are created successfully in vCenter, the status is a `Success`. This status represents the overall or consolidated status of all the folders that were created by NSX in vCenter when this project was created.
   If the overall status next to the toggle is displayed as `Failed`, check the error details to learn the reason for the failure.
   Examples of failure situations:
   - If the vCenter server is down when the folders are being realized, the folder creation can fail. NSX will not be able to communicate with vCenter. In this case, the overall folder status for the project is `Failed`. NSX will attempt to create the folders again at predefined intervals until the status changes to `Success`.
   - If the `cm-inventory` service on the NSX Manager node is down, the folder creation can fail.
     NSX Manager uses this service to retrieve information about the VDS or hosts dynamically from vCenter. When the group members of this service are down, an alarm belonging to the **Clustering** feature is displayed in NSX Manager.
2. Log in to the vSphere Client, open the **Inventory** pane, and verify that the project and VDS folders are created.
   For example, assume that you have added two projects named Project-1, Project-2, and Project-3 in your NSX deployment, and the **Organize NSX Portgroups in vCenter folders** toggle is turned on for these projects. The following screen capture shows that a root folder called `NSX Managed Folders` is created inside the **Datacenter** object. The Project folders and folders are created inside the root folder.
   Each of these project folders will have folders inside them for each of their VPCs and Segments. Each VPC will then have sub-folders for each subnet. Inside Segment and Subnet folders users can find all the NSX port groups mapped to this Segment/Subnet.

   Note:
   A single NSX Segment or Subnet creates one NSX port group per VDS, all with the same name. All those port groups are regrouped under the folder regrouping the Segment or Subnet.



Later, when you add segments in the projects, or add NSX VPCs in the projects, or add subnets in the VPCs, they get organized inside folders in vCenter.

Example: Organizing NSX Port Groups in vCenter Folders

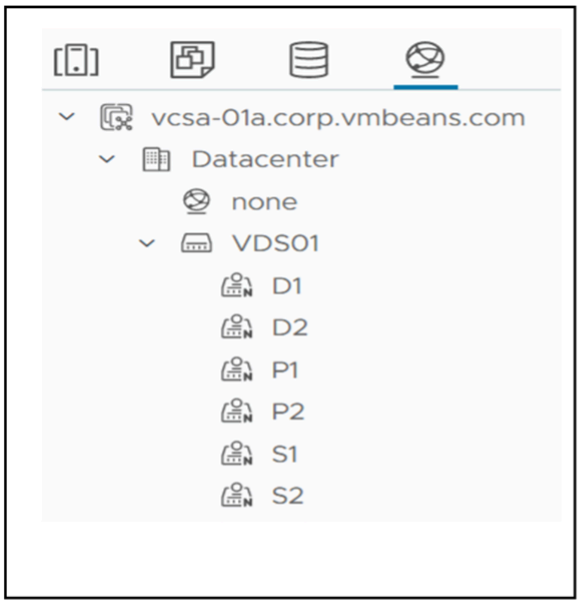
While configuring a project, you can turn on the **Organize NSX Portgroups in vCenter folders** option to organize NSX port groups corresponding to project segments and NSX VPC subnets inside folders on vCenter.

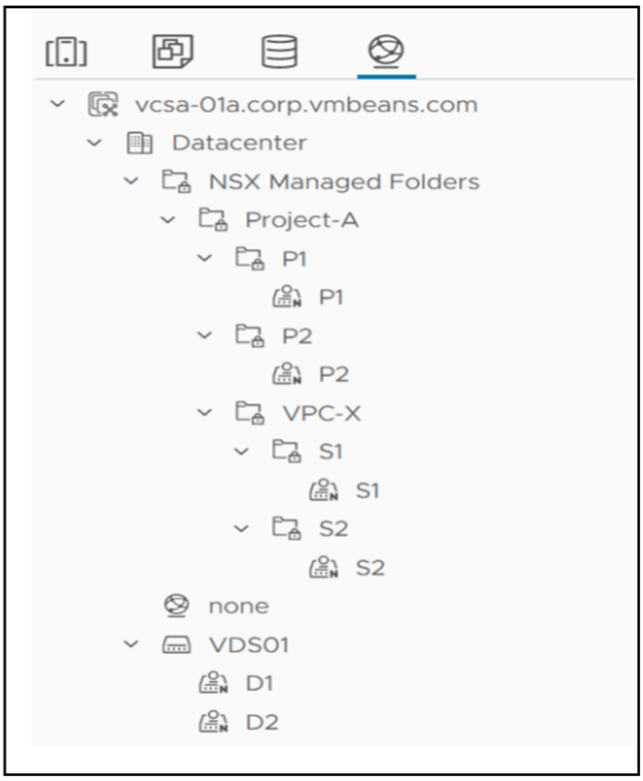
Example:

Assume that your NSX deployment is configured as follows:

- Hosts are attached to a single vSphere Distributed Switch (VDS)
- Segments D1 and D2 are created in the default space
- Segments P1 and P2 are created in Project A
- VPC X is created in Project A
- Subnets S1 and S2 are created in VPC X

In the following figure, the diagram on the left shows the flat organization of the NSX port groups in vCenter when the option is turned off. The diagram on the right shows the hierarchical organization of NSX port groups inside the project and VPC folders when the option is turned on.





The left diagram shows that the NSX port groups for the segments in the default space, segments in the project, and subnets in the NSX VPC are organized in a flat hierarchy under the **VDS** object.

The right diagram shows that a hierarchical folder structure is created in vCenter, as follows:

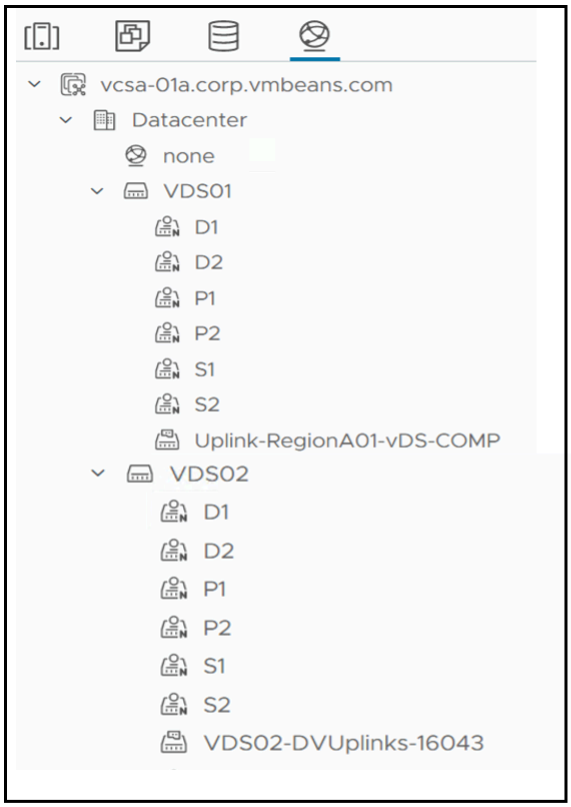
- A root folder named `NSX Managed Folders` is created under the **Datacenter** object.
- *`Project-A`* folder is created inside the root folder.
- A P1 and a P2 segment folder, corresponding to each project segment, is created in the *`Project-A`* folder
- NSX port groups P1 and P2 corresponding to the segments in the project are organized inside the *`Project-A`* segment folder.
- *`VPC-X`* folder is placed inside the *`Project-A`* folder.
- NSX port groups S1 and S2 corresponding to the subnets in the VPC are organized inside the *`VPC-X`* folder.
- An S1 and an S2 subnet folder, corresponding to each VPC subnet, is created in the *`VPC-X`* folder.
- NSX port groups S1 and S2 corresponding to the subnets in the VPC are organized inside the *`VPC-X`* subnet folders.

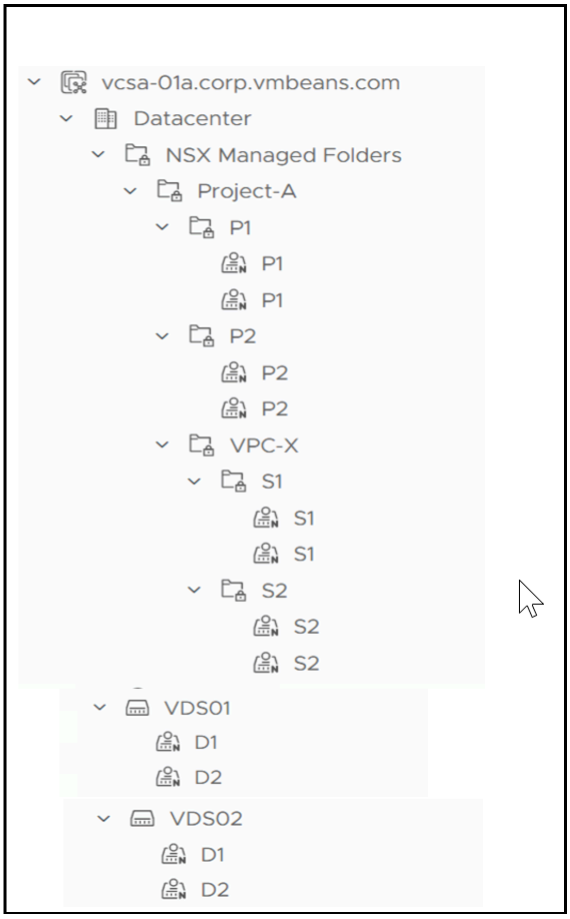
Note:

- The segment and subnet folders are created only for the purpose of grouping the NSX port groups mapped to the same segment or subnet in a multi-VDS deployment. A multi-VDS example is presented below.
- The VDS object continues to remain inside the Datacenter object in VMware vCenter.
- NSX port groups D1 and D2 corresponding to the segments in the default space are not organized in folders. These port groups continue to be placed under the **VDS** object.

Now let us assume that the hosts are connected to multiple VDSs. Below, the figure on the left presents a scenario where the port groups are not organized in folders. A port group for each segment and subnet is present on both VDSs.

The figure on the right presents a scenario where the port-groups are organized in the hierarchical folder structure managed by NSX. In this case, we can see that each segment and subnet folder contains two port groups, mapped to the corresponding NSX segment or VPC subnet, which are created in each of the two VDSs





For more documentation about organizing NSX port groups in vCenter folders, see the *Procedure* and *What to do next* sections in Add an NSX Project.

NSX Project Default Firewall Rules: Licensing Considerations

This topic explains the default state of the **Activate Default Distributed Firewall Rules** toggle in the project under various scenarios.

In the following table, the term "base license" refers to the VMware Cloud Foundation (VCF) license.

For more information about how to turn on or turn off the default distributed firewall rules for a project, see Add an NSX Project.

Sr. No.	Scenario	Default State of the Toggle	Notes
1	You are a new NSX customer and have applied a base license that entitles the system to the NSX networking features, but not distributed firewall security.	Off	This toggle is not editable because the current applied license does not support the configuration of distributed firewall security rules. You need to apply the appropriate security license in the system, and then turn on this toggle to activate the default distributed firewall rules in the project.
2	You are a new NSX customer. On day 0, you have applied a base license that entitles the system to NSX networking features. You have also applied an appropriate security license that entitles the system to distributed firewall security.	On	The default distributed firewall rules are activated for the project. You can turn it off in the project if required.
3	You are a new NSX customer. On day 0, you applied only the base license that entitles the system to the NSX networking features, but not distributed firewall security. You have added some projects in the system, let us say, projects A and B. Later, during day 2 operations, you applied an appropriate security license that entitles the system to distributed firewall security. Now, you added user-defined distributed firewall rules in the existing projects A and B, and also created new projects in the system, let us say, projects C and D.	Off: for pre-existing projects in the system On: for new projects in the system	In this scenario, the term "pre-existing projects" refers to projects that existed in the system before the security license was applied on day 2. In this scenario, they refer to projects A and B. The term "new projects" refers to projects that are added in the system after the security license was applied on day 2. In this scenario, they refer to projects C and D. For pre-existing projects A and B, the system behavior is as follows: This toggle will be in the Off state, by default. The user-defined DFW rules are effective in projects A and B. If you want to activate the default distributed firewall rules in these projects, open these projects in the edit mode, and turn on this toggle manually. However, when turned on, it might impact the behavior of east-west traffic in projects A and B. For new projects C and D, the system behavior is as follows: This toggle will be in the On state by default. That is, for projects C and D, the default distributed firewall rules are activated, by default. If required, you can turn it off so that only the user-defined distributed firewall rules are effective in these projects.

Sr. No.	Scenario	Default State of the Toggle	Notes
4	You are an existing NSX customer with a legacy NSX license that entitles your system to a full DFW access. After the grace period for the legacy license ends, you apply a base license that entitles your system to NSX networking features, and also apply a security license that entitles your system to distributed firewall security.	On	The default distributed firewall rules and user-defined distributed firewall rules continue to run in existing projects that you created before changing to the new license. There is no change in the system behavior. For all new projects that you add after changing the license, this toggle is turned on, by default. You can optionally turn it off, if required.
5	You are an existing NSX customer with a legacy NSX license that entitles your system to a full DFW access. You have added two projects in the system, let us say, projects A and B. After the grace period for the legacy license ends, you apply the base license that entitles your system to only the NSX networking features. A new security license is not applied. Now, you have created two new projects in the system, let us say, projects C and D.	On: for pre-existing projects Off: for new projects	In this scenario, the term "pre-existing projects" refers to projects that were added in the system when the legacy NSX license was valid. In this scenario, they refer to projects A and B. The term "new projects" refers to projects that are added in the system after the base license was applied. In this scenario, they refer to projects C and D. For pre-existing projects A and B, the system behavior is as follows: This toggle is in the On state, by default. You can turn it off, if needed. But, this action is not reversible. That is, you cannot activate the default E-W firewall rules again in projects A and B. The default distributed firewall rules and user-defined distributed firewall rules continue to run in projects A and B. But, you cannot edit these rules. Neither, you can add new distributed firewall rules. But, you can delete the existing user-defined firewall rules. To have a full access to the distributed firewall rules, you need to apply an appropriate security license. For new projects C and D, the system behavior is as follows: This toggle is in the Off state, by default. You cannot turn it on because the current applied license does not entitle the system to the distributed firewall feature.

Sr. No.	Scenario	Default State of the Toggle	Notes
6	You are a new NSX customer and your system has entered an Evaluation mode, which is valid for 60 days.	Off	During the evaluation period of a new NSX deployment, the system is entitled to only the networking features. The security features are not entitled.

For information related to setting the Activate Default Distributed Firewall Rules toggle, see Add an NSX Project.

Add User Role Assignments for an NSX Project

After adding a project, you can assign roles to users in the project. These users can then start configuring networking or security objects in the project.

In a project, you can assign the following roles to users:

- Project Admin
- Security Admin
- Network Admin
- Security Operator
- Network Operator

A Project Admin has full access to all the networking and security objects inside a project. The other project-specific user roles have limited access to objects inside the project as determined by the permissions of those roles.

> **Note:**
> By default, only an Enterprise Admin can add user role assignments in projects. A Project Admin does not have permissions to add user role assignments in projects, unless an Enterprise Admin configures the Project Admin role to do user role assignments.
>
> If the Project Admin is allowed to do user role assignments, it might be perceived as introducing security risks in some NSX environments because it allows a Project Admin to configure role assignments for any user in the system. Therefore, the default behavior is to allow only an Enterprise Admin to add user role assignments in projects.
>
> An Enterprise Admin can do the following steps to grant permissions to the Project Admin role for adding user role assignments:
>
> 1. In the **Default** view, navigate to **System** > **User Management** > **Roles**.
> 2. Next to the **Project Admin** role, click
>
>   
>
>   and then click **Allow Role Assignments**.

For user authentication and authorization, NSX multi-tenancy supports the following identity sources:

- Local users (for example, guestuser1, guestuser2)
- Workspace ONE Access
- Lightweight Directory Access Protocol (LDAP)
- OpenID Connect (See the note after this bulleted list)
- Principal Identity (using certificate or Jason Web Token)

> **Note:**
> OpenID Connect identity source is supported starting in NSX 4.1.2. However, this identity source is supported for user authentication only when you are adding user role assignments from the **User Management** page of the system. If you are adding user role assignments from the **Manage Projects** page, this identity source is currently not supported.

To add user role assignments in projects, the following two methods are available:

Method 1: Add User Role Assignments from the User Management Page	The **User Management** page is available only to the Enterprise Admin. A Project Admin cannot use this page even if an Enterprise Admin has granted permissions to the Project Admin role to do user role assignments in projects.
Method 2: Add User Role Assignments from the Manage Projects Page	The **Manage Projects** page is available to both the Enterprise Admin and the Project Admin. However, a Project Admin can add user roles from this page only when an Enterprise Admin has granted permissions to the Project Admin role to do user role assignments.

Add Role Assignments for an NSX Project from the User Management Page

Use the instructions in this documentation to add user role assignments for a project from the **User Management** page.

User accounts are created. For example:

- Local user accounts are added in the system and they are activated.
- NSX Manager is configured to authenticate users from any of these identity management providers:
  - Workspace ONE Access (vIDM)
  - LDAP-based directory service, for example, Active Directory.
  - OpenID Connect (starting in NSX 4.1.2)

The **User Management** page is available only to the Enterprise Admin. A Project Admin cannot use this page even if an Enterprise Admin has granted permissions to the Project Admin role to do user role assignments in projects.

The following procedure explains the steps for adding role assignments in projects for local user accounts and LDAP user accounts. The steps to add role assignments for vIDM and OpenID Connect user accounts are almost similar, and therefore not covered in this procedure.

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Navigate to **System** > **User Management**.
   The **User Role Assignment** tab is displayed.
3. To add role assignments in projects for a local user account, do these steps:
   a) Next to the local user account name, click the
   
   icon, and then click **Edit**.
   b) Click the link under the **Roles** column.
   The **Set Roles/Scope** dialog box opens.
   c) Click **Add Role**, and then select any one of these roles to assign to the local user:
      - Project Admin
      - Network Admin
      - Security Admin
      - Network Operator
      - Security Operator
   d) Under the **Scope** column, click **Set**.

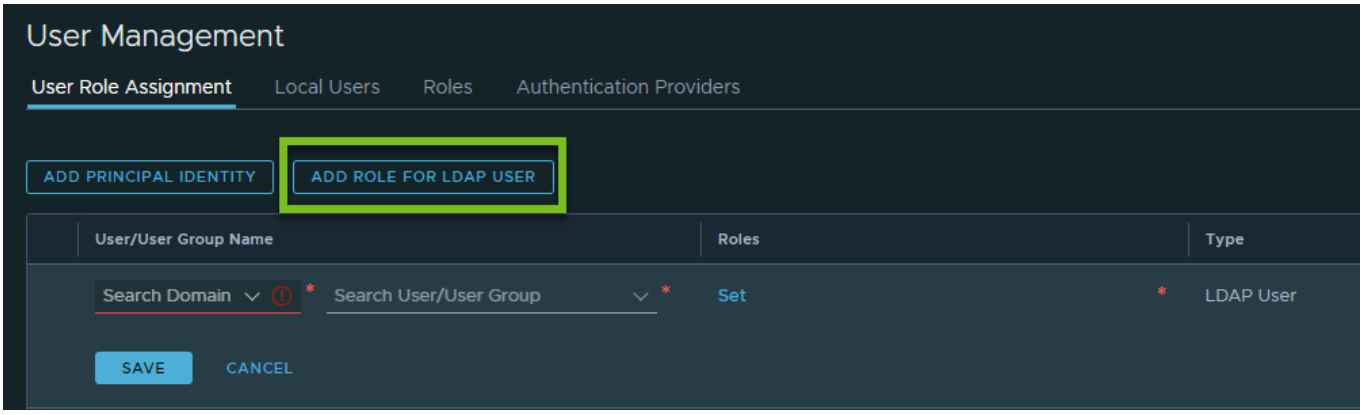
      A project can optionally contain one or more NSX VPCs. If you have assigned the Project Admin role to the user, the scope can be set to one or more projects. If you have assigned any of the following roles to the user, the scope can be set to all projects and VPCs, or to selected projects and VPCs:

      - Network Admin
      - Security Admin
      - Network Operator
      - Security Operator

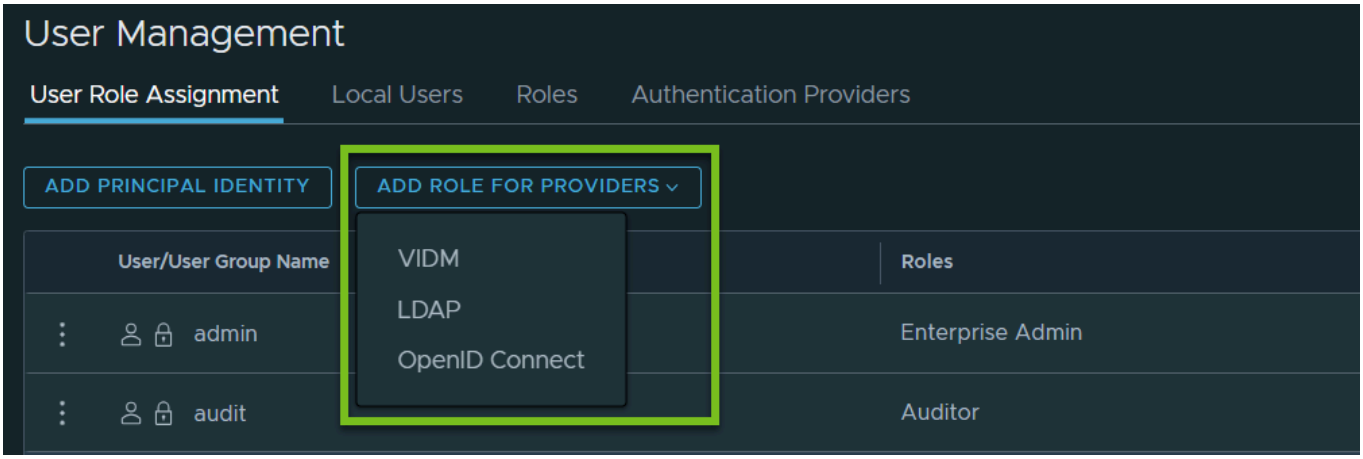
e) Click **Add**, and then click **Apply**.
f) Click **Apply** again to close the **Set Roles/Scope** dialog box.
g) Click **Save** to save the role assignment.

4. To add role assignments in projects for an LDAP user account, do these steps:
   a) Ensure that you are in the **User Role Assignment** tab.
   b) Click **Add Role for LDAP User**.

   For example:



   If you have configured NSX Manager to authenticate users from all supported identity service providers (LDAP, vIDM, and OpenID Connect), the button caption is as shown in the following screen capture.



   c) Select a domain from the drop-down menu.
   d) Enter the first few characters of the user or group name.

      System displays a list of matching user or group names. Select a user or a group name from the list.
   e) Under the **Roles** column, click **Set**.

      The **Set Roles/Scope** dialog box opens.
   f) Click **Add Role** and follow the same process, as explained in steps 3(c) though 3(e), to assign roles to the LDAP user in the project.

Project users can now log in to NSX Manager with their login credentials. Observe that the **Project** drop-down menu shows only those project names that the logged-in user has access to.

Add Role Assignments for an NSX Project from the Manage Projects Page

Use the instructions in this documentation to add user role assignments for a project from the **Manage Projects** page.

User accounts are created. For example:

- Local user accounts are added in the system and they are activated.

- NSX Manager is configured to authenticate users from any of these identity management providers:
  - Workspace ONE Access (vIDM)
  - LDAP-based directory service, for example, Active Directory.

The **Manage Projects** page is available to both the Enterprise Admin and the Project Admin. However, a Project Admin can add user roles from this page only when an Enterprise Admin has granted permissions to the Project Admin role to do user role assignments.

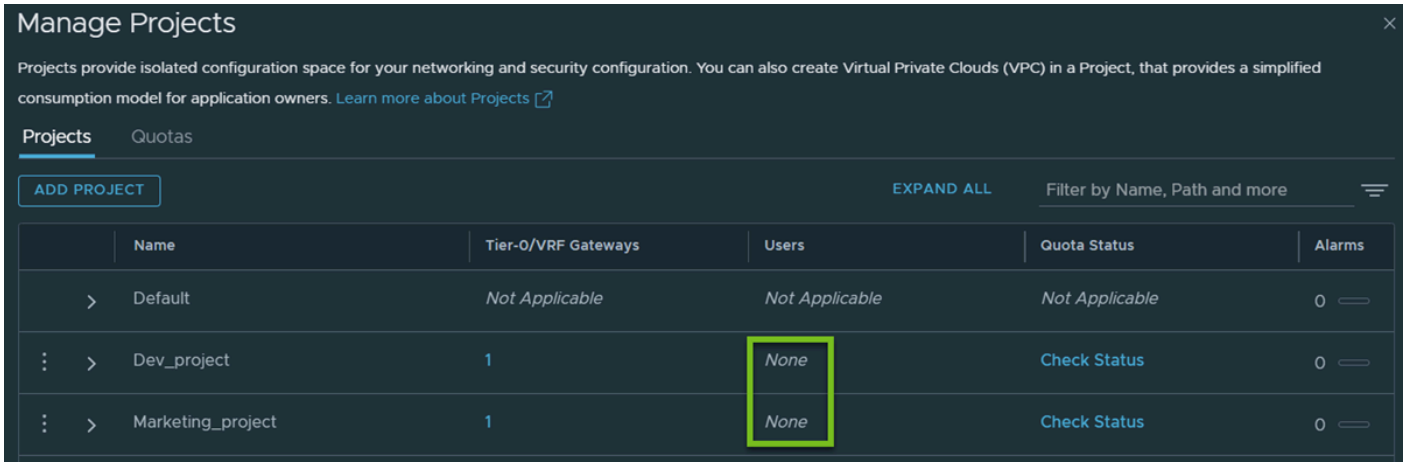
If a project contains NSX VPCs, you can use the **Manage Projects** page to assign roles to users either in the scope of a project or selected NSX VPCs within the project.

In the project scope, you can assign these roles to a user: Project Admin, Network Admin, Security Admin, Network Operator, and Security Operator.

In the VPC scope, you can assign these roles to a user: VPC Admin, Network Admin, Security Admin, Network Operator, and Security Operator.

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Click **Default**, and then click **Manage**.

   For example, the following screen capture shows two projects on the **Manage Projects** page: Dev_project and Marketing_project. No users are currently set for these projects.



3. Next to the project name, click
   
   and then click **Edit**.
4. Under the **Users** column, click **Set**.
5. Click **Add Role Assignment**, and then do the following steps:
   a) Select any one of these options:
      - Local User
      - LDAP User
      - vIDM User
   b) Enter a user name or a group name.

      Note:
      Ensure that the user or group name is entered accurately. The system does not display the list of matching users or groups when you type the first few characters in the text box. This behavior is a known limitation.
   c) Under the **Role** column, click **Set**, and then click **Add Role**.
   d) From the **Select Role** drop-down menu, select a role.
      For example, select the **Project Admin** role. Observe that the scope of this role is to set to `Project`.

      If you select any of the following roles for assigning it to the user, system provides you the option to assign the role in the scope of the project or selected VPCs within the project:

- Network Admin
- Security Admin
- Network Operator
- Security Operator

If you select the **VPC Admin** role, the scope is automatically set to VPCs.

e) Click **Add**, and then click **Apply**.
f) Click **Save** to save the role assignment.

6. Optional: To add more user role assignments in the project, repeat step 5.

Project users can now log in to NSX Manager with their login credentials. Observe that the **Project** drop-down menu shows only those project names that the logged-in user has access to.

Add Quota to an NSX Project

After adding a project, you can add one or multiple quotas to define a maximum limit for the number of objects of a specific type that users can create in the project.

You must be assigned the Enterprise Admin role.

From the **Manage Projects** page in the default space, you can set quota for the following:

- Quota for objects in the project and across all its NSX VPCs.
- Quota for objects only in the project.
- Quota for objects only in the NSX VPCs across the project.

When you specify a quota for an object type in a project, it implies that the total number of user-created objects of that type in the project cannot exceed that limit.

Example: Assume that you have created project-1 and added four NSX VPCs inside this project. You have set a quota of 1000 for the group object in project and all its VPCs. This means that the total number of user-created groups in project-1 and across all the four NSX VPCs in project-1 cannot exceed 1000.

If required, you can set a separate quota of say 600 for the group object in project-1. And set another quota of say 400 for the group object across all NSX VPCs under project-1. This means that the number of user-created groups in project-1 cannot exceed 600, and the total number of user-created groups across all the four NSX VPCs in project-1 cannot exceed 400.

In addition, you can go to the project view, and set a specific quota for the group object in each of the four NSX VPCs.

This documentation focusses on adding quota for objects in projects and NSX VPCs from the default space. To learn about adding quota for objects in an NSX VPC from the project view, see Add Quota to an NSX VPC.

If you do not apply a quota to a project, system-defined maximum limits apply to the number of objects that users can create in that project.

> **Important:**
> The maximum limits that you define in the quota apply only to the creation of objects of a specific type in a project. Objects created in a project are owned by that project. Quotas do not put any constraints on the number of objects that an Enterprise Admin can share with a project through resource shares. For example, if a project has a quota, which defines a limit of 10 groups, it does not restrict the Enterprise Admin from sharing 20 groups with this project through a resource share.
>
> To learn about resource shares, see Sharing NSX Resources.

NSX has a validated maximum scale for a deployment. To know the maximum limits for NSX objects, see the VMware Configuration Maximums portal. The object limits mentioned on this portal apply to the entire system, meaning that the limits apply to objects that are both in and out of the projects. For example, the portal mentions that the maximum limit for segments is 10,000. This limit refers to the maximum number of segments in the system. Unless specified on the portal, there is no separate maximum limit for objects under each project in an NSX deployment.

By limiting the number of objects that users can create in a project, an Enterprise Admin can govern the project resource consumption. For example:

- Ensure that one project does not take too much resources.
- Prevent project users from creating some objects by setting the quota to 0.
- Offer a different level of service between projects.

System allows you to reduce the quota limits of an object even below the current consumption.

For example, assume that for a project, a quota of 20 segments is defined, and 15 segments have been created. Later, the Enterprise Admin decides to reduce the quota to 10 segments for this project. In this case, the consumption of segments has reached 150% of quota. However, existing segments remain in the project until they are deleted. Project users cannot create new segments inside the project until the segment count goes below 10.

Scenario: Set a quota for the same object more than once in the same project	System allows you to set a quota for the same object more than once and apply it to the same project. However, in such a situation, the quota with the lower limit is enforced for that object in the project. For example, consider the following sequence of events: 1. User A creates quota-1 for the group object, and specifies the limit as 100. 2. Quota 1 is applied to project 1. 3. Later, user A creates quota 2 for the group object and specifies the limit as 50. 4. Quota 2 is also applied to project 1. In this case, system enforces the limit of 50 on the group object in project 1.

The following procedure explains the workflow of adding quota for objects in a project from the **Manage Projects** page in the default space.

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Click **Default**, and then click **Manage**.
3. Click the **Quotas** tab, and then click **Add Quota**.
4. Configure the limits for one or multiple objects in the quota.
   a) Enter a name for the quota.
   b) In the **Limits** column, click **Set**.
      The **View Limits** window opens. The objects in this window are organized under several tabs.

      Use the **Networking**, **Security**, and **Inventory** tabs to add limits for:

      - Objects in the project and across all its NSX VPCs.
      - Objects in the project only.

      You can use the **VPCs** tab to add limits for objects only in the NSX VPCs across the project. If you want to add limits for objects inside each NSX VPC of the project, switch to the project view, navigate to **VPCs** > **Quota**, and set the limits over there.
   c) Click the tab that contains the objects that you want to set limits on.
   d) Click **Add Limit**.
   e) From the **Objects** drop-down menu, select an object.

      For example, let us click the **Inventory** tab, and then select the **Group (Project only)** object.
   f) In the **Limit** box, enter an integer, and then click **Save**.
      For example, let us enter `10`.

      Note:
      To prohibit users from creating an object in the project, set the limit to `0` for that object.
   g) Optional: Add limits for other objects, as required.
   h) After adding limits for all objects, click **Apply**.
5. From the **Apply To** drop-down menu, select a project to apply this quota.

   A quota can be applied to multiple projects.

6. Click **Save**.

   Quota

   In the example that we took earlier in this procedure, we defined a limit of 10 for the group object in the project only. Let us assume that this limit is applied to two projects: Project-1 and Project-2.

   This quota puts a limit on creating a maximum of 10 groups in both the projects. After this limit is reached, you cannot add more groups in your project. However, you can edit the limit later or delete it, if required.

   When the number of user-created objects in a project reaches the limit that is set for an object, an alarm is triggered in the system when you try to create and save another object of the same type.

   The alarm due to the `Creation Count Limit Reached` event is visible to both the Enterprise Admin and the Project Admin. For example, to view the alarm in the **Default** view, the Enterprise Admin can navigate to **Home** > **Alarms** and check the details on the **Alarms** page.

   Note:
   For the `Creation Count Limit Reached` event, the threshold value is currently not configurable. System-defined threshold value is used, by default.

After a quota is applied to the project, you can start monitoring the quota status for all the managed projects.

To know more about monitoring quota status, see Monitor Quota Status of an NSX Project.

Monitor Quota Status of an NSX Project

Project users can periodically check the quotas status of their projects to know the percentage of quota that is used, view any alarms that might have been raised, and decide whether any troubleshooting actions are required to resolve those alarms.

1. From your browser, log in to an NSX Manager at `https://`*`nsx-manager-ip-address`*.
2. Click **Default**, and then click **Manage**.
3. Ensure that you are in the **Projects** tab.
4. In the **Quota Status** column, click **Check Status**.
   - If the status changes to `OK`, it means that the number of user-created objects in the project is within the limit.
   - If the status changes to `Error`, it means that the quota has reached (that is, 100% of quota is used).
   - If the status changes to `Not Set`, it means that no quota is defined for the project.

   Use the information that is displayed in the **Alarms** column to know whether any alarms are raised. Alternatively, you can navigate to **Home** > **Alarms** and view the active alarms, if any.

   When the quota limit is reached, a `Creation Count Limit Reached` event is displayed on the **Alarms** page.

   The alarm due to this event is visible to both the Enterprise Admin and the Project Admin. View the details of this event to learn about the recommended actions.

   Note:
   For the `Creation Count Limit Reached` event, the threshold value is currently not configurable. System-defined threshold value is used, by default.

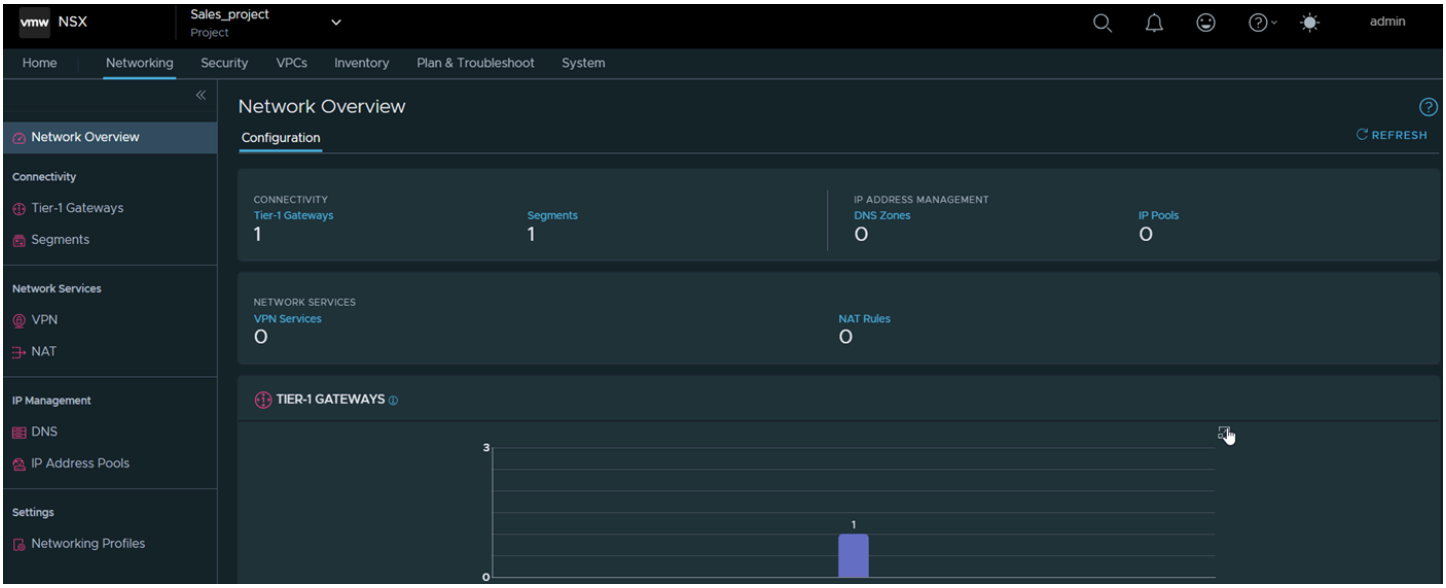
5. To know more about the quota details of a project, click the value (`OK` or `Error`) in the **Quota Status** column.

   The **Quota Status** window displays quota details, such as the limits set for the various object types in the quota, the quota name, the percentage of quota used, and other details.

Networking Within an NSX Project

A subset of NSX networking features are currently supported in projects.

The following screen capture shows the features that are available on the **Networking** page in a project.



Tier-0 gateways and edge clusters in the system are allocated to projects. These objects are owned by the Enterprise Admin in the default space.

Managing Tier-1 Gateways in a Project

- Tier-1 gateways and segments inside a project can connect to tier-0 gateways that are assigned to the project.
- Tier-1 gateways and DHCP are provisioned on edge clusters that are assigned to the project.
- Tier-1 gateways that are provisioned on the edge clusters will benefit from the configuration that an Enterprise Admin has done on the edge cluster, for example, configuration of the failure domain.
- Tier-1 service interfaces can connect to networks that are created within the projects and to networks that are shared with the projects.
- Features such as NAT, Gateway Firewall, and VPN can be configured on the tier-1 gateways that are created inside a project. Tier-1 gateways that are configured in the default space cannot be allocated to projects.

Managing Networks in a Project

- Currently, overlay segments are supported in projects.
- Segments inside a project are realized in the project default transport zone. The project default transport zone is automatically assigned when a project is created and it is the same as the default transport zone of the system.
- DHCP profiles can be created on edge clusters that are allocated to the project.
- Segments can use the DHCP profiles that are shared with the project.
- To provide distributed security to VMs inside a project, these VMs must be connected to the segments inside a project.
- Private IPv4 blocks that are created in a project can be used for adding private subnets in the NSX VPCs within the project.

VPN in an NSX Project

In a multi-tenant NSX environment, a Project Admin can configure L2 VPN and IPSec VPN service on the tier-1 gateways of a project.

To configure VPN on the tier-1 gateway of a project, an edge cluster must be assigned to the project.

The following functionality is supported for VPN service in projects:

- On each tier-1 gateway of a project, you can configure only one IPSec service and one L2 VPN service.
- For configuring route-based IPSec VPN, static-routes are supported. Dynamic routing with virtual tunnel interface (VTI) using BGP is not supported on the tier-1 gateway of a project.
- The workflow for configuring IPSec VPN and L2 VPN service in a project is the same as it is for configuring these services in the default space. For more information, see Adding NSX VPN Services.

- Both pre-shared key authentication and certificate-based authentication are supported for configuring IPSec VPN sessions.
  - An Enterprise Admin can share service certificates and certificate revocation lists (CRLs) with the projects, if required. The Project Admin can use the shared certificates to authenticate IPSec endpoints when certificate-based authentication is configured for IPSec VPN sessions.
  - Alternatively, a Project Admin can create, update, or manage the service certificates and CRLs in the project view, and use them to configure IPSec VPN sessions in the project.
- System-created VPN profiles are shared by default with all the projects in the system. A Project Admin can either use the following default VPN profiles or create new profiles to configure VPN services in the project.
  - IKE profile
  - IPSec tunnel profile
  - DPD profile
  - L2 VPN tunnel profile
  - Compliance suite related profiles

  To learn about adding IKE, IPSec, and DPD profiles, see Adding Profiles. If an Enterprise Admin has created VPN profiles in the default space, they can be shared with specific projects, if required. Shared profiles can be consumed in a read-only mode in the projects.
- An Enterprise Admin can restrict the number of VPN related objects that can be created under a project by defining quotas for various object types. For example, quotas can be defined on these VPN objects:
  - IPSec VPN sessions
  - L2 VPN sessions
  - Local endpoints
  - IPSec VPN services
  - L2 VPN services
  - VPN profiles

  This list is not exhaustive. For the full list of objects, go to **Quotas** tab on the NSX Manager UI.
- Monitoring status of VPN services, VPN sessions, and other VPN statistics is supported in the project view.
- Configuration of L2 VPN and IPSec VPN tunnels between tier-1 gateways of two different projects in the same NSX deployment is supported.
- Configuration of L2 VPN and IPSec VPN tunnels between tier-1 gateways of the same project is supported.

The following points must be kept in mind when configuring VPN service on the tier-1 gateways of a project:

- IPSec local endpoints are realized as loopback IPs on the tier-1 gateway of the project. Local endpoint IP must be unique across the data center. The endpoint IPs are advertised to the tier-0 gateway that is associated with the project, and then pushed to the upstream networks through BGP.
- To allow IPSec packets (IKE UDP port 500 and 4500, ESP) towards tier-1 gateway, the Enterprise Admin must define firewall rules on the tier-0 gateway that is associated with the project. The firewall rules on the tier-0 gateway are not automatically created by the system. However, firewall rules on the tier-1 gateway of the project are auto-created to allow IKE and ESP traffic between the endpoints in the IPSec VPN session.

Route Filtering in NSX Projects

In an NSX environment, multiple projects can be configured with the same tier-0 or a tier-0 VRF gateway. This implies that tier-1 gateways and NSX VPCs in the projects are connected to the same tier-0 or a tier-0 VRF gateway.

To avoid IP address conflicts between the projects, you can filter routes that are advertised out of the specified projects by all the tier-1 gateways or NSX. For instance, prefixes allow you to ensure that only the IP addresses from the external IP blocks are advertised out of the specified projects.

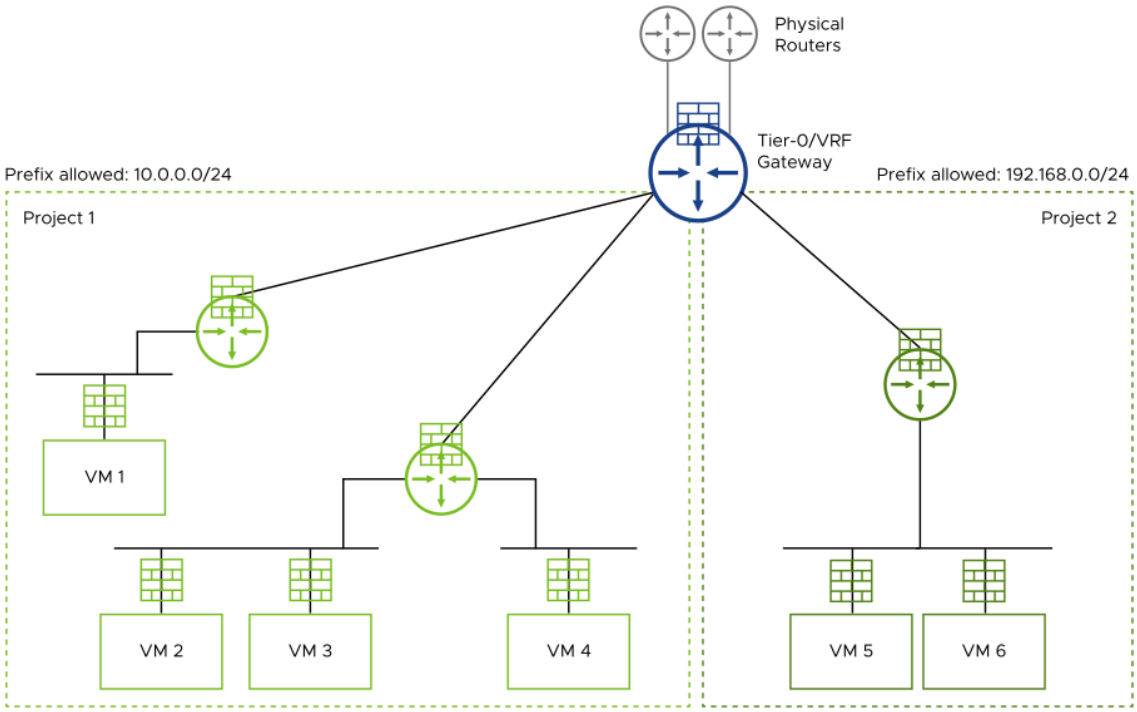
An Enterprise Admin can configure route filtering and it cannot be overwritten at the project level.

First, you need to create the prefix lists that you want to use. Then, in the project route filter, you can define the mapping of the projects to the prefix lists.

Project route filter is supported only using NSX APIs. The UI does not support this feature.

Example

The following diagram shows two projects, Project 1 and Project 2, which are connected to the same tier-0 gateway. Project 1 contains two tier-1 gateways, and Project 2 contains a single tier-1 gateway. For Project 1, the allowed prefix is 10.0.0.0/24, and for Project 2, the allowed prefix is 192.168.0.0/24.



Workflow for Creating a Project Route Filter Using APIs

1. Create a prefix list.
   For example, run the following NSX API:

```
PATCH https://<nsx-mgr>/policy/api/v1/infra/routing-config/prefix-lists/<prefix-list-id>
{
  "prefixes": [
      {
          "network":"172.20.0.0/16",
          "action":"PERMIT"
      },

  ]
}
```

   For more information about this API, see the *NSX API Guide*.
2. Read the prefix lists and verify the details.
   For example, run the following NSX APIs:

```
GET https://<nsx-mgr>/policy/api/v1/infra/routing-config/prefix-lists
GET https://<nsx-mgr>/policy/api/v1/infra/routing-config/prefix-lists/<prefix-list-id>
```

3. Create a project route filter.
   For example, run the following NSX API:

```
PATCH https://<nsx-mgr>/policy/api/v1/infra/routing-config/project-route-filters/<route-filter-id>
{
  "projects_list": ["/orgs/default/projects/project-1"],
  "match_prefix_list" : ["/infra/routing-config/prefix-lists/list-1", "/infra/routing-config/prefix-lists/
list-2"]
}
```

   This API creates a mapping of projects to prefix lists. Prefix lists are applied to all tier-1 gateways in the specified projects.

For more information about this API, see the *NSX API Guide*.

4. Read the project route filter and verify the details.
   For example, run the following NSX API:
   ```
   GET https://<nsx-mgr>/policy/api/v1/infra/routing-config/project-route-filters/<route-filter-id>
   ```
5. Optional: Retrieve a list of all project route filters.
   For example, run the following NSX API:
   ```
   GET https://<nsx-mgr>/policy/api/v1/infra/routing-config/project-route-filters
   ```
6. Optional: Delete a prefix list and a project route filter.
   For example, run the following NSX APIs:
   ```
   DELETE https://<nsx-mgr>/policy/api/v1/infra/routing-config/project-route-filters/<route-filter-id>
   DELETE https://<nsx-mgr>/policy/api/v1/infra/routing-config/prefix-lists/<prefix-list-id>
   ```

Groups in an NSX Project

NSX supports creation of groups when setting up multi-tenancy in your environment.

The **Groups** page in a project displays the following groups:

- Default group created by the system in the project.
- User-created groups that are owned by the project.

If the Enterprise Admin has shared groups with the project, you can optionally choose to view the shared groups.

If you have added NSX VPCs in the project, you can optionally choose to view the following groups inside a project:

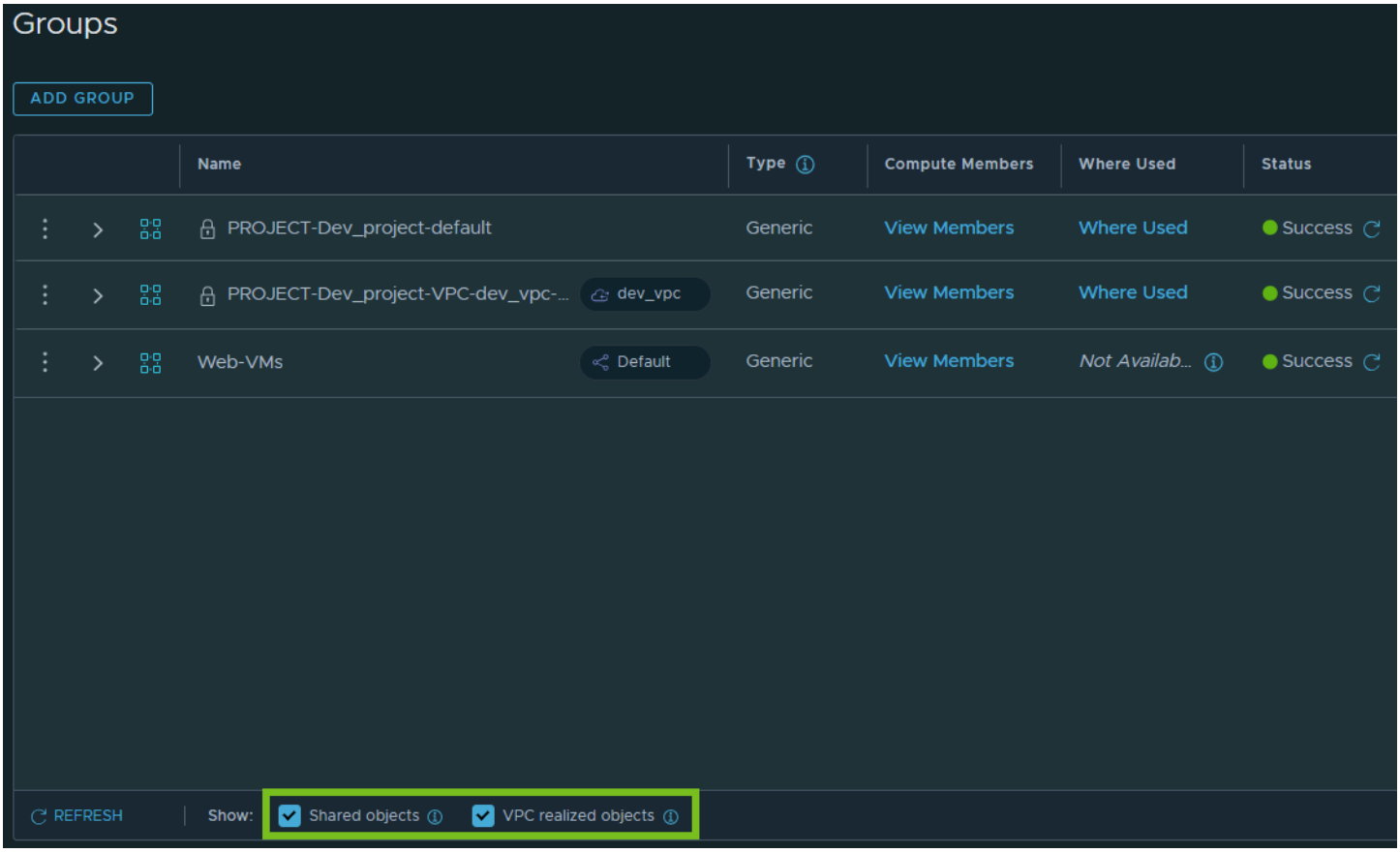
- Default groups created by the system in the NSX VPCs.
- User-created groups in the NSX VPCs.

By default, the groups that are shared with the project, and the groups that are inside the NSX VPCs are not displayed on the **Groups** page.

- To view shared groups, click the **Shared objects** check box.
- To view groups inside NSX VPCs, click the **VPC realized objects** check box.

You can find these check boxes at the bottom of the **Groups** page.

For example, see the following screen capture.



Observe that pill-shaped icons are displayed next to the group name to help you identify the following groups:

- Groups that are owned by the default space and shared with the project.
- Groups that are created in the NSX VPCs, both system-created and user-created VPC groups.

Default Group in a Project

NSX creates a default group of type **Generic** for every project that is added in your environment. This default group represents the project itself.

By default, the system adds all the segments that users have created inside a project to the project's default group. The VM interfaces (VIFs) that are attached to the segments in the project are members of the default group. If the VM is dual-homed, for example, when one interface is connected to a a segment in the default space, and the other interface is connected to a segment in the project, then the VIF on the segment in the default space will not be a member of the project's default group.

The default group helps you to restrict the scope of the firewall rules to a particular project. For example, when adding DFW policies in the default space, the Enterprise Admin can use the default group of the projects in the **Source**, **Destination**, or **Applied To** fields of the firewall rules. However, the firewall policies in the default space cannot consume groups other than the default group of the projects.

The following naming convention is used to identify the default group in a project:

`PROJECT-<Project_Name>-default`

`Project_Name` is replaced with actual value in your system.

User-Created Groups in a Project

Groups of type **Generic** and **IP Addresses Only** are supported in a project.

The following NSX objects are supported for adding statically to a group definition inside a project:

- Segments

- Segment Ports
- VIFs
- Virtual Machines
- Groups

On the **Members** tab of the **Set Members** dialog box, the system displays only those objects that are owned by the project. Objects that are shared with the project are not listed in this dialog box because shared objects cannot be added as members in a project group.

The following NSX objects are supported for adding to dynamic group membership criteria inside a project:

- Virtual Machine
- Segment
- Segment Port
- Group

The following points apply to groups that are created inside the **Default** view (default space) of your NSX environment:

- Dynamic group membership evaluates to all the VMs in the NSX system, including the VMs in the projects and in the NSX VPCs within the projects.
  For example, if a group in the default space includes VMs with the `Web` tag, the effective members of this group will include VMs with the `Web` tag in:
  - Projects
  - NSX VPCs within the projects
  - Outside the projects (default space)

  Note:
  Tags are metadata of the VMs in NSX. If a VM is attached to a segment in a project, both the Project Admin and the Enterprise Admin can assign tags to the VM. For example, let us say that in the default space, the Enterprise Admin has assigned a tag to a VM that is inside project 1. The Project Admin of project 1 can modify the tag that is assigned to this VM because the VM is seen in the project 1 inventory.
- For groups with static membership in the default space, you can add workload VMs that are connected to a project either by explicitly referring to the VMs (Members > Virtual Machines) or by adding the project default groups (`PROJECT-<Project_Name>-default`).
  Other objects created under a project cannot be added as static members to groups inside the default space.

Adding Groups to a Project

The UI workflow for adding a group in a project remains the same as it currently exists for adding groups in the **Default** view (default space) of your NSX deployment.

The only difference is that in the UI, you must first select a project from the **Project** drop-down menu on the application bar at the top, and then navigate to **Inventory** > **Groups** for adding groups in that selected project.

Firewall Policies in an NSX Project

When an NSX project is realized successfully, the system creates default gateway firewall and distributed firewall rules to govern the default behavior of the north-south traffic and east-west traffic for the workloads in the NSX project. For more details about firewall policies in an NSX project, see VMware vDefend Firewall documentation.

Monitoring and Troubleshooting within NSX Projects

NSX supports the monitoring of network traffic within Projects for debugging or troubleshooting purposes. You can also view alarms raised for a Project.

An Enterprise Admin can view alarms within the default space and across all projects while the Project Admin can only view project-specifc alarms.

Traceflow

An Enterprise Admin and the Project Admin can use the Traceflow tool to inspect the path of a packet. While an Enterprise administrator can run a trace across the default space and across all projects, the Project Admin can perform a trace only within a project. Traceflow can be run on VMs and ports that are a part of the same Local Manager site.

To run a trace across projects as an Enterprise Admin, ensure that you select **All Projects** from the project switcher menu. The UI workflow for performing a Traceflow in a project remains the same as it currently exists for the **Default** view (default space) of your NSX deployment.

For more information on Traceflow, see Perform a Traceflow.

To perform a trace as a Project Admin, navigate to **Plan & Troubleshoot** > **Traffic Analysis** > **Traceflow** > **Get Started** in your project space.

Specify the source and destination information according to the traffic type and click **Trace**. The output includes a table listing the observed packets. Details of entities that belong to the default project and encountered during the trace, are hidden from view in the Traceflow output for the project.

Port Mirroring

Only the Enterprise Admin can configure port mirroring from the default space for VMs in a project. This feature is currently unavailable to the Project Admin and only the Enterprise Admin can view the results in the default space.

To configure port mirroring for a project, create appropriate groups in the default space and add your tagged VMs from the project to these groups.

Navigate to **Plan & Troubleshoot** > **Port Mirroring**, select the session type, and set the groups as **Source** and **Destination**.

For more information on port mirroring, see Add a Port Mirroring Session.

IPFIX

Only the Enterprise Admin can configure IPFIX from the default space for VMs in a project. This feature is currently unavailable to the Project Admin and only the Enterprise Admin can view the results in the default space.

To configure IPFIX for a project, create appropriate groups in the default space and add your tagged VMs from the project to these groups.

Navigate to **Plan & Troubleshoot** > **IPFIX**, select the appropriate group under **Applied To** for your IPFIX profile.

For more information on IPFIX, see Network Monitoring.

Features Available for Consumption Under NSX Projects

In a multi-tenant NSX environment, you can configure a subset of NSX features under projects.

This provides flexibility to the tenants while allowing Enterprise Admin and other administrators of the default space to control the overall system life cycle and connectivity.

Table 1054: Features Available Under Projects

NSX Feature	Available Under Projects	Notes
System		
Edge Clusters	Assigned during project creation	During project creation, the Enterprise Admin assigns an edge cluster from the default space to a project.
System Life cycle	Managed by the Enterprise Admin	Platform-wide operations such as install, upgrade, and backup are managed by the Enterprise Admin.
Certificates	Yes	Certificate management tasks are available under projects only for service certificates.

NSX Feature	Available Under Projects	Notes
Networking		
Tier-0 or Tier-0 VRF Gateway	Assigned during project creation	During project creation, the Enterprise Admin assigns a tier-0 gateway or a tier-0 VRF gateway from the default space to a project. The services on the tier-0 gateway are managed by the Enterprise Admin from the default space.
Dynamic Routing (BGP/OSPF)	Managed by the Enterprise Admin	Dynamic routing is configured on the tier-0 gateway or tier-0 VRF gateway by the Enterprise Admin.
EVPN	Managed by the Enterprise Admin	EVPN is configured on the tier-0 gateway or tier-0 VRF gateway by the Enterprise Admin.
Tier-1 Gateway	Yes	
Static Routing	Yes	Static routing on the tier-1 gateway is configured by the Project Admin.
Overlay Segments	Yes	
VLAN Segments	No	
Segment Profiles • SpoofGuard • IP Discovery • MAC Discovery • Segment Security • QoS	Yes	
L2 Bridge	No	
L2 VPN	Yes	On each tier-1 gateway of the project, you can configure only one L2 VPN service.
IPSec VPN (L3 VPN)	Yes	On each tier-1 gateway of the project, you can configure only one IPSec service. For configuring route-based IPSec VPN, static-routes are supported. Dynamic routing with virtual tunnel interface (VTI) using BGP is not supported on the tier-1 gateway of the project.
NAT	Yes	
Load Balancer	No	
DNS Forwarder • DNS Services • DNS Zones	Yes	
IP Address Pools/IP Address Blocks	Yes	
IPv6 Profiles (DAD/ND)	Yes	
Gateway QoS Profiles	Yes	
DHCP and DHCP Relay	Yes	
Security		
Distributed Firewall	Yes	Only for VMs that are connected to segments in a project. Firewall rules that are managed by the Enterprise Admin in the default space have the highest precedence followed by the project policies. DFW policies with Antrea groups in the Sources, or Destinations, or Applied To of the firewall rules are not supported under projects.
Exclusion List	Managed by the Enterprise Admin	Exclusion list excludes a VM from all the firewall application rules.

NSX Feature	Available Under Projects	Notes
Gateway Firewall	Yes	The Enterprise Admin and Project Admin can manage the gateway firewall rules on the tier-1 gateways of the project only within the context of a project. The Project Admin can delete or modify the gateway firewall rules in the project that are created by the Enterprise Admin.
Identity Firewall	No	Identity firewall is not available under projects. Identity firewall rules can be configured only in the default space and these rules can be applied to VMs inside projects.
Distributed IDS/IPS	Yes	
Gateway IDS/IPS Malware Prevention TLS Decryption	No	
TLS Inspection FQDN Filtering URL Filtering	No Yes Yes	FQDN Analysis dashboard is not exposed to the Project Admin. It is available only to the Enterprise Admin.
Firewall Profiles • Session Timer • Flood Protection • DNS Security	Yes	
Inventory		
Services	Yes	
Groups (static and dynamic memberships)	Yes	Kubernetes member types are not available under a project for creating dynamic membership criteria.
Antrea Groups	No	
Context Profiles/L7 Access Profiles	Yes	
Tags	Yes	
Virtual Machines (visibility/tagging)	Yes	Only for VMs that are connected to segments in a project.
Container Clusters	No	Kubernetes resources in Antrea Kubernetes clusters that are registered to NSX are not exposed to the project inventory.
Plan and Troubleshoot		
Traceflow	Yes	Traceflow can only use VMs and ports that are a part of the project. If the destination is an IP that routes to another project, these details are hidden in the Traceflow output.
Antrea Traceflow	No	
Live Traffic Analysis	No	
IPFIX	Managed by the Enterprise Admin	IPFIX is managed centrally by the Enterprise Admin in the default space.
Port Mirroring	Managed by the Enterprise Admin	Port mirroring is configured centrally by the Enterprise Admin in the default space.

NSX Feature	Available Under Projects	Notes
Security Intelligence	No	Security Intelligence features are not exposed to the Project Admin. Only the NSX Enterprise Admin has full access to all the Security Intelligence features. The Security Intelligence features (network flow visualization, micro-segmentation recommendations, and suspicious traffic analysis) are not project-aware. These features work with all the network traffic flow data within your entire on-premises NSX environment. If multi-tenancy is used, that is, if projects are defined in your NSX environment, Security Intelligence displays all the NSX objects that are in the default space and all the NSX objects from all the projects, whether you are using the **Default** view or the **All Projects** view.

Hierarchical API for NSX Projects

NSX Hierarchical API provides users a way to create an entire or a part of an intent by invoking a single API call. The input to the API is expressed in a tree format. Each node in tree can have multiple children of different types.

Hierarchical API is supported for projects. Project users can invoke the Patch API calls, as described in this documentation, to create, update, or delete an entire intent hierarchy, or a part of an intent hierarchy.

Patch API

The API calls in this section are organized in terms of user roles.

Enterprise Admin

Enterprise Admin can run any of these Patch API calls:

```
PATCH https://<nsx-mgr>/policy/api/v1/org-root
PATCH https://<nsx-mgr>/policy/api/v1/orgs/default/projects/<project-id>/infra
```

Example Request Body:

```
PATCH https://<nsx-mgr>/policy/api/v1/org-root
{
 "resource_type": "OrgRoot",
 "children": [{
  "resource_type": "ChildResourceReference",
  "id": "default",
  "target_type": "Org",
  "children": [{
   "resource_type": "ChildProject",
   "Project": {
    "id": "project-1",
    "resource_type": "Project",
    "display_name": "Test HAPI Project",
    "site_infos": [{
     "edge_cluster_paths": [
      "/infra/sites/default/enforcement-points/default/edge-clusters/ec1"
     ],
     "site_path": "/infra/sites/default"
    }],
    "tier_0s": [
     "/infra/tier-0s/vmware"
    ]
   }
  }]
 }]
```

```
}
```

Role based access control is applied to the response payload of this Patch API.

For a detailed information about the `org-root` schema, see the *NSX API Guide*.

Project Admin

Project Admin can run the following Patch API call and use `ChildResourceReference` to specify the children for any node:

```
PATCH https://<nsx-mgr>/policy/api/v1/org-root
```

Alternatively, they can run the following Patch API call:

```
PATCH https://<nsx-mgr>/policy/api/v1/orgs/default/projects/<project-id>/infra
```

For a detailed information about these APIs, see the *NSX API Guide*.

Other Project Users

They can run the following Patch API call:

```
PATCH https://<nsx-mgr>/policy/api/v1/orgs/default/projects/<project-id>/infra
```

However, these users can run the Patch API against this URI only for objects that they are allowed to modify.

Get API

The APIs calls in this section are organized in terms of user roles.

Enterprise Admin

Only an Enterprise Admin can run the Get API call to read the `org-root` properties.

```
GET https://<nsx-mgr>/policy/api/v1/org-root
```

An Enterprise Admin can also run the following GET API calls:

```
GET https://<nsx-mgr>/policy/api/v1/org-root?base_path=/orgs/default
GET https://<nsx-mgr>/policy/api/v1/orgs/default/projects/<project-id>/infra?
filter=Type-
```

For more information about these APIs, see the *NSX API Guide*.

Project Admin

Project Admin can run the following Get API call:

```
GET https://<nsx-mgr>/policy/api/v1/orgs/default/projects/<project-id>/infra?
filter=Type-
```

Other Project Users

Hierarchical Get API call is not allowed.

NSX Virtual Private Clouds

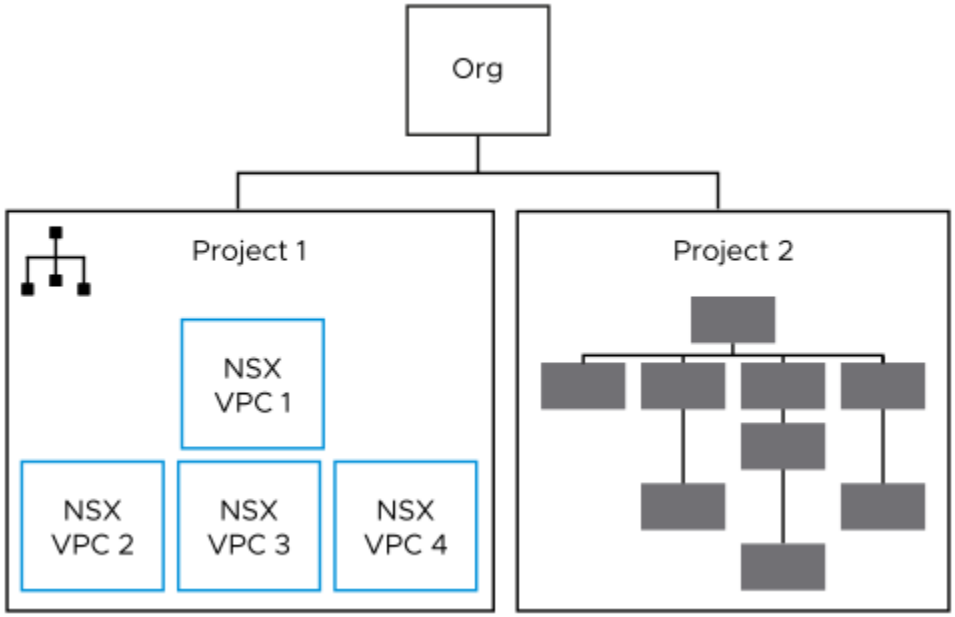
NSX Virtual Private Clouds (VPCs) is an abstraction layer that simplifies setting up self-contained virtual private cloud networks within an NSX project to consume networking and security services in a self-service consumption model.

NSX VPCs represent an additional layer of multi-tenancy within a project. It provides a simplified consumption model of networking and security services, which is aligned to the experience that you would have in a public cloud environment.

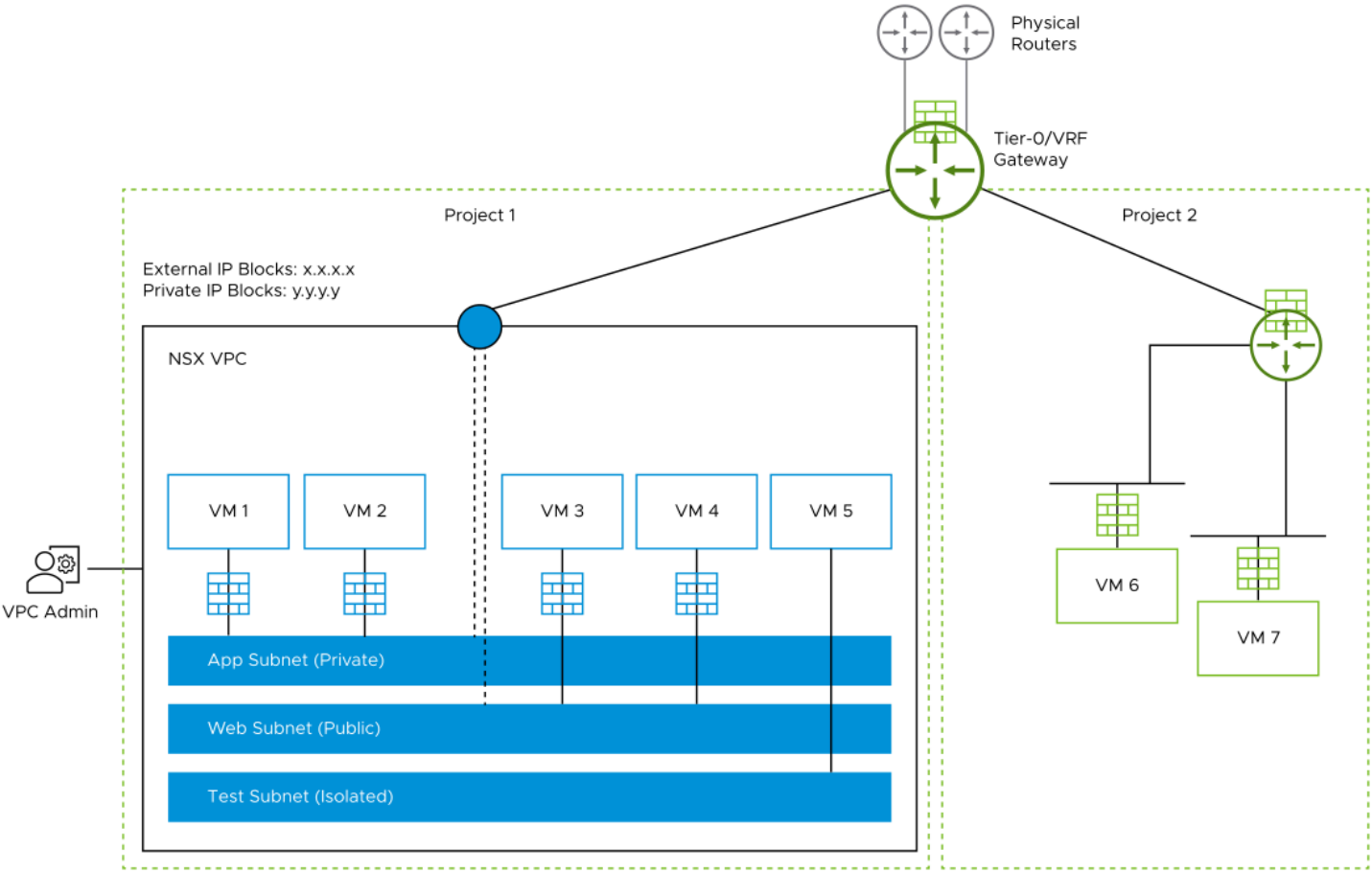
An NSX VPC hides the complexity of the underlying NSX infrastructure, network topology, networking objects, and IP address management from the application owners and offers them a self-service consumption model to run applications in their own private space.

Application owners or DevOps engineers do not need to know about the underlying NSX infrastructure for running applications within their isolated space. They can add subnets (networks) inside the NSX VPC that is assigned to them, and configure security policies to meet their application requirements without having any dependency on the Enterprise Admin.

NSX VPCs are optional under a project. The following diagram shows two projects in the org. Project 1 contains NSX VPCs whereas Project 2 contains no NSX VPCs.



The following diagram shows a sample logical view of an NSX VPC inside project 1. Both projects 1 and 2 are connected to the same tier-0 or tier-0 VRF gateway.



The NSX VPC inside project 1 contains three subnets:

- App subnet (private)
- Web subnet (public)
- Test subnet (isolated)

Each NSX VPC within a project represents an independent routing domain. Subnets within an NSX VPC represent independent layer 2 broadcast domains. Hosts must be prepared with an overlay transport zone before VPC creation. Subnets are realized as overlay segments in the default transport zone of the project. Users that are assigned the VPC Admin role or the Network Admin role in the NSX VPC can add subnets in the NSX VPC. In addition to these two roles, an NSX VPC can also have the following user roles, but with their scope confined to the NSX VPC:

- Security Admin
- Network Operator
- Security Operator

When an NSX VPC is created, the Project Admin specifies the external IP blocks and the private IP blocks to use for creating the subnets in the VPC. The supported subnet access modes are Private, Public, and Isolated. To learn more about subnet access modes, see the *Access Modes for NSX VPC Subnets* section later in this documentation.

When an NSX VPC is created successfully, the system implicitly creates a gateway. However, this implicit gateway is exposed to the Project Admin in a read-only mode and is not visible to the NSX VPC users.

The lifecycle of this implicit gateway is managed by the system. When an NSX VPC is deleted, the implicit gateway is deleted automatically.

In the NSX Policy data model, NSX VPC objects are created under projects at the following path:

```
/orgs/default/projects/<project_id>/vpcs/<vpc-id>
```

The realized implicit gateway is at the following path:

```
/orgs/default/projects/<project_id>/infra/tier-1s/
```

Example of an NSX VPC

Let us assume that your organization has created a project named "Sales" in its NSX environment. The objective of the Project Admin is to provide an isolated networking and security environment for the application developers of the "Order management" and "Analytics" applications in the Sales business unit.

The application developers require an ability to provision networks and configure security policies for these two applications in a self-service consumption model and without any dependency on the Enterprise Admin or the Project Admin.

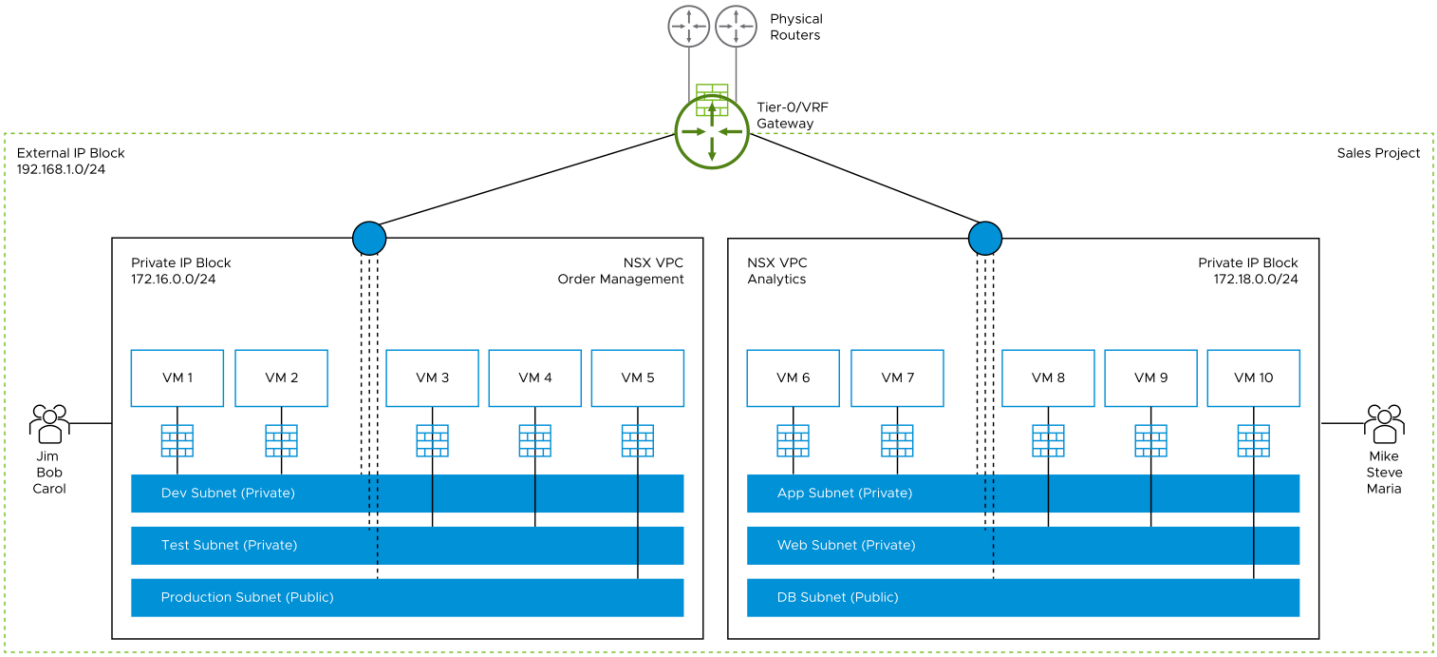
To achieve this objective, the Project Admin can create two NSX VPCs inside the "Sales" project and assign these NSX VPCs to the application developers.

For example:

VPC Name	VPC Users	IP Address Blocks
Order Management	Jim: VPC Admin Bob: Network Operator Carol: Security Operator	Private IPv4 block: 172.16.0.0/24 External IPv4 block: 192.168.1.0/24
Analytics	Mike: VPC Admin Steve: Network Operator Maria: Security Operator	Private IPv4 block: 172.18.0.0/24 External IPv4 block: 192.168.1.0/24

After roles are assigned to the NSX VPC users, these users can add subnets inside the NSX VPC and configure security policies for these workloads. The security policies impact only the workloads within the NSX VPC and not outside the NSX VPC.

For example, the following diagram shows three subnets named as Dev, Test, and Production inside the Order Management NSX VPC, and three subnets named as App, Web, and DB in the Analytics NSX VPC. Workload VMs are attached to all the subnets. The NSX VPCs are attached to the same tier-0 or tier-0 VRF gateway of the Sales project.



High-Level NSX VPC Workflow

At a high level, the NSX VPC workflow is as follows:

1. Project Admin: Adds NSX VPC inside a project and defines basic NSX VPC settings, such as IP assignment, DHCP configuration, edge cluster, and so on.
2. Project Admin: Assigns roles to users in the NSX VPC.
3. Project Admin: Defines quota or limits for the number of objects that users can create within the NSX VPC.
4. VPC Admin or Network Admin: Adds subnets in the NSX VPC. Connects workloads to these subnets based on business requirements.
5. VPC Admin or Security Admin: Adds security policies in the NSX VPC to meet the security requirements of the workloads that are connected to the subnets in the VPC.

Access Modes for NSX VPC Subnets

The supported access modes for subnets in an NSX VPC are:

Private	A private subnet is accessible only within the NSX VPC. Workloads that are attached to a private subnet can communicate with workloads on other private or public subnets within the same NSX VPC. If the IP assignment for the private subnet is set to "automatic", the subnet IP blocks (subnet CIDRs) are automatically assigned from the private IPv4 blocks that are assigned to the NSX VPC. If the IP assignment for the private subnet is set to "manual", the VPC admin can manually assign the subnet CIDR. The CIDR of a VPC subnet cannot overlap with the CIDR of another VPC subnet in the same NSX VPC. However, the subnet IPs can overlap with the subnet in another NSX VPC. It is possible to accomplish this configuration by allocating different private IP blocks with the same IP ranges to different NSX VPCs. Multiple NSX VPCs in a project can share the same private IPv4 block. In this case, the private subnets will be unique across the different VPCs that share the same private IP block. If the **Default Outbound NAT** option is turned on for the NSX VPC, a default SNAT rule is created for the NSX VPC to allow traffic from the workloads on the private subnets to be routed outside the NSX VPC. On similar lines, if this option is turned off, the traffic from the private subnet cannot be routed outside the NSX VPC.
Public	A public subnet is accessible from outside the NSX VPC. This subnet is auto-advertised up to the tier-0 gateway of the project, and it enjoys direct external connectivity, by default. In other words, the IPv4 addresses in the public subnets are reachable both from the project and outside the project.

	If the IP assignment for the public subnet is set to "automatic", the subnet IP blocks (subnet CIDRs) are automatically assigned from the external IPv4 blocks that are specified for the project. If the IP assignment for the public subnet is set to "manual", the VPC admin can manually assign the subnet CIDR.
Isolated	An isolated subnet is not internally connected to the realized implicit gateway. Workloads on an isolated subnet can communicate with each other but cannot communicate with workloads on private or public subnets within the same NSX VPC. In addition, packets from isolated subnets cannot go outside the NSX VPC. A VPC Admin must manually specify the CIDR address of an isolated subnet.

Add an NSX VPC

NSX VPC provides an isolated space for application owners to host applications and consume networking and security objects by using a self-service consumption model.

You must be assigned any one of these roles:

- Project Admin
- Enterprise Admin

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Expand the **Project** drop-down menu, and then select the project where you want to add an NSX VPC.
3. Click the **VPCs** tab, and then click **Add VPC**.
4. Required: Enter a name for the NSX VPC.
5. Select a tier-0 or a tier-0 VRF gateway that the workloads in the NSX VPC can use for north-south connectivity outside this VPC.

   This drop-down menu displays only those tier-0 or tier-0 VRF gateways that were assigned to the project when the project was created.

   If no gateway is selected, the workloads in the NSX VPC will not have north-south connectivity.
6. Configure the **IP Assignment** settings.
   a) In the **External IPv4 Blocks** field, select the IPv4 blocks that the system can use for public subnets in the NSX VPC.

      The external IP blocks that are assigned to the project are available for selection in the NSX VPC. These IPv4 blocks must be routable from outside the NSX VPC. You can assign a maximum of five external IPv4 blocks to each NSX VPC in a project.

      The selected external IPv4 blocks are used for public subnets only when the **IP Assignment** option is set to **Automatic** during the subnet creation.
   b) In the **Private IPv4 Blocks** field, select the IPv4 blocks that the system can use for private subnets in this NSX VPC.

      The IPv4 blocks that are added in the project with visibility set to **Private** are available for selection in the NSX VPC.

      If no IPv4 blocks are available for selection, click

      ,

      and then click **Create New** to add a private IPv4 block.

      The selected private IPv4 blocks are used for private subnets only when the **IP Assignment** option is set to **Automatic** during the subnet creation.

      You must select either external IPv4 blocks or private IPv4 blocks, or both. If neither of them are selected, an error message is displayed when you save the NSX VPC.

c) Under **DHCP**, select any one of these options.

Option	Description
Managed by NSX Policy Management	Default option. The system configures a Segment DHCP server for each subnet in the NSX VPC. The DHCP server dynamically assigns IP addresses to the workload VMs that are connected to the subnets in the NSX VPC.
External	The system uses external or remote DHCP servers to dynamically assign IP addresses to the workload VMs that are connected to the subnets in the NSX VPC. You can specify the IP address of external DHCP servers in the DHCP relay profile that you select for the NSX VPC. **Note:** Workloads on only public VPC subnets can receive IP addresses from the external DHCP server. Currently, workloads on private VPC subnets cannot receive IP addresses from the external DHCP server unless the DHCP server itself is connected to the private or public VPC subnet. In the **DHCP Relay Profile** drop-down menu, select an existing relay profile. If no DHCP relay profile is available, click  to create a new DHCP relay profile. For more information about adding a DHCP relay profile, see Add an NSX DHCP Relay Profile. The configuration of the external DHCP servers is not managed by NSX.
None	System does not configure DHCP for the subnets in the NSX VPC In this case, you must manually assign IP addresses to the workload VMs that are connected to the subnets in the NSX VPC.

7. If DHCP configuration is managed by NSX, you can optionally enter the IP address of the DNS servers.

   When not specified, no DNS is assigned to the workloads (DHCP clients) that are attached to the subnets in the NSX VPC.

8. Configure the **Services Settings**.
   a) By default, the **N-S Services** option is turned on. If required, you can turn it off.

      When this option is turned on, it means that a service router is created for the connected subnets to support centralized services such as N-S firewall, NAT, or gateway QoS profile.

      When this option is turned off, only distributed router is created.

      CAUTION:
      After an NSX VPC is realized in the system, preferably avoid turning off the **N-S Services** option. The reason is that the centralized services will not be enforced on the subnets of the NSX VPC. For example, services such as N-S firewall, NAT, and so on, will not be enforced even if they are configured in the NSX VPC.

   b) From the **Edge Cluster** drop-down menu, select an edge cluster to associate with the NSX VPC.

      If the project does not have edge clusters assigned to it, this drop-down menu will be empty. Ensure that you have allocated at least one edge cluster to the project for it to be available for selection when adding an NSX VPC.

      The selected edge cluster is consumed for running centralized services in the NSX VPC such as NAT, DHCP, and N-S firewall. These services require an edge cluster to be associated with the NSX VPC.

      If **DHCP** is set to **None** and **N-S Services** option is turned off, then edge cluster is optional.

   c) By default, the **Default Outbound NAT** option is turned on. If required, you can turn it off.

      This option is available only when the **N-S Services** option is turned on for the NSX VPC.

When **Default Outbound NAT** is turned on, it means that traffic from the workloads on the private subnets can be routed outside the NSX VPC. System automatically creates one default SNAT rule for the NSX VPC. The translated IP for the SNAT rule is taken from the external IPv4 block of the NSX VPC.

Note:
After an NSX VPC is realized in the system, preferably avoid turning off the **Default Outbound NAT** option. The reason is that the workloads on the private subnets will no longer be able to communicate outside the NSX VPC.

d) Use the **Activate Default E-W Firewall Rules** toggle to specify whether you want to turn on or off the default east-west firewall rules for this NSX VPC.

The default east-west firewall rules allow all outgoing traffic from the workloads that are connected to the subnets in the NSX VPC and drops all incoming traffic to the VPC.

This toggle is editable only when you apply an appropriate security license in your NSX deployment that entitles the system to distributed firewall security feature. This setting only turns on/off the default E-W firewall rules for the NSX VPC. It does not turn off the E-W firewall in the NSX VPC.

For instance, if distributed firewall service is activated in your NSX platform, you can still deactivate the default E-W firewall rules of the NSX VPC. In this case, the system-wide default distributed firewall rules that are configured in the default space and the default distributed firewall rules that are configured in the project will be applied to the NSX VPC.

The following table explains the default state of the **Activate Default E-W Firewall Rules** toggle in the NSX VPC under various scenarios. The term "base license" refers to the VMware Cloud Foundation (VCF) license.

Sr. No.	Scenario	Default State of the Toggle	Notes
1	You are a new NSX customer and have applied a base license that entitles the system to only the NSX networking features.	Off	This toggle is not editable because the current applied license does not support configuration of east-west (distributed) firewall security rules. You need to apply the appropriate security license in the system, and then turn on this toggle to activate the default E-W firewall rules in the NSX VPC.
2	You are a new NSX customer. On day 0, you have applied a base license that entitles the system to NSX networking features. You have also applied an appropriate security license that entitles the system to distributed firewall security.	On	The default E-W firewall rules are activated for the NSX VPC. You can turn it off, if required.

Sr. No.	Scenario	Default State of the Toggle	Notes
3	You are a new NSX customer. On day 0, you applied only the base license that entitles the system to only the NSX networking features. You have added some NSX VPCs in the system, let us say, VPCs A and B Later, during day 2 operations, you applied an appropriate security license that entitles the system to distributed firewall security. Now, you added user-defined E-W firewall rules in the existing VPCs A and B, and also created new VPCs in the project, let us say, VPCs C and D.	Off: for pre-existing VPCs in the project On: for new VPCs in the project	In this scenario, the term "pre-existing VPCs" refers to NSX VPCs that existed in the project before the security license was applied on day 2. In this scenario, they refer to VPCs A and B. The term "new VPCs" refers to NSX VPCs that are added in the project after the security license was applied on day 2. In this scenario, they refer to VPCs C and D. For pre-existing VPCs A and B, the system behavior is as follows: This toggle will be in the Off state, by default. The user-defined E-W rules are effective in VPCs A and B. If you want to activate the default E-W rules in these VPCs, open these VPCs in the edit mode, and turn on this toggle manually. However, when turned on, it might impact the behavior of east-west traffic in VPCs A and B. For new VPCs C and D, the system behavior is as follows: This toggle will be in the On state by default. That is, for VPCs C and D, the default E-W rules are activated, by default. If required, you can turn it off so that only the user-defined E-W rules are effective in these VPCs.
4	You are an existing NSX customer with a legacy NSX license that entitles your system to full DFW access. After the grace period for the legacy license ends, you apply a base license that entitles your system to NSX networking features, and also apply a security license that entitles your system to distributed firewall security.	On	The default E-W firewall rules and user-defined E-W firewall rules continue to run in existing VPCs that you created before changing to the new license. There is no change in the system behavior. For all new VPCs that you add after changing the license, this toggle is turned on, by default. You can turn it off, if required.

Sr. No.	Scenario	Default State of the Toggle	Notes
5	You are an existing NSX customer with a legacy NSX license that entitles your system to full DFW access. You have added two NSX VPCs in the system, let us say, VPCs A and B. After the grace period for the legacy license ends, you apply the base license that entitles your system to only the NSX networking features. A new security license is not applied. Now, you have created two new NSX VPCs in the system, let us say, VPCs C and D.	On: for pre-existing VPCs in the project Off: for new VPCs in the project	In this scenario, the term "pre-existing VPCs" refers to NSX VPCs that were added in the system when the legacy NSX license was valid. In this scenario, they refer to VPCs A and B. The term "new VPCs" refers to NSX VPCs that are added in the system after the base license was applied. In this scenario, they refer to VPCs C and D. For pre-existing VPCs A and B, the system behavior is as follows: This toggle is in the On state, by default. You can turn it off, if needed. But, this action is not reversible. That is, you cannot activate the default E-W firewall rules again in VPCs A and B. The default E-W firewall rules and user-defined E-W firewall rules continue to run in VPCs A and B. But, you cannot edit these rules. Neither, you can add new E-W firewall rules. But, you can delete the existing user-defined firewall rules. To have a full access to the distributed firewall rules, you need to apply an appropriate security license. For new VPCs C and D, the system behavior is as follows: This toggle is in the Off state, by default. You cannot turn it on because the current applied license does not entitle the system to the distributed firewall feature.
6	You are a new NSX customer and your system has entered an Evaluation mode, which is valid for 60 days.	Off	During the evaluation period of a new NSX deployment, the system is entitled to only the networking features. The security features are not entitled.

e) If you want the NSX VPC to be discovered by the Avi Load Balancer Controller, turn on the **Enable for Avi Load Balancer** option.

By default, this option is turned off. When it is turned on, it provides the ability to extend NSX multi-tenancy to Avi Load Balancer Controller and hence enables load balancer configuration in this context.

You can turn on this option only when the following conditions are met:

- At least one private IP address block is assigned to the NSX VPC.
- At least one edge cluster is assigned to the NSX VPC.

NSX VPC requires a private IP block because the load balancer virtual service IP (VIP) needs to be on a private subnet. An edge cluster is required so that the Avi Load Balancer Service Engines can receive IP addresses dynamically from the DHCP server.

To learn more about Avi Load Balancer, see the VMware Avi Load Balancer Documentation.

9. Optionally, attach the following profiles that will be used by the subnets in the NSX VPC:
   - N-S Egress Quality of Service (QoS) profile
   - N-S Ingress QoS profile
   - IP Discovery profile
   - SpoofGuard profile
   - Mac Discovery profile
   - Segment Security profile
   - QoS

   To learn about the purpose of these profiles, see Segment Profiles.
10. In the **Short log identifier** text box, enter a string that the system can use to identify the logs that are generated in the context of the NSX VPC.

    This identifier must be unique across all NSX VPCs. The identifier must not exceed eight alphanumeric characters.

    If the identifier is not specified, the system autogenerates it when you save the NSX VPC. After the identifier is set, you cannot modify it.
11. Optionally, enter a description for the NSX VPC.
12. Click **Save**.

When an NSX VPC is created successfully, the system implicitly creates a gateway. However, this implicit gateway is exposed to the Project Admin in a read-only mode and is not visible to the NSX VPC users.

To view the realized implicit gateway, a Project Admin can do the following steps:

1. Navigate to **Networking** > **Tier-1 Gateways**.
2. Click the **VPC Objects** check box at the bottom of the **Tier-1 Gateways** page.
3. Expand the gateway to view the configuration in a **read-only** mode.
   The following naming convention is used for the implicit gateway:
   `_TIER1-`*`VPC_Name`*

The lifecycle of this implicit gateway is managed by the system. When an NSX VPC is deleted, the implicit gateway is deleted automatically.

If you have enabled the **Default Outbound NAT** option, you can view the system-created default SNAT rule in the NSX VPC. Do these steps:

1. Expand the **Network Services** section of the NSX VPC.
2. Click the count next to **NAT**.
3. Observe the default NAT rule with SNAT action for the private IPv4 block in the NSX VPC. This NAT rule is not editable. If the NSX VPC is assigned multiple private IPv4 blocks, one default NAT rule with SNAT action is created for each private IPv4 block.
   For example:



   The source IP in the SNAT rule is the private IPv4 block of the NSX VPC. In this example, it is 193.0.1.0/24. The translated IP for this SNAT rule is assigned from the external IPv4 block of the NSX VPC. In this example, it is 192.168.1.0.

Add a Subnet in an NSX VPC

A subnet in an NSX VPC represents an independent layer 2 broadcast domain. NSX VPC subnets are realized as overlay segments in the default transport zone of the project.

You must be assigned any one of these roles in the NSX VPC:

- VPC Admin
- Network Admin

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Select the required project from the **Project** drop-down menu, if not already selected.
3. Click the **VPCs** tab and expand the details of the NSX VPC where you want to add a subnet.
4. Expand the **Connectivity** section and click **Set**. If subnets are existing in the NSX VPC, click the subnet count.
   The **Set Subnets** dialog box opens.
5. Click **Add Subnet**.
6. Configure the subnet properties.

Property	Description
Name	Enter a name for the subnet.
Access Mode	Select any one of these access modes: Private, Public, Isolated. To learn more these access modes, see the *Access Modes for NSX VPC Subnets* section in NSX Virtual Private Clouds. By default, private is selected.
IP Assignment	By default, **Automatic** IP assignment is set for private and public subnets. It means that the system will assign an IPv4 CIDR for the subnet automatically. For a public subnet, the CIDR is assigned from the external IPv4 blocks of the NSX VPC. For a private subnet, the CIDR is assigned from the private IPv4 blocks of the NSX VPC. For isolated subnets, only **Manual** IP assignment mode is supported. In **Manual** IP assignment mode, you must enter a valid IPv4 CIDR for the subnet.
Size	This property is applicable only when you select the **Automatic** IP assignment mode. Select a size from the drop-down menu. System reserves four IP addresses for internal use, such as subnet network address, subnet gateway address, subnet broadcast address, DHCP server address. For example, if you select size as `32`, you can attach a maximum of 28 workloads to the subnet.
IP CIDR	This property is applicable only when you select the **Manual** IP assignment mode. Enter the IPv4 subnet address in a CIDR format. For example, 172.16.0.1/24 You can enter only one IPv4 CIDR. If the IPv4 CIDR that you entered is invalid or unavailable for assignment, the system throws an appropriate error message. You must enter a different IPv4 CIDR until the system accepts it.
Description	Optionally, enter a description for the subnet.

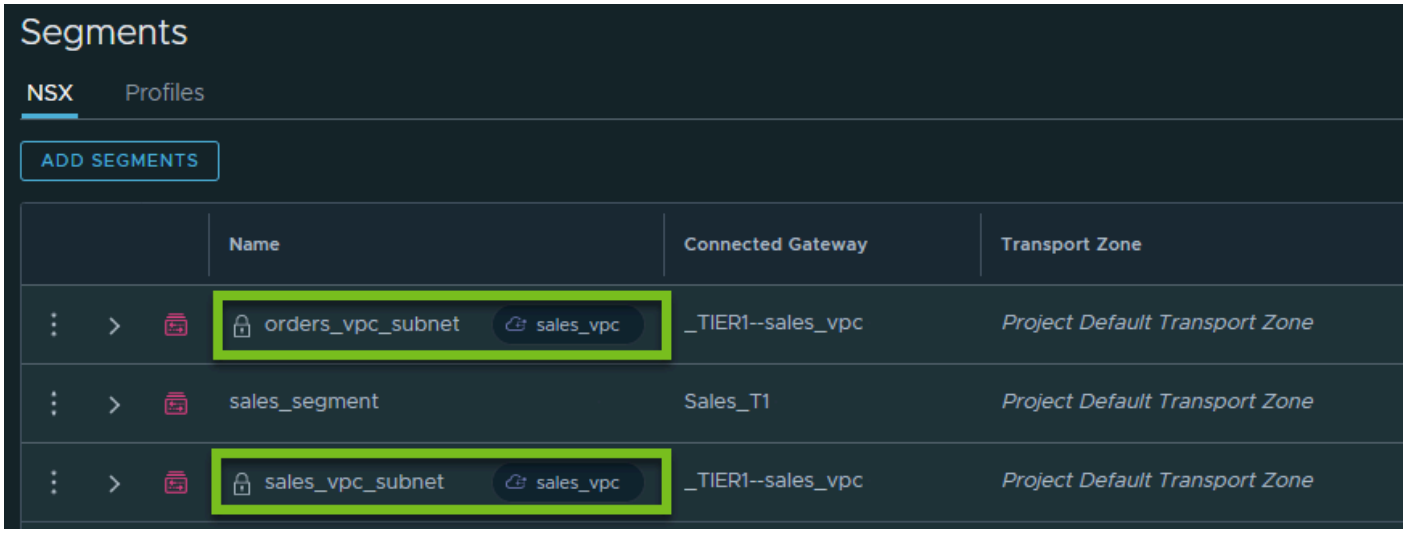
7. Click **Save**.

When a subnet is realized successfully in the NSX VPC, the **Status** column shows `Successful`.

Subnets in an NSX VPC are realized as overlay segments in the default transport zone of the project.

An Enterprise Admin or a Project Admin can view these overlay segments by doing these steps:

1. Ensure that you are in the project view.
2. Navigate to **Networking** > **Segments**.
3. Click the **VPC realized objects** check box at the bottom of the **Segments** page.
   For example:



   This screen capture shows two system-created segments `orders_vpc_subnet` and `sales_vpc_subnet`. These segments represent subnets in the `sales_vpc`, and are displayed in a **read-only** mode on the **Segments** page.

You can now start attaching workload VMs to the subnet by using the vSphere Client.

The following subsections briefly explains some tasks that you can do after a subnet is realized in an NSX VPC.

Set subnet ports

On the **Set Subnets** dialog box, under the **Subnet Ports** column, click **Set**, and then click **Add Subnet Port**.

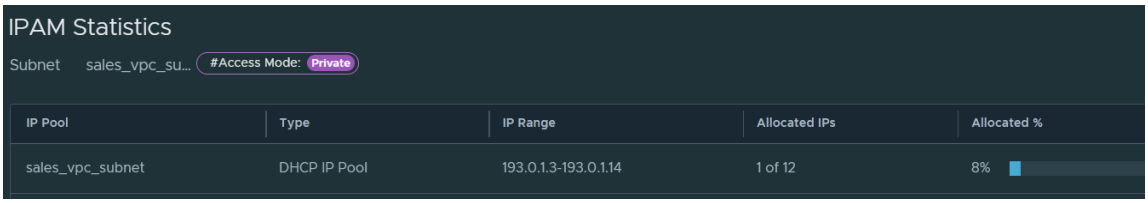
The workflow for adding ports on an NSX VPC subnet is the same as it is for adding ports on an NSX segment. To learn more, see the steps for adding segment ports in Create an NSX Segment.

View IPAM statistics of NSX VPC subnets

Expand the subnet details and click **View IPAM Statistics**. At the time of subnet creation, if DHCP in the NSX VPC is managed by NSX or by external DHCP servers, the **IPAM Statistics** page displays the DHCP IP pool range.

The system automatically allocates 100% of available IP addresses in the subnet to the DHCP pool range. The system reserves four IP addresses from the subnet for its internal use. The DHCP static IP pool is empty and hence it is not displayed on **IPAM Statistics** page. You cannot edit the system-managed DHCP pool configuration of the VPC subnet.

For example:



In this screen capture, the `sales_vpc_subnet` is a private subnet with size 16. The DHCP configuration for this subnet is automatically managed by NSX. The system reserves four IP addresses from the subnet for its internal use, by default. So, the remaining 12 IP addresses are automatically allocated to the DHCP IP pool range (193.0.1.3-193.0.1.14). The IP addresses are dynamically assigned to the workload VMs on the subnet from this DHCP pool range.

If you prefer to manually configure the DHCP pool ranges for dynamic and static IP assignments in the VPC subnet, you can add the subnet in the NSX VPC by running the following NSX API. Currently, the UI does not support manual DHCP configuration of NSX VPC subnets.

```
PUT https://<nsx-mgr>/policy/api/v1/orgs/default/projects/<project-id>/vpcs/<vpc-id>/
subnets/<subnet-id>/
```

Example Request Body:

```
{
    "ipv4_subnet_size": 16,
    "access_mode": "Public",
    "resource_type": "VpcSubnet",
    "dhcp_config": {
        "enable_dhcp": true,
        "static_pool_config": {
            "ipv4_pool_size": 8
        }
    }
}
```

This example payload shows that you have created a public NSX VPC subnet of size 16. You have specified the size of the static IP pool as eight. Therefore, out of the total 12 IP addresses that are available for allocation to the workloads in the NSX VPC subnet, the remaining four IP addresses are allocated to the DHCP IP pool for a dynamic IP assignment.

When you view the **IPAM Statistics** page of this NSX VPC, the system displays the IP address utilization of both the DHCP IP pool and the static IP pool.

At the time of subnet creation, if DHCP in the NSX VPC is none, the **IPAM Statistics** page of the subnet displays only the static IP pool information.

View traffic statistics of NSX VPC subnets	Expand the subnet details and click **View Subnet Statistics**. If you want to view the traffic statistics for the full NSX VPC, and not for a specific subnet in the VPC, expand the NSX VPC details, and then click **View VPC Statistics**.

Add User Role Assignments for an NSX VPC

After adding an NSX VPC in your project, you can assign roles to users in the NSX VPC. These users can then start configuring networking or security objects as required for the workloads, which are running inside the NSX VPC.

In an NSX VPC, you can assign the following roles to users:

- VPC Admin
- Security Admin
- Network Admin
- Security Operator
- Network Operator

A VPC Admin has full access to all the networking and security objects inside an NSX VPC. The other user roles in an NSX VPC have limited access to objects in the NSX VPC as determined by the permissions of those roles.

Note:
By default, only an Enterprise Admin can add user role assignments in NSX VPCs. Project Admin and VPC Admin do not have permissions to add user role assignments in NSX VPCs, unless an Enterprise Admin configures the Project Admin and VPC Admin roles to do user role assignments.

If the Project Admin and VPC Admin roles are allowed to do user role assignments, it might be perceived as introducing security risks in some NSX environments because it allows both these roles to configure role assignments for any user in the system. Therefore, the default behavior is to allow only Enterprise Admin to add user role assignments in NSX VPCs.

An Enterprise Admin can do the following steps to grant permissions to the Project and VPC Admin roles for adding user role assignments:

1. In the **Default** view, navigate to **System** > **User Management** > **Roles**.

2. Next to the **Project Admin** role, click  , and then click **Allow Role Assignments**.
3. Next to the **VPC Admin** role, click  , and then click **Allow Role Assignments**.

For user authentication and authorization, NSX multi-tenancy supports the following identity sources:

- Local users (for example, guestuser1, guestuser2)
- Workspace ONE Access
- Lightweight Directory Access Protocol (LDAP)
- OpenID Connect

> **Note:**
> OpenID Connect identity source is supported starting in NSX 4.1.2. However, this identity source is supported for user authentication only when you are adding user role assignments from the **User Management** page of the system. If you are adding user role assignments from the **Manage Projects** page or the **VPC** page, this identity source is currently not supported.

To add user role assignments in NSX VPCs, the following three methods are available:

Method 1: Add Role Assignments for an NSX VPC from the User Management Page	The **User Management** page is available only to the Enterprise Admin. Project Admin and VPC Admin cannot use this page even if an Enterprise Admin has granted them permissions to do user role assignments.
Method 2: Add Role Assignments for an NSX VPC from the VPC Page	The **VPC** page is available to both the Project Admin and the VPC Admin. However, they can add user roles from this page only when an Enterprise Admin has granted them permissions to do user role assignments.
Method 3: Add Role Assignments for an NSX VPC from the Manage Projects Page	The **Manage Projects** page is available to both the Enterprise Admin and the Project Admin. However, a Project Admin can add user roles from this page only when an Enterprise Admin has granted permissions to the Project Admin role to do user role assignments. To learn more about this method, see Add Role Assignments for an NSX Project from the Manage Projects Page. A VPC Admin cannot access the **Manage Projects** page.

Add Role Assignments for an NSX VPC from the User Management Page

Use the instructions in this documentation to add user role assignments for an NSX VPC from the **User Management** page.

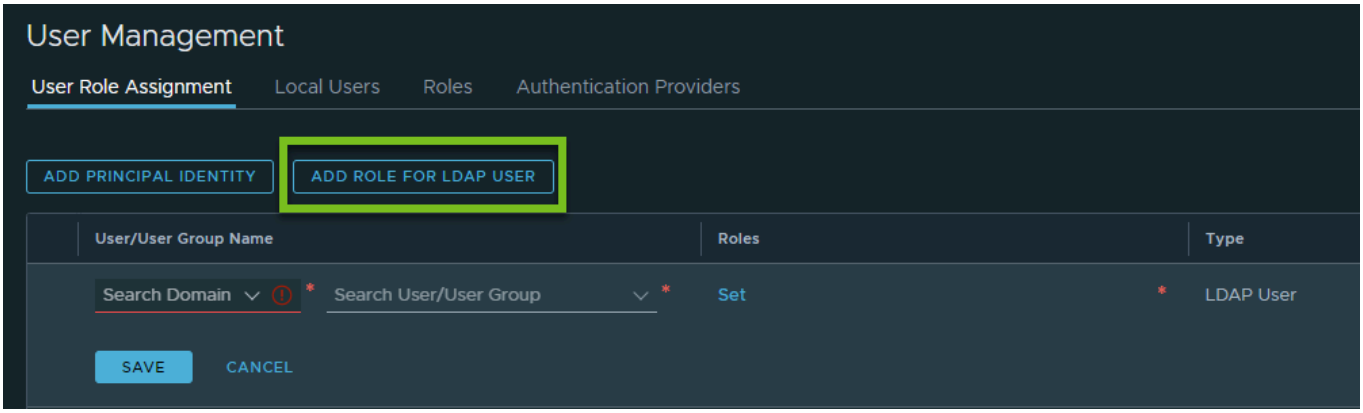
User accounts are created. For example:

- Local user accounts are added in the system and they are activated.
- NSX Manager is configured to authenticate users from any of these identity management providers:
  - Workspace ONE Access (vIDM)
  - LDAP-based directory service, for example, Active Directory.
  - OpenID Connect (starting in NSX 4.1.2)

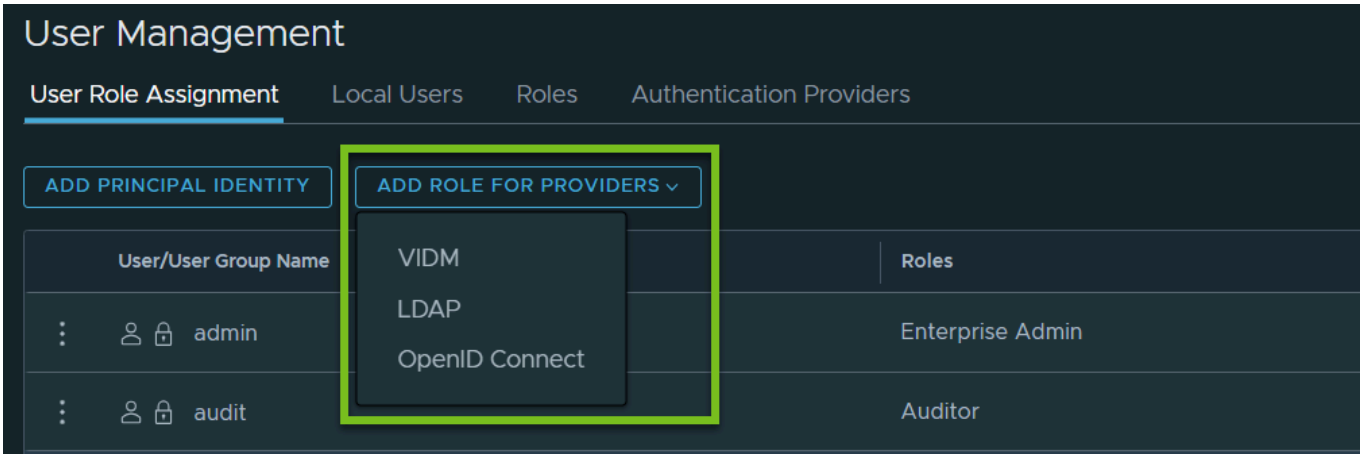
The **User Management** page is available only to the Enterprise Admin. Project Admin and VPC Admin cannot use this page even if an Enterprise Admin has granted them permissions to do user role assignments.

The following procedure explains the steps for adding role assignments in NSX VPCs for local user accounts and LDAP user accounts. The steps to add role assignments for vIDM and OpenID Connect user accounts are almost similar, and therefore not covered in this procedure.

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Navigate to **System** > **User Management**.
   The **User Role Assignment** tab is displayed.
3. To add role assignments in NSX VPCs for a local user account, do these steps:
   a) Next to the local user account name, click
   
   and then click **Edit**.
   b) Click the link under the **Roles** column.
   The **Set Roles/Scope** dialog box opens.
   c) Click **Add Role**, and then select any one of these roles to assign to the local user:
      - VPC Admin
      - Network Admin
      - Security Admin
      - Network Operator
      - Security Operator
   d) Under the **Scope** column, click **Set**.
   You can set the scope to one or more NSX VPCs.
   e) Click **Add**, and then click **Apply**.
   f) Click **Apply** again to close the **Set Roles/Scope** dialog box.
   g) Click **Save** to save the role assignment.
4. To add role assignments in NSX VPCs for an LDAP user account, do these steps:
   a) Ensure that you are in the **User Role Assignment** tab.
   b) Click **Add Role for LDAP User**.



If you have configured NSX Manager to authenticate users from all supported identity service providers (LDAP, vIDM, and OpenID Connect), the button caption is as shown in the following screen capture.



c) Select a domain from the drop-down menu.
d) Enter the first few characters of the user or group name.

   System displays a list of matching user or group names. Select a user or a group name from the list.
e) Under the **Roles** column, click **Set**.

   The **Set Roles/Scope** dialog box opens.
f) Click **Add Role** and follow the same process, as explained in steps 3(c) though 3(e), to assign roles to the LDAP user in the NSX VPC.

Add Role Assignments for an NSX VPC from the VPC Page

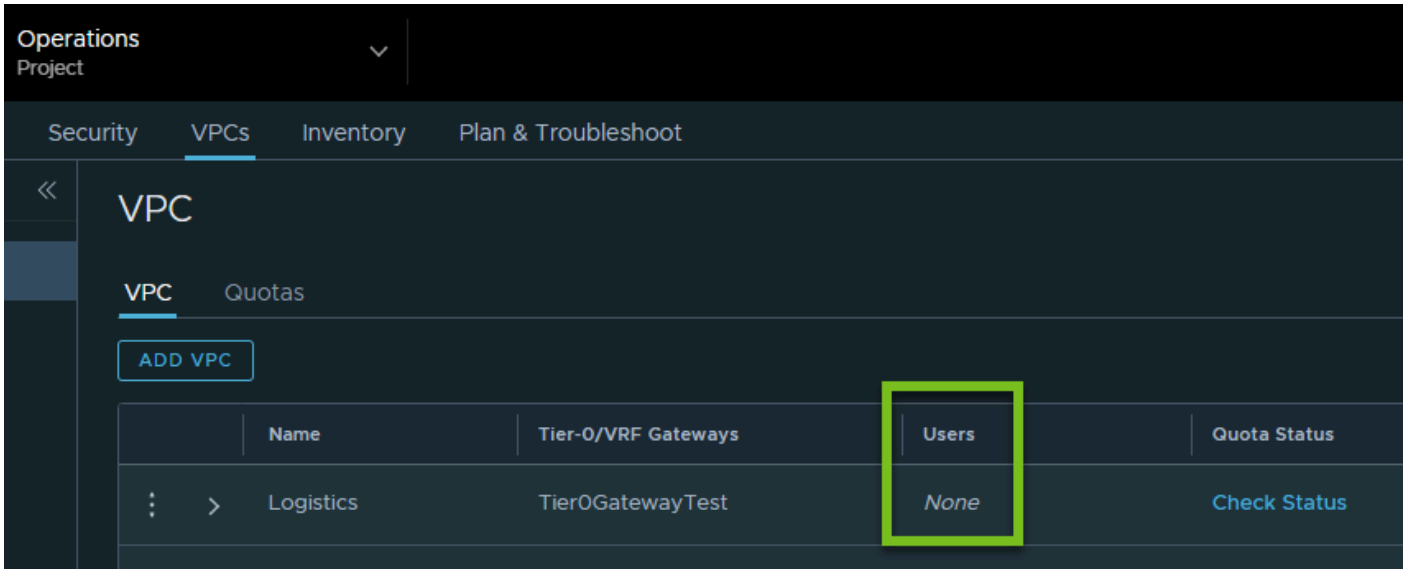
Use the instructions in this documentation to add user role assignments for an NSX VPC from the **VPC** page.

User accounts are created. For example:

- Local user accounts are added in the system and they are activated.
- NSX Manager is configured to authenticate users from any of these identity management providers:
  - Workspace ONE Access (vIDM)
  - LDAP-based directory service, for example, Active Directory.

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Select the required project from the **Project** drop-down menu, if not already selected.
3. Click the **VPCs** tab.

   For example, the following screen capture shows that the Operations project view is open, and no users are currently set for the Logistics VPC.



4. Next to the VPC name, click
   
   and then click **Edit**.
5. Under the **Users** column, click **Set**.
6. Click **Add Role Assignment**, and then do the following steps:
   a) Select any one of these options:
      - Local User
      - LDAP User
      - vIDM User
   b) Enter a user name or a group name.

      Note:
      Ensure that the user or group name is entered accurately. Currently, the system does not display the list of matching users or groups when you type the first few characters in the text box. This behavior is a known limitation.
   c) From the **Select Role** drop-down menu, select a role.
   d) Click **Save** to save the role assignment.
7. Optional: To add more user role assignments in the NSX VPC, repeat step 6.

Add Quota to an NSX VPC

After adding an NSX VPC in a project, you can add one or multiple quotas to define a maximum limit for the number of objects of a specific type that users can create in the NSX VPC.

You must be assigned any one of these roles:

- Project Admin
- Enterprise Admin

When you set a quota for an object type in an NSX VPC, it implies that the total number of user-created objects of that type in the NSX VPC cannot exceed that limit.

For example, assume that you have set a quota for 10 subnets in an NSX VPC. In this case, the total number of subnets that users can add in the NSX VPC cannot exceed 10.

Scenario: Set a quota for the same object	System allows you to set a quota for the same object more than once and apply it to the same NSX VPC. However, in such a situation, the quota with the lower limit is enforced for that object in the NSX VPC. For example, consider the following sequence of events:

more than once in the same NSX VPC	1. User A creates quota 1 for the group object, and specifies the limit as 100. 2. Quota 1 is applied to NSX VPC 1. 3. Later, user A creates quota 2 for the group object and specifies the limit as 50. 4. Quota 2 is also applied to NSX VPC 1. In this case, system enforces the limit of 50 on the group object in NSX VPC 1.

The following procedure explains the workflow of adding quota for objects inside an NSX VPC from the project view. If you want to set limits on the total number of objects that users can create in the NSX VPCs across a project, you must switch to the default space. For more information, see Add Quota to an NSX Project.

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Select the required project from the **Project** drop-down menu, if not already selected.
3. Navigate to **VPCs** > **Quota**.
4. Configure the limits for one or multiple objects in the quota.
   a) Enter a name for the quota.
   b) In the **Limits** column, click **Set**.
      The **View Limits** window opens. By default, you are in the **Connectivity** tab.
   c) Click the tab that contains the VPC objects that you want to set limits on.
   d) Click **Add Limit**.
   e) From the **Objects** drop-down menu, select an object.
      For example, let us click the **Security** tab, and then select the **Group** object.
   f) In the **Limit** box, enter an integer, and then click **Save**.
      For example, let us enter `10`.

      Note:
      To prohibit users from creating an object in an NSX VPC, set the limit to `0` for that object.
   g) Add limits for other objects, as required.
   h) After adding limits for all objects, click **Apply**.
5. From the **Apply To** drop-down menu, select an NSX VPC to apply this quota.
   A quota can be applied to multiple NSX VPCs in the project.
6. Click **Save**.

When the number of user-created objects in an NSX VPC reaches the limit that is set for the object, an alarm is triggered in the system when you try to create and save another object of the same type.

Alarms for the `Creation Count Limit Reached` event are visible to the Project Admin, but not to the VPC Admin.

Note:
For the `Creation Count Limit Reached` event, the threshold value is currently not configurable. System-defined threshold value is used, by default.

Network Address Translation in an NSX VPC

You can add NAT rules with SNAT, DNAT, or Reflexive action in an NSX VPC.

The following functionality is currently not supported for configuring NAT rules in an NSX VPC:

- Destination ports
- Translated ports
- Services
- No SNAT action
- No DNAT action

Workflow for Adding NAT Rules in an NSX VPC

1. Select a project from the **Project** drop-down menu.
2. Click the **VPCs** tab, and then click **VPC**.
3. Expand the VPC details, and under the **Network Services** section, click the count next to **NAT**.
4. Click **Add NAT Rule**.
5. Enter a name for the NAT rule.
6. Select any one of these rule actions:
   – DNAT
   – SNAT
   – Reflexive
7. Depending on the action selected in the previous step, specify the IP address in the required fields.

Action	Required Fields	Notes
DNAT	Destination IP and Translated IP	For destination IP address, only a single IPv4 address is currently supported. A CIDR block or a comma-separated IPv4 address list is not supported. The destination IPv4 address must belong to the external IPv4 block of the NSX VPC and it must be available for allocation.
SNAT	Translated IP	For translated IP address, only a single IPv4 address is currently supported. A CIDR block or a comma-separated IPv4 address list is not supported. The translated IPv4 address must belong to the external IPv4 block of the NSX VPC and it must be available for allocation.
Reflexive	Source IP and Translated IP	For translated IP address, only a single IPv4 address is currently supported. A CIDR block is not supported.

8. By default, rule logging is deactivated. If you require rule logs for troubleshooting purposes, turn on the **Logging** option.
9. Specify a priority value. A lower value means a higher priority. The default is 0.
10. From the **Firewall** drop-down menu, select any one of these options:
    – **Match Internal Address**: This option is the default selection. It means that the N-S firewall rules in the NSX VPC will be applied to the internal address of a NAT rule.
      • For SNAT, the internal address is the original source address before NAT is done.
      • For DNAT, the internal address is the translated destination address after NAT is done.
      • For Reflexive, for egress traffic, the firewall is applied to the original source address before NAT is done. For ingress traffic, the firewall is applied to the translated destination address after NAT is done.
    – **Match External Address**: It means that the N-S firewall rules in the NSX VPC will be applied to the external address of a NAT rule.
      • For SNAT, the external address is the translated source address after NAT is done.
      • For DNAT, the external address is the original destination address before NAT is done.
      • For Reflexive, for egress traffic, the firewall is applied to the translated source address after NAT is done. For ingress traffic, the firewall is applied to the original destination address before NAT is done.
    – **Bypass**: It means that the packet bypasses the N-S firewall rules in the NSX VPC.
11. Click **Save**.

Groups in an NSX VPC

You can create groups in an NSX VPC and use them in firewall policies to meet specific security requirements for the workloads that are running inside the NSX VPC.

Default Groups in an NSX VPC

The system creates a default group for every NSX VPC that is added in your project. The default group helps you to restrict the scope of the firewall rules to a particular NSX VPC.

The following naming convention is used to identify the default group in an NSX VPC:

`PROJECT-<Project_Name>-VPC-<VPC_Name>-default`

Project_Name and *VPC_Name* are replaced with actual values in your system.

This default group represents the NSX VPC itself. The members of this default group are subnets, subnet ports, and VM interfaces (VIFs) that are connected to the NSX VPC subnets. If the VM is dual-homed, for example, when one interface is connected to a VPC subnet and the other interface is connected to a segment in the default space, then the VIF of this VM on the segment will not be a member of the VPC default group.

A typical use case for using the VPC default group is as follows:

Assume that a Project Admin or a user in the default space wants to block traffic to all the VMs inside the NSX VPC. They can use the VPC default group in their firewall policy.

User-Created Groups in an NSX VPC

The following NSX objects are supported for adding statically to a group definition inside an NSX VPC:

- Subnets
- Subnet Ports
- VIFs
- Virtual Machines
- Groups

On the **Members** tab of the **Set Members** dialog box, the system displays only those objects that are owned by the NSX VPC. Objects that are shared with the NSX VPC are not listed in this dialog box because shared objects cannot be added as members in a VPC group.

The following NSX objects are supported for adding to dynamic group membership criteria inside an NSX VPC:

- Virtual Machine
- Subnet
- Subnet Port

When you add a group in an NSX VPC with dynamic criteria based on VM tags, the VMs that are connected to the subnets in the NSX VPC become the effective members of the group.

Groups that are shared with an NSX VPC can be used only in the **Source** or **Destination** fields of the firewall rule, and not in the **Applied To** field of the firewall rule.

Adding Groups in an NSX VPC

1. Select a project from the **Project** drop-down menu.
2. Click the **VPCs** tab.
3. Expand the VPC where you want to add groups.
4. Expand the **Security** section, and then click the count next to **Groups**.
   The **Set VPC Groups** page opens.
5. Now, use the standard procedure to add groups in the NSX VPC.

Firewall Policies in an NSX VPC

When an NSX VPC is realized successfully, the system creates default north-south and east-west firewall rules to govern the default firewall behavior for the workloads in the NSX VPC. For more details about firewall policies in an NSX VPC, see VMware vDefend Firewall documentation.

Features Available for Consumption Under NSX VPCs

NSX VPC exposes a simplified set of features to allow application owners or DevOps engineers to consume NSX networking and security objects in a self-service consumption model.

The following objects are assigned by the Project Admin during an NSX VPC creation and are available for consumption in the VPC.

- Tier-0 or VRF gateway
- Edge cluster
- IP blocks (private, external)
- Various profiles (for example, QoS profile, IP discovery profile, and many more)
- IPAM strategy (DHCP, external, none)

A Project Admin can optionally enable an NSX VPC for Avi Load Balancer.

NSX VPC users can create or modify the following objects in their private VPC space:

Network connectivity	• Subnets (public, private, isolated) • Static routes
Network services	• NAT rules (SNAT, DNAT, reflexive)
Security services	• East-west firewall rules (L4/L7) • North-south firewall rules (L4 only)
Inventory objects	• Groups (static and dynamic memberships) To view the list of NSX objects that are supported for adding statically or dynamically to a group definition, see Groups in an NSX VPC.

Enabling Project Context in NSX Edge Syslog

Log messages generated for objects created in the NSX project contain the project (tenant) context in a label that is appended to every log message. Such a label matches the unique short log identifier that was defined during the project creation.

For example, log messages that are generated in the edge syslog for the centralized services, which are running on the tier-1 gateway of the project, contain the project context.

By default, log messages generated in the edge syslog for the centralized services, which are running on the tier-0/VRF gateways that are consumed by the project, do not contain the project context. The reason is that tier-0/VRF gateways are owned by the default space and they do not belong to any project. When you allocate a tier-0/VRF gateway to a project (say A), it does not prevent you from allocating the same tier-0/VRF gateway to other projects (say B, C, and D) in the system. This is the default NSX behavior.

However, the system supports dedicating a tier-0/VRF gateway to a single project only for logging purposes. When a tier-0/VRF gateway is dedicated to a project, the log messages for the centralized services on this tier-0/VRF gateway will contain the project context in the edge syslog.

A tier-0/VRF gateway that is dedicated to a project (say A) can still be consumed by other projects (say B, C, and D) in the system. However, the context for projects B, C, and D that consume the same tier-0/VRF gateway, will not be stored in the edge syslog unless these projects have a different dedicated tier-0/VRF gateway that is not already dedicated to any other project in the system.

To dedicate a tier-0/VRF gateway to a project, an Enterprise Admin must configure the `dedicated_resources` parameter in the project configuration by using the `project` API. Currently, this functionality is supported only with API. The NSX Manager UI does not support dedicating a tier-0/VRF gateway to a project.

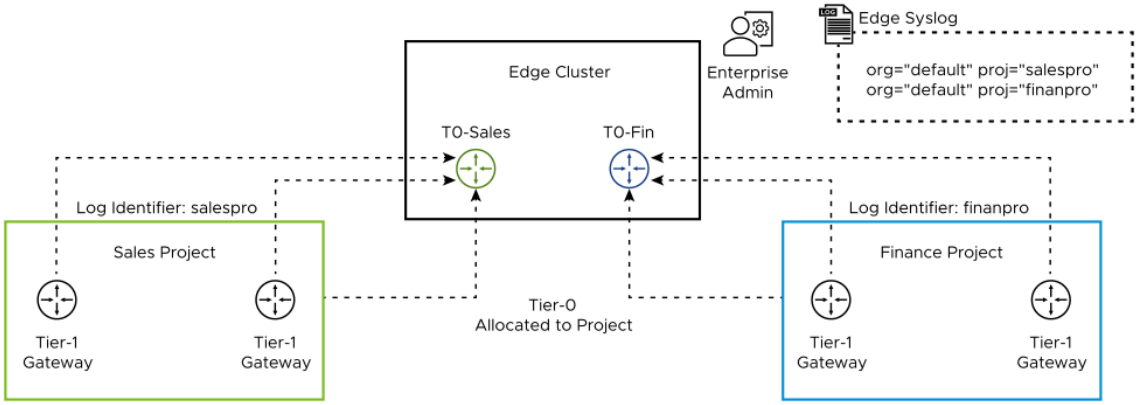
The following validations apply to dedicating a tier-0/VRF gateway in a project:

- To dedicate a tier-0/VRF gateway to a project, it must be allocated to the project. For example, if you have allocated tier-0 gateways X and Y to a project, you cannot dedicate tier-0 gateway Z to this project. To dedicate Z to this project, you must first allocate it to the project.

- A tier-0/VRF gateway can be dedicated to only a single project. However, a project can have multiple tier-0/VRF gateways dedicated to it. For example, assume that project A has two tier-0 gateways X and Y allocated to it. You can dedicate both tier-0 gateways X and Y to project A. However, X and Y cannot be dedicated to any other project in the system. But, X and Y can be consumed by other projects in the system.
- When you remove the `dedicated_resources` configuration from a project, or when you delete the project, the tier-0/VRF gateway that was dedicated to this project can be dedicated to any other project in the system. For example, assume that you have allocated tier-0 gateway X to projects A and B and configured X as the dedicated resource in project A. If required, later, you can remove the `dedicated_resources` configuration from project A and add X as the dedicated resource in project B.

Example

Assume that you have created two projects named *Sales* and *Finance*, as shown in the following diagram. The short log identifier of the Sales project is *salespro* and for the Finance project is *finanpro*. You have configured the `dedicated_resources` parameter to dedicate the *T0-Sales* gateway to the Sales project, and the *T0-Fin* gateway to the Finance project. Both the tier-0 gateways are running on the same edge cluster.



The log messages for the tier-0 services in the edge syslog will contain the following string to help you identify the project context:

```
org="default" proj="salespro"
org="default" proj="finanpro"
```

For example, let's say, you have enabled BGP routing on both the *T0-Sales* and *T0-Fin* gateways. The log messages for both these tier-0 gateways in the edge syslog will have the project context, as follows:

```
nsx-edge> 2023-06-27T03:31:38.717Z Edge NSX 6 ROUTING [nsx comp="nsx edge"
 subcomp="rcpm" s2comp="rcpm-db" level="INFO" org="default" proj="salespro"] BGP
 update request: Bgp Config
nsx-edge> 2023-06-27T03:31:39.717Z Edge NSX 6 ROUTING [nsx comp="nsx edge"
 subcomp="rcpm" s2comp="rcpm-db" level="INFO" org="default" proj="finanpro"] BGP
 update request: Bgp Config
```

API Workflow to Dedicate a Tier-0/VRF Gateway to a Project

For example, assume that you have created a project named *Sales*, and allocated two tier-0 gateways named *T0Sales1* and *T0Sales2* to this project. You want to dedicate *T0Sales1* to this project. To do this, you must use the `project` API, and add the path to the *T0Sales1* gateway in the `dedicated_resources` configuration of the Sales project.

Procedure

1. Retrieve the current project configuration by running the following GET API:
   ```
   GET https://<nsx-mgr>/policy/api/v1/orgs/default/projects/Sales
   ```
2. Copy the response payload of the GET API and paste it in a text editor.
3. In the response payload, add the path to the *T0Sales1* gateway in the `dedicated_resources` configuration, as follows:
   ```
   "tier_0s": [
      "/infra/tier-0s/T0Sales1",
      "/infra/tier-0s/T0Sales2"
   ],
   ```

```
"dedicated_resources": {
   "tier_0s": ["/infra/tier-0s/T0Sales1"]
}
```

To learn more about the `dedicated_resources` parameter in the `project` API, see the *NSX API Guide*.

4. Paste the full updated payload in the request body of the following PATCH API:

   ```
   PATCH https://<nsx-mgr>/policy/api/v1/orgs/default/projects/Sales
   ```

   When this API runs successfully, the Sales project is configured to use *T0Sales1* as the default tier-0 gateway. The log messages for the tier-0 services running on this dedicated gateway will now have the context of the Sales project in the edge syslog.

Sharing NSX Resources

NSX multi-tenancy supports sharing of certain resources (objects) with specific projects or with NSX VPCs inside the projects.

Overview of Resource Share

An Enterprise Admin might want to share resources (objects) with projects so that they are available for consumption inside those projects. Resource sharing avoids the need for recreating the objects again in projects that require them, and thereby saves effort.

Resource sharing is done by creating resource shares. Each resource share is identified by a unique name. In a resource share, you can add the members (objects) that you want to share, and then choose one or more projects with whom you want to share with.

Project users can consume the shared resources in their projects to configure groups, firewall rules, and so on, to meet their networking and security requirements.

Resources can be shared from the default space or from the project view, or both. From the default space, you can share resources with projects or NSX VPCs. From the project view, you can share resources with NSX VPCs within the same project.

Sharing of resources from one project to other projects is currently not supported.

When you share a resource by creating a resource share, the child members of that shared resource are not shared with the target project. However, an Enterprise Admin or a Project Admin can control whether the child members of the shared resources are visible to the project and VPC users. By default, the child members of the shared resources are visible in the project and NSX VPCs. If required, the administrator can turn off the visibility of the child members.

Visibility of child members is applicable to the following resources in the resource share:

- Groups
- Segments

Shared resources are available in a **read-only** mode to the projects or NSX VPCs with which they are shared. In other words, shared resources cannot be modified by users in those projects. When you share resources with a project, the NSX VPCs in that project do not get access to the shared resources automatically. If required, you can share resources with all the NSX VPCs or specific NSX VPCs within a project.

In the default space, the following NSX objects (resources) are currently supported for adding as members in the resource share:

- Groups
- Services
- Context Profiles
- DHCP Profiles
- DAD Profiles
- ND Profiles
- DNS Zones
- IDS Profiles

- IKE Profiles
- IPSec Profiles
- DPD Profiles
- Service Certificates
- CRLs
- Segments
- Segment Profile - IP Discovery (API only)
- Segment Profile - MAC Discovery (API only)
- Segment Profile - QoS (API only)
- Segment Profile - Spoofguard (API only)
- Segment Profile - Segment Security (API only)

Note:

Adding the Segment Profiles to a given share is currently supported only in API. It is best practice to manage a share using Segment Profiles from API (or third party integration) to avoid overwriting shared Segment Profiles. Under the Project, the Segment Profiles are visible in UI and API for consumption by the tenant.

A subset of NSX features are currently supported for consumption in NSX VPCs. If you share resources from the default space with NSX VPCs, but the resources are not supported for consumption in the VPCs, then these resources get propagated to the NSX VPCs. However, VPC users cannot consume those shared resources.

For example, assume that an Enterprise Admin has shared an overlay segment from the default space with a project and all the NSX VPCs within that project. The system propagates the overlay segment to the project and all its NSX VPCs. However, the segment can be consumed only in the project, but not in the NSX VPCs because overlay segment is not supported in NSX VPCs.

The following system-wide user roles can share resources from the default space to projects or NSX VPCs within the projects:

- Enterprise Admin
- Network Admin
- Security Admin

The following user roles in a project can share resources from a project to NSX VPCs within the same project:

- Project Admin
- Network Admin
- Security Admin

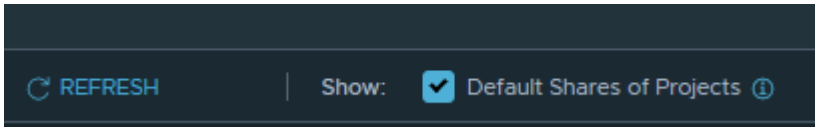
Overview of Default Shares of Project

When you add a new project, no user-created resources exist in that project. A new project has access to only the system-defined NSX resources that are shared by default through the default share. In other words, the default share is created automatically in the default space when NSX is deployed. The resources in the default share are available to all the projects and NSX VPCs in the system. You cannot edit the default share in the UI.

The default share is created by the system for internal use. The members of this share are system-defined resources, such as, Services, BFD Profiles, App IDs, and many more.

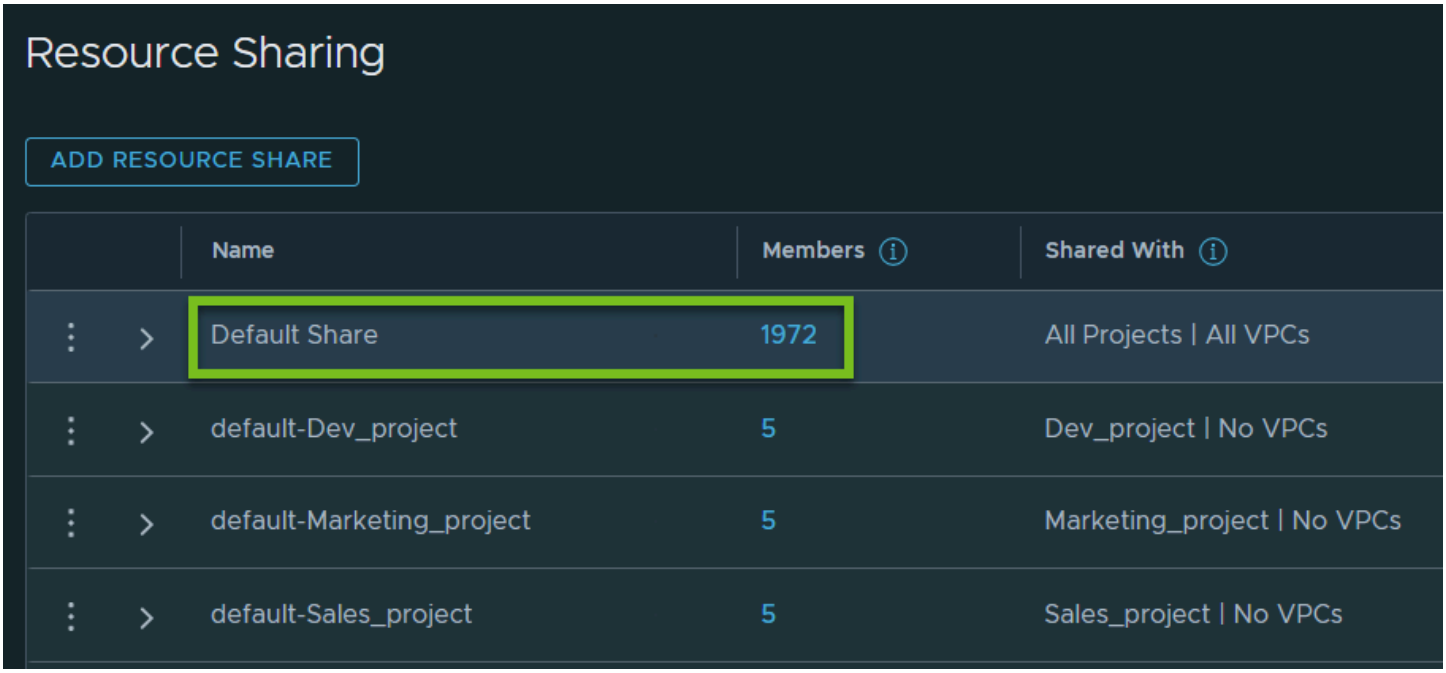
To view the full list of system-defined resources that are included in the default share, do these steps:

1. Ensure that you have selected the **Default** view from the **Project** drop-down menu.
2. Navigate to **Inventory** > **Resource Sharing**.
3. Click the **Default Shares of Projects** check box at the bottom of the **Resource Sharing** page.



4. Next to **Default Share**, click the count in the **Members** column.

For example:



Observe that in addition to the default share, which is available to all the projects and NSX VPCs, the system creates a default share automatically for each project. This project-specific default share is created for internal use of the system. The naming convention of the project-specific default share is:

`default-`*`Project-name`*

For example, the members of the project default share are:

- Tier-0 Gateways (if set during project creation)
- Edge Clusters (if set during project creation)
- Site (if edge cluster is set during project creation)
- Site Enforcement Point (if edge cluster is set during project creation)
- External IPv4 address blocks allocated to the project
- Default overlay transport zone of the system

Tier-0/VRF gateways and edge clusters are managed from the default space and cannot be edited in the project.

If NSX VPCs are added in the project, the system creates a default share automatically for each NSX VPC in the project. This default share contains the private IPv4 address blocks that are allocated to the NSX VPC. This VPC default share is created for internal use of the system.

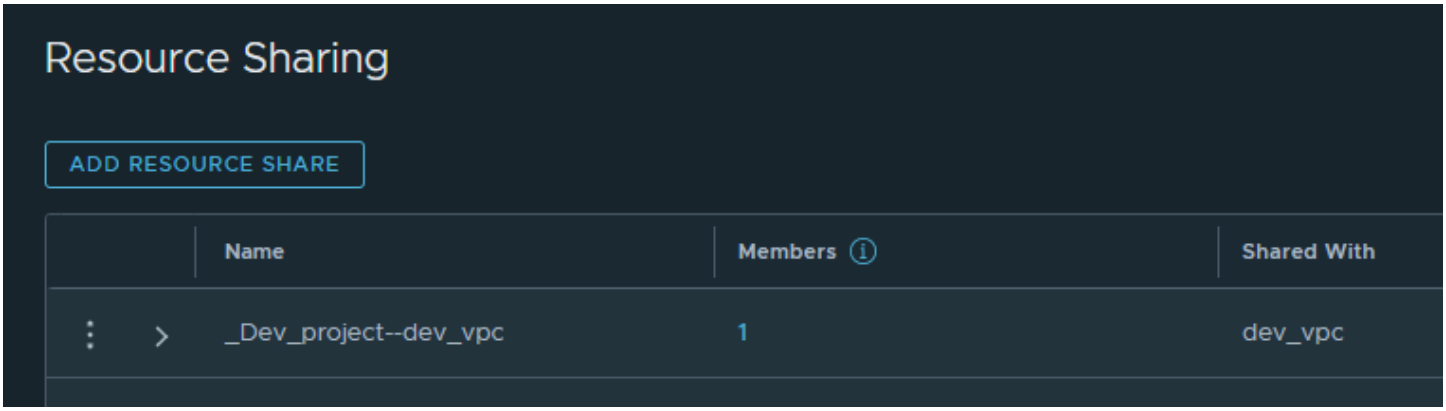
The naming convention of the VPC default share in the project is:

`_Project-name-VPC-name`

To view the VPC default share, do these steps:

1. Switch to the project view where the NSX VPC is added.
2. Navigate to **Inventory** > **Resource Sharing**.
3. Click the **Default Shares of Projects** check box at the bottom of the **Resource Sharing** page.

For example:



Use Case for Sharing Segments with Projects

When a segment is shared with a project, the child objects of the segment such as segment ports are visible in the project only when an Enterprise Admin has enabled the visibility of child members in the resource share. Otherwise, the project does not have visibility into the segment ports. Gateway interfaces of the shared segment are never visible in the project view.

Remember that sharing a segment does not share the VMs that are connected to the segment. Neither does it allow the project users to configure distributed firewall (DFW) policies on the VMs. Sharing a segment allows it to be visible in the project, and to connect the shared segment to the service interface of a tier-1 gateway in the project.

Project users cannot extend the distributed firewall security policy of the project to this shared segment. But they can apply gateway firewall security policies to the service interface of the project tier-1 gateway where the shared segment is connected.

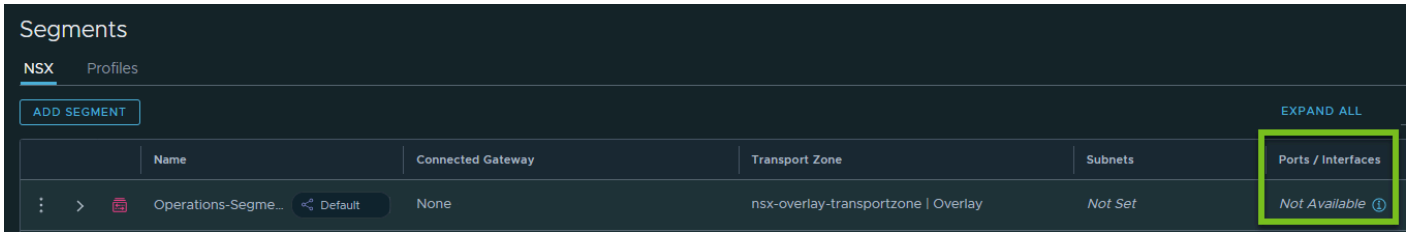
The workflow is as follows:

1. Assume that in the default space, there is a segment named "Operations-Segment". This segment could be an overlay segment or a VLAN segment.
2. The Enterprise Admin adds this segment to a resource share and shares it with project-1.
3. Switch to project-1 in NSX Manager, and connect the shared segment to the service interface of the project tier-1 gateway.
4. While you are in the project-1 view, create a gateway firewall policy and apply it to the tier-1 gateway service interface.

In the following example, you will learn about the information that is displayed on the **Ports/Interfaces** column of the shared segment in the project view when the visibility of child members for the shared segment is either turned on or off.

Example:

- Assume that in the default space, there is an isolated segment named "Operations-Segment" and a tier-0 gateway named "T-0-Operations". You have not added ports on this segment.
- On this tier-0 gateway, add a service interface and connect this interface to the "Operations-Segment".
- The Enterprise Admin adds this segment to a resource share and shares it with project-1. In the resource share, let us assume that the **Allow Viewing Member Hierarchy** toggle is turned off.
- Now, switch to project-1 in NSX Manager, navigate to the **Segments** page, and click the **Shared Objects** check box at the bottom of the page.
- Observe the properties of the shared "Operations-Segment". The **Ports/Interfaces** column shows value as `Not Available`. In other words, child members of the shared segment (that is, segment ports and service interfaces) are not exposed in project-1.



- Now, let us assume that the Enterprise Admin has turned on the **Allow Viewing Member Hierarchy** toggle in the resource share.
  In this case, when you switch to the project-1 view, and go to the **Segments** page, the service interface is still not exposed to project-1. But the **Ports/Interfaces** column of the shared segment now displays the value `0`. This value denotes only the count of ports on the shared segment. In this example, the count is zero because the shared segment contains no ports. The gateway interfaces on the shared segment are never exposed in the project view.

Use Case for Sharing Groups, Services, Context Profiles with Projects

Typically, project users might want to consume NSX objects such as Groups, Services, and Context Profiles that exist in the default space of the system to create firewall rules under their projects. Resource sharing avoids the need for project users to recreate these objects. The shared objects become available to the projects in a **read only** mode, and they cannot be edited inside projects.

Share NSX Resources with Projects or Virtual Private Clouds

Resource sharing avoids the need for recreating the resources (objects) in projects or NSX VPCs that require them, and thereby saves effort.

- Ensure that you have understood the concept of sharing resources. See Sharing NSX Resources.
- You must be assigned any one of these system-wide user roles to share resources from the default space to projects or NSX VPCs:
  - Enterprise Admin
  - Network Admin
  - Security Admin
- You must be assigned any one of these user roles in a project to share resources from the project to NSX VPCs within the same project:
  - Project Admin
  - Network Admin
  - Security Admin

Resources can be shared from the default space or from the project view, or both. From the default space, you can share resources with projects or NSX VPCs. From the project view, you can share resources with NSX VPCs within the same project.

Shared resources are available in a **read-only** mode in the projects or NSX VPCs with which they are shared.

1. From your browser, log in to NSX Manager at https://nsx-manager-ip-address.
2. To share resources from the default space to projects within the organization, do these steps:
   a) Ensure that you are in the **Default** view.
   b) Navigate to **Inventory** > **Resource Sharing**.
   c) To create a resource share, click **Add Resource Share**.
   d) Enter a name for the resource share.
   e) By default, the **Allow Viewing Member Hierarchy** toggle is turned on for the resource share.

      When it is on, it means that the child members of all the objects that you add in this resource share will be visible to users in projects and NSX VPCs where this resource share is available. Visibility of child members is applicable to the following objects in the resource share:

      - Groups
      - Segments

      If you want to turn on or turn off the visibility of the child members for each object individually in the resource share, you must create the resource share by using the API. The UI does not support this granular level of control.

      For example, let us say that in step (g) of this procedure, you have added group A and segment B in the resource share. For group A, you want to turn on the visibility of the group members, but for segment B, you want to turn off the visibility of the child members of this segment, such as ports. To do this mixed configuration, you must create

the resource share by using the API. Later, when you want to edit this resource share, you must preferably edit it with the API. The reason is that from the UI, the toggle gets applied to all the objects in the resource share.

f) Click **Set**.

g) In the **Set Members** window, select the objects you want to add in the resource share.

Note:
Before adding segments in your resource share, make sure that you understand the typical use cases for sharing segments with a project. For more information, see Sharing NSX Resources.

h) Click **Apply**.

i) In the **Shared With** column, click **Set**, and then choose any one of these options:

Option	Description
Share with all Projects	Select this option to share resources with all the projects in your organization.
Share with selected Projects	Select this option to share resources with specific projects in your organization.

j) Optional: Click **Next** and turn on/off the toogle to share resources with all the NSX VPCs or none of the NSX VPCs in the selected projects.

The toggle is editable only when you select the **Share with selected Projects** option in the previous step. If you select the **Share with all Projects** option, the system will share resources with all the NSX VPCs in all the projects, by default. This default setting is currently not editable.

When you are sharing resources from the default space, the system does not allow you to choose specific NSX VPCs in the selected projects. If you want to share resources with specific NSX VPCs, you can switch to each project view and create the resource share in that view. To learn more, see the instructions in step 3.

k) Click **Apply**, and then click **Save**.

3. To share resources from a project to specific NSX VPCs within the same project, do these steps:

a) Select a project from the **Project** drop-down menu.

b) Navigate to **Inventory** > **Resource Sharing**.

c) To create a resource share, click **Add Resource Share**.

d) Enter a name for the resource share.

e) Click **Set**.

f) In the **Shared With** column, click **Set**, and then choose any one of these options:

Option	Description
Share with all VPCs	Select this option to share resources with all the NSX VPCs in the project.
Share with selected VPCs	Select this option to share resources with specific NSX VPCs in the project.

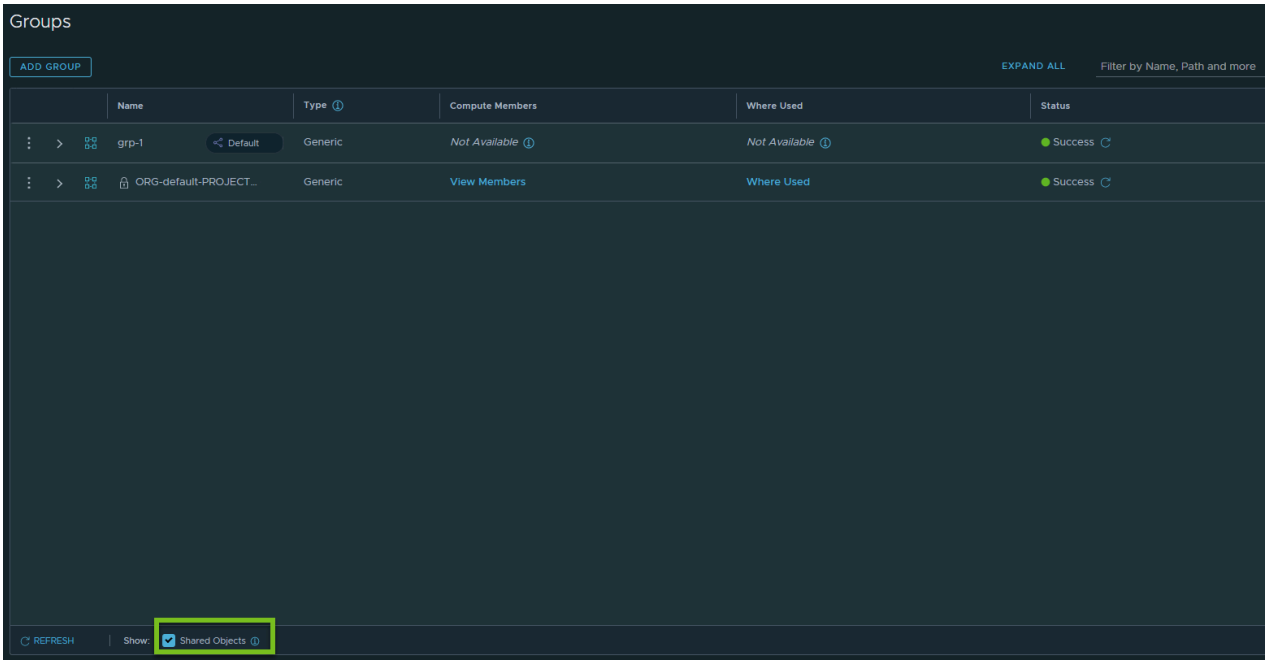
g) Click **Apply**, and then click **Save**.

Resource Share

Let us assume that an Enterprise Admin has shared a group from the default space with project-1 in your organization.

To check whether this group is available for consumption in project-1, do these steps:

1. Log in to NSX Manager with the credentials of any user that has access to project-1.
2. Ensure that **project-1** is selected from the **Project** drop-down menu.
3. Navigate to **Inventory** > **Groups**.
4. Click the **Shared objects** check box at the bottom of the **Groups** page.
   By default, this check box is not selected.
   For example:



5. Observe that the shared group is listed on the **Groups** page.
   The following pill-shaped icon is displayed next to the group name to indicate that the group is owned by the default space and shared with the project.



System Monitoring

You can monitor the health and performance of the NSX environment.

Retrieve System Health Metrics Using APIs

In addition to using the system health dashboard, you can use *NSX* API methods to get extended health metrics for NSX Manager nodes, edge nodes, and host transport nodes.

The following table provides some example API requests to retrieve extended health metrics. For detailed information about each method, see the *NSX REST API Guide*.

API Request Example	Description
`GET https://<nsx-mgr>/api/v1/systemhealth-extended/edge-cluster`	Retrieves health metrics for all edge clusters.
`GET https://<nsx-mgr>/api/v1/systemhealth-extended/edge-cluster?cluster_id={cluster-id}`	Retrieves health metrics for the edge cluster with ID {cluster-id}.
`GET https://<nsx-mgr>/api/v1/systemhealth-extended/edge-transport-nodes`	Retrieves health metrics for all edge transport nodes.
`GET https://<nsx-mgr>/api/v1/systemhealth-extended/edge-transport-nodes?node_id={node-id}`	Retrieves health metrics for the edge transport node with ID {node-id}.
`GET https://<nsx-mgr>/api/v1/systemhealth-extended/host-transport-nodes`	Retrieves health metrics for all host transport nodes. Use the `node_id` query parameter to fetch metrics for a single host transport node.

API Request Example	Description
`GET https://<nsx-mgr>/api/v1/systemhealth-extended/management-cluster`	Retrieves health metrics for the NSX Manager cluster.
`GET https://<nsx-mgr>/api/v1/systemhealth-extended/management-nodes`	Retrieves health metrics from all NSX Manager nodes, along with the high-level status of services on each node. Use the `node_id` query parameter to fetch metrics for a single NSX Manager node.
`GET https://<nsx-mgr>/api/v1/systemhealth-extended/{node_id}/services/{service_name}`	Retrieves health metrics for the {service_name} service on the NSX Manager node with ID {node_id}.
`GET https://<nsx-mgr>/api/v1/systemhealth-extended/metadata`	Retrieves metrics metadata for all the extended health metrics APIs.

Monitor NSX Edge Nodes and Gateways

NSX application supports collection and storage of data up to one year. With this feature, along with the point-in-time data, you can also view time series metrics, such as CPU usage memory, disk usage, packets per second, bytes per second, and packet drop rate, for edge nodes and gateways. You can view the metrics by invoking time series metrics APIs. A few of these metrics are also available on NSX Manager interface along with the point-in-time data.

Ensure that you have deployed NSX Application Platform (NAPP). For more details about deploying NSX Application Platform, see the *Deploying and Managing NSX Application Platform* guide. The Metrics feature is enabled by default when you deploy NSX Application Platform.

If you view statistics without deploying NSX Application Platform or without enabling the Metrics feature, the system displays a message to deploy NSX Application Platform and to enable the Metrics feature.

Starting with NSX 4.1.2, the Physical Port Xstats and verbose metrics are also available in API. Also, the Edge Agent health status, DP thread status, and LR/SR HA status are now available in NAPP.

For more information about APIs and CLIs, see *NSX API Guide* and *NSX Command-Line Interface Reference*.

You can also view time series metrics for VPN. The VPN time series metrics is not available on NSX Manager. You can use time series APIs to view data such as policy based VPN statistics and route based VPN statistics, which includes metadata about number of tunnels configured and how many tunnels are up or down at a certain time.

Using time series metrics, you can monitor the trend in key performance indicators, detect anomalies, perform before and after analysis, and get the historical context which can help in troubleshooting. Based on your role, you can view metrics of only those objects for which you have the authorization.

You can use APIs to retrieve metrics about Edge nodes and gateways. For more information, see Monitoring Security Statistics and the section "How to Use Metrics APIs" in the *Deploying and Managing the VMware NSX Application Platform* guide.

On the NSX Manager interface, the time series metrics is displayed as charts with respective keys and time filter on last one hour, last 24 hours, last one week, last one month, and last one year. You can view the following information for edge nodes:

- CPU cores allocated
- Alarms
  - Edge node - Overall alarm count
  - CPU - Alarms for Datapath CPU and Service CPUs
  - Disk - Overall disk alarms and alarms for each partition
  - Memory - Overall memory alarms and alarms for each memory pool
- Uptime
- CPU
  - System Load - You can view current load average and average system load trend.
  - Datapath CPU - Number of datapath CPU cores and their usage details, which include the average usage of all cores and the highest usage among cores. You can view current utilization, highest utilization for a single core, and

average usage trend. You can also view the average number of packets processed across all available datapath cores for a given edge transport node and which core has processed the highest number of packets.
  - Service CPU - Number of service CPU cores and their usage details, which include the average usage of all cores and highest usage among cores. You can view current utilization, highest utilization for a single core, and average usage trend.
- Memory
  - System Memory - Utilization of RAM on edge node. You can view current utilization and average usage trend.
  - Datapath Memory - Includes heap memory, memory pool, and resident memory. You can view current usage, current pool usage, and average usage trend.
  - Memory Pools - List of all memory pools along with their description and usage values. The memory pools are:

Name	Description
jumbo_mbuf_pool	Packet Pool for Jumbo Frame Used by Ipsec Crypto Device
common_mbuf_pool	Datapath Common Packet Pool
sp_pktmbuf_pool	Datapath Slowpath Packet Pool
fw_mon_msg	Stateful Service Sync Message Pool
vxstt4_frag_q	Vxstt Fragment Pool for Reassembly
pfstatepl3	Stateful Service State Pool
pffqdnippl	Stateful Service FQDN to IP Map Pool
pffqdnsyncpl	Stateful Service FQDN SYNC Pool
pffqdndnpl	Stateful Service FQDN Internal Pool
pfdnsdnpl	Stateful Service FQDN Internal Pool
pfpktpl3	Stateful Service Fragmented Packet Pool
pfsyncmbufpl3	Stateful Service SYNC Pool
pf_fp_rule_node	Stateful Service Rule Node Pool
pf_fp_root_rule_node	Stateful Service Root Rule Node Pool
pf_tb_root_rule_node	Stateful Service Fastpath Root Table Node Pool
pfa_intattr_pl3	Stateful Service Integer Attribute Pool
pfa_attrconn_pl3	Stateful Service Attribute Connection Pool
pfa_ctx_pl3	Stateful Service Context Pool
pfa_key_ace_pl3	Stateful Service Integer Attribute Key Pool
pfa_value_ace_pl3	Stateful Service Integer Attribute Value Pool
lb_pkt_pl3	Load Balancer Temp Packet Cache Pool

- Disk
  - Total disk usage for all ext4 disk partitions and the list of RAM disk and disk partitions. Also, the available free space of each partition. You can view overall GB used, read latency, write latency, current usage, average usage trend by partition, and average disk latency trend.
- Transport node status
- Network interface
  - To view statistics of any network interface, click the graphic icon. You can view cumulative statistics like total packets and dropped packets and time series metrics like network utilization trend and dropped packets trend. You can also view Error chart for a gateway.
- NAT rules statistics

1. To view metrics of an edge node, perform the following steps:
    1. With admin privileges, log in to NSX Manager.
    2. Select **System** > **Fabric** > **Nodes**.

3. Click the **Edge Transport Nodes** tab.
4. Click the edge node that you want to monitor.
5. Click the **Monitor** tab.

   Usage information for CPU, memory, and disk is displayed, as well as the node status, network interfaces, and NAT rule statistics.

2. To view gateway or interface metrics, perform the following steps:
   1. With admin privileges, log in to NSX Manager.
   2. Select **Networking** > **Network Topology**.
   3. Click the required gateway. The system displays a gateway details window.
   4. Under **Statistics**, select the linked gateway or the interface link for which you want to view the metrics.

Dynamic Plugins

A dynamic plugin is a customized plugin that you can create for any supported transport node, such as ESX host, to check the node's health.

A dynamic plugin can be installed at runtime. It performs the following functions:

- Write a system log for the affected transport node.
- Execute a command or CLI through the `run_command()` function.
- Read existing metrics.
- Export data to wavefront (only in a VMware Cloud environment).

You can create a dynamic plugin if you have expertise in managing NSX. The VMware Support can also create a dynamic plugin.

Security management

After you have created the plugin, you must submit it to the VMware engineering team that performs a code review of the plugin. After review completion, the engineering team submits the plugin, its review test result, and code changes in a GIT repository for validation. If the dynamic plugin passes the validation, it is committed and a product build is triggered on the GIT repository to create a new build to publish all the dynamic plugins. The following GIT repository details are used:

- GitLab: `https://github-vcf.devops.broadcom.net/vcf/nsbu-sha-plugin`
- Product: `nsx-sha-plugins`
- Gitreview supporting: Performed by VMware team

You can get the required plugin from the published files of the build. When you upload a signed plugin to the management plane, the management plane uses a public key to verify the signature and confirm that this plugin is valid. After the management plane validates the plugin, it pushes the files of the plugin to the destination hosts through the secured channels between the management plane and the Central Control Plane (CCP) and between CCP and hosts.

When you restart the System Health Agent (SHA) instance, it gets the plugin files from the management plane again. Since all the files are published through secured channels and no temporary files are used, the risk that hackers can modify scripts is prevented.

Also, to prevent risks of harmful code, SHA uses RestrictedPython to check the plugin python script before executing the script.

Version management

A plugin might be based on a command or a tool that is not supported in later versions of NSX, so each dynamic plugin must define the supported NSX version in the `manifest.yml` file. The version should be a REGEX string for all the supported versions. SHA on the host-side checks the version of dynamic plugins and runs only the REGEX matched ones.

Recommended version management policies are:

- Define the supported NSX version of a major release.

Considering that most of the commands and tools do not change between minor releases in the same major release, the following method is the suggested way to define the version for all minor releases.

For example,

```
version: ^2\.5\.[0-9.]+ <== The custom plugin supporting all NSX 2.5 releases
```

- When a new major release is published, the VMware engineering team must review all the submitted dynamic plugins.
- The plugin writer must update the script when the related commands or tools change.

Install a dynamic plugin

The transport node or the edge node on which the plugin is installed must have a minimum of 30-MB memory space. Also, note that you can install only up to 10 plugins. Once the plugin count reaches 10, any further installation of a plugin will fail.

To install the plugin, perform the following tasks:

1. Create the dynamic plugin files in the GIT repository. For more information about the plugin files, see the section *Dynamic plugin files*.
2. Trigger the product build in the GIT repository to generate the zipped package of the dynamic plugin files and download the package.
3. Create the dynamic plugin by using the following API with the POST method.
   `https://<manager_ip>/api/v1/systemhealth/plugins`
4. Upload the plugin zipped package to the management plane using the following API with the POST method. The management plane extracts the uploaded file and perform the required validation.
   `/systemhealth/plugins/<plugin-id>/files/<file-name>/data`

   Note:
   The maximum size of the plugin zipped file is 500k.
5. Create a node group with the required transport nodes as members by using the following API with the POST method.
   `/<manager_ip>/api/v1/ns-groups`
6. Apply the plugin profile to the node group by creating a new service config by using the following API. The service config framework sends the plugin content to the node group.
   `https://<manager_ip>/api/v1/service-configs`

For more information about APIs, see the *NSX API Guide* documentation.

Get the plugin status

Once the dynamic plugin is running, it automatically uploads the status to the management plane through the existing message channel. The management plane aggregates the plugin status information and store it into the database. To get the status of all plugins on each node, use the following API with the GET method.

`https://<manager_ip>/api/v1/systemhealth/plugins/status/<transport_node_id>`

Request Example:

```
GET https://<manager_ip>/api/v1/systemhealth/plugins/status/a257b981-1a1c-4b95-b16c-8646
```

Response Example:

```
{
 "result_count":1,
 "results": [
 {
 "id": "72e1bd4b-6df6-42d0-9c59-a1c31312c9f1",
 "name": "health-check-compute",
 "status": "NORMAL",
 "detail": ""
 }
 ]
 }
```

Uninstall a dynamic plugin

To uninstall a plugin, remove the service config by using the following API.

`https://<manager_ip>/api/v1/service-configs/<service_config_id>`

Other APIs for managing the plugins

The following table lists APIs to manage dynamic plugins. For more information about APIs, see the *NSX API Guide* documentation.

Task	Method	API
Delete a plugin	DELETE	`/systemhealth/plugins/<plugin-id>`
Create a system health profile	POST	`/systemhealth/profiles`
Watch the plugin status	GET	`/systemhealth/plugins/status/<node-id>`

Enable the plugin		Enabling a plugin is a two-step process as follows:
		1. Use the following API to set the `enabled` property to `true` or `false`.
		`https://<manager_ip>/api/v1/systemhealth/profiles/`
		2. Use the following API to apply the SHA profile to the NS group.
		`https://<manager_ip>/api/v1/service-configs`

Change the plugin interval	POST	Changing the plugin interval is a two-step process as follows:
		1. Use the following API to set the `config` property.
		`https://<manager_ip>/api/v1/systemhealth/profiles/`
		2. Use the following API to apply the SHA profile to the NS group.
		`https://<manager_ip>/api/v1/service-configs`

Dynamic plugin files

A dynamic plugin comprises the following files:

- Install specification file
  The install specification file, `manifest.yml`, contains the following information for System Health Agent:
  - Plugin structure
  - Constraints if any
  - How to install and use the plugin
  - Security restrictions for the health check script. For example, permissions the script has and files that the script can access.

The following table lists fields that are specified in a `manifest.yml` file.

Name	Description	Required/Optional	Example
classes	Specifies classes needed in the plugin script. The classes must be specified in the following format. '<module_name>.<class_name>'	Optional	`classes: ['datetime.datetime','dateti`
modules	Specifies modules needed in the plugin script.	Optional	modules: ['random', 'math']

plugin	Specifies the plugin structure as follows:
	config: config file name
	script: script file name
		Required	`plugin:`
			`config:`
			`plugin.cfg.yml`
			`script: plugin.py`

version	Specifies the NSX versions on which this plugin can be installed.	Required	`version: '^3\.1\.[0-9.]+'`
node_type	Specifies NSX node types where this plugin can be installed. The available node types are: • nsx-esx • nsx-bms • nsx-edge	Required	`node_type: ['nsx-esx']`
metrics	Specifies metrics which can be consumed in the plugin script.	Optional	`metrics: ['nsx.host.host-metrics']`
precondition	Specifies precondition for the plugin. The available precondition is wavefront. **Note:** This field is applicable only in a VMware Cloud environment.	Optional	`precondition: ['wavefront']`

Do not use the following built-in modules:

- os
- subprocess
- sys
- multiprocessing
- importlib

The following table lists the interfaces that you must use in place of built-in functions of the respective modules. These interfaces are system provided. You can use them directly without specifying their module/class in the `manifest.yml` file.

Module	Built-in function	Substitute interface
datetime	datetime.date.strftime(self, fmt)	datetime_date_strftime(dt, fmt) :param dt: date instance :param fmt: format string
datetime	datetime.date.today()	datetime_date_today()

Module	Built-in function	Substitute interface
sys	sys.getrecursionlimit()	sys_getrecursionlimit()
sys	sys.getrefcount(object)	sys_getrefcount(object)
sys	sys.getsizeof(object, default)	sys_getsizeof(object, default)
sys	sys.maxsize	sys_maxsize
sys	sys.path	sys_path

Sample of a `manifest.yml` file.

```
# specify classes needed in plugin script
classes: ['datetime.datetime','datetime.date']
# specify modules needed in plugin script
modules: ['random', 'math']
# plugin structure
plugin:
 config: plugin.cfg.yml
 script: plugin.py
# specify nsx versions on which this plugin can be installed
version: '^3\.1\.[0-9.]+'
# specify nsx node type where this plugin can be installed
node_type: ['nsx-esx']
# specify metrics which can be consumed in plugin script
metrics: ['nsx.host.host-metrics']
# specify precondition for plugin
precondition: ['wavefront']
```

- Default profile file

  The default profile file, `plugin.cfg.yml`, is used to configure plugin behavior, such as the execution frequency of the health check script. To change the default configurations, you can create a SHA profile for a specific dynamic plugin and apply it to transport node through NS group by using the management plane to CCP to NestDB channel.

  The following table lists fields that are specified in a `plugin.cfg.yml` file.

Name	Description	Required/Optional	Example
CHECK_INTERVAL	Specifies the default interval in seconds for plugin script execution.	Required	CHECK_INTERVAL: 20
ENABLE	Specifies whether the plugin is enabled by default.	Required	ENABLE: true

Sample of a `plugin.cfg.yml` file.

```
# Required field - default interval (unit: second) between plugin script executions.
CHECK_INTERVAL: 20

# Required field - whether plugin is enabled by default
ENABLE: true

# Plugin user can add other fields as below if needed to control plugin script logic.
EXPORT_DATA: true
```

- Health check script

  A health check script file, `plugin.py`, contains a python script to check the health status of a transport node.

The following table lists system-defined variables and functions that can be used and data that can be read in a `plugin.py` file.

Variable/Data/Function	Description	Type	Example
logger	Writes log information in syslog. The existing system-defined variable, logger, can be used directly in the plugin script. The output log is prefixed with the plugin name and id as shown in the following sample output. `2020-10-28T10:47:43Z nsx-sha: NSX 2101378 - [nsx@6876 comp="nsx-esx" subcomp="nsx-sha" username="root" level="INFO"] [name:hl-esx-002-04] [id:a3eb14f1-d185-4bc7-bfaa-6cf888bbeb22] dynamic plugin - not export data`	Variable	`logger.info("this is a demo log")`
data_store	The existing system-defined dictionary that is used to fetch system-provided data. For example, profile, metric, and host_id.	Variable	`profile = data_store['profile']`
profile	Profile data is a dictionary parsed from the default profile (plugin.cfg.yml) or the effective SHA profile (user-applied through the NSX Manager API) that is read from data_store. It has the following format: {'ENABLE': True, 'CHECK_INTERVAL': 20, 'EXPORT_DATA': True}	Data	`profile = data_store['profile']`

metric	Metric is a dictionary with 'value' and 'timestamp' that is read from data_store. It has the following format:
	data_store['metrics'][<metric_name>]
	Where,
	The first key must be 'metrics'.
	The second key is an existing metric name.
		Data	`metric = data_store['metrics']['nsx.host.host-metrics']`
			`metric is: {`
			`'value':XXX, <== the collected data of the metric`
			`'timestamp': XXX <== timestamp of the data collected`
			`}`
			Note: The first run of plugin might not return a metric as currently a metric is collected asyncly with the plugin running, so the metric might not have been collected in the first run of the plugin.

Variable/Data/Function	Description	Type	Example
host_id	Host_id is an instance of class uuid.UUID that is read from data_store.	Data	`host_id = data_store['host_id']`
run_command	This function runs commands in a list format. It has the following format. `run_command(cmd, timeout=4)` Where, • *cmd*: Commands to be executed. Must be in the list format as in the example. • *timeout*: Timeout for waiting for the command result. Default timeout is 4s. Timeout should not be set larger than 20s. This function returns the command execution result.	Function	`cmd = ['nsxdp-cli', 'ipfix', 'settings', 'granular', 'get', '--dvs-alias', 'nsxvswitch', '--dvport=dafa09ca-33ed-4e04-ae3d-1c53305d5fe6']` `res = run_command(cmd)`
Exportdata	This function exports data to wavefront. Currently, a dynamic plugin supports export to wavefront only. **Note:** This function is applicable only in a VMware Cloud environment. It has the following format: `Exportdata: ExportData(data={}, exports=[], source=host_uuid)` Where, *data*: data to be exported; data should be in dictionary format as example. *exports*: destination list for export. In HL, only support wavefront in destination. It is required. *source*: source string for export. It is useful only for wavefront destination. It is optional, the default value is NSX host_uuid. The function does not return any value.	Function	`Exportdata(data={'esx.plugin {'stats': {'gc-esx-001': data}}}, exports=['wavefront'])`

Sample of a `plugin.py` file.

```
def report_test_data(edge_service_status):
    if edge_service_status == 'running':
        data = 2
    else:
        data = 3
```

```
    # examples on how to report data to wavefront.
    Exportdata(data={'esx.plugin.stats': {'stats': {'esx-dynamic-plugin-001': data}}},
 exports=['wavefront'])


def run():
    # examples on how to write log.
    logger.debug("this is a debug message!")
    logger.info("this is a demo message!")

    # examples on how to use specified module in manifest. Take 'random' as an example.
    s_res = random.randint(1,10)
    logger.warn("random.randint(1,10)=%s", s_res)

    # examples on how to use specified class in manifest. Take 'datetime' and 'date' as an example.
    logger.info('date.ctime(datetime.now()):{}'.format(date.ctime(datetime.now())))

    # examples on how to run cmd via interface run_command
    cmd = ['nsxdp-cli', 'ipfix', 'settings', 'granular', 'get', '--dvs-alias', 'nsxvswitch', '--
dvport=dafa09ca-33ed-4e04-ae3d-1c53305d5fe6']
    c_res = run_command(cmd)
    logger.error("run_command(cmd) res:%s", c_res)

    # examples on how to read existing metrics from data_store
    m_res = data_store['metrics']['nsx.host.host-metrics']
    # examples on how to read effective profile from data_store
    profile = data_store['profile']
    logger.error("data_store['metrics']['nsx.host.host-metrics']:%s, profile:%s", m_res, profile)

    # examples on how to read host_id from data_store
    host_id = data_store['host_id']
    logger.info('host_id:{}'.format(host_id))

    if profile['EXPORT_DATA']:
        report_test_data('running')
        logger.info("dynamic plugin - exported data to wavefront")
    else:
        logger.info("dynamic plugin - not export data ")

    # examples on how to use substitute interfaces for sys.
    logger.info("sys_path:{}".format(sys_path))
    logger.info("sys_getsizeof(1):{}".format(sys_getsizeof(1)))
    logger.info("sys_getrefcount(cmd):{}".format(sys_getrefcount(cmd)))
    logger.info("sys_maxsize:{}".format(sys_maxsize))
    logger.info("sys_getrecursionlimit():{}".format(sys_getrecursionlimit()))

    # examples on how to use substitute interfaces for datetime.
    today = datetime_date_today()
    logger.info("datetime today:{}".format(today))
    logger.info("datetime_date_strftime now:{}".format(datetime_date_strftime(datetime.now(), '%H:%M')))
    logger.info('date.ctime(today):{}'.format(date.ctime(today)))


run()
```

Monitoring NSX Host Transport Nodes

The system collects statistics from each ESX host transport node and exports them to NSX Manager. These statistics can help you monitor the overall health of the host transport nodes.

The collection of health monitoring statistics is activated by default for all the host transport nodes in the system.

Overview of NSX System Health Agent Monitor

Learn about the basic concepts of a SHA monitor, which is a component that is introduced in the System Health Agent (SHA) framework.

SHA is the distributed system health agent in NSX. This framework monitors statistics (metrics) of various NSX components, such as ESX hosts, NSX edges, NSX Manager appliance, and so on. SHA monitors the component statistics, analyzes them, alerts abnormality, and triggers remediation actions when specific conditions are met.

Monitor

A monitor is a component of the SHA framework that monitors various statistics (metrics) about the system health and takes corresponding action. For instance, it exports statistics to a destination for users to view the system health, triggers alarms to notify users, and dumps statistics to a file for offline analysis. A monitor replaces the SHA plug-ins, which were introduced in an earlier NSX release. A monitor provides detailed descriptions of the statistics so that users know the purpose of each statistic, and how to control the monitor when required.

For example, the `esx-obsrv-stats-monitor` contains descriptions of all the observability statistics that are collected from an ESX host transport node.

Currently, a monitor supports exporting statistics to a single destination, which is the NSX Manager.

Monitor profile

A monitor profile helps you to control the monitor by configuring a set of properties.

For example, some configuration properties in the monitor profile are:

- `enable`
- `check_interval`
- `applied_to_group_paths`

Only a few configuration properties are listed here. To learn about all the configuration properties in a monitor profile, see the schema of the SHA monitor profile in the *NSX API Guide* at:

```
GET https://<nsx-mgr>/policy/api/v1/infra/sha/monitors/<monitor-id>/profiles/<profile-id>
```

You can access the monitor and monitor profiles only with NSX APIs. Currently, these components are not exposed in the NSX Manager UI.

To get the list of all monitor IDs, use the following API:

```
GET https://<nsx-mgr>/policy/api/v1/infra/sha/monitors
```

To get the list of all profiles for a specific monitor, use the following API:

```
GET https://<nsx-mgr>/policy/api/v1/infra/sha/monitors/<monitor-id>/profiles
```

> **Note:**
> In a multi-tenant NSX deployment, users in projects and NSX VPCs cannot consume the monitor and monitor profile APIs. These APIs are currently available for consumption only to users in the default space.

Each monitor has a system-created default profile. For instance, the default profile of the `esx-obsrv-stats-monitor` is applied to all the host transport nodes in the system where this monitor runs. When you update the configuration of the default profile, the change affects all the nodes. The default profile cannot be deleted.

You can create a user-defined (custom) profile to override the default profile for a specific group of host transport nodes. If multiple profiles are applied to the monitor on a host transport node, the last applied profile will override the other profiles. The default profile has the lowest priority.

Regardless of whether you are configuring the default profile or a custom profile of the monitor, certain properties in the profile cannot be edited.

For example:

- The default profile of the `esx-obsrv-stats-monitor` is applied to the system-created default group, which contains all the host transport nodes in the system. You cannot edit the `applied_to_group_paths` property of the default profile.
- You cannot deactivate the `esx-obsrv-alarms-monitor` by setting the `enable` property in the profile to false. This monitor reports alarms in the system and is always activated.

Types of Monitors

The SHA framework contains two main types of monitors.

Metric-exporter monitors	These monitors are used to export metrics (statistics) to a destination, such as NSX Manager. For example, `esx-obsrv-stats-monitor`. For a metric-exporter monitor, the system supports only one user-defined (custom) profile.
Standard monitors	These monitors are used to do tasks other than exporting metrics, such as reporting alarms to NSX Manager, dumping statistics to a file, and so on. For example, `esx-obsrv-alarms-monitor, esx-obsrv-segment-stats-file-dump-monitor`. For a standard monitor, there is no restriction on the number of user-defined (custom) profiles that you can create.

A monitor defines a high level category of statistics (metrics) that it controls. The statistics in a monitor are organized in multiple sub-categories. A sub-monitor is defined to control individual sub-categories. By default, the configuration that is defined for the parent monitor is applied to all the sub-monitors in the parent monitor. If required, you can edit the configuration of the sub-monitors.

For example:

The `esx-obsrv-stats-monitor` is a metric-exporter type of monitor. It contains two sub-monitors of metric-exporter type, which are called `esx-obsrv-datapath-traffic-stats-monitor` and `esx-obsrv-datapath-infra-stats-monitor`.

The advantage of categorizing the statistics in a monitor with multiple sub-monitors is that you can control all sub-monitors together with the parent monitor. You don't need to run separate API calls for the individual sub-monitors to configure their properties.

Monitor Statistics of NSX Host Transport Nodes Using APIs

With NSX APIs, you can monitor both point-in-time values and real-time values of host transport node statistics.

The system uses the `esx-obsrv-stats-monitor` to monitor the health of ESX host transport nodes.

Statistics are collected per host transport node. The values of the statistics are not aggregated at the level of each host cluster.

Point-in-time values of the host transport node statistics are collected from each host transport node periodically (default is every 300 seconds) and cached by the system. The last updated cached values are fetched from the host and returned by the API.

For debugging purposes, NSX users can monitor the real-time values of the host transport node statistics. System fetches the statistics on a real-time basis from the data path kernel modules that are running on the host transport nodes.

With NSX APIs, you can view the following types of statistics for host transport nodes:

- Packet statistics (packet_stats)
- Fast path system statistics (fast_path_sys_stats)
- Platform packet statistics (platform_packet_stats)
- Platform CPU usage statistics (platform_cpu_usage_stats)
- Fast path lcore usage statistics (fast_path_lcore_usage_stats)

In the API response, the statistics are organized based on the datapath modules that are collecting the statistics from the host transport nodes. For a description of the statistics, see NSX Host Transport Node Statistics.

The procedure in this documentation explains the API workflow for monitoring the point-in-time values and real-time values of the host transport node statistics. The steps in the procedure mention only the API URIs. For a detailed information about the API schema, parameters, and sample API response or request payload, refer to the *NSX API Guide*.

To learn about using the NSX Central CLI or the ESX host CLI for viewing the transport node statistics, refer to the *NSX Command-Line Interface Reference*.

1. Run the following API to view the descriptions of all the statistics in the `esx-obsrv-stats-monitor`:

   ```
   GET https://<nsx-mgr>/policy/api/v1/infra/sha/monitors/esx-obsrv-stats-monitor
   ```

2. Run the following API to retrieve the list of profiles of the `esx-obsrv-stats-monitor`:

   ```
   GET https://<nsx-mgr>/policy/api/v1/infra/sha/monitors/esx-obsrv-stats-monitor/profiles
   ```

   The API response displays the default profile and a user-defined profile (if created) for this monitor. The default profile is created by the system and it is applied to the system-created default group. The default group contains all the ESX host transport nodes in the system. The default profile is activated by default on all the host transport nodes.

   To view the configuration of the default profile, run the following API:

   ```
   GET https://<nsx-mgr>/policy/api/v1/infra/sha/monitors/esx-obsrv-stats-monitor/profiles/default-profile
   ```

3. Optional: Run the following API to edit the configuration of the default profile:

   ```
   PATCH https://<nsx-mgr>/policy/api/v1/infra/sha/monitors/esx-obsrv-stats-monitor/profiles/default-profile
   ```

   Copy the GET API response of the default profile and paste it in a text editor. Edit the configuration values of the profile. Finally, submit the edited configuration in the request body of this PATCH API.

   Sample Scenario: Let us assume that you want to apply a user-defined profile to a specific set of hosts (say A and B), but apply the default profile to all the remaining hosts in the system. In this case, you must create a custom group, which contains these two host transport nodes (A and B). Groups with only host transport node members can be created using NSX API. The UI does not support this functionality. After you have created the desired group, use this group in the `applied_to_group_paths` property of the user-defined profile.

   Use the same PATCH API, as mentioned earlier, to create the user-defined profile.

   For example, the following API creates a user-defined profile with ID "profile-1":

   ```
   PATCH https://<nsx-mgr>/policy/api/v1/infra/sha/monitors/esx-obsrv-stats-monitor/
   profiles/profile-1
   ```

   When this API is successful, the user-defined profile (profile-1) overrides the default profile on host transport nodes A and B. For all the remaining host transport nodes in the system, the default profile is applied.

4. Optional: Run the following API to read the monitor status on a specific host transport node:

   ```
   GET https://<nsx-mgr>/policy/api/v1/infra/sha/monitors/esx-obsrv-stats-monitor/status?
   transport_node_id=<node-id>
   ```

   To get the *`node_id`* of host transport nodes, run the following API:

   ```
   GET https://<nsx-mgr>/policy/api/v1/infra/sites/default/enforcement-points/default/host-transport-nodes
   ```

5. Run the following API to view the point-in-time values of the host transport node statistics for a specific node:

   ```
   GET https://<nsx-mgr>/policy/api/v1/infra/sites/default/enforcement-points/default/host-transport-nodes/
   <host-tn-id>/statistics/monitor
   ```

   To get the *`node_id`* of the host transport node, use the same API, as mentioned in the previous step.

   This API returns the last updated cached values of the health monitoring statistics for the specified host transport node. By default, only the packet statistics are returned for the host transport node.

   As mentioned earlier in this section, the API can return the following types of health monitoring statistics for a host transport node:

- Packet statistics (packet_stats). This type is the default.
- Fast path system statistics (fast_path_sys_stats)
- Platform packet statistics (platform_packet_stats)
- Platform CPU usage statistics (platform_cpu_usage_stats)
- Fast path lcore usage statistics (fast_path_lcore_usage_stats)

When the `type` query parameter is omitted in the API URI, the API returns only the packet statistics by default.

To view the statistics for a specific type or multiple types together, use the `type` query parameter in the API URI.

Example 1: The following API returns the point-in-time values of the host transport node statistics for a single type:

```
GET https://<nsx-mgr>/policy/api/v1/infra/sites/default/enforcement-points/default/host-transport-nodes/
<host-tn-id>/statistics/monitor?type=platform_cpu_usage_stats
```

Example 2: The following API returns the point-in-time values of the host transport node statistics for multiple types together:

```
GET https://<nsx-mgr>/policy/api/v1/infra/sites/default/enforcement-points/default/host-transport-nodes/
<host-tn-id>/statistics/monitor?type=platform_cpu_usage_stats,fast_path_sys_stats,platform_packet_stats
```

Ensure that the types are specified in a comma-separated list and without any space before and after the commas.

6. Run the following API to view the real-time values of the transport node health statistics for a specific node:

```
GET https://<nsx-mgr>/policy/api/v1/infra/sites/default/enforcement-points/default/host-transport-nodes/
<host-tn-id>/statistics/debug
```

As mentioned earlier in step 5, when the `type` query parameter is omitted in the API URI, the API returns only the packet statistics by default.

To view the statistics for a specific type or multiple types together, use the `type` query parameter in the API URI.

Example 1: The following API returns the real-time values of the host transport node statistics for a single type:

```
GET https://<nsx-mgr>/policy/api/v1/infra/sites/default/enforcement-points/default/host-transport-nodes/
<host-tn-id>/statistics/debug?type=platform_cpu_usage_stats
```

Example 2: The following API returns the real-time values of the host transport node statistics for multiple types together:

```
GET https://<nsx-mgr>/policy/api/v1/infra/sites/default/enforcement-points/default/host-transport-nodes/
<host-tn-id>/statistics/debug?type=platform_cpu_usage_stats,fast_path_sys_stats,platform_packet_stats
```

Monitor NSX Segment Statistics Using APIs

By default, the collection of segment statistics on all the ESX host transport nodes is not activated because these statistics are required only for debugging or troubleshooting purposes.

Activate the collection of segment statistics in the `esx-obsrv-stats-management` monitor.

Do these steps:

1. Run the following NSX API to read the configuration of the default profile of the `esx-obsrv-stats-management` monitor:

   ```
   GET https://<nsx-mgr>/policy/api/v1/infra/sha/monitors/esx-obsrv-stats-management/profiles/default-profile
   ```

   In the API response, observe that the `enable_esx_datapath_per_segment_stats` property is set to false.
2. Copy the GET API response of the default profile and paste it in a text editor. Edit the value of the `enable_esx_datapath_per_segment_stats` to true.
3. Paste the updated profile configuration in the request body of the following PATCH API:

   ```
   PATCH https://<nsx-mgr>/policy/api/v1/infra/sha/monitors/esx-obsrv-stats-management/profiles/default-
   profile
   ```

The default profile activates the collection of real-time statistics for each segment on all the ESX host transport nodes in the system.

You can optionally activate the collection of segment statistics when needed. Segment statistics are collected in real time and they can help you debug issues on a segment when certain anomalies occur in the data path.

Note:

- Collection of segment statistics is currently supported only for overlay segments and not VLAN segments.
- Collection of segment statistics is currently supported only for segments in the default space. If you are using a multi-tenant NSX deployment, you cannot view statistics for segments in NSX projects and subnets in NSX VPCs.

You can either run NSX APIs to fetch the real-time values of segment statistics, or use the NSX Central CLI or the ESX host CLI.

Viewing of segment statistics is currently not supported in the NSX Manager UI.

This documentation explains the procedure of using NSX APIs to monitor the real-time values of segment statistics. To learn about using the NSX Central CLI or the ESX host CLI for monitoring segment statistics, refer to the *NSX Command-Line Interface Reference*.

1. Run the following API to retrieve the real-time statistics of a segment on a specific host transport node:

   ```
   GET https://<nsx-mgr>/policy/api/v1/infra/segments/<segment-id>/statistics?enforcement_point_path=/
   infra/sites/default/enforcement-points/default&stats_type=DATAPATH_STATS%transport_node_id=<host-tn-
   id>&source=realtime
   ```

   In this API URI, replace the *segment-id* and *host-tn-id* with actual values in your NSX environment.
2. Repeat step 1 for other segments on the host transport node, as needed.

Monitor Statistics of NSX Host Transport Nodes in the NSX Manager UI

In the NSX Manager UI, you can monitor the point-in-time values of the host transport node statistics.

The statistics are collected from each host transport node periodically (default is every 300 seconds) and cached by the system. The last updated cached values are fetched from the host and displayed to users.

The NSX Manager UI does not display the real-time values of the host transport node statistics.

Note:
The host transport node statistics are accessible only to NSX users in the default space. If you are using a multi-tenant NSX deployment, users in projects and NSX VPCs cannot access the host transport node statistics.

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Ensure that you are in the **Default** view.
3. Navigate to **System** > **Fabric** > **Hosts** > **Clusters**.
4. Expand a cluster.
5. For the node whose statistics you want to view, click **View Details**.
6. Click the **Datapath Stats** tab.

   For a description of the statistics that are displayed on the **Datapath Stats** tab, see NSX Host Transport Node Statistics.

The UI displays only the packet statistics of the host transport node. In the UI, a subset of health monitoring statistics are displayed compared to the statistics that are returned in the API response.

Observe the last updated date timestamp on the upper-right corner of the screen. The statistics that are displayed on the screen are point-in-time values. The statistics are collected from the host transport node periodically (default is every 300 seconds), and cached by the system. When you click the **Refresh** icon, the last updated cached values are fetched from the host and displayed on the screen.

Dump Statistics of NSX Host Transport Nodes and Segments to Files

The System Health Agent (SHA) framework provides monitors that can dump host transport node statistics and optionally segment statistics over a specified duration to files, which are saved on the ESX host transport nodes. This feature is called the file dump.

This feature is currently supported only with NSX APIs. It is primarily targeted at advanced NSX users or engineers who are interested in studying the historical trend of statistics on a host transport node and delve deeper in to the statistics for debugging data path issues.

The file dump feature uses the following two monitors to dump statistics to files:

- `esx-obsrv-tn-stats-file-dump-monitor`: This monitor dumps ESX host transport node statistics to files.
- `esx-obsrv-segment-stats-file-dump-monitor`: This monitor dumps segment statistics to files. Statistics of only overlay segments are dumped to files. The file dump feature is currently not supported for VLAN segments.

On the ESX host transport nodes, the files are saved at `/var/run/log/nsx-obsrv-stats-filedump/`

The data that is dumped to the files is in a raw format. The system currently does not provide a UI to consume and analyze the raw data in NSX Manager.

You can download the raw files from the hosts and share them with VMware Support for analysis purposes. The raw files are available at the following location on the host transport nodes:

`/var/run/log/nsx-obsrv-stats-filedump/`

The raw files are also saved in the NSX support bundle when you create a support bundle collection request.

The following procedure explains the API workflow for using the file dump feature.

1. To dump host transport node statistics to files, do these steps.
   a) Run the following API to view the configuration of the default profile of the `esx-obsrv-tn-stats-file-dump-monitor`:

      ```
      GET https://<nsx-mgr>/policy/api/v1/infra/sha/monitors/esx-obsrv-tn-stats-file-dump-monitor/profiles/default-profile
      ```

   b) Copy the GET API response of the previous step and paste it in a text editor. Edit the following properties in the default profile:

      - `check_interval`
      - `enable`
      - `disable_after`
      - `file_dump_backup_count_dp_tn_stats`

      To learn more about these configuration properties, go to the documentation of the following API in the *NSX API Guide*:

      ```
      /policy/api/v1/infra/sha/monitors/<monitor-id>/profiles/<monitor-profile-id>
      ```

      Expand the **ShaMonitorProfile** schema. In the **Type** column, click **ShaEsxObsrvTnStatsFileDumpMonitorConfig** to view the property descriptions.
   c) Paste the updated profile configuration in the request body of the following API:

      `PATCH https://<nsx-mgr>/policy/api/v1/infra/sha/monitors/esx-obsrv-tn-stats-file-dump-monitor/profiles/default-profile`

      When this API is successful, the file dump feature is activated for all the host transport nodes in the system.
   d) To verify whether the profile of the `esx-obsrv-tn-stats-file-dump-monitor` has taken effect on a specific host transport node, run the following API:

      ```
      GET https://<nsx-mgr>/policy/api/v1/infra/sha/monitors/esx-obsrv-tn-stats-file-dump-monitor/status?transport_node_id=<node-id>
      ```

      The output of this API displays the monitor profile that is effective on the host transport node and the detailed configuration of that profile.

This API requires you to specify the *node_id* in the API URI. To get the *node_id* of the host transport nodes, run the following API:

```
GET https://<nsx-mgr>/policy/api/v1/infra/sites/default/enforcement-points/default/host-transport-nodes
```

2. To dump statistics of specific segments to files, do these steps.
   a) Set the `enable_esx_datapath_per_segment_stats` property in the profile of the `esx-obsrv-stats-management` monitor to true.

      This action activates the collection of segment statistics from the host transport nodes in the system.

      To learn more about activating the collection of segment statistics, see the *Prerequisites* section in Monitor NSX Segment Statistics Using APIs.
   b) Determine the VNI of the overlay segments whose statistics you want to dump to files.

      You can use any one of these methods to determine the VNI of the segments.

Method 1: Using API	Run the following API: `GET https://<nsx-mgr>/policy/api/v1/infra/realized-state/realized-entities?intent_path=/infra/segments/<segment-id>` In this API URI, replace *segment-id* with the value of the overlay segment ID whose VNI you want to determine.
Method 2: Using NSX Manager UI	Navigate to **Networking** > **Segments** > **NSX**. Expand the details of the overlay segment, and then expand the **Additional Settings** section. Note down the value next to the **Overlay ID (VNI)** field.

   c) Run the following API to view the configuration of the default profile of the `esx-obsrv-segment-stats-file-dump-monitor`:

      ```
      GET https://<nsx-mgr>/policy/api/v1/infra/sha/monitors/esx-obsrv-segment-stats-file-dump-monitor/profiles/default-profile
      ```
   d) Copy the GET API response of the previous step and paste it in a text editor. Edit the following properties in the default profile:

      - `check_interval`
      - `enable`
      - `segment_list`
      - `disable_after`
      - `file_dump_backup_count_dp_segment_stats`

      To learn more about these configuration properties, go to the documentation of the following API in the *NSX API Guide*:

      ```
      /policy/api/v1/infra/sha/monitors/<monitor-id>/profiles/<monitor-profile-id>
      ```

      Expand the **ShaMonitorProfile** schema. In the **Type** column, click **ShaEsxObsrvSegmentStatsFileDumpMonitorConfig** to view the property descriptions.
   e) Paste the updated profile configuration in the request body of the following API:

      ```
      PATCH https://<nsx-mgr>/policy/api/v1/infra/sha/monitors/esx-obsrv-segment-stats-file-dump-monitor/profiles/default-profile
      ```

      When this API is successful, the file dump feature is activated for the specified segments. The segment statistics are dumped to files that are saved on the host transport nodes.
   f) To verify whether the profile of the `esx-obsrv-segment-stats-file-dump-monitor` has taken effect on a specific host transport node, run the following API:

      ```
      GET https://<nsx-mgr>/policy/api/v1/infra/sha/monitors/esx-obsrv-segment-stats-file-dump-monitor/status?transport_node_id=<node-id>
      ```

      The output of this API displays the monitor profile that is effective on the host transport node and the detailed configuration of that profile.

This API requires you to specify the *node_id* in the API URI. To get the *node_id* of the host transport nodes, run the following API:

```
GET https://<nsx-mgr>/policy/api/v1/infra/sites/default/enforcement-points/default/host-transport-nodes
```

Typically, segment statistics are required only for debugging data path issues. Collection of segment statistics is a resource-intensive activity. Therefore, after the debugging is completed, remember to deactivate the collection of segment statistics in the profile of the `esx-obsrv-stats-management` monitor.

NSX Host Transport Node Statistics

This reference documentation describes the various statistics that are collected from the ESX host transport nodes in your NSX deployment.

The first section in this document describes the host transport node statistics that are visible in the NSX Manager UI.

The remaining sections in this document describe the statistics that are collected by the various datapath modules, which are running in the ESX host transport nodes. To view these statistics, you must use either the NSX APIs or the NSX Central CLI. The **Statistic** column in these sections refers to the name of the statistic as seen in the API output. To learn about the API workflow for viewing the host transport node statistics, see Monitor Statistics of NSX Host Transport Nodes Using APIs.

Datapath Stats Tab

This tab displays the aggregate values of statistics for a host transport node in the NSX Manager UI.

Statistic	Description
Broadcast Packets Received	Rate of broadcast packets received by the VDS from the VM. This statistic internally maps to the statistic - rxbcastpkts.
Broadcast Packets Transmitted	Rate of broadcast packets sent by the VDS to the VM. This statistic internally maps to the statistic - txbcastpkts.
Broadcast rate limiting packet drops	Number of ingress or egress packets dropped by broadcast rate limiting. Rate limits are used to protect the network or VMs from events such as broadcast storms. You can configure rate limit values in the NSX Manager UI at **Networking** > **Segments** > **Segment Profile** > **Segment Security**. This statistic internally maps to these detailed statistics: rx_rate_limit_bcast_drops, rx_rate_limit_mcast_drops, tx_rate_limit_bcast_drops, tx_rate_limit_mcast_drops. For more details, see the individual statistic definitions.
DFW	Total packets dropped by the DFW module for various reasons. Click the **NSX Datapath** link in the UI to understand details for the drops.
Datapath L3	Total packets dropped by the Datapath L3 module for various reasons. Click the **NSX Datapath** link in the UI to understand details for the drops.
Datapath System Error	Total packets dropped due to critical internal system errors. If these statistics increment consistently, it means the ESX host is running low on resources. Moving some VMs to other hosts might help to ease the load. This statistic internally maps to these detailed statistics: leaf_rx_system_err_drops, uplink_rx_system_err_drops, pkt_attr_error_drops, tx_dispatch_queue_too_long_drops. For more details, see the individual statistic definitions.

Statistic	Description
Fast Path	Total packets dropped by the Fastpath module for various reasons. Click the **NSX Datapath** link in the UI to understand details for the drops.
Fastpath Flow Hit	Rate of flow table hits in the ENS/flow cache module. This statistic internally maps to the statistic - hits.
Fastpath Flow Miss	Rate of packets that are processed by slowpath because of a flow miss. This statistic does not overlap with the statistic in the next row. This statistic internally maps to the statistic - miss.
Fastpath packet drops	Total number of packets dropped by flow cache fastpath, in the receive or transmit direction to all ports. This statistic internally maps to these detailed statistics: rx_drops ,tx_drops, rx_drops_uplink, tx_drops_uplink, rx_drops_sp, tx_drops_sp. For more details, see the individual statistic definitions.
Firewall flood limit packet drops	Number of packets dropped due to various protocols flood over limit. There is a configured flood limit for different protocols in the kernel interface. This statistic internally maps to these detailed statistics: udp_flood_overlimit_drops, tcp_flood_overlimit_drops, icmp_flood_overlimit_drops, other_flood_overlimit_drops. For more details, see the individual statistic definitions.
Firewall internal error packet drops	Number of packets dropped by the firewall due to internal errors. This statistic internally maps to these detailed statistics: memory_drops, state_insert_drops, l7_attr_error_drops, lb_reject_drops, src_limit_misc. For more details, see the individual statistic definitions.
Firewall malformed packet drops	Number of malformed packets dropped by the firewall. This statistic internally maps to these detailed statistics: fragment_drops, short_drops, normalize_drops, bad_timestamp_drops, proto_cksum_drops. For more details, see the individual statistic definitions.
Firewall packet rejects	Number of packets rejected by the firewall for various reasons. This statistic internally maps to these detailed statistics: rx_ipv4_reject_pkts, tx_ipv4_reject_pkts, rx_ipv6_reject_pkts, tx_ipv6_reject_pkts. For more details, see the individual statistic definitions.
Firewall rule received packet drops	Number of received packets dropped by hitting drop or reject distributed firewall rule. This statistic internally maps to the statistic - match_drop_rule_rx_drops.
Firewall rule transmitted packet drops	Number of transmitted packets dropped by hitting drop or reject distributed firewall rule. This statistic internally maps to the statistic - match_drop_rule_tx_drops.
Firewall state check packet drops	Number of packets dropped due to state related checks. This statistic internally maps to these detailed statistics: icmp_err_pkt_drops, alg_handler_drops, syn_expected_drops, ip_option_drops, syn_proxy_drops, spoof_guard_drops, state_mismatch_drops, strict_no_syn_drops. For more details, see the individual statistic definitions.
Firewall state table full packet drops	Number of packets dropped due to max limit of states reached. For instance, if number of TCP states are higher than the limit, it results in a drop. This statistic internally maps to the statistic - state_limit_drops.

Statistic	Description
Firewall total packet drops	Number of total packets dropped by the firewall for various reasons. This statistic internally maps to these detailed statistics: rx_ipv4_reject_pkts, tx_ipv4_reject_pkts, rx_ipv6_reject_pkts, tx_ipv6_reject_pkts, rx_ipv4_drop_pkts, tx_ipv4_drop_pkts, rx_ipv6_drop_pkts, tx_ipv6_drop_pkts, rx_l2_drop_pkts, tx_l2_drop_pkts. For more details, see the individual statistic definitions.
Hostswitch network mismatch packet drops	Number of unicast, multicast, broadcast packets dropped due to VNI or VLAN tag mismatch. This statistic internally maps to these detailed statistics: vlan_tag_mismatch_rx, vlan_tag_mismatch_tx, vni_tag_mismatch_tx, vlan_tag_mismatch_rx_mcast, vlan_tag_mismatch_tx_mcast, vni_tag_mismatch_tx_mcast. For more details, see the individual statistic definitions.
Hostswitch received forged MAC packet drops	Number of packets dropped as forged drops due to the source MAC of the packet being different from the MAC of the virtual machine adapter. Forged transmits or MAC learning being disabled on the segment causes these drops. Enabling MAC learning or forged transmits on the segment should mitigate the issue. This statistic internally maps to the statistic - forged_transmit_rx_drops.
L3 hop limit packet drops	Number of IPv4 or IPv6 packets dropped due to low TTL (Time-To-Live). Each logical router instance will deduce 1 from TTL value. Use packet capture to determine which packets have low TTL values. This statistic internally maps to these detailed statistics: ttl_ip4_drops, ttl_ipv6_drops. For more details, see the individual statistic definitions.
L3 neighbor unreachable packet drops	Number of IPv4 or IPv6 packets dropped due to failed neighbor resolution. This statistic internally maps to these detailed statistics: arp_hold_pkt_drops, ns_hold_pkt_drops.
L3 no route packet drops	Each logical router instance has its own routing table for route lookups. This statistic increases when IPv4 packets are dropped due to no matching routes for that logical router instance. This statistic internally maps to these detailed statistics: no_route_ipv4_drops, no_route_ipv6_drops. For more details, see the individual statistic definitions.
L3 reverse path forwarding packet drops	Number of IPv4 or IPv6 packets dropped due to reverse path forwarding check failure. Distributed Router may check if the source IP of packets is coming from a valid (reachable) source and may drop the packets based on the configuration. You can change this setting in the NSX Manager UI. This statistic internally maps to these detailed statistics: rpf_ipv4_drops, rpf_ipv6_drops. For more details, see the individual statistic definitions.

Statistic	Description
Mac Learning Table Full	Rate of packet drops due to MAC table update failures at the time of MAC learning either from the central control plane (CCP) or for the packets received from the underlay network. Check if the MAC table is full on the host transport node by using the following command: `$ nsxcli -c "get segment mac-table"` If required, increase the MAC table size. This statistic internally maps to these detailed statistics: mac_tbl_update_full, mac_tbl_lookup_full. For more details, see the individual statistic definitions.
Multicast Packets Received	Rate of multicast packets received by the VDS from the VM. This statistic internally maps to the statistic - rx_mcast_pkts.
Multicast Packets Transmitted	Rate of multicast packets sent by the VDS to the VM. This statistic internally maps to the statistic - tx_mcast_pkts.
Overlay Datapath L2	Total packets dropped by the Overlay Datapath L2 module for various reasons. Click the **NSX Datapath** link in the UI to understand details for the drops.
Overlay Datapath Transmitted to Uplink	Rate of unicast packets flooded to remote VTEPs due to MAC table lookup failure. Large values implies unidirectional L2 flows or MAC table update issues. Check if the MAC table is full on the host transport node by using the following command: `$ nsxcli -c "get segment mac-table"` If required, increase the MAC table size. This statistic internally maps to the statistic - mac_tbl_lookup_flood.
Overlay Unsuccessful Control Plane Assisted Neighbor Resolution	Rate of packet drop due to control plane not being able to successfully assist in the neighbor resolution. The reasons could be CCP has not learnt the IP-MAC mapping yet, or system is running low on packet buffer resource. This statistic internally maps to these detailed statistics: nd_proxy_resp_unknown, arp_proxy_resp_unknown, nd_proxy_req_fail_drops, arp_proxy_req_fail_drops, arp_proxy_resp_drops, nd_proxy_resp_drops. For more details, see the individual statistic definitions.
Overlay received packet drops	Number of packet drops at the VDL2LeafInput due to various reasons. See the other leaf received drop reasons to identify the specific reason for the drops. This statistic internally maps to these detailed statistics: leaf_rx_ref_port_not_found_drops, leaf_rx_drops. For more details, see the individual statistic definitions.
Overlay transmitted packet drops	Total number of drops at VDL2LeafOutput due to various reasons. See the other leaf transmitted drop reasons to identify specific reason for the drops. This statistic internally maps to the statistic - leaf_tx_drops.
Overlay uplink received packet drops	Number of packets dropped at the VDL2UplinkInput due to various reasons. See the other uplink received drop reasons to identify the specific reason for the drops. This statistic internally maps to these detailed statistics: uplink_rx_drops, uplink_rx_guest_vlan_drops, uplink_rx_invalid_encap_drops, mcast_proxy_rx_drops. For more details, see the individual statistic definitions.

Statistic	Description
Overlay uplink transmitted packet drops	Number of total packets drops at the VDL2UplinkOutput due to various reasons. See the other uplink transmitted drop reasons to identify the specific reason for the drops. This statistic internally maps to these detailed statistics: uplink_tx_drops, nested_tn_mcast_proxy_same_vlan_tx_drops, nested_tn_mcast_proxy_diff_vlan_tx_drops, mcast_poxy_tx_drops. For more details, see the individual statistic definitions.
PNIC Received (mbps)	Received megabits per second. This statistic internally maps to the statistic - rxmbps.
PNIC Received (pps)	Received packets per second. This statistic internally maps to the statistic - rxpps.
PNIC Received Drops	Received errors per second. Non-zero value usually indicates the following two cases: 1. PNIC RX ring size is too small and ring can be easily filled up due to a workload spikes. You can consider increasing the ring size. 2. The packet rate is too high for the guest to handle. The guest is not able to pull packets out of the PNIC RX ring, leading to packet drops. This statistic internally maps to the statistic - rxeps.
PNIC Transmitted (mbps)	Transmitted megabits per second. This statistic internally maps to the statistic - txmbps
PNIC Transmitted (pps)	Transmitted packets per second. This statistic internally maps to the statistic - txpps.
PNIC Transmitted Drops	Transmitted errors per second. This statistic internally maps to the statistic - txeps.
PNICs	Number of physical NICs. This statistic internally maps to the statistic - num_pnics.
Packet parsing error drops	Number of IPv6 neighbor discovery (ND) packets which were not correctly parsed. Examine logs for error messages. Do packet captures at the port to identify if the packets are malformed. This statistic internally maps to the statistic - nd_parse_errors.
Slowpath Only	Rate of packets that are always processed in slowpath by design. One example is broadcast packet. This statistic internally maps to the statistic - slowpath.
Spoof guard packet drops	Number of IPv4/IPv6/ARP packets dropped by SpoofGuard. SpoofGuard protects against IP spoofing by maintaining a reference table of VM names/MAC and IP addresses. This will be incremented only if SpoofGuard is enabled on the segment or segment port. This statistic internally maps to these detailed statistics: spoof_guard_ipv4_drops, spoof_guard_arp_drops, spoof_guard_ipv6_drops, spoof_guard_nd_drops, spoof_guard_non_ip_drops. For more details, see the individual statistic definitions.
Switch Security	Total packets dropped by the Switch Security module for various reasons. Click the **NSX Datapath** link in the UI to understand details for the drops.

Statistic	Description
Unknown Tunnel Endpoint	Rate of packets drops for which source outer MAC cannot be learned as incoming GENEVE label is unknown. Large values of this statistic can point to missing remote VTEP updates on the transport node from the control plane. Use the CLI to check the remote VTEP table on the transport node. This statistic internally maps to the statistic - uplink_rx_skip_mac_learn.
VNIC Received (mbps)	Received megabits per second. This statistic internally maps to the statistic - rxmbps.
VNIC Received (pps)	Received packets per second. This statistic internally maps to the statistic - rxpps.
VNIC Received Drops	Received errors per second. Non-zero value usually indicates the following two cases: 1. VNIC RX ring size is too small and ring can be easily filled up due to a workload spikes. You can consider increasing the ring size. 2. The packet rate is too high for the guest to handle. The guest is not able to pull packets out of the VNIC RX ring, leading to packet drops. This statistic internally maps to the statistic - rxeps.
VNIC Transmitted (mbps)	Transmitted megabits per second. This statistic internally maps to the statistic - txmbps.
VNIC Transmitted (pps)	Transmitted packets per second. This statistic internally maps to the statistic - txpps.
VNIC Transmitted Drops	Transmitted errors per second. Non-zero value usually indicates the following: • The packet rate is too high for uplink to handle. • Uplink is not able to pull packets out of network stack's queue, leading to packet drops. This statistic internally maps to the statistic - txeps.
VNICs	Number of virtual NICs. This statistic internally maps to the statistic - num_vnics.
Workload BPDU filter packet drops	Number of packets dropped by BPDU filtering. When the BPDU filter is enabled, traffic to the configured BPDU destination MAC addresses is dropped. This statistic internally maps to the statistic - bpdu_filter_drops.
Workload DHCP not allowed packet drops	Number of DHCPv4 or DHCPv6 packets dropped by DHCP client/server block. This statistic internally maps to these detailed statistics: dhcp_client_block_ipv6_drops, dhcp_server_block_ipv6_drops, dhcp_client_block_ipv4_drops, dhcp_server_block_ipv4_drops, dhcp_client_validate_ipv4_drops. For more details, see the individual statistic definitions.

Statistic	Description
Workload IPv6 RA guard packet drops	Number of IPv6 Router Advertisement packets dropped by RA Guard. The RA Guard feature filters out IPv6 Router Advertisements (ICMPv6 type 134) transmitted from VMs. In an IPv6 deployment, routers periodically multicast Router Advertisement messages, which are used by hosts for autoconfiguration. You can use RA Guard to protect your network against rogue RA messages generated by unauthorized or improperly configured routers connecting to the network segment. You can configure RA Guard in the NSX Manager UI at **Networking** > **Segments** > **Segment Profile** > **Segment Security**. This statistic internally maps to the statistic - ra_guard_drops.
vSwitch	Total packets dropped by the vSwitch module for various reasons. Click the **NSX Datapath** link in the UI to understand details for the drops.
vSwitch Received from Uplink	Rate of packets received from one or more uplinks to the vSwitch which are unknown unicast flooded to other ports in the same broadcast domain by the vSwitch. This statistic increments when packets are unknown unicast flooded in the presence of MAC learning enabled segments or sink ports. Unknown unicast flooding occurs when the destination MAC address of the packet is not found in the vSwitch MAC address table. This statistic increments when a destination MAC ages out from the MAC address table in the presence of MAC learning. This statistic internally maps to the statistic - unknown_unicast_rx_uplink_pkts.
vSwitch Transmitted to Uplink	Rate of packets unknown unicast flooded by the vSwitch to one or more uplinks. The statistic increments when packets are unknown unicast flooded in the presence of MAC learning enabled segments or sink ports. Unknown unicast flooding occurs when the destination MAC address of the packet is not found in the vSwitch MAC address table. This statistic increments when a destination MAC ages out from the MAC address table in the presence of MAC learning. This statistic internally maps to the statistic - unknown_unicast_tx_uplink_pkts.

Module: host_enhanced_fastpath

This datapath module provides host/infrastructure statistics for ENS datapath module. This datapath module is known as `host-fastpath-ens` in the NSX Central CLI.

Statistic	Description
flow_table_occupancy_0_pct	Histogram: Number of flow tables with 0-25% utilization.
flow_table_occupancy_25_pct	Histogram: Number of flow tables with 25-50% utilization.
flow_table_occupancy_50_pct	Histogram: Number of flow tables with 50-75% utilization.
flow_table_occupancy_75_pct	Histogram: Number of flow tables with 75-90% utilization.

Statistic	Description
flow_table_occupancy_90_pct	Histogram: Number of flow tables with 90-95% utilization. If the number of active flows becomes greater than the flow table size, you may see an increase in flow misses, leading to a performance degradation. Flow table occupancy histogram statistics are useful to determine whether flow tables are getting full. Increasing the flow table size doesn't always improve performance, if short-lived connections keep coming in. The flow table might be always full regardless of the flow table size. A large flow table size wouldn't help in this case. EDP has a logic to detect this and automatically enable and disable flow tables to handle such a case.
flow_table_occupancy_95_pct	Histogram: Number of flow tables with 95% utilization. If the number of active flows becomes greater than the flow table size, you may see an increase in flow misses, leading to a performance degradation. Flow table occupancy histogram statistics are useful to determine whether flow tables are getting full. Increasing the flow table size doesn't always improve performance, if short-lived connections keep coming in. The flow table might be always full regardless of the flow table size. A large flow table size wouldn't help in this case. EDP has a logic to detect this and automatically enable and disable flow tables to handle such a case.
flow_table_size	Max size of the flow table in EDP.
hits	Number of flow table hits in the ENS module. This statistic can be used to calculate the flow hit / miss / slowpath rate or calculate hit / miss / slowpath ratio.
insertion_errors	Number of flow table insertion errors. This can happen when a flow table is full (or close to full) and there are hash collisions.
miss	Packets that are processed by slowpath because of a flow miss. This statistic does not overlap with the slowpath statistic that is described later in this table. This statistic can be used to calculate the flow hit / miss / slowpath rate or calculate hit / miss / slowpath ratio.
num_flow_tables	Number of flow tables used for EDP. EDP has one flow table per EDP thread. This is useful to see how many flow tables are created and used.
num_flows	Number of flows in the EDP.
num_flows_created	Number of flows created in the EDP. Use this statistic to calculate the flow creation rate, which will be useful to determine the workload characteristics. Flow table occupancy histogram statistics will tell you whether flow tables are full or not. If the flow creation rate is low and there is no significant changes in num_flows or flow occupancy statistics, this indicates that the traffic is stable and in a steady state. The number of active flows remains stable. If the flow creation rate is high and num_flows is increasing, this means that the number of active flows is increasing. Flow tables will become full eventually if the flow creation rate doesn't drop. If the flow creation rate is high and the average flow size is not small, you should consider increasing the flow table size. Average flow size = hit rate / num_flows_created rate. A small value for the average flow size means that flows are short-lived. Both hits and num_flows_created are accumulated. You can calculate the rate values first to get the average flow size during a specific time period.

Statistic	Description
slowpath	Packets that are always processed in slowpath by design. One example is broadcast packet. This statistic can be used to calculate the flow hit / miss / slowpath rate or calculate hit / miss / slowpath ratio.

Module: host_fastpath_ens_lcore

This datapath module provides the estimated Lcore usage stats for ENS module. Upto 16 Lcores ranked by usage are displayed. If less than 16 Lcores are configured, only Lcores with valid IDs are displayed. This datapath module is known as `host-fastpath-ens-lcore` in the NSX Central CLI.

Statistic	Description
lcorerank01_lcoreid	The ID of rank 1 kernel thread for EDP performance mode in terms of CPU usage. Will be displayed only if ID is valid.
lcorerank01_lcoreusage	CPU usage of rank 1 Lcore. Will be displayed only if ID is valid.
lcorerank02_lcoreid	The ID of rank 2 kernel thread for EDP performance mode in terms of CPU usage. Will be displayed only if ID is valid.
lcorerank02_lcoreusage	CPU usage of rank 2 Lcore. Will be displayed only if ID is valid.
lcorerank03_lcoreid	The ID of rank 3 kernel thread for EDP performance mode in terms of CPU usage. Will be displayed only if ID is valid.
lcorerank03_lcoreusage	CPU usage of rank 3 Lcore. Will be displayed only if ID is valid.
lcorerank04_lcoreid	The ID of rank 4 kernel thread for EDP performance mode in terms of CPU usage. Will be displayed only if ID is valid.
lcorerank04_lcoreusage	CPU usage of rank 4 Lcore. Will be displayed only if ID is valid.
lcorerank05_lcoreid	The ID of rank 5 kernel thread for EDP performance mode in terms of CPU usage. Will be displayed only if ID is valid.
lcorerank05_lcoreusage	CPU usage of rank 5 Lcore. Will be displayed only if ID is valid.
lcorerank06_lcoreid	The ID of rank 6 kernel thread for EDP performance mode in terms of CPU usage. Will be displayed only if ID is valid.
lcorerank06_lcoreusage	CPU usage of rank 6 Lcore. Will be displayed only if ID is valid.
lcorerank07_lcoreid	The ID of rank 7 kernel thread for EDP performance mode in terms of CPU usage. Will be displayed only if ID is valid.
lcorerank07_lcoreusage	CPU usage of rank 7 Lcore. Will be displayed only if ID is valid.
lcorerank08_lcoreid	The ID of rank 8 kernel thread for EDP performance mode in terms of CPU usage. Will be displayed only if ID is valid.
lcorerank08_lcoreusage	CPU usage of rank 8 Lcore. Will be displayed only if ID is valid.
lcorerank09_lcoreid	The ID of rank 9 kernel thread for EDP performance mode in terms of CPU usage. Will be displayed only if ID is valid.
lcorerank09_lcoreusage	CPU usage of rank 9 Lcore. Will be displayed only if ID is valid.
lcorerank10_lcoreid	The ID of rank 10 kernel thread for EDP performance mode in terms of CPU usage. Will be displayed only if ID is valid.
lcorerank10_lcoreusage	CPU usage of rank 10 Lcore. Will be displayed only if ID is valid.
lcorerank11_lcoreid	The ID of rank 11 kernel thread for EDP performance mode in terms of CPU usage. Will be displayed only if ID is valid.
lcorerank11_lcoreusage	CPU usage of rank 11 Lcore. Will be displayed only if ID is valid.
lcorerank12_lcoreid	The ID of rank 12 kernel thread for EDP performance mode in terms of CPU usage. Will be displayed only if ID is valid.
lcorerank12_lcoreusage	CPU usage of rank 12 Lcore. Will be displayed only if ID is valid.
lcorerank13_lcoreid	The ID of rank 13 kernel thread for EDP performance mode in terms of CPU usage. Will be displayed only if ID is valid.

Statistic	Description
lcorerank13_lcoreusage	CPU usage of rank 13 Lcore. Will be displayed only if ID is valid.
lcorerank14_lcoreid	The ID of rank 14 kernel thread for EDP performance mode in terms of CPU usage. Will be displayed only if ID is valid.
lcorerank14_lcoreusage	CPU usage of rank 14 Lcore. Will be displayed only if ID is valid.
lcorerank15_lcoreid	The ID of rank 15 kernel thread for EDP performance mode in terms of CPU usage. Will be displayed only if ID is valid.
lcorerank15_lcoreusage	CPU usage of rank 15 Lcore. Will be displayed only if ID is valid.
lcorerank16_lcoreid	The ID of rank 16 kernel thread for EDP performance mode in terms of CPU usage. Will be displayed only if ID is valid.
lcorerank16_lcoreusage	CPU usage of rank 16 Lcore. Will be displayed only if ID is valid.

Module: host_standard_fastpath

This datapath module provides host/infrastructure statistics for legacy flow cache datapath module. This datapath module is known as `host-fastpath-standard` in the NSX Central CLI.

Statistic	Description
flow_table_occupancy_0_pct	Histogram: Number of flow tables with 0-25% utilization.
flow_table_occupancy_25_pct	Histogram: Number of flow tables with 25-50% utilization.
flow_table_occupancy_50_pct	Histogram: Number of flow tables with 50-75% utilization.
flow_table_occupancy_75_pct	Histogram: Number of flow tables with 75-90% utilization.
flow_table_occupancy_90_pct	Histogram: Number of flow tables with 90-95% utilization. If the number of active flows becomes greater than the flow table size, you may see an increase in flow misses, leading to a performance degradation. Flow table occupancy histogram statistics are useful to determine whether flow tables are getting full. Increasing the flow table size doesn't always improve performance, if short-lived connections keep coming in. The flow table might be always full regardless of the flow table size. A large flow table size wouldn't help in this case. EDP has a logic to detect this and automatically enable and disable flow tables to handle such a case.
flow_table_occupancy_95_pct	Histogram: Number of flow tables with 95% utilization. If the number of active flows becomes greater than the flow table size, you may see an increase in flow misses, leading to a performance degradation. Flow table occupancy histogram statistics are useful to determine whether flow tables are getting full. Increasing the flow table size doesn't always improve performance, if short-lived connections keep coming in. The flow table might be always full regardless of the flow table size. A large flow table size wouldn't help in this case. EDP has a logic to detect this and automatically enable and disable flow tables to handle such a case.
flow_table_size	Max size of the flow table in EDP.
hits	Number of flow table hits in the ENS module. This statistic can be used to calculate the flow hit / miss / slowpath rate or calculate hit / miss / slowpath ratio.
insertion_errors	Number of flow table insertion errors. This can happen when a flow table is full (or close to full) and there are hash collisions.

Statistic	Description
miss	Packets that are processed by slowpath because of a flow miss. This statistic does not overlap with the slowpath statistic that is described later in this table. This statistic can be used to calculate the flow hit / miss / slowpath rate or calculate hit / miss / slowpath ratio.
num_flow_tables	Number of flow tables used for EDP. EDP has one flow table per EDP thread. This is useful to see how many flow tables are created and used.
num_flows	Number of flows in the EDP.
num_flows_created	Number of flows created in the EDP. Use this statistic to calculate the flow creation rate, which will be useful to determine the workload characteristics. Flow table occupancy histogram statistics will tell you whether flow tables are full or not. If the flow creation rate is low and there is no significant changes in num_flows or flow occupancy statistics, this indicates that the traffic is stable and in a steady state. The number of active flows remains stable. If the flow creation rate is high and num_flows is increasing, this means that the number of active flows is increasing. Flow tables will become full eventually if the flow creation rate doesn't drop. If the flow creation rate is high and the average flow size is not small, you should consider increasing the flow table size. Average flow size = hit rate / num_flows_created rate. A small value for the average flow size means that flows are short-lived. Both hits and num_flows_created are accumulated. You can calculate the rate values first to get the average flow size during a specific time period.
slowpath	Packets that are always processed in slowpath by design. One example is broadcast packet. This statistic can be used to calculate the flow hit / miss / slowpath rate or calculate hit / miss / slowpath ratio.

Module: host_net_thread_nioc

This datapath module provides network thread stats related to NIOC. This datapath module is known as `host-net-thread-nioc` in the NSX Central CLI.

Statistic	Description
hist_0_pct	Histogram: Number of threads within 0%-25%
hist_25_pct	Histogram: Number of threads within 25%-50%
hist_50_pct	Histogram: Number of threads within 50%-70%
hist_70_pct	Histogram: Number of threads within 70%-80%
hist_80_pct	Histogram: Number of threads within 80%-85%
hist_85_pct	Histogram: Number of threads within 85%-90%
hist_90_pct	Histogram: Number of threads within 90%-95%
hist_95_pct	Histogram: Number of threads within 95%-97%
hist_97_pct	Histogram: Number of threads within 97%-99%

Statistic	Description
hist_99_pct	Histogram: Number of threads with >99% utilization. Network datapath problems are expressed as three symptoms: packet drops, low throughput, and high latency. While these symptoms are shared by both functional and performance problems, more often than not, they are caused by performance-related problems. It is critical to rule out whether the problem is performance-related or not at an early stage of investigation. In software-defined networking especially built on top of virtualization, CPU is the most critical resource that affects network performance. With faster NICs available in the market, network bandwidth rarely becomes a bottleneck. Packet processing in datapath usually involves a set of threads executed in a pipeline, associated with queues and buffers that hold packets. When any thread in the pipeline from vCPUs to kernel network threads gets overloaded, the corresponding queues and buffers can get full, leading to packet drops. This throttles throughput. It is crucial to monitor CPU usage of kernel network threads in addition to traditional network statistics. Instead of using CPU usage numbers for individual threads, we group them and generate a histogram. Then you can monitor the histogram bins like 90pct, 95pct, 97pct, 99pct, which tell you how many networking threads are getting bottlenecked. The total_CPU statistic is also useful to show how much CPU time is spent processing packets in the kernel.
max_cpu	Maximum thread CPU utilization
min_cpu	Minimum thread CPU utilization
num_threads	Number of threads used for delivering packets from NetIOC packet scheduler to uplink.
total_cpu	Sum of CPU utilization of all network threads in the group. Total CPU is useful to see the overall CPU usage distributions between different thread groups and VMs.

Module: host_net_thread_rx

This datapath module provides network thread stats related to RX. This datapath module is known as `host-net-thread-rx` in the NSX Central CLI.

Statistic	Description
hist_0_pct	Histogram: Number of threads within 0%-25%
hist_25_pct	Histogram: Number of threads within 25%-50%
hist_50_pct	Histogram: Number of threads within 50%-70%
hist_70_pct	Histogram: Number of threads within 70%-80%
hist_80_pct	Histogram: Number of threads within 80%-85%
hist_85_pct	Histogram: Number of threads within 85%-90%
hist_90_pct	Histogram: Number of threads within 90%-95%
hist_95_pct	Histogram: Number of threads within 95%-97%
hist_97_pct	Histogram: Number of threads within 97%-99%

Statistic	Description
hist_99_pct	Histogram: Number of threads with >99% utilization. Network datapath problems are expressed as three symptoms: packet drops, low throughput, and high latency. While these symptoms are shared by both functional and performance problems, more often than not, they are caused by performance-related problems. It is critical to rule out whether the problem is performance-related or not at an early stage of investigation. In software-defined networking especially built on top of virtualization, CPU is the most critical resource that affects network performance. With faster NICs available in the market, network bandwidth rarely becomes a bottleneck. Packet processing in datapath usually involves a set of threads executed in a pipeline, associated with queues and buffers that hold packets. When any thread in the pipeline from vCPUs to kernel network threads gets overloaded, the corresponding queues and buffers can get full, leading to packet drops. This throttles throughput. It is crucial to monitor CPU usage of kernel network threads in addition to traditional network statistics. Instead of using CPU usage numbers for individual threads, we group them and generate a histogram. Then you can monitor the histogram bins like 90pct, 95pct, 97pct, 99pct, which tell you how many networking threads are getting bottlenecked. The total_CPU statistic is also useful to show how much CPU time is spent processing packets in the kernel.
max_cpu	Maximum thread CPU utilization
min_cpu	Minimum thread CPU utilization
num_threads	Number of threads used for delivering packets from NetIOC packet scheduler to uplink.
total_cpu	Sum of CPU utilization of all network threads in the group. Total CPU is useful to see the overall CPU usage distributions between different thread groups and VMs.

Module: host_net_thread_tx

This datapath module provides network thread stats related to TX. This datapath module is known as `host-net-thread-tx`host-net-thread-tx in the NSX Central CLI.

Statistic	Description
hist_0_pct	Histogram: Number of threads within 0%-25%
hist_25_pct	Histogram: Number of threads within 25%-50%
hist_50_pct	Histogram: Number of threads within 50%-70%
hist_70_pct	Histogram: Number of threads within 70%-80%
hist_80_pct	Histogram: Number of threads within 80%-85%
hist_85_pct	Histogram: Number of threads within 85%-90%
hist_90_pct	Histogram: Number of threads within 90%-95%
hist_95_pct	Histogram: Number of threads within 95%-97%
hist_97_pct	Histogram: Number of threads within 97%-99%

Statistic	Description
hist_99_pct	Histogram: Number of threads with >99% utilization. Network datapath problems are expressed as three symptoms: packet drops, low throughput, and high latency. While these symptoms are shared by both functional and performance problems, more often than not, they are caused by performance-related problems. It is critical to rule out whether the problem is performance-related or not at an early stage of investigation. In software-defined networking especially built on top of virtualization, CPU is the most critical resource that affects network performance. With faster NICs available in the market, network bandwidth rarely becomes a bottleneck. Packet processing in datapath usually involves a set of threads executed in a pipeline, associated with queues and buffers that hold packets. When any thread in the pipeline from vCPUs to kernel network threads gets overloaded, the corresponding queues and buffers can get full, leading to packet drops. This throttles throughput. It is crucial to monitor CPU usage of kernel network threads in addition to traditional network statistics. Instead of using CPU usage numbers for individual threads, we group them and generate a histogram. Then you can monitor the histogram bins like 90pct, 95pct, 97pct, 99pct, which tell you how many networking threads are getting bottlenecked. The total_CPU statistic is also useful to show how much CPU time is spent processing packets in the kernel.
max_cpu	Maximum thread CPU utilization
min_cpu	Minimum thread CPU utilization
num_threads	Number of threads used for delivering packets from NetIOC packet scheduler to uplink.
total_cpu	Sum of CPU utilization of all network threads in the group. Total CPU is useful to see the overall CPU usage distributions between different thread groups and VMs.

Module: host_pcpu

This datapath module provides the usage of physical CPUs. This datapath module is known as `host-pcpu`host-pcpu in the NSX Central CLI.

Statistic	Description
hist_0_pct	Histogram: Number of CPUs within 0%-50%
hist_50_pct	Histogram: Number of CPUs within 50%-70%
hist_75_pct	Histogram: Number of CPUs within 75%-85%
hist_85_pct	Histogram: Number of CPUs within 85%-90%
hist_90_pct	Histogram: Number of CPUs within 90%-95%
hist_95_pct	Histogram: Number of CPUs within 95%-100%

Statistic	Description
total_cpu	Total host CPU utilization. Sum of utilization of all physical CPU cores on the host. Network datapath problems are expressed as three symptoms: packet drops, low throughput, and high latency. While these symptoms are shared by both functional and performance problems, more often than not, they are caused by performance-related problems. It is critical to rule out whether the problem is performance-related or not at an early stage of investigation. In software-defined networking especially built on top of virtualization, CPU is the most critical resource that affects network performance. With faster NICs available in the market, network bandwidth rarely becomes a bottleneck. Packet processing in datapath usually involves a set of threads executed in a pipeline, associated with queues and buffers that hold packets. When any thread in the pipeline from vCPUs to kernel network threads gets overloaded, the corresponding queues and buffers can get full, leading to packet drops. This throttles throughput. It is crucial to monitor CPU usage of kernel network threads in addition to traditional network statistics. Instead of using CPU usage numbers for individual threads, we group them and generate a histogram. Then you can monitor the histogram bins like 90pct, 95pct, 97pct, 99pct, which tell you how many networking threads are getting bottlenecked. The total_CPU statistic is also useful to show how much CPU time is spent processing packets in the kernel.

Module: host_uplink

This datapath module provides the usage of physical uplink NICs. This datapath module is known as `host-uplink` in the NSX Central CLI.

Statistic	Description
num_pnics	Number of physical NICs
rx_error_total	Driver stats: received errors total. Usually, this statistic should have a similar value to rx_missed. Non-zero value usually indicates two cases: 1. PNIC RX ring size is too small and ring can be easily filled up due to a workload spikes. You can consider increasing the ring size. 2. The packet rate is too high for the guest to handle. The guest is not able to pull packets out of the PNIC RX ring, leading to packet drops.
rx_missed	Driver stats: received missed. Usually, this statistic should have a similar value to rx_error_total. Non-zero value usually indicates two cases: 1. PNIC RX ring size is too small and ring can be easily filled up due to a workload spikes. You can consider increasing the ring size. 2. The packet rate is too high for the guest to handle. The guest is not able to pull packets out of the PNIC RX ring, leading to packet drops.

Statistic	Description
rxeps	Received errors per second. Non-zero value usually indicates two cases: 1. PNIC RX ring size is too small and ring can be easily filled up due to a workload spikes. You can consider increasing the ring size. 2. The packet rate is too high for the guest to handle. The guest is not able to pull packets out of the PNIC RX ring, leading to packet drops.
rxmbps	Received megabits per second
rxpps	Received packets per second
txeps	Transmitted errors per second
txmbps	Transmitted megabits per second
txpps	Transmitted packets per second

Module: host_vnic

This datapath module provides the usage of virtual NICs. This datapath module is known as `host-vNIC` in the NSX Central CLI.

Statistic	Description
num_vnics	Number of virtual NICs.
rxeps	Received errors per second. Non-zero value usually indicates the following two cases: 1. VNIC RX ring size is too small and ring can be easily filled up due to a workload spikes. You can consider increasing the ring size. 2. The packet rate is too high for the guest to handle. The guest is not able to pull packets out of the VNIC RX ring, leading to packet drops.
rxmbps	Received megabits per second.
rxpps	Received packets per second.
txeps	Transmitted errors per second. Non-zero value usually indicates the following two cases: 1. The packet rate is too high for uplink to handle. 2. Uplink is not able to pull packets out of network stack's queue, leading to packet drops.
txmbps	Transmitted megabits per second.
txpps	Transmitted packets per second.

Module: host_vcpu

This datapath module provides the usage of virtual CPUs. This datapath module is known as `host-vcpu` in the NSX Central CLI.

Statistic	Description
hist_0_pct	Histogram: Number of CPUs within 0%-50%.
hist_50_pct	Histogram: Number of CPUs within 50%-70%.
hist_75_pct	Histogram: Number of CPUs within 75%-85%
hist_85_pct	Histogram: Number of CPUs within 85%-90%.

Statistic	Description
hist_90_pct	Histogram: Number of CPUs within 90%-95%.
hist_95_pct	Histogram: Number of CPUs within 95%-100%. Network datapath problems are expressed as three symptoms: packet drops, low throughput, and high latency. While these symptoms are shared by both functional and performance problems, more often than not, they are caused by performance-related problems. It is critical to rule out whether the problem is performance-related or not at an early stage of investigation. In software-defined networking especially built on top of virtualization, CPU is the most critical resource that affects network performance. With faster NICs available in the market, network bandwidth rarely becomes a bottleneck. Packet processing in datapath usually involves a set of threads executed in a pipeline, associated with queues and buffers that hold packets. When any thread in the pipeline from vCPUs to kernel network threads gets overloaded, the corresponding queues and buffers can get full, leading to packet drops. This throttles throughput. It is crucial to monitor CPU usage of kernel network threads in addition to traditional network statistics. Instead of using CPU usage numbers for individual threads, we group them and generate a histogram. Then you can monitor the histogram bins, which tell you how many networking threads are getting bottlenecked. The total_CPU statistic is also useful to show how much CPU time is spent processing packets in the kernel.
total_cpu	Total vCPU utilization. Sum of CPU utilization of all VMs on the host. Total CPU is useful to see the overall CPU usage distributions between different thread groups and VMs.

Module: fastpath

Fastpath includes flow-cache (FC) and Enhanced network stack (ENS) datapath modules for enhanced datapath packet processing. This datapath module is known as `nsxt-fp` in the NSX Central CLI.

Statistic	Description
rx_bytes	The number of received bytes by flow cache fastpath, in the receive direction from ports.
rx_drops	The number of dropped packets by flow cache fastpath, in the receive direction from ports. This is not applicable for flow cache in non-ENS mode.
rx_drops_sp	Received packets drops when packets are sent to the slowpath from flow cache fastpath. Not applicable for flow cache in non-ENS mode and standard switch mode.
rx_drops_uplink	The number of dropped packets by flow cache fastpath in the receive direction from uplink ports. Not applicable for flow cache in non-ENS mode and standard switch mode.
rx_pkts	The number of received packets by flow cache fastpath in the receive direction from ports.
rx_pkts_sp	Received packets when packets are sent to the slowpath from flow cache fastpath. Not applicable for flow cache in non-ENS mode and standard switch mode.
rx_pkts_uplink	The number of received packets by flow cache faspath in the receive direction from uplink ports.
tx_bytes	The number of transmitted bytes by flow cache fastpath in the transmit direction to ports.

Statistic	Description
tx_drops	The number of dropped packets by flow cache fastpath in the transmit direction to ports.
tx_drops_sp	Transmitted packet drops by fastapth when packets are injected back to flow cache fastpath from slowpath. Not applicable for flow cache in non-ENS mode and standard switch mode.
tx_drops_uplink	The number of dropped packets by flow cache fastpath in the transmit direction to uplink ports.
tx_pkts	The number of transmitted packets by flow cache fastpath in the transmit direction to ports.
tx_pkts_sp	Transmitted packets by fastpath when packets are injected back to flow cache fastpath from slowpath. Not applicable for standard switch mode.
tx_pkts_uplink	The number of transmitted packets by flow cache fastpath in the transmit direction from uplink ports.

Module: switch_security

This datapath module provides stateless L2 and L3 security by checking traffic to the segment and dropping unauthorized packets sent from VMs. In this table, Rx refers to packets received "from" the switch, and Rx refers to packets sent "to" the switch. This datapath module is known as `nsxt-swsec` in the NSX Central CLI.

Statistic	Description
bpdu_filter_drops	Number of packets dropped by BPDU filtering. When the BPDU filter is enabled, traffic to the configured BPDU destination MAC addresses is dropped.
dhcp_client_block_ipv4_drops	Number of IPv4 DHCP packets dropped by DHCP Client Block. DHCP Client Block prevents a VM from acquiring DHCP IP address by blocking DHCP requests. If this is not expected, then you can deactivate DHCPv4 Client Block from the segment security profile of a segment or a segment port. To do this in NSX Manager, navigate to **Networking** > **Segments** > **Segment Profile** > **Segment Security**.
dhcp_client_block_ipv6_drops	Number of IPv6 DHCP packets dropped by DHCP Client Block. DHCP Client Block prevents a VM from acquiring DHCP IP address by blocking DHCP requests. If this is not expected, then you can deactivate DHCPv6 Client Block from the segment security profile of a segment or a segment port. To do this in NSX Manager, navigate to **Networking** > **Segments** > **Segment Profile** > **Segment Security**.
dhcp_client_validate_ipv4_drops	Number of IPv4 DHCP packets dropped because addresses in the payload were not valid. It is possible that a malicious VM on the network might be trying to send invalid DHCP packets, for example, without source IP, client hardware address not matching source MAC, and so on.
dhcp_server_block_ipv4_drops	Number of IPv4 DHCP packets dropped by DHCP Server Block. DHCP Server Block blocks traffic from a DHCP server to a DHCP client. If this is not expected, then you can disable DHCP Server Block from the segment security profile of a segment or a segment port. To do this in NSX Manager, navigate to **Networking** > **Segments** > **Segment Profile** > **Segment Security**.

Statistic	Description
dhcp_server_block_ipv6_drops	Number of DHCPv6 packets dropped by DHCP Server Block. DHCP Server Block blocks traffic from a DHCP server to a DHCP client. If this is not expected, then you can disable DHCPv6 Server block from the segment security profile of a segment or a segment port. To do this in NSX Manager, navigate to **Networking** > **Segments** > **Segment Profile** > **Segment Security**.
nd_parse_errors	Number of IPv6 Neighbor Discovery (ND) packets which were not correctly parsed. Examine logs for error messages. Do packet captures at the port to identify if the packets are malformed.
ra_guard_drops	Number of IPv6 Router Advertisement packets dropped by RA Guard. The RA Guard feature filters out IPv6 Router Advertisements (ICMPv6 type 134) transmitted from VMs. In an IPv6 deployment, routers periodically multicast Router Advertisement messages, which are used by hosts for autoconfiguration. You can use RA Guard to protect your network against rogue RA messages generated by unauthorized or improperly configured routers connecting to the network segment. You can configure RA Guard in the NSX Manager UI at **Networking** > **Segments** > **Segment Profile** > **Segment Security**.
rx_arp_pkts	Number of ARP packets received by the VDS from the VM.
rx_garp_pkts	Number of Gratuitous ARP (GARP) packets received by the VDS from the VM.
rx_ipv4_pkts	Number of IPv4 packets received by the VDS from the VM.
rx_ipv6_pkts	Number of IPv6 packets received by the VDS from the VM.
rx_na_pkts	Number of IPv6 ND (Neighbor Discovery) NA (Neighbor Advertisement) packets received by the VDS from the VM.
rx_non_ip_pkts	Number of non-IP packets received by the VDS from the VM
rx_ns_pkts	Number of IPv6 ND (Neighbor Discovery) NS (Neighbor Solicitation) packets received by the VDS from the VM.
rx_rate_limit_bcast_drops	Number of ingress packets dropped by broadcast rate limiting. Rate limits can be used to protect the network or VMs from events such as broadcast storms. You can configure rate limit values in the NSX Manager UI at **Networking** > **Segments** > **Segment Profile** > **Segment Security**.
rx_rate_limit_mcast_drops	Number of ingress packets dropped by multicast rate limiting. Rate limits can be used to protect the network or VMs from events such as multicast storms. You can configure rate limit values in the NSX Manager UI at **Networking** > **Segments** > **Segment Profile** > **Segment Security**.
rx_unsolicited_na_pkts	Number of unsolicited IPv6 ND (Neighbor Discovery) NA (Neighbor Advertisement) packets received by the VDS from the VM.
rxbcastpkts	Number of broadcast packets received by the VDS from the VM.
rxmcastpkts	Number of multicast packets received by the VDS from the VM.
spoof_guard_arp_drops	Number of ARP packets dropped by SpoofGuard. SpoofGuard protects against malicious ARP spoofing attacks by keeping track of MAC and IP addresses. This statistic will be incremented only when SpoofGuard is enabled on the segment or segment port. (**Networking** > **Segments** > **Segment Profile** > **SpoofGuard**)

Statistic	Description
spoof_guard_ipv4_drops	Number of IPv4 packets dropped by SpoofGuard. SpoofGuard protects against IP spoofing by maintaining a reference table of VM names and IP addresses. This statistic will be incremented only when SpoofGuard is enabled on the segment or segment port. (**Networking** > **Segments** > **Segment Profile** > **SpoofGuard**)
spoof_guard_ipv6_drops	Number of IPv6 packets dropped by SpoofGuard. SpoofGuard protects against IP spoofing by maintaining a reference table of VM names and IP addresses. This statistic will be incremented only when SpoofGuard is enabled on the segment or segment port. (**Networking** > **Segments** > **Segment Profile** > **SpoofGuard**)
spoof_guard_nd_drops	Number of IPv6 Neighbor Discovery (ND) packets dropped by SpoofGuard. SpoofGuard protects against ND Spoofing by filtering out ND packets whose addresses do not match the VM's address. This statistic will be incremented only when SpoofGuard is enabled on the segment or segment port. (**Networking** > **Segments** > **Segment Profile** > **SpoofGuard**)
spoof_guard_non_ip_drops	Number of non-IP packets dropped by SpoofGuard. This statistic will be incremented only when SpoofGuard is enabled on the segment or segment port. (**Networking** > **Segments** > **Segment Profile** > **SpoofGuard**)
tx_arp_pkts	Number of ARP packets sent by the VDS to the VM.
tx_ipv4_pkts	Number of IPv4 packets sent by the VDS to the VM.
tx_ipv6_pkts	Number of IPv6 packets sent by the VDS to the VM.
tx_non_ip_pkts	Number of non-IP packets sent by the VDS to the VM.
tx_rate_limit_bcast_drops	Number of egress packets dropped by broadcast rate limiting. Rate limits can be used to protect the network or VMs from events such as broadcast storms. You can configure rate limit values in the NSX Manager UI at **Networking** > **Segments** > **Segment Profile** > **Segment Security**.
tx_rate_limit_mcast_drops	Number of egress packets dropped by multicast Rate Limiting. Rate limits can be used to protect the network or VMs from events such as multicast storms. You can configure rate limit values in the NSX Manager UI at **Networking** > **Segments** > **Segment Profile** > **Segment Security**.
txbcastpkts	Number of broadcast packets sent by the VDS to the VM.
txmcastpkts	Number of multicast packets sent by the VDS to the VM.

Module: overlay_datapath_l2

This datapath module is responsible for workload connectivity. This datapath module is known as `nsxt-vdl2` in the NSX Central CLI.

Statistic	Description
arp_proxy_req_fail_drops	Number of ARP requests failed to be resent on uplink for datapath based learning when CCP does not have IP-MAC binding leading to ARP suppression failure. Non-zero statistic indicates system is running low on packet buffer resource and continuous increment should be treated as a critical error.
arp_proxy_req_suppress	Number of ARP requests which are suppressed by VDL2 by querying CCP to find IP-MAC binding. These ARP packets are sent out on uplink only when CCP does not know the binding.
arp_proxy_resp	Number of valid IP-MAC binding responses from CCP for each ARP suppression request from this transport node.
arp_proxy_resp_drops	Number of ARP responses corresponding to IP-MAC binding response from CCP that could not be sent to switchport that requested the ARP request.
arp_proxy_resp_filtered	Number of ARP responses generated based on IP-MAC binding response from CCP that are not sent to switch port that initiated the ARP request. This is possible because ARP was requested due to traceflow or the port that initiated the ARP request was no longer present in the transport node.
arp_proxy_resp_unknown	Number of unknown IP-MAC bindings in the control plane for each IP-MAC request from this transport node. On receiving this message, VDL2 module resends the ARP request on uplink to learn IP-MAC binding through datapath.
leaf_rx	For a segment (logical switch), this statistic gets incremented when a workload generated packet is successfully received at VDL2LeafInput (Overlay L2) IOChain. These packets are sent to VDS if there are no other leaf received drops.
leaf_rx_drops	Number of packet drops at the VDL2LeafInput due to various reasons. See the other leaf received drop reasons to identify the specific reason for the drops.
leaf_rx_ref_port_not_found_drops	Number of packet drops at VDL2LeafInput Leaf. This can happen if the trunk port is not part of the segment.
leaf_rx_system_err_drops	Number of packet drops at VDL2LeafInput due to various system errors like memory failure, packet attribute update failures. This generally means that the ESX host is running low on resources. Moving some VMs to other hosts might help ease the load.
leaf_tx	This statistic ncrements when a packet is processed successfully by VDL2LeafOutput (Overlay L2) IOChain for a switch port.
leaf_tx_drops	Total number of drops at VDL2LeafOutput due to various reasons. See the other leaf transmitted drop reasons to identify specific reason for the drops.

Statistic	Description
mac_tbl_lookup_flood	Number of unicast packets flooded to remote VTEPs due to MAC table lookup failure. Large values implies unidirectional L2 flows or MAC table update issues. Check if the MAC table is full on the host transport node by using the following command: `$ nsxcli -c "get segment mac-table"` If required, increase the MAC table size.
mac_tbl_lookup_full	Number of times destination MAC query to control plane for traffic to remote VMs failed as MAC table is already full. Check if the MAC table is full on the host transport node by using the following command: `$ nsxcli -c "get segment mac-table"` If required, increase the MAC table size.
mac_tbl_update_full	Number of MAC table update failures at the time of MAC learning for the packets received from underlay. Check if the MAC table is full on the host transport node by using the following command: `$ nsxcli -c "get segment mac-table"` If required, increase the MAC table size.
mcast_proxy_rx_drops	Number of BUM packets received on the uplink of the MTEP transport node that are dropped while replicating to other VTEPs.
mcast_proxy_tx_drops	Number of BUM packets originated from the workloads in the transport node that are dropped after the replication at the uplink output. This statistic increments if uplink_tx_invalid_state_drops increments or due to system errors like out-of-memory.
nd_proxy_req_fail_drops	Number of ND requests failed to be resent on uplink for datapath based learning when the CCP does not have IP-MAC binding leading to ND suppression failure. Non-zero statistic indicates system is running low on packet buffer resource and continuous increment should be treated as a critical error.
nd_proxy_req_suppress	Number of ND requests suppressed by VDL2 by querying CCP to find IP-MAC binding. These ND packets are sent out on uplink only if CCP does not know the binding.
nd_proxy_resp	Number of valid IP-MAC binding responses from CCP for each ND suppression request from this transport node. These ND responses could be a result of direct CCP response or due to a already cached ND entry in the transport node.
nd_proxy_resp_drops	Number of ND responses corresponding to IP-MAC binding response from CCP that could not be sent to switchport that initiated the ND packet.
nd_proxy_resp_filtered	Number of ND responses generated based on IP-MAC binding response from CCP that are not sent to switch port that initiated the ND request. This is possible because ND was requested due to traceflow or the port that initiated the ND request was no longer present in the transport node.
nd_proxy_resp_unknown	Number of unknown IPv6-MAC bindings in the control plane for each IPv6-MAC request from this transport node. On receiving this message, VDL2 module resends the ND packets on uplink to learn IPv6-MAC binding through datapath.

Statistic	Description
nested_tn_mcast_proxy_diff_vlan_tx_drops	Number of dropped BUM packets replicated towards nested transport node. The nested transport node and this transport node are configured with different transport VLAN ID. Check VTEP GW IP is reachable from this transport node's VTEP VMK interfaces.
nested_tn_mcast_proxy_same_vlan_tx_drops	Number of dropped BUM packets replicated towards nested transport node. The nested transport node and this transport node are configured with the same transport VLAN ID.
uplink_rx	Number of packets which are received at uplink port from the TOR switch. These packets will be sent to VDS when there are no drops at uplink Rx.
uplink_rx_drops	Number of packet drops at the VDL2UplinkInput due to various reasons. See the other uplink received drop reasons to identify the specific reason for the drops.
uplink_rx_filtered	Number of packets sent by the TOR switch, which are filtered at the VDL2 uplink for reasons like IGMP reports from peer ESX transport nodes.
uplink_rx_guest_vlan_drops	Number of packet drops at the VDL2UplinkInput when the guest VLAN tag removal fails for the inner packet due to a system error.
uplink_rx_invalid_encap_drops	Number of packets that are received at uplink from underlay and dropped due to incorrect encapsulation headers. To understand the exact error, perform the packet capture and verify the encapsulation headers (protocol version, checksum, length, and so on) by running the following command: `pktcap-uw --capture UplinkRcvKernel --uplink --ng -o uplink.pcap`
uplink_rx_mcast_invalid_dr_uplink_drops	Number of IP multicast packet drops at the VDL2 uplink input since the vdrPort is not associated with this uplink. This could happen when the TOR switch is flooding the multicast traffic on all the uplinks of the transport node. Check the vdrPort and uplink association by using following command, and then check if the dropped packet was received on the unassociated uplink: `nsxdp-cli vswitch instance list`

uplink_rx_skip_mac_learn	Number of packets for which source outer MAC cannot be learned as incoming GENEVE label is unknown.
	Large values for this statistic can point to missing remote VTEP updates at transport node from the control plane.
	Use the following CLI commands to check the remote VTEP table on the transport node:
	`nsxcli -c "get global-vtep-table"`
	`$ nsxcli -c "get segment vtep-table"`
	A possible workaround can be to restart the local control plane agent (CfgAgent) on the transport node to force a full sync by running the following command:
	`$ /etc/init.d/nsx-cfgagent restart`

uplink_rx_system_err_drops	Number of packet drops at the VDL2UplinkInput due to various system errors like memory failure, packet attribute update failures. This generally means that the ESX host is running low on resources. Moving some VMs to other hosts might help ease the load.

Statistic	Description
uplink_rx_wrong_dest_drops	Number of packets received from underlay and dropped as the destination IP of the packet does not match with any of the VTEPs configured on the host.
uplink_tx	Number of packets sent by the VDS which are received at uplink port's VDL2 IOChain. These packets will be sent to underlay network when there are no drops at uplink Tx.
uplink_tx_drops	Number of total packets drops at the VDL2UplinkOutput due to various reasons. See the other uplink transmitted drop reasons to identify the specific reason for the drops.
uplink_tx_flood_rate_limit	Number of unknown unicast packets flooded on uplinks that are rate limited.
uplink_tx_ignore	Number of packets sent by VDS that are filtered at VDL2 uplink output and not forwarded to underlay. For instance, BUM packets are filtered if there are no VTEPs on the segment to replicate the packets to.
uplink_tx_invalid_frame_drops	Number of packets that are dropped at VDL2 uplink output as either the encap header could not be found or the TSO set on the inner frame could not be performed. This is due to large TCP packets.
uplink_tx_invalid_state_drops	Number of packets that are dropped at VDL2 uplink output due to incorrect transport VLAN configuration. This is due to incorrect uplink profile association at the transport node or if the gateway MAC is not resolved. Use the following procedure to check on an ESX node if the VTEP Gateway IP is reachable from this transport node's VTEP VMK interfaces. 1. Get the Gateway IP by running the following command: `net-vdl2 -l` 2. Get the network stack instance name by running the following command: `esxcfg-vmknic -l` 3. Ping the VTEP Gateway IP by running the following command: `vmkping -I vmk10 -S`
uplink_tx_nested_tn_repl_drops	Number of BUM packets that are dropped at VDL2 uplink output while replicating to nested transport node due to incorrect VTEP association. Use following command to check the source switchport to uplink association: `nsxdp-cli vswitch instance list`
uplink_tx_non_unicast	Number of broadcast or multicast packets replicated to remote VTEPs. Large rate implies that transport node has to replicate these packets to remote VTEPs, which might cause stress on uplink layer transmitted queues.
uplink_tx_teaming_drops	Number of packets that are dropped at VDL2UplinkOutput due to non-availability of the VTEP associated with switchport that originated the traffic. Use the following command to check the workload switchport's uplink association and the teaming status: `nsxdp-cli vswitch instance list`
uplink_tx_ucast_flood	Number of unknown unicast packets flooded at uplink output. Large values implies unidirectional L2 flow or MAC table update issues. Check if the unidirectional flow is expected or if the MAC table is full.

Module: datapath_l3

This datapath module also known as Virtual Distributed Routing (VDR), routes packets on every ESX host. This datapath module is known as `nsxt-vdrb` in the NSX Central CLI.

Statistic	Description
arp_hold_pkt_drops	When Distributed Router is in the process of resolving an IPv4 ARP entry, the packets using this ARP entry will be queued. The number of packets that can be queued is capped per logical router instance. When the cap is reached, the oldest packets will be tail dropped and this statistic will increase by the number of old packets that are dropped.
arpfaildrops (lta)	IPv4 packet drops due to ARP failure.
consumed_icmpv4	Number of IPv4 packets that are destined for a logical routing port IP address of Distributed Router corresponding to a given segment. Keep in mind that statistic will increase after routing the packet from source subnet.
consumed_icmpv6	Number of IPv6 packets that are destined for a logical routing port IP address of Distributed Router corresponding to a given segment. Keep in mind that statistic will increase after routing the packet from source subnet.
drop_route_ipv4_drops	Number of IPv4 packets matching "drop routes". Drop routes are the routes configured to purposely drop the matching packets. If it is not expected, check the routes on ESX host and check the configuration in the management plane.
drop_route_ipv6_drops	Number of IPv6 packets matching "drop routes". Drop routes are the routes configured to purposely drop the matching packets. If it is not expected, check the routes on ESX host and check the configuration in the management plane.
ndfaildrops (lta)	IPv6 packet drops due to Neighbor Discovery failure.
no_nbr_ipv4	No IPv4 ARP entry found in Distributed Router's ARP table.
no_nbr_ipv6	No IPv6 neighbor entry found in Distributed Router's neighbor table.
no_route_ipv4_drops	Each logical router instance has its own routing table for route lookups. This statistic increases when IPv4 packets are dropped due to no matching route existing for that logical router instance.
no_route_ipv6_drops	Each logical router instance has its own routing table for route lookups. This statistic increases when IPv6 packets are dropped due to no matching route existing for that logical router instance.
ns_hold_pkt_drops	When Distributed Router is in the process of resolving an IPv6 neighbor entry, the packets using this neighbor entry will be queued. The number of packets can be queued is capped per logical router instance. When the cap is reached, the oldest packets will be tail dropped and this statistic will increase by the number of old packets that are dropped.

Statistic	Description
pkt_attr_error_drops	Number of packets which failed attribute operation. NSX uses packet attributes to facilitate packet processing. The packet attributes can be allocated, set or unset. In regular cases, the operation won't fail. Some possible reasons for this statistic to increment might be the following: • Packet attribute heap is exhausted. • packet attribute is corrupted.
relayed_dhcpv4_req	Relayed DHCPv4 requests.
relayed_dhcpv4_rsp	Relayed DHCPv4 responses.
relayed_dhcpv6_req	Relayed DHCPv6 requests.
relayed_dhcpv6_rsp	Relayed DHCPv6 responses.
rpf_ipv4_drops	Number of IPv4 packets dropped due to reverse path forwarding check failure. Distributed Router may check if the source IP of packets is coming from a valid (reachable) source and may drop the packets based on the configuration. You can change this setting in the NSX Manager UI. To check the current configuration in the NSX Manager UI, do these steps: 1. Navigate to **Networking** > **Segments**. 2. Edit the segment that is of interest to you. 3. Go to **Additional Settings**. 4. Check the **URPF Mode**.
rpf_ipv6_drops	Number of IPv6 packets dropped due to reverse path forwarding check failure. Distributed Router may check if the source IP of packets is coming from a valid (reachable) source and may drop the packets based on the configuration. You can change this setting in the NSX Manager UI. To check the current configuration in the NSX Manager UI, do these steps: 1. Navigate to **Networking** > **Segments**. 2. Edit the segment that is of interest to you. 3. Go to **Additional Settings**. 4. Check the **URPF Mode**.
rx_arp_req	Number of ARP request packets received by the logical router port of a Distributed Router corresponding to a given segment.
rx_ipv4	Number of IPv4 packets that are coming to a logical routing port of a Distributed Router corresponding to a given segment.
rx_ipv6	Number of IPv6 packets that are coming to a logical routing port of a Distributed Router corresponding to a given segment.
rx_pkt_parsing_error_drops	Number of packet parsing failures for received Distributed Router packets. Distributed Router performs packet parsing for each packet received to read metadata and headers. If you see a high number for this statistic, one possible reason is that the packets are not structured correctly. Monitor if there is any traffic failure and do the packet capture to debug further.
rxgarp (lta)	Gratuitous GARP received on a Distributed Router.

Statistic	Description
ttl_ipv4_drops	Number of IPv4 packets dropped due to low TTL (Time-To-Live). Each logical router instance will deduce 1 from TTL value. Use packet capture to determine which packets have low TTL values. If the TTL is fairly large at the source, possible reasons are too many routing hops on the path or packet is looping, which is rare.
ttl_ipv6_drops	Number of IPv6 packets dropped due to low TTL (Time-To-Live). Each logical router instance will deduce 1 from TTL value. Use packet capture to determine which packets have low TTL values. If the TTL is fairly large at the source, possible reasons are too many routing hops on the path or packet is looping, which is rare.
tx_arp_rsp	Number of ARP request packets sent by the logical router port of a Distributed Router corresponding to a given segment.
tx_dispatch_queue_too_long_drops	Number of packets being tail dropped in the transmit dispatch queue. The transmit dispatch queue holds Distributed Router self generated packets such as ARP packets, NS discovery, and so on. Each packet consumes the packet handling system resources. If too many packets are being queued, limit the queue size and tail drop the packets.
tx_ipv4	Number of IPv4 packets that are going out from a logical router port of a Distributed Router corresponding to a given segment.
tx_ipv6	Number of IPv6 packets that are going out from a logical router port of a Distributed Router corresponding to a given segment.

Module: distributed_firewall

This datapth module provides distributed firewall capability. In this table, Rx refers to packets received by the switchport (sent from the VM), and Tx refers to packets transmitted from the switchport (received by the VM). This datapath module is known as `nsxt-vsip` in the NSX Central CLI.

Statistic	Description
alg_handler_drops	Number of packets dropped due to ALG handling. It tracks the ALG state decoder packet handling error.
bad_offset_drops	Number of packets dropped by bad-offset.
bad_timestamp_drops	Number of packets dropped due to bad timestamp. For instance, ACK packets carrying an old timestamp, packets received with unexpected timestamp should be dropped.
congestion_drops	Number of packets dropped due to congestion. For instance, congestion detected in the queue for a network interface.
fragment_drops	Number of packets dropped due to failed reassembly of fragmented packets. Fragmentation breaks packets into smaller fragments so that the resulting pieces can pass through a link with smaller MTU than the original packet size.
handshake_error_drops	Number of packets dropped due to TCP three-way handshake error. This could happen when both sender and receiver are in SYN sent during three-way handshake. This drop reason falls under TCP state mismatch.

Statistic	Description
icmp_err_pkt_drops	Number of packets dropped due to extra ICMP error response packets. This statistic tracks the extra ICMP error response packets that are dropped.
icmp_error_drops	Number of packets dropped due to sequence failure in ICMP error response to TCP packet. When sequence numbers are out of expected range, it results in a drop.
icmp_flood_overlimit_drops	Number of packets dropped due to ICMP flood overlimit. There is a configured ICMP flood limit in kernel interface.
ignored_offloaded_fpdrops	Number of packets dropped due to flow offloaded to hardware. Flow offloaded to the hardware means that connection tracking is being done by a smartNIC's hardware packet pipeline. In that case, getting a packet in software is unexpected. The packet cannot be processed in software as the software has no up-to-date CT information (for example, states, sequence numbers). So traffic must be dropped. This case can happen because offloading a flow to the hardware takes some time and it can race with packets already enqueued to be delivered to VSIP. This drop reason is for the ENS fastpath packets.
ignored_offloaded_spdrops	Number of packets dropped due to flow offloaded to hardware. Flow offloaded to the hardware means that connection tracking is being done by a smartNIC's hardware packet pipeline. In that case, getting a packet in software is unexpected. The packet cannot be processed in software as the software has no up-to-date CT information (for example, states, sequence numbers). So traffic must be dropped. This case can happen because offloading a flow to the hardware takes some time and it can race with packets already enqueued to be delivered to VSIP. This drop reason is for the IOChain code path, also known as slowpath in this context.
ip_option_drops	Number of packets dropped because IP options is not allowed. If allow_opts in firewall rule is not set, packet hitting that rule is dropped.
l7_alert_drops	L7 rule is present but there is no match. An alert is generated.
l7_attr_error_drops	Number of packets dropped due to failure to set state attributes. This occurs when `attrconn` L7 attributes allocation or modification fails and it results in a drop.
l7_pending_misc	This statistic tracks the packets that are being parsed by DPI currently, rule match is pending. Once L7 rule match happens, corresponding rule action will be taken on the packet.
lb_reject_drops	This statistic tracks drops due to packet rejected by load balancer. Packets are dropped if it matches a load balancer virtual server but no pool member is selected.
match_drop_rule_rx_drops	Number of received packets dropped by hitting drop or reject distributed firewall rule.
match_drop_rule_tx_drops	Number of transmitted packets dropped by hitting drop or reject distributed firewall rule.
memory_drops	Number of packets dropped due to lack of memory. This is a capacity level error.
normalize_drops	Number of packets dropped due to malformed packets. For instance, IP version mismatch, TCP header offset inconsistent with packet description total length

Statistic	Description
other_flood_overlimit_drops	Number of packets dropped due to other protocol flood overlimit. There is a configured flood limit in the kernel interface for other protocols.
pkts_frag_queued_v4_misc	During packet fragmentation, the packet fragments are added to the fragment queue. These packet fragments are not necessarily dropped. Successful packet reassembly means that no fragmented packets are dropped. This statistic tracks IPv4 packets that are added to the fragment queue.
pkts_frag_queued_v6_misc	During packet fragmentation, the packet fragments are added to the fragment queue. These packet fragments are not necessarily dropped. Successful packet reassembly means that no fragmented packets are dropped. This statistic tracks IPv6 packets that are added to the fragment queue.
proto_cksum_drops	Number of packets dropped due to incorrect protocol checksum. This occurs when checksum validation of the packet fails.
rx_ipv4_drop_pkts	Number of received IPv4 dropped packets.
rx_ipv4_pass_pkts	Number of received IPv4 passed packets.
rx_ipv4_reject_pkts	Number of received IPv4 rejected packets.
rx_ipv6_drop_pkts	Number of received IPv6 dropped packets.
rx_ipv6_pass_pkts	Number of received IPv6 passed packets.
rx_ipv6_reject_pkts	Number of received IPv6 rejected packets.
rx_l2_drop_pkts	Number of received layer 2 dropped packets.
seqno_bad_ack_drops	Number of packets dropped due to TCP acknowledging forward more than one window. This drop reason falls under TCP state mismatch.
seqno_gt_max_ack_drops	Number of packets dropped due to TCP sequence number being greater than maximum ACK number. This drop reason falls under TCP state mismatch.
seqno_lt_minack_drops	Number of packets dropped due to TCP sequence number being smaller than minimum ACK number. This drop reason falls under TCP state mismatch.
seqno_old_ack_drops	Number of packets dropped due to TCP acknowledging back more than one fragment. This drop reason falls under TCP state mismatch.
seqno_old_retrans_drops	Number of packets dropped due to TCP retransmission older than one window. This drop reason falls under TCP state mismatch.
seqno_outside_window_drops	Number of packets dropped due to TCP sequence number outside window. This drop reason falls under TCP state mismatch.
short_drops	Number of short packets dropped. Short packets are packets with incorrect length, for instance, packets with malformed `ip_len` value.
spoof_guard_drops	Number of packets dropped due to SpoofGuard check. SpoofGuard is a tool that is designed to prevent virtual machines in your environment from sending traffic with an IP address that it is not authorized to send traffic from.
src_limit_misc	Number of packets hitting source limit. This is related to firewall packet processing. This occurs due to failure of source node insert in Red-Black (RB) tree due to limit reached.

Statistic	Description
state_insert_drops	Number of packets dropped due to state insert failure. This occurs due to duplicate state insert.
state_limit_drops	Number of packets dropped due to maximum limit of states reached. For instance, if number of TCP states are higher than the limit, it results in a drop.
state_mismatch_drops	Number of packets dropped due to state mismatch. There are multiple possible reasons for drop, such as STRICTNOSYN, HANDSHAKE_SYNSENT, SEQ_GT_SEQHI, and so on.
strict_no_syn_drops	Number of packets dropped due to strict enforcement mode with no syn. SYN packet is expected to be seen in strict mode.
syn_expected_drops	Packet matches a load balancer virtual server but it is not a SYN packet. So, the system should not create a state for it. This results in a packet drop. This statistic tracks this drop.
syn_proxy_drops	Number of packets dropped due to synproxy. This is to protect TCP servers from attacks such as SYN FLOOD.
tcp_flood_overlimit_drops	Number of packets dropped due to TCP flood overlimit. There is a configured TCP flood limit in the kernel interface.
tx_ipv4_drop_pkts	Number of transmitted IPv4 dropped packets.
tx_ipv4_pass_pkts	Number of transmitted IPv4 passed packets.
tx_ipv4_reject_pkts	Number of transmitted IPv4 rejected packets.
tx_ipv6_drop_pkts	Number of transmitted IPv6 dropped packets.
tx_ipv6_pass_pkts	Number of transmitted IPv6 passed packets.
tx_ipv6_reject_pkts	Number of transmitted IPv6 rejected packets.
tx_l2_drop_pkts	Number of transmitted layer 2 dropped packets.
udp_flood_overlimit_drops	Number of packets dropped due to UDP flood overlimit. There is a configured UDP flood limit in the kernel interface.

Module: virtual_switch

This Layer 2 datapath module is responsible for providing switching functionality. This module forwards packets within a broadcast domain based on the VLAN and VNI an interface receives a packet on. In this table, Rx refers to packets sent "to" the switch, and Tx refers to packets received "from" the switch. Mcast refers to multicast packets. This datapath module is known as `nsxt-vswitch` in the NSX Central CLI.

Statistic	Description
forged_transmit_rx_drops	Number of packets dropped as forged drops due to the source MAC of the packet being different from the MAC of the virtual machine adapter. Forged transmits or MAC learning being disabled on the segment causes these drops. Enabling MAC learning or forged transmits on the segment should mitigate the issue.
unknown_unicast_rx_pkts	Number of unknown unicast packets received by vSwitch that are flooded to other ports in the same broadcast domain. The statistic increments when packets are unknown unicast flooded in the presence of MAC learning enabled segments or sink ports. Unknown unicast flooding occurs when the destination MAC address of the packet is not found in the vSwitch MAC address table. This statistic increments when a destination MAC ages out from the MAC address table in the presence of MAC learning.

Statistic	Description
unknown_unicast_rx_uplink_pkts	Number of packets received from one or more uplinks to the vSwitch which are unknown unicast flooded to other ports in the same broadcast domain by the vSwitch. The statistic increments when packets are unknown unicast flooded in the presence of MAC learning enabled segments or sink ports. Unknown unicast flooding occurs when the destination MAC address of the packet is not found in the vSwitch MAC address table. This statistic increments when a destination MAC ages out from the MAC address table in the presence of MAC learning.
unknown_unicast_tx_pkts	Number of unknown unicast packets flooded to other ports in the same broadcast domain by the vSwitch. The statistic increments when packets are unknown unicast flooded in the presence of MAC learning enabled segments or sink ports. Unknown unicast flooding occurs when the destination MAC address of the packet is not found in the vSwitch MAC address table. This statistic increments when a destination MAC ages out from the MAC address table in the presence of MAC learning.
unknown_unicast_tx_uplink_pkts	Number of packets unknown unicast flooded by the vSwitch to one or more uplinks. The statistic increments when packets are unknown unicast flooded in the presence of MAC learning enabled segments or sink ports. Unknown unicast flooding occurs when the destination MAC address of the packet is not found in the vSwitch MAC address table. This statistic increments when a destination MAC ages out from the MAC address table in the presence of MAC learning.
vlan_tag_mismatch_rx	Number of unicast and broadcast packets dropped due to a VLAN tag mismatch. These drops occur when the VLAN tag of a packet is not allowed according to the VLAN policy of the segment. Amending the VLAN policy of the segment or sending packets with an allowed VLAN tag can mitigate the issue.
vlan_tag_mismatch_rx_mcast	Number of multicast packets dropped due to a VLAN tag mismatch. These drops occur when the VLAN tag of a packet is not allowed according to the VLAN policy of the segment. Amending the VLAN policy of the segment or sending packets with an allowed VLAN tag can mitigate the issue.
vlan_tag_mismatch_tx	Number of unicast packets dropped due to a VLAN tag mismatch. The host switch locates an entry in its lookup table based on the destination address of the packet. When attempting to forward the packet out of a port, these drops occur when the VLAN tag of a packet is not allowed according to the VLAN policy of the segment. Amending the VLAN policy of the segment or sending packets with an allowed VLAN tag can mitigate the issue.
vlan_tag_mismatch_tx_mcast	Number of multicast packets dropped due to a VLAN tag mismatch. The host switch locates an entry in its lookup table based on the destination address of the packet. When attempting to forward the packet out of a port, these drops occur when the VLAN tag of a packet is not allowed according to the VLAN policy of the segment. Amending the VLAN policy of the segment or sending packets with an allowed VLAN tag can mitigate the issue.

Statistic	Description
vni_tag_mismatch_tx	Number of unicast packets dropped due to a VNI tag mismatch. The host switch locates an entry in its lookup table based on the destination address of the packet. When attempting to forward the packet out of a port, these drops occur when the VNI tag of a packet is not allowed according to the VNI policy of the segment. Moving the destination VM to this overlay segment can fix the issue.
vni_tag_mismatch_tx_mcast	Number of multicast packets dropped due to a VNI tag mismatch. The host switch locates an entry in its lookup table based on the destination address of the packet. When attempting to forward the packet out of a port, these drops occur when the VNI tag of a packet is not allowed according to the VNI policy of the segment. Moving the destination VM to this overlay segment can fix the issue.

Working with Events and Alarms

NSX provides alarms to call your attention to events that can potentially affect performance and system operation. Alarms provide detailed event information such as which component is affected, the type of event, and then recommends a corrective action.

For example, one of the NSX Edge nodes can be experiencing unusually high CPU usage or low available disk space.

> **Note:**
> Alarms are system events with a severity level greater than `LOW`.
>
> If an alarm (for example, Certificate About to Expire) is raised and later a higher-severity alarm (for example, Certificate Expired) is raised about the same issue, the lower-severity alarm is not automatically resolved. You must take the recommended action to resolve the alarm.

Alarm information displays in several locations within the NSX Manager interface. For a complete list of events, see *NSX Event Catalog* in Related Resources.

View Alarm Information

Alarms information is displayed in several locations within the NSX Manager interface. Alarm and event information is also included with other notifications in the Notifications drop-down menu in the title bar.

An alarm can be in one of the following states:

State	Description
Open	Alarm is in an active, unacknowledged state.
Acknowledged	Alarm has been acknowledged by a user. The alarm remains open but no longer appears in the NSX Manager notifications.
Suppressed	Status reporting for this alarm has been disabled by the user for a user-specified duration.

State	Description
Resolved	Alarm has been resolved, whether by the system or through user action. The alarm will continue to appear in the alarm table in the Resolved state for up to eight days, after which it automatically deletes. (The system may delete resolved alarms earlier to accommodate resource needs.) **Note:** If a user changes an alarm state to Resolved but the condition that triggered the alarm is not resolved, a new alarm instance will be instantiated. Also, an event may be resolved for several minutes before the reported state updates in the interface.

Note:
The following steps show how to view alarms from the Home page. However, you can also view alarms from other pages, such as the Tier-0, Tier-1, and Load Balancing pages, among others. See the Alarms columns in the tables on these pages.

1. Navigate to the Home page and click **Alarms**.

   Note:
   A red exclamation mark (**!**) next to the **Alarms** panel label indicates at least one open alarm with a severity of Critical. .

   The Alarms panel appears, displaying along the top graphic dashboards such as Active Alarms, Top Features with the Most Alarms, and Top Events by Occurrence. Below the dashboards is a sortable, filterable list of the current alarms. The table details the following information about each active alarm:

   - Feature affected
   - Event Type
   - Node
   - Entity
   - Severity (Critical, High, Medium)
   - Last Reported Time
   - Alarm State (Open, Suppressed, Resolved, Acknowledged)

   Each row in the Alarms table can be expanded to show more details.

2. Filter the results displayed in the dashboards by clicking the funnel icon in the upper-right corner of the dashboards.
   You can filter by the last 24 hours, last 48 hours, or custom time range, or all open alarms.

3. Filter the results displayed in the table by clicking the filter text box above the table.
   You are prompted to specify a filter: Alarm State, Description, Entity Name, Entity Type, Event Type, Node, and so on.

After viewing an alarm, you can decide on how to respond. See Managing Alarm States.

View Alarm Definitions

Detailed alarm definitions are provided on a separate panel in the Alarms tab. You can open the panel directly or arrive there by clicking the value in the Event Type column in the Alarms table.

Alarms details are displayed in several locations in the NSX Manager. See View Alarm Information.

1. From the Alarms tab.
   a) Navigate to the Home page and click **Alarms**.
      The Alarms panel has two modes, as shown at the top of the panel: **Alarms** and **Alarm Definitions**.
   b) Click **Alarm Definitions**.
      The Alarms tab redisplays to show the table of Alarm definitions.

2. From the Tier-0 Gateways page.
   a) Go to **Networking** > **Connectivity** > **Tier-0 Gateways**.
      The Open Alarms column of the gateway table displays the number of open alarms.
   b) Click the number in the Open Alarms column.
      A dialog opens to display the open alarms in table format.
   c) Click the value in the Event Type column.
      This action moves you to the **Home** > **Alarms** > **Alarm Definitions** panel described above.
3. From the Load Balancing page.
   a) Go to **Networking** > **Network Services** > **Load Balancing**.
      The Open Alarms column of the gateway table displays the number of open alarms.
   b) Click the number in the Open Alarms column.
      A dialog opens to display the open alarms in table format.
   c) Click the value in the Event Type column.
      This action moves you to the **Home** > **Alarms** > **Alarm Definitions** panel described above.
4. After you access the **Alarm Definitions**, expand any definition to view details and user-definable settings.
   Alarm definition details include:

Column	Description
Feature	Displays the component where the alarm is originating, for example: Transport Node.
Event Type	Displays the specific type of error, for example: CPU Usage High.
Severity	Displays the level of alarm: Critical, High, or Medium.
Enabled	Displays whether detection of the Alarm is enabled.
Create Alarms	Displays whether to report the alarm in the interface or API.
Create SNMP Traps	Displays whether the system emits an SNMP trap when the alarm is detected or resolved.

The panel also displays the following:

Item	Description
Description	Describes the condition that triggers the alarm.
Recommended Action	Describes steps you can take to correct the condition.
SNMP OID for Event true	Displays the SNMP Object Identifier for the Event when status is true.
SNMP OID for Event false	Displays the SNMP Object Identifier for the Event when status is false.
Threshold	User-configured threshold for triggering the alarm.
Sensitivity (%)	User-configured sensitivity for triggering the alarm.

Some of fields in an alarm definition can be modified. See Configuring Alarm Definition Settings.

Configuring Alarm Definition Settings

Several settings in an alarm definition can be customized. From the Alarm Definitions page, you can enable or disable an alarm, configure if an event (when true) creates an alarm, create an SNMP trap, set alarm threshold, and set alarm

sensitivity. From the Alarm Definitions page, you can enable or disable detection of an alarm, whether an alarm is reported in the API/user interface, and whether a SNMP trap is emitted when an alarm is detected or resolved.

You can configure the following alarm definition settings:

Setting	Control Type	Description
Enabled	Toggle	Enables or disables detection of the alarm.
Create Alarms	Toggle	Enables or disables whether the alarm is reported in the API/UI.
Create SNMP Traps	Toggle	Enables or disables whether an SNMP trap is emitted when an alarm is detected or resolved.
Threshold	Numerical value	Configures the threshold for triggering the event. This value determines if a single sample is true and triggers an event. • For CPU, disk, and memory alarms, threshold is the percentage usage value to indicate an alarming condition. • For certificate or license expiration alarms, this is the number of days before expiration, including local password expiration.
Sensitivity (%)	Numerical value (percentage)	Configures the sensitivity for triggering the alarm. Sensitivity defines the conditions that trigger an alarm. (The sample size is internally defined and cannot be modified.) If the sample size is ten and sensitivity is set to 80%, then eight or more occurrences in the sample of ten raises the alarms. See the NSX-T Data Center REST API documentation.

1. Navigate to the Home page and click **Alarms**.
   The Alarms panel has two modes, as shown at the top of the panel: **Alarms** and **Alarm Definitions**.
2. Click **Alarm Definitions**.
   The Alarms tab redisplays to show the Alarm Definitions panel.
3. Right-click three vertical dots icon in the leftmost column of an alarm, and select **Edit**.
   The selected alarm definition expands to show the definition details, and puts the configurable settings into edit mode.
4. Modify the settings as desired.
5. Click **Save**.

For details about alarm definitions, see View Alarm Definitions. For details about SNMP traps, see Simple Network Management Protocol (SNMP).

Managing Alarm States

In addition to correcting the underlying causes, you can manage alarms by modifying their states as reported in the Alarms list.

Triggered alarms can be in one of the following states: Open, Acknowledged, Suppressed, or Resolved.

1. Navigate to the Home page and click **Alarms**.
   The Alarms panel has two modes, as shown at the top of the panel: **Alarms** and **Alarm Definitions**.
2. Click the **Alarms** mode, if the panel is not already displayed.
   The **Alarms** panel displays a list of all alarms, including Resolved alarms.

Note:
Resolved alarms continue to be listed for up to eight days after their resolution.

3. Locate the alarm in the table on the page, and select the checkbox in the leftmost column.
4. Click **Action** and select the desired action.
   - If you change the state of an alarm to Acknowledged, this indicates that you are aware of, and have acknowledged, the alarm.
   - If you move an alarm into a Suppressed state, you are prompted to specify the duration in hours. After the specified duration passes, the alarm state reverts to Open. However, if the system determines the condition has been corrected, the alarm state changes to Resolved.
   - You can restore the state of an Acknowledged or Suppressed alarm to Open.
   - You cannot change the state of a Resolved alarm.

   The value in the Alarm State column updates accordingly.

Troubleshooting Communication Alarms Between Your NSX Manager and Host

You may experience a loss of communication alarm between the NSX Manager and the host. Heartbeats are sent every minute from the manager node to the host node. This alarm notifies you of a heartbeat loss which is an indication of the last time NSX Manager and the host communicated.

If the host loses connectivity to the NSX Manager or to the controller service, a full configuration or runtime state synchronization, respectively, gets sent to the host when connectivity is restored.

Alarm information is displayed in several locations within the NSX Manager interface. For a complete list of events, see NSX Event Catalog. For example, alarms display on the Alarms page of the NSX Manager UI. Every alarm contains a recommended action. Use this action and the information in this topic to troubleshoot your alarm.

The following table describes the IP protocol used, channel health check mechanism and related timers, authentication, security, and communication channel health check validations. It also provides insights into the several common issues that lead to degraded fabric health.

To validate the health of the communication channels, use the NSX Manager Health Monitoring table details.

Table 1055: NSX Manager Health Monitoring

Communication	IP Protocol and Port	Channel	Authentication	Health Check
Management Plane Appliance Proxy Hub (MP APH) to host connectivity	TCP:1234	TLS between Manager and host	Mutual TLS authentication based on self-signed or CA certificates between clients (hosts/transport nodes) and server (APH in MP)	To determine the connectivity status between transport node and managers, use `get managers` CLI command on the transport node. Various alarms get raised when communication channels between MP and Host are in a disconnected state. • `management_channel_to_transport_node_down` alarm gets raised when MP and host are disconnected for more than 5 minutes. Alarm description mentions transport node name and IP address: management channel to transport node {transport_node_name} ({transport_node_address}) is down for 5 minutes. • `management_channel_to_tr` alarm gets raised when MP and host are disconnected for more than 15 minutes. Alarm description mentions transport node name and IP address: management channel to transport node {transport_node_name} ({transport_node_address}) is down for 15. • `network_latency_high` alarm gets raised when the latency between MP and host is more than 150 ms for a duration of 5 minutes. Alarm description mentions transport node name and IP address: The average network latency between manager nodes and host {transport_node_name} ({transport_node_address}) is more than 150 ms for 5 minutes. If the host gets detached for any reason the alarms are cleared.
Central Control Plane (CCP) to host (NSX-	TCP: 1235	TLS between CCP and host	Mutual TLS authentication based	To determine the connectivity status

Registering Notification Watchers

You can register a watcher that will receive notifications based on specific criteria.

To register a watcher, you can invoke the following API.

`POST /api/v1/notification-watcher`

After adding a watcher, you must register a notification_id (feature_name.notification_name) with a watcher_id and specify notifications that the watcher should receive. Note that without the registration, the watcher will not receive any notifications. Invoke the following APIs with the required request parameters to register notification_id and to specify notifications. For more information on NSX Notification APIs, see *NSX API Guide*.

- `PUT /api/v1/notification-watchers/<watcher-id>`
- `POST /api/v1/notification-watchers/<watcher-id>/notifications?action=add_uri_filters` with the following request parameters:
  - notification_id: A string identifying feature_name.notification_name to indicate a notification that watcher is interested in receiving for the URI identified by the feature_name.notification_name.
  - uri_filters: Optional list of URIs to filter notifications based on its policy path. When specifying uri_filters, you can also use * as a wildcard character instead of a specific value.
    For example, if the notification_id is group.change_name, the uri_filter pattern is /policy/api/v1/infra/domains/<domain>/groups/<group>. You can specify the pattern as /policy/api/v1/infra/domains/domain1/groups/group2 to get notifications specific to domain1 and group2. Alternatively, you can also specify the pattern /policy/api/v1/infra/domains/domain2/groups/* to get notifications for all groups in domain2 or specify it as /policy/api/v1/infra/domains/*/groups/* to get notifications for all groups in all domains.

The following table lists the feature names and their respective URIs.

Feature Name	Feature Description	Notification Name	Notification Description	URI
group	Notifications supported by NS Group feature.	change_notification	Group notification, <domain> identifies the domain name and <group> identifies group name.	/policy/api/v1/infra/domains/<domain>/groups/<group>
monitoring	Notifications supported by the monitoring feature.	alarm	Alarm notifications. <alarm-id> identifies an alarm instance. A notification is sent whenever an alarm instance is created or deleted and when the alarm instance is updated.	/api/v1/alarms/<alarm-id>
		alarm_status_change_notification	Alarm notifications. <alarm-id> identifies an alarm instance. A notification is sent whenever an alarm instance is created and when the status property value of an alarm instance is updated.	/api/v1/alarms/<alarm-id>
notification	Notifications supported by notification framework.	watcher	Platform notification to convey updates to watcher configuration. <watcher-id> identifies the watcher.	/api/v1/notification-watchers/<watcher-id>
		watcher_notification	Platform notification to convey updates to notifications. <watcher-id> identifies the watcher.	/api/v1/notification-watchers/<watcher-id>/notifications

Feature Name	Feature Description	Notification Name	Notification Description	URI
service_config	Notifications supported by Service Config feature.	change_notification	Service config notification. <domain> identifies the domain name, <policy> identifies the endpoint policy, and <rule> identifies the endpoint rule. This notification is generated when a service config used in endpoint rule is updated or when UPM Profile is updated.	/policy/api/v1/infra/domains/<domain>/endpoint-policies/<policy>/endpoint-rules/<rule>
service_insertion	Notifications supported by Service Insertion module. Currently Service Insertion module supports notifications for Service Profile, Service Instance Runtime, and Policy Groups.	instance_runtime_notification	Service Instance Runtime notification. <service-id> identifies the service, <service-instance-id> identifies the service instance. Notification will be sent for deployed and undeployed operations.	/api/v1/serviceinsertion/services/<service-id>/service-instances/<service-instance-id>/instance-runtimes
		profile_notification	Service Profile change notification. <service-reference> identifies the service name and <service-profile> identifies profile name. Notification will be sent for profile create, update, and delete.	/policy/api/v1/infra/service-references/<service-reference>/service-profiles/<service-profile>
		profile_chain_mapping_notification	Service Profile Chain Mapping notification. <service-reference> identifies the service name and <service-profile> identifies profile name. The notification will be sent when a profile is added or removed as a part of a service chain.	/policy/api/v1/infra/service-references/<service-reference>/service-profiles/<service-profile>/service-chain-mappings
		profile_nsgroups_notification	Service Profile NSGroups notification. <service-reference> identifies the service name and <service-profile> identifies profile name. This notification gets triggered whenever an east-west rule containing nsgroups gets added or deleted with the particular profile.	/policy/api/v1/infra/service-references/<service-reference>/service-profiles/<service-profile>/group-associations
		instance_nsgroups_notification	Service Instance NSGroups notification. <service-id> identifies the service name and <service-instance-id> identifies service instance. This notification gets triggered whenever a north-south rule containing nsgroups gets added or deleted with the particular instance.	/api/v1/serviceinsertion/services/<service-id>/service-instances/<service-instance-id>/group-associations

Using VCF Operations to Monitor the NSX Environment

You can monitor your NSX environment using VCF Operations. Dashboards in VCF Operations let you monitor the system health of NSX Manager nodes and clusters, edge nodes and clusters, and host transport nodes.

Alerts in the Management Pack for NSX

NSX alarm events display as alerts in VCF Operations. For a complete list of alerts, see the NSX Event Catalog.

Viewing NSX Dashboards in VCF Operations

You can access the NSX dashboards under **Infrastructure Operations** > **Dashboards & Reports** > **NSX**.

NSX System Health Dashboard

The NSX System Health dashboard provides an overview of the system status through a collection of widgets. You can hover over a widget to display more information, or double-click a node in a heatmap widget to open a more detailed node-level dashboard.

Widget	Description
NSX Instance	Lists the NSX deployment instances available to be monitored. Select an instance in the list to display the health summary of that instance in the dashboard.
NSX Instance Inventory	Summarizes the inventory of objects in the selected NSX instance.
Manager Cluster	Summarizes the health status and key information about the NSX Manager cluster.
Manager Cluster Critical Alerts	Summarizes all critical alerts affecting the NSX Manager cluster. **Note:** Alerts are also known as "alarms" in NSX Manager.
Manager Cluster Certificates	Shows the certificates used by the NSX Manager cluster and reports whether the certificates are valid or expired.
Manager Nodes	Shows a heatmap view of each node in the NSX Manager cluster. Healthy nodes appear green, and problem nodes appear red in the heatmap. Double-click a node in the heatmap to open the NSX Manager Node Health dashboard.
Edge Clusters	Lists the edge clusters in the selected NSX instance. Select an edge cluster in the list to display the health summary of that cluster in the dashboard.
Edge Cluster & Nodes Critical Alerts	Summarizes all critical alerts affecting the selected edge cluster and nodes in that cluster. **Note:** Alerts are also known as "alarms" in NSX Manager.
Edge Nodes	Shows a heatmap view of each node in the selected cluster. Healthy nodes appear green, and problem nodes appear red in the heatmap. Double-click a node in the heatmap to open the NSX Edge Node Health dashboard.
Host Transport Nodes Critical Alerts	Summarizes all critical alerts affecting the host transport nodes. **Note:** Alerts are also known as "alarms" in NSX Manager.
Host Transport Nodes	Shows a heatmap view of each host transport node. Healthy nodes appear green, and problem nodes appear red in the heatmap. Double-click a node in the heatmap to open the NSX Host TN Health dashboard.

NSX Manager Node Health Dashboard

This dashboard shows the detailed health status of the selected manager node.

You can access the NSX Manager Node Health dashboard by double-clicking a manager node in the heatmap widget of the NSX System Health dashboard. You can also open the dashboard directly from **Infrastructure Operations** > **Dashboards & Reports** > **NSX**.

Hover over widgets in the dashboard to display more information, and double-click metrics, where available, to display detailed time series views. Use the controls in the top right corner of the dashboard to set the range for time series views.

Widget	Description
Manager Node Properties	Summarizes key information and uptime status for the node.
Manager Node Critical Alerts	Summarizes all critical alerts affecting the node. **Note:** Alerts are also known as "alarms" in NSX Manager.
Manager Node Health Indicators	Summarizes key metrics that indicate the health status of the node, including uptime, interface link status, and the overall count of API requests. Double-click a graph to display a detailed time series view.
Manager Node Services	Lists the uptime status of services running on the node.
Manager Node API Requests Top Clients	Shows the clients that make the most API requests to the node, listed in order of call frequency.
Manager Node Utilization	Summarizes utilization metrics for CPU, memory, and disk storage resources in the node. Double-click a graph to display a detailed time series view.

NSX Edge Node Health Dashboard

This dashboard shows the detailed health status of the selected edge node.

You can access the NSX Edge Node Health dashboard by double-clicking an edge node in the heatmap widget of the NSX System Health dashboard. You can also open the dashboard directly from **Infrastructure Operations** > **Dashboards & Reports** > **NSX**.

Hover over widgets in the dashboard to display more information, and double-click metrics, where available, to display detailed time series views. Use the controls in the top right corner of the dashboard to set the range for time series views.

Widget	Description
Edge Node Properties	Summarizes key information for the node.
Edge Node Critical Alerts	Summarizes all critical alerts affecting the node. **Note:** Alerts are also known as "alarms" in NSX Manager.
Edge Node Health Indicators	Summarizes key metrics that indicate the health status of the node. Double-click an indicator metric to display a detailed time series view. The following indicators are monitored: • Node uptime status • Management connection • Controller connection • VTEP status • Tunnels up count • Tunnels down count • PNIC up count • PNIC down count • Total disk space • Storage access • Duration of uptime
Edge Node Interface	Lists the node interface instances and key link and transmission metrics for each instance.
Edge Resource Utilization	Summarizes utilization metrics for CPU, memory, and disk storage resources in the node. Double-click a graph to display a detailed time series view.

NSX Host TN Health Dashboard

This dashboard shows the detailed health status of the selected host transport node.

You can access the NSX Host TN Health dashboard by double-clicking a host transport node in the heatmap widget of the NSX System Health dashboard. You can also open the dashboard directly from **Infrastructure Operations** > **Dashboards & Reports** > **NSX**.

Hover over widgets in the dashboard to display more information, and double-click metrics, where available, to display detailed time series views. Use the controls in the top right corner of the dashboard to set the range for time series views.

Widget	Description
Host TN Key Properties	Summarizes key information for the node.
Host TN Critical Alerts	Summarizes all critical alerts affecting the node. **Note:** Alerts are also known as "alarms" in NSX Manager.
Host TN Health Indicators	Summarizes key metrics that indicate the health status of the node. Double-click an indicator metric to display a detailed time series view. Monitored indicators include: • Node uptime status • Deployment status • Management connection • Controller connection • Tunnels down count • PNIC down count • PNIC status • Agents down count • Uptime duration
Host TN PNICs	Lists the pNIC instances and key link and transmission metrics for each instance.
Host Agent Status	Lists the uptime status of agents running on the node.
Host Transport Node Utilization	Summarizes utilization metrics for system resources in the node. Double-click a graph to display a detailed time series view.

Using VCF Operations for logs or Splunk for System Monitoring

You can monitor your NSX environment using VCF Operations for logs or Splunk.

You can find the NSX Splunk app at https://splunkbase.splunk.com/app/4241.

The VCF Operations for logs content pack has the following alerts:

Alert Name	Description
SysCpuUsage	CPU usage is above 95% for more than 10 minutes.
SysMemUsage	Memory usage is above 95% for more than 10 minutes.
SysDiskUsage	Disk usage for one or more partitions is above 89% for more than 10 minutes.
PasswordExpiry	Password for appliance user account is about to expire or expired.
CertificateExpiry	One or more CA signed certificate is expired.
ClusterNodeStatus	Local edge cluster node is down.
BackupFailure	NSX scheduled backup operation failed.
VipLeadership	NSX Management cluster VIP is down.
ApiRateLimit	Client API reached configured threshold.
CorfuQuorumLost	Two nodes went down in the cluster and lost corfu quorum.

Alert Name	Description
DfwHeapMem	DFW heap memory exceeded configured threshold.
ProcessStatus	Critical process status changed.
ClusterFailoverStatus	SR high availability state changed or active/standby services failover.
DhcpPoolUsageOverloadedEvent	DHCP pool reached configured usage threshold.
FabricCryptoStatus	Edge crypto mux driver is down for failing Known_Answer_Tests (KAT).
VpnTunnelState	VPN tunnel is down.
BfdTunnelStatus	BFD Tunnel status changed.
RoutingBgpNeighborStatus	BGP neighbor status is down.
VpnL2SessionStatus	L2 VPN session is down.
VpnIkeSessionStatus	IKE session is down.
RoutingStatus	Routing(BGP/BFD) is down.
DnsForwarderStatus	DNS forwarder running status is DOWN.
TnConnDown_15min	Transport Node connection to a controller/Manager is down for at least 15 minutes.
TnConnDown_5min	Transport Node connection to controller/Manager is down for at least 5 minutes.
ServiceDown	One or more services are down.
IpNotAvailableInPool	There is no IP available in the Pool or reaches configured threshold.
LoadBalancerError	NSX Load Balancer Service status is ERROR.
LoadBalancerDown	NSXLoad Balancer Service status is DOWN.
LoadBalancerVsDown	VS status: all pool members are down.
LoadBalancerPoolDown	Pool status: all pool members are down.
ProcessCrash	Process or daemon crashes in the datapath or other LB process like dispatcher, etc..

Dashboards

Both the Splunk app and the VCF Operations for logs content pack have the following dashboards.

Table 1056: NSX - Infrastructure

Widget Name	Notes
NSX Manager: Communication Errors	These are all communication log errors on NSX Manager. They are grouped by hostname.
Transport Node - NSX Manager : Communication Errors	Communication errors between NSX transport nodes (vSphere hosts and NSX Edges) and NSX Manager. It is recommended to analyze the hostnames with the highest returned values to look for potential issues.
Transport Node - Controller : Communication Errors	Communication errors between NSX transport nodes (vSphere hosts and NSX Edges) and NSX controllers. It is recommended to analyze the hostnames with the highest returned values to look for potential issues.
Controller: Communication Errors	Communication errors among controllers in a cluster. It is recommended to analyze the hostnames with the highest returned values to look for potential issues.
Configuration Errors	This widget is based on known error patterns generated from various components that create the NSX infrastructure. It is recommended to analyze the hostnames with the highest returned values to look for potential issues.

Widget Name	Notes
Other Errors	Other errors from all NSX components. It is recommended to analyze the hostnames with the highest returned values to look for potential issues. Host agent failures, such as netcpa down or MPA down, may require immediate action.

Table 1057: NSX - Audits

Widget Name	Notes
Logical Switch Audits	Logical switch messages excerpted from the NSX audit log capturing any create, update, or delete events. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of logical switch audit events.
Logical Switch Audit Details	All logical switch audit events. This widget is based on known error patterns generated from various components that create the NSX logical switch infrastructure.
Logical Switch Port Audits	Logical switch port messages excerpted from the NSX audit log capturing any create, update, or delete events. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of logical switch port audit events.
Logical Switch Port Audit Details	All logical switch port audit events. This widget is based on known error patterns generated from various components that create the NSX logical switch infrastructure.
Logical Router Audits	Logical router messages excerpted from the NSX audit log capturing any create, update, or delete events. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of logical router audit events.
Logical Router Audit Details	All logical router audit events. This widget is based on known error patterns generated from various components that create the NSX logical routing infrastructure.
Logical Router Port Audits	Logical router port messages excerpted from the NSX audit log capturing any create, update, or delete events. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of logical router port audit events.
Logical Router Port Audit Details	All logical switch port audit events. This widget is based on known error patterns generated from various components that create the NSX logical router infrastructure.
Firewall Audits	Firewall messages excerpted from the NSX audit log capturing any add section, update section with new rules, or delete section events. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of firewall audit events.
Logical Firewall Audit Details	All firewall audit events. This widget is based on known error patterns generated from the NSX firewall.

Table 1058:

MSX - Logical Switch	Notes
Logical Switch Created	Logical switch messages excerpted from the NSX audit log capturing any create events. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of logical switch audit events.
Logical Switch Updates	Logical switch messages excerpted from the NSX audit log capturing any update events. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of logical switch audit events.
Logical Switch Deleted	Logical switch messages excerpted from the NSX audit log capturing any delete events. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of logical switch audit events.
Logical Switch Audit Details	All logical switch audit events.

MSX - Logical Switch	Notes
Logical Switch - Manager Errors	Logical switch log errors reported to NSX Manager. Errors are grouped by hostname.
Logical Switch - Controller Errors	Logical switch log errors reported from the view of the NSX controllers. Errors are grouped by hostname.
Logical Switch - Transport Node Errors	Logical switch log errors reported for all NSX transport nodes - vSphere hosts and Edge Services Gateways. Errors are grouped by node.

Table 1059: NSX - Logical Router

Widget Name	Notes
Logical Router Create Audit Events	Logical router messages excerpted from the NSX audit log capturing any create events. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of logical router audit events.
Logical Router Update Audit Events	Logical router messages excerpted from the NSX audit log capturing any update events. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of logical router audit events.
Logical Router Delete Audit Events	Logical router messages excerpted from the NSX audit log capturing any delete events. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of logical router audit events.
Logical Router Audit Details	All logical router audit events.
Logical Router - Manager Errors	Logical router log errors reported to NSX Manager. Errors are grouped by hostname.
Logical Router - Controller Errors	Logical router log errors detected by NSX controllers. Errors are grouped by hostname.
Logical Router - Transport Node Errors	Logical router log errors for NSX transport nodes - vSphere hosts and Edge Services Gateways. Errors are grouped by node.

Table 1060: NSX - Distributed Firewall Overview

Widget Name	Notes
Section Create Events	All firewall section create audit events. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of firewall changes.
Section Update Events	All firewall section update audit events. Note - Any create, update, delete activity on a rule raising section update event. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of firewall changes.
Section Delete Events	All firewall section delete audit events. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of firewall changes.
Section Audit Details	Shows all Firewall audit events (who changed what). Note - Any create, update, delete activity on a rule raising section update event.
Firewall - Manager Errors	Firewall log errors reported to NSX Manager. Errors are grouped by hostname.
Firewall - Controller Errors	Firewall log errors reported to NSX controllers. Errors are grouped by hostname.
Firewall - Transport Node Errors	Firewall log errors on NSX transport nodes - vSphere hosts. Errors are grouped by node.

Table 1061: NSX - Distributed Firewall Traffic

Widget Name	Notes
Top Firewall Sources	Top source IP addresses from all firewall rules that are logging data.
Top Firewall Destinations	Top destination IP addresses from all firewall rules that are logging data.
Application Ports Permitted	Measures all in / out connections permitted in the NSX-T environment by destination port. The data is the summation of the specified time range.
Application Ports Denied	All traffic defined by a firewall rule. Data is grouped by application (or destination) port number. This widget displays only data associated with a port. Traffic types, such as ICMP, without an associated port are not displayed.
Top Firewall Sources by bytes - client to server	All firewall traffic, in bytes by source IP address, from the client to a server. Data is only displayed if the firewall is logging its data. The data is the summation of the specified time range.
Top Firewall Destinations by bytes - client to server	All firewall traffic, in bytes by destination IP address, from the client to a server. Data is only displayed if the firewall is logging its data. The data is the summation of the specified time range.
Top Firewall Sources by bytes - server to client	All firewall traffic, in bytes by source IP address, from the server to a client. Data is only displayed if the firewall is logging its data. The data is the summation of the specified time range.
Top Firewall Destinations by bytes - server to client	All firewall traffic, in bytes by destination IP address, from the server to a client. Data is only displayed if the firewall is logging its data. The data is the summation of the specified time range.

Table 1062: NSX - DHCP

Widget Name	Notes
DHCP Create Audit Events	All DHCP create audit events, including new DHCP profiles, static bindings, or IP pools. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of firewall changes.
DHCP Update Audit Events	All DHCP update audit events, including updated DHCP profiles, static bindings, and IP pools. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of firewall changes.
DHCP Delete Audit Events	All DHCP delete audit events, including deleted DHCP profiles, static bindings, or IP pools. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of firewall changes.
DHCP Audit Details	All DHCP audit events.
DHCP - Manager Errors	DHCP log errors reported by NSX Manager.
DHCP - Controller Errors	DHCP log errors reported by NSX controllers.
DHCP - Transport Node Errors	DHCP log errors reported for NSX transport nodes - vSphere hosts and Edge Services Gateways.

Table 1063: NSX - Backup

Widget Name	Notes
Backup Configuration Updates	Total number of times backup configuration was updated.
Backup Failures	Count of all backup failures over time grouped by error code.
Successful Cluster Backups	Total number of cluster and node backups completed successfully.
Failed Cluster Backups	Total number of cluster and node backups failed.
Successful Inventory Backups	Total number of inventory backups completed successfully.
Failed Inventory Backups	Total number of inventory backups failed.

Table 1064: NSX - IPAM

Widget Name	Notes
IPAM Create Events	IPAM messages excerpted from the NSX audit log capturing any create events. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of IPAM audit events.
IPAM Update Events	IPAM messages excerpted from the NSX audit log capturing any update events. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of IPAM audit events.
IPAM Delete Events	IPAM messages excerpted from the NSX audit log capturing any delete events. Note: Events may be duplicated as they are generated in multiple NSX loggers. Thus, this widget will provide a general count of IPAM audit events.
IPAM Audit Details	All IPAM audit events.
IPAM - Manager Errors	IPAM log errors reported by NSX Manager.

Table 1065: NSX - Unified Security Flow Logs

Widget Name	Notes
Top Security Vertical Source	Top source IP addresses from all security verticals that are logging data.
Top Security Vertical Destination	Top destination IP addresses from all security verticals that are logging data.
Application Ports Permitted	Measures all in / out connections permitted in the NSX-T environment by destination port. The data is the summation of the specified time range.
Application Ports Denied	All security flows denied by a security vertical rule. Data is grouped by application (or destination) port number. This widget displays only data associated with a port. Traffic types, such as ICMP, without an associated port are not displayed.
Top Security Vertical Sources by bytes - client to server	All security flows, in bytes by source IP address, from the client to a server. Data is only displayed if the vertical is logging its data. The data is the summation of the specified time range.
Top Security Vertical Destinations by bytes - client to server	All security flows, in bytes by destination IP address, from the client to a server. Data is only displayed if the vertical is logging its data. The data is the summation of the specified time range.
Top Security Vertical Sources by bytes - server to client	All security flows, in bytes by source IP address, from the server to a client. Data is only displayed if the vertical is logging its data. The data is the summation of the specified time range.
Top Security Vertical Destinations by bytes - server to client	All security flows, in bytes by destination IP address, from the server to a client. Data is only displayed if the vertical is logging its data. The data is the summation of the specified time range.

Using VCF Operations for networks for System Monitoring

You can monitor your NSX environment using VCF Operations for networks.

Table 1066: Computed NSX Events in VCF Operations for networks

OID	Event Name	Default Severity	UI Name	Description
1.3.6.1.4.1.6876.100.1.0.80205	NSXTNoUplinkConnectivityEvent	Warning	NSX Tier-1 logical router disconnect event	NSX Tier-1 logical router is disconnected from Tier-0 router. Networks under this router are not reachable from outside and vice versa.
1.3.6.1.4.1.6876.100.1.0.80206	NSXTRoutingAdvertisementEvent	Warning	Routing advertisement disabled	Routing advertisement is disabled for NSX Tier-1 logical router. Networks under this router are not reachable from outside.
1.3.6.1.4.1.6876.100.1.0.80207	NSXTManagerConnectivityDownEvent	Critical	NSX Edge Node has no manager connectivity	NSX Edge Node has lost manager connectivity.
1.3.6.1.4.1.6876.100.1.0.80208	NSXTControllerConnectivityDegradedEvent	Warning	Controller connectivity degraded for NSX Edge Node	NSX Edge Node is not able to communicate with one or more controllers.
1.3.6.1.4.1.6876.100.1.0.80209	NSXTControllerConnectivityDownEvent	Critical	NSX Edge Node has no controller connectivity	NSX Edge Node is not able to communicate with any of the controllers.
1.3.6.1.4.1.6876.100.1.0.80210	NSXTMtuMismatchEvent	Warning	MTU mismatch between NSX Tier-0 and uplink switch/router	The MTU configured on interfaces of Tier-0 logical router does not match with the interfaces of uplink switch/router from same L2 network. This can impact the network performance.
1.3.6.1.4.1.6876.100.1.0.80211	NSXTExcludedVmFlowEvent	Info	One or more VMs excluded from NSX DFW Firewall.	One or more VMs are not protected by NSX DFW firewall. VCF Operations for networks will not receive IPFIX flows for these VMs.
1.3.6.1.4.1.6876.100.1.0.80212	NSXTDoubleVlanTaggingEvent	Warning	Uplink Vlan misconfiguration	Communication is disrupted because VLAN on uplink port of Tier 0 router is different than VLAN on the external gateway.
1.3.6.1.4.1.6876.100.1.0.80213	NSXTNoTzAttachedOnTnEvent	Warning	No transport zone is attached to the transport node.	No transport zone attached to the transport node. VMs might lose connectivity because of this.
1.3.6.1.4.1.6876.100.1.0.80214	NSXTVtepDeleteEvent	Warning	No VTEP available on the transport node.	All vteps are deleted from the transport node. VMs might lose connectivity because of this.
1.3.6.1.4.1.6876.100.1.0.80225	NSXTControllerNodeToControlClusterConnectivityEvent	Critical	NSX controller node has no control cluster connectivity	NSX controller node has lost control cluster connectivity.
1.3.6.1.4.1.6876.100.1.0.80226	NSXTControllerNodeToMgmtPlaneConnectivityEvent	Critical	NSX controller node has no management plane connectivity	NSX controller node has lost management plane connectivity.

OID	Event Name	Default Severity	UI Name	Description
1.3.6.1.4.1.6876.100.1.0.80227	NSXTMPNodeToMgmtClusterConnectivityEvent	Critical	NSX management node has no management cluster connectivity	NSX management node has lost management cluster connectivity.
1.3.6.1.4.1.6876.100.1.0.80246	NSXTHostNodeMgmtConnectivityStatusDownEvent	Warning	NSX Host Node has no manager connectivity	Desynchronization between NSX Manager's State of connectivity with Host Transport Nodes
1.3.6.1.4.1.6876.100.1.0.80247	NSXTEdgeNodeCtlrConnectivityStatusUnknownEvent	Critical	Controller connectivity for NSX Edge Node is Unknown.	NSX Edge Node Controller connectivity is Unknown.
1.3.6.1.4.1.6876.100.1.0.80248	NSXTHostNodeCtlrConnectivityStatusDownEvent	Warning	NSX Host Node has no controller connectivity	NSX Host Node is not able to communicate with any of the controllers.
1.3.6.1.4.1.6876.100.1.0.80249	NSXTHostNodeCtlrConnectivityStatusDegradedEvent	Warning	Controller connectivity degraded for NSX Host Node	NSX Host Node is not able to to communicate with one or more controllers.
1.3.6.1.4.1.6876.100.1.0.80250	NSXTHostNodeCtlrConnectivityStatusUnknownEvent	Warning	Controller connectivity for NSX Host Node is Unknown.	NSX Host Node Controller connectivity is Unknown.
1.3.6.1.4.1.6876.100.1.0.80228	NSXTHostNodePnicStatusDownEvent	Warning	NSX Host Transport Node Pnic Status is 'Down'.	NSX Host Transport Node Pnic Status is 'Down'.
1.3.6.1.4.1.6876.100.1.0.80229	NSXTHostNodePnicStatusDegradedEvent	Warning	NSX Host Transport Node Pnic Status is 'Degraded'	NSX Host Transport Node Pnic Status is 'Degraded'.
1.3.6.1.4.1.6876.100.1.0.80230	NSXTHostNodePnicStatusUnknownEvent	Warning	NSX Host Transport Node Pnic Status is 'Unknown'.	NSX Host Transport Node Pnic Status is 'Unknown'.
1.3.6.1.4.1.6876.100.1.0.80237	NSXTEdgeNodePnicStatusDownEvent	Critical	NSX Edge Transport Node Pnic Status is 'Down'.	NSX Edge Transport Node Pnic Status is 'Down'.
1.3.6.1.4.1.6876.100.1.0.80238	NSXTEdgeNodePnicStatusDegradedEvent	Critical	NSX Edge Transport Node Pnic Status is 'Degraded'.	NSX Edge Transport Node Pnic Status is 'Degraded'.
1.3.6.1.4.1.6876.100.1.0.80239	NSXTEdgeNodePnicStatusUnknownEvent	Critical	NSX Edge Transport Node Pnic Status is 'Unknown'.	NSX Edge Transport Node Pnic Status is 'Unknown'.
1.3.6.1.4.1.6876.100.1.0.80231	NSXTHostNodeTunnelStatusDownEvent	Warning	NSX Host Transport Node Tunnel Status is 'Down'.	NSX Host Transport Node Tunnel Status is 'Down'.
1.3.6.1.4.1.6876.100.1.0.80232	NSXTHostNodeTunnelStatusDegradedEvent	Warning	NSX Host Transport Node Tunnel Status is 'Degraded'.	NSX Host Transport Node Tunnel Status is 'Degraded'.
1.3.6.1.4.1.6876.100.1.0.80233	NSXTHostNodeTunnelStatusUnknownEvent	Warning	NSX Host Transport Node Tunnel Status is 'Unknown'.	NSX Host Transport Node Tunnel Status is 'Unknown'.
1.3.6.1.4.1.6876.100.1.0.80240	NSXTEdgeNodeTunnelStatusDownEvent	Critical	NSX Edge Transport Node Tunnel Status is 'Down'.	NSX Edge Transport Node Tunnel Status is 'Down'.
1.3.6.1.4.1.6876.100.1.0.80241	NSXTEdgeNodeTunnelStatusDegradeEvent	Critical	NSX Edge Transport Node Tunnel Status is 'Degraded'.	NSX Edge Transport Node Tunnel Status is 'Degraded'.

OID	Event Name	Default Severity	UI Name	Description
1.3.6.1.4.1.6876.100.1.0.80242	NSXTEdgeNodeTunnelStatusUnknownEvent	Critical	NSX Edge Transport Node Tunnel Status is 'Unknown'.	NSX Edge Transport Node Tunnel Status is 'Unknown'.
1.3.6.1.4.1.6876.100.1.0.80234	NSXTHostNodeStatusDownEvent	Warning	NSX Host Transport Node Status is 'Down'.	NSX Host Transport Node Status is 'Down'.
1.3.6.1.4.1.6876.100.1.0.80235	NSXTHostNodeStatusDegradedEvent	Warning	NSX Host Transport Node Status is 'Degraded'.	NSX Host Transport Node Status is 'Degraded'.
1.3.6.1.4.1.6876.100.1.0.80236	NSXTHostNodeStatusUnknownEvent	Warning	NSX Host Transport Node Status is 'Unknown'.	NSX Host Transport Node Status is 'Unknown'.
1.3.6.1.4.1.6876.100.1.0.80243	NSXTEdgeNodeStatusDownEvent	Critical	NSX Edge Transport Node Status is 'Down'.	NSX Edge Transport Node Status is 'Down'.
1.3.6.1.4.1.6876.100.1.0.80244	NSXTEdgeNodeStatusDegradedEvent	Critical	NSX Edge Transport Node Status is 'Degraded'.	NSX Edge Transport Node Status is 'Degraded'.
1.3.6.1.4.1.6876.100.1.0.80245	NSXTEdgeNodeStatusUnknownEvent	Critical	NSX Edge Transport Node Status is 'Unknown'.	NSX Edge Transport Node Status is 'Unknown'.
1.3.6.1.4.1.6876.100.1.0.80252	NSXTLogicalSwitchAdminStatusDownEvent	Warning	NSX Logical Switch Admin Status is 'Down'	NSX Logical Switch Admin Status is 'Down'
1.3.6.1.4.1.6876.100.1.0.80253	NSXTLogicalPortOperationalStatusDownEvent	Critical	NSX Logical Port Operational Status is 'Down'	NSX Logical Port Operational Status is 'Down'. This could cause a communication failure between two virtual interfaces (VIFs) that are connected to the same logical switch, for example, you cannot ping one VM from another.
1.3.6.1.4.1.6876.100.1.0.80254	NSXTLogicalPortOperationalStatusUnknownEvent	Warning	NSX Logical Port Operational Status is 'Unknown'	NSX Logical Port Operational Status is 'Unknown'. This could cause a communication failure between two virtual interfaces (VIFs) that are connected to the same logical switch, for example, you cannot ping one VM from another.
1.3.6.1.4.1.6876.100.1.0.80255	NSXTComputeManagerConnectionStatusNotUpEvent	Warning	NSX Compute Manager Connection Status is not up	NSX Compute Manager Connection status is not up
1.3.6.1.4.1.6876.100.1.0.80256	NSXTClusterBackUpDisabledEvent	Warning	NSX Manager backup is not scheduled.	NSX Manager backup is not scheduled
1.3.6.1.4.1.6876.100.1.0.80257	NSXTDFWFirewallDisabledEvent	Critical	NSX DFW Firewall is disabled.	Distributed Firewall is disabled in the NSX Manager
1.3.6.1.4.1.6876.100.1.0.80258	NSXTLogicalPortReceivedPacketDropEvent	Warning	NSX Logical Port Received Packets are getting dropped.	Received packets are getting dropped on the NSX Logical Port and associated entities might get affected

OID	Event Name	Default Severity	UI Name	Description
1.3.6.1.4.1.6876.100.1.0.80259	NSXTLogicalPortTransmittedPacketDropEvent	Warning	NSX Logical Port Transmitted Packets are getting dropped.	Transmitted packets are getting dropped on the NSX Logical Port and associated entities might get affected
1.3.6.1.4.1.6876.100.1.0.80260	NSXTLogicalSwitchReceivedPacketDropEvent	Warning	NSX Logical Switch Received Packets are getting dropped	Received packets are getting dropped on the NSX Logical Switch and associated entities might get affected
1.3.6.1.4.1.6876.100.1.0.80261	NSXTLogicalSwitchTransmittedPacketDropEvent	Warning	NSX Logical Switch Transmitted Packets are getting dropped	Transmitted packets are getting dropped on the NSX Logical Switch and associated entities might get affected
1.3.6.1.4.1.6876.100.1.0.80262	NSXTRxPacketDropOnMPNicEvent	Warning	Received packets are dropping on NSX Management Node's network interface	Received packets are getting dropped on NSX Management Node's network interface. This may impact the network traffic related to NSX management cluster.
1.3.6.1.4.1.6876.100.1.0.80263	NSXTRxPacketDropOnEdgeTnNicEvent	Critical	Received packets are dropping on NSX Edge Node's network interface	Received packets are getting dropped on NSX Edge Node's network interface. This may impact the network traffic of edge cluster.
1.3.6.1.4.1.6876.100.1.0.80264	NSXTRxPacketDropOnHostTnNicEvent	Warning	Received packets are dropping on NSX Host Node's network interface	Received packets are getting dropped on NSX Host Node's network interface. This may impact the network traffic on ESX Host.
1.3.6.1.4.1.6876.100.1.0.80265	NSXTTxPacketDropOnMPNicEvent	Warning	Transmitted packets are dropping on NSX Management Node's network interface	Transmitted packets are getting dropped on NSX Management Node's network interface. This may impact the network traffic related to NSX management cluster.
1.3.6.1.4.1.6876.100.1.0.80266	NSXTTxPacketDropOnEdgeTnNicEvent	Critical	Transmitted packets are dropping on NSX Edge Node's network interface	Transmitted packets are getting dropped on NSX Edge Node's network interface. This may impact the network traffic of edge cluster.
1.3.6.1.4.1.6876.100.1.0.80267	NSXTTxPacketDropOnHostTnNicEvent	Warning	Transmitted packets are dropping on NSX Host Node's network interface	Transmitted packets are getting dropped on NSX Host Node's network interface. This may impact the network traffic on ESX Host.
1.3.6.1.4.1.6876.100.1.0.80402	NSXTMPNodeServiceCmInventoryStatusEvent	Warning	CM Inventory Service has stopped running	CM Inventory Service status has turned to stopped.

OID	Event Name	Default Severity	UI Name	Description
1.3.6.1.4.1.6876.100.1.0.80403	NSXTMPNodeServiceControllerStatusEvent	Warning	Controller Service has stopped running.	Controller Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80404	NSXTMPNodeServiceDataStoreStatusEvent	Warning	DataStore Service has stopped running.	DataStore Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80405	NSXTMPNodeServiceHttpStatusEvent	Warning	HTTP Service has stopped running.	HTTP Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80406	NSXTMPNodeServiceInstallUpgradeStatusEvent	Warning	Install Upgrade Service has stopped running.	Install Upgrade Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80407	NSXTMPNodeServiceLiagentStatusEvent	Warning	Liagent service has stopped running.	Liagent Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80408	NSXTMPNodeServiceManagerStatusEvent	Warning	Manager Service has stopped running.	Manager Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80409	NSXTMPNodeServiceMgmtPlaneBusStatusEvent	Warning	Management Plane Service has stopped running.	Management Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80411	NSXTMPNodeServiceNodeMgmtStatusEvent	Warning	Node Management Service has stopped running.	Node Management Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80412	NSXTMPNodeServiceNodeStatsStatusEvent	Warning	Node Statistics Service has stopped running.	Node Statistics Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80413	NSXTMPNodeServiceNSXMessageBusStatusEvent	Warning	Message Bus Service has stopped running.	Message Bus Client Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80414	NSXTMPNodeServiceNSXPlatformClientStatusEvent	Warning	Platform Client Service has stopped running.	Platform Client Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80415	NSXTMPNodeServiceNSXUpgradeAgentStatusEvent	Warning	Upgrade Agent Service has stopped running.	Upgrade Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80416	NSXTMPNodeServiceNTPStatusEvent	Warning	NTP Service has stopped running.	NTP Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80417	NSXTMPNodeServicePolicyStatusEvent	Warning	Policy Service has stopped running.	Policy Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80418	NSXTMPNodeServiceSearchStatusEvent	Warning	Search Service has stopped running.	Search Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80419	NSXTMPNodeServiceSNMPStatusEvent	Warning	SNMP Service has stopped running.	SNMP Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80420	NSXTMPNodeServiceSSHStatusEvent	Warning	SSH Service has stopped running.	SSH Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80421	NSXTMPNodeServiceSyslogStatusEvent	Warning	Syslog Service has stopped running.	Syslog Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80422	NSXTMPNodeServiceTelemetryStatusEvent	Warning	Telemetry Service has stopped running.	Telemetry Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80423	NSXTMPNodeServiceUIServiceStatusEvent	Warning	UI Service has stopped running.	UI Service status has turned to stopped.
1.3.6.1.4.1.6876.100.1.0.80402	NSXTMPNodeServiceCmInventoryStatusEvent	Critical	CM Inventory Service has stopped	One of the Services of the NSX Management Node, namely CM Inventory Service has stopped running.

OID	Event Name	Default Severity	UI Name	Description
1.3.6.1.4.1.6876.100.1.0.80403	NSXTMPNodeServiceControllerStatusEvent	Critical	Controller Service has stopped	One of the Services of the NSX Management Node, namely Controller Service has stopped running.
1.3.6.1.4.1.6876.100.1.0.80404	NSXTMPNodeServiceDataStoreStatusEvent	Critical	DataStore Service has stopped	One of the Services of the NSX Management Node, namely DataStore Service has stopped running.
1.3.6.1.4.1.6876.100.1.0.80405	NSXTMPNodeServiceHttpStatusEvent	Critical	HTTP Service has stopped	One of the Services of the NSX Management Node, namely HTTP Service has stopped running.
1.3.6.1.4.1.6876.100.1.0.80406	NSXTMPNodeServiceInstallUpgradeEvent	Warning	Install Upgrade Service has stopped	One of the Services of the NSX Management Node, namely Install Upgrade Service has stopped running.
1.3.6.1.4.1.6876.100.1.0.80407	NSXTMPNodeServiceLiagentStatusEvent	Warning	Liagent service has stopped	One of the Services of the NSX Management Node, namely LI Agent Service has stopped running.
1.3.6.1.4.1.6876.100.1.0.80408	NSXTMPNodeServiceManagerStatusEvent	Critical	Manager Service has stopped	One of the Services of the NSX Management Node, namely Manager Service has stopped running.
1.3.6.1.4.1.6876.100.1.0.80409	NSXTMPNodeServiceMgmtPlaneBusStatusEvent	Warning	Management Plane Service has stopped	One of the Services of the NSX Management Node, namely Management Plane Bus Service has stopped running.
1.3.6.1.4.1.6876.100.1.0.80411	NSXTMPNodeServiceNodeMgmtStatusEvent	Critical	Node Management Service has stopped	One of the Services of the NSX Management Node, namely Node Management Service has stopped running.
1.3.6.1.4.1.6876.100.1.0.80412	NSXTMPNodeServiceNodeStatsStatusEvent	Critical	Node Statistics Service has stopped	One of the Services of the NSX Management Node, namely Node Statistics has stopped running.
1.3.6.1.4.1.6876.100.1.0.80413	NSXTMPNodeServiceNSXMessageBusStatusEvent	Warning	Message Bus Service has stopped	One of the Services of the NSX Management Node, namely Message Bus Service has stopped running.
1.3.6.1.4.1.6876.100.1.0.80414	NSXTMPNodeServiceNSXPlatformClientStatusEvent	Critical	Platform Client Service has stopped	One of the Services of the NSX Management Node, namely Platform Client Service has stopped running.

OID	Event Name	Default Severity	UI Name	Description
1.3.6.1.4.1.6876.100.1.0.80415	NSXTMPNodeServiceNSXUpgradeAgentStatusEvent	Critical	Upgrade Agent Service has stopped	One of the Services of the NSX Management Node, namely Upgrade Agent Service has stopped running.
1.3.6.1.4.1.6876.100.1.0.80416	NSXTMPNodeServiceNTPStatusEvent	Critical	NTP Service has stopped	One of the Services of the NSX Management Node, namely NTP Service has stopped running.
1.3.6.1.4.1.6876.100.1.0.80417	NSXTMPNodeServicePolicyStatusEvent	Critical	Policy Service has stopped	One of the Services of the NSX Management Node, namely Policy Service has stopped running.
1.3.6.1.4.1.6876.100.1.0.80418	NSXTMPNodeServiceSearchStatusEvent	Critical	Search Service has stopped	One of the Services of the NSX Management Node, namely Search Service has stopped running.
1.3.6.1.4.1.6876.100.1.0.80419	NSXTMPNodeServiceSNMPStatusEvent	Warning	SNMP Service has stopped	One of the Services of the NSX Management Node, namely SNMP Service has stopped running.
1.3.6.1.4.1.6876.100.1.0.80420	NSXTMPNodeServiceSSHStatusEvent	Critical	SSH Service has stopped	One of the Services of the NSX Management Node, namely SSH Service has stopped running.
1.3.6.1.4.1.6876.100.1.0.80421	NSXTMPNodeServiceSyslogStatusEvent	Critical	Syslog Service has stopped	One of the Services of the NSX Management Node, namely Syslog Service has stopped running.
1.3.6.1.4.1.6876.100.1.0.80422	NSXTMPNodeServiceTelemetryStatusEvent	Warning	Telemetry Service has stopped	One of the Services of the NSX Management Node, namely Telemetry Service has stopped running.
1.3.6.1.4.1.6876.100.1.0.80423	NSXTMPNodeServiceUIServiceStatusEvent	Critical	UI Service has stopped	One of the Services of the NSX Management Node, namely UI Service has stopped running.
1.3.6.1.4.1.6876.100.1.0.80424	NSXTMPNodeServiceClusterManagerStatusEvent	Critical	Cluster Manager Service has stopped	One of the Services of the NSX Management Node, namely Cluster Manager Service has stopped running.

Network Monitoring

The topics in this section show you how to configure monitoring using Internet Protocol Flow Information Export (IPFIX) profiles for the firewall and switches, as well as how to configure an IPFIX collector. You can also learn about other network monitoring options like performing a traceflow.

Add an IPFIX Collector

You can configure IPFIX collectors for firewalls and switches.

1. With admin privileges, log in to NSX Manager.
2. Select **Plan & Troubleshoot** > **IPFIX**.
3. Click the **Collectors** tab.
4. Select **Add New Collector** > **IPFIX Switch** or **Add New Collector** > **IPFIX Firewall**.
5. Enter a name.
6. Enter the IP address and port of up to four collectors. Both IPv4 and IPv6 addresses are supported.
7. Click **Save**.

Add a Firewall IPFIX Profile

You can configure IPFIX profiles for firewalls.

1. With admin privileges, log in to NSX Manager.
2. Select **Plan & Troubleshoot** > **IPFIX**.
3. Click the **Firewall IPFIX Profiles** tab.
4. Click **Add Firewall IPFIX Profile**.
5. Complete the following details.

Setting	Description
Name and Description	Enter a name and optionally a description. **Note:** If you want to create a global profile, name the profile `global`. A global profile cannot be edited or deleted from the UI, but you can do so using NSX APIs.
Active Flow Export Timeout (Minutes)	The length of time after which a flow will time out, even if more packets associated with the flow are received. Default is 1.
Observation Domain ID	This parameter identifies which observation domain the network flows originate from. The default is 0 and indicates no specific observation domain.
Collector Configuration	Select a collector from the drop-down menu.
Applied To	Click **Set** and select a group to apply the filter to, or create a new group. **Note:** If another IPFIX profile exists and applies to the same group, this new profile will take precedence. The **Applied To** setting of the previous profile will be removed.
Priority	This parameter resolves conflicts when multiple profiles apply. The IPFIX exporter will use the profile with the highest priority only. A lower value means a higher priority.

6. Click **Save** and then **Yes** to continue configuring the profile.
7. Click **Save**.

Add a Switch IPFIX Profile

You can configure IPFIX profiles for switches, which can apply to network segments or a group of segment ports..

Flow-based network monitoring enables network administrators to gain insight into traffic traversing a network.

With vSphere Distributed Services Engine, some of the network operations can be offloaded from the server CPU to a Data Processing Unit (DPU also known as SmartNIC). vSphere 8.0 supports NVIDIA BlueField and AMD Pensando DPU devices only.

For more information about VMware vSphere Distributed Services Engine, see *Introducing VMware vSphere® Distributed Services EngineTM and Networking Acceleration by Using DPUs* in the VMware vSphere® product documentation.

Note:

If you want to configure IPFIX on a VDS, create vmknic on the appropriate TCP/IP stack as follows:

- For a DPU-backed VDS, create vmknic on the 'ops' TCP/IP stack. Else, the flow information is not exported to the collector.
- For a non-DPU-backed VDS, create vmknic on the 'defaultTcipip' TCP/IP stack.

Switch IPFIX only supports UDP packets fragmentation and does not support fragmented packets of other supported protocols. So IPFIX will export the correct packets/bytes information for the UDP fragment packets, but for other protocol fragment packets IPFIX will skip these packets and export nothing.

1. With admin privileges, log in to NSX Manager.
2. Select **Plan & Troubleshoot** > **IPFIX**.
3. Click the **Switch IPFIX Profiles** tab.
4. Click **Add Switch IPFIX Profile**.
5. Enter the following details:

Setting	Description
Name and Description	Enter a name and optionally a description. **Note:** If you want to create a global profile, name the profile `global`. A global profile cannot be edited or deleted from the UI, but you can do so using NSX APIs. When a GLOBAL profile is specified, UI does not allow binding of any Segment/Segment Port/Group to this profile. A global Switch IPFIX excludes all VMs on the service subnet of a VPC.
Active Timeout (seconds)	The length of time after which a flow times out, even if more packets associated with the flow are received. Default is 300.
Collector Configuration	Select a collector from the drop-down menu.
Applied To	Select a category: Segment, Segment Port, Groups, or Selected Count. The IPFIX profile applies to the selected object.
Priority	This parameter resolves conflicts when multiple profiles apply. The IPFIX exporter uses the profile with the highest priority only. A lower value means a higher priority.
Observation Domain ID	The observation domain ID identifies which observation domain the network flows originate from. Enter 0 to indicate no specific observation domain.
Tags	Enter a tag to make searching easier.

6. Click **Save** and then **Yes** to continue configuring the profile.
7. Click **Applied To** to apply the profile to an NSGroup. You can select one or more NSGroups.

   Note:

   IPFIX Profile supports NSGroups with member types: Other NSGroups, Segment, and Segment Port. To learn more about NSGroups, see Add a Group.
8. Click **Save**.

IPFIX Monitoring on a vSphere Distributed Switch

Configure IPFIX monitoring for NSX Distributed Virtual port groups, and vSphere Distributed Virtual port groups that are connected to a VDS switch enabled to support NSX networking.

From vSphere, enable IPFIX for Distributed Virtual port groups (vSphere) and from NSX Manager, enable IPFIX for segments (NSX) created on a VDS switch.

vSphere Distributed Services Engine provides the ability to offload some of the network operations from your server CPU to a Data Processing Unit (DPU also known as SmartNIC).

For more information about VMware vSphere Distributed Services Engine, see *Introducing VMware vSphere® Distributed Services EngineTM and Networking Acceleration by Using DPUs* in the VMware vSphere® product documentation.

If you want to configure IPFIX on DPU backed VDS, you must create vmknic on 'ops' TCP/IP stack.

To enable IPFIX monitoring for Distributed Virtual port groups, see the *vSphere Networking* documentation.

To enable IPFIX monitoring for NSX port groups, see Add a Switch IPFIX Profile.

A VDS switch enabled for IPFIX monitoring exhibits the following behavior:

- Only non-uplink ports support IPFIX for bidirectional traffic incoming and outgoing on:
  - Ports, port groups, VMs on vSphere.
  - Segments, segment ports, and groups on NSX
- IPFIX profile samples packets on uplink ports when are coming from or going to non-uplink ports that are IPFIX enabled. For example, consider that *VM-A* and *VM-B* are connected to non-uplink ports (port-1, port-2), where port-1 connected to *VM-A* is IPFIX enabled, and port-2 connected to *VM-B* is not IPFIX enabled. When you send traffic from *VM-A* and *VM-B* traffic to port-1, only packets from *VM-A* are sampled because IPFIX is enabled only on the port that *VM-A* is connected to. IPFIX does not sample packets coming from the port-2 associated to *VM-B* because IPFIX is not enabled on that port.
- Packet count exported to the IPFIX collector is the total count based on a sampling rate, not the sampled packets. For example, IPFIX calculates the count of total packets and exports the info. For 100 incoming packets, IPFIX might sample 9–11 packets. It exports 90 or 110 packets to the IPFIX collector.

Add a Port Mirroring Session

You can use port mirroring to analyze network traffic for debugging or troubleshooting purposes. Port mirroring allows you to copy all network packets or specific packets that are seen on the segment port (or an entire segment) to another segment port.

You can use the NSX Manager UI to create the following port mirror session types:

- Remote L3 Span
- Logical Span - requires a mirror source and destination on the same overlay segment
- Local Uplink Span
- Local Span
- Remote Span Source (RSPAN Source) - requires a corresponding Remote Span Destination session
- Remote Span Destination (RSPAN Destination) - requires a corresponding Remote Span Source session

In addition, you can use the following collections of API calls to manage port mirroring sessions. For more information, see the NSX API Guide.

- `/policy/api/v1/infra/port-mirroring-profiles`
- `/policy/api/vi/infra/.../*monitoring-profile-binding-maps`

> **Note:**
> Port Mirroring is not recommended for monitoring because when used for longer durations performance is impacted.

vSphere Distributed Services Engine provides the ability to offload some of the network operations from your server CPU to a Data Processing Unit (DPU also known as SmartNIC).

For more information about VMware vSphere Distributed Services Engine, see *Introducing VMware vSphere® Distributed Services EngineTM and Networking Acceleration by Using DPUs* in the VMware vSphere® product documentation.

Port Mirroring Restrictions

This feature has the following restrictions:

- A source mirror port cannot be in more than one mirror session.

- For a Local Span session, the mirror session source and destination ports must be on the same host vSwitch. Therefore, if you vMotion the VM that has the source or destination port to another host, traffic on that port can no longer be mirrored.
- For Local Span and RSPAN Destination sessions, normal traffic on mirror destination ports is not allowed.
- On ESX, when mirroring is enabled on the uplink, raw production TCP packets are encapsulated using the Geneve protocol by VDL2 into UDP packets. A physical NIC that supports TSO (TCP segmentation offload) can change the packets and mark the packets with the MUST_TSO flag. On a monitor VM with VMXNET3 or E1000 vNICs, the driver treats the packets as regular UDP packets and cannot handle the MUST_TSO flag, and will drop the packets.
- IPFIX will not sample mirrored packets and Traceflow and Live Traffic Analysis will not trace mirrored packets.
- If you plan to create an RSPAN session, it is recommend that you reserve a few VLANs across your network to be used as RSPAN VLANs. Do not assign segment ports to these VLANs. RSPAN VLANs must be dedicated VLANs and should not be used as switching VLAN or transport VLAN.
- You can create up to 16 RSPAN VLANs for RSPAN Destination sessions.
- You cannot include the same VLAN as a mirror source across multiple RSPAN Destination sessions.
- An RSPAN Destination session will not monitor vNIC ingress traffic.
- You must implement mirror filtering at RSPAN Source and not at RSPAN Destination as RSPAN destination is used to receive mirrored packets.
- The following restrictions apply when upgrading to VCF 9.x:
  - If you have included an RSPAN VLAN as transport, switching, or trunk VLAN, you will receive an alert during the upgrade to update those RSPAN VLANs in RSPAN Source and Destination sessions.
  - If you have configured mirror filtering for RSPAN Destination, you must remove the mirror filtering configuration before starting the upgrade.

1. With admin privileges, log in to NSX Manager.
2. Select **Plan & Troubleshoot** > **Port Mirroring**.
3. From the **Add Session** menu, select a session type.
4. Enter a name and optionally a description.
5. Configure the basic properties of the port mirroring session.

Session Type	Properties
Remote L3 Span	• **Direction** - Select **Bidirectional**, **Ingress**, or **Egress**. • **TCP/IP Stack** - Select **Default** or **Mirror**. To use **Mirror**, you must bind the vmknic to the mirror stack in vSphere. **Note:** If you want to configure ERSPAN on DPU backed VDS, you must create vmknic on 'mirror' TCP/IP stack. The behavior of ERPSAN on AMD Pensando DPU and NVIDIA BlueField DPU is different: • AMD Pensando DPU device supports full offloading. This means that the mirroring is completely handled by hardware (DPU). However, AMD Pensando supports only 6 destination IPs. If it exceeds them, it will use partial offloading. • NVIDIA BlueField DPU device supports partial offloading. This means that the production traffic is handled by hardware (DPU) and mirrored packets are handled by software (ESX on DPU). • **Snap Length** - Specify the number of bytes to capture from a packet. If this parameter is specified, the packet is truncated to the specified length. If not specified, the entire packet is mirrored. Supported range of values is 60–65535. • **Encapsulation Type** - Select **GRE**, **ERSPAN TWO**, or **ERSPAN THREE**. • **GRE Key** - Specify a 32-bit GRE key if encapsulation type is **GRE**. • **ERSPAN ID** - Specify an ERSPAN ID if encapsulation type is **ERSPAN TWO** or **ERSPAN THREE**. Supported range of values is 0–1023. The physical switch uses the ERSPAN ID to forward the mirrored traffic.
Logical Span	• **Direction** - Select **Bidirectional**, **Ingress**, or **Egress**. • **Snap Length** - Specify the number of bytes to capture from a packet. If this parameter is specified, the packet is truncated to the specified length. If not specified, the entire packet is mirrored. Supported range of values is 60–65535.
Local Span / Local Uplink Span	• **Direction** - Select **Bidirectional**, **Ingress**, or **Egress**. • **Host Transport Node** - Select the host transport node that you want to monitor. • **Snap Length** - Specify the number of bytes to capture from a packet. If this parameter is specified, the packet is truncated to the specified length. If not specified, the entire packet is mirrored. Supported range of values is 60–65535.
Remote Span Source	• **Direction** - Select **Bidirectional**, **Ingress**, or **Egress**. • **Host Transport Node** - Select the host transport node that you want to monitor. • **Snap Length** - Specify the number of bytes to capture from a packet. If this parameter is specified, the packet is truncated to the specified length. If not specified, the entire packet is mirrored. Supported range of values is 60–65535. • **Encapsulation VLAN ID** - Specify the VLAN ID used to encapsulate the mirrored packets. • **Preserve Original VLAN ID** - Specify whether to preserve the original VLAN ID in the mirrored packets.
Remote Span Destination	• **Direction** - Select **Ingress**. • **Snap Length** - Specify the number of bytes to capture from a packet. If this parameter is specified, the packet is truncated to the specified length. If not specified, the entire packet is mirrored. Supported range of values is 60–65535.

6. Click **Set** in the **Source** column to specify the source of the packets to be mirrored.

Session Type	Parameters
Remote L3 Span	• Segment • Segment port • Group of virtual machines • Group of virtual network interfaces
Logical Span	• Segment port • Group of virtual machines • Group of virtual network interfaces
Local Span	• Group of virtual network interfaces
Local Uplink Span	• Select a VDS. • Select physical interfaces (up to a maximum of six). • Specify whether to mirror encapsulated packets.
Remote Span Source	• Group of virtual network interfaces
Remote Span Destination	• List of encapsulation VLAN IDs used in corresponding Remote Span Source session. Supported range of values is 1–4094. The list can include up to 16 VLAN IDs.

The following restrictions apply when you select a group of VMs or a group of virtual network interfaces:

- The group can have a maximum of six VMs that are statically added.
- The group can have a maximum of six virtual network interfaces that are statically added.

7. Click **Set** in the **Destination** column to specify the destination for the mirrored packets.

Session Type	Parameters
Local Span / Local Uplink Span	• Group of virtual network interfaces
Remote Span Source	• Select a VDS. • Select physical interfaces (up to a maximum of three).
Remote Span Destination	• Group of virtual network interfaces
Remote L3 Span	• IP Addresses Only group. The group can have a maximum of three IPs.
Logical Span	• Group of virtual machines • Group of virtual network interfaces

The following restrictions apply when you select a group of VMs or a group of virtual network interfaces:

- The group can have a maximum of three VMs that are statically added.
- The group can have a maximum of three virtual network interfaces that are statically added.

8. Optional: Instead of mirroring all the network packets from the source, you can filter the packets that are captured for port mirroring.
   a) Expand the **Advanced Mirroring Filters** section.
   b) Select an **Action**.

Action	Description
Include	Packets that match the filter are mirrored.
Exclude	Packets that do not match the filter are mirrored.

c) Next to **Filters**, click **Set**, and then click **Add Filter**.
d) Specify the filter properties.
Only one filter is supported.

Property	Description
Protocol	The transport protocol that is used to filter the packets. Available options are TCP, UDP.
Source IPs	The source IP address, IP range, or IP prefix that is used to filter the packets.
Source Port	The source port or port range that is used to filter the packets.
Destination IPs	The destination IP address, IP range, or IP prefix that is used to filter the packets.
Destination Port	The destination port or port range that is used to filter the packets.

e) Click **Apply**.

9. Click **Save** to create the port mirroring session.

Port Mirroring on a vSphere Distributed Switch

You can configure port mirroring for port groups, virtual NICs of VMs, and VMs created in NSX and vSphere Distributed Virtual port groups created in vSphere that are connected to a vSphere Distributed Switch (VDS) switch.

In vCenter, configure port mirroring for vSphere Distributed Virtual port groups on a VDS switch.

In NSX Manager, configure port mirroring for segments (in NSX) on a VDS switch.

> **Note:**
> You cannot edit configuration for a segment created in NSX in vCenter. As an admin, you can view the properties of a port mirroring session to know on which switch it is created.

vSphere Distributed Services Engine provides the ability to offload some of the network operations from your server CPU to a Data Processing Unit (DPU also known as SmartNIC).

For more information about VMware vSphere Distributed Services Engine, see *Introducing VMware vSphere® Distributed Services EngineTM and Networking Acceleration by Using DPUs* in the VMware vSphere® product documentation.

If you want to configure ERSPAN on DPU backed VDS, you must create vmknic on 'mirror' TCP/IP stack.

To enable port mirroring on vSphere Distributed Virtual port groups, see the *vSphere Networking* documentation.

The following conditions hold for vSphere port mirroring:

- RSPAN destination will not monitor vNIC ingress traffic.
- RSPAN VLANs should be dedicated VLANs and must not be used as switching VLAN or trunk VLAN.

To enable port mirroring on segments, ports, and groups in NSX from NSX Manager, see Add a Port Mirroring Session.

Uplink Conflict Between Teaming and Remote SPAN

In vSphere, by default, the **Remote SPAN** in a teaming policy is set to `Disallowed`. If you use all the available physical NICs to configure remote SPAN, there are no free uplinks available for the teaming policy to consume. The unavailability of any free uplink means that uplink traffic is not allowed on destination ports, resulting in configuration errors.

However, in NSX, by default, **Normal I/O on Destination Ports** is set to `Allowed`. In NSX, port mirroring configured for NSX port groups on VDS switch allows teaming and port mirroring on destination ports. So, uplink configuration errors do not occur in NSX.

To resolve a uplink conflict when configuring teaming and remote SPAN:

- Ensure that a free uplink is available. For example, on an ESX host with 2 physical NICs, do not assign both these uplinks as destination IP addresses in the remote span port mirroring profile to avoid uplink conflicts in configuration. There must be at least one available uplink that can be configured in teaming profile.

- In vCenter, edit the port mirror configuration profile and set the **Normal I/O on Destination Ports** to `Allowed`.

Perform a Traceflow

Use Traceflow to inspect the path of a packet. Traceflow traces the transport node-level path of a packet. The trace packet traverses the segment overlay, but is not visible to interfaces attached to the segment. In other words, no packet is actually delivered to the test packet's intended recipients.

For a VLAN-backed segment, enable In-band Network Telemetry (INT) by calling the `PUT /api/v1/infra/ops-global-config` API.

Sample request:

URL:

```
PUT https://{{nsx-manager-ip}}/policy/api/v1/infra/ops-global-config
```

Body

```
{
 "display_name": "ops-global-config",
 "in_band_network_telementry": {
 "dscp_value": 2,
 "indicator_type": "DSCP_VALUE"
 },
"path": "/infra/ops-global-config",
"relative_path": "ops-global-config",
 "_revision": 0
}
```

INT cannot be configured if a traceflow request is already in progress. For more information about APIs for In-band Network Telemetry, see NSX. VLAN traceflow supports only TCP/UDP and ICMP packets.

VLAN tracing is not supported on Edge nodes. Attempting to inject a trace packet on an Edge node results in an API validation error. For a VLAN traceflow, if the injected trace packet traverses an Edge node, the resultant trace is incomplete and only observations on ESX nodes are displayed.

> **Note:**
> NSX Traceflow does not work with HCX extended networks.

1. With admin privileges, log in to NSX Manager.
2. Select **Plan & Troubleshoot** > **Traffic Analysis** > **Traceflow** > **Get Started**.
3. Select an IPv4 or IPv6 address type.
4. Select a traffic type.

   For IPv4 addresses the traffic type choices are Unicast, Multicast, and Broadcast. For IPv6 address the traffic type choices are Unicast or Multicast.

   Note: Multicast and broadcast are not supported in a VMware Cloud (VMC) environment.
5. Optional: Select a protocol and provide related information.

Protocol	Parameters
DHCP	Select a DHCP OP code: **Boot Request** or **Boot Reply**.
DHCPv6	Select a DHCP message type: **Solicit**, **Advertise**, **Request**, or **Reply**. **Note:** This option is available only when IPv6 is selected for IP Address.
DNS	Specify an address and select a message type: **Query** or **Response**.

Protocol	Parameters
ICMP	Specify an ICMP ID and a sequence.
ICMPv6	Specify an ICMP ID and a sequence. **Note:** This option is available only when IPv6 is selected for IP Address.
TCP	Specify a source port, a destination port, and TCP flags.
UDP	Specify a source port and a destination port.
ARP	Select an ARP OP Code: **ARP Request** or **ARP Reply** .
NDP	For Unicast traffic, specify the **Destination IP** address. For Multicast traffic, specify the **Destination IP** and **Destination MAC** address. **Note:** This option is available only when IPv6 is selected for IP Address.

For the TCP protocol, note the following:

- The default flag is SYN.
- SYN cannot be combined with RST or FIN.
- If SYN is not selected, you must select ACK or RST.
- ACK cannot be combined with FIN, PSH, or URG.

6. Specify the source and destination information according to the traffic type.

   Note:

   For VLAN tracing across different VLAN segments, ensure that you set the appropriate MAC address using one of the following options:

   - In the NSX Manager user interface, select **IP – Mac** as destination type and select **Layer 2**. Type the destination IP address and the MAC address of the physical gateway that can forward the packet to the destination VLAN network..
   - When using APIs to perform a traceflow, set `routed` as `false` and use the MAC address of the physical gateway that can forward the packet to the destination VLAN network.

For more information about traceflow APIs, see *NSX API Guide*.

Traffic Type	Source	Destination
Unicast	Select a VM or a logical port. For a VM: • Select a VM from the drop-down list. • Select a virtual interface. • The IP address and MAC address are displayed if VMtools is installed in the VM, or if the VM is deployed using OpenStack plug-in (address bindings will be used in this case). If the VM has more than one IP address, select one from the drop-down list. • If the IP address and MAC address are not displayed, enter the IP address and MAC address in the text boxes. For a logical port: • Select an attachment type: **VIF**, **DHCP**, **Edge Uplink**, or **Edge Centralized Service**. • Select a port.	Select a VM, a logical port, or IP-MAC. For a VM: • Select a VM from the drop-down list. • Select a virtual interface. • The IP address and MAC address are displayed if VMtools is installed in the VM or if the VM is deployed using OpenStack plug-in (address bindings will be used in this case). If the VM has more than one IP address, select one from the drop-down list. • If the IP address and MAC address are not displayed, enter the IP address and MAC address in the text boxes. For a logical port: • Select an attachment type: **VIF**, **DHCP**, **Edge Uplink**, or **Edge Centralized Service**. • Select a port. For IP-MAC: • Select the trace type (layer 2 or layer 3). For layer 2, enter an IP address and a MAC address. For layer 3, enter an IP address.
Multicast	Same as above.	Enter an IP Address. It must be a multicast address from 224.0.0.0 - 239.255.255.255.
Broadcast	Same as above.	Enter a subnet prefix length.

7. Optional: Click **Advanced Settings** to see the advanced options.

   In the left column, enter the desired values or input for the following fields:

Field	Description
TTL	The default is 64.
Ethertype	The default is 2048.
Timeout (ms)	The default is 10000.
Frame Size	The default is 128.
Payload Type	Select **Base64**, **Hex**, **Plaintext**, **Binary**, or **Decimal**.
Payload Data	Payload formatted based on selected type.

8. Optional: Click the **Global Settings** tab and enter the INT DSCP values for VLAN-backed entities.

Field	Description
INT DSCP value for Virtual for Network Appliances (VNA)	Enter the required value.
INT DSCP value for VLAN backed entities	Enter the required value.

9. Click **Trace**.

   The output includes a graphical map of the topology and a table listing the observed packets. The first packet listed has the observation type `Injected` and shows the packet that is injected at the injection point.

   You can apply a filter (**All**, **Delivered**, **Dropped**) on the observations that are displayed. If there are dropped observations, the **Dropped** filter is applied by default. Otherwise, the **All** filter is applied.

   The graphical map shows the backplane and router links. Note that bridging information is not displayed.

Simple Network Management Protocol (SNMP)

You can use Simple Network Management Protocol (SNMP) to monitor your NSX components. The SNMP service is not started by default after installation.

The SNMP Framework in NSX enables you to monitor various system entities (such as disk on NSX Edge) and logical entities (such as NSX Edge VPN tunnel) using their SNMP managers. This framework enables NSX verticals and platform to define SNMP MIB objects to be monitored and which can be used to enable their SNMP managers to interact with NSX.

To download the SNMP MIB files, see Knowledge Base article 313538: SNMP MIB module file download. For NSX, download the folder and use the extracted file **VMWARE-NSX-MIB.mib**.

For SNMP configuration, see *Configure SNMP for ESX* in the VMware vSphere product documentation.

1. Log in to the NSX Manager CLI or the NSX Edge CLI.
2. Run the following commands
   - For SNMPv1/SNMPv2:
     ```
     set snmp community <community-string>
     start service snmp
     ```
     The maximum character limit for `community-string` is 64.
   - For SNMPv3
     ```
     set snmp v3-users <user_name> auth-password <auth_password> priv-password <priv_password>

     start service snmp
     ```
     The maximum character limit for `user_name` is 32. Ensure that your passwords meet PAM constraints. If you want to change the default engine id, use the following command:
     ```
     set snmp v3-engine-id <v3-engine-id>

     start service snmp
     ```
     `v3-engine-id` is an even-length hexadecimal string that is 10 to 64 characters long and cannot be all 0s or Fs. NSX supports SHA-1, SHA-2, and AES-128 as the authentication and privacy protocols. You can also use API calls to set up SNMPv3. For more information, see the *NSX API Guide*.
3. To enable the SNMP service to start automatically on reboot on the NSX appliance, run the command: `set service snmp start-on-boot`.

Network Latency Statistics

In a network, latency can accumulate at multiple endpoints in the data path. As a network administrator, you need the ability to monitor the latency of a network to diagnose and troubleshoot performance bottlenecks in the network.

The following network latency statistics can be measured on host transport nodes:

- pNIC to vNIC
- vNIC to pNIC
- vNIC to vNIC
- VTEP to VTEP

In NSX, the following limitations apply to measuring latency statistics:

- Only ESX host transport nodes are supported for measuring network latency in the data plane.
- Edge transport nodes are not supported.
- On VLAN segments, network latency is measured only when the two vNICs belong to VMs on the same ESX host.
- When the VMs are attached to separate segments, network latency is measured only when the data traffic is routed through the distributed router (DR) instance on the ESX host transport nodes. If the data traffic is routed through the DR instance on the edge transport nodes, network latency is not measured.
- Enhanced networking stack (ENS) does not support vNIC-to-pNIC, pNIC-to-vNIC, or vNIC-to-vNIC latency.

- Latency measurement is not supported when an east-west network traffic protection is configured using partner service VMs. Latency monitoring is disabled on the ports of service virtual machine (SVMs) and guest VMs.
- Latency measurement is not supported on the Data Processing Unit (DPU).

You can export the latency data to external network performance monitoring tools and run analytics on the data. The external monitoring tools are also called collectors. By using a collector, you can achieve greater network visibility, optimize network performance, and identify the endpoints in the data path that cause a significant latency in the network.

After the hosts are configured to measure network latency statistics, the network operations agent (netopa) on the hosts periodically polls the data plane. When latency data is available, the agent exports the data at preconfigured intervals to the external collectors.

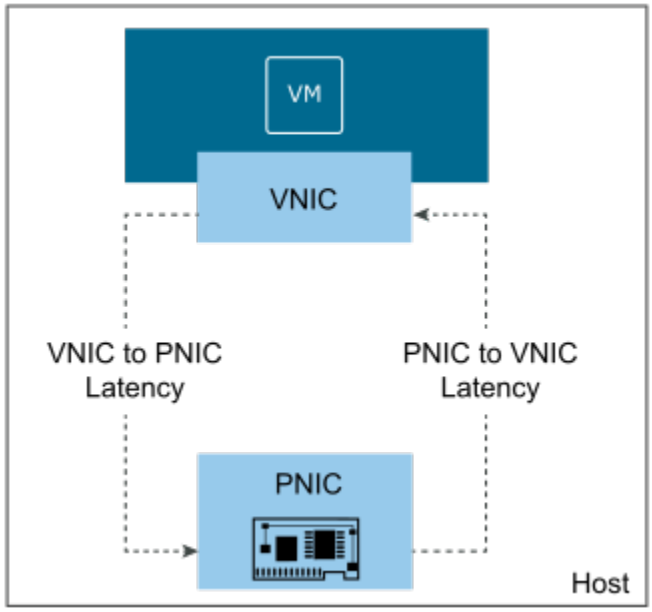
Note:

- The netopa agent can export the network latency statistics only to VCF Operations for networks. Other collector tools are not supported currently.
- For VTEP statistics, VCF Operations for networks 5.0 or later is required. For the pNIC and vNIC statistics, VCF Operations for networks 5.3 or later is required.
- You can configure ESX hosts to measure network latency statistics only by using the NSX REST APIs.

You can measure network latency statistics for both standalone ESX hosts and ESX hosts that are a part of the vCenter cluster. However, VCF Operations for networks supports collecting latency statistics only from vCenter-managed ESX hosts and not from standalone ESX hosts.

pNIC-to-vNIC and vNIC-to-pNIC Latency

When pNIC latency measurement is enabled on a host transport node, vNIC-to-pNIC latency and pNIC-to-vNIC latency are computed for each vNIC on the host transport node.



pNIC-to-vNIC and vNIC-to-pNIC latency statistics are exported to the external collector in the following format:

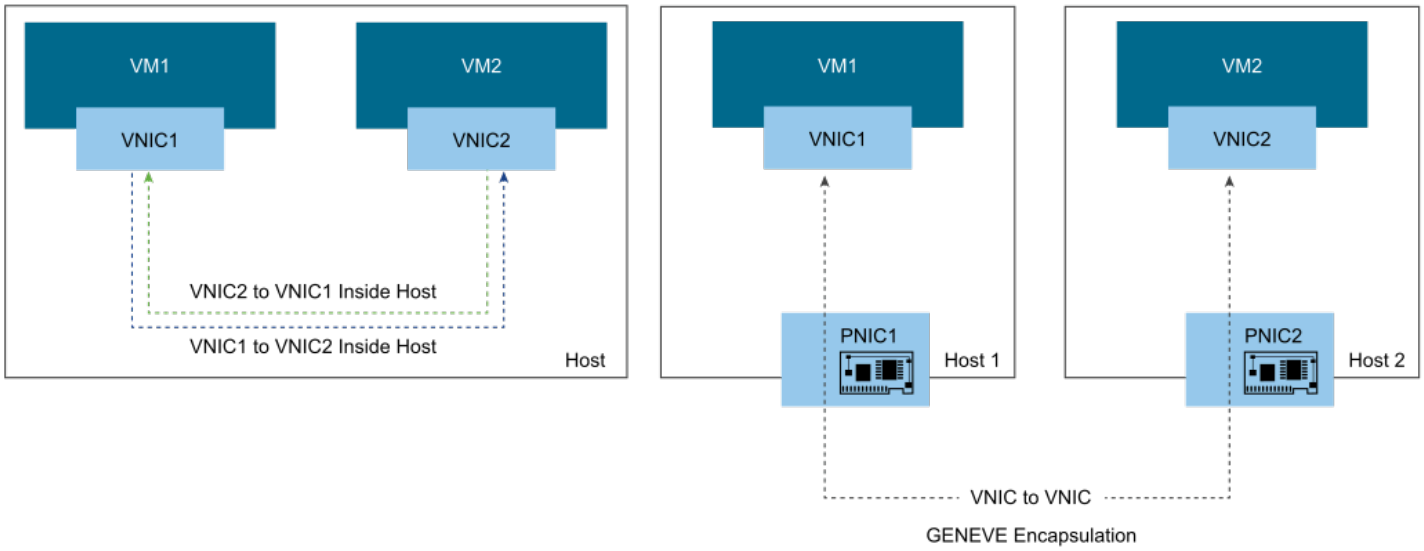
Endpoint1, Endpoint2, Max, Min, Avg

Where:

- *Endpoint1, Endpoint2* can either be the virtual interface ID (VIF ID) or the name of the physical adapter on an ESX host (vmnic).
- *Max*, *Min*, and *Avg* represent the maximum, minimum, and average latency values between the two endpoints in microseconds.

vNIC to vNIC Latency

This latency represents the time taken by the data packet to travel from the source vNIC to the destination vNIC either on the same ESX host or different ESX hosts. If the vNICs are on different ESX hosts, only GENEVE encapsulation protocol is supported in the overlay tunnel between the hosts.



vNIC to vNIC network latency is computed as follows:

- When the source VNIC1 on VM1 and the destination VNIC2 on VM2 are on the same host, a single-trip latency is calculated for each trip and exported to the collector. In other words, latency for each trip from VNIC1 to VNIC2 and from VNIC2 to VNIC1 is computed separately.
- When the source VNIC1 on VM1 and the destination VNIC2 on VM2 are on different hosts, total round-trip latency is calculated, and only a single latency value is exported to the collector. If there is no return traffic from VNIC2 to VNIC1, no network latency is exported to the collector.

> **Note:**
> NSX calculates the vNIC-to-vNIC latency between hosts directly by using the timestamps in the GENEVE encapsulated packets. You do not have to enable pNIC latency measurement on the host and the VTEP-to-VTEP latency. The pNIC-to-vNIC, vNIC-to-pNIC, and VTEP-to-VTEP statistics are independent of the vNIC-to-vNIC statistic.

vNIC-to-vNIC latency statistics are exported to the external collector in the following format:

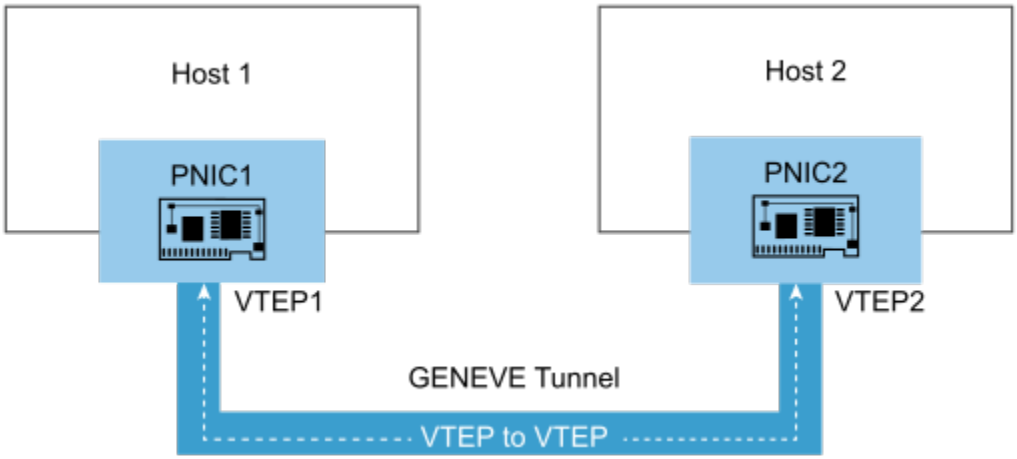
`VIF1, VIF2, Max, Min, Avg`

Where:

- *`VIF1, VIF2`* represent the virtual interfaces or the vNICs.
- *`Max`*, *`Min`*, and *`Avg`* represent the maximum, minimum, and average latency values between the two vNICs in microseconds.

VTEP-to-VTEP Latency

This latency represents the total round-trip time taken by the data packet to travel from the source VTEP to the destination VTEP. To measure VTEP-to-VTEP latency, you must enable latency in the transport zone profile.



To calculate the VTEP-to-VTEP latency between ESX hosts, Bidirectional Flow Detection (BFD) protocol is used. NSX extends the BFD protocol with timestamps in the payload to support latency computation between the VTEPs. BFD packets are transmitted at regular intervals in each tunnel between the hosts to compute the VTEP-to-VTEP latency.

Measure Network Latency Statistics

You can configure ESX hosts in your network to measure: pNIC to vNIC, vNIC to pNIC, vNIC to vNIC, and VTEP to VTEP network latency statistics.

Both vCenter-managed hosts and standalone ESX hosts that you want to configure for measuring network latency statistics must be prepared for NSX. That is, NSX components must be installed on all the ESX hosts in your network.

Configuration is supported only using the NSX REST APIs. The steps in the following procedure list the NSX Policy APIs that you must run to configure the calculation of various network latency statistics. For a detailed information about the API schema, example request, example response, and error messages of all the APIs, you must read the *NSX API Guide*.

1. To compute vNIC to vNIC, pNIC to vNIC, and vNIC to pNIC network latency statistics, do these steps:
   a) Create a group that contains host transport nodes as static members by using the following PUT API:

      ```
      PUT https://<nsx-mgr>/policy/api/v1/infra/domains/<domain-id>/groups/<group-id>
      ```

      In the request payload of this PUT API, specify the host transport node IDs in the `expression` parameter, as shown in the following example:

      Example PUT Request

      ```
      PUT https://<nsx-mgr>/policy/api/v1/infra/domains/default/groups/TNGroup

      {
          "expression": [
              {
                  "paths": [
                      "/infra/sites/default/enforcement-points/default/host-transport-
      nodes/4efdb573-fcce-43ff-8b35-dac583a86239"
                  ],
                  "resource_type": "PathExpression"
              }
          ],
          "extended_expression": [],
          "reference": false,
          "group_type": [],
          "resource_type": "Group",
          "id": "TNGroup",
          "display_name": "TNGroup",
          "path": "/infra/domains/default/groups/TNGroup",
          "relative_path": "TNGroup",
          "parent_path": "/infra/domains/default"
      ```

```
}
```

Observe that in this example request, the `expression` parameter contains a single host transport node ID.

b) Create a latency profile with the following PUT API:

```
PUT https://<nsx-mgr>/policy/api/v1/infra/latency-profiles/<profile-id>
```

By default, vNIC to vNIC latency is measured for all the vNICs on the host transport node.

In the request body of this API, configure the following information:

- Activate or deactivate pNIC latency on the host. When it is activated, pNIC to vNIC and vNIC to pNIC latency are calculated for each vNIC on the host transport node.
- Specify either the sampling rate or the sampling interval, but not both.
- Specify the path to the group that you created in the earlier step.

Example PUT Request

```
PUT https://<nsx-mgr>/policy/api/v1/infra/latency-profiles/profile1
{
  "sampling_rate": 100,
  "pnic_latency_enabled": false,
  "applied_to_group_path": "/infra/domains/default/groups/TNGroup"
}
```

2. To measure VTEP to VTEP latency statistics, enable latency in the BFD health monitoring profile, which is a resource type in the transport zone profile. Run the following PUT API:

```
PUT https://<nsx-mgr>/policy/api/v1/infra/transport-zone-profiles/<tz-profile-id>
```

Export the statistics to an external collector for a deeper network insight and troubleshooting network-specific latency problems.

Export Network Latency Statistics

You can export network latency statistics to external collectors and run analytics on the data. The netopa agent that is running in the ESX hosts can export the network latency statistics only to VCF Operations for networks. Other collector tools are not supported currently.

- In the VCF Operations for networks UI, complete the following tasks in the given order:
  a. Add vCenter as the data source. If you have multiple vCenters added as Compute Managers in your NSX environment, you can add all vCenters as the data source.
  b. Add NSX Manager as the data source.

  For detailed information about adding data sources in VCF Operations for networks, see the VCF Operations for networks documentation.
- Ensure that port 1991 is open on the collector to receive network latency data from the ESX hosts.

You can collect network latency statistics from only vCenter-managed ESX hosts. Collecting latency statistics from standalone ESX hosts is not supported.

You can export network latency statistics by using any one of the following methods:

- Method 1: Use the management plane APIs in NSX.
- Method 2: Enable an optional setting in the VCF Operations for networks UI to collect latency statistics.

1. Method 1: Use the NSX REST APIs.
   a) Ensure that you have configured the ESX hosts to measure network latency statistics.

   For detailed steps, see Measure Network Latency Statistics.
   b) Export the network latency statistics to the collector with the following PUT API:

   ```
   PUT https://<manager-ip>/api/v1/global-configs/OperationCollectorGlobalConfig -d
   '<content>'
   ```

   In the request body of this API, configure the following information:

- Details of external collectors, such as collector IP address, collector port.
- Report interval that controls the frequency at which the netopa agent sends statistics to the collector.

2. Method 2: Enable an optional setting in the VCF Operations for networks UI to collect latency statistics.

   When you add NSX Manager as the data source in VCF Operations for networks, select the **Enable latency metric collection** check box. This option enables VCF Operations for networks to collect latency statistics from the ESX hosts.

   For detailed information about adding NSX Manager as the data source in VCF Operations for networks, see the VCF Operations for networks documentation.

vNIC-to-vNIC latency statistics are exported to the external collector in the following format:

VIF1, VIF2, Max, Min, Avg

Where:

- *VIF1, VIF2* represent the virtual interfaces or the vNICs.
- *Max*, *Min*, and *Avg* represent the maximum, minimum, and average time between the two vNICs in microseconds.

pNIC-to-vNIC and vNIC-to-pNIC latency statistics are exported to the external collector in the following format:

Endpoint1, Endpoint2, Max, Min, Avg

Where:

- *Endpoint1, Endpoint2* can either be the virtual interface ID (VIF ID) or the name of the physical adapter on an ESX host (vmnic).
- *Max*, *Min*, and *Avg* represent the maximum, minimum, and average time between the two endpoints in microseconds.

Checking CPU Usage and Network Latency

Use APIs to monitor CPU usage and network latency.

Checking CPU Usage

These APIs work similar to the `top` command:

- Run the following API to check process status for all appliances:
  ```
  GET /api/v1/systemhealth/appliances/process/status
  ```
- Run the following API to check the process status of a specific appliance.
  ```
  GET https://<nsx-manager-ip>/api/vi/systemhealth/appliances/<appliance-id>/process/status
  ```

Checking Network Latency

- Run the following API to check for network latency values for all appliances:
  ```
  GET /api/v1/systemhealth/appliances/latency/status
  ```
- Run the following API to check for network latency values for a specific appliance:
  ```
  GET https://<nsx-manager-ip>/api/v1/systemhealth/appliances/<appliance-id>/latency/status
  ```

See the latest version of the *NSX API Guide* at https://code.vmware.com/ for API details.

Live Traffic Analysis

Live Traffic Analysis (LTA) provides helpful insight about tracing live traffic and bi-directional packet tracing. Traffic analysis monitors live traffic at a source or between source and destination along with the packet capture. You can identify bad flows between the source and the destination. If the packet counter of the certain flow at the source endpoint is much higher than the packet counter of the certain flow at the destination endpoint, packet drop may occur between two endpoints. Hence the flow is probably a bad flow which you can trace for further analysis. Thus traffic analysis is helpful in troubleshooting virtual network issues. You can find the number of packet enter or leave a port, and the unexpected packet drop.

The following capabilities are supported in an LTA session only with NSX APIs:

- LTA session supports count action for ENS fastpath. For more information, see the *NSX API Guide*.

  Examples: The following API creates an LTA session request with count action.

```
PUT https://<nsx-mgr>/policy/api/v1/infra/livetraces/<livetrace-id>
{
  "src_port_path" : "/infra/segments/ens_segment/ports/default:4907245b-
d4db-41b3-838c-d749fcd5f0d6",
  "actions" : {
    "counter_config" : {
      "trace_type" : "UNI_DIRECTIONAL"
    }
  },
  "timeout" : 10,
  "is_transient" : false
}
```

  The following API retreives the details of an LTA session result.

```
GET https://<nsx-mgr>/policy/api/v1/infra/livetraces/<livetrace-id>/result
```

- LTA session supports datapath statistics action. To specify this action, set the `actions` property to `datapath_stats_config` in the following NSX API:

```
PUT https://<nsx-mgr>/policy/api/v1/infra/livetraces/<livetrace-id>
```

  For more information about this API, see the *NSX API Guide*.

  Examples: The following API creates an LTA session request on a source port with datapath statistics action.

```
PUT https://<nsx-mgr>/policy/api/v1/infra/livetraces/<livetrace-id>
{
  "src_port_path" : "/infra/segments/InfraSegment101/ports/
default:066b6ef4-3f5e-4fb4-b6df-751c006a4a50",
  "filter" : {
    "ip_info" : {
      "src_ip" : "192.168.1.10/32",
      "dst_ip" : "192.168.2.10/32"
     },
    "resource_type" : "FieldsFilterData"
   },
  "actions" : {
    "datapath_stats_config" : {
      "trace_type" : "UNI_DIRECTIONAL"
    }
  },
  "timeout" : 10
}
```

  The following API retrieves the details of an LTA session result.

```
GET https://<nsx-mgr>/policy/api/v1/infra/livetraces/<livetrace-id>/result
```

Limitations

- LTA supports only on the overlay-backed NSX environments.
- LTA is not supported on DPU.
- LTA is not supported on T0 Active/Active setup.
- LTA is not supported on Global Manager.
- LTA cannot observe VMware Cloud components that do not belong to the NSX management domain, such as IGW.

Enhancements

Starting with NSX 4.2, LTA session can capture the real-time debug statistics from the data path kernel-level modules on the ESX host transport nodes. The debug statistics can help advanced NSX users delve deeper into data path issues for real-time debugging purposes. After the LTA session is completed, the debug statistics are reported to the NSX Manager appliance.

Performance Impact

When the LTA count action is enabled on ENS fastpath, it may cause a performance impact on the network. The performance impact observed while using LTA is a reduction of approximately 7% ~ 20% range in PPS and MBPS rates.

Listing some of the scenarios where the performance impact might be observed:

- Traffic inside the ESX host: This refers to the traffic that is contained within a single ESX host, where communication occurs between VMs running on the same hypervisor.
- Traffic across the ESX host: This refers to the traffic that moves between VMs on different hypervisors (ESXs), and the traffic traverses not only within each VM but also through the physical NIC (pNIC) of each hypervisor.
- DFW is enabled along with LTA: This refers to the case where LTA and DFW are enabled at the same time, meaning that both features are active and working together.

Although we have listed three scenarios above, it is important to note that enabling the LTA count action on ENS fastpath may cause a performance impact on various traffic processing paths. The scenarios mentioned are typical examples that illustrate potential performance impacts; and traffic not covered in the listed cases can also be affected. When LTA is enabled, the performance impact may not be limited to just the source and destination ports but can also extend to other network components, such as the pNIC or DFW, that play a role in the data transmission path between these ports.

> **Note:**
> The performance impact is observed only when the LTA session is active and ongoing. The LTA count action is one of the actions that can be appended to an LTA session. Once the LTA session completes (LTA session has a fixed timeout, which is set to a maximum value of 600 seconds (10 mins)), the performance impact will cease. When the LTA session ends, any associated actions, including the LTA count action, will no longer be in effect.

Create a Session

To start a LTA session, click **New session**. For details, see Create a Live Traffic Analysis Session.

Session List

After you create a session, you can view the list of all the active sessions. The session persists only for one hour. The session status can be as follows:

Status	Description
In Progress	The session is collecting the results. Wait for the session to finish.
Realized	Realized session is that session whose intent configuration is realized on the management plane.
Unrealized	Unrealized session is that session whose intent configuration is yet to be realized on management plane.
Partial Finished	Only partial session result is available. Some session results might have lost.
Finished	After the session is finished, you can view the session for further analysis. You can also rerun, duplicate, or delete the session.
Cancelled	The session is canceled by exception.
Invalidated	The session is canceled proactively by the data plane service because of disconnection of the source port or the data plane service is down. Make sure that the source or destination port is not migrated or disconnected during the running LTA session and retry.
Timeout	The session is timed out.

To view the latest status, click the **Refresh** icon. To perform the following tasks, click



and then click the required option.

Option	Description
Rerun	Run an existing session again. The session persists only for one hour.
Delete	Delete an existing session.
Copy Path to Clipboard	Get the path of the LTA configuration. You can use the copied path later. You can filter the LTA sessions using the copied path.

Session Details

After the session is finished, you can click the session ID link and perform the traffic analysis. You can view data under the **Observations** tab and the **Packet Count** tab.

Observations Tab	Description
Delivered	Total number of received observations for the traceflow round.
Dropped	Total number of dropped observations for the traceflow round.
Physical Hop Count	The sequence number is the traceflow observation hop count. The hop count for observations on the transport node that a traceflow packet is injected in can be 0. The hop count is incremented each time a subsequent transport node receives the traceflow packet. The sequence number of 999 indicates that the hop count could not be determined for the containing observation.
Observation Type	The observation type can be Forwarded, Dropped, Delivered, Received, and Injected.
Transport Node	The name of the transport node that observed a traceflow packet.
Component	The name of the component that issued the observation.
Actions	You can view details for certain traceflow observation like MAC details of a logical switch or segment.

Packet Count Tab	Description
Component	The type of the component.
Transport Node	ID of the transport node that observed a traceflow packet.
Packets Received, Forwarded, and Dropped	Number of traceflow packets that were received, forwarded, or dropped.

Create a Live Traffic Analysis Session

You can create a Live Traffic Analysis (LTA) session. Traffic analysis is helpful in troubleshooting virtual network problems.

You can create LTA sessions with count action for ENS fastpath and datapath statistics action. However, these actions are currently supported only with NSX APIs. The NSX Manager UI does not support these actions for creating LTA sessions. To learn more, see Live Traffic Analysis.

The following procedure explains the workflow for creating an LTA session in the NSX Manager UI.

1. With admin privileges, log in to NSX Manager.
2. Navigate to **Plan & Troubleshoot** > **Traffic Analysis** > **Live Traffic Analysis** > **Get Started**.
3. To start a traffic analysis, click **New Session**.
4. System generates the session name. If you wish, you can edit the session name.
5. Select the trace, packet capture and provide related information.

Protocol	Parameters
Trace & Packet Capture Sampling Type	System supports only the *FirstNSampling* type. First *N* number of packets that match the packet filter under the **Advanced Settings** tab are sampled for analysis.
Trace	By default, the toggle is active. Trace generates the packet traceflow. If you select only the source, system generates the traceflow for the packets ingressed from the source (matching the forward filter, if any). If you also select the destination, in addition system generates the traceflow for the packets ingressed from the destination (matching the reverse filter, if any).
Trace Sampling Value	Number of packets to be sampled. Enter the value from 1 through 50.
Packet Capture	Generates the PCAP files with live trace telemetry. By default, the toggle is active.
Packet Capture Sampling Value	Enter the value from 1 through 500.

6. Specify the source and destination information according to the traffic type. Click **Add Destination**, and select a destination to capture.

   Bi-directional trace traces the traffic ingressed from the source and the traffic ingressed from the destination, respectively.

   For example, consider three VMs: *App-VM*, *Web-VM*, and *DB-VM*, and there is a ping traffic from the *App-VM* to the *Web-VM* and a ping traffic from the *App-VM* to the *DB-VM*. If you perform bi-directional trace without any packet filters with source as the *App-VM* and destination as the *Web-VM*, then the traces of the forward direction contains traces for ICMP echo from the *App-VM* to the *Web-VM* and the *DB-VM*, while the traces of the reverse direction contains traces for ICMP reply from the *Web-VM* to the *App-VM*.

   To see the traffic between the source and destination only, specify the proper packet filter.

Traffic Type	Source	Destination
Virtual Machine	For a VM: • Select the name of the virtual machine from the list. • You can select or view the virtual interface and the segment port for the selected VM.	For a VM: • Select the name of the virtual machine from the list. • You can select or view the virtual interface and the segment port for the selected VM.
Port/Interface	For a logical port: • Select an attachment type as **Virtual Interface**, **Edge Uplink**, **Edge Centralized Service**, or **IPSec**. • Select a port. **IPSec** session lists only the route-based VPN sessions.	For a logical port: • Select an attachment type as **Virtual Interface**, **Edge Uplink**, **Edge Centralized Service**, or **IPSec**. • Select a port.

Note:

- You cannot configure a source attachment type as **Edge Uplink** and a destination attachment as **IPSec**, or vice-a-versa.
- When you add **IPSec**, **Edge Uplink**, or **VM** as a destination attachment type, you must configure a source attachment type.
- When you add **IPSec** attachment as an intermediate interface, you might see observations for both destinations- inbound as well as outbound traffic.

7. Optional: Click **Advanced Settings** and view the advanced options.
   Enter the desired values for the following parameters and click **Apply**.

Table 1067:

Option	Description
General Tab	
Session Timeout	Default timeout value is 10 seconds. You can add value between 5 to 300 seconds.
Filters Tab	
IP Type	Select **IPv4** or **IPv6**.
Forward Filters or Reverse Filters	Forward filters formulate the flows of interest for the traffic ingressed from the source. Reverse filters formulate the flows of interest for the traffic ingressed from the destination. You can apply filter based on 5-tuple or plain text as **Fields Filter Data** or **Plain Filter Data**.
Filter Type: Fields Filter Data	For bi-directional LTA session, the system populates the source and destination IPs. Select the protocol type and enter the source and destination IPs and port details. If you change the filter type to plain filter data and set the empty values, then all IP observations are reported for all the traffic in the source and destination VMs. • If you select logical port with **Edge Uplink** attachment, then select filter type as **Fields Filter Data**. If your protocol type is **ESP**, then select the IPSec session name and enter the Service Path Index (SPI) value. SPI is an optional field. • If you select logical port with **IPSec** attachment, then select filter type as **Fields Filter Data** and enter the required values.
Filter Type: Plain Filter Data	Add details for basic and extended filter such as IP address and port number. **Note:** Plain filter data is realized on ESX only.

8. Click **Start Session**.

You can view the status of the created session. After the session is finished, you can view the session for further analysis. You can perform the following tasks:

Option	Description
Download PCAP Files	You can download the PCAP file to your system for further analysis. For the bi-directional LTA sessions, you can download both forward and reverse PCAP files.
Rerun	Run an existing session again. The session persists only for one hour.
Duplicate Session	You can copy the session parameters to create a new session. You can quickly change few options in the new session.
New Trace	You can start a new traffic analysis session again.

East-West Network Introspection

Before VCF 9.0, the East-West Network Introspection feature was tied to NSX security and required an add-on security license. Starting in VCF 9.0, East-West Network Introspection is included as a standard feature and does not require any additional licensing.

The East-West Network Introspection feature uses service insertion architecture to copy network packets and send them to a third-party appliance for network performance monitoring and deep observability pipelines. Only approved VMware by Broadcom partners can register monitoring and observability services for use by East-West Network Introspection.

Note:

In early 2024, VMware by Broadcom announced the end of availability for NSX Network Introspection for Security. The NSX Network Introspection for Security feature refers to the service insertion capability that was previously used to redirect packets to third-party firewalls and security devices. The end-of-availability announcement does not apply to the East-West Network Introspection feature used to copy packets for monitoring and observability. VCF and VMware by Broadcom continue to support East-West Network Introspection for copying packets.

Key Concepts of East-West Network Introspection

East-West Network Introspection copies network packets flowing between guest VMs in an on-premises data center and forwards them to third-party observability services provided by approved partners. This page introduces some key terms and concepts to help you understand the feature.

- **Service:** Approved partners register services with NSX. A service represents the monitoring or observability functionality offered by the partner, the service deployment details (such as the OVF URL of service VMs), and the state of the service.
- **Vendor template:** Defines the traffic copying functionality that a service can perform on network traffic.
- **Service profile:** An instance of a vendor template. An administrator can create a service profile to be consumed by service VMs.
- **Guest VM:** A source or destination of traffic in the network. The incoming or outgoing traffic is copied and sent to the service chain defined for a rule running east-west network services.
- **Service VM (SVM):** A VM that runs the OVA or OVF appliance specified by a service. The VM is connected over the service plane to receive the copied traffic.
- **Service segment:** A segment of a service plane that is associated with a transport zone. You can have only one service segment.
- **Service attachment:** An isolated connection managed by the service plane. Each service attachment is isolated from other service attachments and from the regular L2 or L3 network segments provided by NSX.
- **Service chain:** A logical sequence of service profiles defined by an administrator.
- **Redirection policy:** Ensures that traffic classified for a specific service chain is copied to that service chain based on traffic patterns that match the security group.
- **Service path:** A sequence of service VMs that implement the service profiles of a service chain.

NSX Requirements for East-West Network Introspection

To support East-West Network Introspection, ensure that your NSX environment meets the requirements described on this page.

Deployment Types for the Service

The East-West Network Introspection service is applied to an entire NSX environment. You can deploy the service at a cluster level or on a per-host basis. The type of deployment you choose determines where service VMs run for a particular service.

For both cluster-based and host-based deployments, service VMs can be accessed by all East-West Network Introspection workloads. For example, a workload running on cluster A can use a service VM running on cluster B if there is no better alternative. Therefore, choosing a cluster-based deployment does not limit East-West Network Introspection to that cluster.

Even if you plan a deployment using only VLAN-backed segments, east-west traffic passes through overlay transport zones and overlay-backed segments. East-West Network Introspection applies to all segments in the topology, whether they are backed by overlay or VLAN transport zones.

System Requirements

Verify that your NSX environment meets the following requirements:

- **Communications protocol:** East-West Network Introspection supports IPv4 traffic only. IPv6 traffic is not supported.
- **Overlay transport zone:** Ensure that the host transport nodes hosting guest VMs and service VMs are configured with an overlay transport zone.
- **Service segment:** Create an overlay-backed service segment to be used by the East-West Network Introspection service.
- **Host switch:** Verify that all service segments are backed by the same host switch on each host. Any topology involving multiple host switches does not support East-West Network Introspection. NSX supports service insertion policies only on the VDS switch where the service segment is created.

  Note:

  During a vMotion operation, the guest VM can successfully migrate to another host only if the destination host is configured with an overlay transport zone and there is a single host switch. NSX does not support vMotion of service VMs (SVM) deployed on individual hosts. For cluster-based SVM deployments, do not initiate vMotion of SVMs to avoid loss of traffic.

High-Level Setup Tasks for East-West Network Introspection

Follow these high-level steps to set up East-West Network Introspection.

Each setup task is performed by either the approved partner or the NSX administrator.

Setup Task	Performed By	Setup Method
Step 1: Register service, vendor template, and service manager	Partner	API
Step 2: Deploy a service for East-West Network Introspection	Administrator	API or NSX Manager UI
Step 3: Add a service segment	Administrator	API or NSX Manager UI
Step 4: Add a service profile for the partner service	Administrator	API or NSX Manager UI
Step 5: Add a service chain	Administrator	API or NSX Manager UI
Step 6: Add redirection rules for east-west traffic	Administrator	API or NSX Manager UI

1. Register service, vendor template, and service manager

As the first step of the setup process, the approved partner must register the service with NSX.

2. Deploy a service for East-West Network Introspection

After the approved partner registers the service, the administrator deploys an instance of the service on member hosts of a cluster.

Note:

Administrators may encounter a certificate validation failure when attempting to deploy the service with newer OVF images from the approved partner. By default, NSX Manager only trusts OVF certificates that have been signed by VMware by Broadcom or a Certificate Authority (CA) included in the NSX Manager trust store.

To work around this issue, follow the instructions in Broadcom knowledge base article 424034 to turn off OVF validation in NSX Manager.

1. Log in to NSX Manager with admin privileges.
2. Navigate to **System** > **Service Deployments** > **Deployment** > **Deploy Service**.
3. For **Partner Service**, select the approved partner service.

4. Enter the service deployment name.
5. Select the **Compute Manager** (vCenter) and **Cluster** in which to deploy the service.
6. Select a **Data Store** as the repository for the service VM.
7. In the **Network** column, click **Set** and enter the **Management Network** interface details.
8. For **Deployment Type**, select either **Clustered** or **Host Based**.
9. For **Deployment Template**, select the template that defines attributes for the workload.
10. Click Save.

3. Add a service segment

A service segment is used as a service plane overlay segment in which copied packets from guest VMs are forwarded to the third-party VM for traffic introspection.

1. In NSX Manager, navigate to **Security** > **Network Introspection** > **Service Segment**.
2. Click **Add Service Segment**.
3. For **Name**, enter a name for the segment.
4. For **Transport Zone (Overlay)**, select the overlay transport zone associated with the segment.
5. Leave the **Connected To** setting blank (used for East-West Network Introspection).
6. Click `Save`.

4. Add a service profile for the partner service

A service profile is an instance of a partner vendor template.

1. In NSX Manager, navigate to **Security** > **Network Introspection** > **Service Profiles**.
2. For **Partner Service**, select the service.
3. Click **Add Service Profile**.
4. Enter the service profile name and select the vendor template.
5. Leave unchanged the **Redirection Action** setting, which inherits functionality from the vendor template. For this feature, the redirection action defaults to `COPY`.
6. Click **Save**.

5. Add a service chain

A service chain is a logical sequence of service profiles.

1. In NSX Manager, navigate to **Security** > **Network Introspection** > **Service Chains**.
2. Click **Add Chain**.
3. Enter the service chain name.
4. For **Service Segments**, select the service segment to which you want to apply the service chain.
5. Specify the forward path by clicking **Set Forward Path** and adding the profile in sequence.
6. Specify the reverse path by selecting **Inverse Forward Path** or setting a new service profile for the reverse path.
7. For **Failure Policy**, select **Allow**.
   Since East-West Network Introspection only copies packets for monitoring, traffic should be allowed to continue to the destination VM even if the monitoring service VM fails.
8. Click **Save**.

6. Add redirection rules for east-west traffic

Finally, the administrator adds rules to copy and forward east-west traffic for network introspection.

1. Verify that NSX Manager is set to Policy mode.
2. Navigate to **Security** > **E-W Network Introspection**.
3. Click **Add Policy**. For **Redirect To**, select the registered service to which East-West Network Introspection will send the copied packets.
4. Add one or more rules to define the scope of East-West Network Introspection.
   a. Click **Add Rule**.

b. To define the packets to be copied for network introspection, configure the **Sources**, **Destinations**, and **Services** settings.
c. To apply East-West Network Introspection to a group of VMs, specify that group in the **Applied To** setting. To apply the service to all VMs, select **DFW**.
d. For **Action**, select **Redirect**.

5. Click **Publish**.

Authentication and Authorization

You can log in to NSX Manager using a local user account, a user account managed by VCF Identity Broker, or a user account managed by a directory service such as Active Directory over LDAP or OpenLDAP. Starting in VCF 9.0, VCF Identity Broker allows you to use single sign-on for VMware Cloud Foundation management components. You can also assign roles to user accounts managed by VCF Identity Broker, OpenID Connect, or a directory service to implement role-based access control.

For more information about VCF Identity Broker and VCF single sign-on, see Managing Identity and Access With VCF Single Sign-On.

NSX Manager recognizes only system-generated session identifiers and invalidates session identifiers upon administrator logout or other session termination. Upon successful login, the NSX Manager uses a random number generator to create a random session ID and stores that ID in memory. When clients make requests to the NSX Manager, it only allows clients to authenticate if the session ID they present matches one of the IDs generated by the server. When any user logs out of NSX Manager, the session identifier is immediately destroyed and cannot be reused.

Access to NSX Manager via UI, API and CLI is subject to authentication and authorization. In addition, such access will generate audit logs. This logging is enabled by default and cannot be disabled. Auditing of sessions is initiated at system startup. Audit log messages include the text `audit="true"` in the structured data part of the log message.

Local user passwords on NSX appliances are secured using the default Linux/PAM libraries which store the hashed and salted representation in `/etc/shadow`. NSX Manager uses the SHA512 cryptographic hash algorithm to hash the local user passwords. During authentication, the password entered by the user is obfuscated. Other passwords are encrypted using a random key that is stored in the local file system. For more details, see the VMware Security Hardening Guides or review the SHA512 Ubuntu MAN pages.

Managing Local User Accounts

In addition to the admin user in the Enterprise environment, any user with the Enterprise Admin role can add new local users and remove audit and guest users. To administer NSX Manager, you must log in as admin. Some account management privileges are extended to the Enterprise Admin role.

After installation, the admin account is active. To use other local user accounts, you must activate all other local user accounts including the audit account.

Each NSX appliance has four default local accounts including **admin**, **audit**, and two local guest user accounts. The two default local user accounts are **guestuser1** and **guestuser2**.

- The local guest user accounts, including the cloud_admin and the cloud_audit accounts, have the Auditor role. You can change their role assignments.
- You can change role assignments for the guest users. For example, you can mix read-only and full access permissions by creating a clone Auditor role.
- The admin or the account owners can reset local user account passwords.
- Starting with NSX 4.1, you can add up to 10 guest users with a maximum of 14 total users.
- You can delete the audit user and the guest users. You cannot delete the admin user.

An NSX appliance also has the **root** user account. You cannot log in to the NSX Manager UI as root, and you cannot manage this account through the NSX Manager UI. The root user can log in to an appliance through the CLI, but cannot use the NSX CLI commands. The root user account cannot be renamed, deactivated, or deleted.

The root user has special privileges. You must not log in to an NSX appliance as root and make changes that are not documented in this guide, except when under the guidance of VMware. Changes made by the root user can cause

catastrophic failures. In a production environment, the root password must be secure and made available for privileged access only.

For details on how to manage your local user accounts, including password reset, adding, and deleting users, see Manage Local User Accounts . For additional security-related information about the NSX Manager, see the section "Security" in NSX Manager.

Add a Local User

In addition to admin, users with the Enterprise Admin role can create new local user accounts in the NSX Manager.

- Only a user with the Enterprise Admin role can create new local guest user accounts or replace a deleted audit user account.
- The maximum number of local guest users is 14.
- The maximum number of audit users is one. Guest users always have the default Auditor role.

You can complete this task using the UI, API or CLI. The Cloud admin users do not have privileges to add new local users.

1. From your browser, log in to an NSX Manager at https://<nsx-manager-ip-address>.
2. Select **System > User Management > Local Users > Add**.
   a) To add an audit user, select **Audit User**.
      If an audit user exists, you must first delete that user and retry the step. Only one audit user is available.
   b) To add a guest user, select **Local User**.
      If the maximum number of guest users already exists, then the Add button is inactive. Delete one or more of the guest users and try again.
3. Click **Save**.
4. (Optional) To activate a new local user account, locate the user name and click
   
   .
   a) Select **Activate User**.
   b) Enter a password for the user.
   c) Click **Save**.
   You can choose to activate your local user at a later time, but the user cannot log in until their status is active.

1. The newly added user displays in the Local Users list with the default Auditor role.
2. If you activate the user, the new user must change their password after logging in. Refer them to Manage Local User Accounts for details.

To read more about how to create a custom role or add more roles to a local user, go to step 4 in Manage Local User Accounts . If you plan to activate the user at a later time, refer to Activate a Local User in NSX Manager.

Activate a Local User in NSX Manager

You can activate a local user account from the NSX Manager UI. Initially, only the `admin` account is active. You can authorize additional user access by activating the other local user accounts. As part of the new user activation, you can also optionally change the username and roles.

Familiarize yourself with the password complexity requirements for NSX Manager and NSX Edge. See " NSX Manager Installation" and " NSX Edge Installation" in the *NSX Installation Guide*. Guest users are available on NSX Manager only.

1. From your browser, log in as admin to an NSX Manager at https://<nsx-managr-ip-address>.
2. Select **System** > **Users and Roles** > **Local Users**.
3. Locate the local user name you want to add and click
   
   .
4. Select **Activate**. The guest user and audit accounts are inactive by default and must be activated through the UI and API before using.

   New expiration dates are created after user activation. To view or change password expiration as admin, see Manage Local User Accounts .
5. (Optional) To change the local user name:
   a) Click
      
      for that user and select **Edit**.
   b) Enter the user name changes.
   c) Click **Save**.
6. To change the user role assigned to one or both of the guest users:
   a) On the **User Role Assignment** tab, click
      
      and select **Edit**.
   b) Select the role change from the drop down list.
   c) Click **Save**.
7. (Optional) To deactivate any of the local users, click
   
   for that user, select **Deactivate User**, and click **Deactivate**.

   For more details on customizing roles, see Create or Manage Custom Roles.

Manage Local User Accounts

You can manage the default and new local guest users, through the NSX Manager UI.

Familiarize yourself with the password complexity requirements for NSX Manager and NSX Edge. See " NSX Manager Installation" and " NSX Edge Installation" in the *NSX Installation Guide*. To change password complexity, review Authentication Policy Settings

You cannot deactivate the admin user or change their role assignments. You also cannot change the audit user role assignments. The admin user or any user with the Enterprise Admin role can perform the following tasks unless noted otherwise:

- Starting with NSX 4.1, users with the Enterprise Admin role can add a user. To add users, go to Add a Local User .
- Starting with NSX 4.1, the admin user can delete a user. To delete or remove a user, go to Delete a Local User.
- Activate or deactivate any local user accounts, except for admin.
- Change user role assignments for the two guest users.
- Add a new role, clone an existing role, and edit or delete user-created roles.
- Reset user passwords. In addition, all local users can reset their own passwords if they know their current password.
- Change the user names for any of the local user accounts.
- Change password expiration settings.

The audit and guest users have default read privileges to the NSX environment and are not active by default. Before they can log in to NSX Manager, you must activate the accounts first.

Any changes to the local user accounts is audited.

By default, user passwords expire after 90 days. You can change or deactivate the password expiration for each user.

When a user logs in to NSX Manager, if the password is set to expire within 30 days, the NSX Manager UI displays a password expiration notification. If you set the password expiration to 30 days or less, the notification is always present. The notification includes a **Change Password** link. Click the link to change the user's password.

1. From your browser, log in as admin to an NSX Manager at https://<nsx-manager-ip-address>.
2. Select **System** > **User Management**.
3. To activate a user, select the **Local Users** tab and locate the user name.
   a) Click
   
   .
   b) Select **Activate User**.
   c) Enter a password for the user.
   d) Click **Save**.
4. To change or reset a user password, select the **Local Users** tab and locate the user name.
   a) Click
   
   .
   b) Select **Reset Password**.
   c) Enter the password details.
   d) Click **Save**.
5. (Optional) To add a user role assignment for a guest user in addition to the existing Auditor role, select the **User Role Assignment** tab and locate the user name.
   a) Click
   
   .
   b) Select **Edit**.
   c) Click the highlighted number link under Roles. This is the number of user roles assigned to this user.
   d) From the Set Roles/Scope window, click **Add Role**.
   e) Select the role from the dropdown list.

      Depending on the role you select, you may need to click **Set** to select the scope for this role assignment. If you use projects and want to specify specific access, select the projects to apply to this user role.
   f) To add the new role, click **Add**, then click **Apply**.
   g) Click **Save**.
6. To edit a user role assignment for a guest user, select the **User Role Assignment** tab and locate the user name.
   a) Click
   
   .
   b) Select **Edit**.
   c) Click the highlighted number link under Roles. This is the number of user roles assigned to this user.

      From the Set Roles/Scope window, you can add a role, edit an existing role, or delete a role.
   d) Update this guest user roles, then click **Apply**.
   e) Click **Save**.
7. To delete a role assignment, select the **User Role Assignment** tab and locate the user name.
   a) Click
   
   .
   b) Select **Edit**.
   c) Click the highlighted link under Roles.

      This is the number of user roles of this user.

d) Click

.
e) Select **Delete** and confirm the deletion.
f) Click **Apply**.
g) Click **Save**.

8. (Optional) To change a user name, select the **Local Users** tab and locate the user name.
a) Click

.
b) Select **Edit**.
c) Change the user name.
d) Click **Save** and **Continue**.

If you change the audit user name, you can identify it by the Auditor label containing a padlock.

9. (Optional) To deactivate a user, select the **Local Users** and locate the user name.
a) Click

.
b) Select **Deactivate User** and confirm.
c) Click **Deactivate**.

10. To get the password expiration information, from the **Local Users** tab, expand the row for the user that you want to view.

11. (Optional) To change the password expiration settings, log in to the appliance's CLI as **admin**.
a) To set the password expiration time in days, run the `set user <username> password-expiration <number of days>` command.
```
nsxcli> set user admin password-expiration 120
nsxcli>
```
b) To deactivate password expiration, run the `clear user <username> password-expiration`
```
nsxcli> clear user admin password-expiration
nsxcli>
```

Manage Local User's Password or Name Using the CLI

You can manage NSX Manager user accounts through an NSX appliance's CLI. This topic describes how the admin user manages user account details using the CLI. Alternately, you can also use the UI.

Familiarize yourself with the password complexity requirements for NSX Manager and NSX Edge. See " NSX Manager Installation" and " NSX Edge Installation" in the *NSX Installation Guide*.

The `admin` user can manage passwords, change the name of the admin and other users, and add, delete, or deactivate users. Any user account change is audited.

For extended access, see Manage Local User Accounts .

The audit user has read privileges to the NSX environment and is not active by default unless the audit password is provided during NSX installation. To activate the audit user after installation, use the UI or log in as admin to the CLI and run the `set user audit password` command and provide a new password.

By default, user passwords expire after 90 days. You can change or deactivate the password expiration for each user.

When an NSX Manager user password is within 30 days of expiring, the NSX Manager UI displays a password notification after logging in. The notification includes a **Change Password** link. To change the password, click the link.

For CLI details, see the *NSX Command-Line Interface Reference*.

1. Log in to the CLI of the appliance as `admin`.
2. To change a user password, run the `set user <username> password` command. For example:
```
nsx> set user audit password
New password:
Confirm new password:
nsx>
```
3. To change the name of a user, run the `set user <username> username <new username>` command. For example:
```
nsx> set user admin username admin1
nsx>
```
4. To see a list of existing user names, run the `set user [TAB][TAB]` command. For example:
```
nsx> set user [TAB][TAB]
  admin     Username of user
  audit     Username of user
  root      Username of user
```
5. To get password expiration information, run the `get user <username> password-expiration` command. For example:
```
nsx> get user admin password-expiration
Tue Jun 07 2022 UTC 06:33:29.963
Password expires 90 days after last change
 Current password will expire in 90 days
User will receive warning messages 7 days before password expires.
Password expires 90 days after last change
nsx>
```
6. To set the password expiration time in days, run the `set user <username> password-expiration <number of days>` command. For example:
```
nsx> set user admin password-expiration 120
nsx>
```
7. To deactivate password expiration for users, run the `clear user <username> password-expiration` command. For example:
```
nsx> clear user admin password-expiration
nsx>
```
8. To change the default number of days a user receives a warning message prior to their password expiration, run the `set user <username> password-expiration-warning <password-expiration-warn-days>` command. Default is 7. Range is 1 to 1999. For example:
```
set user admin password-expiration-warning 14
```

Delete a Local User

This section covers different mechanisms to remove local user accounts from different NSX appliances including the NSX Manager and the NSX Edge.

Use the following tables as a guideline for the support and behaviors when deleting local users from the NSX Manager or the NSX Edge.

Table 1068: Operation Support for NSX Appliances

Node Type	UI	API	CLI
NSX Manager	Supported	DELETE /api/v1/node/users/{userid}	`del user <username>`
NSX Edge	Not supported	DELETE /api/v1/transport-nodes/{transport-node-id}/node/users/{userid}	`del user <username>`

For details on tasks and behaviors of the NSX Manager versus the NSX Edge, see the *NSX Appliance Local User Deletion Behaviors* table. Depending on your NSX appliance, refer to the appropriate section to delete a local user. To use the API, refer to "Deleting a Local User in NSX Manager or NSX Edge Using the API." To use the CLI, refer to "Deleting a Local User in NSX Manager or NSX Edge Using the CLI." To use the UI, refer to "Deleting a Local User in NSX Manager Using the UI."

Table 1069: NSX Appliance Local User Deletion Behaviors

Task	NSX Manager	NSX Edge
Delete any local user (root and admin users cannot be removed)	Includes audit, guestuser1, and guestuser2. Also any guest user accounts created by the Enterprise Admin user.	Includes audit, guestuser1, and guestuser2. Once deleted, you will be unable to add these users back onto the NSX Edge.
Recreate default local user accounts after deletion (audit and default guestuser1 and guestuser2 accounts)	You can delete the audit local user, but users deleted from following node types cannot be recovered so plan accordingly: • Autonomous Edge • Cloud Service Manager • Edge • Public Cloud Gateway	Recreation of the local user account is not allowed. To get the default users back on to the NSX Edge , you can redeploy an NSX Edge node or deploy a new NSX Edge node.
Recreate custom local user accounts after deletion	Yes by users with Enterprise Admin role only.	No. This is not allowed on the NSX Edge.
Deletion of root and admin users	No.	No.
Synchronization across other nodes after deletion on one node.	Yes across other nodes in the NSX management cluster.	No. It is not supported.

Deleting a Local User in NSX Manager or NSX Edge Using the API

To delete a specific user on a specific NSX node, a specific cluster, or a specific user ID, refer to the *NSX API Guide* URIs for System Administration > Configuration > Fabric > Nodes > User Management > Users for details. The API commands are also listed in the *Operation Support for NSX Appliances* table. Users deleted from the following node types cannot be recovered:

- Autonomous Edge
- Cloud Service Manager
- Edge
- Public Cloud Gateway

Deleting a Local User in NSX Manager or NSX Edge Using the CLI

To delete local user accounts on an NSX Edge node using the CLI, use the `del user <username>` command. For details on how to delete local user accounts including audit, default guestuser1 and guestuser2 accounts, and custom guest user accounts, refer to the *NSX Command-Line Interface Reference* section on System Administration > Lifecycle Management > Nodes > User Management > Users.

Deleting a Local User in NSX Manager Using the UI

Enterprise Admin users can delete local user accounts from NSX Manager using the UI.

1. To delete local users, from your browser, log in as admin to an NSX Manager at https://<nsx-manager-ip-address>.
2. Select **System > User Management > Local Users**.
3. To remove a local user account, locate the user name you want to remove and click  .
4. From the popup menu, click **Delete**.
5. To confirm, click **Delete** again.
   A message displays that your deletion was successful.

Password Management

This topic provides links to the many options to manage your user passwords using the NSX user interface, CLI, and API.

Resetting User Passwords

There are many ways to reset your user password or have an administrator reset it for you. This topic describes the scenarios and links to the procedures to complete your password resets.

To resolve your password issues, use the following table to find the link to the appropriate procedure.

These password rules are followed in NSX:

- Password changes in the UI may require the user's current password.
- When the admin resets an audit or a local user's password, it results in the immediate expiration of the admin-generated password that forces the user to change the password during login.
- Expired passwords are reset by providing the existing expired password.
- New password must be different from the current and past passwords.

User	Scenario	UI	CLI	API
You are the Admin or the Enterprise Admin	Reset your own forgotten password		If you know the root password, use the NSX CLI. See Resetting the Passwords of an Appliance.	
	Reset any user password (logged in as Admin or Enterprise Admin)	See Manage Local User Accounts . To set a user password during activation, see Activate a Local User in NSX Manager.	Use the NSX CLI (except guest users). See Manage Local User's Password or Name Using the CLI Admin only; remote Enterprise Admin does not have access.	Go to https://code.vmware.com/apis/1163/nsx-t and select your release number. Search under **System Administration > Configuration > Fabric > Nodes > User Management > Users** for API details.

User	Scenario	UI	CLI	API
	Reset expired password (Admin only access)	To reset other user's expired passwords, see Resetting Expired Passwords. Admin cannot reset their own expired password in the UI.	• To reset admin expired password, SSH or log in to the NSX CLI. Requires existing password. • Only admin can reset the Enterprise admin's expired password. • To reset the audit user expired password, SSH or log in to the NSX CLI. See Manage Local User's Password or Name Using the CLI.	Admin can go to https://code.vmware.com/apis/1163/nsx-t and select your release number. Search for **System Administration > Configuration > Fabric > Nodes > User Management** to find the API details.
You are the Audit user	Reset your own forgotten password	Contact your administrator.	Contact your administrator.	Contact your administrator.
	Reset your own password (requires known password)	If you know your password, see Manage Local User Accounts .	Contact your administrator.	Go to https://code.vmware.com/apis/1163/nsx-t and select your release number. Search for **System Administration > Configuration > Fabric > Nodes > User Management** to find the API details.
	Reset audit's expired password (Requires existing password)	Contact your administrator.	SSH or log in to the NSX CLI. Requires existing password. See Manage Local User's Password or Name Using the CLI	Go to the bios and update password.
Guest users	Reset your own forgotten password	Contact your administrator.	Contact your administrator.	Contact your administrator.
	Reset your own password (Requires known password)	If you know the guest user password, see Manage Local User Accounts .		Go to https://code.vmware.com/apis/1163/nsx-t and select your release number. Search for **System Administration > Configuration > Fabric > Nodes > User Management** to find the API details.
	Reset guest user's expired password	Contact Network Admin or administrator.	Contact your Network Admin or administrator.	Contact your Network Admin or administrator.

Note:
If you are an LDAP user, contact the LDAP administrator to reset your password.

Resetting Expired Passwords

In NSX, an admin user can reset your expired password. You must enter the expired user password to complete the password reset in the CLI.

Each user password has an expiration date tracked by NSX. If you log in with an expired password, the error message, `Your password has expired`, appears. A notification also appears if your password is expiring soon. Depending on what release you are using and what user you are, there are different actions to take.

The admin user can reset the expired passwords for their own, as well as the audit and Enterprise Admin user, using the CLI. The admin user can reset the expired passwords for all other local users, but not themselves in the UI.

When the admin resets any local user's expired password using the CLI and UI, the password expiration date does not reset. To extend the password expiration date to a new date or to the default 90 days, the admin must change it using the CLI or API. Another quick way for the admin to reset a local user's password expiration date is to deactivate and then activate the user. For details, see Manage Local User's Password or Name Using the CLI or Manage Local User Accounts .

1. Choose one of the following steps depending on your release and needs:
   - CLI - To reset the admin, Enterprise Admin, or audit user's expired passwords, run the `set user <username> password` command on the appliance's CLI as admin. You must know the current user password to complete the password reset. Guest users cannot be reset using the CLI.
   - UI - The admin user can reset an expired password for other local users in the UI. Admin cannot change their own expired password using the UI. Look for the Notification message about the user password expiration and click on **Change Password**. Admin can also use **System > User Management** to change the passwords. Note, the password expiration date does not reset.
   - API - The admin user can go to Broadcom Developer Portal and select the release number. Go to the All Methods page and search for **System Administration > Configuration > Fabric > Nodes > User Management > Authentication Policy** to find the API details.
2. Once the password reset is complete, the user gets prompted to change their password during login.

Resetting the Passwords of an Appliance

The following procedure applies to NSX Manager and NSX Edge.

> **Note:**
>
> If you have an NSX Manager cluster, resetting the password for any of the local users on one automatically resets the password for the other NSX Managers in the cluster. The synchronization of the password can take a few minutes.
>
> If you have renamed the local user, use the new name in the following procedures.
>
> When you reboot an appliance, the GRUB boot menu does not appear by default. Before you perform this procedure, you must configure GRUB so that the GRUB boot menu displays. For more information about configuring GRUB and changing the GRUB `root` password, see "Configure an Appliance to Display the GRUB Menu at Boot Time" in the *NSX Installation Guide*.

If you know the password for `root` but have forgotten the password for your local users, you can reset it using the following procedure:

1. Log in to the appliance as **root**.
2. To stop the server:
   a. For NSX Edge, run the command `/etc/init.d/nsx-edge-api-server stop`.
   b. Otherwise, run the command `/etc/init.d/nsx-mp-api-server stop`.
3. (Optional) To reset the password for **admin**, run the command `passwd admin`.
4. (Optional) To reset the password for **audit**, run the command `passwd audit`.
5. (Optional) To reset a guest user password, run the command `passwd guestusername`.
6. Run the command `touch /var/vmware/nsx/reset_cluster_credentials`.

7. To restart the server:
   a. For NSX Edge, run the command `/etc/init.d/nsx-edge-api-server start`.
   b. Otherwise, run the command `/etc/init.d/nsx-mp-api-server start`.

If you have forgotten the root user's password, you can reset it using the following procedure. You can then use the above procedure to reset the password for all local users.

1. Connect to the console of the appliance.
2. Reboot the system.
3. When the GRUB boot menu appears, press the left `SHIFT` or `ESC` key quickly. If you wait too long and the boot sequence does not pause, you must reboot the system again.
4. Press `e` to edit the menu.

   Choose the top Ubuntu line. Then enter the user name **root** and the GRUB password for root (not the same as the appliance's user root). For more information, see Broadcom knowledge base article 417635.
5. Press `e` to edit the selected option.
6. Search for the line starting with `linux` and add `systemd.wants=PasswordRecovery.service` to the end of the line.
7. Press `Ctrl-X` to boot.
8. When the log messages stop, enter the new password for root.
9. Enter the password again.

   The boot process continues.
10. After the reboot, you can verify the password change by logging in as root with the new password.

Authentication Policy Settings

You can view or customize your authentication policy settings through the API and CLI.

Some details are important to understand about viewing or changing policy settings.

- This feature allows one password policy only.
- Changes to the password configuration affects new users immediately. After administrators update existing user password configurations current users must follow updated password configuration changes.
- API configuration changes take about 20 seconds to take effect.
- For NSX Edge, changed passwords are not synchronized across the cluster. Both CLI and API changes appear in one node.
- Appliance transport node support includes BCG, autonomous Edge, or Unified Appliances (UA). There is no support for transport node password configuration changes for the local administrator or the auditor user in ESX.
- Privileged Access Management (PAM) supports setting minimum or maximum password length, but not both settings.
- Use of negative numbers for password entries sets the minimum range while positive numbers set the maximum range. To keep the existing default, leave the response empty.
- If modified pre-upgrade, the password policy configuration remains the same post upgrade for existing users. NSX Manager does not enforce the default password policy in this case.
- Authentication policies

  `[API] /api/v1/node/aaa/auth-policy`

  `[API] /api/v1/cluster/<Node-UUID>/node/aaa/auth-policy`

  `[API] /api/v1/transport-nodes/<transport-node-id>/node/aaa/auth-policy`

NSX Manager includes the following CLI and API password complexity and authentication command support. These password policy options now sync across the management cluster nodes. Viewing password details does not require any permissions. Modifying existing password defaults requires admin permissions.

For more information on defaults ranges and other details, see the *NSX Command-Line Interface Reference* and the *NSX API Guide*.

Table 1070: CLI Password Policy Customizable Options

Password Option	CLI Command

View or configure password complexity configuration	`get password-complexity`
	`Wed Jun 08 2022 UTC 12:57:45.325`
	`- minimum 15 characters in length`
	`- maximum 128 characters in length`
	`- minimum 1 lowercase characters`
	`- minimum 1 uppercase characters`
	`- minimum 1 numeric characters`
	`- minimum 1 special characters`
	`- default password complexity rules as enforced by the Linux PAM module`
	`set password-complexity`
	You can change specific parameters using these arguments:
	`Minimum password length (leave empty to not change):`
	`Maximum password length (leave empty to not change):`
	`Lower characters (leave empty to not change):`
	`Upper characters (leave empty to not change):`
	`Numeric characters (leave empty to not change):`
	`Special characters (leave empty to not change):`
	`Minimum unique characters (leave empty to not change):`
	`Allowed similar consecutives (leave empty to not change):`
	`Allowed monotonic sequence (leave empty to not change):`
	`Hash algorithm (leave empty to not change):`
	`Password remembrance (leave empty to not change):`

View authentication policy	`get auth-policy cli`
	`lockout-period Lockout period`
	`max-auth-failures Maximum authentication failures before lockout`

Configure authentication policy	`set auth-policy cli`
	`lockout-period Lockout period`
	`max-auth-failures Maximum authentication failures before lockout`

Table 1071: API Password Policy Customizable Options

Password Option	API Commands

View password complexity and authorization policy	`get /api/v1/node/aaa/auth-policy`
	`{`
	`"_retry_prompt": 3,`
	`"_schema": "AuthenticationPolicyProperties",`
	`"_self": {`
	`"href": "/node/aaa/auth-policy",`
	`"rel": "self"`
	`},`
	`"api_failed_auth_lockout_period": 5,`
	`"api_failed_auth_reset_period": 900,`
	`"api_max_auth_failures": 900,`
	`"cli_failed_auth_lockout_period": 900,`
	`"cli_max_auth_failures": 5,`
	`"digits": -1,`
	`"hash_algorithm": "sha512",`
	`"lower_chars": -1,`
	`"max_repeats": 0,`
	`"max_sequence": 0,`
	`"maximum_password_length": 128,`
	`"minimum_password_length": 12,`
	`"minimum_unique_chars": 0,`
	`"password_remembrance": 5,`
	`"special_chars": -1,`
	`"upper_chars": -1`
	`}`

View Workspace ONE Access(VIDM) authorization policy	`get auth-policy vidm`
	`Wed Jun 08 2022 UTC 12:58:28.357`
	`LB Enabled: False`
	`Enabled: False`
	`Hostname:`
	`Thumbprint:`
	`Client Id:`
	`Node Hostname:`

Configure VIDM authorization policy	`set auth-policy vidm`
	`enabled  Enabled property`
	`hostname  System's network name`
	`lb-extern External Load Balancer Flag For vIDM Wiring`

Configure password complexity and authorization policy • API failed authorization attempts lockout period (in seconds)	`put /api/v1/node/aaa/auth-policy` `lockout_period <lockout-period-arg>`
• Authentication failure lockout reset period	`lockout_reset_period`
• Number of failures allowed before API lockout	`max_auth_failures`
• Number of numeric characters	`digits`
• Hash algorithm	`hash_algorithm`
• Number of lowercase characters	`lower_chars`
• Sequence of same characters	`max_repeats`
• Maximum monotonic character sequence (1234 or DCBA)	`max_sequence`

Password Option	API Commands
• Maximum password length	`maximum_password_length`
• Minimum password length	`minimum_password_length`
• Minimum unique characters	`minimum_unique_chars`
• Minimum number of password reuse	`password_remembrance` • Default value is 5. • If you enter a number, the user cannot reuse that number of previous passwords. For example, if set to 2, the user cannot reuse the last two passwords. • The valid range is 1 to 12.
• Number of special characters	`special_chars`
• Number of uppercase characters	`upper_chars`

Reset password complexity, authentication policy, or both	For node, transport-nodes, and clusters, resets to default password complexity for respective nodes:
	`reset-password-complexity`
	`reset-auth-policies`
	`reset-all`

Integration with LDAP

You can configure NSX Manager to authenticate users using a directory service such as Active Directory over LDAP or OpenLDAP.

If you are using Active Directory (AD), and your AD forest is comprised of multiple subdomains, you should point NSX at your AD Global Catalog (GC) and configure each subdomain as an alternative domain name in NSX. The Global Catalog service usually runs on your primary AD domain controllers, and is a read-only copy of the most important information from all the primary and secondary domains. The GC service runs on port 3268 (plaintext), and 3269 (LDAP over TLS, encrypted).

For example, if your primary domain is "example.com" and you have subdomains "americas.example.com" and "emea.example.com", you should:

1. Configure NSX to use either the LDAP protocol on port 3268 or the LDAPS protocol on port 3269.
2. Add alternative domain names "americas.example.com" and "emea.example.com" in the NSX LDAP configuration.

Users in one of the subdomains must log in using the appropriate domain in their login name. For example, user "john" in the emea.example.com domain, must log in with the username "john@emea.example.com".

LDAP support on a Global Manager (NSX Federation) is identical to a Local Manager. LDAP configuration is not synchronized from Global Manager to Local Managers. Each NSX cluster should be configured separately for LDAP.

LDAP Identity Source

NSX Manager acts as an LDAP client, and interfaces with LDAP servers.

Three identity sources can be configured for user authentication. When a user logs into NSX Manager, the user is authenticated against the appropriate LDAP server of the user's domain. The LDAP server responds back with the authentication results, and the user group information. Once successfully authenticated, the user is assigned the roles corresponding to the groups that they belong to.

When integrating with Active Directory, NSX Manager allows users to log in using their samAccountName, or userPrincipalName. If the @domain portion of the userPrincipalName does not match the domain of the Active Directory instance, then you must also configure an alternative domain in the LDAP configuration for NSX.

In the following example, the domain of the Active Directory instance is "example.com" and a user with a samAccountName "jsmith" has a userPrincipalName of John.Smith@acquiredcompany.com. If you configure

an alternative domain of "acquiredcompany.com", then this user can log in as "jsmith@example.com" using the samAccountName, or as John.Smith@acquiredcompany.com using the userPrincipalName. If the userPrincipalName has no @domain portion, the user won't be able to log in.

Logging in as jsmith@acquiredcompany.com will not work because the samAccountName can only be used with the primary domain.

NSX can only be authenticated to Active Directory or OpenLDAP using LDAP Simple Authentication. NTLM and Kerberos authentication are not supported.

Note:

LDAP Identity Source does not support universal AD groups.

1. Navigate to **System** > **User Management** > **LDAP**.
2. Click **Add Identity Source**.
3. Enter a **Name** for the identity source.
4. Enter the **Domain Name** This must correspond to the domain name of your Active Directory server, if using Active Directory.
5. Select the type: either **Active Directory over LDAP** or **Open LDAP**.
6. Click **Set** to configure LDAP servers. You can add up to three LDAP servers for failover support, to each domain.

Hostname/IP	The hostname or IP address of your LDAP server.
LDAP Protocol	Select the **protocol**: LDAP (unsecured) or LDAPS (secured).
Port	The default port populates based on the selected protocol. If your LDAP server runs on a non-standard port, you can edit this text box to give the port number.
Connection Status	Fill in the mandatory text boxes, including the LDAP server information, then click **Connection Status** to test the connection.
Use StartTLS	If selected, the LDAPv3 StartTLS extension is used to upgrade the connection to use encryption. To determine the use of this option, consult your LDAP server administrator. This option is only available if LDAP protocol is selected.
Certificate	• If you use LDAPS or LDAP + StartTLS, enter the PEM-encoded X.509 certificate of the server in the text box. If you leave this text box blank and click the **Check Status** link, NSX connects to the LDAP server. NSX then retrieves the LDAP server's certificate and asks if you want to trust that certificate. If you verify that the certificate is correct, click **OK**. The certificate text box gets populated with the retrieved certificate. • If your hostname/IP is an L4 Load Balancer VIP, the LDAP servers behind the VIP must present certificates signed by the same certificate authority (CA). You must enter the PEM-encoded X.509 certificate of the CA that signed the certificates. If you do not enter the certificate of the CA, NSX prompts you to accept the certificate of one of the LDAP servers, one which the load balancer randomly selects. If the server presents the full trust chain, including the certificate of the CA that signed the certificates of the other servers in the pool, the LDAP connection works when routed to another server. If the certificate initially presented does not include the CA certificate, the certificate presented by the other LDAP servers gets denied.

	For this reason, you must enter the certificate of the CA that signed all the certificates presented by the different LDAP servers. • If the LDAP servers are behind an L4 Load Balancer VIP, NSX will support certificates of the LDAP servers signed by different CAs if those CAs are subordinate to the same root CA. In this case, you must add the root CA certificate to the certificate field in the NSX LDAP configuration
Bind Identity	Enter the format as user@domainName, or you can specify the distinguished name. For Active Directory, use either the userPrincipalName (user@domainName) or the distinguished name. For OpenLDAP, you must supply a distinguished name. This text box is required unless your LDAP server supports anonymous bind, then it is optional. Consult your LDAP server administrator if you are not sure.
Password	Enter a password for the LDAP server. This text box is required unless your LDAP server supports anonymous bind, then it is optional. Consult your LDAP server administrator.

7. Click **Add**.
8. Enter the **Base DN**.

   To add an Active Directory domain, a base distinguished name (Base DN) is needed. A Base DN is the starting point that an LDAP server uses when searching for users authentication within an Active Directory domain. For example, if your domain name is corp.local the DN for the Base DN for Active Directory is "DC=corp,DC=local".

   All of the user and group entries you intend to use to control access to NSX must be contained within the LDAP directory tree rooted at the specified Base DN. If the Base DN is set to something too specific, such as an Organizational Unit deeper in your LDAP tree, NSX may not be able to find the entries it needs to locate users and determine group membership. Selecting a broad Base DN is a best practice if you are unsure.
9. Your NSX end users can now log in using their login name followed by @ and the domain name of your LDAP server, **user_name@domain_name**.

Assign roles to users and groups. See Add a Role Assignment or Principal Identity.

Integration with VCF Identity Broker for Single-Sign-On Access

You can allow users access to log in to NSX Managers, Local Managers, and Global Managers by connecting VMware NSX to the VCF Identity Broker in vCenter for federated authentication. To configure this capability, it is recommended to use the VMware Cloud Foundation SDDC Manager. For NSX Global Managers and standalone NSX Managers, follow the procedure described in Knowledge Base article 3875053: Procedure to configure Single Sign-On on standalone NSX.

OpenID Connect (OIDC) is a simple identity layer on top of the OAuth 2.0 protocol. OIDC provides a secure way to authenticate users by allowing different organizations to participate in authentication. For example, one organization can verify user identities while another organization decides whether to give users access to various services based on the groups to which that user belongs. OIDC can also obtain basic profile information about the user across applications.

Some organizations involved in an acquisition scenario can use OIDC to allow employees from either company to log into NSX.

For details on how to configure this capability, see Configuring VCF Single Sign-On.

Validate the OIDC authentication to ensure that your user can log in to the NSX Manager and has access to their Enterprise Admin capabilities such as the Tier-0 Gateways menu. For details about log in options, see Logging in to NSX Manager .

NSX API Authentication Using a Session Cookie

This topic describes how to use NSX session-based authentication to generate a JSESSIONID cookie when using the API. Use this method to reduce the number of times you have to enter your username and password. You can use this type of authentication with vIDM and LDAP authentication.

NSX uses several different mechanisms to authenticate NSX users. They include:

- HTTP authentication
- Session-based authentication
- Principal identity or certificate-based authentication
- Single sign on using vIDM and Cloud Service Platform (CSP)

The NSX uses a username and password to generate a session cookie during session creation. Once the session cookie has been created, subsequent API requests might optionally use this session cookie instead of the user name and password credentials. This means that the session state is specific to the particular NSX Manager that generated the session cookie. When clients make requests to the NSX Manager, it only allows clients to authenticate if the session cookie they present matches one of the cookies generated by the manager. When any user logs out of NSX Manager, the session cookie is immediately eliminated from the reverse-proxy of the NSX Manager and cannot be reused. Idle sessions time out automatically or you can delete them using the API.

Access using the API request generates audit log details. This logging is always enabled and cannot be disabled. Auditing of sessions is initiated at system startup. Audit log messages include the text audit="true" in the structured data part of the log message.

Note that if the password contains special characters, such as **'+'** symbol, the API that is used to get the session cookie fails due to an encoding issue. To avoid this issue, use one of the following methods:

Option 1: Encode the username and password separately by using --data-urlencode.

For example,

```
curl -i -k -c cookies.txt -X POST --data-urlencode 'j_username=<username>' --data-
urlencode 'j_password=<password>' https://<NSX-Manager>/api/session/create
```

Replace <*NSX-Manager*>, <*username*>, and <*password*> with the correct details.

Option 2: Manually URL encode the special character and use in the documented cookie session create API call.

For example, convert '+' to URL character '%2B' and use in the password:

```
curl -i -k -c cookies.txt -X POST -d 'j_username=<username>&j_password='#######%2B#######'
https://<NSX-Manager>/api/session/create
```

Replace <*NSX-Manager*> and <*username*> with the correct details. ####### represents the part of the password which is not a special character, whereas %2B is the '+' converted to URL encode.

For more information about failing of the API that is used to get the session cookie, see KB Article.

The following example describes using cURL to create session-based authentication for API calls.

1. To create a new session cookie that authenticates to the NSX Manager and retrieves xsrf from header enter:

   ```
   # curl -i -k -c session.txt -X POST -d 'j_username=admin@mywork.com&j_password=SecretPwsd3c4d' https://
   <nsx-manager>/api/session/create 2>&1 > response.txt
   ```

   In this example, the cURL command authenticates to the manager, places the session cookie in the sessions.txt file and writes all HTTP response headers to the headers.txt file. You will need to use one of the headers from headers.txt, the x-xsrf-token header, to provide in subsequent requests.

   You can also use the standard unicode/URI encoding for the @ in the username.

   An example of the session contents follows:

   ```
   # cat session.txt
         # Netscape HTTP Cookie File
         # https://curl.haxx.se/docs/http-cookies.html
   ```

```
# This file was generated by libcurl! Edit at your own
  risk.
# HttpOnly_172.182.183.135 FALSE / TRUE 0 JSESSIONID CFG588DF6DGF493C0EAEFC62685C42E1
```

2. If you need to create two sessions, change the name of the session.txt files so that both sessions are valid.

```
curl -i -k -c session.txt -X POST -d 'j_username=admin@mywork.com&j_password=SecretPwsd3c4d' https://<nsx-
manager>/api/session/create 2>&1 > response.txt
# curl -i -k -c session2.txt -X POST -d 'j_username= admin@mywork.com&j_password=SecretPwsd3c4d' https://
<nsx-manager>/api/session/create 2>&1 > response2.txt
```

3. For subsequent calls, use the session cookie and xsrf header from the previous step.

```
# curl -k -b session.txt -H "x-xsrf-token: 5a764b19-5ad2-4727-974d-510acbc171c8"
      https://<nsx-manager>/policy/api/v1/infra/segments

      {
        "results" : [ {
          "type" : "ROUTED",
          "subnets" : [ {
            "gateway_address" : "192.168.10.1/24",
            "network" : "192.168.10.0/24"
          } ],
          "connectivity_path" :
        "/infra/tier-1s/test_t1",
          "transport_zone_path" :
        "/infra/sites/default/enforcement-points/default/transport-zones/1b3a2f36-bfd1-443e-
a0f6-4de01abc963e",
          "advanced_config" : {
            "address_pool_paths" : [ ],
            "hybrid" : false,
            "multicast" : true,
            "inter_router" : false,
            "local_egress" : false,
            "urpf_mode" : "STRICT",
            "connectivity" : "ON"
          },
          "admin_state" : "UP",
          "replication_mode" : "MTEP",
          "resource_type" : "Segment",
          "id" : "seg1",
          "display_name" : "seg1",
          "path" : "/infra/segments/seg1",
          "relative_path" : "seg1",
          "parent_path" : "/infra",
          "unique_id" :
        "6573d2c9-f4f9-4b37-b410-71bded8857c3",
          "marked_for_delete" : false,
          "overridden" : false,
          "_create_user" : "admin",
          "_create_time" : 1633331197569,
          "_last_modified_user" : "admin",
          "_last_modified_time" : 1633331252660,
          "_system_owned" : false,
          "_protection" : "NOT_PROTECTED",
          "_revision" : 1
        } ],
        "result_count" : 1,
        "sort_by" : "display_name",
```

```
        "sort_ascending" : true
```

If you use the same session cookie with another node in the cluster, the command fails with the error message:

```
The credentials were incorrect or the account specified has been locked.","error_code":403.
```

When the session expires, NSX Manager responds with a 403 Forbidden HTTP response. You must then obtain a new session cookie and x-xsrf-token.

4. To configure the session expiry setting, use the connection_timeout API command. The default session expiry is 1800 seconds (30 minutes).
   a) Use `GET https://<nsx-mgr>/api/v1/cluster/api-service` to view a snapshot of the current configuration..

```
GET https://<nsx-mgr>/api/v1/cluster/api-service
{
    "session_timeout": 1800,
    "connection_timeout": 30,
    "protocol_versions": [
        {
            "name": "TLSv1.1",
            "enabled": true
        },
        {
            "name": "TLSv1.2",
            "enabled": true
        }
    ],
    "cipher_suites": [
        {
            "name": "TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA",
            "enabled": true
        },
        {
            "name": "TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256",
            "enabled": true
        },
        {
            "name": "TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256",
            "enabled": true
        },
        {
            "name": "TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA",
            "enabled": true
        },
        {
            "name": "TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384",
            "enabled": true
        },
        {
            "name": "TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384",
            "enabled": true
        },
        {
            "name": "TLS_RSA_WITH_AES_128_CBC_SHA",
            "enabled": true
        },
        {
            "name": "TLS_RSA_WITH_AES_128_CBC_SHA256",
```

```
                "enabled": true
            },
            {
                "name": "TLS_RSA_WITH_AES_128_GCM_SHA256",
                "enabled": true
            },
            {
                "name": "TLS_RSA_WITH_AES_256_CBC_SHA",
                "enabled": true
            },
            {
                "name": "TLS_RSA_WITH_AES_256_CBC_SHA256",
                "enabled": true
            },
            {
                "name": "TLS_RSA_WITH_AES_256_GCM_SHA384",
                "enabled": true
            }       ,
            {
                "name": "TLS_ECDSA_WITH_AES_256_GCM_SHA384",
                "enabled": true
            }
        ],
        "redirect_host": "",
        "client_api_rate_limit": 100,
        "global_api_concurrency_limit": 199,
        "client_api_concurrency_limit": 40,
        "basic_authentication_enabled": true,
        "cookie_based_authentication_enabled": true,
        "resource_type": "ApiServiceConfig",
        "id": "reverse_proxy_config",
        "display_name": "reverse_proxy_config",
        "_create_time": 1658339081246,
        "_create_user": "system",
        "_last_modified_time": 1658339081246,
        "_last_modified_user": "system",
        "_system_owned": false,
        "_protection": "NOT_PROTECTED",
        "_revision": 0
    }
```

b) Edit the JSON with the desired values and remove all the underscore fields (_xxxxx), except for `_revision`.

```
 GET https://<nsx-mgr>/api/v1/cluster/api-service
{
    "global_api_concurrency_limit": 199,
    "client_api_rate_limit": 100,
    "client_api_concurrency_limit": 40,
    "connection_timeout": 30,
    "redirect_host": "",
    "cipher_suites": [
      {"enabled": true, "name": "TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384"},
      {"enabled": true, "name": "TLS_RSA_WITH_AES_256_GCM_SHA384"},
      {"enabled": true, "name": "TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256"},
      {"enabled": true, "name": "TLS_RSA_WITH_AES_128_GCM_SHA256"}
      {"enabled": true, "name": "TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384}",
      {"enabled": true, "name": "TLS_RSA_WITH_AES_256_CBC_SHA256"},
      {"enabled": true, "name": "TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA"},
      {"enabled": true, "name": "TLS_RSA_WITH_AES_256_CBC_SHA"},
```

```
            {"enabled": true, "name": "TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256"},
            {"enabled": true, "name": "TLS_RSA_WITH_AES_128_CBC_SHA256"},
            {"enabled": false, "name": "TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA"},
            {"enabled": true, "name": "TLS_RSA_WITH_AES_128_CBC_SHA"},

            {"enabled": true, "name": "TLS_ECDSA_WITH_AES_256_GCM_SHA384"},
          ],
            "protocol_versions": [
              {"enabled": true, "name": "TLSv1.1"},
            {"enabled": true, "name": "TLSv1.2"}
        ]
          "_revision": 0
      }
```

c) Use `PUT https://<nsx-mgr>/api/v1/cluster/api-service` to modify the current configuration.

5. To delete a session cookie, use the /api/session/destroy API command.

```
curl -k -b session.txt -H "x-xsrf-token: `grep -i xsrf response.txt | awk '{print $2}'`" https://<nsx-manager>/api/v1/node/version
```

For example:

```
curl -k -b session.txt -H "x-xsrf-token: `grep -i xsrf response.txt | awk '{print $2}'`" https://<nsx-manager>/api/v1/node/version
```

Response:

```
{
    "module_name" : "common-services",
    "error_message" : "The credentials were incorrect or the account specified has been locked.",
    "error_code" : "403"
}
```

To review the requirements to authenticate users with your session-based supported authentication service, see Integration with VMware Workspace ONE Access or Integration with LDAP.

Add a Role Assignment or Principal Identity

You can assign roles to users or user groups if VMware Workspace ONE Access is integrated with NSX, or if you have LDAP as an authentication provider. You can also assign roles to principal identities.

You must have an authentication provider configured:

- For role assignment for vIDM, verify that a vIDM host is associated with NSX. For more information, see Configure VMware Workspace ONE Access Integration.
- For role assignment for LDAP, verify that you have an LDAP identity source. For more information, see LDAP Identity Source.

A principal is a component or a third-party application such as an OpenStack product. With a principal identity, a principal can use the identity name to create an object and ensure that only an entity with the same identity name can modify or delete the object. A principal identity has the following properties:

- Name
- Node ID - this can be any alphanumeric value assigned to a principal identity
- Certificate
- RBAC role indicating the access rights of this principal

Users (local, remote, or principal identity) with the Enterprise Administrator role can modify or delete objects owned by principal identities. Users (local, remote, or principal identity) without the Enterprise Administrator role cannot modify or delete protected objects owned by principal identities, but can modify or delete unprotected objects.

If a principal identity user's certificate expires, you must import a new certificate and make an API call to update the principal identity user's certificate (see the procedure below). For more information about the NSX API, a link to the API resource is available at https://code.vmware.com.

A principal identity user's certificate must satisfy the following requirements:

- SHA256 based.
- RSA/DSA message algorithm with 2048 bits or above key size.
- It cannot be a root certificate.

You can delete a principal identity using the API. However, deleting a principal identity does not automatically delete the corresponding certificate. You must delete the certificate manually.

Steps to delete a principal identity and its certificate:

1. Get the details of the principal identity to delete and note the `certificate_id` value in the response.
   `GET /api/v1/trust-management/principal-identities/<principal-identity-id>`
2. Delete the principal identity.
   `DELETE /api/v1/trust-management/principal-identities/<principal-identity-id>`
3. Delete the certificate using the `certificate_id` value obtained in step 1.
   `DELETE /api/v1/trust-management/certificates/<certificate_id>`

For LDAP, you configure user groups to user roles mapping information; the groups correspond to the user groups specified in the Active Directory (AD). To grant user permissions on NSX, add that user to the mapped group in AD. Starting in NSX 4.2, a single LDAP identity source can add up to 20 groups to NSX. An error will result if more than 20 groups are attempted.

1. With admin privileges, log in to NSX Manager.
2. Select **System** > **User Management**.
3. To assign roles to users, select **Add** > **Role Assignment for vIDM**.
   a) Select a user or user group.
   b) Select a role.
   c) Click **Save**.
4. To add a principal identity, select **Add** > **Principal Identity with Role**.
   a) Enter a name for the principal identity.
   b) Select a role.
   c) Enter a node ID.
   d) Enter a certificate in PEM format.
   e) Click **Save**.
5. To add a role assignment for LDAP select **Add** > **Role Assignment for LDAP**.
   a) Select a domain.
   b) Enter the first few characters of the user's name, login ID, or a group name to search the LDAP directory, then select a user or group from the list that appears.
   c) Select a role.
   d) Click **Save**.
6. If the certificate for the principal identity expires, perform the following steps. Do not use this procedure to replace Local Manager or Global Manager principal identity certificates. Instead, to replace those certificates refer to Replace NSX Certificates Using API for details.
   a) Import a new certificate and note the certificate's ID. See Import a Self-signed Certificate for NSX.
   b) Call the following API to get the ID of the principal identity.
      `GET https://<nsx-mgr>/api/v1/trust-management/principal-identities`
   c) Call the following API to update the principal identity's certificate. You must provide the imported certificate's ID and the principal identity user's ID.
      For example,

```
POST https://<nsx-mgr>/api/v1/trust-management/principal-identities?action=update_certificate
{
    "principal_identity_id": "ebd3032d-728e-44d4-9914-d4f81c9972cb",
    "certificate_id" : "abd3032d-728e-44d4-9914-d4f81c9972cc"
```

```
}
```

Logging in to NSX Manager

NSX Manager supports single sign (SSO) on several different authentication types and allows users to easily log in.

Starting with NSX 4.1.2, the OpenID Connect (OIDC) OAuth 2.0 workflow involves redirecting a user to an external identity provider (IDP).

This topic describes the logging in and out of NSX supported by these authentication types:

- Local authentication: Users authenticate using the PAM stack on the Unified Appliance.
- LDAP: Users authenticate using the LDAP service (either Microsoft Active Directory or OpenLDAP).
- Workspace ONE Access (Access): Users authenticate using a Access instance. Access is now known as VCF Identity Broker, but the functionality remains the same. Usually the user configures a Access connector that interfaces with their Active Directory.
- OIDC: Users authenticate using a simple security layer built on top of the OAuth2 protocol. Workspace ONE Access, VCF Identity Broker, and the VMware Cloud Services Platform (CSP) all implement OIDC.

The following table describes the login behavior for the supported authentication methods.

When a user accesses NSX and their browser does not have a valid session cookie, the behavior that occurs depends on the configured authentication methods and the deployment type.

Authentication Methods Configured	Deployment Type(s)	Action	Notes
None (local auth only)	On-premise, VMC/AWS in PCI mode	Display local login page.	
LDAP + local auth	On-premise	Display local login page. Users append "@domain" to their login name to specify an LDAP account.	
Access + local auth	On-premise	Redirect to Access login page.	From your browser, log in to an NSX Manager at https://nsx-manager-ip-address/login.jsp?idp=local to log in as a local user. For backwards compatibility, replace with "login.jsp?local=true" to display the local login page.
LDAP + Access + local auth	On-premise	Redirect to Access login page.	From your browser, log in to an NSX Manager at https://nsx-manager-ip-address/login.jsp?idp=local to log in as a local user or an LDAP user. For backwards compatibility, replace with "login.jsp?local=true " to display the local login page.
OIDC + local auth	On-premise	Display welcome page.	User selects an account name or the sign in using another account which redirects to the OIDC provider. Includes sign in with local link. If user selects, from the browser, log in to an NSX Manager at https://nsx-manager-ip-address/login.jsp?idp=local to display the local login page.

Authentication Methods Configured	Deployment Type(s)	Action	Notes
OIDC only (no local auth available to user)	On-premise	Display welcome page.	No sign in with local link. User selects an account name or the sign in using another account which redirects to the OIDC provider.
OIDC + LDAP + local auth	On-premise	Display welcome page.	Includes a sign in with local or LDAP link. From your browser, log in to an NSX Manager at https://nsx-manager-ip-address/login.jsp?idp=local to display the local login page.

Role-Based Access Control

With role-based access control (RBAC), you can restrict system access to authorized users. Users are assigned roles and each role has specific permissions.

To view the built-in and custom roles and their associated permissions, navigate to **System** > **User Management** > **Roles** and expand the row to view details. You can view permissions of all categories by expanding the role's permissions details.

After you have assigned an Active Directory (AD) user a role, if the username is changed on the AD server, you need to assign the role again using the new username.

> **Note:**
> For Security Intelligence RBAC information, see the *Using and Managing Security Intelligence* documentation.

Roles and Permissions

There are four types of permissions. Included in the list are the abbreviations for the permissions that are used in the Roles and Permissions table.

- Full access (FA) - All permissions including Create, Read, Update, and Delete (CRUD)
- Execute (E) - Includes Read and Update
- Read (R)
- None (N)

NSX has the following built-in roles. Role names in the UI can be different in the API.In NSX, if you have permission, you can clone an existing role, add a new role, edit newly created roles, or delete newly created roles.

The following table shows the permissions that each built-in role has for different operations. Also included in the list are the abbreviations for the roles that are used.

- Auditor (A)
- Cloud Admin (CA)(Available in the Cloud environment only)
- Cloud Operator (CO) (Available in the Cloud environment only)
- Cloud Partner Admin (CPA)
- Enterprise Admin (EA)
- GI (Guest Introspection) Partner Administrator (GIPA)
- LB (Load Balancer) Admin (LBA)
- LB Operator (LBO)
- Network Admin (NA)
- Network Operator (NO)
- NETX (Network Introspection) Partner Administrator (NXPA)
- Project Admin (PA) (Refer to note)
- Security Admin (SA)
- Security Operator (SO)

- Support Bundle Collector (SBC)
- VPN Admin (VPNA)
- Cloud Partner Auditor (CPAu)

The Roles and Permissions tables do not include the Project Admin role. The Project Admin role has full access to all configurations in a project. The Network Admin, Network Operator, Security Admin, and Security Operator roles in a project have RBAC permissions only within the scope of the project, and not for the entire NSX system.

RBAC permissions

Table 1072: Roles and Permissions

Operation	EA	A	NA	NO	SA	SO	CA	CO	CPA	LBA	LBO	VPNA	GIPA	NXPA	SBC	CPAu
Networking > Tier-0 Gateways	FA	R	FA	R	R	R	FA	R	R	R	R	R	R	R	N	R
Networking > Tier-1 Gateways	FA	R	FA	R	R	R	FA	R	FA	R	R	R	R	R	N	R
Networking > Network Interface	FA	R	FA	R	R	R	FA	R	FA	R	R	R	R	R	N	R
Networking > Network Static Routes	FA	R	FA	R	R	R	FA	R	FA	R	R	R	R	R	N	R
Networking > Locale Services	FA	R	FA	R	R	R	FA	R	FA	R	R	R	R	R	N	R
Networking > Segments	FA	R	FA	R	R	R	FA	R	FA	R	R	R	R	R	N	R
Networking > Segments > Segment Profiles	FA	R	FA	R	R	R	FA	R	FA	R	R	R	R	R	N	R
Networking > IP Address Pools	FA	R	FA	R	R	R	FA	R	FA	R	R	N	N	N	N	R
Networking > DNS	FA	R	FA	FA	R	R	FA	R	FA	R	R	N	N	N	N	R
Networking > DHCP	FA	R	FA	R	R	R	FA	R	FA	R	R	N	N	N	N	R
Networking > Load Balancing	FA	R	N	N	R	N	FA	R	FA	FA	R	N	N	N	N	R
Networking > NAT	FA	R	FA	R	FA	R	FA	R	FA	R	R	N	N	N	N	R

Operation	EA	A	NA	NO	SA	SO	CA	CO	CPA	LBA	LBO	VPNA	GIPA	NXPA	SBC	CPAu
Networking > VPN	FA	R	FA	R	FA	R	FA	R	FA	N	N	FA	N	N	N	R
Networking > IPv6 Profiles	FA	R	FA	R	R	R	FA	R	FA	R	R	N	N	N	N	R
Security > Distributed Firewall	FA	R	R	R	FA	R	FA	R	FA	R	R	R	R	R	N	R
Security > Gateway Firewall	FA	R	R	R	FA	R	FA	R	FA	N	N	N	N	FA	N	R
Security > Identity Firewall AD	FA	R	FA	R	FA	FA	FA	R	FA	R	R	R	R	R	N	R
Security > Network Introspection	FA	R	R	R	FA	R	FA	R	N	N	N	N	N	FA	N	N
Security > Endpoint Protection Rules	FA	R	R	R	FA	R	FA	R	R	N	N	N	FA	N	N	R
Inventory > Context Profiles	FA	R	R	R	FA	R	R	R	FA	R	R	R	R	R	N	R
Inventory > Virtual Machines	R	R	R	R	R	R	R	R	FA	R	R	R	R	R	N	R
Inventory > Virtual Machines > Create & Assign Tags to VM	FA	R	R	R	FA	R	FA	R	FA	R	R	R	FA	FA	N	R
Inventory > Containers	FA	R	R	R	R	R	N	N	R	N	N	N	N	N	N	R
Inventory > Physical Servers	FA	R	R	R	R	R	R	R	R	R	R	N	N	N	N	R
Plan & Troubleshoot > Port Mirroring	FA	R	FA	R	R	R	FA	R	FA	N	N	N	N	N	N	R
Plan & Troubleshoot > Port Mirroring Binding	FA	R	FA	FA	R	R	FA	R	FA	R	R	R	R	R	N	R

Operation	EA	A	NA	NO	SA	SO	CA	CO	CPA	LBA	LBO	VPNA	GIPA	NXPA	SBC	CPAu
Plan & Troubleshoot > Monitoring Profile Binding	FA	R	FA	FA	R	R	FA	R	FA	R	R	R	R	R	N	R
Plan & Troubleshoot > IPFIX > Firewall IPFIX Profiles	FA	R	FA	R	FA	R	FA	R	FA	R	R	R	R	R	N	R
Plan & Troubleshoot > IPFIX > Switch IPFIX Profiles	FA	R	FA	R	R	R	FA	R	FA	R	R	R	R	R	N	R
Plan & Troubleshoot > Collectors	FA	R	FA	R	R	R	FA	R	FA	R	R	R	R	R	N	R
Plan & Troubleshoot > Traceflow	FA	FA	FA	FA	FA	FA	FA	FA	FA	FA	FA	N	N	N	N	FA
System > Fabric > Hosts > Clusters	FA	R	R	R	R	R	R	R	R	R	R	N	N	N	R	R
System > Fabric > Hosts > Standalone	FA	R	R	R	R	R	R	R	R	R	R	N	N	N	R	R
System > Fabric > Hosts > Unmanaged	FA	R	R	R	R	R	R	R	R	R	R	N	N	N	R	R
System > Fabric > Hosts > Transport Node Profile	FA	R	R	R	R	R	R	R	R	R	R	N	N	N	R	R
System > Fabric > Hosts	FA	R	R	R	R	R	R	R	R	N	N	N	N	N	N	R
System > Fabric > Nodes	FA	R	FA	R	FA	R	R	R	R	R	R	N	N	N	N	R

Operation	EA	A	NA	NO	SA	SO	CA	CO	CPA	LBA	LBO	VPNA	GIPA	NXPA	SBC	CPAu
System > Fabric > Nodes > Edge Transport Nodes	FA	R	R	R	R	R	R	R	R	N	N	N	N	N	N	R
System > Fabric > Nodes > Edge Clusters	FA	R	FA	R	R	R	R	R	R	N	N	N	N	N	N	R
System > Fabric > Nodes > Container Clusters	FA	R	FA	R	R	R	N	N	R	R	R	N	N	N	N	R
System > Fabric > Nodes > Tunnels	R	R	R	R	R	R	R	R	R	R	R	N	N	N	N	R
System > Fabric > Profiles > Uplink Profiles	FA	R	R	R	R	R	R	R	R	R	R	N	N	N	N	R
System > Fabric > Profiles > Edge Cluster Profiles	FA	R	FA	R	R	R	R	R	R	R	R	N	N	N	N	R
System > Fabric > Profiles > Configuration	FA	R	N	N	N	N	R	R	R	N	N	N	N	N	N	R
System > Fabric > Profiles > Node Profiles	FA	R	R	R	R	R	R	R	R	R	R	N	N	N	N	R
System > Fabric > Transport Zones > Add Zones	FA	R	R	R	R	R	R	R	R	R	R	N	N	N	N	R
System > Fabric > Transport Zones > Health Configuration	FA	R	R	R	R	R	R	R	R	R	R	N	N	N	N	R

Operation	EA	A	NA	NO	SA	SO	CA	CO	CPA	LBA	LBO	VPNA	GIPA	NXPA	SBC	CPAu
System > Fabric > Compute Managers	FA	R	R	R	R	R	R	R	R	N	N	N	R	R	N	R
System > Fabric > Settings	FA	N	N	N	N	N	N	N	R	N	N	N	N	N	N	R
System > Certificates	FA	R	N	N	FA	R	N	N	FA	FA	R	FA	N	N	N	R
System > Service Deployments > Service Instances	FA	R	R	R	FA	R	FA	R	R	N	N	N	FA	FA	N	N
System > Support Bundle	FA	N	N	N	N	N	N	N	N	N	N	N	N	N	FA	N
System > Backup	FA	R	N	N	N	N	N	N	R	N	N	N	N	N	N	R
System > Restore	FA	R	N	N	N	N	N	N	R	N	N	N	N	N	N	R
System > Upgrade	FA	R	R	R	R	R	N	N	R	N	N	N	N	N	N	R
System > Migrate	FA	N	N	N	N	N	N	N	R	N	N	N	N	N	FA	R
System > User Mgt > User Role Assignments	FA	R	N	N	N	N	FA	R	FA	N	N	N	N	N	N	R
System > Local Users	FA	R	N	N	N	N	N	N	R	N	N	N	N	N	N	R
System > Roles	FA	R	FA	R	FA	FA	FA	R	FA	R	R	R	R	R	N	R
System > Authentication Providers	FA	R	FA	R	FA	FA	R	R	FA	R	R	R	R	R	N	R
System > Licenses	FA	R	R	R	R	R	N	N	N	N	N	N	N	N	N	N
System > System Administration	FA	R	R	R	R	R	R	R	R	N	N	N	N	N	N	R
Custom Dashboard Configuration	FA	R	R	R	R	R	FA	R	R	R	R	R	R	R	N	R

Create or Manage Custom Roles

Extend the RBAC capabilities provided by NSX and create custom roles that suit your operational requirements. You can clone an existing role and customize it or you can create a role afresh. You can also edit and delete user-created roles.

- Only an Enterprise Administrator can assign the role management feature's permission to a custom role. An Enterprise Administrator can create a custom role to delegate further custom role creation and user role assignment.
- A user assigned with a custom role can only create other custom roles with equal or lower permission sets. A user with a custom role cannot create or assign roles with permissions higher than their own.
- A user assigned with a custom role cannot modify or delete the role assigned to them.

Note:
Custom roles are not supported on Global Manager (Federation).

1. With admin privileges, log in to NSX Manager.
2. Select **System** > **Users and Roles** > **Roles**.
3. Clone an existing role or create one.
   - To clone a role, click
     
     for that role and select **Clone**. Enter a name for the cloned role and specify permissions as per your operational requirements.
   - To create a role, click **Add Role**. Enter a name for the role and update the permissions as per your operational requirements.

     Note:

     Based on the features you select, NSX might suggest additional permissions for the new role definition to be valid. Review the recommendations and click **Apply**.

     When creating a custom role, NSX checks for feature interdependencies. The interdependency check ensures that the user has a minimum of read access to the additional features that are required for the role to be valid.

     For example, if a user creates a role with full access permissions to Gateway Firewall and the **None** access permission to the Networking Gateway feature, the role is invalid. NSX then suggests that the user assign at least read access to the additionally required Networking Gateway feature.
4. (Optional) Edit or delete a user-created role.
   - To edit a user-created role, for example, if you wanted to extend access, click
     
     for that role and select **Edit**. Change the role name, description, and permissions as per your operational requirements.
   - To delete a user-created role, for example, if it was for temporary access, click
     
     for that role and select **Delete**.

Logging User Account Changes

Changes to a user's role assignment are automatically written to syslog and the audit log.

For more information about syslog and the audit log, see Log Messages and Error Codes.

An example of a log message when assigning a role to a vIDM user:

```
2020-09-24T16:05:51.244Z nsxmanager-14663974-1-CertKB-FS NSX 5519 - [nsx@6876 audit="true" comp="nsx-
manager" entId="e3c2af75-9d0f-4020-90cc-f2f00d6af255" level="INFO" reqId="b27711c6-0590-4b39-
b8b6-f0980a0597f0" subcomp="policy" update="true" username="admin"] UserName="admin",
```

```
 ModuleName="AAA", Operation="CreateRoleBinding", Operation status="success", New
 value=[{"name":"test_AU@idfw.local","type":"remote_user","identity_source_type":"VIDM","roles":
[{"role":"auditor"}],"id":"bba634c9-cfbd-4806-a831-e63ec195e1f9","_protection":"UNKNOWN"}]
```

An example of a log message when updating the role of a vIDM user:

```
2020-09-24T16:12:51.217Z nsxmanager-14663974-1-CertKB-FS NSX 5519 - [nsx@6876 audit="true" comp="nsx-
manager" entId="e3c2af75-9d0f-4020-90cc-f2f00d6af255" level="INFO" reqId="973faed4-f4b5-443d-
bd79-7d995c027183" subcomp="policy" update="true" username="admin"] UserName="admin", ModuleName="AAA",
 Operation="UpdateRoleBinding", Operation status="success", New value=["e3c2af75-9d0f-4020-90cc-
f2f00d6af255" {"name":"test_AU@idfw.local","type":"remote_user","identity_source_type":"VIDM","roles":
[{"role":"security_admin"}],"_protection":"UNKNOWN"}]
```

An example of a log message when assigning a role to an LDAP user:

```
2020-09-24T16:06:28.663Z nsxmanager-14663974-1-CertKB-FS NSX 5519 - [nsx@6876 audit="true" comp="nsx-
manager" entId="35e45569-6da6-4dcd-b4a1-75747cdd6cf8" level="INFO" reqId="db27f4ae-25a7-4482-
b3f4-49228d12960b" subcomp="policy" update="true" username="admin"] UserName="admin",
 ModuleName="AAA", Operation="CreateRoleBinding", Operation status="success", New
 value=[{"name":"skrasner@airius.com","type":"remote_user","identity_source_type":"LDAP","identity_source_id":"ldap","r
[{"role":"auditor"}],"id":"dd8d3675-c574-454b-975e-300b65462827","_protection":"UNKNOWN"}]
```

An example of a log message when updating the role of an LDAP user:

```
2020-09-24T16:12:37.449Z nsxmanager-14663974-1-CertKB-FS NSX 5519 - [nsx@6876 audit="true" comp="nsx-manager"
 entId="35e45569-6da6-4dcd-b4a1-75747cdd6cf8" level="INFO" reqId="d7cdd3de-75a1-4d29-9fea-27e1dda4b5e2"
 subcomp="policy" update="true" username="admin"] UserName="admin", ModuleName="AAA",
 Operation="UpdateRoleBinding", Operation status="success", New value=["35e45569-6da6-4dcd-
b4a1-75747cdd6cf8" {"name":"skrasner@airius.com","type":"remote_user","identity_source_type":"LDAP","identity_source_id
[{"role":"network_admin"}],"_protection":"UNKNOWN"}]
```

Certificates in NSX

After you install NSX, the manager nodes and cluster have vCenter certificate authority (CA) certificates.

If you are using NSX Federation, additional certificates are set up to establish trust between the Local Managers and Global Manager. If you are using TLS Inspection, a CA-signed security certificate is required. For details on TLS Inspection and certificates, see the VMware vDefend documentation.

To view details of all the certificates that are installed on the system, navigate to **System** > **Certificates**.

You can perform the following actions for certificates:

- Filter certificates based on their basic parameters (such as name or path) or based on predefined filters (such as expired certificates or used certificates)
- Import certificates
- Create a certificate signing request (CSR)
- Generate self-signed certificates
- Replace self-signed certificates
- Apply certificates to services
- Delete unused certificates
- Import a certificate revocation list (CRL)

To improve the security in the system, it is recommended that you replace the self-signed certificates with CA-signed certificates.

Starting with NSX 4.2, the Certificates page also displays a dashboard that provides a quick glance of total certificates, number of expired certificates, and total used and unused certificates in the system. Also, the following certificates have been consolidated:

- The APH, APH_TN, and CCP certificates have been consolidated into one.
- The API services and MGMT_CLUSTER (aka VIP) certificates have been consolidated into one.

To replace a consolidated certificate, you must follow certain considerations. For more information about these considerations, see Replace NSX Certificates from NSX Manager.

Types of Certificates in NSX

There are three categories of self-signed certificates in NSX.

- Platform Certificates
- NSX Services Certificates
- Principal Identity Certificates

Refer to the following sections for details on each certificate category. Other types of certificates might be present in NSX. Refer to that component or application documentation for details on their certificates.

Many certificates used to support RPC traffic (APH and CCP) have been combined to improve serviceability. As a result, each NSX Manager node has a total of three certificates. You can replace the self-signed internal certificates with CA-signed certificates using ECDSA-based certificates that get generated using AES 256 encryption. For a list of replaceable certificates, refer to Certificates for NSX and NSX Federation. For command details, see the *NSX API Guide*.

> **Note:**
> Though NSX supports secp256k1 keys for all certificates, do not use this key if your environment requires only FIPS-approved cryptographic keys.

Platform Certificates

After you install NSX, navigate to **System** > **Certificates** to view the platform certificates created by the system. By default, these self-signed X.509 RSA 2048/SHA256 certificates get used for internal communication within NSX and for external authentication when you use NSX Manager to access the APIs or the UI.

The internal certificates are not viewable or editable.

When you deploy NSX through the default VMware Cloud Foundation™ (VCF) workflow, the default NSX API and Cluster certificates get replaced with CA certificates signed by the VMware Certificate Authority (VMCA) from vCenter. The API and Cluster certificates might still display in the certificate list, but are not used. Replace the CA-signed certificates using the procedure in the *VCF Administration Guide*. After you perform the replacement, your NSX Manager stores in the UI contain the API and Cluster certificates, the VMCA CA certificates, and the signed certificates by the third-party organization. From then on, the NSX Manager uses the signed certificate from your organization.

Table 1073: Platform Certificates in NSX

Naming Convention in NSX Manager	Purpose	Replaceable?	Default Validity
`API (previously known as tomcat)`	This serves as a server certificate for the API/UI client reaching the NSX Manager HTTPS port (port 443).	Yes. See Replace NSX Certificates Using API	825 days
`Cluster (previously known as mp-cluster)`	This certificate serves as a server certificate for API/UI client reaching the HTTPS port of the cluster VIP (port 443) when a VIP is used for an NSX Manager cluster.	Yes. See Replace NSX Certificates Using API	825 days
Other certificates	Certificates specifically for other purposes including NSX Federation, Cluster Boot Manager (CBM), and Central Control Plane (CCP).	See Certificates for NSX and NSX Federation for details on self-signed certificates auto-configured for NSX and NSX Federation environments.	Varies

NSX Service Certificates

NSX service certificates are user-facing for services such as load balancer, VPN, and TLS Inspection. The policy API manages service certificates. Non-service certificates are used by the platform for tasks such as cluster management. The management plane (MP) or truststore APIs manage non-service certificates.

When adding service certificates using the policy API, the certificate gets sent to the MP/truststore API, but not with the relationship reversed.

NSX service certificates cannot be self signed. You must import them. See Importing Certificates for NSX for instructions.

You can generate a root certificate authority (CA) certificate and a private key based on RSA. CA certificates are able to sign other certificates.

You can use a certificate signing request (CSR) as an NSX service certificate if it is signed by a CA (a local CA or a public CA like Verisign). Once the CSR is signed, you can import that signed certificate into NSX Manager. A CSR can be generated on NSX Manager or outside of NSX Manager. The **Service Certificate** flag is deactivated for CSRs generated on NSX Manager. Therefore, the signed CSRs cannot be used as service certificates, but only as platform certificates.

Platform and NSX service certificates are stored separately within the system and certificates imported as NSX service certificate cannot be used for platform or the reverse.

Principal Identity (PI) Certificates

API requests use PI certificates. PI certificates are one of the NSX Manager authentication mechanisms. Any NSX Manager client can create and use a PI certificate. It is the preferred approach for machine-to-machine communication.

PI for Cloud Management Platforms (CMP), such as Openstack, uses X.509 certificates that are uploaded when onboarding a CMP as a client. For information on assigning roles to Principal Identity and replacing PI certificates, see Add a Role Assignment or Principal Identity

PI for NSX Federation uses X.509 platform certificates for the Local Manager and Global Manager appliances. See Certificates for NSX and NSX Federation for details on self-signed certificates auto-configured for NSX Federation.

Certificates for NSX and NSX Federation

The system creates certificates required for the communication between NSX appliances and external communication, including NSX Federation appliances. Read about the various certificate information.

> **Note:**
> In a federated NSX deployment, certificates originating from other locations will appear in the Certificates pane. Those certificates display with names starting with Site; for example, *Site certificate L=PA,ST=CA,C=US*, *Site certificate UID=369cd66c-...*, or with their UUID only *637a2ebf-84d1-4548-a0ba-51d9420672ff*. If those certificates are expiring, replace them on the originating site where their corresponding private key is stored. You can find that information in the **Used By** field on the UI or in the API. If you replace the certificates on any originating Global Manager or Local Manager, the system automatically synchronizes them across the federated deployment.

The following table reflects certificate details including the span of time that certificates are valid for new deployments only. New certificates are not generated during upgrades, so certificate validity dates reflect a previous NSX version's default certificate expiration date.

To replace existing self-signed certificates with CA-signed certificates, refer to details in Replace NSX Certificates Using API. To read about security compliance events, refer to the NSX Event Catalog.

Table 1074: Certificates for NSX Manager Nodes

Certificate Name	Purpose	Replaceable?	Default Validity
MGMT_CLUSTER site <id>	API server certificate for NSX Manager node	Yes, use service_type=API.	825 days

Certificate Name	Purpose	Replaceable?	Default Validity
	API server certificate used by VIP	Yes, use service_type=MGMT_CLUSTER.	
LocalManager	Local Manager server certificate	Yes, use service_type=LOCAL_MANAGER.	10 years
CBM_CORFU node <id>	Corfu server certificate	Yes, use service_type=CBM_CORFU.	100 years
CBM_CLUSTER_MANAGER node <id>	Corfu client certificate	Yes, use service_type=CBM_CLUSTER_MANAGER.	100 years
APH node <id>	Control Configuration Plane certificate (CCP-APH) for communication with transport nodes	Yes, use service_type=CCP.	10 years
	Appliance Proxy hub certificate (APH - Transport Node) for transport node and intra-cluster communication	Yes, use service_type=APH_TN.	
	Appliance Proxy hub server public key and Asynchronous Replicator (APH - Asynchronous Replicator) for cross-communication in NSX Federation	Yes, use service_type=APH.	

Certificates for NSX Federation Communication

By default, the Global Manager uses self-signed certificates for communicating with internal components, registered Local Managers, and for authentication for NSX Manager UI or APIs.

You can view the external (UI/API) and inter-site certificates in NSX Manager. The internal certificates are not viewable or editable.

> **Note:**
> Do not enable Local Manager external VIP before you register the Local Manager on the Global Manager. When NSX Federation and PKS need to be used on the same Local Manager, complete the PKS tasks to create an external VIP and change the Local Manager certificate **before** you register the Local Manager on Global Manager.

Certificates for Global Managers and Local Managers

After you add a Local Manager into the Global Manager, a trust gets established by exchanging certificates between Local Manager and Global Manager. These certificates also get copied into each of the sites registered with the Global Manager. The certificate used to establish trust with the Global Manager gets generated only when the Local Manager registers with the Global Manager. That same certificate gets deleted if the Local Manager moves out of the NSX Federation environment.

See the following table for a list of all the NSX Federation specific certificates created for each appliance and the certificates these appliances exchange with each other.

Table 1075: Certificates for Global Managers and Local Managers

Naming Convention in the Global Manager or Local Manager	Purpose	Replaceable?	Default Validity
The following are certificates specific to each NSX Federation **appliance.**			
`APH-AR certificate`	• For the Global Manager and each Local Manager. • Used for inter-site communication using the AR channel (Async-Replicator channel).	Yes, use service_type=APH. See Replace NSX Certificates Using API.	10 years
`GlobalManager`	• For the Global Manager. • PI certificate for the Global Manager.	Yes, use service_type=GLOBAL_MANAGER. Refer to Replace NSX Certificates Using API.	10 years
`Cluster certificate`	• For the Global Manager and each Local Manager. • Used for UI/API communication with the VIP of the Global Manager or Local Manager cluster.	Yes, use service_type=MGMT_CLUSTER. Refer to Replace NSX Certificates Using API.	825 days
`API certificate`	• For the Global Manager and each Local Manager. • Used for UI/API communication with individual Global Manager and Local Manager nodes for each of the locations added to the Global Manager.	Yes, use service_type=API. See Replace NSX Certificates Using API.	825 days
`LocalManager`	• Generates only when Local Manager servers are in the NSX Federation environment. The certificate gets deleted if the Local Manager moves out of the NSX Federation environment. • PI certificate for this specific Local Manager.	Yes, use service_type=LOCAL_MANAGER. See Replace NSX Certificates Using API.	10 years
The LM and the GM share their Cluster, API, and APH-AR certificates between them. If a certificate is CA-signed, the CA is synchronized, but that certificate is not.			

Principal Identity (PI) Users for NSX Federation

After you add a Local Manager to the Global Manager, the following PI users with corresponding roles get created.

Table 1076: Principal Identity (PI) Users Created for NSX Federation

NSX Federation Appliance	PI User Name	PI User Role
Global Manager	`LocalManagerIdentity` One for each Local Manager registered with this Global Manager.	Auditor
Local Manager	`GlobalManagerIdentity`	Enterprise Admin

NSX Federation Appliance	PI User Name	PI User Role

	`LocalManagerIdentity`
	One for each Local Manager registered with the same Global Manager. To get a list of all the Local Manager PI users because they are not visible in the UI, enter the following API: command
	`GET https://<local-mgr>/api/v1/trust-management/principal-identities`
		Auditor

Create a Certificate Signing Request File for NSX

Certificate signing request (CSR) is an encrypted text that contains specific information such as, organization name, common name, locality, and country. You send the CSR file to a certificate authority (CA) to apply for a digital identity certificate.

To fill out the CSR file details, gather the information. You must know the FQDN of the server and the organizational unit, organization, city, state, and country.

1. With admin privileges, log in to NSX Manager.
2. Select **System** > **Certificates**.
3. Click the **CSRs** tab.
4. Run the action for generating a self-signed certificate.
   a) In version 9.1.1 and later, click **Generate CSR**.
   b) In version 9.1.0, click **Generate CSR** and select **Generate CSR** or **Generate CA CSR** from the dropdown menu.
5. Complete the file details.

Option	Description
Name	Assign a name for your certificate.
(In version 9.1.1 and later) **Signing CA Certificate**	Set to **Yes** to use the certificate as a CA certificate that can sign other certificates.
Common Name	Enter the fully qualified domain name (FQDN) of your server. For example, test.vmware.com.
Organization Unit	Enter the department in your organization that is handling this certificate For example, IT department.
Organization Name	Enter your organization name with applicable suffixes. For example, VMware Inc.
Locality	Add the city in which your organization is located. For example, Palo Alto.
State	Add the state in which your organization is located. For example, California.
Country/Region	Add your organization location. For example, United States (US).
Algorithm	Set the encryption algorithm for your certificate. • RSA encryption - used for digital signatures and encryption of the message.

Option	Description
	• ECDSA (Elliptic Curve Digital Signature Algorithm) encryption - used for EAL4+ compliance. The performance of this algorithm is more efficient than RSA algorithm.
Key Size	Set the key bits size of the encryption algorithm. • For RSA, the default value, 2048, is adequate unless you specifically need a different key size. Other supported sizes are 3072 and 4096. Many CAs require a minimum value of 2048. Larger key sizes are more secure but have a greater impact on performance. • ECDSA typically uses the Advanced Encryption Standard with 256 bit key in Galois/Counter mode (AES 256 GCM). Other key sizes include 384 and 521 bits.
Subject Alternative Name - DNS Names	Specify the Subject Alternative Names (SANs) secured by the certificate, using DNS names with optional wildcards (for instance, `*.example.com`).
Subject Alternative Name - IP Addresses	Specify the Subject Alternative Names (SANs) secured by the certificate, using IP addresses.
Description	Enter specific details to help you identify this certificate at a later date.

6. Click **Save**.
   A custom CSR appears as a link.
7. Select the CSR then click **Actions** to select one of the following options:
   - **Delete**
   - **Import Certificate for CSR** (available in version 9.1.0)
   - **Self Sign Certificate for CSR**
   - **Download CSR PEM**
     If you selected **Download CSR PEM**, you can save the CSR PEM file for your records and CA submission. Use the contents of the CSR file to submit a certificate request to the CA in accordance with the CA enrollment process. For the other two options, refer to topics Import a Certificate for a CSR for NSX and Create a Self-Signed Certificate for NSX.

The CA creates a server certificate based on the information in the CSR file, signs it with its private key, and sends you the certificate. The CA also sends you a root CA certificate.

Creating Self-signed Certificates for NSX

You can create self-signed service or non-service certificates in NSX Manager. You can also import a signed certificate for NSX-generated CSRs.

Create a Self-Signed Certificate for NSX

You can create a self-signed service or non-service certificate. However, using a self-signed certificate is less secure than using a trusted certificate.

Verify that a CSR is available. See Create a Certificate Signing Request File for NSX.

When you use a self-signed certificate the client user receives a warning message such as, `Invalid Security Certificate`. The client user must then accept the self-signed certificate when first connecting to the server in order to proceed. Allowing client users to select this option provides reduced security than other authorization methods.

1. With admin privileges, log in to NSX Manager.
2. Select **System** > **Certificates**.
3. Run the action for generating a self-signed certificate.
   - In version 9.1.1 and later, click **All Certificates**, and then click the **Generate Certificate** button.
   - In version 9.1.0, click the **CSRs** tab. From your selected CSR, click  and select **Self Sign Certificate for CSR**.

     Note:

     If you have a self-signed CA CSR, NSX Manager always creates a CA CSR.
4. Specify the required certificate details, such as the name and common name for the certificate, among other details.
5. In version 9.1.1 and later, select an option for **Use for**.
   a) Select **Services** to use this certificate for services such as a load balancer and VPN.
   b) Select **NSX manager appliance nodes** to use this certificate for the NSX Manager nodes.
6. In version 9.1.1 and later, to use the certificate as a CA certificate that can sign other certificates, set the **Signing CA Certificate** toggle to **Yes**. This option is available when you have specified to use the certificate for services.
7. In version 9.1.0, choose your **Service Certificate** type.
   a) Toggle the **Service Certificate** button to **Yes** to use this certificate for services such as load balancer, VPN, or TLS Inspection. If you are creating a self-signed CA certificate, Yes is the only choice.
   b) Toggle the **Service Certificate** button to **No** to use this certificate with NSX Manager appliance nodes.
8. Enter the number of days the self-signed certificate is valid.

   The default is 825 days. Even if you change this value for previously generated self-signed certificate, the default value is displayed every time you generate a new certificate.
9. Specify the Subject Alternative Names (SANs) secured by the certificate. You can specify SANs using DNS names with optional wildcards (for instance, `*.example.com`), or using IP addresses.
10. Click **Save**.

The self-signed certificate appears in the **Certificates** tab.

Import a Certificate for a CSR for NSX

You can import a signed certificate for an NSX generated CSR (certificate signing request). You can also use this imported certificate with services such as Load Balancer, VPN, and TLS Inspection. This page provides the steps to import a signed certificate for NSX generated CSR.

- Verify that a CSR is available. See Create a Certificate Signing Request File for NSX.
- NSX generated CSR was used as a CSR for signed certificate.

A self-signed certificate acts as a certificate as well as CA. It is not required to be signed from any external CA, whereas CSR is a certificate signing request that cannot act as CA and must be signed by external CA. There is no support for a self-signed certificate for load balancer.

When you use a self-signed certificate, the client user receives a warning message such as `Invalid Security Certificate`. The client user must then accept the self-signed certificate when first connecting to the server in order to proceed. Allowing client users to select this option provides reduced security than other authorization methods.

1. With admin privileges, log in to NSX Manager.
2. Select **System** > **Certificates**.
3. In version 9.1.1 and later, import the certificate.
   a) Click the **Certificates** tab, and then click **All Certificates**.
   b) Select **Import** and specify the certificate details.
      - For **Import Option**, select **Import certificate generated using NSX CSR**.
      - For **Use for**, specify whether to use the certificate for services or NSX Manager appliance nodes.
      - Browse to the signed certificate file on your computer and add the file.
   c) Click **Save**.
4. In version 9.1.0, import the certificate.
   a) Click the **CSRs** tab.
   b) From a CSR, click  and select **Import Certificate for CSR**.
   c) Browse to the signed certificate file on your computer and add the file.
   d) Choose your **Service Certificate** type.
      - To use this certificate for services such as load balancer, VPN, or TLS Inspection, toggle the **Service Certificate** button to **Yes**.
      - To use this certificate with NSX Manager appliance nodes, toggle the **Service Certificate** button to **No**.
   e) Click **Save**.

The signed certificate appears in the **Certificates** tab.

Importing Certificates for NSX

You can import self-signed or CA-signed certificates for platform or services.

You can also import CA certificates for services such as Load Balancer.

Import a Self-signed Certificate for NSX

You can import a certificate with a private key to replace the default self-signed certificate, after activation.

- Verify that a certificate is available.
- The server certificate must contain the Basic Constraints extension `basicConstraints = cA:FALSE`.
- Ensure that the certificate is signed with a supported algorithm.

You can import self-signed or CA-signed certificates for platform or services using this procedure with the following exceptions:

- A CSR generated on NSX Manager which is self-signed cannot be used as a service certificate, such as the Load Balancer service. If you want to import a CA certificate for the Load Balancer service, see Import a CA Certificate for NSX.
- On a stand-by Global Manager, the UI import operation is deactivated. Use the following REST API command to complete the import of a platform certificate on the standby GM. The display name is optional, but if not used, ensure the last attribute does not contain a trailing comma. Be sure to replace any end of line character with \n. The entire certificate should be provided in a single line. In order to replace newline characters with \n, you can use this command on UNIX based systems to convert each .pem (certificate and key) file to a value that can be passed in a JSON string to the NSX API: `awk 'NF {sub(/\r/, ""); printf "%s\\n",$0;}' certificate-name.pem`. The new format places all the certificate information on a single line with embedded newline characters.
  For example:

```
POST https://<nsx-mgr>/api/v1/trust-management/certificates?action=import
{
    "display_name": "cert_sample",
```

```
    "pem_encoded": "-----BEGIN CERTIFICATE-----
\nMIIC1CCDAbygAwIBAgIUMd1fGNGnvYKtilon2UMBP4rqRAowDQYJKoZIhvcNAQEL
\nBQAwDzENMAsGA1UEAwwETVlDQTAeFw0yMzA5MjYxODMxMzVaFw0zMzA5MjMxODMx
\nMzVaMBYxFDASBgNVBAMMC2NlcnRfc2FtcGxlMIIBIjANBgkqhkiG9w0BAQEFAAOC\nAQ8AMIIBCgKCAQEAzMDsp1EGFPjus/
xnHmacPJYVP0N8iQMb3W8TFFQC5jxdjNzi\ncMIb1YgpI+s3LJoyYCdZKeMcCWDwtgQXMTy9FYJCHKyt86CF0br9U9q9iC+NX93X\n
+/wrWtXY89ESt0NOgj22sKI49EQT9bd0dNWupxapCb98Dyztk0cetIHa7ia1q7un\nXMZ7dofwuWUEUlT8qpyXF84N6bhWQSrXRyeQ
+oZrsq3sAyfnKzbfcs0T3sztWn9M\nR7h8iPkjpJjVV5z1ghAgIDKFXG8RVU8fLgX5srtYV2Ij1II0qYwe/
yGBfj7xsemB\n2lGGPotlbwUE5oPFISJvG9qLOoNKVLvBrxuNnQIDAQABoyEwHzAdBgNVHSUEFjAU
\nBggrBgEFBQcDAQYIKwYBBQUHAwIwDQYJKoZIhvcNAQELBQADggEBAAqPfZWNzG/b
\nBhtN2gDjr0LplfC0yi8K6Ep3exECE5UOUJvHubko4Z6eCZFT8XSrAa6eZQEVe3O3\nwFvpdedCiEpI/
IaFhpRUQDubJMPao7t4Uohz3k3ONMGbIci8dVUcQRQlmxFmx3wf\n0/33fy3b1zIOXqooQF3qUlpjms/
RQOdD80dSlMze8WI7yz9LZt9Zc+sr8ePRi4Xy\ntudO6EYTiWm3CC5BxDDjKpkFCACFRT4zr5HsomHsFeo4hGIHl2zN0+JoGrdrWcta
\nxdl5aQYy79vIMgvz696EKUGePEpJjpyP/wlwzmIY3RvXRKThuVXvg20gi365x8+J\niKbzpCGe0P0=\n-----END CERTIFICATE-----
\n-----BEGIN CERTIFICATE-----\nMIIC/zCCAeegAwIBAgIUUlHXcczsdMpei1ThgeQYpvgzaxMwDQYJKoZIhvcNAQEL
\nBQAwDzENMAsGA1UEAwwETVlDQTAeFw0yMzA5MjYxODI2NDZaFw0zMzA5MjMxODI2\nNDZaMA8xDTALBgNVBAMMBE1ZQ0EwggEiMA0GCSqGSIb3DQEB
\nAoIBAQDbr78t32TUl1qTcDGvVQhiUkktntPO/5/
FRDSIjy9qyNGDrcICDAYzOe79\nceXpOzfUStacEeTXse89q1MJz4ykaU2g6EUN2E4sfoP4KznBlObLHnnlxD482DL4\nbuMA8qCe0soUsGE6uoeFHnS
+NRI3GtJe1MM134JQ/TSNZTv+d93nB4bS2nSK7\nA1fFDRSuj8Ey7a1im8JgykL9ahJ6yxrpk8juEJwII04nHfAG102/8/
YKEZyPWcPX\nYvLZEt/lBVxRPplWfbNIo3zfA09fzb4RMaOSsyBbqTBseL/4fxlnkeu1Rii3ZwcQ
\nL4Wr6mKR1YCievsuXdLK5pWUH+BtAgMBAAGjUzBRMB0GA1UdDgQWBBTnYafa1EXn
\nNPIqTkIO82kdamjDgTAfBgNVHSMEGDAWgBTnYafa1EXnNPIqTkIO82kdamjDgTAP\nBgNVHRMBAf8EBTADAQH/
MA0GCSqGSIb3DQEBCwUAA4IBAQC2Ef
+CPICTdWEKW3e6\nwaObe4Y85CS2wfSBRFvt0yCAUF8yysr3kQx85wdhfDfvidQdrgQIkDKe83J61r5l
\n238wFo9O10RpFWl1csY4hZ19geeTW3L8tABp+f1or1vsAogfVtcaZwmqz/LEaZ0r
\n4JdONE9gq40RgX5R9GPD04k3hKr6HoNHHnBssmNHgo8pLKRv04mx0yQyn45lKvet\ngcInI9j8YLsXGHdeiZ/zXKUgKQdicBw79K/
mQCpgkpaEi3K9mFUFCUU9CiWxiy62\nSN2/SEuOWlb7Kq8VwJUfUn3lKoY9sofr9zsSsh5lhQOKb1uguo8xUF8v6iLuDAjr\n9bcn
\n-----END CERTIFICATE-----\n",
    "private_key": "-----BEGIN RSA PRIVATE KEY-----\nMIIEowIBAAKCAQEAzNDsp1EGFPjus/
xnHmacPJYVP0N8iQMb3W8TFFQC5jxdjNzi\ncMIb1YgpI+s3LJoyYCdZKeMcCWDwtgQXMTy9FYJCHKyt86CF0br9U9q9iC+NX93X\n
+/wrWtXY89ESt0NOgj22sKI49EQT9bd0dNWupxapCb98Dyztk0cetIHa7ia1q7un\nXMZ7dofwuWUEUlT8qpyXF84N6bhWQSrXRyeQ
+oZrsq3sAyfnKzbfcs0T3sztWn9M\nR7h8iPkjpJjVV5z1ghAgIDKFXG8RVU8fLgX5srtYV2Ij1II0qYwe/
yGBfj7xsemB\n2lGGPotlbwUE5oPFISJvG9qLOoNKVLvBrxuNnQIDAQABAoIBAQCE8JH2xIWVYlbh
\np3RwaaDxOWTMMY4PC2SxLegOX8mOIQ2AYv3mxjD6QDCt8I9fNzKT+ZhLuPhAIp/H
\nHfrM7im6aFtycK90qfmYxbarFi/O10kMQGZ2ZjDkBkqZa1qigGHd8CHIp1shRX5M
\nIHPNU9vVAsJ34Mq0s7AA2sFV46X4zyEqHKLi1qVcsj68XJCrKJPTzOXiZWOHL8e0\nx4B8mGKbnWNmrq6styyYi9rzUnucoKL459YkaF/
MEBpou3wvhprkR5Ufr4eNo0YV\nr0KfcEjxZqVT2o6r59+gSZQiCHael2MgslvMUTJOPgZ8tO78RQIHpH8GnNo
+QkzB\nvXDfH2zhAoGBAPbeM7OveieHL37Iu/xY2wtDagSBD5K0VJhP8OOF5G1t1nHNcyWa
\nYa49hTmGJ7bQsw5oGccvvsXCgGzaNbbAQtKlcz9kiXKOpTWV3t+RXtp0IXp8MIG6\nvWYd7yey7FHumHS/wC0h/
REwx10153UpYaFJe2QJHw4yG9BJgN7o4duVAoGBANRT
\n6BMPqV/6P9kJtduU8sZOVv3BbyUIkoBZlw2O7LB1IjIZcEm4By9DEAqCkFMp4gST
\nW6o2eyXKp0oZ1UwqKdESG2LrGePNrmbQp7LvMngyk7CDqczA5gmnlndCy27k/d1Y\nQuWz+WDrqc8EAD7wRBmrwR0p3zCntPFRJPVu
+yfpAoGACkDcYOAu8KlavadUt3xx\nTJx2MM2zeeJniRP461pKTIk9WOixmaQ53mTLvcHmsF8msLh
+KZnAELKtZtgBVx/R\nJrKcgMuKMenezsT0xtBg4i3knhO+aAT7jNw9bKavzg9c4ax9LOK2ghpGjYaJoIIh
\nffNxXoxKb+qA4TvMUHXXu6kCgYAhGeefORzVqqTTiDECx4jFo6bqLoLOSjTUr6Ld
\n6T87DzfCiba4t2jfVFwm1036uRfUUMjEk3PFY3+LDNX05snYHzOHy1Eg84rR2oua
\nWLIMjQ37QbtyAUybirXpZ89hPW/aVw0u1Ez3cCXr8Rq8tSZYvi8ABewWoL6TtGvH\nm4KqKQKBgCfZrv6wpCrS5Ep/
AKQGdPOXCOM8O2+b4e/NJpSIH9Zk5Elg6WAunlCp\ntHyx1pZFq5RboxFw7DsM9eUTakHvGtTJ
+EFHbyc5tKqWKnVbGmDYR6pNRULPEXU9\nhBQ1pzzmwGnO6AyxTxgoY5CosK2Ga1KjsWUXqay2QwIln+E+xxsm\n-----END RSA
 PRIVATE KEY-----\n"
}
```


Note:
In some cases, issues have been reported with certificates that contain \r\n in the PEM format. According to the X509 certificate standard, it is acceptable and NSX Manager allows these certificates to be used and in many cases there are no error results.

1. From your browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address>.
2. With admin privileges, log in to NSX Manager.
3. Select **System** > **Certificates**.
4. Run one of the following import actions.
   - In version 9.1.1 and later, click **All Certificates**, and then click the **Import** button.
   - In version 9.1.0, select **Import** > **Import Certificate**.
5. Enter the certificate details.

Option	Description
(In version 9.1.1 and later) **Import Option**	Select **Import external certificate with private key** to import a certificate with private key.
Name	Assign a name to the certificate.
(In version 9.1.1 and later) **Use for**	• Select **Services** to use this certificate for services such as a load balancer and VPN. • Select **NSX manager appliance nodes** to use this certificate for the NSX Manager nodes.
(In version 9.1.1 and later) **Signing CA Certificate**	Set to **Yes** to use the certificate as a CA certificate that can sign other certificates. This option is available when you have specified to use the certificate for services.
(In version 9.1.0) **Service Certificate**	• Set to **Yes** to use this certificate for services such as a load balancer and VPN. • Set to **No** if this certificate is for the NSX Manager nodes.
Certificate Contents	Browse to the certificate file on your computer and add the file. The certificate must not be encrypted. If it is a CA-signed certificate, be sure to include the whole chain in this order: certificate - intermediate - root. Everything must be provided in a single line.
Private Key	Browse to the private key file on your computer and add the file. Private key is an optional field if imported certificate is based on NSX Manager generated CSR, as a private key exists on the NSX Manager appliance.
Passphrase	In this release, this field is not used.
Description	Enter a description of what is included in this certificate.

6. Click **Save**.

Import a CA Certificate for NSX

You can import a CA certificate from a system external to NSX, for example, to use with the Load Balancer service.

Verify that a CA certificate is available.

1. With admin privileges, log in to NSX Manager.
2. Select **System** > **Certificates**.
3. Run one of the following import actions.
   - In version 9.1.1 and later, click **All Certificates**, and then click the **Import** button.
   - In version 9.1.0, select **Import** > **CA Certificate**.
4. Enter the certificate details.

Option	Description
(In version 9.1.1 and later) **Import Option**	• Select **Import external certificate with private key** to import a certificate with private key. • Select **Import certificate generated using NSX CSR** to import a CA-signed certificate generated through an NSX certificate signing request (CSR). You must first create a CSR file in NSX Manager, submit your certificate request

Option	Description
	to a certificate authority (CA), and obtain the CA-signed certificate. Then use this option to import the CA-signed certificate.
Name	Assign a name to the CA certificate.
(In version 9.1.1 and later) **Use for**	• Select **Services** to use this certificate for services such as a load balancer and VPN. • Select **NSX manager appliance nodes** to use this certificate for the NSX Manager nodes.
(In version 9.1.0) **Service Certificate**	• Set to **Yes** to use this certificate for services such as a load balancer, VPN, or TLS Inspection. • Set to **No** to use the certificate with NSX Manager appliance nodes.
(In version 9.1.1 and later) **Signing CA Certificate**	To generate a signing CA certificate, set the toggle to **Yes**.
Certificate Contents	Browse to the CA certificate file on your computer and add the file.
Private key	For service certificate only. Browse to the private key file on your computer and add the file. Private key is an optional field. Sent to the edge node where the feature is running.
Passphrase	In this release, this field is not used.
Description	Enter a description of what is included in this certificate.

5. Click **Save**.

Set Checks for Certificate Imports for NSX

You can enable or disable the Extended Key Usage (EKU) Extension and the Certificate Revocation List Distribution Point (CDP) validation checks that NSX performs while importing a certificate.

Note: If you have CA-signed certificates without a CDP then you might have problems after upgrade. To avoid this problem you can turn CRL checking off or replace the certificates with certificates that include a CDP.

To set validation checks, use the following API with payload. For more information about the API, see the *NSX API Guide*.

```
PUT https://<manager>/policy/api/v1/infra/security-global-config
{
"crl_checking_enabled": false,
"ca_signed_only": false,
"eku_checking_enabled":false,
"resource_type":"SecurityGlobalConfig",
"_revision": 0
}
```

Where:

- `crl_checking_enabled`: Not enabled by default. If enabled, it checks the CDP specified in the imported CA-signed certificate. Support includes HTTP based CRL-DP only. File or LDAP-based options are not supported.
- `ca_signed_only`: Disabled by default. It allows checks signed by CA only.
- `eku_checking_enabled`: Enabled by default. It checks for EKU Extension in the imported certificate.
- `revision`: The current revision of the resource that must be included in a request. To obtain the value of this parameter issue a GET operation.

Replacing NSX Certificates

You can replace the self-signed or the self-signed CA appliance certificates that have expired or about to expire through the NSX Manager or through the Certificate APIs. Note that you can replace only those certificates that have a private key and are valid. You can replace a certificate by an auto-generated self-signed certificate or by a new certificate.

Note:

In VMware Cloud Foundation 9.0 and later, VMware NSX works in conjunction with VCF non-disruptive certificate (NDC) rotation policies.

VMware NSX relies on connections with vCenter for management and operational activities. When vCenter certificates are rotated, those changes are automatically detected and the new certificates are used without disruption to network connectivity or services. No maintenance window is required to replace certificates.

After the initial VMware Cloud Foundation installation of NSX Manager, there is no further requirement for managing system certificates.

For NDC to work in NSX with the third-party issued certificates, the third-party CA authority must be trusted by NSX.

A TLS logging server does not support the replacement of certificates. When a TLS logging server certificate is about to expire or gets expired, you must perform the steps described in Configure Remote Logging.

It is recommended to run the Certificate Analyzer, Results and Recovery (CARR) script. The goal of running the CARR script is to ensure that certificates, such as transport node (TN) certificates, do not expire within 90 days. For more information, see Broadcom knowledge base article 369034.

To replace transport node certificates that have already expired, see Broadcom knowledge base article 345825.

Replace NSX Certificates from NSX Manager

You can replace self-signed or CA-signed appliance certificates from the NSX Manager. You can only replace certificates that have private key and are valid. You cannot replace a certificate that belongs to a service-certificate category.

You can replace the self-signed certificates for the following service types:

- MGMT_CLUSTER (aka VIP)
- CBM_CLUSTER_MANAGER
- K8S_MSG_CLIENT
- CBM_CORFU
- CCP
- APH_TN
- LOCAL_MANAGER
- GLOBAL_MANAGER
- APH (aka APH_AR)
- API
- WEB_PROXY

In version 9.1.1 and later, you can also use the **Replace Expiring** shortcut action to select certificates that are expired or within 30 days of expiration.

Note:

Some certificates have been consolidated. When you replace such a certificate, ensure that the replacing certificate must either have a wild-card SAN entry that matches all the nodes in the cluster and the VIP or it must have as many SAN entries that match the VIP and the individual node addresses.

Going forward, use 'Apply Certificate' when you want to assign individual certificates to services that have been consolidated previously so that they become individual certificates again or to consolidate certificates that have been previously separated. After the de-consolidation of a certificate, use 'Replace Certificate' to renew it or replace it when it has expired.

1. With admin privileges, log in to NSX Manager.
2. Navigate to **System** > **Certificates**.

   You see a list of all the certificates, including total certificates, certificates that are about to expire, and the certificates that are currently in use. All the certificates are arranged in different groups. You can also filter the certificates as per your requirements.
3. To replace multiple certificates, perform the following steps:
   a) Select the certificates you want to replace, and click **Actions** > **Replace Certificates**. In version 9.1.1 and later, you can also use the **Replace Expiring** shortcut action to select certificates that are expired or within 30 days of expiration.
   b) In the dialog box listing the selected certificates, for each certificate select the required option:

      In version 9.1.1 and later:

      - **Select from an existing certificate**: Replaces the old certificate with an existing certificate specified by you.
      - **Import external certificate with private key**: Replaces the old certificate with a certificate that has a private key.
      - **Import certificate generated using NSX CSR**: Replaces the old certificate with a CA-signed certificate generated through an NSX certificate signing request (CSR). You must first create a CSR file in NSX Manager, submit your certificate request to a certificate authority (CA), and obtain the CA-signed certificate. Then use this option to import the CA-signed certificate.
      - **Generate new self-signed certificate**: Replaces the old certificate with a self-signed certificate generated by NSX. You can either specify the certificate details yourself, or select **Auto-generate self signed certificate** to use details supplied by NSX.

      In version 9.1.0:

      - **Auto-generate Self Signed Certificate**: Replaces the old certificate with an auto-generated self-signed certificate. This is the default option.
      - **Import Certificates**: Imports signed certificates to replace the old certificate. You need to select this option from the drop-down menu.
      - **Generate Self Signed Certificate**: Provides an option to create a self-signed certificate to replace the old certificate. You need to select this option from the drop-down menu.
   c) Click **Save**.
4. To replace PI certificates, perform the following steps:
   a) Select the PI certificate that you want to replace.
   b) From the more option, click **Replace Certificate**.
   c) In the **Replace Certificate** dialog box, select the PI certificate from the drop-down menu.
   d) Click **Save**.

Replace NSX Certificates Using API

After you install NSX, the manager nodes and cluster have self-signed certificates. You can optionally replace the API and MGMT_CLUSTER (aka VIP) self-signed certificates with common CA-signed certificate with a SAN (Subject Alternative Name) that matches the FQDNs and IPs of all the nodes and the VIP for the cluster. You can run only one certificate replacement operation at a time.

- Verify that a certificate is available in the NSX Manager. Note that on a standby Global Manager the UI import operation is deactivated. For details on the import REST API command for a standby Global Manager, refer to Import a Self-signed Certificate for NSX.
- The server certificate must contain the Basic Constraints extension `basicConstraints = CA:FALSE`.
- Verify that the certificate is valid by making the following API call:
  `GET https://<nsx-mgr>/api/v1/trust-management/certificates/<cert-id>?action=validate`
- Have your node ID string available, if needed. For help locating this information using the UI or the CLI, refer to Finding NSX Node IDs for Certificate API Calls.

Note:
Do not use automated scripts to replace multiple certificates at the same time. Errors might occur.

If you are using NSX Federation, you can replace the GM API certificates, GM MGMT_CLUSTER (aka VIP) certificate, LM API certificates, and LM MGMT_CLUSTER (aka VIP) certificates using the following APIs. You can also replace a self-signed certificate used for GM-LM communication.

When you replace the GLOBAL_MANAGER or LOCAL_MANAGER certificate, the site-manager sends these to all the other federated sites, so the communication remains intact.

The cipher suite TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384 can now be used or replaced for communication between:

- NSX nodes within a cluster.
- Within the NSX Federation.
- NSX Manager to NSX Edge.
- NSX Manager to NSX agent.
- NSX Manager REST API communication (external).

You can also replace the GLOBAL_MANAGER and LOCAL_MANAGER certificates. See Certificates for NSX and NSX Federation for details on self-signed certificates auto-configured for NSX Federation.

The API service on each manager and VIP share the same certificate. You can also bulk replace the shared certificate by running the following 'batch replace' API .

```
POST https://{{ip}}/api/v1/trust-management/certificates/action/batch-replace
{
    "certificate_replacements": [
        {
         "old_certificate_id": "fe3d2623-df43-4757-8018-b1d8223a1475"
         "new_certificate_id": "66370296-cacf-45a7-9912-6e2718df87bb"
        }
    ]
}
```

Optionally, you can also replace each API service and the VIP service with their own unique certificates by running the 'apply certificate' API mentioned in the following procedure.

Both the batch replace and apply certificate APIs should be performed after the manager cluster has formed, because the API certificates of the joining manager are replaced by the MGMT_CLUSTER (aka VIP) certificates of the cluster being joined.

1. With admin privileges, log in to NSX Manager.
2. Select **System** > **Certificates**.
3. In the ID column, select the ID of the certificate you want to use and copy the certificate ID from the pop-up window.
   Make sure that when this certificate was imported, the option **Service Certificate** was set to **No**.

   Note: The certificate chain must be in the industry standard order of certificate - intermediate - root.
4. To replace the API certificate of a manager node, use the following API call. To find your Unified Appliance node ID, refer to Finding NSX Node IDs for Certificate API Calls.

   ```
   POST /api/v1/trust-management/certificates/<cert-id>?
   action=apply_certificate&service_type=API&node_id=<node-id>
   ```

   For example:

   ```
   POST https://<nsx-mgr>/api/v1/trust-management/certificates/77c5dc5c-6ba5-4e74-a801-c27dc09be76b?action=apply_certificate&service_type=API&node_id=e61c7537-3090-4149-b2b6-19915c20504f
   ```
5. To replace the certificate of the manager cluster VIP, use the API call:

   ```
   POST /api/v1/trust-management/certificates/<cert-id>?action=apply_certificate&service_type=MGMT_CLUSTER
   ```

For example:

```
POST https://<nsx-mgr>/api/v1/trust-management/certificates/
d60c6a07-6e59-4873-8edb-339bf75711?action=apply_certificate&service_type=MGMT_CLUSTER
```

Ensure the thumbprint value gets updated for each location's Local Manager from within the Global Manager's Location Manager. Otherwise, communication between GM and LMs are disrupted. Tasks like selecting an NSX Edge cluster or requesting a tier-0 BGP summary from the Global Manager UI will not work if the thumbprint is not updated. For more information about the API, see the *NSX API Guide*. This step is not necessary if you did not configure the VIP.

6. Optional: To replace the LOCAL_MANAGER and GLOBAL_MANAGER certificates for NSX Federation use the following API call. The entire NSX Manager cluster (Local Manager and Global Manager) requires one LOCAL_MANAGER certificate and one GLOBAL_MANAGER certificate.

   ```
   POST https://<nsx-mgr>/api/v1/trust-management/certificates/<cert-id>?
   action=apply_certificate&service_type=<service-type>
   ```

   For example:

   ```
   POST https://<local-mgr>/api/v1/trust-management/certificates/77c5dc5c-6ba5-4e74-a801-
   c27dc09be76b?action=apply_certificate&service_type=LOCAL_MANAGER
   ```

   Or

   ```
   POST https://<global-mgr>/api/v1/trust-management/certificates/77c5dc5c-6ba5-4e74-a801-c27dc09be76b?
   action=apply_certificate&service_type=GLOBAL_MANAGER
   ```

7. To replace APH (aka APH_AR) certificates use the following API call. To find your Unified Appliance node ID, refer to Finding NSX Node IDs for Certificate API Calls.

   ```
   POST https://<nsx-mgr>/api/v1/trust-management/certificates/<cert-id>?
   action=apply_certificate&service_type=APH&node_id=<node-id>
   ```

   For example:

   ```
   POST https://<nsx-mgr>/api/v1/trust-management/certificates/77c5dc5c-6ba5-4e74-a801-c27dc09be76b?
   action=apply_certificate&service_type=APH&node_id=93350f42-16b4-cb4e-99e0-fce2d17635a3
   ```

8. To replace the transport node (TN) or the Edge host certificates (called TN certificates), use the following API call. You must generate the private key outside of NSX Manager. If you generate the CSR in the NSX Manager, it is not possible to retrieve the private key, which is required in the API call. If it is a CA-signed certificate, include the whole chain in this order: certificate - intermediate - root. The entire certificate must be provided in a single line. Also replace any end of line character with \n. This is required in the pem_encoded and private key values. To replace newline characters with \n you can use this command on UNIX based systems: `awk '{gsub(/\\n/,"\n")}1' certificate.pem`.

   ```
   POST https://<nsx-mgr>/api/v1/trust-management/certificates/action/replace-host-certificate/<transport-
   node-id>
   {
         "display_name": "cert_name",
         "pem_encoded": "---BEGIN CERTIFICATE---\n<certificate>\n---END CERTIFICATE-----\n",
         "private_key": "---BEGIN RSA PRIVATE KEY---\n<private rsa key>\n---END RSA PRIVATE KEY---\n"
       }
   ```

   For example:

   ```
   POST https://<nsx-mgr>/api/v1/trust-management/certificates/action/replace-host-
   certificate/8e84d532-2cd8-46d8-90c7-04862980f69c
   {
       "display_name": "cert_sample",
       "pem_encoded": "-----BEGIN CERTIFICATE-----
   \nMIIC1DCCAbygAwIBAgIUMd1fGNGnvYKtilon2UMBP4rqRAowDQYJKoZIhvcNAQEL
   \nBQAwDzENMAsGA1UEAwwETVlDQTAeFw0yMzA5MjYxODMxMzVaFw0zMzA5MjMxODMx
   \nMzVaMBYxFDASBgNVBAMMC2NlcnRfc2FtcGxlMIIBIjANBgkqhkiG9w0BAQEFAAOC\nAQ8AMIIBCgKCAQEAzMDsp1EGFPjus/
   xnHmacPJYVP0N8iQMb3W8TFFQC5jxdjNzi\ncMIb1YgpI+s3LJoyYCdZKeMcCWDwtgQXMTy9FYJCHKyt86CF0br9U9q9iC+NX93X\n
   +/wrWtXY89ESt0NOgj22sKI49EQT9bd0dNWupxapCb98Dyztk0cetIHa7ia1q7un\nXMZ7dofwuWUEUlT8qpyXF84N6bhWQSrXRyeQ
   ```

```
+oZrsq3sAyfnKzbfcs0T3sztWn9M\nR7h8iPkjpJjVV5z1ghAgIDKFXG8RVU8fLgX5srtYV2Ij1II0qYwe/
yGBfj7xsemB\n2lGGPotlbwUE5oPFISJvG9qLOoNKVLvBrxuNnQIDAQABoyEwHzAdBgNVHSUEFjAU
\nBggrBgEFBQcDAQYIKwYBBQUHAwIwDQYJKoZIhvcNAQELBQADggEBAAqPfZWNzG/b
\nBhtN2gDjr0LplfC0yi8K6Ep3exECE5UOUJvHubko4Z6eCZFT8XSrAa6eZQEVe3O3\nwFvpdedCiEpI/
IaFhpRUQDubJMPao7t4Uohz3k3ONMGbIci8dVUcQRQlmxFmx3wf\n0/33fy3b1zIOXqooQF3qUlpjms/
RQOdD80dSlMze8WI7yz9LZt9Zc+sr8ePRi4Xy\ntudO6EYTiWm3CC5BxDDjKpkFCACFRT4zr5HsomHsFeo4hGIHl2zN0+JoGrdrWcta
\nxdl5aQYy79vIMgvz696EKUGePEpJjpyP/wlwzmIY3RvXRKThuVXvg20gi365x8+J
\niKbzpCGe0P0=\n-----END CERTIFICATE-----\n-----BEGIN CERTIFICATE-----
\nMIIC/zCCAeegAwIBAgIUUlHXcczsdMpei1ThgeQYpvgzaxMwDQYJKoZIhvcNAQEL
\nBQAwDzENMAsGA1UEAwwETVlDQTAeFw0yMzA5MjYxODI2NDZaFw0zMzA5MjMxODI2\nNDZaMA8xDTALBgNVBAMMBE1ZQ0EwggEiMA0GCSqGSIb3DQEB
\nAoIBAQDbr78t32TUl1qTcDGvVQhiUkktntPO/5/
FRDSIjy9qyNGDrcICDAYzOe79\nceXpOzfUStacEeTXse89q1MJz4ykaU2g6EUN2E4sfoP4KznBlObLHnnlxD482DL4\nbuMA8qCe0soUsGE6uoeFHnS
+NRI3GtJe1MM134JQ/TSNZTv+d93nB4bS2nSK7\nA1fFDRSuj8Ey7a1im8JgykL9ahJ6yxrpk8juEJwII04nHfAG102/8/
YKEZyPWcPX\nYvLZEt/lBVxRPplWfbNIo3zfA09fzb4RMaOSsyBbqTBseL/4fxlnkeu1Rii3ZwcQ
\nL4Wr6mKR1YCievsuXdLK5pWUH+BtAgMBAAGjUzBRMB0GA1UdDgQWBBTnYafa1EXn
\nNPIqTkIO82kdamjDgTAfBgNVHSMEGDAWgBTnYafa1EXnNPIqTkIO82kdamjDgTAP\nBgNVHRMBAf8EBTADAQH/
MA0GCSqGSIb3DQEBCwUAA4IBAQC2Ef
+CPICTdWEKW3e6\nwaObe4Y85CS2wfSBRFvt0yCAUF8yysr3kQx85wdhfDfvidQdrgQIkDKe83J61r5l
\n238wFo9O10RpFWl1csY4hZ19geeTW3L8tABp+f1or1vsAogfVtcaZwmqz/LEaZ0r
\n4JdONE9gq40RgX5R9GPD04k3hKr6HoNHHnBssmNHgo8pLKRv04mx0yQyn45lKvet\ngcInI9j8YLsXGHdeiZ/zXKUgKQdicBw79K/
mQCpgkpaEi3K9mFUFCUU9CiWxiy62\nSN2/SEuOWlb7Kq8VwJUfUn3lKoY9sofr9zsSsh5lhQOKb1uguo8xUF8v6iLuDAjr\n9bcn
\n-----END CERTIFICATE-----\n",
    "private_key": "-----BEGIN RSA PRIVATE KEY-----\nMIIEowIBAAKCAQEAzMDsp1EGFPjus/
xnHmacPJYVP0N8iQMb3W8TFFQC5jxdjNzi\ncMIb1YgpI+s3LJoyYCdZKeMcCWDwtgQXMTy9FYJCHKyt86CF0br9U9q9iC+NX93X\n
+/wrWtXY89ESt0NOgj22sKI49EQT9bd0dNWupxapCb98Dyztk0cetIHa7ia1q7un\nXMZ7dofwuWUEUlT8qpyXF84N6bhWQSrXRyeQ
+oZrsq3sAyfnKzbfcs0T3sztWn9M\nR7h8iPkjpJjVV5z1ghAgIDKFXG8RVU8fLgX5srtYV2Ij1II0qYwe/
yGBfj7xsemB\n2lGGPotlbwUE5oPFISJvG9qLOoNKVLvBrxuNnQIDAQABAoIBAQCE8JH2xIWVYlbh
\np3RwaaDxOWTMMY4PC2SxLegOX8mOIQ2AYv3mxjD6QDCt8I9fNzKT+ZhLuPhAIp/H
\nHfrM7im6aFtycK90qfmYxbarFi/O10kMQGZ2ZjDkBkqZa1qigGHd8CHIp1shRX5M
\nIHPNU9vVAsJ34Mq0s7AA2sFV46X4zyEqHKLi1qVcsj68XJCrKJPTzOXiZWOHL8e0\nx4B8mGKbnWNmrq6styyYi9rzUnucoKL459YkaF/
MEBpou3wvhprkR5Ufr4eNo0YV\nr0KfcEjxZqVT2o6r59+gSZQiCHael2MgslvMUTJOPgZ8tO78RQIHpH8GnNo
+QkzB\nvXDfH2zhAoGBAPbeM7OveieHL37Iu/xY2wtDagSBD5K0VJhP8OOF5G1t1nHNcyWa
\nYa49hTmGJ7bQsw5oGccvvsXCgGzaNbbAQtKlcz9kiXKOpTWV3t+RXtp0IXp8MIG6\nvWYd7yey7FHumHS/wC0h/
REwx10153UpYaFJe2QJHw4yG9BJgN7o4duVAoGBANRT
\n6BMPqV/6P9kJtduU8sZOVv3BbyUIkoBZlw2O7LB1IjIZcEm4By9DEAqCkFMp4gST
\nW6o2eyXKp0oZ1UwqKdESG2LrGePNrmbQp7LvMngyk7CDqczA5gmnlndCy27k/d1Y\nQuWz+WDrqc8EAD7wRBmrwR0p3zCntPFRJPVu
+yfpAoGACkDcYOAu8KlavadUt3xx\nTJx2MM2zeeJniRP461pKTIk9WOixmaQ53mTLvcHmsF8msLh
+KZnAELKtZtgBVx/R\nJrKcgMuKMenezsT0xtBg4i3knhO+aAT7jNw9bKavzg9c4ax9LOK2ghpGjYaJoIIh
\nffNxXoxKb+qA4TvMUHXXu6kCgYAhGeefORzVqqTTiDECx4jFo6bqLoLOSjTUr6Ld
\n6T87DzfCiba4t2jfVFwm1036uRfUUMjEk3PFY3+LDNX05snYHzOHy1Eg84rR2oua
\nWLIMjQ37QbtyAUybirXpZ89hPW/aVw0u1Ez3cCXr8Rq8tSZYvi8ABewWoL6TtGvH\nm4KqKQKBgCfZrv6wpCrS5Ep/
AKQGdPOXCOM8O2+b4e/NJpSIH9Zk5Elg6WAunlCp\ntHyx1pZFq5RboxFw7DsM9eUTakHvGtTJ
+EFHbyc5tKqWKnVbGmDYR6pNRULPEXU9\nhBQ1pzzmwGnO6AyxTxgoY5CosK2Ga1KjsWUXqay2QwIln+E+xxsm\n-----END RSA
 PRIVATE KEY-----\n"
}
```


Use the ID of the transport node for the node_id, not the Unified Appliance (UA) ID. To find your node ID, refer to Finding NSX Node IDs for Certificate API Calls.

For details on the default self-signed certificates configured by the system or for VMware Cloud Foundation certificate requirements, refer to Types of Certificates in NSX.

Finding NSX Node IDs for Certificate API Calls

Use the NSX UI or CLI to find the node IDs required to import or replace certificates in manager nodes and clusters.

There are several types of node IDs required for certificate replacement. The type of certificate replacement you are performing will determine the type of node ID you need to include in your API call.

- To replace the API certificate of a manager node or other NSX Manager certificate, such as APH-AR, use the Unified Appliance node ID.
- To replace the transport node (TN) or the Edge host certificates, use the transport node ID.

Table 1077:

Task	From the UI	From the CLI
To get the Unified Appliance node ID.	1. Go to **System** > **Appliance**. 2. From the NSX Manager tab, click **View Details**. 3. From the details pop-up, select the **UUID** icon to copy to clipboard.	Log in to an NSX Manager node using SSH and run the command: `get cluster status`.
To get the Edge transport node ID.	1. Go to **System** > **Fabric** > **Edges**. 2. Select the ID value of the NSX Edge transport node and copy it to your clipboard.	Log in to an NSX Manager node using SSH and run the command: `get transport-nodes status | find <Edge-IP-or-name>`.
To get the host transport node ID.	1. Go to **System** > **Fabric** > **Hosts** > **Clusters**. 2. Expand the host cluster containing the target node. 3. Select the host transport node by clicking the check box in the first column. 4. Click the menu icon (three dots) and select **Copy ID to Clipboard.**	Log in to an NSX Manager node using SSH and run the command: `get transport-nodes status | find <host-IP-or-name>`.

If required, retrieve all three NSX Manager appliance node IDs and run the API calls for each appliance certificate replacement. For instructions, refer to Replace NSX Certificates Using API.

Automatically Replacing Expiring NSX Certificates

NSX Manager replaces some self-signed appliance certificates before their expiry.

A background task in NSX Manager scans the list of certificates stored in Corfu and identifies all the appliance certificates that have expired or are due to expire within a period of 31 days. This task then constructs and runs a batch certificate replace operation that will replace all of the expired or the expiring certificates.

The automatic replacement task will not perform replacements in case of the following conflicting operations:

- Backup & restore
- Upgrade
- Rollback
- Addition of a new transport node (host or edge)
- Addition of a new manager node

Note:

APH, APH_TN, and CCP certificates are not replaced during the automatic certificate replacement.

You can also manually initiate the replacement of expiring certificates by running the following API.

```
POST /api/v1/trust-management/certificates/action/renew-appliance-certificates
```

By default, the automated certificate replacement performs its first check for expired certificates 10 minutes after proton startup and then repeats the check every 24 hours thereafter.

Turning off Automatic Certificate Replacement

By default, the automatic certificate replacement policy is enabled. To turn off the replacement policy, add the following fields to the /policy/api/v1/infra/security-global-config API.

```
PUT /policy/api/v1/infra/security-global-config
{
   ... (existing properties)
   "automatic_appliance_certificate_renewal_enabled": false,
   "automatic_appliance_certificate_renewal_lead_time": 31  # in days
}
```

Apply a Certificate to an NSX Service

After you add a certificate to NSX, you can associate it with a service by using the **Apply Certificate** option.

When you associate a certificate to a service for the first time, all services in the system are available for selection. The first service applied to a certificate sets the category of that certificate. From now on, only services from the same category of the certificate will be available for selection when you use the **Apply Certificate** option.

You can use the **Apply Certificate** option when you have a unique CA-signed certificate and you want to apply such an individual certificate to the VIP and the individual API services rather than using a single consolidated certificate. Once you have applied an individual certificate to a service, you can always use the **Replace Certificate** option to replace that certificate when it expires.

1. Log in to NSX Manager with admin privilege.
2. Navigate to **System** > **Certificates**. In version 9.1.1 and later, click **All Certificates**.
   You see a list of all the certificates, including total certificates, certificates that are about to expire, and the certificates that are currently in use. All the certificates are arranged in different groups. You can also filter the certificates per your requirements.
3. Select the required certificate from the list.
4. From the more option, click **Apply Certificate**.
5. In the Apply Certificate dialog box, select the required service/entity to apply to the certificate.
6. If a service needs to be applied to a node, the Apply Certificate dialog box displays a **Node** drop-down. Select the required node to apply the service/entity.
7. Click **Save**.

Importing and Retrieving CRLs

You can import a certificate revocation list (CRL) into the NSX Manager and use the API to configure NSX Manager to retrieve a CRL.

Import a Certificate Revocation List

A certificate revocation list (CRL) is a list of subscribers and their certificate status. When a potential user attempts to access a server, the server denies access based on the CRL entry for that particular user. This topic describes how to import a CRL into the NSX Manager.

Verify that a CRL is available.

NSX supports two CRL formats:

- PEM-encoded X.509 CRL - 40 MB maximum size, 500,000 entries
- Mozilla OneCRL - 5 MB maximum size, 10,000 entries

The list contains the following items:

- Revoked certificates and the reasons for revocation
- Dates the certificates are issued
- Entities that issued the certificates
- Proposed date for the next release

1. With admin privileges, log in to NSX Manager.
2. Select **System** > **Certificates**.
3. Click the **CRLs** tab.
4. To browse the *default_public_crl* file, expand that row and click **View Details**.

   You can view the Issuer Name and Serial Numbers details.
5. To import a CRL, click **Import** and add the CRL details.

Option	Description
Name	Assign a name to the CRL.
CRL Bundle	Browse for your PEM or JSON files and select the file for import.
Description	Enter a summary of what is included in this CRL.

6. Click **Save**.

The imported CRL appears as a link.

Configuring NSX Manager to Retrieve a Certificate Revocation List

Using the API, you can configure NSX Manager to retrieve a certificate revocation list (CRL). You can then check the CRL by making an API call to NSX Manager instead of to the certificate authority.

This feature provides the following benefits:

- It is more efficient to have the CRL cached on the server, that is, NSX Manager.
- The client does not need to create any outbound connection to the certificate authority.

The following APIs related to certificate revocation lists are available:

```
GET /api/v1/trust-management
GET /api/v1/trust-management/crl-distribution-points
POST /api/v1/trust-management/crl-distribution-points
DELETE /api/v1/trust-management/crl-distribution-points/<crl-distribution-point-id>
GET /api/v1/trust-management/crl-distribution-points/<crl-distribution-point-id>
PUT /api/v1/trust-management/crl-distribution-points/<crl-distribution-point-id>
GET /api/v1/trust-management/crl-distribution-points/<crl-distribution-point-id>/status
POST /api/v1/trust-management/crl-distribution-points/pem-file
```

You can manage CRL distribution points and retrieve the CRLs stored in NSX Manager. For more information, see the *NSX API Guide*.

Import or Update a Trusted CA Bundle

You can now use a built-in trusted certificate authority (CA) bundle for the TLS Inspection chain of trust to support advanced security applications such as IDS/IPS, URL filtering, malware, and granular App ID.

You can use the built-in CA bundle, `default_trusted_public_ca_bundle`, internally for the TLS inspection and decryption for gateway firewalls.

For external services, TLS Proxy requires a configured trusted CA bundle to validate the certificate that any external service presents to it. You can configure the External_Decryption_Profile.trusted_ca_bundles with one or more CA bundles where each bundle is a list of certificates. You must configure at least one CA bundle. Typically, external services use well known CAs such as Verisign and DigiCert. So, for ease of configuration, NSX includes a built-in default_trusted_public_ca_bundle that contains a list of widely used CA certs, similar to how operating systems come pre-installed with popular CA certs. You can update this bundle or you can create your own CA bundle and use it instead.

You can perform the following tasks in NSX. You can find Trusted CA Bundles by selecting **System** > **Certificates** > **Trusted CA Bundle**.

- Validate TLS inspection and decryption using the default trusted CA bundle.

- View all certificates in the CA bundle including filtering basic details using the **View All Certificates** button.
- Search for expired, expiring, valid, used and unused CA bundles using the **View All Certificates** button.
- Edit CA bundle display name and add or remove certificates from the bundle.
- Export a CA bundle for inclusion on other devices.
- Copy the CA bundle path locally.
- Import a new trusted CA bundle using the **Import CA Bundle** button.

Storage of Public Certificates and Private Keys for Load Balancer or VPN service

Public certificates and private keys are stored on the NSX Managers for load balancer or VPN service. When a load balancer or VPN service is created that requires a private key, NSX Manager sends a copy of the private key to the Edge node where the load balancer or VPN service is running.

Alarm Notification for Certificate Expiration

NSX generates alarms when a certificate is nearing its expiry or if a certificate has already expired. Service certificates generate an alarm only if expiring or expired and in use by a component. Non-service certificates always generate an alarm, whether in use or not.

NSX generates alarms under following events. The defaults are listed below, but are configurable.

- Medium severity alarm starting 30 day before certificate expiry.
- High severity alarm starting 7 days prior to expiry.
- Critical severity alarm after certificate expires.

Certificate Expiry alarms contains details on certificate ID, severity, node, first/last report time, and recommended action.

As a remedial, you must replace the expiring External Platform certificate with a new valid certificate and delete expiring certificate.

Integration of Kubernetes Clusters with Antrea CNI

Antrea is a container network interface (CNI) plugin from VMware that provides network connectivity and security features to pods in container clusters that are based on Kubernetes.

> **Note:**
> Henceforth, this documentation uses the term "Antrea Kubernetes cluster" to mean Kubernetes clusters with Antrea CNI. The term "Kubernetes cluster" is a generic term, which represents Tanzu Kubernetes Grid (TKG) clusters with Antrea CNI, OpenShift clusters with Antrea CNI, or do it yourself (DIY) Kubernetes clusters with Antrea CNI.

The objective is to connect Antrea Kubernetes clusters to the NSX Management Plane and Central Control Plane (CCP). To achieve this integration, you must deploy Antrea NSX Adapter on all the Antrea Kubernetes clusters that you want to integrate to NSX.

Benefits of Integration

The integration of Antrea Kubernetes clusters to NSX enables the following capabilities:

- View Antrea Kubernetes cluster resources in the NSX Manager UI.
- Centrally manage groups and security policies in NSX that reference Antrea Kubernetes clusters and NSX resources (for example, VMs).
- Distribute the security policies to the Kubernetes clusters for enforcement in the cluster by the Antrea CNI.
- Extend the NSX network diagnostic and troubleshooting features to the Antrea Kubernetes clusters, such as collecting support bundles, monitoring logs, and performing Traceflow operations.
- Monitor the runtime state and health status of Antrea Kubernetes cluster components and Antrea Agents in the NSX Manager UI.

All NSX- Antrea integration features can work when Antrea is either a primary or a secondary CNI in a Kubernetes cluster.

Interoperability Requirements

For NSX and Antrea integration, specific interoperability requirements must be met. For more details, see VMware Product Interoperability Matrix.

Antrea CNI in networkPolicyOnly Mode

NSX can be integrated to Antrea Kubernetes clusters in which Antrea CNI is deployed to run in a `networkPolicyOnly` mode. In such a case, Antrea runs as a secondary CNI and does the task of enforcing network policies in the cluster. The native routed CNI (primary CNI) does the IP address management and pod network connectivity tasks.

To set up Antrea CNI to run in a `networkPolicyOnly` mode, you need to deploy VMware Container Networking™ with Antrea™ v1.8 or later in your Kubernetes cluster.

All NSX- Antrea integration features that are discussed in this chapter are supported when Antrea CNI is deployed in a `networkPolicyOnly` mode.

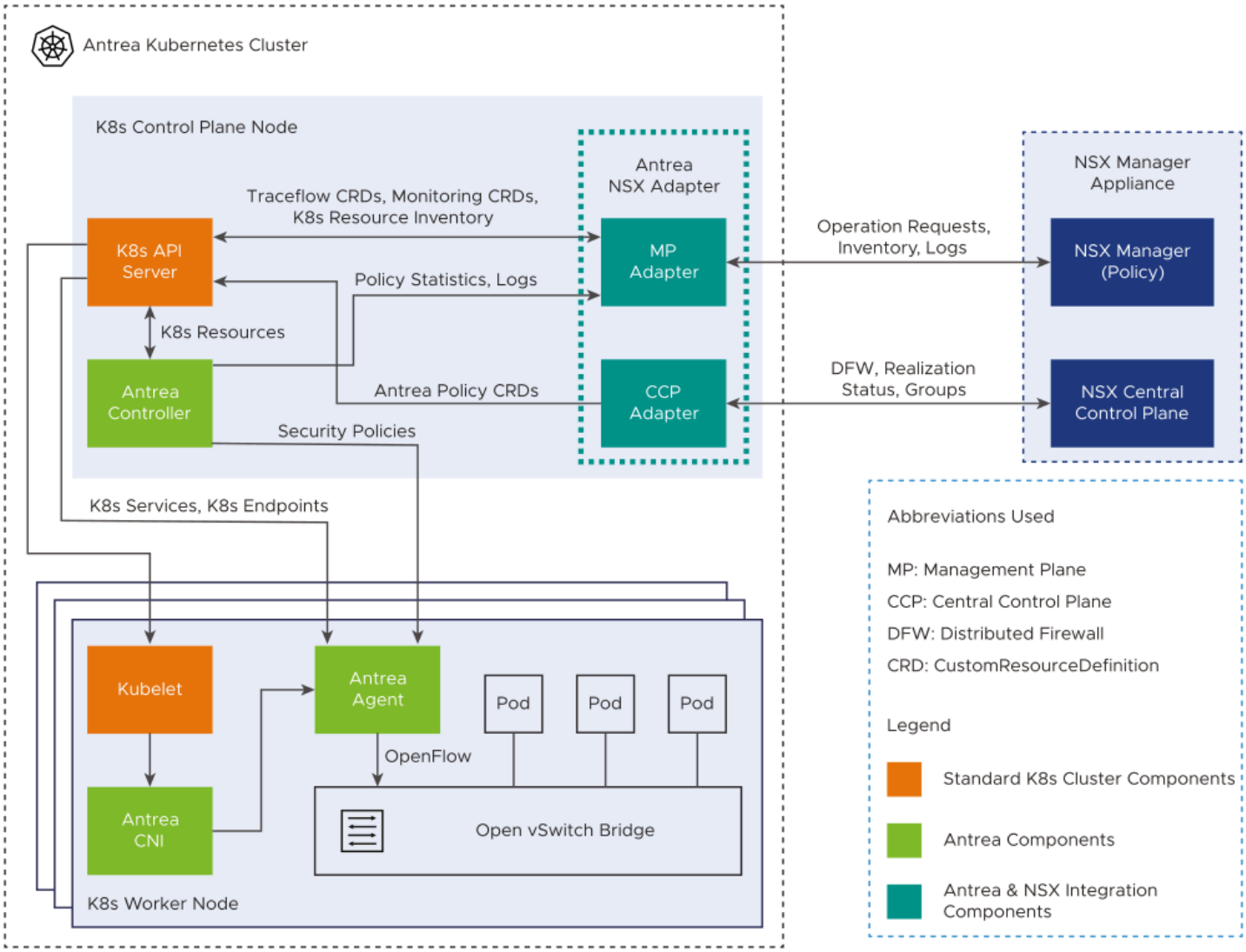
Architecture of Antrea Kubernetes Cluster Integration with NSX

The integration architecture explains the information exchanged between a Kubernetes cluster that uses Antrea CNI and the NSX Manager Appliance, which is deployed in NSX.

This documentation does not explain the functions of Antrea components in a Kubernetes (K8s) cluster. To understand the Antrea architecture and the functions of Antrea components in a Kubernetes cluster, see the Antrea documentation portal at https://antrea.io/docs.

This main objective of this documentation is to understand the functions of the Antrea NSX Adapter that integrates a Kubernetes cluster with Antrea CNI to the NSX Manager Appliance.

Architecture Diagram



Antrea NSX Adapter

This component runs as a pod on one of the Kubernetes Control Plane nodes. Antrea NSX Adapter consists of the following two subcomponents:

- Management Plane Adapter (MP Adapter)
- Central Control Plane Adapter (CCP Adapter)

Management Plane Adapter communicates with the NSX Management Plane (Policy), Kubernetes API Server, and Antrea Controller. Central Control Plane Adapter communicates with the NSX Central Control Plane (CCP) and Kubernetes API Server.

Functions of the Management Plane Adapter

- Watches the Kubernetes resource inventory from Kubernetes API and reports the inventory to NSX Manager. Resource inventory of an Antrea Kubernetes cluster includes resources, such as Pods, Ingress, Services, Network Policies, Namespaces, and Nodes.

- Responds to the policy statistics query from NSX Manager. It receives the statistics from the Antrea Controller API or the statistics that are exported by the Antrea Agent on each K8s worker node, and reports the statistics to NSX Manager.
- Receives troubleshooting operation requests from NSX Manager, sends the requests to Antrea Controller API server, collects the results, and returns the information to NSX Manager. Examples of troubleshooting operations include Traceflow requests, Support Bundle collection requests, log collection requests.
- Watches the runtime state and health status of an Antrea Kubernetes cluster from the Antrea Monitoring CustomResourceDefinition (CRD) objects and reports the status to NSX Manager. The status is reported on a per cluster basis. For example, the health status of the following components is reported to NSX Manager:
  - Management Plane Adapter
  - Central Control Plane Adapter
  - Antrea Controller
  - Antrea Agents

Functions of the Central Control Plane Adapter

- Receives the Distributed Firewall (DFW) rules and groups from NSX Central Control Plane, translates them to Antrea policies, and creates Antrea policy CRDs in K8s API.
- Watches the policy realization status from both K8s network polices and native Antrea policy CRDs and reports the status to NSX Central Control Plane.

Stateless Nature of the Central Control Plane Adapter

The Central Control Plane Adapter is stateless. Each time the adapter restarts or reconnects to K8s API or NSX Manager, it always synchronizes the state with K8s API and NSX Central Control Plane. Resynchronization of the state ensures the following:

- The latest Antrea policies are always pushed to K8s API as native Antrea policy CRDs.
- The stale policy CRDs are removed if the corresponding security policies are deleted in NSX.

Registering an Antrea Kubernetes Cluster to NSX

The registration process involves two roles or personas: NSX administrator and Kubernetes platform administrator. In an organization, both roles might be held by the same person or by different people.

For example, if your organization uses a hosted (managed) Kubernetes service from a public cloud service provider, such as Azure Kubernetes Service, Google Kubernetes Service, and so on, the Kubernetes platform administrator is in the cloud service provider organization, and the NSX administrator is in your organization. In such a scenario, the Kubernetes platform administrator and the NSX administrator can collaborate with each other to exchange information and complete the registration.

For information on registering and integrating an Antrea Kubernetes cluster to NSX, see Enable the Antrea-NSX Adapter for a VKS Cluster.

Viewing Inventory of an Antrea Kubernetes Cluster in NSX Manager

In an NSX environment, you can view the resources of the registered Antrea Kubernetes clusters in **read-only** mode. You cannot modify or delete the Antrea Kubernetes cluster resources in NSX.

For example, in the NSX environment, you cannot modify, add, or remove tags (labels) that are attached to pods.

> **Note:**
> After the Antrea Kubernetes cluster is registered to NSX, the Management Plane Adapter connects with the NSX Management Plane and performs a full synchronization of the Antrea Kubernetes cluster resources in the NSX inventory. The time required to do a full sync operation is directly proportional to the scale of the cluster. Thereafter, only a delta synchronization operation happens at regular predefined intervals. If the Management Plane Adapter fails due to any reason, the resources are not synced with the NSX inventory. Only after the adapter is up again, the resources in the Antrea Kubernetes cluster are compared with the existing objects in the NSX inventory and the difference (delta) is synchronized.

Important:
In a multi-tenant NSX environment, Kubernetes cluster resources are not exposed to the project inventory. In other words, a project administrator cannot view and manage Antrea Kubernetes cluster resources inside a project. You must switch to the **Default** view (default space) to view and manage the inventory of registered Antrea Kubernetes clusters.

View Details of an Antrea Kubernetes Cluster

You can view the list of all Antrea Kubernetes clusters that are registered to NSX, and filter this list based on several filter criteria, such as External ID, Cluster Name, CNI Type, Type of Container Cluster, and some more criteria.

Antrea Kubernetes clusters must be registered to NSX.

For a detailed information about registering an Antrea Kubernetes cluster, see Registering an Antrea Kubernetes Cluster to NSX.

When the inventory of Antrea Kubernetes clusters is available in NSX inventory, cluster administrators can do basic analysis and diagnostic tasks in the NSX Manager UI.

For example:

- View the Kubernetes services that are running in the cluster.
- Verify whether a specific Kubernetes service is up or down.
- Verify the health status of Antrea Agent on each node in the Kubernetes cluster.
- View the pods that are running in the cluster.
- Check whether any pods in the cluster are down.
- Determine which pods in the cluster are associated with a specific Kubernetes service.
- Read the specifications of Antrea cluster network policies (ACNP).

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Navigate to **Inventory** > **Containers** > **Clusters**.
   A list of all container clusters in the NSX inventory is displayed in the table.

   Important:
   NSX Manager UI fetches the information about registered Antrea Kubernetes clusters when you start the NSX Manager application in the browser. If the application UI is already open, it does not fetch the Antrea Kubernetes cluster registration information automatically. This behavior is expected and per the current UI design.

   If you have no Antrea Kubernetes clusters registered to NSX and you register the first cluster while the UI is open, a forced browser refresh is required after navigating to the **Clusters** page. This manual browser refresh is required only once after the first cluster is registered, and not every time after a new Antrea Kubernetes cluster is registered to NSX. If there are existing Antrea Kubernetes clusters registered to NSX, you do not have to force refresh the browser for fetching the newly added clusters. Clicking the **Refresh** link on the **Clusters** page can fetch the updated list.
3. To view the list of only Antrea Kubernetes clusters, filter the table by **CNI Type** as **Antrea**.
4. To view the information about all nodes, Kubernetes services, and pods in a specific Antrea Kubernetes cluster, click the hyperlinked number in the respective columns.
5. Expand a row in the table to view more details about a specific Antrea Kubernetes cluster.
   For example, view the following details:
   - Type of infrastructure for the Antrea Kubernetes cluster (example: vSphere, AWS, Azure, Google, VMware Cloud, and so on).
   - Antrea version installed in the Kubernetes cluster.
   - Number of Antrea cluster network policies in the Kubernetes cluster.
   - Specifications (YAML manifest) of the Antrea cluster network policies. These policies are cluster-scoped.
   - Number of namespaces in the Kubernetes cluster and the details of each namespace.
   - NodePortLocal range. Default range is 61000–62000.
   - NodePort range. Default range is 30000–32767.

- Number of Antrea Egresses, egress IPs, and specifications (YAML manifest) of the Egress resources.
- Number of Antrea IP pools and specifications (YAML manifest) of the IP pools.

This list does not mention all the inventory details that you can view for an Antrea Kubernetes cluster. Check the **Clusters** page in the NSX Manager UI to know more.

Note:

- For Antrea Kubernetes clusters, the **Networking Status** column displays `Not Applicable`. Currently, this column is used only for container clusters that use NSX Container Plugin (NCP) as the CNI.
- When NSX discovers labels on Kubernetes cluster resources, such as nodes, namespaces, pods, services, and so on, these labels are always displayed with a `dis:k8s` prefix in the NSX inventory. Remember that a label in Kubernetes maps to a tag in NSX, whereas a key in Kubernetes maps to a scope in NSX.

View Details of Namespaces in an Antrea Kubernetes Cluster

You can view the list of all namespaces that are created in Antrea Kubernetes clusters, and filter this list based on several criteria, such as ID, Cluster Name, CNI Type, Type of container cluster, and some more criteria.

Antrea Kubernetes clusters must be registered to NSX.

For a detailed information about registering an Antrea Kubernetes cluster, see Registering an Antrea Kubernetes Cluster to NSX.

When the inventory of Antrea Kubernetes clusters is available in NSX inventory, cluster administrators can do basic analysis and diagnostic tasks in the NSX Manager UI.

For example:

- View the number of pods running in each namespace, the number of pods that are down, and so on.
- View the number of Antrea network policies and Kubernetes network policies.
- Read the specifications of Antrea network policies and Kubernetes network policies.
- View the ingress rules that are associated with a namespace.
- View the Kubernetes services that are running in the namespace.

1. From your browser, log in to an NSX Manager at `https://`*`nsx-manager-ip-address`*.
2. Navigate to **Inventory** > **Containers** > **Namespaces**.
   A list of all namespaces across all container clusters is displayed in the table.
3. To view the list of namespaces created only in Antrea Kubernetes clusters, filter the table by **CNI Type** as **Antrea**.
4. Expand a row in the table to view more details about the namespace.
   For example, view the following details:
   - Number of ingress rules and the details of each ingress rule in the namespace.
   - Number of network policies in the namespace. These policies refer to Antrea network policies and Kubernetes network policies, which have their scope limited to a namespace.
   - Specifications (YAML manifest) of the network policies that are associated with the namespace.
   - Number of labels and the details of each label in the namespace.
   - Number of Kubernetes Gateways in the namespace and the specifications (YAML manifest) of the Gateway resources.

   This list does not mention all the inventory details that you can view for a namespace. Check the **Namespaces** page in the NSX Manager UI to know more.

   Note:

   - For Antrea Kubernetes clusters, the **Networking Status** column displays `Not Applicable`. Currently, this column is used only for container clusters that use NSX Container Plugin (NCP) as the CNI.
   - When NSX discovers labels on Kubernetes cluster resources, such as nodes, namespaces, pods, services, and so on, these labels are always displayed with a `dis:k8s` prefix in the NSX inventory. Remember that a label in Kubernetes maps to a tag in NSX, whereas a key in Kubernetes maps to a scope in NSX.

Monitor Health Status of an Antrea Kubernetes Cluster

The overall health status of the Antrea Kubernetes cluster is aggregated or computed from the status of the various Antrea components and displayed in the NSX Manager UI.

Antrea Kubernetes clusters are registered to NSX.

The following Antrea components expose their health status to NSX Manager:

- Antrea Controller
- Antrea Agent
- Management Plane Adapter
- Central Control Plane Adapter

The Monitoring CustomResourceDefinition (CRD) objects in Antrea report the statuses of these Antrea components to NSX Manager after a predefined period called a heartbeat interval. This heartbeat interval is configurable for each cluster. The default value is 60 seconds. The permitted range of values is 60 seconds through 600 seconds. You can modify the default interval by running an NSX API.

To read the heartbeat configuration of a specific Antrea Kubernetes cluster, run the following NSX GET API:

```
GET https://{nsx-mgr-ip}/policy/api/v1/infra/sites/{site-id}/enforcement-points/{enforcementpoint-id}/cluster-control-planes/{cluster-name}/heartbeat-config
```

To update the heartbeat configuration of a specific Antrea Kubernetes cluster, run the following NSX PUT API:

```
PUT https://{nsx-mgr-ip}/policy/api/v1/infra/sites/{site-id}/enforcement-points/{enforcementpoint-id}/cluster-control-planes/{cluster-name}/heartbeat-config
{
   "report_interval": 120,
   ""_revision": 0
}
```

The PUT API body shows 120 as the sample report interval. You can specify any integer value from 60 through 600. The unit of reporting interval is seconds.

The `_revision` parameter describes the current revision of the `heartbeat-config` resource. PUT operation must include the current revision of this resource, which you can obtain by submitting the GET API. If the revision provided in a PUT request is missing or stale, the update operation is rejected.

For a detailed information about all the parameters in the API, including examples of the GET and PUT API responses, see the *NSX API Guide*.

> **Note:**
> If the Antrea components do not send a heartbeat to NSX Manager, the status of that component is shown as Unknown. This status means that status monitoring is not working. However, container networking is working on the node. Existing security policies are still enforced on the pods, but if any new security policies are applied, they are not enforced on the pods.

The following procedure explains the steps to view these statuses in NSX Manager:

- Overall health status of an Antrea Kubernetes cluster.
- Health status of Antrea Agent on each node of the cluster.

1. From your browser, log in to an NSX Manager at `https://`*`nsx-manager-ip-address`*.
2. View the overall health status of an Antrea Kubernetes cluster.
   a) Navigate to **System** > **Fabric** > **Nodes** > **Container Clusters** > **Antrea**.
      A list of all registered Antrea Kubernetes clusters is displayed. The **Status** column displays the overall health status of each cluster.
   b) Click Up or Down in the **Status** column to view more details in a pop-up window.
      The overall health status of the Antrea Kubernetes cluster is computed from the status of the following Antrea components:

- Antrea Controller
- Management Plane Adapter
- Central Control Plane Adapter

If the status of any one component or all three Antrea components is Down, the overall cluster status is Down. Click **Failed/Down** in the pop-up window to view the error message. The cluster status is Up only when the statuses of all the three Antrea components are Up.

The pop-up window also displays the total number of Antrea Agents that are Healthy, Failed, and Degraded. If an Antrea Agent is degraded, it means that the container networking on the node is working. However, new security policies might not be enforced correctly on the node. If an Antrea Agent has failed, it means that the container networking on that node is not working.

To view the status of each individual node in the Antrea Kubernetes cluster, check the Antrea Agent status on each node, as explained in the next step.

3. Check the health status of Antrea Agent on each node of the Antrea Kubernetes cluster.
   a) Navigate to **Inventory** > **Containers** > **Clusters**.
      A list of all container clusters in the NSX inventory is displayed.
   b) Filter this list by **CNI Type** as **Antrea**.
   c) Click the hyperlinked number in the **Nodes** column.
      The **Nodes** window opens. The **Agent Status** column shows whether the Antrea Agent on the node is Up or Down. The **Agent Status** column does not show Degraded as one of the statuses.

Trace the Path of a Packet with Antrea Traceflow

You can start a Traceflow session in NSX Manager to trace the path of packet in an Antrea Kubernetes cluster. Antrea Traceflow currently supports tracing the path of only Unicast traffic. Broadcast and Multicast traffic are not supported.

Antrea Kubernetes cluster is registered to NSX.

> **Note:**
> In a multi-tenant NSX environment, the Antrea Traceflow feature is currently not supported under projects. You must use this feature in the **Default** view (default space) of the NSX environment.

The source of a Traceflow session must be a pod, whereas the destination can be a pod or a service in the same Antrea Kubernetes cluster. You can trace the path of a packet for the following types of traffic in an Antrea Kubernetes cluster:

- Pod to pod traffic on the same node (intra-node traffic)
- Pod to pod traffic between nodes (inter-node traffic)
- Pod to service traffic on the same node
- Pod to service traffic between nodes
- Pod to an arbitrary IP address

> **Note:**
> Currently, Antrea Traceflow does not support tracing the path of a packet that travels between Antrea Kubernetes cluster and NSX overlay network. However, some workaround is possible. For more information, see Workaround to Trace the Packet Flow Between an Antrea Kubernetes Cluster and an NSX Overlay Network.

Traceflow injects a test packet into the Antrea Kubernetes cluster network and monitors the flow of the packet. As the packet flows from the source to destination, observations are collected from various components along the path of the packet. These observations are displayed in the Traceflow output, which shows the various components in the path of the packet.

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Navigate to **Plan & Troubleshoot** > **Traffic Analysis**.

> **Important:**
> NSX Manager UI fetches the information about registered Antrea Kubernetes clusters when you start the NSX Manager application in the browser. If the application UI is already open, it does not fetch the Antrea Kubernetes

cluster registration information automatically. This behavior is expected and per the current UI design. If you have registered the first Antrea Kubernetes cluster after the NSX Manager application is opened, ensure that you refresh the browser after navigating to the **Traceflow Analysis** page. A manual refresh ensures that the **Antrea Traceflow** tab is visible in the UI when you reach step 4 of this procedure.

This manual browser refresh is required only once, and not every time after a new Antrea Kubernetes cluster is registered to NSX.

3. In the **Traceflow** card, click **Get Started**.
4. Click the **Antrea Traceflow** tab.

   This tab is available only when at least one Antrea Kubernetes cluster is registered to NSX. If this tab is not visible, refresh the browser.
5. Specify the configuration settings of the Traceflow session.

Field	Description
Cluster	Select an Antrea Kubernetes cluster from the drop-down menu. Alternatively, enter the first few characters of the cluster name to filter the list, and then select the cluster.
IP Address	Select either **IPv4** or **IPv6**.
Protocol Type	Select any one protocol type: ICMP, TCP, UDP. For ICMP: • Optional: Enter the ICMP identifier. Default is 0. • Optional: Enter the ICMP sequence number. Default is 0. • Optional: Enter the TTL. Default is 64. For TCP, UDP: • Optional: Enter the source port number. Default is 0. • Optional: Enter the destination port number. Default is 0. • (Only for TCP): Add TCP flags, if required. SYN flag is set by default. This flag is required for the Antrea Traceflow. • Optional: Enter the TTL. Default is 64.
Source	Only **Pod** is supported as the source of an Antrea Traceflow. **Note:** Pods that use the host network are currently not supported as the source of an Antrea Traceflow. This limitation is a known behavior. For example, Antrea Agent pods are not supported in the source of an Antrea Traceflow. If you know the pod name, select it directly from the **Pod** drop-down menu. Otherwise, filter the list of pods by doing these steps: 1. Enter the first few characters of the node name, or select a value from the **Node** drop-down menu. 2. Enter the first few characters of the namespace, or select a value from the **Namespace** drop-down menu.
Destination	1. Select the type of destination: **Pod**, **Service**, **IP Address**. 2. If you select **Pod** or **Service** as the destination, use the **Node** and **Namespace** drop-down menus to filter the list of pods or services. 3. If you select **IP Address** as the destination, enter an IP address.

6. Click **Trace**.

The Traceflow observations are displayed in a tabular format. For each observation, the table shows the following information.

Observation Type This column takes the following values.

Observation Type	Description
Delivered	The packet is delivered to destination Pod or Service properly.
Dropped	The packed is dropped by a network policy.
Received	The packet is received from another node in the Kubernetes cluster.
Forwarded	The packet is forwarded to the next logical node or a Kubernetes cluster object.

Component This column shows the components that the test packet had encountered on its path from the source to the destination. Sample component values are: IngressRule, EgressRule, SpoofGuard, Classification, Output, and so on.

Click the component name to view more information in a pop-up window.

Timestamp The date and time for each observation.

Workaround to Trace the Packet Flow Between an Antrea Kubernetes Cluster and an NSX Overlay Network

In NSX 4.1, Antrea Traceflow does not support tracing the path of a packet that travels between Antrea Kubernetes cluster and NSX overlay network. However, some workaround is possible, as explained in this documentation.

Before following the workflow instructions in this documentation, read this important note carefully to ensure that the configuration of the **Protocol Type** and **Destination Port** fields are done correctly in the NSXTraceflow and Antrea Traceflow sessions.

> **Important:**
> By default, the **Protocol Type** of a Traceflow session is set to `ICMP`. When you change it `TCP`, the **Destination Port** is set to `0` by default. Remember to specify the port number of the Kubernetes service or the NSX Load Balancer virtual server. This is because the K8s service and NSX Load Balancer virtual server listen on a specific TCP port. In short, you must specify all the information (IP address, Protocol Type, and Destination Port) accurately in the Traceflow session to ensure that the injected packet is processed by the K8s service or the NSX Load Balancer.

Trace the Path of a Packet from NSX VM to K8s Pod

If you want to trace the path of a packet from a VM in an NSX overlay network to a pod workload in an Antrea Kubernetes cluster, do the following steps:

1. Navigate to **Plan & Troubleshoot** > **Traffic Analysis**, and in the **Traceflow** card, click **Get Started**.
2. In the **Source** area, select **Virtual Machine** from the **Type** drop-down menu.
3. In the **Destination** area, select **IP-Mac**. Click the **Layer 3** option, and then enter any one of these IP addresses:
   - Kubernetes Ingress IP address
   - LoadBalancer Service IP address
   - Node IP address
   - Pod IP address (if pod IP is routable on the underlying network)

NSX Traceflow will trace the path of the packet till it leaves the NSX overlay network.

Currently, the following components are not suported in the **Source** of an Antrea Traceflow:

- Kubernetes Service (LoadBalancer, NodePort)
- Kubernetes Gateway
- Kubernetes Ingress

Therefore, you cannot trace the path of the packet after it reaches the Antrea Kubernetes cluster.

Trace the Path of a Packet from K8s Pod to NSX VM

If you want to trace the path of a packet from a pod in an Antrea Kubernetes cluster to a VM in an NSX overlay network, do the following steps:

1. Navigate to **Plan & Troubleshoot** > **Traffic Analysis**, and in the **Traceflow** card, click **Get Started**.
2. Click the **Antrea Traceflow** tab.
3. In the **Source** area, select **Pod** from the **Type** drop-down menu.
4. In the **Destination** area, select **IP address** from the **Type** drop-down menu, and then enter the IP address of the NSX Load Balancer VM.
   Antrea Traceflow will trace the path of the packet till it leaves the Antrea Kubernetes cluster.
   As explained in the important note earlier in this documentation, esnure that you all the information (IP address, Protocol Type, and Destination Port) accurately in the Antrea Traceflow session
5. Now, you use the NSXTraceflow feature to trace the packet from a tier-0 gateway interface to the destination VM in the NSX overlay network. Configure the **Source** in the NSXTraceflow as follows:
   – **Type**: Port/Interface
   – **Attachment**: Edge Uplink
   – **Port**: Select the tier-0 interface from the drop-down menu.
   Configure the **Destination** in the NSXTraceflow as follows:

   - **Type**: IP-Mac
   - **Layer**: Layer-3 (Network Layer)
   - **IP Address**: Enter the IP address of the NSX Load Balancer VM.

   The trace result will display how the NSX Load Balancer chooses a VM as the destination, and what path the packet has taken from the NSX Load Balancer to the destination VM.

Antrea Groups

You can create Antrea groups and use them as sources or destinations in distributed firewall policies to protect traffic between pods within an Antrea Kubernetes cluster.

> **Note:**
> In a multi-tenant NSX environment, Antrea groups are currently not supported under projects. You can add Antrea groups only in the **Default** view (default space) of the NSX environment.

Antrea groups feature is supported only when your NSX environment has one or more Antrea Kubernetes clusters registered to it. When registered Antrea Kubernetes clusters are detected, NSX Manager shows a separate group type called **Antrea** on the **Add Group** page of the UI. You must select this group type to add Antrea groups.

If your goal is to protect traffic between Antrea Kubernetes clusters and VMs in the NSX overlay network, you must create **Generic** groups with Kubernetes member types in dynamic membership criteria and use these groups in firewall rules. For more information, see Generic Groups with Kubernetes Member Types in Dynamic Membership Criteria.

An Antrea group can include static IP addresses, membership criteria, or both. IP addresses can be pod IP or service IP addresses.

When an Antrea group contains membership criteria, the effective members computed by that membership criteria can only be pods.

> **Note:**
>
> - Effective members are computed for Antrea groups only when the Antrea groups are used in distributed firewall rules.
>   When you add Antrea groups with membership criteria, but do not use these groups in any of the distributed firewall rules, the effective members of these Antrea groups are not computed or evaluated in NSX. In other words, the **Effective Members** page of these Antrea groups is empty.
> - When you add static IP addresses in Antrea groups, effective members are currently not displayed in the UI, regardless of whether the groups are used in distributed firewall rules.

To add membership criteria in an Antrea group, the following member types are currently supported:

- Namespace
- Service
- Pod

Overview of Membership Criteria

You can add Antrea groups with a single membership criterion or multiple criteria. A membership criterion consists of one or multiple conditions. A condition in a membership criterion consists of the following properties:

- Member type
- Either name of the member type or a tag that is attached to the member type
- Tag operator and value (only when tag is used)
- Scope operator and value (only when tag is used)

The conditions in a membership criterion can use the same member type or a mix of different member types. For example, if a membership criterion consists of three conditions, the first two conditions can be based on Pod, whereas the third condition can be based on Namespace. However, some restrictions exist for adding multiple conditions in a membership criterion. See the "Supported and Unsupported Features" section later in this topic.

By default, NSX uses the logical AND operator after each condition in a membership criterion. Other logical operators are not supported to join conditions in a membership criterion.

Examples:

Membership Criteria	Description
Criterion 1: Pod Tag Equals App Scope Equals Servers	Membership criterion consists of only a single condition that is based on Pod. Multiple conditions are not used. The effective members of this Antrea group include all pods with the App tag.
Criterion 2: Pod Tag Equals App Scope Equals Servers Pod Tag Equals DB Scope Equals Servers Pod Tag Equals Web Scope Equals Servers	Membership criterion consists of three conditions. All conditions in the criterion are based on Pod. The effective members of this Antrea group include all pods with the App, DB, and Web tags.
Criterion 3: Namespace Name Equals Production Service Name Equals Cache	Membership criterion consists of two conditions with a mix of Namespace and Service. The effective members of this Antrea group include all pods that are associated with the Cache Service in the Production Namespace.

Joining Membership Criteria with OR, AND Operators

An Antrea group supports multiple membership criteria. To join the criteria, OR and AND operators are available. By default, NSX selects the OR operator to join multiple criteria. AND operator is supported between two criteria only when:

- Both criteria use the same member type.
- Both criteria use a single condition.

Examples:

- Criterion 1, Criterion 2, and Criterion 3 are all based on Pod, and they do not contain multiple conditions. In this case, Criterion 1 and Criterion 2 can be joined with either OR or AND operator. Similarly, Criterion 2 and Criterion 3 can also be joined with either OR or AND operator.
- Criterion 1 is based on Pod, whereas Criterion 2 uses two conditions: one based on Service and the other based on Namespace. In this case, only OR operator is supported for joining Criterion 1 and 2. AND operator is not allowed.
- Criterion 1 and Criterion 2 are based on Pod, whereas Criterion 3 uses two conditions: one based on Service and the other based on Namespace. In this case, Criterion 1 and Criterion 2 can be joined with either AND or OR operator. However, Criterion 2 and Criterion 3 can be joined only with OR operator. AND operator is not allowed.

Supported and Unsupported Features

The following table lists the member types, tag operator, and scope operator that are supported for adding membership criteria in Antrea groups.

Member Type	Object Attribute	Tag Operator	Scope Operator	Example Criteria
Namespace	Name	Equals	Not applicable	Namespace Name Equals Production
Namespace	Tag	Equals Not Equals	Equals	Namespace Tag Equals DB Scope Equals Servers
Service	Name	Not supported	Not supported	Service Name Equals Cache
Pod	Tag	Equals Not Equals	Equals	Pod Tag Equals App Scope Equals Servers

- The following tag operators are currently not supported for Namespace and Pod member types:
  - Contains
  - Starts With
  - Ends With
- In a membership criterion, a condition based on Service must be combined with a condition based on the Name attribute of a Namespace. In other words, a criterion with only the Service member type is not allowed.

  Example:

Supported	Not Supported
Criterion: Service Name Equals My-Service Namespace Name Equals Staging	Criterion: Service Name Equals My-Service

- In a membership criterion, a condition based on Service cannot be combined with a condition based on Pod. However, you can add conditions based on Service and Pod in two separate membership criteria and join them with an OR operator.

  Example:

Supported	Not Supported
Criterion 1: Service Name Equals My-Service OR Criterion 2: Pod Tag Equals DB Scope Equals Servers	Criterion: Service Name Equals My-Service Pod Tag Equals DB Scope Equals Servers

- For adding members statically in an Antrea group, only IP addresses are supported. Namespace, Pod, and Service cannot be added statically as members in an Antrea group.
- When you are adding an Antrea group, NSX shows an information message if you try to change the group type from **Antrea** to **Generic**, or from **Antrea** to **IP Addresses Only**. On confirming the change, all the membership criteria in the group are lost. Only the IP addresses are retained in the group.

  After an Antrea group is realized (saved) in NSX, you cannot change the group type. The **Generic** and **IP Addresses Only** group types are dimmed.

Workaround for Kubernetes Version = 1.20

Antrea group criterion "Namespace Name Equals `Value`" works with Kubernetes version = 1.21.

Kubernetes 1.21 or later automatically adds a special label to all namespaces, and criterion "Namespace Name Equals `Value`" internally uses this special label. However, for Kubernetes version = 1.20, a workaround is required. You must register the Antrea Controller webhook on namespace creating events. When the Antrea Controller webhook is called, Antrea Controller adds a special label to the new namespace, so criterion "Namespace Name Equals `Value`" can use this label.

Note:
Antrea Controller webhook is effective only for new namespaces that you create after registering the webhook. In other words, existing namespaces, such as `kube-system` and `default` do not get the special label, and criterion "Namespace Name Equals *`Value`*" does not work with these namespaces.

Add an Antrea Group

You can add static IP addresses, membership criteria, or both in Antrea groups, and then use these groups as the source or destination of the distributed firewall policies that you create to protect traffic within an Antrea Kubernetes cluster.

At least one Antrea Kubernetes cluster is registered to NSX.

1. From your browser, log in to an NSX Manager at `https://`*`nsx-manager-ip-address`*.
2. Navigate to **Inventory** > **Groups**.

   Note:
   NSX Manager UI fetches the information about registered Antrea Kubernetes clusters when you start the NSX Manager application in the browser. If the application UI is already open, it does not fetch the Antrea Kubernetes cluster registration information automatically. This behavior is expected and per the current UI design. If you have registered the first Antrea Kubernetes cluster after the NSX Manager application is opened, ensure that you refresh the browser after navigating to the **Groups** page. A manual refresh ensures that the **Antrea** group type option is visible in the UI when you reach step 5 of this procedure.

   This manual browser refresh is required only once, and not every time after a new Antrea Kubernetes cluster is registered to NSX.
3. Click **Add Group**.
4. Enter a name and optionally a description for the group.
5. Click **Set** and select **Antrea** as the group type.

   An Antrea group can include membership criteria, static IP addresses, or both. Depending on your requirements, perform steps 6 or 7 or both.
6. To add a membership criterion, click **Add Criterion**.
   a) In the **Criterion** pane, select the member type on which you want to define the condition.

      The supported member type are: Namespace, Service, and Pod.
   b) Specify the properties of the condition, such as Name or Tag, Tag operator, Scope operator, as required.
   c) Optional: To add more than one condition in a membership criterion, click the plus icon in the upper-right corner of the **Criterion** pane, and define the properties of the condition.

      In a membership criterion, NSX joins all the conditions with the AND operator, by default. OR operator is not supported.
   d) Optional: To add multiple criteria, click **Add Criterion** again.

      To join membership criteria, AND and OR operators are available. By default, NSX selects the OR operator to join two criteria. AND operator is supported between two criteria only when:

      - Both criteria use the same member type.
      - Both criteria use a single condition.

      For more information about what is supported and not supported for adding membership criteria, see Antrea Groups.
7. To add static IP addresses in the group, click **IP Addresses**, and enter IP values in the text box.

   If you want to import IP values from a TXT or a CSV file, click **Actions** > **Import** . The values in the file must be separated with commas. The allowed values are IP addresses, IP ranges, or IP addresses in a CIDR format. You can also do a combination of both actions. That is, enter values in the text box and import values from a file. However, the total number of IP values in the text box must not exceed the maximum limit that is displayed on the **IP Addresses** tab.

8. Click **Apply**, and then click **Save**.

The Antrea group is saved in NSX and the status changes to Success.

Note:

- Effective members are computed for Antrea groups only when the Antrea groups are used in distributed firewall rules.
  When you add Antrea groups with membership criteria, but do not use these groups in any of the distributed firewall rules, the effective members of these Antrea groups are not computed or evaluated in NSX. In other words, the **Effective Members** page of these Antrea groups is empty.
- When you add static IP addresses in Antrea groups, effective members are currently not displayed in the UI, regardless of whether the groups are used in distributed firewall rules.

Add an Antrea Group Based on Pods

Assume that you want to add an Antrea group that contains all pods running the Revenue, Sales, and Metrics financial applications across all the namespaces in the Antrea Kubernetes cluster.

Consider that the following tags are attached to pods in the Antrea Kubernetes cluster.

Tag	Scope
RevenueApp	Finance
SalesApp	Finance
MetricsApp	Finance

Create a membership criterion with three conditions based on the Pod member type as follows:

Criterion:

Pod Tag Equals RevenueApp Scope Equals Finance

Pod Tag Equals SalesApp Scope Equals Finance

Pod Tag Equals MetricsApp Scope Equals Finance

By default, NSX uses the AND operator after each condition. When this Antrea group is used in a distributed firewall rule, the effective pod members for this group are computed.

After the distributed firewall policy is realized, go to the **Add Group** page. Click **View Members** for this Antrea group, and verify that the effective pod members are displayed on the **Effective Members** page.

Generic Groups with Kubernetes Member Types in Dynamic Membership Criteria

You can create groups with Kubernetes member types (resources) in dynamic membership criteria to match traffic entering into or leaving from Antrea Kubernetes clusters.

You can then use these generic groups in distributed firewall rules or gateway firewall rules to secure traffic between VMs in the NSX environment and pods in Antrea Kubernetes clusters.

This feature requires Antrea- NSX Interworking version that is available with VMware Container Networking™ with Antrea™ 1.6.0 or later. See the 1.6.0 release notes.

This documentation uses the phrase "Kubernetes member types" to refer to "Kubernetes resources" that you can use to configure dynamic membership criteria.

Currently, generic groups with Kubernetes member types support only dynamic membership criteria. You cannot add Kubernetes member types statically in a generic group definition.

Note:
To evaluate the effective members of a generic group that contains dynamic membership criteria defined using Kubernetes member types, NSX considers the resource inventory that is reported by Kubernetes clusters

with Antrea CNI. The resource inventory that is reported by Kubernetes clusters with NCP CNI is ignored for evaluating the effective group members.

Kubernetes Member Types in Membership Criteria

In a generic group, Kubernetes member types are available on the **Membership Criteria** page only when at least one Antrea Kubernetes cluster is registered to your NSX environment.

Note:
In a multi-tenant NSX environment, Kubernetes cluster resources are not exposed to the project inventory. Therefore, inside a project, you cannot create generic groups with Kubernetes member types in dynamic membership criteria.

The following table lists the Kubernetes member types that are available in the **Default** view of NSX 4.1 onwards to add dynamic membership criteria in a generic group. Each Kubernetes member type belongs to either cluster scope or namespace scope.

Member Type	Scope
Kubernetes Cluster	Cluster
Kubernetes Namespace	Namespace
Kubernetes Node	Cluster
Kubernetes Service	Namespace
Kubernetes Ingress	Namespace
Kubernetes Gateway	Namespace
Antrea Egress	Cluster
Antrea IP Pool	Cluster

Naming Conventions Used for Conditions With Kubernetes Member Types

The following table explains the naming conventions that are used in this documentation to represent the different conditions that you can add in dynamic membership criteria, which are based on Kubernetes member types.

Naming Convention for Condition	Meaning
Kubernetes Cluster condition	Conditions in the dynamic membership criteria are based on the Kubernetes Cluster member type.
Kubernetes Namespace condition	Conditions in the dynamic membership criteria are based on the Kubernetes Namespace member type.
Resource condition	Conditions in the dynamic membership criterion are based on any of these Kubernetes member types: • Kubernetes Service • Kubernetes Ingress • Kubernetes Gateway • Antrea Egress • Antrea IP Pool • Kubernetes Node
Cluster-scoped resource condition	Conditions in the dynamic membership criteria are based on any of these Kubernetes member types: • Antrea Egress • Antrea IP Pool • Kubernetes Node

Naming Convention for Condition	Meaning
Namespace-scoped resource condition	Conditions in the dynamic membership criteria are based on any of these Kubernetes member types: • Kubernetes Service • Kubernetes Ingress • Kubernetes Gateway

Overview of Membership Criteria with Kubernetes Member Types

A criterion can have one or more conditions. The conditions can use the same Kubernetes member type or a mix of different Kubernetes member types. However, some restrictions apply to adding multiple conditions with mixed member types in a membership criterion. See the *Restrictions for Using Kubernetes Member Types in a Criterion* section later in this documentation.

By default, NSX uses the logical AND operator after each condition in a membership criterion. Other logical operators are not supported to join the conditions in a membership criterion.

To join criteria, OR and AND operators are available. By default, NSX selects the OR operator to join two criteria. AND operator is supported between two criteria only when:

- Both criteria use the same Kubernetes member type.
- Both criteria use a single condition.

The following restrictions apply to adding multiple conditions:

- A maximum of five conditions with the same Kubernetes member type is supported in a single membership criterion. For example, in a criterion, you can add a maximum of five Kubernetes Service conditions.
- A maximum of 15 conditions with mixed Kubernetes member types are supported in a single membership criterion. For example, in a criterion, you can add a maximum of 15 conditions with a mix of Kubernetes Namespace conditions and Kubernetes Ingress conditions.
- A maximum of 35 conditions with mixed member types are supported in a generic group.

A group can have a maximum of five membership criteria. However, the total number of criteria that you can add in a generic group is determined by the number of conditions in each criterion. See the following examples.

Example 1

A generic group with three membership criteria and a total of 35 conditions:

- Criterion 1 has 15 conditions with mixed member types.
- Criterion 2 has 15 conditions with mixed member types.
- Criterion 3 has 5 conditions with the same member type.

Example 2

A generic group with four membership criteria and a total of 35 conditions:

- Criterion 1 has 15 conditions with mixed member types.
- Criterion 2 has 14 conditions with mixed member types.
- Criterion 3 has four conditions with the same member type.
- Criterion 4 has two conditions with the same member type.

Example 3

A generic group with five membership criteria and a total of 22 conditions:

- Criterion 1 has 10 conditions with mixed member types.
- Criterion 2 has three conditions with the same member type.
- Criterion 3 has four conditions with the same member type.
- Criterion 4 has three conditions with the same member type.
- Criterion 5 has two conditions with mixed member types.

Because this group has reached the limit of five criteria, you cannot add another membership criterion. However, you can add more conditions, if required, in any of the five criteria until you don't exceed the following upper limits mentioned earlier:

- A maximum of five conditions with the same member type in a single criterion.
- A maximum of 15 conditions with mixed member types in a single criterion.
- A total of 35 conditions in the generic group.

Restrictions for Using Kubernetes Member Types in a Criterion

The following table provides a high-level summary of the restrictions or validations that apply to using Kubernetes member types in a single membership criterion. For examples of validations, see the *Validations for Dynamic Grouping with Kubernetes Member Types* section later in this documentation.

Member Type	Restrictions for Using the Member Type in a Criterion	Supported Attributes	Tag Operator	Scope Operator
Kubernetes Cluster	Cannot be used alone in a criterion. Only one cluster condition is allowed in a criterion. Must be mixed with at least one Kubernetes resource condition. Optionally, can be mixed with Kubernetes Namespace conditions and Kubernetes resource conditions.	Name	Not Supported	Not Supported
Kubernetes Namespace	Cannot be used alone in a criterion. Cannot be mixed with cluster-scoped resource conditions. Must be mixed with namespace-scoped resource conditions. Optionally, can be mixed with a Kubernetes Cluster condition.	Name Tag	Equals - one tag can be selected	Equals
Antrea Egress	Can be used alone in a criterion. Optionally, can be mixed with a Kubernetes Cluster condition.	Name Tag	Equals - one tag can be selected	Equals
Antrea IP Pool	Can be used alone in a criterion. Optionally, can be mixed with a Kubernetes Cluster condition.	Name Tag	Equals - one tag can be selected	Equals
Kubernetes Ingress	Can be used alone in a criterion. Optionally, can be mixed with only Kubernetes Namespace conditions or only Kubernetes Cluster condition, or both.	Name Tag	Equals - one tag can be selected	Equals
Kubernetes Gateway	Can be used alone in a criterion. Optionally, can be mixed with only Kubernetes Namespace conditions or only Kubernetes Cluster condition, or both.	Name Tag	Equals - one tag can be selected	Equals

Member Type	Restrictions for Using the Member Type in a Criterion	Supported Attributes	Tag Operator	Scope Operator
Kubernetes Service	Can be used alone in a criterion. Optionally, can be mixed with only Kubernetes Namespace conditions or only Kubernetes Cluster condition, or both.	Name Tag	Equals - one tag can be selected	Equals
Kubernetes Node	Can be used alone in a criterion. Only a single node condition is allowed. Optionally, can be mixed with a Kubernetes Cluster condition. Cannot be mixed with Kubernetes Namespace conditions or Kubernetes resource conditions.	IP Address Pod CIDR	Not supported See Note after this table.	Not supported

Note:
When you create a generic group with Kubernetes Node member type by using the API, only the Matches operator is allowed. This operator can take only the * value. The * wildcard character will match all the nodes in the K8s cluster (if the Kubernetes Node condition is mixed with a Kubernetes Cluster condition), or it will match nodes across all clusters (if the Kubernetes Node condition is used alone).

Validations for Dynamic Grouping with Kubernetes Member Types

Validation 1

A membership criterion can have a maximum of one Kubernetes Cluster condition. To match a single Kubernetes cluster by name, use the Equals operator and enter the cluster name.

Note:
Kubernetes cluster name must be unique.

If you want to match multiple clusters in a single Kubernetes Cluster condition, you can use one of these operators:

- In
- Starts With
- Ends With

Example:

Matches Single K8s Cluster	Matches Multiple K8s Clusters
Criterion: Kubernetes Cluster Name Equals ClusterA	Criterion: Kubernetes Cluster Name In ClusterA,ClusterB,ClusterC A maximum of five comma-separated values are allowed. The values must not be separated with spaces.

Validation 2

A membership criterion with a Kubernetes Cluster condition can be mixed with any one of the Kubernetes resource conditions. If you add Kubernetes Namespace conditions too in the same criterion, then resource conditions must be limited only to namespace-scoped resource conditions.

Examples:

- A membership criterion with a Kubernetes Cluster condition and three Kubernetes Ingress conditions is valid.

- A membership criterion with a Kubernetes Cluster condition and two Antrea Egress conditions is valid.
- A membership criterion with a Kubernetes Cluster condition and three Antrea IP Pool conditions is valid.
- A membership criterion with a Kubernetes Cluster condition, one Kubernetes Namespace condition, and three Kubernetes Gateway conditions is valid.

Validation 3

A membership criterion must include at least one Kubernetes resource condition. A membership criterion is invalid if it contains:

- Only Kubernetes Cluster condition
- Only Kubernetes Namespace condition
- Only Kubernetes Cluster condition and Kubernetes Namespace condition

Examples:

- A membership criterion with one Kubernetes Cluster condition, one Kubernetes Namespace condition, and one Kubernetes Ingress condition is valid.
- A membership criterion with one Kubernetes Cluster condition, two Kubernetes Namespace conditions, and three Kubernetes Ingress conditions is valid.

Validation 4

A membership criterion can have only a single Kubernetes Node condition. Optionally, a Kubernetes Node condition can be mixed with only a Kubernetes Cluster condition.

A Kubernetes Node condition cannot be mixed with Kubernetes Namespace conditions or Kubernetes resource conditions.

A membership criterion with only a Kubernetes Node condition can stand alone. However, a group with only a Kubernetes Node condition will match nodes of all Antrea Kubernetes clusters that are registered to NSX.

Tag operator and Scope operator are currently not supported for the Kubernetes Node condition.

Kubernetes Node condition supports the following two properties.

Property	Description
IP Address	Matches the internal IP address of all nodes of specified Antrea Kubernetes clusters.
Pod CIDR	Matches the Pod CIDR of all nodes of specified Antrea Kubernetes clusters.

Validation 5

If a membership criterion includes Kubernetes Cluster condition and Kubernetes Namespace conditions, then it must include at least one namespace-scoped Kubernetes resource condition. You cannot mix any of the following cluster-scoped Kubernetes resource conditions in the same criterion:

- Antrea Egress
- Antrea IP Pool
- Kubernetes Node

Examples:

- A membership criterion with one Kubernetes Cluster condition, two Kubernetes Namespace conditions, and Kubernetes Gateway conditions is valid.
- A membership criterion with one Kubernetes Cluster condition, four Kubernetes Namespace conditions, and three Kubernetes Service conditions is valid.
- A membership criterion with one Kubernetes Cluster condition, one Kubernetes Namespace condition, and one Antrea Egress condition is invalid because Antrea Egress is a cluster-scoped resource.

Validation 6

A membership criterion must include at least one Kubernetes resource condition. A resource condition can stand alone in a criterion. However, if you add multiple resource conditions in a criterion, then all resource conditions must be of the same member type.

> **Note:**
> Kubernetes Cluster condition and Kubernetes Namespace conditions are used for defining the scope of a criterion. They are not Kubernetes resource conditions, and hence they are not limited by this validation rule.

Examples:

- A membership criterion with five Kubernetes Service conditions is valid.

- A membership criterion with one Kubernetes Cluster condition, three Kubernetes Namespace conditions, and four Kubernetes Service conditions is valid.
- A membership criterion with one Kubernetes Cluster condition, three Kubernetes Namespace conditions, four Kubernetes Service conditions, and three Kubernetes Ingress conditions is invalid. The reason is that you mixed resource conditions of two different member types (Kubernetes Service and Kubernetes Ingress) in the same criterion.
  However, you can create separate criteria with a resource condition based on a single member type and then join both criteria with an OR operator, as shown below:
  Criterion 1:
  One Kubernetes Cluster condition + three Kubernetes Namespace conditions + four Kubernetes Service conditions
  OR
  Criterion 2:
  One Kubernetes Cluster condition + three Kubernetes Namespace conditions + three Kubernetes Ingress conditions

Validation 7

In a single membership criterion, conditions based on NSX member types cannot be mixed with conditions based on Kubernetes member types. However, you can have a group with one criterion based on only NSX members types, and other criterion based on only Kubernetes member types, and then join both criteria with an OR operator.

Example:

Valid	Invalid
Criterion 1: Virtual Machine conditions OR Criterion 2: Kubernetes Cluster condition + Kubernetes Gateway conditions	Criterion: NSX Segment condition + Segment Port condition AND Kubernetes Cluster condition + Kubernetes Gateway conditions

Effective Members for Conditions with Kubernetes Member Types

From an NSX point of view, the effective members for groups with Kubernetes member types are either discrete IP addresses, IP ranges, or list of IP addresses and ports.

The following table provides some examples.

Example Group Definition	Use Case	Description	Effective Members
Kubernetes Cluster condition AND Kubernetes Node condition (based on IP address)	Traffic from Antrea Kubernetes cluster to NSX	Matches all node IP addresses in specified clusters	IP addresses
Kubernetes Cluster condition AND Kubernetes Node condition (based on Pod CIDR)	Traffic from Antrea Kubernetes cluster to NSX	Matches Pod CIDRs of all nodes in specified clusters	Pod CIDRs
Kubernetes Cluster condition AND Antrea Egress condition (based on name)	Traffic from Antrea Kubernetes cluster to NSX	Matches Egress by name in specified clusters	Egress IP addresses
Kubernetes Cluster condition AND Antrea Egress condition (based on tag) AND More Antrea Egress conditions (based on tag)	Traffic from Antrea Kubernetes cluster to NSX	Matches Egress by tags in specified clusters.	Egress IP addresses

Example Group Definition	Use Case	Description	Effective Members
Kubernetes Cluster condition AND Antrea IP Pool condition (based on name)	Traffic from Antrea Kubernetes cluster to NSX Traffic from NSX to Antrea Kubernetes cluster	Matches IP pool by name in specified clusters	IP ranges
Kubernetes Cluster condition AND Antrea IP Pool condition (based on tag) AND More Antrea IP Pool conditions (based on tags)	Traffic from Antrea Kubernetes cluster to NSX Traffic from NSX to Antrea Kubernetes cluster	Matches IP pools by tags in specified clusters	IP ranges
Kubernetes Cluster condition AND Kubernetes Namespace condition AND Kubernetes Ingress condition (based on name)	Traffic from NSX to Antrea Kubernetes cluster	Matches Ingress by name in specified clusters and namespace	Ingress IP addresses
Kubernetes Cluster condition AND Kubernetes Namespace condition AND Kubernetes Gateway condition (based on name)	Traffic from NSX to Antrea Kubernetes cluster	Matches Gateway by name in specified clusters and namespace	Gateway IP addresses
Kubernetes Cluster condition AND Kubernetes Namespace condition AND Kubernetes Ingress condition (based on tag) AND More Kubernetes Ingress conditions (based on tags)	Traffic from NSX to Antrea Kubernetes cluster	Matches Ingress by tags in specified clusters and namespace	Ingress IP addresses
Kubernetes Cluster condition AND Kubernetes Namespace condition AND Kubernetes Gateway condition (based on tag) AND More Kubernetes Gateway conditions (based on tags)	Traffic from NSX to Antrea Kubernetes cluster	Matches Gateway by tags in specified clusters and namespace	Gateway IP addresses
Kubernetes Cluster condition AND Kubernetes Namespace condition AND Kubernetes Service condition (based on tag and type=LoadBalancer) AND More Kubernetes Service conditions (based on tags and type=LoadBalancer)	Traffic from NSX to Antrea Kubernetes cluster	Matches Service (LoadBalancer) by tags in specified clusters and namespace	LoadBalancer Ingress IP addresses
Kubernetes Cluster condition AND Kubernetes Namespace condition AND Kubernetes Service condition (based on name,type=ClusterIP and NodePortLocal feature enabled)	Traffic from NSX to Antrea Kubernetes cluster	Matches Service (ClusterIP with NodePortLocal enabled) by name in specified clusters and namespace.	Node IP addresses, NodePortLocal range

Distributed Firewall Policies for Securing Traffic Within an Antrea Kubernetes Cluster

You can create Antrea groups in your NSX environment and use these groups in the distributed firewall policies (security policies) to secure traffic between pods within an Antrea Kubernetes cluster.

> **Note:**
> In a multi-tenant NSX environment, Antrea groups are currently not supported under projects. Therefore, you cannot create security policies under projects to secure traffic between pods within an Antrea Kubernetes cluster. You must create the security policies in the **Default** view (default space) of the NSX environment.

A VMware vDefend policy can be applied to multiple Antrea Kubernetes clusters. However, the distributed firewall (DFW) policy can secure traffic between pods within a single Antrea Kubernetes cluster. Pod-to-pod traffic between Antrea Kubernetes clusters is currently not protected.

When a vDefend policy is applied to one or more Antrea Kubernetes clusters, the Antrea network plug-in enforces this security policy at the Antrea Controller of each Kubernetes cluster. In other words, the enforcement point of the security policy is the Antrea Controller of each Antrea Kubernetes cluster.

If your goal is to protect traffic between pods in an Antrea Kubernetes cluster and VMs on hosts in the NSX environment, see Firewall Policies for Securing Traffic Between Antrea Kubernetes Clusters and VMs in an NSX Network.

Security Policy Features Supported for Antrea Kubernetes Clusters

- Only Layer 3 and 4 security policies can be applied to Antrea Kubernetes clusters. Rules in the following firewall categories are supported: Emergency, Infrastructure, Environment, and Application.
- Sources, Destinations, and Applied To of a rule can contain only Antrea groups.
- Applied To is supported at both policy level and rule level. If both are specified, Applied To at the policy level takes precedence.
- Services, including raw port and protocol combination, are supported. However, the following constraints apply:
  - Only TCP and UDP services are supported. All other services are not supported.
  - In raw port and protocol combinations, TCP and UDP service types are supported.
  - Only destination ports are supported.
- Policy statistics and rule statistics are supported. Rule statistics are not aggregated for all the Antrea Kubernetes clusters to which the security policy is applied. In other words, rule statistics are displayed for each Antrea Kubernetes cluster.

Security Policy Features Not Supported for Antrea Kubernetes Clusters

- Layer 2 (Ethernet) rules based on MAC addresses are not supported.
- Layer 7 rules based on Context Profiles are not supported. For example, rules based on application ID, FQDN, and so on.
- Antrea groups with IP addresses are not supported in the Applied To of the security policy and firewall rules.
- Time-based scheduling of rules is not supported.
- Antrea groups are not supported in a firewall exclusion list. (**Security** > **Distributed Firewall** > **Actions** > **Exclusion List**).
- Negating or excluding the Antrea groups that you have selected in the sources or destinations of a firewall rule is not supported.
- Identity firewall is not supported.
- Global groups created for an NSX Federated environment cannot be used in security policies that are applied to Antrea Kubernetes clusters.
- Advanced policy configuration does not support the following settings:
  - TCP Strict
  - Stateful

Add a Distributed Firewall Policy to Secure Traffic Within an Antrea Kubernetes Cluster

To secure traffic between pods in an Antrea Kubernetes cluster, you can create distributed firewall policies (security policies) in NSX and apply them to one or more Antrea Kubernetes clusters.

- Antrea Kubernetes clusters are registered to NSX.
- Apply an appropriate security license in your NSX deployment that entitles the system to configure distributed firewall security policies.

> **Note:**
> This documentation uses the term "Antrea Kubernetes cluster" to mean Kubernetes clusters with Antrea CNI. The term "Kubernetes cluster" is a generic term, which represents Tanzu Kubernetes Grid (TKG) clusters with Antrea CNI, OpenShift clusters with Antrea CNI, or do it yourself (DIY) Kubernetes clusters with Antrea CNI.
>
> The UI use the term "Antrea container cluster" for a few UI fields or labels. In the *Procedure* section of this documentation, the term "Antrea container cluster" is retained for those UI fields or labels. For all free-form text, the term "Antrea Kubernetes cluster" is used.

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Click the **Security** tab, and then under **Policy Management**, click **Distributed Firewall**.

   The **Category Specific Rules** page is displayed.

   > **Note:**
   > NSX Manager UI fetches the information about registered Antrea Kubernetes clusters when you start the NSX Manager application in the browser. If the application UI is already open, it does not fetch the Antrea Kubernetes cluster registration information automatically. This behavior is expected and per the current UI design. If you have registered the first Antrea Kubernetes cluster after the NSX Manager application is opened, ensure that you refresh the browser after navigating to the **Category Specific Rules** page. A manual refresh ensures that the Antrea-specific UI elements are visible in the UI when you reach step 4 of this procedure.
   >
   > This manual browser refresh is required only once, and not every time after a new Antrea Kubernetes cluster is registered to NSX.

3. Select the category in which you want to create the security policy.

   Layer 2 (Ethernet) firewall rules based on MAC addresses are currently not supported for Antrea Kubernetes clusters. Categories in NSX correspond to Tiers in Antrea. Security policies are enforced in the Antrea Kubernetes clusters in the following descending order of precedence:

   - Emergency category (highest precedence)
   - Infrastructure category
   - Environment category
   - Application category (lowest precedence)

   Within a category, firewall rules are processed top-down in the order in which the rules are set. Categories provide a means of organizing rules. For instance, multiple user roles (personas) can create security policies without overriding or conflicting the policies of each other. For example, a security administrator can create policies in the Emergency category for specific quarantine or allow rules. An application developer can create policies in the Application category to secure traffic between specific pods in an application. A network administrator can create policies in the Infrastructure category to define access rules for shared services, such as DHCP, DNS, Active Directory, and so on.

4. Click **Add Policy** and specify the policy configuration settings.
   a) Enter a unique name for the policy.
   b) By default, the policy is applied to distributed firewall. Next to **Applied To**, click the edit icon.

      The **Set Applied To** page opens.
   c) Select the **Antrea Container Clusters** option.
   d) Required: Select at least one cluster to decide the span or scope of enforcement of the security policy.

      The span of the policy can either be a single Antrea Kubernetes cluster or multiple Antrea Kubernetes clusters.

e) Optional: Limit the span of the policy by selecting Antrea groups.

When you select Antrea groups in the **Applied To** of a policy, this configuration is used for all the rules in the policy. To specify a different set of Antrea groups for each rule in the policy, skip this step, and specify the **Applied To** while adding the rules in the policy.

Note:
Antrea groups with IP addresses must not be used in the **Applied To** of the policy because NSX cannot compute effective pod members from the IP addresses.

f) Optional: Click the gear icon at the extreme right corner to specify advanced configuration settings of the policy.

For security policies that are applied to Antrea Kubernetes clusters, **TCP Strict** and **Stateful** settings are dimmed. These settings are currently not supported.

Only **Locked** and **Comments** settings are supported. By default, a policy is not locked. To prevent multiple users from making changes to the policy, turn on the **Locked** option.

g) Click **Publish**.

You can add multiple policies and then publish all of them together.

The policy status initially changes to In Progress, and after it is successfully realized in the Antrea Kubernetes clusters, the status changes to Success. If the policy realization fails due to any reason, click the `Failed` status to view the errors in a pop-up window.

5. Select the check box next to the policy name and click **Add Rule**. Enter a rule name.

By default, the **Sources**, **Destinations**, **Services**, and **Applied To** columns of the rule show `Any`.

Note:
Context Profiles are currently not supported for rules that are applied to Antrea Kubernetes clusters.

6. Specify the rule settings.

Antrea data path does the filtering on the **Applied To** group members. Only Groups which can be resolved to Pods can be specified in the **Applied To** field. Setting **Applied To** and **Sources** filters incoming traffic to Pods from the Sources. Setting **Applied To** and **Destinations** filters outgoing traffic from Pods to Destinations.

a) In the **Applied To** column, click the edit icon, and select the Antrea groups to which you want to apply the rule.

If no groups are selected, Any is used.

Note:

- Antrea groups with IP addresses must not be used in the **Applied To** of the rule because NSX cannot compute effective pod members from the IP addresses.
- If you specify Antrea groups in the **Applied To** of both the policy and the rule, the groups in the **Applied To** of the policy take precedence over the groups in the **Applied To** of the rule.

b) In the **Sources** or **Destinations** column, click the edit icon, and select one or more Antrea groups.

The following constraints apply to specifying rule sources and destinations:

- Only Antrea groups can be selected. Groups with NSX members cannot be used in a rule. In other words, a rule cannot contain a mix of Antrea groups and groups of type **Generic**, **IP Addresses Only**.
- When groups are selected in the **Sources** column, the **Destinations** column is not applicable. You can add destinations in the **Applied To** of the rule. In the **Applied To** column, click the edit icon, and select the Antrea groups to which you want to apply the rule. If no groups are selected, **Any** is used.
- When groups are selected in the **Destinations** column, the **Sources** column is not applicable. You can add sources in the **Applied To** of the rule. In the **Applied To** column, click the edit icon, and select the Antrea groups to which you want to apply the rule. If no groups are selected, **Any** is used.

c) Optional: In the **Services** column, click the edit icon, and select the services.

If no services are selected, Any is used.

The following constraints apply to specifying services:

- Only TCP and UDP services are supported. All other services are not supported.
- Raw port and protocol combination supports only TCP and UDP service type.
- Only destination ports are supported. Source ports are not supported.

d) From the **Action** drop-down menu, select one of these options.

Option	Description
Allow	Allows all L3 traffic with the specified source, destination, and protocol to pass through the current firewall context. Packets that match the rule, and are accepted, traverse the Kubernetes cluster as if the firewall is not present.
Drop	Drops packets with the specified source, destination, and protocol. Dropping a packet is a silent action with no notification to the source or destination. Dropping the packet causes the connection to be retried until the retry threshold is reached.
Reject	Rejects packets with the specified source, destination, and protocol. Rejecting a packet is a more graceful way to deny a packet, as it sends a destination unreachable message to the sender. If the protocol is TCP, a TCP RST message is sent. ICMP messages with administratively prohibited code are sent for UDP, ICMP, and other IP connections. A benefit of the reject action is that the sending application is notified after only one attempt that the connection cannot be established.

e) Click the toggle button to turn on or turn off the rule.
By default, the rule is turned on.

f) Optional: Click the gear icon to configure other rule settings.

Rule Setting	Description
Logging	By default, logging is turned off. The firewall logs are included in the Antrea Agent logs. When you create a support bundle request for Antrea Kubernetes cluster, and select the nodes from the cluster, the support bundle includes the Antrea Agent logs for those nodes.
Direction	Refers to the direction of traffic from the perspective of the destination pod. Rule direction is read-only in the following cases, and it cannot be edited: • When sources are specified in the rule, the direction is In. • When destinations are specified in the rule, the direction is Out. Rule direction is editable when the sources and destinations are set to Any. In this case, the default direction is In-Out. However, you can change the direction to In or Out.
Comments	Enter any notes about the rule, if required. These comments are not propagated to the Antrea Kubernetes cluster. Therefore, the rule comments will not appear as annotations in the Antrea Cluster Network Policy specifications.

7. Click **Publish** to push the rules to the Antrea Kubernetes clusters.
You can add multiple rules and then publish them together.

Note:
After the security policy is realized in the Antrea Kubernetes clusters, you cannot edit the **Applied To** of the policy. That is, NSX does not allow you to change the span of the security policy from **Antrea Container Clusters** to **DFW** or **Groups**.

The following results occur in the Antrea Kubernetes clusters:

- Antrea network plug-in creates a cluster network policy corresponding to each distributed firewall policy that is applied to the Antrea Kubernetes clusters.

- If the rules contain sources, corresponding ingress rules are created in the Antrea Cluster Network Policy.
- If the rules contain destinations, corresponding egress rules are created in the Antrea Cluster Network Policy.
- If the rules contain Any-Any configuration, Antrea Controller in the cluster splits the Any-Any rule into two rules: One ingress rule with Any to Any, and another egress rule with Any to Any.

Note:
Antrea network plug-in does not prevent you from updating or deleting the Antrea cluster network policies from the kubectl command line. But, you must avoid doing it. The reason is that the security policies are managed by NSX. Therefore, the Central Control Plane Adapter in the Antrea Kubernetes cluster immediately overwrites the policy changes that are made from the kubectl command line. In other words, NSX is the source of truth for the policies. The changes made to these cluster network policies through the kubectl command line are not displayed in NSX Manager.

Example 1: Filtering traffic from Pods in some Namespaces to an external IP.

1. Create a Group "test-namespaces"
   – Type: Antrea
   – Criteria: Namespace Tag Equals "SomeK8sLabelValue" Scope Equals "SomeK8sLabel"
2. Create a Group "test-ip-addresses"
   – Type: Antrea
   – Criteria: None
   – IP Addresses: 192.168.100.0/24 192.168.200.0/24
3. Create a Security Policy "test-policy"
   – Set the Policy-level Applied-to and select some Antrea Container Clusters.
4. Add a new Rule to the Policy
   – Set the Rule-level Applied-to and select Group "test-namespaces".
   – Set the Destinations and select Group "test-ip-addresses".
   – Set the Action to "Allow"/"Drop"/"Reject".

Example 2: For Pods in Namespace A, only allow traffic from Namespace B, drop traffic originated from elsewhere.

1. Create a Group "namespace-a"
   – Type: Antrea
   – Criteria: Namespace Name Equals "A"
2. Create a Group "namespace-b"
   – Type: Antrea
   – Criteria: Namespace Name Equals "B"
3. Create a Security Policy "allow-from-b"
   – Set the Policy-level Applied-to and select some Antrea Container Clusters.
4. Add a new Rule to the Policy
   – Set the Rule-level Applied-to and select Group "namespace-a".
   – Set Action to "Drop".
   – Click the gear icon in this rule to see advanced settings. Set Direction to "In".
5. Insert a new Rule at the top of the Policy
   – Set the Rule-level Applied-to and select Group "namespace-a".
   – Set the Sources and select Group "namespace-b".
   – Set Action to "Allow".
   – Ensure that this rule is before the above "Drop" rule.

Example 3: For Pods in Namespace B, only allow traffic to Namespace A, drop traffic targeting elsewhere.

1. Create a Group "namespace-a"
   – Type: Antrea
   – Criteria: Namespace Name Equals "A"

2. Create a Group "namespace-b"
   – Type: Antrea
   – Criteria: Namespace Name Equals "B"
3. Create a Security Policy "allow-to-a"
   – Set the Policy-level Applied-to and select some Antrea Container Clusters.
4. Add a new Rule to the Policy
   – Set the Rule-level Applied-to and select Group "namespace-b".
   – Set Action to "Drop".
   – Click the gear icon in this rule to see advanced settings. Set Direction to "Out".
5. Insert a new Rule at the top of the Policy
   – Set the Rule-level Applied-to and select Group "namespace-b".
   – Set the Destinations and select Group "namespace-a".
   – Set Action to "Allow".
   – Ensure that this rule is before the above "Drop" rule.

Comparing Examples 2 and 3

- When example 2 "allow-from-b" is enabled and example 3 "allow-to-a" is disabled:
  – Pods in Namespace A can accept traffic only from Pods in Namespace B.
  – Pods in Namespace B can send traffic to anywhere including to Namespace A.
  – This is because applied-to = Group "namespace-a", so the filtering happens at Pods in Namespace A.
- When example 2 "allow-from-b" is disabled and example 3 "allow-to-a" is enabled:
  – Pods in Namespace A can accept traffic from anywhere including from Namespace B.
  – Pods in Namespace B can send traffic only to Namespace A.
  – This is because applied-to = Group "namespace-b", so the filtering happens at Pods in Namespace B.
- When both example 2 "allow-from-b" and example 3 "allow-to-a" are enabled:
  – Pods in Namespace A can accept traffic only from Pods in Namespace B.
  – Pods in Namespace B can send traffic only to Namespace A.
  – For traffic from Namespace B to A, it will be filtered twice, first by "allow-to-a" when Pods in B send the traffic, then by "allow-from-b" when Pods in A receive the traffic.

After the security policies are successfully realized in the Antrea Kubernetes clusters, you can do the following optional tasks:

- Verify that the Antrea cluster network policies are shown in the Kubernetes clusters. Run the following kubectl command in each Antrea Kubernetes cluster:

  ```
  $ kubectl get acnp
  ```

  Note:
  The `priority` parameter in the Antrea cluster network policies shows a float value. This result is expected. NSX Manager UI does not display the priority of the distributed firewall policies. NSX internally assigns an integer value to the priority of each policy. This integer value is assigned from a large range. But, Antrea network plug-in assigns a smaller float number (absolute value) to the priority of Antrea cluster network policies. Therefore, the NSX priority values are internally normalized to smaller float numbers. However, the order in which you add the policies in a distributed firewall category is preserved for the Antrea cluster network policies.

  You can also view the details of the Antreacluster network policies in the NSX inventory. In NSX Manager, navigate to, **Inventory** > **Containers** > **Clusters**. Expand the cluster name and click the number next to **Cluster Network Policies** to view the details of the policies, including the YAML specifications.
- View policy statistics by using the NSX API:

  ```
  GET https://{nsx-mgr-ip}/api/v1/infra/domains{domain-id}/security-policies/{security-policy-name}/
  statistics?container_cluster_path=/infra/sites/{site-id}/enforcement-points/{enforcement-point-id}/cluster-
  control-planes/{cluster-name}
  ```
- View runtime rule statistics in the UI:
  a. In NSX Manager, navigate to **Security** > **Distributed Firewall**.
  b. Expand the policy name, and then click the graph icon at the extreme right corner of each rule.
  c. Select the Kubernetes cluster from the drop-down menu to view the rule statistics for each Kubernetes cluster.

The statistics of the rule are computed separately for each Kubernetes cluster where the rule is enforced. The statistics are not aggregated for all the Kubernetes clusters and displayed in the UI. The rule statistics are computed every minute.

Example: Add a Distributed Firewall Policy to Secure Traffic Within an Antrea Kubernetes Cluster

In this example, your goal is to create a distributed firewall policy in NSX to secure pod-to-pod traffic in the Enterprise Human Resource application, which is running in a single Antrea Kubernetes cluster.

- Antrea Kubernetes cluster is registered to NSX.
- Apply an appropriate security license in your NSX deployment that entitles the system to configure distributed firewall security policies.

Let us assume that the pod workloads in the Antrea Kubernetes cluster are running Web, App, and Database microservices of the Enterprise Human Resource application. You have added Antrea groups in your NSX environment by using pod-based membership criteria, as shown in the following table.

Antrea Group Name	Membership Criteria
HR-Web	Pod Tag Equals Web Scope Equals HR
HR-App	Pod Tag Equals App Scope Equals HR
HR-DB	Pod Tag Equals DB Scope Equals HR

Your objective is to create a security policy in the Application category with three firewall rules, as follows:

- Allow all traffic from HR-Web group to HR-App group.
- Allow all traffic from HR-App group to HR-DB group.
- Reject all traffic from HR-Web to HR-DB group.

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Click the **Security** tab, and then under **Policy Management**, click **Distributed Firewall**.
   The **Category Specific Rules** page is displayed.
3. Make sure that you are in the **Application** category.
4. Click **Add Policy** and enter a policy name.
   For example, enter `EnterpriseHRPolicy`.
5. In the **Applied To** of the policy, select the Antrea Kubernetes cluster where the pod workloads of the Enterprise Human Resource application are running.
6. Publish the policy.
7. Select the policy name, and click **Add Rule**.
   Configure three firewall rules, as shown in the following table.

Rule Name	Rule ID	Sources	Destinations	Services	Applied To	Action
Web-to-App	1022	HR-Web	N/A	Any	HR-App	Allow
App-to-DB	1023	HR-App	N/A	Any	HR-DB	Allow
Web-to-DB	1024	HR-Web	N/A	Any	HR-DB	Reject

   The rule IDs in the table are only sample values for this example. The rule IDs can vary in your NSX environment.
8. Publish the rules.

When the policy is realized successfully, the following results occur in the Antrea Kubernetes cluster:

- An Antrea cluster network policy (ACNP) is created.

- Rules 1022, 1023, and 1024 are enforced in the Kubernetes cluster in that order.
- For each firewall rule, a corresponding ingress rule is created in the cluster network policy.

Firewall Policies for Securing Traffic Between Antrea Kubernetes Clusters and VMs in an NSX Network

You can create generic groups with Kubernetes member types in dynamic membership criteria to match traffic entering into or leaving from Antrea Kubernetes clusters.

You can then use these generic groups in distributed firewall rules or gateway firewall rules to secure traffic between VMs in the NSX environment and pods in Antrea Kubernetes clusters.

> **Important:**
> In a multi-tenant NSX environment, Kubernetes cluster resources are not exposed to the project inventory. Therefore, inside a project, you cannot create generic groups with Kubernetes member types in dynamic membership criteria. You must create the groups in the **Default** view (default space) of NSX, and then consume them in the distributed firewall rules or gateway firewall rules under the default space.

Prerequisites

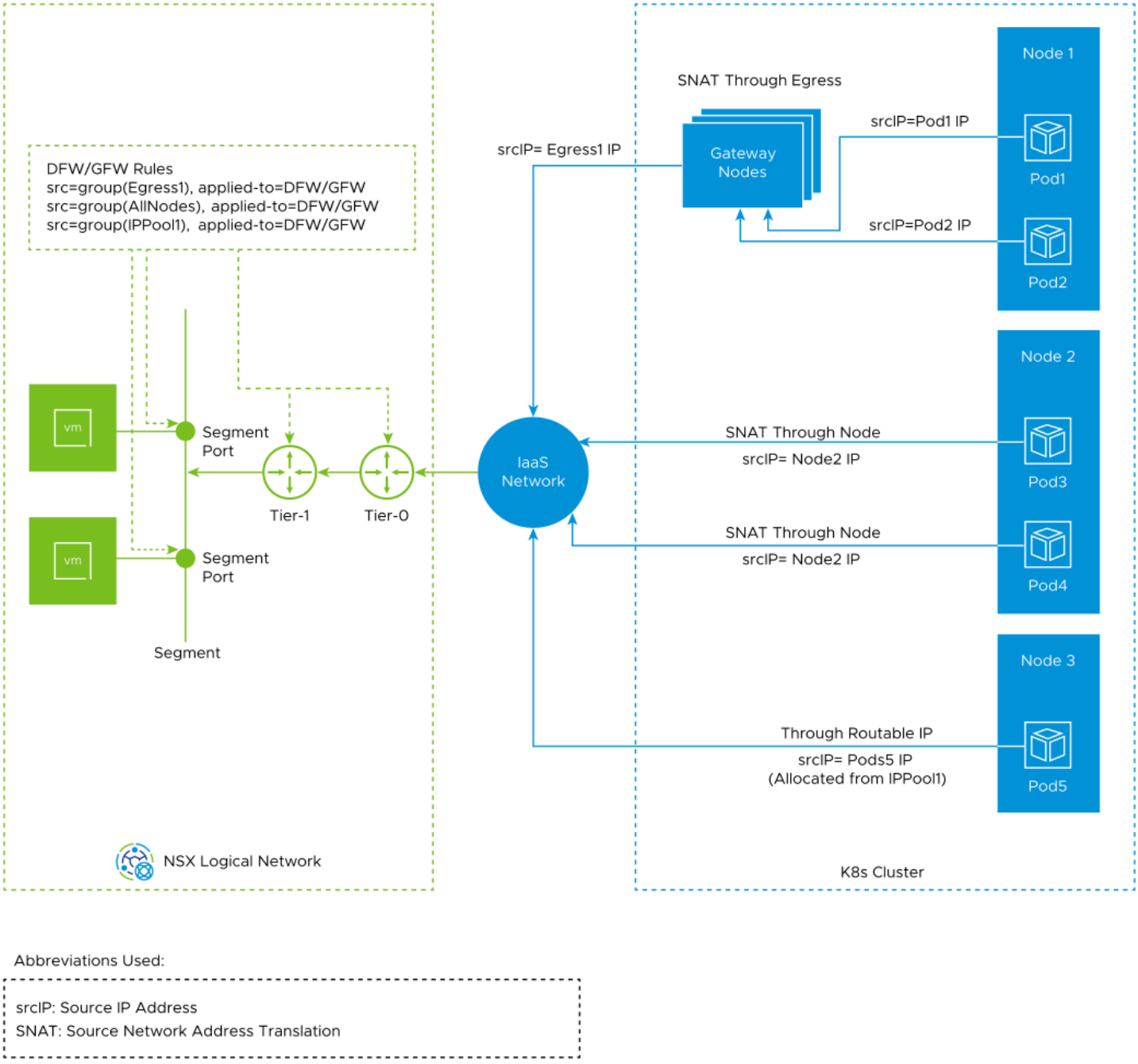
- Antrea Kubernetes clusters are registered to NSX.
- Apply an appropriate security license in your NSX deployment that entitles the system to configure distributed firewall security policies.

Securing Traffic from Antrea Kubernetes Clusters to VMs in NSX

To match traffic from Antrea Kubernetes clusters to NSX, you can add generic groups based on the following Kubernetes member types (resources) in the firewall rules:

- Kubernetes Node
- Antrea Egress
- Antrea IP Pool

The following diagram shows the typical traffic flows from the Antrea Kubernetes cluster to the VMs in the NSX logical network.



Abbreviations Used:

srcIP: Source IP Address
SNAT: Source Network Address Translation

As shown in this traffic flow diagram, to match traffic that is leaving the Antrea Kubernetes cluster, the following scenarios are possible:

Scenario 1: SNAT through Egress (gateway nodes)

You can deploy an Egress resource in your Antrea Kubernetes cluster. In the manifest file of the Egress resource, select the grouping criteria of pods to which the Egress resource applies to. Either you can specify the egress IP manually in the manifest file or Antrea CNI can allocate an egress IP from an ExternalIPPool custom resource.

Important:
The egress IP addresses must be routable on the physical network and they must be reachable from all the nodes in the K8s cluster.

Antrea CNI sends the outgoing traffic from the selected pods to the gateway nodes. The gateway nodes perform a source address translation (SNAT) to translate the source IP (pod IP) to the egress IP.

For example, in the preceding traffic flow diagram, outgoing traffic from pod1 and pod2, which are running on node1, is sent to the gateway nodes. The gateway node translates the pod IP addresses to egress1 IP.

To learn more about Egress resource, see the Egress Guide on the Antrea documentation portal.

The cluster nodes that can be used as gateway nodes depend on the node network topology, which is controlled by your physical network administrator or the IaaS network administrator. The gateway nodes have the following two characteristics:

- The physical network allows the nodes to have a routable IP as a source IP to access the external network.
- Optionally, a physical firewall on the physical gateway is configured to filter outgoing traffic from the gateway nodes.

Consider the following examples:

- Examp1e 1: Assume that your Antrea Kubernetes cluster has three nodes: node1, node2, and node3. Let us say that node1 and node2 in the cluster are special nodes because they are assigned two network interfaces. One interface for K8s cluster communication and another interface for connecting to the external network. Node3 is assigned only one network interface for cluster communication. In this scenario, only node1 and node2 can be used as gateway nodes.
- Example 2: Assume that all three nodes in the K8s cluster are assigned only one network interface, and the network administrator has configured the physical firewall to allow only node-1 and node-2 MAC addresses to visit the Internet. In this scenario, node1 and node2 can be used as gateway nodes.
- Example 3: Assume that all the three nodes can visit the Internet. The network administrator has configured a physical firewall on the physical gateway, and the firewall wants to filter traffic from a specific egress IP. In this scenario, all three nodes can be used as gateway nodes.

The following paragraph explains an implementation-specific limitation:

Any physical host or a VM that is outside the Antrea Kubernetes cluster cannot be used as a gateway node for tunnelling the pod's outgoing traffic. The reason is that Antrea Agent maintains the egress IP addresses, and this agent runs as a pod on each node of the K8s cluster. Therefore, the physical host or a VM that you want to use as a gateway node must be a part of the K8s cluster.

Scenario 2: SNAT through nodes

In this scenario, pods are not explicitly selected in the manifest file of any Egress resource.

The pod's outgoing traffic is masqueraded to the node IP, and then the traffic is sent out to the IaaS network. For example, in the preceding traffic flow diagram, an SNAT rule translates the source IP (pod3 IP, pod4 IP) to the node2 IP, and then the pod's outgoing traffic is sent out to the IaaS network.

Scenario 3: Routable pods bridged directly to the IaaS network

Routable pods mean that pods are assigned routable IP addresses on the underlay network. Usually, pods are assigned private IP addresses from the PodCIDR property, which is configured on the node. When pods want to access a network that is external to the K8s cluster, the traffic requires an SNAT operation to translate the source IP (pod IP) to a routable IP.

If your Antrea Kubernetes cluster is deployed on the NSX overlay network with the Tanzu Kubernetes Grid Service (TKGS), then there is a mechanism for the Tanzu Kubernetes cluster (TKC) nodes to request routable subnets from NSX, and use this routable subnet as the node's PodCIDR property. Then, the pods can get routable IP addresses. To learn more about configuring TKC with a routable pods network, see the *VMware vSphere with Tanzu* documentation.

If your Antrea Kubernetes cluster is not deployed on the NSX overlay network, the Kubernetes administrator can ensure that a routable subnet or a routable IP range is reserved in the underlay network. The administrator can configure the subnet or an IP range in the manifest file of the IPPool custom resource. Then, Antrea CNI can allocate routable IP addresses to the pods.

For example, in the preceding traffic flow diagram, pod5 that is running on node5 gets a routable IP from an IPPool custom resource.

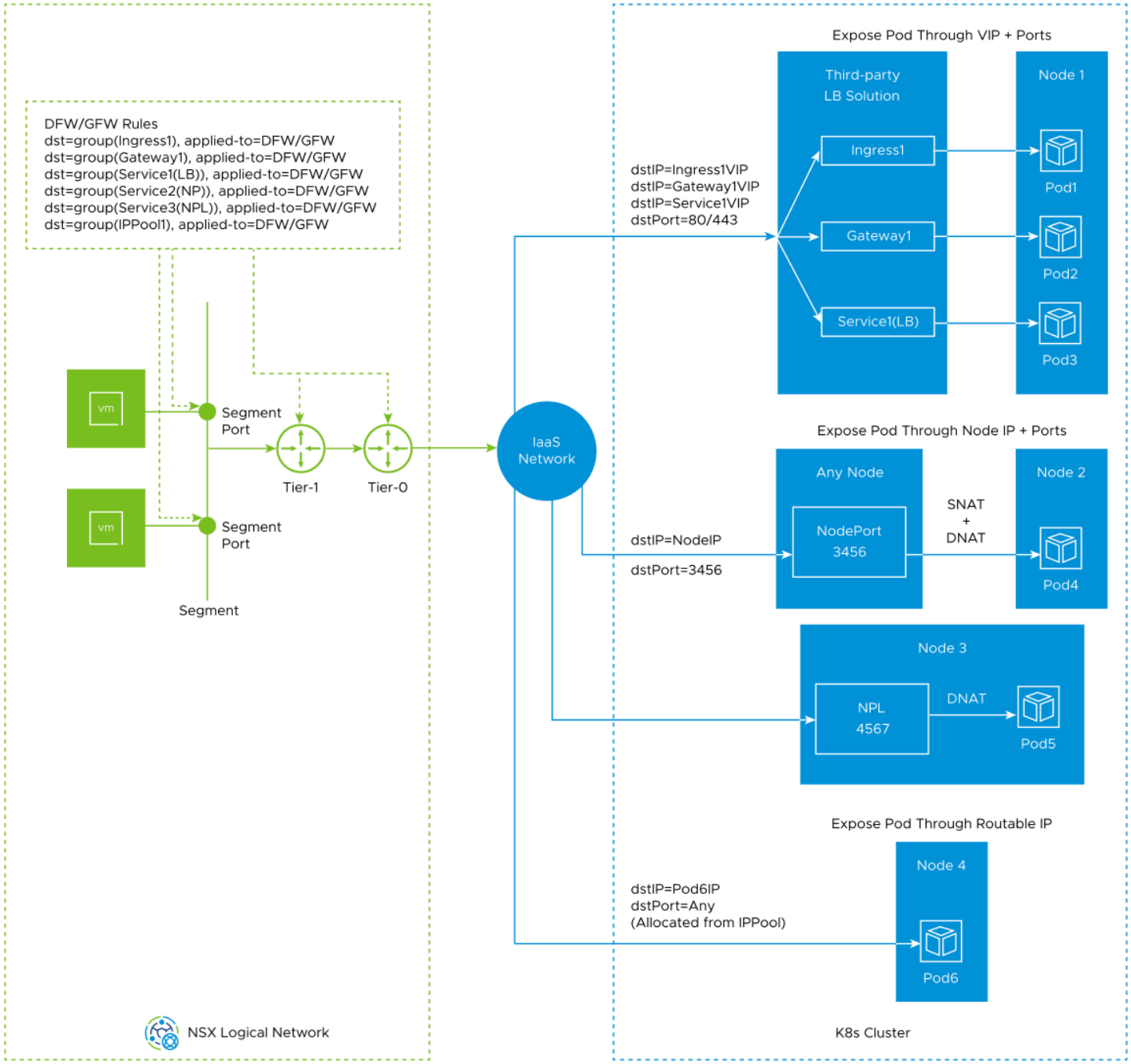
To learn more about IPPool custom resource, see the Antrea IPAM Guide on the Antrea documentation portal.

Securing Traffic from VMs in NSX to Antrea Kubernetes Clusters

To match traffic from NSX to Antrea Kubernetes Clusters, you can add generic groups based on the following Kubernetes member types (resources) in the firewall rules:

- Kubernetes Service
- Kubernetes Ingress
- Kubernetes Gateway
- Antrea IP Pool

The following diagram shows the typical traffic flows from VMs in the NSX logical network to the Antrea Kubernetes cluster.



Abbreviations Used:

dstIP: Destination IP Address
dstPort: Destination Port
VIP: Virtual IP Address
SNAT: Source Network Address Translation
DNAT: Destination Network Address Translation
NPL: NodePortlLocal

As shown in this traffic flow diagram, to match traffic that is entering into the Antrea Kubernetes cluster, the following scenarios are possible:

Scenario 1: Expose pods to external traffic through virtual IP and ports	You can expose pods to external traffic by implementing a third-party load balancer solution with these native Kubernetes resources: • Ingress: It functions as a layer 7 load balancer. An Ingress resource will provide a virtual IP address (VIP) and expose some ports (TCP/UDP). • Gateway: It functions as a layer 4 and layer 7 load balancer. A Gateway resource will provide a VIP and expose some ports (TCP/UDP). • Service of type LoadBalancer: It functions as a layer 4 load balancer. The service will provide a VIP and expose some ports (TCP/UDP). **Note:** Although Ingress and Gateway resources can function as a layer 7 load balancer, for distributed firewall rules and gateway firewall rules in NSX, they appear as layer 3 or layer 4 load balancer by using IP addresses and ports to match the incoming traffic.
Scenario 2: Expose pods to external traffic through node IP and ports	• Kubernetes service of type NodePort: It triggers all the nodes in the K8s cluster to open a port on the node for each service port. The service becomes reachable through node IP and port. The node performs SNAT and DNAT on the traffic. • Kubernetes service of type ClusterIP with NodePortLocal feature enabled: It requires you to enable the NodePortLocal feature on the Antrea Agent and add annotation in the manifest file of the Kubernetes service. Antrea CNI recognizes the annotation and opens a port for each pod on the node where the pod is running. NodePortlLocal avoids opening a port on all the nodes of the K8s cluster, thereby saving available port ranges. It also avoids an SNAT operation, and preserves the original client IP. To learn more about NodePortLocal feature, see the NodePortLocal Guide on the Antrea documentation portal.
Scenario 3: Expose pods to external traffic through routable pod IP addresses	Antrea supports assigning routable IP addresses to pods by deploying the IPPool custom resource in your Antrea Kubernetes cluster. The pods are allocated IP address from this pool and they are bridged directly to the IaaS network.

Supported IaaS Networks

Antrea Kubernetes clusters can be deployed on the following IaaS platforms:

- vSphere-based on-premises data center: as Tanzu Kubernetes clusters, which are created by using the Tanzu Kubernetes Grid Service. The clusters are managed by Tanzu Kubernetes Grid (TKG) 2.0 and vSphere.
- VMware Cloud: as Tanzu Kubernetes clusters that are managed by TKG 2.0 and VMC SDDC.
- Public clouds: as managed Kubernetes clusters on the public cloud platforms.
- Physical servers: as Kubernetes clusters on bare-metal servers.
- OpenShift:
  - Antrea CNI deployed in OpenShift clusters, and OpenShift is deployed in installer provisioned infrastruture (IPI) mode or user provisioned infrastructure (UPI) mode.
  - Infrastructure providers can be any one of the following: vSphere, Amazon Web Services (AWS), Azure, OpenStack, Google Cloud Platform (GCP), bare metal.

Important:
The IaaS network is responsible for traffic between Antrea Kubernetes clusters and VMs. The traffic between different sites, such as on-premises data centers, VMware Cloud, and public cloud might involve an IaaS-specific VPN. In all cases, there might be SNAT operations applied by the IaaS network. The NSX administrator must ensure that the source or destination IP addresses are routable so that they can be used in firewall rules to protect traffic between VMs and Kubernetes clusters.

Recommended Approaches to Apply Firewall Policies

Both distributed firewall and gateway firewall allow you to specify generic groups with Kubernetes member types in the sources and destinations of the firewall rules. So, you have the flexibility to decide whether to reference the groups in the distributed firewall or gateway firewall.

VMware provides the following recommendation:

- Use vDefend Gateway Firewall to allow or block traffic from Antrea Kubernetes clusters to VMs in an NSX network.

This approach helps you to filter the traffic at the earliest point when traffic enters into the NSX overlay network.

- Use vDefend Distributed Firewall to allow or block traffic from VMs in an NSX network to Antrea Kubernetes clusters. This approach helps you to filter the traffic at the earliest point when traffic leaves the VMs in an NSX overlay network.

Summary of Kubernetes Member Types for Usage in Firewall Policies

The following table summarizes the Kubernetes member types that you can use in NSX generic groups to match traffic in firewall rules.

Member Type	Scope	Usage in Firewall Policy
Kubernetes Cluster	Cluster	Appears as an AND condition in the dynamic group criteria to match resources from specific clusters.
Kubernetes Namespace	Namespace	Appears as an AND condition in the dynamic group criteria to match resources from specific namespaces.
Antrea Egress	Cluster	Match outgoing traffic from Antrea Kubernetes clusters where source IP = egress IP
Antrea IP Pool	Cluster	Match outgoing traffic from Antrea Kubernetes clusters where source IP is in IP ranges. Match traffic entering into Antrea Kubernetes clusters where destination IP is in IP ranges.
Kubernetes Node	Cluster	Match outgoing traffic from Antrea Kubernetes clusters where source IP is among the cluster node IP addresses.
Kubernetes Ingress	Namespace	Match traffic entering into Antrea Kubernetes clusters where destination IP is in Ingress IP addresses.
Kubernetes Gateway	Namespace	Match traffic entering into Antrea Kubernetes clusters where destination IP is in Gateway IP addresses.
Kubernetes Service (type=LoadBalancer)	Namespace	Match traffic entering into Antrea Kubernetes clusters where destination IP is in LoadBalancer IP addresses.
Kubernetes Service (type=NodePort)	Namespace	Match traffic entering into Antrea Kubernetes clusters where destination IP + port is in node IP addresses + NodePort range.
Kubernetes Service (type=ClusterIP)	Namespace	None
Kubernetes Service (type=ClusterIP) and NodePortLocal feature enabled	Namespace	Match traffic entering into Antrea Kubernetes clusters where destination IP + port is in node IP addresses + NodePortLocal range.

Converting Kubernetes Network Policies to vDefend Distributed Firewall Policies

The Antrea NSX Adapter synchronizes the Kubernetes network policies of the registered Antrea Kubernetes clusters to the NSX inventory. However, these K8s network policies cannot be updated or managed in the NSX environment.

If you want to edit the K8s network policies in NSX, you can import them to your NSX environment. This import functionality is available starting in NSX 4.2, and it is supported only with NSX API.

Overview of Importing K8s Network Policies

The import operation converts the K8s network policies to vDefend Distributed Firewall policies with an equivalent traffic behavior. After the K8s network policies are converted to vDefend Distributed Firewall policies, NSX becomes the source of truth for managing the converted DFW policies. You can then edit the DFW policies by using either NSX Manager UI or NSX API.

This conversion is a one-way operation. You cannot convert the vDefend Distributed Firewall policies back to K8s network policies.

> **Important:**
> If your NSX inventory contains Kubernetes network policies that are managed by the NSX Container Plugin (NCP), the import functionality is not supported. The K8s network policies that you want to import to NSX must be managed by the Antrea CNI.

Each imported K8s network policy is converted to either one or two vDefend Distributed Firewall policies. One DFW policy will contain all the allow rules (if ingress or egress rules exist in the K8s network policy), and the other DFW policy will contain the default drop traffic rule. The DFW policy with the default drop rule is always generated.

The system sets the span (scope) of the vDefend Distributed Firewall policy to the Antrea Kubernetes cluster. In addition, the span of the DFW policy is further limited to the Antrea group that contains the effective pod members of the Kubernetes namespace.

The converted DFW policies are placed in the Application tier of NSX. The import API appends the converted DFW policies after other existing Antrea policies in the Application tier. The imported policies are enforced after existing Antrea policies in NSX but before the Antrea cluster network policies (ACNP) and Antrea network policies (ANP) are enforced, which are defined natively in the Kubernetes cluster. The original traffic behavior within the K8s namespace is retained after the conversion to vDefend Distributed Firewall policies.

The Antrea CNI realizes the converted DFW policies as Antrea cluster network policies in the Kubernetes cluster. These Antrea cluster network policies are now managed by NSX, and they can be edited only in the NSX environment. If you try to edit the ACNP configuration by using the kubectl command line, the Antrea NSX Adapter will overwrite or revert the ACNP changes to the original policy definition as it exists in NSX.

After the K8s network policies are imported successfully to NSX, the system automatically deletes the original Kubernetes network policies from the K8s inventory. The system also tags the converted DFW policies with the name of the K8s cluster, original K8s network policy, and the K8s namespace. These tags can help you to search the converted DFW policies in the NSX Manager UI. Alternatively, you can run the `kubectl get acnp -o wide` command on the kubectl command line to view the tags (that is, labels in K8s) in the corresponding realized ACNP.

Prerequisites for Importing K8s Network Policies

- Antrea Kubernetes cluster must be registered to NSX 4.2 or later.
- Import functionality requires Antrea- NSX interworking version that is available with VMware Container Networking™ with Antrea™ 1.9.0 or later.
- Apply an appropriate security license in your NSX deployment that entitles the system to configure distributed firewall security policies.
- You are assigned either the Enterprise Admin role or the Security Admin role in the default space of your NSX environment.
- Verify that the Antrea NSX Adapter has successfully reported the Kubernetes network policies to the NSX inventory.
  For example, to verify in the NSX Manager UI, do these steps:
  1. Navigate to **Inventory** > **Containers** > **Namespaces**.
  2. If required, filter the list of namespaces with **CNI Type** as **Antrea**.
  3. Expand a namespace, and then click the count of network policies to verify whether the Kubernetes network policies are synchronized to the NSX inventory.
     To check whether NSX is reading all the Kubernetes network policies in the namespace, you can compare the count in the UI with the number of policies that are retrieved by the following kubectl command. The count must be the same.
     ```
     kubectl get networkpolicies -n <namespace>
     ```

Mapping of K8s Network Policy Fields to vDefend Distributed Firewall Policy Fields

As explained earlier, when a Kubernetes network policy is imported to NSX, the system converts this network policy to one or two DFW policies. One DFW policy contains all the allow rules and the other DFW policy contains the default drop traffic rule. The system creates Antrea groups according to the specification of the K8s network policy. The Antrea groups are used in the **Sources**, **Destinations**, and **Applied To** fields of the converted DFW policies.

The following table explains the mapping of the fields in the Kubernetes network policy to the fields in the vDefend Distributed Firewall policy.

Field in K8s Network Policy	NSX Resource	Description
K8s Network Policy itself	Either one or two DFW policies	• All rules in the `spec.ingress` or `spec.egress` of the K8s network policy are added in the DFW "allow" policy. That is, the rule action for all the rules in this DFW policy is set to **allow**. • A DFW policy with default drop rule is created with a rule for dropping either In traffic, or Out traffic, or In & Out traffic according to values in the `spec.policyTypes` list of the K8s network policy. For example, if the `spec.policyTypes` list of the K8s network policy specification contains only `egress`, the DFW "drop" policy will have a default drop rule for egress traffic (traffic direction: Out). • If the Kubernetes network policy does not contain any ingress and egress rules, a DFW policy with only the default drop rule is created.

`spec.podSelector`
and
`metadata.namespace`
	Group of type Antrea	Both `spec.podSelector` and `metadata.namespace` are used to convert to an Antrea group.
		The Antrea group that is created is referenced in the **Applied To** of the DFW "allow" policy and DFW "drop" policies.
		If `spec.podSelector` is not specified, the Antrea group will contain only the namespace.

`spec.ingress[*]`
and
`spec.egress[*]`
	Firewall rules in the DFW "allow" policy	Each rule in the `spec.ingress` and `spec.egress` section of the K8s network policy is converted to an vDefend Distributed Firewall rule.
		These DFW rules are added in the DFW "allow" policy.

`spec.ingress[*].from` • `podSelector` • `namespaceSelector` • `ipBlock`	Groups of type Antrea	• The pod and namespace selectors in the `ingressfrom` section are converted to Antrea groups with dynamic membership criteria. • The IP CIDR ranges in the `ipBlock` selector are converted to static IP address members in the Antrea group. • These Antrea groups are referenced in the **Sources** field of the DFW rules.

Field in K8s Network Policy	NSX Resource	Description
`spec.egress[*].to` • `podSelector` • `namespaceSelector` • `ipBlock`	Groups of type Antrea	• The pod and namespace selectors in the `egressto` section are converted to Antrea groups with dynamic membership criteria. • The IP CIDR ranges in the `ipBlock` selector are converted to static IP address members in the Antrea group. • These Antrea groups are referenced in the **Destinations** field of the DFW rules.

`spec.ingress[*].ports`
• `protocol`
• `port`
and
`spec.egress[*].ports`
• `protocol`
• `port`
	Service entries in the DFW rule	• The layer 4 protocol and the target port (including range of target ports) in the ingress and egress rule specification of the K8s network policy are converted to **Service** entries in the DFW rules.
		• The target ports or port range in the K8s rule specification always map to destination ports in the **Service** entry.

Examples of Field Mappings

This section includes a few examples to help you understand the mapping of the fields in the Kubernetes network policy to the fields in the DFW policy when a K8s network policy is imported to NSX.

Example 1

K8s network policy specification:

```
apiVersion: v1
kind: Namespace
metadata:
  name: my-ns1
---
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: knp1
  namespace: my-ns1
spec:
  podSelector: {}
  policyTypes:
    - Egress
  egress:
    - to:
        - ipBlock:
            cidr: 8.8.8.8/32
      ports:
        - protocol: UDP
          port: 53
```

This K8s network policy selects all the pods in the my-ns1 namespace and allows egress traffic (outgoing connections) from any pod in this namespace to the 8.8.8.8/32 CIDR on UDP port 53. All other egress traffic from the pods in this namespace is dropped.

When this K8s network policy is imported to NSX, two DFW policies are created. One DFW policy contains a single allow rule and the other DFW policy contains a single drop rule.

The `spec.podSelector` section is converted to an Antrea group with a dynamic membership criterion. The criterion in the group selects all the pods in the my-ns1 namespace. This Antrea group is referenced in the **Applied To** field of both the DFW allow policy and the DFW drop policy.

The `ipBlock` selector in the `spec.egress.to` section is converted to an Antrea group with a static IP address member 8.8.8.8/32. This Antrea group is referenced in the **Destinations** field of the DFW allow rule. Let us refer to this group as Group-1.

The `spec.egress.ports` section is converted to a **Service** entry with UDP protocol and destination port 53.

The configuration of the DFW allow rule in NSX is as follows.

Sources	Destinations	Services	Context Profiles	Rule Applied To	Rule Action	Rule Direction
N/A	Group-1	UDP Source: Any, Destination: 53	N/A	Any	Allow	Out

The configuration of the DFW drop rule in NSX is as follows.

Sources	Destinations	Services	Context Profiles	Rule Applied To	Rule Action	Rule Direction
Any	Any	Any	N/A	Any	Drop	Out

Example 2

K8s network policy specification:

```
apiVersion: v1
kind: Namespace
metadata:
  name: my-ns2
---
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: knp2
  namespace: my-ns2
spec:
  podSelector: {}
  policyTypes:
    - Ingress
    - Egress
```

This Kubernetes network policy selects all the pods in the my-ns2 namespace and drops all ingress traffic and egress traffic from these pods. This policy is basically creating a default drop all ingress and egress traffic for the namespace.

When this K8s network policy is imported to NSX, only one DFW drop policy is created. The converted DFW policy contains a single firewall rule with drop action.

The `spec.podSelector` section is converted to an Antrea group with a dynamic membership criterion. The criterion in the group selects all the pods in the my-ns2 namespace. This Antrea group is referenced in the **Applied To** field of the DFW drop policy.

The configuration of the DFW drop rule in NSX is as follows.

Sources	Destinations	Services	Context Profiles	Rule Applied To	Rule Action	Rule Direction
Any	Any	Any	N/A	Any	Drop	In_Out

Naming Convention of Converted DFW Policies and Antrea Groups

System uses the following naming convention for the converted DFW allow and drop policies:

DFW allow policy	*<cluster_name>*-*<namespace>*-*<K8s_networkpolicy_name>*-*<K8s_networkpolicy_uuid>*-allow
DFW drop policy	*<cluster_name>*-*<namespace>*-*<K8s_networkpolicy_name>*-*<K8s_networkpolicy_uuid>*-drop

The values of *cluster_name*, *namespace*, and *K8s_networkpolicy_name* are truncated to 12 bytes each.

System uses the following naming convention for the Antrea groups:

Groups referenced in the Applied To field of DFW allow and drop policies	*<cluster_name>*-*<namespace>*-*<K8s_networkpolicy_uuid>*
Groups referenced in the Sources field of DFW rules	*<cluster_name>*-*<namespace>*-*<K8s_networkpolicy_uuid>*-rule[rule index]-from-*<peer index>*
Groups referenced in the Destinations field of DFW rules	*<cluster_name>*-*<namespace>*-*<K8s_networkpolicy_uuid>*-rule[rule index]-to-*<peer index>*

The values of *cluster_name* and *namespace* are truncated to 12 bytes each.

To understand how the system assigns values to the rule index and peer index in the Antrea group name, consider the following example of a K8s network policy, which contains three ingress rules.

Example

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: knp3
  namespace: my-ns3
spec:
  podSelector: {}
  policyTypes:
    - Ingress
  ingress:
    - from:
        - namespaceSelector:
            matchExpressions:
            - key: namespace
              operator: In
              values: ["test-ns1"]
        - podSelector:
            matchExpressions:
            - key: app
              operator: In
              values: ["nginx"]
      ports:
        - protocol: TCP
          port: 80
    - from:
        - namespaceSelector:
            matchExpressions:
            - key: namespace
              operator: In
              values: [""]
      ports:
        - protocol: TCP
          port: 443
    - from:
        - namespaceSelector:
```

```
            matchExpressions:
            - key: namespace
              operator: In
              values: ["test-ns4"]
      ports:
        - protocol: TCP
          port: 8080
          endPort: 8090
```

Let us consider the snippet of the first `ingress.from` rule, as follows:

```
ingress:
    - from:
        - namespaceSelector:
            matchExpressions:
            - key: namespace
              operator: In
              values: ["test-ns1"]
        - podSelector:
            matchExpressions:
            - key: app
              operator: In
              values: ["nginx"]
      ports:
        - protocol: TCP
          port: 80
```

The `from` section in this ingress rule contains two elements. One element selects pods by using the namespace selector and the other element selects pods by using the pod selector. We refer to these two elements as peers.

The system creates one Antrea group for each peer in the rule. The converted DFW rule has rule index 0, and this rule contains two Antrea groups in the **Sources** field of the rule. One Antrea group name has peer index 0 and the other group name has peer index 1, as follows:

`<`*`cluster_name`*`>-my-ns3-`*`knp3`*`-rule[0]-from-0`

`<`*`cluster_name`*`>-my-ns3-`*`knp3`*`-rule[0]-from-1`

Now, consider the snippet of the second and third `ingress.from` rules, as follows:

```
ingress:
    - from:
        - namespaceSelector:
            matchExpressions:
            - key: namespace
              operator: In
              values: [""]
      ports:
        - protocol: TCP
          port: 443
    - from:
        - namespaceSelector:
            matchExpressions:
            - key: namespace
              operator: In
              values: ["test-ns4"]
      ports:
        - protocol: TCP
          port: 8080
          endPort: 8090
```

The `from` section in each of these two ingress rules contains only a single element that selects pods by using the namespace selector. The converted DFW rules have rule index 1 and 2 respectively, and each rule contains a single Antrea group in the **Sources** field of the rule. In this case, peer index is not appended to the names of Antrea groups. The group names are as follows:

`<cluster_name>-my-ns3-knp3-rule[1]-from`

`<cluster_name>-my-ns3-knp3-rule[2]-from`

High-level Import Workflow Using API

1. Ensure that you have met the prerequisites for importing the Kubernetes network policies, as explained earlier in this documentation.
2. Run the following kubectl command to view the list of Kubernetes network policies in a given namespace:

   ```
   kubectl get networkpolicies -n <namespace>
   ```

   For example:

   ```
   kubectl get networkpolicies -n test-ns5

   NAME     POD-SELECTOR   AGE
   k-np10   app=demo       82m
   k-np11   app=myapp      82m
   k-np12   app=demo       82m
   k-np9    app=myapp      82m
   ```

   In this example, the `test-ns5` namespace contains four Kuberentes network policies. In this procedure, we will import the "k-np9" and "k-np11" policies to NSX.
3. Run the following kubectl command to retrieve the ID of each Kubernetes network policy that you want to import.

   ```
   kubectl get networkpolicies <policy-name> -n <namespace> -o yaml
   ```

   For example, to retrieve the IDs of "k-np9" and "k-np11" network policies, run these commands:

   ```
   kubectl get networkpolicies k-np9 -n test-ns5 -o yaml
   kubectl get networkpolicies k-np11 -n test-ns5 -o yaml
   ```

   In the output of both these kubectl commands, note the ID that you see in the `metadata.uid` field. In our K8s cluster, the policy IDs are as follows:

   - For k-np9: `e5a59ae6-cc0e-42a5-80bd-f6fa13b5b70d`
   - For k-np11: `84b850fb-69ad-4e95-a563-a95ce6b70557`

   These policy IDs are only for example purposes. They can be different in your K8s cluster. You will require these policy IDs to run the import API, which is explained in the next step.
4. Run the following NSX API to import the Kubernetes network policies:

   For example:

   ```
   POST https://<nsx-mgr>/policy/api/v1/infra/import-k8s-np-to-dfw?on_error=ABORT  -k
   {
       "network_policy_ids" : ["e5a59ae6-cc0e-42a5-80bd-f6fa13b5b70d", "84b850fb-69ad-4e95-a563-
   a95ce6b70557"],
       "sequence_number_upper" : 1000,
       "sequence_number_lower" : 2000
   }
   ```

   The `network_policy_ids` parameter is mandatory, whereas the `sequence_number_upper` and `sequence_number_lower` parameters are optional.

   In this example, the IDs of K8s network policies (k-np9 and k-np11) are specified for importing to NSX.

   For a detailed information about these request parameters, API example request, and API example response, see the *NSX API Guide*.

The `on_error` query parameter determines the action that the API must take when an error occurs. The following table explains the valid values of this query parameter.

Value	Description
ABORT	This value is the default. If an error occurs while importing the K8s network policies, the system will prevent all converted DFW policies and Antrea groups from being committed. The import operation ends prematurely and none of the K8s network policies are converted. The API response returns the error message for each K8s network policy that has resulted in an error. Example: Assume that you have specified UUIDs of two K8s network policies, say knp1 and knp2 in the API request body. The knp1 policy contains the unsupported SCTP protocol in an egress rule specification. When you run the import API, the conversion of knp1 network policy throws an error and the import operation ends prematurely. The system does not convert the next K8s network policy (knp2), even if this network policy is valid.
CONTINUE	If an error occurs while importing the current K8s network policy, system skips this policy and continues to import the next K8s network policy. The API response returns the error message for each K8s network policy that has been skipped during the import operation. **CAUTION:** A change in the traffic behavior can be expected if the imported K8s network policies and the skipped K8s network policies are applied to the same pod. Example: Continuing with the same example, as mentioned in the previous row. When the import of knp1 network policy throws an error, the system continues importing the knp2 network policy, and converts this network policy successfully. The API response returns the same error message for the knp1 network policy, which failed during the conversion.

The K8s network policies that you import in a single API request can belong to different registered Antrea Kubernetes clusters. Or they can belong to multiple namespaces within a single Antrea Kubernetes cluster.

If a namespace includes multiple K8s network policies, we recommend that you import them in a single API request. The reason is that K8s network policies inside a namespace can be related to each other. This practice helps in ensuring that the traffic behavior does not change after the network policies are imported to NSX.

5. After the import is successful, go to the NSX Manager UI and view the configuration of the Antrea groups and DFW policies.
6. Optional: View the realized Antrea cluster network policies, check the ACNP specifications, and ClusterGroup specifications by running these kubectl commands:

   ```
   kubectl get acnp
   kubectl get acnp <acnp-id> -o yaml
   kubectl get cg <cg-id> -o yaml
   ```

7. Optional: Verify that the K8s network policies that are imported successfully to NSX are not seen in the K8s cluster by running the following kubectl command:

   ```
   kubectl get networkpolicies -n <namespace>
   ```

   For our example, the command is as follows:

   ```
   kubectl get networkpolicies -n test-ns5

   NAME      POD-SELECTOR   AGE
   k-np10    app=demo       84m
   ```

```
k-np12    app=demo        84m
```

Observe that the imported Kubernetes network policies (k-np9 and k-np11) are no longer seen in the K8s cluster.

Unsupported Kubernetes Network Policy Features

Some features in Kubernetes network policies are currently not supported for conversion to vDefend Distributed Firewall policies and Antrea groups. The API response displays an appropriate error message when the conversion fails due to any of the following unsupported features.

- K8s network policies that use layer 4 port names for pods are not converted. DFW policies support only port numbers.
- K8s network policies that contain SCTP protocol are not converted. vDefend Distributed Firewall policies do not support SCTP traffic.
- K8s network policies can have a large number of `matchLabels` or `matchExpressions` in the `podSelector` or `NetworkPolicyPeer` section. However, dynamic membership criterion in an Antrea group can support a maximum of 15 conditions with mixed member types and 5 conditions with the same member type. When this maximum limit is exceeded in a group membership criterion, the conversion fails.
- Kubernetes network policies that contain `matchExpressions` with the `DoesNotExist` operator are not converted to Antrea groups. The `DoesNotExist` operator in Kubernetes maps to the `NotEquals` value for the scope operator in NSX. However, the scope operator in an Antrea group definition currently does not support this value.
- Kubernetes network policies that contain `matchExpressions` with the `In` operator must contain only a single value for the conversion to succeed. Multiple values in the `In` operator are currently not supported for conversion to Antrea groups.

Behavior with Different Antrea NSX Adapter Versions

You can upgrade Antrea NSX Adapter and NSX separately, and in any order. The new changes in the Antrea NSX Adapter are compatible with NSX versions prior to 4.2.

Consider the following scenario:

Your NSX environment is at version 4.2, and it has multiple Antrea Kubernetes clusters registered to it. Some Kubernetes clusters have the new version of Antrea NSX Adapter (say v0.15), whereas some clusters have an old version of Antrea NSX Adapter (say v0.11). In this case, the old version of Antrea NSX Adapter won't delete the original Kubernetes network policies automatically from the Kubernetes cluster after the conversion. The Kubernetes administrator or the namespace administrator needs to delete the original Kubernetes network policies manually. Note that the conversion to DFW policies is still performed. The converted DFW allow policies and the default drop policies are realized as Antrea cluster network policies in the Kubernetes cluster, and they are evaluated by the Antrea CNI before the original Kubernetes network policies. If the original Kubernetes policies are not deleted from the Kubernetes cluster, the traffic behavior defined in the converted DFW policies might conflict with the original Kubernetes network policies.

Deregister an Antrea Kubernetes Cluster from NSX

Use the `deregisterjob.yaml` file that is included with the `antrea-interworking-`*`version`*`.zip` file to deregister an Antrea Kubernetes cluster from NSX.

- In addition to the VMware Cloud Foundation license, which entitles your NSX environment to networking features, you must have a security license that enables full access to the distributed firewall feature.
- Open the `deregisterjob.yaml` file in a text editor and replace the image URLs with any one these container images that are hosted on VMware Harbor Registry.
  - `projects.registry.vmware.com/antreainterworking/interworking-debian:`*`version`*
  - `projects.registry.vmware.com/antreainterworking/interworking-ubuntu:`*`version`*
  - `projects.registry.vmware.com/antreainterworking/interworking-photon:`*`version`*

  For *`version`* information, see the *VMware Container Networking with Antrea* release notes at https://docs.vmware.com/en/VMware-Container-Networking-with-Antrea/index.html.

  If your Kubernetes cluster does not have Internet access, you can find the container images in the `antrea-interworking-`*`version`*`.zip` file, which you downloaded before registering the Kubernetes cluster. If the interworking deployment is running in the Kubernetes cluster, the container images are already loaded on the cluster nodes. The deregister job and the interworking deployment use the same container images.

1. Run the following kubectl command to submit the `deregisterjob.yaml` file to the Kubernetes API server.

   ```
   $ kubectl apply -f deregisterjob.yaml
   ```

   This job takes some time to complete. The following actions happen in the background:

   - Deletes the interworking deployment.
   - Deletes the resources of the Antrea Kubernetes cluster from the NSX inventory. The resources include pods, services, ingress rules, nodes, and network policies.
   - Removes references to the Antrea Kubernetes cluster in the distributed firewall policies and groups (if any).
   - Deletes the Antrea custom resources, which were managed by NSX from the Kubernetes cluster. These custom resources include: Traceflow, ClusterNetworkPolicy, ClusterGroup, and Tier.

2. Check the status of the deregister job by running the following kubectl command:

   ```
   kubectl get job -o wide deregister -n vmware-system-antrea
   ```

   Wait for the job to complete. You might have to run this command a few times to check the status of the job.

3. After the deregister job is completed, run the following kubectl command to delete the `vmware-system-antrea` namespace and the role-based access control (RBAC) resources.

   ```
   kubectl delete -f interworking.yaml --ignore-not-found
   ```

   The `ignore-not-found` flag is used in this command to avoid the `Resource Not Found` error in the command output if some resources are not found for deletion.

   RBAC resources, such as ServiceAccount, ClusterRole, and ClusterRoleBinding are deleted.

Verify that the Antrea Kubernetes cluster is not shown in the NSX inventory.

1. In the NSX Manager UI, navigate to **Inventory** > **Containers** > **Clusters**.
2. Observe that the Antrea Kubernetes cluster is not shown in the inventory.

Optional: After deregistering the Antrea Kubernetes cluster, delete the principal identity (PI) user and the self-signed certificate.

- To delete the PI user account, navigate to **System** > **User Management** > **User Role Assignment**. Next to the PI user name, click

  , and then click **Delete**.
- To delete the self-signed certificate, navigate to **System** > **Certificates**. Next to the certificate name, click

  , and then click **Delete**.

If required, you can re-register the same Antrea Kubernetes cluster. However, before re-registering the cluster, ensure that you have run the following kubectl command:

```
kubectl delete -f interworking.yaml --ignore-not-found
```

To re-register the same Antrea Kubernetes cluster, do any one of the following:

- If you want to reuse the same PI user account and self-signed certificate for re-registering the Antrea Kubernetes cluster, do not delete PI user account and the self-signed certificate from NSX. In this case, no changes are required in the `bootstrap-config.yaml`.
- If you want to use a new PI user account and self-signed certificate for re-registering the Antrea Kubernetes cluster, delete the old PI user account and the self-signed certificate. Begin the process by creating a self-signed certificate and use this new certificate to create a PI user account. Edit the `tls.crt` and `tsl.key` arguments in the `bootstrap-config.yaml` with the information of this new PI user.

Clean up Antrea Data from NSX

If you destroy the Antrea Kubernetes cluster or delete the Antrea- NSX Interworking Adapter before running the `deregisterjob.yaml` file, some Antrea data is retained in the NSX inventory.

- You must have the user name and password of the NSX Enterprise Admin user.

- You must be able to connect to NSX Manager UI and API.

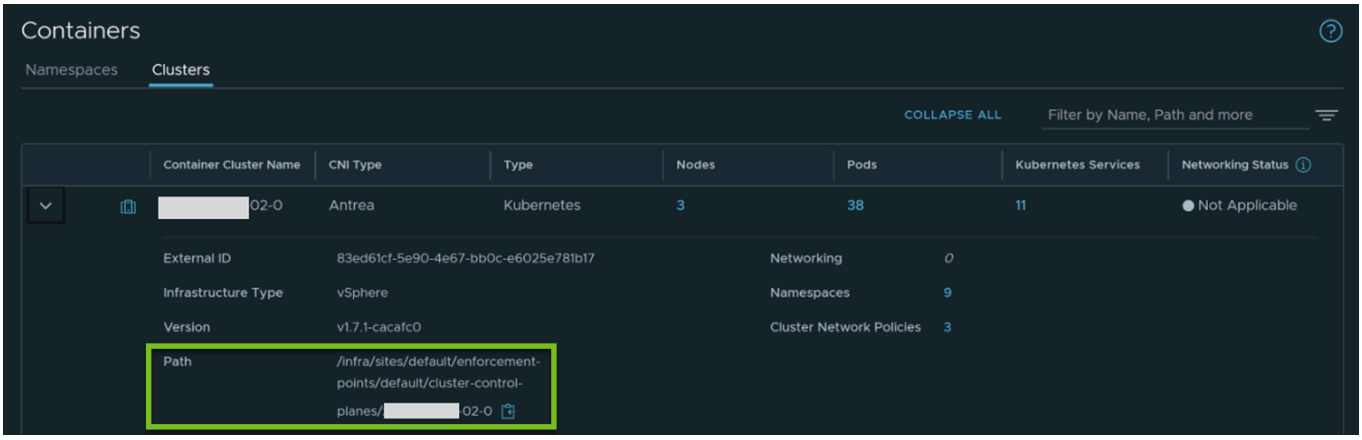
You can use the instructions in this documentation to clean up the Antrea data from NSX.

Starting with VMware Container Networking™ with Antrea™ version 1.7.0, the `antreansxctl` command-line utility provides the `cluster-cleanup` command to clean up leftover Antrea data in the NSX inventory.The `antreansxctl` utility is a Linux-only executable. So, you require a Linux machine to run this utility.

To learn about the usage of this `cluster-cleanup` command, see the *antreansxctl Command-Line* documentation in the *VMware Container Networking with Antrea Installation Guide.*

In VMware Container Networking™ with Antrea™ versions prior to 1.7.0, the `antreansxctl` command-line utility is unavailable. In that case, you can run a curl command, as explained in the following procedure. The curl command calls an NSX API to delete the leftover Antrea data from the NSX inventory.

1. Find the path of the Antrea Kubernetes cluster in the NSX Manager UI.
   a) From your browser, log in to an NSX Manager at https://*nsx-manager-ip-address*.
   b) Navigate to **Inventory** > **Containers** > **Clusters**.
   c) Expand the cluster to be deleted and copy the text that you see next to the **Path** field.
   For example:



2. Run the following curl command to delete the Antrea data from NSX:

```
curl -k -u '{AdminUserName}:{AdminPassword}' \
-X DELETE -H "X-Allow-Overwrite: true" \
https://{NSX-Mgr-IP}/policy/api/v1{Path}?cascade=true
```

   In this command:

   - Replace *{AdminUserName}*, *{AdminPassword}*, and *{NSX-Mgr-IP}* with their actual values as applicable to your NSX environment.
   - Replace *{Path}* with the text that you copied in step 1.

   For example:

```
curl -k -u 'Admin:Password123' \
-X DELETE -H "X-Allow-Overwrite: true" \
https://192.168.1.1/policy/api/v1/infra/sites/default/enforcement-points/default/cluster-control-planes/
cluster-sales?cascade=true
```

Upgrade Antrea- NSX Interworking Deployment in an Antrea Kubernetes Cluster

If you have upgraded the VMware Container Networking version to a new release, the Antrea- NSX interworking deployment in your registered Antrea Kubernetes cluster must be upgraded together.

1. The VMware Container Networking version in your Antrea Kubernetes cluster, which is registered to NSX, must be upgraded successfully.
2. Complete the following prerequisite steps.
   – Determine the Antrea version from the Kubernetes cluster.

- Download the Antrea- NSX interworking zip file.
- Import the container images to container registry.
  Make sure to edit the `interworking.yaml` and `deregisterjob.yaml` files, and update the image URLs to the imported image, or to the online image URL.

1. Run the following kubectl commands to delete the register job and the antrea-interworking deployment.

   ```
   kubectl delete job register -n vmware-system-antrea --ignore-not-found
   kubectl delete deployment antrea-interworking -n vmware-system-antrea --ignore-not-found
   ```

   Note:
   Do not delete the `vmware-system-antrea` namespace, configmaps, and secrets in this namespace.

2. Run the following kubectl command to submit the `interworking.yaml` file to the Kubernetes API server, and trigger the upgrade.

   ```
   kubectl apply -f interworking.yaml
   ```

   Note:
   Only the `interworking.yaml` must be submitted. The `bootstrap-config.yaml` file is not required for the upgrade process.

3. Run the following kubectl command to list all the pods in the `vmware-system-antrea` namespace.

   ```
   kubectl get pods -o wide -n vmware-system-antrea
   ```

   Observe that the status of the register-xxx pod is `Running`. Because the Antrea Kubernetes cluster is already registered to NSX, the register-xxx pod skips the registration process and the status soon changes to `Completed`. The old interworking-yyy pod status changes to `Terminating`, and the new interworking-zzz pod status changes to `Running`.

   When the new interworking-zzz pod status is `Running`, and the ready containers are `4/4`, there is no need to restart the containers, and the upgrade is successful.

   ```
   #Example output:

   NAME                             READY   STATUS    RESTARTS   AGE   IP          NODE             NOMINATED
    NODE   READINESS GATES
   interworking-7764988ddd-wnvcg    4/4     Running   0          29s   192.168.x.y example-node-10 <none>
        <none>
   ```

4. Run the following kubectl command to verify that the new interworking-zzz pod is using the new image and the image URLs are the same as expected.

   Make sure to replace the "interworking-7764988ddd-wnvcg" pod name with the actual pod name that you see in the output of the `kubectl get pods` command of the previous step.

   ```
   kubectl get pods -o yaml interworking-7764988ddd-wnvcg -n vmware-system-antrea | grep image:

   # Example output:
       image: vmware.io/antrea/interworking:0.11.0
       image: vmware.io/antrea/interworking:0.11.0
       image: vmware.io/antrea/interworking:0.11.0
       image: vmware.io/antrea/interworking:0.11.0
       image: vmware.io/antrea/interworking:0.11.0
       image: vmware.io/antrea/interworking:0.11.0
       image: vmware.io/antrea/interworking:0.11.0
       image: vmware.io/antrea/interworking:0.11.0
   ```

Restoring Antrea Kubernetes Clusters from an NSX Backup

Use this documentation to understand the behavior of the restore operation from an NSX backup that contains Antrea Kubernetes clusters, which are registered to NSX.

When you restore an NSX backup, the following behavior occurs:

- All existing K8s resources of the registered Antrea Kubernetes clusters, for example, pods, services, namespaces, and so on, are not restored to their previous status when the NSX backup was taken. There is no impact to the Kubernetes cluster and it will continue to work.
- When NSX is restored from the backup file, DFW policies that were applied to Antrea K8s clusters return to their previous status (that is, the status when the backup was taken). However, K8s resources that are seen by the DFW rules, such as Ingress, Antrea Egress, services, namespaces, and so on, are at their latest (current) status. This causes inconsistency.

  A resynchronization is automatically done after the NSX restore is completed. The resynchronization ensures that all K8s resources in the NSX inventory return to their current status, and all restored NSX DFW policies are realized to the Kubernetes cluster.
- If you delete an Antrea Kubernetes cluster after the backup is done, and then restore NSX from this backup, some orphan Management Plane resources are detected in NSX. You must clean-up the orphan resources manually, such as DFW policies, cluster control plane node, and Principal Identity.
- If you register a new Antrea Kubernetes cluster to NSX after the backup is done, and then restore NSX from this backup, the new Antrea Kubernetes cluster cannot connect to NSX Manager. The reason is that Principal Identity (PI) user and cluster control plane node of this new Antrea Kubernetes cluster are missing.

  In this case, do the following steps:
  a. Deregister the Antrea Kubernetes cluster from NSX to ensure a clean-up. For more information, see Deregister an Antrea Kubernetes Cluster from NSX.
  b. Add the PI user again in NSX.
  c. Register the Antrea Kubernetes cluster to NSX again.
     For more information, see Prerequisites for Registering an Antrea Kubernetes Cluster to NSX.

Troubleshooting Antrea to NSX Integration Issues

Use this section to learn about collecting logs from the Antrea Kubernetes cluster nodes and resolving problems that you might encounter when integrating Antrea Kubernetes clusters to NSX.

Collect Support Bundles for an Antrea Kubernetes Cluster

You can use the Support Bundle feature in NSX to collect log files from Antrea Kubernetes cluster nodes for detailed troubleshooting and diagnostic purposes.

Antrea Kubernetes clusters are registered to NSX.

NSX can upload the support bundles to an NSX Manager node from where you triggered the support bundle collection request, or it can upload the support bundles to a remote file server that you specified in the request. If the support bundles are uploaded to an NSX Manager node, you can download them to your local computer.

> **Note:**
> This documentation uses the term "Antrea Kubernetes cluster" to mean Kubernetes clusters with Antrea CNI. The term "Kubernetes cluster" is a generic term, which represents Tanzu Kubernetes Grid (TKG) clusters with Antrea CNI, OpenShift clusters with Antrea CNI, or do it yourself (DIY) Kubernetes clusters with Antrea CNI.
>
> The UI use the term "Antrea container cluster" for a few UI fields or labels. In the *Procedure* section of this documentation, the term "Antrea container cluster" is retained for those UI fields or labels. For all free-form text, the term "Antrea Kubernetes cluster" is used.

A support bundle for an Antrea Kubernetes cluster contains log files for the following components:

- Antrea Controller
- Antrea Agent
- Management Plane Adapter
- Central Control Plane Adapter
- Open vSwitch

Supported and Unsupported Features	• From an NSX Manager node, you can start only a single support bundle collection request. But, you can collect support bundles for multiple Antrea Kubernetes clusters with a single collection request.

- If you are using an NSX Manager cluster with three Management nodes, you can start a separate support bundle collection request simultaneously from each NSX Manager node. However, the Antrea Kubernetes cluster that you select in each collection request must be different.

  For example, assume that you have started a support bundle collection request on NSX Manager node A. In this collection request, you selected cluster nodes 1 and 2 from Antrea Kubernetes cluster X. Simultaneously, if you start a second collection request on NSX Manager node B for the cluster nodes 3 and 4 in the Antrea Kubernetes cluster X, one of these two collection requests will fail. You must wait for the first collection request to complete before triggering the second request for the same Antrea Kubernetes cluster.
- Collection of support bundles from NSX Manager Central CLI is currently not supported for Antrea Kubernetes clusters.

1. From your browser, log in to an NSX Manager at `https://nsx-manager-ip-address`.
2. Navigate to **System** > **Support Bundle**.

   The **Request Bundle** page opens.

   Important:
   NSX Manager UI fetches the information about registered Antrea Kubernetes clusters when you start the NSX Manager application in the browser. If the application UI is already open, it does not fetch the Antrea Kubernetes cluster registration information automatically. This behavior is expected and per the current UI design. If you have registered the first Antrea Kubernetes cluster after the NSX Manager application is opened, ensure that you refresh the browser after navigating to the **Request Bundle** page. A manual refresh ensures that you can select Antrea Kubernetes clusters as the target nodes in the next step of this procedure.

   This manual browser refresh is required only once, and not every time after a new Antrea Kubernetes cluster is registered to NSX.
3. Select the target nodes to include in the support bundle request.

   The available types of nodes are:

   - Antrea Container Clusters
   - Management Nodes
   - Edges
   - Hosts

   A single support bundle request can include a mix of different types of nodes in the NSX environment. For example, you can select nodes from Antrea Kubernetes clusters, NSX Manager nodes, and NSX Edge nodes in the same collection request. However, the scope of this procedure is to explain the workflow of creating a support bundle collection request for only Antrea Kubernetes clusters.
4. From the **Type** drop-down menu, select **Antrea Container Clusters**.
5. From the **Container Cluster** list, select the name of a cluster.

   If the list has several clusters to select from, enter the first few characters of the cluster name. System filters the list and displays only the cluster names that match the characters you have entered.

   All nodes in the selected cluster are displayed in the **Available** list.
6. Select one or multiple nodes from the cluster and click the right arrow to move them to the **Selected** list.

   To select nodes from multiple clusters in a single collection request, repeat steps 4 and 5 for each cluster.
7. Optional: In the **Log age (days)** text box, keep the default value or enter the specific number of days' worth of logs that you want the support bundle to include. Specify the log age as a number of days.
8. Optional: To upload the support bundle to a remote file server, specify the file server settings.
   a) Enter an IP address or the host name of the remote file server.
   b) Enter the file transfer protocol and port number. Default port number is 22.
   c) Enter the user name and password to access the remote file server.
   d) Enter the path to the destination folder where the support bundle file is to be uploaded.

   When remote file server settings are not specified, the support bundle is uploaded to the NSX Manager node from where you triggered the support bundle collection request.

9. Click **Start Bundle Collection**.

   The runtime details of the collection request are displayed on the **Status** page. The collection process takes a few minutes. The time taken to create the support bundle depends on the number of log files to collect from each node in the container cluster.

10. After the collection process is complete, click **Download**.

    The support bundle file is saved on your local computer. If you had specified remote file server settings, the **Download** button is not displayed in the UI.

A support bundle collection request generates a single tape archive (TAR) file with the following file naming convention: `nsx_support_archive_datestamp_timestamp.tar`

Support bundle collection request can fail in the following situations:

- If the Antrea NSX Adapter on a Kubernetes cluster fails when the support bundle request is in progress, the collection of logs fails for that Kubernetes cluster.
- If the NSX Manager Appliance fails or is not reachable when the support bundle request is in progress, the collection of logs fails. Until the connectivity issue to the NSX Manager is resolved, you can use the native command line tool of Antrea (antctl) to collect log files from the Antrea Kubernetes clusters.

Partial Success Scenario	Consider that you selected 10 nodes from a single Antrea Kubernetes cluster for the support bundle collection. During the collection process, log files were collected successfully from five nodes in the Antrea Kubernetes cluster, but were not collected for the remaining five nodes. In other words, the collection request succeeded partially. In this situation, the collection request status is `Successful` and the support bundle file (TAR) contains logs for the five successful nodes.

1. Extract the TAR file. The following files are displayed.

File Name	Description
`manifest.json`	This file contains a summary of the collection request results and the properties of the collection request. For example, it contains information about: • The nodes for which the collection succeeded. • The nodes for which the collection failed. • The cluster IDs and node IDs that were used in the collection request.
`nsx_antrea_`*`cluster-id`*`.tgz`	A single `.tgz` archive file is created for each Antrea Kubernetes cluster in the support bundle.

2. Extract the `nsx_antrea_`*`cluster-id`*`.tgz` file. The following files are displayed.

File Name	Description
`adapters.tar.gz`	This archive file contains the log files of the Management Plane Adapter and the Central Control Plane Adapter.
`agent_`*`node_name`*`.tar.gz`	This archive file contains the log files of the Antrea Agent and Open vSwitch. One archive file is generated for each Antrea Kubernetes cluster node in the collection request. On extracting this archive file, you can view the following files: • `agentinfo` file • Agent logs at `/logs/agent` • Open vSwitch logs at `/logs/ovs` • OpenFlow dump • IPtables • Route dump

File Name	Description
`clusterinfo`	This file is generated for each Antrea Kubernetes cluster in the support bundle request. The file contains information about the following Kubernetes resources that are collected from the Kubernetes API server: • Pods • Nodes • Deployments • ReplicaSets • DaemonSets
`controller.tar.gz`	This archive file contains the log files of the Antrea Controller. On extracting this archive file, you can view the following files: • `controllerinfo` file • Controller logs at `/logs/controller`

Antrea Kubernetes Cluster Status is Down

If the status of the Antrea Kubernetes cluster is down, follow the steps in this documentation either to determine the cause of this issue and recover from it, or collect the support bundle.

The cluster control plane node is down. The Antrea Kubernetes cluster is disconnected from the Central Control Plane (CCP).

In the NSX Manager UI, navigate to **System** > **Fabric** > **Nodes** > **Container Clusters** > **Antrea**. If required, filter the list of clusters on the **Antrea** page with the **External ID** field.

Click the **Status** column of the problematic cluster. If all the components are down, the possible causes are:

- The Kubernetes cluster is deleted.
- Network connectivity issue with the CCP.
- The adapters are crashed or deleted for some reason.
- The client certificate of the adapters is incorrect.
- The version of the adapters is incompatible with the CCP.

If only the Central Control Plane Adapter is down, the CCP Adapter might have crashed.

1. If the Kubernetes cluster is deleted, clean up the leftover registration and inventory data in NSX. See Clean up Antrea Data from NSX.
2. Get the kubectl and kubeconfig access for the Kubernetes cluster. Use kubectl to retrieve the node name on which the interworking pod is running. Start an SSH session to the node and use the curl or nc command to connect to every NSX Manager IP on ports 1234 and 1235. If the connection cannot be established, the cause is network connectivity issue with the CCP.
   Example of the curl command:

   Ensure that you replace *`NSX-Manager-IP`* with the IP address of NSX Manager in your environment.

   ```
   curl -v NSX-Manager-IP:1235

   Trying NSX-Manager-IP...
   Connected to NSX-Manager-IP (NSX-Manager-IP) port 1235 (#0)
   ...
   Empty reply from server
   Connection #0 to host NSX-Manager-IP left intact
   curl: (52) Empty reply from server
   ```

   Example of the nc command:

   ```
   nc -v NSX-Manager-IP 1235 < /dev/null

   Ncat: Version 7.50 (https://nmap.org/ncat)
   ```

```
Ncat: Connected to NSX-Manager-IP:1235.
Ncat: 0 bytes sent, 0 bytes received in 0.37 seconds.
```

3. Use kubectl to check whether all containers of the interworking pod in the vmware-system-antrea namespace are up.

   If any container is down, use kubectl to get logs of the crashed containers and check the error message. This step can help you identify failure due to any of these reasons:

   - The adapters are crashed or deleted for some reason.
   - CCP Adapter is crashed.

   Example of the kubectl command for getting the interworking pod:

   ```
   kubectl get pod -o wide -l app=antrea-interworking -n vmware-system-antrea
   ```

   Note down the interworking pod name.

   Example of the kubectl command for getting the detailed state of the interworking pod:

   Ensure that you replace *pod-name* with the actual pod name.

   ```
   kubectl get pod -o yaml pod-name -n vmware-system-antrea
   ```

   Example of the kubectl command for getting container logs:

   Ensure that you replace *pod-name* with the actual pod name.

   ```
   kubectl logs pod-name -c mp-adapter -n vmware-system-antrea > mp-adapter.log
   kubectl logs pod-name -c ccp-adapter -n vmware-system-antrea > ccp-adapter.log
   kubectl logs pod-name -c tn-proxy -n vmware-system-antrea > tn-proxy.log
   kubectl logs pod-name -c election-runner -n vmware-system-antrea > election-runner.log
   ```

   If the vmware-system-antrea namespace is missing or the interworking pod is missing, the adapters might have been deleted from the Kubernetes cluster without running the deregistration steps. You can clean up the leftover registration data and inventory from the system, and then register the Kubernetes cluster again. The cluster ID will be different after reregistering the cluster. If there is any Antrea policy applied to the cluster, you must apply the policy again after reregistering the cluster.

   For instructions about cleaning up the leftover registration data, see Clean up Antrea Data from NSX.

   For instructions about registering an Antrea Kubernetes cluster to NSX, see Registering an Antrea Kubernetes Cluster to NSX.

4. Use kubectl to get nsx-proxy container logs from the interworking pod, and check the error messages.

   This step can help you identify failure due to any of these reasons:

   - The client certificate of the adapters is incorrect.
   - The version of the adapters is incompatible with the CCP.

   For example commands, see step 3.

5. If the Management Plane Adapter is up, use the support bundle feature in NSX to collect log files for the Antrea Kubernetes cluster.

   For more information, see Collect Support Bundles for an Antrea Kubernetes Cluster.

Backing Up and Restoring NSX Manager or Global Manager

If an NSX Manager or a Global Manager appliance becomes inoperable or if you want to restore your environment to a previous state, you can restore from a backup. NSX Managers are called Local Managers if they are managed with a Global Manager using NSX Federation.

You can restore an NSX configuration back to the captured state in any of the backups. While the appliance is inoperable, the data plane is not affected but you cannot make configuration changes.

Note the following considerations:

- For NSX backup authentication to connect to the SFTP backup server, you can use the user name and password method or have the NSX Manager use an SSH private key.
- You must restore the same version you have backed up from your NSX appliance, to the new NSX appliances.
- The key types supported are: ECDSA, RSA, and ED25519.
- Storing libraries during backup is not supported in NSX backup. During restore NSX will always have default version libraries. If you are using NSX Application Platform, you must upload Kubernetes tools if requested.
- NSX Manager or Global Manager restore can use the same IP or a different IP address.
  - If you use an NSX Manager or Global Manager IP address to restore, then you must use the same IP address as in the backup.
  - If you use an NSX Manager or a Global Manager FQDN to restore, then you must use the same FQDN as in the backup. Use a lowercase FQDN only for backup and restore.
- Backups on Global Managers are possible only on active Global Managers.
- If both the active and standby Global Manager fail or only the active Global Manager fail, you need to:

Issue	Solution	Result
Both the active and standby Global Manager fail.	1. Delete both active and standby Global Managers. Ensure there are no Global Manager appliances up in any other clusters. 2. Deploy a new Global Manager with the same IP address/FQDN as old active Global Manager. 3. Restore the active Global Manager from backup. If any Local Managers are present, they sync to the new active Global Manager. 4. Deploy a new Global Manager on another site and onboard it to the restore Global Manager.	• The active Global Manager syncs with any Global Manager on the network. • The standby Global Manager syncs with the active Global Manager.
The active Global Manager fails and the standby Global Manager is up.	If standby Global Manager is in a good state, it automatically becomes the active Global Manager.	Standby Global Manager becomes active Global Manager. A new standby Global Manager must be manually added for backup. After the new active Global Manager is online, the Local Manager syncs up and ensures configuration replication.

Configure Backups

By default, VCF configures NSX Manager backups to the SDDC Manager appliance. You can use the VCF Ops workflows to change the destination of backups to an external SFTP server. For more information about changing the backup destination, see *Reconfigure SFTP Backups for SDDC Manager and NSX Manager* in the section Backup and Restore of VMware Cloud Foundation.

- Verify that the SFTP server is running the supported OS and the supported SFTP software. The following table displays the supported and tested software for backup, although other software versions might work.

Currently supported OS	Specifically tested version	SFTP software version
CentOS	8.4	OpenSSH_8.0p1
	7.9 or 7.7	OpenSSH_7.4p1
RHEL	8.4	OpenSSH_8.0p1
	7.9 or 7.7	OpenSSH_7.4p1
Ubuntu	20.04	OpenSSH_8.2p1
	18.04	OpenSSH_7.6p1

Currently supported OS	Specifically tested version	SFTP software version
Windows	Windows Server 2019	OpenSSH_10.0p2

- Verify that the SFTP server is ready for use and is running SSH and SFTP, using the following commands:
  - `$ ssh backup_user@sftp_server`
  - `$ sftp backup_user@sftp_server`
- Verify the required hashed ECDSA host key is present on the backup server. See Find the SSH Fingerprint of a Remote Server.
- Ensure that the directory path where you want to store your backups exists and that you have read/write permissions to that directory.You cannot use the root directory (/).
- If you are using the SSH private key option, ensure:
  - The SFTP backup server configuration includes an SSH public key linked to one of the users (in ~/.ssh/authorized_keys in a Linux server).
  - To complete the NSX Manager configuration, you have the corresponding private key.
  - You verify that you are storing the private key in a place other than the configuration.
- If you have multiple NSX deployments, you must use a different directory for storing the backup of each deployment.
- If your NSX Manager or your Global Manager appliance has the DNS server access set to "publish_fqdns": true, you must configure that setting on the new NSX Manager or new Global Manager appliance before restore. Follow instructions at "Configuring NSX Manager for Access by the DNS Server" in the *NSX Installation Guide*.

Perform the following steps to configure backups:

Before backups can occur, you must configure a backup file server. After you configure a backup file server, you can start a backup at any time or schedule recurring backups. Administrators can choose to connect to the SFTP backup server using an SSH private key or a password-based authentication.

1. From a browser, log in with admin privileges to the NSX Manager or Global Manager at https://<manager-ip-address>.
2. Select **System** > **Backup & Restore**.
3. Click **Edit** under the **SFTP Server** label to configure your SFTP server.
4. Enter the FQDN or IP address of the backup file server.

   The protocol text box is already filled in. SFTP is the only supported protocol.
5. Change the default port if necessary. The default TCP port is 22.
6. In the **Directory Path** text box, enter the absolute directory path where the backups will be stored.

   The directory must already exist and cannot be the root directory (`/`). Do not use path drive letters or spaces in directory names as they are not supported. If the backup file server is a Windows machine, you must use the forward slash when you specify the destination directory. For example, if the backup directory on the Windows machine is `c:\SFTP_Root\backup`, specify `/SFTP_Root/backup` as the destination directory.

   The path to the backup directory can contain only the following characters: alphanumerics (a-z , A-Z, 0-9), underscore (_) , plus and minus sign (+ -), tilde and percent sign (~ %), forward slash (/), and period (.).

   The backup process generates a name for the backup file that can be long. On a Windows server, the length of the full path name of the backup file can exceed the limit set by Windows that can cause backups to fail. To avoid this issue, see Knowledge Base article 312385: NSX backups are failing on Windows backup server.
7. Choose which authentication method you want to use to log into the backup file server.
   a) To enter a user name and password that authenticates to the backup file server, select **Password** and enter the required information.
   b) To use an SSH private key to authenticate against the backup file server, select **SSH Private Key** and enter the required information.

   If you edit the backup configuration, you do not need to re-enter the password or SSH Private Key.

   Support also includes RSA for SSH private key generation using key sizes 1024-bits, 2048-bits, and 4096-bits. 4096-bits is recommended.

8. You can leave the **SSH Fingerprint** blank and accept or reject the fingerprint that is provided by the server after you click **Save** in a later step. If necessary, you can retrieve the SSH fingerprint by using this API: `POST /api/v1/cluster/backups?action=retrieve_ssh_fingerprint`.

   For more details, see Find the SSH Fingerprint of a Remote Server.

9. Verify that one of the required host keys from the list of ECDSA, RSA, and ED25519 is present on the backup server by running the following command: `#ssh-keyscan -t ecdsa,rsa,ed25519 <backup server IP/FQDN> | ssh-keygen -lf -.`

   ```
   #ssh-keyscan -t ecdsa,rsa,ed25519  ftpserver.corp.local | ssh-keygen -lf -
   3072 SHA256:j2ct7Sd0YLesHw/n3LOTvOWyq35t6Celj0QH9gg56p4 ftpserver.corp.local (RSA)
   256 SHA256:2yhk6DD/HG+fv1P6sG0ZPJ2UipXpCHXB9EfSlMg6LtE ftpserver.corp.local (ECDSA)
   256 SHA256:WiGm7uZono12UX9xJu2vmqz1w8+gJ3Yy6Uv4aNmvaxM ftpserver.corp.local (ED25519)
   ```

   The command returns the already configured keys on the SFTP server. For example, if ECDSA and RSA keys are configured on the SFTP server then the command will return only those two keys. Use any of the keys returned by the command to configure backup. If the command does not return the required key, then configure it on the backup server. Contact the OS vendor if you need guidance for that configuration.

   NSX supports the following host keys:

   - ecdsa-sha2-nistp256
   - ecdsa-sha2-nistp384
   - ecdsa-sha2-nistp521
   - ssh-ed25519
   - rsa-sha2-512
   - rsa-sha2-256

10. Enter a passphrase.

    Important:

    You will need this passphrase to restore a backup. If you forget the passphrase, you cannot restore any backups.

11. Click **Save**.

The Backup and Restore page refreshes with the newly configured SFTP server updated.

After you successfully configure a backup file server, you can click **Start Backup** to manually start a backup immediately. Or, to schedule recurring automatic backups see Start or Schedule Backups. To see a list of available backups if you cannot access an NSX Manager or Global Manager appliance, see Listing Available Backups for details.

Start or Schedule Backups

After you configure your SFTP file server you can start a backup at any time or schedule recurring backups.

Complete the instructions at Configure Backups.

When you set up recurring backups, the system automatically backs up the inventory if there is an inventory change, such as the addition or removal of a Transport Node. This feature is not available for manual backups. You can also trigger backups for configuration changes. You can optionally select both options for recurring backups.

Inventory backups do not get collected for Global Manager.

1. To schedule a backup, click **Edit** under the **Schedule** label on the **System** > **Backup and Restore**.
   a) Click the **Recurring Backup** toggle.
   b) To set a weekly schedule, click **Weekly** and set the days and time of the backup.
   c) To set up to a 24 hour interval, click **Interval** and set the interval between backups.
2. To trigger an unscheduled full configuration backup when the system detects any user, runtime, or non-configuration related changes, click the **Detect NSX configuration change** toggle.

   For Global Manager, this setting triggers a backup when the system detects any changes in the database, such as the addition or removal of a Local Manager, tier-0 gateway, or DFW policy.

You can specify a time interval for detecting database configuration changes. The valid range is 5 minutes to 1,440 minutes (24 hours). This option can potentially generate a large number of backups. Use it with caution.

If you selected **Start Backup**, you see a progress bar of the in-progress backup.

When the manual or scheduled backup completes, the backup gets listed in the Backup History section of the page. The **Last Backup Status** label indicates whether the backup was successful and lists the timestamp, node, and cluster details of the appliance backed up. If the backup fails, you can see an error message.
To restore a backup follow instructions at Restore a Backup.

Remove Old Backups

Backups can accumulate on the backup file server and consume a large amount of storage. You can run a script that comes with NSX to automatically delete old backups.

You can find the Python script `nsx_backup_cleaner.py` in the directory `/var/vmware/nsx/file-store` on NSX Manager. To access this file, you must log in as root. Typically, you schedule a job on the backup file server to run this script periodically to clean up old backups. The script works only for subfolders named cluster-node-backups and inventory-summary. The script fails if any other subfolders are present in the backup directory other than cluster-node-backups and inventory-summary.

The following usage information describes how to run the script:

```
nsx_backup_cleaner.py -d backup_dir [-k 1] [-l 5] [-h]
Or
nsx_backup_cleaner.py --dir backup_dir [--retention-period 1] [--min-count 5] [--help]

Required parameters:
    -d/--dir: Backup root directory
    -k/--retention-period: Number of days need to retain a backup file

Optional parameters:
    -l/--min-count: Minimum number of backup files to be kept, default value is 100
    -h/--help: Display help message
```

The age of a backup is calculated as the difference between the backup's timestamp and the time the script is run. If this value is larger than the retention period, the backup is deleted if there are more backups on the disk than the minimum number of backups.

If you have backed up multiple release versions, then for each release version, the script retains the minimum number of backup files specified by `-l/--min-count`.

For more information about setting up the script to run periodically on a Linux or Windows server, see the comments at the beginning of the script.

Listing Available Backups

The backup file server stores backups from all the NSX Manager or Global Manager nodes. To get the list of backups so that you can find the one you want to restore, you must run the `get_backup_timestamps.sh` script.

> **Note:**
> Use the same key size for backup and restore. If the key size is different at time of backup and restore, the backup does not appear in the list. Starting in NSX 3.2.1, support includes key size 256-bit, 384-bit, and 521-bit. In 3.2.0, support includes only 256-bit key size.

The script can be found on each NSX Manager or Global Manager appliance at `/var/vmware/nsx/file-store/get_backup_timestamps.sh`. You can run this script on any Linux machine or on the NSX appliance. As a best practice, copy this script after installing NSX to a machine that is not an NSX Manager or Global Manager so that you can run this script even if all the NSX Manager or Global Manager nodes become inaccessible. If you need to restore a backup but have no access to this script, you can install a new NSX Manager or Global Manager node and run the script there.

You can copy the script to another machine or to the backup file server by logging in to the NSX Manager or Global Manager as admin and running a CLI command. For example:

```
nsxmgr-1> copy file get_backup_timestamps.sh url scp://admin@server1/tmp/
admin@server's password:
nsxmgr-1>
```

The script is interactive and prompts you for the information that you specified when you configured the backup file server. You can specify the number of backups to display. Each backup is listed with a timestamp, the NSX Manager or Global Manager node's IP address or FQDN if the NSX Manager or Global Manager node is set up to publish its FQDN, and the node ID. For example,

```
admin@host1:/home/admin# ./get_backup_timestamps.sh
Enter file server ip:
10.10.10.20
Enter port:
22
Enter directory path:
/home/nsx/backups
Enter number of latest backup or press Enter to list all backups:

root@10.10.10.20's password:
Latest backups:
[Backup timestamp; IP address/FQDN; Node id]
2019-01-22;09:16:43 nsxmgr.example.com 41893642-597b-915f-5117-7da576df4ff2
2019-01-22;09:14:42 nsxmgr.example.com 41893642-597b-915f-5117-7da576df4ff2
2019-01-22;09:13:30 nsxmgr.example.com 41893642-597b-915f-5117-7da576df4ff2
2019-01-22;09:01:52 10.10.10.77 35163642-6623-8f6d-7af0-52e03f16faed
2019-01-22;09:00:33 10.10.10.77 35163642-6623-8f6d-7af0-52e03f16faed
```

Restore a Backup

Restoring a backup restores the state of the network at the time of the backup. In addition, the configurations maintained by NSX Manager or Global Manager appliances are also restored. For NSX Manager, any changes, such as adding or deleting nodes, that were made to the fabric since the backup was taken, are reconciled. NSX Managers are called Local Managers (LM) when they are federated with a Global Manager (GM).

- Verify that you have the login credentials (username and password or the SSH private key matching the public key stored on the backup file server) for the backup file server.
- Verify that you have the SSH fingerprint of the backup file server. Support includes key size 256-bit, 384-bit, and 521-bit. Ensure whatever key size is used at time of backup is used at time of restore.
- Verify that you have the passphrase of the backup file.
- Identify which backup you want to restore by following the procedure in Listing Available Backups. Take note of the IP or FQDN of the NSX appliance that took the backup.
- Ensure the network setup where you are performing the restore has the same set of network connectivity as the system on which you performed the backup. For example, the same VIPs, DNS, NTP communication,and so on. If network connectivity is not same, fix the inconsistencies before adding a second or third node to the restored system.
- Perform a federated restore when both the active and standby Global Managers are down. If this is not the case, see Backup and Restore in NSX Federation.
- Familiarize yourself with the Management Plane upgrade process as part of restoring a backup during an upgrade. For details, see *Backup and Restore During Upgrade* in the *NSX Upgrade Guide*.

  Note:

  DNS entries (name servers and search domains) are not retained when you restore from a backup.To redeploy in a VMware Cloud Foundation (VCF) deployment using an OVF file, you must use FQDNs for the NSX Manager VM names.

You must restore the backup to a new NSX Manager or Global Manager appliance. Follow the instructions for your specific case.

- If you had a cluster of the NSX Manager appliance when the backup was taken, the restore process restores one node first and then prompts you to add the other nodes. You can add the other nodes during the restore process or after the first node is restored. See the following detailed steps.
- If you had a cluster of Global Manager appliances, you can only restore one node using the restore process. You must create the cluster after the restore of the first node completes. For instructions on restoring a lost active Global Manager, a lost standby Global Manager, or a lost Local Manager, see Backup and Restore in NSX Federation.

Important:
If any nodes in the appliance cluster are still available, you must power them off before you start the restore.

1. If any nodes in the appliance cluster are still available, you must power them off before you start the restore.
2. Install one new appliance node on which to restore the backup.
   - If the backup listing for the backup you are restoring contains an IP address, you must deploy the new NSX Manager or Global Manager node with the same IP address. Do not configure the node to publish its FQDN.
   - If the backup listing for the backup you are restoring contains an FQDN, you must configure the new appliance node with this FQDN and publish the FQDN. Only lowercase FQDN is supported for backup and restore.

     Note:
     Until the FQDN is configured and published, the Restore button for the backup is disabled in the newly deployed NSX Manager or Global Manager UI.

     Use this API to publish the NSX Manager or Global Manager FQDN.

     Example request:

     ```
     PUT https://<nsx-mgr OR global-mgr>/api/v1/configs/management

     {
      "publish_fqdns": true,
      "_revision": 0
     }
     ```

     See the *NSX API Guide* for API details.

     In addition, if the new manager node has a different IP address than the original one, you must update the DNS server's forward and reverse lookup entries for the manager node with the new IP address.

   After the new manager node is running and online, you can proceed with the restore.
3. From a browser, log in with admin privileges to the NSX Manager or Global Manager at https://<manager-ip-address>.
4. Select **System** > **Backup & Restore**.
5. To configure the backup file server, click **Edit**.

   Do not configure automatic backup if you are going to perform a restore.
6. Enter the IP address or FQDN.
7. Change the port number, if necessary.

   The default is 22.
8. In the **Directory Path** text box, enter the absolute directory path where the backups are stored.

   The path to the backup directory can contain only the following characters: alphanumerics (a-z , A-Z, 0-9), underscore (_) , plus and minus sign (+ -), tilde and percent sign (~ %), forward slash (/), and period (.).

   Avoid using path drive letters or spaces in directory names; they are not supported. If the backup file server is a Windows machine, you must use the forward slash when you specify the destination directory. For example, if the backup directory on the Windows machine is c:\SFTP_Root\backup, specify /SFTP_Root/backup as the destination directory.

9. To log in to the server, enter the user name and password or the SSH private key, depending on your backup authentication scheme.
10. You can leave the SSH Fingerprint blank and accept or reject the fingerprint provided by the server after you click **Save** in a later step. If necessary, you can retrieve the SSH fingerprint by using this API: `POST /api/v1/cluster/backups?action=retrieve_ssh_fingerprint`.
11. Enter the passphrase that was used to encrypt the backup data.
12. Click **Save**.
13. Select a backup.
14. Click **Restore**.
15. The restore process prompts you to take action, if necessary, as it progresses.

    Note:
    If you are restoring a Global Manager appliance, the following steps do not appear.After restoring the first Global Manager node, you must manually join the other nodes to form the cluster.If you are restoring a multi-site network, see the "Limitations" section of the NSX Multisite topic.

    a) Confirm CM/VC Connectivity: If you want to restore existing compute managers, ensure that they are registered with the new NSX Manager node and available during the restore process.
    b) If you have deleted or added fabric nodes or transport nodes, you are prompted to take certain actions, for example, log in to a node and run a script. If you have created a logical switch or segment since the backup, the logical switch or segment will not appear after the restore.
    c) If the backup has information about a manager cluster, you are prompted to add other nodes. If you decide not to add nodes, you can still proceed with the restore and manually add other nodes to form the cluster after the restore of this node completes.
    d) If there are fabric nodes that did not discover the new manager node, you are provided a list of them.
    e) Storing libraries during backup is not supported in NSX backup. If you are using NSX Application Platform, you must upload Kubernetes tools if requested.

    A progress bar displays the status of the restore operation noting the step the restore process is on. During the restore process, services on the manager appliance get restarted and the control plane becomes unavailable until restore completes.

    After the restore operation is finished, the **Restore Complete** screen shows the result of the restore, the timestamp of the backup file, and the start and end time of the restore operation. Any segments created after the backup was taken are not restored.

    If the restore operation fails, the screen displays the step where the failure occurred, for example, `Current Step: Restoring Cluster (DB)` or `Current Step: Restoring Node`. If either cluster restore or node restore fails, the error might be transient. In that case, there is no need to click **Retry**. You can restart or reboot the manager and the restore operation continues.

    You can also determine if there was a cluster or node restore failure by selecting the log files. Run `get log-file syslog` to view the system log file and search for the strings `Cluster restore failed` and `Node restore failed`.

    To restart the manager, run the `restart service manager` command.

    To reboot the manager, run the `reboot` command.

    Note:

    If you added a compute manager after the backup, and you try to add the compute manager again after the restore, you get an error message indicating that registration failed. Click the **Resolve** button to resolve the error and successfully add the compute manager. For more information, see Add a Compute Manager, step 4. If you want to remove information about NSX that is stored in a vCenter, follow the steps in Remove NSX Extension from vCenter.

    If the vCenter was registered with custom ports in the backup, you must manually open all the custom ports on the restored manager appliances.

16. If you have only one node deployed, after the restored manager node is up and functional, you can deploy additional nodes to form a cluster.

    See the *NSX Installation Guide* for instructions.

17. If you had other manager cluster VMs that you powered down in Step 1, delete them after the new manager cluster is deployed.

Operations and Management

You may need to change the configuration of the appliances you've installed, for example, adding licenses, certificates, and changing passwords. There are also routine maintenance tasks that you should perform, including running backups. Additionally, there are tools to help you find information about the appliances that are part of the NSX infrastructure and the logical networks created by NSX, including remote system logging, traceflow, and port connections.

View the Usage and Capacity of Categories of Objects

You can view the usage and capacity of different categories of objects. By default, an alarm is generated when the current inventory a category of objects reaches a certain level.

You can set thresholds for each object category. If a threshold is reached, an alarm will be generated. You can see alarms by navigating to **Home** > **Alarms**.

To see the usage and capacity of different categories of objects, click one of the following tabs:

- **Networking** > **Network Overview** > **Capacity**
- **Security** > **Security Overview** > **Capacity**
- **Inventory** > **Inventory Overview** > **Capacity**
- **System** > **System Overview** > **Capacity**

You can also navigate to **Plan & Troubleshoot** > **Consolidated Capacity** to see all the object categories on one page.

Capacity Information

On each capacity page, for each category of objects, the following information is displayed:

- Current Inventory - The number of objects that have been successfully created or configured. A color-coded bar is displayed to indicate the usage percentage. If usage is below the minimum capacity threshold, the color is green. If usage is at or above the minimum capacity threshold but below the maximum capacity threshold, the color is orange. If usage is at or above the maximum capacity threshold, the color is red.
- Maximum Capacity.
- Minimum Capacity Threshold - This is the usage level at which the usage bar mentioned above will show an orange color. You can change this value. The default is 70%.
- Maximum Capacity Threshold - This is the usage level at which the usage bar mentioned above will show a red color. You can change this value. The default is 100%.

If you change the minimum or maximum capacity threshold, you can click **Revert** to go back to the last saved value. You can click **Reset Values** to restore the default values for all the object categories.

Object Categories

The networking capacity page shows the following object categories:

- Segments
- System-wide DHCP Ranges
- Tier-0 Gateways
- Tier-1 Gateways
- DHCP Server Instances
- System-wide NAT Rules
- Prefix Lists
- Tier-1 Gateways with NAT Enabled

- Segment Ports
- Distributed Port Groups

The security capacity page shows the following object categories:

- Introspection Rules N-S Tier-1
- Service Chains
- Active Directory Groups (Identity Firewall)
- Saved Firewall Rules Configuration
- Introspection Policies E-W
- Introspection Policies N-S Tier-0
- System-wide Firewall Rules
- System-wide Endpoint Protection Enabled Virtual Machines
- Introspection Rules N-S Tier-0
- Introspection Rules E-W
- Distributed Firewall Sections
- Introspection Policies N-S Tier-1
- System-wide Firewall Sections
- Active Directory Domains (Identity Firewall)
- System-wide Endpoint Protection Enabled Hosts
- Distributed Firewall Rules
- Service Paths

The inventory capacity page shows the following object categories:

- Groups
- vSphere Clusters Prepared for NSX
- Services
- Hypervisor Hosts
- Groups based on IP Addresses
- System-wide Geo IP Groups

The system capacity page shows the following object categories:

- Compute Managers
- System-wide Edge(s) / VNA(s)
- Edge/VNA Clusters

Configure the Login Window with a User Agreement Banner

You can configure a custom login message that requires a user agreement as part of the login process for NSX.

If activated (or enabled), this agreement displays at every login and is changed only by users with Enterprise admin roles. For Federation, each Global and Local Manager instance requires its own configuration. If you have DoD or other security compliance requirements, this feature provides such options.

1. From your browser, log in with admin privileges to an NSX Manager or Global Manager at `https://<nsx-manager-ip-address>`.
2. Navigate to **System** > **General System Settings**.
3. To add or remove banner settings, click the **User Interface** tab and click **Edit** for **Login Consent Settings**.
4. To activate or deactivate this option, toggle the Login Consent **On** or **Off**.
5. To require explicit user consent before the user logs in, select **Yes**.
6. To create a consent message that displays on the login page, enter the consent message title.
   This message starts with the phrase **I agree to …** and can be up to 255 characters.

7. If required, add a custom consent message description.

   The description limit is 64,000 characters. Users must select the title to view this description. HTML links are not supported.

Configure a Node Profile

You can configure settings -- such as time zone, NTP servers, SNMP, syslog servers, DNS servers, and search domains -- to apply to all NSX Manager, NSX Edge, and Virtual Network Appliance (VNA) nodes.

This profile represents a collection of time zone, NTP servers, SNMP configuration and syslog servers. By default, the node profile is applied to all nodes, unless the node is configured to not accept such configuration from the NSX Manager. To prevent a node from accepting the node profile, use the CLI command `set node central-config disabled` on that node.

1. With admin privileges, log in to NSX Manager.
2. Select **System** > **Fabric** > **Profiles**.
3. Click the **Node Profiles** tab.
4. Under **General Settings**, click **Edit** to configure the time zone, NTP servers, search domains, and DNS servers.
5. Under **Syslog Servers**, click **Add Syslog Server** to add a Syslog server.
   a) Enter the FQDN or IP address of the Syslog server.
   b) Specify a port number.
   c) Select a protocol.

      The available protocols are **TCP**, **UDP**, and **LI (Aria Operations for Logs)**.
   d) Select a log level.

      The available levels are **Emergency**, **Alert**, **Critical**, **Error**, **Warning**, **Notice**, **Information**, and **Debug**.

      When you choose a level, you will also see the logs for all the previous levels (starting with the **Emergency** level). For example, if you choose **Emergency**, you will see **Emergency**-level logs. If you choose **Critical**, you will see logs for **Emergency**, **Alert** and **Critical**. If you choose **Information**, you will see logs for **Emergency**, **Alert**, **Critical**, **Error**, **Warning**, **Notice**, and **Information**. If you choose **Debug**, you will see messages for all log levels.
6. In the **SNMP Polling** section, under **v2c**, click **Add SNMP v2c Community** to add an SNMPv2c community.
   a) Enter a name for the community.
   b) Enter a **Community String** value.

      This value is used for authentication.
7. In the **SNMP Polling** section, under **v3**, click **Add SNMP v3 User** to add an SNMPv3 user.
   a) Enter a user name.
   b) Enter an authentication password.

      You can click the icon on the right to show or hide the password.
   c) Enter a private password.

      You can click the icon on the right to show or hide the password.
8. In the **SNMP Traps** section, under **v2c**, click **Add SNMP v2c Trap Configuration** to add an SNMPv2c trap configuration.
   a) Enter a FQDN or IP address.
   b) Specify a port number.
   c) Enter a name for the community.
   d) Enter a **Community String** value.

      This value is used for authentication.

9. In the **SNMP Traps** section, under **v3**, click **Add SNMP v3 Trap Configuration** to add an SNMPv3 trap configuration.
   a) Enter a FQDN or IP address.
   b) Specify a port number.
   c) Enter a user name.

Verify that the profile configurations are applied to the NSX Manager, NSX Edge, and VNA nodes. Log in to the NSX Manager, NSX Edge, and VNA nodes with **admin** privileges, and run the following commands:

- `get clock`
- `get ntp-server`
- `get logging-servers`
- `get snmp v2-targets`
- `get snmp v3-targets`
- `get snmp v2-configured`
- `get snmp v3-configured`
- `get snmp v3-engine-id`
- `get snmp v3-protocols`
- `get snmp v3-users`

For more information about these commands including examples, see the NSX CLI Guide.

Error situations

If the node profile configurations are not applied successfully, then there are two possibilities:

- The central configuration was not synchronized with the remote node due to connectivity issues between NSX Manager and the remote node. In this case, you cannot do anything from the central configuration side.
- The central configuration was synchronized with the remote node, but the command to apply the central configuration failed to run. In this case, you can check syslog on the remote node.
  In the logs, search for the `subcomp="central_node_config_update"` string to look for any errors.
  For example, the syslog exporter configuration might fail if the host name specified cannot be resolved to IP addresses, or if a second VCF Operations for logs server is being configured.
  The following example logs show the error messages:
  Log example 1:

```
2020-05-18T22:56:06.485Z vmw-svc.nsxmanager-sb-36265022-1-rhel NSX 24904 -
 [nsx@6876 comp="nsx-manager"
subcomp="central_node_config_update" username="root" level="INFO"]
No change in timezone 2020-05-18T22:56:07.184Z vmw-svc.nsxmanager-sb-36265022-1-
rhel NSX 24904 -
[nsx@6876 comp="nsx-manager" subcomp="central_node_config_update" username="root"
 level="INFO"]
No change in NTP configuration 2020-05-18T22:56:07.210Z vmw-svc.nsxmanager-
sb-36265022-1-rhel NSX 24904 -
[nsx@6876 comp="nsx-manager" subcomp="central_node_config_update" username="root"
 level="INFO"]
Updating Syslog configuration 2020-05-18T22:56:08.826Z vmw-svc.nsxmanager-
sb-36265022-1-rhel NSX 24904 -
[nsx@6876 comp="nsx-manager" subcomp="central_node_config_update" username="root"
 level="WARNING"]
Failed to add syslog exporter {"port": 514, "exporter_name": "264aa005-dfb0-4942-
a1c4-f749bfc1a2c4",
"protocol": "TCP", "level": "ERR", "server": "vikas.2020.com"}, response:
{#012  "error_code": 36569,#012  "error_message": "Error modifying firewall rule
 due
to invalid hostname.",#012  "module_name": "node-services"#012}, status: 400, err:
400 Client Error: Bad Request for url: http://localhost:7441/api/v1/node/services/
syslog/exporters
```

  Log example 2:

```
2020-05-18T22:56:08.839Z vmw-svc.nsxmanager-sb-36265022-1-rhel NSX 24904 -
[nsx@6876 comp="nsx-manager" subcomp="central_node_config_update" username="root"
 level="WARNING"]
Failed to add syslog exporter {"port": 514, "exporter_name": "f4e088d4-4b45-42fe-
ba1d-7f98838c7f61",
"protocol": "LI", "level": "INFO", "server": "loginsight.vmware.com"},
response: {#012  "error_code": 36400,#012  "error_message": "Maximum number of
 loginsight
servers exceeded",#012  "module_name": "node-services"#012}, status: 400, err:
400 Client Error: Bad Request for url: http://localhost:7441/api/v1/node/services/
syslog/exporters
```

Log example 3:

```
2020-05-18T22:56:10.639Z vmw-svc.nsxmanager-sb-36265022-1-rhel NSX 24904 -
[nsx@6876 comp="nsx-manager" subcomp="central_node_config_update" username="root"
 level="WARNING"]
Failed to add syslog exporter {"port": 514, "exporter_name": "d0dc1797-b5dc-42ba-
b07d-fe107dd70111",
"protocol": "UDP", "level": "INFO", "server": "logging.vmware.com"},
response: {#012  "error_code": 36569,#012  "error_message": "Error modifying
 firewall
rule due to invalid hostname.",#012  "module_name": "node-services"#012}, status:
 400,
err: 400 Client Error: Bad Request for url: http://localhost:7441/api/v1/node/
services/syslog/exporters
```

Checking the Realized State of a Configuration Change

When you make a configuration change, NSX Manager typically sends a request to another component to implement the change. For the supported Layer 3 entities listed on this page, if you make the configuration change using the API, you can track the status of the request to see if the change is successfully implemented.

The configuration change that you initiate is called the desired state. The result of implementing the change is called the realized state. If NSX Manager implements the change successfully, the realized state will be the same as the desired state. If there is an error, the realized state will not be the same as the desired state.

For supported Layer 3 entities, when you call an API to make a configuration change, the response will include the parameter `request_id`. You can use the parameters `request_id` and the `entity_id` to make an API call to find out the status of the request.

This feature supports the following entities and APIs:

```
EdgeCluster
    POST /edge-clusters
    PUT /edge-clusters/<edge-cluster-id>
    DELETE /edge-clusters/<edge-cluster-id>
    POST /edge-clusters/<edge-cluster-id>?action=replace_transport_node

LogicalRouter
    POST /logical-routers
    PUT /logical-routers/<logical-router-id>
    DELETE /logical-routers/<logical-router-id>
    POST /logical-routers/<logical-router-id>?action=reprocess
    POST /logical-routers/<logical-router-id>?action=reallocate

LogicalRouterPort
    POST /logical-router-ports
    PUT /logical-router-ports/<logical-router-port-id>
    DELETE /logical-router-ports/<logical-router-port-id>

StaticRoute
```

```
    POST /logical-routers/<logical-router-id>/routing/static-routes
    PUT /logical-routers/<logical-router-id>/routing/static-routes/<static-route-id>
    DELETE /logical-routers/<logical-router-id>/routing/static-routes/<static-route-id>

BGPConfig
    PUT /logical-routers/<logical-router-id>/routing/bgp

BgpNeighbor
    POST /logical-routers/<logical-router-id>/routing/bgp/neighbors
    PUT /logical-routers/<logical-router-id>/routing/bgp/neighbors/<bgp-neighbor-id>
    DELETE /logical-routers/<logical-router-id>/routing/bgp/neighbors/<bgp-neighbor-id>
    POST /logical-routers/<logical-router-id>/routing/bgp/neighbors/<bgp-neighbor-id>

BGPCommunityList
    POST /logical-routers/<logical-router-id>/routing/bgp/community-lists
    PUT /logical-routers/<logical-router-id>/routing/bgp/community-lists/<community-list-id>
    DELETE /logical-routers/<logical-router-id>/routing/bgp/community-lists/<community-list-id>

AdvertisementConfig
    PUT /logical-routers/<logical-router-id>/routing/advertisement

AdvertiseRouteList
    PUT /logical-routers/<logical-router-id>/routing/advertisement/rules

NatRule
    POST /logical-routers/<logical-router-id>/nat/rules
    PUT /logical-routers/<logical-router-id>/nat/rules/<rule-id>
    DELETE /logical-routers/<logical-router-id>/nat/rules/<rule-id>

DhcpRelayService
    POST /dhcp/relays
    PUT /dhcp/relays/<relay-id>
    DELETE /dhcp/relays/<relay-id>

DhcpRelayProfile
    POST /dhcp/relay-profiles
    PUT /dhcp/relay-profiles/<relay-profile-id>
    DELETE /dhcp/relay-profiles/<relay-profile-id>

StaticHopBfdPeer
    POST /logical-routers/<logical-router-id>/routing/static-routes/bfd-peers
    PUT /logical-routers/<logical-router-id>/routing/static-routes/bfd-peers/<bfd-peers-id>
    DELETE /logical-routers/<logical-router-id>/routing/static-routes/bfd-peers/<bfd-peers-id>

IPPrefixList
    POST /logical-routers/<logical-router-id>/routing/ip-prefix-lists
    PUT /logical-routers/<logical-router-id>/routing/ip-prefix-lists/<ip-prefix-list-id>
    DELETE /logical-routers/<logical-router-id>/routing/ip-prefix-lists/<ip-prefix-list-id>

RouteMap
    POST /logical-routers/<logical-router-id>/routing/route-maps
    PUT /logical-routers/<logical-router-id>/routing/route-maps/<route-map-id>
    DELETE /logical-routers/<logical-router-id>/routing/route-maps/<route-map-id>

RedistributionConfig
    PUT /logical-routers/<logical-router-id>/routing/redistribution
RedistributionRuleList
```

```
    PUT /logical-routers/<logical-router-id>/routing/redistribution/rules

BfdConfig
    PUT /logical-routers/<logical-router-id>/routing/bfd-config

MplsConfig
    PUT /logical-routers/<logical-router-id>/routing/mpls

RoutingGlobalConfig
    PUT /logical-routers/<logical-router-id>/routing

IPSecVPNIKEProfile
    POST /vpn/ipsec/ike-profiles
    PUT /vpn/ipsec/ike-profiles/<ike-profile-id>
    DELETE /vpn/ipsec/ike-profiles/<ike-profile-id>

IPSecVPNDPDProfile
    POST /vpn/ipsec/dpd-profiles
    PUT /vpn/ipsec/dpd-profiles/<dpd-profile-id>
    DELETE /vpn/ipsec/dpd-profiles/<dpd-profile-id>

IPSecVPNTunnelProfile
    POST /vpn/ipsec/tunnel-profiles
    PUT /vpn/ipsec/tunnel-profiles/<tunnel-profile-id>
    DELETE /vpn/ipsec/tunnel-profiles/<tunnel-profile-id>

IPSecVPNLocalEndpoint
    POST /vpn/ipsec/local-endpoints
    PUT /vpn/ipsec/local-endpoints/<local-endpoint-id>
    DELETE /vpn/ipsec/local-endpoints/<local-endpoint-id>

IPSecVPNPeerEndpoint
    POST /vpn/ipsec/peer-endpoints
    PUT /vpn/ipsec/peer-endpoints/<peer-endpoint-id>
    DELETE /vpn/ipsec/peer-endpoints/<peer-endpoint-id>

IPSecVPNService
    POST /vpn/ipsec/services
    PUT /vpn/ipsec/services/<service-id>
    DELETE /vpn/ipsec/services/<service-id>

IPSecVPNSession
    POST /vpn/ipsec/sessions
    PUT /vpn/ipsec/sessions/<session-id>
    DELETE /vpn/ipsec/sessions/<session-id>

DhcpServer
    POST /dhcp/servers
    PUT /dhcp/servers/<server-id>
    DELETE /dhcp/servers/<server-id>

DhcpStaticBinding
    POST /dhcp/servers/static-bindings
    PUT /dhcp/servers/<server-id>/static-bindings/<binding-id>
    DELETE /dhcp/servers/<server-id>/static-bindings/<binding-id>

DhcpIpPool
```

```
    POST /dhcp/servers/ip-pools
    PUT /dhcp/servers/<server-id>/ip-pools/<pool-id>
    DELETE /dhcp/servers/<server-id>/ip-pools/<pool-id>

DnsForwarder
    POST /dns/forwarders
    PUT /dns/forwarders/<forwarder-id>
    DELETE /dns/forwarders/<forwarder-id>
```

You can call the following APIs to get the realized states:

```
EdgeCluster
Request - GET /edge-clusters/<edge-cluster-id>/state?request_id=<request-id>
Response - An instance of EdgeClusterStateDto which will inherit ConfigurationState. If the edge cluster is
 deleted then the state will be unknown and it will return the common entity not found error.

LogicalRouter / All L3 Entites - All L3 entities can use this API to get realization state
Request - GET /logical-routers/<logical-router-id>/state?request_id=<request-id>
Response - An instance of LogicalRouterStateDto which will inherit ConfigurationState. Delete operation of
 any entity other than logical router can be covered by getting the state of logical router but if the logical
 router itself is deleted then the state will be unknown and it will return the common entity not found error.

LogicalServiceRouterCluster - All L3 entities which are the part of services can use this API to get the
 realization state
Request - GET /logical-routers/<logical-router-id>/service-cluster/state?request_id=<request-id>
Response - An instance of LogicalServiceRouterClusterState which will inherit ConfigurationState.

LogicalRouterPort / DhcpRelayService / DhcpRelayProfile
Request - GET /logical-router-ports/<logical-router-port-id>/state?request_id=<request-id>
Response - An instance of LogicalRouterPortStateDto which will inherit ConfigurationState.

IPSecVPNIKEProfile / IPSecVPNDPDProfile / IPSecVPNTunnelProfile / IPSecVPNLocalEndpoint /
 IPSecVPNPeerEndpoint / IPSecVPNService / IPSecVPNSession
Request - GET /vpn/ipsec/sessions/<session-id>/state?request_id=<request-id>
Response - An instance of IPSecVPNSessionStateDto which will inherit ConfigurationState. If the session
 is deleted then the state will be unknown and it will return the common entity not found error. When
 IPSecVPNService is disabled, IKE itself is down and it does not respond. It will return unknown state in such
 a case.

DhcpServer
Request - GET /dhcp/servers/<server-id>/state?request_id=<request-id>
Response - An instance of ConfigurationState.

DhcpStaticBinding
Request - GET /dhcp/servers/<server-id>/static-bindings/<binding-id>/state?request_id=<request-id>
Response - An instance of ConfigurationState.

DhcpIpPool
Request - GET /dhcp/servers/<server-id>/ip-pools/<pool-id>/state?request_id=<request-id>
Response - An instance of ConfigurationState.

DnsForwarder
Request - GET /dns/forwarders/<forwarder-id>/state?request_id=<request-id>
Response - An instance of ConfigurationState.
```

For more information about the APIs, see the *NSX API Guide*.

View Network Topology

View the network topology of your NSX environment for an overview of the entities, services, and the underlying fabric in your network. The graphical representation of the network topology is helpful when you are verifying your network configuration or troubleshooting errors.

Table 1078: Entities and Services

Entity	Service
Network Services	Load Balancer, NAT, and Distributed Firewall
Tier-0	L2 VPN service, IPSec VPN service, NAT rules, DNS forwarder, Gateway firewall rules, DHCP
Tier-1	Load Balancer, L2 VPN service, IPSec VPN service, NAT rules, DNS forwarder, Gateway firewall rules, DHCP
Segment	DHCP, Metadata proxy
Edge Transport Node	Switches, uplink interfaces, host configuration, uplinks, PortGroup connectivity
VirtualMachine	Underlying host cluster and host transport nodes
Pod	None
VRF	NAT rules, Gateway firewall rules, DHCP

1. From your browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address>.
2. Select **Networking** > **Network Topology**.
3. Navigate through the network topology to see more information:
   - Network connectivity - Zoom in to view more details about the logical entities. You can point to an entity to view its logical path in the network and view the services configured on it.
   - Service configuration - Click a service to view more configuration details.
     For the IPSec VPN service, NSX displays a visual representation of configuration information like local and remote endpoints and also indicates whether the sessions are policy-based or rule-based. You can then click a session to view session-specific details like tunnel status, among others.
   - Fabric view - Double-click an object to see the fabric view for that object and the parent object in its logical path.
     – For VMs and Containers, fabric view displays the host cluster name, host transport node details, and the configuration view for the host transport node.
     – For a physical server, fabric view displays details of the host transport node and the configuration view for the node.
     – For a tier-0 and tier-1 gateways, transit gateways, and Virtual Network Appliance (VNA) clusters, the fabric view displays the distributed router(DR), Edge cluster, Edge nodes where the service router(SR) is realized with HA status of the SR. You can also view configuration details of the edge node where the SR is realized for the gateway. For a tier-0 gateway, the uplink interfaces are also displayed.
     Click an Edge cluster to see a list of Edge transport nodes that are members of the cluster. You can click a node and expand **Tunnels** for additional information on the Edge tunnels BFD status using filters. Use the **Configuration** option to download routing or forwarding tables.
     Zoom in for the **View Edge Node Configuration** option to see a visual representation of the Edge node configuration.
   - Click **Export** on the tool bar to save the topology to a PDF file.
   - Apply filters to focus on specific objects. See Filter by Object Attributes for more details about filters.

Search for Objects

You can search for objects using various criteria throughout the NSX inventory.

The search results are sorted by relevance and you can filter these results based on your search query.

Note:
If you have special characters in your search query that also function as operators, then you must add a leading backslash. The characters that function as operators are: +, -, =, &&, ||, <, >, !, (,), {, }, [,], ^, ", ~, ?, :, /, \.

1. With admin privileges, log in to NSX Manager.
2. On the homepage, click the magnifying glass icon and enter a search pattern for an object or object type.

   You can also select a recent query or a search query that you saved.

   As you enter a search pattern, the search feature provides assistance by showing the applicable keywords.

Search	Search Query
Objects with Logical as the name or property	Logical
Exact logical switch name	display_name:LSP-301
Names with special characters such as, !	Logical\!

   All the related search results are listed and grouped by resource type in different tabs.

   You can click the tabs for specific search results for a resource type.
3. Optional: In the search bar, click the save icon to save your refined search criteria.
4. In the search bar, click the

   icon or **Advanced** tab to open the advanced search column where you can refine your search.
5. Specify one or more criteria to refine your search.
   - Name
   - Resource Type
   - Description
   - ID
   - Created by
   - Modified by
   - Tags
   - Creation Date
   - Modified Date

   You can also view your recent search results and saved search criteria.
6. Optional: Click **Clear All** to reset your advanced search criteria.

Filter by Object Attributes

When viewing objects in NSX Manager, you can filter the objects by one or more of their attributes. For example, when viewing details of Tier 0 gateways you can choose to filter by **Status** and view only those gateways that are `Down`.

The following types of filters are available:

- Predefined filters – A list of commonly used filters that you can apply to your objects.
- Text-based filter – A filter based on the attribute value that you enter. This filter is applicable only to the **Name**, **Tag**, **Path**, and **Description** attributes of the objects.
- Attribute-value pairs – An attribute drop-down menu that you can use to specify attribute-value pairs for filtering.

You can either use multiple attributes of an object or multiple values of a single attribute to filter objects. The AND operator is applied when you select multiple attributes whereas the OR operator is used when you specify multiple values of a single attribute.

1. From your browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address>.
2. Navigate to the tab that displays the objects you want to view.
3. Specify the attributes that you want to use to filter the objects.
   - Click
     
     and select from a list of predefined filters.
   - Enter a value for the **Name**, **Tag**, **Path**, or **Description** attributes.
   - Select an attribute from the drop-down menu and specify its value. For example, **Status**: `Down`

   Objects satisfying your filter criteria are displayed.
4. Optional: Click **Clear** to reset your filters.

Add a Compute Manager

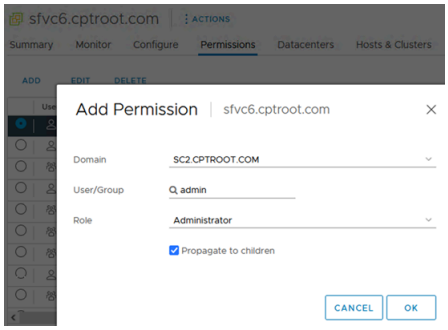
A compute manager, for example, vCenter, is an application that manages resources such as hosts and VMs.

- Verify that you are using the supported vSphere version.
- Verify that you have IPv4 communication with vCenter. For a dual-stack deployment, verify that you have IPv4 communication with vCenter.
- Verify that you use the recommended number of compute managers. See VMware Configuration Maximums.
- Decide on the hashing algorithm type you want to use for stamping the NSX Manager thumbprint in the computer manager extension. SHA1 and SHA256 algorithm types are supported. The default is SHA1. If you use SHA256 there might be communication issues between the WCP component in vCenter and NSX Manager.
  - To set the hashing algorithm, run the API:

```
PUT https://<nsx-mgr>/api/v1/fabric/compute-managers/thumbprint-hashing-algorithm

{
    "hashing_algorithm_type": "SHA1"
}
```

- For Day-0 operations, provide the credentials of a vCenter user.
  - These credentials will be used only on Day-0 for NSX to communicate with vCenter. NSX will not store these credentials in its database.
  - You can provide the credentials of a vCenter administrator, or create a role and a user specifically for NSX and provide this user's credentials. Go to the **Administration** > **Global Permissions** tab. Add global permissions to the newly created user and role and select **Propagate to Children**.



  - Create a custom role with the following vCenter privileges:

Global	Cancel task
Extension	Register extension
Extension	Unregister extension
Extension	Update extension
Host	Configuration.Maintenance
Host	Configuration.NetworkConfiguration

Host	Local Operations.Create virtual machine
Host	Local Operations.Delete virtual machine
Host	Local Operations.Reconfigure virtual machine
Network	Assign network
Permissions	Reassign role permissions
Resource	Assign vApp to resource pool
Resource	Assign virtual machine to resource pool
Sessions	Message
Sessions	Validate session
Sessions	View and stop sessions
Scheduled task	Select all privileges
Tasks	Select all privileges
vApp	Select all privileges
Virtual Machine.	Configuration
Virtual Machine	Guest Operations
Virtual Machine	Provisioning
Virtual Machine	Inventory

For Day-0 operations, NSX uses the credentials of a vCenter user to communicate with vCenter.

For all Day-N operations, NSX automatically creates and uses a secure service account for the vCenter instance.

- The service account created for the vCenter instance will have all the privileges required to execute the vCenter workflow. By the principle of least privilege, the service account will have the minimum privilege required to perform its duties.
- The lifecycle of the service account follows the lifecycle of the vCenter instances connected to NSX.
- When you upgrade VCF 5.x domains to VCF 9.x, the upgrade process automatically converts the old service accounts used with vCenter to new service accounts. No user intervention is required.

By default, trust is enabled between NSX and vCenter with full access, so that services running in vCenter can establish trusted communication with NSX. This access level gives vCenter complete access to NSX. The vCenter user's role will be mapped to the Enterprise Admin role in NSX.

VMware NSX works in conjunction with VCF non-disruptive certificate (NDC) rotation policies for Machine SSL certificate changes. After the initial VCF installation of NSX Manager, no further action is required for Machine SSL certificate management as NDC handles rotation automatically. However, root CA certificate rotation remains disruptive. To import a new root CA certificate after rotation, the administrator must edit the compute manager registration in NSX and then save the changes.

1. From your browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.
2. Select **System** > **Fabric** > **Compute Managers** > **Add Compute Manager**.
3. Complete the compute manager details.

Option	Description
Name	Type the name to identify the vCenter instance.
FQDN or IP Address	Type the FQDN or IP address of the vCenter instance.
User Name	Type the user name to log in to the vCenter instance.
Password	Type the password.
HTTPS Port	The default port is 443. If you use another port, verify that the port is open on all the NSX Manager appliances.

Option	Description
VPC Access in vSphere Client	Defines representation and consumption of VPCs from default Project in the vCenter instance: • Full Access - VPC and subnet available for configuration from vCenter. This option is enabled by default. • Read Only - VPC and subnet available in vCenter for read only access from vCenter • Disabled - None of the VPC objects in vCenter network are represented as port group under relevant VDS
Description	Enter a description.
Tag	Enter a tag.

4. Click **Save**.
5. Click **Confirm** to accept the server provided thumbprint.
6. If the progress icon changes from **In progress** to **Not registered**, perform the following steps to resolve the error instead of editing Compute Manager.
   a) Select the error message and click **Resolve**. One possible error message is the following:`Extension already registered at CM <vCenter name> with id <extension ID>`
   b) (Optional) If prompted, enter the vCenter credentials.
   c) Click **Resolve**.

      If an existing registration exists, it will be replaced.

It takes some time for NSX to register the compute manager with vCenter and for the connection status to appear as `UP`.

You can click the compute manager's name to view the details, edit the compute manager, or manage tags that apply to the compute manager.

After vCenter is successfully registered, the vCenter instance's Trusted Root CA certificate can be seen on **System** > **Certificates** tab under the miscellaneous category.

Unregistering a compute manager from NSX deletes the service account for that NSX deployment in the VC instance.

Note:

After you successfully register the vCenter instance, do not power off and delete all Global Manager VMs without first unregistering that vCenter instance. Otherwise, you will not be able to register that vCenter instance again when you deploy a new Global Manager node. An error message will state that the vCenter instance is already registered with another Global Manager. To resolve this error, use the **Resolve** action on the Compute Managers page.

Note:

After a vCenter (VC) compute manager is successfully added, it cannot be removed if you successfully performed any of the following actions:

- Transport nodes are prepared using VDS that is dependent on the VC.
- Service VMs deployed on a host or a cluster in the VC using NSX service insertion.
- You use the NSX Manager UI to deploy Edge VMs or NSX Manager nodes on a host or a cluster in the VC.

If you try to perform any of these actions and you encounter an error (for example, installation failed), you can remove the VC if you have not successfully performed any of the actions listed above.

If you have successfully prepared any transport node using VDS that is dependent on the VC or deployed any VM, you can remove the VC after you have done the following:

- Unprepare all transport nodes. If uninstalling a transport node fails, you must force delete the transport node.

- Undeploy all service VMs, all NSX Edge VMs, and all NSX Manager nodes. The undeployment must be successful or in a failed state.
- If an NSX Manager cluster consists of nodes deployed from the VC (manual method) and nodes deployed from the NSX Manager UI, and you had to undeploy the manually deployed nodes, then you cannot remove the VC. To successfully remove the VC, ensure that you re-deploy an NSX Manager node from the VC.

This restriction applies to a fresh installation of NSX as well as an upgrade.

Replace Compute Manager

To replace a compute manager (vCenter) with another compute manager (vCenter), determine if the existing compute manager has NSX Manager cluster or NSX Edge nodes registered to it and third-party services running on the NSX cluster.

Before you can replace the vCenter, you must delete any appliances or objects registered with the existing vCenter and then deploy them on the new vCenter.

1. Add a new compute manager. See *Add a Compute Manager*.
2. Replace NSX Manager cluster deployed from the **Add Appliance** wizard in the NSX Manager UI.

   Note:

   Do not follow this procedure to replace NSX Manager appliances deployed from vCenter.

   a) Go to **System** > **Appliances** > **Add NSX Appliance**.
   b) Verify whether the status of NSX Manager is stable. If not, ensure the status of NSX Manager is stable before proceeding with the next steps.
   c) Delete an existing NSX Manager on the old compute manager.
   d) Verify status of cluster by running the command, `get cluster status`.
   e) On the new compute manager, deploy a new NSX Manager.
   f) Once the new NSX Manager is fully operational and has joined the NSX Manager cluster, verify the cluster status.
   g) You can also view the status by running the command, `get cluster status`.
   h) Repeat these steps till all the existing NSX Manager are replaced by new ones and the cluster status is stable.
3. Add new NSX Edge nodes.
   a) To add Edge node, see *Create an NSX Edge Transport Node*.
   b) When creating a new NSX Edge node,
      1. In the **Name and Descritpion** window, enter a new Host Name /FQDN.
      2. In the **Configure Deployment** window, select a different compute manager.
   c) Verify NSX Edge node is deployed.
   d) Repeat the steps for all the NSX Edge nodes.
   e) All NSX Edge nodes are registered with the new compute manager.
   f) Go to **System** > **Fabric** > **Nodes** > **Edge Clusters**.
   g) Select the NSX Edge cluster and replace old NSX Edge nodes and add new NSX Edge nodes to the cluster. See *Replacing an NSX Edge Transport Node in an NSX Edge Cluster*.
   h) Click **Save** to save the NSX Edge cluster configuration.
4. Deploy third-party services.

   Any third-party services registered with the existing vCenter must be uninstalled from the old vCenter and reinstalled on the new vCenter.

   a) On the old compute manager, uninstall services introspecting east-west traffic. See *Uninstall an East-West Traffic Introspection Service*.
   b) On the old compute manager, uninstall services introspecting north-south traffic. See *Uninstall a North-South Traffic Introspection Service*.
   c) Install third-party services on the new compute manager.

      See *Deploy a Service for East-West Traffic Introspection*.

See *Deploy a Service for North-South Traffic Introspection*.

The old vCenter is replaced by the new vCenter and all appliances and objects are running on the new vCenter.

Configuring Active Directory and Event Log Scraping

Active Directory is used in creating user-based Identity Firewall rules.

If using event log scraping, make sure that NTP is configured correctly across all devices that will be using log scraping, for more information see Time Synchronization between NSX Manager, VMware Workspace ONE Access, and Related Components.

The domain account must have Active Directory read permission for all objects in the domain tree. The event log reader account must have read permissions for security event logs. See Enable Windows Security Log Access for the Event Log Reader.

Windows 2008 is not supported as an Active Directory server or RDSH Server OS.

You can register one or more Windows domains with an NSX Manager. NSX Manager gets group and user information, and the relationship between them from each domain that it is registered. NSX Manager also retrieves Active Directory (AD) credentials.

Once the Active Directory is synced to the NSX Manager, you can create security groups based on user identity, and create identity-based firewall rules.

Scale limits for Active Directory, Event Log Scraping, and IDFW can be found on the VMware Configuration Maximums page.

> **Note:**
> For Identity Firewall rule enforcement, Windows Time service should be **on** for all VMs using Active Directory. This ensures that the date and time is synchronized between Active Directory and VMs. AD group membership changes, including enabling and deleting users, do not immediately take effect for logged in users. For changes to take effect, users must log out and then log back in. AD administrator's should force a logout when group membership is modified. This behavior is a limitation of Active Directory.

1. With admin privileges, log in to NSX Manager.
2. Navigate to **System** > **Identity Firewall AD**.
3. Click **Add Active Directory**.
4. Enter the name of the active directory.
5. Enter the **NetBios Name** and **Base Distinguished Name**.

   To retrieve the netBIOS name for your domain, enter `nbtstat -n` in a command window on a Windows Workstation that is part of a domain, or on a domain controller. In the NetBIOS Local Name Table, the entry with a <00> prefix and type Group is the NetBIOS name.

   A base distinguished name (Base DN) is needed to add an Active Directory domain. A Base DN is the starting point that an LDAP server uses when searching for users authentication within an Active Directory domain. For example, if your domain name is corp.local the DN for the Base DN for Active Directory would be "DC=corp,DC=local".
6. Set the **Delta Synchronization Interval**, if necessary. A delta synchronization updates local AD objects that have changed since the last synchronization event.

   Any changes made in Active Directory are NOT seen on NSX Manager until a delta or full synchronization has been performed.
7. Set the **LDAP Server**. See Add an LDAP Server for more information.
8. Optional: Set the **Event Log Server**. Enter the Host IP or FQDN, user name and password, then click **Apply**.
9. Next to **Organization Units To Sync**, click **Sync all organization units and domains** or **Select organization units to sync**.

   Groups that are moved out of the selected OrgUnits are not updated during a selective sync. Deleted groups are removed in a full sync, when all groups are updated.

Option	Description
Sync all organization units and domains	Full sync of all organization units is performed.
Select organization units to sync	Individually select organization units. If the parent is selected, the child units inside of the parent are automatically selected. You can also select all of the organization units by selecting the top **Organization Units** box, and then unselect the specific units you do not want to include in the sync. Only the selected organization units which are created and changed since the last delta sync will be updated during a selective sync. Note that if users and groups are in different organization units, you must select organization units that contain users.

10. Click **Save**.
11. The Active Directory screen appears, in a read only mode.
12. To edit an Active Directory:
    1. Click the three-dot menu
    ()
    next to the Active Directory, and click **Edit**.
    2. You can now perform two actions: **Sync Delta**, or **Sync All**. For more information, see Synchronize Active Directory.

Enable Windows Security Log Access for the Event Log Reader

Read-only security access is used by event log scraper in IDFW.

The domain account must have Active Directory read permission for all objects in the domain tree. The event log reader account must have read permissions for security event logs.

After creating a new user account you must enable read-only security log access on a Windows 2008 and later server-based domain section, to grant the user read-only access

Members of the Event Log Readers group are granted permissions to read the event logs on the local computer. You must perform these steps on one Domain Controller of the domain, tree, or forest.

1. Navigate to **Start** > **Administrative Tools** > **Active Directory Users and Computers** > **.**
2. In the navigation tree, expand the node that corresponds to the domain for which you and to enable security log access.
3. Under the expanded node, select the **Builtin** node.
4. Double-click **Event Log Readers** in the list of groups.
5. Select the **Members** tab in the Event Log Readers Properties dialog box.
6. Click **Add**, and select the user or group you want to add to the Event Log Readers Group.
   The **Select Users, Contacts, Computers, or Groups** dialog appears.
7. Click **OK** to close all open dialog boxes.

Add an LDAP Server

LDAP (Lightweight Directory Access Protocol) server configuration and functionality is only for use with Identity Firewall. LDAP provides a central place for authentication, meaning that when you configure a connection to your LDAP server, the user records are stored in your external LDAP server.

The domain account must have AD read permission for all objects in the domain tree.

When there is a cluster of NSX Managers, all nodes need to be able to reach the LDAP server.

1. With admin privileges, log in to NSX Manager.
2. Navigate to **System** > **Identity Firewall AD**.
3. Click LDAP Server **Set**.
4. Click **Add LDAP Server**.
5. Enter the **Host** name of the LDAP server.
6. Select the active directory the LDAP server is connected to from the **Connected to (Directory)** drop-down menu.
7. Optional: Select the **protocol**: LDAP (unsecured) or LDAPS (secured).
8. If LDAPS was selected, select the SHA-256 Thumbprint suggested by NSX Manager, or enter a SHA-256 Thumbprint.
9. Enter the **Port** number of the LDAP server.

   For local domain controllers, the default LDAP port 389 and LDAPS port 636 are used for the Active Directory sync, and should not be edited from the default values.
10. Enter the **username** and **password** of an Active Directory account with a minimum of read-only access to the Active Directory domain.
11. Click **ADD** and **APPLY**.
12. To verify that you can connect to the LDAP server, click **Test Connection**.

Synchronize Active Directory

Active Directory objects can be used to create security groups based on user identity, and identity-based firewall rules.

> **Note:**
> Do not enable Distributed Intrusion Detection Service (IDS) in an environment that is using Distributed Load Balancer. NSX does not support using IDS with a Distributed Load Balancer.

You can register an entire AD (Active Directory) domain to be used by IDFW (Identity Firewall), or you can synchronize a subset of a large domain. Once a domain is registered, NSX synchronizes all AD data required by IDFW. Selective sync is used for large active directory domains.

Selective sync allows you to selectively choose organizational units so that you do not have to sync the entire domain. Only the selected organization units which are created and changed since the last delta sync will be updated during a selective sync. Groups that are moved out of the selected organization units are not updated during a selective sync. Configuration maximums still apply selective sync. Deleted groups are removed in a full sync, when all groups are updated. To specify organization units for synchronization, see Configuring Active Directory and Event Log Scraping.

> **Note:**
> Use the API to connect an AD domain with more than 500 OUs. The UI does not support showing an AD Domain with more than 500 OUs.

If you use the API to manually end a full sync after it is has begun, the sync stats will not be updated correctly.

Scale limits for Active Directory and IDFW can be found on the VMware Configuration Maximums page.

> **Note:**
> IDFW relies on the security and integrity of the guest operating system. There are multiple methods for a malicious local administrator to spoof their identity to bypass firewall rules. User identity information is provided by the Guest Introspection Agent inside guest VMs. Security administrators must ensure that NSX Guest Introspection Agent is installed and running in each guest VM. Logged-in users should not have the privilege to remove or stop the agent.

1. With admin privileges, log in to NSX Manager.
2. Navigate to **System** > **Identity Firewall AD**.
3. Click the three button menu
   ()
   next to the Active Directory that you want to synchronize, and select one of the following:

Option	Description
Sync All	Full sync of all data is performed from the Active Directory, regardless of the state of sync on NSX.
Sync Delta	Perform a delta synchronization, where local AD objects that have changed since the last synchronization are updated. A full sync of all data is not performed. Deleted groups are removed during Sync All, when all groups are updated.

4. Click **Save**.
5. Click **View Sync Status** to see the current state of the Active Directory, the previous synchronization state, the synchronization status, and the last synchronization time.

Remove NSX Extension from vCenter

When you add a compute manager, NSX Manager adds its identity as an extension in vCenter. If you remove the compute manager, the extension in vCenter will be removed automatically. If the extension is not removed for some reason, you can maually remove the extension with the following procedure.

1. Log in to the MOB at `https://<vCenter hostname or IP address>/mob`.
2. Click the **content** link, which is the value for the **content** property in the Properties table.
3. Click the **ExtensionManager** link, which is the value for **extensionManager** property in the Properties table.
4. Click the **UnregisterExtension** link in the Methods table.
5. Enter `com.vmware.nsx.management.nsxt` in the **value** text field.
6. Click the **Invoke Method** link on the right hand side of the page below the Parameters table.
   The method result says `void` but the extension will be removed.
7. To make sure the extension is removed, click the **FindExtension** method on the previous page and invoke it by entering the same value for the extension.
   The result should be `void`.

Managing the NSX Manager Cluster

You can reboot an NSX Manager if it becomes inoperable. You can also change the IP address of an NSX Manager.

In a production environment, it is highly recommended that the NSX Manager cluster has three members to provide high availability. If you delete an NSX Manager and deploy a new one, the new NSX Manager can have the same or a different IP address.

> **Note:**
> The primary NSX Manager node is the node that you create first, before you create a manager cluster. This node cannot be deleted. After you deploy two more manager nodes from the primary manager node's UI to form a cluster, only the second and the third manager nodes have the option (from the gear icon) to be deleted. For information about removing and adding a manager node, see Change the IP Address of an NSX Manager.

View the Configuration and Status of the NSX Manager Cluster

You can view the configuration and status of the NSX Manager cluster from the NSX Manager UI. You can get additional information using the CLI.

1. From your browser, log in with admin privileges to an NSX Manager at https://*nsx-manager-ip-address*.
2. Select **System** > **Overview**
   The status of the NSX Manager cluster is displayed.
3. To see additional information about the configuration, run the following CLI command:

```
manager1> get cluster config
Cluster Id: 18807edd-56d1-4107-b7b7-508d766a08e3
Cluster Configuration Version: 3
Number of nodes in the cluster: 3

Node UUID: 43cd0642-275c-af1d-fe46-1f5200f9e5f9
Node Status: JOINED
    ENTITY                              UUID                                    IP ADDRESS      PORT
   FQDN
    HTTPS                               5c8d01f1-f3ee-4f94-b517-a093d8fbfad3    10.160.71.225   443
   ychin-nsxmanager-ob-12065118-1-F5
    CONTROLLER                          06fd0574-69c0-432e-a8af-53d140dbef8f    10.160.71.225   -
   ychin-nsxmanager-ob-12065118-1-F5
    CLUSTER_BOOT_MANAGER                da8d535e-7a0c-4dd8-8919-d88bdde006b8    10.160.71.225   -
   ychin-nsxmanager-ob-12065118-1-F5
    DATASTORE                           3c9c4ec1-afef-47bd-aadb-1ed6a5536bc4    10.160.71.225   9000
   ychin-nsxmanager-ob-12065118-1-F5
    MANAGER                             eb5e8922-23bd-4c3a-ae22-d13d9195a6bc    10.160.71.225   -
   ychin-nsxmanager-ob-12065118-1-F5
    POLICY                              f9da1039-08ad-4a20-bacc-5b91c5d67730    10.160.71.225   -
   ychin-nsxmanager-ob-12065118-1-F5

Node UUID: 8ebb0642-201e-6a5f-dd47-a1e38542e672
Node Status: JOINED
    ENTITY                              UUID                                    IP ADDRESS      PORT
   FQDN
    HTTPS                               3757f155-8a5d-4b53-828f-d67041d5a210    10.160.93.240   443
   ychin-nsxmanager-ob-12065118-2-F5
    CONTROLLER                          7b1c9952-8738-4900-b68b-ca862aa4f6a9    10.160.93.240   -
   ychin-nsxmanager-ob-12065118-2-F5
    CLUSTER_BOOT_MANAGER                b5e12db1-5e0d-4e33-a571-6ba258dceb2e    10.160.93.240   -
   ychin-nsxmanager-ob-12065118-2-F5
    DATASTORE                           bee1f629-4e23-4ab8-8083-9e0f0bb83178    10.160.93.240   9000
   ychin-nsxmanager-ob-12065118-2-F5
    MANAGER                             45ccd6e3-1497-4334-944c-e6bbcd5c723e    10.160.93.240   -
   ychin-nsxmanager-ob-12065118-2-F5
    POLICY                              d5ba5803-b059-4fbc-897c-3aace8cf1219    10.160.93.240   -
   ychin-nsxmanager-ob-12065118-2-F5

Node UUID: 2e7e0642-df4a-b2ec-b9e8-633d1469f1ea
Node Status: JOINED
    ENTITY                              UUID                                    IP ADDRESS      PORT
   FQDN
    HTTPS                               bce3cc4c-7d60-45e2-aa7b-cdc75e445a14    10.160.76.33    443
   ychin-nsxmanager-ob-12065118-3-F5
    CONTROLLER                          ced46f5c-9e52-4b31-a1cb-b3dead991c71    10.160.76.33    -
   ychin-nsxmanager-ob-12065118-3-F5
    CLUSTER_BOOT_MANAGER                88b70d31-3428-4ccc-ab57-55859f45030c    10.160.76.33    -
   ychin-nsxmanager-ob-12065118-3-F5
    DATASTORE                           fb4aec3c-cae3-4386-b5b9-c0b99b7d9048    10.160.76.33    9000
   ychin-nsxmanager-ob-12065118-3-F5
```

```
    MANAGER                          82b07440-3ff6-4f67-a1c9-e9327d1686ad       10.160.76.33    -
   ychin-nsxmanager-ob-12065118-3-F5
    POLICY                           61f21a78-a56c-4af1-867b-3f24132d53c7       10.160.76.33    -
   ychin-nsxmanager-ob-12065118-3-F5
```

4. To see additional information about the status, run the following CLI command:

```
manager1> get cluster status
Cluster Id: 18807edd-56d1-4107-b7b7-508d766a08e3
Group Type: DATASTORE
Group Status: STABLE

Members:
    UUID                                     FQDN                                     IP
 STATUS
    43cd0642-275c-af1d-fe46-1f5200f9e5f9     ychin-nsxmanager-ob-12065118-1-F5        10.160.71.225
 UP
    8ebb0642-201e-6a5f-dd47-a1e38542e672     ychin-nsxmanager-ob-12065118-2-F5        10.160.93.240
 UP
    2e7e0642-df4a-b2ec-b9e8-633d1469f1ea     ychin-nsxmanager-ob-12065118-3-F5        10.160.76.33
 UP

Group Type: CLUSTER_BOOT_MANAGER
Group Status: STABLE

Members:
    UUID                                     FQDN                                     IP
 STATUS
    43cd0642-275c-af1d-fe46-1f5200f9e5f9     ychin-nsxmanager-ob-12065118-1-F5        10.160.71.225
 UP
    8ebb0642-201e-6a5f-dd47-a1e38542e672     ychin-nsxmanager-ob-12065118-2-F5        10.160.93.240
 UP
    2e7e0642-df4a-b2ec-b9e8-633d1469f1ea     ychin-nsxmanager-ob-12065118-3-F5        10.160.76.33
 UP

Group Type: CONTROLLER
Group Status: STABLE

Members:
    UUID                                     FQDN                                     IP
 STATUS
    7b1c9952-8738-4900-b68b-ca862aa4f6a9     ychin-nsxmanager-ob-12065118-2-F5        10.160.93.240
 UP
    ced46f5c-9e52-4b31-a1cb-b3dead991c71     ychin-nsxmanager-ob-12065118-3-F5        10.160.76.33
 UP
    06fd0574-69c0-432e-a8af-53d140dbef8f     ychin-nsxmanager-ob-12065118-1-F5        10.160.71.225
 UP

Group Type: MANAGER
Group Status: STABLE

Members:
    UUID                                     FQDN                                     IP
 STATUS
    43cd0642-275c-af1d-fe46-1f5200f9e5f9     ychin-nsxmanager-ob-12065118-1-F5        10.160.71.225
 UP
    8ebb0642-201e-6a5f-dd47-a1e38542e672     ychin-nsxmanager-ob-12065118-2-F5        10.160.93.240
 UP
```

```
    2e7e0642-df4a-b2ec-b9e8-633d1469f1ea       ychin-nsxmanager-ob-12065118-3-F5          10.160.76.33
 UP

Group Type: POLICY
Group Status: STABLE

Members:
    UUID                                       FQDN                                       IP
 STATUS
    43cd0642-275c-af1d-fe46-1f5200f9e5f9       ychin-nsxmanager-ob-12065118-1-F5          10.160.71.225
 UP
    8ebb0642-201e-6a5f-dd47-a1e38542e672       ychin-nsxmanager-ob-12065118-2-F5          10.160.93.240
 UP
    2e7e0642-df4a-b2ec-b9e8-633d1469f1ea       ychin-nsxmanager-ob-12065118-3-F5          10.160.76.33
 UP

Group Type: HTTPS
Group Status: STABLE

Members:
    UUID                                       FQDN                                       IP
 STATUS
    43cd0642-275c-af1d-fe46-1f5200f9e5f9       ychin-nsxmanager-ob-12065118-1-F5          10.160.71.225
 UP
    8ebb0642-201e-6a5f-dd47-a1e38542e672       ychin-nsxmanager-ob-12065118-2-F5          10.160.93.240
 UP
    2e7e0642-df4a-b2ec-b9e8-633d1469f1ea       ychin-nsxmanager-ob-12065118-3-F5          10.160.76.33
 UP
```

Update API Service Configuration of the NSX Manager Cluster

You can modify the API service properties of the NSX Manager cluster, such as TLS protocol version, cipher suites, and so on.

Starting in 4.2, TLS 1.1 ciphers are deactivated by default, but can be activated by the user using the following procedure. The supported ciphers for TLS 1.1 are:

- TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA
- TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA
- TLS_RSA_WITH_AES_128_CBC_SHA
- TLS_RSA_WITH_AES_256_CBC_SHA

The supported ciphers for TLS 1.2 are:

- TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA
- TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256
- TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
- TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA
- TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384
- TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384
- TLS_RSA_WITH_AES_128_CBC_SHA
- TLS_RSA_WITH_AES_128_CBC_SHA256
- TLS_RSA_WITH_AES_128_GCM_SHA256
- TLS_RSA_WITH_AES_256_CBC_SHA
- TLS_RSA_WITH_AES_256_CBC_SHA256
- TLS_RSA_WITH_AES_256_GCM_SHA384
- TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384

The supported ciphers for TLS 1.3 are:

- TLS_AES_128_GCM_SHA256
- TLS_AES_256_GCM_SHA384
- TLS_CHACHA20_POLY1305_SHA256
- Starting in NSX 4.2, TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384

The following procedure explains the workflow of running the NSX API service calls to activate the TLS 1.1 protocol, and to activate or deactivate the cipher suites in the API service configuration. TLS 1.1 is deactivated by default. You can use this same procedure to deactivate other TLS versions as needed. Note that NSX supports a minimum and maximum version, so support of TLS 1.1 and TLS 1.3, but not TLS 1.2 is not supported. For example, TLS 1.1 and TLS 1.2 or TLS 1.2 and TLS 1.3 can be supported.

For a detailed information about the API schema, example request, example response, and error messages of the NSX API service, read the *NSX API Guide*.

1. Run the following GET API to read the configuration of the NSX API service:

   ```
   GET https://<NSX-Manager-IP>/api/v1/cluster/api-service
   ```

   The API response contains the list of cipher suites and TLS protocols. Note that the TLS 1.0 support is not listed.

   ```
   curl -u admin:${PASSWORD} -i -k https://$IP/api/v1/cluster/api-service  "protocol_versions" : [ {
       "name" : "TLSv1.1",
       "enabled" : false
     }, {
       "name" : "TLSv1.2",
       "enabled" : true
     }, {
       "name" : "TLSv1.3",
       "enabled" : true
     } ],
   ```

2.

   ```
   curl -u admin:${PASSWORD} -i -k https://$IP/api/v1/cluster/api-service
   {
     "global_api_concurrency_limit": 199,
     "client_api_rate_limit": 100,
     "client_api_concurrency_limit": 40,
     "connection_timeout": 30,
     "redirect_host": "",
     "cipher_suites": [
       {"enabled": true, "name": "TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384"},
       {"enabled": true, "name": "TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384"},
       {"enabled": true, "name": "TLS_RSA_WITH_AES_256_GCM_SHA384"},
       {"enabled": true, "name": "TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256"},
       {"enabled": true, "name": "TLS_RSA_WITH_AES_128_GCM_SHA256"}
       {"enabled": true, "name": "TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384}",
       {"enabled": true, "name": "TLS_RSA_WITH_AES_256_CBC_SHA256"},
       {"enabled": true, "name": "TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA"},
       {"enabled": true, "name": "TLS_RSA_WITH_AES_256_CBC_SHA"},
       {"enabled": true, "name": "TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256"},
       {"enabled": true, "name": "TLS_RSA_WITH_AES_128_CBC_SHA256"},
       {"enabled": false, "name": "TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA"},
       {"enabled": false, "name": "TLS_RSA_WITH_AES_128_CBC_SHA"}
       {"enabled": false, "name": "TLS_RSA_WITH_AES_128_CBC_SHA"}
     ],
     "protocol_versions": [
       {"enabled": false, "name": "TLSv1.1"},
       {"enabled": true, "name": "TLSv1.2"}
       {"enabled": true, "name": "TLSv1.3"}]
   }
   ```

3. Activate or deactivate the TLS 1.1 protocol.
   a) To activate a TLS version, set `TLSv1.1` to `enabled = true`, for example. To deactivate a TLS version, set your TLS version to `enabled = false`.
   b) Run the following PUT API to send the changes to the NSX API server:
      `PUT https://<NSX-Manager-IP>/api/v1/cluster/api-service`
4. Activate or deactivate the cipher suites.
   a) Set one or more cipher names to `enabled = true` or `enabled = false` depending on your requirements.
   b) Run the following PUT API to send the changes to the NSX API server:
      `PUT https://<NSX-Manager-IP>/api/v1/cluster/api-service`
5. Verify the activation is complete.

The API service on each NSX Manager node restarts after it is updated using the API. There might be a delay of up to a minute between the time the API call completes and when the new configuration comes into effect. The changes in the API service configuration are applied to all the nodes in the NSX Manager cluster.

Shut Down and Power On the NSX Manager Cluster

If you need to shut down the NSX Manager cluster, use the following procedure.

1. To shut down an NSX Manager cluster, shut down one manager node at a time. You can log in to the command-line interface (CLI) of a manager node as `admin` and run the command `shutdown`, or shut down the manager node VM from vCenter.

   Make sure that the VM is powered off in vCenter before proceeding to the next one. If only one manager node remains, it is no longer possible to login to the UI.
2. To power on an NSX Manager cluster, power on one manager node VM at a time in vCenter.

Reboot an NSX Manager

You can reboot an NSX Manager with a CLI command to recover from critical errors.

If you need to reboot multiple NSX Managers, you must reboot them one at a time. Wait for the rebooted NSX Manager to be online before rebooting another.

1. Log in to the CLI of the NSX Manager.
2. Run the following command.

```
nsx-manager> reboot
Are you sure you want to reboot (yes/no): y
```

Resize an NSX Manager Node

You can change the memory and CPU resources of an NSX Manager node in a cluster.

- Verify that the new size satisfies the memory and CPU resources system requirements for a manager node. .
- Familiarize yourself with how to run CLI commands. For more information, see the*NSX Command-Line Interface Reference*. Also familiarize yourself with how to change the memory and CPU resources of a VM. For more information, see the vSphere documentation.

Note that in normal operating conditions all three manager nodes must have the same CPU and memory resources. A mismatch of CPU or memory between NSX Managers in an NSX Manager cluster should only be done when transitioning from one size of NSX Manager to another size.

If you have configured resource allocation reservation for the NSX Manager VMs in vCenter, you might need to adjust the reservation. For more information, see the vSphere documentation.

Resizing Effort	Form Factors	Description
Less effort	Small, Medium or Large	NSX requires that two managers are available at all times. If you have cluster VIP (virtual IP) configured, there will be a brief outage when the VIP switches to another node in the cluster. You can access the other two nodes directly during the outage if the VIP-assigned node is shut down for resizing. If you have deployed a load balancer for the manager nodes, health checks will be triggered when a manager goes offline. The load balancer should direct traffic to another node.

- Resize an NSX manager node (Applicable to Small, Medium or Large form factor only. Do not use Small for Global Manager.)

  Change the CPU and memory of the existing NSX Manager nodes. You must make the change to one manager at a time so that two managers are available at all times. For form factor details, see *Prerequisites*.

  Note:
  This procedure does not support the resizing of NSX Managers from or to the Extra Large form factor.

  This procedure remains the same if the NSX Manager nodes are deployed manually and the cluster is formed using `join` node CLI or if the NSX Manager nodes are deployed and the cluster is formed using NSX Manager UI.

  1. Log in to a manager's CLI as **admin** and run the `shutdown` command.
  2. From NSX Manager UI, verify that the state of the manager cluster is DEGRADED.
  3. From VMware vSphere, change the memory and/or CPU resources of the manager VM that was shut down.
  4. From vSphere, power on the VM. From NSX Manager UI, wait for the state of the manager cluster to be STABLE.
  5. Repeat steps 1 to 4 for the other two manager VMs.

Change the IP Address of an NSX Manager

You can change the IP address of an NSX Manager in an NSX Manager cluster. This section describes several approaches.

Familiarize yourself with how to deploy an NSX Manager into a cluster. For more information, see the *NSX Installation Guide*.

For example, if you have a cluster consisting of Manager A, Manager B, and Manager C, you can change the IP address of one or more of the managers in the following ways:

- Scenario A:
  - Manager A has IP address 172.16.1.11.
  - Manager B has IP address 172.16.1.12.
  - Manager C has IP address 172.16.1.13.
  - Add Manager D with a new IP address, for example, 192.168.55.11.
  - Remove Manager A.
  - Add Manager E with a new IP address, for example, 192.168.55.12.
  - Remove Manager B.
  - Add Manager F with a new IP address, for example, 192.168.55.13.
  - Remove Manager C.
- Scenario B:
  - Manager A has IP address 172.16.1.11.
  - Manager B has IP address 172.16.1.12.
  - Manager C has IP address 172.16.1.13.
  - Add Manager D with a new IP address, for example, 192.168.55.11.
  - Add Manager E with a new IP address, for example, 192.168.55.12.

  - Add Manager F with a new IP address, for example, 192.168.55.13.
  - Remove Manager A, Manager B, and Manager C.
- Scenario C:
  - Manager A has IP address 172.16.1.11.
  - Manager B has IP address 172.16.1.12.
  - Manager C has IP address 172.16.1.13.
  - Remove Manager A.
  - Add Manager D with a new IP address, for example, 192.168.55.11.
  - Remove Manager B.
  - Add Manager E with a new IP address, for example, 192.168.55.12.
  - Remove Manager C.
  - Add Manager F with a new IP address, for example, 192.168.55.13.

The first two scenarios require additional virtual RAM, CPU, and disk for the additional NSX Managers during this IP address change.

Scenario C is not recommended because it temporarily reduces the number of NSX Managers and a loss of one of the two active managers during the IP address change will have an impact on the operations of NSX. This scenario is for a situation where additional virtual RAM, CPU and disk are not available and an IP address change is required.

> **Note:**
> If you are using the cluster VIP feature, you must either use the same subnet for the new IP addresses or deactivate the cluster VIP during the IP address changes because the cluster VIP requires all NSX Managers to be in the same subnet.

1. If the NSX Manager you want to remove was deployed manually, perform the following steps.
   a) Run the following CLI command to detach the NSX Manager from the cluster.
      ```
      detach node <node-id>
      ```
   b) Delete the NSX Manager VM.
2. If the NSX Manager you want to delete was deployed automatically through the NSX Manager UI, perform the following steps.
   a) From your browser, log in with administrator privileges to an NSX Manager at https://*nsx-manager-ip-address*.
      This NSX Manager must not be the one that you want to delete.
   b) From the **Systems** tab, click **NSX Management Nodes**.
      The status of the NSX Manager cluster is displayed.
   c) For the NSX Manager that you want to delete, click the gear icon and select **Delete**.
3. Deploy a new NSX Manager.

Replacing an NSX Edge Transport Node in an NSX Edge Cluster

You can replace an NSX Edge transport node in an NSX Edge cluster using the NSX Manager UI or the API.

The ACL (Access Control List) rules configured on the NSX Edge node to be replaced will be lost. The rules will not be present on the new node.

Replace an NSX Edge Transport Node Using the NSX Manager UI

The following procedure describes replacing an NSX Edge transport node in an NSX Edge cluster that has both VM NSX Edge and Bare Metal NSX Edge transport nodes using the NSX Manager UI. You can replace an VM NSX Edge with Bare Metal NSX Edge or the other way around. You can replace the Edge transport node regardless of whether it is running or not.

- Familiarize yourself with the procedure to install an NSX Edge node, join the Edge node with the management plane, and create an NSX Edge transport node. For more information, see the *NSX Installation Guide*.
- Both VM NSX Edge and Bare Metal NSX Edge transport nodes must have the same VLAN connectivity to the physical Top of Rack (TOR) switches.

1. If you want the new NSX Edge transport node to have the same configurations as the NSX Edge transport node to be replaced, make the following API call to find the configurations:

   ```
   GET https://<nsx-manager-IP>/api/v1/transport-nodes/<tn-id>
   ```

   ```
   An example output of a Bare Metal NSX Edge transport node.
   {
     "node_id": "cd15d368-569b-11ed-8143-b07b25e93f64",
     "host_switch_spec": {
     "host_switches": [
      {
      "host_switch_name": "nsxHostSwitch",
      "host_switch_id": "809299a2-c090-4543-8747-d200e12cd2ea",
      "host_switch_type": "NVDS",
      "host_switch_mode": "STANDARD",
      "host_switch_profile_ids": [
      {
      "key": "UplinkHostSwitchProfile",
      "value": "57da58fa-bce6-448b-8db3-874ceff59656"
      },
      {
      "key": "LldpHostSwitchProfile",
      "value": "9e0b4d2d-d155-4b4b-8947-fbfe5b79f7cb"
      }
      ],
     "pnics": [
      {
      "device_name": "fp-eth1",
      "uplink_name": "lag-0"
      },
      {
      "device_name": "fp-eth3",
      "uplink_name": "lag-1"
      },
      {
      "device_name": "fp-eth5",
      "uplink_name": "lag-2"
      },
      {
      "device_name": "fp-eth7",
      "uplink_name": "lag-3"
      },
      {
      "device_name": "fp-eth0",
      "uplink_name": "Uplink3"
      },
      {
      "device_name": "fp-eth2",
      "uplink_name": "Uplink4"
      },
      {
      "device_name": "fp-eth4",
      "uplink_name": "Uplink5"
      },
      {
      "device_name": "fp-eth6",
      "uplink_name": "Uplink6"
       }
   ```

```
],
"is_migrate_pnics": false,
"ip_assignment_spec": {
"ip_pool_id": "82f8ae96-992b-45c6-8376-777b82bfeb1d",
"resource_type": "StaticIpPoolSpec"
},
"cpu_config": [],
"transport_zone_endpoints": [
 {
"transport_zone_id": "15897bda-802f-4481-b9fd-4e5cc1ef084b",
 "transport_zone_profile_ids": [
{
"resource_type": "BfdHealthMonitoringProfile",
"profile_id": "52035bb3-ab02-4a08-9884-18631312e50a"
 }
 ]
 },
 {
 "transport_zone_id": "4a237a28-050e-4499-a241-0eb0c9dad97f",
 "transport_zone_profile_ids": [
 {
 "resource_type": "BfdHealthMonitoringProfile",
"profile_id": "52035bb3-ab02-4a08-9884-18631312e50a"
 }
  ]
  }
],
"pnics_uninstall_migration": [],
"vmk_uninstall_migration": [],
"not_ready": false
}
],
"resource_type": "StandardHostSwitchSpec"
},
"maintenance_mode": "DISABLED",
"node_deployment_info": {
"deployment_type": "PHYSICAL_MACHINE",
"node_settings": {
"hostname": "w1-hs2-m2716.eng.vmware.com",
"enable_ssh": true,
"allow_ssh_root_login": false,
"enable_upt_mode": false
    },
    "resource_type": "EdgeNode",
    "external_id": "cd15d368-569b-11ed-8143-b07b25e93f64",
    "ip_addresses": [
        "10.196.145.177"
    ],
    "id": "cd15d368-569b-11ed-8143-b07b25e93f64",
    "display_name": "w1-hs2-m2716.eng.vmware.com",
    "tags": [],
    "_revision": 1
},
"is_overridden": false,
"failure_domain_id": "4fc1e3b0-1cd4-4339-86c8-f76baddbaafb",
"resource_type": "TransportNode",
"id": "cd15d368-569b-11ed-8143-b07b25e93f64",
```

```
        "display_name": "w1-hs2-m2716.eng.vmware.com",
        "tags": [],
        "_create_time": 1666946274614,
        "_create_user": "admin",
        "_last_modified_time": 1666946708328,
        "_last_modified_user": "admin",
        "_system_owned": false,
        "_protection": "NOT_PROTECTED",
        "_revision": 1
    }
```


2. Follow the procedure in the *Create an NSX Edge Transport Node* topic in the *NSX Installation Guide*.

   If you want this NSX Edge transport node to have the same configurations as the NSX Edge transport node to be replaced, use the configurations obtained in step 1. For example, in the API output from step 1, you can make a note of the host switch specifications, node deployment details and configure the new NSX Edge transport node using the same configuration.
3. In NSX Manager, select **System** > **Fabric** > **Nodes** > **Edge Clusters**.
4. Select an NSX Edge cluster by clicking the checkbox in the first column.
5. SSH into the NSX Edge nodes where Tier-0 is hosted.
6. Run `get logical router`. Check the VRF ID for Tier-0 Service Router (SR) on all NSX Edge nodes in the NSX Edge Cluster.
7. If the VRF id of the Tier-0 SR is 1, run `vrf 1`.
8. To check the output of the service router, run `get high-availability status`.
9. Enable maintenance mode on one of NSX Edge nodes that has Tier-0 SR in Standby. In the Edge CLI console, run `set maintenance-mode enabled`.

   This NSX Edge node could have Tier-1 SRs in Active state. Putting NSX Edge node into maintenance mode triggers a HA failover and all Tier-1 or Tier-0 SRs on this NSX Edge Node go into Standby state on this NSX Edge node. This might cause traffic disruption for the active SRs on this NSX Edge node because of failover of Tier-1 or Tier-0 SRs.
10. Ensure that Bare Metal NSX Edge transport node is not a part of any other cluster.
11. Click **Actions** > **Replace Edge Cluster Member**.

    It is recommended that you place the transport node being replaced in maintenance mode. If the transport node is not running, you can safely ignore this recommendation.
12. Select the VM NSX Edge transport node to be replaced from the dropdown list.
13. Select the Bare Metal NSX Edge transport node replacement node from the dropdown list.
14. Click **Save**.
15. Verify that the Bare Metal NSX Edge transport node has moved into existing Edge VM Cluster.
16. To verify that Tier-0 and Tier-1 gateways have moved from NSX Edge VM (in maintenance mode) to Bare Metal NSX Edge transport node, run `get logical router`.
17. Repeat the previous steps to move another VM NSX Edge with Bare Metal NSX Edge transport node.
18. Verify E-W and N-S connectivity from the workloads connected to Tier-1 or Tier-0 LRs.

If you are running an NSX version earlier than 3.1.3, after replacing the NSX Edge transport node, you might see the alarm `All BGP/BFD sessions are down`. To resolve the issue, follow the workaround instructions in Knowledge Base article 318313: NSX tier-0 logical router in an A/A topology, Internal BGP (iBGP) session are down between the service routers.

Replacing a VM NSX Edge VM with Bare Metal NSX Edge node does not automatically rebalance the Tier-1 gateways across NSX Edge nodes. You need to manually reconfigure each Tier-1 gateway.

Replace an NSX Edge Transport Node Using the API

The following procedure describes replacing an NSX Edge transport node in an NSX Edge cluster using the NSX API. You can replace the Edge transport node regardless of whether it is running or not.

- Familiarize yourself with the procedure to install an NSX Edge node, join the Edge node with the management plane, and create an NSX Edge transport node. For more information, see the *NSX Installation Guide*.
- Ensure the new NSX Edge node that will use to replace the old NSX Edge node is ready. For more information, see the *NSX Installation Guide*.

1. If you want the new NSX Edge transport node to have the same configurations as the old NSX Edge transport node to be replaced, make the following API call to find the configurations:

   ```
   GET https://<nsx-manager-IP>/api/v1/transport-nodes/<tn-id>
   ```


   ```
   {
       "node_id": "250175b8-223b-11ed-826e-b07b25e93f64",
       "host_switch_spec": {
           "host_switches": [
               {
                   "host_switch_name": "nsxHostSwitch",
                   "host_switch_id": "809299a2-c090-4543-8747-d200e12cd2ea",
                   "host_switch_type": "NVDS",
                   "host_switch_mode": "STANDARD",
                   "host_switch_profile_ids": [
                       {
                           "key": "UplinkHostSwitchProfile",
                           "value": "57da58fa-bce6-448b-8db3-874ceff59656"
                       },
                       {
                           "key": "LldpHostSwitchProfile",
                           "value": "9e0b4d2d-d155-4b4b-8947-fbfe5b79f7cb"
                       }
                   ],
                   "pnics": [
                       {
                           "device_name": "fp-eth0",
                           "uplink_name": "Uplink1"
                       },
                       {
                           "device_name": "fp-eth2",
                           "uplink_name": "Uplink2"
                       },
                       {
                           "device_name": "fp-eth4",
                           "uplink_name": "Uplink3"
                       },
                       {
                           "device_name": "fp-eth6",
                           "uplink_name": "Uplink4"
                       }
                   ],
                   "is_migrate_pnics": false,
                   "ip_assignment_spec": {
                       "ip_pool_id": "82f8ae96-992b-45c6-8376-777b82bfeb1d",
                       "resource_type": "StaticIpPoolSpec"
                   },
                   "cpu_config": [],
                   "transport_zone_endpoints": [
   ```


```
                {
                    "transport_zone_id": "15897bda-802f-4481-b9fd-4e5cc1ef084b",
                    "transport_zone_profile_ids": [
                        {
                            "resource_type": "BfdHealthMonitoringProfile",
                            "profile_id": "52035bb3-ab02-4a08-9884-18631312e50a"
                        }
                    ]
                },
                {
                    "transport_zone_id": "4a237a28-050e-4499-a241-0eb0c9dad97f",
                    "transport_zone_profile_ids": [
                        {
                            "resource_type": "BfdHealthMonitoringProfile",
                            "profile_id": "52035bb3-ab02-4a08-9884-18631312e50a"
                        }
                    ]
                }
            ],
            "pnics_uninstall_migration": [],
            "vmk_uninstall_migration": [],
            "not_ready": false
        }
    ],
    "resource_type": "StandardHostSwitchSpec"
},
"maintenance_mode": "DISABLED",
"node_deployment_info": {
    "deployment_type": "PHYSICAL_MACHINE",
    "node_settings": {
        "hostname": "w1-hs2-m2716.eng.vmware.com",
        "enable_ssh": true,
        "allow_ssh_root_login": false
    },
    "resource_type": "EdgeNode",
    "external_id": "250175b8-223b-11ed-826e-b07b25e93f64",
    "ip_addresses": [
        "10.196.145.177"
    ],
    "id": "250175b8-223b-11ed-826e-b07b25e93f64",
    "display_name": "w1-hs2-m2716.eng.vmware.com",
    "tags": [],
    "_revision": 3
},
"is_overridden": false,
"failure_domain_id": "4fc1e3b0-1cd4-4339-86c8-f76baddbaafb",
"resource_type": "TransportNode",
"id": "250175b8-223b-11ed-826e-b07b25e93f64",
"display_name": "w1-hs2-m2716.eng.vmware.com",
"tags": [],
"_create_time": 1661187299037,
"_create_user": "admin",
"_last_modified_time": 1661255498968,
"_last_modified_user": "admin",
"_system_owned": false,
"_protection": "NOT_PROTECTED",
"_revision": 3
```

```
}
```

2. Note the transport node ID of the node to be replaced "55120a1a-51c6-4c20-b4a3-6f59662c9f6a".
3. Prepare a new NSX Edge Transport Node that will replace the old NSX Edge node. See the *Create an NSX Edge Transport Node* in the *NSX Installation Guide*.

   Note the following configuration points while preparing the new NSX Edge node:

   - Do not use the same IP address of the old NSX Edge node if it is running, `"ip_addresses": ["10.161.68.92"]`.
   - Do not use the same TEP IP address if the old NSX Edge node is running
4. Confirm UUID of the new Edge Transport Node. Run the API mentioned in step 1.
5. Make an API call to retrieve the member index of the transport node that has to be replaced.

```
GET https://<nsx-manager-IP>/api/v1/edge-clusters
 ....
     {
       "resource_type": "EdgeCluster",
       "description": "",
       "id": "9a302df7-0833-4237-af1f-4d826c25ad78",
       "display_name": "Edge-Cluster-1",
...
       "members": [
         {
           "member_index": 0,
           "transport_node_id": "55120a1a-51c6-4c20-b4a3-6f59662c9f6a"
         },
         {
           "member_index": 1,
           "transport_node_id": "890f0e3c-aa81-46aa-843b-8ac25fe30bd3"
         }
       ],
```

6. Make an API call to replace a transport node in an NSX Edge cluster. The `member_index` must match the index of the transport node to be replaced.
   For example, the transport node `TN-edgenode-01a` (73cb00c9-70d0-4808-abfe-a12a43251133) has failed and is replaced by transport node `TN-edgenode-03a` (890f0e3c-aa81-46aa-843b-8ac25fe30bd3) in NSX Edge cluster `Edge-Cluster-1` (9a302df7-0833-4237-af1f-4d826c25ad78).

```
POST http://<nsx-manager-IP>/api/v1/edge-clusters/9a302df7-0833-4237-af1f-4d826c25ad78?
action=replace_transport_node
{
    "member_index": 0,
    "transport_node_id" : "890f0e3c-aa81-46aa-843b-8ac25fe30bd3"
}
```

If you are running an NSX version earlier than 3.1.3, after replacing the NSX Edge transport node, you might see the alarm "All BGP/BFD sessions are down." To resolve the issue, follow the workaround instructions in Knowledge Base article 318313: NSX tier-0 logical router in an A/A topology, Internal BGP (iBGP) session are down between the service routers.

Managing Resource Reservations for an Edge VM Appliance

NSX uses vSphere resource allocation to reserve resources for NSX Edge appliances. You can tune the CPU and memory resources reserved for NSX Edge to ensure optimal use of resources on an NSX Edge.

For maximum performance NSX Edge VM appliance must be assigned 100% of the available resources. If you customize resources allocated to the NSX Edge VM, turn back the allocation later to 100% to get maximum performance.

For auto-deployed NSX Edge appliances, you can change the resource allocation from NSX Manager. However, if an NSX Edge appliance is deployed from vSphere, you can only manage resource reservations for that NSX Edge VM from vSphere.

As per the resource requirements of the Edge VM deployed in your environment, there are two ways to manage reservations:

- Default values assigned to give 100% resource reservations.
- Custom values assigned to give 0–100% resource reservations.

Default Reservations

Assumes the NSX Edge set to the **High** priority. The level of priority importance defines the number of vCPU shares and memory assigned to the NSX Edge. To assign custom values, you can change the relative priority assigned to the NSX Edge.

Resource constraints for different form factors set with **Normal** priority:

Form Factor	Number of vCPUs	vCPU Shares (Normal priority)	RAM (GB)
Small	2	2000	4
Medium	4	4000	8
Large	8	8000	32
XLarge	16	16000	64

You can tune reservations of an NSX Edge appliance by considering two parameters:

- Relative priority assigned to a VM
- Pre-assigned resource constraints for a VM form factor

Custom Reservations

Assign relative priority for an NSX Edge appliance. You can change the relative importance of an NSX Edge appliance to assign the following resource requirements:

For an Extra Large form factor, the releative importance of CPU Reservation Priority, vCPUs and Memory are listed in the following table:

Relative Importance	vCPUs	CPU Share value	Memory (shares per MB configured virtual machine memory)
Extra High	16	34000	64
High	8	32000	32
Normal	4	16000	8
Low	2	8000	4

For example, a High relative importance to an NSX Edge appliance deployed in a medium form factor assigns the following vCPU and memory shares:

- 4 (vCPUs) X 8000 (vCPU share value) = 32000 shares of vCPU
- 20 (GB RAM) X 1000 = 20000 shares of memory

> **Note:**
> Before assigning a CPU value in MHz to guarantee the allocated CPU cycles for an NSX EdgeVM, ensure that the relative importance is set to Low. If the relative importance is set to Normal or High with a custom CPU value in MHz, the VM deployment might face issues due to resource constraints.

Tune Resource Reservations for an NSX Edge Appliance

You can tune resource reservations on an NSX Edge VM appliance. By default, 100% resources are allocated to an NSX Edge VM. Flexibility to change resource reservations avoids the need to add additional capacity to the vCenter and the need to reduce current reservations on other non-Edge VMs.

- Verify that the cluster has sufficient capacity to avoid failures.

1. With admin privileges, log in to NSX Manager.
2. Select **System** > **Fabric** > **Nodes** > **Edge Transport Nodes**.
3. Select the NSX Edge transport node.
4. Click **Actions** > **Change Edge VM Resource Reservations**.
5. In the **Change Edge VM Resource Reservations** window, you can customize the existing resource allocation applied to the Edge transport node.

Action	Description
CPU Reservation Priority	Low - 2000 shares Normal - 4000 shares High - 8000 shares Extra High - 10000 shares
Memory Reservation (%)	Reservation percentage is relative to the pre-defined value in the form factor. 100 indicates 100% of memory is reserved for the NSX Edge VM. If you enter 50, it indicates that 50% of memory is assigned to the Edge transport node.
CPU Reservation (MHz)	Enter CPU reservation in MHz. The maximum amount of MHz is equal to the number of vCPUs multiplied by the normal CPU operation rate of the physical CPU core. **Note:** If the MHz value entered exceeds the maximum CPU capacity of the physical CPU cores, the NSX Edge VM might fail to start even though the allocation was accepted.

6. Click **Save**.

   If changes made to the resource reservations do not take effect, you might need to reboot the NSX Edge VM from vCenter.

   The NSX Edge VM appliance autostarts on ESX host reboot provided the NSX Edge cluster has vSphere HA turned off. For more details on vSphere HA, see the vSphere documentation.

Redeploy an NSX Edge Node

You might want to redeploy an NSX Edge node when you experience problems with the node or want to change node configuration settings that are unavailable in edit mode.

What Happens When You Redeploy an Edge Node

The redeployment workflow performs the following operations:

1. Places the existing edge node in maintenance mode and then powers off the edge node.
2. Deletes the existing edge node VM.
3. Deploys a new edge node VM with the configuration settings that you specified.

The placement of the existing gateway and services on the node remain unchanged.

If the NSX Edge node is redeployed, the new node is upgraded to a VM hardware version compatible with the ESX host version. After redeploying an edge node, check all the other nodes in that edge cluster to ensure that VM hardware versions are consistent. VM hardware versions compatible with ESX hosts are listed in Knowledge Base article 312100: ESX hosts and compatible virtual machine hardware versions list.

All logs collected for an existing NSX Edge node are deleted when you redeploy the node. It is recommended that you generate and download the support bundle for the node before starting the redeployment workflow.

Redeployment Methods

You can redeploy an edge node using either the NSX Manager UI or the NSX API. Use the appropriate method:

- To redeploy a VM node that was auto-deployed through NSX Manager, use the NSX Manager UI or the API.
- To redeploy a node that was manually deployed through vSphere Client, use the API.

Prerequisites

- Ensure that the node shows **Success** as its Configuration State.
- Ensure connectivity between the NSX Edge node and NSX Manager is down if the existing NSX Edge node was manually deployed through vSphere Client. If connectivity is Up, then NSX does not allow the existing NSX Edge node to be replaced with a new one.
- If possible, collect the support bundle for the node.

Redeploy an NSX Edge Node Using the NSX Manager UI

This redeployment method is supported for NSX Edge VM nodes that were auto-deployed through NSX Manager.

1. From a browser, log in with admin privileges to NSX Manager.
2. Select **System > Fabric > Nodes > Edge Transport Nodes**.
3. In the list, select the edge node that you want to redeploy.
4. Select **Actions > Redeploy Edge** to open the Redeploy Edge Node workflow.
   The Redeploy Edge Node workflow is identical to the Add Edge Node workflow that was originally used to auto-deploy the node. The editable fields in the Redeploy Edge Node workflow are pre-filled with the existing configuration settings of the node. You can modify these settings as needed.
5. Proceed through the workflow steps and either accept the existing settings or modify the settings as needed. See Create an NSX Edge Cluster and Add Edge Nodes for detailed information on each configuration setting.

   Important:

   You must reenter the CLI credentials for the node VM, as the Credentials fields are left blank for security. Also, do not modify the Compute Manager setting unless you intend to change the compute manager for the node. Once you modify the setting, the configuration mappings associated with the original compute manager are cleared and cannot be retrieved, even if you re-select the original compute manager.
6. After finishing the Redeploy Edge Node workflow, allow the node redeployment to proceed and reach completion.
7. Verify the node configuration status and connection status on the **Edge Transport Nodes** page. Alternatively, you can verify the transport node status by running the `Get edge-cluster-status | get managers | get controllers | get host-switch` CLI command.
8. Perform the postrequisites described later on this page.

Redeploy an NSX Edge Node Using the API

Redeployment through the API is recommended in the following scenarios:

- You need to recover an NSX Edge node that has been lost due to storage corruption.
- You need to change the node's form factor by scaling up or down, upgrade the hardware version, or update management connectivity.
- You need to convert a manually deployed edge node to a system-managed node.

1. (NSX Edge node manually deployed through vSphere Client) Open an SSH session and connect to the NSX Edge console.
2. Verify the logical routes configured on the NSX Edge node through the CLI console, `get logical-routers`.
3. Power off NSX Edge node.
4. Verify that the NSX Edge node is disconnected from NSX Manager by running the API command:
   `GET api/v1/transport-nodes/<edgenode>/state`
   a. If the `node_deployment_state` value is `MPA Disconnected`, you can proceed to redeploy the NSX Edge node.
   b. If `node_deployment_state` is `Node Ready`, NSX Manager displays an error `78006 - Manager connectivity to Edge node must be down. Else replacement/redeployment of hardware is not allowed`.
5. Alternatively, view the state of connectivity between NSX Edge node and NSX Manager from the **Edge Transport Node** page. A disconnected NSX Edge node displays the following system message, `Configuration Error, Edge VM MPA Connectivity is down`.
6. Retrieve the placement parameters and management network settings for the node.
   a. For an auto-deployed node, run `GET /<NSX-Manager-IPaddress>/api/v1/transport-nodes/<edgenode>`. Copy the output payload of this API.

```
 "resource_type": "EdgeNode",
            "id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
            "display_name": "Edge_TN2",
            "description": "EN",
            "external_id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
            "ip_addresses": [
                "10.170.94.240"
            ],
            "_create_user": "admin",
            "_create_time": 1600106319056,
            "_last_modified_user": "admin",
            "_last_modified_time": 1600106907312,
            "_system_owned": false,
            "_protection": "NOT_PROTECTED",
            "_revision": 2
        },
        "is_overridden": false,
        "failure_domain_id": "4fc1e3b0-1cd4-4339-86c8-f76baddbaafb",
        "resource_type": "TransportNode",
        "id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
        "display_name": "Edge_TN2",
        "_create_user": "admin",
        "_create_time": 1600106319399,
        "_last_modified_user": "admin",
        "_last_modified_time": 1600106907401,
        "_system_owned": false,
        "_protection": "NOT_PROTECTED",
        "_revision": 1
    }
```

   b. For a manually deployed node, retrieve the placement parameters and management network settings from vCenter.

7. You can choose from one of these redeployment scenarios:

Choice	Actions
Redeploy a manually deployed NSX Edge node with a system-managed NSX Edge node (deployed through NSX Manager API)	Perform the following in the API command, `/api/v1/transport-nodes/<transport-node-id>?action=redeploy` • Paste the payload in the body of the redeploy API. • Verify the `deployment_config` section references the compute manager, datstore, and network details, where you want to redeploy the node. Ensure these values are consistent with the values used in the `node_settings` section. • Add login passwords in the `deployment_config` section. NSX Manager redeploys the NSX Edge node based on the details provided in the `deployment_config` section.
Change the placement of the existing NSX Edge node	Perform the following in the API command, `/api/v1/transport-nodes/<transport-node-id>?action=redeploy` • Paste the payload in the body of the redeploy API. • In the `deployment_config` section, reference the new compute manager, datastore, network, CPU, memory, or latency sensitivity details. Confirm the VLAN and MTU configuration on the new cluster and host.

An example of the payload for `POST https://<manager-ip>/api/v1/transport-nodes/<transport-node-id>?action=redeploy`:

```
    {
        "node_id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
        "host_switch_spec": {
            "host_switches": [
                {
                    "host_switch_name": "nsxvswitch_overlay",
                    "host_switch_id": "c0a4a83e-c8b8-4324-a4d7-dbbc07b30b53",
                    "host_switch_type": "NVDS",
                    "host_switch_mode": "STANDARD",
                    "host_switch_profile_ids": [
                        {
                            "key": "UplinkHostSwitchProfile",
                            "value": "f9a2a2fa-b49d-498f-abaf-2fdc81917716"
                        },
                        {
                            "key": "LldpHostSwitchProfile",
                            "value": "9e0b4d2d-d155-4b4b-8947-fbfe5b79f7cb"
                        }
                    ],
                    "pnics": [
                        {
                            "device_name": "fp-eth0",
                            "uplink_name": "uplink1"
                        }
                    ],
                    "is_migrate_pnics": false,
                    "ip_assignment_spec": {
                        "ip_pool_id": "647d9b0d-0143-4903-91f5-930d9ab011e8",
                        "resource_type": "StaticIpPoolSpec"
                    },
                    "cpu_config": [],
                    "transport_zone_endpoints": [
                        {
                            "transport_zone_id": "0b33b078-6438-4d9b-a1ec-33211fd36822",
```

```
                    "transport_zone_profile_ids": [
                        {
                            "resource_type": "BfdHealthMonitoringProfile",
                            "profile_id": "52035bb3-ab02-4a08-9884-18631312e50a"
                        }
                    ]
                },
                {
                    "transport_zone_id": "a0133574-48de-4e3a-9407-7db1a68bae41",
                    "transport_zone_profile_ids": [
                        {
                            "resource_type": "BfdHealthMonitoringProfile",
                            "profile_id": "52035bb3-ab02-4a08-9884-18631312e50a"
                        }
                    ]
                }
            ],
            "vmk_install_migration": [],
            "pnics_uninstall_migration": [],
            "vmk_uninstall_migration": [],
            "not_ready": false
        }
    ],
    "resource_type": "StandardHostSwitchSpec"
},
"transport_zone_endpoints": [],
"maintenance_mode": "DISABLED",
"node_deployment_info": {
    "deployment_type": "VIRTUAL_MACHINE",
    "deployment_config": {
        "vm_deployment_config": {
            "vc_id": "cc82da39-b119-4869-a7fe-a54621cb4d3d",
            "compute_id": "domain-c9",
            "storage_id": "datastore-14",
            "host_id": "host-12",
            "compute_folder_id": "group-v5",
            "management_network_id": "network-16",
            "hostname": "EdgeSmallFactor",
            "data_network_ids": [
                "5638c577-e142-4a50-aed3-a7079dc3b08c",
                "5638c577-e142-4a50-aed3-a7079dc3b08c",
                "5638c577-e142-4a50-aed3-a7079dc3b08c"
            ],
            "search_domains": [
                "eng.vmware.com",
                "vmware.com"
            ],
            "enable_ssh": true,
            "allow_ssh_root_login": true,
            "reservation_info": {
                "memory_reservation": {
                    "reservation_percentage": 100
                },
                "cpu_reservation": {
                    "reservation_in_shares": "HIGH_PRIORITY",
                    "reservation_in_mhz": 0
                }
```

```
                },
                "resource_allocation": {
                    "cpu_count": 4,
                    "memory_allocation_in_mb": 8192
                },
                "placement_type": "VsphereDeploymentConfig"
            },
            "form_factor": "MEDIUM",
            "node_user_settings": {
                 "cli_username": "admin",
                    "root_password":"Admin!23Admin",
                    "cli_password":"Admin!23Admin"
            }
        },
        "node_settings": {
            "hostname": "EdgeSmallFactor",
            "search_domains": [
                "eng.vmware.com",
                "vmware.com"
            ],
            "enable_ssh": true,
            "allow_ssh_root_login": true
        },
        "resource_type": "EdgeNode",
        "id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
        "display_name": "Edge_TN2",
        "description": "EN",
        "external_id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
        "ip_addresses": [
            "10.170.94.240"
        ],
        "_create_user": "admin",
        "_create_time": 1600106319056,
        "_last_modified_user": "admin",
        "_last_modified_time": 1600106907312,
        "_system_owned": false,
        "_protection": "NOT_PROTECTED",
        "_revision": 2
    },
    "is_overridden": false,
    "failure_domain_id": "4fc1e3b0-1cd4-4339-86c8-f76baddbaafb",
    "resource_type": "TransportNode",
    "id": "9f34c0ea-4aac-4b7f-a02c-62f306f96649",
    "display_name": "Edge_TN2",
    "_create_user": "admin",
    "_create_time": 1600106319399,
    "_last_modified_user": "admin",
    "_last_modified_time": 1600106907401,
    "_system_owned": false,
    "_protection": "NOT_PROTECTED",
    "_revision": 1
}
```


Note:

If the old node is an NSX Edge node auto-deployed through NSX Manager UI, then you do not need to provide login credentials in the `node_user_settings` section in the API payload. For more information on the payload details, see the NSX API Guide.

8. In NSX Manager, verify the **Configuration Status** of the new NSX Edge node.
9. Alternatively, verify the status of the newly prepared NSX Edge node by running the API command, `Get api/v1/transport-nodes/<node-id>/state`.
10. Verify logical router configurations are migrated to the new NSX Edge node, by running the `get logical-routers` CLI command.
11. Verify the TEP address remain same on the replaced NSX Edge node.
12. Verify NSX Edge cluster status is Up. API is `GET api/v1/edge-clusters/<cluster>`. If NSX is configured to use NSX Federation, verify the intersite status us Up.
13. Check NSX Edge node and cluster state API to verify status is Up.
14. Troubleshoot error messages if needed:
    a. `(78006) NSX Manager connectivity to Edge node must be down. Else replacement of hardware is not allowed`: Ensure NSX Edge node is not connected to NSX Manager.
    b. `(16064) Deployment configuration is missing`: In the redeploy API, enter details for the `deployment_config` section.
    c. `(16066) Login password is missing`: Provide login credentials.
    d. `(15019) Insufficient resources on node to be allocated to load balancer pool`: The form factor size of the new NSX Edge node might be smaller than the form factor of the old NSX Edge node. The new form factor might not have enough resources to be allocated to load balancer pool.
15. Perform the postrequisites described in the next section.

Postrequisites

- If you want to bring up a replaced node VM as part of your network, ensure that the replaced node is disconnected from the network. Then, run `del nsx` to completely delete NSX VIBs on the node. After you run `del nsx` on the host, old entries of logical routers, VTEP IP addresses, uplink IP addresses are released.
- Check the redeployed node and all the other nodes in that edge cluster to ensure that their VM hardware versions are consistent.
- After you redeploy a NSX Edge node, several security parameters are set to their default values. Reconfigure these parameters based on your environment.
  - `set auth-policy minimum-password-length <password-length-arg>`
    Set the minimum number of characters that passwords must have. The smallest value that can be set is 8
    For example, `nsx> set auth-policy minimum-password-length 12`
  - `set user <node-username> password-expiration <password-expiration-arg>`
    Set number of days the user's password is valid after a password change.
    Where, *`<username>`* is the Username of user,
    `<password-expiration>` is the number of days password valid after change (1 - 9999)
    For example, `nsx> set user audit password-expiration 120`
  - `set auth-policy cli max-auth-failures <auth-failures-arg>`
    Set the number of failed CLI authentication attempts that are allowed before the account is locked. If set to 0, account lockout is disabled.
    Where, *`<auth-failures>`* is the number of authentication failures to trigger lockout
    For example, `nsx> set auth-policy cli max-auth-failures 5`
  - `set banner`
    Set the security banner or message of the day.
    For example, `nsx> set banner`
    `Enter TEXT message. End with 'Ctrl-D'`
  - `reset dataplane hugepage`
    Reset the hugepage-related boot time option to factory default.
    For example

```
nsx-edge-1> reset dataplane hugepage
 0000:0b:00.0 already bound to driver vfio-pci, skipping
 0000:1b:00.0 already bound to driver vfio-pci, skipping
 0000:13:00.0 already bound to driver vfio-pci, skipping
```

```
INFO: Config was written to: /config/vmware/edge/config.json
Generating grub configuration file ...
Found linux image: /vmlinuz-3.14.17-nn4-server
Found initrd image: //initrd.img-3.14.17-nn4-server
File descriptor 4 (/tmp/ffinvYglp (deleted)) leaked on lvs invocation. Parent PID 32203: /bin/sh
done
INFO: Updated grub. Please reboot to take effect.
```

Configure Advanced NSX Edge Parameters to Improve Performance

To improve NSX Edge node performance, you can configure the Coalescing Scheme and Coalescing Param advanced configuration parameters, which are available when you prepare an NSX Edge transport node. Call the NSX Edge transport node API to configure the Coalescing configuration.

- Ensure Coalescing Scheme and Coalescing Param configuration is enabled in vSphere. See vSphere documentation.
- VMXNET3 driver is running on VMs.

Coalescing Scheme and Coalescing Param configuration define the virtual network interrupt rate in interrupts per second. Coalescing can reduce the number of interrupts, thus potentially decreasing CPU utilization. Though this could increase network latency, many workloads are not impacted by additional network latency of anywhere from a few hundred microseconds to a few milliseconds, and the reduction in virtual networking overhead can potentially allow more virtual machines on an ESX host. For more information, see vSphere documentation.

Supported Coalescing Schemes: Decide the type of Coalescing Scheme you want to configure and the Coalescing Parameter value you want to set to configure virtual network interrupts on the Edge node.

- `rbc`: Is rate-based coalescing scheme. It supports values in the range of 100 to 100000 interrupts per second. The default value vSphere takes is 4000 interrupts per second.
- `static`: It sets the number of packets before interrupting the CPU.
- `adapt`: Is an adaptive scheme where vSphere decides the interrupt behavior depending upon the load. If the load is lower, the number of interrupts are higher and if the load is higher, the number of interrupts are lower.
- `disabled`: Is used to disable coalescing scheme and param on the NSX Edge node.

To know more about the coalescing scheme and parameters, see the topic *Virtual Network Interrupt Coalescing* .

There are a couple of ways to configure Coalescing Scheme and Param configuration.

1. To configure Coalescing Scheme and Param, you can configure the NSX Edge node in a couple of ways:
   - (API only) Deploy the NSX Edge node by making an API call.
   - Deploy the NSX Edge node from the NSX Manager and call the Redeploy transport node API to update the Coalescing Scheme and Param configuration as part of the NSX Edge node settings.
2. Deploy the NSX Edge node transport node API to deploy the node with Coalescing Scheme and Param.

```
POST https://<nsxManagerIp>/api/v1/transport-nodes/
 {
  "host_switch_spec":{
  "host_switches":[
      ....
      ....
    ],
   "resource_type":"StandardHostSwitchSpec"
   },
   "maintenance_mode":"DISABLED",
   "node_deployment_info":{
    ....
    ....
  },
 "node_settings":{
 "hostname":"edgetwo.com",
```

```
"enable_ssh":true,
"allow_ssh_root_login":true,
"enable_upt_mode":false,
"advanced_configuration":[
{"key": "advanced-config:coalescingScheme",
 "value": "rbc"
      },
      {
  "key": "advanced-config:coalescingParams",
   "value": "4000"
    }
   ]
  },
"resource_type":"EdgeNode",
"display_name":"edge-two",
 "tags":[

],
"_revision":0
},
"is_overridden":false,
"failure_domain_id":"4fc1e3b0-1cd4-4339-86c8-f76baddbaafb",
"resource_type":"TransportNode",
"id":"30b425c5-85ca-4402-9705-88b077d08a06",
"display_name":"edge-nsx-2"
}
```

3. If you already deployed NSX Edge node from NSX Manager, call the redeploy API to edit the settings.

```
POST https://<nsxManagerIp>/api/v1/transport-nodes/<edge-id>?action=redeploy
 {
  "transport_node":{
 "resource_type": "TransportNode",
 "display_name": "edge-node5",
 "host_switch_spec": {
 "resource_type": "StandardHostSwitchSpec",
  "host_switches": [
    {
     "host_switch_profile_ids": [
       {
        "value": "e331116d-f59e-4004-8cfd-c577aefe563a",
        "key": "UplinkHostSwitchProfile"
        }
        ],
      "host_switch_name": "nsxvswitch",
      "pnics": [
       {
       "device_name": "fp-eth0",
       "uplink_name": "uplink1"
       }
       ]
       },
      "transport_zone_endpoints": [
       {
        "transport_zone_id": "e14c6b8a-9edd-489f-b624-f9ef12afbd8f"
          }
       ],
       "node_id": "8538f119-ba45-4fb1-9cf1-ee849e4cf168",
```

```
        "node_deployment_info": {
        "resource_type": "EdgeNode",
        "id": "8538f119-ba45-4fb1-9cf1-ee849e4cf168",
        "display_name": "edge-node5",

      "node_settings":{
              "hostname":"edgetwo.com",
              "enable_ssh":true,
              "allow_ssh_root_login":true,
              "enable_upt_mode":false,
              "advanced_configuration":[
                 {
                    "key": "advanced-config:coalescingScheme",
                     "value": "rbc"
                 },
                 {
                    "key": "advanced-config:coalescingParams",
                     "value": "4000"
                 }
              ]
           },


        "ip_addresses": [
          "192.168.110.37",
          "192.168.110.38"
        ]
      }
  }
```

The Coalescing configuration will fail in the following cases:

- Coalescing scheme is not set and coalescing params is set, then coalescing params will be ignored.
- Coalescing scheme is set and coalescing params is not set then NSX will validate and fail this configuration.

Edit NSX Edge VM Restart Policy

You can set the restart policy for an NSX Edge VM deployed from the vSphere Client at a cluster level using the `VM Override` property.

- NSX Edge node must be deployed. See the *NSX Installation Guide* for more details.

Restart priority ensure that NSX Edge VM is restarted first and virtual network infrastructure is up and running before the worload sends North-South traffic to the NSX Edge VM.

> **Note:**
> When you deploy an NSX Edge node from NSX Manager, redeploy an NSX Edge or upgrade an NSX Edge, restart priority of NSX Edge VM is automatically set to Highest.

To validate NSX Edge restart policy is set to Highest, go to the

1. With admin privileges, log in to the vSphere Client.
2. At the cluster lever, vSphere HA is enabled.
3. To verify the NSX Edge restart policy, perform the following steps:
   a) Select a cluster.
   b) Select **Configure** > **vSphere Availability**.
   c) Verify that **vSphere HA** is turned on.
   d) In the **Edit Cluster Settings** window, expand **Host Failure Response** section and verify that **Failure Response** is set to **Restart VMs** and **Default VM Restrart Priority** is set to **Highest**.

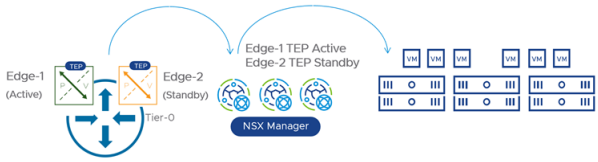
When an NSX Edge node fails, it is restarted as per the configured restart priority. If you set the priority to Highest, NSX Edge node comes back up before other hosts in the cluster.

Verify Failover of NSX Edge VTEP Active Standby Nodes

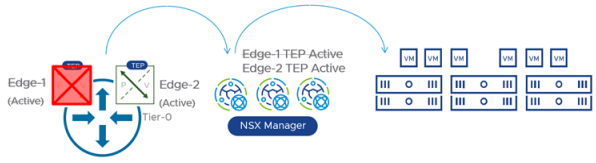
In NSX Federation environment when we use stretched overlay segment for each Segment we have one Active and one Standby SR running on the NSX Edge Node.

When an active NSX Edge Node fails, the NSX Control Plane ensures the traffic failover to the standby NSX Edge node and respectively all transport nodes forward traffic to the new active NSX Edge node.

Review a topology where you have created L2 stretched overlay segments across multiple sites in your NSX Federation environment, each segment relies on two NSX Edge nodes that are in an Active/Standby configuration for cross site traffic. For example, as shown in image 1, NSX Edge 1 is Active and NSX Edge 2 is Standby node. Respectively VTEP of the Active NSX Edge will act as active L2 forwarder for all cross site traffic from all ESX Transport Node.



To ensure HA functionality for NSX Edge nodes, each NSX Edge node communicates the HA state to the central control plane so it knows who is the Active NSX Edge. In turn, the Control Plane communicates, VTEP group membership and HA State information received from NSX Edge node to all transport nodes that host these stretched segments.



The output displays the HA state of both NSX Edge nodes. And the stale record is False, which indicates that the HA state is accurate.

To view the HA state of the Active Edge VTEP after a failover follow the procedure:

> **Note:**
> This procedure is applicable to all NSX stretched Segments used in Federation.

1. Copy the UUID of the stretched segment that is attached to a Tier-1 or Tier-0 gateway.
2. Define which is the Active Edge for that specific T1 Gateway where the segment is connected.
3. You can also use the following API to get the Active/Standby Edge for the specific Segment.

   `/infra/segments/<segment-id>/inter-site-forwarder/site-span-info`

4. Connect to the Edge with user: admin password.
5. Run `get vtep-group` to know the VTEP groups present on the NSX Edge node.
6. Verify whether the output of the `get vtep-group` command has the stretched segment UUID.
7. Copy the VTEP-Group ID corresponding to the segment.
8. In the NSX Manager node CLI terminal, run `get vtep-group <vtep-group-ID> vteps-staleness-status`.

```
VNI       IP                  MAC            LABEL       Segment      TransportNode-Id
   TN-Connection   HA-STATE  STALE-RECORD

26625     172.20.1.151     00:0c:29:9e:64:5e      0x18001    172.20.1.0
 32330174-32bc-11ee-8063-000c299e6454        true          ACTIVE     False
26625     172.20.1.152     00:0c:29:ea:8e:aa      0xFC01     172.20.1.0   914d0362-32bc-11ee-
ba27-000c29ea8ea0        true        STANDBY     False
```

9.
10. Run `get vtep-group` to know the VTEP groups present on the NSX Edge node.
11. Verify whether the output of the `get vtep-group` command has the stretched segment UUID.
12. Copy the `VTEP-Group ID` corresponding to the segment.
13. In the NSX Manager node CLI terminal, run `get vtep-group <vtep-group-ID> vteps-staleness-status`.

```
VNI           IP                   MAC            LABEL       Segment      TransportNode-Id
       TN-Connection   HA-STATE  STALE-RECORD
26625     172.20.1.151     00:0c:29:9e:64:5e      0x18001    172.20.1.0
 32330174-32bc-11ee-8063-000c299e6454        true          ACTIVE     False
26625     172.20.1.152     00:0c:29:ea:8e:aa      0xFC01     172.20.1.0   914d0362-32bc-11ee-
ba27-000c29ea8ea0        true        STANDBY     False
```

    The output displays the HA state of both NSX Edge nodes. And the stale record is False, which indicates that the HA state is accurate.

14. Verify that one of the NSX Edge nodes is active and the other one is in Standby mode.
15. If the active NSX Edge goes down, the HA state mode changes. The Standby node becomes the Active node.
16. Run `get vtep-group <vtep-group-ID> vteps-staleness-status`.

```
 VNI        IP                  MAC              LABEL       Segment        TransportNode-Id
       TN-Connection   HA-STATE  STALE-RECORD
 26625   172.20.1.151    00:0c:29:9e:64:5e       0x18001    172.20.1.0
 32330174-32bc-11ee-8063-000c299e6454        true          ACTIVE      True
 26625   172.20.1.152    00:0c:29:ea:8e:aa       0xFC01     172.20.1.0    914d0362-32bc-11ee-
ba27-000c29ea8ea0        true         ACTIVE     False
```

    In the output, NSX Edge with `172.20.1.151` address has gone down and hence the `Stale-Record` is `True`.

The NSX Control Plane ensures that stale entries of failed NSX Edge nodes are correctly recorded in the output of the VTEP groups.

Adding and Removing an ESX Host Transport Node to and from vCenters

You can move an ESX host transport node from one vCenter (VC) to another, and also from one NSX Manager cluster to another.

Scenario 1: VC1 connected to NSX Manager cluster 1, and VC2 connected to NSX Manager cluster 2

Assuming ESX1, an ESX host transport node, is in VC1, you can move it to VC2 by performing the following steps:

1. Uninstall NSX from ESX1.
2. Move ESX1 to VC2.

3. Apply a transport node profile to ESX1.

Scenario 2: Both VC1 and VC2 connected to NSX Manager cluster

Assuming ESX1, an ESX host transport node, is in VC1, you can move it to VC2 by performing the following steps:

1. Uninstall NSX from ESX1.
2. Move ESX1 to VC2.
3. Apply a transport node profile to ESX1.

Scenario 3: VC1 connected to NSX Manager cluster 1

Assuming ESX1, an ESX host transport node, is in VC1, you can move it to NSX Manager cluster 2 as a standalone host by performing the following steps:

1. Uninstall NSX from ESX1.
2. Add ESX1 to NSX Manager cluster 2.

> **Note:**
>
> The user interface displays a warning, if at any point, your host transport node goes out of sync with the Management Plane. To initiate the resync operation, select **Actions** > **Sync Transport Node** for your host transport node.

Changing the Distributed Router Interfaces' MAC Address

All logical router interfaces in NSX and NSX-V setups have the same MAC address (02:50:56:56:44:52). Starting with NSX 3.0.2, you can change this address in NSX to avoid issues when migrating VMs from an NSX-V setup to an NSX setup.

Changing the MAC address involves making two API calls.

If you have not created any transport node, make the following GET API call. For example:

```
GET https://10.40.79.126/api/v1/global-configs/RoutingGlobalConfig

Response:
{
  "l3_forwarding_mode" : "IPV4_ONLY",
  "logical_uplink_mtu" : 1500,
  "vdr_mac" : "02:50:56:56:44:77",
  "vdr_mac_nested" : "02:50:56:56:44:52",
  "allow_changing_vdr_mac_in_use" : true,
  "resource_type" : "RoutingGlobalConfig",
  "id" : "49b261fe-f4e4-46ad-958c-da9cb4271e32",
  "display_name" : "49b261fe-f4e4-46ad-958c-da9cb4271e32",
  "_create_user" : "system",
  "_create_time" : 1595313890595,
  "_last_modified_user" : "admin",
  "_last_modified_time" : 1595465694142,
  "_system_owned" : false,
  "_protection" : "NOT_PROTECTED",
  "_revision" : 14
}
```

Take the response of the call, change the `vdr_mac` value, and use it to make the following PUT API call. For example:

```
PUT  https://10.40.79.126/api/v1/global-configs/RoutingGlobalConfig
{
  "l3_forwarding_mode" : "IPV4_ONLY",
  "logical_uplink_mtu" : 1500,
  "vdr_mac" : "02:50:56:56:44:99",
```

```
  "vdr_mac_nested" : "02:50:56:56:44:53",
  "allow_changing_vdr_mac_in_use" : true,
  "resource_type" : "RoutingGlobalConfig",
  "id" : "49b261fe-f4e4-46ad-958c-da9cb4271e32",
  "display_name" : "49b261fe-f4e4-46ad-958c-da9cb4271e32",
  "_create_user" : "system",
  "_create_time" : 1595313890595,
  "_last_modified_user" : "admin",
  "_last_modified_time" : 1595465694142,
  "_system_owned" : false,
  "_protection" : "NOT_PROTECTED",
  "_revision" : 14
}

Response:
{
  "l3_forwarding_mode" : "IPV4_ONLY",
  "logical_uplink_mtu" : 1500,
  "vdr_mac" : "02:50:56:56:44:99",
  "vdr_mac_nested" : "02:50:56:56:44:53",
  "allow_changing_vdr_mac_in_use" : true,
  "resource_type" : "RoutingGlobalConfig",
  "id" : "49b261fe-f4e4-46ad-958c-da9cb4271e32",
  "display_name" : "49b261fe-f4e4-46ad-958c-da9cb4271e32",
  "_create_user" : "system",
  "_create_time" : 1595313890595,
  "_last_modified_user" : "admin",
  "_last_modified_time" : 1595466163148,
  "_system_owned" : false,
  "_protection" : "NOT_PROTECTED",
  "_revision" : 15
}
```

If you have already created transport nodes, make the same GET and PUT API calls, except that for the PUT call, set the parameter `allow_changing_vdr_mac_in_use` to `true`.

Configuring Appliances

Some system configuration tasks must be done using the command line or API.

For complete command line interface information, see the *NSX Command-Line Interface Reference*. For complete API information, see the *NSX API Guide*.

Table 1079: System configuration commands and API requests.

Task	Command Line (NSX Manager and NSX Edge)	API Request (NSX Manager only)
Set system timezone	`set timezone <timezone>`	`PUT https://<nsx-mgr>/api/v1/node`
Set NTP Server	`set ntp-server <ntp-server>`	`PUT https://<nsx-mgr>/api/v1/node/services/ntp`
Set a DNS server	`set name-servers <dns-server>`	`PUT https://<nsx-mgr>/api/v1/node/network/name-servers`
Set DNS Search Domain	`set search-domains <domain>`	`PUT https://<nsx-mgr>/api/v1/node/network/search-domains`

Note:
The recommended method to configure an NTP server for all appliances is to configure a node profile. See Configure a Node Profile. If you configure an NTP server individually on an appliance, be sure to configure the same NTP server on all the appliances.

Configuring NTP on Appliances and Transport Nodes

Some features require that NTP be configured on all the components in the NSX environment.

It is highly recommended that you configure NTP for all components, regardless of the features you plan to use.

Note:
Change NTP time configuration in small increments. Do not change the time in a one large jump. If you change the time in a large increment, you must restart the appliance for correct functionality.

To configure NTP on an appliance, see Configuring Appliances.

To configure NTP for all Manager and Edge nodes, see Configure a Node Profile.

To configure NTP for an ESX host, see the topic *Synchronize ESX Clocks with a Network Time Server*, in VSphere Security.

By default, different NTP services run in different Linux distributions. In all releases, run the `timedatectl` to view the synchronization status. Only one NTP service can be running at a time.

The following Linux distributions are supported with their default NTP services:

Linux Server	NTP Client
RHEL/CentOS 7.6	chronyd.service
RHEL/CentOS 7.7	chronyd.service
RHEL/CentOS 8.0	chronyd.service
RHEL/CentOS 8.2	chronyd.service
Ubuntu 18.04	systemd-timesyncd
Ubuntu 20.04	systemd-timesyncd
SLES12 SP4	systemd-timedated

Standards Compliance

NSX is configured to use FIPS 140-3 validated cryptographic modules by default to comply with FIPS requirements. The modules are validated to FIPS 140-3 standards by the NIST Cryptographic Module Validation Program (CMVP).

All exceptions to FIPS compliance can be retrieved using the compliance report. See View Compliance Status Report for more information.

The following validated modules are used:

- VMware VPN Crypto Module 2.0, Certificate #4881
- IKE: Rambus Safezone 2.0, Certificate #4898
- VMware Openssl 3.5.5, Certificate **#**4861
- VMware BouncyCastle 2.0.0, Certificate #4986
- VMware BoringCrypto 6.0, Certificate #4694

You can find more information about the cryptographic modules that VMware has validated against the FIPS 140 standard here: https://www.vmware.com/security/certifications/fips.html.

View Compliance Status Report

You can view a compliance report for NSX features. You can use the report to configure your NSX environment to adhere to your IT policies and industry standards.

The compliance report includes information about each non-compliant configuration.

Table 1080: Compliance Report Information

Compliance Report Column	Description	Example
Non Compliance Code	Code to identify the type of non-compliance.	72301
Description	Description of the type of non-compliance.	Certificate is not CA signed.
Compliance Standard	The compliance standard or program that the resource has violated.	EAL4+
Resource By	Name or ID of the affected resource.	nsx-manager-1
Resource Type	Type of resource affected.	CertificateComplianceReporter
Affected Resources	Number of affected resources. The number can be 0 if there are non-compliant configurations present, but the feature is not used.	1

You can also retrieve the report using the API: `GET /policy/api/v1/compliance/status`.

1. From your browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address>.
2. From the **Home** page, click **Monitoring Dashboards** > **Compliance Report**.

Compliance Status Report Codes

You can find the information about the meaning of the compliance report codes in the compliance status report.

For a complete list of events, see the NSX Event Catalog.

Table 1081: Compliance Report Codes

Code	Description	Compliance Status Source	Remediation
72001	Encryption is deactivated.	Reports this status if a VPN IPSec Profile configuration contains `NO_ENCRYPTION`, `NO_ENCRYPTION_AUTH_AES_GMAC_128`, `NO_ENCRYPTION_AUTH_AES_GMAC_192`, or `NO_ENCRYPTION_AUTH_AES_GMAC_256` encryption_algorithms. This status affects IPSec VPN session configurations which use the reported non-compliant configurations.	Add a VPN IPSec Profile that uses compliant encryption algorithms and use the profile in all VPN configurations. See Add IPSec Profiles.
72011	BGP messages with neighbor bypass integrity check. No message authentication defined.	Reports this status if BGP neighbors has no confgured password. This status affects the BGP neighbor configuration.	Configure a password on the BGP neighbor and update the tier-0 gateway configuration to use the password. See Configure BGP.

Code	Description	Compliance Status Source	Remediation
72012	Communication with BGP neighbor uses weak integrity check. MD5 is used for message authentication.	Reports this status if MD5 authentication is used for the BGP neighbor password. This status affects the BGP neighbor configuration.	No remediation available as NSX supports only MD5 authentication for BGP.
72021	SSL version 3 used for establishing secure socket connection. It is recommended to run TLS v 1.1 or higher and fully deactivate SSLv3 that have protocol weaknesses.	Reports this status if SSL version 3 configuration is in the load balancer client SSL profile, load balancer server SSL profile, or load balancer HTTPS monitor. This status affects the following configurations: • Load balancer pools that are associated with HTTPS monitors. • Load balancer virtual servers that are associated with load balancer client SSL profiles or server SSL profiles.	Configure an SSL profile to use TLS v1.1 or later and use this profile in all load balancer configurations. See Add an SSL Profile.
72022	TLS version 1.0 used for establishing secure socket connection. Run TLS v1.1 or higher and fully deactivate TLS v1.0 that have protocol weaknesses.	Reports this status if the load balancer client SSL profile, load balancer server SSL profile, or load balancer HTTPS monitor includes the TLS v1.0 configuration. This status affects the following configurations: • Load balancer pools that are associated with HTTPS monitors. • Load balancer virtual servers that are associated with load balancer client SSL profiles or server SSL profiles.	Configure an SSL profile to use TLS v1.1 or later and use this profile in all load balancer configurations. See Add an SSL Profile.
72023	Weak Diffie-Hellman group is used.	Reports this error if a VPN IPSec Profile or VPN IKE Profile configuration includes the following Diffie-Hellman groups: 2, 5, 14, 15 or 16. Groups 2 and 5 are weak Diffie-Hellman groups. Groups 14, 15, and 16 are not weak groups, but are not FIPS-compliant. This status affects IPSec VPN session configurations which use the reported non-compliant configurations.	Configure the VPN Profiles to use Diffie-Hellman group 19, 20, or 21. See Adding Profiles.
72025	Quick Assist Technologies (QAT) running on Edge node is non-FIPS compliant.	QAT is a set of hardware accelerated services provided by Intel for cryptography and compression.	To turn off QAT usage, use the NSX CLI. For details, see "Intel QAT Support for IPSec VPN Bulk Cryptography" in the *NSX Installation Guide*.
72026	Bouncy-castle FIPS module is not ready.		Check to ensure your applications are running and check your logs.

Code	Description	Compliance Status Source	Remediation
72200	Insufficient true entropy available.	Reports this status if a pseudo random number generator generates the entropy rather than relying on hardware-generated entropy. NSX Manager node does not use hardware-generated entropy because is does not have the required hardware acceleration support to create sufficient true entropy.	Use newer hardware to run the NSX Manager node. Most recent hardware supports this feature. **Note:** If the underlying infrastructure is virtual, you will not get true entropy.
72201	Entropy source unknown.	Reports this status when no entropy status is available for the indicated node.	This error is an internal communication error preventing the NSX Manager from being able to determine the source of entropy. Open a service request with VMware.
72301	Certificate is not CA signed.	Reports this status when one of the NSX Manager certificates is not CA signed. NSX Manager uses the following certificates: • Syslog certificate. • API certificates for the individual NSX Manager nodes. • Cluster certificate used for the NSX Manager VIP.	Install CA-signed certificates. Refer Certificates in NSX.
72302	Certificate is missing Extended Key Usage (EKU) information.	EKU extensions present in the CSR should also be present in the server certificates.	EKU needs to match the intended usage. For example, SERVER when connecting to a server, CLIENT when used as a client certificate on a server.
72303	Certificate has been revoked.		Validity of the certificate is in a terminal state and cannot recover.
72304	Certificate is not RSA.		Ensure your certificate public key is for an RSA certificate.
72305	Certificate is not Elliptic Curve.	Reports this status when your certificate fails EAL4+ compliance.	Replace your non-compliant certificates with CA-signed certificates. See Replace NSX Certificates Using API.
72306	Certificate is missing Basic Constraints.		The certificate does not appear to be valid for the selected purpose or may be missing necessary fields or values..
72307	Unable to fetch the CRL.		
72308	The CRL is invalid.		Check CRL to determine why it has been marked invalid.
72309	The Corfu Certificate Expiry Check has been deactivated.		

Code	Description	Compliance Status Source	Remediation
72310	Some Compute Manager does not have RSA certificate or the certificate key length is less than 3072 bits.	When a compute manager (for example, vCenter) is connected to the NSX Manager, it passes its certificate to NSX Manager. If the certificate is not an RSA type, or if the certificate is an RSA type, but the certificate's RSA key size is less then 3072 bits, it triggers this error.	Delete the compute manager from NSX Manager. Replace the compute manager's certificate with an RSA certificate with key size of at least 3072 bits.
72401	Gateway TLS Inspection is non-FIPS compliant.		
72402	System not FIPS compliant.		
72403	Principal Identity presence detected in system. Remove all the Principal Identities for EAL4 compliance.		
72404	OIDC end-point presence detected in system. Remove all the OIDC end-points for EAL4 compliance.		
72501	Configured user passwords are not compliant. Users must use high-quality passwords of length at least 16 characters.	Reports when user passwords fail EAL4+ compliance.	The password for local users (except for the root user) must be at least 16 characters. In other words, this means that the admin user password must be at least 16 characters. This is in addition to the password complextity check that is required when modifying the password.
72502	Unable to get synchronized users.		
72503	SSH service for Manager appliance found to be in running state.		
72504	Non-admin active user accounts detected.	When any user other than admin is active in the NSX Manager.	Deactivate all other users other than admin.
73000	Error occurred while gathering platform compliance data.		

Common Criteria Compliance

NSX is designed to be EAL4+ compliant in accordance with the Common Criteria Certification Program.

For more information about Common Criteria, see the Common Criteria Portal.

If your environment is not compliant with EAL4+, NSX will raise alarms. For more information about the alarms, see the "Security Compliance Events" section in NSX Event Catalog.

NSX supports the following security functional requirements:

Requirement	Description	Auditable
FAU_GEN.1	Audit data generation	Yes
FAU_SAR.1	Audit review	No

Requirement	Description	Auditable
FAU_STG.1	Protected audit trail storage	No
FCS_CKM.1/TLS	Cryptographic key generation (for asymmetric keys)	No
FCS_COP.1/TLS.HMAC	Cryptographic operation (hashing)	No
FCS_COP.1/TLS	Cryptographic operation (TLS)	No
FCS_CKM.2/TLS	Cryptographic key distribution (TLS)	Yes
FCS_CKM.4	Cryptographic key destruction	No
FCS_RNG.1/OSSL	Random number generation (OpenSSL)	No
FCS_RNG.1/BC	Random number generation (Bouncy Castle)	No
FDP_IFC.1	Subset information flow control	No
FDP_IFF.1	Simple security attributes	Yes
FIA_AFL.1	Authentication failure handling	Yes
FIA_SOS.1	Verification of secrets	Yes
FIA_UAU.2	User authentication before any action	Yes
FIA_UID.1	Timing of identification	Yes
FMT_SMR.1	Security roles	Yes
FMT_SMF.1	Specification of management functions	Yes
FMT_MOF.1	Management of security functions behavior	No
FMT_MSA.1	Management of security attributes	No
FMT_MSA.3	Static attribute initialisation	No
FPT_TDC.1	Inter-TSF basic TSF data consistency	Yes
FTP_ITC.1	Inter-TSF trusted channel	Yes

For each requirement, events that are auditable will be logged. For more information about event logging, see Log Messages and Error Codes.

Configure an External Entropy Source

You can configure an external entropy source to generate random numbers for cryptographic operations. The external entropy source can be used to satisfy one of the requirements of EAL4.

- The ESX host where NSX Manager is running is configured to activate the correct entropy source. For more information, see *Controlling ESX Entropy*.

After the ESX host is configured with an external entropy source, NSX will automatically obtain entropy data from ESX. No configuration on NSX is required.

The external entropy will impact the functions related to random number generation and cryptographic operation mentioned in *Common Criteria Compliance*.

1. Log in to vCenter from vSphere Client.
2. Browse to the NSX Manager VM.
3. Shut down the VM.
4. Right-click the VM and select **Edit Settings**.
5. Select **Advanced Parameters**.
6. Set the **isolation.tools.getEntropy.disable** parameter to `FALSE`.
7. Click **OK**.

Collect Support Bundles

You can collect support bundles on registered cluster and fabric nodes and download the bundles to your machine or upload them to a file server.

If you choose to download the bundles to your machine, you get a single archive file consisting of a manifest file and support bundles for each node. If you choose to upload the bundles to a file server, the manifest file and the individual bundles are uploaded to the file server separately.

Collect Support Bundles in Version 9.1.1 and Later

1. From your browser, log in as a local admin user to an NSX Manager at `https://nsx-manager-ip-address/login.jsp?local=true`.
2. Select **System** > **Support Bundle**
3. Select the target nodes.
   The available types of nodes are **Management**, **Edge**, **Host**, and **Virtual Network Appliance**.
4. (Optional) To configure more options for the support bundles, expand **Additional Configuration**.
   a. Specify log age in days to exclude logs that are older than the specified number of days.
   b. Toggle the switch that indicates whether to include or exclude files with sensitive data.

      Note:

      Core files and audit logs might contain sensitive information such as passwords or encryption keys.

   a. Toggle the switch that indicates whether to upload the bundles to a remote file server. If you toggle to **Yes**, specify the following settings:

Setting	Description
Hostname or IP Address	FQDN or IP address of the remote file server.
Username	User login to the remote file server.
Password	Password credentials for the user login.
Destination Path	Path to the directory location for the support bundles.
Port	Port number to use for the file uploads.
Protocol	Network protocol to use for the file uploads. Choose between Secure Copy Protocol (SCP) and Secure File Transfer Protocol (SFTP).
SSH Fingerprint	Fingerprint for the SSH connection. Leave this setting blank to fetch the SSH fingerprint automatically.
Manager Upload	Set the toggle to specify which node performs the upload: • Specify **No** to have each target node (excluding host nodes) upload its support bundle directly to the remote file server. • Specify **Yes** to have NSX Manager upload the support bundles of target nodes to the remote file server. **Note:** NSX Manager always performs the upload of a host node's support bundle, regardless of this toggle setting.

5. Click **Start Bundle Collection** to start collecting support bundles.
   Depending on how many log files exist, each node might take several minutes.
6. Monitor the status of the collection process.
   The **Status** tab shows the progress of collecting support bundles.
7. Click **Download** to download the bundle if the option to send the bundle to a file remote server was not set.

   Note:

   The bundle collection might fail for a manager node if there is not enough disk space. If you encounter an error, check whether older support bundles are present on the failed node. Log in to the NSX Manager UI of the failed manager node using its IP address and initiate the bundle collection from that node. When prompted by NSX Manager, either download the older bundle or delete it.

Collect Support Bundles in Version 9.1.0

1. From your browser, log in as a local admin user to an NSX Manager at `https://nsx-manager-ip-address/login.jsp?local=true`.
2. Select **System** > **Support Bundle**
3. Select the target nodes.
   The available types of nodes are **Management Nodes**, **Edges**, and **Hosts**.
4. (Optional) Specify log age in days to exclude logs that are older than the specified number of days.
5. (Optional) Toggle the switch that indicates whether to include or exclude core files and audit logs.

   Note:

   Core files and audit logs might contain sensitive information such as passwords or encryption keys.
6. (Optional) Select the check box to upload the bundles to a remote file server.
7. Click **Start Bundle Collection** to start collecting support bundles.
   Depending on how many log files exist, each node might take several minutes.
8. Monitor the status of the collection process.
   The status tab shows the progress of collecting support bundles.
9. Click **Download** to download the bundle if the option to send the bundle to a file remote server was not set.
   The bundle collection might fail for a manager node if there is not enough disk space. If you encounter an error, check whether older support bundles are present on the failed node. Log in to the NSX Manager UI of the failed manager node using its IP address and initiate the bundle collection from that node. When prompted by the NSX Manager, either download the older bundle or delete it.

Understanding Support Bundle File Paths

Know the different files collected when you download a support bundle and the features that store logs in these file paths.

NSX Node Support Bundle

File Path	Features Using the File Path
/var/log/syslog	All
/var/log/nsx-audit.log	All
/var/log/policy/policy.log	All
/var/log/policy/policy_restart.log	All
/var/log/policy/policy-tomcat-wrapper.log	All
/var/log/proton/activity-stats.log	All
/var/log/proton/gc.log	All
/var/log/proton/localhost_access_log.txt	All
/var/log/proton/nsxapi.log	All

File Path	Features Using the File Path
/var/log/proton/proton_restart.log	All
/var/log/proton/proton-tomcat-wrapper.log	All
/var/log/proton-tomcat/nsxapi.log	All
/var/log/proxy/gc.log	AAA (Auth)
/var/log/proxy/keystore.log	AAA (Auth)
/var/log/proxy/localhost.log	AAA (Auth)
/var/log/proxy/localhost_access_log.txt	AAA (Auth)
/var/log/proxy/proxy-tomcat-wrapper.log	AAA (Auth)
/var/log/proxy/reverse-proxy.log	AAA (Auth)
/var/log/phonehome-coordinator/localhost_access_log.txt	Alarm Event Framework, Telemetry, Notification Framework, Capacity Dashboard
/var/log/phonehome-coordinator/phonehome-audit-log.txt.0	Alarm Event Framework, Telemetry, Notification Framework, Capacity Dashboard
/var/log/phonehome-coordinator/phonehome-coordinator.log	Alarm Event Framework, Telemetry, Notification Framework, Capacity Dashboard
/var/log/cbm/cbm.log	Clustering
/var/log/cm-inventory/localhost_access_log.txt	CM-inventory (Compute Manager)
/var/log/vmware/top-cpu.log	Common - Appliance Health
/var/log/vmware/top-mem.log	Common - Appliance Health
/var/log/stats	Common - IP, memory, thread stats
/var/log/cloudnet/nsx-ccp-transaction.log	Controller
/var/log/cloudnet/nsx-ccp.log	Controller
/var/log/core	Cores generated
/var/log/corfu/corfu.9000.log	Corfu DB
/var/log/corfu/corfu-compactor-audit.log	Corfu DB
/var/log/corfu-nonconfig/corfu.9040.log	Corfu DB
/var/log/corfu-nonconfig/nonconfig-corfu-compactor-audit.log	Corfu DB
/var/log/cloudnet/nsx-ccp.log	DNS, Upgrade, IPDiscovery, Groups
/var/log/cm-inventory/cm-inventory.log	Endpoint Security, Deployment, NSX Edge Install, Vmotion, Inventory, NCP/Openshift/PKS-windows, Install/Host Upgrade, Install
/var/log/async-replicator-service/ar.log	Federation (Local Manager)
/var/log/proton/nsx-common-dashboard.log	NCP/Openshift/PKS-windows
/var/log/search/elasticsearch_index_indexing_slowlog.log	NCP/Openshift/PKS-windows
/var/log/search/search-policy.log	NCP/Openshift/PKS-windows
/var/log/migration-coordinator/migration-coordinator.log	Overlay adoption
/var/log/policy/localhost_access_log.txt	Policy realization
/var/run/log/nsxaVim.log	Stateless Hosts/ Transport Node Profile
/var/log/search/elasticsearch.log	UI
/var/log/intelligence-upgrade-coordinator/localhost_access_log.txt	Upgrade
/var/log/upgrade	Upgrade
/var/log/resume-upgrade.log	Upgrade

File Path	Features Using the File Path
/var/log/upgrade-coordinator/localhost_access_log.txt	Upgrade
/var/log/upgrade-coordinator/upgrade-coordinator.log	Upgrade
/var/log/nvpapi/api_server.log	Upgrade, BnR
/var/log/upgrade-coordinator/localhost_access_log.txt	Upgrade Coordinator

NSX Edge Support Bundle

File Path	Features Using the File Path
/var/log/syslog	All
/controller/data/data_dump	DHCP
/controller/falcon/falcon_dump	DHCP
/controller/span/config_span_graph_dump	DHCP
/controller/span/dag_span_graph_dumpContainer_Dfw_Canary	DHCP
/controller/mediator/mediator_dump	DHCP
/var/log/cloudnet/nsx-ccp.log	DHCP
/var/log/proton/nsxapi.log	DHCP
/edge/dns_fdr-all	DNS
/edge/dns-forwarder	DNS
/edge/service_binding	DNS
/var/log/join_mp.log	Edge Install, Install, Upgrade
/edge/fw-if-ruleset/	Load Balancer
/edge/Lb_UUID/	Load Balancer
/edge/lb-*	Load Balancer
/var/log/lb/<LB_UUID>/logs/error.log	Load Balancer
/var/log/lb/<LB_UUID>/logs/ acess_vipid.log	Load Balancer
/var/log/lb/<LB_UUID>/logs/nginx.conf	Load Balancer
/var/log/lb/<LB_UUID>/lbconf_gen.log	Load Balancer
/var/log/core/	Load Balancer, L3VPN, L2VPN, Edge HA, Upgrade
/var/log/nsx-cli/nsxcli.log	L3VPN,L2VPN,Edge HA, Upgrade
/var/log/rcpm/frr-config.log	L3VPN, Upgrade
/var/log/rcpm/frr-reload.log	L3VPN, Upgrade
/var/log/vmware/top-cpu.log	NCP/Openshift/PKS-windows
/var/log/vmware/top-mem.log	NCP/Openshift/PKS-windows
/var/log/frr/frr.log	Openstack, Upgrade
/opt/vmware/nsx-nestdb/bin/nestdb-cli	Openstack, Upgrade
/var/log/nginx/access.log	Openstack
/var/log/upgrade	Upgrade
/var/log/nvpapi/api_server.log	Upgrade, BnR

NSX Application Platform Support Bundle

The following support bundles include log information for the features that NSX Application Platform hosts. These features are Security Intelligence, Network Detection and Response, Malware Prevention for vDefend, and NSX Metrics.

File Path	Features Using the File Path
/var/log/napp/supportbundle/platform_service_dbg.log	NSX Application Platform
/var/log/napp/supportbundle/<K8s_worker_node_name>/kafka*.log	NSX Application Platform
/var/log/napp/supportbundle/<K8s_worker_node_name>/zookeeper*.log	NSX Application Platform
/var/log/napp/supportbundle/<K8s_worker_node_name>/ spark-app-rawflow-driver*.log	NSX Application Platform
/var/log/napp/supportbundle/<K8s_worker_node_name>/authserver*.log	NSX Application Platform
/var/log/napp/supportbundle/<K8s_worker_node_name>/cluster-api*.log	NSX Application Platform
/var/log/napp/supportbundle/<K8s_worker_node_name>/common-agent*.log	NSX Application Platform
/var/log/napp/supportbundle/<K8s_worker_node_name>/projectcontour*.log	NSX Application Platform
/var/log/napp/supportbundle/<K8s_worker_node_name>/routing-controller*.log	NSX Application Platform
/var/log/napp/supportbundle/<K8s_worker_node_name>/trust-manager*.log	NSX Application Platform
/var/log/napp/supportbundle/<K8s_worker_node_name>/upgrade-coordinator*.log	NSX Application Platform
/var/log/napp/supportbundle/<K8s_worker_node_name>/postgresql*.log	NSX Application Platform
/var/log/napp/supportbundle/<K8s_worker_node_name>/visualization*.log	Security Intelligence Visualization
/var/log/napp/supportbundle/<K8s_worker_node_name>/recommendation*.log	Security Intelligence Recommendation
/var/log/napp/supportbundle/<K8s_worker_node_name>/rawflowcorrelator*.log	Security Intelligence
/var/log/napp/supportbundle/<K8s_worker_node_name>/ overflowcorrelator*.log	Security Intelligence
/var/log/napp/supportbundle/<K8s_worker_node_name>/*druid*.log	Security Intelligence
/var/log/napp/supportbundle/<K8s_worker_node_name>/nsx-config*.log	Security Intelligence
/var/log/napp/supportbundle/<K8s_worker_node_name>/redis*.log	Security Intelligence
/var/log/napp/supportbundle/<K8s_worker_node_name>/minio*.log	Security Intelligence
/var/log/napp/supportbundle/<K8s_worker_node_name>/nsx-ndr*.log	Network Detection and Response
/var/log/napp/supportbundle/<K8s_worker_node_name>/sa-*.log	Malware Prevention for vDefend
/var/log/napp/supportbundle/<K8s_worker_node_name>/malware*.log	Malware Prevention for vDefend

File Path	Features Using the File Path
/var/log/napp/supportbundle/<K8s_worker_node_name>/metrics*.log	NSX Metrics

NSX Global Manager Support Bundle

File Path	Features Using the File Path
/var/logdata-log.log	Federation
/var/log/async-replicator- ar.log	Federation
/var/log/connections.log	Federation
/var/log/gm.log	Federation
/var/log/nvpapi/api_access.log	Federation
/var/log/global-manager/gmanager.log	Grouping

Diagnose NSX Using ODS Runbooks

The Online Diagnostic System (ODS) is a diagnostics capability of NSX in VCF Networking. ODS executes runbooks that follow built-in procedures step by step, collecting the data and visibility you need to understand network behavior and how to troubleshoot issues in your environment. Each runbook concludes with a report and might return artifacts that you can review in NSX or export for deeper analysis.

Predefined runbooks ship with NSX. You can also add dynamic runbooks from packages that have been tested and certified by Broadcom to troubleshoot targeted issues.

You can use APIs to perform the following runbook operations:

- Invoke a runbook to initiate debugging at runtime
- Check the debugging report
- Download artifacts generated by the runbook
- Manage the lifecycle of a runbook

Predefined runbooks

The following table lists the predefined runbooks.

Note:

You cannot modify a predefined runbook. You can invoke edge-specific runbooks on both edge transport nodes and virtual network appliance (VNA) nodes. ODS is also supported for Unified Appliance (UA).

Runbook	Description	Node
AdfCollect	This runbook gathers ESX datapath performance data through ADF collector. This runbook takes the following input arguments: • advanced: Whether advanced performance tools are enabled. • cycle: Number of ADF collection cycles. • interval: Waiting interval between consecutive ADF collection cycles in seconds.	ESX
BgpNeighborState	This runbook diagnoses various flows that can cause the BGP neighbor to be down. The runbook also collects the following supporting artifacts for offline debugging. • BGP and BFD packet capture. • Ping and traceroute results. This runbook takes the following input arguments: • Peer IP: This is a mandatory argument and takes the BGP neighbor IP address. • Logical Router name: This is a mandatory argument and takes the logical router name configured by user. • Packet capture: This is an optional argument. If this is set to `True`, the runbook will capture BGP/BFD packets. By default this is set to `False`.	Edge
ControllerConn	This runbook can identify controller connectivity issues caused by controller failure, proxy failure, underlay network outage, or FQDN resolution failure. This runbook takes the IP address of the ESX host and the IP address of controller as input arguments.	ESX
CorfuServer	This runbook checks the Corfu server layout stability, infra ping and disk IO condition, and compactor health. To override the default look back days or hours of log events, this runbook takes lookback days and lookback hours as input arguments.	UA
DistributedMps	This runbook verifies the health of the Malware Prevention Service (MPS) pipeline and diagnoses any issues encountered with MPS, such as protection not working on a particular VM or certain files not being scanned .Input argument for this runbook is the VM UUID of the VM to be diagnosed for protection.	ESX
DuplicateVtepDetectorProvider	This runbook detects any duplicate IP or label in VTEPs. This runbook does not have any input argument.	UA
EdgeDpBfd	This runbook triages NSX Edge BFD issues. This runbook takes the source IP and destination IP of BFD session as input arguments. It also takes capture as an optional argument to ensure that only BFD packets specific to the session are filtered and captured. In addition to executing at runtime, this runbook is triggered when a BFD session goes down.	Edge
EdgeHealth	This runbook performs following functions: • Triages common edge down issues occurring during deployments and upgrades. • Suggests workarounds to get edge node back into healthy state and resume normal functioning. This runbook does not have any input argument.	Edge

Runbook	Description	Node
EdgeIDPS	This runbook checks and retrieves stats for IDS/IPS signature present in NSX Edge. This runbook takes signature ID as an input argument.	Edge
EdgePerformanceCollect	This runbook serves as an entrypoint runbook that orchestrates performance diagnostics, depending on whether the NSX Edge node is deployed as a virtual machine (VM) or bare metal. This runbook triggers the appropriate downstream runbooks: 1. `EdgePerformanceDiagnosis` (for VM and bare metal nodes) 2. `AdfCollect` (for VM form factor only) This runbook takes the following input arguments: • UUID of NSX Edge node (`edge_uuid`) -- required for VM and bare metal nodes	UA
EdgePerformanceDiagnosis	This runbook performs first-level analysis of performance impacts based on physical port drop counters and historical data. It covers common cases of performance impact and can perform analysis on CPU usage, packet drop patterns, and other performance metrics. It evaluates whether an issue is current or requires historical logs for diagnosis. This runbook does not have any input arguments. This runbook is triggered by the following events: • `EdgePerformanceCollect` orchestration runbook • RX buffer overflow	Edge
EdgeRouting	This runbook checks multiple aspects of a given logical router's health and provides a list of edge processes that are not in good health. The runbook provides troubleshooting directions for the common north/south routing issues. The runbook gathers various pieces of information and reports the status of the routing protocols, tunnels, and ports. It will do a health check for the routing stack and various daemons used by routing on the Edge. For the provided destination IP it will check the RIB, FIB, and ARP tables. It will also run a ping and traceroute using the source and destination IP. This runbook takes the following input arguments: • Destination/peer IP: This is a mandatory argument. • Source IP: This is a mandatory argument. The source IP is used for the ping and traceroute and must be an IP from the interface the destination IP is expected to be directed to. • Logical Router (LR) name: This is a mandatory argument. It is the logical router name as configured in the system.	Edge

Runbook	Description	Node
EsxDpConnIssuesDebug	This runbook debugs ESX datapath connectivity issues by identifying the area of packet loss and categorizing the issues as follows: • Underlay Network Drop • Source Host Drop • Destination Host Drop The runbook also performs the following checks: • Run basic configuration checks on the transport nodes involved. • Run port status checks on the VM VIFs involved. This runbook only supports debugging connectivity between two endpoints on the same overlay segment, VLAN segment, or VPC subnet, that is, the same Layer 2 broadcast domain. It does not debug routed (cross-subnet) connectivity issues. This runbook takes the following input arguments: • Source VIF ID • Source VIF IP (optional) • Source VIF MAC (optional) • Destination VIF ID • Destination VIF IP (optional) • Destination VIF MAC (optional) • ip_type_for_auto_discovery: IP family type (IPv4 or IPv6) to auto-discover if a user does not provide source and destination VIF ip addresses.	UA
IdpsDpStatus	This runbook checks the status of the IDPS/IDS module (in context engine) and IDPS engine. It also compares loaded profiles, rules, signatures, and captured events in both IDPS module and IDPS engine module. This runbook does not have any input argument.	ESX

Runbook	Description	Node
(In version 9.1.1 and later) IPSecVPNSession	This runbook checks the status of a VPN session and provides diagnostic guidance for session failures, connectivity issues, and packet drops. IPSecVPNSession can analyze both policy-based and route-based VPN sessions. This runbook provides context-aware diagnostics for the following failure conditions: • Peer Not Responding — Verifies the local endpoint IP in the namespace and datapath, checks routing reachability to the peer IP, and performs a local-to-peer ping test. • No Proposal Chosen (IKE) — Displays the configured IKE profile and provides Phase 1 proposal comparison guidance. • No Proposal Chosen (IPsec) — Displays the configured IPsec tunnel profile and provides Phase 2 proposal comparison guidance. • Authentication Failure — For PSK sessions, highlights possible PSK mismatch and checks syslogs for remote/local ID mismatches. For certificate-based sessions, searches syslogs for ID mismatches and recommends using the UI Download Config option. • TS Unacceptable — Displays configured local and remote traffic selectors with detailed mismatch troubleshooting steps. • Negotiation Not Started — Checks the IKE connection initiation mode (Respond Only or On Demand) and provides mode-specific recommendations. • Configuration Failed — Retrieves the session configuration failure reason, lists common causes (for example, exceeding 1024 tunnels, invalid configuration, resource exhaustion), and refers to the NSX Edge and IKED logs. IPSecVPNSession also provides appropriate guidance for generic conditions (for example, deactivated session/service, service router not active on current NSX Edge). This runbook takes the following input arguments. Required: • `local_ip` address of VPN local endpoint • `peer_ip` address of VPN peer endpoint Optional: • `local_nw_ip` address of local network for traffic analysis, taken from the session configuration • `remote_nw_ip` address of remote network for traffic analysis, taken from the session configuration • `pkt_capture` flag to capture packets at given interface • `intf_uuid` UUID of interface (for example, uplink or loopback)	Edge
LspStaleInfo	This runbook fetches the stale logical ports. This runbook does not have any input argument.	UA
MacAddressInfo	This runbook diagnoses issues related to a specific MAC address and checks and remediates performance issues on a host-switch. It performs the following functions: • For a given MAC address and host-switch, this runbook obtains MAC address details (such as MAC, VLAN, VNI, and portID) from the host-switch MAC address table and provides a diagnosis in case a MAC is not present. • For a given static MAC address used by a VNIC, it also retrieves the VLAN and VNI information of the VNIC port. This runbook takes the host-switch name and mac-address as input arguments.	ESX

Runbook	Description	Node
NCPPendingPod	This runbook debugs a pod stuck in a pending state. This runbook takes the namespace and name of the pending pod as input arguments.	ESX
NxgiPlatform	This runbook diagnoses issues in NXGI platform datapath that powers various features, such as Distributed MPS, Endpoint Protection, IDFW, IDS, and Intelligence. This runbook takes functionality as an input argument. This is the specific NXGI dependent functionality (one from MPS, IDFW, IDS, EPP, and Intelligence with EPP being the default) that needs to be checked or debugged.	ESX
NxgiPlatformUA	This runbook diagnoses issues in management plane for NXGI platform, which powers various features like Distributed MPS, Endpoint Protection, IDFW, IDS, Intelligence. This runbook can be used in conjunction with the NxgiPlatform runbook which runs on transport nodes. This runbook takes transport node ID as an optional input argument.	UA
OspfNbrState	This runbook triages ospf neighbor state issues by diagnosing various logical flows. This runbook takes the neighbor IP address as an input argument.	Edge
OverlayTunnel	This runbook can identify overlay tunnel failures, such as gateway configuration error tunnel missing, tunnel down. This runbook takes IP address of source VTEP and IP address of destination VTEP as input arguments.	ESX
PimConfigCheck	This runbook provides more details about what exactly failed when edge throws routing_config_error. It also helps in triaging the issue. This runbook does not have any input argument.	Edge
PimMrouteState	This runbook triages multicast traffic loss which can happen due to various reasons. Given a source IP of the multicast traffic sender and group IP of multicast traffic, this runbook will help to identify the root cause of the problem and collect supporting artifacts for offline debugging. This runbook takes source IP, group IP, and traffic direction as input arguments.	Edge
PnicPerf	This runbook can identify pNIC TX and RX performance issues. This runbook takes the ID of the logical port to be diagnosed as an input argument.	ESX
PortBlock	This runbook can identify causes for a port's blockage, such as a DVPort could be blocked because of incorrect LSP/LS configuration.	ESX
PortStatusInfo	This runbook identifies issues on a port that is connected to vSwitch. This runbook takes the VIF ID as an input argument to retrieve port details.	ESX
VifInfo	This runbook gets a detailed information about a virtual interface, which can be used as input to other runbooks. This runbook takes VIF as an input argument.	ESX

Steps to debug at runtime

To debug at runtime, perform the following steps:

Step 1: Fetch a list of predefined runbook.

Run the following API to fetch a list of predefined runbooks.

```
GET https://<nsx-mgr>/policy/api/v1/infra/sha/pre-defined-runbooks
```

This API returns a list of predefined runbooks along with the following information:

- Configuration details.
- Node type on which the runbook is supported.
- General details of the runbook such as id, name, and path.
- Parameter details if any required at the time of invoking the runbook.

Example Response:


```
{
 "results": [
     {
         "version": {
             "major": 1,
             "minor": 0
         },
         "default_config": {
             "enabled": true,
             "timeout": 300,
             "threshold_number": 5,
             "throttle_cycle": 10
         },
         "supported_node_types": [
             "nsx-esx"
         ],
         "parameters": [
             {
                 "name": "advanced",
                 "optional": true,
                 "parameter_type": "BOOLEAN",
                 "default_value": "False"
             },
             {
                 "name": "cycle",
                 "optional": false,
                 "parameter_type": "INTEGER",
                 "max": "20",
                 "min": "1"
             },
             {
                 "name": "interval",
                 "optional": true,
                 "parameter_type": "INTEGER",
                 "max": "300",
                 "min": "1"
             }
         ],
         "resource_type": "OdsPredefinedRunbook",
         "id": "00000000-0000-4164-6643-6f6c6c656374",
         "display_name": "AdfCollect",
         "path": "/infra/sha/pre-defined-
runbooks/00000000-0000-4164-6643-6f6c6c656374",
         "relative_path": "00000000-0000-4164-6643-6f6c6c656374",
         "parent_path": "/infra",
         "remote_path": "",
         "unique_id": "069a4c7b-532a-4926-a402-6a7986a306b2",
         "realization_id": "069a4c7b-532a-4926-a402-6a7986a306b2",
         "owner_id": "f2a2a9e1-2578-435d-877d-47e01eb04954",
         "origin_site_id": "f2a2a9e1-2578-435d-877d-47e01eb04954",
         "marked_for_delete": false,
```

```
            "overridden": false,
            "_create_time": 1669600880454,
            "_create_user": "system",
            "_last_modified_time": 1669600880454,
            "_last_modified_user": "system",
            "_system_owned": false,
            "_revision": 0
        },
        {
            "version": {
                "major": 1,
                "minor": 0
            },
            "default_config": {
                "enabled": true,
                "timeout": 60,
                "threshold_number": 5,
                "throttle_cycle": 3
            },
            "supported_node_types": [
                "nsx-esx"
            ],
            "resource_type": "OdsPredefinedRunbook",
            "id": "0000436f-6e74-726f-6c6c-6572436f6e6e",
            "display_name": "ControllerConn",
            "path": "/infra/sha/pre-defined-
runbooks/0000436f-6e74-726f-6c6c-6572436f6e6e",
            "relative_path": "0000436f-6e74-726f-6c6c-6572436f6e6e",
            "parent_path": "/infra",
            "remote_path": "",
            "unique_id": "3914cbe4-41b4-45f1-9dad-2bd96a2de0d8",
            "realization_id": "3914cbe4-41b4-45f1-9dad-2bd96a2de0d8",
            "owner_id": "f2a2a9e1-2578-435d-877d-47e01eb04954",
            "origin_site_id": "f2a2a9e1-2578-435d-877d-47e01eb04954",
            "marked_for_delete": false,
            "overridden": false,
            "_create_time": 1669600880493,
            "_create_user": "system",
            "_last_modified_time": 1669600880493,
            "_last_modified_user": "system",
            "_system_owned": false,
            "_revision": 0
        },
        {
            "version": {
                "major": 1,
                "minor": 0
            },
            "default_config": {
                "enabled": true,
                "timeout": 120,
                "threshold_number": 5,
                "throttle_cycle": 5
            },
            "supported_node_types": [
                "nsx-esx"
            ],
```

```
            "parameters": [
                {
                    "name": "src",
                    "optional": false,
                    "parameter_type": "COMPOUND"
                },
                {
                    "name": "dst",
                    "optional": false,
                    "parameter_type": "COMPOUND"
                }
            ],
            "resource_type": "OdsPredefinedRunbook",
            "id": "0000004f-7665-726c-6179-54756e6e656c",
            "display_name": "OverlayTunnel",
            "path": "/infra/sha/pre-defined-
runbooks/0000004f-7665-726c-6179-54756e6e656c",
            "relative_path": "0000004f-7665-726c-6179-54756e6e656c",
            "parent_path": "/infra",
            "remote_path": "",
            "unique_id": "3597af28-e670-456e-8347-4d1a53a5cb90",
            "realization_id": "3597af28-e670-456e-8347-4d1a53a5cb90",
            "owner_id": "f2a2a9e1-2578-435d-877d-47e01eb04954",
            "origin_site_id": "f2a2a9e1-2578-435d-877d-47e01eb04954",
            "marked_for_delete": false,
            "overridden": false,
            "_create_time": 1669600880518,
            "_create_user": "system",
            "_last_modified_time": 1669600880518,
            "_last_modified_user": "system",
            "_system_owned": false,
            "_revision": 0
        },
        {
            "version": {
                "major": 1,
                "minor": 0
            },
            "default_config": {
                "enabled": true,
                "timeout": 120,
                "threshold_number": 5,
                "throttle_cycle": 5
            },
            "supported_node_types": [
                "nsx-esx"
            ],
            "parameters": [
                {
                    "name": "lsp",
                    "optional": false,
                    "parameter_type": "STRING"
                }
            ],
            "resource_type": "OdsPredefinedRunbook",
            "id": "00000000-0000-0000-506e-696350657266",
            "display_name": "PnicPerf",
```

```
            "path": "/infra/sha/pre-defined-
runbooks/00000000-0000-0000-506e-696350657266",
            "relative_path": "00000000-0000-0000-506e-696350657266",
            "parent_path": "/infra",
            "remote_path": "",
            "unique_id": "53f29b77-dcf5-4561-85ec-8f35280e3f3a",
            "realization_id": "53f29b77-dcf5-4561-85ec-8f35280e3f3a",
            "owner_id": "f2a2a9e1-2578-435d-877d-47e01eb04954",
            "origin_site_id": "f2a2a9e1-2578-435d-877d-47e01eb04954",
            "marked_for_delete": false,
            "overridden": false,
            "_create_time": 1669600880553,
            "_create_user": "system",
            "_last_modified_time": 1669600880553,
            "_last_modified_user": "system",
            "_system_owned": false,
            "_revision": 0
        },
        {
            "version": {
                "major": 1,
                "minor": 0
            },
            "default_config": {
                "enabled": true,
                "timeout": 60,
                "threshold_number": 5,
                "throttle_cycle": 3
            },
            "supported_node_types": [
                "nsx-esx"
            ],
            "parameters": [
                {
                    "name": "vif",
                    "optional": false,
                    "parameter_type": "STRING"
                }
            ],
            "resource_type": "OdsPredefinedRunbook",
            "id": "00000000-0000-0050-6f72-74426c6f636b",
            "display_name": "PortBlock",
            "path": "/infra/sha/pre-defined-
runbooks/00000000-0000-0050-6f72-74426c6f636b",
            "relative_path": "00000000-0000-0050-6f72-74426c6f636b",
            "parent_path": "/infra",
            "remote_path": "",
            "unique_id": "6f411a92-30c0-4838-9758-c00220cb5fab",
            "realization_id": "6f411a92-30c0-4838-9758-c00220cb5fab",
            "owner_id": "f2a2a9e1-2578-435d-877d-47e01eb04954",
            "origin_site_id": "f2a2a9e1-2578-435d-877d-47e01eb04954",
            "marked_for_delete": false,
            "overridden": false,
            "_create_time": 1669600880572,
            "_create_user": "system",
            "_last_modified_time": 1669600880572,
            "_last_modified_user": "system",
```

```
            "_system_owned": false,
            "_revision": 0
        }
    ],
    "result_count": 5,
    "sort_by": "display_name",
    "sort_ascending": true
}
```

In this example response, the following predefined runbooks are returned.

- AdfCollect
- ControllerConn
- OverlayTunnel
- PnicPerf
- PortBlock

  If a runbook requires a parameter at the time of invocation, the parameter details are specified in the *parameters* key. For example, the Overlay Tunnel runbook requires two parameters, `source` and `destination` that are local and remote VTEP IPs of the tunnel to be diagnosed.

Step 2: Get parameter details of the runbook.

Run the following API to fetch the parameter details of a runbook that you want to invoke.

`https://{{MANAGER_IP}}/policy/api/v1/infra/sha/pre-defined-runbooks/0000004f-7665-726c-6179-54756e6e656c/help`

```
{
 "summary": "Runbook to diagnose overlay tunnel issues.",
 "parameter_info": [
     {
         "summary": "IP address of source VTEP",
         "parameter": {
             "name": "src",
             "optional": false,
             "parameter_type": "COMPOUND"
         }
     },
     {
         "summary": "IP address of destination VTEP",
         "parameter": {
             "name": "dst",
             "optional": false,
             "parameter_type": "COMPOUND"
         }
     }
 ]
}
```

Note that the `parameter name` is mapped to `key` and `parameter type` is mapped to `value` in a runbook's invocation API.

Step 3: Invoke the runbook, with option to collect artifacts.

- Run the following API to invoke a runbook.

`POST https://<nsx-mgr>/policy/api/v1/infra/sha/runbook-invocations/<invoke-name>`

Example Request: Invoking the Overlay Tunnel runbook without artifacts.

```
POST https://<nsx-mgr>/policy/api/v1/infra/sha/runbook-invocations/OverlayTunnel
{
 "runbook_path": "/infra/sha/pre-defined-
runbooks/0000004f-7665-726c-6179-54756e6e656c",
 "target_node": "6c7a9374-459d-46b2-9ea6-c63b37c7cc38",
 "arguments": [
     {
```

```
            "key": "src",
            "value": "192.168.0.11"
        },
        {
            "key": "dst",
            "value": "192.168.0.10"
        }
    ]
}
```

The *target_node* can be a host node, an edge node, or a Unified Appliance node.

The ID of a host or an edge can be obtained through the following API.

`https://{{MANAGER_IP}}/api/v1/transport-nodes`

The appliance ID list can be obtained from cluster API. The valid ID is the external ID of appliance with manager role.

`https://{{MANAGER_IP}}/api/v1/cluster/nodes`

- To have the runbook gather artifacts during execution, include the `advanced` parameter as an argument in the runbook invocation.
  Example Request: Invoking the ADF Collector runbook with artifacts

```
POST https://{{mgr-ip}}/policy/api/v1/infra/sha/runbook-invocations/AdfCollect
{
"runbook_path": "/infra/sha/pre-defined-
runbooks/00000000-0000-4164-6643-6f6c6c656374",
"target_node": "{{target-node}}",
"arguments": [
    {
        "key": "advanced",
        "value": "{{advanced-mode}}"
    },
    {
        "key": "cycle",
        "value": "{{cycle-count}}"
    },
    {
        "key": "interval",
        "value": "{{interval-in-sec}}"
    }
]
}
```

  If the `advanced` parameter is set to false at the time of invocation, the runbook collects topology info, net-stats, NSX DP stats and uplink artifacts. If the `advanced` parameter is set to true, the runbook also runs advanced performance tools such as vmkstats (available only on the physical machine).
- Run the following API to download the artifacts.
  `GET https://<nsx-mgr>/policy/api/v1/infra/sha/runbook-invocations/<invoke-name>/artifact`

If the runbook generated the artifacts, the API returns a bundled file, else it returns an error message. Save the binary response to a tar.gz file. This file contains a runbook invocation report (in JSON) as well as a tar file for the performed ADF collection.

Step 4: Check the report of the invoked runbook.

Run the following API to get the report of the invoked runbook.

`GET https://<nsx-mgr>/policy/api/v1/infra/sha/runbook-invocations/<invoke-name>/report`

Example Request:

`GET https://<nsx-mgr>/policy/api/v1/infra/sha/runbook-invocations/OverlayTunnel/report`

Example Response:

```
{
 "invocation_id": "70527fed-1e5e-4fed-a880-28cde04a66b1",
 "target_node": "6c7a9374-459d-46b2-9ea6-c63b37c7cc38",
 "timestamp": 1662469099,
 "sys_info": {
     "host_name": "sc2-10-185-4-158.eng.vmware.com",
     "os_name": "VMkernel",
     "os_version": "7.0.3"
 },
 "result_message": "Tunnel 192.168.0.11 -> 192.168.0.10 is in up state",
 "recommendation_message": "No changes required as the tunnel is in up state.",
 "step_details": [
     {
         "step_id": 1,
         "action_summary": "Check the status of tunnel for the given source/
destination VTEPs.",
         "action_result": "Tunnel 192.168.0.11 -> 192.168.0.10 is in state up"
     }
 ],
 "status": {
     "request_status": "SUCCESS",
     "operation_state": "FINISHED"
 }
}
```

The response returns some metadata information, such as timestamp and system details like host name and operating system. The report also returns the result of debugging, remediation suggestion if any along with steps executed for debugging with action summary and action result of each step. If debugging is interrupted for any reason, the `operation_state` field will hold a value to define the reason for the interruption. In case the runbook invocation does not succeed, the report provides error details and does not show debugging related fields.

Changing a runbook configuration

A transport node (TN) group is first created and then a runbook profile is bound to it. If you change the configuration of a runbook, it is changed at all nodes to which the runbook profile is bound. You can configure whether a runbook is enabled or not, the debugging timeout, and frequency at which the runbook can be invoked using the throttle cycle mechanism. The throttle cycle mechanism specifies the number of times a runbook can be executed within a specified time.

Note that a runbook can have only one profile, but a node might have multiple runbook profiles based on the TN groups to which it belongs. In this case, the profile with the highest priority is applied on the node.

To change a runbook configuration, perform the following steps:

Step 1: Create a TN group of ESX.

Run the following API to create a TN group.

`PATCH https://<nsx-mgr>/policy/api/v1/infra/domains/default/groups/<group-name>`

Example Request:

```
PATCH https://<nsx-mgr>/policy/api/v1/infra/domains/default/groups/<group-name>
{
 "expression": [
     {
         "paths": [
             "/infra/sites/default/enforcement-points/default/host-transport-nodes/
TN1"
         ],
         "resource_type": "PathExpression"
```

```
        }
    ],
    "extended_expression": [],
    "reference": false,
    "group_type": [],
    "resource_type": "Group"
}
```

Step 2: Change the configuration of the runbook profile and bind the TN group with the profile.

Run the following API to change the configuration,

`PATCH https://<nsx-mgr>/policy/api/v1/infra/sha/runbook-profiles/<profile-name>`

Example Request:

```
PATCH https://<nsx-mgr>/policy/api/v1/infra/sha/runbook-profiles/<profile-name>
{
 "runbook_path": "/infra/sha/pre-defined-
runbooks/0000004f-7665-726c-6179-54756e6e656c",
 "applied_to_group_path": "/infra/domains/default/groups/tngroup2",
 "config": {
     "enabled": true,
     "timeout": 120,
     "threshold_number": 2,
     "throttle_cycle": 6
 }
}
```

In this example, the `throttle_cycle` is 6 minutes and the t`hreshold_number` is 2, which means that within 6 minutes, the runbook can be invoked no more than two times.

For complete information about ODS APIs, see *Security Intelligence & NSX Application Platform API Guide*.

Install Dynamic Runbooks

In addition to predefined runbooks, the Online Diagnostic System (ODS) feature also supports dynamic runbooks to debug NSX at runtime.

Download the dynamic package from the VMware site https://support.broadcom.com/group/ecx/downloads.

Dynamic runbooks are not dependent on NSX release and you can install them anytime. Dynamic runbooks address the following issues:

- As development and maintenance of predefined runbooks follow the NSX release cycle, it prevents debugging of issues emerging at live sites in between release cycles. With dynamic runbooks, any debugging requirements can be implemented in between release cycles as dynamic runbooks can be developed and installed anytime.
- If you upgrade NSX partially, there could be an incompatibility between a runbook and the API that calls it. For example, you upgrade the transport node, without upgrading the Unified Appliance (UA), and the upgrade installs a new version of runbook on the transport node. In this case, UA might not be able to invoke the newer version of the runbook with the older API as the API might be outdated.

Only the VMware team can create dynamic runbooks and you can download them for installing them on your system. Note that runbooks do not support data backup and restore.

To install the dynamic runbooks, you need to define a group called the dynamic runbook instance. The dynamic runbook instance defines all the installation node groups where runbooks are to be installed. If a new node is added to a group, the runbook will be automatically installed on the new node.

With NSX 4.2, the following runbook is available for download. For more information about the runbook, download the package from the following VMware site https://support.broadcom.com/group/ecx/downloads.

Table 1082:

Runbook	Description
Hyperbus runbook	The dynamic hyperbus runbook can check for the following hyperbus issues and provide remediation suggestion: • Hyperbus connection • Configurations of hyperbus connection • vmknic • Status of hyperbus connection This runbook takes vif-id as the input argument.

To install dynamic runbooks, perform the following steps:

1. Obtain the path of the transport nodes to be included in the installation node group.

   ```
   GET https://10.180.85.179/policy/api/v1/infra/sites/default/enforcement-points/default/host-transport-nodes
   ```

2. Create the installation node group. The group can be a policy group, transport node group, or UA node group.

   ```
   PATCH https://<nsx-mgr>/policy/api/v1/infra/domains/default/groups/<group-name>
   ```

   ```
   {
       "expression":
      [
           {
               "paths": [
                   "/infra/sites/default/enforcement-points/default/host-transport-nodes/TN1"
               ],
               "resource_type": "PathExpression"
           }
       ],
       "extended_expression": [],
       "reference": false,
       "group_type": [],
       "resource_type": "Group"
   }
   ```

3. Create a dynamic runbook instance that includes all the installation node groups.

   ```
   POST https://{{MANAGER_IP}}/policy/api/v1/infra/sha/dynamic-runbook-instances/<dynamic-instanace-id>
   ```

   ```
   {
       "applied_to_group_paths": [
           "/infra/domains/default/groups/TNGroup1"  [Policy Group with node path]
       ],
       "applied_to_nodes": [
           "1e6314a2-a268-4a3a-bcae-4f23b2536ea8"  [The node id can be a host/edge/UA node id]
       ],
       "applied_to_all_appliances": true   [All UA cluster]
   }
   ```

4. Upload the runbook package to the dynamic runbook instance.

   ```
   POST https://{{MANAGER_IP}}/policy/api/v1/infra/sha/dynamic-runbook-instances/<dynamic-instanace-id>/file
   ```

   The dynamic runbook data is applied to the target nodes specified this group.

5. Query the installation state.

```
GET https://10.180.85.179/policy/api/v1/infra/sha/dynamic-runbook-instances/<dynamic-instanace-id>/state
```

```
{
    "runbook_name": "Example",
    "management_state": "INSTALL_FINISHED",
    "install_state": [
        {
            "node_id": "084583fa-3ae7-4d0c-98f1-6fc4cb5044b1",
            "node_name": "TN301",
            "version": {
                "major": 1,
                "minor": 0
            },
            "status": "INSTALL_FINISHED"
        }
}
```

6. Check the details of the installed runbooks by using the following API.

```
GET https://{{MANAGER_IP}}/policy/api/v1/infra/sha/dynamic-runbooks
```

7. Invoke the runbook by using the steps mentioned in the topic Diagnose NSX Using ODS Runbooks.

To extend a dynamic runbook instance to additional nodes, run the following API.

```
PATCH https://10.180.85.179/policy/api/v1/infra/sha/dynamic-runbook-instances/<dynamic-instanace-id>
```

```
{
"applied_to_group_paths":["/infra/domains/default/groups/<tn-group-name>"],
"applied_to_nodes":["<target-edge-id>","<target-ua-id>"]
}
```

To uninstall a runbook, run the following API.

```
DELETE https://10.180.85.179/policy/api/v1/infra/sha/dynamic-runbook-instances/<dynamic-instanace-id>/file
```

To uninstall a dynamic runbook instance, run the following API .

```
DELETE https://10.180.85.179/policy/api/v1/infra/sha/dynamic-runbook-instances/<dynamic-instanace-id>
```

Testing End-To-End Workflow Tracing

Network workflow tracing provides access to internal NSX workflow. Tracing helps users to analyze the logs, monitor whether the NSX Policy Manager, CCP, and host daemons are running properly, and optimize the network traffic where needed.

Verify that your NSX host clusters are connected to the ESX host.

The workflow tracing includes management and configuration APIs that collect execution time, API input or output parameters, and judgment conditions. Workflow tracing at a granular level is available in applications such as NAT and Edge. The API enables tracing, which can be stored locally or forwarded to a third-party distributed tracing like Jaeger.

ESX host VM shares network information with Jaeger when an NSX segment is created. The tracing dashboard contains the following information:

- Trace the internal traffic within the segment.
- Identify any specific operations that could affect system performance.
- Troubleshoot and analyze reported errors.

Select a destination for tracing output and add network tracing settings to the application or machine configuration file to enable network tracing. When network tracing is enabled, you can capture calls from NSX Policy Manager, CCP, and host daemons. Use end-to-end workflow tracing to connect and disconnect to VIF and install or uninstall the transport node.

1. In the vSphere Client UI, navigate to the host clusters.
2. Select the VM in the datacenter.
3. Right-click and select **Edit Settings** from the drop-down menu.
4. For Network adapter 2, select the **Connected** option.

   The VM is connected to the NSX segment.
5. Open your tracing tool and view the end-to-end trace workflow.

   The trace workflow starts from the configuration agent to NSX NestDb, ending at CCP.

Log Messages and Error Codes

NSX components write to log files in the directory `/var/log`. On NSX appliances, NSX syslog messages conform with RFC 5424. On ESX hosts, syslog messages conform with RFC 3164.

Viewing Logs

On NSX appliances syslog messages are in `/var/log/syslog`.

On NSX appliances, you can run the following NSX CLI command to view the logs:

```
get log-file <auth.log | controller | controller-error | http.log | kern.log | manager.log | node-mgmt.log |
  syslog> [follow]
```

The log files are:

Name	Description
auth.log	Authorization log
controller	Controller log
controller-error	Controller error log
http.log	HTTP service log
kern.log	Kernel log
manager.log	Manager service log
node-mgmt.log	Node management log
nsx-audit-write.log	NSX audit write log
nsx-audit.log	NSX audit log
syslog	System log

NSX also sends audit (audispd) logs directly to the remote syslog server. These messages are logged in `/var/log/audit/audit.log` but not in the local syslog file. Audit logs record security-related events in the operating system.

On hypervisors, you can use Linux commands such as `tac`, `tail`, `grep`, and `more` to view the logs.

Each syslog message has the component (`comp`) and sub-component (`subcomp`) information to help identify the source of the message.

NSX produces logs with facility `local6`, which has a numerical value of 22.

The audit log is part of syslog. An audit log message can be identified by the string `audit="true"` in the `structured-data` field. You can configure an external log server to receive log messages. You can also access audit logs using the API `/api/v1/administration/audit-logs`. The file `nsx-audit.log` contains syslog messages with `audit="true"` in the `structured-data` field. The file `nsx-audit-write.log` contains syslog messages with both `audit="true"` and `update="true"` in the `structured-data` field.

Each syslog and audit log message contains a timestamp generated by an NTP server, if configured, or by the system clock. An example of an audit log message:

```
<182>1 2020-05-05T00:29:02.900Z nsx-manager1 NSX 147324 - [nsx@6876 audit="true" comp="nsx-manager"
 level="INFO" reqId="fe75651d-c3e7-4680-8753-9ae9d92d7f0c" subcomp="policy" username="admin"]
 UserName="admin", ModuleName="AAA", Operation="GetCurrentUserInfo", Operation status="success"
```

An example of a syslog message:

```
2023-01-25T21:58:42.173Z w4-mydevice-220.eng.vmware.com NSX 1463241016 - [nsx@6876 comp="nsx-controller"
 level="INFO" subcomp="corfu-cluster"] Received maintenance mode status: {da3a0242-bf6d-ee98-224c-
b9799cb5e29f=MAINTENANCE_MODE_OFF}
```

An API call can come from NSX Manager, a policy API client, or an NSX node. All API calls are subject to authentication and authorization, and will generate audit logs. This logging is enabled by default and cannot be disabled. An audit log that is associated with an API call has the following information:

- An entity ID parameter `entId` to identify the object of the API.
- A request ID parameter `req-id` to identify a specific API call.
- An external request ID parameter `ereqId` if the API call contains the header `X-NSX-EREQID:<string>`.
- An external user parameter `euser` if the API call contains the header `X-NSX-EUSER:<string>`.

An audit log message from an API call will have the following additional fields. Note that node API (NAPI) and CLI audit logs will not have these fields.

- An `update` flag that shows whether the API operation is a read (GET) or write (PUT/POST/DELETE/...) operation.
- An `operation name` field that shows the name of the API operation.
- An `operation status` field that shows whether the API operation succeeded or failed.
- A `new value` field that shows all parameter values of the API request.

NSX does not have the concept of a privileged mode. API calls from all sources and users are audited.

An example of login and logout syslog messages showing a successful login, a failed login, and logins from 2 different devices (note the different IP addresses):

```
2020-07-07T16:33:20.339Z svc.nsxmanager NSX 1513 SYSTEM [nsx@6876 audit="true" comp="nsx-manager" level="INFO"
 subcomp="http"] UserName="admin@10.166.61.56", ModuleName="ACCESS_CONTROL", Operation="LOGIN", Operation
 status="success"

2020-07-07T16:33:58.779Z svc.nsxmanager NSX 1513 SYSTEM [nsx@6876 audit="true" comp="nsx-manager" level="INFO"
 subcomp="http"] UserName="admin", ModuleName="ACCESS_CONTROL", Operation="LOGOUT", Operation status="success"

2020-07-07T16:50:21.301Z svc.nsxmanager NSX 1513 SYSTEM [nsx@6876 audit="true" comp="nsx-manager" level="INFO"
 subcomp="http"] UserName="admin@10.166.61.80", ModuleName="ACCESS_CONTROL", Operation="LOGIN", Operation
 status="success"

2020-07-07T16:43:20.339Z svc.nsxmanager NSX 1513 SYSTEM [nsx@6876 audit="true" comp="nsx-manager" level="INFO"
 subcomp="http"] UserName="admin@10.166.61.56", ModuleName="ACCESS_CONTROL", Operation="LOGIN", Operation
 status="failure"
```

An example of a syslog message of a policy API call:

```
<182>1 2020-07-06T18:09:14.210Z svc.nsxmanager NSX 2326 FABRIC [nsx@6876 audit="true"
 comp="nsx-manager" entId="68d5a9d0-4691-4c9c-94ed-64fd1c96150f" level="INFO" reqId="4c2335aa-
c973-4f74-983f-331a4f7041ca" subcomp="manager" update="true" username="admin"] UserName="admin",
 ModuleName="TransportZone", Operation="CreateTransportZone", Operation status="success", New
 value=[{"transport_type":"OVERLAY","host_switch_name":"nsxvswitch","host_switch_mode":"STANDARD","nested_nsx":false,"i
transportzone-1307","_protection":"UNKNOWN"}]
```

An example of syslog messages of CLI access:

```
2020-07-07T16:36:41.783Z svc.nsxmanager NSX 21018 - [nsx@6876 comp="nsx-manager" subcomp="cli"
 username="admin" level="INFO"] NSX CLI started (Manager, Policy, Controller) for user: admin
2020-07-07T16:36:53.469Z svc.nsxmanager NSX 21018 - [nsx@6876 comp="nsx-manager" subcomp="cli"
 username="admin" level="INFO"] NSX CLI stopped for user: admin
```

An example of a syslog message when a user runs a CLI command (in this example, `set user admin password-expiration 100`):

```
<182>1 2020-07-22T20:51:49.017Z manager2 NSX 1864 - [nsx@6876 comp="nsx-manager" subcomp="cli"
 username="admin" level="INFO" audit="true"] CMD: set user admin password-expiration 100 (duration: 2.185s),
 Operation status: CMD_EXECUTED
```

An example of a syslog message of an NAPI call:

```
<182>1 2020-07-21T21:01:38.803Z manager2 NSX 4690 - [nsx@6876 comp="nsx-manager" subcomp="node-mgmt"
 username="admin" level="INFO" audit="true"] admin 'GET /api/v1/node/services/syslog/exporters' 200 731 ""
 "PostmanRuntime/7.26.1" 0.004588
```

An example of a syslog message of a CLI command:

```
<182>1 2020-07-21T20:54:40.018Z manager2 NSX 16915 - [nsx@6876 comp="nsx-manager" subcomp="cli"
 username="admin" level="INFO" audit="true"] CMD: set logging-server 1.1.1.1 proto udp level info (duration:
 4.356s), Operation status: CMD_EXECUTED
```

RFC 5424 and RFC 3164 define the following severity levels:

Severity Level	Description
0	Emergency: system is unusable
1	Alert: action must be taken immediately
2	Critical: critical conditions
3	Error: error conditions
4	Warning: warning conditions
5	Notice: normal but significant condition
6	Informational: informational messages
7	Debug: debug-level messages

All logs with a severity of emergency, alert, critical, or error contain a unique error code in the structured data portion of the log message. The error code consists of a string and a decimal number. The string represents a specific module.

Failure to Access a Log File or Remote Log Server

If NSX fails to access or write messages to a log file, an alarm will be generated. The possible errors are:

- A local log file is missing.
- A local log file's permission or ownership setting prevents NSX from writing to the file.
- NSX is unable to send log messages to a third-party remote log server. Note that an alarm will not be raised if NSX fails to send logs to the VCF Operations for logs agent.

The alarm can be resolved through the alarm framework.

Log Message Formats

For more information about RFC 5424, see https://tools.ietf.org/html/rfc5424. For more information about RFC 3164, see https://tools.ietf.org/html/rfc3164.

RFC 5424 defines the following format for log messages:

```
<facility * 8 + severity> version UTC-TZ hostname APP-NAME procid MSGID [structured-data] msg
```

A sample log message:

```
<187>1 2016-03-15T22:53:00.114Z nsx-manager NSX - SYSTEM [nsx@6876 comp="nsx-manager" errorCode="MP4039"
 subcomp="manager"] Connection verification failed for broker '10.160.108.196'. Marking broker unhealthy.
```

Error Codes

For a list of error codes, see NSX Error Codes.

Configure Remote Logging

You can configure NSX appliances, NSX Edges, and hypervisors to send log messages to a remote log server.

- Familiarize yourself with the CLI command `set logging-server`. For more information, see the *NSX Command-Line Interface Reference*.
- If you specify the secure protocol TLS or LI-TLS, the server and client certificates must be stored in `/image/vmware/nsx/file-store` on each NSX appliance. Note that certificates in the file store are needed only if the exporter is configured using NSX CLI. If you use the API, then there is no need to use the file store. Once you complete the syslog exporter configuration, you must delete all certificates and keys from this location to avoid potential security vulnerabilities. Note that a TLS logging server does not support the replacement of certificates. When a TLS logging server certificate is about to expire or has expired, you must perform the following steps:
  – Delete the existing TLS logging server by using an API or CLI.
  – Put the new certificate under `/image/vmware/nsx/file-store` and create a new TLS logging server with new certificate again.

  To configure a secure connection to a log server, verify that the server is configured with a CA-signed certificates. For example, if you have a VCF Operations for logs server `vrli.prome.local` as the log server, you can run the following command from a client to see the certificate chain on the server:

  ```
  echo -n | openssl s_client -connect vrli.prome.local:443  | sed -ne '/^Certificate chain/,/^---/p'
  ```

  For example:

  ```
  root@caserver:~# echo -n | openssl s_client -connect vrli.prome.local:443  | sed -ne '/^Certificate
   chain/,/^---/p'
  depth=2 C = US, L = California, O = GS, CN = Orange Root Certification Authority
  verify error:num=19:self signed certificate in certificate chain
  Certificate chain
  0 s:/C=US/ST=California/L=HTG/O=GSS/CN=vrli.prome.local
    i:/C=US/L=California/O=GS/CN=Green Intermediate Certification Authority
  1 s:/C=US/L=California/O=GS/CN=Green Intermediate Certification Authority
    i:/C=US/L=California/O=GS/CN=Orange Root Certification Authority
  2 s:/C=US/L=California/O=GS/CN=Orange Root Certification Authority
    i:/C=US/L=California/O=GS/CN=Orange Root Certification Authority
  ---
  DONE
  ```

Remote logging is supported on NSX Manager, NSX Edge, and hypervisors. You must configure remote logging on each node individually.

If host clusters are activated for Distributed Malware Prevention for vDefend service, you can log in to the Malware Prevention for vDefend service virtual machine (SVM) on each host and configure remote logging on the SVM. This functionality is supported starting in NSX 4.1.2. For more information, see *Configure Remote Logging on an NSX Malware Prevention Service Virtual Machine*.

For the protocol parameter, the options are UDP, TCP, LI, and the secure protocols TLS and LI-TLS. Note that LI and LI-TLS are VCF Operations for logs protocols. An VCF Operations for logs log server supports all the protocols. The protocols LI and LI-TLS can only be used if the log server is VCF Operations for logs. The benefit of using LI or LI-TLS is that they optimize network usage. If the log server is VCF Operations for logs, using LI or LI-TLS is recommended. If LI cannot be used, TCP has the advantage of being more reliable, whereas UDP has the advantage of requiring less system and network overhead.

1. To configure remote logging on an NSX appliance or an NSX Edge:
   a) Run the following command to configure a log server and the types of messages to send to the log server. Multiple facilities or message IDs can be specified as a comma delimited list, without spaces.

   ```
   set logging-server <hostname-or-ip-address[:port]> proto <proto> level <level> [facility <facility>]
    [messageid <messageid>] [serverca <filename>] [clientca <filename>] [certificate <filename>] [key
    <filename>] [structured-data <structured-data>]
   ```

   You can run the command multiple times to add multiple configurations. For example:

   ```
   set logging-server 192.168.110.60 proto udp level info facility local6 messageid SYSTEM,FABRIC
   set logging-server 192.168.110.60 proto udp level info facility auth,user
   ```

To forward multi-tenancy logs to the remote logging server, specify the project and the VPC in the `structured-data` parameter. For example,

```
set logging-server <server-ip>  proto udp level err structured-data proj=<project-shortId>
set logging-server <server -ip  proto udp level err structured-data vpc=<vpc-shortId>
```

To forward only audit logs to the remote server, specify `audit="true"` in the `structured-data` parameter. For example:

```
set logging-server <server-ip> proto udp level info structured-data audit="true"
```

All NSX logs use facility `local6`. You should use the `messageid` and `structured-data` filters only when the facility filter is not set or when `local6` is included in the specified facilities.

b) Configure secure remote logging:

- To configure secure remote logging using the protocol LI-TLS, specify the parameter `proto li-tls`. For example:

```
set logging-server vrli.prome.local proto li-tls level info messageid
 SWITCHING,ROUTING,FABRIC,SYSTEM,POLICY,HEALTHCHECK,SHA,MONITORING serverca intermed-ca-full-
chain.crt
```

If the configuration is successful, you will get a prompt without any text. To see the content of the server certificate chain (intermediate followed by root), log in as **root** and run the following command:

```
keytool -printcert -file /image/vmware/nsx/file-store/intermed-ca-full-chain.crt
```

For example,

```
root@nsx1:~# keytool -printcert -file /image/vmware/nsx/file-store/intermed-ca-full-chain.crt
Certificate[1]:
Owner: CN=Green Intermediate Certification Authority, O=GS, L=California, C=US
Issuer: CN=Orange Root Certification Authority, O=GS, L=California, C=US
Serial number: 3e726e7fbb3b0a7a6b4edd767f867fd2
Valid from: Sun Mar 15 00:00:00 UTC 2020 until: Mon Mar 17 00:00:00 UTC 2025
Certificate fingerprints:
 MD5:  94:C8:9F:92:56:60:EB:DB:ED:4B:11:17:33:27:C0:C9
 SHA1: 42:9C:3C:51:E8:8E:AC:2E:5E:62:95:82:D7:22:E0:FB:08:B8:64:29
 SHA256:
 58:B8:63:3D:0C:34:35:39:FC:3D:1E:BA:AA:E3:CE:A9:C0:F3:58:53:1F:AD:89:A5:01:0D:D3:89:9E:7B:C5:69
Signature algorithm name: SHA256WITHRSA
Subject Public Key Algorithm: 4096-bit RSA key
Version: 3
Certificate[2]:
Owner: CN=Orange Root Certification Authority, O=GS, L=California, C=US
Issuer: CN=Orange Root Certification Authority, O=GS, L=California, C=US
Serial number: 3e726e7fbb3b0a7a6b4edd767f867fd1
Valid from: Mon Mar 16 07:16:07 UTC 2020 until: Fri Mar 10 07:16:07 UTC 2045
Certificate fingerprints:
 MD5:  ED:AC:F1:7F:88:05:83:2A:83:C0:09:03:D5:00:CA:7B
 SHA1: DC:B5:3F:37:DF:BD:E0:5C:A4:B7:F4:4C:96:12:75:7A:16:C7:61:37
 SHA256:
 F2:5B:DE:8A:F2:31:9D:E6:EF:35:F1:30:6F:DA:05:FF:92:B4:15:96:AA:82:67:E3:3C:C1:69:A3:E5:27:B9:A5
Signature algorithm name: SHA256WITHRSA
Subject Public Key Algorithm: 4096-bit RSA key
Version: 3
```

The logs for both successful and failure conditions are in `/var/log/loginsight-agent/liagent_2020-MM-DD-<file-num>.log`. If the configuration is successful, you can view the VCF Operations for logs configuration with the following command:

```
cat /var/lib/loginsight-agent/liagent-effective.ini
```

For example,

```
root@nsx1:/image/vmware/nsx/file-store# cat /var/lib/loginsight-agent/liagent-effective.ini
; Dynamic file representing the effective configuration of VMware Log Insight Agent (merged server-
side and client-side configuration)
;     DO NOT EDIT THIS FILE BY HAND -- YOUR CHANGES WILL BE OVERWRITTEN
```

```
; Creation time: 2020-03-22T19:41:21.648800

[server]
hostname=vrli.prome.local
proto=cfapi
ssl=yes
ssl_ca_path=/config/vmware/nsx-node-api/syslog/bb466082-996f-4d77-b6e3-1fa93f4a20d4_ca.pem
ssl_accept_any_trusted=yes
port=9543
filter={filelog; nsx-syslog; pri_severity <= 6 and ( msgid == "SWITCHING" or msgid == "ROUTING" or
 msgid == "FABRIC" or msgid == "SYSTEM" or msgid == "POLICY" or msgid == "HEALTHCHECK" or msgid ==
 "SHA" or msgid == "MONITORING" )}

[filelog|nsx-syslog]
directory=/var/log
include=syslog;syslog.*
parser=nsx-syslog_parser

[parser|nsx-syslog_parser]
base_parser=syslog
extract_sd=yes

[update]
auto_update=no
```

- To configure secure remote logging using the protocol TLS, specify the parameter `proto tls`. For example:

  ```
  set logging-server vrli.prome.local proto tls level info serverca Orange-CA.crt.pem clientca Orange-
  CA.crt.pem certificate gc-nsxt-mgr-full.crt.pem key gc-nsxt-mgr.key.pem
  ```

  Note the following:

  - For the `serverCA` parameter, only the root certificate is required, not the full chain.
  - If `clientCA` is different from `serverCA`, only the root certificate is required.
  - The certificate should hold the full chain of the NSX Manager (they should be NDcPP compliant - EKU, BASIC and CDP (CDP - this check can be ignored)).

  You can inspect the content of each certificate with the `keytool` command. For example,

  ```
  keytool -printcert -file /image/vmware/nsx/file-store/Orange-CA.crt.pem
  keytool -printcert -file gc-nsxt-mgr-full.crt.pem
  ```

  Example output:

  ```
  root@gc3:~# keytool -printcert -file /image/vmware/nsx/file-store/Orange-CA.crt.pem
  Owner: CN=Orange Root Certification Authority, O=GS, L=California, C=US
  Issuer: CN=Orange Root Certification Authority, O=GS, L=California, C=US
  Serial number: 3e726e7fbb3b0a7a6b4edd767f867fd1
  Valid from: Mon Mar 16 07:16:07 UTC 2020 until: Fri Mar 10 07:16:07 UTC 2045
  Certificate fingerprints:
   MD5:  ED:AC:F1:7F:88:05:83:2A:83:C0:09:03:D5:00:CA:7B
   SHA1: DC:B5:3F:37:DF:BD:E0:5C:A4:B7:F4:4C:96:12:75:7A:16:C7:61:37
   SHA256:
   F2:5B:DE:8A:F2:31:9D:E6:EF:35:F1:30:6F:DA:05:FF:92:B4:15:96:AA:82:67:E3:3C:C1:69:A3:E5:27:B9:A5
  Signature algorithm name: SHA256WITHRSA
  Subject Public Key Algorithm: 4096-bit RSA key
  Version: 3
  root@gc3:~#

  root@gc3:/image/vmware/nsx/file-store# keytool -printcert -file gc-nsxt-mgr-full.crt.pem
  Certificate[1]:
  Owner: CN=gc.prome.local, O=GS, L=HTG, ST=California, C=US
  Issuer: CN=Green Intermediate Certification Authority, O=GS, L=California, C=US
  ```

```
Serial number: bdf43ab31340b87f323b438a2895a075
Valid from: Mon Mar 16 07:26:51 UTC 2020 until: Wed Mar 16 07:26:51 UTC 2022
Certificate fingerprints:
       MD5:  36:3C:1F:57:96:07:84:C0:6D:B7:33:9A:8D:25:4D:27
       SHA1: D1:4E:F9:45:2D:0D:34:79:D2:B4:FA:65:28:E0:5C:DC:74:50:CA:3B
       SHA256:
 3C:FF:A9:5D:AA:68:44:44:DD:07:2F:DD:E2:BE:9C:32:19:7A:03:D5:26:8D:5F:AD:56:CA:D2:6C:91:96:27:6F
Signature algorithm name: SHA256WITHRSA
Subject Public Key Algorithm: 4096-bit RSA key
Version: 3
Certificate[2]:
Owner: CN=Green Intermediate Certification Authority, O=GS, L=California, C=US
Issuer: CN=Orange Root Certification Authority, O=GS, L=California, C=US
Serial number: 3e726e7fbb3b0a7a6b4edd767f867fd2
Valid from: Sun Mar 15 00:00:00 UTC 2020 until: Mon Mar 17 00:00:00 UTC 2025
Certificate fingerprints:
       MD5:  94:C8:9F:92:56:60:EB:DB:ED:4B:11:17:33:27:C0:C9
       SHA1: 42:9C:3C:51:E8:8E:AC:2E:5E:62:95:82:D7:22:E0:FB:08:B8:64:29
       SHA256:
 58:B8:63:3D:0C:34:35:39:FC:3D:1E:BA:AA:E3:CE:A9:C0:F3:58:53:1F:AD:89:A5:01:0D:D3:89:9E:7B:C5:69
Signature algorithm name: SHA256WITHRSA
Subject Public Key Algorithm: 4096-bit RSA key
Version: 3
Certificate[3]:
Owner: CN=Orange Root Certification Authority, O=GS, L=California, C=US
Issuer: CN=Orange Root Certification Authority, O=GS, L=California, C=US
Serial number: 3e726e7fbb3b0a7a6b4edd767f867fd1
Valid from: Mon Mar 16 07:16:07 UTC 2020 until: Fri Mar 10 07:16:07 UTC 2045
Certificate fingerprints:
       MD5:  ED:AC:F1:7F:88:05:83:2A:83:C0:09:03:D5:00:CA:7B
       SHA1: DC:B5:3F:37:DF:BD:E0:5C:A4:B7:F4:4C:96:12:75:7A:16:C7:61:37
       SHA256:
 F2:5B:DE:8A:F2:31:9D:E6:EF:35:F1:30:6F:DA:05:FF:92:B4:15:96:AA:82:67:E3:3C:C1:69:A3:E5:27:B9:A5
Signature algorithm name: SHA256WITHRSA
Subject Public Key Algorithm: 4096-bit RSA key
Version: 3
```

Examples of successful logging in `/var/log/syslog`:

```
<182>1 2020-03-22T21:54:34.501Z gc3.prome.local NSX 5187 - [nsx@6876 comp="nsx-manager"
 subcomp="node-mgmt" username="admin" level="INFO"] Successfully created CA PEM file /config/
vmwarensx-node-api/syslog/92a78d8a-acfd-4515-b05a-2927b70ae920_ca.pem for logging server
 vrli.prome.local:6514
<182>1 2020-03-22T21:54:36.269Z gc3.prome.local NSX 5187 - [nsx@6876 comp="nsx-manager"
 subcomp="node-mgmt" username="admin" level="INFO"] Successfully created client CA PEM file /config/
vmwarensx-node-api/syslog/92a78d8a-acfd-4515-b05a-2927b70ae920_client_ca.pem for logging server
 vrli.prome.local:6514
<182>1 2020-03-22T21:54:36.495Z gc3.prome.local NSX 5187 - [nsx@6876 comp="nsx-manager"
 subcomp="node-mgmt" username="root" level="INFO"] cert issuer = /C=US/L=California/O=GS/CN=Green
 IntermediateCertification Authority
<182>1 2020-03-22T21:54:36.514Z gc3.prome.local NSX 5187 - [nsx@6876 comp="nsx-manager"
 subcomp="node-mgmt" username="root" level="INFO"] cert subject = /C=US/ST=California/L=HTG/O=GS/
CN=gc.promelocal
<182>1 2020-03-22T21:54:36.539Z gc3.prome.local NSX 5187 - [nsx@6876 comp="nsx-manager"
 subcomp="node-mgmt" username="root" level="INFO"] certificate trust check succeeded. status: 200,
 result:{'status': 'OK'}
<182>1 2020-03-22T21:54:36.612Z gc3.prome.local NSX 5187 - [nsx@6876 comp="nsx-manager"
 subcomp="node-mgmt" username="root" level="INFO"] Certificate already exists, skip import
```

```
<182>1 2020-03-22T21:54:37.322Z gc3.prome.local NSX 5187 - [nsx@6876 comp="nsx-manager"
 subcomp="node-mgmt" username="admin" level="INFO"] Successfully created certificate PEM file /
config/vmwarensx-node-api/syslog/92a78d8a-acfd-4515-b05a-2927b70ae920_cert.pem for logging server
 vrli.prome.local:6514
<182>1 2020-03-22T21:54:38.020Z gc3.prome.local NSX 5187 - [nsx@6876 comp="nsx-manager"
 subcomp="node-mgmt" username="admin" level="INFO"] Successfully created key PEM file /config/
vmwarensx-node-api/syslog/92a78d8a-acfd-4515-b05a-2927b70ae920_key.pem for logging server
 vrli.prome.local:6514
```

Examples of logging failure in `/var/log/syslog`:

```
<182>1 2020-03-22T21:33:30.424Z gc3.prome.local NSX 5187 - [nsx@6876 comp="nsx-manager"
 subcomp="node-mgmt" username="admin" level="INFO"] Successfully created client CA PEM file /config/
vmwarensx-node-api/syslog/76332782-1ec6-483a-95d4-2adeaf2ef112_client_ca.pem for logging server
 vrli.prome.local:6514
<182>1 2020-03-22T21:33:30.779Z gc3.prome.local NSX 5187 - [nsx@6876 comp="nsx-manager"
 subcomp="node-mgmt" username="root" level="INFO"] cert issuer = /C=US/L=California/O=GS/CN=Green
 IntermediateCertification Authority
<182>1 2020-03-22T21:33:30.803Z gc3.prome.local NSX 5187 - [nsx@6876 comp="nsx-manager"
 subcomp="node-mgmt" username="root" level="INFO"] cert subject = /C=US/ST=California/L=HTG/O=GS/
CN=gc.promelocal
<179>1 2020-03-22T21:33:30.823Z gc3.prome.local NSX 5187 - [nsx@6876 comp="nsx-
manager" subcomp="node-mgmt" username="root" level="ERROR" errorCode="NODE10"]
 Certificate trust check failed. status:200, result: {'error_message': 'Certificate
 CN=gc.prome.local,O=GS,L=HTG,ST=California,C=US was not verifiably signed by
 CN=gc.prome.local,O=GS,L=HTG,ST=California,C=US: certificate does not verifywith supplied key',
 'status': 'ERROR'}
<179>1 2020-03-22T21:33:30.824Z gc3.prome.local NSX 5187 - [nsx@6876 comp="nsx-manager"
 subcomp="node-mgmt" username="admin" level="ERROR" errorCode="NODE10"] Failed to create certificate
 PEM file config/vmware/nsx-node-api/syslog/76332782-1ec6-483a-95d4-2adeaf2ef112_cert.pem for
 logging server vrli.prome.local:6514
<182>1 2020-03-22T21:33:31.578Z gc3.prome.local NSX 5187 - [nsx@6876 comp="nsx-manager"
 subcomp="node-mgmt" username="admin" level="INFO"] Successfully deleted CA PEM file /config/
vmwarensx-node-api/syslog/76332782-1ec6-483a-95d4-2adeaf2ef112_ca.pem
<182>1 2020-03-22T21:33:32.342Z gc3.prome.local NSX 5187 - [nsx@6876 comp="nsx-manager"
 subcomp="node-mgmt" username="admin" level="INFO"] Successfully deleted client CA PEM file /config/
vmwarensx-node-api/syslog/76332782-1ec6-483a-95d4-2adeaf2ef112_ca.pem
<182>1 2020-03-22T21:33:32.346Z gc3.prome.local NSX 16698 - [nsx@6876 comp="nsx-cli" subcomp="node-
mgmt" username="admin" level="INFO" audit="true"] CMD: set logging-server vrli.prome.local prototls
 level info serverca Orange-CA.crt.pem clientca Orange-CA.crt.pem certifi
cate gc-nsxt-mgr.crt.pem key gc-nsxt-mgr.key.pem (duration: 6.365s), Operation status: CMD_EXECUTED
```

You can check if the certificate and private key match with the following command. For example:

```
diff  <(openssl x509 -in certs/gc-nsxt-mgr.crt.pem -pubkey -noout) <(openssl rsa -in private/gc-
nsxt-mgr.key.pem -pubout)
```

If the certificate and private key match, the output will be `writing RSA key`. Any other output means they do not match. For example, if the certificate and private key match, you will see:

```
root@caserver:~/server-certs# diff  <(openssl x509 -in certs/gc-nsxt-mgr.crt.pem -pubkey -noout)
 <(openssl rsa -in private/gc-nsxt-mgr.key.pem -pubout)
writing RSA key
```

Example of a corrupt private key:

```
root@caserver:~/server-certs# diff  <(openssl x509 -in certs/gc-nsxt-mgr.crt.pem -pubkey -noout)
 <(openssl rsa -in private/gc-nsxt-mgr-corrupt.key.pem -pubout)
unable to load Private Key
140404188370584:error:0D07209B:asn1 encoding routines:ASN1_get_object:too long:asn1_lib.c:147:
140404188370584:error:0D068066:asn1 encoding routines:ASN1_CHECK_TLEN:bad object
 header:tasn_dec.c:1205:
140404188370584:error:0D07803A:asn1 encoding routines:ASN1_ITEM_EX_D2I:nested asn1
 error:tasn_dec.c:386:Type=RSA
```

```
140404188370584:error:04093004:rsa routines:OLD_RSA_PRIV_DECODE:RSA lib:rsa_ameth.c:119:
140404188370584:error:0D07209B:asn1 encoding routines:ASN1_get_object:too long:asn1_lib.c:147:
140404188370584:error:0D068066:asn1 encoding routines:ASN1_CHECK_TLEN:bad object
 header:tasn_dec.c:1205:
140404188370584:error:0D07803A:asn1 encoding routines:ASN1_ITEM_EX_D2I:nested asn1
 error:tasn_dec.c:386:Type=PKCS8_PRIV_KEY_INFO
140404188370584:error:0907B00D:PEM routines:PEM_READ_BIO_PRIVATEKEY:ASN1 lib:pem_pkey.c:141:
1,14d0
< -----BEGIN PUBLIC KEY-----
< MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICCgKCAgEAv3yH7pZidfkLrEP3zVa9
< EcOKXlFFjkThZRZMfguenlm8s6QHYVvuUX8IRB48Li3/DUfOj0bzaPWktpv+Q2P0
< N/j4LoX2RzjV/DPxYfLP6GMNMc21L3s9ruBeWUthtUP8khCWd2d2rZ09cUZVl0P9
< kIYBb5RMFC7Z1OUtH3bKdepEf+sXz3DaKZ/WySzYq9x86QDaA3ABO3Q0i7txBscI
< FvXuMDOMQaC3pPp9FWO6IPRAWB57wahLJv6K5qGIfwubSBFg53grT4snf1lDZAhZ
< 9hz5JgGr80GVyWyb7rgigpl9iUWAZx8U9De9XoxmvBN5iEGTIuKGaEgICL176crb
< RMkhjnCqNHI+z6sQvpYJ7U0zZc72eBIWoHUkcWWk3eU6Oy4OiyW6jYuXG7hZYlly
< nSkme3mZUWJKvcoX05+3zeCP623/HzE7X2sNyWFjzeF3XEvauZrIbsJh/xp2ShDa
< uKKEY0gUGhLtCa3TpV9l8d6tFWVy8XjVjdjoVt4s7MfUo/airVmRykfsWrKyNUOQ
< qRZvSbqjt8pm+3bSvKdXX4ul7ptPG2GF20ETWHPwjk2JwQpGhR9zK8fsKzvm6hXi
< kq76zI4FefuVps3e1r39+0F+p6d6i2oUoo24sC1iSePTDhU74efVp6iv8HmnDgYX
< Ylm6Kusr0JT5TJFDfASmrj8CAwEAAQ==
< -----END PUBLIC KEY-----
```

Example of a valid private key and certificate but they are not made for each other:

```
root@caserver:~/server-certs# diff  <(openssl x509 -in certs/gc-nsxt-mgr.crt.pem -pubkey -noout)
<(openssl rsa -in private/vrli.key.pem -pubout)
writing RSA key
2,13c2,13
< MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICCgKCAgEAv3yH7pZidfkLrEP3zVa9
< EcOKXlFFjkThZRZMfguenlm8s6QHYVvuUX8IRB48Li3/DUfOj0bzaPWktpv+Q2P0
< N/j4LoX2RzjV/DPxYfLP6GMNMc21L3s9ruBeWUthtUP8khCWd2d2rZ09cUZVl0P9
< kIYBb5RMFC7Z1OUtH3bKdepEf+sXz3DaKZ/WySzYq9x86QDaA3ABO3Q0i7txBscI
< FvXuMDOMQaC3pPp9FWO6IPRAWB57wahLJv6K5qGIfwubSBFg53grT4snf1lDZAhZ
< 9hz5JgGr80GVyWyb7rgigpl9iUWAZx8U9De9XoxmvBN5iEGTIuKGaEgICL176crb
< RMkhjnCqNHI+z6sQvpYJ7U0zZc72eBIWoHUkcWWk3eU6Oy4OiyW6jYuXG7hZYlly
< nSkme3mZUWJKvcoX05+3zeCP623/HzE7X2sNyWFjzeF3XEvauZrIbsJh/xp2ShDa
< uKKEY0gUGhLtCa3TpV9l8d6tFWVy8XjVjdjoVt4s7MfUo/airVmRykfsWrKyNUOQ
< qRZvSbqjt8pm+3bSvKdXX4ul7ptPG2GF20ETWHPwjk2JwQpGhR9zK8fsKzvm6hXi
< kq76zI4FefuVps3e1r39+0F+p6d6i2oUoo24sC1iSePTDhU74efVp6iv8HmnDgYX
< Ylm6Kusr0JT5TJFDfASmrj8CAwEAAQ==
---
> MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICCgKCAgEAqvsjay7+o7gCW7szT3ho
> bC34XX2l6u5Jl4/X/pUDI/YHmIf06bsZ1r/l4bTL4Q7BM6+9MI6UYEE7DxUoINGO
> o4FEEQE32KWVFe3gw3homHU39q4pQjsJsxTcTE3oDMlIY0nWJ0PRUst3DffyUH1L
> W0NUN9YdN+fAl2Uf021iuDqVy9V8AH3ON6fu+QCA8nt71ZkzeTxSA0ldpl2NA17F
> rD8rm05wxnV7WtuV7V8PstISiClzhHgZRMl+B0r30OitnyAzEGLaRT3//PKfe0Oe
> HCdxGMlrUtMqxIItJahEsqvMufyqNYecVscyXLHPelizKCsQfy8cO8LnznG8VAdc
> YILSn3uYGZap6aF1SgVxsvZicwvlYnssmgE13Af0nScmfM96k9h5joHVEkWK6O8v
> oT5DGG1kVL2Qly97x0b6EnzUorzivv5zJMKvFcOektR8HdMHQit5uvmMRY3S5zow
> FtvfSDfWxxKyTy6GBrpP+8F+Jq91yGy/qa9lhKBzT2lg+rJp7T8k7/Nm9Tjyx7jL
> EqgEKZEL4chxpo8ucF98hbvXWRuaFHC2iDzGuUmuS1FfjVvHTuIbEMQfjapLZrHx
> 8jHfOP/PL+6kPbvNZZ2rTpczuEoGTQFFW9vX48GzIEyMeR6QWpPR0F7r4xak68P5
> 2PJmMveinDhU35IqWEXHAwcCAwEAAQ==
```

c) To view the logging configuration, run the following command:

```
get logging-server
```

For example,

```
nsx> get logging-servers
```

```
192.168.110.60 proto udp level info facility local6 messageid SYSTEM,FABRIC
192.168.110.60 proto udp level info facility auth,user
```

d) To clear the remote logging configuration, run the following command:

```
clear logging-servers
```

2. To configure remote logging on an ESX host:
   a) Run the following commands to configure syslog and send a test message:

```
esxcli network firewall ruleset set -r syslog -e true
esxcli system syslog config set --loghost=udp://<log server IP>:<port> --log-level=info
esxcli system syslog reload
esxcli system syslog mark -s "This is a test message"
```

   b) You can run the following command to display the configuration:

```
esxcli system syslog config get
```

Add Syslog Servers for NSX Nodes

You can use the **Node Profiles** page in NSX Manager to add syslog servers for NSX Manager and NSX Edge nodes.

By default, the node profile is applied to all nodes, unless the node is configured to not accept such configuration from the NSX Manager. To prevent a node from accepting the node profile, use the CLI command `set node central-config disabled` on that node.

1. From your browser, log in with **admin** privileges to an NSX Manager at `https://`*`nsx-manager-ip-address`*.
2. Select **System** > **Fabric** > **Profiles**.
3. Click the **Node Profiles** tab.
4. Click **All NSX Nodes** in the **Name** column.
5. In the **Syslog Servers** section, click **Add** to add a Syslog server.
   a) Enter the FQDN or IP address of the Syslog server.
   b) Specify a port number.
   c) Select a protocol.
      The available protocols are **TCP**, **UDP**, and **LI** (Log Insight).
   d) Select a log level.
      The available levels are **Emergency**, **Alert**, **Critical**, **Error**, **Warning**, **Notice**, **Information**, and **Debug**.

      When you choose a level, you will also see the logs for all the previous levels (starting with the **Emergency** level). For example, if you choose **Emergency**, you will see **Emergency**-level logs. If you choose **Critical**, you will see logs for **Emergency**, **Alert** and **Critical**. If you choose **Information**, you will see logs for **Emergency**, **Alert**, **Critical**, **Error**, **Warning**, **Notice**, and **Information**. If you choose **Debug**, you will see messages for all log levels.
   e) Click **Add**.
6. Repeat step 5 to add more syslog servers, if required.

Log Message IDs

In a log message, the message ID field identifies the type of message. You can use the `messageid` parameter in the `set logging-server` command to filter which log messages are sent to a logging server.

Table 1083: Log Message IDs

Message ID	Examples
FABRIC	Host node Host preparation Edge node Transport zone Transport node Uplink profiles Cluster profiles Edge cluster
SWITCHING	Logical switch Logical switch ports Switching profiles switch security features
ROUTING	Logical router Logical router ports Static routing Dynamic routing NAT
FIREWALL	Firewall rules Firewall rule sections
FIREWALL-PKTLOG	Firewall connection logs Firewall packet logs
GROUPING	IP sets Mac sets NSGroups NSServices NSService groups VNI Pool IP Pool
DHCP	DHCP relay
SYSTEM	Appliance management (remote syslog, ntp, etc) Cluster management Trust management Licensing User and roles Task management Install Upgrade (NSX Manager, NSX Edge and host-packages upgrades) Realization Tags
MONITORING?	SNMP Port connection Traceflow
-	All other log messages.

Troubleshooting Syslog Issues

If logs are not received by the remote log server, perform the following steps.

- Verify the remote log server's IP address.
- Verify that the `level` parameter is configured correctly.
- Verify that the `facility` parameter is configured correctly.

- If the protocol is TLS, set the protocol to UDP to see if there is a certificate mismatch.
- If the protocol is TLS, verify that port 6514 is open on both ends.
- Remove the message ID filter and see if logs are received by the server.
- Restart the rsyslog service with the command `restart service syslog`.

To learn more about how to configure NSX appliances and hypervisors to send log messages to a remote logging server, see Configure Remote Logging.

Configure Serial Logging on an Appliance VM

You can configure serial logging on an appliance VM to capture log messages when the VM crashes.

1. Log in to the VM as `root`.
2. Edit `/etc/default/grub`.
3. Find the parameter `GRUB_CMDLINE_LINUX_DEFAUL` and append `console=ttyS0 console=tty0`.
4. Run the command `update-grub2`.
5. Verify that the `/boot/grub/grub.cfg` file has the change made in step 3.
6. Power off the VM.
7. Edit the VM's configuration (.vmx) file and add the following lines:

```
serial0.present = "TRUE"
serial0.fileType = "file"
serial0.fileName = "serial.out"
serial0.yieldOnMsrRead = "TRUE"
answer.msg.serial.file.open = "Append"
```

8. Power on the VM.

If a kernel panic occurs in the VM, you can find the file `serial.out` containing log messages at the same location as that of the `.vmx` file.

Firewall Audit Log Messages

Firewall configuration changes are audited. Below are examples of audit log messages related to these changes.

Distributed firewall changes

Adding a firewall section (`SecurityPolicy-1`) with a rule (`Rule1_1`):

```
<182>1 2020-08-11T21:58:50.319Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager"
 level="INFO" reqId="2aff6b4f-3d4f-4d62-a639-61291f7e879e" splitId="a5mxlu78" splitIndex="1
 of 2" subcomp="policy" update="true"] UserName="admin", ModuleName="DfwSecurityPolicy",
 Operation="PatchSecurityPolicyForDomain", Operation status="success", Old
 value=[{"precedence":10,"category":"Application","resource_type":"CommunicationMap","id":"SecurityPolicy-1","display_n
infra/domains/default/security-policies/SecurityPolicy-1","relative_path":"SecurityPolicy-1","parent_path":"/
infra/domains/default","unique_id":"895eeac5-641b-4306-be7f-
a43fdd969ee5","marked_for_delete":false,"overridden":false,"_create_user":"admin","_create_time":1597183130247,"_last_m
 New value=["default"
 "SecurityPolicy-1" {"resource_type":"SecurityPolicy","id":"SecurityPolicy-1","display_name":"SecurityPolicy-1","path":
infra/domains/default/security-policies/SecurityPolicy-1","children":[{"Rule":
{"action":"ALLOW","resource_type":"Rule","id":"Rule1_1","display_name":"Rule1_1","path":

<182>1 2020-08-11T21:58:50.320Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager"
 level="INFO" reqId="2aff6b4f-3d4f-4d62-a639-61291f7e879e" splitId="a5mxlu78" splitIndex="2 of 2"
 subcomp="policy" update="true"] "/infra/domains/default/security-policies/SecurityPolicy-1/rules/
Rule1_1","marked_for_delete":false,"overridden":false,"sequence_number":10,"sources_excluded":false,"destinations_exclu
```

["ANY"],"destination_groups":["ANY"],"services":["ANY"],"profiles":["ANY"],"logged":false,"scope":
["ANY"],"disabled":false,"direction":"IN_OUT","ip_protocol":"IPV4_IPV6","_protection":"UNKNOWN"},"resource_type":"Child
["ANY"],"_protection":"UNKNOWN"}]

<182>1 2020-08-11T21:58:50.404Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager"
 entId="Rule1_1" level="INFO" reqId="2aff6b4f-3d4f-4d62-a639-61291f7e879e" splitId="E993J2LF"
 splitIndex="1 of 2" subcomp="policy" update="true"] UserName="admin", ModuleName="DfwSecurityPolicy",
 Operation="UpdateSecurityRule", Operation status="success", Old value=[{"sequence_number":10,"source_groups":
["ANY"],"destination_groups":["ANY"],"services":["ANY"],"action":"ALLOW","logged":false,"scope":
["ANY"],"disabled":false,"direction":"IN_OUT","resource_type":"CommunicationEntry","id":"Rule1_1","display_name":"Rule1
infra/domains/default/security-policies/SecurityPolicy-1/rules/
Rule1_1","relative_path":"Rule1_1","parent_path":"/infra/domains/default/security-policies/
SecurityPolicy-1","unique_id":"2024","marked_for_delete":false,"overridden":false,"_create_user":"admin","_create_time"
 New value=["default" "SecurityPolicy-1"
 "Rule1_1" {"action":"ALLOW","resource_type":"Rule","id":"Rule1_1","display_name":"Rule1_1","path":

<182>1 2020-08-11T21:58:50.404Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager" entId="Rule1_1"
 level="INFO" reqId="2aff6b4f-3d4f-4d62-a639-61291f7e879e" splitId="E993J2LF" splitIndex="2 of 2"
 subcomp="policy" update="true"] "/infra/domains/default/security-policies/SecurityPolicy-1/rules/
Rule1_1","marked_for_delete":false,"overridden":false,"sequence_number":10,"sources_excluded":false,"destinations_exclu
["ANY"],"destination_groups":["ANY"],"services":["ANY"],"profiles":["ANY"],"logged":false,"scope":
["ANY"],"disabled":false,"direction":"IN_OUT","ip_protocol":"IPV4_IPV6","_protection":"UNKNOWN"}]

<182>1 2020-08-11T21:58:50.466Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager" level="INFO"
 reqId="2aff6b4f-3d4f-4d62-a639-61291f7e879e" splitId="iMHW1shi" splitIndex="1 of 2" subcomp="policy"
 update="true"] UserName="admin", ModuleName="Policy", Operation="PatchInfra", Operation status="success", New
 value=[{"enforce_revision_check":true} {"resource_type":"Infra","children":[{"children":[{"SecurityPolicy":
{"resource_type":"SecurityPolicy","id":"SecurityPolicy-1","display_name":"SecurityPolicy-1","path":"/
infra/domains/default/security-policies/SecurityPolicy-1","children":[{"Rule":
{"action":"ALLOW","resource_type":"Rule","id":"Rule1_1","display_name":"Rule1_1","path":"/infra/domains/
default/security-policies/SecurityPolicy-1/rules/
Rule1_1","marked_for_delete":false,"overridden":false,"sequence_number":10,"sources_excluded":false,"destinations_exclu
["ANY"],"destination_groups":["ANY"],"services":["ANY"],"profiles":["ANY"],"logged":false,"scope":
["ANY"],"disabled":false,"direction":"IN_OUT","ip_protocol":"IPV4_IPV6","_protection":"UNKNOWN"},"resource_type":"Child

<182>1 2020-08-11T21:58:50.466Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager" level="INFO"
 reqId="2aff6b4f-3d4f-4d62-a639-61291f7e879e" splitId="iMHW1shi" splitIndex="2 of 2" subcomp="policy"
 update="true"] :false,"overridden":false,"sequence_number":10,"category":"Application","stateful":true,"locked":false,
["ANY"],"_protection":"UNKNOWN"},"resource_type":"ChildSecurityPolicy","marked_for_delete":false,"mark_for_override":fa

Updating a rule (from `Rule1_1` to `Rule1_1_updated`) in a section (`SecurityPolicy-1`):

<182>1 2020-08-11T22:22:06.303Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager"
 level="INFO" reqId="6aadd8de-d157-4479-b84c-8410dd48c2aa" splitId="mJ7hQGhg" splitIndex="1
 of 2" subcomp="policy" update="true"] UserName="admin", ModuleName="DfwSecurityPolicy",
 Operation="PatchSecurityPolicyForDomain", Operation status="success", New value=["default"
 "SecurityPolicy-1" {"resource_type":"SecurityPolicy","id":"SecurityPolicy-1","display_name":"SecurityPolicy-1","path":
infra/domains/default/security-policies/SecurityPolicy-1","unique_id":"895eeac5-641b-4306-be7f-
a43fdd969ee5","children":[{"Rule":
{"action":"ALLOW","resource_type":"Rule","id":"Rule1_1","display_name":"Rule1_1_updated","path":"/infra/
domains/default/security-policies/SecurityPolicy-1/rules/
Rule1_1","unique_id":"2024","marked_for_delete":false,"overridden":false,"rule_id":2024,"sequence_number":10,"sources_e
["ANY"],"destination_groups":["ANY"],"services":["ANY"],"profiles":["ANY"],"logged":false,"scope":
["ANY"],"disabled":false,"direction":"IN_OUT","ip_protocol":"IPV4_IPV6","is_default":false,"_protection":"UNKNOWN","_re

```
<182>1 2020-08-11T22:22:06.303Z manager1 NSX 22164 - [nsx@6876 audit="true"
 comp="nsx-manager" level="INFO" reqId="6aadd8de-d157-4479-b84c-8410dd48c2aa"
 splitId="mJ7hQGhg" splitIndex="2 of 2" subcomp="policy" update="true"]
 "ChildRule","marked_for_delete":false,"mark_for_override":false,"_protection":"UNKNOWN"}],"marked_for_delete":false,"c
["ANY"],"is_default":false,"_protection":"UNKNOWN","_revision":0}]

<182>1 2020-08-11T22:22:06.324Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager"
 entId="Rule1_1" level="INFO" reqId="6aadd8de-d157-4479-b84c-8410dd48c2aa" splitId="JKVilI6n"
 splitIndex="1 of 2" subcomp="policy" update="true"] UserName="admin", ModuleName="DfwSecurityPolicy",
 Operation="UpdateSecurityRule", Operation status="success", Old value=[{"sequence_number":10,"source_groups":
["ANY"],"destination_groups":["ANY"],"services":["ANY"],"action":"ALLOW","logged":false,"scope":
["ANY"],"disabled":false,"direction":"IN_OUT","resource_type":"CommunicationEntry","id":"Rule1_1","display_name":"Rule1
infra/domains/default/security-policies/SecurityPolicy-1/rules/
Rule1_1","relative_path":"Rule1_1","parent_path":"/infra/domains/default/security-policies/
SecurityPolicy-1","unique_id":"2024","marked_for_delete":false,"overridden":false,"_create_user":"admin","_create_time"
 New value=["default" "SecurityPolicy-1"
 "Rule1_1" {"action":"ALLOW","resource_type":"Rule","id":"Rule1_1","display_name":"Rule1_1_updated","path":

<182>1 2020-08-11T22:22:06.324Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager" entId="Rule1_1"
 level="INFO" reqId="6aadd8de-d157-4479-b84c-8410dd48c2aa" splitId="JKVilI6n" splitIndex="2 of 2"
 subcomp="policy" update="true"] "/infra/domains/default/security-policies/SecurityPolicy-1/rules/
Rule1_1","unique_id":"2024","marked_for_delete":false,"overridden":false,"rule_id":2024,"sequence_number":10,"sources_e
["ANY"],"destination_groups":["ANY"],"services":["ANY"],"profiles":["ANY"],"logged":false,"scope":
["ANY"],"disabled":false,"direction":"IN_OUT","ip_protocol":"IPV4_IPV6","is_default":false,"_protection":"UNKNOWN","_re

<182>1 2020-08-11T22:22:06.363Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager" level="INFO"
 reqId="6aadd8de-d157-4479-b84c-8410dd48c2aa" splitId="9MtbEpd8" splitIndex="1 of 2" subcomp="policy"
 update="true"] UserName="admin", ModuleName="Policy", Operation="PatchInfra", Operation status="success", New
 value=[{"enforce_revision_check":true} {"resource_type":"Infra","children":[{"children":[{"SecurityPolicy":
{"resource_type":"SecurityPolicy","id":"SecurityPolicy-1","display_name":"SecurityPolicy-1","path":"/
infra/domains/default/security-policies/SecurityPolicy-1","unique_id":"895eeac5-641b-4306-be7f-
a43fdd969ee5","children":[{"Rule":
{"action":"ALLOW","resource_type":"Rule","id":"Rule1_1","display_name":"Rule1_1_updated","path":"/infra/
domains/default/security-policies/SecurityPolicy-1/rules/
Rule1_1","unique_id":"2024","marked_for_delete":false,"overridden":false,"rule_id":2024,"sequence_number":10,"sources_e
["ANY"],"destination_groups":["ANY"],"services":["ANY"],"profiles":["ANY"],"logged":false,"scope":
["ANY"],"disabled":false,"direction":"IN_OUT","ip_protocol":"IPV4_IPV6","is_default":false,"_protection":"UNKNOWN","_re

<182>1 2020-08-11T22:22:06.363Z manager1 NSX 22164 - [nsx@6876 audit="true"
 comp="nsx-manager" level="INFO" reqId="6aadd8de-d157-4479-b84c-8410dd48c2aa"
 splitId="9MtbEpd8" splitIndex="2 of 2" subcomp="policy" update="true"]
 0},"resource_type":"ChildRule","marked_for_delete":false,"mark_for_override":false,"_protection":"UNKNOWN"}],"marked_f
["ANY"],"is_default":false,"_protection":"UNKNOWN","_revision":0},"resource_type":"ChildSecurityPolicy","marked_for_del
```

Deleting a rule (`Rule1_2`) from a section (`SecurityPolicy-1`):

```
<182>1 2020-08-11T22:12:24.444Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager"
 entId="Rule1_2" level="INFO" reqId="1a58e753-460c-443f-8a28-0d40d8af9b76" subcomp="policy"
 update="true" username="admin"] UserName="admin", ModuleName="DfwSecurityPolicy",
 Operation="DeleteSecurityRule", Operation status="success", Old value=[{"sequence_number":20,"source_groups":
["ANY"],"destination_groups":["ANY"],"services":["ANY"],"action":"ALLOW","logged":false,"scope":
["ANY"],"disabled":false,"direction":"IN_OUT","resource_type":"CommunicationEntry","id":"Rule1_2","display_name":"Rule1
infra/domains/default/security-policies/SecurityPolicy-1/rules/
Rule1_2","relative_path":"Rule1_2","parent_path":"/infra/domains/default/security-policies/
SecurityPolicy-1","unique_id":"2026","marked_for_delete":false,"overridden":false,"_create_user":"admin","_create_time"
 New value=["default" "SecurityPolicy-1" "Rule1_2"]
```

```
<182>1 2020-08-11T22:12:24.463Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager"
 level="INFO" reqId="1a58e753-460c-443f-8a28-0d40d8af9b76" splitId="hoDI5YJQ" splitIndex="1
 of 2" subcomp="policy" update="true"] UserName="admin", ModuleName="DfwSecurityPolicy",
 Operation="PatchSecurityPolicyForDomain", Operation status="success", New value=["default"
 "SecurityPolicy-1" {"resource_type":"SecurityPolicy","id":"SecurityPolicy-1","display_name":"SecurityPolicy-1","path":
infra/domains/default/security-policies/SecurityPolicy-1","unique_id":"895eeac5-641b-4306-be7f-
a43fdd969ee5","children":[{"Rule":{"resource_type":"Rule","id":"Rule1_2","path":"/infra/domains/default/
security-policies/SecurityPolicy-1/rules/
Rule1_2","marked_for_delete":true,"overridden":false,"sources_excluded":false,"destinations_excluded":false,"logged":fa

<182>1 2020-08-11T22:12:24.463Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager" level="INFO"
 reqId="1a58e753-460c-443f-8a28-0d40d8af9b76" splitId="hoDI5YJQ" splitIndex="2 of 2" subcomp="policy"
 update="true"] ["ANY"],"is_default":false,"_protection":"UNKNOWN","_revision":0}]


<182>1 2020-08-11T22:12:24.497Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager" level="INFO"
 reqId="1a58e753-460c-443f-8a28-0d40d8af9b76" splitId="mxpzQHfF" splitIndex="1 of 2" subcomp="policy"
 update="true"] UserName="admin", ModuleName="Policy", Operation="PatchInfra", Operation status="success", New
 value=[{"enforce_revision_check":true} {"resource_type":"Infra","children":[{"children":[{"SecurityPolicy":
{"resource_type":"SecurityPolicy","id":"SecurityPolicy-1","display_name":"SecurityPolicy-1","path":"/
infra/domains/default/security-policies/SecurityPolicy-1","unique_id":"895eeac5-641b-4306-be7f-
a43fdd969ee5","children":[{"Rule":{"resource_type":"Rule","id":"Rule1_2","path":"/infra/domains/default/
security-policies/SecurityPolicy-1/rules/
Rule1_2","marked_for_delete":true,"overridden":false,"sources_excluded":false,"destinations_excluded":false,"logged":fa
<182>1 2020-08-11T22:12:24.497Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager"
 level="INFO" reqId="1a58e753-460c-443f-8a28-0d40d8af9b76" splitId="mxpzQHfF" splitIndex="2
 of 2" subcomp="policy" update="true"] true,"locked":false,"lock_modified_time":0,"scope":
["ANY"],"is_default":false,"_protection":"UNKNOWN","_revision":0},"resource_type":"ChildSecurityPolicy","marked_for_del
```

Deleting a section (`SecurityPolicy-1`) that contains a rule (`Rule1_1`):

```
<182>1 2020-08-11T22:24:24.898Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager"
 entId="SecurityPolicy-1" level="INFO" reqId="724b5494-10cd-4124-a431-56ba7d922bbf" splitId="4WIXz9qL"
 splitIndex="1 of 2" subcomp="policy" update="true" username="admin"] UserName="admin",
 ModuleName="DfwSecurityPolicy", Operation="DeleteSecurityPolicyForDomain", Operation status="success", Old
 value=[{"precedence":10,"category":"Application","resource_type":"CommunicationMap","id":"SecurityPolicy-1","display_n
infra/domains/default/security-policies/SecurityPolicy-1","relative_path":"SecurityPolicy-1","parent_path":"/
infra/domains/default","unique_id":"895eeac5-641b-4306-be7f-
a43fdd969ee5","marked_for_delete":false,"overridden":false,"_create_user":"admin","_create_time":1597183130247,"_last_m
{"sequence_number":10,"source_groups":["ANY"],"destination_groups":["ANY"],"services":
["ANY"],"action":"ALLOW","logged":false,"scope":
["ANY"],"disabled":false,"direction":"IN_OUT","resource_type":"CommunicationEntry","id":"Rule1_1","display_name":"Rule1

<182>1 2020-08-11T22:24:24.898Z manager1 NSX 22164 - [nsx@6876 audit="true"
 comp="nsx-manager" entId="SecurityPolicy-1" level="INFO" reqId="724b5494-10cd-4124-
a431-56ba7d922bbf" splitId="4WIXz9qL" splitIndex="2 of 2" subcomp="policy" update="true"
 username="admin"] "/infra/domains/default/security-policies/SecurityPolicy-1/rules/
Rule1_1","relative_path":"Rule1_1","parent_path":"/infra/domains/default/security-policies/
SecurityPolicy-1","unique_id":"2024","marked_for_delete":false,"overridden":false,"_create_user":"admin","_create_time"
 New value=["default" "SecurityPolicy-1"]

<182>1 2020-08-11T22:24:24.938Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager"
 level="INFO" reqId="724b5494-10cd-4124-a431-56ba7d922bbf" subcomp="policy" update="true"]
 UserName="admin", ModuleName="Policy", Operation="PatchInfra", Operation status="success", New
 value=[{"enforce_revision_check":true} {"resource_type":"Infra","children":[{"children":[{"SecurityPolicy":
{"resource_type":"SecurityPolicy","id":"SecurityPolicy-1","path":"/infra/domains/default/security-policies/
SecurityPolicy-1","marked_for_delete":true,"overridden":false,"locked":false,"_protection":"UNKNOWN"},"resource_type":"
```

Gateway firewall changes

Note that log messages for a tier-0 gateway and a tier-1 gateway are similar.

Adding a section (`T1-Policies`) with a rule (`myT1_Rule1`) for a tier-1 gateway (`myT1`):

```
<182>1 2020-08-11T22:31:26.800Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager"
 level="INFO" reqId="c2790fbd-db29-46d3-9a0e-1003455ee9ea" splitId="Ta8faYzQ" splitIndex="1
 of 2" subcomp="policy" update="true"] UserName="admin", ModuleName="PolicyEdgeFirewall",
 Operation="PatchGatewayPolicyForDomain", Operation status="success", Old
 value=[{"precedence":10,"category":"LocalGatewayRules","resource_type":"CommunicationMap","id":"T1-
Policies","display_name":"T1-Policies","path":"/infra/domains/default/gateway-
policies/T1-Policies","relative_path":"T1-Policies","parent_path":"/infra/domains/
default","unique_id":"a73c1345-6b4e-43e0-
b4ee-9a91c7ba9df6","marked_for_delete":false,"overridden":false,"_create_user":"admin","_create_time":1597185086789,"_l
 New value=["default" "T1-Policies" {"resource_type":"GatewayPolicy","id":"T1-Policies","display_name":"T1-
Policies","path":"/infra/domains/default/gateway-policies/T1-Policies","children":[{"Rule":
{"action":"ALLOW","resource_type":"Rule","id":"myT1_Rule1","display_name":"myT1_Rule1","path":

<182>1 2020-08-11T22:31:26.801Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager"
 level="INFO" reqId="c2790fbd-db29-46d3-9a0e-1003455ee9ea" splitId="Ta8faYzQ" splitIndex="2 of
 2" subcomp="policy" update="true"] "/infra/domains/default/gateway-policies/T1-Policies/rules/
myT1_Rule1","marked_for_delete":false,"overridden":false,"sequence_number":10,"sources_excluded":false,"destinations_ex
["ANY"],"destination_groups":["ANY"],"services":["ANY"],"profiles":["ANY"],"logged":false,"scope":["/infra/
tier-1s/
myT1"],"disabled":false,"direction":"IN_OUT","ip_protocol":"IPV4_IPV6","_protection":"UNKNOWN"},"resource_type":"ChildR

<182>1 2020-08-11T22:31:26.878Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-
manager" level="INFO" reqId="c2790fbd-db29-46d3-9a0e-1003455ee9ea" splitId="aZfgiFKt"
 splitIndex="1 of 2" subcomp="policy" update="true" username="admin"] UserName="admin",
 ModuleName="PolicyEdgeFirewall", Operation="PatchGatewayRule", Operation status="success",
 Old value=[{"sequence_number":10,"source_groups":["ANY"],"destination_groups":
["ANY"],"services":["ANY"],"action":"ALLOW","logged":false,"scope":["/infra/tier-1s/
myT1"],"disabled":false,"direction":"IN_OUT","resource_type":"CommunicationEntry","id":"myT1_Rule1","display_name":"myT
infra/domains/default/gateway-policies/T1-Policies/rules/
myT1_Rule1","relative_path":"myT1_Rule1","parent_path":"/infra/domains/default/gateway-policies/T1-
Policies","unique_id":"2028","marked_for_delete":false,"overridden":false,"_create_user":"admin","_create_time":1597185
 New value=["default" "T1-Policies"
 "myT1_Rule1" {"action":"ALLOW","resource_type":"Rule","id":"myT1_Rule1","display_name":"myT1_Rule1","path":

<182>1 2020-08-11T22:31:26.878Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager"
 level="INFO" reqId="c2790fbd-db29-46d3-9a0e-1003455ee9ea" splitId="aZfgiFKt" splitIndex="2 of 2"
 subcomp="policy" update="true" username="admin"] "/infra/domains/default/gateway-policies/T1-Policies/rules/
myT1_Rule1","marked_for_delete":false,"overridden":false,"sequence_number":10,"sources_excluded":false,"destinations_ex
["ANY"],"destination_groups":["ANY"],"services":["ANY"],"profiles":["ANY"],"logged":false,"scope":["/infra/
tier-1s/myT1"],"disabled":false,"direction":"IN_OUT","ip_protocol":"IPV4_IPV6","_protection":"UNKNOWN"}]

<182>1 2020-08-11T22:31:26.890Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager"
 level="INFO" reqId="c2790fbd-db29-46d3-9a0e-1003455ee9ea" splitId="0s7tdCjN" splitIndex="1 of 2"
 subcomp="policy" update="true"] UserName="admin", ModuleName="Policy", Operation="PatchInfra", Operation
 status="success", New value=[{"enforce_revision_check":true} {"resource_type":"Infra","children":
[{"children":[{"GatewayPolicy":{"resource_type":"GatewayPolicy","id":"T1-Policies","display_name":"T1-
Policies","path":"/infra/domains/default/gateway-policies/T1-Policies","children":[{"Rule":
{"action":"ALLOW","resource_type":"Rule","id":"myT1_Rule1","display_name":"myT1_Rule1","path":"/
infra/domains/default/gateway-policies/T1-Policies/rules/
myT1_Rule1","marked_for_delete":false,"overridden":false,"sequence_number":10,"sources_excluded":false,"destinations_ex
["ANY"],"destination_groups":["ANY"],"services":["ANY"],"profiles":["ANY"],"logged":false,"scope":["/infra/
tier-1s/
myT1"],"disabled":false,"direction":"IN_OUT","ip_protocol":"IPV4_IPV6","_protection":"UNKNOWN"},"resource_type":"ChildR
```

```
<182>1 2020-08-11T22:31:26.890Z manager1 NSX 22164 - [nsx@6876 audit="true"
 comp="nsx-manager" level="INFO" reqId="c2790fbd-db29-46d3-9a0e-1003455ee9ea"
 splitId="0s7tdCjN" splitIndex="2 of 2" subcomp="policy" update="true"]
 "UNKNOWN"}],"marked_for_delete":false,"overridden":false,"sequence_number":10,"category":"LocalGatewayRules","stateful
```

Updating a rule (from `myT1_Rule1` to `myT1_Rule1_Updated`) in a section (`T1-Policies`):

```
<182>1 2020-08-11T22:36:19.410Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager" level="INFO"
 reqId="a17fcbdc-1aed-4526-93e9-40a3730eeb7f" splitId="BiHDjsY8" splitIndex="1 of 2" subcomp="policy"
 update="true"] UserName="admin", ModuleName="PolicyEdgeFirewall", Operation="PatchGatewayPolicyForDomain",
 Operation status="success", New value=["default" "T1-Policies" {"resource_type":"GatewayPolicy","id":"T1-
Policies","display_name":"T1-Policies","path":"/infra/domains/default/gateway-policies/
T1-Policies","unique_id":"a73c1345-6b4e-43e0-b4ee-9a91c7ba9df6","children":[{"Rule":
{"action":"ALLOW","resource_type":"Rule","id":"myT1_Rule1","display_name":"myT1_Rule1_Updated","path":"/
infra/domains/default/gateway-policies/T1-Policies/rules/
myT1_Rule1","unique_id":"2028","marked_for_delete":false,"overridden":false,"rule_id":2028,"sequence_number":10,"source
["ANY"],"destination_groups":["ANY"],"services":["ANY"],"profiles":["ANY"],"logged":false,"scope":["/infra/
tier-1s/
myT1"],"disabled":false,"direction":"IN_OUT","ip_protocol":"IPV4_IPV6","is_default":false,"_protection":"UNKNOWN","_rev

<182>1 2020-08-11T22:36:19.410Z manager1 NSX 22164 - [nsx@6876 audit="true"
 comp="nsx-manager" level="INFO" reqId="a17fcbdc-1aed-4526-93e9-40a3730eeb7f"
 splitId="BiHDjsY8" splitIndex="2 of 2" subcomp="policy" update="true"]
 "ChildRule","marked_for_delete":false,"mark_for_override":false,"_protection":"UNKNOWN"}],"marked_for_delete":false,"c

<182>1 2020-08-11T22:36:19.430Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-
manager" level="INFO" reqId="a17fcbdc-1aed-4526-93e9-40a3730eeb7f" splitId="HqttDMqz"
 splitIndex="1 of 2" subcomp="policy" update="true" username="admin"] UserName="admin",
 ModuleName="PolicyEdgeFirewall", Operation="PatchGatewayRule", Operation status="success",
 Old value=[{"sequence_number":10,"source_groups":["ANY"],"destination_groups":
["ANY"],"services":["ANY"],"action":"ALLOW","logged":false,"scope":["/infra/tier-1s/
myT1"],"disabled":false,"direction":"IN_OUT","resource_type":"CommunicationEntry","id":"myT1_Rule1","display_name":"myT
infra/domains/default/gateway-policies/T1-Policies/rules/
myT1_Rule1","relative_path":"myT1_Rule1","parent_path":"/infra/domains/default/gateway-policies/T1-
Policies","unique_id":"2028","marked_for_delete":false,"overridden":false,"_create_user":"admin","_create_time":1597185
 New value=["default" "T1-Policies"
 "myT1_Rule1" {"action":"ALLOW","resource_type":"Rule","id":"myT1_Rule1","display_name":"myT1_Rule1_Updated","path":

<182>1 2020-08-11T22:36:19.430Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager"
 level="INFO" reqId="a17fcbdc-1aed-4526-93e9-40a3730eeb7f" splitId="HqttDMqz" splitIndex="2 of 2"
 subcomp="policy" update="true" username="admin"] "/infra/domains/default/gateway-policies/T1-Policies/rules/
myT1_Rule1","unique_id":"2028","marked_for_delete":false,"overridden":false,"rule_id":2028,"sequence_number":10,"source
["ANY"],"destination_groups":["ANY"],"services":["ANY"],"profiles":["ANY"],"logged":false,"scope":["/infra/
tier-1s/
myT1"],"disabled":false,"direction":"IN_OUT","ip_protocol":"IPV4_IPV6","is_default":false,"_protection":"UNKNOWN","_rev

<182>1 2020-08-11T22:36:19.443Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager" level="INFO"
 reqId="a17fcbdc-1aed-4526-93e9-40a3730eeb7f" splitId="fMYsYjV5" splitIndex="1 of 2" subcomp="policy"
 update="true"] UserName="admin", ModuleName="Policy", Operation="PatchInfra", Operation status="success", New
 value=[{"enforce_revision_check":true} {"resource_type":"Infra","children":[{"children":[{"GatewayPolicy":
{"resource_type":"GatewayPolicy","id":"T1-Policies","display_name":"T1-Policies","path":"/infra/domains/
default/gateway-policies/T1-Policies","unique_id":"a73c1345-6b4e-43e0-b4ee-9a91c7ba9df6","children":[{"Rule":
{"action":"ALLOW","resource_type":"Rule","id":"myT1_Rule1","display_name":"myT1_Rule1_Updated","path":"/
infra/domains/default/gateway-policies/T1-Policies/rules/
myT1_Rule1","unique_id":"2028","marked_for_delete":false,"overridden":false,"rule_id":2028,"sequence_number":10,"source
["ANY"],"destination_groups":["ANY"],"services":["ANY"],"profiles":["ANY"],"logged":false,"scope":["/infra/
tier-1s/
myT1"],"disabled":false,"direction":"IN_OUT","ip_protocol":"IPV4_IPV6","is_default":false,"_protection":"UNKNOWN","_rev
```

```
<182>1 2020-08-11T22:36:19.443Z manager1 NSX 22164 - [nsx@6876 audit="true"
 comp="nsx-manager" level="INFO" reqId="a17fcbdc-1aed-4526-93e9-40a3730eeb7f"
 splitId="fMYsYjV5" splitIndex="2 of 2" subcomp="policy" update="true"]
 0},"resource_type":"ChildRule","marked_for_delete":false,"mark_for_override":false,"_protection":"UNKNOWN"}],"marked_f
```

Delete a rule (`myT1_Rule2`) from a section (`T1-Policies`):

```
<182>1 2020-08-11T22:38:03.262Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager"
 entId="myT1_Rule2" level="INFO" reqId="ccb8d0bb-0fe2-415a-9979-a1a3a80a7038" subcomp="policy" update="true"
 username="admin"] UserName="admin", ModuleName="PolicyEdgeFirewall", Operation="DeleteGatewayRule",
 Operation status="success", Old value=[{"sequence_number":20,"source_groups":["ANY"],"destination_groups":
["ANY"],"services":["ANY"],"action":"ALLOW","logged":false,"scope":["/infra/tier-1s/
myT1"],"disabled":false,"direction":"IN_OUT","resource_type":"CommunicationEntry","id":"myT1_Rule2","display_name":"myT
infra/domains/default/gateway-policies/T1-Policies/rules/
myT1_Rule2","relative_path":"myT1_Rule2","parent_path":"/infra/domains/default/gateway-policies/T1-
Policies","unique_id":"2029","marked_for_delete":false,"overridden":false,"_create_user":"admin","_create_time":1597185
 New value=["default" "T1-Policies" "myT1_Rule2"]

<182>1 2020-08-11T22:38:03.280Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager" level="INFO"
 reqId="ccb8d0bb-0fe2-415a-9979-a1a3a80a7038" splitId="GlUhKvqu" splitIndex="1 of 2" subcomp="policy"
 update="true"] UserName="admin", ModuleName="PolicyEdgeFirewall", Operation="PatchGatewayPolicyForDomain",
 Operation status="success", New value=["default" "T1-Policies" {"resource_type":"GatewayPolicy","id":"T1-
Policies","display_name":"T1-Policies","path":"/infra/domains/default/gateway-policies/
T1-Policies","unique_id":"a73c1345-6b4e-43e0-b4ee-9a91c7ba9df6","children":[{"Rule":
{"resource_type":"Rule","id":"myT1_Rule2","path":"/infra/domains/default/gateway-policies/T1-Policies/rules/
myT1_Rule2","marked_for_delete":true,"overridden":false,"sources_excluded":false,"destinations_excluded":false,"logged"

<182>1 2020-08-11T22:38:03.280Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager" level="INFO"
 reqId="ccb8d0bb-0fe2-415a-9979-a1a3a80a7038" splitId="GlUhKvqu" splitIndex="2 of 2" subcomp="policy"
 update="true"] false,"_protection":"UNKNOWN","_revision":0}]

<182>1 2020-08-11T22:38:03.295Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager" level="INFO"
 reqId="ccb8d0bb-0fe2-415a-9979-a1a3a80a7038" splitId="xnO9T8NE" splitIndex="1 of 2" subcomp="policy"
 update="true"] UserName="admin", ModuleName="Policy", Operation="PatchInfra", Operation status="success", New
 value=[{"enforce_revision_check":true} {"resource_type":"Infra","children":[{"children":[{"GatewayPolicy":
{"resource_type":"GatewayPolicy","id":"T1-Policies","display_name":"T1-Policies","path":"/infra/domains/
default/gateway-policies/T1-Policies","unique_id":"a73c1345-6b4e-43e0-b4ee-9a91c7ba9df6","children":[{"Rule":
{"resource_type":"Rule","id":"myT1_Rule2","path":"/infra/domains/default/gateway-policies/T1-Policies/rules/
myT1_Rule2","marked_for_delete":true,"overridden":false,"sources_excluded":false,"destinations_excluded":false,"logged"

<182>1 2020-08-11T22:38:03.295Z manager1 NSX 22164 - [nsx@6876 audit="true"
 comp="nsx-manager" level="INFO" reqId="ccb8d0bb-0fe2-415a-9979-a1a3a80a7038"
 splitId="xnO9T8NE" splitIndex="2 of 2" subcomp="policy" update="true"]
 false,"lock_modified_time":0,"is_default":false,"_protection":"UNKNOWN","_revision":0},"resource_type":"ChildGatewayPo
```

Deleting a section (`T1-Policies`) that contains a rule (`myT1_Rule1_Updated`):

```
<182>1 2020-08-11T22:41:30.726Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager"
 entId="T1-Policies" level="INFO" reqId="d751343c-32ab-46ee-b176-752b8ae1ec0d" splitId="WZc3oxDG"
 splitIndex="1 of 2" subcomp="policy" update="true" username="admin"] UserName="admin",
 ModuleName="PolicyEdgeFirewall", Operation="DeleteGatewayPolicy", Operation status="success", Old
 value=[{"precedence":10,"category":"LocalGatewayRules","resource_type":"CommunicationMap","id":"T1-
Policies","display_name":"T1-Policies","path":"/infra/domains/default/gateway-
policies/T1-Policies","relative_path":"T1-Policies","parent_path":"/infra/domains/
default","unique_id":"a73c1345-6b4e-43e0-
b4ee-9a91c7ba9df6","marked_for_delete":false,"overridden":false,"_create_user":"admin","_create_time":1597185086789,"_l
{"sequence_number":10,"source_groups":["ANY"],"destination_groups":["ANY"],"services":
["ANY"],"action":"ALLOW","logged":false,"scope":["/infra/tier-1s/
myT1"],"disabled":false,"direction":"IN_OUT","resource_type":"CommunicationEntry","id":"myT1_Rule1","display_name":"myT
```

```
<182>1 2020-08-11T22:41:30.726Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager" entId="T1-
Policies" level="INFO" reqId="d751343c-32ab-46ee-b176-752b8ae1ec0d" splitId="WZc3oxDG" splitIndex="2 of
 2" subcomp="policy" update="true" username="admin"] "/infra/domains/default/gateway-policies/T1-Policies/
rules/myT1_Rule1","relative_path":"myT1_Rule1","parent_path":"/infra/domains/default/gateway-policies/T1-
Policies","unique_id":"2028","marked_for_delete":false,"overridden":false,"_create_user":"admin","_create_time":1597185
 New value=["default" "T1-Policies"]

<182>1 2020-08-11T22:41:30.733Z manager1 NSX 22164 - [nsx@6876 audit="true" comp="nsx-manager"
 level="INFO" reqId="d751343c-32ab-46ee-b176-752b8ae1ec0d" subcomp="policy" update="true"]
 UserName="admin", ModuleName="Policy", Operation="PatchInfra", Operation status="success", New
 value=[{"enforce_revision_check":true} {"resource_type":"Infra","children":[{"children":[{"GatewayPolicy":
{"resource_type":"GatewayPolicy","id":"T1-Policies","path":"/infra/domains/default/gateway-policies/T1-
Policies","marked_for_delete":true,"overridden":false,"locked":false,"_protection":"UNKNOWN"},"resource_type":"ChildGat
```

Monitoring Logs

NSX logging is important and useful for troubleshooting. NSX Manager supports Logging Monitor to control the quality and quantity of logs for internal tests and product environment. The default mode of Logging Monitor is set to the product environment. Logging Monitor leverages the System Health Agent (SHA) framework to keep a check on the log generation rate, infer log file retention time, and generate log duration alarms to indicate abnormal log rotation rates and potential log spews. By monitoring the logs, NSX Manager can find issues before valuable logs are flushed and suggest preventive actions that you can take.

Logging Monitor checks log duration in runtime. It periodically monitors the logging rate (over the last *LOG_RATE_PERIOD* which has a default value of 1 hour) and calculates the estimated duration based on log rotation settings of each log. When the estimated duration is less than desired duration for several threshold (which is configurable) times, Logging Monitor gives an indication of potential log spew and raises log duration alarms for you to take proper actions. Meanwhile, Logging Monitor also generates a report with more details about the problem. In the product environment mode, Logging Monitor raises only one alarm for all abnormal logs.

The mode and log rate period are set in the variables *LOG_MONIOTR_MODE* and *LOG_RATE_PERIOD* respectively that are defined in the `sha_config.yml` file.

You can also provide a backup disk and notify to SHA using a plugin profile. The SHA plugin will move the oldest compressed log file of the log to the backup disk when the current total size hits a predefined percentage (50%) of the maximum size. The backup disk should be a directory to which SHA can move the log files.

Note the following conditions for different platforms:

- On UA and edge nodes, ensure that the directory can be written by “nsx-sha” user.
- On ESX, the sub-directory under `/var/run/log` can be used to set up the backup directory.

The SHA plugin moves the oldest log file to the backup disk and also performs the following functions:

- Monitors the log files size of a given log.
- Calculates the maximum size based on the rotation configuration.
- Renames the oldest log file with a timestamp and move it to the backup disk when the total size hits the predefined percentage (50%).

Files that are moved to the backup disk will not be considered in log duration monitor.

Customer Experience Improvement Program

NSX participates in VMware’s Customer Experience Improvement Program (CEIP).

Details regarding the data collected through CEIP and the purposes for which it is used by VMware are set forth at the Trust & Assurance Center at https://www.vmware.com/solutions/trustvmware/ceip-products.html.

To join or leave the CEIP for NSX, or edit program settings, see Edit the Customer Experience Improvement Program Configuration.

Edit the Customer Experience Improvement Program Configuration

When you install or upgrade NSX Manager, you can decide to join the CEIP and configure data collection settings.

- Verify that the NSX Manager is connected and can synchronize with your hypervisor.
- Verify that NSX is connected to a public network for uploading data.

You can also edit the existing CEIP configuration to join or leave the CEIP program, define the frequency and the days the information is collected, and proxy server configuration.

1. With admin privileges, log in to NSX Manager.
2. Select **System** > **General Settings** > **Customer Program**.
3. Click **Edit** in the Customer Experience Improvement Program section.
4. In the Edit Customer Experience Program dialog box, select the **Join the VMware Customer Experience Improvement Program** check box.
5. Toggle the **Schedule** switch to disable or enable the data collection.
   The schedule is enabled by default.
6. Optional: Configure the data collection and upload recurrence settings.
7. Click **Save**.
8. If you configured the recurrence settings in step 6, you must restart the telemetry service on NSX Manager. If you have an NSX Manager cluster, you must do this on every manager in the cluster.
   a) SSH to NSX Manager and log in as **admin**.
   b) Run the following command to restart the telemetry service:

```
restart service telemetry
```

Find the SSH Fingerprint of a Remote Server

Some tasks that involve communication with a remote server require that you provide the SSH fingerprint for the remote server. The SSH fingerprint is derived from a host key on the remote server.

To connect using SSH, the NSX Manager and the remote server must have a host key type in common. Support includes ECDSA, RSA, and ED25519 key types. The default location of this key is `/etc/ssh`.

Having the fingerprint for a remote server helps you confirm that you are connecting to the correct server, protecting you from man-in-the-middle attacks. You can ask the administrator of the remote server to provide the SSH fingerprint of the server. Or you can connect to the remote server to find the fingerprint. Connecting to the server over the console is more secure than over the network.

1. Log in to the remote server as root.
   Logging in using a console is more secure than over the network.
2. Verify that the required hashed ECDSA, RSA, and ED25519 host keys are present on the backup server by running `ssh-keyscan -t ecdsa,rsa,ed25519 <backup server IP/FQDN> | ssh-keygen -lf -.`

```
#ssh-keyscan -t ecdsa,rsa,ed25519  ftpserver.corp.local | ssh-keygen -lf -

Output:
3072 SHA256:j2ct7Sd0YLesHw/n3LOTvOWyq35t6Celj0QH9gg56p4 ftpserver.corp.local (RSA)
256 SHA256:2yhk6DD/HG+fv1P6sG0ZPJ2UipXpCHXB9EfSlMg6LtE ftpserver.corp.local (ECDSA)
256 SHA256:WiGm7uZono12UX9xJu2vmqz1w8+gJ3Yy6Uv4aNmvaxM ftpserver.corp.local (ED25519)
```

   If the command does not return ECDSA, RSA, or ED25519 keys, then configure the required key on the backup server. Contact the OS vendor if you need guidance for that configuration.
3. Locate ECDSA, RSA, ED25519 keys. Locate the ECDSA key. The default location of the key is `/etc/ssh/`.

```
$ ls -al /etc/ssh/*pub
-rw-r--r-- 1 root root  93 Apr  8 18:10 /etc/ssh/ssh_host_ecdsa_key.pub
-rw-r--r-- 1 root root 393 Apr  8 18:10 /etc/ssh/ssh_host_rsa_key.pub
```

```
-rw-r--r-- 1 root root 393 Apr  8 18:10 /etc/ssh/ssh_host_ed25519_key.pub
```

Configuring an External Load Balancer

You can configure an external load balancer to distribute traffic to the NSX Managers in a manager cluster.

An NSX Manager cluster does not require an external load balancer. The NSX Manager virtual IP (VIP) provides resiliency in the event of a Manager node failure but has the following limitations:

- VIP does not perform load balancing across the NSX Managers.
- VIP requires all the NSX Managers to be in the same subnet.
- VIP recovery takes about 1 - 3 minutes in the event of a Manager node failure.

An external load balancer can provide the following benefits:

- Load balance across the NSX Managers.
- The NSX Managers can be in different subnets.
- Fast recovery time in the event of a Manager node failure.

Authentication Methods When Accessing NSX Manager

The following authentication methods are supported by NSX Manager. For more information about the authentication methods, see the *NSX API Guide*.

- HTTP Basic Authentication
- Session-Based Authentication
- Authentication using an X.509 certificate and a Principal Identity
- Authentication in VMware Cloud on AWS

The session-based authentication method (used when you access NSX Manager from a browser) requires source-IP persistence (all requests from the client must go to the same NSX Manager). The other methods do not require source-IP persistence (requests from the client can go to different NSX Managers).

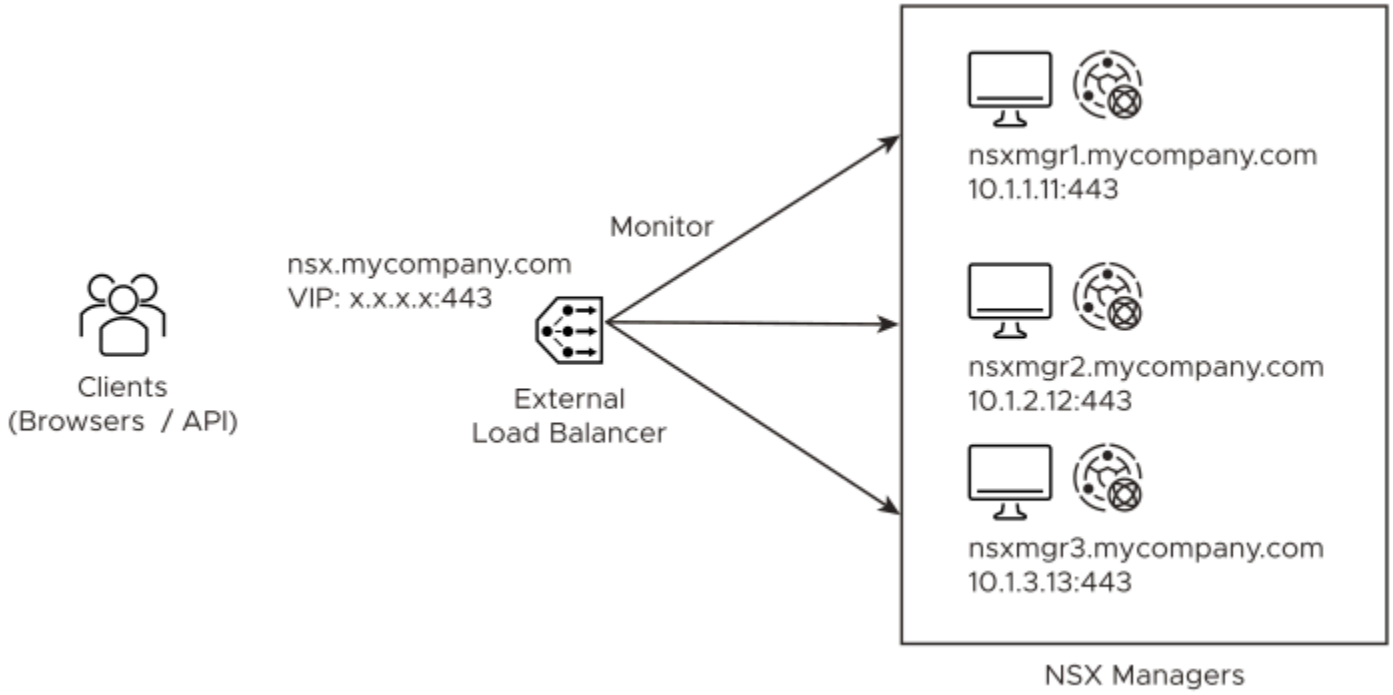
Recommendations

- Create a single VIP on the load balancer with source-IP persistence configured to handle all the authentication methods.
- If you have applications or scripts that might generate a lot of requests to NSX Manager, create a second VIP without source-IP persistence for these applications or scripts. Use the first VIP for browser access to NSX Manager only.

The VIP must have the following configurations:

- Type: Layer4-TCP
- Port: 443
- Pool: NSX Manager Pool
- Persistence: Source-IP persistence for the first VIP. None for the second VIP (if present).

Example of an external load balancer configuration:



NSX Manager's Certificate

The clients access NSX Manager using FQDN name (for example, nsx.mycompany.com). This FQDN is resolved to the load balancer's VIP. To avoid any certificate mismatch, each NSX Manager must have a certificate that is valid for the VIP's FQDN name. Therefore, you must configure each NSX Manager with a SAN certificate that is valid for its own name (for example, nsxmgr1.mycompany.com) and the VIP's FQDN.

Monitoring the Health of NSX Managers

The load balancer can check that each NSX Manager is running with the following API:

```
GET /api/v1/reverse-proxy/node/health
```

The request headers are:

- Header 1
  - Name: Content-Type
  - Value: application/json
- Header 2
  - Name: Accept
  - Value: application/json

A response indicating that the NSX Manager is running will be:

```
"healthy" : true
```

Note that the format of the response is `"healthy"<space>:<space>true`.

Configure Proxy Settings

To route and monitor all internet-bound HTTP/HTTPS traffic through a reliable HTTP/HTTPS Proxy, you can configure proxy settings for your NSX Manager environment.

For HTTPS proxy, you must have a proxy server self-signed certificate available in NSX Manager. See Creating Self-signed Certificates for NSX to create or import an existing self-signed certificate.

Do not set up proxy configuration on an NSX Edge, since transport nodes require direct Internet connectivity for FQDN analysis and for URL filtering.

1. With admin privileges, log in to NSX Manager.
2. Select **System** > **General Settings**.
3. To create an optional proxy configuration the first time, click the highlighted button on the Internet Proxy Server page or continue to the next step.
4. To turn on proxy server configuration, toggle the **Proxy Enabled** to Yes.
5. In the **Scheme** column, select **HTTP** or **HTTPS**.
6. In the **Host** field, enter an IP address or FQDN.
7. In the **Port** field, enter a port number.
   In general, the default port is 3128, but you can configure a different port if needed. This number should be unique and not already in use by other services on the network. This port must allow incoming traffic from the NSX components that need to use the proxy server.
8. Optional: In the **Username** field, enter a user name.
9. Optional: In the **Password** field, enter a password.
10. Optional: In the **Test Connection URL**, enter any web URL for confirmation of configuration success.
    For example: https://www.broadcom.com
11. If you selected HTTPS, in the **Certificate** drop-down, select a proxy certificate trusted by NSX.
12. Click **Save**.

Back up and restore NSX configured in vCenter

Back up and restore NSX Manager that is configured in vCenter from the vSphere Client.

> **Note:**
> When you restore NSX Manager, use the `nsx-unified-appliance-<releaseversion.buildversion>.ova` file.

1. From your browser, log in with admin privileges to a vCenter (version 7.0.3 or later) at https://< vCenter>.
2. Click **Login** to **Launch vSphere Client**.
3. In the vSphere Client, from the main menu list, click NSX. The NSX page is displayed.
4. From the vSphere Client, accesss the NSX Manager UI.
5. Configure a backup location and perform backup. See Configure Backups.
6. If the NSX Manager is deleted for some reason, connectivity between the NSX Manager and vCenter is lost.
7. With admin privileges, log in to vCenter, at https://< vCenter>.
8. Install NSX Manager. See *Install NSX Manager and Available Appliances* topic in the *NSX Installation Guide*.
9. Configure a backup location. See Configure Backups.
10. Restore the backup. See Restore a Backup.

Restore is complete. You can access the NSX UI from vCenter.

Scale Out NSX Manager

Use the information in this section to deploy a cluster of three NSX Manager nodes, each running on a separate ESX host.

Installing an NSX Manager Cluster

As an administrator, your first task will be to install NSX in setting up your NSX environment.

NSX Manager is the application that you use to administer your NSX environment. You can use the NSX Manager UI, API or CLI to manage workloads and NSX Edge nodes. In a production environment, for fault tolerance, deploy a cluster of three NSX Manager nodes, each running on a separate ESX host.

NSX Manager can be deployed on an ESX host and this section covers the use of vSphere Client to deploy the NSX Manager virtual appliances OVA/OVF on an ESX host.

As an admin, your first task is to install NSX Manager as part of setting up your NSX environment.

Make sure that you have the supported vSphere version. See vSphere support.

After you install the NSX Manager cluster, register a vCenter instance as the compute manager. See Add a Compute Manager.

Install NSX Manager and Available Appliances

You use the vSphere Client to deploy NSX Manager virtual appliances. The same OVF file can be used to deploy different types of appliances: NSX Manager, and Global Manager for NSX Federation.

- Verify that the system requirements are met. See System Requirements.
- Verify that the required ports are open. See Ports and Protocols.
- Verify that a datastore is configured and accessible on the ESX host.
- Verify that you have the IP address and gateway, DNS server IP addresses, domain search list, and the NTP Server IP or FQDN for the NSX Managerto use.
- Create a management VDS and target VM port group in vCenter. Place the NSX appliances onto this management VDS port group network. See Prepare a vSphere Distributed Switch for NSX.
  Multiple management networks can be used as long as the NSX Manager nodes has consistent connectivity and recommended latency between them.

  Note:
  If you plan to use Cluster VIP, all NSX Manager appliances should belong to same subnet.
- Plan your NSX Manager IP and NSX Manager Cluster VIP addressing scheme.

  Note:
  Verify that you have the hostname for NSX Manager to use. The Hostname format must be `nsx-manager-fqdn@domain-name.com`. This format is required if NSX installation is dual stack (IPv4, IPv6) and/or if planning to configure CA-signed certificates.

1. Locate and download the NSX OVA file from the **My Downloads** panel on the Broadcom Support page. You can either copy the download URL or download the OVA file.
2. In the vSphere Client, select the ESX host or ESX host cluster on which to install NSX.
3. Right-click Host and select **Deploy OVF template** to start the installation wizard.
4. Enter the download OVA URL or navigate to the OVA file, and click **Next**.
5. Enter a name and a location for the NSX Manager VM, and click **Next**.

   The name you enter appears in the vSphere and vCenter inventory.
6. Select a compute resource for the NSX Manager appliance, and click **Next**.
   - To install on a ESX host managed by vCenter, select a host on which to deploy the NSX Manager appliance.
   - To install on a standalone ESX host, select the host on which to deploy the NSX Manager appliance.
7. Review and verify the OVF template details, and click **Next**.
8. Specify the deployment configuration size, and click **Next**.
   The Description panel on the right side of the wizard shows details of the selected configuration.

9. Specify storage for the configuration and disk files.
   a) Select the virtual disk format.
   b) Select the VM storage policy.
   c) Specify the datastore to store the NSX Manager appliance files.
   d) Click **Next**.
10. Select a destination network for each source network.
11. Select the port group or destination network for the NSX Manager.
12. Configure IP Allocation settings.
    a) For IP allocation, specify `Static - Manual`.
    b) For IP protocol, select `IPv4` or `IPv6`.

    Note:
    You can ignore the IP Allocation settings. You can select either IPv4 or IPv6. It would not impact ingress or egress network traffic of NSX Manager.
13. Click **Next**.

    The following steps are all located in the Customize Template section of the Deploy OVF Template wizard.
14. In the Application section, enter the system root, CLI admin, and audit passwords for the NSX Manager. The `root` and `admin` credentials are mandatory fields.

    Your passwords must comply with the password strength restrictions.

    - At least 12 characters
    - At least one lower-case letter
    - At least one upper-case letter
    - At least one digit
    - At least one special character
    - At least five different characters
    - Default password complexity rules are enforced by the following Linux PAM module arguments:
      - `retry=3`: The maximum number of times a new password can be entered, for this argument at the most 3 times, before returning with an error.
      - `minlen=12`: The minimum acceptable size for the new password. In addition to the number of characters in the new password, credit (of +1 in length) is given for each different kind of character (other, upper, lower and digit).
      - `difok=0`: The minimum number of bytes that must be different in the new password. Indicates similarity between the old and new password. With a value 0 assigned to `difok`, there is no requirement for any byte of the old and new password to be different. An exact match is allowed.
      - `lcredit=1`: The maximum credit for having lower case letters in the new password. If you have less than or 1 lower case letter, each letter will count +1 towards meeting the current `minlen` value.
      - `ucredit=1`: The maximum credit for having upper case letters in the new password. If you have less than or 1 upper case letter each letter will count +1 towards meeting the current `minlen` value.
      - `dcredit=1`: The maximum credit for having digits in the new password. If you have less than or 1 digit, each digit will count +1 towards meeting the current `minlen` value.
      - `ocredit=1`: The maximum credit for having other characters in the new password. If you have less than or 1 other characters, each character will count +1 towards meeting the current minlen value.
      - `enforce_for_root`: The password is set for the root user.

    Note:
    For more details on Linux PAM module to check the password against dictionary words, refer to the man page.

    For example, avoid simple and systematic passwords such as `VMware123!123` or `VMware12345`. Passwords that meet complexity standards are not simple and systematic but are a combination of letters, alphabets, special characters, and numbers, such as `VMware123!45`, `VMware 1!2345` or `VMware@1az23x`.

15. In the Optional parameters section, leave the password fields blank. It avoids the risk of compromising passwords set for VMC roles by a user who has access to the vCenter. When deploying VMC for NSX, this field is used internally to set passwords for the Cloud Admin and Cloud Operator roles.
16. In the Network Properties section, enter the hostname of the NSX Manager.

    Note:
    The host name must be a valid domain name. Ensure that each part of the host name (domain/subdomain) that is separated by dot starts with an alphabet character. Also, NSX accepts only latin alphabets that do not have an accent mark, as in í, ó, ú, ý.

    Important:
    If you plan to install NSX in dual stack (IPv4 and IPv6) and/or if you plan to configure CA-signed certificates, then enter a Hostname with valid domain name to NSX Manager VMs and Cluster VIP (if configured).
17. Select a **Rolename** for the appliance. The default role is **NSX Manager**.
    - To install an NSX Manager appliance, select the **NSX Manager** role.
    - To install a Global Manager appliance for a NSX Federation deployment, select the **NSX Global Manager** role. See Getting Started with NSX Federation for details.
18. Enter a default gateway, management network IP address (required), and management network netmask (required).

    Note:
    Entering a default gateway is optional. However, you cannot configure it after deploying NSX Manager.
19. In the DNS section, enter DNS Server list and Domain Search list.
20. In the Services Configuration section, enter NTP Server IP or FQDN.

    Optionally, you can enable SSH service and allow root SSH login. But, it is not recommended to allow root access to SSH service.
21. Verify that all your custom OVF template specification is accurate and click **Finish** to begin installation.

    The installation might take 7-8 minutes.
22. From the vSphere Client, verify that the VM is powered on. Open the VM console to track the boot process of the node.
23. After the VM node boots a second time, log in to the CLI as admin and run the `get interface eth0` command to verify that the IP address was applied as expected.
24. Enter the `get services` command after waiting for about 5 minutes to verify that all default services are running.

    The following services are not required by default and do not start automatically.
    - `liagent`
    - `migration-coordinator`: This service is used only when running migration coordinator. See the *NSX Migration Guide* before starting this service.
    - `snmp`: For information on starting SNMP see *Simple Network Management Protocol* in the *NSX Administration Guide*.
    - `nsx-message-bus`: This service is not used in NSX 3.0 and later releases.
25. After deployment, verify that the NSX Manager UI comes up by accessing the following URL, `https://nsx-manager-ip` or `https://nsx-manager-fqdn`.
26. Verify that your NSX Manager,or Global Manager node has the required connectivity.

    Perform the following tasks:
    - Ping your node from another machine.
    - The node can ping its default gateway.
    - The node can ping the hypervisor hosts that are in the same network using the management interface.
    - The node can ping its DNS server and its NTP Server or FQDN.
    - If you enabled SSH, make sure that you can SSH to your node.

If connectivity is not established, make sure that the network adapter of the virtual appliance is in the proper network or VLAN.

27. Troubleshooting OVA failures:

    Note:
    During deployment, if you entered incorrect configuration details, delete the appliance and redeploy with correct configuration.

    a) Verify that the datastore chosen for deployment is mounted on all the hosts that are members of a cluster. Redeploy and choose ESX host instead of vCenter to bypass vCenter cluster related checks.
    b) If a proxy is enabled on vCenter, edit the file /etc/sysconfig/proxy and add the line `.*.domainname` to bypass the proxy for ESX hosts. See Knowledge Base article 321922: Unable to deploy OVF using vSphere Client in vCenter when an HTTPS Proxy is configured.
    c) If the deployment of the appliance through the OVF tool results in the error `ovf descriptor not found`, view the file contents in terminal `cat -A <filepath/filename>` and remove hidden formatting characters. Then try again.

28. Troubleshooting issues related to bringing up the appliance. SSH to NSX Manager CLI as admin and run the following commands to troubleshoot:
    a) Run `get configuration` and verify `hostname/name-server/search-domain/ntp` settings are correct.
    b) Run `get services` and verify all required services are running (other than `nsx-message-bus`, `snmp`, `migration-coordinator`). If these services are not running, try restarting the service by running `restart service <service-name>`.
    c) Run `get cluster status` and verify all manager cluster components are up. If any component is down, try restarting the service associated to the component by running `restart service <associated-component-service-name>`.
    d) Run `get core-dumps` to verify no core dumps generated in `/var/log/core` or `/image/core`. If you find any core dumps, contact VMware Support.
    e) Run `get filesystem-status` to verify that no disk partition is full, especially those partitions that are consumed by NSX.
    f) Alternatively, you can run API commands to know the node and service status.

    ```
    GET api/v1/node/status
    GET api/v1/node/services
    ```

- Log in to the NSX Manager from a supported web browser. See Log In to the Newly Installed NSX Manager.
- Deploy NSX Manager Nodes to Form a Cluster from the UI.

Log In to the Newly Installed NSX Manager

After you install NSX Manager, log in to the NSX Manager UI to perform additional tasks.

Verify that NSX Manager is installed. See Install NSX Manager and Available Appliances.

After you install NSX Manager, you can join the Customer Experience Improvement Program (CEIP) for NSX. See Customer Experience Improvement Program in the *NSX Administration Guide* for more information about the program, including how to join or leave the program later.

1. From a browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.

   The EULA appears.

2. Read and accept the EULA terms.
3. Select whether to join the VMware's Customer Experience Improvement Program (CEIP).
4. Click **Save**.

Deploy NSX Manager Nodes to Form a Cluster from the UI

Forming an NSX Manager or Global Manager cluster provides high availability and reliability. Deploying nodes using the UI is supported only on ESX hosts managed by vCenter that is added as a compute manager.

- Verify that an NSX Manager or Global Manager node is installed. See Install NSX Manager and Available Appliances.
- Verify that compute manager is configured. See Add a Compute Manager.
- Verify that the system requirements are met. See Planning and Preparation.
- Verify that the required ports are open. Refer to VMware Ports and Protocols for more details.

For deploying additional NSX Manager nodes on a vCenter that is not added as a compute manager, see Install NSX Manager on ESX Using the Command-Line OVF Tool and Deploy NSX Manager Nodes to Form a Cluster from the UI.

When you deploy a new node from the UI, the node connects to the first deployed node to form a cluster. All the repository details and the password of the first deployed node are synchronized with the newly deployed node. The first node is known as the orchestrator node because it contains the original copy of the VIBs and installation files required to prepare hosts of the cluster. The orchestrator node also help identify the node on which the Upgrade-Coordinator is running. When new nodes are added to the cluster, NSX uses the repository IP to synchronize the repository of VIBs and installation files on the new nodes of the cluster.

To create an NSX Manager cluster, deploy two additional nodes to form a cluster of three nodes total.

> **Note:**
>
> Data is replicated to all the active NSX Manager nodes of the cluster. So, when the NSX Manager cluster is stable, every NSX Manager node contains the same data.

1. From a browser, log in with admin privileges to the NSX Manager or Global Manager at https://<manager-ip-address>.
2. Deploy an appliance.
   - From NSX Manager, select .
   - From Global Manager, select .
3. Enter the appliance information details.

Option	Description
Host Name or FQDN	Enter a name for the node. **Note:** If you plan to deploy NSX Manager in dual stack mode (IPv4 and IPv6) and if you plan to configure NSX Manager with CA signed certificates, you must set a FQDN with valid domain name.
IP Type	Select the IP type. The appliance can have IPv4 address only, dual stack address (both IPv4 and IPv6), or IPv6 address only. Note that if you have deployed the first appliance node with IPv4 address only or dual stack address (both IPv4 and IPv6), you cannot deploy additional appliance nodes with IPv6 address only.
Management IPv4/Netmask	Enter an IPv4 address to be assigned to the node. This field is not displayed if you have selected **IP Type** as **IPv6**.
Management Gateway IPv4	Enter a gateway IPv4 address to be used by the node. This field is not displayed if you have selected **IP Type** as **IPv6**.
Management IPv6/Netmask	Enter an IPv6 address to be assigned to the node. This option appears when **IP Type** is set to **Both IPv4 and IPv6** or **IPv6**.

Option	Description
Management Gateway IPv6	Enter a gateway IPv6 address to be used by the node. This option appears when **IP Type** is set to **Both IPv4 and IPv6** or **IPv6**.
DNS Servers	Enter DNS server IP addresses to be used by the node.
NTP Server	Enter an NTP server IP address to be used by the node.
Node Size	Select the form factor to deploy the node from the following options: • Small (4 vCPU, 16 GB RAM, 300 GB storage) • Medium (6 vCPU, 24 GB RAM, 300 GB storage) • Large (12 vCPU, 48 GB RAM, 300 GB storage) • Extra Large (24 vCPU, 96 GB RAM, 400 GB storage) For Global Manager select size: • **Medium** GM appliance for deployments up to four locations and 128 hypervisors across all locations • **Large** or **Extra Large** GM appliance for deployments with higher scale Do not use **Small** GM appliance for scale deployment. For details see the VMware Configuration Maximums tool.

4. Enter the configuration details.

Option	Description
Compute Manager	Select the vCenter to provision compute resources for deploying the node.
Compute Cluster	Select the cluster the node is going to join.
(Optional) Resource Pool	Select a resource pool for the node from the drop-down menu.
(Optional) Host	Select a host for the node from the drop-down menu.
Datastore	Select a datastore for the node files from the drop-down menu.
Virtual Disk Format	• For NFS datastores, select a virtual disk format from the available provisioned policies on the underlying datastore. – With hardware acceleration, **Thin Provision**, **Thick Provision Lazy Zeroed**, and **Thick Provision Eager Zeroed** formats are supported. – Without hardware acceleration, only **Thin Provision** format is supported. • For VMFS datastores, **Thin Provision**, **Thick Provision Lazy Zeroed**, and **Thick Provision Eager Zeroed** formats are supported. • For vSAN datastores, you cannot select a virtual disk format because the VM storage policy defines the format. – The vSAN storage policies determine the disk format. The default virtual disk format for vSAN is Thin Provision. You can change the vSAN storage policies to set a percentage of the virtual disk that must be thick-provisioned. By default, the virtual disk for an NSX Manager or Global Manager node is prepared in the **Thin Provision** format. **Note:** You can provision each node with a different disk format based on which policies are provisioned on the datastore.
Network	Click **Select Network** to select the management network for the node.

5. Enter the access and credentials details.

Option	Description
Enable SSH	Toggle the button to allow an SSH login to the new node.
Enable Root Access	Toggle the button to allow root access to the new node.
System Root Credentials	Set the root password and confirm the password for the new node.

Option	Description
	Your password must comply with the password strength restrictions. • At least 12 characters • At least one lower-case letter • At least one upper-case letter • At least one digit • At least one special character • At least five different characters • Default password complexity rules are enforced by the following Linux PAM module arguments: – `retry=3`: The maximum number of times a new password can be entered, for this argument at the most 3 times, before returning with an error. – `minlen=12`: The minimum acceptable size for the new password. In addition to the number of characters in the new password, credit (of +1 in length) is given for each different kind of character (other, upper, lower and digit). – `difok=0`: The minimum number of bytes that must be different in the new password. Indicates similarity between the old and new password. With a value 0 assigned to `difok`, there is no requirement for any byte of the old and new password to be different. An exact match is allowed. – `lcredit=1`: The maximum credit for having lower case letters in the new password. If you have less than or 1 lower case letter, each letter will count +1 towards meeting the current `minlen` value. – `ucredit=1`: The maximum credit for having upper case letters in the new password. If you have less than or 1 upper case letter each letter will count +1 towards meeting the current `minlen` value. – `dcredit=1`: The maximum credit for having digits in the new password. If you have less than or 1 digit, each digit will count +1 towards meeting the current `minlen` value. – `ocredit=1`: The maximum credit for having other characters in the new password. If you have less than or 1 other characters, each character will count +1 towards meeting the current minlen value. – `enforce_for_root`: The password is set for the root user. **Note:** For more details on Linux PAM module to check the password against dictionary words, refer to the man page. For example, avoid simple and systematic passwords such as `VMware123!123` or `VMware12345`. Passwords that meet complexity standards are not simple and systematic but are a combination of letters, alphabets, special characters, and numbers, such as `VMware123!45`, `VMware 1!2345` or `VMware@1az23x`.
Admin CLI Credentials and Audit CLI Credentials	Select the **Same as root password** check box to use the same password that you configured for root, or deselect the check box and set a different password.

6. Click **Install Appliance**.

   The new node is deployed. You can track the deployment process in the page for NSX Manager, the for Global Manager, or the vCenter for either. Do not add additional nodes until the installation is finished and the cluster is stable.

7. Wait for the deployment, cluster formation, and repository synchronization to finish.

   The joining and cluster stabilizing process might take from 10 to 15 minutes. After the node boots (as part of appliance bringup) and is back up, log in to the first deployed NSX Manager CLI as an admin. Run `get cluster status` to view the status. Verify that the first node and second node are members of the cluster and the status for every cluster service group is `UP` and cluster status `STABLE` before making any other cluster changes.

   Note:

   - If you reboot the first deployed NSX Manager node, when the deployment of a new node is in progress, the new node might fail to register with the cluster. It displays the `Failed to Register` message on the new node's thumbnail. To redeploy the node manually on the cluster, delete and redeploy the failed node again.
   - If a node deployment fails, you cannot reuse the same IP address to deploy another node until the failed node is deleted.
   - NSX Manager nodes that are removed from the cluster should be powered off or deleted. Do not reuse it in your deployments.

8. After the VM node boots a second time, log in to the CLI as admin and run the `get interface eth0` command to verify that the IP address was applied as expected.

9. Verify that your NSX Manager,or Global Manager node has the required connectivity.

   Perform the following tasks:

   - Ping your node from another machine.
   - The node can ping its default gateway.
   - The node can ping the hypervisor hosts that are in the same network using the management interface.
   - The node can ping its DNS server and its NTP Server or FQDN.
   - If you enabled SSH, make sure that you can SSH to your node.

   If connectivity is not established, make sure that the network adapter of the virtual appliance is in the proper network or VLAN.

10. If your cluster has only two nodes, add another appliance.

    - From NSX Manager, select and repeat the configuration steps.
    - From Global Manager, select and repeat the configuration steps.

    Important:
    If the orchestrator node (first NSX Manager node deployed) goes down or becomes unreachable while additional nodes in the cluster have not finished replicating with first node fully, further operations on the cluster will fail.
    To avoid this, bring up the first node up successfully and then delete and redeploy the additional nodes that had failed to finish replication of data with the first node.

    Note:
    NSX Manager nodes that are removed from the cluster should be powered off or deleted. Do not reuse it in your deployments.

11. Before adding or removing an existing node from the cluster, place all Security Intelligence and NDR SaaS agents in maintenance mode.

```
PUT https://nsx-manager-ip/policy/api/v1/infra/sites/agents/intelligence/maintenance
 {
 "enable": true
 }
```

12. After adding or removing nodes from the cluster, take all the Security Intelligence and NDR SaaS agents out of maintenance mode.

```
PUT https://nsx-manager-ip/policy/api/v1/infra/sites/agents/intelligence/maintenance
 {
 "enable": false
 }
```

Configure NSX Edge. See Install an NSX Edge on ESX Using the vSphere GUI.

Install NSX Manager on ESX Using the Command-Line OVF Tool

If you prefer to automate or use CLI for the NSX Manager installation, you can use the VMware OVF Tool, which is a command-line utility.

- You can download the latest OVF tool from the Broadcom Support portal.
- Verify that the system requirements are met. See Planning and Preparation.
- Verify that the required ports are open. Refer to VMware Ports and Protocols for more details.
- Verify that a datastore is configured and accessible on the ESX host.
- Verify that you have the IP address and gateway, DNS server IP addresses, domain search list, and the NTP Server IP or FQDN for the NSX Manager to use.
- Create a management VDS and target VM port group in vCenter. Place the NSX appliances onto this management VDS port group network. See Prepare a vSphere Distributed Switch for NSX. Multiple management networks can be used as long as the NSX Manager nodes has consistent connectivity and recommended latency between them.

  Note:

  If you plan to use Cluster VIP, all NSX Manager appliances should belong to same subnet.
- Plan your NSX Manager IP and NSX Manager Cluster VIP addressing scheme.

  Note:

  Verify that you have the hostname for NSX Manager to use. The Hostname format must be nsx-manager-fqdn@domain-name.com. This format is required if NSX installation is dual stack (IPv4, IPv6) and/or if planning to configure CA-signed certificates.
- Verify that the system requirements are met. See System Requirements.
- Verify that the required ports are open. See Ports and Protocols.
- Verify that a datastore is configured and accessible on the ESX host.
- Verify that you have the IP address and gateway, DNS server IP addresses, domain search list, and the NTP Server IP or FQDN for the NSX Managerto use.
- Create a management VDS and target VM port group in vCenter. Place the NSX appliances onto this management VDS port group network. See Prepare a vSphere Distributed Switch for NSX.
  Multiple management networks can be used as long as the NSX Manager nodes has consistent connectivity and recommended latency between them.

  Note:
  If you plan to use Cluster VIP, all NSX Manager appliances should belong to same subnet.
- Plan your NSX Manager IP and NSX Manager Cluster VIP addressing scheme.

  Note:
  Verify that you have the hostname for NSX Manager to use. The Hostname format must be `nsx-manager-fqdn@domain-name.com`. This format is required if NSX installation is dual stack (IPv4, IPv6) and/or if planning to configure CA-signed certificates.

By default, `nsx_isSSHEnabled` and `nsx_allowSSHRootLogin` are both disabled for security reasons. When they are disabled, you cannot SSH or log in to the NSX Manager command line. If you enable `nsx_isSSHEnabled` but not `nsx_allowSSHRootLogin`, you can SSH to NSX Manager but you cannot log in as root.

1. Run the `ovftool` command with the appropriate parameters.
   The process depends on whether the host is standalone or managed by vCenter.
   - For a standalone host:

Note:
On a standalone host, if you enter an incorrect role in the `nsx_role` property, then the appliance is deployed in the `NSX Manager` role.

– Windows example:

```
C:\Program Files\VMware\VMware OVF Tool>ovftool \
--name=<nsxmanager>
--X:injectOvfEnv
--X:logFile=ovftool.log
--sourceType=OVA
--vmFolder='Folder-in-VC'
--allowExtraConfig
--datastore=<datastore>
--net:"<network-name-of-OVF>=<network-name>"
--acceptAllEulas
--skipManifestCheck
--noSSLVerify
--diskMode=thin
--quiet
--hideEula
--powerOn
--prop:nsx_ip_0=10.196.176.81
--prop:nsx_netmask_0=255.255.252.0
--prop:nsx_gateway_0=10.196.179.253
--prop:nsx_dns1_0=10.142.7.1
--prop:nsx_domain_0=eng.yourcompany.com
--prop:nsx_ntp_0=10.128.243.14
--prop:nsx_isSSHEnabled=True
--prop:"nsx_passwd_0=<password>"
--prop:"nsx_cli_passwd_0=<password-cli>"
--prop:"nsx_cli_audit_passwd_0=<password-cli-audit>"
--prop:nsx_hostname=<hostname>
--prop:mgrhostname01="nsx-manager-02@yourcompany.com"
--prop:nsx_allowSSHRootLogin=True
--prop:nsx_role="NSX Manager"
--X:logFile=/root/ovftool/<ovf-folder>.log
--X:logLevel=trivia
--ipProtocol=IPv4
--ipAllocationPolicy="fixedPolicy"
<nsx-unified-applicance>.ova
'vi://admin@vsphere.local:password@vc-or-ip.eng.yourcompany.com/<datacenter>/host/
Install/10.196.6.78/'
```

Note:
The above Windows code block uses the backslash (\) to indicate the continuation of the command line. In actual use, omit the backslash and put the entire command in a single line.

Note:
In the above example, 10.168.110.51 is the IP address of the host machine where NSX Manager is to be deployed.

Note:

In the above example, --deploymentOption is set to the default size Medium. To learn about other supported sizes, see Planning and Preparation.

– Linux example:

```
mgrformfactor="small"
ipAllocationPolicy="fixedPolicy"
mgrdatastore="QNAP-Share-VMs"
```

```
mgrnetwork="Management-VLAN-210"

mgrname01="nsx-manager-01"
mgrhostname01="nsx-manager-01"
mgrip01="192.168.210.121"

mgrnetmask="255.255.255.0"
mgrgw="192.168.210.254"
mgrdns="192.168.110.10"
mgrntp="192.168.210.254"
mgrpasswd="<password>"
mgrssh="<True|False>"
mgrroot="<True|False>"
logLevel="trivia"

mgresxhost01="192.168.110.113"

ovftool
--name=<nsxmanager>
--X:injectOvfEnv
--X:logFile=ovftool.log
--sourceType=OVA
--vmFolder='Folder-in-VC'
--allowExtraConfig
--datastore=<datastore>
--net:"<network-name-of-OVF>=<network-name>"
--acceptAllEulas
--skipManifestCheck
--noSSLVerify
--diskMode=thin
--quiet
--hideEula
--powerOn
--prop:nsx_ip_0=10.196.176.81
--prop:nsx_netmask_0=255.255.252.0
--prop:nsx_gateway_0=10.196.179.253
--prop:nsx_dns1_0=10.142.7.1
--prop:nsx_domain_0=eng.yourcompany.com
--prop:nsx_ntp_0=10.128.243.14
--prop:nsx_isSSHEnabled=True
--prop:"nsx_passwd_0=<password>"
--prop:"nsx_cli_passwd_0=<password-cli>"
--prop:"nsx_cli_audit_passwd_0=<password-cli-audit>"
--prop:nsx_hostname=<hostname>
--prop:mgrhostname01="nsx-manager-02@yourcompany.com"
--prop:nsx_allowSSHRootLogin=True
--prop:nsx_role="NSX Manager"
--X:logFile=/root/ovftool/<ovf-folder>.log
--X:logLevel=trivia
--ipProtocol=IPv4
--ipAllocationPolicy="fixedPolicy"
<nsx-unified-applicance>.ova
'vi://admin@vsphere.local:password@vc-or-ip.eng.yourcompany.com/<datacenter>/host/
Install/10.196.6.78/'
```

The result should look something like this:

```
Opening OVA source: nsx-<component>.ova
```

```
The manifest validates
Source is signed and the certificate validates
Opening VI target: vi://root:<password>@<esxi-IP-address>
Deploying to VI: vi://root:<password>@<esxi-IP-address>
Transfer Completed
Powering on VM: NSX Manager
Task Completed
Completed successfully
```

- For a host managed by vCenter:
  - Windows example:

```
C:\Users\Administrator\Downloads>ovftool
--name=<nsxmanager>
--X:injectOvfEnv
--X:logFile=ovftool.log
--sourceType=OVA
--vmFolder='Folder-in-VC'
--allowExtraConfig
--datastore=<datastore>
--net:"<network-name-of-OVF>=<network-name>"
--acceptAllEulas
--skipManifestCheck
--noSSLVerify
--diskMode=thin
--quiet
--hideEula
--powerOn
--prop:nsx_ip_0=10.196.176.81
--prop:nsx_netmask_0=255.255.252.0
--prop:nsx_gateway_0=10.196.179.253
--prop:nsx_dns1_0=10.142.7.1
--prop:nsx_domain_0=eng.yourcompany.com
--prop:nsx_ntp_0=10.128.243.14
--prop:nsx_isSSHEnabled=True
--prop:"nsx_passwd_0=<password>"
--prop:"nsx_cli_passwd_0=<password-cli>"
--prop:"nsx_cli_audit_passwd_0=<password-cli-audit>"
--prop:nsx_hostname=<hostname>
--prop:mgrhostname01="nsx-manager-02@yourcompany.com"
--prop:nsx_allowSSHRootLogin=True
--prop:nsx_role="NSX Manager"
--X:logFile=/root/ovftool/<ovf-folder>.log
--X:logLevel=trivia
--ipProtocol=IPv4
--ipAllocationPolicy="fixedPolicy" <nsx-unified-applicance>.ova
'vi://admin@vsphere.local:password@vc-or-ip.eng.yourcompany.com/<datacenter>/host/
Install/10.196.6.78/'
```

Note:
The above Windows code block uses the backslash (\) to indicate the continuation of the command line. In actual use, omit the backslash and put the entire command in a single line.

Note:

In the above example, --deploymentOption is set to the default size Medium. To learn about other supported sizes, see Planning and Preparation.

  - Linux example:

```
mgrformfactor="small"
ipAllocationPolicy="fixedPolicy"
mgrdatastore="QNAP-Share-VMs"
mgrnetwork="Management-VLAN-210"

mgrname01="nsx-manager-01"
mgrhostname01="nsx-manager-01"
mgrip01="192.168.210.121"

mgrnetmask="255.255.255.0"
mgrgw="192.168.210.254"
mgrdns="192.168.110.10"
mgrntp="192.168.210.254"
mgrpasswd="<password>"
mgrssh="<True|False>"
mgrroot="<True|False>"
logLevel="trivia"

vcadmin="administrator@vsphere.local"
vcpass="<password>"
vcip="192.168.110.151"
mgresxhost01="192.168.110.113"

ovftool
--name=<nsxmanager>
--X:injectOvfEnv
--X:logFile=ovftool.log
--sourceType=OVA
--vmFolder='Folder-in-VC'
--allowExtraConfig
--datastore=<datastore>
--net:"<network-name-of-OVF>=<network-name>"
--acceptAllEulas
--skipManifestCheck
--noSSLVerify
--diskMode=thin
--quiet
--hideEula
--powerOn
--prop:nsx_ip_0=10.196.176.81
--prop:nsx_netmask_0=255.255.252.0
--prop:nsx_gateway_0=10.196.179.253
--prop:nsx_dns1_0=10.142.7.1
--prop:nsx_domain_0=eng.yourcompany.com
--prop:nsx_ntp_0=10.128.243.14
--prop:nsx_isSSHEnabled=True
--prop:"nsx_passwd_0=<password>"
--prop:"nsx_cli_passwd_0=<password-cli>"
--prop:"nsx_cli_audit_passwd_0=<password-cli-audit>"
--prop:nsx_hostname=<hostname>
--prop:mgrhostname01="nsx-manager-02@yourcompany.com"
--prop:nsx_allowSSHRootLogin=True
--prop:nsx_role="NSX Manager"
--X:logFile=/root/ovftool/<ovf-folder>.log
--X:logLevel=trivia
--ipProtocol=IPv4
--ipAllocationPolicy="fixedPolicy" <nsx-unified-applicance>.ova
```

```
'vi://admin@vsphere.local:password@vc-or-ip.eng.yourcompany.com/<datacenter>/host/
Install/10.196.6.78/'
```

The result should look something like this:

```
Opening OVA source: nsx-<component>.ova
The manifest validates
Source is signed and the certificate validates
Opening VI target: vi://administrator@vsphere.local@<esxi-IP-address:port>/
Deploying to VI: vi://administrator@vsphere.local@<esxi-IP-address:port>/
Transfer Completed
Powering on VM: NSX Manager
Task Completed
Completed successfully
```

2. You can also run the OVF tool in Probe mode to view contents of a source. OVA and OVF packages can be probed among a list of other supported source types. You can use the information returned by the Probe mode to configure deployments.

   ```
   $> \ovftool --allowExtraConfig <OVA path or URL>
   ```

3. For optimal performance, reserve memory for the node.

   Set the reservation to ensure that NSX Manager has sufficient memory to run efficiently. See Planning and Preparation.

4. From the vSphere Client, verify that the VM is powered on. Open the VM console to track the boot process of the node.

5. After the VM node boots a second time, log in to the CLI as admin and run the `get interface eth0` command to verify that the IP address was applied as expected.

6. Enter the `get services` command after waiting for about 5 minutes to verify that all default services are running.

   The following services are not required by default and do not start automatically.

   - `liagent`
   - `migration-coordinator`: This service is used only when running migration coordinator. See the *NSX Migration Guide* before starting this service.
   - `snmp`: For information on starting SNMP see *Simple Network Management Protocol* in the *NSX Administration Guide*.
   - `nsx-message-bus`: This service is not used in NSX 3.0 and later releases.

7. After deployment, verify that the NSX Manager UI comes up by accessing the following URL, `https://nsx-manager-ip` or `https://nsx-manager-fqdn`.

8. Verify that your NSX Manager,or Global Manager node has the required connectivity.

   Perform the following tasks:

   - Ping your node from another machine.
   - The node can ping its default gateway.
   - The node can ping the hypervisor hosts that are in the same network using the management interface.
   - The node can ping its DNS server and its NTP Server or FQDN.
   - If you enabled SSH, make sure that you can SSH to your node.

   If connectivity is not established, make sure that the network adapter of the virtual appliance is in the proper network or VLAN.

- Log in to the NSX Manager from a supported web browser. See Log In to the Newly Installed NSX Manager.

  Note:
  NSX Manager nodes that are removed from the cluster should be powered off or deleted. Do not reuse the same NSX Manager again in your environment.

- If deploying second and third NSX Manager nodes as OVA/OVF, join the manager nodes to first deployed manager node to create NSX Manager Cluster. see Form an NSX Manager Cluster Using the CLI.

Form an NSX Manager Cluster Using the CLI

Forming an NSX Manager or Global Manager cluster provides high availability and reliability. If all or atleast one of the three NSX Manager appliances is deployed as an OVA/OVF using vCenter, use the `join` command to join the NSX Manager nodes and create a cluster.

- To create an NSX Manager cluster, deploy three NSX Manager nodes from the OVF tool CLI or vCenter UI.
- To create a Global Manager cluster, deploy three nodes to create the cluster. However, if your Global Manager has NSX 3.0.0 installed, deploy only one node, and do not form a cluster. See Install the Active and Standby Global Manager.

1. Open an SSH or console session to the first deployed NSX Manager or Global Manager node and log in with the administrator credentials.
2. On the first deployed node, run the following commands.
   a) Run the `get certificate api thumbprint` command.
      The command output is a string that is unique to this node.
   b) Run the `get cluster config` command to get the cluster ID of the first deployed node.

```
mgr-first> get cluster config
Cluster Id: 7b50abb9-0402-4ed5-afec-363587c3c705
Cluster Configuration Version: 0
Number of nodes in the cluster: 1

...
```

3. Open an SSH or console session to the new node and log in with the administrator credentials.
4. On the new node that is joining the cluster, run the `join` command.

   Provide the following information about the first deployed node in the `join` command:

   - IP address
   - Cluster ID
   - User name
   - Password
   - Certificate thumbprint

```
mgr-new> join <Manager-IP> cluster-id <cluster-id> username <Manager-username> password <Manager-password>
 thumbprint <Manager-thumbprint>
```

   The joining and cluster stabilizing process might take from 10 to 15 minutes. Run `get cluster status` to view the status. Verify that the status for every cluster service group is `UP` before making any other cluster changes.
5. Add the third node to the cluster.

   Repeat step 4 on the third node.
6. Verify the cluster status on the web interface.
   - On NSX Manager, log in to the NSX Manager web interface and select **System** > **Appliances**.
   - On Global Manager, log in to the Global Manager web interface and select **System** > **Global Manager Appliances**.

Create a transport zone. See Create Transport Zones.

Verify NSX Manager Clustering

Verify whether individual nodes are listed as members of the NSX Manager cluster.

Log in to any one of the nodes and perform one of these steps to verify whether all nodes are listed as members of the NSX Manager cluster.

1. To verify NSX Manager cluster status from CLI, perform these steps:
   a) Log in as an admin to any one of the NSX Manager nodes.
   b) run 'get cluster status' to view the correct number of manager nodes are listed as members of the cluster and cluster status is STABLE.
2. To verify NSX Manager cluster status from UI, perform these steps:
   a) Go to **System ? Appliances**. Verify the Cluster status is `STABLE` and a correct number of nodes are shown to be part of the cluster.
3. To verify NSX Manager cluster status from API, run these commands:
   - `GET /api/v1/cluster/status`
   - `GET api/v1/cluster/nodes or GET api/vi/cluster/<node-id>`

Configure an Appliance to Display the GRUB Menu at Boot Time

You must configure an NSX appliance to display the GRUB menu at boot time if you want to reset the root password of the appliance.

> **Important:**
> If the configuration is not performed after deploying the appliance and you forget the root password, resetting it is not possible.

1. Log in to the VM as root.
2. In the `/etc/default/grub` file, set the GRUB_TIMEOUT_STYLE to `menu` or `countdown`.
   - If this option set to `menu`, then GRUB will display the menu and then wait for the timeout set by `GRUB_TIMEOUT` to expire before booting the default entry. Pressing a key interrupts the timeout.
   - If this option is set to `countdown`, then before displaying the menu, GRUB will wait for the timeout set by `GRUB_TIMEOUT` to expire. If ESC or F4 are pressed, or SHIFT is held down during that time, it will display the menu and wait for input. It will show a one-line indication of the remaining time.
3. In the `/etc/default/grub` file, change the value for the parameter `GRUB_TIMEOUT`.
   ```
   GRUB_TIMEOUT=4
   ```
4. Optional: Generate a new password by running the following command:
   ```
   grub-mkpasswd-pbkdf2
   ```
5. Optional: In the `/etc/grub.d/40_custom` file, replace the existing GRUB password.
6. Update the GRUB configuration.
   ```
   update-grub
   ```

Configure GRUB Menu Using CLI or API

You must configure an NSX appliance to display the GRUB menu at boot time if you want to reset the root password of the appliance.

You can use CLI or API commands to set the GRUB timeout value and password. You can follow these commands post deployment of NSX.

> **Important:**
> If the configuration is not performed after deploying the appliance and you forget the root password, resetting it is not possible.

1. Using CLI to set GRUB menu:
   a) Log in to the NSX command line interface.
   b) Run `set grub menu timeout <value>`.

      Where `<value>` is time in seconds. The default timeout value is 4.

c) Run `set grub user root password <newpassword>`.

OR

d) Run `set grub user root password`

```
Enter password:<newpassword>
Confirm password:<newpassword>
```

2. Using API to set GRUB menu:

a) Use the GET API to retrieve GRUB menu values.

```
GET https://<nsx-mgr>/api/v1/node/grub

Example Response:
{
  "timeout": 4,
  "users": [
    {
      "username": "root"
    }
  ]
}
```

b) Set the GRUB timeout value.

```
PUT https://<nsx-mgr>/api/v1/node/grub { "timeout": 4 }
Example Response:
{
  "timeout": 4
}
```

c) Set the GRUB menu password.

```
PUT https://<nsx-mgr>/api/v1/node/grub/root { "password": "Str0ng_Pwd!Wins$" }
Example Response:
{
  "username": "root"
}
```

3. Get GRUB timeout value.

```
get grub menu timeout
GRUB Menu Timeout = 4
```

Configure a Virtual IP Address for a Cluster

To provide fault tolerance and high availability to NSX Manager nodes, assign a virtual IP address (VIP) to the NSX cluster.

NSX Manager nodes of a cluster become part of an HTTPS group to service API and UI requests. The leader node of the cluster assumes ownership of the set VIP of the cluster to service any API and UI request. Any API and UI request coming in from clients is directed to the leader node.

> **Note:**
>
> When assigning Virtual IP, all the NSX Manager VMs in the cluster must be configured in the same subnet. But if external load balancer is used to configure cluster VIP, then NSX Managers and VIP can belong to a different subnet.

If the leader node that owns VIP becomes unavailable, NSX elects a new leader. The new leader owns the VIP. It sends out a gratuitous ARP packet advertising the new VIP to MAC address mapping. After a new leader node is elected, new API and UI requests are sent to the new leader node.

Failover of VIP to a new leader node of the cluster might take a few minutes to become functional. If the VIP fails over to a new leader node because the previous leader node became unavailable, reauthenticate the NSX Manager credentials so that API requests are directed to the new leader node.

> **Note:**
> VIP is not designed to serve as a load-balancer and you cannot use it if you enable the vIDM **External Load Balancer Integration** from **System** > **Users** > **Configuration**. Do not set up a VIP if you want to use the External Load Balancer from vIDM. See Configure VMware Workspace ONE Access Integration in the *NSX Administration Guide* for more details.
>
> **Important:**
> If you reset the cluster VIP, then vIDM configurations that are using the VIP is cleared. You will need to reconfigure vIDM configuration with the new VIP.

1. From a browser, log in with admin privileges to an NSX Manager at https://<nsx-manager-ip-address> or https://<nsx-manager-fqdn>.
2. Go to **System** > **Appliances**.
3. In the Virtual IP field, click **Set Virtual IP**.
4. Enter the IPv4 and/or IPv6 address to use as VIP for the cluster.

   Ensure that VIP is part of the same subnet as the other management nodes. When you deploy a NSX Manager in a dual stack environment (IPv4 and IPv6), use a valid FQDN address and use the same FQDN address for both IPv4 and IPv6 addresses.
5. Click **Save**.
6. To verify the cluster status and the API leader of the HTTPS group, enter the NSX Manager CLI command `get cluster status verbose` in the NSX Manager console or over SSH.
   The following is an example output:

```
Group Type: HTTPS
Group Status: STABLE

Members:
 UUID                                   FQDN                  IP               STATUS
 cdb93642-ccba-fdf4-8819-90bf018cd727   nsx-manager       192.196.197.84        UP
 51a13642-929b-8dfc-3455-109e6cc2a7ae   nsx-manager       192.196.198.156       UP
 d0de3642-d03f-c909-9cca-312fd22e486b   nsx-manager       192.196.198.54        UP

Leaders:
 SERVICE                                LEADER                                    LEASE VERSION

 api                                    cdb93642-ccba-fdf4-8819-90bf018cd727            8
```

7. Verify that the VIP is working correctly.

   From a browser, log in to the NSX Manager using the virtual IP address assigned to the cluster at https://<VIP-address>.

Any API requests to NSX are redirected to the virtual IP address of the cluster, which is owned by the leader node. The leader node then routes the request forward to the other components of the appliance.

> **Note:**
> If you deploy NSX Manager in dual stack mode (IPv4, IPv6) and/or if you plan to configure NSX Manager deployment with CA-signed certificates, associate the VIP IP address with DNS name. Also, configure reverse and forward proxy for the VIP IP address on the DNS server. Then access VIP from https://<VIP-dns-name>.

Verify Appliance Proxy Hub on all NSX Manager Nodes are Connected

Appliance Proxy Hub (APH) acts as a communication channel between NSX Manager and a transport node. It runs as a service on NSX Manager and provides secure connection between a transport node and NSX Manager.

Before you set a virtual IP (VIP) address for the cluster nodes, ensure the APH running on each NSX Manager cluster nodes are connected to each other. If APH on all cluster nodes are not connected, you might face issues when configuring the VIP for the cluster.

1. To verify APH on all the cluster nodes are connected, go to one of the NSX Manager nodes and run the `GET /api/v1/messaging/cluster-connection/status` call.

```
{
    "results": [
      {
        "address": "ssl-tcp://192.161.1.52:38522",
        "conn_status": "Connected",
        "node_id": "e85ceb93-df30-43fa-84d7-32d88f68a2ba",
        "node_type": "APPLIANCE_PROXY_HUB"
      },
      {
        "address": "ssl://192.161.1.51:1234",
        "conn_status": "Connected",
        "node_id": "46cc9a95-6194-4d84-94bd-0744fb46e225",
        "node_type": "APPLIANCE_PROXY_HUB"
      }
    ]
```

2. Repeat Step 1 on the remaining cluster nodes.
3. Wait for 60 seconds before you set the cluster VIP for the cluster.

When APH on all cluster nodes are connected to each other, you can proceed to configure VIP for the cluster.
Set up the cluster VIP address.

Troubleshooting Issues

This chapter provides information about troubleshooting issues related to NSX installation and configuration.

Troubleshooting Installation to check for Basic Infrastructure Services

This section provides information about troubleshooting installation issues.

Basic Infrastructure Services

The following services must be running on the appliances and hypervisors, also on vCenter if it is used as a compute manager.

- NTP
- DNS

Make sure that firewall is not blocking traffic between NSX components and hypervisors. Make sure that the required ports are open between the components.

To flush the DNS cache on the NSX Manager, SSH as root to the manager and run the following command:

```
root@nsx-mgr-01:~# /etc/init.d/resolvconf restart
[ ok ] Restarting resolvconf (via systemctl): resolvconf.service.
```

You can then check the DNS configuration file.

```
root@nsx-mgr-01:~# cat /etc/resolv.conf
# Dynamic resolv.conf(5) file for glibc resolver(3) generated by resolvconf(8)
#     DO NOT EDIT THIS FILE BY HAND -- YOUR CHANGES WILL BE OVERWRITTEN
nameserver 192.168.253.1
search mgt.sg.lab
```

Log in as root user and run `su admin` to launch `nsxcli` on NSX manager. As an admin user, `nsxcli` is default prompt.

Check DNS servers using following `nsxcli` command:

```
get name-servers
```

```
198.10.10.1
198.10.10.2
198.10.10.3
```

Checking Communication from Host to Controller and Manager

On an ESX host using NSX CLI commands:

```
esxi-01.corp.local> get managers
- 192.168.110.19   Connected

esxi-01.corp.local> get controllers
 Controller IP    Port     SSL         Status       Is Physical Master   Session State  Controller FQDN
 192.168.110.16   1235   enabled     connected             true               up               NA
```

On an ESX host using host CLI commands:

```
[root@esxi-01:~] esxcli network ip  connection list | grep 1235
tcp         0       0  192.168.110.53:42271                      192.168.110.16:1235   ESTABLISHED     67702
  newreno  nsx-proxy
[root@esxi-01:~]
[root@esxi-01:~] esxcli network ip  connection list | grep 5671
tcp         0       0  192.168.110.253:11721             192.168.110.19:5671   ESTABLISHED   2103688  newreno
  mpa
tcp         0       0  192.168.110.253:30977             192.168.110.19:5671   ESTABLISHED   2103688  newreno
  mpa
```

Host Registration Failure

If NSX uses the wrong IP address, host registration will fail. This can happen when a host has multiple IP addresses. Trying to delete the transport node leaves it in the Orphaned state. To resolve the issue:

- Go to **Fabric > Nodes > Hosts**, edit the host and remove all IP addresses except the management one.
- Click on the errors and select **Resolve**.

Configuration Error when Deploying an Edge VM

After deploying an Edge VM, NSX Manager shows the VM's status as **configuration error**. The manager log has a message similar to the following:

```
nsx-manager NSX - FABRIC [nsx@6876 comp="nsx-manager" errorCode="MP16027" subcomp="manager"] Edge
 758ad396-0754-11e8-877e-005056abf715 is not ready for configuration error occurred, error detail is NSX Edge
 configuration has failed. The host does not support required cpu features: ['aes'].
```

Restarting the edge datapath service and then the VM should resolve the issue.

Force Removing a Transport Node

You can remove a transport node that is stuck in the Orphaned state by making the following API call:

```
DELETE https://<NSX Manager>/api/v1/transport-nodes/<TN ID>?force=true
```

NSX Manager will not do any validations as to whether you have any active VMs running on the host. You are responsible for deleting the N-VDS and VIBs. If you have the node added through Compute Manager, delete the Compute Manager first and then delete the node. The transport node will be deleted as well.

Troubleshoot OVA Deployment and Appliance Bringup

Troubleshooting OVA Failures

Note:
During deployment, if you entered incorrect configuration details, delete the appliance and redeploy with correct configuration.

- Verify that the datastore chosen for deployment is mounted on all the hosts that are members of a cluster. Redeploy and choose ESX host instead of vCenter to bypass vCenter cluster related checks.
- If proxy enabled on vCenter, edit file `/etc/sysconfig/proxy` and add line `.*.domainname` to bypass proxy for ESX hosts. See Knowledge Base article 321922: Unable to deploy OVF using vSphere Client in vCenter when an HTTPS Proxy is configured.
- If the deployment of the appliance through the OVF tool results in the error `ovf descriptor not found`, view the file contents in terminal `cat -A <filepath/filename>` and remove hidden formatting characters. Then try again.

Troubleshooting Issues Related to Bringing Up the Appliance

SSH to NSX Manager CLI as admin and run following commands to troubleshoot.

- Run `get configuration` and verify `hostname/name-server/search-domain/ntp` settings are correct.
- Run `get services` and verify all required services are running (other than `nsx-message-bus`, `snmp`, `migration-coordinator`). If these services are not running, try restarting the service by running `restart service <service-name>`.
- Run `get cluster status` and verify all manager cluster components are up. If any component is down, try restarting the service associated to the component by running `restart service <associated-component-service-name>`.
- Run `get core-dumps` to verify no core dumps generated in `/var/log/core` or `/image/core`. If you find any core dumps, contact VMware Support.
- Run `get filesystem-stats` to verify that no disk partition is full, especially those partitions that are consumed by NSX.
- Alternatively, you can run API commands to know the node and service status.

  ```
  GET api/v1/node/status
  GET api/v1/node/services
  ```

Troubleshoot NSX Manager Cluster

You must configure a three-node NSX Manager cluster and only one of the nodes can fail at any given time for the cluster to self-heal.

For an NSX Manager cluster to self-recover from a node failure, the majority of nodes must not have failed (the number of active nodes must be greater than the failed nodes) otherwise the entire cluster becomes unavailable. It means that all write operations are blocked, all clustering related API/CLIs fail. However, local API/CLI commands continue to work.

NSX Manager logs are written into `/var/log/syslog`, directory.

Admin CLI

To activate Admin CLI, log in as an admin to a NSX Manager. But if you login as root, you can run singleton admin CLI directly from the root shell using the `su admin -c <cmd-to-run>` command. You can switch on interactive mode of admin CLI using the `su admin` command and then run admin commands.

Manually deployed NSX Manager failed to join the NSX Manager Cluster

A NSX Manager that is manually deployed in vCenter using OVF or OVA file failed to join the NSX Manager cluster.

You used an incorrect thumbprint value to join the new NSX Manager node to the existing NSX Manager cluster. If NSX VIP is configured, then use the cluster thumbprint command. If external VIP or no VIP is configured, use the API thumbprint command. Try registering the NSX Manager node with the cluster using correct thumbprint.

You can manually deploy NSX Manager by running one of the following commands.

- Run the `get certificate cluster thumbprint` command and then run `Join management-plane <cluster-vip> username <manager-username> password <manager-pwd> thumbprint <cluster-thumbprint>` command. The procedure to join the management node with the cluster is complete. Alternatively, you can use the NSX Manager thumbprint, as shown in the following steps to join the NSX Manager with the cluster.
- Or run the `get certificate api thumbprint` command and then run the `Join management-plane <manager-ip> username <manager-username> password <manager-pwd> thumbprint <manager-thumbprint>` command.

NSX Manager Cluster status Degraded As Datastore-related Components Are Down

NSX Manager status is degraded due to DATASTORE component and/or CORFU_NONCONFIG component down on one or more manager nodes member of NSX Cluster.

1. SSH to NSX Manager CLI terminal as an admin.
2. To identify the manager node with components down, run `get cluster status`.
3. Verify underlying datastore is available with recommended disk access latency. To get and fix the disk access latency numbers, see Knowledge Base article 316654: Storage latency causes NSX Manager cluster instability.

   Note:
   Datastore outage can cause NSX Manager appliance VMs to go into read-only mode. Linux does not provide a usable utility to recover from this error. Follow the KB, Recover NSX Manager upon storage outage, if appliance reboot does not fix the issue. Verify backend datastore has fully recovered before rebooting the affected NSX Manager VM.
4. To verify no disk partition consumed by NSX is full or close-to-full, run ‘get filesystem-stats'.

   Note:
   Datastore specific logs can be found at `var/log/corfu/corfu.9000.log` and `/var/log/corfu/tanuki.log`.
5. Clean up disk space and / or reboot all NSX Manager nodes (after storage issues are resolved) to remove the read-only mode.
6. If components continue to be down, contact VMware Support.

   Important:
   With the datastore down, the NSX Manager continues to participate in the cluster if its network connectivity is up. This state might result in the management plane and control plane to become unavailable. If datastore issue cannot be resolved, replace the impacted NSX Manager by powering off the impacted NSX Manager (as long as cluster quorum is maintained by having majority number of nodes up).
7. Remove NSX Manager node from existing NSX Manager Cluster. On the NSX Manager, run `detach node <node-id>`.
8. Verify the problematic node is not a member of the cluster anymore by running the `get cluster status`.
9. Deploy a new NSX Manager node and join it to the existing Manager Cluster.

Manager and HTTPS Services Are Down Frequently Due to Incorrect NAT Configuration

If you incorrectly configured NAT, the Manager and HTTPS service might go down frequently.

1. Do not use 0.0.0.0/0, x.0.0.0/8, x.x.0.0/16 subnet ranges in the NAT rules for SNAT Translated IP section (SNAT rule) and for Destination Network section (DNAT rule).
2. Fix the incorrect NAT configuration.
3. Configure NAT rule where IP address range and count does not exceed 256 for SNAT Translated IP section (SNAT rule) and for Destination Network section (DNAT rule).

NSX Manager is Slow To Load And Tasks Fail

NSX Manager is slow to load and tasks fail with message `server is overloaded` or `too many requests`.

This issue occurs upon hitting the nsx manager incoming API rate limit. The solution is to either limit the number of incoming API requests to NSX Manager or modify the default values for Client API rate limit or Client API concurrency limit or Global API concurrently limit from API or CLI.

1. To view current default values, run `get cluster api-service` as admin or `GET API /api/v1/cluster/api-service`.
2. To set a new API rate value, run `set cluster api-service client-api-rate-limit` or `PUT /api/v1/cluster/api-service` with updated values.
3. To view logs of incoming API requests, go to `/var/log/proxy` directory, `reverse-proxy.log`.

   Note:
   Try to avoid configuring the API rate limits. Instead, design your API client to gracefully deal with situations where limits are exceeded.

NSX Manager UI is not loading even when NSX Clustering is up

NSX Manager is slow to load and tasks fail with message `server is overloaded` or `too many requests`.

1. SSH to one of the NSX Manager that is a member of the cluster.
2. Run admin CLI `get cluster status`.
3. If cluster status is stable, run `get services` or `get service <service-name>` and verify that the following services are running: `http`, `manager`, `search`, `ui-service`.
4. To restart or start the stopped service, run `restart | start service <service-name>`.

NSX Manager cluster is DOWN or UNAVAILABLE if all nodes part of the NSX Manager cluster is down or majority nodes are down

NSX Manager is down or unavailable if majority of the nodes in the cluster are down.

NSX Manager UI will fail to load with following error `Some appliance components are not functioning properly. Component health: POLICY:UNKNOWN, MANAGER:UNKNOWN, SEARCH:UNKNOWN, NO` and clustering related commands will fail using the CLI and API.

1. SSH to each of the affected NSX Manager nodes and run following commands:
   a) Run `get file-system-stats` and verify /config and /image is not 100% full.
   b) Run `get core-dumps` to verify no cores have gotten generated in NSX Manager.
   c) Verify there was no datastore outage. See NSX Manager Cluster status Degraded As Datastore-related Components Are Down.
   d) Check logs for out-of-memory errors. See `/var/log/proton/proton-tomcat-wrapper.log`
2. To restore clustering and UI, any two nodes in a three node cluster must be up. If you are not able to bring any failed node back up, but if there is a healthy node available, then do one of the following steps to restore clustering:
   - Deploy a new manager node (as 4th member node), join the existing cluster and then detach one of the failed nodes using CLI cmd `detach node <node-uuid>` or API `POST /api/v1/cluster/<node-uuid>?action=remove_node`. The commands should be executed from one of the healthy nodes. Alternatively, you can follow the next bulleted point to deactivate the cluster.

- (Optional) Run the `deactivate cluster` command on active node such that you end up with single node cluster. Now continue to add the new additional nodes to make a 3-member NSX Manager cluster.

  Note:
  NSX Manager nodes that are removed from the cluster should be powered off and deleted.

Troubelshoot NSX Appliance or NSX Clustering issues Using APIs

Use APIs to troubleshoot NSX appliance and clustering issues.

Verify Cluster Status

```
GET https://<nsx-manager>/api/v1/cluster/status
```

```
GET https://<nsx-manager>/api/v1/cluster/<node-id>/status
```

```
GET/api/v1/reverse-proxy/node/health
```

Verify CPU Usage

```
GET /api/v1/systemhealth/appliances/process/status
```

```
GET https://<nsx-manager-ip>/api/vi/systemhealth/appliances/<appliance-id>/process/status
```

Verify Network Latency

```
GET https://<nsx-manager>/api/v1/systemhealth/appliances/latency/status
GET https://<nsx-manager-ip>/api/v1/systemhealth/appliances/<appliance-id>/latency/status
```

Troubleshooting Host Transport Nodes

Host Fails to Install as Transport Node

When many hosts are are prepared simultaneously (in non vLCM way) some hosts may fail in the internal step of downloading nsx lcp bundle to the host. Error seen is "Failed to download NSX components on host.

1. Check the host connectivity and ensure that the "/tmp" folder has enough space.
2. If it is fine then check if there are many hosts getting prepared at this time. Wait for host prep of other hosts to complete.
3. Follow the usual error resolution steps from the UI or API to re trigger the operation on the failed host.

Accessing the NSX CLI Terminal

After logging into the root shell of an ESX host, you can do one of the following:

- Use a singleton NSX CLI directly from the root shell by running the `nsxcli -c <cmd-to-run>` command.
- Go into interactive mode by running the `nsxcli` command.
  The NSX Syslog directory on the ESX host is `/var/log/nsx-syslog`.

Transport Node Installation Failure Due To Pending Reboot

Transport Node install task fails due to pending reboot.

This issue occurs if host is pending reboot from previous installation or upgrade of vibs on the host.

Run following CLI on the host as root to verify.

```
vim-cmd hostsvc/hostsummary|grep -i reboot requireSecureBoot = <unset>, rebootRequired = true
```

If *`rebootRequired`* is set to `true`, reboot the host and then try installation of NSX modules again on the host.

Unable to Reach Transport Node Due to Incorrect Credentials As Host is in Orphaned State

- Host is not reachable. Cannot complete login due to an incorrect username or password.
- The Transport Node Apply Task fails with error `Node already exists`.
- Host is in orphaned state.

This issue occurs due to a race condition under heavy traffic load. Run the API (deprecated) `GET /api/v1/transport-nodes/<TN-UUID>/status` or `GET api/v1/infra/sites/<site-id>/enforcement-points/<enforcementpoint-id>/host-transport-nodes/<host-transport-node-id>/state | status`, where default values for *`enforcementpoint-id`* and *`site-id`* is `default` to review the transport node shows transport nodes status unknown and node deployment status shows as `Failed`.

All these cases occur for host TN that did not get cleaned correctly upon initiation of NSX removal task therefore is still registered with NSX Manager.

In this case, the GET transport node api and GET transport node status api will fail but GET transport node state api works and will show failure message `Failed to uninstall the software on host....`

To fix the existence of stale entry, you must forcefully remove NSX from the host and also run the following api to delete stale host entries in the setup.

a) (NSX Manager UI) On **Hosts** page, select the **Force Delete** option and click **Remove NSX**.
b) (API) To forcefully delete NSX, run the API, `https://{{MPIP}}/api/v1/transport-nodes/<Transport-Node-UUID>?force=true&unprepare_host=false`.
c) (API) To remove stale entries, run the API, `https://{{nsx-mgr-ip}}/api/v1/transport-nodes?action=clean_stale_entries`.

Transport node profile Fails to Prepare Transport Nodes Due to Stale Objects

Transport node profile (TNP) fails to prepare transport nodes due to stale objects in TNP.

NSX displays the following error when TNP fails to prepare hosts into transport nodes: `The requested object <uuid> could not be found. Object identifiers are case sensitive. Please make the appropriate changes and reapply the configuration.`

This issue can occur if any configuration specified in the TNP does not exist in NSX Manager or vCenter.

1. Search the UUID in NSX Manager to find out the object being referred to in the error.
2. Edit the Transport Node Profile to remove the referenced stale object (could be TNP itself, IP Pool, Profile etc) and then retrigger host configuration.

Transport Node Creation is in Partial Success State Due to Logical Switch or Segment Full-sync Realization Error

Transport Node has `partial success` status with error `LogicalSwitch full-sync realization query skipped`.

This issue can occur if uplink profile associated with host has teaming defined with no active uplinks or if host switch uplinks is down or if failure in creation of VTEP.

Run transport-node state API to view more info on failure details using one of the following API command:

- (deprecated) `GET api/v1/transport-nodes/<uuid>/state`
- `GET api/v1/infra/sites/<site-id>/enforcement-points/<enforcementpoint-id>/host-transport-nodes/<host-transport-node-id>/state`, where default values for enforcementpoint-id and site-id is `default`.

1. Edit the uplink profile associated with the transport node and ensure active uplinks are selected for each teaming profile defined.
2. Verify PNIC link status is up by running CLI on the host: `esxcli network nic | esxcli network nic get -n <vmnic-name>`.
3. If uplink profile is active-active, verify correct number of VTEPS got created (rather than just one).

   Run `net-vdl2 -l |more` to verify VTEP count is correct, is assigned a valid IP Address, Gateway and state of each VTEP interface is `UP`.

Transport Node status is degraded when its interface is down

Transport Node status is degraded when its interface is down.

The NSX Manager declares a transport node to be in a degraded state if any of its interface that is used by host switch is `Down`.

Transport Node status is Disconnected or Unknown

ESX host prepared as a Transport Node status goes into `Unknown` or `Disconnected` state due to lost connection to NSX Manager. NSX displays the following error: `Heart beating between NSX management node and host <uuid> is down`.

Host infrastructure services being down due to ESX disks being full or memory leak may result in this condition. If ESX.

If ESX disks are full or if there is a memory leak, it can cause certain processes to crash and cause the transport node to go into `Disconnected` state. When you run admin cli `get managers`, NSX might return active manager node if the crash occurred post successful manager registration. When you run admin cli `get controllers`, NSX gives error `Failed to get controller list`.

1. Run admin cli `get core-dumps` to see if any cores got generated (in `/var/core or /image/core`) due to service crash.
2. If core-dump happens, run cmd `esxtop` to see which NSX process is consuming too much memory and `df -h` to verify disk partitions used by NSX is not full or close to full.
3. Run `/etc/init.d/nsx-proxy | nsx-nestdb status` to get status of infrastructure services on the host.
4. Clean up the disk space, then start any stopped infrastructure services on the host by issuing command `/etc/init.d/<service-name> start` (as a temporary workaround).
5. Open a support case with VMware if you see any cores.

Transport Node is Down as Agent Service is Down

The agent service is `Down`.

1. To view the host agent status in UI, go to **Host** > **Transport Node** > **View Details** > **Monitor** tab.
2. To view the status of three host agent services (`nsx-cfgagent`, `nsx-opsagent`, `nsx-nestdb`) in CLI, run `/etc/init.d/<service-name> status`.
3. To view the status of agents in API, call these APIs:

   (deprecated) `GET api/v1/transport-nodes/<uuid>/status`

   `GET api/v1/infra/sites/<site-id>/enforcement-points/<enforcementpoint-id>/host-transport-nodes/<host-transport-node-id>/status`

   The default value for `enforcementpoint-id` and `site-id` is `default`.
4. Restart service that is down by running the following CLI command: `/etc/init.d/<service-name> start`.

Transport Node Connectivity to Controller is Down

The issue is seen when a transport node's connectivity to NSX Manager is `Up` but its controller is `Down`. When you run `get managers`, NSX returns active manager node, while `get controllers` does not return any active controller for this transport node, which is in Connected state and its session state is also `Up`.

1. Verify transport node is not in NSX Maintenance Mode via admin cli `get maintenance-mode`.
2. Call one of the following API:
   a) (Deprecated) `GET API/v1/transport-nodes/<uuid>/status`
   b) `GET api/v1/infra/sites/<site-id>/enforcement-points/<enforcementpoint-id>/host-transport-nodes/<host-transport-node-id>/status`, where default values for *<site-id>* and *<enforcementpoint-id>* is default.
3. Verify FQDN property (used by transport nodes to talk with NSX Manager or controller) is set by running API: `GET /api/v1/configs/management` and view value for `publish_fqdns`.
   If FQDN is set, verify the controller FQDN is reachable and FQDN value is being used to by the transport node to talk to controller by first running ICMP ping to controller FQDN followed by admin cli `get controllers` to verify controller FQDN value is getting populated correctly.
4. Verify host agent services are running by following host agent troubleshooting step mentioned above.
5. Verify `controller.xml` file exists and contains Host transport node as its member: `/etc/vmware/nsx/controller-info.xml`.
6. If host is in NSX Maintenance Mode, run admin cli `set maintenance-mode false` or API to take host out of NSX maintenance mode: `POST /api/v1/transport-nodes/<node-id>?action=exit_maintenance_mode`.
7. If FQDN set and ICMP ping works for controller FQDN then try unsetting and setting the FQDN property again by running API `PUT /api/v1/configs/management` with value for `publish_fqdns` as `false` and then run the API again with value `true`.
8. Start agent services on the host (if any stopped) by running the CLI command `etc/init.d/<service-name> start`.
9. If `controller.xml` file has incorrect data, restart `nsx-proxy` service on the host to trigger re-creation of file.

Unable to power on VMs or vMotion VMs on Transport Node

The Virtual Distributed L2 (VDL2) component is down. This component must be up for NSX to successfully complete VM operations that are attached to a segment on the transport node.

NSX displays the following error message: `Currently connected network interface" 'Network adapter 1' uses network 'VM_NETWORK:vdl2 down)`

1. SSH to host and run following command to verify status of vdl2 component, `net-dvs | grep "component.vdl2"`
   `com.vmware.common.opaqueDvs.status.component.vdl2 = down , propType = RUNTIME`
2. Run `net-vdl2 -l` to verify VTEP interface is assigned valid IP Address, Gateway and status of each interface is UP.
3. Run `esxcfg-vswitch -l` to verify minimum MTU of minimum 1600 bytes (minimum 1700 bytes is recommended) is set up on the VDS switch used by NSX and uplinks assigned to VTEP interfaces are UP.
4. To view host switch information, run one of the following transport node state API:
   a. (deprecated) `GET api/v1/transport-nodes/<uuid>/state`
   b. `GET api/v1/infra/sites/<site-id>/enforcement-points/<enforcementpoint-id>/host-transport-nodes/<host-transport-node-id>/state`, where default values for enforcementpoint-id and site-id is ‘default’ or `GET api/v1/transport-nodes/<uuid>/state` (deprecated).

1. Ensure that the configuration details entered in these fields are correct:
   - VTEP IP Pool
   - VTEP VLAN
   - VDS MTU
   - Status of assigned PNIC (must be Up)

2. If VTEP pool is configured using DHCP, verify that DHCP server is assigning valid IP addresses to the VTEP pool.

Transport Node Tunnels Down

ESX hosts that are prepared as transport nodes have their tunnel status as `Down`.

1. If all tunnels are down, verify existence of valid VTEP on the host by running the `net-vdl2 -l | more` command.
2. Verify these configuration details are correct: VTEP state is UP, VTEP count is correct, and a valid VTEP addresses is assigned.
3. To find VTEP BFD tunnel sessions that are down on the ESX transport node, run the CLI command `nsxdp-cli bfd sessions list | grep down`.
4. For sessions that are down, validate IP connectivity between the TEPs using ICMP ping.

   Note:
   The TEP interfaces on ESX are instantiated on the vxlan netstack and on the tunnel VRF for NSX Edge nodes. Therefore, initiate the ping from within the vxlan netstack on ESX and if you have more than one TEP, be sure to specify the source IP address or interface used for the ping.
5. Run `ping ++netstack=vxlan <remote address> -I vmk10 .`
6. If ping fails, verify the underlay connectivity of the network.
7. If ping is successful, verify Fabric MTU is configured correctly on the underlay network and within NSX. Use ICMP with the don't fragment bit to test proper delivery for large packets.
8. Run `ping ++netstack=vxlan <remote-vtep-ip=-address> -I vmk10 -d -s 1600 .`

Installation Fails Due to Insufficient Space in Bootbank on ESX Host

NSX installation might fail if there is insufficient space in the bootbank or in the alt-bootbank on an ESX host.

On the ESX host, you might see a similar log (`esxupdate.log`) message:

```
20**-**-**T13:37:50Z esxupdate: 5557508: BootBankInstaller.pyc:
ERROR: The pending transaction requires 245 MB free space,
however the maximum supported size is 239 MB.^@
```

Unused VIBs on the ESX host can be relatively large in size. These unused VIBs can result in insufficient space in the bootbank or in the alt-bootbank when installing the required VIBs.

- Uninstall the VIBs that are no longer required and free up additional disk space.

For more information on deleting the unused VIBs, see Knowledge Base article 324231: NSX VIBs fail to install/upgrade, due to insufficient space in bootbank on ESX host.

NSX Agent on ESX Transport Nodes Times Out Communicating with NSX Manager

In a large-scale environment with many transport nodes and VMs on ESX hosts, NSX agents, which run on ESX hosts, might time out when communicating with NSX Manager.

Some operations, such as when a VM vnic tries to attach to a logical switch, fail. The `/var/run/log/nsx-opsagent.log` has messages such as:

```
level="ERROR" errorCode="MPA41542"] [MP_AddVnicAttachment] RPC call [0e316296-13-14] to NSX management plane
 timout
2017-05-15T05:32:13Z nsxa: [nsx@6876 comp="nsx-esx" subcomp="NSXA[VifHandlerThread:-2282640]" tid="1000017079"
 level="ERROR" errorCode="MPA42003"] [DoMpVifAttachRpc] MP_AddVnicAttachment() failed: RPC call to NSX
 management plane timout
```

In a large-scale environment, some operations might take longer than usual and fail because the default timeout values are exceeded.

1. Increase the NSX agent timeout (seconds) value.
   a) On the ESX host, stop the NSX ops agent with the following command:

      On NSX 2.3 or later releases:

      ```
      /etc/init.d/nsx-opsagent stop
      ```

      On NSX 2.1 or previous releases:

      ```
      /etc/init.d/nsxa stop
      ```

   b) Edit the file `/etc/vmware/nsx-opsagent/nsxa.json` and change the `vifOperationTimeout` value from 25 seconds to, for example, 55 seconds.

      ```
      "mp" : {
          /* timeout for VIF operation */
          "vifOperationTimeout" : 25,
      ```

      Note:
      This timeout value must be less than the hostd timeout value that you set in step 2.

   c) Start the NSX ops agent with the following command:

      ```
      /etc/init.d/nsx-opsagent start
      ```

2. Increase the hostd timeout (seconds) value.
   a) On the ESX host, stop the hostd agent with the following command:

      ```
      /etc/init.d/hostd stop
      ```

   b) Edit the file `/etc/vmware/hostd/config.xml`. Under `<opaqueNetwork>`, uncomment the entry for `<taskTimeout>` and change the value from 30 seconds to, for example, 60 seconds.

      ```
      <opaqueNetwork>
          <!-- maximum message size allowed in opaque network manager IPC, in bytes. -->
          <!-- <maxMsgSize> 65536 </maxMsgSize> -->
          <!-- maximum wait time for opaque network response -->
          <!-- <taskTimeout> 30 </taskTimeout> -->
      ```

   c) Start the hostd agent with the following command:

      ```
      /etc/init.d/hostd start
      ```

Troubleshooting NSX Edge Nodes

Admin CLI is activated when you log in to an NSX Edge using admin credential. As a root user, you can run singleton admin CLI directly from the root shell `su admin -c <cmd-to-run>` or go into interactive mode of admin CLI `su admin` and then run admin commands.

Accessing the NSX Edge CLI Terminal

Admin CLI is activated if you login to NSX Edge as admin.

As a root user, you can:

- Use a singleton NSX Edge CLI directly from the root shell by running the `su admin -c <cmd-to-run>` command.
- Go into interactive mode by running the `su admin` command.

NSX Edge MPA Connectivity Down

The NSX Edge MPA connectivity Down due to infrastructure service crash.

NSX Edge node disks being full or memory leak can cause certain processes to crash and lead to this failure. Admin cli `get managers` may return active manager node (if crash occurred post successful manager registration) and admin cli `get controller` will give error `Failed to get controller list`.

1. Run admin cli `get diagnosis config` or `GET API /api/v1/transport-nodes/{transport-node-id}/node/diagnosis` to diagnose failures related to health of NSX Edge nodes that are caused when services go down.
2. Run admin cli `get cores-dumps` to see if any cores got generated (in /var/core or /image/core) due to service crash. If core dump is seen, run cmd `top -o %MEM` as root to see which nsx process is consuming too much memory and admin cli `get filesystem-stats` to verify if partitions used by nsx is not full or close to full.
3. Run root cli `/etc/init.d/nsx-proxy | nsx-nestdb status` to get status of infrastructure services running on the NSX Edge node.
4. Clean up the disk space, then start any stopped infrastructure services on the host by issuing command `/etc/init.d/<service-name> start` (as a temporary workaround). Open support case with VMware if any cores are seen.

NSX Edge Status DOWN or DEGRADED As BFD Tunnel(s) are Down

NSX Edge

NSX Edge status DOWN or DEGRADED due to BFD tunnel(s) to remote NSX Edge down.

Between two NSX Edge, one BFD session is run on management interface, and one or more BFD session on each VTEP interface. An NSX Edge considers its peer as unreachable only when all BFD sessions to that edges (management one and all VTEP ones) are down.

1. To get information about NSX Edge VTEP devices, run admin cli `get host-switch`.
2. To verify physical port status, run `get physical-port <vtep device>`. Then on edge-1> `get phy fp-eth0`.

```
Physical Port
ADMIN_STATUS : up <----------------- should be "up"
DRIVER : net_vmxnet3
DUPLEX : full
ID : 0
LINK : up <----------------- should be "up"
```

3. Run admin cli `get diagnosis topology` and `get edge-cluster status` to verify Edge is healthy with edge cluster High Availability State `Up`, edge node status `Up`, admin status `Up`. Then verify if VTEP State `Up` and status of BFD healthcheck sessions.

```
Interface          : nsx-edge-vtep
       Device              : fp-eth0
       Session             : 71.23.54.3:71.23.54.1
       Status              : Unreachable
       Interface           : nsx-edge-vtep.1
       Device              : fp-eth1
       Session             : 71.23.54.4:71.23.54.2
       Status              : Unreachable
```

   If status is unreachable, or Neighbor Signal Down, validate IP connectivity using ICMP Ping.

   For all other status, check the BFD error code explanation in the guide. See View Bidirectional Forwarding Detection Status.

4. Since TEP interfaces reside on the tunnel VRF for Edges therefore initiate ping from the tunnel VRF 0 on edges and if you have more than one TEP, specify the source IP address or interface used for the ping.
5. Run admin cli ‘get logical-routers' to get tunnel vrf followed by ping.

```
vrf 0
ping 71.23.47.8 source 71.23.46.1 repeat 3
```

6. Run admin cli `get neighbor` to check if ARP is getting resolved for the BFD session.
7. Run admin cli `get interface` to check status of interface with BFD tunnel down.
8. If any of the status is unreachable, verify underlay wiring is correct.
9. If ICMP ping is working yet VTEP status is unreachable, verify the VTEP IP addresses are not already in use.

NSX Edge Router High Availability Status is Down

NSX Edge

Edge Router High Availability Status is `Down`. NSX Edge status in UI may also show as `DOWN`.

- After you run the CLI `get edge-cluster status`, the Edge Node Status shows as `UP` but Routing Status shows as `Down`.

```
Edge Node Id                : 9e60f8c7-c0ac-42ef-8854-5466cd0cc7eb
Edge Node Status            : Up (Routing Down)
Admin State                 : Up
Service Status              :
Datapath Config Channel : Up
Datapath Status Channel : Up
Routing Status Channel : Up
Routing Status          : Down
```

- In the NSX Manager UI, navigate to **Networking** > **Tier-0 Logical Routers** > **Logical Router** > **Overview** > **High Availability State** shows as `Down`.
- In the Edge CLI, run `get logical-routers` ->`vrf (Tier0 SR)` ->`get high-availability status` of SR shows as `Down`.

This issue occurs if all BFP and/or BGP sessions for the router goes down.

1. To troubleshoot BFD sessions, see the previous described troubleshooting cases related to BFD sessions. see NSX Edge status DOWN or DEGRADED due to BFD tunnels between Edge and ESX down and NSX Edge Status DOWN or DEGRADED As BFD Tunnel(s) are Down.
2. To troubleshoot BGP sessions, follow these steps:
   a) Within Tier0 SR vrf (get logical-routers; vrf x), run admin cli, `get bgp neighbor summary`.
   b) If there is no connection established, ping bgp neighbor addresses and source addresses (T0 upinks/interfaces) from inside as well as outside SR vrf to verify the interfaces are correctly setup up as BGP peers on the TOR and BGP neighbors are up and accessible.
   c) If connection status is established, then run cli `get bgp neighbor <neighbor-ip> advertised-routes` followed by `get route <ip-address> | get route connected | get route bgp` to view if BGP routes are getting advertised.
   d) Run `get logical-routers` to view logical routers ID of Tier-0 Service Router followed by `get logical-router <logical-router-id> interfaces stats` to view if TX or RX drops are seen on the Service Router interfaces.
   e) Run admin cli `get diagnosis topology` to view status of edge uplink interfaces, bgp peers.

NSX Edge Node Status Down Due to PNIC Bond Status Down

NSX Edge node status is Down because the PNIC bond status is down.

This issue can occur if one or more Edge interfaces used by host-switch are down.

1. Verify if uplinks used by fast path interfaces are not down on the physical server hosting the uplinks.
2. If fast-path components are down, verify if dataplane service is running by running admin cli `get service dataplane`.
3. Verify status of Edge VTEP Device via admin cli `get host-switch` to find out vtep device name.
4. Run `get physical-port <vtep device>` .

```
edge-1> get phy fp-eth0
```

```
Physical Port
ADMIN_STATUS : up <----------------- should be "up"
DRIVER : net_vmxnet3
DUPLEX : full
ID : 0
LINK : up <----------------- should be "up"
```

5. If VTEP device `admin_status` or link is down, fix this infrastructure issue.
6. If dataplane service is stopped, start by running `start | restart service dataplane`.
7. After resolving the issue, run admin cli `get logical-routers` on the edge transport node to make sure edge is functional again.

NSX Edge Transport Node Connectivity to Controller is Down

NSX Edge Transport Node connectivity to controller is down.

This issue is seen when connectivity to manager is up but connectivity to controller is down. Admin cmd ‘get managers’ returns active manager node while cmd 'get controllers’ does not return any active controller for this transport node with status connected and/or session-state UP.

1. Verify transport node is not in NSX Maintenance Mode using admin cli `get maintenance-mode` or run the API, `GET api/v1/transport-nodes/<tn-uuid>| state | status`.
2. Verify if FQDN property (used by transport-nodes to talk with NSX Manager/Controller) is set by running API, `GET /api/v1/configs/management` and view value for `publish_FQDNS`.
3. If FQDN set, verify the controller FQDN is reachable and FQDN value is being used to by TN to talk to controller by first running ICMP ping to controller FQDN followed by admin cli `get controllers` to verify controller FQDN value is getting populated correctly.
4. Verify node agent services are running by following node agent troubleshooting step outlined before.
5. If edge transport node is in NSX Maintenance Mode, run admin cli `set maintenance-mode false` or API `POST /api/v1/transport-nodes/<node-id>?action=exit_maintenance_mode` to take node out of NSX Maintenance Mode
6. If FQDN is set and ICMP ping works for controller FQDN then try unsetting and setting the FQDN property again by running API `PUT /api/v1/configs/management` with value for `publish_fqdns` false followed by true.
7. Verify that agent services are running on the NSX Edge node by running `get edge diagnosis config`. If any service shows as failed, restart by running admin cli `start service <service-name>` or root cli `etc/init.d/<service-name> start`.

NSX Edge node status is Down As Controller Is Unavailable

NSX Edge node status is Down due to controller unavailability.

NSX Edge node status is Down with controller connectivity status ‘unavailable’ because agent service down.

The `get manager` and `get controller` commands will return correct values but `GET api/v1/transport-nodes/<tn-uuid>/status` shows agent-status is `Down`. The error is due to the service `nsx-opsagent` is not running.

1. Run admin cli `get status nsx-opsagent` to verify the service status.
2. Start or restart the service by running admin cli, `restart | start service nsx-opsagent`

Transport Node Failed as Named Teaming Defined with No Active Uplink

Transport Node `configuration has failed` due to named teaming defined with no active uplink.

This issue can occur if uplink profile associated with transport-node has named teaming defined with no active uplinks.

1. Make sure active uplinks are selected for any teaming policies defined.
2. Edit the uplink profile associated with the transport node by either removing the teaming with no active uplink or assigning a valid uplink to teaming policy defined.

NSX Edge Transport Node goes into NSX Maintenance Mode On HA Failover

NSX Edge Transport Node goes into NSX Maintenance Mode automatically upon HA failover.

NSX Edge Transport Nodes may go automatically into NSX Maintenance Mode if there are issues with its datapath or with the usage of heap memory.

To view edge node high availability status, status changes and reasons for it, run admin cli `get edge-cluster status` and `get edge-cluster history state`. If the Edge STATE would be DOWN, implying either datapath process is not running or physical link is down or vtep tunnels are down.

1. Run admin cli `get diagnois config` and `get service dataplane` to verify core services are up.
2. Run admin cli `get diagnosis topology` to view detailed edge config state.
3. Run admin cli `get host-switch` to get vtep-device name and physical-port-name.
4. Run admin cli `get physical-port <port-name>` to view status of host switch port followed by `get physical-port <interface-name> stats` and look for rx_misses (ingress buffer) or tx_drops (egress buffer) counter to determine occurrence of packet loss. Packet loss may be seen if edge is flooded with traffic rate higher rate higher than the datapath CPUs can process. Packets are first held in input/egress buffer and gets dropped if buffers are full. To check the current buffer size configuration, use the CLI `get dataplane | fing ring`.
5. If dataplane service is stopped, start by issuing cmd `start service dataplane` (as a temporary workaround).
6. If host switch port is down, start by issuing cmd `set physical-port fp-eth0 state up` (as a temporary workaround).
7. If packet loss is seen or issue with host switch status|state, file a ticket with VMware Support Desk.

   You can also try to change the rx/tx buffer configuration (to enhance edge interface traffic management capacity) using the CLI `set dataplane ring-size <rx/tx> <size>`. The supported buffer size range is `128-4096` bytes and dataplane service needs to be restarted in order to make the new configuration effective resulting in a downtime of about `60` seconds.

   a) For example, `set dataplane ring-size rx 2048`. Restart dataplane service for the change to take effect.
   b) `set dataplane ring-size tx 2048`. Restart dataplane service for the change to take effect.

```
restart service dataplane
get dataplane | find ring
Bfd_ring_size     : 512
Lacp_ring_size    : 512
Learning_ring_size : 512
Livetrace_ring_size: 512
Rx_ring_size      : 2048
Slowpath_ring_size : 512
Tx_ring_size      : 2048
```

VMotion of NSX Edge VM Fails Due to ESX Running Out of Resources

VMotion of NSX Edge VM may fail in vCenter due to ESX running out of resources from a shared buffer pool.

This issue can occur for an NSX Edge of XL VM form factor of 16 vCPU cores with error `reservation failed`.

To fix the issue, increase the P2M buffer slots for all the virtual machines on the host. For more information, see Knowledge Base article 318766: Configuring P2M Buffer size for virtual machines.

State Not Consistent of NSX BFD BGP or HA Functionality

State Not Consistent of NSX BFD BGP or HA Functionality

Pings are not consistent. BFD status changes occur. Edge HA failovers occur consistently.

1. Run admin cli `Get edge-cluster history state`.

   The CLI showcases history of edge cluster HA transition and reason. The only acceptable state is Active. Anything else means edge node is down and no service is able to function. For the purpose of identifying events leading to such HA state, focus on the timestamp and "Reason" line.

2. Run admin cli `get diagnosis topology` to view status of all attached interfaces and config.

NSX Edge Node Status Down and BFD Tunnel Data is Missing

This issue can occur if NSX Edge dataplane service goes down due to memory leak in certain NSX Edge processes. In this case, certain NSX Edge CLIs may fail with `error encountered`.

1. Run admin cli `get service dataplane` to view the status of service.
2. If status is down, run cli `start service dataplane`.
3. Once service has started, run the following CLIs to view the CPU and Memory consumption by NSX Edge dataplane module.
4. Get dataplane cpu stats from `/var/log/vmware/top-cpu.log` and `top-mem.log` to watch for values going above 60% usage.

NSX Edge status DOWN or DEGRADED due to BFD tunnels between Edge and ESX down

NSX Edge

1. Run admin cli `get bfd-sessions` to find VTEP BFD tunnel sessions that are down on the edge transport node. As a guard against VTEP misconfiguration, when edge loses all its BFD to hypervisor, it brings down itself.
2. For down sessions, validate IP connectivity between the TEPs using ICMP ping.

   Note:
   The TEP interfaces reside on the tunnel VRF for Edges and vxlan netstack for ESX. Therefore initiate ping from the tunnel VRF on edges and if you have more than one TEP, be sure to specify the source IP address or interface used for the ping.

3. Run admin cli ‘get logical-routers' to get tunnel vrf.

```
vrf 0
 ping 48.13.47.8 source 48.13.46.1 repeat 3
```

4. ESX: `ping ++netstack=vxlan <remote-vtep-ip=-address> -I vmk10 -d -s 1600`
5. As shown in the CLI, the ping should be initiated by specifying both the remote and local IP address relevant to the BFD session. While not required to test simple connectivity, specify the payload size that’s 100 bytes less than the TEP’s configured MTU, and the *`dfbit`* set to enable to prevent the underlay network from fragmenting the packet. Testing with bigger payloads will validate that your underlay network has been properly setup to support your NSX Geneve overlay configuration.
6. Validate if ARP is getting resolved for neighbor VTEP address.

```
edge-1(vrf)> get neighbor
Logical Router
UUID    : 736a80e3-23f6-5a2d-81d6-bbefb2786666
VRF     : 0
LR-ID   : 0
Name    :
Type    : TUNNEL
Neighbor
   Interface : 4d9091fe-b971-5d3c-9201-4cb9c7f455fe
   IP        : 202.1.1.2 <------------ peer TN VTEP IP
   MAC       : 00:50:56:a6:7d:9b <---- resolved
   State     : reach <---------- ARP reachable state
```

```
Timeout   : 37
```

7. Run `get interface` cmd followed by `get logical-router interface <uuid> status` to get VTEP interface status.

```
Interface    : ac80718b-72d3-5028-bb07-8f3c4ea2231a
 Ifuid        : 258
 Name         :
 Fwd-mode     : IPV4_AND_IPV6
 Internal name : uplink-258
 Mode         : lif
 Port-type    : uplink
 IP/Mask      : 71.23.46.1/24
 MAC          : 00:50:56:b8:2c:c4
 VLAN         : 2046
 Access-VLAN  : untagged
 LS port      : d31578e5-bc91-5466-97c1-8e4a6aa1b2e8
 Urpf-mode    : PORT_CHECK
 DAD-mode     : LOOSE
 RA-mode      : RA_INVALID
 Admin        : up
 Op_state     : up
 Enable-mcast : True
 MTU          : 8800
 arp_proxy
```

8. Run `get bfd-session stats` to look for RX drops and TX misses counter values.
9. If ICMP ping fails or ARP is not reachable, verify your underlay connectivity and peer host TEP interface address is valid. If large packet MTU ping fails, fix the NSX Fabric and/or underly infrastructure MTU to correct values.
10. Validate Edge TEP address is not in use by another transport node and validate Edge TEP VLAN and Host TEP VLAN are not using the same VLAN and uplink.

NSX Feature and Edition Information

The NSX Feature and Edition Guide is no longer being maintained for NSX 9.0 and beyond. For detailed information on feature entitlement, please see the following links.

VCF Networking features in VCF 9.0.x will be listed in the VCF Feature Comparison Datasheet.

NSX Feature and Edition Guide 4.2.x

https://techdocs.broadcom.com/us/en/vmware-cis/nsx/vmware-nsx/4-2/feature-and-edition-guide.html

NSX Feature and Edition Guide 4.1.2

https://techdocs.broadcom.com/us/en/vmware-cis/nsx/vmware-nsx/4-1-2/feature-and-edition-guide.html

NSX Feature and Edition Guide 4.1.1

https://techdocs.broadcom.com/us/en/vmware-cis/nsx/vmware-nsx/4-1-1/feature-and-edition-guide.html

NSX Feature and Edition Guide 4.1.0

https://techdocs.broadcom.com/us/en/vmware-cis/nsx/vmware-nsx/4-1/feature-and-edition-guide.html

NSX Feature and Edition Guide 4.0.0

https://techdocs.broadcom.com/us/en/vmware-cis/nsx/vmware-nsx/4-0/feature-and-edition-guide.html

Product offerings for VMware NSX-T Data Center 3.2.x (319112)

https://knowledge.broadcom.com/external/article?articleNumber=319112

Product offerings for VMware NSX-T Data Center 3.1.x (319118)

https://knowledge.broadcom.com/external/article?articleNumber=319118

Product offerings for VMware NSX-T Data Center 3.0.0 (315188)

https://knowledge.broadcom.com/external/article?articleNumber=315188

Product Guide: https://www.vmware.com/content/dam/digitalmarketing/vmware/en/pdf/product/vmware-product-guide.pdf

Related Resources

Table 1084: VMware NSX Related Resources

Resource List
NSX Events Catalog
Application IDs for NSX Landing page
App Dictionary
Configuration Maximums
Ports and Protocols
Product Interoperability Matrix
Operations Design Guide

Table 1085: VMware NSX API and CLI

Resource List
REST API Reference Guide
Global Manager REST API Guide
Autonomous Edge REST API Guide
CLI Reference Guide
Security Intelligence and Application Platform APIs
Malware Prevention API

Table 1086: VMware NSX Related Products

Product List
VMware NSX Application Platform
Security Intelligence
VMware vDefend Firewall
VMware vDefend Advanced Threat Prevention
VMware vDefend Network Detection and Response